Hironaka: canonical principalization, divisor sums and centre geometry

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Verification scope of the complete chapters

The continuation retains complete upstream statements and mathematical proofs. The 357 complete synchronized chapters retain all 1,465 original proofs. Their 4,317 distinct declaration references comprise 4,074 local bindings, 241 verified upstream-only bindings and 2 explicitly unresolved historical attachment names. The unresolved names are not counted as verified. Local bindings use `lean` metadata; verified upstream-only facts use `upstreamlean`, and unresolved historical names use `upstreamunbound`. The current published source revision is `09dc511c4a14836a0287748d1a63f9f467f677d3` in MingchenXia/archon-nakamaye, corresponding to the accepted native checkpoint `db4e4065403b4ab8b077fac41985825931e5cc02`. The per-checkpoint audit reports identify all earlier fixed source revisions and the complete proof, binding and namespace provenance. The full document also retains the earlier original extension chapters.

1 Prerequisites from the original API

These results use the embedded resolution theorems already proved in the library. The following pointers identify the earlier results referred to in the proofs below.

1.1 Pure-dimensional embedded resolution

See `Hironaka.exists_embeddedResolution_pure_of_isCompact` and its compact-ambient and relatively compact variants in `API` and `API2`.

1.2 Embedded resolution with an SNC boundary

See `Hironaka.exists_embeddedResolution_boundary_of_isCompact` and the corresponding variants in `API2`.

1.3 Embedded resolution in mixed dimensions

See `Hironaka.exists_embeddedResolution_analytic_of_isCompact` and its boundary form in `API3`.

2 Dimension parts and genuine global irreducible components

Let A be a closed analytic subset of a complex manifold M of dimension n. The global irreducible components are the closures of the connected components of the regular locus, as in [2], II.5.3. A set has pure geometric dimension k when every one of its regular points has dimension k. This characterization uses actual local manifold dimensions; it assigns no extraneous dimension certificate. Write $A^{(k)} = \overline{\mathrm{Reg}_k(A)}$.

Lemma 2.1 (Pure geometric dimension of a global component)**.** (II.5.3) *Every global irreducible component of A has a unique pure geometric dimension $k \leq n$.*

Proof. A global component B is the closure of the connected component C through some regular point c of A, say of dimension k. All points of C are regular points of A of dimension k, and they are also regular points of B of dimension k. In particular B has a regular point of dimension k. The closure B is an irreducible analytic set. Pure dimensionality of an irreducible analytic set therefore implies that every regular point of B has dimension k. The local dimension at c is at most n. If B also has pure dimension j, its regular point c gives $j = k$, proving uniqueness. □

Lemma 2.2 (Closed unions of global components)**.** (II.5.3) *The union of any subfamily of the global irreducible components of A is closed in M.*

Proof. Every global component is closed, and their family is locally finite. Any subfamily remains locally finite. A locally finite union of closed sets is closed: at a point outside the union, choose a neighborhood meeting only finitely many members, and remove those finitely many closed sets from the neighborhood. The resulting neighborhood misses the entire union. □

Theorem 2.3 (The irreducible-component description of dimension parts)**.** (II.5.3) *For every $k \in \mathbb{N}$, the dimension part $A^{(k)}$ is exactly the union of the global irreducible components of A of pure geometric dimension k. No compactness, empty-interior, or positive-dimension assumption is required.*

Proof. Let S be that union. If c is a regular point of A of dimension k, the closure of its connected component in the regular locus is a global component of pure dimension k containing c. Hence $\mathrm{Reg}_k(A) \subset S$. The preceding lemma makes S closed, so $A^{(k)} \subset S$.

Conversely, let B be a global component of pure dimension k, and write $B = \overline{C}$ for the regular-locus component through a regular point c of A of dimension j. That same point is regular of dimension j on B, so pure dimension k gives $j = k$. Every point of C is then regular of dimension k on A, and therefore $B = \overline{C} \subset \overline{\mathrm{Reg}_k(A)} = A^{(k)}$. Taking the union gives $S \subset A^{(k)}$ and the claimed equality. □

Lemma 2.4 (Geometric purity of the dimension part)**.** *Every regular point of $A^{(k)}$ has local manifold dimension k.*

Proof. The regular points of dimension k on $A^{(k)}$ are dense in $A^{(k)}$. At a regular point y of dimension j on that set, choose a neighborhood where all points of $A^{(k)}$ are regular of dimension j. Density supplies a regular point of dimension k inside this neighborhood. Uniqueness of local manifold dimension at that point gives $j = k$. □

Lemma 2.5 (Component membership in a dimension part)**.** *A global irreducible component B of A is contained in $A^{(k)}$ if and only if it has pure geometric dimension k.*

Proof. The reverse implication follows from the union description. For the forward implication, write $B = \overline{C}$ where C is the regular-locus component through a regular point c of A of dimension j. Since $c \in B \subset A^{(k)}$, every neighborhood of c meets $\mathrm{Reg}_k(A)$. Take a neighborhood where all points of A are regular of dimension j. A point of $\mathrm{Reg}_k(A)$ in it has both dimensions k and j, so $j = k$. The component-purity lemma gives pure dimension k for B. □

Theorem 2.6 (Pure local decompositions of a dimension part)**.** (II.4.24, II.5.3) *At every point of $A^{(k)}$ there is a finite local analytic decomposition of $A^{(k)}$ whose pieces all have dimension k. More precisely, from any local analytic decomposition $A \cap U = \bigcup_i A_i$ with piece dimensions d_i, retain exactly the indices with $d_i = k$. They give such a decomposition on the same U, with the same holomorphic equations for each retained piece, dense regular loci and pairwise relatively thin intersections.*

Proof. The local stratum-closure theorem identifies $A^{(k)} \cap U = \bigcup_{d_i=k} A_i$. The selected index set is finite, so choose a bijection from a finite consecutive index set onto it. Transport the retained pieces and their equations along this bijection. Their union is the displayed dimension part. The neighborhood is unchanged, hence remains open, contains the chosen point and lies in its preferred chart. Holomorphicity of every equation, the equation descriptions of each piece, purity and density of its regular locus remain exactly the corresponding properties of the original piece. The assigned dimension is k because the index is selected. Distinct selected indices remain distinct original indices by injectivity of the bijection, so the relatively thin-intersection property is inherited as well. Finally any point of $A^{(k)}$ lies in A; existence of the original analytic local decomposition at that point supplies the asserted decomposition of $A^{(k)}$. □

3 Transversality of the resolved set and the accumulated divisor

Lemma 3.1 (Density excludes a containing coordinate hyperplane)**.** *Let C be a subset of a complex manifold and $E = [H_1, \dots, H_s]$ a finite family of subsets. Assume that C has only normal crossings with E and*

$$C \subset \overline{C \setminus |E|}, \qquad |E| = \bigcup_i H_i.$$

Then E is transverse to C: in the normal crossings coordinates at every point of C, the coordinate defining any member of E through that point is not among the coordinates defining C.

Proof. Fix $x \in C$ and normal crossings coordinates on an open neighbourhood U of x. Write $C \cap U = \{z_j = 0 : j \in S_0\}$ and, for every member H_i through x, write $H_i \cap U = \{z_{a_i} = 0\}$. If $a_i \in S_0$, then $C \cap U \subset H_i$. Density gives a point of $(C \setminus |E|) \cap U$, which would belong to $H_i \subset |E|$, a contradiction. Thus $a_i \notin S_0$ for every such member, which is precisely transversality. No connectedness or irreducibility is required. □

Lemma 3.2 (The initial boundary is the only additional obstruction)**.** *Let $f : Y \to X$ be holomorphic, A closed, $B \subset X$, and E a finite family on Y such that*

$$A \subset \overline{A \setminus B}, \qquad |E| \subset \mathrm{Exc}(f) \cup f^{-1}(B).$$

If the strict transform A' has only normal crossings with E, then E is transverse to A'. In particular, this applies to an admissible sequence of (X, E_0) with centres over $G \cup (A \cap |E_0|)$, whenever $f^{-1}(G) \subset \mathrm{Exc}(f)$ and $A \subset \overline{A \setminus |E_0|}$.

Proof. Put $D = f^{-1}(A) \setminus \mathrm{Exc}(f)$, so $A' = \overline{D}$. Fix $y \in D$ and a neighbourhood V of y. The exceptional locus has open complement, and f is a local biholomorphism at y. Consequently $f(V \setminus \mathrm{Exc}(f))$ is a neighbourhood of $f(y)$. Density off B provides a point $f(w) \in A \setminus B$ with $w \in V \setminus \mathrm{Exc}(f)$. Then $w \in D \subset A'$, and the support inclusion gives $w \notin |E|$. Every neighbourhood of y therefore meets $A' \setminus |E|$. Taking closures gives $A' \subset \overline{A' \setminus |E|}$, and Lemma 3.1 applies.

For the final assertion, the accumulated support lies in $f^{-1}(|E_0| \cup G)$, since $A \cap |E_0| \subset |E_0|$. The assumed inclusion over G puts that support inside $f^{-1}(|E_0|) \cup \mathrm{Exc}(f)$, giving the required hypotheses. □

Lemma 3.3 (Transversality after an admissible sequence)**.** *Let $f : Y \to X$ be an admissible blowup sequence of (X, E_0), with centres over F and accumulated divisor E. Suppose the members of E_0 are closed, $A \subset X$ is closed,*

$$A \subset \overline{A \setminus (|E_0| \cup F)},$$

and the strict transform A' has only normal crossings with E. Then E is transverse to A'.

Proof. The support of the accumulated divisor lies in $f^{-1}(|E_0| \cup F)$. Density survives passage to the strict transform, so

$$A' \subset \overline{A' \setminus f^{-1}(|E_0| \cup F)} \subset \overline{A' \setminus |E|}.$$

Apply Lemma 3.1. □

Theorem 3.4 (Embedded resolution with a transverse boundary)**.** *Let W be a compact analytic subset of a complex manifold of pure dimension $d < n$. Its embedded resolution can be chosen with all the conclusions of Theorem 1.1, and with the accumulated divisor transverse to the smooth strict transform. If E_0 is a simple normal crossings divisor and $W \subset \overline{W \setminus |E_0|}$, the same strengthening holds for Theorem 1.2, retaining its centres over $W_{\mathrm{sing}} \cup (W \cap |E_0|)$. These statements apply in particular to every such W in a compact ambient manifold.*

Proof. For the empty initial divisor the regular points of dimension d are dense in W and avoid $W_{\mathrm{sing}} = W \setminus \mathrm{Reg}_d(W)$. Therefore $W \subset \overline{W \setminus W_{\mathrm{sing}}}$, and Lemma 3.3 applies to the sequence of Theorem 1.1.

For the boundary form, put $B = |E_0|$, which is closed. We claim that $W \subset \overline{W \setminus (W_{\mathrm{sing}} \cup B)}$. Given $x \in W$ and an open neighbourhood U of x, density off B gives $y \in (W \setminus B) \cap U$. The open set $U \setminus B$ contains y, so density of $\mathrm{Reg}_d(W)$ gives $z \in \mathrm{Reg}_d(W) \cap (U \setminus B)$. This proves the claim. Since

$$B \cup \big(W_{\mathrm{sing}} \cup (W \cap B)\big) = W_{\mathrm{sing}} \cup B,$$

Lemma 3.3 applies to the existing boundary resolution. Its conclusion strengthens the normal crossings conclusion without changing the support conditions, dimensions, or position of the centres. □

Theorem 3.5 (Transverse resolution without equidimensionality)**.** *Let W be compact analytic with empty interior, allowing components of different dimensions. The embedded resolutions of Theorem 1.3, with or without an initial simple normal crossings divisor E_0 satisfying $W \subset \overline{W \setminus |E_0|}$, can be chosen with the accumulated divisor transverse to W'. All dimension-part, disjointness, exceptional-support and isomorphism conclusions are retained.*

Proof. The boundary theorem supplies an admissible sequence with centres over $W_{\mathrm{sing}} \cup (W \cap |E_0|)$, with $f^{-1}(W_{\mathrm{sing}}) \subset \mathrm{Exc}(f)$, and with W' having normal crossings with E. Apply Lemma 3.2 with $G = W_{\mathrm{sing}}$. This argument does not use equidimensionality. For the empty initial divisor, the old theorem gives $|E| = \mathrm{Exc}(f)$. Apply the same lemma with $B = \varnothing$. These operations retain the original witnesses and every stated conclusion about their dimension parts. □

Theorem 3.6 (Transverse embedded resolution on relatively compact open sets)**.** *The transverse conclusions of Theorem 3.4 also hold on any open set $U \subset X$ contained in a compact set, without requiring W compact. In particular, every compact subset $K \subset X$ has an open neighbourhood U with compact closure on which the boundary resolution has this transverse conclusion.*

Proof. Use the existing embedded resolutions on U. In the empty-boundary case the accumulated support is exactly the exceptional locus, so Lemma 3.2 applies with $B = \varnothing$.

For the boundary case put $W_U = W \cap U$ and $E_{0,U} = E_0|_U$. The inclusion $U \hookrightarrow X$ is an open embedding, so pulling back the closure of any set equals the closure of its pullback. Hence density of $W \setminus |E_0|$ in W gives density of $W_U \setminus |E_{0,U}|$ in W_U. The old theorem supplies a sequence with centres over $(W_{\mathrm{sing}} \cap U) \cup (W_U \cap |E_{0,U}|)$, and the entire preimage of $W_{\mathrm{sing}} \cap U$ is exceptional. Lemma 3.2 therefore gives transversality to the accumulated divisor, retaining every existing conclusion of the local resolution.

A complex manifold is locally compact. Cover a compact K by finitely many open neighbourhoods with compact closures and take their union U. Its closure is contained in the finite union of those compact closures, so it is compact. Applying the preceding result on $U \subset \overline{U}$ completes the neighbourhood form. □

Remark 3.7. The density argument supplies the noncontainment needed for simultaneous normal crossings in Bierstone–Milman's Theorem 1.6(2). It does not change the locus over which the boundary resolution may blow up, and does not treat an initial divisor with general, nonsimple normal crossings.

4 Divisor exponents of genuine ideal powers

The monomial factor in Włodarczyk's Step 2 is a product of principal divisor ideals, with multiplicities. The following local algebra records how those multiplicities change when an ideal is replaced by a genuine ideal power. It is an extension of the order and divisor-exponent calculus, not a construction of the global decomposition of every principalizing divisor into the members of a prescribed simple normal crossings family.

Lemma 4.1 (Successor thresholds for ideal powers)**.** *Let $\mathfrak{a}$ be an ideal of holomorphic germs on a complex manifold M, and let $k \geq 1$ and $e \geq 0$ be integers. Then*

$$\operatorname{cosupp}(\mathfrak{a}^k, ke+1) = \operatorname{cosupp}(\mathfrak{a}, e+1).$$

Proof. If every germ of $\mathfrak{a}_x$ has order at least $e+1$, then every product of k such germs has order at least $k(e+1)$. Finite sums and holomorphic multiples preserve this lower bound. Since $ke+1 \leq k(e+1)$, every germ of $(\mathfrak{a}^k)_x$ has order at least $ke+1$. Conversely, if a germ $g \in \mathfrak{a}_x$ has order at most e, then multiplicativity of the order gives $\operatorname{ord}_x(g^k) \leq ke$. The germ g^k belongs to $(\mathfrak{a}^k)_x$, so this stalk does not have order at least $ke+1$. These two implications prove the equality. □

Lemma 4.2 (Divisibility thresholds for ideal powers)**.** *For a subset $H \subset M$, a point $x \in M$, $k \geq 1$ and $e \geq 0$, the ideal $\mathfrak{a}^k$ has order at least $ke+1$ along H near x if and only if $\mathfrak{a}$ has order at least $e+1$ along H near x.*

Proof. Apply Lemma 4.1 at each point of H in a neighbourhood of x. The equality of the two cosupports identifies the two neighbourhood conditions in both directions. □

Lemma 4.3 (A power of a nonzero ideal stalk is nonzero)**.** *If $\mathfrak{a}_x$ contains a holomorphic germ which does not vanish identically near x, then $(\mathfrak{a}^k)_x$ has the same property for every integer $k \geq 0$.*

Proof. Choose such a germ g. Its power g^k lies in $(\mathfrak{a}^k)_x$. At each point where g is nonzero, g^k is nonzero; these points occur arbitrarily close to x. This also covers $k = 0$, since $g^0 = 1$. □

Theorem 4.4 (Divisor exponents scale with ideal powers)**.** (Włodarczyk 2005, Step 2) *Let $x \in H \subset M$, and let $\mathfrak{a}$ be an ideal of holomorphic germs whose stalk at x is nonzero. For every integer $k \geq 0$,*

$$e_H(\mathfrak{a}^k, x) = k\, e_H(\mathfrak{a}, x).$$

Proof. Put $e = e_H(\mathfrak{a}, x)$. Nonvanishing makes this exponent finite: $\mathfrak{a}$ has order at least e along H near x, and it does not have order at least $e+1$ there. Products of germs give divisibility of $\mathfrak{a}^k$ to order ke, so its exponent is at least ke. For $k \geq 1$, Lemma 4.2 rules out divisibility to order $ke+1$. Lemma 4.3 ensures that its exponent is also finite, hence is at most ke. For $k = 0$, the ideal is the unit ideal and its germ 1 has order zero at every point of H; its exponent is therefore zero. □

Lemma 4.5 (Successor thresholds along a submanifold)**.** *Let N be a submanifold of M. For $k \geq 1$ and $e \geq 0$,*

$$\operatorname{cosupp}(\mathfrak{a}^k|_N, ke+1) = \operatorname{cosupp}(\mathfrak{a}|_N, e+1).$$

Consequently the corresponding neighbourhood divisibility conditions along $N \cap H$ agree.

Proof. The inclusion from right to left follows by multiplying germs vanishing to order $e+1$ on N, and using $ke+1 \le k(e+1)$. For the reverse inclusion, fix $x \in N$ and a holomorphic retraction r onto N near x. A germ g has order at least q on N at x if and only if $g \circ r$ has ambient order at least q. If g fails to have restricted order $e+1$, then $g \circ r$ has ambient order at most e, and its k-th power has ambient order at most ke. Since $(g^k) \circ r = (g \circ r)^k$, the germ $g^k \in \mathfrak{a}_x^k$ fails to have restricted order $ke+1$. This proves the reverse inclusion. Applying the equality pointwise near x proves the divisibility assertion. □

Lemma 4.6 (Restricted nonvanishing survives powers)**.** *If $\mathfrak{a}_x$ contains a holomorphic germ whose restriction does not vanish identically on N near x, then $(\mathfrak{a}^k)_x$ has the same property, for every $k \ge 0$.*

Proof. Choose the germ g in the assumption. Then g^k belongs to $(\mathfrak{a}^k)_x$ and is nonzero at each point of N where g is nonzero. Such points occur arbitrarily close to x. □

Theorem 4.7 (Divisor exponents of powers along a submanifold)**.** (Włodarczyk 2005, Step 2) *Let N be a submanifold, $x \in N \cap H$, and suppose $\mathfrak{a}_x|_N$ is nonzero. For every $k \ge 0$,*

$$e_{H|_N}(\mathfrak{a}^k, x) = k\, e_{H|_N}(\mathfrak{a}, x).$$

Proof. Let $e = e_{H|_N}(\mathfrak{a}, x)$. A local holomorphic retraction onto N and nonvanishing ensure that this exponent is finite. Products give restricted divisibility of $\mathfrak{a}^k$ to order ke. For $k \ge 1$, Lemma 4.5 excludes restricted divisibility to order $ke+1$, since it would imply divisibility of $\mathfrak{a}$ to order $e+1$. Restricted nonvanishing survives powers, so the exponent of $\mathfrak{a}^k$ is finite and equals ke. When $k = 0$, its unit germ has order zero on N, which gives the asserted equality. □

Lemma 4.8 (An invertible divisor ideal has nonzero stalks)**.** *Every stalk of an invertible divisor ideal contains a holomorphic germ which does not vanish identically near that point.*

Proof. Near a point x, choose a principal generator g for the ideal. Invertibility includes the fact that g vanishes identically on no nonempty open subset of that neighbourhood. Every open neighbourhood of x therefore contains a point where g is nonzero. The germ g belongs to the ideal stalk at x and has the asserted property. □

Corollary 4.9 (Multiplicities of powers of an actual divisor ideal)**.** *For an effective divisor D on M, $x \in H$, and $k \ge 0$,*

$$e_H(\mathcal{O}(-D)^k, x) = k\, e_H(\mathcal{O}(-D), x).$$

Proof. Apply Theorem 4.4 to the actual invertible ideal $\mathcal{O}(-D)$; its holomorphic nonzero stalk members are supplied by Lemma 4.8. □

4.1 Removing a genuine connected smooth component

The next results construct one global multiplicity and a residual effective divisor from the actual invertible ideal. The component is assumed connected: a possibly disconnected member of a simple normal crossings list need not carry one global multiplicity.

Lemma 4.10 (Local constancy of an ideal exponent on a smooth carrier)**.** *Let $W \subset M$ be a closed submanifold and $\mathfrak{a}$ a coherent ideal nonzero at every stalk. The function $x \mapsto e_W(\mathfrak{a}, x)$ on W is locally constant.*

Proof. At each $x \in W$, the regularity of W and nonvanishing of $\mathfrak{a}$ in a neighbourhood give local constancy of the divisor exponent on W. Restrict this neighbourhood to the subspace W. Thus every point of W has a neighbourhood on which the displayed function equals its value at that point. □

Lemma 4.11 (A connected carrier supplies one actual multiplicity)**.** *Let W be a nonempty connected closed submanifold of fixed dimension and $\mathfrak{a}$ a coherent ideal nonzero at every stalk. There is $a \in \mathbb{N}$ such that, for every $x \in W$ and $r \geq 0$,*

$$\mathfrak{a} \text{ has order at least } r \text{ along } W \text{ near } x \quad \Longleftrightarrow \quad r \leq a.$$

Proof. A locally constant function on a connected space is constant: each nonempty fibre is both open and closed, so its complement is empty. Choose $p \in W$ and put $a = e_W(\mathfrak{a}, p)$. By Lemma 4.10, $e_W(\mathfrak{a}, x) = a$ for every $x \in W$. Nonzero stalks ensure that the finite exponent characterizes divisibility, proving the displayed equivalence. □

Lemma 4.12 (A smooth hypersurface has empty interior)**.** *If W is a closed smooth submanifold of dimension d in an n-dimensional complex manifold and $d+1 = n$, then W has empty interior.*

Proof. An interior point of W has an open neighbourhood contained in W; in the ambient chart it is therefore regular of dimension n. But every point of W is regular of dimension d. The dimension of a regular germ is unique, so $d = n$, contrary to $d + 1 = n$. □

Theorem 4.13 (Actual residual after removing a connected smooth component)**.** (Włodarczyk 2005, Step 2) *Let D be an effective divisor on an n-dimensional complex manifold M, and let W be a nonempty connected closed smooth submanifold of dimension d with $d + 1 = n$. There exist $a \geq 0$ and an effective divisor R on M such that*

$$\begin{aligned} e_W(\mathcal{O}(-D), x) &= a \qquad\qquad (x \in W), \\ \mathcal{O}(-D) &= \mathcal{I}_W^a \mathcal{O}(-R), \\ \mathcal{O}(-R) &= (\mathcal{O}(-D) : \mathcal{I}_W^a), \end{aligned}$$

and W is not contained in the support of R.

Proof. The ideal $\mathcal{O}(-D)$ is coherent and has nonzero stalks. Lemma 4.11 therefore supplies the constant finite exponent a. At every $x \in W$, all germs of $\mathcal{O}(-D)_x$ vanish to order at least a on W. The Hadamard bridge gives the global inclusion $\mathcal{O}(-D) \subset \mathcal{I}_W^a$, including the stalks off W.

The smooth hypersurface ideal $\mathcal{I}_W$ is invertible: locally on W it is generated by one flattening coordinate, and off W by 1; its local generator vanishes identically on no open subset because W has empty interior. Its power $\mathcal{I}_W^a$ is invertible as well. The colon ideal $\mathfrak{r} = (\mathcal{O}(-D) : \mathcal{I}_W^a)$ is invertible. Indeed, for local generators u and v of these two nested invertible ideals, the inclusion gives $u = qv$ with q holomorphic. The germ q generates the colon ideal: cancellation by v proves one inclusion, and multiplication by v proves the other. Nonvanishing of u on every nonempty open set forces the same nonvanishing for q. Thus $\mathfrak{r} = \mathcal{O}(-R)$ for an effective divisor R, and the same cancellation yields $\mathcal{O}(-D) = \mathcal{I}_W^a \mathcal{O}(-R)$.

If W were contained in the support of R, every germ of the coherent ideal $\mathcal{O}(-R)$ would vanish on W near its base point: membership propagates to nearby stalks, and each such germ vanishes at the points of W. Hence $\mathcal{O}(-R) \subset \mathcal{I}_W$, and the product identity would give $\mathcal{O}(-D) \subset \mathcal{I}_W^{a+1}$. At any $p \in W$ this implies divisibility to order $a + 1$, contradicting the characterization of the exponent a. Therefore W is not contained in the support of R. □

Corollary 4.14 (Powers of the actual component factorization)**.** *Under the hypotheses of Theorem 4.13, for each $k \geq 0$ the same actual a and residual R satisfy*

$$\mathcal{O}(-D)^k = \mathcal{I}_W^{ak} \mathcal{O}(-R)^k.$$

Proof. Construct a and R by the preceding theorem, then take the k-th power of its whole ideal identity. Products of coherent ideals commute and associate, so the k-th power of a product is the product of the k-th powers, and $(\mathcal{I}_W^a)^k = \mathcal{I}_W^{ak}$. The exponent zero gives the unit ideal on both sides. □

Definition 4.15 (Finite product of component powers)**.** For a finite ordered list $A = ((W_1, a_1), \dots, (W_s, a_s))$ of subsets of M and nonnegative integers, put

$$P(A) = \mathcal{I}_{W_1}^{a_1} \cdots \mathcal{I}_{W_s}^{a_s}, \qquad P(\varnothing) = \mathcal{O}_M.$$

Lemma 4.16 (Coherence of a finite component product)**.** *If every W_i is a closed submanifold, then $P(A)$ is coherent.*

Proof. The vanishing ideal of each closed submanifold is coherent. Powers and products of coherent ideals are coherent. Induction on the list, starting from the coherent unit ideal, proves the claim. □

Theorem 4.17 (Extraction along a finite family of connected hypersurfaces)**.** *Let $E = (W_1, \dots, W_s)$ be a finite list of nonempty connected closed smooth hypersurfaces in M, and D an effective divisor. There exist nonnegative integers $a_1, \dots, a_s$ and an effective divisor R such that*

$$\mathcal{O}(-D) = \mathcal{I}_{W_1}^{a_1} \cdots \mathcal{I}_{W_s}^{a_s} \mathcal{O}(-R), \qquad \operatorname{Supp}(R) \subset \operatorname{Supp}(D),$$

and no W_i is contained in $\operatorname{Supp}(R)$*. The list may be empty and may contain repeated members.*

Proof. Induct on the length. For the empty list take $R = D$; the product is the unit ideal. For $E = (W, E')$, Theorem 4.13 produces a and D_1 with $\mathcal{O}(-D) = \mathcal{I}_W^a \mathcal{O}(-D_1)$ and $W \not\subset \operatorname{Supp}(D_1)$. The induction hypothesis applied to D_1 and E' produces A' and R with $\mathcal{O}(-D_1) = P(A')\mathcal{O}(-R)$, $\operatorname{Supp}(R) \subset \operatorname{Supp}(D_1)$, and no member of E' contained in $\operatorname{Supp}(R)$. Associativity of coherent ideal products gives $\mathcal{O}(-D) = (\mathcal{I}_W^a P(A'))\mathcal{O}(-R)$. A zero of every germ of $\mathcal{O}(-D_1)$ is a zero of every germ of its product with $\mathcal{I}_W^a$, so $\operatorname{Supp}(D_1) \subset \operatorname{Supp}(D)$. These inclusions give the asserted support inclusion and also rule out $W \subset \operatorname{Supp}(R)$. The members of E' are handled by the induction hypothesis. The argument does not require the members to be distinct, so repeated members satisfy the same conclusion. □

Remark 4.18 (Scope of the multiplicity calculation)**.** Each exponent extracted from a single connected component is defined by neighbourhood divisibility of stalks. In the finite theorem, a_i is the exponent of the intermediate divisor after the preceding members have been removed; these exponents follow the order of the list. They are not asserted to be the exponents of the original D for every list position. In particular a repeated member has exponent zero after its first removal. For a smooth hypersurface the local exponent agrees with the largest power of its vanishing ideal dividing the stalk. Constructing a global identity $\mathcal{O}(-D) = \prod_i \mathcal{I}_{E_i}^{a_i}$ for a general simple-normal-crossings-supported divisor D, and then identifying $D = \sum_i a_i E_i$, remains an additional obligation. The finite extraction theorem leaves an actual residual factor. It does not prove this factor is the unit ideal from an SNC support assumption, nor does it refine disconnected SNC members into their connected components.

5 A genuine global analytic extension of a regular order stratum

Bierstone–Milman[4], Section 13, proof of Theorem 13.4 (printed page 300), extends a local centre by taking appropriate dimensional components of genuine global analytic superlevel sets. The following producer proves that extension step for an actual ideal order superlevel already regular of one dimension on the selected open set. It does not assume a global extension, or assert that this global extension is already smooth away from that open set.

Lemma 5.1 (Dimension selection preserves a regular trace)**.** *Let A be closed, and suppose every point of $A \cap U$ is regular of complex dimension q in A. Then $A^{(q)} \cap U = A \cap U$, where $A^{(q)} = \overline{\operatorname{Reg}_q A}$.*

Proof. Closedness gives $A^{(q)} \subset A$. Conversely, each point of $A \cap U$ belongs to $\mathrm{Reg}_q A$ by the assumed actual regularity, and therefore to its closure. Intersect these two containments with U. No compactness or global pure-dimensionality of A is needed. □

Theorem 5.2 (Produced global analytic extension of an actual order stratum)**.** *Let I be a coherent holomorphic germ ideal on a complex manifold, N a closed smooth submanifold, and $\mu \geq 1$. Put $A = \mathrm{cosupp}_N(I, \mu)$. Suppose $A \cap U$ is regular of complex dimension q at every point, where U is open. There is a genuine global analytic pure-q subset $T \subset A$ with $T \cap U = A \cap U$. It is exactly the union of the actual q-dimensional global components selected from A, and every point of its trace on U is regular of dimension q in T.*

Proof. The derivative-ideal order theorem identifies the actual superlevel with $N \cap V(\mathcal{D}_N^{\mu-1} I)$. Coherence of the derivative ideal and analyticity of the closed smooth N make A a global analytic set. Take $T = A^{(q)}$. Actual dimension-part analyticity and purity give the first assertions, and closedness of A gives $T \subset A$. The preceding lemma gives the exact trace equality. A regular point of A is regular of the same dimension in its actual dimension part: the regular coordinate neighbourhood agrees with that part. Finally the global dimension-part identification proved from genuine global components gives $T = \bigcup\{B : B \text{ is a global component of } A \text{ of pure dimension } q\}$. Thus this is an internally constructed analytic extension; no extension or component labels are supplied by the caller. Its singularities away from U remain for the separate recursive smoothification step of Lemma 13.7. □

6 Actual comparison cocycles for a common blowup centre

The universal property of the analytic blowup supplies unique comparison maps. For gluing, these maps must be produced simultaneously with their cocycle identities. This section proves that step for a family of genuine blowups of the same centre; alignment of centres of independently chosen local canonical towers and analytic manifold descent remain separate obligations.

Theorem 6.1 (The actual holomorphic comparison cocycle)**.** *Let $(\pi_i : Y_i \to X)_{i \in I}$ be any family of actual blowups along the same closed smooth centre C. There are bijective holomorphic maps $f_{ij} : Y_i \to Y_j$ satisfying*

$$\pi_j f_{ij} = \pi_i, \qquad f_{ii} = \mathrm{id}, \qquad f_{jk} f_{ij} = f_{ik} \quad (i, j, k \in I).$$

No comparison maps or cocycle identities are required as inputs.

Proof. For each pair (i, j), the universal property of the actual blowup Y_j applied to π_i produces a unique holomorphic map f_{ij} over X. The comparison in the opposite direction is an inverse: both composites are holomorphic maps of the respective blowup over X, and uniqueness identifies each with the identity. Consequently every f_{ij} is bijective. Choose these actual maps simultaneously over all pairs. The identity is itself a holomorphic comparison from Y_i to itself, so uniqueness gives $f_{ii} = \mathrm{id}$. The composite $f_{jk} f_{ij}$ is holomorphic, and $\pi_k f_{jk} f_{ij} = \pi_j f_{ij} = \pi_i$. It is therefore the unique comparison from Y_i to Y_k, which gives the displayed cocycle. This argument applies to arbitrary index sets and needs neither finiteness nor compactness. □

Theorem 6.2 (Biholomorphic comparison data with the actual cocycle)**.** *Under the same geometric hypotheses, there are actual biholomorphisms $\Phi_{ij} : Y_i \simeq Y_j$ with inverse Φ_{ji}, commuting with the actual blowup projections, whose underlying maps satisfy the identity and triple-overlap cocycle.*

Proof. Use the produced maps f_{ij} and f_{ji} as the forward and inverse maps. The cocycle with indices (i, j, i), followed by the identity result, gives $f_{ji} f_{ij} = \mathrm{id}_{Y_i}$; the indices (j, i, j) give the other inverse equality. Both maps are holomorphic by the preceding theorem, so these data define an actual biholomorphism Φ_{ij}. Its projection, identity, inverse and cocycle assertions are precisely the already proved equalities of those actual maps. □

7 Canonical elementary principalization with compact support

Canonical principalization can retain its marked-ideal centre rule while exporting all the elementary geometric conclusions. Compactness of the ambient manifold is unnecessary when the ideal's zero locus is compact. The compatibility established here is with open restrictions and biholomorphisms. General smooth morphisms require additional invariant and blowup base-change theorems.

Theorem 7.1 (Canonical marked resolution with compact cosupport)**.** *Let $\mathfrak{m}$ lie in the domain of a resolution invariant with well-ordered values and have compact cosupport on a complex manifold. There exists a resolution which is a canonical segment with nonempty centres. The invariant is upper semicontinuous along canonical segments, every open restriction is canonical, and canonical resolutions after a biholomorphism are isomorphic, stage by stage, over that biholomorphism.*

Proof. Along a canonical segment the transformed cosupport is compact: it is closed and contained in the inverse image of the initial compact cosupport under a proper composition of blowups. Upper semicontinuity gives a maximum on every nonempty transformed cosupport. Its maximum locus is a nonempty smooth admissible centre with empty interior. The invariant strictly decreases above that centre and stays unchanged elsewhere. The compact-cosupport, well-founded driver therefore produces a resolving segment using only nonempty maximum loci.

Upper semicontinuity is the invariant's second property. Its open restriction property identifies every nonempty restricted centre with the restricted maximum locus; centres missing an open set restrict to empty centres. For two canonical resolving segments related by a biholomorphism, induction gives corresponding maximum loci and the uniqueness of each blowup gives isomorphisms of their stages. Their lengths are equal: if one were longer, its prefix corresponding to the completed shorter resolution would have empty cosupport, and a nonempty next maximum locus could not exist. The resulting stage isomorphisms therefore extend through both complete towers. □

Theorem 7.2 (Canonical principalization relative to an SNC boundary)**.** *Let $\mathfrak{a}$ be a coherent ideal on a complex manifold, nonzero at every point, with compact zero locus, and let E_0 be an SNC boundary. There is an admissible principalization f obtained from the canonical resolution of $(X, X, E_0, \mathfrak{a}, 1)$. Its effective divisor D satisfies $f^*\mathfrak{a} = \mathcal{O}(-D)$, and its accumulated boundary E is SNC. The support of D is both $f^{-1}Z(\mathfrak{a})$ and the union of the new members of E; f is an isomorphism off $Z(\mathfrak{a})$. The same tower has the open-restriction and biholomorphism compatibility of the preceding theorem.*

Proof. The boundary is transverse to the whole manifold, so the marked ideal $(X, X, E_0, \mathfrak{a}, 1)$ lies in the invariant's domain. Its cosupport is exactly $Z(\mathfrak{a})$: order at least one is precisely vanishing of every member of the stalk. Apply the preceding theorem to the canonical invariant. Every centre of the resulting test tower has normal crossings with the accumulated boundary and lies over the initial cosupport, proving admissibility over $Z(\mathfrak{a})$.

The controlled-transform identity at mark one identifies the total pullback ideal, at the terminal empty cosupport, with the tower's effective exceptional divisor ideal. The tower transform preserves SNC. Induction on its blowups shows that the support of its exceptional divisor is the union of the new boundary members. Pullback of coherent ideals identifies that support with $f^{-1}Z(\mathfrak{a})$. The exceptional locus is contained in this support, and a proper composition of blowups is an isomorphism over the complement of its exceptional image. This yields the asserted isomorphism off $Z(\mathfrak{a})$. All compatibility clauses belong to this very tower, since it was selected by the canonical marked-ideal producer. □

8 Actual component sums in canonical principalization

Definition 8.1 (Finite sum of genuine connected boundary components)**.** For an effective divisor D and a finite SNC boundary E, a finite component sum consists of a distinct finite list F of actual connected components of members of E, still SNC, and a weighted list A with underlying list F. Each member is a closed smooth hypersurface and

$$D = \sum_{(C,a)\in A} a[C], \qquad \operatorname{divExp}(\mathcal{I}_D, C, x) = a \quad (x \in C).$$

The sum is the actual sum of effective Cartier divisors constructed by ideal products, and the equality is equality of those divisors. A coefficient may be zero when a boundary component merely meets the support of D.

Theorem 8.2 (Complete sum on a finite connected SNC list)**.** *Let F be a distinct finite SNC list of closed connected smooth hypersurfaces. Every effective divisor supported in $|F|$ is its actual weighted component sum, with weights equal to the original divisor exponents at every point of each component.*

Proof. The complete factorization producer constructs the weighted list A, proves $\mathcal{I}_D = \prod_{(C,a)\in A} \mathcal{I}_C^a$, and computes its coefficients as the original exponents. The supplied actual closed codimension-one geometry constructs each Cartier divisor $[C]$. The genuine finite sum has exactly this ideal product. Equality of the invertible ideal determines the effective divisor, giving the required sum equality and retaining the computed exponents. Nothing about factorization or coefficient identification is required from the caller. □

Theorem 8.3 (Complete component sum for compact support)**.** *Let D have compact support contained in an arbitrary finite SNC boundary E. Then D has a finite component sum in the preceding sense, even if the ambient manifold and the original boundary members are noncompact or disconnected. This holds also in dimension zero.*

Proof. In positive ambient dimension, the actual connected-component family of E is locally finite. Compactness selects a finite distinct list of its members meeting $\operatorname{Supp} D$. The coordinate refinement theorem makes that list SNC and each member a closed connected smooth hypersurface. The union of the selected list agrees with the original boundary on the compact support, so it contains the support of D. Apply the preceding complete sum producer. In dimension zero, no SNC member can contain a point: its adapted singleton coordinate would lie in the empty coordinate index set. Thus $\operatorname{Supp} D$ is empty, its actual ideal is the unit ideal, and D is the empty sum. Empty ambient spaces and empty lists are covered by the same argument. □

Theorem 8.4 (Canonical principalization with the actual divisor sum)**.** *Canonical principalization of a coherent ideal with nonzero stalks and compact zero locus, relative to any initial SNC boundary, has its actual effective divisor equal to a finite sum of genuine connected components of the newly created boundary. Its coefficients are the original divisor exponents. The same tower retains admissibility, the complete pullback ideal identity, both support identities, the isomorphism off the zero locus, canonicality, open restriction, and biholomorphism compatibility.*

Proof. Construct the actual canonical principalization tower with all these conclusions. Its projection is proper, so the preimage of the compact zero locus is compact. This is exactly the support of its effective divisor. The newly created boundary is the sublist obtained by dropping the transforms of the initial members; a sublist of an SNC list is SNC, and its union is the actual divisor support. Apply the compact-support sum producer to that divisor and that sublist. Append its genuine component sum to the conclusions of the very same tower. This uses no compactness of the initial boundary and no assumption identifying product coefficients. □

9 Connected components of a simple normal crossings boundary

The smooth members of a simple normal crossings boundary need not be connected or irreducible, as in Bierstone–Milman[5], Definition 2.1. Multiplicities are therefore assigned to their genuine connected components. The global component theorem of Demailly[2], (II.5.3), supplies local finiteness; compactness is used only when selecting a finite refinement.

Definition 9.1 (The connected boundary components)**.** For an ordered finite family $E = [H_1, \dots, H_s]$ of closed subsets of a complex manifold, let $\mathcal{C}(E)$ be the family of all nonempty connected components of its members. Equivalently, its members are the sets $\mathrm{Conn}_{H_i}(x)$, where $x \in H_i$; each set is included once in the family. Then

$$\bigcup_{C \in \mathcal{C}(E)} C = \bigcup_i H_i.$$

For an SNC family in ambient dimension zero, $\mathcal{C}(E)$ is empty.

Proof. Every connected component of H_i is contained in H_i. Conversely, if $x \in H_i$, the component $\mathrm{Conn}_{H_i}(x)$ contains x, and is one of the specified members. These two containments prove the identity, including when the family is empty. In ambient dimension zero, the simple normal crossings condition admits no member through any point: its defining coordinate singleton would be a singleton in the empty coordinate set. All members are therefore empty, and the connected component family is empty as well. □

Lemma 9.2 (The actual geometry of a boundary component)**.** *Let E be a simple normal crossings boundary in a complex manifold of dimension $m+1$. Every $C \in \mathcal{C}(E)$ is a nonempty closed connected complex submanifold of dimension m. It is a component of some H_i, and at every $x \in C$ there is an open neighbourhood U with $H_i \cap U = C \cap U$.*

Proof. The normal crossings coordinates exhibit H_i locally as one coordinate hyperplane; exchanging its defining coordinate with the last coordinate exhibits dimension m. Thus H_i is analytic and all its points are regular. A connected component of H_i is closed: its closure is connected, is contained in the closed set H_i, and hence remains in the same maximal connected component. The regular locus of H_i is all of H_i, so its global analytic components are precisely these already closed connected components. At a regular point the connected component contains a sufficiently small connected coordinate neighbourhood inside H_i. This gives an ambient open neighbourhood on which C and H_i agree, and transports the hyperplane flattening to C. The point defining the component supplies its nonemptiness, and its connectedness is the defining maximality property. □

Lemma 9.3 (Local finiteness and compact selection)**.** *For a simple normal crossings boundary E, the family $\mathcal{C}(E)$ is locally finite. For every compact set K, only finitely many $C \in \mathcal{C}(E)$ meet K.*

Proof. The components of each H_i are its global analytic components, so the global component theorem makes that family locally finite. At an arbitrary point, intersect the finitely many open neighbourhoods supplied for the original members H_i. The resulting open neighbourhood meets only finitely many of their components, and hence only finitely many members of $\mathcal{C}(E)$. To prove the compact assertion, choose such a neighbourhood at each point of K. A finite subcover exists. A component meeting K meets one of these finitely many neighbourhoods, and therefore belongs to the finite union of their finite meeting families. □

Lemma 9.4 (Normal crossings survive finite component refinement)**.** *Let F be a finite list of distinct members of $\mathcal{C}(E)$, where E is simple normal crossings. Then F is simple normal crossings.*

Proof. Every member of F is closed by the component geometry lemma. Fix a point x and normal crossings coordinates for E. For each component in F through x, choose its original member H_i and an open neighbourhood on which the two agree. Intersect these finitely many neighbourhoods and restrict the coordinate chart to the intersection. The selected components through x are now precisely the corresponding coordinate hyperplanes. Two such components with the same original member and containing x are the same connected component. Since the list has no repetitions, their indices are equal. Distinct original members through x have distinct defining coordinates by the original simple normal crossings condition. The selected hyperplanes are therefore distinct, which is the required condition for F. □

Theorem 9.5 (Finite refinement near a compact set)**.** *For a simple normal crossings boundary E and a compact set K, there is a finite list F, with no repetitions, whose members are exactly those connected boundary components meeting K. All its members are nonempty closed connected smooth hypersurfaces, F is simple normal crossings, and there is an open neighbourhood $U \supset K$ such that*

$$U \cap |F| = U \cap |E|, \qquad K \cap |F| = K \cap |E|.$$

If $|E| \subset K$, then $|F| = |E|$.

Proof. List once each of the finitely many components meeting K. The component geometry and finite refinement lemmas give the stated smoothness and normal crossings. The inclusion $|F| \subset |E|$ follows from component containment. The union of the omitted components is closed. Indeed, at a point outside that union choose an open neighbourhood meeting only finitely many components, and remove those finitely many closed omitted components from the neighbourhood. The result is an open neighbourhood disjoint from the entire omitted union. No omitted component meets K. Its closed union therefore has an open complement U containing K. If $x \in U \cap |E|$, choose an original member containing x. Its connected component through x cannot be omitted, and thus appears in F. This gives the support equality on U, and hence on K. If the original support is contained in K, every nonempty component meets K, so the same argument gives the global support equality. □

Lemma 9.6 (Divisibility depends on the local component)**.** *Let C be the connected component of H_i through a point of an SNC boundary. At every $x \in C$, a holomorphic germ ideal has the same local divisibility exponents along C and along H_i. In particular, their maximal divisorial exponents at x are equal.*

Proof. Choose an open neighbourhood U of x on which H_i and C agree. For each integer r, the defining local condition says that the points of the carrier near x lie in the order-r cosupport. Intersecting its neighbourhood with U makes the two conditions identical. The sets of admissible powers therefore agree. Their maximal exponent, with the same totalized convention when all powers are admissible, is equal. □

Lemma 9.7 (Componentwise values glue locally constantly)**.** *For any function a on $\mathcal{C}(E)$, the function on H_i assigning to x the value of a on the connected component of H_i through x is locally constant.*

Proof. Fix $x \in H_i$ and its connected component C. Choose an ambient open neighbourhood U on which H_i and C agree. Every point of $H_i \cap U$ belongs to C, and its connected component in H_i is therefore the same set C. The assigned value on this neighbourhood is consequently constant. These relative neighbourhoods prove local constancy at every point of H_i. □

Theorem 9.8 (Multiplicities on genuine boundary components)**.** *Let D be an effective divisor and E a simple normal crossings boundary. There is a function $a : \mathcal{C}(E) \to \mathbb{N}$ such that, for every $C \in \mathcal{C}(E)$, $x \in C$ and $r \in \mathbb{N}$, the ideal of D is locally divisible by $\mathcal{I}_C^r$ at x if and only if $r \leq a(C)$. On every original member H_i, the induced function assigning $a(\mathrm{Conn}_{H_i}(x))$ to x is locally constant, and characterizes divisibility along H_i by the same inequality.*

Proof. An invertible divisor ideal is coherent and has a nonzero stalk member at every point. Its divisorial exponent along a smooth carrier is locally constant. Each C is nonempty and connected, so this locally constant function is constant on C. Choose its value as $a(C)$. The actual maximal-exponent divisibility characterization gives the asserted equivalence at every point. Choice of these values over the component family gives the single function a; there is no requirement that different connected components of one original member receive the same value. The component-function gluing lemma proves local constancy on the original member. The local divisibility lemma transfers its characterization from the connected component to that original member at each of its points. □

Theorem 9.9 (Actual residual factorization after compact refinement)**.** *Suppose the support of a simple normal crossings boundary E is contained in a compact set. For every effective divisor D, there is a finite connected component refinement F with $|F| = |E|$, a list of coefficients b_C, and an effective residual divisor R such that*

$$\mathcal{O}(-D) = \left(\prod_{C \in F} \mathcal{I}_C^{b_C} \right) \mathcal{O}(-R), \qquad \operatorname{Supp}(R) \subset \operatorname{Supp}(D), \qquad C \not\subset \operatorname{Supp}(R) \quad (C \in F).$$

The coefficients in this factorization are extracted in order from successive intermediate residual divisors.

Proof. Take the finite refinement of the preceding theorem. Each member is a nonempty connected closed smooth hypersurface, so the finite connected hypersurface residual theorem applies. Its induction successively removes the maximal coefficient of the next member using the actual invertible colon ideal. At each step the complete ideal factors as the chosen hypersurface power times the new residual. Associating these equalities produces the displayed product factorization. Support inclusion at each step preserves noncontainment already obtained for earlier members, and gives the final support inclusion. Empty refinements give the empty product and residual D, as required. □

Remark 9.10. On a noncompact manifold, a single disconnected smooth member can have infinitely many connected components. The full component family is locally finite; its finite selection near a compact set is not a global finite refinement unless the original support is contained in that compact set. An equality with a divisor sum additionally requires proving the residual ideal is the unit ideal under the appropriate support hypothesis.

10 The actual current reduced ideal and residual order

Lemma 10.1 (Order reverses ideal inclusion)**.** *If $\mathfrak{a} \subset \mathfrak{b}$ are ideals of germs, then* $\operatorname{ord}_x(\mathfrak{b}|_N) \leq \operatorname{ord}_x(\mathfrak{a}|_N)$.

Proof. If every member of $\mathfrak{b}_x$ belongs to the qth power of the maximal ideal on N, the same holds for every member of $\mathfrak{a}_x$. Each integer occurring in the supremum defining the order of $\mathfrak{b}$ therefore occurs in the supremum for $\mathfrak{a}$. Taking suprema proves the assertion, including infinite orders. □

Lemma 10.2 (Current strict transform lies in the actual cosupport)**.** *Let W be closed and $W \subset \overline{W \cap O}$ for an open O. Let the marked ideal $(X, X, E_0, \mathcal{I}_W, 1)$ be in the invariant domain. For every actual test tower T whose centres avoid $T^{-1}O$, its current strict transform S is contained in its actual current cosupport. No smoothness of W is required.*

Proof. The actual current ambient carrier is the whole manifold and its mark is one, so its cosupport is the zero locus of its coherent controlled ideal and is closed. Density transports to $S \subset \overline{S \cap T^{-1}O}$. Every point of S maps into W, since W is closed. On $T^{-1}O$ the actual cosupport is exactly the inverse image of W: restriction of the recorded tower to O has only empty centres

and is a local biholomorphic pullback. Thus $S \cap T^{-1}O$ is in the current cosupport. Closure gives the result. □

Lemma 10.3 (The controlled ideal vanishes on the current strict transform)**.** *Under the preceding hypotheses, the actual current controlled ideal satisfies* $\mathfrak{a}_T \subset \mathcal{I}_S$.

Proof. The previous lemma puts S in the zero locus of $\mathfrak{a}_T$. For a germ $g \in (\mathfrak{a}_T)_x$, coherence gives a local presentation by holomorphic generators and expresses g as a holomorphic linear combination near x. The same expression makes g a member of the ideal at every sufficiently nearby point. It consequently vanishes on S near x, which is precisely membership in $(\mathcal{I}_S)_x$. □

Lemma 10.4 (Residual ideal outside the divisor)**.** *For a coherent ideal on the whole ambient manifold, its nonmonomial part has exactly the same germs as the ideal at a point outside the divisor.*

Proof. At such a point every condition indexed by a divisor member through that point is empty, so the constant germ 1 belongs to the monomial part. A residual germ multiplied by 1 therefore belongs to the original ideal; restriction to the whole manifold changes no coherent ideal. Conversely every original germ belongs to its colon by the monomial part, by ideal closure under holomorphic multiplication. This proves both inclusions. □

Lemma 10.5 (Removing the monomial factor preserves vanishing on a dense set)**.** *Let a well-formed nonzero marked ideal have the whole manifold as its ambient carrier, and let $n > 0$. If S is in its ideal's zero locus and $S \subset \overline{S \setminus |E|}$, then its actual nonmonomial ideal is contained in $\mathcal{I}_S$.*

Proof. The residual ideal is coherent, by the actual SNC monomial-factorization theorem and nonzero germs. Its zero locus is therefore closed. At a point of $S \setminus |E|$, the preceding lemma identifies its residual germs with original germs, all of which vanish at that point. Hence the residual zero locus contains $S \setminus |E|$ and by density contains S. The same coherent-germ propagation argument as above gives inclusion in $\mathcal{I}_S$. □

Theorem 10.6 (Actual residual ideal inside the current reduced ideal)**.** *Under the current-strict-transform hypotheses, suppose additionally that $n > 0$ and every initial boundary member is disjoint from O. Then the actual residual ideal of the current controlled transform satisfies $\mathcal{N}(\mathfrak{a}_T) \subset \mathcal{I}_S$.*

Proof. Every actual centre lies over $X \setminus O$, because the recorded tower avoids O. The actual admissible-sequence producer and the boundary-support induction put every current boundary member over either the initial boundary or $X \setminus O$. Both miss O. Density transports to $S \subset \overline{S \cap T^{-1}O}$, and this dense part misses the whole current boundary. Thus $S \subset \overline{S \setminus |E_T|}$. The invariant domain is preserved by the actual test tower; in particular the current ideal is coherent and nonzero and its current divisor is SNC. Its zero locus contains S by the preceding current-ideal lemma. Apply the residual-ideal density lemma. □

Theorem 10.7 (Fresh reduced order is bounded by the actual residual order)**.** *With the hypotheses of the preceding theorem, at every current point y,*

$$\operatorname{ord}_y(\mathcal{I}_S) \leq \operatorname{ord}_y \mathcal{N}(\mathfrak{a}_T).$$

The right hand side is the actual finite first coordinate used by the canonical general-step invariant. In particular the freshly initialized reduced order is finite.

Proof. For every integer q such that all germs of $\mathcal{I}_S$ vanish to order q, the proved inclusion makes all residual germs vanish to order q. The actual domain and nonzero-germ theorem identify this condition with q being at most the finite residual order of the current marked ideal. Take the supremum over those integers q. □

This comparison proves only the first order-coordinate bound. It does not identify the higher coefficient-ideal coordinates after reinitialization, prove their strict lexicographic descent, or exclude boundary-only branches of the old controlled cosupport from every canonical maximum. The literal current-singular-locus condition for an initially singular set remains open.

11 Effective divisors and finite component sums

Effective divisors on a complex manifold are represented by their invertible ideal sheaves. The following arithmetic uses these actual ideals, so that a component factorization yields an equality of effective divisors, including multiplicities, rather than only an equality of supports.

Definition 11.1 (Addition and zero for effective divisors)**.** The zero divisor has ideal $\mathcal{O}_X$. For effective divisors D and G, define $D+G$ by $\mathcal{O}_X(-D-G) = \mathcal{O}_X(-D)\mathcal{O}_X(-G)$.

The unit ideal is invertible, with generator 1. Products of invertible ideals are invertible: on the intersection of their generator neighborhoods the product generator is the product of the two generators. Its nonvanishing on every nonempty open subset follows by successively avoiding the zero sets of the two generators. Thus both operations produce effective divisors.

Lemma 11.2 (Commutative additive arithmetic)**.** *Effective divisors form a commutative additive monoid. Two effective divisors are equal if and only if their invertible ideals are equal.*

Proof. The representation of a divisor consists of its ideal and the proposition that the ideal is invertible, so equality of the ideals gives equality of the divisors. Commutativity of products of ideals follows by interchanging the two factors of each product generator. Associativity follows by expanding finite holomorphic combinations twice and collecting their products. The unit ideal is neutral because an ideal absorbs holomorphic combinations of its members, while every member is its own product with 1. Invertible ideals are coherent, so these identities apply to all effective divisors and prove the monoid laws. □

Lemma 11.3 (Natural multiples and their support)**.** *For every $a \in \mathbb{N}$ and every effective divisor D, $\mathcal{O}_X(-aD) = \mathcal{O}_X(-D)^a$. The support of 0 is empty, the support of $D+G$ is $\operatorname{Supp} D \cup \operatorname{Supp} G$, and for $a > 0$ the support of aD is the support of D.*

Proof. The ideal identity follows by induction on a, using the unit ideal at $a = 0$ and multiplication by $\mathcal{O}_X(-D)$ at the successor step. At a point outside the zero loci of two ideals choose a nonzero value from each; their product is a product generator with nonzero value. At a point of either zero locus every product generator, and hence every holomorphic combination of them, vanishes. This proves the union identity for support. The unit ideal contains 1, so has empty zero locus. The support of a positive multiple follows by induction and the union identity. □

Lemma 11.4 (A smooth hypersurface defines an effective divisor)**.** *A closed smooth hypersurface H defines an effective divisor $[H]$ with ideal $\mathcal{I}_H$ and support H; the empty hypersurface is permitted.*

Proof. Flattening coordinates identify H locally with the vanishing of one coordinate. Hadamard division shows that this coordinate generates $\mathcal{I}_H$ there. Off the closed set H the generator is 1. A hypersurface has empty ambient interior, since a regular point of an open subset of the ambient manifold has dimension n, whereas a regular point of H has dimension $n-1$. Thus its local generator is identically zero on no nonempty open subset. This proves invertibility. Members of $\mathcal{I}_H$ vanish at every point of H, and off H the germ of 1 is a member; hence its support is exactly H. □

Theorem 11.5 (Actual finite component sum)**.** *For a finite list $A = ((H_1, a_1), \ldots, (H_s, a_s))$ of closed smooth hypersurfaces with natural coefficients, the effective divisor $\sum_{i=1}^{s} a_i[H_i]$ has ideal $\mathcal{I}_{H_1}^{a_1} \cdots \mathcal{I}_{H_s}^{a_s}$. Consequently an equality of an effective divisor's ideal with this product gives equality of that divisor with this finite sum.*

Proof. Construct each $[H_i]$ by the preceding lemma. Starting with the zero divisor, form the finite sum using addition and natural multiplication. Induction on the list, with the power identity for each summand and the product identity for addition, gives the asserted ideal. Equality of ideals then gives equality of effective divisors. This argument permits the empty list, zero coefficients and repeated hypersurfaces. □

12 Removing the residual of an SNC-supported divisor

Włodarczyk's Step 2 defines the monomial part using the irreducible components of the boundary divisors. Here the boundary is a finite list of nonempty connected closed smooth hypersurfaces with simple normal crossings. The actual residual produced in Theorem 4.17 is shown to be the unit ideal when the original divisor is supported on that boundary. No residual factorization or unit property is an input to this argument. Disconnected prescribed boundary members require component refinement; this theorem does not assign one multiplicity to such a member.

Lemma 12.1 (A point on exactly one coordinate hyperplane)**.** *For $j \in \{1, \ldots, n\}$ and $r > 0$, there is a point $z \in B(0, r)$ with $z_j = 0$ and $z_i \neq 0$ for every $i \neq j$.*

Proof. Let $v_j = 0$ and $v_i = 1$ for $i \neq j$, and put $\delta = r/(\|v\| + 1)$ and $z = \delta v$. Then $\delta > 0$ and $\|z\| = \delta\|v\| < \delta(\|v\| + 1) = r$. The asserted coordinate values follow immediately. When $n = 1$, this construction gives $z = 0$, which satisfies the same conditions. □

Lemma 12.2 (A nonempty supported divisor has a local boundary component)**.** *Let R be an effective divisor on a complex n-manifold, and let E be a finite SNC list. Suppose* $\operatorname{Supp} R \subseteq \bigcup_{W \in E} W$*. If $x \in \operatorname{Supp} R$, there is $W \in E$ containing x such that W is contained in* $\operatorname{Supp} R$ *in a neighbourhood of x.*

Proof. Invertibility supplies a holomorphic generator g of the actual ideal of R on an open neighbourhood U of x. At each $y \in U$, $y \in \operatorname{Supp} R$ if and only if $g(y) = 0$: the generator itself belongs to the ideal, and every other member is a holomorphic multiple of it. Choose an adapted SNC coordinate map ψ with $\psi(x) = 0$. Each member of E through x is a single coordinate hyperplane in this chart. Closedness of the finitely many other members lets us shrink the chart to avoid all of them.

The germ $G = g \circ \psi^{-1}$ has zero locus contained in the union of coordinate hyperplanes. The existing holomorphic coordinate factorization theorem therefore gives

$$G(z) = H(z) \prod_{j=1}^{n} z_j^{a_j}$$

near zero, with $H(0) \neq 0$ and $a_j \in \mathbb{N}$. This is a factorization of the genuine generator, obtained by the analytic Nullstellensatz and the primality of coordinate germs. It is not assumed as a divisor certificate. Only containment of the zero locus is needed for this local coordinate factorization.

Since $g(x) = 0$, some $a_j > 0$. Otherwise every factor at zero is one, so $g(x) = H(0) \neq 0$. This argument also rules out the support case in ambient dimension zero, where the coordinate product is empty. Take a sufficiently small point z from Lemma 12.1 for this j. The displayed factorization gives $G(z) = 0$, so $\psi^{-1}(z) \in \operatorname{Supp} R$. Support containment puts that point in a member W of E; our shrinking ensures $x \in W$. Its adapted coordinate equation is $z_k = 0$ for a single k. All coordinates of z except j are nonzero, hence $k = j$. Thus W is precisely the jth coordinate hyperplane locally. The same factorization vanishes at every nearby point of that

hyperplane because $a_j > 0$. Translating back to the manifold proves the local containment of W in the actual support. □

Theorem 12.3 (No boundary component means empty support)**.** *Let E be a finite SNC list whose members are nonempty connected closed smooth submanifolds. If an effective divisor R satisfies* $\operatorname{Supp} R \subseteq \bigcup_{W \in E} W$ *and $W \not\subseteq \operatorname{Supp} R$ for every $W \in E$, then* $\operatorname{Supp} R = \varnothing$.

Proof. Suppose $x \in \operatorname{Supp} R$. The preceding lemma produces a member W through x locally contained in the support. Since the ideal of R is coherent and its cosupport of mark one equals its zero locus, this is the genuine condition $\mathrm{IsDivNear}(I_R, W, 1, x)$. The ideal of an effective divisor has a nonzero stalk at every point. The actual divisor exponent is locally constant on a smooth carrier, and connectedness makes it a single integer a on all of W. Its divisibility characterization at x gives $1 \leq a$. At each $y \in W$ the same characterization now gives $\mathrm{IsDivNear}(I_R, W, 1, y)$; evaluating this neighbourhood condition at y puts y in the mark-one cosupport, hence in $\operatorname{Supp} R$. Consequently $W \subseteq \operatorname{Supp} R$, contrary to the hypothesis. There can therefore be no such x. □

Theorem 12.4 (The actual residual ideal is the unit ideal)**.** *Under the hypotheses of Theorem 12.3, the complete germ ideal of R equals the holomorphic unit ideal: $I_R = \mathcal{O}_M$.*

Proof. The support is the zero locus of the actual coherent ideal. Empty zero locus means that at every x there is a member b of the ideal with $b(x) \neq 0$. The inverse of b is holomorphic near x. Every holomorphic germ f is the holomorphic multiple $(f/b)b$, so it belongs to the original ideal. Conversely every ideal member is holomorphic by coherence. These two inclusions identify every stalk with the holomorphic unit stalk, proving equality of the germ ideals. □

Theorem 12.5 (A genuine monomial factorization of an SNC-supported divisor)**.** (Włodarczyk 2005, Step 2) *Let E be a finite SNC list of nonempty connected closed smooth hypersurfaces, and let D be an effective divisor with* $\operatorname{Supp} D \subseteq \bigcup_{W \in E} W$. *There is a list $A = ((W_1, a_1), \ldots, (W_s, a_s))$, with underlying list E and nonnegative integer exponents, such that*

$$I_D = \prod_{i=1}^{s} I_{W_i}^{a_i}$$

as complete germ ideals at all original points.

Proof. The existing finite connected-component extraction theorem constructs A and an actual invertible residual R, together with $I_D = (\prod_i I_{W_i}^{a_i}) I_R$, $\operatorname{Supp} R \subseteq \operatorname{Supp} D$, and $W \not\subseteq \operatorname{Supp} R$ for every $W \in E$. The support hypothesis on D therefore applies to R. Theorem 12.4 identifies I_R with the unit ideal. Coherence of the genuine finite product and multiplication by the unit ideal remove this factor from the full ideal equality. The empty boundary case yields $I_D = \mathcal{O}_M$ by the same argument. The exponents here are the actual exponents of the successive intermediate divisors in the ordered extraction; no identification with all exponents of the original divisor is asserted by this theorem. □

13 Fresh maximal multiplicity centres

Lemma 13.1 (The reduced ideal has order one at a regular point)**.** *If W is regular at x of dimension $d < n$, then its actual vanishing ideal has order one at x.*

Proof. Every member of its vanishing ideal vanishes at x, so the order is at least one. Choose a local holomorphic flattening of W to $\mathbb{C}^d \times \{0\}$. Its dth coordinate is an actual holomorphic germ in $\mathcal{I}_{W,x}$. A coordinate of a local biholomorphism has a nonzero first directional derivative: composing the derivatives of the chart and its inverse gives the identity. It consequently does not belong to the square of the point ideal, by the forward Taylor-order theorem. The vanishing ideal therefore does not have order at least two. Every integer in its defining order supremum is at most one, proving the asserted equality. □

Lemma 13.2 (High multiplicity lies in the current singular locus)**.** *Let W be closed, $d < n$, and $\mu \geq 2$. The actual cosupport of $(\mathcal{I}_W, \mu)$ is contained in $W \setminus \operatorname{Reg}_d W$.*

Proof. The cosupport for mark μ is contained in the cosupport for mark one, which equals W. If a point there were regular of dimension d, the preceding lemma would give order one, whereas cosupport membership gives order at least $\mu \geq 2$. This is impossible. □

Theorem 13.3 (Actual fresh maximal-multiplicity centre)**.** *Let W be compact analytic, with regular points of dimension $d < n$ dense in W, and let E be SNC. Suppose some point of W has reduced ideal order at least two. Then the actual maximum multiplicity $\mu \geq 2$ and a nonempty smooth centre C are produced internally, with*

$$C \subset W \setminus \operatorname{Reg}_d W, \qquad \operatorname{ord}_c \mathcal{I}_W = \mu \quad (c \in C).$$

This C is admissible for the actual marked ideal $(X, X, E, \mathcal{I}_W, \mu)$; in particular it has normal crossings with the given boundary. The reduced order is at most μ on all of W.

Proof. Cartan coherence and empty interior put $(X, X, E, \mathcal{I}_W, 1)$ in the actual domain. The reduced order is upper semicontinuous by its closed derivative-ideal superlevel sets, so it attains a maximum on compact W. At the point attaining this maximum, initialize instead with empty boundary and mark one. The previous current-reduced-order comparison for the empty tower bounds its order by the finite nonmonomial order of this actual fresh initialization. The maximum is therefore a finite integer μ. The given high-order point forces $\mu \geq 2$.

Changing the mark to μ preserves well-formedness, dimension and nonzero germs. Its cosupport is a nonempty closed subset of compact W: it contains the point of maximum reduced order. The actual canonical invariant attains a maximum on this compact cosupport. Its own admissibility theorem produces the nonempty smooth maximum centre C with normal crossings with E. The preceding high-multiplicity lemma puts C inside $W \setminus \operatorname{Reg}_d W$. Its points have order at least μ by admissibility and at most μ by maximality, hence exactly μ. No centre-position or constant-multiplicity data is assumed. □

Theorem 13.4 (Actual blowup with exact exceptional multiplicity)**.** *Under the preceding hypotheses there are an actual produced centre C, its blowup π, and the actual maximum multiplicity $\mu \geq 2$, such that C is nonempty, admissible, and contained in the present singular locus. The actual controlled ideal $\mathfrak{a}'$ is coherent and satisfies*

$$\pi^* \mathcal{I}_W = \mathcal{I}_D^{\mu} \mathfrak{a}', \qquad D = \pi^{-1} C,$$

where $\mathcal{I}_D$ denotes the actual exceptional invertible ideal. At every point of D, some germ of $\mathfrak{a}'$ does not belong to the vanishing ideal of D. The next strict transform is actually analytic and compact, and the actual accumulated boundary is SNC.

Proof. Produce μ and C by the preceding theorem. Since $C \subset W$ and W has empty interior, the actual blowup-existence theorem applies. Hadamard divisibility produces $\mathcal{I}_W \subset \mathcal{I}_C^{\mu}$; pulling back and taking the colon by the invertible exceptional power gives the stated actual factorization and coherence.

At a point z of the exceptional divisor the order at πz is exactly μ. The exceptional-ideal pushforward theorem says that if every controlled germ at z belonged to the exceptional ideal, the original order at πz would be at least $\mu + 1$. This contradicts the proved constant multiplicity. Its contrapositive produces the required germ. Every germ vanishing on D belongs to the exceptional ideal, by the actual SNC blowup coordinate theorem, so the produced germ also fails membership in the vanishing ideal of D.

The actual test blowup preserves well-formedness, proving that the accumulated boundary is SNC. The actual analytic strict-transform theorem and properness of the blowup give analyticity and compactness of the next strict transform. □

The missing hypersurface step is the actual identity between the reduced strict-transform ideal and the μ-controlled ideal after blowing up this produced centre. Exact exceptional multiplicity is supplied by the actual nondivisibility producer above; reducedness of the whole residual ideal remains to be proved. The maximal-multiplicity centre and blowup are genuine producers, but no finite restarted tower or general G8 completion is asserted here.

14 The genuine locally finite divisor sum

Let D be an actual effective divisor supported on an SNC boundary E. The members of E may be disconnected, and the ambient manifold need not be compact. Multiplicities belong to the genuine connected carriers $\mathcal{C}(E)$, rather than to the original list members. Their local finiteness and the intrinsic monomial identity give an actual divisor sum on a neighbourhood of every point.

Lemma 14.1 (Powers depend on the whole stalk)**.** *Suppose $I_x = J_x$ as complete holomorphic ideal stalks. Then $(I^r)_x = (J^r)_x$ for every $r \geq 0$. In particular, if two closed sets agree near x, all powers of their actual vanishing ideals agree at x.*

Proof. For $r = 0$, both powers are the holomorphic unit ideal. If the assertion holds for r, a generator of $(I^{r+1})_x$ is a product of a member of $(I^r)_x$ with a member of I_x. Both factors belong to their corresponding J stalks, so the product belongs to the generating family for $(J^{r+1})_x$. This also holds for every finite holomorphic linear combination, and hence gives inclusion of the whole product stalk. Interchanging I and J gives equality. For vanishing ideals, equality of the sets near x makes the eventual vanishing condition identical for every holomorphic germ, giving the required original stalk equality. □

Lemma 14.2 (Support of an actual finite hypersurface sum)**.** *For a finite weighted family $A = ((C_i, b_i))$ of closed smooth hypersurfaces, the actual effective sum $G = \sum_i b_i C_i$ satisfies $\operatorname{Supp} G \subseteq \bigcup_i C_i$ and $I_G = \prod_i I_{C_i}^{b_i}$.*

Proof. The genuine hypersurface divisor has ideal I_{C_i} and support C_i. A zero multiple is the zero divisor and has empty support; a positive multiple retains support C_i. Actual divisor addition is multiplication of invertible ideals. Its support is the union of the two supports: at a point outside both zero loci, each coherent factor contains a unit germ, whose product is a unit; at a point in either zero locus, every product germ vanishes. Induction on the finite list proves the support containment. The same induction, using the actual power identity for natural multiples, proves the stated complete ideal identity. □

Lemma 14.3 (One actual coefficient on every genuine component)**.** *In positive ambient dimension there is a single function $a : \mathcal{C}(E) \to \mathbb{N}$ such that, for every $C \in \mathcal{C}(E)$ and every original point $y \in C$, $e_C(I_D, y) = a(C)$. The family of the actual carriers is locally finite.*

Proof. The actual connected-component exponent theorem supplies $a(C)$ and the equivalence $\mathrm{IsDivNear}(I_D, C, r, y) \iff r \leq a(C)$ at every point of C. Invertibility gives a nonzero germ in each stalk of I_D, so the genuine finite divisor exponent characterizes exactly the same divisibility thresholds. Taking $r = a(C)$ and $r = e_C(I_D, y)$ gives the two inequalities and therefore equality. The local finiteness theorem applies to the genuine connected boundary carriers without any compactness assumption. □

Lemma 14.4 (The complete stalk dictionary for a finite local refinement)**.** *Let F be a finite list of distinct genuine components of E containing every component through x. Assign to $C \in F$ the actual original coefficient $a(C)$. Then the entire intrinsic monomial stalk of I_D at x equals the actual product stalk $\prod_{C \in F} I_C^{a(C)}$.*

Proof. Every original member H_i through x has a genuine component C through x. It belongs to F. The two carriers agree on a neighbourhood of x, so their actual vanishing-ideal powers have the same whole stalk. Their genuine divisor exponents are equal by the component-to-parent germ bridge, and are $a(C)$. Conversely, every $C \in F$ through x has an original parent H_i with the same germ at x. Thus the definition of the intrinsic monomial stalk, as simultaneous membership in all the relevant powered vanishing ideals, is unchanged by replacing E with F.

Form the actual effective sum $G = \sum_{C \in F} a(C)C$. The preceding support lemma makes it SNC-supported, and its full ideal is the actual component-power product. The genuine weighted SNC exponent calculation identifies $e_C(I_G, x) = a(C)$ for every C through x, including crossing points and zero coefficients. Hence the intrinsic monomial stalks of I_D and I_G relative to F are identical. The actual SNC-supported divisor identity gives $I_G = \mathcal{M}(I_G, F)$. Combining these complete stalk identities proves the claim. No ideal factorization of D was assumed in this argument. □

Theorem 14.5 (Positive local finite products with original coefficients)**.** *In every ambient dimension, there is one actual coefficient function $a : \mathcal{C}(E) \to \mathbb{N}$ with $a(C) = e_C(I_D, y)$ for all $y \in C$. The carrier family is locally finite. For every x there are an open neighbourhood U and a finite list of distinct actual carriers with positive coefficients, $A = ((C_i, a(C_i)))$, such that for every $y \in U$ and every holomorphic germ represented by g,*

$$g \in (I_D)_y \quad \Longleftrightarrow \quad g \in \left(\prod_i I_{C_i}^{a(C_i)} \right)_y .$$

Proof. First suppose the ambient dimension is positive. Select the single genuine function a above. Local finiteness gives an open neighbourhood U of x meeting only finitely many genuine boundary components. Enumerate those components once to obtain F. Their genuine geometry and the finite-refinement theorem make F SNC. At any $y \in U$, every genuine component containing y meets U and therefore belongs to F. The preceding dictionary applies at every such y. Since D is actually SNC-supported, $I_D = \mathcal{M}(I_D, E)$, which proves the desired whole-stalk equality with the full weighted product on F.

Delete exactly those pairs whose coefficient is zero. Every deleted factor is $I_C^0 = \mathcal{O}$. The remaining products are coherent, so multiplication by the holomorphic unit ideal leaves the entire actual product ideal unchanged. Induction on the weighted list verifies this without changing any stalk. The retained coefficients are positive, remain values of the same global function a, and the retained carriers remain distinct.

In dimension zero, the SNC coordinate condition permits no member through any point. The genuine component family and the support of D are empty. A coherent ideal with empty zero locus is the unit ideal. Choose the unique function on the empty component family, U equal to the ambient manifold, and the empty product. All assertions follow, including the vacuous original-point coefficient statement. □

Corollary 14.6 (The actual local divisor sum)**.** *With the same genuine global function a, each local list A defines an actual effective divisor $G_U = \sum_{(C,b) \in A} bC$. At every point $y \in U$, the complete stalks of I_D and I_{G_U} are equal. Thus these sums give the locally finite effective-divisor dictionary with the original divisor's actual multiplicities, without compactness.*

Proof. Each retained carrier is an actual closed smooth hypersurface, so it defines a genuine invertible hypersurface ideal and effective divisor. The genuine finite-sum ideal identity is $I_{G_U} = \prod_{(C,b) \in A} I_C^b$. Substitute this identity in the whole-stalk equivalence of the theorem. All coefficients and carrier conditions remain those produced there. In dimension zero the local list is empty and the actual sum is the zero divisor. □

15 Actual stabilization of growing canonical sequences

Bierstone–Milman[5], Section 7, and Włodarczyk[3] use local finiteness of the recorded centres to obtain eventual local stabilization. This section proves that bridge for genuine growing canonical sequences. It does not assume, or assert as a consequence of local finiteness alone, that such a globally resolving sequence has already been constructed on every noncompact manifold.

Definition 15.1 (The actual next centre and its proper image)**.** Let s be a growing sequence $X = X_0 \leftarrow X_1 \leftarrow \cdots$ of actual recorded blowup towers T_i. Choose the actual centre $C_i \subset X_i$ and blowup $\pi_i : X_{i+1} \to X_i$ from its one-step extension, so that $T_{i+1} = T_i \cdot (C_i, \pi_i)$. Put $B_i = T_i(C_i) \subset X$.

Proof. The growing sequence supplies an actual one-step extension at every index. Choice selects its centre, and the same witness provides the map, its blowup property, and the exact equality of recorded towers. The definition of B_i is the image of that actual centre, not an exceptional-set surrogate. □

Lemma 15.2 (Counting actual centre intersections)**.** *For $K \subset X$, write $q_i(K)$ for the number of recorded centres meeting the inverse image of K in T_i. Then*

$$q_{i+1}(K) = q_i(K) + \mathbf{1}_{B_i \cap K \neq \varnothing}.$$

In particular $q_i(K)$ is nondecreasing.

Proof. Rewrite T_{i+1} using the actual one-step witness. The recursive definition of the count adds one precisely when $C_i \cap T_i^{-1}(K)$ is nonempty. This is equivalent to nonemptiness of $B_i \cap K$, by taking the image of a witness in one direction and its actual preimage witness in the other. Each increment is either zero or one, so induction on indices gives monotonicity. □

Theorem 15.3 (Compact sets eventually meet no later centre)**.** *Let R be a resolution invariant with per-sequence local finiteness, and let every T_i be canonical for an actual marked ideal in its domain. For every compact $K \subset X$, there is N such that $B_i \cap K$ is empty for all $i \geq N$. In particular only finitely many indices have centres whose images meet K.*

Proof. Per-sequence local finiteness bounds all $q_i(K)$ by an integer m. Their range is a nonempty finite subset of $\{0, \ldots, m\}$, so its maximum is attained at some N. Monotonicity and maximality force $q_i(K) = q_N(K)$ for every $i \geq N$. The count identity then forces every later increment to be zero, which says exactly $B_i \cap K = \varnothing$. The meeting indices are contained in $\{0, \ldots, N-1\}$, so they form a finite set. □

Theorem 15.4 (A locally finite family of closed proper centre images)**.** *Every B_i is closed, and the indexed family $(B_i)_{i \in \mathbb{N}}$ is locally finite on the base manifold X.*

Proof. The actual centre is a closed smooth submanifold by the blowup property. The actual tower projection is proper, hence closed, so B_i is closed. Every point of a complex manifold has a compact neighbourhood K. Only finitely many B_i meet that neighbourhood, by the preceding theorem. This proves local finiteness of the indexed family, including repeated or empty images; finiteness counts actual indices rather than distinct sets. □

Theorem 15.5 (An open neighbourhood misses the whole actual tail)**.** *For every compact $K \subset X$ there are N and an open $U \supset K$ such that $C_i \cap T_i^{-1}(U) = \varnothing$ for every $i \geq N$.*

Proof. Choose N so all B_i with $i \geq N$ miss K. This tail family remains locally finite. Since every member is closed, its union is closed: near any point only finitely many members need to be removed. Let U be the complement of that union. It is open and contains K. Any point of $C_i \cap T_i^{-1}(U)$ would have image in both U and the omitted tail union, a contradiction. □

Theorem 15.6 (The later actual blowups are isomorphisms over that neighbourhood)**.** *For the same N, U, every actual later map $\pi_i : X_{i+1} \to X_i$ is an isomorphism over $T_i^{-1}(U)$. Thus its restriction over U has empty centre, and every finite later composite stabilizes there up to actual isomorphism.*

Proof. The actual blowup is an isomorphism off C_i. Since $T_i^{-1}(U)$ misses C_i, restriction of that actual isomorphism to $T_i^{-1}(U)$ remains bijective and avoids the exceptional locus. The one-step tower witness retains the actual map and blowup proof. Successive restricted maps therefore compose to isomorphisms over U; the restricted-centre trace is empty at all those steps. □

Definition 15.7 (The actual finite tail tower)**.** For every N, j, the actual steps of the growing sequence construct a finite tower $T_{N,j} : X_N \leftarrow \cdots \leftarrow X_{N+j}$ with $T_N \cdot T_{N,j} = T_{N+j}$ as an equality of recorded towers.

Proof. For $j = 0$ take the empty tower on X_N. For $j + 1$ append to $T_{N,j}$ the actual chosen next blowup at index $N + j$. By the one-step witness this is a genuine blowup and has last manifold X_{N+j+1}. Associativity of appending and the induction equality give $T_N \cdot T_{N,j+1} = T_{N+j} \cdot (C_{N+j}, \pi_{N+j}) = T_{N+j+1}$. Thus the construction retains the actual intermediate spaces, centres and maps, not just their composite. □

Theorem 15.8 (The entire actual restricted tail is empty)**.** *The compact-tail neighbourhood U may be chosen so that, for every finite j, the restriction of the actual $T_{N,j}$ to $T_N^{-1}(U)$ has only empty centres.*

Proof. Induct through the actual steps of $T_{N,j}$. The empty tail has no centres. At the next step, the append equality of recorded towers identifies $T_N \circ T_{N,j}$ with the original projection T_{N+j}. Consequently the inverse image of $T_N^{-1}(U)$ at that step equals $T_{N+j}^{-1}(U)$. The actual next centre misses this set by the open-tail theorem. This proves that all actual tail centres avoid the starting open preimage. The genuine restricted-tower calculus identifies such avoidance exactly with having empty centre at every recorded step of the restriction. □

Remark 15.9 (The remaining noncompact existence obligation)**.** To produce a globally resolving sequence one must still glue genuine local canonical towers and prove a scheduling rule that eventually resolves every point. The local driver shrinks open bases around compact sets. Its independently chosen output towers are not already compatible on overlaps. Moreover a global maximum of the invariant need not be attained on a noncompact cosupport, so the compact dependent-choice driver cannot simply be reused. Neither a caller-supplied compatible-tower certificate nor the local finiteness of a given sequence discharges this remaining obligation.

16 Literal centre positions for initially smooth sets

Lemma 16.1 (The current ideal for an initially smooth set)**.** *Let W be a closed smooth submanifold, and let $O \subset X$ be open with $W \subset \overline{W \cap O}$. Every actual multiple test tower of $(X, X, E_0, \mathcal{I}_W, 1)$ whose centres avoid the preimages of O has smooth current strict transform at every prefix. Moreover its current controlled ideal is exactly $\mathcal{I}_{W_j}$.*

Proof. Induct on the recorded tower. The initial equality and smoothness are immediate. At a step after prefix T, induction gives a smooth current $S = W_T$ and ideal $\mathcal{I}_S$. The ambient carrier stays the whole manifold and the mark stays one, so the current cosupport is exactly S. Thus admissibility puts the actual centre C inside S.

Density transports to the strict transform: $S \subset \overline{S \cap T^{-1}O}$. The actual centre avoids $T^{-1}O$, so $S \subset \overline{S \setminus C}$. The geometric strict transform $\overline{\pi^{-1}(S \setminus C)}$ therefore equals the library strict transform, including possible codimension-one components of the centre. The coordinate blowup

theorem for a smooth submanifold containing the centre makes that transform smooth and proves the actual ideal identity

$$\pi^*\mathcal{I}_S = \mathcal{O}(-\pi^{-1}C)\mathcal{I}_{S'}.$$

Its controlled transform with mark one is exactly $\mathcal{I}_{S'}$, by cancellation of the invertible exceptional factor. Strict-transform composition identifies S' with the strict transform for the extended prefix. This proves both induction claims without assuming the current centre containment or current ideal equality. □

Theorem 16.2 (Literal centre position for an initially smooth set)**.** *Under the preceding assumptions, if all centres lie over $W \cap |E_0|$, then every current strict transform W_j is smooth and every recorded centre satisfies $C_j \subset W_j \cap |E_j|$.*

Proof. Apply Lemma 16.1 at each prefix to obtain smoothness and equality of the actual current cosupport with W_j. Admissibility puts C_j in W_j. Its image belongs to $|E_0|$ by the assumed original bad locus. A point outside the current boundary has image outside the initial boundary, by the actual boundary-transform induction. Consequently every point of C_j belongs to $|E_j|$. These are exactly the smooth-current alternative in the literal classical centre condition. □

Theorem 16.3 (Actual boundary resolution with literal smooth-current centres)**.** *Let W be a compact smooth analytic set, regular of fixed dimension $d < n$ at every point. Let E_0 be SNC and assume $W \subset \overline{W \setminus |E_0|}$. There is an actual finite admissible resolution tower over $W \cap |E_0|$ whose final strict transform is smooth of dimension d, has normal crossings with its SNC accumulated divisor, and is transverse to that divisor. At every recorded step the current strict transform is smooth and the actual centre is contained in its intersection with the actual current divisor. Thus this existence theorem satisfies the literal smooth-current alternative at every step, with no centre-containment assumption.*

Proof. The dense pure-dimensional regular locus and $d < n$ give empty interior. Cartan coherence, nonzero germs, and the SNC initial boundary construct the domain of the actual canonical invariant for $(X, X, E_0, \mathcal{I}_W, 1)$. Its cosupport is the given compact W. The actual compact-cosupport driver supplies a finite canonical resolution. Take $O = X \setminus |E_0|$. It misses all initial boundary members, the set W is regular of dimension d there, and $W \cap O$ is dense in W. The first-centre construction for the original boundary existence theorem therefore produces an actual prefix tower whose centres avoid the preimages of O and whose final strict transform has normal crossings with the accumulated divisor.

Every centre lies over W, by the actual cosupport-transport producer, and over $|E_0|$, because it avoids O. Intersect these two proved containments to obtain admissibility over $W \cap |E_0|$. Well-formedness of the actual marked transform gives the SNC output boundary. The earlier density transversality theorem applies, since adjoining $W \cap |E_0|$ to the bad set $|E_0|$ changes nothing. Finally apply Theorem 16.2 to this produced tower: it proves the exact current-stage centre condition for every recorded step. The existence consumer carries actual geometric data, rather than accepting a caller-supplied centre-position certificate. □

The initially singular case of the literal centre-position obligation remains open. Its controlled ideal can retain exceptional branches away from the strict transform, so the preceding smooth-input ideal identity does not extend to that case by dropping smoothness.

17 The coefficients are exponents of the original divisor

A sequential extraction produces genuine nonnegative coefficients, but their meaning as exponents of the original divisor needs a separate proof. Here an actual finite product of powers of distinct SNC members is first put in a genuine coordinate normal form. Its original divisor exponent on each member is then computed. This calculation identifies the coefficients produced by extraction, including zero coefficients and points where several members meet.

Lemma 17.1 (A power of a genuinely principal stalk)**.** *Suppose an ideal I is generated at x by a holomorphic germ u. For every integer $a \geq 0$, the actual stalk I_x^a is generated by u^a.*

Proof. For $a = 0$, both are the holomorphic unit ideal generated by one. Inductively $I^{a+1} = I^a I$, and products of two genuinely principal stalks are generated by the product of their generators. Indeed every generator of the product ideal is a product of a holomorphic multiple of u^a with a holomorphic multiple of u; finite holomorphic linear combinations remain multiples of u^{a+1}. Conversely u^a and u belong to their actual factors, so u^{a+1} belongs to the product and all its holomorphic multiples do as well. □

Lemma 17.2 (The genuine finite product of principal stalks)**.** *Let $A = ((W_i, a_i))$ be a finite list, and suppose each actual power $(I_{W_i}^{a_i})_x$ is generated by a holomorphic germ v_i. Then the complete stalk of the actual product ideal is generated by $\prod_i v_i$.*

Proof. Induct on the ordered list A. The empty product is the holomorphic unit ideal, generated by one. For a nonempty list, the first power ideal is generated by v_1, and the induction hypothesis gives the generator $\prod_{i>1} v_i$ of the tail product. The genuine principal-stalk product calculation from the preceding proof supplies the generator $v_1 \prod_{i>1} v_i$ of the full product. The equality is membership in the complete actual product stalk, including finite holomorphic combinations. □

Lemma 17.3 (The actual weighted SNC product has a coordinate generator)**.** *In positive ambient dimension, let $A = ((W_i, a_i))$ be a finite weighted list whose underlying members are distinct and SNC. At every original point x, there are centred adapted coordinates ψ and distinct coordinate indices $\sigma(i)$ for the members through x, such that near x the actual ideal $\prod_i I_{W_i}^{a_i}$ is generated by*

$$m = \prod_{i : x \in W_i} (\psi_{\sigma(i)})^{a_i}.$$

Proof. Choose an actual adapted SNC chart at x. Closedness and finiteness permit an open shrinking avoiding every member not through x. For each member through x, Hadamard's lemma identifies its actual vanishing ideal with the principal ideal of its corresponding coordinate function $u_i = \psi_{\sigma(i)}$ throughout the chart. Lemma 17.1 identifies its actual a_ith power with $(u_i^{a_i})$. For a member not through x, its vanishing ideal is the unit ideal on the chosen shrinking: every holomorphic germ vacuously vanishes on that member near the point. One generates all its powers. The preceding product lemma gives a genuine generator of the entire product. Factors for inactive members are one and disappear, leaving the displayed coordinate monomial. Distinctness ensures that the coordinate assigned to a list member is unambiguous; SNC gives distinct coordinates for the active members. This produces the full nearby stalk presentation from the actual vanishing ideals, rather than assuming a monomial presentation. □

Theorem 17.4 (The original exponent of a genuine weighted SNC product)**.** *For every pair $(W, a) \in A$ and every original point $x \in W$, the actual exponent of the complete weighted SNC product along W is a:*

$$e_W \left(\prod_i I_{W_i}^{a_i}, x \right) = a.$$

Proof. The preceding theorem produces the actual coordinate monomial generator near x. It is divisible by u_W^a, so its exponent is at least a. To exclude divisibility by u_W^{a+1}, keep $u_W = 0$ and vary the other active coordinates to small nonzero values. Such points of W occur arbitrarily close to x. At these points the factors belonging to other active members are units; the remaining coordinate power has order exactly a because a coordinate has order one and orders of nonzero holomorphic products add. The genuine generator lies in the product ideal, so that ideal cannot have order at least $a + 1$ along W near x. This is the established principal-monomial exponent calculation, applied to the actual presentation just produced. Its lower and upper bounds give

equality, also when $a = 0$. In dimension zero, there is no point of an SNC member by Lemma 18.4, so the quantified membership case cannot occur. No connectedness of a member is needed for this weighted-product calculation. □

Theorem 17.5 (Produced coefficients belong to the original divisor)**.** (Włodarczyk 2005, Step 2) *Let F be a finite list of distinct nonempty connected closed smooth SNC hypersurfaces, and let D be an effective divisor supported on $\bigcup_{W\in F} W$. There is a weighted list A with underlying list F such that*

$$I_D = \prod_{(W,a)\in A} I_W^a, \qquad e_W(I_D, x) = a \quad ((W, a) \in A,\ x \in W).$$

Both the actual factorization and the coefficient identification are produced from these geometric hypotheses.

Proof. The genuine extraction and residual-unit theorem constructs the finite weighted list A and the full original ideal product equality. Its underlying list is exactly F, so distinctness and SNC pass to the underlying weighted list. Apply the preceding theorem to this actual product at every original point of every member. Substitute the proved ideal equality to obtain the exponent of the original D, rather than of an intermediate residual. This proves the two asserted conclusions jointly. An empty boundary gives the empty weighted list and the unit ideal; there are then no coefficient instances to check. Zero coefficients are permitted and mean that the original divisor has exponent zero on that member. □

18 The intrinsic monomial ideal for a disconnected SNC boundary

The members of a finite SNC list need not be connected. Their divisor exponents are locally constant on their smooth carriers, and can differ between connected components. The intrinsic monomial ideal uses these actual stalk exponents and remains globally defined. The following argument identifies it with the actual ideal of every effective divisor supported on the boundary, without assuming any global multiplicity for a disconnected member. It implements the monomial/nonmonomial decomposition of Włodarczyk's Step 2 at the level of genuine ideals.

Lemma 18.1 (The intrinsic monomial ideal is invertible)**.** *Let $n > 0$, let E be a finite SNC list, and let $\mathfrak{a}$ be a coherent ideal with a nonzero stalk at every point. Its intrinsic monomial ideal $\mathcal{M}(\mathfrak{a})$ is invertible.*

Proof. The existing genuine monomial generator theorem supplies, near each x, an open set U and a holomorphic function m generating every stalk of $\mathcal{M}(\mathfrak{a})$ on U. At each $y \in U$, points where m is nonzero occur arbitrarily close to y. If $V \subseteq U$ is a nonempty open set, choose $y \in V$; the latter property supplies $z \in V$ with $m(z) \neq 0$. Thus m vanishes identically on no nonempty open subset of U. These are exactly the local generator and nonvanishing properties of an invertible holomorphic ideal. □

Lemma 18.2 (The intrinsic nonmonomial ideal is the actual colon)**.** *For every germ ideal $\mathfrak{a}$ and finite boundary list E,*

$$\mathcal{N}(\mathfrak{a}) = (\mathfrak{a} : \mathcal{M}(\mathfrak{a})).$$

Proof. Both sides consist, at every point, of holomorphic germs f such that fm belongs to $\mathfrak{a}$ for every germ $m \in \mathcal{M}(\mathfrak{a})$. The product $fm = mf$ identifies the two membership conditions, and hence the complete germ ideals. □

Lemma 18.3 (The nonmonomial part of an effective divisor is invertible)**.** *If $n > 0$ and E is a finite SNC list, the intrinsic nonmonomial ideal of an effective divisor D is an actual invertible ideal.*

Proof. The ideal I_D is invertible, hence coherent with nonzero stalks. The first lemma makes $\mathcal{M}(I_D)$ invertible, and the existing divisor-exponent and Hadamard calculus gives $I_D \subseteq \mathcal{M}(I_D)$ at all original points. The actual colon of two nested invertible ideals is invertible: locally their generators have a holomorphic quotient, and cancellation against the nonzero divisor generator proves that this quotient generates the entire colon stalk. Apply that theorem and the preceding colon identity. □

Lemma 18.4 (An SNC boundary has no point in dimension zero)**.** *On a complex zero-dimensional manifold, every member of a finite SNC list is empty, and the intrinsic monomial ideal of any ideal is the holomorphic unit ideal.*

Proof. If x belonged to a member, an adapted SNC chart would cut that member out by a set of exactly one coordinate. There are no coordinates in dimension zero, so such a coordinate set cannot exist. Thus there is no active member at any point. The definition of the intrinsic monomial ideal consequently imposes only holomorphicity on its germs, which is the unit ideal. □

Theorem 18.5 (The genuine nonmonomial residual is the unit ideal)**.** (Włodarczyk 2005, Step 2) *Let E be a finite SNC list on a complex manifold of any dimension, and let D be an effective divisor with* $\operatorname{Supp} D \subseteq \bigcup_{W\in E} W$*. Then its complete intrinsic nonmonomial ideal is the unit ideal:*

$$\mathcal{N}(I_D) = \mathcal{O}_M.$$

No connectedness of the members is required.

Proof. First suppose $n > 0$. The nonmonomial ideal is invertible by the previous lemma, so it defines an actual effective divisor R. The genuine factorization $I_D = \mathcal{N}(I_D)\mathcal{M}(I_D)$ implies $\operatorname{Supp} R \subseteq \operatorname{Supp} D$: if all germs of its first factor vanish at a point, all germs of the product vanish there as well. If $x \in \operatorname{Supp} R$, the actual local component producer gives $W \in E$ through x with W locally contained in $\operatorname{Supp} R$. The existing maximal-exponent extraction theorem supplies a germ $f \in \mathcal{N}(I_D)_x$ which does not belong to $(I_W)_x$. Coherence propagates membership of this same germ f to nearby points. At every such point in W, local support containment therefore gives $f = 0$. Since f is holomorphic, this says precisely $f \in (I_W)_x$, a contradiction. Thus R has empty support, and a coherent ideal with empty zero locus is the unit ideal, as proved by dividing by a nonvanishing stalk member.

When $n = 0$, the preceding lemma shows that the boundary union is empty, so the actual support of D is empty and $I_D = \mathcal{O}_M$. Every holomorphic germ then belongs to the nonmonomial ideal: its product with any holomorphic monomial-ideal germ still belongs to $\mathcal{O}_M$. Conversely nonmonomial-ideal germs are holomorphic by definition. This proves the asserted equality in dimension zero too. □

Theorem 18.6 (The actual supported ideal equals its intrinsic monomial ideal)**.** *For every finite SNC list E and every effective divisor D with* $\operatorname{Supp} D \subseteq \bigcup_{W\in E} W$*, there is equality of the complete original germ ideals*

$$I_D = \mathcal{M}(I_D).$$

Proof. For positive ambient dimension, substitute the proved unit residual into the genuine factorization $I_D = \mathcal{N}(I_D)\mathcal{M}(I_D)$. The monomial ideal is coherent, so multiplication by the holomorphic unit ideal leaves it unchanged. In dimension zero, the boundary is empty, hence the supported ideal and its monomial ideal are both the unit ideal. This proves the equality at all original stalks in every dimension. The statement permits disconnected members: their actual locally constant exponents are retained in $\mathcal{M}(I_D)$, rather than replaced by unjustified globally fixed integers. □

19 Positions of centres at the current stage

The literal obligation in Bierstone–Milman [4], Theorem 1.6(1), is a statement about each current strict transform: a centre is contained in its singular locus, or the current transform is smooth and the centre is contained in its intersection with the accumulated divisor. The results in this section distinguish the part forced by a marked-ideal tower from the additional producer needed for that literal obligation.

Lemma 19.1 (Transport away from the accumulated divisor)**.** *Let $\mathfrak{m} = (X, N, E_0, \mathfrak{a}, d)$ have closed smooth N and coherent $\mathfrak{a}$, and let T be a multiple test blowup. At a point y outside the accumulated divisor $|E_T|$, membership in the current cosupport is equivalent to membership of $T(y)$ in the original cosupport. In particular, for $\mathfrak{m} = (X, X, E_0, \mathcal{I}_W, 1)$ with W closed analytic, this is equivalent to $y \in W_T$, where W_T is the actual strict transform. Thus* $\operatorname{cosupp}(\mathfrak{m}_T) \subset W_T \cup |E_T|$.

Proof. Induct on the recorded tower. There is nothing to transform at the initial stage. Write a final step as $\pi : Z \to Y$ with centre C. The new divisor contains $\pi^{-1}C$, so $z \notin |E_{T\pi}|$ implies $\pi(z) \notin C$. It also implies $\pi(z) \notin |E_T|$: if $\pi(z) \in H \in E_T$ and $\pi(z) \notin C$, then $z \in \pi^{-1}(H \setminus C)$ and hence belongs to the recorded strict transform of H. Off $\pi^{-1}C$ the controlled transform agrees, locally up to a unit, with the pullback of the old ideal; the blowup is a local biholomorphism. Consequently its cosupport is the preimage of the old cosupport there. Apply the induction hypothesis.

For the final specialization, the original cosupport is exactly W. The exceptional locus of the tower projection is contained in $|E_T|$, by induction using the exceptional-locus composition formula. Away from that locus the strict transform of a closed set is its inverse image. This proves the equivalence. For any point of the cosupport, either it belongs to $|E_T|$ or the equivalence puts it in W_T. □

Lemma 19.2 (Regularity and singularity at an unramified point)**.** *Let $f : Y \to X$ be holomorphic between complex manifolds of the same dimension, $W \subset X$ closed, and $y \notin \operatorname{Exc} f$. For every d,*

$$\operatorname{Reg}_d(W_f, y) \quad \Longleftrightarrow \quad \operatorname{Reg}_d(W, f(y)).$$

Proof. The holomorphic inverse function theorem gives open neighbourhoods of y and $f(y)$ and a holomorphic inverse section s through y. Restricting s to its open domain gives an actual holomorphic map of complex manifolds. The chain rule applied to $f \circ s = \mathrm{id}$ shows that both differentials are bijective. The inverse section therefore transports flattening coordinates in both directions; this identifies regularity of $f^{-1}W$ at y with regularity of W at $f(y)$. The exceptional locus is closed, and on its open complement the strict transform of closed W has the same germ as $f^{-1}W$. Replace that germ in the equivalence. This also transports the corresponding singular loci. □

Definition 19.3 (The recorded-stage predicate)**.** For a predicate $P(T, C)$ on an actual prefix tower and its next centre, StageCentres(P, T) is true for the identity tower. For an extended tower $T\pi$ with recorded centre C, it means StageCentres(P, T) and $P(T, C)$. Thus it checks every actual recorded centre and its own prefix projection and accumulated boundary. No centre-containment evidence is built into the definition.

Theorem 19.4 (Every recorded centre outside the boundary)**.** *For every actual multiple test tower of $(X, X, E_0, \mathcal{I}_W, 1)$, each recorded centre C_j is contained in $W_j \cup |E_j|$. If in addition all centres lie over $W \setminus \operatorname{Reg}_d(W) \cup (W \cap |E_0|)$, then*

$$C_j \setminus |E_j| \subset W_j \setminus \operatorname{Reg}_d(W_j).$$

These are properties of every recorded step, not merely of the final support.

Proof. At a recorded step admissibility says $C_j \subset \operatorname{cosupp} \mathfrak{m}_j$. Apply Lemma 19.1. For the second assertion a point $c \in C_j \setminus |E_j|$ is in W_j and is unramified for the prefix projection. Its image lies in the prescribed original bad set. It cannot lie in the initial boundary: outside the accumulated divisor the image is outside the initial boundary, by the same induction used in Lemma 19.1. Thus its image is singular of dimension d. Lemma 19.2 shows that c is singular in the current strict transform. Induct on the tower to obtain the stated property at every prefix. □

Theorem 19.5 (Compact pure-dimensional resolution with recorded positions)**.** *A compact pure-dimensional analytic set W of dimension $d < n$ has an actual finite test tower, admissible over its singular set, with smooth strict transform of dimension d having normal crossings with the accumulated divisor. Every recorded centre belongs to the current strict transform or the current divisor, and its part outside that divisor is in the current singular locus.*

Proof. Initialize the actual vanishing ideal with mark one and empty boundary. Cartan coherence and the pure-dimensional empty-interior theorem put it in the domain. Its compact cosupport equals W. The canonical compact-cosupport driver produces a finite actual resolution tower. Truncate it before its first centre meeting the regular open set, using the existing identity-principle construction. The resulting actual tower has smooth normal-crossings strict transform, and all its centres avoid the preimage of that regular open set. Thus they lie over the original singular set, which gives admissibility with that bad locus. Apply Theorem 19.4 to the recorded tower. No property of an exceptional centre inside the divisor is silently inferred here. □

Lemma 19.6 (A fresh canonical maximum detects a nonsmooth current set)**.** *Let W be a compact analytic set of pure dimension $d < n$. If W is not a closed smooth submanifold, the actual canonical invariant, initialized on $(X, X, [], \mathcal{I}_W, 1)$, has a nonempty smooth admissible maximum locus C contained in the singular locus of the present W.*

Proof. Cartan coherence and empty interior put the actual vanishing ideal in the domain of the canonical invariant. Its initial cosupport is W. Upper semicontinuity on the compact nonempty set W produces a maximum, and the admissibility theorem produces a closed smooth centre $C \subset W$. Suppose that C meets the regular locus. On the open complement of the singular locus, regularity has constant dimension d; restriction to that open set and invariance under local biholomorphisms make all its cosupport points have the same invariant. Since one belongs to the global maximum, all do. Hence C and W agree on that open set. The regular locus is dense in W, and C is closed, so $W \subset C$. The analytic filling argument in the smooth manifold C, with the pure-dimensional density hypothesis, shows that W and C have equal germs at every point of W. Therefore W is a closed smooth submanifold, contrary to hypothesis. Thus C contains no regular point of W, proving $C \subset \operatorname{Sing} W$. No premise asserts this centre containment: it is produced from the actual canonical maximum and the geometry of its initial cosupport. □

Theorem 19.7 (A genuine singular blowup and the next current ideal)**.** *Let V be a compact irreducible analytic subvariety of dimension $d < n$ which is not smooth. There is a nonempty smooth centre $C \subset \operatorname{Sing} V$ and an actual blowup $\pi : Y \to X$ along C. The actual current strict transform is the carrier of a compact irreducible analytic subvariety V' of dimension d. The new boundary $[\pi^{-1}C]$ has simple normal crossings, and the freshly initialized marked ideal $(Y, Y, [\pi^{-1}C], \mathcal{I}_{V'}, 1)$ belongs to the actual domain of the canonical invariant.*

Proof. Apply Lemma 19.6 to V. The centre has empty interior because it is contained in V and $d < n$. Demailly's explicit blowup construction therefore supplies the manifold Y and map π. Its exceptional image is contained in $C \subset \operatorname{Sing} V$. Choose a regular point of V, which exists for an irreducible analytic subvariety. It is outside that image, so V is not contained in the exceptional image. The strict-transform irreducibility and dimension theorem then constructs V' with the stated carrier and dimension. The strict transform is closed inside $\pi^{-1}V$, which is compact by properness of the actual blowup; hence V' is compact.

The centre is admissible for $(X, X, [], \mathcal{I}_V, 1)$, so its actual marked transform is well formed and its boundary $[\pi^{-1}C]$ is SNC. The present subvariety V' is analytic of dimension $d < n$, so its actual vanishing ideal is coherent and nonzero at every point. These properties, together with the SNC boundary transverse to the whole ambient manifold, construct the domain membership of the freshly initialized ideal. This statement constructs one honest geometric step and its continuation data; it does not assert a bound on repeated fresh initialization. □

The remaining literal producer. None of the preceding results closes the obligation on centres contained in $|E_j|$. It is insufficient to identify the controlled cosupport with the strict transform. For example, the reduced curve ideal $(y^2 - x^3, z)$ in $\mathbb{C}^3$, blown up at the origin, has controlled ideal $(x(y'^2 - x), z')$ in the x-chart, whereas its strict transform has ideal $(y'^2 - x, z')$. The controlled ideal has an additional exceptional branch $\{x = z' = 0\}$, and extracting divisorial common factors does not remove it. This calculation illustrates the missing implication; it is not a claimed counterexample to the actual maximum centre.

A dynamic-current-ideal route can apply the fresh-centre producer to each actual current analytic strict transform, with the actual accumulated boundary. To turn this into finite embedded resolution one must prove an invariant of these freshly initialized current ideals that strictly decreases under those steps, remains compatible with restriction and boundary history, and stops precisely at the smooth normal-crossings locus. The existing decrease theorem concerns controlled transforms of one initial marked ideal and does not prove this reinitialization statement. The second alternative, for smooth current W_j, further needs centres contained in $W_j \cap |E_j|$; initial marked-ideal principalization by itself can instead choose the whole smooth W_j. Both obligations remain explicit.

20 Actual analytic superlevels of coherent ideal-order tuples

The analytic semicontinuity used in Bierstone–Milman[4], Section 13, is stronger than topological upper semicontinuity. Here the superlevels are constructed from genuine coherent ideal orders, including infinite order, before any dimensional component is selected. This proves the finite global ideal-order tuple layer. It does not identify an arbitrary enhanced resolution invariant with such a tuple.

Definition 20.1 (Constructed lexicographic order locus)**.** Fix a closed smooth submanifold N and a finite list of coherent ideals with natural thresholds $L = ((I_1, \mu_1), \dots, (I_s, \mu_s))$. Define $A_\varnothing = N$ and recursively

$$A_{(I,\mu)::L} = \operatorname{cosupp}_N(I, \mu + 1) \cup \big(\operatorname{cosupp}_N(I, \mu) \cap A_L\big).$$

Proof. This finite recursion uses actual order cosupports, with the zero threshold giving all of N. No superlevel analyticity or extension certificate is supplied as data. □

Lemma 20.2 (The actual lexicographic dictionary)**.** *The constructed locus equals*

$$\{x \in N : (\mu_1, \dots, \mu_s) \le_{\text{lex}} (\operatorname{ord}_x(I_1|_N), \dots, \operatorname{ord}_x(I_s|_N))\},$$

where orders lie in $\mathbb{N} \cup \{\infty\}$ *and the thresholds are finite.*

Proof. For an empty list both sides are N. For a nonempty list, the actual order theorem identifies cosupport membership with the inequality on the first extended-natural order. Strictly exceeding a finite threshold μ is equivalent to order at least $\mu + 1$, also for infinite order. Lexicographic comparison of two nonempty lists says either the first entry strictly exceeds its threshold, or it equals its threshold and the tail comparison holds. Equivalently it says that the first entry strictly exceeds its threshold, or is at least its threshold and the tail comparison holds. The recursive formula and induction give this exact equivalence, with membership in N in either branch. □

Theorem 20.3 (Produced global analytic tuple superlevels)**.** *Every constructed order-tuple superlevel is a global analytic subset, without compactness or nonvanishing assumptions on the ideals.*

Proof. The closed smooth N is analytic. At a positive threshold r the actual cosupport is $N \cap V(\mathcal{D}_N^{r-1} I)$, with a coherent derivative ideal, so it is analytic. At threshold zero, actual holomorphicity makes cosupport exactly N, so it is analytic as well. Global analytic sets are closed under finite intersection and union. Induction through the displayed recursion proves the assertion. A zero ideal, hence infinite order, causes no failure of derivative coherence or of this argument. □

Theorem 20.4 (Actual global pure extension of a regular tuple trace)**.** *If the actual order-tuple superlevel A is regular of dimension q at every point of its trace on an open U, there is an internally produced global analytic pure-q set $T \subset A$, with $T \cap U = A \cap U$, regular of dimension q on that trace. It equals the union of A's actual global components of geometric pure dimension q.*

Proof. Construct $T = A^{(q)} = \overline{\mathrm{Reg}_q A}$ using the preceding actual global analytic set. Dimension-part analyticity, geometric purity, closedness and the actual global-component identification give all global assertions. The regular-trace lemma gives exact trace equality, and a regular coordinate neighbourhood in A is also a regular neighbourhood in its dimension part. Singularities of T outside U are not excluded: they remain the separate auxiliary-chain smoothification problem of Bierstone–Milman, Lemma 13.7. □

Theorem 20.5 (The actual canonical invariant's first-entry superlevels)**.** *For positive ambient dimension, let $\mathfrak{m}$ be an actual marked ideal in the invariant domain and let T be any genuine admissible test tower. The locus in N_T on which the first entry of the constructed canonical invariant in positive induction dimension is at least μ is a global analytic subset. It equals the internally constructed one-entry order-tuple locus for the actual nonmonomial ideal $\mathcal{N}_{N_T}(I_T)$ and threshold μ.*

Proof. The actual transform remains in the invariant domain: its ideal is coherent, its carrier is closed smooth, the genuine transformed boundary is SNC and transverse, and the ideal is nonzero along its carrier. The nonmonomial coherence theorem, proved by actual local monomial division, therefore gives coherence of $\mathcal{N}_{N_T}(I_T)$. At every point of N_T, its finite natural order used by the constructed invariant equals its extended-natural ideal order. Indeed membership in every actual order cosupport characterizes both orders; equality follows from all finite threshold comparisons.

By the definitions of the genuine dimension-inductive invariant, its first entry is exactly this nonmonomial order. Apply the actual one-entry lexicographic dictionary to identify the stated locus with the produced tuple locus, then apply its global analyticity theorem. This selects no tuple as additional caller data and uses no analyticity assumption on the canonical invariant. The higher companion/history entries, the monomial second-entry superlevels, and the Hilbert–Samuel auxiliary-chain invariant of Lemma 13.7 remain separate obligations. □

21 Actual analytic superlevels of the canonical boundary count

The maximal-order step of Włodarczyk[3], proof of Proposition `pr:resol`, Step 1a, uses the number of original boundary members through a point. Its superlevel is globally analytic for a genuine analytic boundary. This supplies another actual canonical invariant layer needed before the analytic extension construction of Bierstone–Milman[4], Section 13.

Lemma 21.1 (Actual finite intersection dictionary)**.** *For a finite list E and $s \in \mathbb{N}$,*

$$\{x : s \le \#\{i : x \in E_i\}\} = \bigcup_{S \subset \{1,\dots,|E|\},\ |S|=s} \ \bigcap_{i \in S} E_i.$$

Indices retain the actual original list occurrences. No choice of intersections or superlevel certificate is part of the input.

Proof. At a point of count at least s, the finite set of indices through that point has a subset S of cardinality exactly s. The point lies in every member indexed by S, so it lies in the displayed union. Conversely, membership in a displayed intersection puts all s indices of S into the actual index set through the point; cardinality monotonicity proves the count bound. For $s = 0$, the empty intersection is the whole ambient space. For $s > |E|$, both sides are empty. Repeated list members do not invalidate this indexed combinatorial identity. □

Theorem 21.2 (Produced analytic boundary-count superlevels)**.** *If every actual boundary member is analytic, each boundary-count superlevel is globally analytic in every ambient dimension, without compactness.*

Proof. Finite intersections of the genuine analytic members are analytic, including the empty intersection, the whole ambient space. The exact dictionary expresses the superlevel as a finite union of those actual intersections. Finite unions are analytic by products of their actual local defining equations. Thus the analytic structure is produced from the original members, rather than inferred from closedness or topological upper semicontinuity. □

Theorem 21.3 (Actual transformed original-member level superlevels)**.** *Let $\mathfrak{J}$ be a genuinely well formed marked ideal in positive ambient dimension, and T a genuine admissible test tower of its actual homogenized coefficient marked ideal $\overline{\mathfrak{J}}$. The locus where the genuine original-member level s_T is at least s is globally analytic.*

Proof. The actual homogenized coefficient marked ideal is well formed. Its genuine transform along T therefore has an actual SNC boundary. Taking the original number of its members gives the genuine list of original strict transforms; this sublist remains SNC. In positive dimension, every actual SNC member is a closed smooth hypersurface, as follows by the genuine singleton member-intersection regularity theorem. Hence each is analytic. The definition of s_T is exactly the actual boundary count of this list, so apply the produced analytic-superlevel theorem. No analytic presentation of the level or transformed members is requested from the caller. □

Corollary 21.4 (Actual masked StepOne first-entry analytic superlevels)**.** *Under the same genuine geometric inputs, the first-entry superlevels of the actual masked StepOne invariant are globally analytic.*

Proof. At special points the actual invariant is (s_T, ∞); at every other point it is $(s_T, \text{actual lower value})$. Consequently its first entry equals s_T at every point. Substitute this identity into the preceding theorem. This connection uses the actual invariant constructor, not a supplied tuple or an analyticity certificate. The lower/history entries and the full auxiliary-chain invariant still require separate analyticity and smoothification producers before general noncompact global resolution can be claimed. □

22 Genuine analytic monomial-invariant superlevels

The monomial step in Włodarczyk[3], Proposition `pr:resol`, Step 2b, uses the maximum of finitely many feasible boundary-subset keys. Actual divisor exponents are locally constant on the corresponding smooth boundary traces. These facts give analytic, rather than merely topological, superlevels for that invariant.

Lemma 22.1 (Actual finite-maximum threshold dictionary)**.** *For fixed actual weights a_i, a finite set S of member indices and mark d, the genuine monomial invariant γ satisfies*

$$v \le \gamma(S, a, d) \quad \Longleftrightarrow \quad \exists K \subset S :\ d \le \sum_{i \in K} a_i \text{ and } v \le \mathrm{key}(K, a).$$

Here v is a genuine nonbottom key value; the key is the actual lexicographic/colexicographic key used by the monomial algorithm.

Proof. The invariant is the maximum of the actual finite image of the feasible subsets. A nonbottom lower bound implies that this finite family is nonempty. Its actual maximum is attained by one of its feasible subsets, which witnesses the right side. Conversely the key of any feasible subset is at most the actual maximum; transitivity proves the lower bound. No optimal or feasible subset is supplied by the caller. □

Lemma 22.2 (Produced local analytic monomial superlevel)**.** *Let I be coherent and nonzero along a genuine closed smooth N, and let the genuine SNC boundary E be transverse to N, in positive ambient dimension. Near each $x \in N$, the actual locus $\{y \in N : v \le \gamma_N(I, E, d, y)\}$ agrees with an internally produced global analytic finite union of actual member intersections.*

Proof. The genuine exponent local-constancy theorem and closedness of missing members give one neighbourhood on which only the members through x can occur and each occurring member carries its actual original exponent at x. Freeze those genuine exponents as a. Select, from the actual finite feasible subsets of members through x, those whose actual key is at least v. For each selected K, take the genuine closed smooth trace $N \cap \bigcap_{i\in K} E_i$. These are analytic by SNC transversality; their finite union is the produced analytic set.

At any nearby $y \in N$, its actual invariant equals the fixed-weight invariant on the actual members through y. The preceding finite-maximum dictionary says the threshold holds exactly when one selected feasible K is contained in that member set, which says precisely that y belongs to the corresponding actual intersection. Both containments use the same actual neighbourhood and all actual stalk exponents. The chosen finite analytic set can depend on x; no globally constant coefficient on a disconnected original boundary member is assumed. □

Theorem 22.3 (Global analytic monomial superlevels)**.** *Every actual monomial-invariant superlevel on N is globally analytic, including the bottom threshold, without compactness.*

Proof. At the bottom threshold the locus is exactly N, hence analytic. For a nonbottom threshold, genuine upper semicontinuity of the monomial invariant and closedness of N prove that the actual superlevel is closed. At each point in that locus, the preceding producer supplies an actual analytic set agreeing with it on a neighbourhood. Restrict that set's actual holomorphic presentation to the common neighbourhood to present the actual superlevel. At a point outside the closed locus, a neighbourhood misses it and the constant function 1 presents the empty germ. These actual presentations prove global analyticity; topological upper semicontinuity is used only for closedness, never for analyticity itself. □

Theorem 22.4 (Actual canonical superlevels at monomial thresholds)**.** *For the genuine constructed canonical invariant along any actual test tower of a marked ideal in its invariant domain, every superlevel on the actual transformed carrier at a monomial threshold $(0, \operatorname{inl} v)$ is globally analytic. It equals the union of the actual first-entry superlevel at 1 and the genuine monomial-invariant superlevel at v.*

Proof. The actual transform remains in the invariant domain, so the genuine closed smooth, coherent, SNC, transverse and nonzero-along hypotheses needed above are internally available. At a point of positive nonmonomial order, the first entry strictly exceeds 0, so the lexicographic canonical value exceeds the monomial threshold regardless of its companion entry. At order zero, the canonical value is exactly $(0, \operatorname{inl}\gamma)$, so the comparison is precisely $v \le \gamma$. This proves the exact union dictionary. Both actual members of that union are globally analytic by the two genuine producers, and finite union proves the result. No tuple, coefficient or enhanced-invariant analyticity certificate is an input. Companion/history thresholds and the auxiliary Hilbert–Samuel chain engine of Bierstone–Milman, Lemma 13.7, remain separate open obligations. □

23 Actual analytic special loci and maximal second-entry thresholds

Włodarczyk[3], Proposition `pr:resol`, Step 1aa and Step 1ba, assigns the maximal second entry precisely to the genuine member-intersection locus and the codimension-one part. The following construction proves analytic superlevels at these thresholds without assuming analyticity of a lower invariant. It is one part of the enhanced-invariant analyticity required by Bierstone–Milman[4], Theorem 13.4 and Lemma 13.7; the remaining companion/history thresholds and the global auxiliary-chain scheduler remain open.

Lemma 23.1 (Smooth selected actual components)**.** *Let H be an actual closed complex submanifold, possibly disconnected and with different dimensions on different components. For any set A, the union of the actual global irreducible components of H contained in A is a closed complex submanifold, hence analytic.*

Proof. Every point of H is regular, so its actual regular-locus components are its connected components. Each is closed, since H is closed. The actual analytic-component family is locally finite; its selected subfamily therefore has closed union. At a point in that union, choose the actual component containing it. The regular-patch theorem supplies an open neighbourhood on which H is contained in this component. The selected union lies in H and contains this component, so their germs agree. Transfer the actual regular patch of H to this union. No finiteness of the global component family is used. □

Theorem 23.2 (Actual member locus as an analytic component union)**.** *Let N be genuinely closed and smooth, let the finite SNC list E be transverse to N in positive ambient dimension, and let $A \subset N$ be analytic. The actual member locus is*

$$\bigcup_{S\subset\{0,\dots,|E|-1\}} \bigcup\{B : B \text{ is an actual global component of } N \cap \bigcap_{i\in S} E_i,\ B \subset A\}.$$

In particular it is globally analytic, without a maximal-level or compactness hypothesis.

Proof. For a point of the actual member locus choose exactly its actual member indices S. The defining neighbourhood implication says that an open piece of the actual component of $N\cap\bigcap_{i\in S} E_i$ lies in A. Analytic continuation along the connected regular-locus component puts the whole component in A; its closedness identifies it with the actual global component.

Conversely, suppose a point lies in a selected component of one such intersection. Every index in S is an actual member through the point. A genuine regular neighbourhood of that component identifies its germ with the whole intersection. Hence any sufficiently nearby point of N lying on all members through the original point lies in this selected component, and therefore in A. This is exactly the defining member-locus condition. Each intersection is genuinely closed smooth by SNC transversality. The preceding producer makes each selected component union analytic, and the actual index set is finite, so its finite union is analytic. Empty intersections and repeated original members cause no change to these actual indexed conditions. □

Lemma 23.3 (Actual maximal second-entry dictionary)**.** *For a well-formed positive-mark maximal-order ideal and an actual test tower of its homogenized coefficient ideal, at every point y of the actual transformed cosupport,*

$$(s,\infty) \le \mathrm{stepOneInvM}(y) \quad\Longleftrightarrow\quad s < \mathrm{level}(y) \text{ or } (s = \mathrm{level}(y) \text{ and } y \text{ is special}).$$

The lower invariant is the actual per-sequence structure; neither a canonical-segment nor a lower-analyticity certificate is assumed.

Proof. The genuine Step 1 value has first entry equal to the actual level. At special points its second entry is the top element. At every non-special point of the actual cosupport, the actual lower-mask membership theorem proves that its lower value is not top. Thus the lexicographic comparison is exactly the stated disjunction. This explicitly excludes the undefined-mask junk top value. □

Theorem 23.4 (Actual Step 1 maximal second-entry superlevels)**.** *Every actual superlevel of the genuine masked Step 1 invariant at* (s, ∞) *on the actual transformed cosupport is globally analytic, in positive ambient dimension, along any actual test tower of a well-formed positive-mark maximal-order ideal.*

Proof. Internally transform the genuine well-formed ideal. Its actual cosupport is analytic by the derivative-ideal construction; the actual original members retain SNC transversality by sublist restriction. For $s \geq 1$ the dictionary identifies the superlevel with

$$(\text{cosupp} \cap \{\text{level} \geq s+1\}) \ \cup \ (\text{memberInterLocus} \cap \{\text{level} \geq s\}).$$

The second union member already lies in the cosupport. For $s = 0$ it identifies the superlevel with the actual positive-level cosupport union the actual codimension-one part along the transformed carrier. The latter is genuinely closed smooth by the maximal-order theorem, hence analytic. The former level superlevels and the produced member locus are analytic. Finite intersections and finite unions prove both cases. Empty cosupports, excessive levels, disconnected members, and zero-dimensional carrier components are included; the ambient dimension assumption is explicit and never inferred by vacuity. □

24 All actual Step 1 thresholds over the zero-dimensional invariant

The canonical induction uses the genuine constant zero-dimensional invariant as its lower base. This permits all Step 1 thresholds, including finite lower second entries, to be handled without a supplied analytic presentation. The construction implements the dimension-zero base of Włodarczyk[3], Proposition `pr:resol`, Step 1. General positive-order companion/history thresholds and the BM13.7 auxiliary-chain scheduler remain open.

Lemma 24.1 (The actual finite lower value is the genuine constant)**.** *Let the lower invariant be the actual per-sequence invariant* $R_0 = \text{ofDimZero}(\Lambda, l_0)$*. At a nonspecial point of the actual cosupport along any actual test tower of the homogenized coefficient ideal of a well-formed positive-mark maximal-order ideal, the genuine historical masked lower value equals the finite value* l_0*.*

Proof. The actual nonspecial lower-mask theorem excludes the junk value ∞, and hence the point lies in its actual historical mask. Unfold the actual masked lower-value definition on this mask. The actual invariant of the lower marked ideal along the actual restricted tail is definitionally the constant l_0; no lower ideal, history, or restricted tower is supplied as a certificate. □

Theorem 24.2 (Whole actual Step 1 thresholds at the lower base)**.** *For the genuine lower base* R_0*, every whole actual Step 1 superlevel at* (s, α) *on the actual transformed cosupport is globally analytic, for arbitrary* $\alpha \in \Lambda \cup \{\infty\}$*. In particular this holds for the actual canonical base* $\text{invFamily}(0)$ *with its natural-number zero value.*

Proof. At a nonspecial cosupport point, the preceding lemma makes the second entry exactly l_0. At a special point it is top. If $\alpha \leq l_0$, the superlevel is precisely the actual cosupport intersected with the actual level superlevel $\{s(y) \geq s\}$. If $\alpha \not\leq l_0$, only special points can contribute at level exactly s, while higher levels always contribute; the entire locus is the already produced actual (s, ∞) superlevel. Both cases are analytic by the genuine derivative-ideal cosupport producer and

the preceding actual geometric superlevel theorems. Unfolding the actual canonical recursion identifies its zero-dimensional base with ofDimZero($\mathbb{N}, 0$).

The comparison includes top, an empty cosupport, empty or repeated original members, excessive level thresholds, disconnected geometry, and zero-dimensional carrier components. Positive ambient dimension is explicit. This proves the whole maximal-order Step 1 base layer; it does not identify arbitrary full positive-order canonical companion thresholds or construct a global noncompact resolving sequence. □

25 All actual canonical superlevels in carrier dimension at most one

The genuine low-dimensional geometry supplies a complete analytic base for the canonical invariant, without an enhanced-invariant presentation certificate. It also proves the local constancy at the zero-dimensional last-member base of Bierstone–Milman[4], Lemma 13.7. The higher-dimensional simultaneous chain engine remains open.

Lemma 25.1 (Actual zero-dimensional analytic geometry)**.** *An actual analytic set of geometric dimension at most zero consists entirely of genuine regular zero-dimensional points and is discrete. The statement includes ambient dimension zero and the empty set.*

Proof. Express the actual analytic set as the union of its actual global regular-locus components. The dimension bound forces the generating regular point of each component to have dimension zero, and every point in that regular-locus component has the same genuine dimension. Its preconnectedness and the actual zero-dimensional regular patch force it to lie in the singleton of its generating point. The closed singleton also contains its closure. Thus any point of the analytic set is that actual generating regular point. Its genuine flattening gives an open neighbourhood meeting the whole set precisely in that singleton, which proves discreteness. □

Theorem 25.2 (An actual infinite whole-cosupport iteration)**.** *A genuine dimension-at-most-one invariant domain internally produces a growing sequence of actual global test towers starting at the empty tower. Every next tower is an actual blowup of the entire preceding transformed cosupport, with the actual map, blowup proof and exact tower equality retained. This theorem makes no termination or local-finiteness assertion.*

Proof. Use as the state the actual pair of a last-stage manifold and a finite tower, together with its true geometric test-tower proof. In each state, internally produce the genuine transformed invariant-domain data. Its whole cosupport is an actual globally admissible centre with empty interior, so the actual blowup existence theorem produces a next state and the exact one-step extension equality. Choose a next state from these actual witnesses and iterate this function from the reflexive tower. The actual iterate equations give the growing-sequence successor data and retain the exact whole-cosupport choice at every step. Every iterate has a genuine test-tower proof from its state. In particular, no preconstructed global tower or compatibility certificate is an input. The further order-countdown and compact-set local finiteness proofs remain separate obligations. □

Theorem 25.3 (All subsets and all functions on the actual zero-dimensional set)**.** *Every subset of an actual zero-dimensional analytic set is a genuine closed complex submanifold. Every function is locally constant on that actual analytic carrier, without a constancy certificate.*

Proof. An arbitrary subset of the genuine closed discrete carrier is closed. At each subset point, its actual singleton neighbourhood makes its germ agree with the original carrier, so transfer the genuine zero-dimensional regular patch. The same singleton neighbourhood proves that any function has its value at that point on every nearby carrier point. No finite global carrier assumption is used. □

Lemma 25.4 (Actual nonzero coherent cosupport drops dimension)**.** *The actual cosupport of a marked ideal in the genuine dimension-at-most-k invariant domain has geometric dimension at most $k-1$ (natural truncated subtraction). In particular, in carrier dimension at most one it has geometric dimension at most zero.*

Proof. The nonzero-along condition makes the actual positive-order cosupport thin in the genuine smooth carrier. Indeed, if an open carrier piece were contained in cosupport, choose a genuine nonvanishing germ from the nonzero-along condition. A real local coherent generator presentation expresses it as a finite holomorphic combination. Every actual local generator vanishes on that open carrier piece because its actual order is at least the positive mark. The expression then vanishes there too, contradicting the actual frequent nonvanishing witness. Apply the genuine thin analytic dimension-drop theorem to the analytic smooth carrier and its dimension bound at most one. The same dimension-drop proof works for every k, using its true smooth carrier dimension bound. The actual cosupport itself is analytic by coherent derivative ideals. □

Theorem 25.5 (Whole actual one-dimensional canonical thresholds)**.** *Along every actual test tower of a marked ideal in the genuine dimension-at-most-one invariant domain, every entire superlevel of the actual constructed canonical invariant* invFamily(1) *on its actual transformed cosupport is globally analytic. All genuine value thresholds, including every companion/history entry, are covered.*

Proof. Internally produce the genuine transformed invariant-domain data along the actual test tower. Its true cosupport is analytic and of geometric dimension at most zero by the preceding producer. The entire actual invariant superlevel is an actual subset of that carrier, so the genuine closed-submanifold subset producer gives analyticity. No upper-semicontinuity or historical-start compatibility certificate is needed because every subset of this actual closed discrete geometric carrier is analytic. No value is replaced or discarded. Higher-dimensional canonical superlevels, the global auxiliary chain, and the noncompact G3 resolving sequence remain open. □

Theorem 25.6 (Actual zero-dimensional centres with a genuine SNC boundary)**.** *Every actual subset of a genuine zero-dimensional analytic set has normal crossings with any genuine SNC boundary. Consequently, every actual subset of the cosupport of a genuine dimension-at-most-one invariant domain is a global admissible centre. No finite carrier or global maximal value is used.*

Proof. At a singleton centre use the actual centred SNC coordinate system for the boundary. Take every ambient coordinate as the singleton's vanishing index set. A point maps to zero precisely when it is the centre point, by the actual inverse coordinate maps and the injectivity of the preferred chart. The boundary coordinate equations and one-coordinate cardinalities are unchanged. At a point of an arbitrary subset of the zero-dimensional analytic carrier, the genuine singleton neighbourhood makes its germ agree with that singleton. Normal crossings transfers under that actual germ equality. For a genuine one-dimensional invariant domain, internally produce the analytic zero-dimensional cosupport. The preceding closed-submanifold subset producer, its actual SNC boundary, and the given subset relation then give all three actual admissibility clauses. □

Theorem 25.7 (A genuine global cosupport blowup step in carrier dimension one)**.** *A genuine marked ideal in the dimension-at-most-one invariant domain admits an actual global blowup along its entire cosupport, and the resulting one-step tower is an actual geometric test tower. This includes the empty cosupport and does not assert termination or local finiteness of its iteration.*

Proof. The preceding producer gives actual global admissibility of the entire cosupport. Its interior is empty by the genuine invariant-domain nonzero condition. Apply the actual analytic blowup existence construction to this closed smooth centre, and use the geometric test-tower step equivalence with the reflexive prefix and this actual admissibility proof. No global maximum, tower compatibility certificate, or termination bound is assumed. □

26 Actual blowup fibres on a smooth carrier of dimension at most one

We use the actual monomial charts of the analytic blowup and the actual strict-transform equations, as in Bierstone–Milman[5], Section 2. The genuine fibre theorem and the actual constructed invariant's per-sequence finiteness internally produce a globally locally finite and locally resolving whole-cosupport iteration in carrier dimension at most one. General higher-dimensional G3 and an analytic final direct-limit space remain open.

Lemma 26.1 (Actual singleton strict-transform fibre in nested flat coordinates)**.** *Suppose an actual blowup along C has actual centred nested coordinates at x, flattening C to the zero plane and N to the k-plane with $k \leq 1$. Then the actual fibre over x in the actual strict transform of N has at most one point.*

Proof. If two points lie in that actual strict-transform fibre, apply the genuine local monomial strict-transform theorem to each point. Each has a monomial index j strictly below k and a local inverse W taking its value in the k-plane. Hence $k = 1$ and $j = 0$ for both points. Choose one actual holomorphic lift of that common monomial chart, retaining its true projection equality. The actual local inverse theorem recovers each point as the lift of its actual W-value. Since the projection is x and the original coordinates are centred, the j-coordinate of this value is zero: the true monomial projection leaves that coordinate unchanged. All other coordinates vanish by membership in the k-plane. Thus both local inverse values are zero, and both points equal the same actual monomial lift of zero. If $k = 0$, the required inequality $j < k$ is impossible, so the fibre is empty. □

Theorem 26.2 (Actual blowup injectivity on a smooth curve carrier)**.** *Let $C \subseteq N$ be an actual centre whose every point is genuinely regular of dimension zero. Let N be an actual smooth carrier whose every point has genuine dimension at most one. For any actual blowup along C, its actual projection is injective on the actual strict transform of N. The centre may be infinite and disconnected and the carrier may have mixed zero- and one-dimensional components.*

Proof. For two strict-transform points with the same projection, first consider a projection outside C. The actual isomorphism of the blowup off its centre makes the points equal. Over C, the common projection is a point of N. Use its genuine regular dimension, at most one, and the genuine regular dimension zero of C to internally produce actual centred nested flattening coordinates for $C \subseteq N$. Apply the preceding actual fibre theorem. All choices are generated by the genuine submanifold geometry; no fibre injectivity or compatibility certificate is an input. □

Theorem 26.3 (Genuine dimension-one invariant domains produce strict-carrier injectivity)**.** *For a genuine dimension-at-most-one invariant domain, every subset of the actual cosupport is genuinely regular of dimension zero at its points. Every actual admissible blowup is therefore injective on its actual strict carrier, and every actual geometric test-tower projection is injective on its actual transformed carrier.*

Proof. The internally produced zero-dimensional analytic cosupport has a genuine singleton neighbourhood at each of its points. Any actual subset containing that point has the same germ, so transfer the actual regular dimension zero. The genuine invariant-domain smooth carrier has pointwise dimension at most one, and an admissible centre is a subset of this actual cosupport. The preceding monomial producer gives injectivity for one actual blowup. For a finite actual test tower, induct on its recorded blowups. Internally produce the genuine domain data at its preceding stage. The last blowup is injective on the new actual strict carrier. Its projection carries that strict carrier into the preceding actual carrier, by continuity and closedness of the latter. Apply the induction hypothesis to the equal composite projections and then the last-blowup injectivity. The empty tower is the identity. □

Theorem 26.4 (Actual restricted cosupport stays a subsingleton)**.** *If an open subset of the initial genuine dimension-at-most-one domain meets its actual cosupport in at most one point, the actual cosupport along the restriction of every actual test tower to that open set is a subsingleton.*

Proof. The genuine restriction dictionary identifies the restricted cosupport with the actual trace of the transformed cosupport. Its projection lies in the initial cosupport and in the given open subset. Thus any two of its points have equal projected initial points by the actual initial subsingleton property. They lie in the actual transformed carrier, where the preceding internally produced tower-projection injectivity makes them equal. Equality of their open-subset representatives follows from subtype injectivity. Neither a canonical-prefix assumption nor a singleton-fibre certificate is needed. □

Theorem 26.5 (Whole-cosupport prefixes are genuinely canonical on each isolated branch)**.** *For a genuine dimension-at-most-one domain, a finite actual test tower whose every centre is the entire preceding actual cosupport restricts to a genuinely canonical tower on each initial open singleton-cosupport neighbourhood. This holds for any actual value function on those restricted towers.*

Proof. At every restricted prefix the preceding theorem gives a true subsingleton cosupport. Every function attains its maximum at every point of a subsingleton set, so its actual maximum locus equals that entire set, including the empty case. The actual restriction dictionary identifies the next actual centre trace with this entire restricted cosupport. Induct on the recorded tower: the empty tower is canonical, and every successor centre is the actual maximum locus of the actual restricted value function. No comparison of global and restricted invariant values or empty-step deletion is used. □

Theorem 26.6 (The actual whole-cosupport iteration is finite over each compact set)**.** *Along any actual whole-cosupport growing sequence of genuine test towers of a dimension-at-most-one domain, the actual numbers of blowup centres meeting the inverse image of a given compact set are bounded.*

Proof. The initial actual cosupport is closed and discrete. Its intersection with the given compact set is therefore compact and discrete, hence finite. For each of these finitely many actual points choose its genuine open singleton-cosupport neighbourhood. The actual restriction of the growing sequence is a growing sequence of genuine test towers, and the preceding theorem proves all of its prefixes genuinely canonical for the actual constructed dimension-one invariant of the restricted marked ideal. The real per-sequence local-finiteness theorem of that invariant bounds the centres meeting the compact singleton in this restricted sequence. The genuine finite-tail restriction count theorem, at the initial stage, sums these finitely many actual bounds and covers precisely the initial cosupport over the compact set. Every later transformed cosupport lies over that initial cosupport, so this gives the stated bound for all actual prefixes. No global maximum and no global canonical-prefix property are assumed for the whole-cosupport sequence. □

Theorem 26.7 (An actual locally resolving global whole-cosupport sequence in carrier dimension one)**.** *In every ambient dimension every genuine dimension-at-most-one invariant domain internally produces an actual global growing sequence of test towers blowing up their entire preceding actual cosupport. For every compact subset of the initial manifold, all sufficiently late actual transformed cosupports miss its inverse image. Thus this sequence really resolves locally and its actual centres meet each compact initial subset only finitely often. This is a carrier-dimension-one result; general higher-dimensional G3 and the construction of an analytic final direct-limit space remain open.*

Proof. Internally construct the actual whole-cosupport growing sequence from the genuine domain data. For each compact set the preceding producer bounds its actual prefix centre counts. These natural counts are nondecreasing, so their finite range has a maximum attained at some stage.

All later counts are equal to that maximum. Their actual one-step recurrence increments by one exactly when the actual whole-cosupport centre at that stage meets the inverse image of the compact set. Therefore no such intersection can occur at a later stage. Since the actual next centre is the entire actual transformed cosupport, every later cosupport itself misses that inverse image. Around each base point choose a genuine compact neighbourhood. Its eventual cosupport emptiness bounds every meeting index of the actual centre image family by the corresponding stage number, giving indexed local finiteness even with repetitions or empty steps. This supplies actual local resolving behaviour, rather than merely a locally finite family or a preconstructed tower certificate. In ambient dimension zero, the genuine regular carrier dimensions are at most the ambient dimension, so the domain is genuinely dimension zero. Its actual cosupport is empty throughout every test tower. The actual whole-cosupport iteration therefore has empty centre images and resolves every compact set at stage zero. □

Theorem 26.8 (One actual open resolving tail around every compact set)**.** *The genuine dimension-at-most-one domain internally produces an actual global whole-cosupport growing sequence such that every compact initial subset has one actual open neighbourhood on which every sufficiently late actual prefix is a geometric resolution of the actual restricted marked ideal.*

Proof. Use the internally constructed indexed locally finite, genuinely locally resolving sequence. Each actual centre image is closed: its centre is the actual analytic cosupport, and its actual prefix projection is proper. For a compact initial subset, take a stage after which its inverse image misses every actual cosupport. The union of all later actual centre images is closed, because it is a subfamily of the actual indexed locally finite closed family. Its complement is one actual open neighbourhood containing that compact subset. No later actual cosupport projects into this complement. The genuine transformed-cosupport restriction dictionary therefore makes the cosupport of each sufficiently late actual restricted prefix empty. The actual test-tower restriction theorem supplies its genuine geometric test-tower proof, so each such actual finite prefix is a real resolution over this one neighbourhood. No open-tail compatibility or final-limit manifold certificate is assumed. □

27 Actual local branches of a general normal crossings carrier

A globally self-intersecting hypersurface can have normal crossings without admitting a global list of smooth members. The local notion below describes its carrier by actual coordinates. Local smooth branches and their complete divisor-ideal dictionary are then produced from those coordinates. No global labelling or descent of blowup sequences is asserted.

Definition 27.1 (A general normal crossings carrier)**.** A closed set H has normal crossings if, at every ambient point x, there are centred holomorphic coordinates ψ on a neighbourhood and a set S of distinct coordinate indices such that

$$y \in H \quad \Longleftrightarrow \quad \text{there exists } j \in S \text{ with } \psi_j(y) = 0$$

throughout their actual domain. The set S is allowed to be empty. The definition imposes no global smoothness, connectedness, or irreducibility on H.

Lemma 27.2 (The actual coordinate hyperplanes are SNC)**.** *For a finite set S of coordinate indices in $\mathbb{C}^n$, the list of distinct actual hyperplanes $\{z_j = 0\}$, $j \in S$, is SNC. Their traces in every open subset are SNC as well.*

Proof. Each hyperplane is closed because a coordinate projection is continuous. At any point w, translate the coordinates by w. For a hyperplane through w its coordinate w_j is zero, so the translated jth coordinate vanishes exactly on that hyperplane. The assigned singleton coordinate sets are distinct because the indices in S are distinct. Translation is a genuine holomorphic

equivalence with inverse translation by w, and is centred at zero. These are precisely the actual SNC coordinate conditions. Passing to an open subset preserves closedness relative to that subset and restricts the same adapted coordinates. The construction also applies to $n = 0$, when S and the list are empty. □

Lemma 27.3 (The actual flattening map is a biholomorphism)**.** *Let $\psi = \Phi \circ e$ be actual holomorphic flattening coordinates, where e is the preferred chart and Φ is a local holomorphic equivalence whose source lies in the actual chart target. The map ψ is a biholomorphism from its open actual domain onto Φ's open target.*

Proof. On the actual source define the map by $\Phi(e(y))$. On the actual target define its inverse by $e^{-1}(\Phi^{-1}(w))$. The source containment ensures that this inverse is in the chart source and in the flattening domain. The two inverse equations are exactly the actual chart and local holomorphic equivalence inverse equations. Both maps are holomorphic on their specified open domains, being compositions of holomorphic chart maps and holomorphic equivalence maps. Complex holomorphic maps there are smooth of every order. Thus these actual mutually inverse maps give the stated biholomorphism. □

Theorem 27.4 (Actual smooth coordinate branches)**.** *For every point x of a manifold with a normal crossings carrier H, there is an actual open neighbourhood U and a finite SNC list B of closed smooth hypersurfaces in U, such that*

$$H \cap U = \bigcup_{C \in B} C.$$

The list is the inverse image of distinct coordinate hyperplanes under the actual coordinate biholomorphism. In particular it is produced from the geometric coordinate condition rather than supplied as input.

Proof. Choose the actual coordinates and S from the definition at x, and let U be their actual flattening domain. It is open and contains x. For each $j \in S$ take the actual set $C_j = \{y \in U : \psi_j(y) = 0\}$. Under the preceding actual biholomorphism it is the inverse image of the corresponding coordinate hyperplane in the open target. The coordinate-hyperplane lemma and biholomorphic invariance produce the SNC condition on the whole of U. Each branch is relatively closed and smooth. The defining coordinate equation gives the stated carrier equality at every point of U. Empty S gives the empty branch list and an empty carrier on U. □

Lemma 27.5 (The actual original divisor and exponents on an open subset)**.** *For an actual effective divisor D and an open subset U, its actual restriction D_U has ideal $I_{D_U} = \iota^* I_D$ and support $\iota^{-1} \operatorname{Supp} D$, where $\iota : U \to M$ is inclusion. For every point $y \in U$ and every germ represented by a function g on U, membership in $I_{D_U,y}$ is precisely membership of the extension of g in $I_{D,y}$. For every carrier $C \subset U$ and $y \in C$, its genuine divisorial exponent satisfies*

$$e_C(I_{D_U}, y) = e_{\iota(C)}(I_D, \iota(y)).$$

Proof. An open inclusion is holomorphic and has bijective differential at every point. Pullback therefore preserves actual invertibility, giving the genuine effective divisor D_U. The coherent pullback zero-locus identity gives the asserted actual support.

For complete stalk membership, a function on U may be extended by zero outside U: near an original point of U this is exactly its holomorphic germ. Finite holomorphic linear combinations of pulled-back original ideal germs extend to the corresponding original combinations, and every original germ restricts back. Coherence identifies these spans with the complete original ideal stalk, proving the equivalence for every g.

Actual marked cosupports restrict under open inclusion, for every mark. The inclusion identifies neighbourhood filters at y with their restriction from M. Since the original image of C and

C have exactly matching membership at points of U, divisibility near y along C is equivalent to divisibility near $\iota(y)$ along $\iota(C)$ at every mark. Both actual invertible divisor ideals have nonzero germs in each stalk, so their genuine finite exponents are determined by these identical divisibility thresholds. This proves the claimed original exponent equality. The image carrier need not be globally closed in M; the argument uses its actual germ at a point of the open subset. □

Theorem 27.6 (The full local divisor dictionary for general normal crossings)**.** *Suppose the actual effective divisor D is supported on a general normal crossings carrier H. At every original point x, produce the open subset U and actual SNC branch list B above. The genuine component family $\mathcal{C}(B)$ is locally finite, and there is one function a on this whole family such that*

$$a(C) = e_{\iota(C)}(I_D, \iota(y)) \qquad (y \in C).$$

Every point of U has an open neighbourhood V in U and a finite list of distinct actual carriers $A = ((C_i, a(C_i)))$ with positive coefficients, satisfying at every $y \in V$ the complete original stalk dictionary

$$g \in (I_{D_U})_y \quad \Longleftrightarrow \quad g \in \left(\prod_i I_{C_i}^{a(C_i)} \right)_y \qquad (g : U \to \mathbb{C}).$$

The left-hand side also equals membership of the extension of g in the actual original ideal of D. No compactness or global smooth boundary list is assumed.

Proof. Produce the actual SNC list B from the original coordinate condition, and construct the actual restricted divisor D_U. Its support is contained in $\iota^{-1}H = \bigcup B$ by original support containment and the actual carrier equality. Apply the genuine locally finite SNC divisor-product theorem to this actual divisor and actual branch list. It produces a, local finiteness, and the positive finite lists with the complete stalk equivalence. The preceding exponent transport identifies every produced coefficient with the exponent of the original divisor on the image of the genuine carrier, at every original carrier point. The complete open-stalk equivalence supplies the final original ideal interpretation. All choices are produced from the original geometric normal crossings condition and the actual divisor; no factor or coefficient dictionary is an incoming premise. □

These are genuine local branches and actual divisor-ideal identities. For a global normal crossings boundary whose branches have monodromy or self-intersections, compatibility and descent of canonical blowup sequences still require a separate proof. The local dictionary does not assert that global resolution theorem or close that remaining G9 obligation.

28 Actual normal crossings branch compatibility

The local normal crossings branch construction produces genuine SNC lists on actual open coordinate domains. The following identities and actual-sequence constructions remove dependence on branch labels and make actual divisor transforms commute with open restriction. These are compatibility ingredients; comparison and descent of canonical centres under changes of coordinates remain separate obligations.

Lemma 28.1 (Intrinsic ideals ignore branch labels)**.** *If two finite boundary lists E, F have the same members, the actual monomial and nonmonomial ideals of every ideal I agree for these lists.*

Proof. The monomial ideal consists of holomorphic germs g such that, for every member H through the given point y, the germ g lies in $I_H^{e_H(I,y)}$. This condition depends only on membership in the family of sets, not on the index or ordering. For each index in F, its member occurs at some index in E, so the latter condition implies the former with the identical carrier, exponent and power ideal. The reverse implication uses the reverse membership equality. This proves equality of every complete ideal stalk. The nonmonomial ideal is the actual colon $(I : \mathcal{M}_E(I))$,

so this proved equality of monomial ideals proves equality of their nonmonomial ideals as well. The argument requires no SNC, coherence or nonvanishing premise. □

Theorem 28.2 (Relabelling a genuine admissible blowup sequence)**.** *Let $f : Y \to X$ be an actual finite admissible blowup sequence with initial list E_0, accumulated final list E, and centres over Z. For every permutation F_0 of E_0, there is an actual final list F which is a permutation of E, such that the same actual f is an admissible blowup sequence from F_0, with centres over Z and final list F.*

Proof. Induct on the actual finite blowup sequence. The empty sequence is the identity; take $F = F_0$. For the last genuine blowup $\pi : Y \to Y_1$ with actual centre C, induction relabels the preceding sequence with final list F_1, a permutation of E_1. For each index in F_1, choose the identical actual member in E_1 and assign its adapted coordinate set. Keep the actual centre coordinates and the actual holomorphic chart unchanged. Every adapted equation and singleton-coordinate condition is then the identical old condition, proving normal crossings of the same centre C with F_1. Keep the same actual blowup, preceding projection, and containment of the centre over Z. Their actual step constructor gives a genuine admissible sequence with final list

$$F = \big(\overline{\pi^{-1}(H \setminus C)}\big)_{H \in F_1} +\!\!+ [\pi^{-1}C].$$

Applying the strict-transform set function to the permutation of E_1, F_1 and adjoining the identical exceptional member proves that this actual F is a permutation of the actual final E. The projection and every geometric centre and blowup were preserved in the construction. No comparison of independently selected canonical centres is asserted. □

Lemma 28.3 (Actual finite admissible sequences preserve SNC)**.** *An actual finite admissible blowup sequence starting at an SNC list has an SNC accumulated final list.*

Proof. Induct on the actual sequence. For the identity the accumulated list is the given initial SNC list. At a genuine last blowup, induction makes the preceding accumulated list SNC, and actual admissibility makes its centre have normal crossings with that list. The single-blowup SNC coordinate theorem therefore proves SNC for the list of actual strict transforms followed by the actual exceptional preimage. This is exactly the accumulated list in the finite-sequence construction, proving the assertion. □

Theorem 28.4 (Actual divisor blowup transforms commute with open restriction)**.** *For an actual blowup $\pi : Y \to X$ with genuine smooth centre, an actual effective divisor D on X, and an open subset $U \subset X$, let $V = \pi^{-1}U$ and $\pi_U : V \to U$ be the actual restricted blowup. There is an equality of genuine effective divisors*

$$\pi_U^*(D|_U) = (\pi^* D)|_V.$$

Proof. The original smooth-centre blowup and its actual open restriction are genuine blowups, hence their exceptional loci have empty interior. Their actual pullback ideals are therefore invertible and give the stated effective divisors. Write the actual inclusions as $\iota_U : U \to X$ and $\iota_V : V \to Y$. Their maps satisfy the actual commuting square $\iota_U \circ \pi_U = \pi \circ \iota_V$. Pullback composition for the coherent ideal I_D gives

$$\pi_U^*(\iota_U^* I_D) = (\iota_U \circ \pi_U)^* I_D = (\pi \circ \iota_V)^* I_D = \iota_V^*(\pi^* I_D).$$

These are the complete actual divisor ideals, not just equations of supports or chosen generators. An effective divisor is determined by its invertible ideal, so the ideal equality proves the asserted actual divisor equality. □

Theorem 28.5 (Actual transformed divisors retain the full branch dictionary)**.** *Let D be an actual effective divisor supported on the initial SNC list of a genuine finite admissible sequence $f : Y \to X$. Its actual total transform D' has ideal f^*I_D, and its genuine connected final-boundary component family is locally finite. One actual coefficient function on these carriers equals the exponent of D' at every point of each carrier. Near every point of Y there is a finite positive list of distinct genuine carriers whose ideal-power product agrees with the complete ideal stalk of D' at every point of that neighbourhood.*

Proof. Construct D' by induction on the actual finite blowup sequence. For the identity use D; coherence gives the actual identity pullback equality. For a genuine last blowup π, pull back the preceding actual effective divisor. Its exceptional locus has empty interior, so this pullback ideal is invertible. Coherent pullback composition identifies its ideal with the pullback of the original I_D under the actual composite projection. Its support is the preimage of the preceding actual support. At every actual step, a preimage point on an old boundary member either lies over the centre, hence belongs to the actual exceptional preimage, or belongs to the preimage of that member minus the centre, hence to its actual strict transform. Induction therefore puts $f^{-1}\bigcup E_0$ inside the union of the final accumulated boundary. This proves the required support containment of D'. The preceding SNC induction supplies a genuine SNC final list. Apply the genuine SNC locally-finite divisor theorem to this actual D' and actual final list. Its produced coefficients and positive finite lists give precisely all asserted original D'-exponent and complete-stalk identities. This assumes an actual geometric sequence, but supplies no factorization or coefficient certificate and makes no claim that a canonical sequence has yet been chosen for a general normal crossings carrier. □

29 Reducedness of the actual hypersurface transform

The maximum-multiplicity centre construction produces exact exceptional multiplicity. The remaining hypersurface issue concerns the entire residual ideal, including its reducedness at points of the exceptional divisor. The following local analytic argument detects a repeated factor on the same genuine regular piece as the tests excluding the exceptional divisor. The general current-stage centre assertion of [4], Theorem 1.6(1), is a separate objective.

Lemma 29.1 (Reducedness across a divisor without a common factor)**.** *Let f, e be holomorphic germs at the origin of $\mathbb{C}^{m+1}$, with $f \neq 0$. Suppose no prime factor of f divides e. On a neighbourhood of the origin, suppose that at every x with $e(x) \neq 0$, the actual principal ideal germ $(f)_x$ is radical. Then the original germ f is squarefree.*

Proof. The actual holomorphic germ ring is factorial. Suppose a prime p occurs at least twice in f. Its finite maximum exponent gives $f = p^k b$, where $k \geq 2$ and $p \nmid b$. Choose holomorphic representatives of these original germs; the identity holds on a neighbourhood of the origin. Generic regular differential avoidance for the original prime, with the simultaneous tests e and b, gives points arbitrarily near the origin at which $p(x) = 0$, $e(x)b(x) \neq 0$, and the differential of the same representative p is nonzero in a fixed direction. Choose such a point inside the neighbourhood where the factorization and the off-divisor radicality assertions hold.

Since $b(x) \neq 0$, its reciprocal is holomorphic near x. Thus $p^k = b^{-1}f$ belongs to $(f)_x$. Radicality implies $p \in (f)_x$, so $p = af$ for a holomorphic germ a at x. On the other hand, $f = p^k b$ and $k \geq 2$ imply both $f(x) = 0$ and $df_x = 0$. Differentiating $p = af$ gives $dp_x = 0$, contrary to the selected nonzero differential. Hence every prime has exponent at most one and f is squarefree. If $g^r \in (f)$ for $r > 0$, squarefree divisibility implies $f \mid g$, which proves radicality of the principal ideal. □

Lemma 29.2 (A nonexceptional equation cannot be supported only on the divisor)**.** *Let f be a nonzero actual holomorphic germ at the origin, and suppose the coordinate germ z_j does not divide it. If the zero set of f is contained in $\{z_j = 0\}$ near the origin, then $f(0) \neq 0$.*

Proof. If $f(0) = 0$, its germ is a nonunit. Actual factoriality supplies a prime factor p of f. The coordinate prime cannot be associated to p, since otherwise it would divide f. Therefore $p \nmid z_j$. Genuine prime avoidance produces zeros of p arbitrarily near the origin with $z_j \neq 0$. Choose such a point inside a neighbourhood on which the actual factorization $f = pb$ and the given zero-set containment both hold. There $f = 0$ while $z_j \neq 0$, a contradiction. □

Lemma 29.3 (A whole-ideal witness excludes the coordinate factor)**.** *Let an actual ideal stalk R_0 be generated by a holomorphic germ f. If R_0 has an actual member outside the vanishing ideal of the coordinate hyperplane $\{z_j = 0\}$, then z_j does not divide f.*

Proof. If $f = z_j b$ as holomorphic germs, then f vanishes on that hyperplane near the origin. Every holomorphic multiple of f, and every finite holomorphic linear combination of such multiples, also vanishes there. Thus the entire original stalk R_0 lies in the actual hyperplane vanishing ideal. This contradicts the supplied actual member. □

Corollary 29.4 (The actual coordinate-divisor criterion)**.** *In the preceding lemma, let $e(z) = z_j$. It suffices to assume that the actual coordinate germ z_j does not divide f. Then f is squarefree.*

Proof. The coordinate germ is prime: its nonzero coordinate derivative rules out a factorization into two nonunits, and the actual factorial germ ring turns irreducibility into primeness. A prime factor of f dividing z_j would therefore be associated to z_j. This would make z_j divide f, contrary to the hypothesis. Apply the preceding lemma. □

Lemma 29.5 (The full ideal of a squarefree equation)**.** *For a nonzero squarefree actual holomorphic germ f at the origin, the entire vanishing ideal of its zero germ is the principal ideal (f).*

Proof. The actual radical Nullstellensatz and factoriality give a nonzero squarefree germ r with the same zero germ and with full vanishing ideal (r). The reduced-equation unit-transition theorem applied to the two actual squarefree equations gives $f = ur$ with u a holomorphic unit. Thus f and r divide each other, their principal ideals are equal, and the whole vanishing ideal is (f). □

Lemma 29.6 (Invertibility of the actual controlled hypersurface ideal)**.** *Let $\pi : Y \to X$ be an actual blowup along a closed smooth centre C, and let I be an invertible holomorphic ideal with $I \subseteq I_C^{\mu}$. The actual controlled transform $R = (\pi^* I : I^{\mu}_{\pi^{-1}C})$ is invertible.*

Proof. The exceptional locus of an actual blowup has empty interior. The pullback of the invertible ideal is therefore an actual invertible ideal: its local nonzero generator cannot vanish identically after this pullback, since the blowup is locally biholomorphic off the exceptional divisor. The exceptional divisor ideal and each of its powers are invertible. The given centre divisibility, pulled back through the actual blowup, places $\pi^* I$ inside the exceptional power. The actual colon of nested invertible ideals is invertible: divide their local generators, then use density of the nonzero denominator locus to cancel it as germs. The separate coherent colon-product identity supplies the usual factorization when needed below. □

Lemma 29.7 (The actual off-centre controlled-ideal dictionary)**.** *For any actual blowup $\pi : Y \to X$ along C, any closed set W, any $\mu \geq 0$, and any y with $\pi(y) \notin C$, the entire stalk of the actual controlled transform of (I_W, μ) is the actual reduced vanishing ideal of $\pi^{-1}W$ at y.*

Proof. The exceptional divisor ideal contains 1 at y: the original centre ideal contains 1 near $\pi(y)$, and actual pullback preserves this membership. Induction on μ gives 1 in every exceptional power. Thus a member g of the actual colon belongs to the original pullback ideal by testing the colon condition against 1. The genuine holomorphic inverse of the blowup off its centre identifies the complete pullback ideal stalk with $I_{\pi^{-1}W,y}$. Conversely, if g belongs to this vanishing ideal, multiplying it by any holomorphic member of the exceptional power preserves its vanishing on $\pi^{-1}W$. The same genuine inverse puts each such product in the original pullback ideal, which is

exactly the actual colon condition. This proves both directions for every holomorphic germ, not just for a chosen generator or zero locus. □

Lemma 29.8 (Support equality across a smooth divisor)**.** *Let R be an actual invertible ideal on a complex manifold of dimension $m+1$, let E be a closed smooth hypersurface, and let S be closed with $S \subseteq \overline{S \setminus E}$. Suppose the whole ideal stalks R_x and $I_{S,x}$ agree for $x \notin E$, and that at each $x \in E$ some actual member of R_x lies outside $I_{E,x}$. Then the actual zero locus of R is exactly S.*

Proof. The coherent zero locus of R is closed, and contains $S \setminus E$ by the off-divisor whole-stalk equality. Density therefore puts all of S in this zero locus. Conversely, away from E the off-divisor equality and closedness of S give the required containment. Suppose x belongs to the zero locus and to E, but not to S. In a neighbourhood missing S, choose an actual nondegenerate local generator f for R and genuine flattening coordinates for the smooth hypersurface E. Nondegeneracy makes the coordinate pullback of f a nonzero germ. The actual member outside I_E excludes the exceptional coordinate factor, since the complete principal stalk is preserved in both directions by these holomorphic coordinates. At any nearby point outside E, the actual ideal is the unit ideal, because that point is outside S. Hence the zero germ of f is contained in the exceptional coordinate hyperplane. The preceding nonexceptional equation lemma makes $f(x) \neq 0$, contrary to x being in the original zero locus. This proves the converse containment. □

Theorem 29.9 (The entire reduced ideal across a smooth divisor)**.** *Under the preceding geometric hypotheses, $R = I_S$ as entire holomorphic ideal sheaves, including all stalks on E.*

Proof. The support equality and coherence make every actual member of R vanish on S near its base point, giving $R \subseteq I_S$. Away from E the other inclusion is already the actual off-divisor stalk equality. At $x \in E$, choose the same genuine generator and flattening coordinates. The original whole-ideal witness excludes the exceptional coordinate from the pulled-back generator f. At every nearby point outside that coordinate hyperplane, the principal stalk of f is the actual vanishing ideal of the original S in these coordinates. It is radical: a positive power vanishing on S forces the germ itself to vanish there. Both the generator and arbitrary test germs transport through the same actual local holomorphic inverse; thus this is the whole principal stalk, not merely its zero locus. The coordinate-divisor reducedness theorem therefore proves that f is squarefree at the original point x. Its entire zero-germ vanishing ideal is (f). By the support equality, an arbitrary germ in $I_{S,x}$ vanishes on that same zero germ. It is therefore a holomorphic multiple of f in the coordinates. Pull the multiple back through the actual coordinate inverse. This gives membership in the original complete stalk R_x. Since this holds at every x, the two ideal sheaves are equal. □

Lemma 29.10 (The off-centre dictionary for the actual strict transform)**.** *For a closed W and an actual blowup π along C, at each $y \notin \pi^{-1}C$ the whole controlled ideal stalk, with any mark, equals $I_{\mathrm{strictTransform}(\pi,W),y}$.*

Proof. The exceptional locus lies in $\pi^{-1}C$. On the open complement of that preimage the closed strict transform and $\pi^{-1}W$ have the same membership germ: inverse-image points already belong to the set whose closure defines the strict transform, and the closed strict transform lies in $\pi^{-1}W$. Their whole vanishing ideal stalks agree. Apply the actual off-centre controlled-ideal dictionary above. □

Theorem 29.11 (Actual strict ideal after a maximum-order hypersurface blowup)**.** *Let $W \subset X$ be an analytic hypersurface in an $(m+1)$-dimensional complex manifold, with $W \subset \overline{\mathrm{Reg}_m W}$ and actual invertible vanishing ideal. Let E be an SNC boundary and $\mu \geq 2$ an upper bound on the actual orders of I_W on W. For any actual admissible centre of (X, X, E, I_W, μ) and its actual blowup $\pi : Y \to X$, the controlled ideal is the entire reduced vanishing ideal of $S =$ $\mathrm{strictTransform}(\pi, W)$.*

Proof. Admissibility puts the centre in the order-μ locus. Order is one at each regular point, so the centre lies in $W \setminus \operatorname{Reg}_m W$. The original upper bound and this lower bound make the actual order exactly μ at every centre point. Hadamard centre divisibility gives $I_W \subset I_C^{\mu}$, and the invertibility theorem gives an actual invertible controlled ideal R.

The updated boundary, including $D = \pi^{-1}C$, is SNC, hence its original last member D is a smooth hypersurface. Since the regular locus of W avoids C and is dense, $W \subset \overline{W \setminus C}$. The geometric strict transform $\overline{\pi^{-1}(W \setminus C)}$ is therefore the actual library strict transform S. Its defining dense subset lies in $S \setminus D$, so $S \subset \overline{S \setminus D}$. The actual off-centre dictionary supplies $R_y = I_{S,y}$ off D.

Exact order μ excludes order $\mu + 1$ at the image of every point of D. The actual exceptional pushforward theorem consequently produces, at each such point, an element of the controlled ideal which is not in I_D. This is a whole germ membership assertion, including at points where its value vanishes. The reducedness-across-the-divisor theorem now applies with all its inputs internally supplied, and gives $R = I_S$ as complete ideal sheaves. □

Theorem 29.12 (Internally produced next reduced hypersurface ideal)**.** *For a compact analytic hypersurface W satisfying the preceding density and invertibility conditions, an SNC boundary, and an actual point of order at least two, there exist an internally chosen finite maximum order $\mu \geq 2$, a nonempty admissible centre in the current singular locus with constant order μ, and an actual blowup such that the next current hypersurface is analytic and compact, has invertible vanishing ideal, and satisfies*

$$\pi^* I_W = I^{\mu}_{\pi^{-1}C} I_S, \qquad \text{controlledTransform}(I_W, \mu) = I_S.$$

The transformed old boundary together with the actual new divisor is SNC.

Proof. The actual compact upper semicontinuous reduced order function attains its finite maximum. Use the actual canonical maximum locus of the marked invariant to produce its nonempty admissible centre; the maximum-order centre construction supplies the global bound and exact order there. Resolve the actual smooth centre by the actual blowup constructor. Apply the preceding whole-ideal theorem; its invertibility and the ordinary coherent controlled-product identity give the displayed factorization with the actual I_S. Analyticity and compactness of S are the actual blowup strict-transform theorems. The actual admissible transform supplies the updated SNC list. □

The first lemma's radicality is local. In the actual blowup producers it follows from the original reduced ideal through the blowup's off-centre biholomorphism; no reduced-transform certificate is assumed. The arbitrary pure codimension-one local equation producer and the finite restarted multiplicity descent remain separate open requirements. The general G8 assertion in higher codimension remains open.

30 Original reduced ideals of arbitrary pure hypersurfaces

The local equations and Cartan coherence theorems of [2], Chapter II, give the complete original ideal of a pure analytic hypersurface. The finite local collection used below consists of the actual global irreducible components meeting a neighbourhood. Reducible, disconnected and noncompact hypersurfaces are allowed.

Lemma 30.1 (Actual finite original component equations)**.** *Let W be analytic in a complex manifold of dimension $m + 1$, with $W \subseteq \overline{\operatorname{Reg}_m W}$. At every ambient point x there is a nonzero holomorphic germ whose zero set is the entire original centred chart germ of W.*

Proof. The actual global irreducible components of W are locally finite and cover W. Choose an open neighbourhood U of x meeting only finitely many of them. A global component is the

closure of an actual connected component of a regular locus, generated at a point $c \in \operatorname{Reg}_k W$. The original regular density implies that every point of $W \setminus \operatorname{Reg}_m W$ is nonregular in every dimension: a regular flattening of dimension k gives a relatively open regular-k piece, and density gives a regular-m point in that piece, forcing $k = m$ by regular dimension uniqueness. Therefore c is regular of dimension m. The closure of its regular-locus component is an actual closed irreducible analytic subvariety of dimension m, with the same genuine generating regular point. Its codimension-one local-equation theorem supplies a nonzero holomorphic equation for the entire component chart germ at x, including when x lies outside that component.

Apply this theorem to each of the finitely many components meeting U and take the finite product of their equations. Holomorphicity and all individual chart-germ equalities hold on a common neighbourhood. The holomorphic germ ring is a domain, so the product germ is nonzero. Its zero set is the union of those component zero sets. Local finiteness and the actual global-component cover identify that union with the whole original chart germ of W. If no component meets U, the product is 1 and the original germ is empty. All components and equations are produced from the original analytic set. □

Lemma 30.2 (Actual reduced generators of every nearby stalk)**.** *A nonzero holomorphic equation f near the origin internally produces a smaller neighbourhood and a holomorphic equation r generating the complete vanishing ideal of $\{f = 0\}$ at every point of that neighbourhood. Their zero sets agree there. This includes empty zero germs and dimension zero.*

Proof. Factoriality and the radical Nullstellensatz produce a reduced germ r with the original zero germ and original complete vanishing ideal generated by r at the origin. Local Cartan coherence supplies finitely many holomorphic generators s_i generating all nearby original ideal stalks on one neighbourhood. At the origin each s_i belongs to (r), so choose a holomorphic quotient b_i with $s_i = b_i r$ as germs. Shrink once so that all finitely many quotient identities hold, all their functions are holomorphic, and the original zero-set equality holds. Every nearby ideal germ is a finite holomorphic combination of the s_i, and hence of r. Conversely every multiple of r vanishes on the original zero set. These two tests give the whole stalk equality at every point of the same neighbourhood. The construction of the neighbourhood uses only holomorphic germs and coherence. □

Lemma 30.3 (An original chart equation gives the entire nearby chart ideal)**.** *Suppose the centred chart germ of a set W at x equals the zero germ of a nonzero holomorphic equation f. Then a smaller centred chart neighbourhood has a holomorphic function r generating every original vanishing-ideal stalk there, and its zero set equals the original chart germ there. The neighbourhood lies inside the actual chart target.*

Proof. Apply the preceding reduced-generator theorem to f. The original chart-germ equality holds on a neighbourhood, and implies equality of all vanishing-ideal stalks at nearby points. Intersect with that neighbourhood and with the translate of the open actual chart target, which contains the centred origin. A smaller ball lies in the intersection. Restricting the holomorphic reduced equation and the all-stalk generation tests to this ball proves every assertion. □

Lemma 30.4 (Empty interior gives a nondegenerate original generator)**.** *Let W have empty interior. If at every original ambient point its entire centred chart germ has a nonzero holomorphic equation, then the complete original vanishing ideal of W is invertible.*

Proof. At an arbitrary point choose the original equation and the preceding principal chart neighbourhood. Compose its reduced equation with the actual centred chart map to get a holomorphic function h on an original manifold neighbourhood U. The chart and its inverse transport each complete original ideal stalk in both directions: a germ vanishing on W transports to a chart ideal germ, is a multiple of the reduced equation there, and its holomorphic quotient transports back to a multiple of h. Conversely every multiple of h vanishes on W near that point. The

same actual chart identities give $\{h = 0\} \cap U = W \cap U$. If h vanished on a nonempty open set $Z \subseteq U$, then $Z \subseteq W$; each point of Z would lie in the empty interior of W, a contradiction. Thus the generator is nondegenerate on U in the sense required by an invertible ideal. These neighbourhoods at every original point prove invertibility of the entire original ideal. □

Theorem 30.5 (The actual original pure hypersurface ideal is invertible)**.** *For every analytic W with $W \subseteq \overline{\mathrm{Reg}_m W}$ in a complex manifold of dimension $m + 1$, its complete reduced vanishing ideal is invertible, including at singular points. Compactness and irreducibility are unnecessary.*

Proof. The actual finite component equations produce an entire original nonzero chart equation at every point. Original pure codimension-one regular density implies that W has empty interior: a nonempty open subset of W would meet its dimension-m regular locus, whereas a regular flattening of dimension $m < m + 1$ has empty interior. Apply the preceding all-stalk invertibility criterion to these two internally proved original geometric facts. Every original local equation and reduced generator, including at a singular point, is produced in the argument. □

31 Actual hypersurface maximum-multiplicity phases

This section proves the literal current-centre alternative of Bierstone–Milman[4], Theorem 1.6(1), for compact hypersurfaces. It uses fresh maximum reduced-multiplicity phases and the actual controlled transform of marked ideals as in Bierstone–Milman[5], Section 2. The full higher-codimension Hilbert–Samuel invariant assertion and noncompact locally finite resolution are separate conclusions. This section follows the fresh multiplicity phases used in the proof of embedded resolution for a hypersurface. It concerns the actual current reduced vanishing ideal, not the original marked ideal's nonmonomial part alone. The geometric and analytic lemmas below identify those ideals on every actual recorded prefix of one fixed maximum-order phase.

Lemma 31.1 (Regular density survives a singular-centre blowup)**.** *Let W be closed analytic in a complex manifold, with W contained in the closure of its dimension-d regular locus. Let C be a closed smooth centre contained in W minus that regular locus, and let pi be its actual blowup. Then its actual strict transform S is contained in the closure of its own dimension-d regular locus.*

Proof. The centre misses the entire regular locus, so W is contained in the closure of W minus C. Therefore S is the actual geometric strict transform, the closure of pi inverse-image of W minus C. Fix a point z of that defining inverse-image set and an arbitrary open neighbourhood U of z. Since pi(z) is outside C, z is outside the actual exceptional locus and the image of U intersected with its open complement is a neighbourhood of pi(z). Original regular density supplies a point of Reg_d W in that image, lifted to a point y of U outside the exceptional locus. Its image is outside C since it is regular. The original regular flattening transports through the actual local biholomorphic inverse of pi at y; the local strict-transform membership germ is the inverse-image germ, so y is in Reg_d S. Thus every defining point is in the closure of Reg_d S. Taking closures proves the claimed density on all S, including points of the new divisor. No current purity or transformed regular-density certificate is supplied. □

Lemma 31.2 (The actual derivative maximal-order bound)**.** *For the actual complete vanishing ideal of an analytic set W, a bound mu on its actual order on W implies that its mu-th iterated derivative ideal contains 1 at every ambient point. Conversely this derivative-unit condition for any actual ideal implies the ambient order bound. Under an actual admissible blowup with I contained in the mu-th centre ideal power, the genuine derivative-transform inclusion preserves this unit condition for the actual controlled transform.*

Proof. Containing 1 after mu derivatives prevents order at least mu+1: the iterated Giraud inclusion would force that 1 to vanish to order one. Conversely absence of the order-(mu+1) locus gives actual derivatives of order at most mu whose values do not vanish at each point.

Their generated coherent derivative ideal therefore contains 1 locally. Under the actual blowup, the derivative transform inclusion places the pullback of this iterated derivative ideal inside the mu-th derivative ideal of the actual controlled transform. Pulling back the actual unit and taking mark zero gives the new unit. These are the actual previously proved derivative inclusions and local Taylor witnesses, not a newly introduced bound-preservation certificate. □

Theorem 31.3 (Whole current ideals on actual maximum-order test towers)**.** *Let W be an analytic hypersurface in dimension m+1, with actual invertible vanishing ideal and dense m-dimensional regular locus, and let E be an SNC boundary. Let mu be at least two and bound the actual order of I_W at every point of W. On every actual admissible test tower of the original marked ideal (X,X,E,I_W,mu), every current strict transform is analytic, has dense m-dimensional regular locus and an actual invertible vanishing ideal. Its actual marked transform ideal is exactly its complete reduced vanishing ideal. Its order is at most mu everywhere. Every recorded centre is contained in the singular locus of the actual current strict transform and has order exactly mu.*

Proof. Outside the closed W its actual ideal contains 1, so the given bound extends to the whole ambient space. The derivative maximal-order criterion supplies its actual derivative-unit condition. Induct on the actual recorded blowup tower. The empty tower gives the original data. At a successor prefix, genuine admissibility and the preceding entire ideal equality put its next centre in the order-mu locus of the actual current reduced ideal. As its current order is bounded by mu everywhere, it has exact order mu on that centre. Order one at regular points and mu at least two put the entire centre in the current singular locus, including points on the accumulated boundary.

Apply the previously proved actual one-step whole strict-ideal theorem to that current hypersurface, its actual SNC boundary, its true order bound and the literal next blowup. It proves the next controlled ideal is the complete reduced strict ideal and is invertible. The actual derivative-unit transform maintains the global mu bound; regular density is preserved by the preceding singular-centre lemma. Analyticity comes from the actual blowup theorem. The actual strict-transform composition identity identifies this successor with the strict transform by the original prefix projection. All induction assertions refer to the true recorded prefix and its actual objects. No equality or centre-position data for later stages is supplied by the caller. □

Theorem 31.4 (A genuinely constructed finite multiplicity-drop phase)**.** *Suppose additionally W is compact and has an actual point of order at least two. There exist an internally chosen finite maximum mu, at least two, and a genuinely constructed finite canonical resolution tower of (X,X,E,I_W,mu) such that its entire recorded centres lie in the current strict-transform singular loci with exact current order mu, and its final current reduced ideal has order strictly less than mu everywhere. Its final strict transform is analytic and compact, with dense m-dimensional regular locus, invertible vanishing ideal and actual SNC boundary.*

Proof. Choose the actual finite maximum of the original reduced order internally by upper semi-continuity and compactness. The original pure hypersurface geometry supplies the invariant-domain data for the genuine marked ideal at this mu. Its cosupport is closed and contained in compact W, so the actual compact-support canonical resolution constructor gives an actual finite tower with final cosupport empty. The whole-current-ideal and recorded-centre induction theorem applies to this tower, hence identifies the final cosupport as the order-mu locus of the actual final reduced vanishing ideal. Its emptiness says every actual order is strictly below mu. The actual proper projection gives compactness; the preceding theorem supplies all the remaining geometric conclusions. This is a produced multiplicity drop for a true canonical tower, not an assumed terminating tower or a post-reinitialization comparison. □

Lemma 31.5 (Order one of the actual invertible reduced ideal gives regularity)**.** *Let W be closed in dimension m+1 and have an actual invertible vanishing ideal. At a point x of W where that ideal has order less than two, W is regular of dimension m.*

Proof. Choose the actual invertible local generator f on one neighbourhood U of x, generating every original nearby ideal stalk. It vanishes at x. Its actual preferred-chart differential is nonzero: otherwise every test germ in I_W has the form a f, whose value and first differential vanish at x. The actual Taylor criterion puts every such germ in the square of the point ideal, forcing order at least two, a contradiction. The actual local zero set of f is W: membership in W makes the generator vanish; outside closed W the original ideal contains 1, so local generation prevents f from vanishing. Transfer this equation to the actual chart. It is an actual one-equation holomorphic presentation and its single nonzero differential is linearly independent. The actual holomorphic inverse function theorem gives a flattening to the codimension-one plane, which transports back to a genuine RegularAtDim witness for W at x. No regularity, local equation or post-transform ideal certificate is input. □

Lemma 31.6 (Actual singular-centre phases preserve every original regular open)**.** *Let O be an open subset on which the original W is dimension-m regular at each of its points. Every actual maximum-order hypersurface test tower as above has all its centres avoiding the original inverse-images of O.*

Proof. Induct on the literal recorded tower. The empty tower has no centres. At a successor, the preceding centres avoid O, so the actual prefix projection is locally biholomorphic at each point over O. Its strict transform has the actual inverse-image membership germ there. Hence its regularity is equivalent to the original regularity through that actual inverse, with the same dimension. A point in its next singular centre over O would therefore be both regular and nonregular, a contradiction. This proves avoidance for the actual next centre and completes induction. The hypothesis refers only to the original open set and the original W, not to current-stage position data. □

General higher-codimension centre positions require a further invariant argument. The original pure hypersurface ideal is produced by the preceding codimension-one local-equation construction.

Lemma 31.7 (Actual boundary and centre positions through concatenation)**.** *The actual accumulated boundary of a recorded blowup tower depends only on its original boundary list, not on the original ideal or its mark. Let T be a tower over W and S a tower over its actual final strict transform. If the recorded centres of each are singular in the respective current strict transform and have normal crossings with their actual current boundary, then the concatenated tower has these same literal properties relative to W and the original boundary.*

Proof. For the boundary identity, induct on the recorded tower. At a successor, the actual boundary consists precisely of the geometric strict transforms of its preceding members and the actual exceptional preimage of its centre. Equality of the preceding lists therefore gives equality of the entire successor lists, including ordering and multiplicity. For concatenation, induct on S. The empty tail is T. At each actual next step of S, strict transform composition identifies its current strict transform of the final transform of W with the strict transform of W by the concatenated prefix. The genuine boundary composition identity and the preceding ideal-independent boundary identity identify the two current boundary lists. Thus its actual singular containment and normal-crossing witness are exactly the required witnesses for that concatenated prefix. The preceding induction assertion preserves all earlier recorded centres. □

Theorem 31.8 (Genuine finite multiplicity restarts to a smooth hypersurface)**.** *For a compact analytic hypersurface W in dimension m+1 with invertible actual reduced ideal, dense m-dimensional regular locus, and SNC boundary, a finite tower exists whose final actual strict transform is smooth of dimension m and whose every actual centre lies in the current strict transform's singular locus with normal crossings with the accumulated boundary. It avoids every original open set on which W is dimension-m regular.*

Proof. First assume an integer b bounds the actual original order on W. Induct strongly on b, universally over the actual ambient manifold and the original geometric data. If no point of W has order at least two, the actual order-one lemma proves regularity at every point, and the empty tower suffices. Otherwise choose the actual attained maximum mu internally and apply the genuinely constructed finite multiplicity-drop phase. The attained original maximum satisfies mu at most b. Its final complete reduced ideal has order strictly below mu everywhere, hence is bounded by the integer mu-1, strictly less than b. Its actual final hypersurface is analytic and compact, with invertible actual reduced ideal and dense m-dimensional regular locus, and its actual final boundary is SNC. Apply the induction hypothesis to precisely these actual produced objects. Concatenate the two actual finite towers. The preceding composition lemma preserves the literal singular containment and normal-crossing condition at every recorded centre, and identifies the final actual strict transform. The actual marked-domain and compactness maximum argument supplies an original finite bound internally when needed; the resulting tower's existence does not assume a terminating sequence. Finally, any such singular-centre tower avoids every original regular open by induction: above that open the prior prefix is genuinely biholomorphic, so current regularity transports from the original W and excludes the next singular centre. This proves actual avoidance for the entire constructed tower. □

Corollary 31.9 (Natural irreducible hypersurface input)**.** *For an actual compact closed irreducible analytic subvariety V of dimension m in a smooth ambient manifold of dimension m+1, the preceding finite smoothing tower exists without a supplied invertibility hypothesis.*

Proof. Original irreducible regular-density gives dimension-m regular points dense in V. The pure hypersurface local-equation theorem internally constructs an actual nonzero equation at every original point, including singular points. Its reduced all-nearby-stalk and empty-interior argument proves that the original complete ideal I_V is invertible. Apply the finite-restart theorem to these produced original geometric facts. No current transform equation, post-transform invertibility or terminating sequence is supplied. □

Lemma 31.10 (Density outside the original singular part and boundary)**.** *If dimension-m regular points are dense in closed W and W is dense off a closed initial boundary B, then W is dense off B union its original nonregular part.*

Proof. At any y in W outside B, intersect an arbitrary neighbourhood with the open complement of B. Regular density supplies a dimension-m regular point in that intersection. Thus every such y lies in the closure of regular W outside B. The latter set is contained in W outside B union its nonregular part. Taking closures and using the assumed density off B gives the conclusion. □

Lemma 31.11 (The intrinsic singular part of every produced prefix)**.** *For closed W with dense dimension-d regular locus, an actual recorded tower whose centres lie in each current dimension-d nonregular part has dense dimension-d regular locus at every actual prefix. Consequently each such centre consists of actual current points nonregular in every dimension, not just dimension d.*

Proof. Induct on the actual recorded tower. The empty tower has the original regular density. At the next literal centre apply the genuine singular-centre regular-density lemma to the actual current closed strict transform, its induction-produced density and its actual centre. Strict-transform composition identifies its conclusion with the next actual prefix. This proves actual density at every prefix. To identify its dimension-d nonregular part intrinsically, suppose a point there were regular of dimension k. Its actual regular flattening gives a neighbourhood whose current-set points are regular of dimension k. Density supplies a dimension-d regular point in this same neighbourhood. Uniqueness of regular dimension gives k=d, contradicting nonregularity in dimension d at the original point. Conversely a point nonregular in every dimension is nonregular in dimension d. This proves the exact set equality and applies it to every actual centre; no later-stage purity certificate is assumed. □

Theorem 31.12 (Literal hypersurface embedded centre conditions)**.** *Suppose also the original hypersurface is dense off the original SNC boundary. There is an actual finite admissible embedded resolution whose final strict transform is smooth and transverse to the final SNC boundary. Each actual recorded centre is contained in the singular part of its current strict transform, or that current strict transform is already smooth and the centre is contained in its intersection with the actual current boundary.*

Proof. Construct the actual finite smoothing tower above, whose centres lie over the original singular part. Its final accumulated boundary is contained in the inverse image of the original boundary union original singular part by the true admissible boundary-support theorem. The preceding original density and actual strict-transform density theorem therefore make its actual final smooth strict transform dense off its entire actual accumulated boundary. Apply the already proved genuine smooth-input boundary-resolution tower producer to these actual objects. It constructs an actual finite tail with final normal crossings and transversality, and supplies the literal smooth-current and current boundary containment at every recorded tail centre. Concatenate the actual towers. For a smoothing prefix use its literal current singular containment and the preceding intrinsic dictionary, whose regular density is produced along that actual prefix. For every tail prefix, actual strict transform composition and the ideal-independent actual boundary identity transport precisely its smooth-current and boundary-intersection witnesses to the concatenated original prefix. The actual blowup constructors and normal-crossing witnesses give admissibility throughout. The final strict transform and boundary identities transport the tail's smoothness, dimension, normal-crossings and transversality conclusions. No current-stage centre condition is an input of this existence theorem. □

Corollary 31.13 (Natural compact irreducible hypersurface embedded resolution)**.** *For an actual compact irreducible analytic hypersurface V, with an original SNC boundary off which V is dense, the full embedded resolution and the literal per-stage centre condition above are produced without any supplied original invertibility, initial multiplicity bound, current ideal equality or terminating-sequence certificate. In particular the empty-boundary case needs no boundary-density certificate.*

Proof. Use actual irreducible regular density for the original dimension-m regular locus. The pure hypersurface local-equation, all-nearby-stalk reduced-generator and empty-interior producer constructs the original complete invertible vanishing ideal from that geometric density. Apply the genuine finite-restart and boundary-append theorem to these original facts. For the empty boundary the density assertion is the original closed set contained in its own closure. All later ideal equalities, geometric regularity, finite descent, actual boundary compatibility and recorded centre positions come from that theorem's internal constructors. □

Theorem 31.14 (Natural compact pure hypersurface embedded resolution)**.** *Let W be any compact analytic set of pure dimension m in a complex manifold of dimension $m+1$, including a reducible or disconnected set. Let the original boundary E be SNC and suppose W is dense off its union. There is a finite actual admissible embedded resolution tower with final smooth dimension-m strict transform, normal crossings and transversality with the final boundary, and an isomorphism outside*

$$(W \setminus \operatorname{Reg}_m W) \cup (W \cap \bigcup E).$$

At every literal recorded prefix, either its next actual centre lies in the intrinsic singular locus of the current strict transform, or that entire current strict transform is smooth and the centre lies in its intersection with the actual accumulated boundary. Every centre has normal crossings with that current boundary. Original invertibility, original equations, a multiplicity bound, finite descent and current-stage centre conditions are all internally produced.

Proof. Original pure codimension-one regular density and analyticity produce the complete original invertible vanishing ideal by the actual finite component equations, reduced all-stalk generators and empty-interior argument. Apply the finite-multiplicity-restart and smooth boundary-tail

construction to that original ideal and the given original geometric boundary data. The constructed phases successively lower the actual maximum order of the entire reduced current strict ideal; strong integer induction gives a finite smooth-current tower. Its produced regular density identifies every phase centre with the intrinsic current singular locus. The actual smooth-current boundary-tail producer gives the second centre alternative on each of its recorded prefixes. Composition of the actual strict transforms and actual boundary lists transports both alternatives to all prefixes of the concatenated tower, and transports the final smoothness, normal crossings and transversality. The produced admissible sequence is an isomorphism off the displayed original bad set. Reducibility needs no additional ideal input because the original pure hypersurface ideal producer already applies to all locally finite actual global components. □

32 The actual current embedding dimension

The rank of the complete reduced current ideal detects singular points even when that ideal has order one. The following construction extends the exact Jacobian criterion of [2], II.4.31, to every embedding-dimension superlevel and to its actual global maximum locus. The finite-minor presentation agrees with the support description in [4], Proposition 10.1; the present construction uses the entire original reduced ideal at every nearby stalk. It is a first geometric layer for the current-centre condition in [4], Theorem 1.6(1). It does not assert that this first value decreases under arbitrary singular-centre blowups; normal flatness and the higher Hilbert–Samuel entries remain necessary.

Definition 32.1 (Embedding dimension of the complete original ideal)**.** For a set A in an n-dimensional complex manifold and x in the ambient manifold, let $e(A,x)$ be the dimension of the common kernel, in the actual preferred chart at x, of the differentials of all holomorphic germs in the complete vanishing ideal $I_{A,x}$.

Lemma 32.2 (An actual chart-independent finite value)**.** *The same value is the dimension of that actual common kernel in any chart containing x. It lies between 0 and n. At a genuine regular point of dimension d it equals d.*

Proof. The actual chart-transition derivative is an invertible linear map. The chain rule carries the common kernel for all original ideal germs to the common kernel in the other chart. Restricting this invertible derivative gives a linear isomorphism between the two common kernels. This proves equality of their dimensions. The common kernel is a subspace of $\mathbb{C}^n$, so its dimension is at most n. At a regular point the actual flattening identifies it with the image of the differential of the dimension-d parametrization. That differential is injective and its range has dimension d. □

Lemma 32.3 (The actual complete-ideal rank dictionary)**.** *Let holomorphic functions $s_1,\dots,s_N$ generate the entire ideal of A at every point of an open neighbourhood U and cut out A there. At $x \in A \cap U$, in every genuine chart containing x,*

$$\dim \operatorname{span}\{ds_i(x) : 1 \le i \le N\} + e(A,x) = n.$$

Proof. Each s_i is an original ideal germ, so a vector in the kernel of every original ideal differential lies in every $\ker ds_i(x)$. Conversely, an arbitrary ideal germ has a finite holomorphic expression $\sum_i a_i s_i$ near x. Since all $s_i(x) = 0$, its differential is $\sum_i a_i(x) ds_i(x)$. Thus the common kernel of all actual ideal differentials is exactly the common kernel of these finitely many $ds_i(x)$. Chart independence identifies its dimension with $e(A,x)$. Rank-nullity for the linear map having these covectors as its rows gives the displayed equality: its rank equals the dimension of their span. □

Lemma 32.4 (Every actual embedding-dimension superlevel is analytic)**.** *Let A be an arbitrary analytic set. For every integer $q \ge 0$, the actual whole locus $\{x \in A : e(A,x) \ge q\}$ is analytic, without pure dimensionality, a supplied equation family or a constant-rank certificate.*

Proof. If $q > n$ the actual subspace dimension bound makes the locus empty. Otherwise, at an arbitrary ambient point, produce actual Cartan generators $s_1, \dots, s_N$ of the entire original ideal on one open neighbourhood. They cut out A on that neighbourhood. At a point of A, differentiating an arbitrary holomorphic ideal combination gives a linear combination of the ds_i, since all s_i vanish at that point. Thus the actual common kernel of all ideal differentials is exactly the common kernel of these finitely many differentials. Rank-nullity gives

$$e(A, x) + \dim \operatorname{span}\{ds_i(x)\} = n.$$

Consequently $e(A, x) \geq q$ if and only if the latter rank is less than $n - q + 1$, equivalently all $(n - q + 1)$-minors of the actual Jacobian vanish. In the true chart this whole locus is therefore cut out by the original generators and the finite family of these minors. Their derivatives and determinants are holomorphic on the same neighbourhood, so this is a genuine local analytic presentation. The chart-independent dimension equality proves it represents the whole stated intrinsic locus. For $q = 0$ the excessive-size minors are automatically zero and the locus is precisely A; empty generator families and $n = 0$ are included in the same finite presentation. □

Lemma 32.5 (The entire pure singular locus is the first strict superlevel)**.** *Let A be analytic and pure of dimension $d \leq n$. At each $x \in A$, regularity of dimension d is equivalent to $e(A, x) \leq d$. Hence $A \setminus \operatorname{Reg}_d A$ is exactly $\{x \in A : e(A, x) \geq d + 1\}$.*

Proof. At a regular-d point the preceding actual tangent-space theorem gives $e(A, x) = d$. Conversely if $e(A, x) \leq d$, choose actual all-stalk Cartan generators at x. The true kernel dictionary and rank-nullity give rank at least $n - d$ for their differentials, hence a nonzero $(n - d)$-minor. The actual pure-dimensional Jacobian criterion then produces a flattening of the entire set A at this same point of dimension d. It uses all original generators to identify the selected equations' smooth zero set with A, by regular density and the holomorphic identity theorem. Negating this exact equivalence gives the displayed whole singular-locus equality. □

Theorem 32.6 (An internally produced singular maximum locus)**.** *If a pure analytic set A of dimension $d \leq n$ has a singular point, its actual embedding dimension has an internally attained global maximum $q > d$. Its entire maximum locus is nonempty, analytic, and contained in the actual current singular locus of A. No compactness, attained-maximum or selected-component hypothesis is needed.*

Proof. The actual values on A lie in the finite set $\{0, \dots, n\}$. A singular point makes their image nonempty, so this finite image has a maximum q attained at an actual point of A. The singular point's value is at least $d+1$, hence $q > d$. Since every actual value on A is at most q, the maximum locus is exactly the internally produced superlevel $\{x \in A : e(A, x) \geq q\}$. Its analyticity is the preceding genuine finite-minor producer. At each point of the locus $q > d$, so the exact singular-threshold theorem places that point in $A \setminus \operatorname{Reg}_d A$. This is a produced analytic maximum locus; it does not assume that the locus is smooth or claim a resolution/descent theorem from this first value alone. □

33 Whole maximal-order Step 1 thresholds on carriers of dimension at most two

All dimensions below are genuine complex analytic dimensions. The construction uses actual SNC intersections and the identity theorem on the genuine global components, as in Demailly[2], II.5.3. It treats the full actual masked Step 1 value, including its actual historical lower entry, rather than a caller-supplied tuple. This does not yet establish the full general invariant's dynamic companion/history gluing or the higher-dimensional global sequence.

Lemma 33.1 (The genuine remainder of a smooth analytic carrier)**.** *For a closed smooth analytic set H and an analytic set A, let H^- be the union of the genuine global components of H which are not contained in A. Then H^- is closed and smooth, equals H minus the union of the components contained in A, and $A \cap H^-$ is thin in H^-. If H has dimension at most one, $A \cap H^-$ is analytic and has dimension zero.*

Proof. The regular locus of H is H, so its regular connected components are its actual connected components. Each is closed, and at every one of its points a genuine open neighbourhood meets H only in that component. The genuine global component family is locally finite. Every subfamily's union is therefore closed and has, near each of its points, the same germ as H, proving smoothness. Two such components meeting at a point are equal, by the connected-component equality at a common point. Coverage by the actual global components then proves the stated difference identity. If an open O meets H^- and $O \cap H^- \subseteq A$, choose the actual unselected component B through a point of that intersection. Its nonempty relatively open trace is contained in A. Analytic continuation on its connected regular component makes the whole component, hence its closure, contained in the closed analytic A, contradicting its unselected property. Thus every such trace is empty, which is exactly thinness. If H has dimension at most one, the remaining union has the same bound by regular submanifold inclusion. Its intersection with A is analytic, and genuine thin analytic dimension drop makes this intersection zero-dimensional. □

Lemma 33.2 (Genuine dimension of a transverse boundary trace)**.** *Let the closed smooth carrier N have pointwise dimension at most two, and let an actual SNC divisor E be transverse to N. Each actual trace $N \cap E_i$ is closed smooth and has dimension at most one.*

Proof. At a point of the trace, genuine simultaneous adapted coordinates flatten the trace and N to coordinate planes of dimensions k_C and k_N and make E_i the hyperplane with coordinate $s_i = 0$, where $s_i < k_N$. The true dimensions of the planes are their ambient-truncated dimensions. Since the trace lies in E_i, its free coordinate plane cannot contain coordinate s_i: otherwise perturb the actual coordinate value in that direction by a small complex scalar. The actual coordinate target is open, so this perturbation remains in the target. The actual inverse chart then produces a trace point whose s_i coordinate is nonzero, a contradiction. Hence $\min(k_C, n) \leq s_i < \min(k_N, n)$. Actual uniqueness of the regular dimension and the genuine dimension bound on N give $\min(k_N, n) \leq 2$, proving that the trace dimension is at most one. □

Lemma 33.3 (Actual isolated maximal-order remainder)**.** *For a coherent positive-mark ideal of maximal order along a genuine closed smooth carrier N of pointwise dimension at most two, its actual cosupport minus its actual codimension-one part is closed analytic and has dimension zero.*

Proof. The genuine codimension-one part is closed and relatively open in the actual cosupport. Its complement in the cosupport is closed; near any of its points it agrees with the analytic cosupport on the open complement of that closed codimension-one part, so it is analytic. At a regular point of this remainder, choose the actual regular dimension of N, at most two. If that dimension is zero, regular submanifold inclusion forces remainder dimension zero. If it is one or two, the genuine maximal-contact filling and codimension-one remainder theorem makes the remainder dimension strictly less than respectively zero or one. The former case is impossible and the latter forces dimension zero. □

Lemma 33.4 (Actual finite unions in analytic dimension zero)**.** *A finite indexed union of genuine closed analytic sets of dimension at most zero is analytic and has dimension at most zero, including an empty union.*

Proof. The actual zero-dimensional analytic sets are closed and discrete, and every point is regular of dimension zero. Their finite union is analytic by finite analytic union and is closed and discrete. At a point of the union, choose an open neighbourhood in which the union is that singleton, and a member containing the point. The member and union have identical germs there, so its genuine

zero-dimensional regularity transfers to the union. Uniqueness of regular dimension excludes all other regular dimensions. Induction on the finite index set proves the general assertion; the empty union has no regular points and its analytic presentation is the constant unit equation. □

Theorem 33.5 (Actual zero-dimensional envelope for all nonspecial points)**.** *For an actual maximal-order well-formed positive-mark Step 1 datum on a carrier of pointwise dimension at most two, and any actual geometric test tower of its actual coefficient marked ideal, there is a genuinely closed analytic zero-dimensional set Z containing every actual nonspecial point of its transformed cosupport.*

Proof. Internally produce the actual transformed coherence, SNC divisor, transversality, maximal order and dimension bound. At level zero, nonspecial points lie in the actual isolated remainder of the preceding lemma. At positive level, choose an actual original boundary member through the point and its actual smooth trace H_i on the transformed carrier. If its genuine global component is contained in the actual cosupport A, a true local component neighbourhood shows that the intersection of all original members through the point, on the carrier, has germ contained in A. The point is special, a contradiction. Hence it lies in $A \cap H_i^-$. Each H_i^- is actual closed smooth of dimension at most one, and its intersection with A is thin by analytic continuation. Genuine thin analytic dimension drop makes this closed analytic intersection zero-dimensional. Take the union of the actual level-zero remainder with the finitely many actual original-member remainders. Each is genuinely closed and discrete; a finite union remains closed and discrete, and is analytic. At each point its singleton germ gives actual regular dimension zero. This is the required actual envelope. No global finite component family or history certificate is an assumption. □

Theorem 33.6 (Every whole actual Step 1 superlevel is analytic in carrier dimension two)**.** *Let the lower invariant be any genuine constructed sequence-resolution invariant. Under the preceding actual Step 1 hypotheses, every superlevel of the whole actual masked Step 1 value, with finite or infinite lower threshold and all actual historical entries retained, is globally analytic.*

Proof. Fix the whole threshold (s, α). Its actual superlevel is the union of the already internally produced whole threshold (s, ∞) superlevel and the points of the former superlevel outside the latter. Special points have infinite lower entry and hence cannot lie in the difference. Actual cosupport nonspecial points lie in the produced zero-dimensional envelope. Every subset of a genuine closed zero-dimensional analytic set is closed smooth and analytic, so this actual difference is analytic. Union with the actual top superlevel proves the assertion. This argument uses the literal actual historical value; it neither assumes local constancy nor replaces that value with a proxy tuple. □

Corollary 33.7 (Actual canonical lower-invariant application)**.** *The preceding complete superlevel conclusion applies directly to Step 1 with the actual canonical dimension-one lower invariant used in the constructed dimension-two recursion.*

Proof. Instantiate the genuine lower invariant by the already constructed canonical dimension-one invariant and retain its literal actual value function. All geometric data and the zero-dimensional envelope were produced independently of any assumption on this invariant's thresholds. The preceding theorem therefore gives exactly its actual Step 1 superlevels. □

34 Intrinsic branches on genuine normal-crossing chart overlaps

Normal-crossing coordinate descriptions need not share a global branch labelling. Their local branches nevertheless match uniquely. The argument uses actual holomorphic germ factoriality and the radical Nullstellensatz, as in [2], Chapter II. It compares the original coordinate functions themselves, without assuming a correspondence of their branches.

Lemma 34.1 (A first-order holomorphic equation is prime)**.** *If a holomorphic germ f at 0 vanishes at 0 and has nonzero complex differential there, then its genuine holomorphic germ is prime.*

Proof. The germ is not a unit because its value is zero. If $f = ab$ and both factors are nonunits, their values at zero are both zero. Choose actual holomorphic representatives of the factors. Their product equals f on a neighbourhood, and the product rule gives $df_0 = a(0)db_0 + b(0)da_0 = 0$, a contradiction. Thus the germ is irreducible. Actual factoriality of the holomorphic germ ring makes every irreducible germ prime. □

Lemma 34.2 (Actual coordinates have distinct prime branches)**.** *Let F be an actual local holomorphic equivalence centred at zero, with $F(0) = 0$. Every coordinate function F_i is a prime holomorphic germ, and germs for distinct coordinates are not associated.*

Proof. Differentiate the actual identity $F \circ F^{-1} = \mathrm{id}$ near zero. It makes dF_0 surjective. Each coordinate row is nonzero, since a preimage of the ith coordinate direction has ith image coordinate one. The preceding lemma proves primeness. If $F_i = uF_j$ for a unit germ u, choose its actual holomorphic representative. Differentiating at zero gives $d(F_i)_0 = u(0)d(F_j)_0$, since $F_j(0) = 0$. Applying this identity to a preimage under dF_0 of the ith coordinate direction gives $1 = 0$ when $i \neq j$. This excludes association. □

Lemma 34.3 (Equal zero unions produce a unique branch correspondence)**.** *Let $(f_i)_{i\in I}$ and $(g_j)_{j\in J}$ be finite actual holomorphic equations whose germs are prime and pairwise nonassociated within each family. If their zero unions agree as germs at zero, there is a unique bijection $\sigma : I \to J$ for which f_i and $g_{\sigma(i)}$ are associated.*

Proof. Fix i. The product $G = \prod_{j\in J} g_j$ vanishes on the zero germ of f_i, because this germ lies in the first union and hence in the second. The radical Nullstellensatz places the germ of G in the radical of (f_i). Since f_i is prime, its principal ideal is prime and hence radical, so f_i divides G. Primeness selects a factor g_j divisible by f_i. Irreducibility of both germs makes them associated. Two possible j's would be associated to one another, contradicting the second family's distinctness. Thus every i has a unique match. Reversing the two families gives a unique match for every j. The two match maps are inverse by the uniqueness just proved, so the first is a bijection, uniquely determined by association. If one index set is empty, its zero union is empty; the same primeness argument prohibits any branch in the other family, and the resulting bijection is empty. □

Lemma 34.4 (Associated original equations have actual holomorphic units)**.** *If two actual holomorphic equations f, g have associated germs at zero, there is an actual holomorphic function u, nonzero at zero, such that $f = ug$ on a neighbourhood of zero.*

Proof. Association supplies a unit of the actual holomorphic germ ring with $f = ug$ as germs. Choose an actual holomorphic representative of that unit. Evaluation at zero is nonzero because the germ is a unit. Equality of the actual germs is equality of their representatives on a neighbourhood, and gives the asserted original equation there. □

Theorem 34.5 (Actual normal-crossing flattenings agree branch by branch)**.** *Let F and G be two actual centred local holomorphic equivalences in $\mathbb{C}^n$, both mapping zero to zero. Let $S, T \subseteq \{1, \ldots, n\}$ be finite sets of distinct coordinates. If*

$$\bigcup_{i\in S} \{F_i = 0\} = \bigcup_{j\in T} \{G_j = 0\}$$

as actual germs at zero, their original branch germs have a uniquely produced bijection. Corresponding coordinate functions differ by actual holomorphic units on one common neighbourhood of zero, and $|S| = |T|$. The same assertion holds for maps centred at any common point q, with equations $F_i(z + q)$ and $G_j(z + q)$ at zero.

Proof. The actual local inverse identities produce prime, pairwise distinct coordinate germs in both families by the preceding coordinate lemma. Apply the zero-union correspondence theorem to the selected coordinates. Its internally produced bijection matches the entire original selected branch germs. Apply the preceding original-unit lemma to each pair; intersect the finitely many neighbourhoods on which those equations hold to get all equations on one neighbourhood. These holomorphic units make their zero germs equal. Equality of finite cardinalities follows from that actual bijection. For a common point q, translate the source by q before this argument; its derivative is the identity and the original inverse derivative remains surjective. Dimension zero yields two empty selected coordinate sets and is included. □

Corollary 34.6 (Original manifold branches and intersections agree on overlaps)**.** *At a point of an actual general normal-crossing carrier, any two actual adapted flattenings have uniquely matching original coordinate branch germs. Their local branch counts agree. Under the produced matching, every finite intersection of original branches agrees as a germ.*

Proof. Write both flattenings after the actual chart at the point, and translate that chart centre to zero. Translation and each actual local inverse give centred local holomorphic equivalences. The actual chart and its inverse identify both selected zero unions with the original carrier germ. Apply the preceding theorem. Transport each matched zero-set germ through the actual chart back to the original manifold; continuity and the local inverse identities preserve neighbourhood-germ equality. For a finite intersection, intersect the finitely many neighbourhoods on which the matched branch membership tests agree. On this common neighbourhood all tests, and hence their conjunctions, agree. Equality of the actual finite branch counts is already supplied by the produced bijection, independently of the chosen flattening. □

Remark 34.7. This intrinsic correspondence does not imply that an ordered resolution algorithm ignores initial labels. An invariant using a colexicographic tie-break on ordered boundary indices can choose different centres under a branch permutation. Canonical descent requires an invariant or a first-blowup construction whose dependence on original unordered branches has itself been proved, followed by genuine compatibility of subsequent blowups. The correspondence above supplies the original overlap geometry.

35 Intrinsic highest-multiplicity strata of normal-crossing carriers

These statements concern the geometry of an original general normal-crossing carrier. They produce an intrinsic first centre without choosing an ordering of its local branches. They do not assert that this centre is contained in the cosupport of an additional marked ideal.

Definition 35.1 (Intrinsic local branch multiplicity)**.** For an actual normal-crossing carrier H in a complex manifold of dimension n, let $b_H(x)$ be the number of distinct coordinate hyperplanes in an actual centred flattening of its germ at x.

Proof. An actual flattening exists by the normal-crossing condition. Any two flattenings have the uniquely produced branch bijection of the overlap theorem, so the numbers agree. Thus the definition is independent of the flattening. Its values lie between zero and n. Since the flattening sends x to zero, x lies in H exactly when its selected coordinate set is nonempty, equivalently when $b_H(x) > 0$. □

Lemma 35.2 (The exact nearby branch count)**.** *Fix an actual flattening F at x with selected coordinate set S. On a neighbourhood of x, the original local branch multiplicity at y equals the number of coordinates $i \in S$ for which $F_i(y) = 0$.*

Proof. The flattening identifies H with the selected hyperplane union on its whole open domain. Fix a point y in that domain and write $S_y = \{i \in S : F_i(y) = 0\}$. Every coordinate indexed by $S \setminus S_y$ is nonzero at y. The finitely many open nonvanishing neighbourhoods have an intersection on which none of these coordinates contributes to the union. Translate the actual coordinate map by $F(y)$. It has an actual translated inverse and is centred at y. For $i \in S_y$, its coordinate i equals the original F_i, since $F_i(y) = 0$. Thus these coordinates describe exactly the original carrier germ at y. Express this actual local holomorphic coordinate equivalence in the preferred chart at y using the two actual chart maps and their inverses. It is an actual centred flattening with selected set S_y. The defining overlap independence gives $b_H(y) = |S_y|$. The argument holds for every y in the domain of the original flattening, so that whole domain is the required neighbourhood of x. □

Lemma 35.3 (Closed intrinsic multiplicity superlevels)**.** *For every nonnegative integer r, the intrinsic set $C_r = \{x : b_H(x) \geq r\}$ is closed. For $r > 0$ it is contained in H.*

Proof. At a point with $b_H(x) < r$, take an actual flattening with $|S| = b_H(x)$. On its domain the preceding exact count is the cardinality of a subset of S, so it remains strictly less than r. Thus the complement of C_r is open and C_r is closed. If $r > 0$, every point of C_r has positive branch multiplicity and hence lies in H. □

Theorem 35.4 (The genuine highest-multiplicity centre)**.** *Suppose $r > 0$ bounds the original branch multiplicity everywhere. Then $C_r = \{x : b_H(x) = r\}$ is an actual closed smooth submanifold. At each of its points it has dimension $n - r$, and an actual original normal-crossing chart expresses it as the intersection of all of the r local original branches. These intersections agree as actual germs on every chart overlap, independently of any branch ordering.*

Proof. The upper bound identifies C_r with the closed superlevel above. At a point of C_r, take a flattening with selected set S of size r. The exact nearby-count theorem says a nearby point has count at least r exactly when every coordinate in S vanishes. Thus the actual centre germ is precisely the coordinate subspace cut out by S. The original flattening has an invertible differential. Its distinct selected coordinate rows are linearly independent: any linear relation can be evaluated on the inverse image of each corresponding coordinate basis vector to make its coefficient zero. These r holomorphic functions cut out the whole centre germ. The holomorphic implicit function theorem therefore gives an actual regular flattening to the dimension-$(n - r)$ coordinate plane; $r \leq n$ follows from the selected-set cardinality. Together with closedness this proves the entire centre is an actual closed smooth submanifold. The overlap theorem matches every local original branch uniquely; its finite-intersection conclusion therefore identifies the whole centre germ on every overlap. No matching or smooth-centre certificate is assumed. If the level is empty, closed smoothness is vacuous and the pointwise dimension assertion has no instances. □

Corollary 35.5 (A nonempty intrinsic highest stratum)**.** *If H is nonempty, there is a positive integer $r \leq n$ attained by b_H and bounding it everywhere, so its intrinsic highest stratum is a nonempty actual closed smooth centre. No compactness is required.*

Proof. The nonempty range of b_H is contained in the finite set $\{0, \ldots, n\}$ and therefore has a largest attained value r. A point of H has positive branch multiplicity, hence $r > 0$. Choose a point attaining r; it lies in the highest stratum. Apply the preceding theorem with the actual global upper bound. In dimension zero the normal-crossing carrier must be empty, so the nonemptiness hypothesis does not arise. □

Remark 35.6. An actual blowup of this intrinsic smooth centre is available from the proved geometric existence theorem. Using it to lower the number of strict original branches requires its genuine coordinate-chart calculation and compatibility with any accumulated exceptional boundary. A marked-ideal resolution that must preserve the complement of its original cosupport

cannot use this centre unless its required cosupport containment has also been proved. The ordered monomial invariant retains its colexicographic tie-break on branch indices. The intrinsic branch count and its centre do not prove that this ordered canonical sequence is independent of an arbitrary initial permutation of those indices.

36 Faithful finite jets of the entire holomorphic germ ring

Hilbert–Samuel values concern the entire reduced current local ring, rather than a selected equation or an order-one substitute. In the convention of [4], Remark 1.3, the value at ℓ is $\dim_{\mathbb{C}} \mathcal{O}/(I + \mathfrak{m}^{\ell+1})$. The following genuine finite-jet construction establishes that this dimension is finite for every original ideal. It does not assert that these values have a semicoherent transformation presentation or supply a normally flat centre.

Lemma 36.1 (Literal point powers and actual local-ring powers)**.** *Let $R = \mathcal{O}_{\mathbb{C}^n,0}$ be the entire genuine holomorphic germ ring and $\mathfrak{m}$ its actual maximal ideal. For every function f holomorphic near zero and every $q \geq 0$, its actual germ lies in $\mathfrak{m}^q$ if and only if f belongs to the qth power of the original point ideal at zero.*

Proof. The actual maximal ideal consists exactly of germs with value zero. For one direction, an original point-power expression is a finite holomorphic sum of products of original point-ideal functions. Passing every actual representative to its holomorphic germ carries sums, coefficients and products to the same sums, coefficients and products in R. Induction on q therefore places the resulting germ in $\mathfrak{m}^q$. At $q = 0$ both powers are the whole holomorphic ring.

For the other direction, let J_q consist of germs having an original representative in that point power. This is an actual ideal: the power is closed under finite sums and holomorphic coefficients, and membership is invariant under equality on a neighbourhood. The ideal J_0 is R. Every element of $\mathfrak{m}$ has a holomorphic representative vanishing at zero and hence belongs to the original point ideal. If $\mathfrak{m}^q \subseteq J_q$, products of these representatives lie in the $(q+1)$st original point power. Since J_{q+1} is an ideal, the full product ideal $\mathfrak{m}^q\mathfrak{m}$ is contained in it. This proves $\mathfrak{m}^q \subseteq J_q$ by induction. Equality of two actual germs is neighbourhood equality of their original representatives, so membership transfers to the stated original f. □

Lemma 36.2 (The exact whole-jet kernel)**.** *An actual holomorphic germ lies in $\mathfrak{m}^q$ exactly when all its actual iterated complex derivatives of degrees less than q vanish at zero.*

Proof. The preceding dictionary identifies membership with the genuine original point power. The convergent holomorphic Taylor theorem identifies this power with vanishing of every derivative below degree q: the forward implication differentiates products, and the reverse implication expresses the convergent Taylor remainder as a sum of degree-q coordinate products with holomorphic coefficients. The original point-power derivative theorem therefore gives both implications. In the original Euclidean chart the preferred chart is the identity, so these derivatives are precisely the unmodified derivatives of the same original representative at zero. □

Definition 36.3 (The actual finite linear jet)**.** Give R its actual complex algebra structure through constant germs. The qth jet of a germ is the finite tuple $(D^j f(0))_{0 \leq j < q}$, with each entry an algebraic complex multilinear map on $(\mathbb{C}^n)^j$. These are the derivatives of any genuine original holomorphic representative.

Lemma 36.4 (Well-defined linearity and the exact kernel)**.** *The actual finite jet is a complex linear map. Its kernel is exactly $\mathfrak{m}^q$, considered as a complex subspace of R.*

Proof. Representatives of the same germ agree on an open neighbourhood, so every iterated derivative agrees at zero. Thus the finite tuple is well defined on the entire original germ ring. Every holomorphic representative is analytic near zero and hence has all complex derivatives there.

The iterated derivative of $f+g$ is $D^j f + D^j g$, and the derivative of a constant scalar multiple cf is $cD^j f$. Constant-germ multiplication is exactly this same scalar multiplication, proving complex linearity. The jet vanishes precisely when each actual derivative below degree q vanishes. The preceding whole-jet kernel theorem identifies this condition with the actual maximal-ideal power, including $q = 0$. □

Theorem 36.5 (Finite dimension of every actual point quotient)**.** *For every $q \geq 0$, the entire actual quotient $R/\mathfrak{m}^q$ is a finite-dimensional complex vector space. For every original ideal I of R, the actual quotient $R/(I + \mathfrak{m}^q)$ is also finite dimensional.*

Proof. Each degree-j multilinear-map space on finite-dimensional $\mathbb{C}^n$ with values in $\mathbb{C}$ is finite dimensional: repeated currying identifies it with iterated finite-dimensional spaces of linear maps. There are only q entries, so the whole finite jet target is finite dimensional. The actual linear map's first isomorphism theorem and its proved exact kernel identify $R/\mathfrak{m}^q$ with its genuine image, a subspace of that finite-dimensional target. This proves finite dimension without a supplied Taylor-polynomial basis or a finiteness certificate. Since $\mathfrak{m}^q \subseteq I + \mathfrak{m}^q$, the actual quotient map $R/\mathfrak{m}^q \to R/(I + \mathfrak{m}^q)$ is surjective. Its image is thus finite dimensional, proving the second conclusion. At $q = 0$ the quotient is zero; at $n = 0$ the same real jet construction includes the degree-zero value and the higher zero multilinear entries. □

Definition 36.6 (Actual Hilbert–Samuel point values)**.** For an original ideal $I \subseteq R$, define its actual point value at ℓ by

$$H_I(\ell) = \dim_{\mathbb{C}} R/(I + \mathfrak{m}^{\ell+1}).$$

The preceding theorem proves this dimension is finite for every original ideal; no finite-dimensional quotient hypothesis is part of the definition.

37 Actual finite historical predicates on genuine curve components

The following producers address the dynamic positive-order history seam of the constructed general invariant. Every exception set is constructed from actual analytic sets and the genuinely locally finite global component family. No finite global component list, constant-history certificate, or given zero-dimensional exception is assumed. The final consumer uses the actual coherent nonmonomial ideals at all heads of a finite geometric tower. Full General2 companion descent and G3 remain open.

Lemma 37.1 (Locally finite analytic unions and their actual dimensions)**.** *A genuinely locally finite family of closed analytic subsets has an analytic union. If each member has dimension at most zero, so does its union.*

Proof. The locally finite union is closed. At any point choose a genuine open neighbourhood meeting only finitely many indexed members. On this same neighbourhood the full union agrees exactly with that finite union, whose analytic equations come from actual finite analytic union. Transfer its analytic presentation using this local equality; outside the closed union use the unit equation. In the zero-dimensional case the finite union has actual dimension zero by the proved finite-union producer. Every regular point of the full union transfers to that finite union by the same open local equality, so its regular dimension is zero. This permits infinite, empty and repeated families whenever their actual indexed family is locally finite. □

Lemma 37.2 (A proper analytic intersection on a genuine curve component)**.** *Let A be analytic of dimension at most one, B one of its genuine global components, and P analytic. If $B \not\subseteq P$, then $B \cap P$ is closed analytic and has dimension at most zero.*

Proof. The actual B is an analytic closure of a connected component C of the regular locus of A, is contained in A, and inherits its dimension bound by analytic dimension monotonicity. If a nonempty relatively open trace of B were contained in P, choose a genuine point of C in that open set using its density in B. The corresponding nonempty relatively open trace of C is contained in P. Analytic continuation on C makes all of C, and then its closure B, contained in the closed P, a contradiction. Thus the actual intersection is thin in B. Genuine thin analytic dimension drop gives dimension at most zero; intersection of the actual analytic sets provides analyticity. □

Theorem 37.3 (Internally produced exception for finite analytic predicates)**.** *For an actual analytic A of dimension at most one and a finite indexed family of actual analytic predicates P_i, there is a constructed closed analytic zero-dimensional $Z \subseteq A$ such that, for every actual global component B and every $x \in B \setminus Z$,*

$$x \in P_i \quad \Longleftrightarrow \quad B \subseteq P_i$$

for all indices i.

Proof. For each i, take the union of $B \cap P_i$ over precisely the genuine components B not contained in P_i. This actual subfamily is locally finite: inject its component subtype into the genuinely locally finite full global-component family, then take actual subsets. Every intersection is analytic and zero-dimensional by the preceding lemma. The actual union producer therefore makes it analytic and zero-dimensional. Take the finite union over i, retaining its actual inclusion in A. At $x \in B \setminus Z$, if $B \subseteq P_i$, membership is immediate. If $B \not\subseteq P_i$, membership in P_i would place x in the corresponding actual intersection, hence in Z, contradicting the hypothesis. This proves the literal predicate dictionary. □

Lemma 37.4 (Actual zero-dimensional singular envelope of an analytic curve)**.** *For an actual analytic A of dimension at most one, an internally constructed analytic zero-dimensional $S \subseteq A$ contains all its zero-dimensional components and all its nonregular points. Every point of $A \setminus S$ is genuinely regular of dimension one.*

Proof. The actual global components have internally witnessed regular dimensions zero or one. Their exact geometric dictionary with the dimension parts therefore gives $A = A_0 \cup A_1$. Both parts are analytic and genuinely pure of their selected dimensions. Take $S = A_0 \cup (A_1 \setminus \operatorname{Reg}_1 A_1)$. The first part has dimension zero by purity; the second is analytic and zero-dimensional by the actual pure singular-locus producer. The finite union remains analytic and zero-dimensional. Away from its closed A_0 part, A has the same germ as A_1. Outside the second part the point is actually regular of dimension one in A_1, hence also in A. Both parts are subsets of A, so this is an actual exception producer. □

Theorem 37.5 (Actual finite head-tail nonmonomial history predicates)**.** *Let $\mathfrak{m}$ be an actual dimension-two invariant-domain datum, T an actual finite geometric test tower, and o any natural order. On the current actual cosupport there is a constructed analytic zero-dimensional exception such that, on each genuine global component outside it, every actual historical nonmonomial comparison $\nu_r \geq o$ and $\nu_r \geq o+1$ is constant, where ν_r is the order at the actual image under the tail to head r. The actual head-stage cosupport conditions in these predicates are retained.*

Proof. For every $0 \leq r \leq \operatorname{length} T$, split the actual tower into its literal head and tail. The head is an actual geometric test tower and its transformed datum is in the genuine invariant domain. Its nonmonomial ideal is coherent by the actual domain producer, so the already proved actual canonical first-entry superlevel is analytic. Intersect this actual locus with the genuine analytic head-stage cosupport, then pull it back by the genuine holomorphic tail projection. Use the two thresholds o and $o+1$ at each head; the index set is genuinely finite. The current actual cosupport is analytic and has dimension at most one by the actual positive-mark, nonzero ideal and carrier-dimension-two dimension-drop producer. Apply the preceding internally constructed

finite-predicate exception to this actual family and this actual cosupport. Its literal membership dictionary gives the claimed comparisons and retains every actual history stage and map. □

Corollary 37.6 (Literal actual order-regime run on curve components)**.** *In the preceding actual situation the produced exception can be chosen so that every point outside it is genuinely regular of dimension one and any two points of the same genuine current global component outside it have exactly the same literal trailing order-o regime run along T.*

Proof. Each actual regime predicate is membership in the actual carrier together with equality of its nonmonomial order to o. Equality is the conjunction of the comparisons at o and the negation of the comparison at $o+1$, so the actual finite predicate dictionary makes all head regime predicates identical at the two actual points. Induct on the genuine tower: a reflexive tower has run zero; at a literal blowup step the actual previous-stage predicate decides whether the run is zero or the previous actual run plus one. The next head-tail predicates reduce definitionally to the previous tower's head-tail predicates, so induction gives equality. Apply this proved structural identity to the produced actual predicate dictionary. This is an equality of the genuine history function, not an assumed constant-run certificate. Take the union with the internally produced singular envelope to ensure actual dimension-one regularity outside the exception. Complete lower companion values and their analytic descent across higher-order boundary points are still to be proved. □

38 Actual order under holomorphic submersions

This chapter produces the first order layer of smooth functoriality from an actual holomorphic submersion between manifolds of possibly different dimensions. It does not assume a section or an equality of invariants. Compatibility of the full higher invariant and of its selected blowup sequence remains a separate obligation.

Theorem 38.1. *Let $f : Y \to X$ be a holomorphic submersion at y. There is a map $s : X \to Y$ that is holomorphic on a neighbourhood of $f(y)$, with $s(f(y)) = y$ and $f \circ s = \mathrm{id}$ on a neighbourhood of $f(y)$.*

Proof. Choose the genuine submersion charts ϕ at y and ψ at $f(y)$, and the complex linear equivalence L for which $\psi \circ f \circ \phi^{-1} = \mathrm{pr}_1 \circ L$. Put $v = \mathrm{pr}_2 L(\phi(y))$ and define $s(z) = \phi^{-1}(L^{-1}(\psi(z), v))$. The original identity at y gives $L(\phi(y)) = (\psi(f(y)), v)$, so $s(f(y)) = y$. The open chart targets and continuity give a neighbourhood of $f(y)$ on which every expression stays in its actual chart domain. Chart holomorphicity and the affine complex linear slice prove that s is holomorphic there. On that neighbourhood the submersion chart equation gives $\psi(f(s(z))) = \psi(z)$; injectivity on the actual chart source then gives $f(s(z)) = z$. This produces all section properties internally. □

Theorem 38.2. *For an actual ideal of holomorphic germs I on X, at a point where f is a holomorphic submersion, and every natural q, $I_{f(y)} \subseteq \mathfrak{m}_{f(y)}^q$ if and only if $(f^*I)_y \subseteq \mathfrak{m}_y^q$.*

Proof. For the forward direction, each actual pullback generator is $a \circ f$ with $a \in I_{f(y)}$. The existing genuine pullback theorem for powers of point ideals places it in $\mathfrak{m}_y^q$. Closure under finite holomorphic combinations gives the entire pullback ideal inclusion. Conversely, for each original $a \in I_{f(y)}$, its actual pullback $a \circ f$ belongs to $(f^*I)_y$, hence to $\mathfrak{m}_y^q$. Compose with the internally produced holomorphic section through y. Pullback of point-ideal powers gives $(a \circ f) \circ s$ in $\mathfrak{m}_{f(y)}^q$; the genuine local identity $f \circ s = \mathrm{id}$ and locality of germ membership identify this germ with a. Both directions apply to the original full ideal, not just one generator. □

Theorem 38.3. *For the same original data, $\mathrm{ord}_y(f^*I) = \mathrm{ord}_{f(y)} I$ in $\mathbb{N} \cup \{\infty\}$.*

Proof. The actual order is the supremum of the natural q for which the full stalk lies in the q-th point-ideal power. The preceding equivalence identifies these indexing predicates exactly. Their suprema are equal, including infinite order and the zero ideal. □

Theorem 38.4. *For a holomorphic submersion f and a coherent original ideal I, every marked cosupport satisfies* $\operatorname{cosupp}(f^*I, q) = f^{-1}\operatorname{cosupp}(I, q)$.

Proof. Coherence supplies holomorphicity of every original ideal germ. At each actual source point, apply the preceding pointwise order equivalence to the genuine submersion at that point. Unfolding the actual marked cosupport gives equality of the two sets by extensionality. No compactness, equal-dimensional inverse, or section certificate is an input. □

39 Actual coherent controlled transforms across order-mask boundaries

The constructed general invariant forms its literal companion on an open order mask. The following producers use the actual globally defined companion and coefficient ideals, with their actual colon-controlled transforms, and prove their exact restriction identities. In particular the extension is the same ideal used by the actual invariant on the mask, not an independently supplied ideal or extension certificate.

Lemma 39.1 (Coherence of the actual colon along a carrier without inclusion)**.** *If the genuine ideals I, $\mathcal{I}_N$, and L are coherent, their actual colon along N is coherent, without any inclusion of $I|_N$ in $L|_N$.*

Proof. By its actual definition the colon along N is $(I + \mathcal{I}_N) : L$. The actual restricted ideal is coherent by coherent sum, and the actual colon of two coherent ideals is coherent by the already proved Oka colon theorem. No invertible quotient interpretation or restricted inclusion is used in this coherence conclusion. □

Lemma 39.2 (Actual blowup coherence and nonzero-along preservation)**.** *For an actual blowup along $C \subseteq N$, a genuine closed smooth carrier N and coherent I, the actual colon-controlled transform I' is coherent. If I is genuinely nonzero along N at every point, I' is genuinely nonzero along the actual strict transform N' at every point. The centre need not lie in the cosupport of this candidate marked ideal.*

Proof. The actual strict transform of the closed smooth N is closed smooth. Its vanishing ideal is coherent. The actual pullback of I is coherent, as are the actual pullback centre ideal and its actual mark power. The actual transformed ideal is exactly the colon along N' of these ideals, so the preceding producer proves coherence. Away from the actual centre, the actual blowup is locally biholomorphic, and its actual controlled transform is nonzero along N' by the genuine off-centre transform identity. At a point over the centre, suppose every germ of the transformed coherent ideal vanished on N' near that point. Actual finite coherent generators make this vanishing true for every germ at nearby points. Genuine points of the actual strict transform lying over $N \setminus C$ accumulate at this point, by the actual closure definition. Choose one such nearby point. Its off-centre nonzero germ contradicts the produced local vanishing. This proves the actual nonzero property without any candidate admissibility hypothesis. □

Theorem 39.3 (Entire actual candidate transforms along a true carrier tower)**.** *Let T be an actual geometric test tower for $\mathfrak{m}$. Any genuine candidate $\mathfrak{j}$ with the same initial carrier, a closed smooth carrier and coherent ideal has an actual globally coherent colon-controlled transform at every prefix of T and exactly the same transformed carrier. If its initial ideal is genuinely nonzero along this carrier, its actual entire transformed ideal remains so. No test-tower condition for the candidate is assumed.*

Proof. The actual carrier recursion depends only on the previous carrier and literal centre and map, so induction proves equality of the transformed carriers. If the initial boundaries also agree, the entire transformed boundaries agree by the identical map-and-append recursion; initial carrier equality alone is not used to assert boundary equality. Each actual centre of the true test tower is contained in the transformed carrier of $\mathfrak{m}$, hence in the candidate's identical carrier. Apply the preceding actual blowup producers at the literal step. Induction retains genuine closed smoothness, coherence, and nonzero along at every step. Reflexive towers and empty centres are included without removing any literal tower steps. □

Theorem 39.4 (Exact open restriction of the entire actual controlled transform)**.** *In the preceding situation, for any actual open U, the literal transformed restriction of the candidate along the actual restricted tower equals the open pullback of its globally defined actual controlled transform, as marked ideals: carrier, divisor, ideal and mark all agree.*

Proof. Induct on the actual tower. The genuine one-step restriction equality uses only coherence of the input and output controlled ideals; both have been internally produced at every actual prefix by the preceding theorem. The true open-preimage restriction keeps the literal centre preimage and literal restricted blowup map. Apply that actual one-step equality and the inductive marked-ideal equality. The result is an exact equality, including the actual ambient germs, not just equality of supports or stalk orders. □

Lemma 39.5 (Actual invariant domain for entire candidates)**.** *If the candidate is in the actual dimension-k invariant domain and shares the genuine initial carrier and boundary with the well-formed template, its actual entire controlled transform along the true template test tower is in the same actual invariant domain, even without candidate admissibility.*

Proof. Use the produced actual closed smooth carrier, coherent ideal and nonzero along property. The genuine transformed carrier and boundary agree exactly with the template's, so its actual SNC and transversality conclusions apply. The candidate's actual initial carrier dimension bound transfers to the template's equal initial carrier; genuine strict-transform dimension preservation then gives this bound at every literal stage and equality of carriers returns it to the candidate. The actual mark is unchanged and positive. These are precisely the real invariant-domain fields. □

Lemma 39.6 (Genuine nonzero-along coefficient companion)**.** *For a genuine coherent positive-mark ideal nonzero along its carrier, its actual coefficient-of-homogenized ideal is genuinely nonzero along that carrier. Consequently this actual marked construction preserves the true invariant domain.*

Proof. Choose an actual original ideal germ g nonzero at arbitrarily nearby carrier points. The genuine homogenized ideal contains the original ideal, so it contains the same actual germ. The coefficient marked sum has its actual index-zero summand, since the mark is positive. This summand contains the actual power g^e for its actual coefficient-sum exponent e, and the actual sum therefore contains this genuine power germ. It is nonzero at every one of the same nearby carrier points, including when $e = 0$. This proves genuine nonzero along internally. Coherence, positive mark, and equality of the actual carrier and boundary follow from the actual coefficient construction. Retaining the same real dimension bound produces all invariant-domain fields. □

Theorem 39.7 (The literal order-mask companion has a true coherent entire transform)**.** *For an actual invariant-domain datum, an actual true head tower and an actual true tail, and a positive order o, the actual globally defined companion formed at that head has a coherent, genuinely nonzero-along, entire controlled transform along the tail. Its pullback to the actual order-mask preimage equals exactly the literal transformed companion used by the constructed general invariant on that actual restricted tail.*

Proof. The true head-stage datum lies in the actual invariant domain. The actual companion, formed from its actual nonmonomial and monomial ideals, has the same carrier, is coherent, has positive mark, and is genuinely nonzero along its carrier by the actual companion-domain producer; maximal order outside the mask is unnecessary. Apply the same-carrier entire transform producer along the actual true tail. At the initial mask the literal companion equals the restriction of this global companion by the actual companion/open pullback theorem. The exact open restriction identity then identifies its literal transformed marked ideal with the open pullback of the produced actual entire controlled transform. Thus the literal invariant uses this exact entire ideal on the mask. □

Theorem 39.8 (The actual coefficient companion also extends with its literal transform)**.** *The actual coefficient-of-homogenized companion used by literal masked Step 1 has a controlled transform in the genuine invariant domain along the same true tail. Its carrier, divisor, ideal and mark restrict exactly to the actual coefficient companion transformed along the literal restricted tail.*

Proof. The actual global companion is well formed. Its actual homogenized coefficient ideal is coherent with the same carrier, and positive mark, by the proved actual coefficient construction. Its genuine nonzero along property and real invariant-domain fields are internally produced by the preceding power-germ argument. Apply the actual entire-domain producer along the true tail. Coefficient-of-homogenization commutes with actual open pullback, since the actual companion is coherent, its actual carrier is closed smooth, and its actual mark is positive. Combine this initial exact identity, the actual companion/open identity, and the exact entire transform restriction theorem. This identifies the literal restricted coefficient transform with the produced actual entire marked ideal. Full General2 whole-superlevel descent and the general G3 sequence are still open; coherence alone does not assert global maximal order. □

40 Actual coordinate-monomial representatives and finite ideal jet columns

Work in the entire actual holomorphic local ring $R = \mathcal{O}_{\mathbb{C}^n,0}$ with its original constant-germ complex algebra structure and actual maximal ideal $\mathfrak{m}$. This is the next genuine producer for the literal Hilbert–Samuel point values of [4], Remark 1.3. The words used below may repeat variables and may represent the same commutative monomial. They form a finite spanning family; no independence or supplied Taylor-representative certificate is asserted.

Definition 40.1 (Actual finite coordinate words)**.** Let z_i denote the actual germ of the original coordinate function. For $q \geq 0$, use the finite index set of pairs (j, w) with $j < q$ and $w : \{0, \dots, j-1\} \to \{0, \dots, n-1\}$. Associate the genuine product $z_w = \prod_{k<j} z_{w(k)}$ and let P_q be their complex linear span in R. The empty word represents 1. If $q = 0$, the index set is empty and $P_q = 0$.

Proof. There are finitely many degrees $j < q$; for each one, there are finitely many maps between the two finite coordinate index sets. Their finite disjoint union is therefore finite. The products and span are formed in the same entire original ring. This includes $n = 0$ and the unique empty word of degree zero. □

Lemma 40.2 (Original Hadamard decomposition of every whole germ)**.** *Every actual germ $g \in R$ has genuine whole germs h_i such that $g = g(0) + \sum_{i<n} z_i h_i$.*

Proof. Choose a genuine original holomorphic representative f of g. The function $f - f(0)$ vanishes on the zero-dimensional coordinate plane. The original holomorphic coordinate-plane Hadamard lemma therefore produces original holomorphic functions H_i on a neighbourhood of zero with $f - f(0) = \sum_i z_i H_i$ there. Pass these same original functions to actual germs. Germ formation

preserves every finite sum and product and sends the constant $f(0)$ to the constant germ $g(0)$. Rearranging the resulting actual ring identity gives the stated equality. No decomposition is an input. □

Lemma 40.3 (Coordinate products raise the genuine finite span)**.** *The constant* 1 *belongs to* P_{q+1}*, and multiplication by any original coordinate sends* P_q *into* P_{q+1}*.*

Proof. The empty word has length zero, hence lies among words of length less than $q+1$. If a word has length $j < q$, prefixing a coordinate gives a word of length $j+1 < q+1$, whose actual product is $z_i z_w$. Multiplication by z_i is an actual complex linear map on R. Thus this generator calculation extends by linear-span induction to every element of P_q. □

Theorem 40.4 (Internally produced monomial representatives)**.** *For every* $g \in R$ *and* $q \geq 0$*, there is an actual* $p \in P_q$ *with* $g - p \in \mathfrak{m}^q$*.*

Proof. Induct on q. For $q = 0$, choose $p = 0$ since $\mathfrak{m}^0 = R$. For the next degree use the proved original Hadamard decomposition $g = g(0) + \sum_i z_i h_i$. The induction hypothesis internally produces $p_i \in P_q$ with $h_i - p_i \in \mathfrak{m}^q$. Set $p = g(0) + \sum_i z_i p_i$. The empty word and the preceding coordinate-span lemma put p in P_{q+1}. Each z_i is in the actual maximal ideal, because the original coordinate vanishes at zero. Thus every genuine product $z_i(h_i - p_i)$ lies in $\mathfrak{m}\mathfrak{m}^q = \mathfrak{m}^{q+1}$. Their finite sum equals $g - p$ in the actual ring, proving the claim. The argument includes the zero-coordinate case, when the sum is empty. □

Theorem 40.5 (Actual whole finite-jet image)**.** *The image of the true* q*th complex linear jet of the entire ring is exactly the complex span of the jets of this finite family of actual coordinate words.*

Proof. For an arbitrary actual germ, the preceding theorem produces a finite monomial representative with remainder in the exact kernel $\mathfrak{m}^q$. Its jet therefore equals the jet of that representative, which lies in the span of the actual word jets by linearity. Conversely each word jet is in the actual linear-map range, and that range is a complex subspace. This gives both inclusions, not merely a dimension bound. □

Theorem 40.6 (Actual finite Macaulay columns for a whole generated ideal)**.** *For a finite family of actual original germs* G_i*, the jet image of the entire ring ideal they generate equals the complex span of the actual finite columns* $j_q(z_w G_i)$*, where* w *has length less than* q*.*

Proof. Every element of the actual ring ideal has a finite sum expression $\sum_i a_i G_i$ with whole-ring coefficients a_i. The internally produced representative theorem replaces every a_i by $p_i \in P_q$ with remainder in $\mathfrak{m}^q$. The maximal-ideal power is an actual ideal, so multiplying each remainder by the actual G_i and summing remains in $\mathfrak{m}^q$. The proved exact jet kernel removes this remainder. Complex linearity and span induction on every p_i put the jet of $\sum_i p_i G_i$ into the stated finite column span. Conversely every actual word product $z_w G_i$ belongs to the entire generated ideal. Its jet lies in the ideal's actual image, a complex subspace, so the finite column span is contained in that image. No supplied jet-image or polynomial-representative certificate is used. □

Theorem 40.7 (Internally generated columns of every original ideal)**.** *For every original actual ideal* $I \subseteq R$ *there is a finite family of original functions* G_i*, holomorphic near zero, whose entire germs generate* I*. For every* q*, the whole jet image of* I *is exactly the complex span of the finitely many actual columns* $j_q(z_w[G_i])$ *above.*

Proof. The genuine Weierstrass-division Noetherian theorem applies to the entire original germ ring. It internally provides a finite family of actual germs generating the given I. Every one of those germs has a genuine original holomorphic representative; choose these finitely many original functions. Replacing each generating germ by its equal original representative preserves the entire generated ring ideal. Apply the preceding whole-ideal column theorem for every q to this same

internally produced family. Thus original representative existence, ideal generation and exact finite jet columns are outputs, not additional hypotheses. □

Theorem 40.8 (Exact whole quotient rank)**.** *Let V_q be the actual whole finite-jet image of R and let W_q be the image of any original ideal I. Then* $\dim_{\mathbb{C}} R/(I+\mathfrak{m}^q) = \dim_{\mathbb{C}} V_q - \dim_{\mathbb{C}} W_q$.

Proof. Restrict the genuine jet map to its actual image V_q, and follow it by the quotient map to V_q/W_q. Both maps are surjective. A whole germ maps to zero precisely when its jet equals the jet of some actual member of I. By the proved exact kernel, this holds precisely when that germ is an element of $I+\mathfrak{m}^q$. The actual first isomorphism theorem therefore identifies the entire original quotient with V_q/W_q. The true finite-dimensional quotient dimension formula gives $\dim(V_q/W_q)+\dim W_q = \dim V_q$. Viewing W_q inside V_q or inside the original jet target gives the same dimension by the injective actual subtype map. Rearranging in the natural numbers gives the asserted subtraction, including the whole-ideal and zero-degree cases. □

Definition 40.9 (Actual finite Macaulay linear map)**.** The actual finite Macaulay map sends complex coefficient vectors indexed by (w,i) to $\sum_{w,i} c_{w,i} j_q(z_w G_i)$. Its actual linear-map range is exactly the proved whole ideal jet image.

Proof. This is the genuine linear combination map of the finite column family. Its image is exactly the complex span of those original columns, by the linear-span definition and the finite linear combination universal property. The preceding whole-generated-ideal theorem identifies that span with the true image of the entire original ring ideal. □

Theorem 40.10 (Literal Hilbert–Samuel value from actual Macaulay rank)**.** *For the entire actual ideal generated by the original finite family G_i, its literal value at ℓ is* $H_I(\ell) = \dim V_{\ell+1} - \operatorname{rank} M_{\ell+1,G}$.

Proof. Unfold the literal whole-ring quotient defining $H_I(\ell)$. Apply the preceding exact quotient-rank theorem with $q=\ell+1$. The actual range equality for the original Macaulay map replaces $\dim W_q$ by its genuine range dimension. Thus every term is the rank of an actual finite complex linear map, with no supplied presentation or transformation certificate. For any given original ideal, the preceding internal Noetherian generator producer supplies original functions and their exact whole ideal equality. Applying the same rank identity for every ℓ gives that same original family as an output, together with the literal rank formula for all its Hilbert–Samuel values. No generating family is required as an extra input. □

Remark 40.11. These are literal whole-ring monomial representatives and whole-ideal jet columns. Turning local coherent generators into holomorphically varying matrix columns and exact analytic Hilbert–Samuel rank thresholds is the next producer. A normally flat smooth centre, a semicoherent Hilbert–Samuel transformation presentation, higher-entry descent and finite higher-codimension resolution are not consequences yet. General G8 remains open.

41 The actual blowup of the highest original branch stratum

These statements concern the strict transform of the original carrier. They do not count the new exceptional component as an original branch. They construct the next original-branch value by actual monomial charts, without identifying an ordered invariant under arbitrary permutations.

Lemma 41.1 (Coordinates adapted to the whole intrinsic highest stratum)**.** *Let H be an actual normal-crossing carrier with all original local branch counts at most r. At a point x of its highest stratum C_r, there are actual centred coordinates in which C_r is the plane of dimension $n-r$ and H is the union of precisely the last r coordinate hyperplanes. These coordinates are produced from the original flattening, without a supplied simultaneous-flattening witness.*

Proof. Take an actual original flattening with selected set S. The intrinsic highest-stratum theorem gives $|S| = r$ and expresses the entire centre on that chart domain as the simultaneous vanishing of the selected coordinates. Since $|S| \leq n$, there is a permutation of the n indices sending S onto the last r indices: choose bijections on S and its complement to these two blocks and combine them. The resulting linear coordinate permutation is a holomorphic linear isomorphism. Postcompose the actual flattening with this map, and precompose its actual inverse with the inverse permutation. The new maps have the same actual source, a permuted open target and both inverse identities. Their centre value is zero. The selected-coordinate union is now the union of the last r coordinate hyperplanes, and the selected-coordinate intersection is precisely the dimension-$(n-r)$ plane. This produces both whole-set dictionaries on one actual open domain. □

Lemma 41.2 (The intrinsic positive highest stratum admits an actual blowup)**.** *If $r > 0$ bounds all original branch counts, the actual closed smooth highest stratum C_r has empty interior and its complex blowup exists.*

Proof. The entire centre is closed and regular of dimension $n - r$ at every one of its points. At any hypothetical interior point, the centre is the whole ambient manifold on a neighbourhood and hence regular of dimension n there. Uniqueness of the dimension of an actual regular flattening would give $n = n - r$. At a point of the centre, the actual branch bound gives $r \leq n$, so this contradicts $r > 0$. Thus the centre has empty interior. The proved complex blowup existence theorem applies to this actual closed smooth centre, producing its manifold, projection and genuine blowup-along-centre property. An empty highest level also has empty interior and is covered by the same existence theorem. □

Lemma 41.3 (The genuine local strict-original-carrier calculation)**.** *Consider a genuine monomial chart of the blowup of C_r, with pivot j among the last r coordinates and its actual lift inverse. On this chart the strict transform of H is exactly the union of the remaining last-coordinate hyperplanes, with the pivot omitted. The exceptional hypersurface is the pivot-coordinate zero set.*

Proof. Choose the produced branch-adapted coordinates at the base point. The actual blowup universal property supplies a monomial local inverse W with pivot $j \geq n - r$ and an actual holomorphic lift inverse on its open domain. In those coordinates the monomial projection sends the pivot coordinate to itself, each other normal coordinate to its product with the pivot, and each tangential coordinate to itself. The centre preimage is therefore exactly $W_j = 0$.

On the base chart the original carrier is a finite union of the actual coordinate zero sets. Local equality on a base neighbourhood pulls back to local equality on the blowup by continuity of the actual projection. Membership in the closure of the preimage outside the centre depends only on this local equality. Closure commutes with this finite union, so its strict transform is the finite union of the branch strict transforms. The pivot branch has no points outside the centre in this chart and its strict transform misses the chart. For a remaining normal coordinate i, outside $W_j = 0$ its pulled back zero equation is exactly $W_i = 0$. Every point of $W_i = 0$ in the chart is a limit of points there with $W_j \neq 0$, because the coordinates i and j are distinct and the actual chart has an open model domain. Thus its genuine strict transform is the whole $W_i = 0$ set. Conversely any such limit remains in that zero set by continuity. These are the two directions of the whole strict-original-carrier dictionary; no transform membership certificate is an input. □

Lemma 41.4 (Actual centred strict-original coordinates and the lower count)**.** *At every blowup point above the centre, the original strict carrier has an actual centred normal-crossing flattening with at most $r - 1$ branches. The exceptional equation is simultaneously a coordinate hyperplane with a different index from all of these branches.*

Proof. The actual monomial local inverse has injective differential, as follows by differentiating its genuine lift inverse. Express it in the preferred chart at the blowup point, apply the holomorphic inverse function theorem and subtract its centre value. This produces actual centred

coordinates. Among the $r-1$ remaining normal coordinates, retain exactly those vanishing at the point. The finitely many other coordinates are nonzero on one smaller open neighbourhood, so they contribute no branch there. For each retained coordinate, subtraction of the centre value changes nothing. The preceding whole strict-transform dictionary thus gives a genuine centred NC flattening of the original strict carrier. Its selected set has at most $r-1$ elements. Since the point is above the centre, the pivot coordinate vanishes there; its translated equation is still the actual exceptional equation, and the pivot is distinct from each remaining selected index. □

Theorem 41.5 (A genuine global first step with smaller original branch count)**.** *If H is nonempty, its internally produced maximum $r>0$ and the blowup of its whole intrinsic maximum stratum produce an actual strict normal-crossing carrier H' with all original branch counts at most $r-1$. The construction requires neither compactness nor a global ordering of the original branches.*

Proof. Produce the attained positive actual maximum and its smooth whole centre, then its genuine blowup. The strict carrier is closed by its closure definition. Above the centre use the preceding produced centred chart and lower count. Away from the centre the actual blowup is a local biholomorphism and the strict carrier equals the preimage of H: the original carrier is closed, so local closure outside the centre gives exactly this preimage. Compose an actual original flattening at the image point with the actual local inverse coordinates of this biholomorphism. The resulting centred chart gives the whole strict-carrier union with the same original selected indices, hence the same intrinsic count. The image point is outside the global maximum stratum, so that integer count is strictly less than r, equivalently at most $r-1$. These two cases prove actual NC everywhere and the claimed global strict-original count bound. The chosen maximum and centre were intrinsic; no assertion about the old ordered algorithm was used. □

Remark 41.6. The new exceptional branch can restore the total boundary count to r; the decrease asserted here counts only strict original branches. Compatibility with previously accumulated exceptional boundary requires a further simultaneous-coordinate producer. For an additional marked ideal, this centre need not be contained in its cosupport. Consequently this first geometric step alone does not supply the required marked-ideal admissibility, cosupport-complement isomorphism or canonical global sequence descent.

42 Actual derivative ideals under maps of different dimensions

We prove the genuine chain-rule inclusion on entire holomorphic germ ideals for arbitrary source and target dimensions. This is a necessary first derivative layer of smooth functoriality. The converse under a submersion, its coefficient and homogenization consumers, and the selected canonical tower remain open.

Lemma 42.1. *Let $f: Y^m \to X^n$ be holomorphic and let e' and e be actual atlas charts at y and $f(y)$. The real chart map $T = e \circ f \circ (e')^{-1}$ is holomorphic on a neighbourhood of $e'(y)$.*

Proof. The actual chart inverse is continuous and holomorphic near $e'(y)$ and sends that point to y. The original holomorphic map sends y to $f(y)$. Continuity gives a neighbourhood contained in the true source-chart target and whose image under $f \circ (e')^{-1}$ stays in the true target-chart source. On that neighbourhood, compose the two holomorphic atlas maps with the original map. The ordinary differentiability characterization in the model vector spaces proves the claimed holomorphicity, with no same-dimension hypothesis. □

Definition 42.2. For $v \in \mathbb{C}^m$ and $j \in \{1,\dots,n\}$, set $c_j(p) = (DT(e'(p))v)_j$ using the actual original chart map.

Proof. This formula uses the actual Frechet derivative of the genuine map between the original model spaces and coordinate evaluation in its target. Its source and target dimensions are independent; no matrix or transport certificate is input. □

Lemma 42.3. *Every actual coefficient c_j is a holomorphic germ at y.*

Proof. Choose a genuine open neighbourhood on which T is holomorphic, using the preceding lemma. The holomorphic derivative theorem makes $z \mapsto DT(z)v$ holomorphic there. Compose with complex linear coordinate evaluation in $\mathbb{C}^n$ and the original source atlas chart. This proves actual scalar germ holomorphicity, including an empty target coordinate family when $n = 0$. □

Lemma 42.4. *For every original holomorphic germ a at $f(y)$ and $v \in \mathbb{C}^m$, near y one has $\partial_v^{e'}(a \circ f) = \sum_{j=1}^{n} c_j \, (\partial_j^e a) \circ f$.*

Proof. On one true neighbourhood all original charts and the original germ are holomorphic and their arguments stay in the actual chart domains. The germ $(a \circ f) \circ (e')^{-1}$ equals $(a \circ e^{-1}) \circ T$ there, by the actual inverse identity of e. Apply the genuine Frechet chain rule. Expand the true vector $DT(e'(p))v$ in the standard finite coordinate basis of $\mathbb{C}^n$ and use complex linearity of the derivative of $a \circ e^{-1}$. The resulting finite sum is exactly the stated formula for the original chart derivatives. □

Lemma 42.5. *Every actual member a of the derivative ideal $\mathcal{D}(I)_{f(y)}$ pulls back to a genuine germ of $f^*\mathcal{D}(I)$ at y.*

Proof. Derivative-ideal membership itself gives holomorphicity of a at the original point. Compose with the actual holomorphic map. The defining genuine pullback family contains $a \circ f$ with coefficient one, which supplies the required span relation. No generator equality is an input. □

Theorem 42.6. *For any actual ideal I of holomorphic germs, $\mathcal{D}(f^*I) \subseteq f^*\mathcal{D}(I)$.*

Proof. The original pullback ideal is the span of the actual functions $a \circ f$ with $a \in I$. Those original germs already belong to $\mathcal{D}(I)$, so all germs of f^*I satisfy the inclusion. For a chart derivative of an arbitrary actual pullback member, write that member as a real finite holomorphic combination of those generators. The actual product rule separates derivatives of coefficients and derivatives of pulled-back original generators. The first terms use $I \subseteq \mathcal{D}(I)$. For the second terms apply the preceding actual chain rule, whose genuine coefficients are holomorphic. Each original chart derivative of a is an actual member of $\mathcal{D}(I)$. Actual finite-sum, product and germ locality properties conclude the inclusion on the whole ideal. □

Theorem 42.7. *For every natural r, $\mathcal{D}^r(f^*I) \subseteq f^*\mathcal{D}^r(I)$.*

Proof. Induct on r. At zero both sides are the literal original pullback ideal. For the successor, derivative monotonicity applies to the actual induction inclusion. Apply the preceding one-step theorem to the real original iterated derivative ideal, whose members are holomorphic by the actual derivative-ideal construction. Compose these inclusions. No higher invariant or canonical functoriality identity is assumed. □

43 Actual entire companion curve loci in dimension two

These producers start from the actual global coefficient companion and its literal controlled transform. No maximal-order condition outside the actual order mask, caller-supplied exceptional set, or special-locus certificate is assumed. They provide the global curve geometry required for subsequent General2 superlevel descent; that complete descent and G3 remain open.

Lemma 43.1 (The actual component germ at an actual regular point)**.** *If B is an actual global component of an analytic A, and $x \in B$ is an actual regular point of A, then A and B have the same germ at x.*

Proof. Write $B = \overline{C}$, where C is the genuine connected component of the actual regular locus through its actual generating regular point. An actual regular flattening at x has a connected flat-ball neighbourhood whose trace on A lies in the component of the regular locus through x. Since $x \in \overline{C}$, this open neighbourhood meets C. The common regular point identifies the two genuine connected components. Thus x belongs to C and on a smaller actual open neighbourhood the whole A lies in $C \subseteq B$. Conversely $B \subseteq A$ follows from the true analytic component theorem. This gives equality of the actual germs. □

Lemma 43.2 (A regular cosupport curve has an actual surface carrier)**.** *For an actual dimension-two invariant-domain marked ideal, an actual regular dimension-one point of its positive-mark cosupport is an actual regular dimension-two point of its carrier.*

Proof. The genuine closed smooth carrier has an actual regular dimension $j \leq 2$ at the point. Inclusion of the real cosupport forces $1 \leq j$. If $j = 1$, the actual analytic filling lemma applied to the dimension-one cosupport and this regular carrier makes the whole carrier lie in the cosupport on a neighbourhood. This contradicts actual coherent nonzero along: choose a real nonzero germ, express it using the actual finite coherent generators, and shrink so that every generator belongs to the actual ideal at each nearby point. Positive cosupport order makes every such generator zero on the carrier there. Their actual finite holomorphic combination makes the chosen germ zero at all nearby carrier points, contradicting its genuine frequent nonzero points. Hence $j = 2$. No global maximal-order condition is used. □

Lemma 43.3 (A genuine boundary-containing curve is in the member locus)**.** *Let an actual marked ideal be in the dimension-two invariant domain and let F be a finite list of genuine members of its actual SNC boundary. If an actual global cosupport component B contains an actual regular curve point x and B is contained in one member of F through x, then x belongs to the actual member-intersection locus for F.*

Proof. The preceding germ theorem makes B an actual regular dimension-one set at x. Its containing member has a genuine closed smooth trace H on the carrier, by the actual SNC transversality theorem. The true adapted coordinate dimension-drop producer bounds this trace by one. Actual regular dimension monotonicity and $B \subseteq H$ force the trace to have dimension exactly one at x. The true analytic filling lemma therefore gives $B = H$ as germs at x. Since B lies in the real cosupport, the whole trace H lies in that cosupport near x. Every nearby carrier point on all members of F through x lies in this one chosen member, hence in H and in the real cosupport. This is precisely the actual member locus. □

Theorem 43.4 (An internally produced zero-dimensional exception to specialness)**.** *An actual dimension-two invariant-domain ideal and any finite list F of its genuine boundary members internally produce an analytic zero-dimensional Z contained in the actual cosupport. Outside Z, a point of zero actual F count belongs to the actual codimension-one part, and a point of positive actual F count belongs to the actual member locus.*

Proof. The actual coherent nonzero positive-mark ideal internally yields an analytic cosupport of dimension at most one. Its actual dimension-zero part and the genuine singular locus of its pure dimension-one part give an analytic zero-dimensional regular-curve exception. Apply the proved finite analytic-predicate producer to the actual members of F, which are analytic by their genuine SNC geometry. Its exception is the locally finite union of the actual proper intersections of global cosupport components with those members. Union the two produced zero-dimensional sets to obtain Z. At a cosupport point outside Z the actual cosupport is regular of dimension one and the carrier is regular of dimension two, giving the actual codimension-one part directly by its definition. At positive actual count choose an actual indexed member through the point and an actual global cosupport component containing it. The finite-predicate dictionary forces that whole component into the chosen member. The preceding real trace filling producer gives

the actual member locus. Infinite global component families are handled by their genuine local finiteness; they are not replaced by a finite global family. Repeated or empty original members are allowed. □

Theorem 43.5 (Entire actual companion special locus)**.** *For a genuine dimension-two datum, a true head tower, a true tail tower, and positive order, the entire transformed actual coefficient companion has an internally produced analytic zero-dimensional Z such that every point of its real cosupport outside Z is an actual Step 1 special point.*

Proof. The actual entire coefficient-companion invariant-domain producer supplies all genuine domain fields, without companion admissibility or maximal order outside the mask. Take F to be precisely the actual first original-length members of its transformed boundary. Every such member is genuinely in the actual SNC boundary. Apply the preceding producer. Its actual count is the literal Step 1 level. The two actual alternatives are precisely the zero-level codimension-one and positive-level member-locus alternatives of the actual Step 1 special predicate. □

Theorem 43.6 (The literal masked companion has the same actual special predicate)**.** *On the actual order-mask preimage the literal restricted companion's special predicate equals the actual entire companion's special predicate. Consequently all literal masked nonspecial cosupport points lie in the preimage of an internally produced global analytic zero-dimensional Z.*

Proof. Use the proved exact marked-ideal equality between the literal masked coefficient transform and the restriction of its actual global transform. The actual initial companion boundary is the true open restriction of the global companion boundary, so its length is exactly unchanged. Taking that many original members in the transformed equality commutes with the actual list map; hence literal original members are exactly the open preimages of the entire original members, including duplicates. The true boundary-count preimage theorem gives equality of literal levels. Actual coherent open cosupport restriction and regular-point transport give equality of the codimension-one parts. The genuine subtype open embedding maps the actual neighbourhood filter onto the ambient filter, so the actual member-locus preimage theorem gives equality of the member loci. Substituting these exact dictionaries into the real special predicate proves both directions. Pull back the preceding internally constructed Z, and use the actual coherent cosupport pullback identity to exclude every nonspecial point outside it. The global Z is retained, not an unrelated analytic set living only on the mask. These producers do not yet assert complete General2 superlevel descent or the full global resolution sequence. □

44 The original normal-crossing carrier with accumulated boundary

Definition 44.1 (A simultaneous original carrier and boundary)**.** A closed original normal-crossing carrier H and a finite list E of closed hypersurfaces are boundary adapted if, at every point x, there are actual centred holomorphic coordinates on an open neighbourhood, a finite selected coordinate set S, and coordinate sets B_i for the members of E. On the whole neighbourhood, H is the union of the coordinate hyperplanes indexed by S. For each E_i through x, B_i is a singleton and E_i is its coordinate hyperplane on that same neighbourhood. The singletons for the through members are distinct and disjoint from S. There is no condition on the coordinate sets of members not through x. In particular this definition is meaningful in dimension zero and for empty through families.

Lemma 44.2 (The actual empty-boundary producer and geometric consumers)**.** *Every actual original NC carrier is boundary adapted with the empty list. Any boundary adapted pair produces an actual original NC carrier and an actual SNC boundary. Its intrinsic highest original-branch stratum has only normal crossings with that actual boundary.*

Proof. For the empty list use the actual original coordinates and selected set; there are no boundary equations or distinctness conditions. For a general pair, discarding the boundary equations gives an actual NC flattening for the entire carrier. Discarding the union equation gives one common actual coordinate system for every through boundary member; their singleton equations and distinctness give precisely SNC. At a point of the highest original stratum, the intrinsic count theorem identifies the whole stratum on the original chart with the intersection of all selected hyperplanes. Those very equations, together with the existing singleton boundary equations on the same domain, give actual normal-crossing coordinates for the whole centre and the boundary. No centre coordinate system is supplied separately. □

Lemma 44.3 (Joint coordinates adapted to the whole highest stratum)**.** *Suppose all original branch counts are at most r. At a point of its highest stratum there are actual centred coordinates in which the whole centre is the dimension-$(n-r)$ plane, the entire original carrier is the union of the last r coordinate hyperplanes, and every through old boundary member is one of the distinct first $n-r$ coordinate hyperplanes. These coordinates and boundary indices are internally produced from the simultaneous original pair.*

Proof. Take actual simultaneous coordinates and their selected set S. The intrinsic highest-stratum dictionary gives $|S| = r$ and the centre as the whole intersection of those selected equations. Choose a permutation sending S onto the last r indices and its complement onto the first $n - r$ indices. Postcomposition with that holomorphic linear isomorphism, and precomposition of the actual inverse with its inverse, gives an actual coordinate map on the same open source. Transport each old boundary coordinate set by this permutation. Singletons and their distinctness are preserved. Disjointness from S places every through old boundary singleton in the first block. The original union and the centre intersection are transported on the whole actual domain, producing all claimed dictionaries together. □

Lemma 44.4 (Actual joint transformed coordinates over the centre)**.** *In the genuine blowup of the highest original stratum, every point above the centre has one actual centred coordinate system simultaneously expressing the whole original strict carrier, every through strict old boundary member and the new exceptional member. The original selected set has at most $r-1$ indices. All boundary equations are distinct singleton coordinates disjoint from that selected set.*

Proof. Use the produced joint base coordinates. The genuine blowup supplies a monomial local inverse W, pivot j in the last block, and its actual holomorphic lift inverse. Differentiating those inverse identities gives an invertible differential for W, so its actual recentred local coordinates exist. In the uncentred monomial chart, the original strict carrier is exactly the union of the last-block hyperplanes other than the pivot, by the proved genuine finite-union strict-closure calculation. A through old boundary member has index in the first block. Its pivot is distinct, and its original coordinate is unchanged by the monomial projection. The actual strict-transform calculation therefore gives exactly that same coordinate hyperplane. The new exceptional member is exactly the pivot hyperplane.

Every through strict old member maps to a through old base member, because it lies in the preimage of that closed member. Its coordinate thus vanishes at the blowup point. The pivot also vanishes because the point is above the centre. Subtracting the centre value preserves all these equations. Retain among the remaining original indices exactly those whose coordinate vanishes at the point, and shrink to one open neighbourhood on which the finitely many other coordinates remain nonzero. The entire original union is then precisely the union of this retained set, whose size is at most $r-1$. All old boundary indices are tangential and mutually distinct; the new pivot is normal and is omitted from the retained original set. Hence every through boundary singleton is distinct and disjoint from the entire retained original set on the same actual coordinate domain. □

Theorem 44.5 (A genuine boundary-preserving original-count step)**.** *The genuine highest-original-stratum blowup of a boundary adapted pair produces a boundary adapted pair consisting of the whole strict original carrier and the list of strict old boundary members followed by the actual new exceptional member. The new list is genuinely SNC and every original branch count is at most $r-1$. For a nonempty original carrier, the positive attained maximum and the entire step are internally produced without compactness or global original labels.*

Proof. All transformed original and old boundary sets are closed by their strict-closure definition, and the exceptional member is closed by continuity of the actual projection and closedness of the actual centre. Above the centre use the jointly produced coordinates of the preceding lemma. Off the centre use the actual blowup local biholomorphism to transport one original simultaneous coordinate system. On its actual open inverse domain, strict transforms of the closed original and old boundary sets equal their preimages. Thus the same original selected set and the same old coordinate singletons give the entire strict original carrier and every through strict old member together. The exceptional member does not pass through the point, so assign its unused coordinate set the empty set. Through singleton cardinality, distinctness and disjointness are inherited from the actual original pair. These two cases give a boundary adapted pair at every actual point; its boundary consumer proves SNC.

Above the centre the genuine original-count chart has at most $r-1$ indices. Off it the intrinsic count is unchanged and the image point is outside the highest stratum, so its integer count is strictly below r. This proves the global bound. If the original carrier is nonempty, the actual finite intrinsic count image first produces its positive attained maximum and smooth whole centre, and the proved complex blowup existence theorem produces the genuine blowup. Applying the preceding result to these actual objects produces the whole step. □

Remark 44.6. These are statements about actual simultaneous geometric carriers and boundary equations. They do not assert that the total branch count decreases, that a separately marked ideal contains the centre in its cosupport, or that the old ordered monomial algorithm is independent of its colexicographic branch-label tie-break. Canonical marked-ideal global sequence descent remains a separate obligation.

45 Actual holomorphic Macaulay columns and point-value rank loci

The preceding whole-ring producers compute literal Hilbert–Samuel values using the rank of actual finite columns. The next obligation is that these same columns vary holomorphically for an original holomorphic family, and that their rank loci describe the entire original point quotient. In the convention of [4], Remark 1.3, the value at ℓ is the dimension of the actual quotient by $I+\mathfrak{m}^{\ell+1}$. No matrix, quotient dimension, point-value equality, or normally flat centre is supplied as a certificate below.

Lemma 45.1 (Faithful finite coordinate evaluation)**.** *Evaluate each degree-j jet on every j-tuple of original coordinate basis vectors. This defines an actual injective complex linear map from the whole finite multilinear jet target into the finite scalar coordinate array.*

Proof. Evaluation on a fixed coordinate tuple is complex linear, so the finite array is complex linear. Two actual multilinear maps that agree on all basis tuples agree on arbitrary tuples: expand each argument in its genuine finite coordinate basis and apply multilinearity, obtaining the same finite sum of values on basis tuples. This proves injectivity on every degree and hence on the whole finite tuple. Degree zero uses the unique empty tuple; zero ambient dimension has no positive-degree coordinate tuples and those actual positive-degree multilinear maps are zero. No symmetry basis or independence of commutative coordinate words is asserted. □

Lemma 45.2 (Actual translated whole germs and derivatives)**.** *Let G_i be an original finite holomorphic family on an open $U \subseteq \mathbb{C}^n$. At each $p \in U$, its original functions $z \mapsto G_i(p+z)$ are holomorphic near zero and define genuine whole germs. Their actual finite jets are the unmodified derivatives $D^j G_i(p)$ at the original point. The finite coordinate-word germs times these original translated germs are represented by $z \mapsto z_w G_i(p+z)$.*

Proof. Addition by p is an actual holomorphic translation with inverse subtraction by p. The open set U gives a genuine neighbourhood of zero whose image lies in U, proving holomorphicity of the same original translated functions. The original derivative translation identity gives $D^j(G_i(p+\cdot))(0) = D^j G_i(p)$. The actual finite-jet representative identity identifies these with the actual germ jets. Germ formation preserves all original finite coordinate products and multiplication by the original translated function, proving the whole word-times-generator representative identity, including the empty word. □

Lemma 45.3 (Original partial-derivative columns are holomorphic)**.** *For each fixed word w, degree j and actual coordinate tuple v, the scalar entry*

$$p \longmapsto D_z^j\big(z_w G_i(p+z)\big)\big|_{z=0}(v)$$

is holomorphic on U. Thus the genuine finite coordinate array of the actual Macaulay map varies holomorphically on the same original open set.

Proof. On the genuine open set $D = \{(p,z) : p+z \in U\}$ consider the original joint function $F(p,z) = z_w G_i(p+z)$. The coordinate product is holomorphic, addition is complex linear, and G_i is holomorphic on U, so this joint function is holomorphic on D. Holomorphic functions are analytic; their actual iterated Fréchet derivatives are analytic on the same open set. Restrict the derivative at $(p,0)$ to the actual directions $(0,v_k)$ and evaluate. Fixed multilinear evaluation is a continuous complex linear map, and the map $p \mapsto (p,0)$ is complex linear, so the resulting scalar function is holomorphic on U. The genuine affine slice $z \mapsto (p,z)$ and the iterated chain rule identify this derivative with the actual partial derivative displayed above. Locally restrict to the open domain before using the chain rule, so no global differentiability outside D is assumed. The finite entry set is the actual finite word and coordinate-tuple set. □

Theorem 45.4 (Literal point values from the same original matrix)**.** *At every $p \in U$, let I_p be the entire ring ideal generated by the actual translated original germs. The literal point value of I_p at ℓ equals the fixed entire jet-image dimension minus the rank of the same original holomorphic finite column matrix at p.*

Proof. The genuine arbitrary-family Macaulay theorem applies to the actual translated germs, and gives the literal whole quotient value using its actual map range. Compose with the proved injective finite coordinate array map. Restricting this injection to the actual range is an actual linear equivalence onto the composed range, so it preserves dimension. The translated representative identity identifies every resulting column with the original holomorphic column just proved. Thus the actual matrix is produced internally from those same original functions, and its rank is the dimension of the whole ideal jet image. The ambient whole-jet range dimension is fixed and independent of p. □

Theorem 45.5 (Actual finite point-value superlevel equations)**.** *For every finite degree ℓ and natural threshold r, the actual point-value superlevel for this original family is the common zero locus of finitely many original holomorphic minors on U.*

Proof. Write d for the fixed whole-jet image dimension at degree $\ell+1$. If $r > d$, the literal finite quotient value cannot reach r, so the superlevel is empty and has the constant-one equation. Otherwise the proved literal value formula identifies the threshold with matrix rank at most $d-r$, equivalently less than $d-r+1$. A matrix has rank less than this integer exactly when all its

square minors of that size vanish. For the reverse implication, send each genuine scalar column v to the actual complex-bilinear coordinate functional $x \mapsto \sum_a v_a x_a$. Evaluation on the genuine original coordinate directions recovers every v_a, proving that this real linear map is injective even for the empty coordinate set. It preserves the dimension of the actual column span. Independent functionals and actual coordinate row selection produce a nonzero minor if the rank is at least the prescribed size. Transposing these genuine evaluation minors gives exactly the original column-matrix minors, with identical determinant. There are finitely many row and column selections, with repetitions permitted and their determinants automatically zero. Each determinant is a genuine finite sum of products of the original holomorphic entries, hence holomorphic. This gives both directions for the entire superlevel, including $r = 0$, empty generator families, zero ideals and unit ideals. □

Theorem 45.6 (Full nearby coherent ideal dictionary)**.** *An actual coherent ideal has genuine finite Cartan generators on one neighbourhood of any point. Read them in a genuine chart and translate at each nearby point. Their whole ring ideal equals the translated entire original coherent stalk ideal at every nearby point; consequently the preceding actual finite column equations compute its entire quotient values.*

Proof. Choose the actual coherent generator neighbourhood and the actual chart; intersect their open domains. At every nearby point, coherence identifies membership in the entire original ideal with finite holomorphic span of the same original generators. The genuine translated chart and its inverse carry original holomorphic functions, coefficients, sums and products to their identical germ-ring operations. Form the actual whole chart-stalk ideal from all original membership germs, using the actual chart and translation. The genuine finite-span dictionary sends every such original member into the generated whole ring ideal; conversely every original generator is an actual member by the original coherent span identity. Taking whole ring spans proves equality of these entire actual ideals, rather than an equality of selected jets. This actual finite-span germ dictionary therefore identifies the entire translated coherent ideal with the generated ring ideal, in both directions. No stalk equality or Cartan dictionary is an extra input beyond the original coherence. Apply the preceding true whole-ring quotient/rank formula and the internally constructed minors to these same generators throughout this genuine neighbourhood. □

Remark 45.7. The faithful finite scalar array, actual holomorphic column producer and full coherent dictionary must be proved before claiming these local rank loci. Intrinsic comparison between different nonlinear original charts requires an actual local holomorphic germ-ring isomorphism preserving the maximal ideal and its powers; local biholomorphic point-power transport provides the route, but chart-independent global Hilbert–Samuel loci will not be claimed without that proof. A normally flat smooth centre, genuine Hilbert–Samuel transformation presentation, higher-entry descent and finite higher-codimension embedded resolution remain subsequent obligations. General G8 is still open.

46 Whole actual General2 analytic superlevels

The construction below consumes the literal canonical invariant in carrier dimension at most two. Its finite history and actual entire coherent companion ideals produce the component dictionaries internally. The proof uses actual locally finite global components and exact whole-set equalities. It does not infer analyticity from upper semicontinuity or from analytic traces on open order masks alone. Higher-dimensional superlevels and G3 remain open.

Lemma 46.1 (The actual complement of a proper analytic subset is dense in its component)**.** *For an actual global component B of an analytic set and a genuine analytic Z not containing B, the closure of $B \setminus Z$ is exactly B.*

Proof. The actual component B is analytically irreducible and closed analytic. The actual intersection $B \cap Z$ and actual closure $\overline{B \setminus Z}$ are both analytic and contained in B. Their union is exactly B: membership in Z gives the first part, and nonmembership gives the second by the real closure inclusion. Actual analytic irreducibility makes one part all of B. The first alternative would imply $B \subseteq Z$, contrary to the real hypothesis. Thus the second is all of B. This argument includes every dimension and uses no supplied density certificate. □

Theorem 46.2 (Actual component selection and the zero-dimensional remainder)**.** *Let L be a genuinely closed subset of an analytic A, and let Z be analytic of dimension at most zero. If membership in L agrees at any two points of $B \setminus Z$ for every actual global component B of A, then*

$$L = \bigcup_{B \in \mathrm{globalComponents}(A),\ B \subseteq L} B \ \cup\ (L \cap Z),$$

and this entire set is analytic.

Proof. A point of L outside Z lies on an actual global component B. The membership dictionary puts every point of $B \setminus Z$ in L. Since this complement is nonempty, B is not contained in Z. The preceding actual density producer and genuine closedness of L fill the whole B into L. A point of L on Z lies in $L \cap Z$. The opposite inclusion is literal from selection and intersection. The selected family is a genuinely locally finite subfamily of the actual global components, so its actual analytic union is analytic. Every subset of the actual analytic zero-dimensional Z is closed smooth by its actual singleton flattenings, so $L \cap Z$ is analytic. Taking the genuine finite union proves whole-set analyticity. The component dictionary is a general descent lemma input; the next producers establish it for the actual invariant rather than assuming it. □

Lemma 46.3 (Finite literal history and actual companion exceptions)**.** *Fix a positive actual first-entry threshold q and a genuine canonical tower for the constructed dimension-two invariant. The actual ideals, actual historical comparisons and actual finite boundary members internally produce an analytic zero-dimensional Z such that, on every actual current global component outside Z, the comparisons of the current order with q and $q+1$, the literal order-q trailing run, and membership in every current boundary member agree. For each literal split, all nonspecial cosupport points of its actual masked coefficient companion also map into Z.*

Proof. The actual invariant domain persists along the true test tower, so the actual current cosupport is analytic of dimension at most one. Its actual coherent nonmonomial ideal supplies the two genuine order superlevels. Apply finite analytic predicates to their traces and to the actual finite current SNC boundary. The actual finite head-tail order predicate producer separately supplies the literal trailing-run dictionary, including its internally constructed regular-curve exception. For each $r \leq \operatorname{length} T$, the literal head and tail are true test towers. Apply the actual entire companion curve-locus producer at positive q to obtain its actual global zero-dimensional special exception Z_r, whose exact literal mask identity is already proved. Union these finitely many actual Z_r with the actual order, boundary and history exceptions. Genuine finite-union dimension-zero producers prove that this whole Z is analytic and zero-dimensional. Only literal tower length is finite; the actual global component family is allowed to be infinite. □

Lemma 46.4 (The actual generic positive-order value on a current component)**.** *For every actual positive-first-entry threshold, membership in its entire constructed-invariant superlevel agrees at any two points of an actual current global component outside an internally produced analytic zero-dimensional Z.*

Proof. Compare the actual current order with q. The genuine order predicate dictionaries make both points either strictly below q, strictly above q, or exactly at q. In the first two cases the literal lexicographic definition fixes superlevel membership immediately. In the equality case the actual history dictionary makes their literal trailing runs the same r. The true canonical-tail theorem

internally produces their actual mask points, actual maximal-order companions, actual canonical restricted tails and actual companion cosupport membership. Actual Step 1 equivalence supplies membership in the actual transformed coefficient-companion cosupport. Since both points avoid the actually constructed Z_r, both are actual special points of this literal masked companion.

The real masked Step 1 definition at a special point is exactly (s, ∞). Its literal level is the actual boundary count of the first initial-companion-length members of the transformed boundary. Initial companion boundaries equal the actual head boundary, and identical real map-and-append recursion makes the entire transformed boundaries equal the actual current boundary. Exact mask restriction identifies the literal original list with its actual preimages. The actual boundary membership dictionary makes the counts of this same literal prefix equal at the two points, including repeated indexed members. Thus the actual lower values, then actual general values, are equal in this equality case. These are values of the true construction, not substituted generic values. □

Theorem 46.5 (Every actual positive-first-entry General2 superlevel is analytic)**.** *For every positive first entry and every second-entry threshold, the entire actual constructed General2 superlevel on the current actual cosupport is analytic along every genuine canonical tower. The actual positive superlevel also has the exact selected-component and zero-dimensional-remainder equality displayed in the descent theorem, with its exception internally produced from the literal history.*

Proof. The actual constructed general invariant's proved upper semicontinuity on the true closed cosupport gives genuine closedness of the entire superlevel. The preceding actual finite-history producer supplies its real component membership dictionary and actual analytic zero-dimensional exception. Apply the exact actual component-selection descent theorem. It returns the entire superlevel as the actual locally finite selected component union and the actual zero-dimensional remainder, proving analyticity across all order-mask boundaries. No bounded global order, compactness, finite global component family or global-maximal-order assumption is needed. □

Theorem 46.6 (Whole General2 superlevels at first entry zero)**.** *All actual General2 superlevels with first-entry threshold zero are analytic on the actual current cosupport.*

Proof. Split the actual second threshold into its two genuine sum alternatives. For the monomial alternative, the proved real canonical monomial-superlevel dictionary gives a whole analytic locus on the actual carrier; intersect with the actual analytic cosupport. For the companion alternative, every positive-order point has strictly larger first entry and qualifies, whereas every zero-order point has a literal monomial second entry, strictly below any companion alternative. Thus the whole superlevel is exactly the actual cosupport intersected with the genuine first-entry superlevel at one. Both factors are actual analytic sets produced from actual ideals. The second alternative includes its actual top value. □

Theorem 46.7 (All literal constructed dimension-two thresholds)**.** *For every genuine dimension-two invariant-domain datum, every actual canonical tower and every literal value of the constructed value type* ValTy(2)*, its entire superlevel on the actual transformed cosupport is analytic.*

Proof. Destructure the genuine lexicographic value into its actual first and second entries. Natural first entry is either zero or positive. Apply the respective actual whole-superlevel theorem. The invariant appearing here is literally the constructed family at dimension two, including its real masked Step 1 history and recursively constructed dimension-one lower invariant. The result gives no analyticity theorem for higher-dimensional families and does not assert a noncompact global resolution sequence. □

47 An actual finite separation tower for the original NC carrier

The strict carrier in this section is the actual closure of the preimage off the differential exceptional locus. The construction stops at original branch count at most one. All centres have original count at least two; the Cartier centre-removal case is never used in this tower.

Lemma 47.1 (Actual original density off a higher highest stratum)**.** *Let H be a genuine original NC carrier whose branch counts are at most $r \geq 2$, and let C be its whole highest stratum. Then $H \subseteq \overline{H \setminus C}$. Consequently the centre-removed geometric strict transform in any genuine blowup of C equals its actual differential-exceptional-locus strict transform.*

Proof. A point of H outside C already belongs to $H \setminus C$. At a point of C, the produced actual centred coordinates express H as the union of the last r coordinate hyperplanes and C as their intersection. There are two distinct indices i, j in that block since $r \geq 2$. For any neighbourhood of the original point, the actual coordinate inverse is continuous at zero and has an open target domain. Hence for sufficiently small nonzero complex t, the vector te_j lies in that target and its actual inverse lies in the chosen neighbourhood. Its ith coordinate is zero, so it belongs to the actual original carrier; its jth coordinate is nonzero, so it lies outside the whole centre. Every neighbourhood thus meets $H \setminus C$, proving the actual relative density. The proved genuine blowup strict-transform comparison applies to this produced density and gives the stated exact equality. No density or strict transform dictionary is an input. □

Lemma 47.2 (Actual smooth terminal carrier and transverse boundary)**.** *If a boundary adapted pair has all original branch counts at most one, its whole original carrier is a closed smooth submanifold, regular of dimension $n-1$ at each of its points, and is transverse to its actual SNC boundary. The list obtained by appending that entire original carrier to the boundary is SNC.*

Proof. At a point of the carrier, the actual intrinsic count is positive, so it is exactly one. The intrinsic count-one superlevel is exactly the whole carrier, including all its connected components, because positivity is equivalent to actual carrier membership. The genuine highest-superlevel regularity theorem therefore produces its actual regular coordinates of dimension $n-1$ everywhere on the carrier; it is closed by the original pair hypothesis. Alternatively in the actual simultaneous coordinates its selected original set has exactly one index. The whole union equation is then that single hyperplane equation, hence also its singleton coordinate-subspace equation. The existing through boundary singletons are disjoint from it and mutually distinct, giving actual transversality on the same domain. Appending the original carrier adds precisely that new singleton at points of the carrier and no through member elsewhere, so the actual whole appended list satisfies SNC. If the original carrier is empty, all regularity and transversality claims are vacuous and the appended empty member creates no through coordinate condition. Dimension zero is included: positive original count there is impossible. □

Definition 47.3 (A genuine intrinsic higher separation step)**.** A current carrier H, current finite boundary E and current centre C form an intrinsic higher separation step when they have an actual boundary adapted pair, an attained whole global original maximum $r \geq 2$, and C is exactly that whole highest original stratum. This is a geometric condition on one current centre, not a finite tower or a termination certificate.

Lemma 47.4 (A genuinely produced higher step and exact original transform)**.** *A boundary adapted pair whose original counts are not all at most one internally produces its attained maximum $r \geq 2$, its nonempty whole closed smooth centre of codimension r, a genuine blowup, and an actual transformed pair with every original count at most $r-1$. Its original carrier is exactly the differential strict transform; the current accumulated boundary is exactly the strict old list followed by the new exceptional preimage.*

Proof. The failure of the count-one bound gives an actual point of count greater than one and hence a nonempty original carrier. The genuine finite intrinsic-count image produces an attained positive maximum r, bounding that point, so $r \geq 2$. Its whole stratum is nonempty, closed and regular of dimension $n - r$ at every centre point, by the actual maximum-stratum producer. Its entire centre lies in the original carrier because its count is positive. The actual simultaneous pair coordinates flatten the centre using the original selected indices; all through old boundary singletons are disjoint from those indices, so the entire centre is transverse to the actual SNC boundary. The genuine blowup and whole transformed-pair producer then give the actual projection, true accumulated boundary list and smaller original count. The preceding actual relative-density theorem identifies this centre-removed original strict carrier with the differential strict carrier. The old boundary uses exactly its genuine centre-removed strict transforms, as required by the accumulated-boundary operation; no replacement of those boundary definitions is made. □

Theorem 47.5 (A bounded actual tower with literal higher centres)**.** *For a boundary adapted pair with all original counts bounded by b, there is an actual finite blowup tower of length at most* $\max(b-1, 0)$ *whose whole differential strict original carrier has count at most one and is boundary adapted with its literal accumulated boundary. Every actual recorded centre is an intrinsic higher separation step for the actual current differential strict carrier and actual current accumulated boundary. In particular the boundary at each actual centre stage is SNC and every centre has codimension at least two and lies in the current strict original carrier.*

Proof. Induct on the actual integer bound b, uniformly in the current manifold, original carrier and current finite boundary. If all actual counts are at most one, take the real empty tower. Its projection is the identity, its strict original carrier is the same closed carrier, and its boundary list is unchanged. It has length zero and no centre conditions. This covers an empty carrier and the base case $b = 0$.

Otherwise apply the preceding produced higher step. Its attained maximum satisfies $2 \leq r \leq b$. The new actual original count is at most $r - 1 \leq b - 1$. Apply the induction hypothesis to the actual blowup manifold and produced transformed pair, then prepend the actual first blowup to the resulting real tail tower. The exact first-step differential strict-transform dictionary and the genuine holomorphic strict-composition theorem identify every tail current carrier with the actual differential strict transform under its whole prefix. The actual accumulated-boundary append identity identifies every tail boundary with the literal original boundary transformed along that whole prefix. Thus all actual tail centre conditions transport to the concatenated tower; the first centre has the produced intrinsic higher-step property. Tower lengths add, and the tail bound gives at most $1+\max(b-2, 0) = b-1$, since this case has $b \geq 2$. The final actual pair and count-one bound are transported by those same exact dictionaries. This is an actual finite construction, with no caller-supplied candidate tower, termination certificate or global ordering of original branches. □

Theorem 47.6 (A genuine global NC separation and complement isomorphism)**.** *Every actual original NC carrier has a genuine separation tower of length at most* $\max(n-1, 0)$ *starting with empty boundary. Its final whole differential strict carrier is smooth, transverse to the actual accumulated SNC boundary, and appending that carrier gives a genuine SNC list. More generally an initial boundary adapted pair has the same conclusion with its true accumulated old boundary. The tower projection is an isomorphism over the complement of the original carrier.*

Proof. The genuine actual count is always at most n, so apply the bounded tower producer with $b = n$. Its actual terminal pair and count-one bound give smoothness, pointwise dimension, transversality and the SNC appended list by the terminal-pair theorem. For an original NC carrier alone, its empty-boundary pair is internally produced first.

Each literal centre lies in its actual current differential strict carrier. Closedness of the original carrier and continuity of the actual prefix projection put that entire strict carrier inside the

preimage of the original carrier. Thus every real recorded centre lies over the original carrier. Restricting the actual tower to its open complement makes every centre empty; the resulting true tower has no differential exceptional point there. The actual exceptional image therefore lies in the original carrier. The genuine blowup sequence is surjective and injective away from its exceptional locus, so the proved blowup-sequence isomorphism theorem gives the stated isomorphism over that whole open complement. No compactness, connectedness or finite global component family is required. □

Remark 47.7. The output is an actual intrinsic geometric separation tower, with literal current carriers and accumulated boundary at every stage. This does not imply cosupport admissibility for a separately marked ideal. The old ordered invariant retains its colexicographic label tie-break, and its canonical marked-ideal sequence is not identified with this geometric tower. That full marked-ideal general-NC descent remains open.

48 Actual nonlinear germ algebras and intrinsic Hilbert–Samuel loci

Remark 1.3 of [4] uses the literal Hilbert–Samuel function of the actual local ring modulo the actual ideal. The preceding Macaulay producers give all finite point values in one original chart. Here genuine nonlinear chart transitions internally identify the entire local algebras, original ideal and actual maximal-ideal powers. The resulting single intrinsic function has actual global analytic superlevels. No Hilbert–Samuel chart-equality, germ-algebra isomorphism, ideal-map equality or analytic-locus certificate is assumed by the final manifold producers.

Lemma 48.1 (Actual nonlinear composition on the entire holomorphic germ algebra)**.** *An actual holomorphic map near zero, sending zero to zero, induces an actual complex-algebra homomorphism on the entire holomorphic germ rings by original function composition.*

Proof. The original map is continuous at zero and fixes zero, so it sends the genuine neighbourhood filter into itself. Pull back every genuine holomorphic representative along that actual map. The true complex chain rule shows that the resulting original function is holomorphic near zero. Eventual equality of representatives is preserved by the same filter map, so the operation is well defined on entire germs. Original composition preserves zero, one, addition, multiplication and all actual constant functions; therefore this is a homomorphism of the actual complex algebras. Its value on every original representative is exactly that composed representative, with no jet truncation and no selected function subring. □

Lemma 48.2 (Entire germ algebra of a genuine nonlinear local biholomorphism)**.** *A genuine local holomorphic coordinate equivalence fixing zero internally produces a complex-algebra isomorphism of the entire holomorphic germ ring.*

Proof. Its actual source and target are open and contain zero. Thus the original coordinate map and its actual inverse are holomorphic near zero. Their actual local inverse identities hold on those genuine neighbourhoods, and the inverse also fixes zero by the original left-inverse identity. Compose whole germs with these two actual maps. For every actual holomorphic representative, the two composites agree with the same original function on the genuine source or target neighbourhood. Equality of entire germs then proves the two algebra maps inverse. This constructs the isomorphism from the actual coordinate geometry; no algebra isomorphism is supplied. □

Lemma 48.3 (Actual point powers and entire finite quotient algebras)**.** *The internally constructed germ algebra isomorphism maps the actual maximal ideal and every genuine power onto their actual counterparts. For every whole original ideal and every natural degree it induces an*

actual complex linear equivalence of the entire point quotient algebras, hence preserves the literal Hilbert–Samuel point value.

Proof. An actual ring isomorphism preserves units and nonunits in both directions. The genuine maximal ideal of a local ring is exactly the nonunits; hence its whole ideal image is the target maximal ideal. Ideal images under the same ring isomorphism preserve actual ideal multiplication, so induction on the natural exponent gives equality for every power, including exponent zero. Ideal images also preserve sums. Consequently the whole sum $I + \mathfrak{m}^q$ maps exactly to the actual sum of the whole original ideal image and target point power. Quotienting the constructed algebra isomorphism gives an actual algebra equivalence between these entire quotients. Its genuine underlying complex linear equivalence preserves finite dimension. Taking $q = \ell+1$ proves equality of the literal actual quotient dimensions. Neither a point-value equality nor a quotient-dimension certificate enters the construction. □

Lemma 48.4 (Actual translated nonlinear transitions preserve the whole original stalk)**.** *At an actual point lying in two genuine original chart sources, translate both charts to zero. Their actual nonlinear transition is a genuine local holomorphic equivalence fixing zero, and its internally constructed germ algebra map sends the entire original chart-stalk ideal onto the entire original chart-stalk ideal in the other chart.*

Proof. Let e, e' be the two original charts and y their common point. The actual transition is $z \mapsto e'(e^{-1}(z + e(y))) - e'(y)$, with inverse $w \mapsto e(e'^{-1}(w + e'(y))) - e(y)$. Its genuine source is the translation of e's actual target intersected with the inverse image of e''s actual source; define its target symmetrically. Both are open and contain zero. The original manifold holomorphic chart and inverse-chart maps give holomorphicity of these actual functions on these actual domains. Original chart inverse identities prove that they map these domains into one another and are genuine local inverses. The same identities prove that zero maps to zero. For every whole original holomorphic membership function, composing its translated representative in the second chart with this transition agrees with its translated representative in the first chart on that actual neighbourhood. Thus the algebra map carries each genuine original membership germ to exactly the corresponding original germ. Taking the whole ring spans of all original membership germs and using the inverse transition proves equality of the entire original ideals. No finite jet, chosen generator ideal, or caller stalk-map certificate replaces them. □

Theorem 48.5 (One intrinsic whole-ideal Hilbert–Samuel point function)**.** *Define the original point value using the actual preferred chart at that point and the entire original stalk ideal. The same function is computed in every actual original chart containing the point, for every finite degree simultaneously.*

Proof. The preferred chart contains its own point. Construct the actual translated nonlinear transition from any other original chart containing the point to that preferred chart. The whole original ideal-map identity and the actual maximal-ideal-power identity identify the two entire finite quotient algebras. The proved actual quotient linear equivalence preserves their literal dimensions for each degree. This proves equality with the single preferred-chart function, rather than defining a separate value for each chosen chart or accepting chart-equality as an assumption. □

Theorem 48.6 (Actual global analytic superlevels of the whole Hilbert–Samuel function)**.** *For any original coherent ideal, natural degree and natural threshold, the superlevel of this same intrinsic whole-ideal point function is an actual global complex analytic subset of the original manifold.*

Proof. At an arbitrary original point, coherence internally selects one actual Cartan family on a genuine neighbourhood. The preceding whole-stalk Macaulay producer constructs finitely many actual holomorphic minors on the actual coordinate neighbourhood, with an exact point-value superlevel identity in the fixed original chart throughout that neighbourhood. The proved nonlinear

chart invariance identifies those values at every nearby point with the same intrinsic preferred-chart function. Thus the same actual minors, on that same genuine coordinate neighbourhood, give the full local zero-locus identity for the intrinsic superlevel. This is exactly a complex analytic presentation at the arbitrary point, so the global superlevel is analytic. Empty generator families, zero and unit ideals, zero ambient dimension and threshold zero are covered by the earlier actual quotient and finite-minor producers; no compactness, constant rank, or selected maximum is needed. □

Remark 48.7. These results construct genuine nonlinear coordinate invariance and actual analytic loci for each finite value of the complete original Hilbert–Samuel function. A normally flat smooth centre, actual transformation of that function, its higher-entry invariant refinement and finite higher-codimension resolution descent are still required. General G8 remains open. No regularity or normal-flatness of a maximum locus is claimed.

49 Actual dimension-three Step 1 lower analytic neighbourhoods

This section keeps the literal Seq masked Step 1 value with its actual constructed dimension-two lower invariant. Every lower domain, actual restricted canonical tail and actual cosupport dictionary below is produced from the original dimension-three domain and genuine canonical tower. The result proves whole-superlevel analyticity on the actual nonspecial point's lower patch for thresholds whose first entry is its current level. Descent at higher-level and special boundaries, whole General3 thresholds and the general noncompact terminating sequence remain open.

Definition 49.1 (The literal lower patch at an actual current point)**.** For original marked datum J, actual tower T and its current point x, put $s = \mathrm{level}(J, T, x)$ and $r = \mathrm{regimeRun}(\mathrm{stepRegime}(J, s), T, x)$. The mask is the literal lower mask of J at the actual image of x in the head at r. The patch is its actual open preimage under the tail. The lower datum is the actual masked lower marked ideal on that mask; the lower tail is the actual restriction of that same literal tail. These are abbreviations of the actual construction and contain no domain, canonicity or analyticity certificate.

Lemma 49.2 (Actual lower domain and the whole same-level cosupport dictionary)**.** *Let J satisfy the real dimension-three invariant domain and actual maximal-order condition, and let T be canonical for its literal Seq masked Step 1 value with lower family* $\mathrm{invFamily}(2)$*. For every actual nonspecial point x of its current cosupport, the actual patch contains x, its literal lower datum satisfies the genuine dimension-two domain, and the literal restricted tail is canonical for the actual constructed dimension-two invariant. On that whole patch, the actual lower cosupport is exactly the actual original current cosupport at level $s = \mathrm{level}(J, T, x)$.*

Proof. The actual original/coefficient-companion canonical bridge internally produces the true test towers, true coefficient-companion canonical tower, and equality of their actual transformed cosupports. The actual canonical Step 1 tail theorem applies using the real original coherence, nonzero, smooth carrier, dimension and maximal order. At x it produces literal mask membership, the real lower dimension-two domain, true lower canonical tail and actual lower cosupport membership. Thus x belongs to its true open patch. The real trailing-run head theorem puts the actual head image in its true regime and cosupport. The actual constructed dimension-two invariant supplies a true test tower for the canonical lower tail. The genuine masked lower cosupport theorem then identifies, at every point of this whole patch, lower cosupport membership with current coefficient-companion cosupport membership and equality of the actual level to s. The exact head-tail append identity and the actual canonical cosupport bridge give the stated original-current-cosupport dictionary. No lower domain or canonical tail is assumed. □

Lemma 49.3 (The actual whole Step 1 value on a fixed lower cosupport)**.** *At every point of the actual lower cosupport in x's literal patch, the actual whole Step 1 value is exactly* $(s, \mathrm{invFamily}(2))$ *evaluated on that fixed actual lower datum and its true restricted tail.*

Proof. Apply the genuine canonical-tail fixed-lower-value theorem. Its all-shorter-towers lower-data input is internally supplied by the same actual canonical-tail theorem at each true shorter tower, using unchanged original domain and actual maximal order. The actual point x is in the current regime of level s, with its literal run r by definition. The preceding genuine lower-cosupport dictionary supplies the actual lower cosupport point. The fixed-lower-value theorem transports through the actual empty intervening prefix and the real member-restriction or contact comparison. Consequently even a point with a different literal entry time has precisely the invariant of this fixed actual lower datum. The lower value used here is the true recursively constructed invariant; no substitute value or caller-chosen tuple appears. □

Theorem 49.4 (Exact full threshold equality on the actual patch)**.** *For every actual lower threshold* $\alpha \in \mathrm{WithTop}(\mathrm{ValTy}(2))$*, the preimage on the entire literal patch of the actual whole Step 1 superlevel at* (s, α) *is exactly the actual lower cosupport superlevel at* α*.*

Proof. Every point of the actual patch has current level at most s, by the actual lower-mask level theorem and the true head image's regime. If it belongs to the whole Step 1 superlevel at first entry s, the genuine lexicographic first component of Step 1 forces its level at least s. Thus its level equals s, and the preceding actual cosupport dictionary puts it in the real lower cosupport. Substitute the exact fixed-lower value; lexicographic comparison with equal first components becomes exactly the actual lower comparison. Conversely a lower cosupport point belongs to the true original current cosupport, and the same literal fixed-lower-value identity turns its lower comparison into the whole Step 1 comparison. This proves equality of actual whole sets on the patch, including all points off the lower cosupport. □

Theorem 49.5 (All lower thresholds are actually analytic near the nonspecial point)**.** *Every lower threshold at first entry equal to the actual level of the nonspecial point gives an actual analytic preimage superlevel on its whole actual open patch, which contains the point. In particular the whole Step 1 superlevel is analytic in a genuine neighbourhood of x.*

Proof. Use the exact preceding whole-set equality and the internally produced lower dimension-two domain and true lower canonical tail. If α is a finite actual dimension-two value, the complete actual constructed General2 analytic-superlevel producer proves that exact lower set analytic. If α is top, every literal lower invariant value is a finite embedded element of its WithTop type, hence no point qualifies; the set is exactly empty and therefore analytic. The actual patch contains x by the internally produced mask membership. Taking this literal open patch supplies the neighbourhood conclusion. Ambient dimension is genuinely positive where the canonical-tail construction requires it; empty patches cannot satisfy the actual current-point premise. Empty towers and repeated empty steps are handled by their literal head-tail identities. No statement about special or higher-level boundary descent, whole General3 analyticity or global termination is deduced from this neighbourhood theorem. □

50 Actual whole NC multiplicity under biholomorphic geometry

Intrinsic original branch cardinality must be transported before highest centres can be aligned. The statements below use actual whole carrier coordinates and actual biholomorphisms; no branch matching, count equality, maximum identity or naturality certificate is supplied.

Lemma 50.1 (An actual full-coordinate reader under a local biholomorphism)**.** *A holomorphic map with bijective differential at a point internally produces, from any actual original centred holomorphic coordinate map, an actual coordinate map at that point. It reads the entire same*

original coordinate vector, and maps its whole genuine domain into the whole original coordinate domain.

Proof. Let e be the actual preferred chart at the source point z, and e' the preferred chart at its image. On the genuine open set $N = e.\text{target} \cap e^{-1}(g^{-1}(e'.\text{source}))$, read the actual map as $F = e' \circ g \circ e^{-1}$. This open set contains $e(z)$, and the manifold holomorphic coordinate formula gives actual holomorphicity of F there. The genuine manifold derivative formula identifies its derivative with the original bijective manifold differential. The actual holomorphic inverse-function theorem thus produces a genuine local holomorphic equivalence Ψ with actual source contained in N and actual function equal to F on that source. Compose it with the original coordinate equivalence Φ, using the genuine composite source $\Psi.\text{source} \cap \Psi^{-1}(\Phi.\text{source})$. This source lies in the actual preferred chart target and contains $e(z)$. Actual chart inverse identities show at every point of its whole flat domain that the image lies in the original flat domain and that the entire coordinate vector is exactly $\Phi(e'(g(y)))$. Its centre value is the original centre value. No hyperplane or branch correspondence is a hypothesis. □

Lemma 50.2 (Internally produced entire NC carrier and simultaneous boundary)**.** *An actual NC carrier pulls back to an actual NC carrier under every holomorphic map with bijective differential everywhere. Its entire original selected-coordinate union equation is produced internally. An actual boundary-adapted pair likewise pulls back to an actual boundary-adapted pair with the literal preimage boundary list.*

Proof. At an arbitrary source point apply the actual preceding reader to the original coordinate map at its actual image. The whole original NC union equation pulls back to the union of exactly the same finite selected hyperplanes on the whole actual new domain. The original closed carrier pulls back to a closed carrier by continuity. For a boundary-adapted pair use the same actual reader simultaneously for its whole original union and all original boundary equations. A through preimage member is precisely a through original member, so the genuine singleton sizes, disjointness from original selected indices and through-member injectivity are unchanged. The literal preimage list has the same length; its actual indexed elements are the preimages of the original indexed elements. This proves every pair field on a genuine common domain, without assuming local matching of branches or boundary equations in the new coordinates. Empty selected sets, zero ambient dimension, empty finite boundary and disconnected manifolds are included. □

Theorem 50.3 (Genuine intrinsic original branch multiplicity invariance)**.** *The intrinsic original branch count of the actual pulled-back whole carrier at z equals the intrinsic original count at $g(z)$. In particular it is invariant under every actual biholomorphism of whole NC carriers, independently of all initial coordinate branch labels or coordinate choices.*

Proof. Choose the original actual centred coordinates and selected finite set at $g(z)$. The entire pulled-back union coordinates were just produced with exactly that same selected set. The genuine intrinsic cardinality theorem, based on unique actual coordinate-germ overlap bijections, identifies the two intrinsic counts with the cardinality of this same actual set. Thus they are equal. Arbitrary proofs of the NC property do not change this intrinsic number. In the biholomorphic case the original map has bijective differential by its actual holomorphic inverse and the two derivative chain rules; therefore the preceding actual local proof applies at every point. No count equality or branch permutation is a supplied certificate. □

Theorem 50.4 (Actual whole superlevels and unique attained maximum alignment)**.** *Under an actual biholomorphism, the entire original branch-count value sets, every actual superlevel, all uniform bounds and the count-at-most-one stopping condition agree by genuine preimage. Every nonempty original carrier internally has one unique attained positive whole global maximum. Its entire highest centre is therefore uniquely determined and corresponds by the exact biholomorphic preimage, without a supplied maximum or centre alignment certificate.*

Proof. The pointwise actual count identity gives exact equality for every superlevel. Actual surjectivity gives equality of the two whole value sets and both directions for every uniform bound and stopping condition. The genuine finite count-image theorem internally produces a positive attained maximum for a nonempty original carrier. If r and s are two attained bounds, evaluate the s bound at an actual r-attaining point and the r bound at an actual s-attaining point: $r \le s$ and $s \le r$, hence $r = s$. Surjectivity transports attainment as well as the bound, so the maximum is the same on the actual pulled-back whole carrier. The entire maximum stratum is its actual superlevel at that same internally produced integer, giving the exact whole-centre preimage identity. The empty-carrier case has identically zero count and an invariant terminal condition; it requires no positive maximum. These statements make no assertion that restriction to an arbitrary open subset preserves a global maximum. □

Remark 50.5. The produced whole-union and pair dictionaries, intrinsic count identity and unique actual highest-centre correspondence are the mathematical prerequisites for arbitrary genuine blowup-choice comparison of the finite geometric separation tower. Actual analytic blowup uniqueness and its already proved comparison cocycle will then be applied to these genuinely aligned centres. The subsequent finite choice-comparison construction remains a separate new obligation. The old marked-ideal ordered invariant retains its colexicographic label tie-break, and extra marked cosupport admissibility is not a consequence of these geometric identities. Full canonical marked-ideal general-NC G9 remains open.

51 The first actual Hilbert–Samuel value and current singular position

The literal convention is $H_I(\ell) = \dim_{\mathbb{C}} R/(I + \mathfrak{m}^{\ell+1})$ ([4], Remark 1.3). At a point of a reduced analytic set, $H_I(1)$ therefore equals one plus the actual embedding dimension. The whole original ideal, properness, complete cotangent image and genuine pure singular threshold are produced from original analytic geometry; no caller tangent-value identity is assumed.

Lemma 51.1 (All original first coordinate words)**.** *The actual words of length less than two are exactly the empty word and the single original coordinate words. Their genuine finite index set has cardinality* $1 + n$*.*

Proof. The actual degree is a natural number strictly below two, hence zero or one. A tuple on the empty index has a unique value; a tuple on the singleton index is determined by its sole genuine original coordinate. Send the empty word to the one-point summand and a singleton word to that original coordinate. The inverse uses the actual empty tuple and actual constant singleton tuple. Their genuine inverse identities follow from those two finite tuple facts, including zero ambient dimension. Thus the actual word index is equivalent to a one-point set disjoint union the n coordinates, which has cardinality $1 + n$. □

Lemma 51.2 (Entire first-jet range from genuine original affine functions)**.** *The actual first scalar coordinate jet on the entire holomorphic germ ring is surjective onto the whole genuine* $(1 + n)$*-coordinate array. Consequently the entire actual first-jet range has dimension* $1 + n$*.*

Proof. Given an arbitrary whole array, take its genuine empty-word value a and singleton values b_i. The same original affine holomorphic function $z \mapsto a + \sum_i b_i z_i$ has value a and actual derivative b_i on coordinate direction i at zero. Its true degree-zero and degree-one iterated derivatives are exactly those same genuine value and derivative. Thus its entire actual germ realizes the arbitrary array. This is an internal representative producer; no first-jet-surjectivity certificate is assumed. The earlier faithful coordinate array is injective on the entire multilinear jet target and hence on the true range, and the new affine producer makes its image the entire scalar array. The induced genuine linear equivalence proves the actual range dimension, including $n = 0$. □

Lemma 51.3 (The entire first ideal image is the original cotangent span)**.** *For actual original finite holomorphic generators vanishing at the original point, the entire ideal first-jet image equals the faithful coordinate embedding of the span of their unmodified original differentials.*

Proof. Embed each actual complex linear functional into the genuine first scalar array by empty-word value zero and its original values on each coordinate direction. Genuine basis evaluation makes this linear map injective. The internally proved whole ideal-image Macaulay formula spans all actual empty/singleton word-times-generator first jets. An empty word gives zero constant part and the actual generator differential. A singleton word $z_i G_j(p+z)$ has value zero at zero, and its genuine first derivative is $G_j(p)$ times the original coordinate differential plus $z_i(0)$ times the original generator differential. Both terms vanish by the real original zero value and true coordinate value. In the entire original germ ring, both the singleton coordinate and translated vanishing generator belong to the actual maximal ideal. Their real product belongs to its genuine square, which is exactly the kernel of the entire first-jet map. This gives the same genuine vanishing without a selected derivative formula. Therefore all nonempty word columns vanish and every original generator differential occurs as an empty-word column. This proves equality with the entire ideal image, in both directions. Injectivity of the actual differential-coordinate map preserves its whole span dimension. No selected-derivative or whole-image equality is supplied as a premise. □

Lemma 51.4 (First literal whole-ideal point value from genuine differentials)**.** *For the actual whole ideal of the original vanishing family,* $H_I(1) = 1 + n - \dim \operatorname{span}\{dG_i(p)\}$.

Proof. The actual entire first-jet range dimension is $1+n$. The actual entire ideal first-jet image has exactly the dimension of the full original differential span. Substitute both genuine image dimensions into the proved actual entire quotient/rank formula at $\ell = 1$, whose truncation exponent is exactly two. This gives the literal whole quotient value, including the empty family and zero ideal. Vanishing at the original point makes the whole family ideal proper; the unit-ideal case is deliberately not asserted under this genuine vanishing hypothesis. □

Theorem 51.5 (Actual complete reduced stalk value equals embedding dimension plus one)**.** *At every genuine point of an original reduced analytic set, the entire actual reduced ideal is internally proper and its first actual intrinsic Hilbert–Samuel value is exactly embedding dimension plus one.*

Proof. Every original member of the whole vanishing ideal vanishes at the genuine point, since its original eventual vanishing condition holds at that point itself. In the actual translated preferred chart its genuine germ value at zero is the same original value by the true chart inverse identity. All original membership germs therefore lie in the actual maximal ideal; taking their whole ring span proves actual properness internally. Original analyticity supplies one genuine Cartan family for the complete vanishing ideal near the point. Its exact original membership and zero-locus dictionaries show internally that every generator vanishes at this actual point and that their entire translated ring ideal is the whole reduced stalk ideal. The previous actual cotangent-image computation thus applies without a caller selected-Jacobian equality. The genuine whole tangent-space Cartan dictionary identifies the common kernel of these same original differentials with the actual reduced tangent space. Its actual rank-nullity formula says that differential-span dimension plus actual embedding dimension equals n. Combining it with the literal whole quotient value $1+n$ minus that actual span dimension gives precisely embedding dimension plus one. The ideal and all generator/differential dictionaries are internally produced from the original reduced analytic set. □

Theorem 51.6 (Exact current pure singular threshold of the same intrinsic value)**.** *For an original pure-d reduced analytic set, its genuine singular locus is exactly the points of the set where the same actual intrinsic* $H(1)$ *is at least* $d+2$.

Proof. The previously proved actual complete reduced-ideal embedding-dimension criterion identifies pure singular points exactly with embedding dimension at least $d+1$. At every point of the actual set the new entire stalk-value identity is $H(1) = 1 + \text{embdim}$. Adding one converts that genuine threshold exactly to $d+2$, in both directions. Outside the actual set the intersection condition excludes points, so no unit-stalk exception is silently treated as a proper stalk. Empty pure sets satisfy the identity vacuously; zero-dimensional regular points give $H(1) = 1$ and are excluded from threshold two. The same intrinsic function and its genuine analytic superlevel, not a separately chosen chart function, occur in this actual current-singularity dictionary. □

Remark 51.7. This is a genuine first-value bridge for the modified current embedded invariant. It supplies exact original singular position data; it does not produce a normally flat smooth centre or a Hilbert–Samuel decrease after an actual blowup. General higher-codimension G8 remains open until those actual geometry and higher-entry descent producers are constructed.

52 Actual General3 nonspecial local analytic loci

The invariants below are the literal constructed family at dimension three and its literal Seq masked Step 1 companion history. Actual lower domain and canonical-tail data are internally produced. This section concerns positive actual first order and nonspecial actual companion points, with Step 1 first threshold equal to the point's actual level. Special and higher-level boundaries, whole General3 analyticity and the general noncompact terminating sequence remain open.

Definition 52.1 (The actual positive-order companion patch)**.** At actual current point x, let q be the real nonmonomial order and r its literal order-q trailing run. Take the genuine head order mask, its actual tail preimage, the literal companion formed at that head and the actual restricted tail. These are the actual literal definitions. The nested lower patch is the true Step 1 lower mask preimage for this companion and its actual tail. The literal threshold at a companion point p and lower value α is $(q, \mathrm{inr}(\mathrm{level}(K, S, p), \alpha))$ in ValTy3.

Lemma 52.2 (The actual companion domain and whole cosupport dictionary)**.** *For actual original domain3, a canonical tower of actual invFamily3, and a current cosupport point of positive real order q, its actual companion patch contains the point. The literal companion has real domain3 and maximal order; its literal restricted tail is canonical for actual Seq masked Step 1 with lower family invFamily2. On the whole actual patch, companion cosupport is exactly actual current original cosupport at order q.*

Proof. The constructed dimension-three invariant internally supplies a true original test tower. Its actual general canonical-tail producer supplies mask membership, real companion domain, actual maximal order, canonical restricted Step 1 tail and actual companion cosupport membership at the point. The literal Step 1 bridge internally produces the actual companion test tower. The original head and tail are genuine test towers by the actual head-tail splitting theorem. Apply the actual companion cosupport dictionary and the exact whole head-tail transform identity. This gives the claimed whole-patch equality at every point, with no supplied lower domain or compatibility certificate. □

Lemma 52.3 (Exact actual general value and whole companion threshold)**.** *On the whole actual companion patch, every threshold $(q, \mathrm{inr}\,\beta)$ of actual invFamily3 agrees exactly with the true companion cosupport superlevel at β for its literal Seq Step 1 value. Here β may be any actual WithTop Step 1 value.*

Proof. The real order-mask propagation theorem bounds the current actual order by q at every original cosupport point of this genuine patch. Indeed the actual head mask excludes the genuine order-$q+1$ locus and the true original and companion test tails propagate that exclusion. If a

point qualifies for a threshold with first entry q, the actual lexicographic first entry forces order at least q; thus its order is exactly q. The preceding actual cosupport dictionary puts it in the literal companion cosupport. The genuine canonical-tail fixed-companion value theorem, whose actual general lower data are internally produced, then identifies its actual invFamily3 value with $(q, \operatorname{inr} M)$ for the true literal Seq Step 1 value of this fixed companion and tail. Lexicographic comparison and equal first entries yield precisely $\beta \leq M$. Conversely a companion cosupport point is in the true original current cosupport, and the same actual fixed-value identity turns its companion comparison into the general comparison. This proves exact equality of whole sets on the patch, not merely equality on a chosen level stratum. It includes the actual top companion threshold, whose whole set is empty on this patch. □

Lemma 52.4 (A genuine ambient open neighbourhood from two nested opens)**.** *For an actual open U in a complex manifold and actual open V in U, its actual image under inclusion is an open W of the original manifold. Every analytic preimage of an original set on V gives the corresponding analytic preimage on this actual W.*

Proof. The real inclusion of open U is an open map, so its actual image of V is open. Every point of this image has a unique lift to U and then to V, obtained from its actual image membership. Both lifts are holomorphic: the open-subtype lifting theorem applies twice to the actual inclusion, which is holomorphic. Their underlying original point is unchanged. Holomorphic preimage of the real analytic set on V is therefore analytic on W; the exact unchanged-point identity makes that preimage exactly the original set's true preimage on W. No analyticity of an arbitrary image, closed-phase argument or descent certificate is assumed. □

Theorem 52.5 (Actual General3 local analytic thresholds at a nonspecial companion)**.** *For an actual domain3 datum, actual canonical invFamily3 tower and actual positive-order current cosupport point, suppose its literal companion point is genuinely nonspecial. There is a real ambient open neighbourhood on which all actual General3 thresholds $(q, \operatorname{inr}(s, \alpha))$, with s the actual companion point's current Step 1 level and any literal lower threshold α, have analytic whole-set preimages.*

Proof. The real internally produced mask membership gives the actual companion point; thus its nonspecialness premise is not vacuous. Its actual domain3, maximal order and canonical Seq Step 1 tail were internally produced above. Apply the actual dimension-three lower-patch producer to this real companion point. It gives a true open neighbourhood in the companion patch and analytic entire Step 1 superlevels at first entry equal to the point's actual level, for every real lower threshold. The exact preceding General3/companion threshold dictionary identifies those entire preimage sets with the actual invFamily3 superlevel preimages on the nested patch. The preceding genuine two-open transfer constructs a real ambient open containing the original point and preserves every one of these actual analytic preimages. The same open works for all lower thresholds. No global maximum or compactness enters; special and higher-level boundary descent, whole General3 analyticity and global termination are not inferred. □

53 Actual Jacobian pivots of a holomorphic submersion

The input is a genuine holomorphic submersion of possibly different dimensions, with original atlas charts through the two original points. The local section, the derivative right inverse and the nonzero holomorphic Jacobian pivot are constructed internally. Derivative-ideal equality and the complete selected canonical tower under smooth base change are subsequent obligations.

Lemma 53.1. *Let $f : Y^m \to X^n$ be holomorphic and a submersion at y, and let e' and e be original atlas charts through y and $f(y)$. For the actual chart map $F = e \circ f \circ (e')^{-1}$, the complex derivative $DF(e'(y))$ has a complex continuous linear right inverse.*

Proof. Use the genuine submersion charts to obtain an actual holomorphic local section s through the original point, with $s(f(y)) = y$ and $f \circ s = \mathrm{id}$ near $f(y)$. Put $S = e' \circ s \circ e^{-1}$. The actual inverse-chart continuity, the original point identities and holomorphicity of s give a neighbourhood of $e(f(y))$ on which all arguments stay in their true chart domains. There $F \circ S = \mathrm{id}$ and $S(e(f(y))) = e'(y)$. Both chart maps are complex differentiable at those points. Differentiate the actual neighbourhood identity and apply the complex chain rule. The derivative $DS(e(f(y)))$ is the required right inverse. No equality of Jacobians or chosen inverse is a hypothesis. □

Lemma 53.2. *The derivative of the actual chart map F at $e'(y)$ is surjective.*

Proof. For any vector v in the original target model, apply the internally produced right inverse to v. The differentiated actual section identity shows that its image under $DF(e'(y))$ is exactly v. □

Lemma 53.3. *There are actual original source-coordinate columns $c : \{1, \dots, n\} \to \{1, \dots, m\}$ for which the square matrix $A_{ij} = (DF(e'(y))\epsilon_{c(j)})_i$ has nonzero complex determinant.*

Proof. Let L_i be the ith target coordinate functional composed with the actual derivative. A linear relation between the L_i vanishes after evaluation at each preimage of a target coordinate vector. Surjectivity supplies those preimages, and evaluation gives every coefficient zero. Thus the actual rows are linearly independent. The finite original-coordinate evaluation map is injective. The resulting rectangular matrix has row rank n, and its columns span the whole n-dimensional coordinate space. Select a genuine finite basis from those original columns and reindex it by $\{1, \dots, n\}$. The square matrix of that basis has nonzero determinant. If $n = 0$, the chosen empty matrix has determinant one; no positivity or nonempty column assumption is needed. □

Theorem 53.4. *Some actual original-coordinate Jacobian minor is a holomorphic scalar germ at y, is nonzero there and stays nonzero on a genuine neighbourhood of y. For any actual finite matrix of scalar holomorphic germs its determinant is holomorphic, including the empty matrix.*

Proof. Keep the actual columns supplied by the previous lemma. Every matrix entry is one of the genuine chart chain coefficients for an original source coordinate vector, so is holomorphic at y. Expand the square determinant as its finite signed sum of products. Closure of holomorphic germs under products, sums and complex constants makes this actual determinant holomorphic. Holomorphicity gives continuity; because its value at the original point is nonzero, an open neighbourhood of zero's complement pulls back to a genuine neighbourhood of y where this same internally selected minor stays nonzero. The empty determinant is the constant one, so the zero-dimensional target is included. □

54 Actual finite NC separation under arbitrary genuine blowup choices

The choices in this section are actual individual blowup data over actual higher highest strata. No candidate finite tower, termination certificate, centre alignment or naturality certificate is a producer input. The finite procedures themselves and all stage comparisons are constructed internally. The old ordered marked-ideal label tie-break is a separate obligation.

Lemma 54.1 (Unique genuine lift over the actual aligned highest centre)**.** *For entire centres identified by a genuine base biholomorphism as exact whole preimages, arbitrary actual blowups internally produce one unique holomorphic comparison lift, which is a genuine biholomorphism commuting with the original projections.*

Proof. The preceding genuine intrinsic-count theorem and actual maximum uniqueness identify the two attained maxima and their whole centre preimages. Compose the target blowup projection

with the inverse base biholomorphism. The actual biholomorphic blowup theorem makes this a genuine blowup of the very same source centre, using the actual whole vanishing-ideal pullback identity in the universal property. Actual analytic blowup uniqueness produces the comparison and reverse comparison; each composite is the identity by genuine uniqueness. Thus the lift is an actual biholomorphism. Any two holomorphic lifts with the same base square also factor the source projection through this same transported actual target blowup, so universal uniqueness forces their equality. No lift or centre correspondence is supplied. □

Lemma 54.2 (Whole transformed original and literal boundary dictionary)**.** *The internally produced step lift carries the entire actual strict original carrier, the literal strict old boundary list and the new exceptional preimage to their true counterparts. The same holds for every actual prefix of stage-isomorphic geometric towers, independently of any marked ideal coherence or cosupport admissibility.*

Proof. The genuine commuting projection square gives exact equality of the preimages of each original set off its exact whole centre. The actual step biholomorphism is a homeomorphism, so it commutes with the closure of that whole preimage. This yields equality for the entire original geometric strict transform and every actual old boundary member. The same square and centre identity give equality of the entire exceptional preimage. Exact list maps and append identities identify the literal new finite boundary list. At all performed steps the actual original count is at least two, so relative density identifies the produced geometric original with the true differential strict original. An actual biholomorphism has no exceptional differential point, since its derivative and that of its inverse compose in both orders to the identity. Thus its differential strict transform of any closed carrier is its entire preimage. Apply true holomorphic strict composition to both sides of the projection square to identify the entire differential original under every prefix. Induction on the actual recorded stage maps proves the full-prefix original and boundary dictionaries. Only the literal boundary field of its tracking marked datum is involved; its ideal/coherence is not used to obtain these geometric equalities. □

Definition 54.3 (Actual higher blowup choice data and internal existence)**.** A higher blowup choice assigns to each actual NC carrier, actual integer $r \geq 2$ and its true global count bound one actual manifold, projection and genuine blowup proof over the literal whole count-r superlevel. These are individual geometric blowup data, with no finite tower or termination/naturality property. Such choices exist internally by the already constructed genuine highest-stratum blowup.

Theorem 54.4 (A real finite procedure for every actual blowup choice)**.** *Every such individual geometric choice and every actual boundary-adapted pair with original count bounded by b internally give a recorded finite tower of length at most* $\max(b-1,0)$*. Every actual current centre is its whole attained original maximum of count at least two. Its literal final differential strict original carrier has count at most one and is adapted to its literal actual accumulated boundary.*

Proof. Define the actual procedure by natural-number recursion on b, uniformly in the current manifold, original carrier and literal finite boundary. At count at most one use the real reflexive tower; this includes $b = 0$. Otherwise the finite actual count image internally produces a unique attained higher maximum $2 \leq r \leq b$. Invoke the actual individual choice on that literal carrier, integer and bound. The existing actual transformed-pair producer, not the choice, proves the entire strict-original/boundary dictionary and count at most $r - 1 \leq b - 1$. Recurse on this genuinely produced pair and smaller bound, then concatenate the actual first blowup with the actual tail. The exact strict-composition and literal boundary-append identities transport each recorded prefix centre and terminal original pair. The length estimate is the same actual natural-number estimate as before. The recursion therefore internally supplies a real finite tower and its conclusions for every choice. No termination property or preselected finite output is required of a choice. □

Theorem 54.5 (Actual arbitrary-choice comparison of the finite procedures)**.** *For any two individual geometric choices and an actual biholomorphism of initial whole carrier and finite boundary, the internally constructed finite procedures have a genuine stagewise tower isomorphism over that initial biholomorphism. In particular they have the same actual length and terminal stopping, and their entire actual original carriers and literal accumulated boundaries correspond under the produced maps.*

Proof. Induct on the same actual original bound b for the two procedures. The intrinsic count identity makes their terminal tests equivalent. If terminal, both actual outputs are reflexive and the given initial map is their genuine stage map. Otherwise their internally selected attained maxima agree by genuine maximum uniqueness and their entire centres agree by exact preimage. The two arbitrary choices therefore produce genuine blowups of aligned centres. The preceding unique actual lift gives their first stage map. Its actual whole transformed pair dictionary gives the true initial relation for the two smaller procedures; their genuine transformed original counts have bound $b-1$. Apply the induction hypothesis to these actual smaller outputs and concatenate the real first-stage and tail isomorphisms. This constructs the stagewise isomorphism internally. The true tower isomorphism projection and length theorems, together with the proved whole original/boundary dictionaries, give the stated literal outputs. This proof does not use a caller-supplied tower isomorphism or a naturality property of the individual choices. □

Lemma 54.6 (Unique full-stage comparisons and actual cocycle)**.** *Over a fixed base biholomorphism, two stagewise comparisons of the same actual towers have the same last-stage biholomorphism. Consequently the internally produced choice comparisons have genuine identity, inverse and triple cocycle under composition of the actual base maps.*

Proof. Induct on the actual stage-isomorphism construction. At the reflexive stage the map is the fixed base map. At a blowup stage the induction hypothesis identifies the two preceding actual stage maps. Inverting the last stage constructor gives the same actual centre correspondence and commuting projection square. Unique holomorphic lift of that single genuine blowup then identifies the two last maps. Actual identity and inverse stage isomorphisms and actual composition of stage isomorphisms are already proved for recorded towers. Comparing them with the internally produced maps by this uniqueness gives the identity, inverse and triple cocycle. The cocycle is a conclusion of genuine geometry, never a property demanded of a choice. □

Theorem 54.7 (Internally produced natural geometric NC separation)**.** *The internally existing actual choices give a real finite geometric NC separation producer, unique up to the genuine stage comparison under arbitrary individual blowup choices and actual biholomorphisms. Its final whole differential strict original is smooth, transverse to its literal accumulated SNC boundary, and appending it gives SNC; the projection is an isomorphism over the whole original complement.*

Proof. Take the internally produced individual choice and the actual bound n. The real procedure gives its actual finite length bound and literal higher-centre stages. The existing terminal count-one smoothness/transversality/whole appended-SNC consumers apply to its actual final pair. Every actual centre lies in its real current original strict carrier, hence over the original closed carrier; the genuine complement-isomorphism theorem applies. The preceding arbitrary-choice comparison and uniqueness/cocycle apply to any other actual choices and original biholomorphic pair. Empty original carrier and dimension zero terminate at the genuine reflexive tower; disconnected manifolds and arbitrary finite old boundary need no compactness or globally finite component family. □

Remark 54.8. These statements concern the unordered intrinsic geometric NC separation procedure. They do not identify it with the old canonical marked-ideal algorithm, whose colexicographic label tie-break and additional marked cosupport admissibility remain separate. Full canonical marked-ideal G9 remains open until those additional obligations are genuinely proved.

55 Actual dimension-three Step 1 curve envelope and special thresholds

All loci below are genuinely constructed from actual coherent ideals, actual smooth carriers and literal original SNC members. The main original dimension-three canonical consumer internally produces its companion domain and actual test history. A closed analytic dimension-one envelope is a geometric reduction; arbitrary closed subsets of that envelope are not inferred analytic. Finite lower-threshold gluing across this curve, whole General3 analyticity and global termination remain open.

Lemma 55.1 (Dimension of the actual maximal-order remainder)**.** *In a genuine closed smooth carrier of dimension at most three, for a coherent ideal of positive actual maximal order, the true cosupport minus its actual codimension-one part has dimension at most one.*

Proof. The bound concerns the true remainder itself; its separately proved analytic structure is used later in constructing the envelope. At each regular point of the remainder, take the actual local carrier dimension supplied by smoothness. If it is zero, monotonicity already bounds the remainder dimension by zero. Otherwise the genuine maximal-order tangent-direction theorem bounds the remainder's regular dimension strictly below one less than this actual carrier dimension. The true carrier bound three then gives dimension at most one, including variable-dimensional carrier components. No lower-dimension exception or thinness certificate is assumed. □

Lemma 55.2 (Actual transverse boundary traces and unselected remainder)**.** *Every actual transverse SNC member trace on a smooth dimension-at-most-three carrier has dimension at most two. Its actual cosupport intersection with the union of genuine trace components not contained in cosupport is analytic of dimension at most one.*

Proof. Read the actual carrier, singleton member trace and indexed boundary member in the same genuine adapted coordinate chart. The member's coordinate lies among the carrier's free coordinates. If the trace retained this coordinate as free, a small actual perturbation in this coordinate would remain in the flattened trace but have nonzero member coordinate, contradicting its exact flattening identity. Thus the trace has at most one less genuine free coordinate than the carrier, hence at most two. Genuine locally finite smooth component subfamilies are closed and smooth. Analytic continuation on a trace component shows that an open piece of its intersection with cosupport would force the entire true component into cosupport, contradicting its actual unselected definition. This internally proves thinness. Genuine thin dimension drop gives dimension at most one. No component-absence certificate is an input. □

Lemma 55.3 (Dimension preserved by actual finite analytic unions)**.** *A finite union of actual analytic sets with dimension at most d is analytic and has dimension at most d.*

Proof. Finite products of actual generators give union analyticity. At a regular point of a binary union, take the closure of its genuine connected regular component. This is an actual analytic irreducible set and that point retains its real regular dimension on the closure. Intersecting the closure with each actual analytic member expresses it as a union of two genuine analytic subsets. True irreducibility therefore forces the entire closure into one original member. Actual dimension monotonicity bounds its regular dimension by that member's bound. Finite induction proves the indexed union, with the empty union having no regular points. □

Theorem 55.4 (Internally constructed actual nonspecial curve envelope)**.** *For a true well-formed maximal-order datum with carrier dimension at most three and a literal true test tower for its actual barMarked datum, there is a closed analytic set of dimension at most one containing every genuine nonspecial point of the actual current cosupport.*

Proof. The actual barMarked datum and literal true tower internally produce current well-formedness, coherence, positive transformed order, actual maximal order and current carrier dimension bound. Take the genuine cosupport remainder from the first lemma and, for each actual indexed original boundary member, the true unselected-component trace intersection from the second lemma. Their actual finite union with the remainder is the claimed analytic envelope. If the current level is zero, failure of true specialness puts the point in the codimension-one remainder. If the level is positive, the literal boundary count supplies an actual incident original member. Were its true connected trace component selected, analytic continuation and the exact memberInterLocus component dictionary would make the point special. Its nonspecialness forces membership in the actual unselected trace intersection. Empty/repeated indexed members and disconnected carriers are handled by these actual dictionaries. □

Theorem 55.5 (Actual original General3 companion envelope)**.** *Original domain3 and a literal canonical constructed invFamily3 tower at a positive-order current point internally produce an actual closed analytic curve envelope for nonspecial points of the true companion on its entire literal order-mask tail preimage.*

Proof. The real General3 canonical-tail producer internally supplies actual mask membership, true companion domain3/maximal order and actual canonical Seq Step1 restricted tail. Its genuine Step1 bridge gives the actual barMarked test tower. The preceding producer then applies to that actual companion, with all geometric inputs supplied internally by original actual data. □

Theorem 55.6 (Actual original special and higher-level top thresholds)**.** *Under the same original actual domain3 and canonical tower, every entire General3 threshold (q,inr(s,top)), for arbitrary actual natural s, is analytic on the entire literal order-mask preimage. No nonspecialness hypothesis is used.*

Proof. The exact whole General3/companion cosupport threshold dictionary rewrites the entire original set as the true companion Step1 top-lower-entry threshold. Actual companion domain and canonical-tail data internally supply its real barMarked true test tower and exact original/barMarked current cosupport bridge. The genuine top-threshold producer constructs analytic derivative cosupport, true codimension-one and actual member-intersection loci, together with actual indexed level superlevels, proving this entire Step1 set analytic for every s. This includes actual special and higher-level points without extending the nonspecial current-level theorem to arbitrary finite thresholds. □

56 Actual fixed-order whole-quotient frames on an internally produced pivot patch

These are genuine frames for the entire point quotients at one fixed truncation order, constructed from original holomorphic functions. The patch is internally chosen where their actual finite Macaulay rank attains its maximum on the given nonempty original open domain. A single nonzero minor at an arbitrary point does not imply constant rank on a neighbourhood. No density, maximal-Hilbert–Samuel centre or normal-flatness for all relative powers is asserted here.

Definition 56.1 (The fixed actual whole scalar-jet space)**.** For fixed q, let J_q be the genuine entire first-q scalar-coordinate jet map and $S_q = \operatorname{range} J_q$. This is a fixed finite-dimensional complex subspace of the actual coordinate-word array space, independent of any point or ideal. Its actual kernel is the entire maximal-ideal power $\mathfrak{m}^q$.

Proof. The real scalar-coordinate evaluation is injective on the entire original multilinear jet target. Hence its composite with the actual whole-germ jet has exactly that jet's kernel. The previous actual point-power/kernel theorem identifies this kernel with the complete maximal-ideal power. The actual range is a subspace of a finite coordinate array, so is finite-dimensional.

No independence of repeated word coordinates or artificial full-array surjectivity is asserted for $q > 2$. □

Lemma 56.2 (Actual selected columns and the complete ideal image)**.** *For original holomorphic family G at p, let $C(p)$ be its true word-times- generator Macaulay matrix. Given actual row and column selections of size k, let F select those rows, $B(p)$ combine those selected actual columns, and $K(p) = FB(p)$. Every selected and unselected actual column belongs to S_q; the complete column image equals the whole original translated ideal jet image.*

Proof. The actual matrix entry dictionary is precisely $J_q(z^\alpha G_i(p + z))$. Thus every genuine column is in the fixed range S_q. Finite linear combination of the selected columns gives $B(p)$ and direct finite-sum evaluation gives $FB(p) = K(p)$. The genuine whole-ideal jet-image/Macaulay formula supplies ALL original ideal membership germs, not only a selected family of derivatives. Consequently the full matrix image is exactly that entire ideal image; selected columns span a subspace of it. This applies equally to empty selections and empty families. □

Theorem 56.3 (Internally attained maximal rank and a real nonempty open pivot patch)**.** *For a nonempty original open domain on which the actual original finite family is holomorphic, the real Macaulay rank attains an actual maximum k. Internally chosen row and column selections give a nonzero actual k-minor at a maximizing point. On an internally constructed nonempty open pivot patch, this minor stays nonzero and the ENTIRE actual matrix rank equals k at every point.*

Proof. All genuine matrix ranks are natural numbers bounded by the fixed finite array dimension, so their image on the nonempty original domain is a nonempty finite set. Choose its actual maximum and a genuine point realizing it. The true rank/minor dictionary produces an actual k-square minor with nonzero determinant at that point (for $k = 0$ its true empty determinant is one). Original holomorphic matrix entries give holomorphic finite minors; continuity on the original open domain makes its nonzero locus open. Intersect with that domain to obtain the real nonempty patch. At every point of it, the actual nonzero minor forces rank at least k, and the internally chosen GLOBAL maximum on that domain forces rank at most k. Thus rank is exactly k. The upper bound is indispensable; the proof does not infer it from a single arbitrary nonzero minor. No compactness or caller rank/attainment/patch certificate is assumed. □

Lemma 56.4 (Actual holomorphic pivot projection and its complete kernel)**.** *On that genuine patch, the actual endomorphism $P(p) = 1 - B(p)K(p)^{-1}F$ restricts to a surjection from S_q to the fixed space $L = S_q \cap \ker F$. It is identity on L, and its kernel inside S_q is precisely the ENTIRE actual ideal jet image. Its unmodified original finite array entries are holomorphic on the real pivot patch.*

Proof. The actual determinant is nonzero, so $K(p)$ has its genuine inverse. Original matrix entries are holomorphic and the adjugate/determinant formula expresses every inverse entry as a genuine finite polynomial divided by the same nonzero original determinant, hence holomorphic. Finite products/sums give the actual entries of P. The identity $FB = K$ yields $FP = 0$; all actual columns of B belong to S_q, hence $P(S_q) \subseteq S_q \cap \ker F$. On that fixed intersection, $Fv = 0$ gives $Pv = v$, proving surjectivity. The inverse identity gives $PB = 0$. Conversely $Pv = 0$ expresses $v = BK^{-1}Fv$, so its kernel is exactly the selected column span. The invertible pivot makes those k selected columns independent. They lie in the actual complete image, and on the real patch the actual complete image has the same dimension k. The finite-dimensional inclusion and equal actual dimensions identify selected image with ENTIRE ideal image. This proves the claimed full kernel, with no selected-image equality premise. □

Theorem 56.5 (Genuine entire point quotients have fixed local frame coordinates)**.** *For EVERY point of the internally produced pivot patch, $P(p)J_q$ gives genuine coordinates identifying the literal*

ENTIRE point quotient $R/(I_p + \mathfrak{m}^q)$ with the SAME fixed complex space L. Its actual dimension is $\dim S_q - k$. The original finite projection entries vary holomorphically on the patch.

Proof. The entire scalar jet surjects onto its actual range S_q by definition; P surjects from that range onto L. Their composite therefore surjects onto L. If $PJ_q g = 0$, the complete kernel theorem supplies an original entire ideal member a with $J_q g = J_q a$. Thus $g - a$ belongs to the genuine kernel $\mathfrak{m}^q$, proving $g \in I_p + \mathfrak{m}^q$. Conversely an actual ideal member maps into the complete ideal image and is killed by P, while every actual maximal-ideal-power member is killed by J_q. This proves the exact ENTIRE kernel. The genuine first-isomorphism theorem produces a linear equivalence from the literal ring-ideal quotient to L. Because F restricts to a surjection on S_q (using the actual invertible pivot columns), true rank-nullity gives $\dim L = \dim S_q - k$. At $q = 0$, $S_q = 0$ and the actual quotient by $\mathfrak{m}^0 = R$ is zero. Empty original families have $k = 0$ and give the actual entire truncated ambient quotient; unit ideals give a zero quotient. No full-array dimension is substituted for the actual possibly redundant whole-jet range. □

Theorem 56.6 (Coherence alone internally produces whole intrinsic quotient frames)**.** *At every original base point of a coherent ideal on a genuine complex manifold, there is an original open neighbourhood and an internally chosen nonempty open subpatch on which its ENTIRE intrinsic fixed-q point quotients have the same actual finite complex frame model. The actual preferred chart point quotient is transported by genuine nonlinear original chart algebra equivalences. The subpatch need not contain the initial base point.*

Proof. Original coherence internally produces one genuine Cartan family, its original holomorphic neighbourhood and the ALL-membership whole-stalk ideal dictionary in the initial original chart. Move that neighbourhood into the actual chart coordinates: the real target/preimage intersection is open and nonempty at the base point, and the entire original family is holomorphic there. Apply the internally attained maximum/pivot/frame producer. Pull its genuine nonempty open patch back through the original chart; it stays inside the Cartan neighbourhood and is nonempty because every chosen coordinate point belongs to the true target. At every point of this subpatch the ALL-membership dictionary identifies the fixed-chart reduced whole ideal with the same actual translated family ideal, so the preceding entire quotient frame applies. The actual translated nonlinear overlap between the preferred point chart and this fixed original chart internally induces a full germ algebra equivalence; it maps ALL original membership germs and preserves every genuine maximal-ideal power. Its actual quotient algebra equivalence transports the preferred ENTIRE point quotient into those same frame coordinates. No caller ideal-map equality, quotient/frame isomorphism, matrix-rank constant or analytic chart certificate is supplied. The actual fixed-q Hilbert–Samuel value is constant on this chosen genuine open patch by the real quotient dimension. □

Remark 56.7. This supplies actual finite-truncation quotient coordinates from genuine holomorphic data. The internally maximizing rank gives a minimal fixed-q Hilbert–Samuel value on that original Cartan patch; it is not a construction of a maximal-Hilbert–Samuel resolution centre. Density would require actual analytic component/identity arguments and is not deduced. Full normal flatness still requires the real smooth-centre relative ideal filtration and all degrees; actual higher-HS blowup transformation and termination remain OPEN.

57 Literal Step 1 histories on the actual curve envelope

All historical predicates below are genuine pullbacks along literal tail projections of actual headstage loci. The finite indexed family is internally analytic. Actual curve components and genuine analytic dimension drop produce the exception, without a supplied compatibility or fixed-run certificate. This freezes actual history but does not yet extend literal member/contact lower ideals across their masks. Whole General3 finite-lower-threshold descent and global termination remain open.

Definition 57.1 (Actual head-level and whole top-threshold pullbacks)**.** At the literal head before the last r stages, pull back by the real literal tail projection the actual level-at-least-s locus, and the actual head barMarked cosupport superlevel at literal Seq Step1(invFamily2) threshold (s,top).

Lemma 57.2 (Internally analytic literal historical predicates)**.** *For an actual well-formed positive maximal-order datum and a true literal barMarked test tower, every actual historical predicate above is analytic.*

Proof. Literal head/tail splitting internally makes the actual head a true barMarked test tower. Actual transformed old-boundary SNC and transversality produce analytic indexed level superlevels. The genuine whole Step1 top-threshold producer internally gives analytic derivative cosupport, true codimension-one and member-intersection loci for the actual head. The true literal tail projection is holomorphic. Genuine holomorphic preimage preserves the actual analytic equations of both full loci, proving the historical pullbacks for every actual r and s. □

Lemma 57.3 (Exact actual head regime dictionary)**.** *At a genuine current barMarked cosupport point, the actual head stepRegime s is exactly membership in historical level-at-least-s, nonmembership in level at least s+1, and nonmembership in the whole top-threshold pullback.*

Proof. The real head/tail tower splitting and actual coherent transformed ideal internally project every genuine current cosupport point to actual head cosupport. There the actual Step1 top threshold is precisely level greater than s or level equal s and specialness; the genuine masked lower value is never junk top at an actual nonspecial cosupport point. Membership in level at least s and exclusion of level at least s+1 force true level equality. Under this equality, excluding the actual top threshold is exactly actual nonspecialness. These are precisely the literal definition of stepRegime, which proves both directions without a head-support or predicate certificate. □

Theorem 57.4 (The genuine curve's own current level and actual run freeze)**.** *For an actual analytic dimension-at-most-one set Z contained in true current barMarked cosupport, internally produce an analytic dimension-zero S subset Z. Every point of Z minus S is genuinely regular of dimension one, and on each true global component of Z, at points outside S the point's actual current level and literal regimeRun at its own actual current level agree.*

Proof. The true indexed boundary count is bounded by the original finite list length, since origMembers is its actual take. Use the actual finite index of head positions r at most the literal tower length, thresholds s from zero through original boundary length plus one, and two predicate types: actual level and whole top threshold. The preceding producer internally proves every one analytic. Genuine locally finite global components and their proper analytic intersections yield an actual dimension-zero predicate exception on Z, outside which all predicate memberships are precisely the component's whole containment. Thus all actual points on the same component outside that exception have equal memberships in every literal head predicate. The genuine zero-dimensional part of Z and actual pure curve singular locus give another actual analytic dimension-zero exception. Combine the two actual exceptions.

At the literal full-head position zero, compare the predicate at each point's own true current level. Both levels lie in the internally bounded index. This gives both level inequalities and hence their actual equality. For each actual head, apply the exact regime dictionary at this common true level: the finite family includes thresholds s and s+1 and the actual whole top threshold at s, so the real head predicates agree. The literal regimeRun recursion therefore agrees by induction on the actual tower. This is a true computed run identity, not a compatible-history input. The regular-envelope exception also makes every remaining actual point regular of dimension one. □

Theorem 57.5 (Actual General3 original inputs produce the whole curve/history seam)**.** *Original actual domain3 and literal canonical invFamily3 tower at an actual positive-order current cosupport point internally produce the genuine companion curve Z, actual analytic dimension-zero exception S subset Z, containment of every nonspecial companion cosupport point in Z, actual*

curve regularity outside S and the real own-level/current-run identity on every true component outside S.

Proof. The true original canonical-tail producer internally supplies all actual companion domain3/maximal-order and canonical restricted-tail data. Its true Step1 bridge gives the genuine barMarked test tail and exact current cosupport identity. The preceding dimension-three curve-envelope theorem internally produces analytic Z of dimension at most one, contained in actual companion cosupport and containing every true nonspecial point. Its exact bridge makes this Z a subset of real barMarked current cosupport. Apply the preceding actual finite-history producer to this internally constructed Z. It returns S and all literal history/regularity conclusions. No curve, compatibility, fixed run or analytic exception is supplied by the caller. Empty or zero-dimensional Z is absorbed in S and yields the honest empty regular complement. Arbitrary noncompact/disconnected carriers, repeated indexed empty boundary entries and literal empty blowup stages preserve the actual dictionaries. □

Remark 57.6. A closed arbitrary subset of the true curve is not inferred analytic. The actual lower member/contact datum must still be extended coherently across its literal mask and its original value identified there before finite lower threshold boundary gluing is complete. Whole General3 and G3 remain open.

58 Entire derivative ideals under a genuine holomorphic submersion

The input is the original holomorphic map $f : Y^m \to X^n$, a genuine submersion at every point, and an original ideal of holomorphic germs. Jacobian pivots, inverse coefficients and the entire ideal comparison are constructed internally. This is a prerequisite for full canonical smooth functoriality; higher invariant equality and blowup base change remain separate.

Lemma 58.1. *For an actual square matrix A of scalar holomorphic germs with $\det A(y) \neq 0$, the actual coefficients $B_{ij} = \det(A)^{-1} \operatorname{adj}(A)_{ij}$ are holomorphic at y. Where $\det A \neq 0$, the actual matrices satisfy $AB = 1$.*

Proof. The determinant is a finite signed sum of products of the original entries, hence holomorphic. Its nonzero value makes its reciprocal holomorphic near the original point. Each adjugate entry is the determinant of the original matrix with one row replaced by a coordinate vector; each replacement entry is either an original holomorphic entry or a constant. Thus every adjugate entry, and its product with the reciprocal determinant, is holomorphic. The algebraic identity $A \operatorname{adj}(A) = \det(A)1$, followed by scalar multiplication by $\det(A)^{-1}$, gives $AB = 1$ at each point where the same actual determinant is nonzero. The empty matrix is included. □

Lemma 58.2. *Let e, e' be original charts through $f(y), y$ and let a be holomorphic at $f(y)$. Internally choose actual source columns c_i of a nonzero Jacobian pivot. Every target coordinate derivative has the genuine germ identity*

$$(\partial_j^e a) \circ f = \sum_i B_{ij}\, \partial_{c_i}^{e'}(a \circ f).$$

All the actual coefficients B_{ij} are holomorphic at y.

Proof. The actual submersion derivative has a right inverse, so the internally constructed original-coordinate Jacobian minor is nonzero at y and remains nonzero near y. Keep these actual columns on this neighbourhood and construct B as above. The genuine chart chain rule, on a further neighbourhood where the original charts and a are valid, says $\partial_{c_i}^{e'}(a \circ f) = \sum_k A_{ki}(\partial_k^e a) \circ f$. Multiply by B_{ij} and sum over i. Exchange the two finite sums and use $\sum_i A_{ki} B_{ij} = \delta_{kj}$ from

the actual matrix identity. Only the jth target derivative remains. Every neighbourhood used is produced from the original point and actual holomorphic germs; no inverse or chain identity is supplied as an extra hypothesis. □

Lemma 58.3. *For every original ideal member a at $f(y)$, every original chart e through $f(y)$ and every target direction v, the actual germ $(\partial_v^e a) \circ f$ belongs to $\mathcal{D}(f^*I)$ at y.*

Proof. Choose an actual source chart through y. The holomorphic germ $a \circ f$ is an original generator of f^*I, and every actual source coordinate derivative of this generator belongs to $\mathcal{D}(f^*I)$ by its definition. The preceding internal recovery identity puts each pulled target coordinate derivative in that same entire ideal, since its coefficients are holomorphic. Write the actual target vector as the finite sum of its coordinate vectors with coefficients v_j. Complex linearity of the actual derivative gives $\partial_v^e a = \sum_j v_j \partial_j^e a$ on the original chart. Closure of the entire span under finite sums, constants and germ equality puts its pullback in $\mathcal{D}(f^*I)$ as required. □

Theorem 58.4. *For every ideal I of holomorphic germs on X and every genuine holomorphic submersion $f : Y \to X$, the entire original germ ideals satisfy $f^*\mathcal{D}(I) = \mathcal{D}(f^*I)$.*

Proof. At any original point y, a member of $f^*\mathcal{D}(I)$ is a finite holomorphic combination of pullbacks of actual members of $\mathcal{D}(I)$. Each such actual member is itself a finite holomorphic combination of members of I and original-chart directional derivatives of members of I. Pull this combination back along the original holomorphic map. Its coefficients remain holomorphic; pulled ideal members belong to f^*I and hence its derivative ideal, and pulled directional derivatives belong to that derivative ideal by the previous lemma. Transitivity of the actual entire finite spans proves $f^*\mathcal{D}(I) \subseteq \mathcal{D}(f^*I)$. The genuine cross-dimensional chain and product rules already prove the opposite inclusion for every holomorphic map. Apply both inclusions at every point and every scalar germ and use extensionality of the original germ ideal. □

Theorem 58.5. *For every integer $r \geq 0$, the actual entire iterated ideals satisfy $f^*\mathcal{D}^r(I) = \mathcal{D}^r(f^*I)$.*

Proof. Induct on r. The zero case is the original ideal itself, so reflexivity suffices. Every member of the actual iterated derivative ideal is holomorphic, by the defining finite holomorphic span (and the original hypothesis for $r = 0$). Apply the entire first derivative equality to this actual iterated ideal. Replace its pullback using the induction hypothesis and unfold the successor definition of the original iteration. This is equality of all members of the entire original ideals at all points; no selected generator space is substituted. □

59 Actual smooth coordinate-plane relative ideal powers

This supplies the entire normal ideal and finite relative truncations of an actual smooth coordinate plane, from genuine holomorphic functions. All scalar coefficients are whole holomorphic germs on the actual tangential plane. No kernel, power equality, finite-generation or frame certificate is an input. Finite generation is not flatness or freeness. Full smooth-centre normal flatness, all-degree relative ranks and actual higher Hilbert–Samuel blowup descent remain open.

Definition 59.1 (Original tangential projection and inclusion)**.** For $n = d+t$, let $\pi : \mathbf{C}^{d+t} \to \mathbf{C}^d$ be the actual first-d coordinate projection and $i : \mathbf{C}^d \to \mathbf{C}^{d+t}$ the actual first-d coordinate inclusion with the last t coordinates zero. Both are original continuous complex linear maps. Their actual composite πi is identity; $i\pi$ is identity precisely on the original coordinate plane $\mathbf{C}^d \times 0$.

Proof. Construct π from the finite actual coordinate continuous linear functionals and the genuine finite-product/PiLp linear equivalence. Construct i as the finite sum of the actual source coordinate functionals times the corresponding actual ambient coordinate vectors. Evaluate each

coordinate to obtain the projection/inclusion formulas and the two true identities. In the normal coordinates, $i\pi$ is zero, so its fixed-point condition is exactly the actual plane equations; in the tangential coordinates it preserves the original value. Continuous complex linear maps are everywhere holomorphic, including $d = 0$, $t = 0$ and empty ambient dimension. □

Lemma 59.2 (Entire original germ algebra retraction)**.** *Let $R_d = \mathcal{O}_{\mathbf{C}^d,0}$ and $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$. Actual composition gives $\pi^* : R_d \to R$ and $i^* : R \to R_d$, genuine entire complex germ algebra homomorphisms, with $i^*\pi^* = \mathrm{id}$. The trace is surjective and the pullback injective. Use π^* as the real R_d-algebra structure on R. Every original normal-coordinate germ z_{d+j} has trace zero.*

Proof. Both actual linear maps fix zero and are holomorphic, so their real nonlinear composition maps on ALL holomorphic representatives induce entire germ algebra homomorphisms. On any original holomorphic representative the composite acts by $F \circ \pi \circ i = F$ at every point, proving the retraction. Any entire R_d germ is the trace of its genuine pullback; applying trace to an equality of pullbacks proves injectivity. Each actual normal coordinate vanishes identically after i, giving trace zero as an equality of entire germs. The algebra action is literally multiplication by this actual pullback. □

Lemma 59.3 (Actual relative Hadamard decomposition of every whole germ)**.** *Every $g \in R$ has internally constructed entire holomorphic germs $h_j \in R$ with*

$$g = \pi^* i^* g + \sum_{j=0}^{t-1} z_{d+j} h_j.$$

Proof. Choose an actual holomorphic representative F of the ENTIRE germ g. The real function $H = F - F \circ i \circ \pi$ is holomorphic near zero, since the true projection/inclusion preserve zero. At every point of the actual coordinate plane, the genuine identity $i\pi z = z$ makes $H(z) = 0$. Actual coordinate-plane Hadamard division internally constructs holomorphic coefficient functions for ALL normal coordinates, with the tangential coefficients identically zero. Pass their original holomorphic representatives to entire germs. The actual finite sum split between first d and last t coordinates discards the zero tangential coefficients and leaves exactly the displayed genuine normal sum. The original composition dictionary identifies the remaining base term with $\pi^* i^* g$. No decomposition or finite-generation data is supplied. □

Theorem 59.4 (The entire reduced plane ideal is exactly the whole trace kernel)**.** *Let J be the entire ring ideal generated by the actual t normal coordinate germs. Then $J = \ker i^*$ and membership is exactly vanishing near zero on the actual coordinate plane, for an original holomorphic representative. The true quotient R/J is isomorphic as an entire complex germ algebra to R_d.*

Proof. Every normal coordinate has trace zero; since trace is an actual ring homomorphism, its kernel contains their whole generated ideal. Conversely if an ENTIRE germ has trace zero, its actual relative Hadamard decomposition has zero base term and expresses it as a true finite sum of those coordinate multiples. This lies in the full ideal J and proves both ideal inclusions. If an actual representative vanishes on the plane near zero, composing with i gives zero near zero because i is continuous and fixes zero; hence its whole trace is zero. Conversely trace zero gives a representative F whose $F \circ i$ is zero near zero. For ambient points near zero on the plane, π continuity sends them into that true neighbourhood and $i\pi z = z$, so $F(z) = 0$. This proves the genuine all-representative geometric dictionary. Finally the actual trace is surjective, and its real ring/complex-algebra first-isomorphism theorem gives the entire plane quotient, using the proved kernel equality. Thus the ideal is reduced, without assuming reducedness or supplying an abstract ideal-kernel certificate. □

Theorem 59.5 (Internally constructed all-degree relative normal-word representatives)**.** *For every actual natural q and every ENTIRE $g \in R$, there is a finite R_d-linear combination p of actual*

normal-coordinate words of length strictly below q, with $g - p \in J^q$. Here the scalar action is actual π^*; coefficients are entire tangential holomorphic germs, not point scalars.

Proof. Induct on the actual q. At $q = 0$ choose zero; the real zeroth power is the entire ring. For $q + 1$, apply the actual relative Hadamard decomposition to g and the induction hypothesis to each internally produced ENTIRE coefficient h_j. Write $h_j = p_j + r_j$, with actual $r_j \in J^q$. Set $p = \pi^* i^* g + \sum_j z_{d+j} p_j$. The base term is its actual R_d coefficient times the genuine degree-zero word. Prefixing each actual normal word by z_{d+j} raises its degree by one, so the actual finite scalar span is the required below-q+1 span. Multiplication by a normal coordinate commutes with the true R_d action, so this extends from generating words to their entire scalar span. The true remainder is $\sum_j z_{d+j} r_j$; actual ideal power multiplication puts each summand in $J\,J^q = J^{q+1}$ and finite ideal addition puts the whole remainder there. This is a genuine manufactured all-power representative, not a caller Taylor or normal-ideal-power equality certificate. □

Theorem 59.6 (True entire relative power quotients are finite tangential modules)**.** *For every q, the literal entire ring quotient R/J^q is a finite R_d module. More generally, for EVERY whole original ring ideal I, the literal quotient $R/(I + J^q)$ is a finite R_d module. No freeness or flatness is inferred.*

Proof. The actual normal-word index is finite: a degree in $\{0, \ldots, q - 1\}$ and a word of that degree in the actual t normal coordinates. Its real finite R_d span is therefore finitely generated. Every ENTIRE quotient member has an actual whole-germ representative, and the preceding internally constructed representative differs from that whole germ by an actual J^q member. The real quotient map sends both to the same class. Thus the actual image of the finite normal-word span is the ENTIRE quotient module. Taking its finite image of generators proves genuine module finiteness. The identical representative works for $I + J^q$, since the actual power is contained in this actual sum ideal. At $q = 0$ both literal quotients are zero; at $t = 0$ all positive truncations have the actual degree-zero word alone and recover the tangential whole germ ring or its true quotient. At $d = 0$ these are genuine finite point truncations. No relative monomial independence, free graded pieces, normal-flat centre or all-degree constant rank is presumed or claimed. □

60 Actual positive-level member lower ideals beyond the literal mask

These producers use the actual transformed barMarked ideal and actual original indexed SNC members. Genuine globally locally finite smooth components remove exactly the components contained in the real cosupport. Coherence and the actual cosupport dictionary internally prove nonzero along the remaining carrier. The resulting entire datum restricts exactly to literal positive-level lowerMarked. Full controlled tail extension, zero-level contact extension, whole General3 finite thresholds and general global termination remain open.

Lemma 60.1 (Actual unselected carrier has the original local germ)**.** *At every point of the actual remainingComponents of a closed smooth H relative to an analytic A, that remaining carrier and H have the same genuine local germ.*

Proof. The real global component families of a smooth H are locally finite, and their selected union is closed and smooth. The actual unselected union is H minus that selected union. A point of the unselected union lies outside the closed selected union, hence a genuine open neighbourhood avoids the latter. On that neighbourhood the unselected union and H are exactly equal. This proves the actual eventual membership equivalence, without a local-germ certificate. □

Lemma 60.2 (Thin genuine cosupport internally gives nonzero along)**.** *For a coherent actual ideal I on a carrier H, if its genuine cosupport is thin in H, every point of H has an actual ideal germ frequently nonzero on H.*

Proof. Suppose no actual ideal member is frequently nonzero along H at such a point. Take the actual finite coherent generators near it. Each vanishes on H on a genuine neighbourhood; their finite span then makes every actual ideal member vanish on H near every nearby point. The genuine order dictionary with the zero maximal-ideal-power witness puts every nearby H point in the actual cosupport. A genuine open neighbourhood of the original H point is therefore contained in that cosupport on H, contradicting true thinness. Thus an actual ideal germ is frequently nonzero along H. No nonzero-along or order certificate is assumed by the actual member producer below. □

Definition 60.3 (The entire actual positive-level member remainder datum)**.** For a true actual marked ideal J and actual member index set S, form the literal memberRestrict J S and replace its actual carrier $H = N \cap H_S$ by the actual remainingComponents of H relative to J.cosupp. Keep its entire actual coefficient ideal, actual positive coefficient mark and actual deleted-member boundary. For a literal head test tower and point, memberLowerExtension uses its true head barMarked transform and the actual membersThrough among literal original members; it has no caller-provided lower datum.

Theorem 60.4 (Internally genuine entire member lower domain)**.** *A genuine nonempty actual SNC member intersection drops the carrier dimension by at least one. Its entire actual remainder datum belongs to that lower invariant domain. Original true domain and a real literal barMarked test tower, at a true positive-level point, internally supply this genuine lower domain for memberLowerExtension; all member nonemptiness, smoothness, coherence, transversality and nonzero-along fields are internally produced.*

Proof. True well formedness gives an actual closed smooth SNC intersection H and real transversality to the actual boundary with those indexed members deleted. A genuine selected/unselected global-component subfamily is closed and smooth; its same local H germ preserves transversality in the actual coordinate family. The actual coefficient ideal is coherent and its mark is genuinely positive. Actual nonempty member intersection dimension drop and regular inclusion give the true lower carrier dimension bound. The real coefficient cosupport restriction dictionary identifies cosupport on H with J.cosupp intersect H. On the remaining union the same local H germ preserves actual orders, giving exactly J.cosupp intersect remainingComponents. Genuine analytic continuation on each actual unselected component proves this intersection thin. The preceding finite-generator argument internally proves nonzero along every remainder point, completing every literal IsInvDomain field.

For a literal head of a true barMarked tower, real transform well formedness, coherence, positive coefficient mark and actual dimension bound come from the original domain. The actual positive indexed boundary count gives a genuine original member through the point; membersThrough in the full transformed boundary is therefore internally nonempty. Apply the preceding construction. No lower domain, coherence, thinness or nonzero-along conclusion certificate is an input to this actual original producer. □

Theorem 60.5 (Exact equality with literal lowerMarked on the original mask)**.** *At true positive level s+1 the pullback of this entire memberLowerExtension by the literal original maskA inclusion equals the actual literal lowerMarked at s+1, in every marked-ideal field.*

Proof. Their genuine full coefficient ideals, positive marks and deleted boundaries are literally identical, so only the actual carriers must be compared. The remaining carrier lies in H. Conversely let an actual mask point lie in H. Membership in H puts it on all actual original members through the chosen point; the true mask excludes every original member missing the chosen point. Thus the two points have exactly the same actual original incidence pattern and true boundary count s+1. If it lay in a selected genuine H component, an actual smooth component neighbourhood would put the whole local H germ in J.cosupp. The incidence pattern makes this precisely the true member-intersection locus at the mask point, and its positive count makes it special at level

s+1. The literal mask excludes exactly that specialness, a contradiction. It therefore lies in the actual remaining carrier. These two real inclusions prove the actual carrier pullback equality and thus equality of the entire literal marked ideals, without changing the original mask or lower definition. □

Theorem 60.6 (Original actual General3 canonical input produces the member extension)**.** *Original true domain3 and a literal canonical constructed invFamily3 tower at an actual positive-order current point internally produce, at every actual positive-level nonspecial companion point, the entire genuine domain2 member extension on the real head at its actual regimeRun, actual head-image membership in the original lower mask and its exact equality with the literal lowerMarked on that actual mask.*

Proof. The real original canonical-tail producer internally supplies actual companion domain3, maximal order and genuine canonical Seq Step1 tail. Its true bridge gives the actual barMarked test tower. Compute the literal own-level regimeRun of the chosen actual nonspecial companion point. If it is zero, the current point is its own actual head image and has its current regime. If positive, the genuine run recursion gives that exact regime at its real initial head image. Thus the actual head count is the positive current count. Split the actual test tower at this computed run, internally obtaining a genuine head test tower. The preceding real member producer gives its entire domain2 datum and the exact original mask/lowerMarked identity. Neither a head, run, lower datum, lower domain nor equality certificate is supplied by the caller. □

Remark 60.7. This is an initial lower-datum extension with exact original mask equality. It does not yet construct its entire controlled transform along the literal remaining tail or identify that full transform with the original lower value. That real tail problem and the true level-zero contact extension remain OPEN, as do whole General3 finite-threshold descent and general global termination.

61 Actual NC separation on an open set after deleting genuine empty steps

A global highest stratum can miss an entire open region. Its actual restricted blowup is then an isomorphism with an empty centre, while the independent local procedure may already have stopped. Therefore full restricted tower lengths and unfiltered boundary lists need not agree. The comparison below internally records every actual empty step and deletes exactly these steps and their actual empty exceptional entries. It assumes no compressed tower, maximum history, termination or naturality certificate.

Lemma 61.1 (Actual intrinsic original count on an open manifold)**.** *The literal open preimage of a whole boundary-adapted NC pair is a genuine boundary-adapted pair. Every actual original branch count and entire count-r superlevel on the open manifold is the corresponding literal original trace. The same dictionary holds under a genuine biholomorphism onto that open set.*

Proof. In the actual inherited open chart, the derivative of the open inclusion is the identity. Hence it has no exceptional differential point. The genuine local coordinate reader constructs actual pulled-back flat coordinates for the entire original carrier and every literal indexed boundary member. The same finite set of actual normal branch coordinates computes both intrinsic counts, giving equality at every point. Whole superlevel equality follows pointwise for every natural r. Compose this dictionary with the actual biholomorphic local-base dictionary; no branch labels or count identity are supplied. □

Lemma 61.2 (Actual highest-centre trace and drop dichotomy)**.** *At an actual global higher step with internally attained maximum r, the whole current centre trace on the actual open preimage is empty or nonempty. If it is empty, all actual local original counts are at most r-1. If nonempty,*

the internally attained local maximum equals r and the entire local highest centre is the literal global centre trace.

Proof. Every local count is at most r by the genuine original count dictionary. If the whole trace is empty, no actual point has count at least r, so its integer count is at most r-1. If nonempty, a true trace point has count at least r and therefore exactly r. The actual local count image thus has attained maximum r. The genuine maximum uniqueness theorem identifies the internally selected local maximum with r. Exact pointwise count transport identifies their entire highest centres. This includes disconnected opens omitting every global highest component and uses no compactness or preassigned drop history. □

Lemma 61.3 (True empty-centre blowup and whole closed carrier)**.** *A genuine blowup of an actual empty centre internally gives a biholomorphism with its original base. Every whole closed original and old boundary is its entire preimage under this map, and the new actual exceptional boundary is empty.*

Proof. The identity map is a genuine blowup of the empty centre, by its true analytic universal property. Actual blowup uniqueness compares the given blowup and this identity in both directions. Their composites are identities by the same universal uniqueness, so the given projection is genuinely biholomorphic. Its actual derivative is bijective everywhere. The differential strict transform of a closed carrier is therefore its entire preimage. Its geometric strict transform off the empty centre is the same closed preimage. The whole exceptional centre preimage is literally empty. Thus the local original and its intrinsic count are unchanged under the real isomorphism. □

Definition 61.4 (Actual stage compression with only empty source steps)**.** A stage compression relates two actual recorded towers over a genuine base biholomorphism. Its reflexive stage is that base map. A matched stage appends actual source and target blowups with exact whole centre preimage and a true commuting projection square. A silent source stage appends a genuine source blowup whose actual centre is empty, keeps the real target prefix, and records its actual biholomorphic projection square. No other steps are omitted.

Lemma 61.5 (Actual projection, concatenation and retained lengths)**.** *Actual stage compressions commute with full projections and concatenate along actual recorded heads and tails. Genuine source tower isomorphisms transport them by their actual stage maps. The retained source-stage count equals the actual target length and is at most the uncompressed source length.*

Proof. Induct on the last recorded constructor. At the reflexive stage all equalities are identities. A matched stage adds the true projection square after the previous projection equality and adds one retained stage on both sides. A silent stage adds only its actual source projection and no retained stage. For concatenation induct on the tail compression: its matched constructor appends paired real blowups to the already concatenated shorter comparisons; its silent constructor appends only its real empty source step. For source tower isomorphisms, invert the actual last-stage isomorphism and transport the whole centre and commuting square. An empty source centre remains empty under an actual biholomorphism. This gives the transported compression in both constructors. In particular the actual restriction-append tower isomorphism transports a comparison of restricted head and restricted tail to the literal restriction of their entire concatenation. □

Lemma 61.6 (Literal boundary dictionary after deleting actual empty entries)**.** *Let clean(E) delete exactly the empty sets from a literal finite boundary list. Actual stage compression carries the whole differential strict original and cleaned literal accumulated boundaries to their true counterparts, when the original whole carriers and cleaned boundary lists agree under its real base map and the original literal source boundary members are closed. Actual intermediate boundary closedness is derived internally from the recorded tower.*

Proof. An actual biholomorphism preserves nonemptiness, so clean commutes with its literal preimage list map. For every geometric strict-transform operation F, F(empty) is empty. Finite-list induction therefore gives clean(map F E)=clean(map F(clean E)), even if some nonempty entry transforms to empty. At a matched actual step, the true square and actual homeomorphism closure transport identify every whole old geometric strict boundary and the entire new exceptional preimage. The cleaned lists agree by these actual finite-map identities. At a silent actual empty step, closed old boundaries transform to their entire biholomorphic preimages; the appended actual empty exceptional entry is deleted by clean. Induction proves the entire boundary dictionary. The real differential strict original follows the actual squares by true holomorphic strict composition and the absence of exceptional points of the actual stage biholomorphisms. No ideal coherence or marked admissibility is used for these literal geometric equalities. □

Theorem 61.7 (Internally compared actual global and independent local procedures)**.** *For a true original boundary-adapted NC pair, any actual open set and arbitrary independent individual genuine blowup choices, the literal restriction of the actual global finite procedure and the actual local finite procedure internally have a real stage compression. Their initial true count bounds may differ. The produced final map is a genuine biholomorphism commuting with the entire original projections, with whole differential strict-original and cleaned literal accumulated-boundary dictionaries.*

Proof. Induct on the actual global count bound b, uniformly in the actual global ambient, current open, genuine biholomorphism from the local base to that open and arbitrary independent local bound b'. This permits the same local procedure to wait through a genuinely empty global step. If b is zero or the whole global count is at most one, the true local count dictionary makes both procedures reflexive; the actual local-base map is the comparison. Otherwise read the actual first global blowup from the transparent procedure equation and test its entire actual centre trace. If empty, its genuine restricted projection gives the real empty-step biholomorphism. It identifies the unchanged actual local base with the new true global open preimage. Apply the induction hypothesis to the real smaller global tail and the same local procedure at bound b'. Append the actual silent head comparison and transport through the genuine restriction-append isomorphism. If the trace is nonempty, its actual point forces the internally computed local maximum to equal r. Then the local count is at least two, b' is positive, and its true procedure equation performs its actual chosen first blowup. Genuine universal uniqueness produces the lift of the two actual aligned blowups. Their whole strict-original dictionary supplies the real smaller-pair relation. The compression induction itself uses no initial boundary alignment; the final cleaned-boundary consumer uses only the literal initial list dictionary. Apply the induction hypothesis to the actual global and local tails, at bounds b-1 and b'-1, then append the genuinely matched head and actual tail comparisons. Exact projection and whole boundary consumers prove the claimed literal outputs. No candidate compressed tower or naturality property is assumed. □

Lemma 61.8 (Unique actual final comparison after genuine empty-step omission)**.** *The final map of the internally produced stage compression is unique over its actual base map. It is independent of the internal construction of its actual drop/match history.*

Proof. Every target step of the real local procedure has a genuinely nonempty centre, since its selected higher maximum is actually attained. Induct on the actual source last step for two comparisons over the same base. If its centre is empty, neither comparison can be matched: the biholomorphic preimage of the nonempty target centre would be nonempty. Both are silent. Induction identifies the preceding real maps and their actual projection squares identify the last maps. If the source centre is nonempty, neither comparison is silent. Invert the two genuinely matched last steps. Induction identifies their preceding maps; true analytic single-blowup universal uniqueness identifies their last lifts. At the reflexive stage the map is the actual base map. Thus the genuine comparison is unique without a caller uniqueness or compression certificate. □

Theorem 61.9 (Actual natural geometric separation on every open set)**.** *Internally existing actual choices and the intrinsic dimension bound give a real geometric NC separation on every open set, uniquely compared with the literal restriction of the global natural separation after deleting exactly its actual empty-centre steps. Its entire final original and cleaned accumulated boundary correspond under the produced biholomorphism.*

Proof. The actual open coordinate reader internally supplies the true pulled-back pair and dimension bound. Use the genuinely existing default individual choices and the internally proved bound n on both actual manifolds. The preceding producer constructs the genuine compression and whole terminal dictionary; its actual uniqueness consumer applies because the local procedure's genuine higher centres are nonempty. Empty original, dimension zero and empty open are included. A count-one open region may inherit genuine empty global steps, which are deleted as recorded isomorphisms rather than claimed nonexistent. Disconnected and noncompact opens need no globally finite component family. □

Remark 61.10. This is restriction naturality of the intrinsic unordered geometric NC procedure. It does not eliminate the old marked colexicographic label tie-break or prove arbitrary additional marked-ideal cosupport admissibility. The full canonical marked-ideal G9 obligation remains separate.

62 Entire coefficient ideals under actual smooth pullback

The map, original coherent ideal, carrier and boundary are the actual inputs. All products, powers and finite weighted sums below are the original entire germ ideals. No pullback algebra or coefficient comparison certificate is an input. The coefficient equality does not assert blowup base change or complete canonical smooth functoriality.

Lemma 62.1. *For an actual holomorphic map $f : Y^m \to X^n$ and original coherent ideals I, J, the entire ideals satisfy $f^*(IJ) = (f^*I)(f^*J)$.*

Proof. A generator of the product is a product of original ideal members. Pull a finite holomorphic expression for a product-ideal member back along the actual map: its coefficients remain holomorphic, and each original product pulls back to the product of its two actual pulled members. The resulting entire finite span gives the first inclusion. For the reverse inclusion, write each actual pulled member as its genuine finite holomorphic combination of original pulled ideal members. Multiply both sums and distribute. Each term is a holomorphic coefficient times the pullback of a product of actual members of I and J. Coherence makes those original members holomorphic and their product a genuine product-ideal member. Transitivity of the original entire finite spans gives the reverse inclusion. Apply membership extensionality. The construction uses neither dimension equality nor a local inverse. □

Lemma 62.2. *The actual unit ideal pulls back to the actual unit ideal. For every original coherent ideal I and every $q \geq 0$, $f^*(I^q) = (f^*I)^q$.*

Proof. Every pulled unit-ideal member is holomorphic, so lies in the genuine target unit ideal. Conversely every target holomorphic germ is itself its coefficient times the pullback of the original constant one; hence it belongs to the entire pulled unit ideal. For powers induct on q. The genuine zero power is precisely this unit ideal. The genuine successor power is the product of the previous power with the original ideal, and both are coherent. Apply the actual product equality and replace the previous power using induction. This covers the actual zero and unit ideals and literal zeroth powers. □

Lemma 62.3. *For any actual finite family I_i of coherent original ideals, $f^*(\sum_{i<r} I_i) = \sum_{i<r} f^*I_i$, as entire germ ideals.*

Proof. A member of the original finite sum is a genuine finite holomorphic combination of members of the actual summands. Pull this whole expression back; every pulled summand member is an original generator of its entire pulled summand and hence of the target finite sum. Span transitivity gives the first inclusion. Conversely a member of a pulled summand is a genuine finite holomorphic combination of pullbacks of its original members. Each original member is a member of the original full finite sum by its defining span and holomorphicity from coherence. Thus this whole pulled summand is contained in the pullback of that finite sum. Span transitivity gives the opposite inclusion. With no summands the same empty-span argument applies. □

Definition 62.4. The actual marked pullback of (N, E, I, d) along $f : Y^m \to X^n$ is the literal marked ideal $(f^{-1}N, (f^{-1}E_i)_i, f^*I, d)$ on Y.

Proof. Use the original inverse image of the carrier, the original ordered boundary list mapped by inverse image, the entire original germ-ideal pullback and the same original mark. These are actual data of the target marked-ideal type, and their four projections are exactly the displayed data. This definition does not supply or presume centre admissibility or canonical tower equality. □

Theorem 62.5. *For an original marked ideal with coherent I and a genuine holomorphic submersion $f : Y^m \to X^n$, the actual coefficient marked ideal commutes with this literal marked pullback, including all original carriers, ordered boundary data, marks and entire ideal members.*

Proof. The original coefficient ideal is the finite marked sum of actual iterated derivative ideals $\mathcal{D}^i I$, with original marks $d - i$ and the original weighted exponents obtained by multiplying the other marks. The genuine submersion already gives equality of ENTIRE pulled iterated derivative ideals. Each original iterated derivative ideal is coherent. Apply the proved entire power equality to its actual weighted power and the entire finite-sum equality to the original finite weighted family. Replace each actual iterated pullback using the derivative equality. The result is precisely the original coefficient ideal of the literal target marked data. Both actual constructions retain the inverse-image carrier and ordered boundary and have the same numerical mark, because the original source marks are unchanged. Extensionality of the actual marked records gives the stated entire marked-ideal equality. No threshold, coefficient-space or selected generator equality is used as a hypothesis. □

Theorem 62.6. *For an original ambient marked ideal $N = X$ with coherent I, the cosupport of its actual pulled coefficient marked ideal is exactly the inverse image of its original coefficient cosupport under the genuine holomorphic submersion.*

Proof. The original finite weighted sum of derivative powers is coherent, by the actual coherent finite-sum and power constructors. The previous result identifies the entire target coefficient ideal with the genuine pullback of this same original coherent ideal and identifies its mark and full ambient carrier. The actual original-point section of the genuine submersion gives equality of all entire germ order thresholds for this original coefficient ideal, so its marked cosupport is exactly the real inverse image. Substitute the true coefficient-ideal equality. The empty weighted family and literal zero mark retain the original definitions throughout; no positivity assumption is added. □

63 Actual whole smooth-centre relative ideals in original nonlinear coordinates

The input is actual original regularity of the smooth centre at an original point. The flattening, entire germ algebra equivalence, whole vanishing-ideal and all-power dictionaries, base algebra and finite relative quotient modules are produced internally. A caller does not supply a flattening, ideal map, power identity or finite-generation certificate. Module finiteness is not normal flatness or freeness; higher-Hilbert–Samuel transformation and full G8 termination remain open.

Lemma 63.1 (Actual original flattening is genuinely recentered at zero)**.** *Any actual local holomorphic equivalence Φ at original coordinate p with $\Phi(p) = 0$ gives a true zero-centred local holomorphic equivalence $\Psi(z) = \Phi(z + p)$, with inverse $\Psi^{-1}(w) = \Phi^{-1}(w) - p$ and the actual translated original source and original target. It internally induces the whole complex germ algebra equivalence $E = \Psi^*$.*

Proof. The true translated source is open by continuity of addition and contains zero because the original source contains p. Keep the genuine original open target. Holomorphicity of both true maps follows by composition with actual translation and subtraction. The actual original inverse identities and cancellation of the translations prove both new inverse identities and both source/target mapping properties. Its real value at zero is zero, so the previous genuine nonlinear whole-germ construction applies to this actual zero-centred equivalence. This is not a germ-isomorphism input certificate. □

Lemma 63.2 (ALL original flattened coordinate germs match the true entire map)**.** *Let e be the genuine preferred chart at the original point x and $p = e(x)$. For EVERY actual original ambient coordinate j, the entire germ $E(z_j)$ equals the true translated preferred-chart germ of the original function $y \mapsto \Phi(e(y))_j$.*

Proof. The actual composition dictionary reads $E(z_j)$ as the whole original germ of $z \mapsto \Phi(z+p)_j$. The genuine preferred-chart germ of the original flattened coordinate reads $z \mapsto \Phi(e(e^{-1}(z+p)))_j$. On the true original open chart target near p, the actual inverse chart identity makes those original functions identical. Translating that real neighbourhood to zero proves equality of the ENTIRE germs, not just their point values or jets. □

Theorem 63.3 (Actual smooth centre supplies its complete normal ideal dictionary)**.** *In ambient dimension $d + t$, take a genuine original flattening of C at x with its actual source inside the preferred-chart target, value zero at x, and actual centre/coordinate-plane membership dictionary. Its actual entire map E above satisfies*

$$E(J) = I_{C,x},$$

where J is the COMPLETE coordinate-plane normal ideal and $I_{C,x}$ is the ENTIRE preferred-chart stalk ideal formed from ALL original vanishing germs.

Proof. Use the genuine flattening, its true source inside the original target, its real zero-centre value and actual bidirectional centre/plane membership dictionary. Every normal flattened coordinate vanishes on the original centre near x, so its ENTIRE preferred-chart germ is an actual generator of the WHOLE original vanishing stalk ideal. The preceding real coordinate dictionary and genuine ideal map of the full coordinate span prove $E(J) \subseteq I_{C,x}$.

Conversely, take ANY literal original holomorphic vanishing germ. Actual Hadamard division in this very original flattening internally gives an entire finite original holomorphic coefficient family, tangential coefficients identically zero, and an actual neighbourhood equality of that original function with the finite sum of normal flattened-coordinate multiples. Read the actual entire sum/product germ equality in the original preferred chart; the real first-d/last-t split discards the zero coefficients. Every remaining normal-coordinate whole germ is in $E(J)$ by the previous dictionary, so the entire sum lies there. The WHOLE original ideal is the full span of ALL these membership germs; ideal closure proves the other inclusion. No selected generator, whole ideal identity or kernel conclusion is supplied. Zero codimension and zero ambient dimension give the actual zero normal ideal and agree with the genuine open-centre vanishing stalk. □

Lemma 63.4 (All actual powers and entire original quotient algebras transport)**.** *For every natural q, $E(J^q) = I_{C,x}^q$. For every whole original ideal I, the genuine full quotient by $I + I_{C,x}^q$ is transported from the entire quotient by $E^{-1}(I) + J^q$. No ideal-power or quotient equality is input.*

Proof. The actual algebra equivalence is a ring equivalence, so genuine ideal maps preserve multiplication and EVERY natural power. Apply this true map-power law to the already manufactured full ideal equality. The actual inverse ideal is the full preimage of I under the true ring map E, and actual surjectivity identifies its entire image back with I. Actual ideal-map preservation of sums and powers then proves the whole sum ideal equality. The genuine ring/complex-algebra quotient-equivalence theorem transports the literal ENTIRE quotients using exactly that proved equality. At $q = 0$ the actual sum ideals are top and both whole quotients are zero. □

Theorem 63.5 (Actual transported whole base algebra produces finite relative quotients)**.** *Use the actual composite $R_d \xrightarrow{\pi^*} R \xrightarrow{E} R$ as the whole tangential germ algebra on the original preferred stalk. Then for EVERY q and whole original I, the literal original quotient $R/(I+I^q_{C,x})$ is a finite module over this actual ENTIRE base germ ring.*

Proof. Take any ENTIRE original stalk germ g and its actual full inverse $E^{-1}g$. The previous relative producer internally gives a finite whole tangential normal-word representative modulo the actual J^q. Apply the actual entire algebra map E to this finite sum and whole remainder. The real composite base algebra makes each transported coefficient literally the original R_d scalar action. Thus g has the internally produced finite span of actual transported normal words as its representative, with remainder in $E(J^q) = I^q_{C,x}$. This is a genuine all-power original representative, not a finite-module certificate. The actual finite word index and original quotient map send this finite entire-base span onto EVERY member of the literal quotient by $I + I^q_{C,x}$, because the true remainder is in that actual sum ideal. Genuine finite span/image generators yield Module.Finite. The exact base action and full ideal transport are constructed internally; no flat or free graded module conclusion is asserted. □

Theorem 63.6 (Actual closed smooth centres produce the relative germ geometry at every point)**.** *Actual* RegularAtDim(C, x, d) *in ambient dimension $d+t$ internally produces the genuine original flattening, real source and bidirectional centre/plane dictionary, whole normal ideal and every power identity, every entire original quotient transport and actual finite modules over the internally transported whole tangential germ algebra. In particular, an actual closed smooth centre, with possibly different dimensions on its components, supplies at every original member point these data in that point's actual dimension and codimension. No caller supplies the flattening, ideal, power, quotient or module conclusion.*

Proof. First unpack original regularity: it supplies a genuine original flattening, its source inside the true preferred-chart target, zero image of the original point, and the real bidirectional centre/plane dictionary. Apply the entire vanishing-stalk theorem to this actual flattening. All actual powers and entire original quotients then transport by the genuine map-power and quotient constructions. The actual composite base algebra and finite whole-word representative producer give every relative finite module. All conclusion data have therefore been constructed from original regularity.

The genuine closed smooth centre structure supplies at each actual member point its actual regular dimension d with $d \leq n$. Put $t = n - d$; their natural sum is the ambient dimension. Apply this actual regular-point producer in precisely that original ambient space. All flattening, entire ideal and power transports, base action and finite-generation evidence are therefore supplied internally from the ORIGINAL centre geometry. No uniform component dimension, compactness or connectivity is assumed. The empty centre conclusion is the honest universal statement over no points. This is the local relative algebra foundation; producing an actual normal-flat selected resolution centre and proving higher-HS transformation/descent are still subsequent obligations. □

64 Entire member controlled transforms along the genuine original tail

The positive-level datum is the actual member remainder constructed above. Its original test tail need not be admissible for this smaller carrier. The following arguments use the genuine strict transforms of the original SNC intersection to construct every domain field of its entire controlled transform. They then identify all fields with the literal lower transform on the original mask at every actual stage. They do not assert that the entire candidate tail is canonical, or complete the level-zero contact and deeper companion descent.

Lemma 64.1 (Strict transforms of actual closed component partitions)**.** *Let a closed set H be the disjoint union of closed A and B. Along any genuine blowup tower T, their actual strict transforms satisfy*

$$A_T = H_T \cap T^{-1}A,$$

and at every point of A_T its actual local germ equals that of H_T. In particular this holds for the genuine remaining and selected smooth components of the original member intersection.

Proof. For one blowup, preimages commute with unions and removing the centre does too; closure of a finite union is the union of the closures. Induction proves $H_T = A_T \cup B_T$. Each strict transform of an initially closed set is closed and lies over that set, by continuity of the actual projection and the closure in the strict-transform definition. Since A and B are disjoint, these two preimage containments make A_T and B_T disjoint. The first inclusion of the displayed identity follows from monotonicity and the preimage containment. Conversely, a point of H_T over A belongs to A_T or B_T; the latter case would put its image in the disjoint set B. The complement of the closed B_T is a neighbourhood of every A_T point, and on it H_T and A_T are equal. This is the asserted genuine local-germ equality, without a chosen component chart. □

Lemma 64.2 (Actual member and deleted-boundary geometry at every stage)**.** *Let J be a true well-formed marked ideal of dimension at most k, S a nonempty set of actual indexed boundary members, and T a genuine J-test tower. The strict transform of its actual remainder carrier is closed and smooth, has dimension at most $k-1$, and is transverse to the actual boundary of the entire transformed remainder datum. That boundary is SNC.*

Proof. At each literal stage the genuine original test tower supplies well formedness of J_T, dimension at most k, and its full transformed SNC boundary. The actual strict transform of $H = N \cap H_S$ is exactly N_T intersected with the strict transforms of the indexed original members in S. Those are precisely the members at the original indices embedded in the current index set. This embedded set is nonempty. The genuine simultaneous-coordinate intersection therefore is closed and smooth, drops dimension by at least one, and is transverse to the full current boundary with those actual indices deleted. By the preceding partition result the actual remainder transform has that same local smooth germ at each of its points. Regularity and transversality transport through this actual germ equality, and regular inclusion gives its stated dimension bound.

The boundary transformation only maps the existing list by genuine strict transform and appends the new exceptional member. Consequently the transformed initial deleted list is a sublist of the transformed full list at every stage. Its original members are exactly the strict transforms of initial members whose indices are outside S. Its later exceptional suffix is the same as the full boundary's suffix: both begin empty, and at the next stage both are mapped by the same strict-transform map and receive the same appended exceptional member. Each of these original retained indices is outside the embedded S; each exceptional suffix index is beyond all original indices and also outside it. Thus every actual candidate member belongs to the actual deleted current boundary. The sublist proves SNC, and this real membership inclusion proves transversality using the genuine simultaneous coordinates already obtained. No candidate admissibility or centre-containment assertion is used. □

Lemma 64.3 (Coherent controlled ideal and true nonzero along a smooth strict carrier)**.** *For an actual blowup, an actual coherent ideal with nonzero germs along its closed smooth carrier has coherent controlled transform and nonzero germs along the entire transformed carrier whenever that actual strict carrier is closed and smooth. No containment of the centre in the candidate carrier is required. The actual member remainder has these properties along every literal prefix of the original test tower.*

Proof. The actual transformed ideal is the colon of its actual pulled-back ideal, restricted to the genuine strict carrier, by the pulled-back centre ideal to the original mark. Coherence of the pulled-back ideal, actual smooth carrier vanishing ideal, actual smooth centre vanishing ideal and its actual power prove coherence by the general coherent-colon theorem; no restricted inclusion hypothesis is needed.

Outside the literal centre the actual blowup is biholomorphic, and the genuine controlled-transform dictionary carries an original germ frequently nonzero along the carrier to such a transformed germ. At a remaining transformed carrier point, suppose no actual ideal germ is frequently nonzero. Actual coherence and its finite generators then make every transformed ideal member vanish on the transformed carrier near that point. The actual strict-carrier closure definition supplies arbitrarily nearby points above the original carrier outside the centre. At such a point the preceding nonzero-germ producer gives a germ frequently nonzero on the transformed carrier; its frequent nonzero points can also be chosen in the neighbourhood where every actual ideal member vanishes, a contradiction. This proves true nonzero along the entire actual strict carrier.

Induct along the literal original tower. The original remainder-domain producer supplies initial coherence and nonzero along. The preceding geometric lemma internally supplies each actual smooth strict carrier. The one-step argument supplies the next actual coherence and nonzero fields. Combining those with its actual SNC boundary, transversality, positive unchanged mark and genuine lower dimension bound gives the entire lower invariant domain at every original prefix. □

Theorem 64.4 (Exact controlled-transform identity on the entire original mask tail)**.** *At a genuine positive-level head, let R be the entire actual member lower extension and U the literal original mask. At every literal stage T of the genuine remaining original test tail, its actual controlled transform is in the lower invariant domain and*

$$(R|_U)_{T|_U} = R_T|_{T^{-1}U}.$$

Here $R|_U$ is exactly the original literal lowerMarked, so the equality is the entire marked-ideal dictionary for its literal masked tail, including its true ideal, carrier, indexed boundary and mark.

Proof. At the initial stage the exact member-extension mask identity identifies $R|_U$ with literal lowerMarked in all its fields. For the next actual blowup, old and new candidate coherence have already been internally proved. Genuine open pullback commutes with its actual ideal restriction, pulled-back centre power and coherent colon. The exact one-step open controlled-transform theorem therefore gives equality of the actual ideals. Actual carrier strict transform, the indexed boundary map and appended exceptional member, and the unchanged mark satisfy the same open dictionary, yielding full marked-ideal equality. Inducting proves the displayed identity at every genuine stage. The preceding producer simultaneously proves the entire lower domain at all those stages. This uses the actual original tower, never a caller-supplied admissible candidate tower or transformed-ideal equality certificate. □

Theorem 64.5 (Original General3 data produce every literal member-tail domain and dictionary)**.** *Original true domain3 and its actual canonical constructed invFamily3 tower, at an actual positive-order point and genuine positive-level nonspecial companion point, internally produce the actual regimeRun head member extension, its entire domain2 controlled transform at every literal*

remaining prefix, and its exact equality on the original mask with every literal lowerMarked tail transform. No lower datum, lower domain, smooth strict-carrier data, candidate admissibility or whole-tail equality is supplied as a certificate.

Proof. The genuine original canonical-tail producer supplies actual companion domain3, maximal order and the literal canonical Step1 tail. Its true bridge supplies the genuine barMarked test tower. Compute the actual own-level regimeRun and split that literal tower. The actual run-head producer gives the true positive head level, original-mask head image, internally nonempty member set and entire initial domain2 extension. The split test-tower identity supplies the genuine remaining tail of the actual transformed head barMarked. Apply the preceding entire-tail producer to each literal prefix of that tail and its original mask. Its exact initial equality is with the original positive-level lowerMarked; the exact transformed dictionary is consequently the actual one used by the literal constructed lower value at every stage. The construction uses only the original canonical data and computed true objects. □

Remark 64.6. The actual entire lower domain and mask dictionary are produced. Canonicality of this entire candidate tower, the deeper General2 companion-history extension, the true level-zero contact extension, whole General3 finite-threshold descent and general global termination remain OPEN.

65 Entire original NC ideal on the actual natural finite separation

The original divisor is allowed arbitrary normal crossings, without a global finite family of its smooth branches. Its complete vanishing ideal, the actual finite separation tower, its whole total support and its original intrinsic coefficients are produced internally. The ordered marked-ideal colexicographic algorithm and admissibility for an unrelated additional ideal remain separate.

Lemma 65.1 (Actual nonzero whole original coordinate product)**.** *At every original point of a genuine NC carrier, a real flattening chart internally gives a nonzero holomorphic product germ whose entire zero germ is the original carrier chart germ. Every original factor is an actual normal coordinate of that flattening; an empty branch set gives the unit product.*

Proof. Write e for the actual chart, q=e(x), Phi for the genuine NC flattening and S for its actual finite set of distinct original coordinates. Its shifted map F(z)=Phi(z+q) is holomorphic near zero. The genuine holomorphic local inverse satisfies F composed with its shifted inverse equal to the identity near F(0). Differentiate that true identity. The differential of F is surjective, since the derivative of the inverse is its right inverse. Composing each actual coordinate projection with this surjective differential gives a nonzero linear map. Each coordinate F_j has value zero at zero and nonzero derivative; its entire holomorphic germ is therefore prime and nonzero. The whole germ algebra is an integral domain, so the product over S is nonzero. If S is empty it is 1. Near zero, the actual inverse chart lands in the actual flat domain, and the NC definition identifies membership in the original chart germ with vanishing of at least one precisely indexed original coordinate. The finite product vanishes exactly under that condition. This proves the whole zero-germ dictionary, not a supplied equation or a generator certificate. □

Theorem 65.2 (Internally invertible complete original NC vanishing ideal)**.** *The entire original vanishing ideal of every genuine NC carrier is invertible, in every ambient dimension. Its actual carrier has empty interior. In dimension zero the carrier is empty and its whole ideal is the unit ideal.*

Proof. If x belongs to the actual interior, the original open neighbourhood pulls back through the actual inverse chart and translation to a neighbourhood of zero. The whole original coordinate product then vanishes throughout that neighbourhood, hence has zero holomorphic germ,

contradicting the preceding nonzero product. Thus the actual interior is empty. In positive dimension, apply the proved whole original-equation reduction theorem to the internally produced nonzero equation at every original point and this genuine empty interior. It produces the reduced local generator and equality with every nearby stalk of the complete original vanishing ideal; its generator is nondegenerate. These internally produced neighbourhoods give actual invertibility. In dimension zero the finite coordinate index type is empty. The NC membership dictionary therefore excludes every point from the carrier. Its vanishing ideal is the unit ideal, generated by the holomorphic nonzero constant 1 on the whole ambient manifold. □

Lemma 65.3 (Whole original and finite boundary full-preimage dictionary)**.** *For a real blowup projection pi, an actual closed carrier K, a finite actual closed boundary E and an actual centre C contained in K, the whole preimage of K together with the boundary union equals the union of geoStrict(pi,C,K), every geoStrict(pi,C,E_i), and the entire new exceptional pi inverse C. The full preimage of K alone equals its geometric strict transform together with this whole new exceptional set.*

Proof. The identity K=(K minus C) union C gives the full-preimage decomposition. Each nonexceptional preimage point belongs to its closure, which remains in the closed full preimage of K by actual continuity. All points of the new exception lie in that full preimage because C is contained in K. This proves both inclusions for K. For any old E_i, a full-preimage point away from C belongs to its geometric strict transform, and a point above C belongs to the whole new exceptional set. Conversely its geometric strict closure remains inside the closed full preimage of E_i. Take the actual finite union and use the already proved new-exception inclusion in the full preimage of K. No condition C contained in E_i is required. □

Lemma 65.4 (Entire support at every actual generated prefix)**.** *At every prefix of the actual generated higher-stratum procedure with empty initial boundary, the entire inverse image of the original H equals the current whole differential strict original plus the union of every actual accumulated new boundary member.*

Proof. The procedure internally gives the actual higher-centre condition at every recorded step. Such a centre has true original branch count at least two, is contained in the current strict original and has codimension at least two. The actual highest-centre density theorem identifies the geometric original strict closure with its differential strict transform. Holomorphic strict composition identifies the whole current original at the next real prefix. Every actual boundary member is closed, by induction using its genuine strict closure or its actual closed centre preimage. Induct on the true recorded tower. At its reflexive prefix the support identity is H=H union empty. At a real next blowup, pull back the preceding whole equality and apply the actual closed carrier/finite-boundary step dictionary. Its terms are exactly the literal next accumulated boundary and whole next strict original. This proves the identity at every actual prefix; no history or tower support certificate is supplied to the producer. □

Theorem 65.5 (Actual whole original total divisor on the finite separation)**.** *From only a genuine original NC carrier, internally existing choices and the intrinsic dimension bound produce a real finite tower of length at most n-1. Its final original is a true smooth hypersurface, transverse to its entire actual accumulated SNC boundary. The entire original pulled-back vanishing ideal is an invertible effective divisor. Its whole support equals the union of precisely the actual accumulated new boundary and the whole final original. The final projection is a real isomorphism over the original complement.*

Proof. Use the internally invertible whole original ideal as the actual ideal datum, with degree one and empty initial boundary. Actual NC charts produce the true empty-boundary adapted pair. The internally natural finite geometric procedure uses genuine default individual choices and the internally proved dimension bound, and gives the asserted real terminal geometry, all genuine

higher centres, finite length and complement isomorphism. At each genuine single blowup the differential exceptional locus has empty interior, so true pullback preserves invertibility. Induct along the actual recorded tower, starting from the internally invertible original ideal. Invertibility gives actual coherence at each step; genuine holomorphic pullback composition identifies the iterated ideal with the COMPLETE original ideal pulled back by the real tower projection. Thus its true whole total ideal is invertible, without importing any project resolution statement. The actual coherent pullback zero-locus theorem and original closed vanishing-ideal zero-locus equality identify its whole support with the full inverse image of H. The preceding actual prefix identity identifies this support with every real new boundary member plus the entire final strict original. Appending that original to the boundary gives the actual finite SNC list. None of invertibility, support equality, finite tower or termination is an input. □

Theorem 65.6 (Entire pulled-back original ideal and actual original coefficients)**.** *The entire pulled-back original ideal on this actual natural tower equals its intrinsic monomial ideal on the actual terminal SNC list; its complete nonmonomial residual is the unit ideal. In positive ambient dimension, the genuine connected components of that list form an internally locally finite family, with coefficients equal to the divisor exponents of this SAME whole original pulled-back ideal at every actual component point. At each actual point an internally produced finite list of positive components and their actual original powers gives equality of the complete ideal at EVERY nearby stalk. In dimension zero the positive component-product list is empty and the ideal is the unit ideal; the actual accumulated boundary is empty and the terminal boundary list is the singleton containing the empty original carrier.*

Proof. Apply the proved SNC-supported whole-ideal monomial factorization to the actual total divisor and actual terminal SNC list. The exact preceding support identity supplies its support containment internally; the conclusion is an equality of complete germ ideals. The actual SNC-supported residual-unit theorem applied to the same total divisor identifies its whole nonmonomial residual with the unit ideal at every stalk. In positive dimension, apply the genuine original-coefficient locally finite component product theorem to that same divisor and list. Genuine SNC geometry produces its actual connected component family and local finiteness. The theorem constructs the global coefficient of each actual component from its intrinsic original divisor exponent, proves that value at all its points and internally selects a finite nearby component list. It deletes exactly components of coefficient zero, leaving true positive integer powers and complete nearby stalk equality for their actual product. No coefficient, local finiteness or factorization certificate is a hypothesis. In dimension zero the original carrier is empty and the natural procedure is reflexive. The actual accumulated boundary list is empty, while the terminal boundary list is the singleton containing the empty final original carrier. Its entire union and total support are empty. The positive component-product list is empty, and the original whole ideal and pullback are the unit ideal, represented by the empty product. □

Remark 65.7. These are actual whole-ideal outputs for an arbitrary initial normal-crossings carrier. They do not assert that a geometric highest centre is admissible for an arbitrary unrelated marked ideal, or identify this unordered procedure with the old ordered colexicographic marked algorithm. Full G9 remains open.

66 Actual entire lower-order predicates and finite prefix history on curves

Throughout this section the ambient dimension is positive. The entire member controlled transform supplies a genuine domain2 ideal at all literal stages without requiring its original tail to be canonical for the entire candidate. Its actual nonmonomial ideals produce the first-entry order predicates below. Component minima replace any assumed finite order range or global order

bound. The second entry, deeper companion history, level-zero contact and whole General3/global termination remain open.

Definition 66.1 (Actual nonmonomial order predicate)**.** For a genuine current marked ideal J, its order predicate at q is the actual cosupport of its genuine nonmonomial ideal with mark q. For a literal prefix of the entire member candidate tail, pull that actual predicate back by the true remaining projection to the endpoint. These are actual coherent ideal order predicates, not supplied presentations of an invariant.

Lemma 66.2 (Entire lower first-entry predicates and exact literal masked trace)**.** *The whole-tail domain producer internally makes the genuine nonmonomial ideal coherent at every literal candidate stage. All its q-th order predicates are analytic. On the actual carrier they equal order at least q, and on the literal original mask this actual order equals the first entry of the original constructed invFamily2 lower value. This equality holds at every genuine prefix, with no entire-candidate canonical hypothesis.*

Proof. True domain fields give actual smooth carrier, genuine SNC/transverse boundary, coherence and nonzero along. The actual nonmonomial coherence theorem therefore produces its coherent ideal. The true along-carrier derivative-ideal order producer makes its q-th cosupport analytic; q=0 is the actual analytic carrier. The nonzero germ and genuine Krull argument make the actual order finite at all carrier points, identifying this cosupport with actual natural order at least q. The constructed invariant's first entry is literally this order of the full transformed nonmonomial ideal. The preceding entire member-tail marked-ideal identity identifies the masked transform with the actual open pullback of the entire transform. Genuine open nonmonomial-part locality and order locality identify their actual orders exactly, completing the literal first-entry and threshold trace dictionary in both directions. □

Lemma 66.3 (Actual attained component minima produce a zero-dimensional order exception)**.** *Let J be an actual invariant-domain ideal and g a genuine holomorphic map. On an actual analytic curve Z contained in the preimage of J's real carrier, there is a genuine analytic zero-dimensional Q contained in Z such that on each actual global component B of Z, outside Q, the actual natural order of J at g(x) is constant. Each constant is internally an actually attained minimum on that B. All natural thresholds consequently have identical truth values at any two such points. No finite range or global maximum is assumed.*

Proof. Each actual global component B is nonempty: it is the closure of the genuine regular component generated by an actual regular point. The actual natural orders on B therefore form a nonempty subset of the naturals. Its least value q_B is attained by a genuine point of B, and q_B is no larger than the order at every B point. This is an actual minimum, not a bound supplied by the caller.

The actual excess locus B intersected with g's preimage of J's order predicate at $q_B + 1$ is analytic. The minimizing point is outside it, so it is a proper intersection with the actual irreducible component. Analytic continuation and actual dimension drop make it zero-dimensional. These excess loci are subsets of the real globally locally finite component family of Z, and hence are locally finite. Their actual union Q is analytic and zero-dimensional by the locally finite analytic union theorem, and is contained in Z. At a point of B outside Q, its actual order is at least q_B by minimality and below $q_B + 1$ by exclusion of the genuine excess locus. It thus equals q_B. This proves constancy and all-threshold equivalence without any compactness or finite component/order assumption. □

Theorem 66.4 (Finite genuine prefix history has one analytic zero-dimensional exception)**.** *For an actual entire member tail T and an actual analytic curve Z contained in its endpoint carrier, one genuine analytic zero-dimensional Q contained in Z makes every actual prefix first-entry order, read by the literal remaining projection, constant on each real global component outside Q. All actual prefix natural thresholds share that same component dictionary.*

Proof. The true member-tail producer gives domain2 at every literal prefix. The endpoint carrier lies over each actual prefix carrier: split the genuine geometric tower and use its actual strict-transform/preimage containment. The genuine remaining projection is holomorphic. Apply the preceding actual component-minimum producer to this projection and actual prefix ideal for each of the finitely many prefix indices from zero through the true length of T. Take the finite union of these actual exceptions. Finite analytic unions of zero-dimensional sets are analytic and zero-dimensional and remain in Z. A point outside this union lies outside every individual exception, giving actual order constancy and all-threshold truth equality simultaneously at all literal prefixes. The original full transform and original literal masked prefixes share the exact first-entry dictionary of the previous lemma. □

Theorem 66.5 (Original actual General3 data internally construct the lower-order history curve)**.** *Original true domain3 and its constructed canonical invFamily3 tower, at an actual positive-order point and positive-level nonspecial companion point, internally construct an actual analytic curve A in the entire member-tail endpoint carrier, containing that actual lower cosupport point, and a genuine analytic zero-dimensional exception Q in A. Outside Q on each actual global component, all actual literal prefix lower first-entry orders and thresholds have the genuine common component dictionary. The original literal masked first entries are exactly those of the actual entire candidate on their mask.*

Proof. The actual companion/canonical producer gives its true domain3 and literal Step1 canonical test tower. Compute the actual own-level regimeRun and genuine head/member tail as in the preceding producer; its true positive-level member extension has domain2 and exact marked-ideal mask identity at every stage. The original genuine nonspecial curve-envelope theorem internally constructs an actual analytic curve Z containing every nonspecial companion cosupport point. Intersect Z with the actual closed smooth endpoint candidate carrier; this genuine analytic intersection A has dimension at most one by analytic dimension monotonicity.

The genuine original Step1 canonical-tail theorem puts the chosen point in the original literal lower cosupport on its own mask. The actual full member mask equality and true coherent open cosupport dictionary put its underlying point in the entire candidate endpoint cosupport, hence in its actual carrier. It is already in Z, so it belongs to A. Apply the actual finite prefix-history producer to A and the true original tail, producing Q internally. The exact literal first-entry dictionary follows at every masked prefix from the preceding actual open order identity. No curve, exception, minimum, lower-domain, canonical-candidate or analytic-threshold certificate is supplied. □

Remark 66.6. The actual lower first-entry order history now has genuine global analytic predicates and an internally constructed discrete curve exception. This does not assert analyticity of arbitrary closed curve subsets or of the whole second entry. Deeper General2 companion/history descent, the actual level-zero contact extension, whole General3 finite-threshold gluing and general global termination remain OPEN.

67 Actual marking-one admissibility of the original NC separation

The input is the genuine original NC carrier H. The ideal is its COMPLETE vanishing ideal, the carrier of the marked datum is the whole ambient manifold, the original boundary is empty and the mark is one. Every ideal below is its actual controlled transform, rather than a reinitialized strict-carrier ideal.

Lemma 67.1 (Original actual NC datum in the whole invariant domain)**.** *The actual original marking-one datum lies in the invariant domain of ambient dimension n, including actual coherence and nonzero along the entire ambient.*

Proof. The original whole NC vanishing ideal is internally invertible, hence coherent. Its entire zero locus is H because H is closed. The actual finite coherent local generators exhibit that zero locus as an analytic subset, so H is analytic. The internally proved empty interior implies that at every point some actual local defining equation is frequently nonzero; otherwise all finitely many local equations vanish on a common neighbourhood and that neighbourhood lies in H. Off H the constant one is an actual ideal germ and is nonzero. These are the genuine nonzero-along-ambient germs. The whole ambient is closed smooth of dimension n, the empty boundary is SNC and transverse to it, and the unchanged mark is the positive integer one. Thus every actual domain field is produced, also when n is zero and the ideal is the unit ideal. □

Lemma 67.2 (True strict closure lies in the actual transformed cosupport)**.** *For any true well-formed marked datum, with arbitrary marking and its actual closed smooth carrier, an actual admissible blowup and an actual set K contained in its actual cosupport, the entire geometric strict closure of K is contained in the actual transformed cosupport. If the carrier is the whole ambient, the whole transformed ideal is the genuine controlled ideal.*

Proof. At a point above K outside the literal centre, the actual blowup is locally biholomorphic. The genuine off-centre controlled-cosupport dictionary therefore puts that point in the transformed cosupport. Actual admissibility supplies coherence and closed smooth geometry of the whole transformed datum. Its actual cosupport is closed: its finite coherent generators and actual smooth carrier express the genuine fixed-order threshold as an analytic subset of that closed carrier. This closed-cosupport theorem applies to every marking, including zero. Hence that closed cosupport contains the closure of the entire nonexceptional inverse image of K. That closure is precisely its geometric strict transform. When the ambient carrier is the whole space, its genuine strict transform remains the whole space, and the actual marked ideal is literally the controlled colon of the actual pulled-back ideal by the pulled-back centre ideal to the marking power. □

Theorem 67.3 (Every genuine higher-stratum step is marking-one admissible)**.** *The actual internally generated finite original NC separation is a genuine test tower for the original marking-one datum. At every literal prefix its whole current differential strict original lies in the cosupport of the SAME actual controlled original ideal, and every next actual centre is admissible. Every prefix has the entire invariant-domain data, without any caller cosupport, order, domain, tower or termination certificate.*

Proof. Induct jointly on the actual recorded higher-stratum tower, proving test-tower admissibility and the whole current strict-original cosupport inclusion. At the reflexive stage, the entire original vanishing-ideal zero locus is H and its marking-one cosupport is exactly H. At the next actual step, the actual procedure has already internally supplied its true highest centre, of attained original multiplicity at least two, with closed smooth geometry and actual normal crossings with the entire accumulated boundary. That centre is contained in the whole current original, which the induction places in the true current controlled cosupport. It is consequently an admissible centre for this SAME current ideal. This proves the next test step and supplies its true coherent transformed datum. The preceding closure lemma puts the entire geometric strict current original inside its actual next controlled cosupport. Genuine coordinate density away from a multiplicity-at-least-two centre identifies that geometric closure with the differential strict transform. Actual holomorphic strict composition identifies this with the next whole original strict transform along the true prefix projection. The two induction assertions therefore hold at the next literal stage. All real prefixes inherit the internally proved test-tower assertions, and true stability of the invariant domain gives all their whole domain fields. The original procedure supplies its higher-centre history internally; no such history is a hypothesis of this original-NC consumer. □

Theorem 67.4 (Same whole original controlled ideal and actual exceptional factor)**.** *The whole actual original controlled ideal at every generated prefix equals the literal iterated marking-one controlled transform. The COMPLETE original pullback is the product of the actual accumulated*

exceptional divisor ideal and this SAME whole controlled ideal. No transformed-ideal certificate is supplied. Each of these actual controlled ideals is invertible.

Proof. At the base the controlled ideal is literally the original whole ideal and the accumulated exceptional factor is the unit ideal. At a real next step the ambient carrier remains the whole space. The actual marked ideal is therefore the actual pulled-back ideal colon the genuine exceptional ideal, so its literal iteration equals the actual marked transform. True admissibility and the Hadamard centre-order bridge put the actual current whole ideal inside the centre ideal to marking one. Genuine single-blowup factorization identifies its actual pullback with the product of the new exceptional ideal and the whole next controlled ideal. Pull back the preceding entire factorization; true holomorphic pullback commutes with these actual coherent ideal products. The accumulated exceptional divisor is, by its actual definition, the old exceptional divisor pulled back and multiplied by the new exceptional ideal. Associating these actual products proves the entire next factorization. Invertibility holds initially by the original producer. At the next step its actual pullback and genuine exceptional ideal are invertible, and the proved whole-ideal inclusion makes their actual colon invertible. Induction proves invertibility of the SAME current controlled ideal at every actual prefix. □

Theorem 67.5 (Actual terminal controlled ideal on the augmented SNC boundary)**.** *On the actual finite terminal separation, the SAME whole original controlled ideal is an invertible effective divisor supported on the union of all actual accumulated exceptional members and the ENTIRE final original strict carrier. On this actual augmented SNC list its whole intrinsic monomial ideal equals it, and its COMPLETE nonmonomial residual is the unit ideal at every stalk.*

Proof. Form the actual effective divisor from the true whole terminal controlled ideal and its internally proved invertibility. Every point of its zero locus is a zero of the product of that ideal with the actual accumulated exceptional factor. The preceding whole factorization identifies this product with the COMPLETE original ideal pulled back by the real projection. The actual total support identity consequently contains the controlled support in precisely the union of all true accumulated exceptional members and the whole final original. The original finite-separation geometry proves that appending this whole original to its genuine finite boundary gives an actual SNC list. Apply the true SNC-supported whole-ideal factorization and residual-unit theorems to this SAME actual divisor and that actual augmented list. Their support hypothesis has just been internally produced, so whole monomial and complete unit-residual equalities follow without a caller support certificate. □

Remark 67.6. The terminal augmented list includes the entire final original; it differs from the literal marked datum boundary, which records only actual exceptional members. Thus these results do not claim empty terminal marking-one cosupport or a completed marked resolution. In dimension zero the accumulated boundary is literally empty and the augmented terminal list is literally the singleton containing the empty set; its whole union is empty and all ideals are unit. No centre containment is asserted for an unrelated additional marked ideal, and no equality with the old ordered colexicographic algorithm is asserted. Full canonical marked-ideal G9 remains OPEN.

68 Actual relative normal Taylor coefficients and whole ideal-power kernels

Write $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$, $B = \mathcal{O}_{\mathbf{C}^{d},0}$, $J = (z_{d+1}, \dots, z_{d+t})$ and use the genuine tangential projection and inclusion. The actual base algebra is pullback from this projection. Normal Taylor coefficients are entire holomorphic germs on the actual plane, not point evaluations. The source analytic input is the genuine convergent one-variable Taylor remainder used in [4] and in the normal-coordinate expansion of maximal contact; the proof below uses the previously established actual holomorphic

Taylor remainder, rather than formal power-series or kernel certificates. Smooth-centre transport uses the preceding original-regularity producer. Arbitrary original ideal quotients need not be free or flat; normal-flat centre selection and higher-HS descent remain OPEN.

Lemma 68.1 (Entire original iterated derivative germs)**.** *Every fixed list of actual ambient vectors defines a complex-linear map on entire holomorphic germs by the original iterated directional derivative. On each original holomorphic representative its value is exactly that original derivative germ.*

Proof. Take an original holomorphic representative on a genuine open neighbourhood. Complex holomorphicity of the derivative follows by differentiating on that open set, and induction gives holomorphicity of every actual iterated partial derivative. Two representatives agree on a genuine neighbourhood; there their derivatives agree, and the same induction proves eventual equality of their iterated derivatives. Hence the entire output germ is independent of the chosen original representative. The original derivative is additive and commutes with constant complex scalars on a common actual holomorphic neighbourhood. Induction proves both identities for every fixed derivative list. Passing those real identities to whole germs proves complex linearity. □

Lemma 68.2 (Normal derivatives preserve the genuine entire tangential scalar action)**.** *For every original base germ $a \in B$, ambient germ $g \in R$ and list of normal coordinate vectors, the entire iterated derivative of $(\pi^* a)g$ equals $(\pi^* a)$ times the entire corresponding derivative of g.*

Proof. Take actual holomorphic representatives on a common neighbourhood. The actual projection kills every normal basis vector. Its complex chain rule therefore makes the derivative of $a \circ \pi$ in each of these actual normal directions zero throughout that neighbourhood. The true product rule gives the first normal derivative identity. All subsequent derivatives stay holomorphic on the same neighbourhood; induction applies the same zero-derivative and product-rule identities. Passing to the whole germs proves the formula for every original entire a and g. This is genuine base-module linearity, not merely complex linearity of point coefficients. □

Definition 68.3 (Actual finite relative normal Taylor coefficient map)**.** For $q \in \mathbf{N}$, index by all ordered normal coordinate words w of length $r < q$. Define

$$T_q(g)_w = \frac{1}{r!}\, i^*(\partial_{w_1} \cdots \partial_{w_r} g) \in B.$$

This is a genuine B-linear map from the entire original ring R to the finite actual product of whole base germ rings. The ordered index may repeat monomials; its image retains the actual derivative symmetries. No independence of the redundant word family is asserted.

Lemma 68.4 (Actual recursive normal powers equal the complete ring ideal powers)**.** *An entire original germ belongs to J^q exactly when an actual holomorphic representative has a recursive normal-power expression of degree q near zero. The expression uses only genuine normal-coordinate factors, with holomorphic original coefficients; it is independent of representative.*

Proof. At degree zero both sides contain every holomorphic germ. In a recursive expression of degree $q+1$, the tangential coefficients are identically zero. Read its actual finite sum and products as entire germs, split tangential and normal coordinates, and use the already proved degree-q inclusion. Each remaining term is a genuine normal coordinate in J times a member of J^q, so the entire sum belongs to J^{q+1}.

Conversely the actual recursive-expression germs form an ideal: zero and addition use the real recursive expressions, and multiplying by an arbitrary whole original holomorphic germ multiplies its actual coefficient functions. At degree one each actual normal coordinate has its genuine one-term recursive expression. For the induction step, $J^{q+1} = JJ^q$. A normal coordinate times any actual degree-q expression has its literal next-degree recursive expression. Finite span induction over all genuine generators of J, with ideal closure under arbitrary entire multipliers, proves the

whole product ideal inclusion. Eventual equality of original representatives preserves the actual recursive expression, completing both directions. □

Lemma 68.5 (Vanishing actual normal derivatives produce a genuine normal-power expression)**.** *If all original normal coordinate derivatives of orders $< q$ of an actual holomorphic function vanish on the genuine coordinate plane near zero, then that original function has an actual recursive normal-power expression of degree q near zero. Only normal derivatives are assumed.*

Proof. Induct on the number of normal coordinates, uniformly in q. With none, positive q includes the empty derivative, so the original function is zero near zero and has the genuine zero expression; $q = 0$ is holomorphicity. Otherwise let z_j be the first normal coordinate and let p_j kill it. The actual convergent Taylor theorem gives, on a genuine neighbourhood,

$$f(z) = \sum_{a<q} \frac{z_j^a}{a!}(\partial_j^a f)(p_j z) + z_j^q h(p_j z, z_j)$$

with a jointly holomorphic true remainder h. Every other normal basis vector is fixed by p_j. The true linear chain rule identifies its iterated derivatives of $(\partial_j^a f) \circ p_j$ with the corresponding derivatives of f with a copies of the direction j appended. On the smaller plane with $z_j = 0$ these vanish below order $q - a$, by the actual original hypothesis. The induction hypothesis gives a genuine degree-$(q-a)$ expression in the remaining normal coordinates for each coefficient. Allowing also z_j preserves that expression; multiplication by z_j^a raises its true degree to q. The last term has its direct degree-q expression because the remainder is genuinely holomorphic. Finite sum, actual constant scalars and the true original Taylor neighbourhood equality complete the result. □

Theorem 68.6 (Exact entire normal-power kernel and genuine coefficient uniqueness)**.** *The complete kernel of the actual B-linear map T_q is exactly J^q, restricted to the genuine base scalar action. Consequently two entire original germs have the same actual relative normal Taylor coefficient tuple if and only if their difference belongs to the whole ideal J^q.*

Proof. A normal derivative lowers a recursive normal power by at most one. This follows directly from the actual product rule for each normal-coordinate factor and ideal closure: differentiating that factor removes at most one true normal factor, and differentiating its original holomorphic coefficient removes none. Iterate to see that derivatives of order $r < q$ still have at least one genuine normal factor and vanish on the actual plane. Thus every actual T_q coefficient is the zero entire base germ.

Conversely suppose the whole coefficient tuple is zero. Each original normal derivative restricted to the actual plane is therefore zero on a genuine base neighbourhood; its factorial is nonzero. There are only finitely many ordered words below q, so these neighbourhoods have a genuine common intersection. Pull it back by the actual tangential projection. Every nearby actual plane point is the inclusion of its true projection, so all those original derivatives vanish there simultaneously. Every finite normal-index list below q is one of the actual indexed words. Apply the preceding true normal-only Taylor theorem, then the complete recursive-power dictionary, to obtain membership of the entire original germ in J^q. The equality-of- coefficients assertion now follows from real base linearity applied to the whole difference. It concerns the uniquely determined actual coefficient tuple; the redundant ordered-word scalar representation is not unique. □

Theorem 68.7 (Actual all-degree whole quotient presentation by finite normal words)**.** *For every q, the entire original quotient R/J^q is genuinely isomorphic as a B-module to the actual image of T_q. That actual image is the finite B-span of the internally computed coefficient tuples of genuine normal words below degree q.*

Proof. The exact whole kernel and the true quotient/range theorem give the genuine module isomorphism. Take any entire original germ. The previous actual relative Hadamard producer gives a finite whole-base normal-word representative whose remainder belongs to the actual J^q. The exact kernel sends that entire remainder to zero. Genuine base linearity sends the representative to its actual finite base linear combination of the computed word tuples. Thus the entire range is contained in that actual finite span; the reverse inclusion follows because each genuine word tuple is the image of its actual whole word germ. This proves the exact range dictionary and finite module presentation internally, retaining all genuine relations of the ordered word family rather than declaring a redundant family to be a basis. □

Theorem 68.8 (Whole original ideal columns and literal entire relative quotient)**.** *For every whole original ideal $I \subset R$, genuine Noetherianity internally produces original holomorphic generators, simultaneously for all q. The entire image of I under T_q is exactly the finite B-span of the actual normal-word-times-generator tuples. The literal whole quotient $R/(I + J^q)$ is genuinely isomorphic to the actual coefficient-image module modulo this entire actual ideal-image submodule.*

Proof. Choose genuine finite original whole generators using actual local Noetherianity and actual holomorphic representatives of those germs. Every entire ideal member is an actual finite generator combination with whole ambient coefficient germs. Apply the true all-degree normal-word representative producer to each such coefficient. Multiplying its genuine remainder in J^q by the whole original generator keeps it in J^q, so the exact actual Taylor kernel discards it. Genuine base linearity identifies its remaining image with the finite actual normal-word-times-generator column span. Conversely every one of those actual columns comes from a literal original ideal member, proving the other inclusion. The genuine second quotient theorem with the proved complete kernel identifies the whole quotient by $I + J^q$ with precisely the coefficient range modulo that whole ideal image. The generator family is original and fixed for every q; neither an input matrix nor an ideal-image equality is supplied. The same original smooth-centre flattening internally transports these complete ideal/power/relative presentation dictionaries to actual centre stalks. No normal-flatness or free-quotient conclusion follows for arbitrary I. □

Theorem 68.9 (Original regular centre produces actual whole-power Taylor kernels)**.** *Actual original regularity of a centre at a point internally supplies genuine nonlinear holomorphic flattening, its entire germ algebra isomorphism E and the whole original vanishing stalk $K = E(J)$. Over the internally transported entire tangential algebra, the genuine maps $g \mapsto T_q(E^{-1}g)$ are base-linear and have complete kernel K^q, simultaneously for every q.*

Proof. Apply the preceding original-regularity producer. It constructs true local coordinates from the actual geometric regularity and proves equality of the ENTIRE original vanishing stalk with the image of the genuine normal ideal. Define the transported base action by $a \cdot g = E(\pi^* a)g$. Since the genuine inverse germ algebra isomorphism carries this actual product to $(\pi^* a)E^{-1}g$, the proved entire tangential linearity of T_q gives base linearity of the displayed actual transported map. It vanishes precisely when $E^{-1}g \in J^q$, by the complete normal-power kernel. Surjectivity and injectivity of the actual whole germ isomorphism identify this with $g \in E(J^q) = E(J)^q = K^q$. The selected coordinates and complete stalk equality are produced from the original regularity; neither is an input. The same coordinates serve every degree. This transports genuine normal coefficients and whole powers, with no flatness assertion for any quotient by an additional original ideal. □

69 Whole original homogenization under actual smooth pullback

The actual map is a genuine holomorphic submersion of arbitrary dimensions. The input ideal is the original coherent ideal; the mark, carrier and ordered boundary are its literal original data. No

tangent, homogenization, coefficient, cosupport or invariant equality certificate is supplied. Full canonical smooth functoriality and genuine blowup base change remain subsequent obligations.

Lemma 69.1. *For every original coherent ideal I and mark $\mu \geq 0$, the original entire tangent ideal $T(I,\mu) = \mathcal{D}^{\mu-1}I$ commutes with actual smooth pullback.*

Proof. The actual tangent ideal is precisely the original iterated derivative ideal at the natural-number index $\mu - 1$. Apply the already internally proved ENTIRE iterated derivative pullback equality to that actual index. Equality includes all whole germ members. Natural subtraction retains the genuine index zero when $\mu = 0$; no positive-mark or selected-generator premise is introduced. □

Lemma 69.2. *Every literal original summand $\mathcal{D}^i I\, T(I,\mu)^i$ of the true homogenization pulls back to $\mathcal{D}^i(f^*I)\, T(f^*I,\mu)^i$ as an ENTIRE ideal.*

Proof. Every original iterated derivative ideal is coherent, and so is the actual tangent ideal. Its genuine ith power is coherent. The actual holomorphic pullback preserves the ENTIRE product and that ENTIRE power. Substitute the true entire iterated derivative equality for the first factor and the preceding true tangent equality for the second. The actual target product is exactly the displayed original homogenization summand. The unit zeroth power is preserved by the genuine constant-one span argument already proved. □

Theorem 69.3. *The COMPLETE original homogenized ideal satisfies $f^*\mathcal{H}(I,\mu) = \mathcal{H}(f^*I,\mu)$ under the genuine submersion.*

Proof. The original definition is the entire finite span of the actual product summands indexed by $i < \mu$. These are coherent by genuine iterated derivative, power and product coherence. Read that defining whole span as the actual finite sum ideal and apply the already proved entire finite-sum pullback equality. Replace every whole pulled summand by the preceding true equality. Finite-span extensionality yields exactly the original target homogenized ideal, with no chosen coefficient system or order assumption. At zero mark both finite families are genuinely empty, so the same entire-span proof applies. □

Theorem 69.4. *The actual original homogenized marked datum commutes with the literal cross-dimensional marked pullback, including carrier, ordered boundary, mark and EVERY ideal member.*

Proof. Original homogenization retains the original carrier, ordered boundary and mark and replaces only its entire ideal by the genuine homogenized ideal. Literal pullback takes the actual preimage carrier and ordered preimage list and retains the same numerical mark. These three actual fields therefore agree on both sides. The preceding theorem identifies the fourth field, the COMPLETE original ideal. Extensionality of the actual marked record proves equality of the WHOLE data. No well-formedness or canonical-domain conclusion is inferred from these fields. □

Theorem 69.5. *The original coefficient marked datum of the original homogenized marked datum also commutes with genuine smooth pullback as WHOLE marked data. For actual ambient original carrier, the homogenized and homogenized-coefficient cosupports are exactly their original preimages.*

Proof. The original homogenized ideal is genuinely coherent, by its actual finite coherent summand construction. First use the WHOLE homogenized marked-data equality, and then the already proved actual coefficient marked-data equality for that SAME original coherent homogenized ideal. This proves the stated entire composite marked equality. For ambient carrier, its actual homogenized carrier remains the whole original manifold. The genuine original-point section of the submersion preserves every whole germ order threshold for that SAME original coherent

homogenized ideal, so its marked cosupport is exactly the original preimage. The corresponding actual coefficient-cosupport producer applies to that SAME original homogenized marked datum, proving the other whole preimage identity. Zero marks and empty weighted families retain the literal definitions; no caller coherence of a transformed ideal or threshold equality is assumed. □

70 Actual member companion history and entire controlled extensions

Throughout, the ambient dimension is positive. The original genuine parent test tower supplies the member candidate's entire strict carrier and boundary geometry, but need not be a test or canonical tower for that candidate. The actual ideal, all cosupport and order predicates, and every companion below are constructed from that same candidate. Second-entry descent, level-zero contact and whole General3/global termination remain open.

Definition 70.1 (Actual member prefix cosupport and companion models)**.** For the entire actual member candidate I along the genuine parent tower T, pull each actual prefix cosupport back to the endpoint by its true remaining projection. At a literal prefix and positive natural q form the actual companion ideal of that prefix I, and its actual coefficient-of-homogenization barMarked. These use the whole original nonmonomial and monomial ideals, not a supplied presentation or an unrelated extension.

Lemma 70.2 (Actual prefix predicates and exact original masked order runs)**.** *Every actual prefix cosupport predicate is analytic. For all natural q, the OrdRegime of the actual whole member candidate and that of its literal original masked lower ideal agree at every corresponding prefix point. Their genuine regimeRun along the actual whole and original restricted tower is equal. No candidate testness or candidate canonicality is an input.*

Proof. The genuine member whole-tail theorem gives its true smooth carrier and coherent positive-mark nonzero ideal at each actual prefix. The derivative ideal cosupport producer thus gives an actual analytic prefix cosupport. The remaining projection is holomorphic, so its true inverse image is analytic. At each actual prefix the full original masked marked ideal is exactly the open pullback of the entire candidate. Actual coherent open cosupport locality and actual nonmonomial-order locality identify the two genuine OrdRegime conjuncts pointwise. Induct through the actual tower recursion: the last-step run is the preceding run plus one precisely when its genuine image satisfies the regime, and is zero otherwise. The actual predicate equality and induction hypothesis give equal runs, including empty and repeated steps. The original parent test predicate is inherited by actual prefixes; it internally supplies the needed whole-domain and full mask identity at each induction stage. □

Lemma 70.3 (One genuine discrete history exception fixes all actual order runs)**.** *On an actual analytic curve in the endpoint member carrier there is one actual analytic zero-dimensional exception such that, outside it on each true global component, the whole actual OrdRegime run agrees for every natural q, including the run computed at each point's own actual endpoint natural order. No finite order range or global order bound is assumed.*

Proof. The previous attained-minimum producer gives one genuine zero-dimensional exception for all prefix natural orders, simultaneously for every threshold. Apply the real finite analytic-predicate producer to the genuine pulled-back prefix cosupports, obtaining another actual zero-dimensional exception. Their actual union is analytic, zero-dimensional and contained in the curve. At two points of any real global component outside that union, the cosupport conjunct and natural-order conjunct of every actual prefix OrdRegime agree, for every q. Induction through the true tower recursion, reading those prefix images by the genuine remaining projections, identifies the actual runs. The actual endpoint orders also agree, so runs evaluated at each point's

own actual endpoint order agree. This gives a genuine common head for the next companion construction; it does not assert that the candidate is canonical. □

Lemma 70.4 (True member geometry preserves an actual ideal with the same carrier and boundary)**.** *The actual companion and barMarked constructed at any literal member prefix and positive natural q have genuine domain2 full controlled transforms on the parent tail, coherent and nonzero along at every literal stage. All smooth carrier, SNC/transverse, dimension and positive-mark fields are internally provided by the genuine member candidate geometry and actual companion/coefficient ideal producers.*

Proof. The carrier and boundary recursions depend only on their true initial sets, and hence coincide at every actual stage. The member producer supplies actual closed smooth strict carriers, actual SNC/transverse boundary and actual dimension bound at every genuine prefix. Induct through the literal parent tail. Pullback of the actual coherent ideal is coherent; the actual colon by the true positive-mark exceptional power along the genuine smooth strict carrier is coherent. Actual nonzero along is preserved off the centre by the true local blowup inverse, and at exceptional points by strict-transform accumulation and the actual finite-generator nonzero argument. Neither step uses containment of the centre in the candidate carrier or cosupport. The genuine mark is unchanged along the controlled transform. Combining these actual ideal facts with the already produced true geometric fields gives the full domain at every actual prefix. For the open identity use the actual pullback/colon open locality with the true coherent before and after ideals, then induct through all real stages. All hypotheses used here are proved from actual member geometry when the ideal is a companion or barMarked. □

Theorem 70.5 (Actual companion and coefficient transforms extend the literal masked ideals)**.** *For each literal member prefix and each positive natural q, its actual whole companion and barMarked have genuine positive-mark coherent nonzero full controlled tail transforms with actual domain2 at every literal stage. Their whole marked transforms restrict exactly to the corresponding ideals constructed from the original literal masked member prefix, including their true carrier, boundary, generated ideal and mark.*

Proof. The whole-prefix true member domain produces the actual monomial and nonmonomial ideals. Their actual power/sum definition produces the genuine companion domain for q positive; its nonmonomial summand supplies nonzero along, and the actual mark is positive in either branch. The genuine homogenization/coefficient producer then makes barMarked coherent and nonzero along, using an actual power of an original nonzero germ in its index-zero coefficient summand. Both have literally the same true carrier and boundary as the prefix member ideal. Apply the preceding actual same-geometry induction to the real remaining parent tail.

At the original mask, the genuine marked prefix is the actual open pullback of the whole member prefix. True whole monomial/nonmonomial ideal locality, actual ideal powers/sums and coefficient/homogenization locality identify the constructed masked companion and barMarked with the pullbacks of these same whole generated ideals. The preceding entire open-transform identity then identifies their full controlled transforms on the literal masked tail, at every stage. Thus the whole extension is attached to the actual original masked generated ideal, rather than merely supplying an independent coherent ideal. For the actual order mask this is exactly the literal compAt and barMarked controlled transform dictionary as well. The true run identity selects the same literal companion head. □

Theorem 70.6 (Original General3 data internally supply the actual next-companion history)**.** *The original constructed General3 canonical inputs at a true positive-level nonspecial companion point internally supply the actual member history curve, an actual zero-dimensional exception, all-order common runs on its genuine global components outside that exception, and the genuine entire next companion and coefficient transforms with their exact original literal masked dictionary.*

No lower-domain, history, candidate canonicality, centre containment or coherence certificate is supplied by the caller.

Proof. The preceding original General3 member-history producer internally supplies the true companion, genuine own-level head and parent tail, actual nonempty member family, chosen entire lower cosupport point, analytic history curve and attained-minimum prefix history. Apply the actual finite cosupport-history producer on that internally constructed curve and take the genuine union of its exceptions. All actual order runs now agree componentwise outside the exception, including at the actual endpoint order. The original true parent head/tail testness and internally produced member domain give the actual next companion/bar entire-transform producers and original masked identities at each literal prefix and positive mark. The previous actual mask dictionary identifies these with the same original constructed lower ideals. This is the precise next companion/history seam; the deeper invariant value and its boundary descent are not claimed here. □

Remark 70.7. Deeper General2 value descent across all actual history changes, level-zero contact, whole General3 threshold gluing and general global G3 termination remain OPEN. Constancy of the actual first-entry/run history does not imply candidate canonicality or analyticity of an arbitrary closed curve subset.

71 Actual relative normal monomial bases of entire normal-power quotients

Let $B = \mathcal{O}_{\mathbf{C}^d,0}$, $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$ and $J = (z_{d+1}, \dots, z_{d+t})$. Use the actual tangential pullback as the entire base algebra. This is the usual genuine normal-coordinate polynomial expansion in the smooth-centre charts of [4]. All derivatives below are actual holomorphic germ derivatives; the complete Taylor kernel was produced analytically in the preceding extension. Only R/J^q is asserted free. A quotient by an additional original ideal need not be free or flat. Actual relative ideal matrices, fixed-degree free patches, all-degree normal-flat centre selection and higher HS descent remain subsequent obligations; general G8 remains OPEN.

Lemma 71.1 (Actual entire normal derivations and genuine polynomial evaluation)**.** *Each genuine normal basis vector defines a B-derivation on the ENTIRE original holomorphic germ ring. Its derivative of the original normal coordinate z_j is δ_{ij}. Evaluation of the actual polynomial ring $B[Z_1, \dots, Z_t]$ at these original coordinate germs is a genuine entire base-algebra map. The actual derivative of its value is exactly the value of the true formal polynomial partial derivative. Its whole plane trace is the original polynomial constant coefficient.*

Proof. Take actual original holomorphic representatives of two whole germs on a common neighbourhood. The original complex product rule gives the true Leibniz identity throughout that neighbourhood, hence on entire germs. The previous tangential-linearity theorem supplies actual B-linearity, and the constant function one has zero derivative. The actual normal coordinate is an original linear coordinate functional; its genuine derivative evaluated at the actual normal basis vector is the Kronecker delta everywhere. These are the true derivation fields, not supplied certificates.

Evaluate a polynomial by its entire base coefficients and actual normal germs. On constants the actual derivative is zero because the tangential projection kills the true normal basis vector. On Z_j it is the proved Kronecker delta. Induct on the original polynomial, with true addition and multiplication by Z_j, and apply the actual Leibniz rule. The genuine formal partial derivative satisfies exactly these original operations, proving the whole equality. The actual entire trace fixes every base coefficient and kills every normal coordinate, so the same genuine polynomial induction identifies it with the whole constant coefficient. □

Lemma 71.2 (Actual normal-power membership forces every lower polynomial coefficient to vanish)**.** *For every ORIGINAL polynomial $P \in B[Z]$, actual membership of its entire coordinate evaluation in J^q forces its coefficient at each genuine exponent α of total degree $|\alpha| < q$ to be the zero WHOLE base germ.*

Proof. Induct on q, uniformly for every actual polynomial and exponent. Degree zero gives no exponent satisfying the strict inequality. For positive q and $\alpha = 0$, take the actual whole trace: $J^q \subset J$ and the actual entire plane kernel is J, so the original constant coefficient is zero. If $\alpha \neq 0$, choose a genuine coordinate i with $\alpha_i > 0$ and put $\beta = \alpha - e_i$. The actual original normal derivative lowers J^q to J^{q-1}, by the already proved true derivative/power producer. Its value is the true evaluated formal derivative, by the preceding actual whole equality. Since $|\beta| = |\alpha| - 1 < q - 1$, induction makes the original β coefficient of this derivative zero. The genuine polynomial coefficient formula is α_i times the original α coefficient. This actual positive integer is nonzero in the entire base germ ring: evaluating a claimed zero constant germ would give the corresponding nonzero complex number. The original holomorphic germ ring is a domain. Therefore that entire original coefficient is zero. All derivatives and ideal-power facts are internally produced; the coefficient equality is not an assumption. □

Definition 71.3 (Finite actual normal exponent family and genuine monomial combination)**.** Index by genuine finitely supported t-tuples α with $|\alpha| < q$. This is internally finite: each actual entry is bounded by the actual total degree, hence is $< q$. The genuine quotient combination map sends an entire base coefficient tuple (a_α) to the original whole germ class $\sum_\alpha \pi^* a_\alpha z^\alpha \pmod{J^q}$.

Proof. Encode each actual exponent by its actual entries in the finite type $\mathrm{Fin}(q)^t$. Equality of those genuine entries gives equality of the original finitely supported exponent, proving injectivity and finite indexing internally. True polynomial evaluation identifies each original monomial Z^α with the genuine normal-coordinate product. The actual quotient map is base-linear, and its finite combination is therefore a real base-linear map on the entire original coefficient tuples. At $q = 0$ the index is empty. When $t = 0$ and $q > 0$, its sole exponent is zero. □

Theorem 71.4 (Actual monomials below degree q form a true entire relative quotient basis)**.** *The genuine combination map is bijective, hence internally gives a true B-basis of the ENTIRE original quotient R/J^q indexed by those actual exponents. Its basis vectors are precisely the original monomial classes. In particular this literal whole quotient is a finite free base module.*

Proof. Suppose an actual coefficient combination is the zero quotient class. The entire corresponding original finite polynomial evaluates to a true member of J^q. Every exponent in the real combination has degree $< q$. The preceding actual separation theorem makes each original polynomial coefficient zero as a WHOLE base germ. The genuine coefficient at an indexed exponent is exactly the corresponding entry of the actual tuple, since the actual exponent family has no repetitions. Thus the tuple is zero. Base linearity applied to the difference proves injectivity.

The preceding actual relative Hadamard producer supplies every entire germ a finite real normal-word representative modulo J^q. Count each actual normal index in a word. Its counts form a genuine exponent of degree equal to that word's actual length, hence $< q$. Commutativity of the genuine entire germ ring identifies the original word product with its actual exponent monomial. Therefore every true representative lies in the real monomial span, and the actual finite quotient combination is surjective. The internally proved bijection gives the actual linear equivalence and transported standard finite basis. This is an original basis producer, not a supplied independence, spanning or rank certificate. The zero quotient and empty basis at $q = 0$ are included, and the $t = 0$ case has the genuine one-vector base basis for $q > 0$. □

Theorem 71.5 (Original regular centre transports all true normal-power quotient bases)**.** *Actual original regularity internally constructs genuine nonlinear flattening, the true entire centre ideal K and its genuine transported base algebra. For every q, it then supplies an actual finite base*

basis of the whole original centre-power quotient R/K^q, by the true transported original normal monomials. True closed smooth centre geometry internally supplies the dimension at each actual centre point and this same producer. This does not assert freeness or flatness after adjoining any other original ideal.

Proof. Use the actual regular-centre producer to select the genuine flattening and entire germ algebra isomorphism E, and to prove the ENTIRE original ideal equality $K = E(J)$ with every whole power equality. The transported base action is multiplication by $E(\pi^* a)$. Combine the actual transformed normal monomials with this genuine entire base action. The result is exactly the class of E applied to the original finite evaluated polynomial. If that class vanishes, its whole normal-power membership pulls back through the genuine inverse to actual J^q membership; the internally proved original coefficient separation gives injectivity. For any original germ g, apply the actual untransported surjectivity to $E^{-1}g$ and map its whole power remainder through E. This proves true transported surjectivity. Its actual linear equivalence transports the standard finite coefficient basis to the genuine monomial basis. Its vectors are exactly the classes of $E(z^\alpha)$, by the original whole quotient map. All coordinate, ideal, power, base and basis data have been internally produced from original geometry, simultaneously in all q. The resulting base need not be constant across differently dimensioned components; original regularity gives the actual dimension at the chosen point. All-degree normal-flatness of a singular space along a selected centre still requires controlling its additional original ideal, which this result does not presume. □

72 Actual positive-level member companion predicates and histories

The ambient dimension is positive. The parent is a genuine domain-three marked ideal and its tower is an actual parent test tower. The entire member candidate, its literal prefix companions and their coefficient ideals are the same generated ideals as in the actual masked dictionary. They need not be canonical or of maximal order on the entire ambient space. This distinction is essential for the level-zero contact branch, which remains open here.

Definition 72.1 (Actual positive special and companion history predicates)**.** For an actual marked ideal J and tower R, let its positive special threshold at s be the points of level at least s which are genuinely special. For a member candidate along T, form its actual companion of positive mark q at literal head i; read its positive special threshold at one after a further literal head j of the remaining tail, pulled back by the true remaining projection to the endpoint. The count history predicate is the corresponding pulled-back original-member count threshold; the companion mark one is used for this predicate because its actual initial and transformed boundaries do not depend on the positive companion mark. The actual component history property records attained endpoint minima, their constancy off the exception, all whole member order runs, and the positive-special/count and positive regime run equalities for each component's own positive minimum.

Lemma 72.2 (Exact positive special dictionary without maximal order)**.** *For s positive the genuine special threshold equals the actual member locus of the current coefficient cosupport intersected with the original-member count threshold at s.*

Proof. At a point with level at least s positive the level-zero disjunct of the actual special definition is impossible. Its remaining disjunct is precisely positive level together with membership in the actual member locus. Conversely membership in that member locus and level at least s gives the positive-level special disjunct. This proves both inclusions without any maximum-order or canonical-tower hypothesis. □

Lemma 72.3 (Actual positive special and count analyticity)**.** *Current true well-formed coefficient geometry gives every original-member count superlevel analytically. Current true domain geometry*

also gives every positive special threshold analytically. For the actual member prefix companion and the actual parent tail all of this current domain geometry is internally produced, for every positive companion mark.

Proof. Take the actual original-member sublist of the current SNC boundary. It is SNC and transverse to the current closed smooth carrier. Each member is analytic, by the actual smooth SNC singleton-intersection theorem. The finite indexed intersection dictionary therefore makes every count threshold analytic. Current coherence, closed smooth carrier and positive mark give genuine analytic equations for the current entire cosupport. The actual member locus is a finite union of selected genuine global components of the smooth SNC intersections. Analytic continuation selects these components and their real local finiteness gives closed smooth selected unions, hence analytic member locus. Its intersection with the count threshold is analytic and equals the positive special threshold by the preceding dictionary. Actual member geometry and the coherent controlled-transform induction provide the whole companion and coefficient domains on every actual parent tail. Apply the preceding argument to those internally produced domains. No entire maximal-order premise is introduced. □

Lemma 72.4 (Actual companion count geometry and prefix predicates)**.** *The genuine original-member count for every actual prefix companion equals that of its actual member prefix model, independently of q. All literal positive special and count history predicates are analytic at the endpoint.*

Proof. The actual companion and coefficient operation preserve the entire initial boundary. The boundary controlled-transform recursion uses only that initial boundary and the identical actual tower maps and exceptional appends. Hence the full transformed boundaries and their actual original-member takes agree. Counting their indexed members gives the stated equality. Split the genuine parent test tower at each literal head, then split its remaining actual tail again. The preceding whole companion-domain producer applies at that real head and further real prefix. Its positive special and count predicates are analytic there; the actual remaining projection is holomorphic, so their actual inverse images are analytic at the final endpoint. □

Theorem 72.5 (Component-own-order special and count histories)**.** *For a genuine analytic curve Z in the endpoint entire member carrier there is an analytic zero-dimensional exception E contained in Z. Each genuine global component B has an actually attained minimum endpoint natural order q which equals the endpoint order everywhere on B outside E. If q is positive, then for all literal companion heads and all further literal prefix images, both the actual original-member counts and the actual positive-special predicates agree at any two points of B outside E. All positive-level actual stepRegime runs on these companion tails agree as well. The whole member order runs agree for all natural marks, including the point-own-order run. No global order bound, finite range, or common exception for all companion marks is asserted.*

Proof. The actual attained-minimum order producer gives a discrete analytic order exception and, on each real nonempty global component B, its true attained minimum q. Its proper analytic excess at q+1 is the source of the order exception, so no upper bound on orders is needed.

For a component with q positive consider the finite family of the preceding actual positive-special predicates at that fixed q, indexed by all literal heads and further literal prefixes. Include the actual count predicates at all thresholds from zero to the relevant finite original-member-list length. For each predicate P use the empty set if B is contained in P and B intersected with P otherwise. In the latter case real analytic continuation excludes an open component trace and genuine analytic dimension drop gives dimension at most zero. These actual excesses are all contained in B. For each finite index the true locally finite component family therefore restricts to a locally finite excess family, whose union is analytic and zero-dimensional. Their finite union has the same properties and lies in Z. Components with q zero contribute an empty family. Unite this exception with the order exception and the actual all-order-run history exception.

Outside the union each positive special predicate has exactly its whole- component truth value. Likewise every finite count threshold has its whole- component truth value. The count is bounded by the actual list length, so threshold equality at the two attained counts forces count equality. At any positive level s, equality of counts and positive-special predicates gives literal equivalence of the actual regime condition level=s and not special, even at prefix images whose count is zero. The true tower run recursion then gives equality of stepRegime runs. The existing genuine all-order history producer gives equality of all order runs and their own-order evaluation on the same final exception. The component mark q is its actual minimum; a single finite family is used for this q on this B, and local finiteness of B makes the varying-component construction analytic without any finite assumption on the set of marks. □

Theorem 72.6 (Original General3 data produce positive next-companion histories)**.** *The original constructed General3 canonical domain data, at a genuine positive-order companion point which is nonspecial and has positive original level, internally provide the actual member curve, its genuine discrete exception, and the preceding component-own-order positive-special/count history for the actual next companions. The member family, lower domain, curve, attained orders, whole companions and original literal mask identity are produced from those original data, without caller certificates for any of them.*

Proof. The original companion producer gives its true domain-three marked ideal, actual canonical StepOne tail, maximal order on its original order mask and whole cosupport dictionary. Its own-level run gives the genuine parent head and tail. Positive original level internally gives a nonempty actual member family. The true curve-envelope and entire member-domain producers then give an actual analytic curve containing the chosen point and lying in the entire member endpoint carrier. Apply the preceding genuine component-own-order history construction there. Its companion models are precisely the same actual literal member-prefix companions whose full controlled marked ideals already agree with the original masked companion and coefficient ideals. Thus the history is attached to the actual original lower construction. No canonicality of the whole extension or maximum order outside the original mask is used. □

Remark 72.7. The level-zero codimension-one contact branch, deeper finite lower-value history descent, whole General3 threshold gluing and general global G3 termination remain OPEN. A closed subset of a curve is never identified as analytic merely from closedness. The exception above varies the predicate mark with each component's actually attained order and does not control positive special loci at every natural companion mark simultaneously.

73 Actual original ideal matrices in the entire distinct normal monomial basis

Let $B = \mathcal{O}_{\mathbf{C}^d,0}$, $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$, and J the original normal coordinate ideal. The preceding actual finite free basis is used; no caller matrix, kernel, frame, rank or flatness certificate is supplied. Every statement uses entire original base germs. The ideal I below is the entire original ideal, not merely a chosen collection of jets.

Lemma 73.1 (Actual entire monomial coordinates and full power kernel)**.** *For every q the genuine basis gives a surjective entire B-linear map $c_q : R \longrightarrow B^{A_q}$ with complete kernel J^q. Its original normal monomial classes give the true standard coordinate vectors. Every whole germ admits a genuine lower-degree monomial representative with entire original base coefficients, modulo the whole J^q.*

Proof. Compose the actual whole quotient map with the inverse of the internally proved monomial linear equivalence. Its inverse is genuine and the quotient map is surjective, so the composition is surjective. Its full kernel is exactly the kernel of that original quotient map, namely the entire

power. The actual basis vector identity identifies coordinates of the original normal monomial with its standard coefficient vector. Apply the genuine monomial combination surjectivity to the class of any original whole germ. Its finite polynomial evaluation equals the genuine finite sum of original normal monomials times the entire base coefficients. Equality of quotient classes is precisely the true whole-power remainder membership. □

Lemma 73.2 (True original ideal images are entire finite monomial-generator column spans)**.** *If the original entire ideal is generated by actual germs $G_1, \dots, G_r$, its whole image under c_q is exactly the B-span of the genuine finite columns $c_q(z^\alpha G_i)$ for $\alpha \in A_q$.*

Proof. For an arbitrary entire multiplier a, take the genuine monomial representative $p = \sum_\alpha \pi^* a_\alpha z^\alpha$ modulo J^q. Then $(a-p)G_i$ is still in the entire original J^q and has zero true coordinate tuple. Thus $c_q(aG_i) = \sum_\alpha a_\alpha c_q(z^\alpha G_i)$. The actual finite ideal generating-family dictionary expresses every whole original ideal germ as a finite sum of genuine multipliers times G_i. The preceding equality gives the forward inclusion in the genuine column span. Conversely every original $z^\alpha G_i$ is in the entire original ideal, and the image is a whole base submodule. It therefore contains the complete column span. □

Theorem 73.3 (Original arbitrary ideals internally supply genuine entire relative matrices)**.** *Every original whole ideal I internally supplies finitely many ORIGINAL holomorphic generator representatives and, for every q, its actual finite matrix of distinct normal-monomial coordinate columns. The matrix entries are entire base germs and internally have genuine holomorphic representatives on a common actual base neighbourhood. Its actual column image is the whole coordinate image of I.*

Proof. The actual holomorphic germ ring is Noetherian. Select a true finite original ideal generating family from this theorem and choose actual original holomorphic representatives for each generator. For every q, apply the preceding column equality. Index rows by the distinct actual bounded exponents and columns by the actual exponent-generator pairs. The entry in a row is the corresponding entire base coordinate of that original monomial-generator product. Genuine finite matrix multiplication is precisely its whole column combination; hence its range is exactly the complete original ideal image. Every actual whole base germ has an original holomorphic representative. Choose these for the finite actual entries and intersect their actual holomorphic neighbourhoods. This constructs genuine common-neighbourhood representatives internally; no point-value replacement or supplied representative/kernel hypothesis is made. □

Theorem 73.4 (Literal whole original quotients are the genuine actual matrix cokernels)**.** *The literal entire original quotient $R/(I+J^q)$ is truly B-linearly isomorphic to the full finite coordinate module modulo the range of the actual original ideal matrix.*

Proof. Compose c_q with the true quotient by its entire original ideal image. This composition is surjective. A germ maps to zero exactly when its tuple is the actual tuple of a whole original ideal germ. Subtract that original germ: the genuine full coordinate kernel is J^q. Hence this complete composition kernel is exactly the whole $I + J^q$, in both directions. The original first isomorphism theorem identifies its literal whole quotient with the actual full coordinate cokernel. The preceding matrix range equality identifies the same genuine cokernel with the actual original matrix cokernel. At $q = 0$ the actual row family is empty and both quotients are zero. Empty generator families, zero/unit ideals and zero normal or tangential dimensions are handled by the literal definitions. □

Theorem 73.5 (Actual fixed-degree whole quotient freeness after a genuine nonzero base localization)**.** *For each actual q and whole original ideal I, an internally produced nonzero whole base germ b makes the actual localized module $(R/(I + J^q))_b$ free over the actual base localization B_b. Its rank equals the genuine fraction-field rank of the same entire original quotient. This produces no common denominator for all degrees and asserts no all-degree normal-flat centre.*

Proof. The literal whole original quotient is finite over the actual entire base ring by the proved original normal-power representative producer. The original base holomorphic germ ring is Noetherian, so this true quotient is finitely presented. Localize the actual quotient at the nonzero base germs. The true base domain has its genuine fraction field; every module over that field has a real basis, internally constructed by the usual basis existence theorem. The genuine finite-presentation localization theorem lifts this actual fraction-field basis to the localization at the powers of one true base element from the same nonzero-divisor family. This proves actual whole module freeness and equality of its actual finite rank. The selected denominator is nonzero as a whole germ. No caller generic freeness, rank, basis or denominator certificate is an input. □

Remark 73.6. No arbitrary original I quotient is asserted free or flat. True fixed-degree open freeness patches require genuine holomorphic minor producers. All-degree normal-flatness additionally requires a genuine graded Noetherian/generic freeness producer yielding a common actual analytic open set, rather than an intersection of separately chosen degree-wise patches. Higher HS descent and general G8 remain OPEN.

74 Actual divisor exponents under genuine holomorphic submersions

All maps in this chapter are genuine holomorphic submersions between arbitrary smooth complex manifolds of dimensions m and n. The whole original ideal is coherent. Its actual divisibility exponents and original indexed monomial data are produced without caller neighbourhood-surjectivity, divisibility, exponent or invariant certificates. The original gamma uses its ambient dimension in the first coordinate; the cross-dimensional statement retains the genuine source dimension m, rather than asserting an incorrect dimension-independent gamma equality. Higher invariants and canonical blowup base change remain open.

Lemma 74.1. *At an actual submersion point y, the image of the entire neighbourhood filter is exactly the neighbourhood filter at f(y).*

Proof. The genuine submersion charts internally produce a holomorphic local section s through y, with f composed with s equal to the identity near f(y). Its holomorphy at f(y) gives continuity and hence a map of that neighbourhood filter into the neighbourhood filter at y. Applying f and using the actual local section identity proves the reverse filter inclusion. Continuity of f at y supplies the forward inclusion. Both inclusions give the exact equality; no openness or neighbourhood-image certificate is input. □

Theorem 74.2. *For every original set H, every degree e and every y, actual IsDivNear of the entire pulled-back ideal along the actual preimage of H is equivalent to original IsDivNear along H at f(y). Their original natural divisibility exponents are equal, including the zero fallback when no first failure exists.*

Proof. The already proved genuine submersion order dictionary identifies every actual pulled-back cosupport with its true preimage. The definition of IsDivNear is eventual containment of H in that whole cosupport. Rewrite its neighbourhood filter using the internally proved exact image-filter equality, then use the eventual-map dictionary. This proves the whole predicate equivalence in every degree. Function extensionality gives equality of the degree predicates. Unfold the literal original divExp definition, including its dependent first-failure search and fallback, and rewrite by this equality. No nonzero, hypersurface, positive dimension or bounded-order hypothesis is needed for this exact predicate statement. □

Theorem 74.3. *The entire original indexed exponent function of the pulled-back ideal and actual pulled-back boundary list equals the original indexed exponent function at f(y), at every natural index, with zero beyond the original list length.*

Proof. The actual map of the finite list preserves its length. At an index within that length its literal get is the true preimage of the same original member. Apply the preceding internally produced divExp equality to that actual member. At every remaining index both literal original definitions give zero. Function extensionality yields the whole indexed-function equality. Empty lists, repeated empty members and zero exponents are retained literally. □

Theorem 74.4. *The actual original gamma of the entire pullback in dimension m equals the literal original MonomialInvariant.inv with bound m, the original boundary indices through f(y), the entire original indexed exponent function, and the original mark. In equal dimensions this is exactly the original gamma at f(y).*

Proof. A pulled-back list member contains y exactly when its original member contains f(y); the genuine pure memberIdx map dictionary therefore gives equality of the original finite index sets. Unfold the literal gamma, then rewrite this actual index-set equality and the complete exponent-function equality. The dimension parameter remains m in the literal key. When m equals n the rewritten expression is the original gamma by definition. This does not replace the original invariant or suppress its dimension normalization. □

Theorem 74.5. *The literal original whole markedPullback record has the same exact monomial dictionary with the original ideal, actual original indexed boundary, unchanged original mark and genuine source dimension bound.*

Proof. Unfold only the literal markedPullback ideal, boundary and mark fields. They are the whole genuine ideal pullback, the literal preimage map of the entire original list, and the unchanged original mark. The preceding actual gamma dictionary applies directly. The original carrier field remains its entire actual preimage; no canonicality or admissibility of a pulled-back tower is assumed or claimed. □

75 Actual original exponents on the genuine finite NC separation

Every ideal in this section is the SAME complete original vanishing ideal or its actual whole total or marking-one controlled transform. The finite boundary coefficients are constructed from the REAL recorded highest strata. No local equation, coefficient, factorization or global bound is supplied.

Lemma 75.1 (Whole reduced original NC product dictionary)**.** *The actual flattening chart for any genuine original NC carrier supplies the COMPLETE nearby original vanishing ideal, generated by the product of precisely its actual distinct through original coordinates, each to exponent one. The empty original coordinate set supplies the unit product. The actual ambient order of this complete ideal is exactly the intrinsic original branch count, so every natural order threshold is the corresponding real count threshold.*

Proof. The true flattening and inverse give surjective derivative. Each actual shifted coordinate germ is therefore prime and nonzero. If two such germs are associated, their differentials at zero are scalar multiples. Composing the surjective flattening differential with distinct coordinate projections contradicts that relation; hence distinct original coordinates give nonassociated primes. The product is squarefree: a prime cannot divide two distinct factors, so no prime square divides the product. Its whole zero germ is the actual original carrier, by the genuine NC membership dictionary. The actual analytic vanishing-ideal theorem identifies the entire vanishing germ ideal of a holomorphic equation with the radical of its principal ideal. For a squarefree product that principal ideal is radical: if its product divides a positive power of a germ, each distinct prime divides the germ, and their coprime product divides it. Thus its complete original vanishing ideal germ is exactly the principal ideal of the actual whole product. Translate this genuine germ identity through the actual chart inverse. Coherence of the complete original ideal and actual

holomorphic coordinate product extends the stalk identity to equality at every stalk of a genuine nearby open neighbourhood. The empty product is one, and the same proof covers points outside the carrier and ambient dimension zero. The true coordinate-product order theorem puts this product in the actual point ideal to power the number of through coordinates, and excludes the next power. Complete principal generation makes every ideal member belong to the first power; the product itself is a complete ideal member and excludes the next power. The definition of the whole ideal order therefore gives exactly this coordinate count. The actual chart-overlap count dictionary identifies it with the intrinsic original count at the same point. □

Lemma 75.2 (Whole original and old-boundary ideals on actual joint blowup charts)**.** *For the actual highest centre of original count r at least two, the complete current original ideal pulls back to the complete next strict-original ideal times the genuine new exceptional ideal to exponent r. Every actual old boundary ideal pulls back to its own complete geometric strict ideal, with no additional exceptional factor.*

Proof. Use the actual simultaneous original and old-boundary chart at a true centre point. The centre normal coordinate set S has cardinal r. Every through old boundary is a distinct tangential coordinate, outside S. On each genuine monomial blowup chart with pivot j in S, that original pivot coordinate pulls back to t, every other original coordinate i in S pulls back to t times its actual strict coordinate, and every tangential boundary coordinate pulls back unchanged. The complete original product therefore has exact order r on the whole centre, by the preceding whole-ideal order dictionary. Actual Hadamard division by the r-th centre ideal power and the true controlled factorization give the whole pullback as the exceptional ideal to power r times an actual invertible quotient R. Off the exceptional divisor, the genuine biholomorphic dictionary identifies R with the complete reduced strict-original ideal. The strict original is the closure of its off-exceptional portion, by the true NC strict-closure dictionary. Exact order r at every centre point makes R have an actual stalk member not vanishing on the exceptional hypersurface; otherwise the original ideal would have order at least r+1 there. The genuine reduced-off-smooth-divisor theorem now makes R the COMPLETE reduced strict ideal everywhere, including every exceptional intersection. Thus the whole original pullback identity has exponent r, not just a support equality.

Each actual old boundary generator is its unchanged tangential coordinate. The genuine geometric strict-closure dictionary identifies its full inverse-image hyperplane with its actual strict boundary, including points above the centre, since t can vary through nonzero values while the tangential coordinate remains zero. Its whole reduced ideal is generated by that same tangential coordinate. Old boundary members absent at the original chart point have unit ideals locally, and remain absent near the actual lifted point. Outside the centre the actual blowup is biholomorphic and transports the complete ideal identities. These genuine local charts cover the whole actual blowup, proving the entire ideal equalities globally. □

Theorem 75.3 (Internally recorded exact total and controlled weights)**.** *At every literal prefix of the ACTUAL generated original NC separation, an internally computed finite weight list has precisely the literal current boundary carriers. The whole original total pullback equals the complete current strict-original ideal times their actual ideal powers. The SAME whole controlled original ideal has the analogous complete product with each recorded new exceptional weight r-1 instead of r. Every previously recorded boundary index retains its recorded weight through all subsequent true stages.*

Proof. Compute the lists recursively on the actual recorded tower. Initially both lists are empty. At each actual higher step the entire internally attained original maximum determines its unique integer r. Map every previous carrier by its actual geometric strict transform, retain its actual weight, and append the whole genuine new exceptional carrier with weights r and r-1 respectively. No list or coefficient certificate is input. The first lemma identifies the whole initial ideal with

its complete original product. Induct along the true tower. Pullback commutes with all actual coherent finite ideal products and powers. The preceding joint-chart lemma transports each old boundary ideal without a new exceptional factor and the current original with exactly the new factor of power r. This proves the total product at the next stage. For the actual marking-one controlled ideal the true admissible factorization divides its whole pullback by precisely one new exceptional ideal. Genuine invertible whole-ideal cancellation therefore leaves exponent r-1 while retaining all old weights and the complete next strict-original factor. This proves the controlled product at the next actual stage. Cancellation here is an actual local ideal argument: choose true nondegenerate holomorphic generators l,a,b for the invertible common factor and the two remaining complete ideals. Equality of product ideals gives la=c lb with an actual holomorphic c, and conversely lb=d la. The nonzero locus of l is dense in its genuine neighbourhood, so continuity cancels l in both identities. Thus a=c b and b=d a as actual germs, proving equality of the complete ideals. The genuine boundary map-and-append definition gives the entire literal carrier-list identity, not just an equality of unions. The same induction proves these assertions for every literal prefix of the internally generated tower. □

Theorem 75.4 (Exact intrinsic original exponents at all actual component points)**.** *Every actual original strict local branch has exponent one in the SAME whole original total and controlled ideals. At every point of every actual recorded boundary member, its intrinsic original total exponent is the internally recorded r and its intrinsic original controlled exponent is r-1. This remains true on every actual connected component and at intersections with other members. It does not assume connectedness of an entire recorded member.*

Proof. Use the real simultaneous pair chart at an actual point. Its genuine distinct original coordinates and boundary coordinates give pairwise nonassociated prime germs. The preceding entire finite-product identity is therefore the product of those true primes to exactly their recorded powers. Along any one actual member, the other coordinate factors do not change its prime multiplicity: their prime germs are nonassociated and therefore not divisible by that prime. Actual derivative order on the dense portion where the other factors are nonzero is exactly the displayed exponent. Divisibility by its displayed coordinate power holds on the entire genuine neighbourhood. A strictly higher neighbourhood order along the member would also hold at those dense points, contradicting that exact derivative calculation. These two actual neighbourhood thresholds therefore compute the intrinsic divisor exponent even at intersection points. Thus the intrinsic divisor exponent, defined from the actual original ideal's order near the member, is exactly the recorded coefficient at every member point. Original strict local coordinates have recorded coefficient one. Restricting to a true connected component preserves this whole local hypersurface germ and hence the same intrinsic exponent; no global connectedness or finite global original-branch family is required. □

Theorem 75.5 (Internally bounded noncompact whole original orders)**.** *At every actual prefix, all actual recorded total weights are at most n and all actual controlled weights are at most n-1. At every actual ambient point, the order of the SAME complete original total and controlled ideals is finite and at most n squared, without compactness or a caller order bound. The final strict-original weight is one and the actual boundary length is at most n-1.*

Proof. Each actually performed highest centre is nonempty, so a true centre point and the actual original branch-count bound imply 2 at most r at most n. The internally recorded weights are consequently r at most n and r-1 at most n-1. At every actual point the real simultaneous pair chart uses distinct normal coordinates for the original branches and distinct tangential coordinates for the through boundary members, with these two coordinate sets disjoint. The complete local product has true derivative order equal to the sum of its actual powers: all smaller total derivatives vanish, and differentiating each coordinate its displayed number of times gives a nonzero factorial product. The actual nearby WHOLE ideal identity makes this the order of the SAME true ideal.

For a direct global estimate, there are at most n original coordinate factors, each of weight one, and at most the actual finite boundary length through boundary factors, each of total weight at most n (the controlled weights are smaller). Starting with the empty boundary, the actual procedure appends exactly one literal boundary entry per step. Its genuine length is at most n-1, and the same length bound holds for every literal prefix. Consequently the actual sum is at most n+n(n-1)=n squared in positive dimension. This uses neither compactness nor a supplied finite-value/global-order certificate. In dimension zero the tower is reflexive, both original ideals are unit, and their true order is zero. □

Remark 75.6. These internally computed whole-ideal exponents and bounds are the termination foundation for the actual remaining monomial marking-one resolution. Its finite Cartier division steps are a subsequent construction. No compact monomial driver, compact cosupport, unrelated marked ideal or old ordered colexicographic canonicality is assumed. Full canonical marked-ideal G9 remains OPEN.

76 Literal relative normal associated graded of an original ideal

Let B be the entire tangential holomorphic germ ring, R the original ambient holomorphic germ ring and J the actual ideal of the normal coordinates. Fix an arbitrary original whole ideal $I \subseteq R$. No caller kernel, finite relation, flattening, basis or generic-freeness certificate is an input to the following constructions.

Theorem 76.1 (Actual original normal monomials and literal graded layers)**.** *The actual normal monomial z^α belongs to $J^{|\alpha|}$. The literal degree-q layer is the entire B-submodule $L_q = \operatorname{image}(J^q \longrightarrow R/(I + J^{q+1}))$. It is precisely the quotient of the image of J^q in R/I by the image of J^{q+1}, with the true original representative dictionary. The associated graded module is the actual direct sum $\bigoplus_q L_q$.*

Proof. Each normal coordinate belongs to its genuine generated ideal; its kth power belongs to the kth ideal power. Multiply these actual memberships and use the finite product identity $\prod_i J^{\alpha_i} = J^{\sum_i \alpha_i}$. Use the literal quotient map and submodule image to define L_q. Its representatives are precisely original germs in J^q. Two such germs have the same layer class exactly when their difference lies in $I + J^{q+1}$. Passing first to R/I gives the same dictionary, since a representative in J^q maps to the next power image exactly when its difference from some actual J^{q+1} germ lies in I. The first isomorphism theorem gives the actual layer identification and hence the literal direct sum. This includes $q = 0$ and zero/unit original ideals. □

Theorem 76.2 (Genuine homogeneous normal polynomial surjection)**.** *Evaluate an actual degree-q homogeneous normal polynomial in the original normal coordinates and then take its actual class in L_q. This is a true B-linear surjection onto L_q. Summing these maps after the genuine polynomial homogeneous decomposition gives a true surjection $B[Z_1, \ldots, Z_t] \twoheadrightarrow \bigoplus_q L_q$. Its whole kernel is exactly the condition that the evaluated qth homogeneous part belongs to $I + J^{q+1}$ for every q.*

Proof. Homogeneous polynomial support has degree q, so actual monomial membership proves that its entire evaluation belongs to J^q. Given an original representative $g \in J^q$, use the genuine distinct normal-monomial representative modulo J^{q+1}. Its evaluated polynomial also lies in J^q, because the actual remainder is in $J^{q+1} \subseteq J^q$. Whole polynomial coefficient separation forces every coefficient of degree less than q to vanish as an entire base germ. All its remaining degrees are exactly q, giving the required homogeneous representative. The true finite homogeneous decomposition of a polynomial and the actual universal property of the direct sum construct the full map. Surjectivity on each literal layer implies surjectivity on the true direct sum. Equality to zero in a layer is precisely whole membership in $I + J^{q+1}$, proving the asserted complete kernel condition. □

Theorem 76.3 (True original initial symbols and the complete homogeneous kernel ideal)**.** *For every actual $g \in I \cap J^q$, its true degree-q normal initial symbol is the qth homogeneous part of its actual polynomial representative modulo J^{q+1}. Let H be the ideal generated by all these actual homogeneous symbols. Then H is homogeneous and is the COMPLETE whole kernel of the preceding polynomial surjection.*

Proof. The initial symbol is homogeneous by the actual polynomial projection, and its evaluation differs from g by a true element of J^{q+1}. It therefore maps to zero. Multiplication of homogeneous polynomials corresponds to multiplication of their original evaluated germs: an actual error in $I + J^{q+1}$ times a degree-r representative in J^r lies in $I + J^{q+r+1}$. Decompose every multiplier into its actual finite homogeneous parts to obtain closure of the complete kernel under arbitrary polynomial multiplication. Addition and entire base scalar multiplication are actual linear operations. This proves an actual ideal, and stability under all homogeneous projections proves homogeneity. Conversely, a homogeneous P of degree q in the kernel has actual evaluation $g + h$ with $g \in I$ and $h \in J^{q+1}$. Since its evaluation and h belong to J^q, the entire original g lies in $I \cap J^q$. Whole coefficient separation modulo J^{q+1} identifies P with the actual initial symbol of g. The finite homogeneous decomposition then expresses every complete kernel element in the ideal generated by these actual symbols. Both inclusions are proved with whole germs and actual original ideal memberships. □

Theorem 76.4 (Internally finite actual homogeneous original relations)**.** *An actual finite family of original germs $g_i \in I \cap J^{q_i}$ internally supplies homogeneous relations generating the complete kernel H. Thus the literal original associated graded module has a true finite homogeneous polynomial presentation, valid in all degrees simultaneously.*

Proof. The genuine entire base holomorphic germ ring is Noetherian. The true polynomial ring in the finite actual normal coordinate family is therefore Noetherian by Hilbert's basis theorem. Its actual ideal H is finitely generated. Apply finite extraction to its proved span by all actual original initial symbols. This chooses a finite SUBSET of that actual symbol family, rather than replacing it by caller relations or arbitrary nonhomogeneous polynomial generators. Choose the original germ and degree witness for each of these finitely many actual symbols. The complete kernel equality then identifies the polynomial quotient by these genuine finite homogeneous relations with the literal original direct sum of layers. No generic-freeness or flatness result is inferred from finite presentation alone. □

Remark 76.5. A genuine nonzero common denominator making this entire graded module free, and a true uniform analytic neighbourhood realizing the required original stalks, remain subsequent producer obligations. Separate degree-wise localizations and finite presentation do not settle all-degree normal flatness. Higher HS descent and general G8 remain OPEN.

77 Actual original masked member companion values at positive order

The parent domain-three marked ideal and parent test tower are genuine. Its entire member extension need not be canonical on the whole space. Canonicality below belongs to the actual original masked lower ideal and is internally supplied by the original constructed General3 canonical data. Component orders may be unbounded. Zero-order monomial descent, the original StepOne level-zero contact branch, whole General3 and general global G3 remain open.

Lemma 77.1 (Entire generated member companion ideals give the exact open predicates)**.** *For every actual member prefix companion of positive mark and the real remaining parent tail, its construction on an arbitrary original open set and its actual order mask have the exact traces of the whole companion's current cosupport, original-member count and genuine special predicate.*

Both special disjuncts are included in this exact locality statement. No whole maximal-order, candidate canonicality or candidate testness is an input.

Proof. The genuine full companion/coefficient marked-transform identities give the current ideal, carrier, boundary and mark as the actual open pullbacks. The original member list length is preserved by pullback. Taking that length of the whole current boundary commutes with mapping by inverse image, so the entire original-member lists correspond. The indexed count of a pullback list equals its original count at the image point.

The actual whole current ideal is coherent; its germs are genuinely holomorphic. Open locality of vanishing to the true order identifies the current cosupport with the exact trace. The actual regular-point dimension predicate and entire cosupport locality likewise identify both regularity conditions defining the codimension-one part. For the actual member locus, the open embedding maps each point's neighbourhood filter exactly, and the current original-member lists are the genuine inverse-image lists; its entire local containment condition therefore pulls back in both directions. Apply these identities to the two disjuncts defining the actual special predicate. This transports codimension-one membership without asserting that the whole codimension-one part is globally closed or analytic. The actual order mask is just the corresponding original open set, so the same proof gives its literal compAt dictionary. □

Lemma 77.2 (True entire domain-two companion cosupport has a discrete nonspecial exception)**.** *Every actual whole member prefix companion of positive mark has an analytic zero-dimensional subset of its actual current coefficient cosupport outside which all actual cosupport points are genuinely special. Whole maximal order is unnecessary.*

Proof. The actual controlled-extension producer internally supplies the whole current coefficient ideal's domain two. True coherence and nonzero along give analytic cosupport of dimension at most one. Its genuine singular curve envelope and finite actual original-member predicates yield an analytic zero-dimensional exception. At an actual regular cosupport point outside it, nonzero along excludes current carrier dimension one, and actual dimension monotonicity forces current carrier dimension two. If its original-member count is zero, these real regularities give precisely codimension-one special membership. If the count is positive, each member through the point contains its whole genuine cosupport component, outside the predicate exception. Actual analytic continuation and the smooth transverse SNC trace then give the member-locus germ condition and hence positive-level special membership. These are exactly the actual special disjuncts. The argument only claims membership off the true discrete exception, without global analyticity of the codimension-one part. □

Theorem 77.3 (A true locally finite component-own companion special exception)**.** *For a genuine analytic curve in the endpoint member carrier, there is one analytic zero-dimensional exception giving the already constructed real component-own attained order and count/run history, and such that at each literal head of a component's positive-order companion every actual current coefficient cosupport point on that component outside the exception is special. Each component uses its own actual attained order, not all marks.*

Proof. Start with the actual previous history exception, which gives every real component B its actual minimum endpoint order q and all-order common runs. If q is positive, form the preceding actual discrete companion exception for each of the finitely many literal heads at this q. Intersect each such exception with the actual B; this is analytic and has dimension at most zero by genuine analytic dimension monotonicity. It lies in B. For a zero component order use the empty set. For each head the family of these real intersections is locally finite because the actual global component family is locally finite. Hence its union is analytic and zero-dimensional, even when the q values vary without any global bound. The finite union over heads and union with the previous actual history exception remains analytic, zero-dimensional and contained in the curve. Restricting the old history to the smaller complements preserves it. If a point on B outside this

union lies in the actual coefficient cosupport, it avoids the genuine nonspecial exception for that same B, q and head, and so is special. □

Definition 77.4 (The exact positive masked value dictionary)**.** Let I be the actual remaining member ideal and L its restriction to an original open U; restrict the same true parent tower T to U. The positive masked value dictionary on a genuine curve Z away from E means the following. For every actual global component B of Z and every positive attained minimum q of I at the endpoint, each actual cosupport point of L on B outside E has literal constructed General2 value $(q, \mathrm{inr}(\ell_r, \infty))$, where r is its actual order-q run in the whole T and ℓ_r is the count of the genuine whole member prefix companion of mark q along the real remaining tail. Any two such points have equal entire General2 values and equal comparisons with every General2 threshold. This property is an output; it is never an assumed canonicality, value or history certificate.

Lemma 77.5 (The literal masked positive General2 value is the actual companion count with top)**.** *For the actual original masked member ideal and its actual canonical restricted tail, each component's positive attained whole endpoint order q and true common order run r give its literal constructed General2 value, at every actual masked cosupport point off the discrete exception, as*

$$(q, \mathrm{inr}(\ell_r, \infty)),$$

where the count is exactly the actual whole member-prefix companion's original-member count. Consequently those literal values, and all their threshold comparisons, agree at any two such points of the component. The whole member extension is never assumed canonical.

Proof. The actual entire mask identity and nonmonomial open locality identify the literal masked ideal's endpoint natural order with the whole candidate's true order. The previously proved actual order-run identity gives the same r. The literal split of the restricted original tower is compared with the restriction of the real whole split by its genuine head and tail biholomorphisms, whose final map is the identity on underlying points. Actual coherent whole controlled-transform locality identifies the prefix marked ideals under the head comparison. For the companion coefficient ideal, every genuine prefix is coherent by the whole controlled-extension producer and its actual open restriction identity. Induct along the structural tail comparison: at a blowup, the inverse image of the true centre agrees, and pullback of the actual colon by its exceptional ideal commutes with both biholomorphisms, using the actual coherence before and after that blowup. This gives the entire current marked identity and coherence on the other tail without requiring it to be a test tower of the unmasked companion. The same prefix induction proves open restriction of this coherent tail: at each blowup the genuine inverse-image centre and the actual coherent colon give the full open marked identity. Thus its fresh actual order mask has exactly the trace predicates proved above. Companion, coefficient and original order-mask constructions therefore identify the actual literal companion and coefficient tail with the same original open restriction of the actual whole companion generated at that real head; the genuine tail comparison identifies its current cosupport, count and special predicate.

The actual canonical masked General2 lower data internally supply the literal positive-order companion domain, its maximum order on the true original order mask, its canonical StepOne tail and actual cosupport point. The preceding full predicate dictionary and the actual component-own special exception make that literal companion point genuinely special. The actual StepOne value at a special point is exactly its true count and top, and the literal positive-order General2 definition embeds precisely that value with its genuine q. The head comparison preserves actual original-member lists, so the count is the actual whole companion count. The actual common order run and finite count history give equal counts at two component points, hence equality of the literal values and every threshold comparison. Only the actual original masked ideal is canonical in this argument. □

Theorem 77.6 (Original General3 data produce the actual masked positive lower-value dictionary)**.** *The original domain-three marked ideal and its literal constructed General3 canonical tower, at an actual positive-order companion point which is nonspecial and has positive original level, internally produce the actual member curve, real discrete exception and exact original masked General2 positive-order value dictionary above, together with the full marked equality between the restricted entire member ideal and that original lower ideal. No caller lower domain, masked canonicality, minimum order, history, flattening or whole maximal-order certificate is supplied.*

Proof. The original constructed General3 companion producer gives its true domain, actual maximum order on the original order mask, canonical StepOne tail and actual cosupport dictionary. At the actual nonspecial positive-level point, its genuine own-level run and canonical lower-data producer yield the real parent split, nonempty true member family, and actual original masked lower ideal with its domain two and canonical restricted tail. The original entire member/curve producer supplies the same whole ideal and genuine analytic curve containing the point. Apply the preceding actual component-own special-history and literal value construction to these internally produced data. Their mask and tower are the original lower construction, and every whole companion ideal is identified with the literal constructed one by its full marked-transform dictionary. This gives the claimed actual masked positive lower values and every threshold comparison, without extending canonicality to the whole space. □

Remark 77.7. A zero attained member order is not covered by the positive-value assertion. Its actual monomial value histories remain separate. The original StepOne level-zero contact branch, across-mask lower history gluing, whole General3 analytic thresholds and general global G3 termination remain OPEN. Genuine codimension-one regularity off a discrete exception does not imply that the whole codimension-one part is analytic without maximum order.

78 Whole invertible ideal quotients under genuine submersions

Let $f : Y \to X$ be a genuine holomorphic submersion of arbitrary smooth complex manifolds, with source dimension m and target dimension n. We use the original germ ideals, original colon ideal, original natural powers and literal marked pullback. No openness, quotient, factorization or controlled-transform certificate is supplied. The general higher-invariant and canonical blowup base-change goals remain open.

Lemma 78.1. *Suppose the original function h vanishes identically on no nonempty open subset of $U \subseteq X$. Then $h \circ f$ vanishes identically on no nonempty open subset of $f^{-1}U$.*

Proof. Take a nonempty open $W \subseteq f^{-1}U$ and an actual point $w \in W$. The internally proved equality of entire image neighbourhood filters says that $f(W)$ is a neighbourhood of $f(w)$: its preimage contains the genuine neighbourhood W. Choose an open neighbourhood V of $f(w)$ contained in this actual image. Then $V \subseteq U$. The original nonvanishing condition supplies $z \in V$ with $h(z) \neq 0$. Since $z \in f(W)$, take its actual preimage $y \in W$. This gives $(h \circ f)(y) \neq 0$. No image-openness hypothesis or nonvanishing witness is an input. □

Theorem 78.2. *The whole pullback of an original invertible germ ideal L is invertible, in the genuine source dimension.*

Proof. At $f(y)$ choose the actual open set U and nonzero local holomorphic generator h from the original invertibility definition. The genuine submersion is holomorphic and continuous. Therefore $f^{-1}U$ is an open neighbourhood of y, and $h \circ f$ is holomorphic there. The preceding lemma proves its required nonvanishing on every nonempty open subset. An arbitrary germ in the whole pullback is a finite holomorphic combination of pulled-back original ideal germs. Express each original germ as a multiple of h and compose the actual span expression with f. Transitivity of

actual holomorphic spans shows that the entire pullback is generated by $h \circ f$. Conversely h is in the original ideal, so its actual pullback generates germs of the pullback ideal. These prove every clause of original invertibility. □

Theorem 78.3. *For an original coherent ideal I contained in an original invertible ideal L, the entire original colon ideal obeys $f^*(I : L) = (f^*I : f^*L)$.*

Proof. The original coherent colon theorem internally produces coherence of $Q = (I : L)$. Its two genuine inclusions give the full original ideal equality $I = LQ$. Invertibility also gives coherence of L. The actual whole product pullback theorem therefore gives $f^*I = (f^*L)(f^*Q)$. The actual pullback of Q is coherent, and the preceding theorem makes f^*L invertible. Cancellation of this genuine invertible local generator in the original colon definition gives $((f^*L)(f^*Q) : f^*L) = f^*Q$. Combining these equalities proves the whole colon dictionary. The proof never assumes a caller-produced factorization or quotient. □

Theorem 78.4. *For every natural q, if the original coherent ideal I is contained in L^q, then $f^*(I : L^q) = (f^*I : (f^*L)^q)$. In particular the power zero and empty-divisor cases use the actual original unit ideal.*

Proof. The original invertible power construction makes L^q invertible for every natural q. Apply the preceding full colon theorem to this actual power and the given original inclusion. The whole original coherent-power pullback dictionary identifies $f^*(L^q)$ with $(f^*L)^q$. Rewrite that genuine denominator ideal. The power-zero definition and actual unit pullback are included in the same argument. □

Theorem 78.5. *Replacing an original marked ideal's ideal by its actual colon quotient commutes with the literal entire marked pullback. Its original support, literal ordered boundary list and original mark are retained. For every natural order threshold, the ambient cosupport of the whole pulled-back colon ideal is precisely the preimage of the original ambient cosupport.*

Proof. The ideal field is exactly the internally proved whole colon equality. The other three original marked fields are unchanged by the quotient, and their actual pullback definitions give the same preimage support, mapped ordered list and original mark on both sides. Destruct the original record to obtain the complete record equality. For the cosupport statement the original quotient is coherent by the genuine coherent-colon theorem. Apply the actual submersion whole-order cosupport dictionary to that entire quotient, then rewrite its pullback by the colon theorem. This proves the statement for every natural threshold, including zero, without an assumed order or cosupport certificate. □

79 Internally produced entire standard normal monomial bases over the true fraction field

Let K be a field and $P = K[Z_1, \ldots, Z_t]$, with an actual well-founded monomial order. Every finite family below is internally produced from the whole polynomial ideal. No caller Groebner family, leading-cover, relation, basis or freeness certificate is supplied. The original analytic consumer uses the genuine fraction field of the entire tangential germ ring and the actual scalar extension of its internally proved whole original initial ideal.

Theorem 79.1 (Internal finite actual leading-monomial generators)**.** *Every entire polynomial ideal $H \subseteq P$ internally supplies a finite family G of ACTUAL nonzero polynomials belonging to H such that the leading exponent of every actual nonzero $f \in H$ dominates the leading exponent of one actual $g \in G$ coordinatewise.*

Proof. Form the actual monomial ideal generated by the monic monomials of the leading exponents of ALL actual nonzero original polynomials in H. The true finite-variable polynomial ring over the field is Noetherian. Finite SUBSET extraction therefore selects finitely many of those ACTUAL monomial generators spanning the same whole leading ideal. Choose for each the ACTUAL nonzero original polynomial in H witnessing its selected leading exponent, and form their actual finite image G. The monic leading monomial of every actual nonzero $f \in H$ lies in this same whole leading ideal. The genuine monomial-ideal membership theorem says that each support exponent of that monomial dominates an actual selected leading exponent. Its actual singleton support gives the claimed whole leading coverage. Zero and unit ideals are literal cases; no nonzero ideal hypothesis is used. □

Theorem 79.2 (Genuine finite polynomial division and complete ideal reduction)**.** *The internally produced finite G gives actual division of EVERY entire polynomial into a genuine whole G combination plus a remainder supported on precisely the standard exponents α not dominating any leading exponent of G. A whole ideal polynomial has zero remainder. In particular, H is the actual whole ideal span of this internally produced finite family.*

Proof. Each ACTUAL nonzero $g \in G$ has a genuine nonzero leading coefficient; over the true field this coefficient is a unit. The actual well-founded monomial division theorem constructs a finite polynomial coefficient combination and a remainder whose EVERY actual support exponent avoids each selected leading exponent. The combination lies in the actual ideal H, since all actual generators do. If the input lies in H, so does the whole remainder. A nonzero remainder has its true leading exponent in its actual support. Internal leading coverage would make it dominate one selected leading exponent, contradicting the actual remainder support property. Hence its entire polynomial is zero. This proves both the complete whole reduction statement and equality of actual generated ideals. □

Theorem 79.3 (The entire standard ORIGINAL monomial quotient basis)**.** *The ORIGINAL monomial classes indexed by ALL actual standard exponents form a genuine entire K-basis of P/H, including its actual infinite index family when it is infinite. This actual basis is internally produced from H, and hence proves actual whole quotient freeness with no caller basis.*

Proof. Map a true finitely supported coefficient function on the ENTIRE actual standard exponent set to its original monomial polynomial and then to its actual quotient class. Whole division proves surjectivity: the genuine remainder is supported on that true set and gives its actual finite coefficient function. For injectivity, a polynomial supported on standard exponents and lying in H must vanish: if it were nonzero its genuine leading exponent would be an actual support exponent, and actual internal leading coverage contradicts standardness. The actual coefficient embedding by the original exponent inclusion is injective, so the entire coefficient function is zero. Thus this actual monomial combination is a linear equivalence. Transport the genuine finitely supported single-vector basis through it; each vector is the literal ORIGINAL monomial quotient class. This proves the actual whole basis, including zero-dimensional variables, zero/unit ideals and empty or infinite standard exponent sets. □

Theorem 79.4 (Actual original normal fraction-field standard basis consumer)**.** *For an arbitrary WHOLE original holomorphic ideal I, its actual internally produced entire homogeneous normal initial ideal, extended through the GENUINE entire base fraction-field map, internally supplies the actual finite leading generators and the entire ORIGINAL normal standard monomial basis of that genuine polynomial quotient.*

Proof. The true entire tangential holomorphic germ ring is a domain and has its genuine fraction field. Map the SAME actual whole original initial ideal through the actual coefficient extension homomorphism. Apply the preceding internal finite leading generator and whole original standard basis producers to this actual ideal, using the actual well-founded degree-lex monomial order on

the finite actual normal coordinate family. No kernel, relation, Groebner or vector-space basis is supplied by a caller. This is actual FRACTION-FIELD freeness. Constructing a nonzero COMMON original base denominator with a true localized whole basis, and actual uniform analytic spread to original nearby stalks, remain subsequent obligations. □

Remark 79.5. A true fraction-field standard basis does not by itself prove uniform localized generic freeness or normal flatness. Clearing actual finite membership coefficients and actual leading coefficients by one genuine nonzero original base denominator, with literal whole quotient transport, and uniform analytic spread still require real producers. General G8 and higher HS descent remain OPEN.

80 One genuine original base denominator for the entire normal polynomial quotient

Let B be an actual domain, $K = \mathrm{Frac}(B)$ its genuine fraction field, and $H \subset B[Z_1, \ldots, Z_t]$ an arbitrary whole ideal. Fix an actual well-founded monomial order. Every family and denominator below is internally produced from H; no leading-family, Groebner, basis, freeness or localization result certificate is supplied by a caller.

Theorem 80.1 (Actual original lifts of the internally produced field leading family)**.** *The actual fraction-field extension H_K internally supplies finitely many nonzero original whole polynomials $h_i \in H$ whose leading exponents are exactly those of its internally produced actual field leading family. They therefore give complete leading coverage of every actual nonzero member of H_K after coefficient extension.*

Proof. The genuine polynomial ring $K[Z]$ is the localization of $B[Z]$ at the image under the constant-polynomial map of the ACTUAL nonzero elements of B. Every actual selected field generator $g_i \in H_K$ therefore has a genuine original ideal numerator $h_i \in H$ and an actual nonzero base denominator $s_i \in B$, with $g_i C(s_i) = h_i$ after the actual coefficient map. A nonzero constant multiplier over the field preserves the entire actual support and leading exponent. The original coefficient map is injective, so it preserves support and leading exponents as well. Thus each actual h_i is nonzero and has precisely the leading exponent of g_i. Choose these actual membership witnesses internally for the finite field family. The original zero and unit ideal cases are included literally; an empty selected family is allowed. □

Theorem 80.2 (A single genuine nonzero original leading coefficient denominator)**.** *There is an internally produced nonzero $b \in B$ such that every actual lifted leading coefficient is a genuine unit in the actual localization B_b, and the actual coefficient map $B_b \longrightarrow K$ is injective. The same b is used for the entire polynomial quotient, without dependence on a degree or truncation parameter.*

Proof. Form the actual finite product b of the leading coefficients of the actual nonzero original lifts. Each factor is nonzero in the actual domain B, so the true product is nonzero; the empty product is 1. Each factor divides b, and the image of b is a unit in the ACTUAL localization away from b. Hence every factor becomes a genuine unit there. The actual map into K is constructed by the universal localization lift of the original fraction map, since the actual nonzero image of b is a unit in the true field. A localized element mapping to zero can be represented by an original numerator divided by a power of b. Its original numerator maps to zero in K, and injectivity of the original domain fraction map makes it zero. The actual localized element is therefore zero. This proves true map injectivity and actual support preservation of coefficient extension. □

Theorem 80.3 (The entire actual localized original standard monomial quotient basis)**.** *For this SAME actual internally produced b, the ORIGINAL monomial classes indexed by the entire field-standard exponent family form a genuine B_b-basis of $B_b[Z]/H_b$, where H_b is the actual coefficient*

extension of the SAME original ideal H. In particular this actual whole polynomial quotient is free over B_b.

Proof. Map the actual original lifts into $B_b[Z]$. Their leading coefficients are actual units, so genuine well-founded polynomial division expresses every whole polynomial as a finite combination of these actual whole H_b members plus a remainder. Preservation of each actual leading exponent makes EVERY actual remainder support exponent field-standard. Hence the actual standard monomial coefficient map onto the whole quotient is surjective. For injectivity, a standard-supported polynomial lying in H_b maps under the actual injective coefficient map to a standard-supported polynomial in H_K: extension of the original ideal commutes with the actual composite coefficient maps. The already proved whole field standard-kernel separation makes that entire field polynomial zero. Injectivity of coefficient extension makes the original localized polynomial zero, and the actual coefficient embedding makes its entire finitely supported coefficient function zero. Thus the actual standard coefficient map is a linear equivalence. Transport the genuine single-vector coefficient basis through it. Each vector is its LITERAL original monomial quotient class; the entire, possibly infinite index set is retained. □

Theorem 80.4 (Actual original holomorphic normal ideal common-denominator consumer)**.** *For every WHOLE original holomorphic ideal I, its SAME internally proved whole normal initial ideal has an internally produced genuine nonzero entire tangential base germ b for which its whole coefficient-localized normal polynomial quotient has the actual original standard monomial basis.*

Proof. Take B to be the genuine entire tangential holomorphic germ domain and H to be the actual internally constructed whole original normal initial ideal. Its original ideal is used literally, and its fraction extension is the same one used in the earlier field-standard producer. Apply the three actual producers above with the actual finite normal coordinate family and the actual well-founded degree-lex order. All witnesses are internally obtained from this whole original ideal. The nonzero b is an entire base germ; no point-value nonvanishing hypothesis or caller flatness certificate is used. □

Remark 80.5. This chapter proves a genuine SAME-denominator basis of the WHOLE coefficient- localized polynomial quotient. A literal identification with the localization of the actual original associated module, a degree-compatible localized basis of EACH actual original graded piece, freeness of ALL actual original $R/(I + J^q)$ truncations on that same localization, and uniform analytic spread to original nearby stalks are distinct subsequent producer obligations. Whole ungraded freeness is never treated as all-degree graded freeness. General G8, uniform analytic normal flatness and higher HS descent remain OPEN.

81 Actual zero-order masked member monomial values

The whole remaining-member datum is the genuine controlled extension of the original data; its testness or canonicality is not assumed. Its actual current ideal, smooth carrier and finite current boundary are produced from the original domain-three datum and genuine parent test tower. Component orders and boundary exponents may be unbounded. The actual General2 value at order zero reads this same current ideal's intrinsic monomial invariant. The original StepOne level-zero contact branch and across-mask General3 history gluing remain separate open problems.

Lemma 81.1 (True component connectedness and actual along-exponent constancy)**.** *Every actual global analytic component is preconnected. If a genuine analytic curve lies in the smooth carrier of an actual well-formed ideal nonzero along that carrier, then on any actual curve component contained in an actual current boundary member its intrinsic divisor exponent along the smooth carrier is constant. No connectedness or constant-exponent certificate is supplied by the caller.*

Proof. By its literal definition a global component is the closure of the connected component of the actual regular-point set through a genuine regular point. Connected components are preconnected; closure preserves preconnectedness. This proves preconnectedness of the whole actual component, including its singular points. Its containment in the genuine curve places it in the real smooth carrier. If it is also contained in one actual boundary member, every point lies in the real trace of that member on the carrier. The genuine coherent along-exponent theorem applies there: the true SNC and transversality supply regularity of that trace, the true smooth carrier supplies its local retraction, and true nonzero along supplies finite intrinsic exponents. Consequently the actual natural-valued exponent is locally constant on the entire component. Restrict this function to the component. Its local constant neighbourhoods make it continuous into the discrete space of natural numbers. The actual preconnected component has a preconnected image in this discrete space, which contains at most one point. Thus any two component points have the same actual exponent. This uses neither finiteness of the exponent range nor a bound nor connectivity of the complement of a discrete exception. □

Theorem 81.2 (Finite genuine boundary incidence produces a monomial history exception)**.** *For any actual well-formed ideal nonzero along its true carrier, and a genuine analytic curve of dimension at most one contained in that carrier, there is an analytic zero-dimensional subset of the curve outside which the entire actual monomial invariant is constant on each genuine global curve component. All current boundary entries and intrinsic along-exponents are the actual ones of this same ideal.*

Proof. Each actual current boundary member is a closed smooth hypersurface and hence analytic. Apply the true finite analytic-predicate exception producer to the finite indexed current list. For every component B it removes precisely the proper intersections with members that do not contain B. Analytic continuation and dimension drop show each such intersection is analytic of dimension at most zero; true global component local finiteness makes the union analytic and zero-dimensional. After the finite union over the actual current indices, all points of B off this genuine exception have exactly the same indexed member set: the members containing B. For each such index, the preceding lemma makes its intrinsic along-exponent constant on the whole B. Indices outside the common member set have no effect on any actual feasible subset or its key. Thus the true finite feasible subsets and their actual weighted keys are the same at two points, so their actual maximum defining the entire monomial invariant is the same, including the bottom case. Empty or repeated entries are treated by the true indexed list and its predicates; no replacement boundary or hidden distinctness assumption is introduced. □

Definition 81.3 (The actual zero-order masked dictionary)**.** For the real remaining-member datum I, same parent tower T and original open U, let L be its actual open restriction. The zero-order masked dictionary on a true curve Z away from E means: for each actual global component B of Z on which zero is an attained endpoint nonmonomial order, every actual masked cosupport point on B outside E has literal constructed General2 value

$$(0, \mathrm{inl}(\gamma_T)),$$

where the entire γ_T is the intrinsic monomial invariant of the SAME whole current member datum at that underlying endpoint point. Any two such masked points have equal literal entire values and equal comparisons with every General2 threshold. This is an output dictionary, never an input ideal, coefficient, canonicality or finite-range certificate.

Theorem 81.4 (Actual member data internally produce the zero-order literal values)**.** *The original domain-three parent, nonempty actual member selection, genuine parent test tower, any original open U and a genuine endpoint curve of dimension at most one in the actual endpoint member carrier internally produce one analytic dimension-zero history exception and the exact zero-order masked dictionary. The whole member candidate and its restricted tower are not assumed test or canonical.*

Proof. The true controlled member-extension producer gives the SAME whole endpoint member ideal its actual domain two, hence true coherence, smooth carrier, actual SNC boundary and transversality, and nonzero along the carrier. Apply the preceding actual monomial-history producer to this true endpoint datum. Independently the genuine earlier order-history producer gives a dimension-zero exception and each real curve component its actual attained minimum natural endpoint order, constant off that exception. Union the two actual exceptions. Their true analytic dimension bounds and containment in the original curve are preserved. Restricting the earlier order/count/run history to the smaller complements preserves all those actual histories.

If zero is attained on a component, its produced minimum must equal zero: it is nonnegative and at most that actual attained value. Thus the true whole nonmonomial endpoint order is zero at every component point off the combined exception. The entire actual open marked-transform identity and genuine nonmonomial open locality give order zero for the literal masked current ideal at the same point. Its actual constructed General2 zero-order definition is exactly the zero first entry and the monomial alternative. Actual monomial-invariant pullback along the open inclusion identifies this entire masked monomial value with the intrinsic invariant of the same whole current ideal. This uses true current coherence and the entire marked identity, not candidate-testness or candidate-canonicality. The actual whole monomial-history constancy gives equality at two component points, and therefore equality of all threshold comparisons of their literal General2 values. □

Theorem 81.5 (Original General3 data produce both literal member lower alternatives)**.** *At an actual nonspecial positive-level companion point of the original constructed General3 canonical tower, the original domain-three datum internally produces its true member curve and one analytic zero-dimensional exception on that curve. On every actual component it gives both the exact positive-order dictionary already proved and the exact zero-order dictionary above, together with the FULL equality of the restricted whole remaining member datum and the original literal lower marked ideal. Both dictionaries use the SAME true curve, mask, ideal and restricted parent tail.*

Proof. Start with the genuine original positive-value consumer. It internally produces the actual companion, own-level split, selected nonempty original member family, same whole member ideal and original mask, true analytic curve, positive-value exception and full initial marked equality with the literal lower ideal. The original companion-domain producer and actual canonical StepOne bridge internally provide the true parent test tower, its head and tail, and the actual transformed parent domain three. Apply the preceding zero-order producer on this SAME curve, mask and actual parent tail. Union its actual exception with the positive-value exception. Both dictionaries and the previously produced actual histories restrict to this smaller complement, since all their point hypotheses are on complements. Return these two exact literal dictionaries and the unchanged full marked equality. Each component's genuine attained minimum is a natural number, so it is zero or positive; no finite spectrum or global maximum is required for these two alternatives. This identifies the actual member lower General2 alternatives on the true curve away from a genuine discrete exception. It does not fill across different original StepOne levels or the level-zero contact branch, and therefore does not assert whole General3 superlevel analyticity or noncompact global termination. □

Remark 81.6. All boundary indices, coefficients, ideals, masks and towers above are actual constructed data. No global connectedness of a boundary member, finite coefficient range, compactness or caller canonical candidate is assumed. The original StepOne level-zero contact branch, across-mask history gluing, whole General3 analytic superlevels and general global G3 remain OPEN.

82 Actual finite marking-one completion for the original NC ideal

The original input is a genuine closed NC carrier H in an arbitrary complex manifold of dimension n. Its original marked datum is the SAME complete vanishing ideal with whole ambient, empty original boundary and mark one. The genuine NC separation and its historical r-1 weights are produced by the preceding section. No tower, coefficient, order, admissibility, compactness or termination certificate is an input to the original consumer. The following Cartier steps are actual blowups, including literal empty centres.

Lemma 82.1 (Genuine Cartier blowups and literal indexed strict boundary transport)**.** *For an actual member C of a genuine finite SNC boundary, its complete vanishing ideal is invertible and the identity projection satisfies the TRUE blowup universal property along C. The whole ambient has geometric strict transform the whole ambient. The selected member has literal geometric strict transform empty; every different literal original SNC index retains its whole carrier. The actual marked boundary is the entire strict-list image followed by the actual new exceptional carrier C, with all empty historical entries retained. In dimension zero every actual SNC member is empty and the same statements hold by the genuine empty blowup.*

Proof. At a true member point, the SNC singleton coordinate dictionary and a genuine coordinate permutation flatten the member to one coordinate hyperplane. Actual single-coordinate Hadamard division identifies its COMPLETE vanishing ideal with its nondegenerate coordinate generator. Off the closed member its ideal is unit. This also proves its empty interior and genuine smooth closed hypersurface structure. The actual identity pullback of that complete ideal is itself invertible. Given any holomorphic map with invertible centre ideal, its unique lift through the identity is that very map; both its holomorphicity and projection equality are literal. Thus the identity is a genuine blowup in the universal-property definition, not a surrogate isomorphism.

The true geometric strict transform is the closure of the actual off-centre inverse image. For the centre itself this off-centre set is empty. For the whole ambient its closure is whole ambient because C has empty interior. For different literal SNC indices A and C, every point of A admits the SAME simultaneous true coordinates. If the point is outside C, it already lies in A minus C. At a point in both, the actual coordinate singleton injection makes their two coordinates distinct. Hold the A coordinate zero and perturb the C coordinate through nonzero values within the genuine coordinate image. The true local inverse transports these points into A minus C, tending to the original point. Hence all of A is in its off-C closure. The reverse inclusion follows from true closedness of A. This includes empty members without any nonempty argument. The literal map-and-append marked transform now gives exactly the stated entire list; no empty entry is removed. If n=0, an actual through singleton would have cardinal one inside Fin 0, impossible. All members are empty, their ideals unit, and the actual empty-centre universal property proves the same identities. □

Lemma 82.2 (Actual marking-one Cartier control divides precisely one whole ideal factor)**.** *Let the true current whole ideal be the COMPLETE product L to positive power k times a genuine invertible whole ideal J, where L is the actual selected SNC hypersurface vanishing ideal. Its genuine marking-one Cartier blowup is admissible and its SAME whole controlled ideal is L to power k-1 times J. The mark remains one, its carrier remains whole ambient, and its literal new boundary is genuinely SNC. No ideal quotient, cosupport or admissibility certificate is supplied to the actual original consumer.*

Proof. The actual entire product inclusion in L follows from k positive and genuine coherent ideal multiplication. Every ideal member therefore vanishes on the selected actual centre, so has actual point order at least one there. This proves the true centre is contained in the SAME marked cosupport. Its actual smooth closed hypersurface structure and genuine common SNC

coordinates give all remaining admissibility fields. The genuine controlled-transform factorization gives whole identity pullback I equals L times the actual mark-one controlled ideal. The actual exceptional ideal on this identity blowup is the COMPLETE vanishing ideal of C, namely L. Meanwhile genuine coherent powers and products give I equals L times L to power k-1 times J. Both remaining ideals and L are genuinely invertible: actual controlled invertibility follows from the true centre order inclusion, and the explicit finite product is invertible by its actual local nondegenerate generators. The preceding real local cancellation lemma therefore identifies the COMPLETE controlled ideal with that exact product. This divides one whole factor and leaves every other complete factor unchanged. Genuine well-formed marked transform gives its actual SNC boundary, whole carrier and unchanged mark. For C empty the complete L is unit, so both identities still hold literally; the real empty Cartier step appends an empty exceptional boundary entry. □

Lemma 82.3 (Every actual unprocessed component remains available in the literal history)**.** *After any actually performed Cartier step along a present boundary member, every whole carrier of the actual finite original component pool remains a member of the actual transformed boundary. Repeating a real Cartier centre replaces its previous literal carrier by empty and appends the SAME carrier as its actual new exceptional member. All other actual pool members retain their entire carriers. Every intermediate literal boundary is SNC.*

Proof. Take a real pool member A occurring in the actual current list. If A equals the selected actual centre C, the new appended exceptional member is A, so availability is retained even when its previous entries become empty. If A is different, choose its actual list index and a true centre index. Their indices must be different, since their carriers differ. The preceding true indexed strict dictionary retains its whole A in the image list. Actual SNC injection forbids two nonempty repeated copies from surviving simultaneously; empty copies may remain and are retained literally. Genuine admissibility and actual blowup well-formedness give the entire next list SNC, so the same argument applies again at the NEXT actual stage. It requires no boundary compression, permutation, globally connected component or finite global original branch family. Literal empty centres simply append another empty entry and preserve every other actual closed carrier. □

Theorem 82.4 (The genuine original terminal Cartier step removes its whole reduced strict factor)**.** *At the real terminal NC separation, perform the genuine Cartier blowup along the ENTIRE actual terminal strict original Hfinal. Its SAME marking-one controlled ideal becomes exactly the whole product of the actual historical r-1 boundary powers. All those actual whole historical carriers occur in the literal new SNC boundary. This step is admissible, including Hfinal empty. The SAME recorded actual weight list has precisely the current original boundary carriers, length precisely the genuine separation length, and each coefficient is between one and n-1.*

Proof. The genuine internally generated separation has terminal original count at most one and the TRUE simultaneous original/accumulated-boundary dictionary. Its entire strict original is therefore an actual closed smooth hypersurface (empty allowed), transverse to the entire actual accumulated boundary. The true augmented list consisting of that boundary followed by ENTIRE Hfinal is SNC. The preceding actual original exponent producer identifies the SAME complete current controlled ideal with the historical boundary product times the COMPLETE reduced ideal of Hfinal to power one. This whole identity, invertibility, actual augmented coordinates and the second lemma internally prove genuine marked admissibility of Hfinal and remove precisely its whole factor. The mark-one ideal is now the complete historical boundary product. The actual identity blowup erases Hfinal as a geometric strict carrier; this is a Cartier completion step AFTER separation stops, and is not claimed to preserve a count-one original differential strict carrier. All earlier boundary indices and the newly appended Hfinal index are different literal indices in the true augmented SNC list. The first lemma therefore keeps every historical

whole carrier in the actual transformed boundary, even if an empty historical carrier equals empty Hfinal. The true new exceptional member is ENTIRE Hfinal. No comparison with an unrelated marked ideal is made. □

Theorem 82.5 (Internally terminating actual finite Cartier division of the complete component product)**.** *A true finite list of actual present SNC carriers with natural powers defines its entire component-power ideal internally. Its whole-ambient marking-one datum admits an internally constructed actual Cartier test tower of length exactly the sum of those powers, after which its SAME complete controlled ideal is unit. Every real intermediate boundary is literal and SNC. Zero powers add zero steps; positive powers on empty carriers add their genuine empty steps and retain the corresponding literal history.*

Proof. The actual complete component ideal is defined as the coherent finite product of those whole vanishing ideals to the supplied natural powers. Actual SNC smoothness internally makes every such ideal invertible, including empty carriers with unit ideals. Their powers and full product are therefore actually invertible and coherent, and genuinely nonzero along the entire ambient. Whole-ambient smoothness, true SNC and transverse full ambient give all actual marking-one domain fields.

Induct structurally on the ACTUAL finite list. Its empty product is unit and the genuine reflexive tower has length zero. For an actual head carrier C of power k and its true complete tail ideal J, induct on the natural k. At k=0 the exact complete product is J and no step is added. At k+1 the true current ideal is L to power k+1 times J. Actual component availability and the second lemma give the genuine admissible identity Cartier blowup whose SAME ideal is L to power k times J. The third lemma retains every needed actual carrier in its new literal SNC boundary. Apply the induction on k to the TRUE new marked datum, then apply the finite-list induction to the actual tail. Genuine blowup-tower concatenation and true marked-transform composition give the SAME entire controlled ideal at every stage and actual admissibility of every performed centre. Length is k plus the actual tail sum. All induction parameters are actual finite lists and natural exponents; neither a caller termination bound nor a candidate resolution certificate occurs. At the end all genuine powers have been divided, giving the COMPLETE unit ideal. □

Theorem 82.6 (Genuine finite marked resolution of the same original NC ideal without compactness)**.** *Only the genuine original NC carrier H internally constructs a finite actual test tower resolving its SAME whole original vanishing ideal with marking one. Its final complete controlled ideal is unit and its actual cosupport empty. The Cartier tail has length at most 1+(n-1) squared; the COMPLETE tower, including its real preceding separation, has length at most 1+n(n-1). Every literal prefix lies in the actual marked domain and has genuine SNC boundary. No compactness or caller global-order/termination certificate is used. The entire completion projection equals the genuine separation projection, preserving the real original-complement isomorphism. In dimension zero the COMPLETE tower has exactly one step and its literal final boundary is [empty], rather than an empty list. No old ordered/colex canonical driver identification is claimed.*

Proof. Consume the ACTUAL internally generated separation and its proven original marking-one test-tower property. Its whole original controlled factorization and actual history list come from the preceding original-exponent producer, not hypotheses. The fourth lemma gives one REAL terminal original Cartier step. Its actual marked datum has precisely the full historical product. The fifth lemma, applied to that internally computed weight list and actual literal carrier coverage, constructs the remaining TRUE finite Cartier test tower and complete unit endpoint. Genuine concatenation makes it a test tower for the SAME original datum, not a reinitialization or replacement ideal. The mark remains one and its carrier whole ambient. The whole unit ideal has actual order zero everywhere, so its genuine marked cosupport is empty.

Every actually recorded higher centre has 2 at most r at most n, internally from its attained original count and real nonempty centre. Every controlled historical weight r-1 is thus at most

n-1. The actual list length equals the real separation length and is at most n-1. Its actual sum is at most (n-1) squared, so the actual tail length is at most 1+(n-1) squared. Adding the genuine separation length gives at most (n-1)+1+(n-1) squared, which equals 1+n(n-1) in positive dimension. For n=0 there are no actual higher centres, the original carrier is empty and all historical lists are empty. The REAL one empty Cartier step has unit endpoint and length one, so the same full bound holds. Empty original and disconnected or noncompact ambient require no finite-value spectrum, compact cosupport or common compact neighbourhood: the actual natural recursion terminates by its explicit internally bounded finite historical sum. Actual original-domain preservation along the produced true test tower proves every literal prefix has all marked-domain fields and therefore actual SNC boundary. Literal empty or repeated history is never suppressed. □

Remark 82.7. This produces an elementary actual marking-one resolution for the SAME original NC vanishing ideal. It does not identify this tower with the old ordered/colexicographic invariant driver, claim naturality under forgetting that driver's labels, or prove admissibility for an arbitrary unrelated ideal. These canonical ordered and arbitrary-ideal obligations remain OPEN.

83 Actual monomial ordering under arbitrary dimensional smooth pullback

For a genuine holomorphic submersion $f : Y^m \to X^n$, the existing whole divisor dictionary identifies every original index and exponent. Its monomial key still uses the genuine ambient dimension: a key with support K starts with $n - |K|$ on X and $m - |K|$ on Y. We prove that this dimension change preserves and reflects the complete original ordering. All index cardinality bounds are produced from the actual SNC coordinates. No equality of keys in different dimensions is asserted. Higher invariants, canonical blowup base change and complete smooth functoriality remain open.

Lemma 83.1 (Actual submersion dimension inequality)**.** *A genuine submersion at an actual point of Y^m internally gives $n \leq m$.*

Proof. The actual submersion coordinates give a continuous complex linear isomorphism $L : \mathbb{C}^m \simeq \mathbb{C}^n \times F$. Compose its linear map with first projection. For every v, $L^{-1}(v, 0)$ is a genuine preimage, so this actual linear map is surjective. The finite-rank inequality for this map gives $\dim_{\mathbb{C}} \mathbb{C}^n \leq \dim_{\mathbb{C}} \mathbb{C}^m$. Their genuine ranks are n and m. There is no ambient nonemptiness assumption: the actual point supplies it. □

Lemma 83.2 (Entire original key ordering)**.** *If $|K|, |L| \leq n \leq m$, then for arbitrary actual exponent functions a, b the strict and weak comparisons of original keys of K, L agree in dimensions n and m.*

Proof. Write the complete original lexicographic key as $(n - |K|, \sum_K a, K)$, with the last entry colexicographically ordered. Because both cardinalities are at most n, subtraction does not truncate. Thus the strict comparison, and equality, of the first coordinates agree after replacing n by m. The actual exponent sums and colex entries remain the same. Expanding the two original lexicographic comparisons proves strict equivalence. Weak comparison is the negation of the reverse strict comparison in the actual linear order, which proves the weak case. □

Lemma 83.3 (Same genuine top subset in both dimensions)**.** *For a genuine finite member set S with $|S| \leq n \leq m$ and $d \leq \sum_S a$, one actual feasible subset attains the original invariant in both dimensions and dominates every actual feasible key in both.*

Proof. Take the actual maximizing subset of the original finite feasible family in dimension n. Every feasible subset is contained in S, so the genuine cardinality bounds needed by the preceding

lemma hold. The whole family of key inequalities therefore transports to dimension m. Taking the actual maximum of its key image shows that the same subset attains the dimension-m invariant. Feasibility, sums and colex tie breaking are unchanged, and original key injectivity makes the subset unique in either dimension. No maximizing subset is supplied by a caller. □

Lemma 83.4 (Whole invariant ordering, including the bottom value)**.** *For two arbitrary real member sets of cardinality at most $n \leq m$, with arbitrary exponent functions and marks, the entire original invariants have the same weak comparisons and equality in both dimensions.*

Proof. If either feasible family is empty, the original invariant is literally bottom, in both dimensions. Empty families do not depend on dimension; the corresponding bottom/nonbottom comparisons therefore agree. If both are nonempty, internally take their actual common top subsets from the preceding lemma. Rewrite the actual maxima as their keys and use the whole key-order equivalence. Equality follows by the two weak comparisons. Zero marks, zero dimensions, empty member sets and bottom values are included in the same argument. □

Lemma 83.5 (Actual original gamma ordering under smooth pullback)**.** *For an actual coherent ideal and actual SNC boundary on X, the original gamma values of its genuine pullback at any two actual points of Y have exactly the original gamma ordering and equality at their images.*

Proof. Use the actual submersion point to produce $n \leq m$. The genuine SNC coordinate family injects the actual member indices at each image point into distinct coordinate hyperplanes, and hence internally bounds both cardinalities by n. The previously proved whole divisor dictionary identifies all original member indices and the entire exponent function of the pullback. Apply the preceding ordering lemma to these exact original data. This proves every comparison and equality without assuming any pullback value, member-cardinality or ordering certificate. □

Lemma 83.6 (Literal original marked maximum locus)**.** *For a surjective genuine holomorphic submersion and an actual coherent marked ideal whose marked ambient locus is the whole manifold, with genuine SNC boundary, the original gamma maximum locus on its actual controlled cosupport pulls back to the literal maximum locus of the original marked pullback.*

Proof. The genuine whole marked ambient locus remains whole under preimage. Whole original ambient order preservation identifies the actual pulled-back cosupport with the preimage of the original cosupport. If an image point is in the original maximum locus, every point of the pulled cosupport has a smaller or equal gamma by the complete ordering equivalence. In the reverse direction, lift every original cosupport point using actual surjectivity. Its lift lies in the actual pulled cosupport, and the same ordering equivalence gives the required original inequality. Membership and all inequalities give equality of the entire literal sets, including empty cosupports and unattained maxima. Surjectivity is real geometry; no maximum certificate is an input. For a nonsurjective submersion, the pointwise comparison statement remains valid without this global claim. □

84 Actual head incidence and whole positive-member mask agreement

The inputs are the genuine well-formed maximal-order datum, its true literal barMarked test tower, and an actual analytic curve envelope. The last theorem internally constructs these from the original domain-three datum and the constructed canonical invariant. No incidence, mask agreement, lower domain, candidate canonicality or gluing certificate is supplied. Original Step1 level-zero contact and complete General3 descent remain OPEN.

Definition 84.1 (Literal historical original-member predicate)**.** At an actual head index r and an actual index j of its original-member list, take the inverse image of that whole member under the actual literal tail projection. Both indices refer to the actual finite tower and actual lists.

Lemma 84.2 (Internally analytic true head-member incidence)**.** *For a true barMarked test tower of a well-formed datum in positive ambient dimension, every literal historical original-member predicate is analytic.*

Proof. Split the actual tower at its literal head and tail. The head is genuinely a barMarked test tower, so its actual whole transformed boundary is SNC by well-formed marked-transform preservation. Each entry of the actual original list is an entry of that boundary, since it is its literal take. SNC gives the complete closed smooth hypersurface structure of that entry, including empty entries; hence its entire carrier is analytic. The true literal tail projection is holomorphic. Actual analytic-preimage preservation therefore proves the stated entire historical predicate analytic. No closure of an arbitrary curve subset or supplied analytic predicate is used. □

Definition 84.3 (Whole literal positive-member alignment at every actual head)**.** Two actual terminal points are aligned when, at every literal head r from zero through the actual tower length, their projected original-member incidences agree, their indexed levels and membersThrough Finsets agree, their entire maskA Opens agree, their entire memberLowerExtension marked records agree, and for every positive level their actual lowerMarked records are heterogeneously equal over those equal entire Opens. This is an output dictionary, never an input certificate.

Theorem 84.4 (True finite incidence internally produces whole mask and marked-record agreement)**.** *On any actual analytic set Z of dimension at most one, internally produce an analytic dimension-zero E contained in Z. Any two points on the SAME genuine global component of Z outside E have whole headIncidenceAligned along the actual given barMarked test tower.*

Proof. Use the true finite dependent index: r belongs to Fin of tower length plus one, and j belongs to Fin of the actual original-member list length at that head. This dependent Sigma is finite even when head lengths vary. The preceding lemma proves every actual predicate analytic. The genuine curve predicate producer takes proper analytic intersections on uncontained genuine global components. Local finiteness of the true component family and the finite predicate union produce an actual analytic dimension-zero exception E. Outside E each predicate contains a point precisely when it contains its whole component. Thus any two component points outside E have the SAME incidence at every literal indexed head member. Taking a genuine list index for each member gives equality of membership for every original member.

The actual level is the cardinality of the finite indexed incidence filter, so both levels agree, including repeated empty indices. The membersThrough filter requires membership in the original list and membership of the point in that same entry; its entire Finset therefore agrees. The literal maskA definition uses only the original members through the reference point and its actual level: its excluded special-point predicate is evaluated at the varying point, with that SAME reference level. Equal incidence and equal level consequently give equality of the entire sets before interior, hence of their entire Opens. This proves a whole-open equality, not an overlap or germ statement. The memberLowerExtension definition uses only the fixed true head barMarked record and that actual selected Finset, so every field of its whole marked record agrees. At any positive level lowerMarked is exactly the pullback of the same memberRestrict record to that actual mask. Equal Finsets and equal whole Opens give heterogeneous equality of these entire dependent marked records, including carrier, ideal, boundary and mark. No choice of tangent direction or zero-level contact identity is asserted. □

Theorem 84.5 (Actual own-level and literal run freeze together with all whole positive masks)**.** *Assume the actual positive-mark maximal-order datum and a true barMarked test tower. For an actual analytic curve envelope Z inside its true current barMarked cosupport, internally produce an analytic dimension-zero E subset Z. Every point outside E is genuinely regular of dimension one; any two points on one genuine global component outside E have equal actual current level, equal literal regimeRun at their OWN current level, and whole headIncidenceAligned at every literal head.*

Proof. The previously proved genuine historical level and whole top-threshold predicate producer internally constructs its analytic dimension-zero history exception S and regular-curve exception on the actual Z. The preceding theorem internally constructs the incidence exception Q on the SAME actual Z and SAME actual tower. Their union is genuinely analytic and dimension zero by the true finite analytic dimension-zero union theorem. Both are subsets of Z. A point outside the union is outside each exception, so the actual regularity, own-current-level equality, computed literal-run equality and all whole incidence/mask/record agreements hold simultaneously. There is no inference from closedness of a curve subset to analyticity. □

Theorem 84.6 (Original constructed General3 produces its actual companion history and whole masks)**.** *From ONLY the original domain-three datum, its actual canonical invFamily3 tower, an actual current cosupport point and positive actual order, internally construct the genuine companion, literal companion tail, analytic curve envelope Z of dimension at most one containing every actual nonspecial companion-cosupport point, and the exception E of the preceding theorem. Its genuine component points outside E have the actual current-level/run equality and all whole original head masks and positive marked-record agreements for that SAME companion and SAME literal tail.*

Proof. The true original companion domain and cosupport dictionary internally produce the actual domain-three companion, its actual maximal order and the literal canonical masked Step1 companion tail. The actual Step1 bridge internally proves that tail is a genuine barMarked test tower and that its barMarked current cosupport equals the actual companion current cosupport. The genuine original companion curve-envelope producer constructs Z analytic of dimension at most one, inside the latter true cosupport and containing every actual nonspecial companion-cosupport point. The cosupport identity puts this SAME Z inside the true barMarked cosupport. Apply the preceding theorem to these internally produced data. Its returned E and dictionaries are therefore about the literal original companion, actual tail and actual masks, without any caller-supplied lower datum, incidence, candidate canonicality or compatibility certificate. Original level-zero contact, complete cross-mask deeper history descent and whole General3/G3 termination remain OPEN. □

85 Actual noncompact ordered monomial tail with marking one

The original consumer starts with the SAME true original NC ideal, actual internally generated separation and actual terminal-original Cartier step. The whole current marked ideal is exactly its internally recorded boundary power product with marking one. All lists below retain every literal empty or repeated historical entry. No maximum, centre, coefficient, value, termination or canonical-segment certificate is an original input. The old ordered monomial invariant itself is used unchanged, including its colex tie break. Initial general NC separation is not identified with the old general invariant driver; that separate obligation and unrelated ideals remain OPEN.

Lemma 85.1 (Whole actual indexed boundary product and literal intrinsic exponents)**.** *Let F be an actual finite list of closed carriers and natural powers whose ENTIRE literal carrier list is SNC. The complete component-power ideal P internally defines its whole-ambient marking-one marked datum, including all actual well-formed and nonzero-along fields. When n is positive, at every actual point real centred SNC coordinates generate its COMPLETE stalk by the product of the through coordinates to their actual listed powers. At a point of its ith actual member, the intrinsic divisor exponent of P along that actual member is precisely its listed power. The true cosupport is exactly the union of the actual positive-power members. An empty member's exponent can be replaced by zero without changing ANY stalk of P or any intrinsic exponent on any actual through member; its literal boundary entry is retained.*

Proof. The previous actual SNC producer gives each complete member ideal genuine invertibility and coherence, including empty members with unit ideal. Finite whole powers and products are

consequently invertible and coherent. A real nondegenerate local generator makes every stalk nonzero; full ambient smoothness, true SNC and full-ambient transversality give the genuine marked domain. At a true point choose its SAME entire centred SNC chart. Every through member has its true singleton coordinate, and the singleton map is injective. An absent closed member has unit ideal near the point. Actual single-coordinate Hadamard division gives COMPLETE coordinate generation for every through member ideal, at EVERY nearby stalk. True powers and ideal multiplication, by actual indexed-product generation, identify P with the entire product of these coordinate powers. The resulting principal generator is a real nonzero monomial; unit factors from absent members do not alter its ideal. Genuine coordinate divisibility and nondivisibility prove its intrinsic exponent along each through member is exactly that coordinate's listed power, including power zero. This uses the actual COMPLETE ideal, not a supplied monomial factorization or exponent certificate.

A marking-one coherent ideal's cosupport is its true zero locus. The real whole product generator vanishes precisely when a through positive-power coordinate vanishes. This gives the stated union, including empty products, zero powers and empty carriers. For an empty member its COMPLETE vanishing ideal is unit at every stalk; every natural power of it remains unit. Replacing its exponent by zero therefore changes neither the whole product ideal nor any intrinsic exponent computed from that SAME ideal. The literal entry is not deleted; the actual SNC index list is unchanged. □

Lemma 85.2 (The unchanged marking-one invariant has an actual singleton formula)**.** *In positive dimension, at every actual point the old monomial invariant gamma for P with marking one is exactly the maximum of the singleton keys for its actual through positive-power indices, or bottom when there are none. The singleton key is the unchanged lexicographic/colex key (n-1, actual power, singleton index). No maximizing subset or invariant value is a hypothesis.*

Proof. The preceding COMPLETE principal stalk dictionary reads the genuine gamma as the actual combinatorial invariant of the real through-member index set and listed exponents. True SNC injection bounds its through cardinality by n. If the feasible family is empty, all through powers are zero and both maxima are bottom. Otherwise its actual finite maximum is internally attained by a feasible subset I. Feasibility for mark one means its natural-power sum is at least one; hence some actual index j in I has positive power. Its singleton is feasible. The genuine minimal-cardinality theorem for the old key gives card I at most one. The empty subset has sum zero and is infeasible, so card I is exactly one. Its true singleton witness therefore gives the same maximum as the finite singleton family. Conversely every such singleton is a real feasible subset and its key is bounded by the old maximum. The unchanged key definition supplies its first coordinate n-1, listed power second coordinate and colex singleton last coordinate. In dimension zero actual SNC has no through member; the genuine reflexive resolution for that case is produced in the fifth node below. □

Theorem 85.3 (Actual finite boundary internally produces the unique whole maximum centre)**.** *In positive dimension the actual finite indexed family of nonempty positive-power carriers internally selects its unique maximum singleton key. If it is empty, the SAME whole P is unit and the actual cosupport empty. Otherwise the actual maximum centre of the old gamma on the whole cosupport is EXACTLY the entire selected actual carrier C. It is nonempty, genuinely closed smooth Cartier, and admissible for the SAME current marked datum. No compactness, maximum existence or centre certificate is supplied.*

Proof. Filter the actual finite index type by positive power and actual carrier nonemptiness. An empty filtered family makes each positive-power carrier empty, so the preceding true whole unit-factor dictionary makes P unit. For a nonempty family finite maximum-image existence internally chooses a maximizing singleton key. The old key determines its subset, so two actual maximizing singleton indices must be equal. This gives a unique literal index j, with actual whole carrier C

and actual positive coefficient k. Its true nonemptiness supplies a point x of C. Every through positive index at every point belongs to the global filtered family; the preceding singleton formula bounds gamma everywhere by the selected key. At EVERY point of C, its selected singleton occurs, hence gamma equals that key and the point lies in the actual marking-one cosupport. Conversely if a cosupport point has the global maximum value, its real singleton formula has an internally attained index whose key equals the selected key. Key injectivity identifies that index with j, so the point lies in C. Thus the whole maxLocus is C, not merely a chartwise trace or a chosen connected component. Actual SNC member geometry gives its genuine Cartier blowup; its positive complete factor gives true cosupport inclusion, while genuine common SNC coordinates give all remaining marked admissibility fields. The existence and uniqueness of the whole centre are produced internally without compactness. □

Theorem 85.4 (True maximum Cartier control and actual coefficient descent)**.** *Perform the true identity blowup along the internally produced actual old maximum C. The SAME whole controlled ideal has precisely the literal strict old boundary product followed by the new actual C with exponent k-1. Normalize the selected old empty carrier's coefficient to zero, keeping that literal empty entry. All other literal indices retain their whole carriers and their actual through exponents. The new actual carrier has intrinsic exponent k-1 at every point of it. The entire new boundary is genuinely SNC. The natural sum of all actual listed coefficients decreases by EXACTLY one.*

Proof. The selected real positive factor is L to power k, L the COMPLETE actual vanishing ideal of C. Commutativity and associativity of real coherent whole ideal products split it from the entire remaining product. Actual Cartier admissibility and the preceding controlled division theorem give precisely L to power k-1 times that same remaining whole product. The selected old geometric strict carrier is empty; every different literal true SNC index retains its whole carrier. The actual new exceptional carrier is C. Thus the literal map-and-append boundary is exactly the claimed indexed carrier list, including every earlier empty or repeated entry. The selected old empty factor has unit ideal and is assigned coefficient zero using the WHOLE unit factor dictionary; no original definition or boundary entry is changed. All other old coefficients remain literal, and the new coefficient is k-1. Real admissible-transform well-formedness proves this whole new list SNC. Its own true COMPLETE stalk dictionary gives the stated intrinsic exponents, including new coefficient zero; no intrinsic exponent is claimed at a point of an empty carrier. The complete new coefficient sum is the old sum minus k plus k-1. Since k is genuinely positive, new sum plus one equals old sum. There is no caller descent or termination certificate. □

Theorem 85.5 (Actual noncompact old-gamma canonical marking-one monomial tail)**.** *Only the actual finite SNC indexed product internally constructs a genuine finite test resolution that is a canonical segment for the UNCHANGED old gamma tower invariant. Its SAME whole controlled endpoint ideal is unit. Its length is at most the actual initial natural coefficient sum. Every real centre is internally nonempty and exactly the actual gamma maximum locus; every prefix has the true marked domain and literal SNC boundary. All empty or repeated historical entries remain literal; inactive empty factors do not force an empty canonical centre.*

Proof. Induct on the actual natural coefficient sum, with the entire actual list and real SNC proof quantified in the induction. If the internally filtered family is empty, the genuine whole product is unit and the true reflexive tower is a resolved canonical segment of length zero. Otherwise the preceding actual singleton-centre producer supplies the TRUE old gamma maximum, its nonemptiness and actual universal-property identity Cartier blowup. The true coefficient-drop producer supplies the SAME controlled datum with its entire literal new boundary and strictly smaller actual natural coefficient sum. Apply the induction to that real new product. Concatenate the genuine first step and its internally produced recursive tower. Actual marked-transform composition and true old towerInv composition show every real centre is the UNCHANGED gamma maximum for the SAME original product along the actual prior prefix. This proves actual

IsCanonicalSegment internally, rather than accepting it as a hypothesis. True concatenation gives the test resolution and unit endpoint. Length is one plus the recursive length, bounded by one plus its actual new sum, equal to the old sum. Actual domain preservation on the genuine test tower gives every literal prefix domain and SNC boundary. The recursion is on actual natural arithmetic; no compact ambient, compact cosupport, global finite invariant spectrum or external well-founded/canonical certificate is used. In dimension zero every actual SNC carrier is empty, so the filtered family is empty and the old-gamma segment is reflexive. □

Theorem 85.6 (SAME original NC controlled ideal has its actual ordered old-gamma completion)**.** *The genuine original NC carrier alone internally constructs an ordered old- gamma monomial tail for its SAME controlled ideal immediately after its actual terminal-original Cartier step. The tail is a real old-gamma canonical segment, has complete unit endpoint and length at most (n-1) squared. Appending the real terminal-original step and SAME actual separation gives a genuine original marking-one resolution with full length at most 1+n(n-1), preserving all literal boundary history. This does not assert that the prior NC separation is canonical for the old general invariant driver.*

Proof. The earlier original producer supplies the true separating tower, original marked testness, internally recorded actual r-1 coefficients, and COMPLETE whole controlled factorization. The genuine terminal-original Cartier step removes precisely the whole reduced strict-original factor; its actual new boundary is the entire old historical carrier list followed by that actual terminal original. Form the indexed product from the real recorded weight list followed by this real new member of coefficient zero. Its WHOLE ideal is exactly the SAME actual current controlled ideal: the final zero-power factor is unit. Its literal carrier list is the entire actual SNC boundary, including an empty terminal original or empty historical entries. No coefficient or carrier dictionary is an input to this consumer.

Apply the preceding internally terminating actual old-gamma producer to this real current product. The genuine original coefficients are at most n-1 and their actual list length is at most n-1, so their sum is at most (n-1) squared. The appended actual zero-power factor adds nothing. True marked equality identifies the produced old-gamma canonical segment with that SAME original transformed datum. Append the real original Cartier step and original separation to obtain a genuine test resolution of the SAME original ideal and unit endpoint. Its full length bound follows by adding the one actual terminal-original step and the actual separation length at most n-1; dimension zero has real separation zero, old-gamma tail zero and precisely the retained one empty original Cartier step. Genuine prefix-domain preservation retains the entire literal SNC history. This closes the noncompact finite ordered MONOMIAL-tail construction for this actual original ideal, while leaving its earlier general-driver comparison and unrelated arbitrary marked ideals explicitly OPEN. □

86 Actual positive-member values across all reference masks

All reference masks, original lower marked records and their canonical restricted tails below are internally produced from the genuine canonical Step1 input. The final consumer starts with the original domain-three datum and its constructed invFamily3 canonical tower. The finite index is an actual head and a subset of its actual finite boundary indices, never a finite value spectrum. Original Step1 level-zero contact remains OPEN.

Lemma 86.1 (Actual global components persist in intermediate analytic subsets)**.** *If B is a genuine global component of an analytic set Z, with B contained in an intermediate set A contained in Z, then B is a genuine global component of A. No component-identification certificate is an input.*

Proof. Write B as the closure of the actual connected regular component C of Z through a genuine regular point c. At every point of C, true regular-component geometry gives an open

neighbourhood on which Z lies inside C. Because C is contained in B, and B is contained in A contained in Z, the germs of A and Z agree there. Thus every point of C is regular for A, with the actual same dimension. True preconnectedness puts C inside the actual regular component D of A through c. Conversely, near c the actual A equals Z and lies in C contained in B. The actual analytic continuation theorem for a regular component puts all of D inside the analytic B. Taking closures gives B contained in closure D contained in B, so B is literally closure D and belongs to the actual global component family of A. □

Lemma 86.2 (Equal actual member selection gives the entire original mask)**.** *At one literal head, equality of the actual membersThrough Finsets gives equality of its entire maskA Opens. All original-member incidences and the true head levels are determined by that actual Finset.*

Proof. For any original head member W, its real list membership gives an actual index i in the whole transformed boundary. The literal membersThrough filter contains i exactly when W passes through the reference point. Equality of Finsets consequently gives membership equality for each W. The actual indexed level filters on the original list agree, hence their cardinalities agree, including actual empty or repeated historical entries. In the literal maskA definition the reference point enters only through these incidences and this head level. The special-point predicate is evaluated at the varying point with the same level. Therefore the whole sets before interior agree, and so do their entire Opens. The same selection also fixes the full memberLowerExtension record, and the positive lowerMarked records are equal after transport over these actual equal Opens. These identities are algebraic consequences of actual original definitions, not a supplied mask or value certificate. □

Definition 86.3 (The original positive lower values agree across basepoints)**.** On each genuine global component of Z, two points outside the exception that are nonspecial and of positive actual Step1 level have equal ORIGINAL lowerValue at their own computed regimeRun head and tail, equal whole actual masked Step1 value, and identical original lower-value superlevel comparisons at every threshold. This is an output dictionary, never a caller hypothesis.

Theorem 86.4 (Internally produced finite mask configurations glue all actual positive-member lower values)**.** *Let K be a genuine domain-three positive-mark maximal-order datum, R its actual canonical masked Step1 tower for invFamily2, and Z an actual analytic curve envelope inside its true current cosupport. Internally produce an analytic dimension-zero E contained in Z, outside which the original positiveAcrossMaskValues dictionary holds on every actual component. No lower canonicality, value, mask or configuration-family certificate is provided by a caller.*

Proof. The genuine canonical Step1 bridge internally gives the true barMarked test tower and its actual current cosupport identity. Apply the proven whole HeadIncidence history producer to the actual Z. Its true dimension-zero exception makes own levels, actual regimeRun, all historical member selections and whole masks agree between component points outside that exception.

Internally enumerate the finite dependent configurations: r ranges from zero through the true tower length, and S ranges over Finsets of the actual whole transformed boundary indices at that head. A configuration is present only if a real current cosupport point is nonspecial of positive level, has actual computed run r, and has actual membersThrough S. For every present configuration choose such an actual point internally. Its genuine original canonical lower-data producer proves the actual positive lower domain, canonical restricted tail and current cosupport dictionary. True member extension and whole tail producers internally give the actual entire member remainder I, actual mask U, and full identity of I restricted to U with this SAME original lower marked record. The parent head and tail are true test towers; no whole-candidate canonicality is assumed.

For each present configuration let A be Z intersected with the actual whole member endpoint carrier. This is genuinely analytic of dimension at most one. The true whole member domain

and masked canonical tail internally produce both previously proved positive-order and zero-order value dictionaries on A, including component-own attained natural minima and their true dimension-zero exceptions. No common order bound or finite spectrum is used. Add the actual A itself to the finite analytic predicate family on Z. Its true predicate exception ensures that if any component point outside the exception belongs to A, its entire component B lies in A. The first lemma then identifies that SAME B as an actual global component of A.

Take the finite union of all actual configuration value exceptions, the actual A-predicate exception and the genuine HeadIncidence history exception. Each is a genuine analytic dimension-zero subset of Z. This is an actual finite union indexed by finite heads and finite boundary subsets, not by orders or values, so the whole union E is analytic of dimension zero.

Take any two actual nonspecial positive-level points z and w of one genuine component B outside E. HeadIncidence gives the SAME computed head index and actual member selection, hence their configuration is present. Its internal reference point has precisely that head and selection. The second lemma and literal marked definitions give whole U equality and exact original lower record identities at z, w and that reference point. Actual canonical lower cosupport data and the true whole member extension put z and w in the actual A and in the SAME fixed masked lower cosupport. The A-predicate dictionary gives B contained in A, and the first lemma gives B as a genuine global component there.

Use the actual attained natural minimum of the whole member endpoint order on B, produced by the existing true order history. If it is zero, the literal invFamily2 value is its zero-order monomial branch and the true whole gamma dictionary is constant outside its actual exception. If it is positive, the true original canonical masked lower value equals its actual positive companion special branch and is constant by the proved positive dictionary. Both cases give equal entire invFamily2 values for z and w on the SAME fixed original mask, at every threshold. Whole mask/record identity transports the literal original lowerValue definitions to these fixed-mask values. The genuine actual fixed-lower Step1 dictionary, or its literal nonspecial value definition with equal own level and computed run, then gives equality of the entire original masked Step1 values. Thus all varying-reference original positive lower values and their complete superlevel comparisons agree. No arbitrary closed curve subset is declared analytic and no zero-contact or complete General3 threshold claim is made. □

Theorem 86.5 (Original constructed General3 internally supplies the across-mask value producer)**.** *Only the original domain-three datum, its actual invFamily3 canonical tower, an actual current cosupport point and positive original order internally produce the SAME actual companion, literal canonical companion tail, analytic dimension-at-most-one nonspecial envelope Z and an analytic dimension-zero E subset Z satisfying positiveAcrossMaskValues.*

Proof. The genuine original companion dictionary produces its actual domain-three companion, true maximal order and actual canonical masked Step1 tail. Its actual curve-envelope producer gives Z analytic of dimension at most one, inside true companion current cosupport and containing every actual nonspecial companion-cosupport point. Apply the preceding producer to these internally constructed data. Its actual values refer to this SAME literal companion and tail, original masks and old lowerMarked records. Nothing is reinitialized or supplied as a caller value, family or compatibility certificate. The resulting positive-member across-mask value gluing is distinct from original Step1 level-zero contact, which still depends on real tangent/contact histories; that obligation and whole General3/G3 remain OPEN. □

87 True mixed-discrete ambient trace residuals with the WHOLE accumulated boundary

The singleton NEW-exceptional dictionary alone is insufficient for a real chronological history. Retain every OLD exceptional position, and prove the genuine whole accumulated comparison

for the factor mechanism in [4, Remark 12.6, pp. 293–295]. No indexed count is used as a distinct-germ count.

Lemma 87.1 (Genuine monomiality is monotone under actual boundary family inclusion)**.** *For actual ordered families $E \subset F$ in the membership sense, genuine monomiality of ANY actual ideal A along N with respect to E implies monomiality with respect to F. No SNC, coherence, rank or family bijection is a premise; all original positions and multiplicities remain.*

Proof. A genuine F-monomial-part germ satisfies the actual ideal-power constraint for EVERY F position. For each actual E position, membership inclusion chooses a real F position with the SAME actual member set. Its actual vanishing ideal and intrinsic exponent are consequently the SAME, so its constraint supplies the actual E constraint. This constructs membership in the smaller-family monomial part, not equality of indexed counts. The given genuine monomial restriction then supplies the required relative membership in A. This is precisely actual monomiality for the larger family by definition. □

Theorem 87.2 (A genuine whole monomial factor does not change the whole residual)**.** *For actual smooth N, SNC E transverse to N, actual coherent ideals A, B nonzero along EVERY point of N, and genuine monomiality of A with respect to E, one has the WHOLE actual equality*

$$\mathcal{N}_N(\mathrm{Res}_N(AB), E) = \mathcal{N}_N(B, E).$$

No desired factorization, residual equality, invertibility, generator, exponent or quotient certificate is supplied.

Proof. The genuine parts theorem independently constructs $R = \mathcal{N}_N(B, E)$ and $M = \mathcal{M}_N(B, E)$, their real coherence, nonzero and monomial geometry and actual BOTH factor directions for B. The true residual has no factor along ANY actual boundary member. Genuine relative monomial product geometry constructs the actual coherent nonzero monomial $H = AM$. Multiply the two actual factor directions by the SAME A, using real finite product spans. Actual associativity and commutativity identify $A(RM) = R(AM)$. The normalized input has BOTH real relative directions with RH. The genuine greatest-factor uniqueness theorem therefore identifies its actual monomial part with that of H; BOTH true residual factor implications give equality with R along N. Actual colon normalization and the genuine off-closed-carrier unit germs give WHOLE ideal equality at every ambient point. No cancellation or factorization is assumed as a desired output. □

Lemma 87.3 (Actual discrete centres internally have normal crossings with the whole real SNC family)**.** *A genuine regular-zero centre has actual normal crossings with ANY real ambient SNC ordered family, before any chosen blowup or residual comparison.*

Proof. At each genuine centre point, actual zero-dimensional regularity supplies an open isolating that point in the centre. A singleton actual point has normal crossings with the ENTIRE given SNC family. On the true isolating neighborhood the actual centre has that SAME germ, in BOTH directions. The genuine local normal-crossings comparison therefore gives whole-centre normal crossings. No desired admissibility or coordinate certificate is input. □

Theorem 87.4 (ONLY original mixed-discrete geometry proves the WHOLE accumulated trace versus raw TOTAL residual)**.** *ONLY original smooth N, genuine SNC ordered OLD family E transverse to N, singleton ambient SNC H, actual density of $N \setminus H$, ANY discrete $C \subset N$ and its SAME actual blowup internally construct the true new ordered family $E' = \mathrm{Strict}(E) +\!\!+ [D]$ with its actual SNC and carrier transversality. For the TRUE normalized ambient strict trace I' and actual normalized raw TOTAL P' one has the WHOLE equality*

$$\mathcal{N}_{N'}(I', E') = \mathcal{N}_{N'}(P', E').$$

Global $C \subset H$, desired next geometry and any residual comparison certificate are absent. EVERY actual old position, including empty and repeated positions, remains in the SAME ordered family.

Proof. The preceding genuine discrete-centre theorem first supplies actual NC with the ENTIRE original E. The actual same-map accumulated blowup geometry therefore constructs SNC E' and genuine strict-carrier transversality, with every original strict member followed by the real new exceptional member. Independently the original mixed-centre trace producer supplies the actual true trace coherence/nonzero geometry and its greatest NEW-exceptional factor D_t, with the genuine WHOLE $I' = \mathrm{Res}_{N'}(D_t J)$ for independent raw all-power STRICT J.

Independently construct the actual empty-old-boundary zero-mark raw record. Its real SNC/transverse/coherent/nonzero principal geometry supplies the true singleton greatest factor D_r and raw singleton residual. The independent all-power STRICT comparison identifies that residual with THIS SAME J, giving $P' = \mathrm{Res}_{N'}(D_r J)$ with genuine coherence/nonzero of J. Both D_t, D_r are genuinely monomial along $[D]$, and D actually occurs in the real E'. The first lemma therefore supplies true monomiality of BOTH factors along the ENTIRE accumulated family, without identifying indexed counts or deleting positions.

Apply the genuine whole monomial-product residual theorem separately to these two independently verified actual factorizations, for the actual smooth carrier and real whole E'. Both whole residuals equal $\mathcal{N}_{N'}(J, E')$, so they are equal as asserted. This is an ALL-accumulated-boundary statement, not a singleton substitution. Literal chronological $I_i = D_i J_i$, exponent evolution, the separate τ_D phase and full sharp Σ/termination remain next consumers. □

88 Literal finite ambient strict traces have the genuine chronological factor dictionary

Use the EXISTING literal iterated ambient strict transform $H_T = \mathrm{strictAlong}(T, H)$, and the independently constructed chronological all-power relative STRICT ideal J_T. These two objects are not identified. The intended genuine dictionary of [4, Remark 12.6, pp. 293–295] is $I_T = \mathrm{Res}_{N_T}(D_T J_T)$.

Lemma 88.1 (Original dimension-one tests internally supply actual discrete centres)**.** *Define* actualDiscreteCentres(T) *by literal tower recursion: the empty tower satisfies it; an actual step satisfies it exactly when its prefix does and EVERY point of its actual centre is genuinely regular of dimension zero. For ANY ORIGINAL dimension-one invariant domain and ANY genuine ORIGINAL test tower, this actual geometric property is proved. No discreteness or selected-centre certificate is supplied.*

Proof. Induct on the SAME actual original test. Genuine original domain preservation makes the current cosupport analytic of dimension at most zero at EVERY prefix. Each real admissible centre lies in that actual cosupport. True zero-dimensional regularity supplies an open isolating each such point; real centre containment and actual point membership make the centre germ the SAME singleton germ, hence genuinely regular zero. This constructs the literal recursive property, retaining every actual empty step. The genuine original-test template theorem independently supplies the SAME geometric history when it is needed by the final consumer. □

Theorem 88.2 (Original smooth density constructs all true ambient trace geometry at every discrete-history position)**.** *ONLY original smooth W, singleton ambient SNC H, true density of $W \setminus H$, an ACTUAL geometric-template test T and its genuine discrete centres internally give the following. The EXISTING literal $H_T = \mathrm{strictAlong}(T, H)$ is an actual ambient SNC singleton; the actual current carrier N_T has genuine density away H_T. The TRUE $I_T = \mathrm{Res}_{N_T} \mathcal{I}_{H_T}$ is coherent, nonzero at EVERY actual carrier point and has BOTH relative principal germ directions there. No desired current trace, density or generator certificate is supplied.*

Proof. At the empty history, original genuine SNC generators and normalization give the true trace dictionary; true density supplies nonzero, and original vanishing coherence supplies trace coherence. For an actual step, genuine geometric-template well formedness gives actual smooth

current carrier and centre containment. The recursive discrete-history condition supplies real regular-zero centre geometry. Apply the independently proved mixed-centre whole trace producer to the ACTUAL current ambient member and the genuine current density already constructed by induction. It constructs actual new trace coherence/nonzero/principal germs and NEW density. Its real ambient SNC pair has the actual strict member as a genuine sublist, so that member is itself a true SNC singleton. The defining EXISTING strictAlong step equation identifies exactly this same ambient set, not a saturation support or independently chosen substitute. Thus induction supplies all assertions at every literal position, including empty centres. □

Theorem 88.3 (The TRUE ambient trace residual along ALL accumulated exceptionals equals the independently chronological STRICT ideal)**.** *Under the preceding ONLY original geometric inputs and the SAME genuine history T, let E_T be its literal WHOLE accumulated exceptional list. Then as WHOLE actual germ ideals*

$$\mathcal{N}_{N_T}(I_T, E_T) = J_T,$$

where J_T is the independently defined genuine chronological pullback then NEW-exceptional all-power-saturation recursion. No desired ideal, factor, value, history comparison or no-old-factor certificate is input.

Proof. At the empty history the actual boundary is empty, and true normalization idempotence identifies its residual with the original relative member ideal, which is the independent chronological initial record. Suppose the whole identity has been proved at a true prefix. The previous theorem supplies the actual current ambient member geometry and density. Actual whole geometric-template well formedness supplies current SNC E_T and actual carrier transversality, while the actual next centre is genuinely discrete and contained in the current carrier. The genuine ALL-old-family mixed comparison identifies the next ambient trace residual with the raw TOTAL residual of the current actual ambient member.

The actual pullback of its normalized true trace and the actual pullback of the unnormalized ambient vanishing ideal have BOTH genuine relative-span directions on the actual next carrier; actual holomorphic composition and strict-carrier membership supply that whole normalization equality. The genuine whole raw TOTAL residual transport theorem then replaces the current trace by ITS actual whole residual before pullback. Substitute the induction identity with the independently chronological current J_T. Original chronological geometry and its independently proved actual no-old-exceptional-factor theorem supply ALL coherent/nonzero/principal and factor-exclusion inputs internally. Thus the genuine ALL-accumulated raw-to-next-STRICT theorem identifies the next residual with the literal next chronological recursion. Actual carrier and ordered-list dictionaries match the SAME geometric template before any substitution. This proves the whole identity at the next actual position, with every old and empty entry retained. No invariant rebasing or canonicality is invoked. □

Theorem 88.4 (ONLY original smooth-curve tests give EVERY literal true ambient member factor $I_i = D_i J_i$)**.** *ONLY original closed smooth curve W, real ambient SNC ordered family E, actual density of W away the WHOLE family, ANY genuine ORIGINAL mark-one product test T and ANY original member position i internally give the TRUE $I_{i,T} = \mathrm{Res}_{N_T} \mathcal{I}_{\mathrm{strictAlong}(T,E_i)}$. Its ACTUAL greatest monomial part $D_{i,T}$ along the SAME WHOLE accumulated exceptional list is coherent, nonzero and monomial, and*

$$I_{i,T} = \mathrm{Res}_{N_T}(D_{i,T} J_{i,T})$$

as WHOLE actual ideals. Here $J_{i,T}$ is independently chronological relative STRICT, not ambient trace by definition.

Proof. Original smooth curve geometry, actual ambient SNC/coherence and genuine whole-family density internally construct the original mark-one product's true dimension-one domain. The first

theorem supplies actual discrete centres from the SAME original test, and original-test template transport supplies the genuine geometric history with identical carrier, list, centres and maps. Whole-family density implies genuine density away the chosen actual member; actual SNC sublist geometry supplies its genuine singleton SNC. The second and third theorems therefore apply to this actual original member and SAME history, constructing true trace coherence/nonzero and the whole residual/chronological equality. The genuine greatest-factor theorem applies to this ACTUAL trace on the independently verified smooth current carrier and whole accumulated SNC/transverse family. It internally constructs the asserted ACTUAL factor geometry and true two factor directions. Substitute the proved whole chronological equality; actual normalization idempotence and the essential OUTER carrier restriction yield the displayed whole identity.

This supplies the actual finite-history $I_i = D_i J_i$ dictionary. It does not erase D_i, identify supports of I_i and J_i, give the remaining exponent evolution or execute the separate τ_D phase. The parabola counterexample remains valid; full sharp Σ preservation and general termination remain OPEN, even when curve FIRST Z has been erased. □

89 The ACTUAL chronological FIRST DISTINCT-defect has genuine finite analytic equations

Every actual chronological ideal is defined by genuine iterated NEW-exceptional saturation. The product is the literal carrier-relative finite product, whose coherence, nonzero and true multiplicity SUM were independently proved. FIRST Z is defined using the TRUE quotient classes of proper ACTUAL IDEAL germs. Raw indexed counts are never substituted for DISTINCT classes. Equality with the different composite raw-TOTAL residual, the modified centre/decrease driver and sharp termination remain OPEN.

Definition 89.1 (Actual proper ideal-germ quotient classes and literal dynamic FIRST defect)**.** For a genuine finite current ideal family, identify positions exactly when ALL their actual holomorphic ideal-germ predicates agree at the true point. A position is proper exactly when its actual ideal germ does not contain one. Define t as the cardinality of the image of these proper positions in the TRUE quotient, and define FIRST Z on the carrier by $t < \nu$, with ν the actual ordinary order of the genuine full relative product. Apply the same definitions to EVERY actual original chronological position on the SAME geometric history; no branch/count certificate is included.

Proof. Equality of actual full predicates is a real equivalence relation. The proper finite-position set, true quotient image and literal product order are independent actual definitions. Coherence proves proper iff literal zero-locus incidence: one cannot vanish at the point, while a true nonzero ideal germ at the point supplies a real reciprocal unit. Actual mark-one cosupport/order dictionaries then identify proper positions with positive TRUE factor orders, without identifying distinct classes with positions. □

Theorem 89.2 (True dynamic quotient collisions and factor multiplicities recognize the literal FIRST defect)**.** *For genuine coherent nonzero relatively principal current ideals on a smooth carrier, the actual full product order dominates the TRUE distinct proper count. Equality holds exactly when EVERY proper actual factor has order one AND the quotient map is injective on those proper positions. Consequently the LITERAL FIRST defect occurs exactly when some actual factor has order at least two, OR two distinct proper positions have the SAME actual full ideal germ. All actual multiplicities are retained.*

Proof. The independently proved true product SUM dominates the proper-position cardinality, which dominates the TRUE quotient-image cardinality. Equality forces equality in both real bounds. The actual finite sum equality forces each genuine nonnegative factor order to equal its zero-or-one proper indicator; equality of image and source cardinalities forces genuine quotient

injectivity. Conversely these two facts yield the actual equality. If no factor has order at least two, every proper actual order is one; strict count deficit is therefore precisely a real quotient collision. The reverse follows from the same independently established equality criterion. No indexed/DISTINCT substitution or difference-of-USC argument is used, even at bad points. □

Theorem 89.3 (Genuine pair-dimension geometry gives the entire dynamic finite analytic formula)**.** *If the true carrier is everywhere regular of dimension $d+1$, the LITERAL FIRST defect equals the finite union of actual factor mark-two cosupports and, over DISTINCT position pairs, the true local-dimension-at-least-d loci of their literal carrier/zero-support intersections. This SAME entire set is genuinely analytic and hence closed.*

Proof. A high factor is exactly actual mark-two cosupport by the real order predicate. Without any high factor, each proper actual factor has order one. The independently proved CURRENT relative-principal reduced-ideal and pair-dimension equivalence therefore replaces an ACTUAL ideal-germ collision by its genuine literal pair-dimension condition in BOTH directions. Conversely a real pair-dimension point belongs to that closed analytic pair, hence to the true carrier and both actual proper supports; absent a high factor the same true geometric equivalence gives an actual collision. Thus the finite formula holds at EVERY ambient point, including outside the carrier and unit/empty-position cases. Genuine coherent derivative ideals make high-factor cosupports analytic. Real coherent zero sets and carrier analyticity make each literal pair analytic; the already proved local-dimension-lower locus theorem supplies its actual dimension locus analyticity. Genuine finite unions establish the SAME whole FIRST Z analytic, without supplying its closed/analytic presentation as input. □

Theorem 89.4 (Only original finite-history inputs produce the actual dynamic FIRST analytic locus)**.** *ONLY original smooth closed carrier of fixed positive pure dimension, genuine ambient SNC, whole-boundary density and an ACTUAL geometric finite history internally supply the current ideals and common carrier needed above. The LITERAL dynamic FIRST Z of these ACTUAL chronological ideals is analytic on the WHOLE actual ambient manifold. No current reduced presentation, desired branch relation or analytic/count certificate is input.*

Proof. Genuine geometric-template history independently supplies the current closed smooth carrier, including the empty-original-list case. Induct on the REAL blowups, using actual centre containment and true strict-transform regularity to transport the original fixed pure dimension to EVERY current carrier point. Original-only finite chronological geometry supplies EACH actual position's real coherence/nonzero/relative generator, identifying its carrier with the SAME template BEFORE transporting these facts. The actual ordered family equals the genuine finite list of these positions, so its relative product is the independently defined chronological full product. Apply the preceding actual finite analytic formula. This proves dynamic FIRST Z analyticity for the defined history, not yet modified centre selection or original-invariant decrease under that new history. FIRST Z remains distinct from full sharp Σ; composite original raw-TOTAL factor transport, recursive intersections, exceptional τ_D and sharp termination remain separate OPEN mathematical producers. □

90 Actual chronological saturation does not reintroduce OLD exceptional common factors

The ideals here are the proved literal iterated NEW-exceptional saturations. We now prove actual absence of common factors for EVERY accumulated boundary member. This is independent geometry, not a desired strict-history premise. Comparison with the composite original raw-TOTAL factorization remains OPEN.

Lemma 90.1 (True old-member strict trace geometry produces an actual TOTAL germ outside its old strict member ideal)**.** *Suppose the genuine old coherent ideal has, at EVERY actual carrier/old-member point, a real germ outside that actual restricted member ideal. For a real blowup with centre inside the actual smooth transverse carrier and NC with the ENTIRE old SNC list, its actual mark-zero TOTAL has such a genuine germ outside EVERY old strict-member restricted ideal on the new actual carrier. No bound on the old ideal's order along the centre is needed.*

Proof. Real whole-list NC and transversality identify the geometric strict transform of the literal old member/carrier intersection with the actual new carrier intersected with that SAME old strict member. Actual old germ exclusion, through BOTH true restricted-member/trace-vanishing directions, gives a real old germ not vanishing on that actual trace at every point off the centre. Genuine coherent continuation along the real blowup therefore constructs an actual TOTAL transformed germ not vanishing on the actual strict trace. The independently proved real whole accumulated SNC/transverse geometry and true trace-vanishing dictionary turn it into genuine restricted-member exclusion. Only the OLD-member branch is used, so no NEW-member order-bound hypothesis is imported or supplied. □

Theorem 90.2 (The true next NEW-exceptional saturation has no common factor from ANY actual accumulated member)**.** *For a genuine coherent nonzero relatively principal old ideal on a smooth transverse carrier, with actual no-common-factor witnesses for its old list, the independent actual next chronological STRICT has, at EVERY true new carrier point, a genuine ideal germ outside EVERY incident member of the precise whole ordered old strict list followed by the actual new exceptional member. All required centre containment and WHOLE old-list NC are genuine geometric inputs.*

Proof. The actual chronological step theorem internally constructs real next TOTAL coherence/nonzero and the WHOLE equality of next STRICT with its independently defined NEW singleton greatest-factor residual. Split the actual numeric index into a real old position or the genuine appended NEW position, retaining each original order. For an old position the first producer gives a genuine TOTAL germ outside that actual old strict member. Genuine coherent total inclusion into its actual residual and the proved whole STRICT equality place that SAME witness in next STRICT, preserving the true nonmembership. For the NEW position actual maximal-exponent geometry independently supplies a real residual germ outside the actual NEW member. Its whole equality with next STRICT gives the true witness. No desired no-old-factor certificate, order bound along the centre or renamed residual is assumed. □

Theorem 90.3 (Every genuine finite chronological history preserves actual absence of all accumulated exceptional common factors)**.** *For genuine initial coherent/nonzero/relative-principal data and actual old no-common-member germs, EVERY real geometric template test history gives actual no-common-factor witnesses for EVERY incident actual final ordered exceptional member, at EVERY actual final carrier point. It applies to EVERY actual finite prefix and retains every empty step.*

Proof. Induct on the literal history. At identity use genuine old witnesses. At a real successor, the independent finite-history theorem has ALREADY constructed the actual coherent/nonzero/principal prior ideal and whole geometric field equalities. Real template admissibility supplies actual centre containment and NC with its ENTIRE prior list; transfer these only through proved actual field equalities. Apply the second producer using the genuine previous no-factor induction conclusion. Its actual next ideal, strict carrier and ordered list are literally those in the defined chronological record. Thus the real witnesses persist for every position without deletion or relabelling. □

Theorem 90.4 (Only original data supply actual ALL-exceptional exclusion for every member's genuine finite chronological history)**.** *ONLY original closed smooth carrier, genuine ambient SNC and actual whole boundary density construct, at EVERY true finite geometric prefix, real no-common-factor witnesses for EVERY accumulated exceptional member of EACH original position's actual chronological ideal. No desired old-factor or history dictionary is supplied.*

Proof. The independently proved original-only finite-history producer internally supplies the true initial relative member ideal with actual wellformedness, nonzero and relative principal germs. Its real initial exceptional list is EMPTY, so initial no-member exclusion is vacuous by the actual empty index type, not a caller certificate. Apply the preceding genuine history theorem to the SAME actual geometric template tower. It constructs all real final witnesses for the original ordered accumulated list. This proves absence of reintroduced OLD common factors for the actual iterated NEW-exceptional saturations; the genuine composite raw-TOTAL/ALL-exceptional factor transport, dynamic DISTINCT driver/decrease, recursive intersections, exceptional τ_D and sharp termination remain separate OPEN obligations. □

91 Genuine accumulated exponents vanish and chronological ideals are their actual residuals

The actual no-common-member germs are now converted into the TRUE intrinsic exponents and whole monomial/nonmonomial dictionaries. These are the actual iterated NEW-exceptional histories. Composite original raw-TOTAL factor transport and dynamic sharp-driver comparison remain OPEN.

Lemma 91.1 (Actual coherent germ persistence forces every genuine incident exceptional exponent to be zero)**.** *For a real coherent ideal on a carrier transverse to a true SNC family, actual no-common-member witnesses imply that its genuine divExpAlong is ZERO at every actual carrier/member point, for every real indexed member. No zero-exponent field is supplied.*

Proof. Suppose the actual maximal exponent were positive. Its independently proved actual divisibility neighborhood therefore has order at least one along EVERY nearby carrier/member point. Choose the genuine ideal germ outside that member ideal at the actual point. True coherence continues that SAME germ as an actual ideal member nearby. The genuine first-order vanishing dictionary forces it to vanish on the entire nearby literal carrier/member intersection. BOTH real restricted-member/trace-vanishing directions then put the original germ in the true restricted member ideal, contradicting its actual exclusion. Thus the actual exponent is zero by genuine geometry, not a formal restatement of a no-factor field. □

Theorem 91.2 (The WHOLE actual monomial ideal is unit and the true nonmonomial ideal is precisely the carrier normalization)**.** *For that genuine coherent ideal and actual smooth closed carrier, its ENTIRE actual intrinsic monomial part is the holomorphic unit ideal. Its ENTIRE actual nonmonomial part is exactly restrictOn of the original ideal to the real carrier, in BOTH full holomorphic germ directions.*

Proof. At every actual ambient point, genuine monomial membership always requires holomorphicity. Conversely every real holomorphic germ satisfies each actual incident constraint: the first producer proves its true exponent zero, the actual power is the real unit ideal, and its genuine restricted span contains the given holomorphic germ. Outside the carrier there are no constraints. This gives the whole monomial ideal equality before using any quotient law. The actual nonmonomial part is DEFINED independently as the colon of the true carrier-normalized ideal by that actual monomial ideal. The proved whole unit identity and true coherent colon-by-unit theorem therefore give its ENTIRE equality with the genuine carrier normalization. Normal carrier-vanishing germs remain; no ambient principal identity is inferred. □

Theorem 91.3 (Only original inputs give actual all-prefix exponent and residual dictionaries for each chronological member)**.** *ONLY original smooth closed carrier, true ambient SNC and whole-boundary density give at EVERY true finite geometric prefix and EVERY original position: actual incident accumulated exponents zero, whole actual monomial ideal unit, whole actual nonmonomial ideal equal to the true carrier normalization, and at EVERY actual carrier point BOTH actual nonmonomial membership directions equal the SAME chronological ideal's full germ predicate. No no-factor, exponent, residual or history certificate is input.*

Proof. The independent original-only finite-history producer internally supplies actual wellformedness, coherent ideal and every genuine relative principal generator. The independently proved old/new no-factor history gives real exclusion witnesses for the ENTIRE accumulated ordered list. Apply the first two producers to these actual objects. At a real carrier point, expand any true restricted span of chronological ideal germs. Actual relative principal dictionaries substitute each into the SAME genuine principal span, which is therefore again an actual original ideal germ. The converse uses genuine ideal inclusion into its true carrier normalization. Composing with the proved whole nonmonomial identity gives BOTH displayed membership directions. This supplies a real residual dictionary for the actual chronological objects, without identifying them with residuals of a different composite raw TOTAL. Composite factor transport, dynamic DISTINCT max driver/decrease, recursive intersections, exceptional τ_D and sharp termination remain OPEN. □

92 A genuine next chronological relative STRICT step from the actual first STRICT ideal

Here the second STRICT is literally the arbitrary NEW-exceptional saturation of the actual pullback of the independently constructed first STRICT. No desired next ideal or generator is input. When accumulated exceptional geometry is claimed, the real next centre is required to have NC with the ENTIRE actual prior exceptional list, in addition to lying in the true carrier. We do not select that centre, or identify this next STRICT with the residual of the accumulated original raw TOTAL pullback.

Lemma 92.1 (Real relative principal germs pull through the actual blowup to its genuine strict carrier)**.** *A genuine coherent ideal relatively principal along a real closed smooth carrier has genuinely relatively principal mark-zero TOTAL transform along the actual strict carrier of a real blowup with centre in that carrier. The real new principal generator is internally constructed at every actual strict-carrier point.*

Proof. The actual strict carrier lies over the original closed carrier by genuine closure and continuity. Obtain the true old relative generator at that actual projected point. Its real ideal membership pulls back to actual membership of the genuine composed function. Each old arbitrary ideal germ is a true finite multiple of that generator ON the old carrier; compose its real coefficients with the genuine holomorphic projection. Continuity pulls its actual neighborhood equation back, and the entire new strict carrier maps into the old carrier. Thus the equation holds on the true new carrier, including all actual old normal-vanishing germs. Expand actual pullback spans and substitute these genuine relative relations. The reverse direction uses the real composed generator membership. The proved whole mark-zero TOTAL identity supplies exactly the displayed carrier restriction. No ambient principal assertion or chosen desired pullback dictionary is input. □

Theorem 92.2 (The actual next chronological STRICT is coherent, nonzero, relatively principal and has its genuine normalized NEW-exceptional factor)**.** *For a real coherent nonzero relatively principal ideal on the actual smooth carrier and a genuine blowup with centre inside it, define the next STRICT by the TRUE pullback followed by ALL powers of the actual NEW exceptional ideal. That actual ideal is WHOLE equal to its independently formed single-new-exceptional residual, is*

coherent and nonzero along the real new strict carrier, and is genuinely relatively principal there. Its actual mark-zero TOTAL has the true normalized NEW-exceptional-power factor at EVERY ambient point.

Proof. Use the literal mark-zero input with EMPTY auxiliary boundary. Its actual cosupport is the entire old carrier, so the genuine smooth centre is auxiliary-admissible; no accumulated-boundary conclusion is deduced from that empty-list auxiliary geometry. True transformed wellformedness gives the real strict carrier and NEW singleton exceptional geometry. Independent actual coherent transform and true accumulation off the centre give nonzero. The first lemma internally supplies all real new TOTAL relative generators. The independently proved actual whole singleton saturation theorem and real factor theorem therefore give actual next STRICT equality, coherence, nonzero and normalized factor. Genuine greatest-factor division of the true TOTAL principal generator constructs each real next STRICT principal generator. Proved whole ideal equality gives the true next membership/span dictionary. No next STRICT/coherence/nonzero/principal certificate is assumed. □

Theorem 92.3 (Real accumulated exceptional geometry requires and uses actual whole-list centre NC)**.** *If the genuine old carrier is transverse to its actual SNC exceptional list, and the real next centre lies in that carrier AND has NC with that ENTIRE list, the actual next strict carrier is transverse to the precise ordered old strict list followed by the actual new exceptional member, and that whole list is SNC.*

Proof. Form the literal mark-zero old-carrier/actual-exceptional-list/actual-coherent- ideal record. Genuine SNC/transversality and coherence prove wellformedness. Real zero-mark cosupport is the entire old carrier, so true centre containment and the actual whole-list NC hypothesis prove genuine marked admissibility. The actual wellformed transform theorem now gives precisely both claimed geometric conclusions on the SAME real projection. No boundary NC is inferred merely from centre containment or from the auxiliary empty boundary. □

Theorem 92.4 (Only original inputs give the true first-to-second chronological STRICT consumer on the SAME actual first blowup)**.** *ONLY the genuine original compact smooth fixed-positive-dimensional carrier, SNC, whole-boundary density and nonempty FIRST Z construct the SAME first blowup. For EVERY original member and EVERY actual next blowup whose true centre lies in its actual carrier and has NC with its ENTIRE accumulated exceptional list, the genuine next chronological STRICT of its ACTUAL first STRICT has the preceding coherence, nonzero, principal and normalized-factor properties, together with precise true accumulated exceptional SNC and transversality. No desired second STRICT dictionary is supplied.*

Proof. Internally construct the same actual first total record and true raw-total STRICT, whole equality/coherence/nonzero, and actual relative principal residual generators. Apply the second producer to these real original objects and the actual next projection. Apply the third to the independently verified prior SNC/transversality and the REAL whole-list centre NC input. This constructs a true chronological consumer without selecting the next centre or assuming its desired transformed ideal. It does not yet compare with saturation by ALL accumulated exceptionals of the original composite raw TOTAL, prove dynamic DISTINCT driver admissibility/decrease, recursive intersections, exceptional τ_D or termination; these remain OPEN. □

93 Literal ORIGINAL full-product composite TOTAL stripped along ALL exceptional members

The per-member composite raw-TOTAL dictionary is established. The modified sharp first coordinate uses the entire true product, so it needs its OWN whole residual comparison, retaining genuine multiplicities and all original positions. Prove the true original product normalization before its finite-history factor induction; no product residual certificate is assumed.

Lemma 93.1 (True finite relative product of original restricted ideals is the whole restricted original product)**.** *For ANY genuine smooth closed carrier and actual finite list of coherent original ideals, the relative finite product of their actual carrier restrictions equals the WHOLE carrier restriction of their literal original ambient ideal product. Empty lists and all carrier-normal germs are retained.*

Proof. Induct on the genuine finite list. For the empty list the actual holomorphic unit ideal and its true carrier restriction are equal in both full germ directions. For a nonempty list, the first source restriction has BOTH genuine restricted-span relations with its original ideal. The induction equality supplies BOTH such relations for the true tail product. Multiply these actual finite span relations, using independently proved coherence of both original factors and actual restricted extensions. Normalization on the closed carrier gives WHOLE equality of the true relative product and the restriction of the literal ambient product. No choice of generators or equality of orders is substituted. □

Theorem 93.2 (Actual full-product original zero-mark history residual is the independently chronological full product)**.** *ONLY original smooth carrier, true original ambient SNC boundary and genuine density outside its ENTIRE boundary internally give, on EVERY real finite geometric-template history, the following WHOLE ideal equality. The intrinsic ALL-accumulated-exceptional nonmonomial part of the actual zero-mark TOTAL history of the ORIGINAL full ambient member product restricted to the carrier is the independently defined actual chronological full member product. Genuine multiplicities, empty-family unit, every ordered boundary position and actual empty steps remain.*

Proof. Original ambient SNC geometry independently gives genuine full-product coherence and nonzero on the actual carrier by true whole-boundary density. Construct its true initial restricted zero-mark record. For the reflexive history, the actual empty-boundary nonmonomial colon is exactly its carrier restriction; real restriction idempotence and the preceding whole product normalization identify it with the actual chronological initial product.

At a real next step, independently actual raw-history geometry gives its whole coherent nonzero smooth/SNC/transverse record. Genuine template field identities supply centre containment and NC with ALL old members. Apply true one-step intrinsic nonmonomial TOTAL transport BEFORE using the induction equality. That equality identifies the old raw residual with the independently constructed chronological full product. Genuine whole chronological geometry, prime-derived excluded product germs and actual NEW-saturation/product transport already identify its next TOTAL ALL-boundary residual with the true next chronological full product. The actual zero mark and exact raw geometric fields finish the full original-history equality, without declaring the original raw tower canonical for an earlier invariant. □

Theorem 93.3 (WHOLE intrinsic residual of literal composite ORIGINAL full-product TOTAL)**.** *Under ONLY the preceding original inputs, pull the literal ORIGINAL full ambient member product back by the ACTUAL composite tower projection. Its intrinsic nonmonomial part along the SAME actual final strict carrier, computed with respect to the ENTIRE actual ordered accumulated exceptional boundary, equals the independently defined actual chronological full member product in BOTH entire ambient germ directions. No caller product factor, residual, canonical history or comparison certificate is supplied.*

Proof. The independent literal composite projection theorem identifies the actual original restricted zero-mark full-product history with the WHOLE carrier restriction of the ORIGINAL product pullback. Its nonmonomial part is the actual chronological full product by the preceding genuine finite-history induction. Coherent pullback supplies BOTH true restricted span relations between the literal raw pullback and its actual carrier restriction, so actual intrinsic nonmonomial restricted congruence removes only the redundant outer restriction inside that construction. It

retains ALL normal-vanishing germs of the resulting WHOLE ideal. This is the required original full-product composite residual dictionary, not a comparison of earlier ORIGINAL complete invariant values or their histories. That comparison, controlled/test-prefix extension, modified decrease, recursive intersection and exceptional tauD phases and sharp termination remain OPEN; FIRST Z differs from full sharp Sigma. □

94 Literal composite ORIGINAL TOTAL pullback and the true ALL-exceptional residual

The actual mark-zero history and the separately defined chronological ideals have now been compared. Here independently identify that history with the literal pullback by its ACTUAL composite projection. This is a WHOLE relative-ideal statement; the real outer carrier normalization is retained before intrinsic monomial division.

Lemma 94.1 (Actual carrier normalization commutes with a genuine contained pullback)**.** *For a genuine holomorphic map carrying a closed target carrier into a smooth source carrier and any coherent source ideal, restricting its pullback AFTER the source carrier restriction gives exactly the SAME whole ideal as restricting the direct pullback to the target carrier. No submersion, desired transformed dictionary or equality of zero sets is assumed.*

Proof. The true source ideal and its actual restricted extension have both restricted-span inclusions on the source carrier; the latter is coherent by actual smooth-carrier vanishing-ideal coherence. Compose their actual finite holomorphic span relations with the genuine map. Real carrier containment keeps every resulting relation on the actual target carrier. This gives BOTH restricted pullback directions, with their original coefficients and generators. Genuine normalization on the closed target carrier gives the WHOLE equality, including its actual off-carrier units. □

Theorem 94.2 (Actual zero-mark history is the literal composite raw TOTAL on its true strict carrier)**.** *Take any actual initial smooth carrier, original coherent ideal and original ordered boundary, and the initial zero-mark record with its true restricted ideal. Along EVERY genuine geometric-template test history, its ACTUAL transformed WHOLE ideal is exactly the restricted extension, on the SAME actual final strict carrier, of the ORIGINAL ideal pulled back by the literal composite tower projection. Every actual empty step and ordered exceptional position is retained. No raw-TOTAL history or composite factorization certificate is supplied.*

Proof. Induct on the actual tower. The reflexive projection is the identity and genuine coherent pullback by the identity recovers the actual initial restricted ideal. Before a real next step, independently proved entire- tower geometry supplies actual smoothness of the current carrier and coherence of the literal composite pullback. Its actual strict transform is closed smooth and maps into that current carrier by genuine strict- transform containment. The original mark has remained zero. Actual zero-mark division by the true zero power gives exactly the normalized next raw pullback. Substitute the REAL preceding whole equality, apply the preceding normalization theorem, and use genuine coherent pullback composition. The result is precisely the pullback by the literal NEW composite projection, restricted on its SAME strict carrier. No position is compressed and no centre admissibility for the candidate ideal is assumed. □

Theorem 94.3 (Literal composite ORIGINAL member TOTAL stripped along ALL actual accumulated members equals chronological STRICT)**.** *ONLY the original smooth carrier, genuine original ambient SNC boundary, true density off the ENTIRE original boundary and a real finite geometric- template history internally give the following for EACH original boundary position. Pull its ORIGINAL ambient member ideal back by the ACTUAL composite projection, take its*

intrinsic nonmonomial part on the SAME actual strict carrier with respect to the ENTIRE actual ordered accumulated exceptional list: this WHOLE ideal equals the independently constructed chronological NEW-exceptional-saturation member ideal. All ambient normal germs and empty positions remain; no desired composite factor, residual or history comparison is input.

Proof. The original member vanishing ideal is coherent by genuine original SNC geometry. Apply the actual composite raw-TOTAL formula to its original restricted zero-mark record. Actual finite original-member geometry independently supplies the whole chronological ideal, and the proved finite TOTAL-history residual dictionary identifies it with the actual raw-history nonmonomial part. Finally the genuine monomial and nonmonomial constructions depend only on the TRUE restricted ideal: both actual restricted-span directions between the raw pullback and its carrier-normalized extension are proved from coherent generators. Thus the outer restriction in the raw-history formula can be removed INSIDE the intrinsic nonmonomial construction without deleting any normal germs from its whole ideal. This gives the literal composite-projection ALL-exceptional dictionary. Earlier ORIGINAL full invariant/presentation comparison, modified decrease, recursive intersection and exceptional tauD phases and sharp termination remain OPEN. FIRST Z still differs from the full sharp defect Sigma. □

95 The genuine ALL-boundary positive-mark controlled residual is the ACTUAL next chronological full product

The actual finite-colon saturation comparison and real controlled relative principality are independently proved. This chapter connects them to the TRUE nonmonomial part of the actual controlled transform, retaining the original positive mark and the ENTIRE accumulated boundary. It does not identify the reselected current invariant with an earlier original history.

Lemma 95.1 (Actual TOTAL germs enter the genuine controlled transform and retain OLD strict-trace exclusion)**.** *For an actual coherent old marked ideal and REAL blowup, EVERY actual normalized TOTAL germ belongs to its ACTUAL controlled transform: test EVERY true finite denominator-power germ and multiply the same real normalized raw-pullback relation. If the old ideal has genuine excluded germs along all old members and the centre is REAL admissible, true old strict-trace continuation therefore supplies controlled excluded germs. Consequently the controlled transform's WHOLE accumulated monomial part equals its NEW-singleton monomial part. No NEW-exceptional exclusion is assumed for the controlled transform.*

Proof. Unfold actual TOTAL and controlled-colon membership. Actual coherence of the real denominator power proves EVERY denominator germ holomorphic; multiplication preserves the true carrier-normalized numerator relation. Thus TOTAL is contained in the SAME actual controlled ideal in full ambient germs. Independently true old strict-intersection continuation gives an actual TOTAL excluded germ along each old strict trace. Its actual controlled membership retains the SAME excluded germ and SAME trace. Genuine transformed coherence and the fixed-member continuation theorem force exactly EACH OLD incident exponent zero. The actual ordered-list insertion dictionary makes all old monomial constraints zero-power unit constraints; the NEW constraint remains EXACTLY the singleton one. Prove BOTH complete monomial germ directions, including off-carrier unit cases. □

Theorem 95.2 (Genuine controlled geometry identifies the WHOLE ALL-boundary nonmonomial part with the independently defined next relative STRICT)**.** *For an actual coherent nonzero relatively principal old marked ideal with smooth carrier and SNC/transverse old boundary, genuine old-member exclusion and REAL admissibility internally prove that the WHOLE nonmonomial part of the ACTUAL geometric controlled transform for the ENTIRE accumulated list equals the genuine next raw-pullback relative STRICT. The true original mark is retained. No desired residual, controlled-principal or saturation certificate is input.*

Proof. FIRST construct genuine transformed wellformedness/coherence and true nonzero-along geometry. The independently proved actual source/Hadamard/ invertible-along construction supplies real controlled relative generators. Only THEN the genuine singleton theorem identifies actual NEW-divisor all-power saturation with the true NEW-singleton nonmonomial residual; this step is not valid for arbitrary nonprincipal ideals. The preceding actual old-trace producer proves the full accumulated monomial part equals that singleton denominator. Thus the FULL nonmonomial colon equals the same actual singleton residual at EVERY ambient point. Finally the independently proved real finite-colon saturation invariance identifies this genuine saturation with the independent raw-pullback relative STRICT. All objects are those of the SAME actual geometric transform and mark. □

Theorem 95.3 (ONLY original chronological geometry internally gives the real current full-product controlled residual dictionary)**.** *ONLY original smooth closed carrier/SNC/whole-boundary density, an ACTUAL finite geometric history, and a REAL admissible blowup of its actual CURRENT full-product mark-one record internally prove that this genuine controlled transform's WHOLE ALL-accumulated-boundary nonmonomial ideal is EXACTLY the independently constructed next chronological full product. No current ideal/generator/exclusion/residual dictionary is supplied.*

Proof. Original-only finite-history geometry independently provides the actual current carrier and entire ordered boundary, actual full-product coherence, nonzero and real relative generators. Genuine relative member primality supplies its EVERY-old-member excluded germs. Apply the preceding genuine controlled transform producer to THOSE internally constructed objects and the SAME actual admissible centre/map. Real whole STRICT/product commutation and literal chronological recursion identify its residual with the ACTUAL next product, preserving every original position and actual empty step. This supplies a true positive-mark transform/residual bridge. Composite ORIGINAL raw-TOTAL factor transport, comparison with an earlier original invariant/history, modified decrease, recursive intersection/exceptional cleanup and sharp termination remain OPEN. FIRST Z is not full sharp Sigma. □

96 Actual controlled relative principality from genuine source equations and admissibility

A literal saturation comparison does NOT make an arbitrary controlled ideal relatively principal. This chapter proves actual controlled principality from genuine old relative generators and real admissibility, before an intrinsic positive-mark residual comparison is attempted.

Lemma 96.1 (A real invertible-along denominator and true restricted divisibility construct the actual quotient generator)**.** *Suppose the ACTUAL carrier-normalized numerator has a real relative principal generator. A denominator genuinely invertible ALONG that carrier, and actual restricted inclusion of the numerator into it, internally construct a true holomorphic quotient generator and BOTH full relative colon directions. No quotient generator, cancellation, principal quotient or desired ideal dictionary is input.*

Proof. Choose the genuine invertible-along local denominator equation and its real nonzero points in EVERY relative carrier neighborhood. The actual numerator generator belongs to its true normalized ideal. Its real restricted divisibility gives a finite span of ACTUAL denominator germs. Use their genuine cofactors to construct a holomorphic function whose product with the SAME denominator equation equals that SAME numerator generator on the carrier. Actual multiplication and germ congruence prove this constructed cofactor belongs to the real relative colon.

For EVERY actual quotient germ, test the SAME denominator equation. The true numerator principal dictionary expresses the resulting product as a holomorphic multiple of the SAME

numerator generator. Substitute the actual constructed factor relation. Real invertible-along nonzero cancellation then gives the desired relative span of the true cofactor. Conversely multiply that actual quotient-generator relation by arbitrary true denominator germs. Thus BOTH full relative germ directions hold, retaining every carrier-normal germ. □

Theorem 96.2 (ONLY genuine original relative source equations and REAL admissibility internally supply actual positive-mark controlled relative generators)**.** *For an actual coherent marked ideal with smooth closed carrier, true old relative generators at EVERY carrier point, a REAL admissible centre and its REAL blowup, the ACTUAL geometric transform has genuine relative principal generators at EVERY point of its SAME actual strict carrier. The actual mark and ordered boundary are retained. No controlled ideal, new generator or principality certificate is assumed.*

Proof. Real strict-carrier projection maps the actual new carrier into the true old carrier. Compose the independently supplied old relative generator and its ACTUAL cofactor dictionary with the real projection; genuine normalized TOTAL geometry supplies BOTH new relative numerator directions. Original ideal coherence, actual smooth-centre admissibility and the true Hadamard/order lemma give restricted inclusion into the ACTUAL centre-ideal power. Pull this real inclusion to the SAME strict carrier using the actual holomorphic projection and the genuine pullback-power identity. The real blowup's actual exceptional power is independently invertible ALONG this carrier, with its true nonzero clause. Apply the first producer to THOSE genuine numerator and denominator objects. Unfold the ACTUAL geometric transform colon: the constructed function and BOTH full germ directions are EXACTLY its true relative generator dictionary. This supplies the missing principality input; intrinsic all-boundary residual, original modified-history comparison and sharp termination remain OPEN. □

97 Actual finite controlled division preserves the literal NEW-exceptional relative saturation

The actual geometric transform divides a genuine raw pullback by a FIXED exceptional power, whereas chronological relative STRICT uses ALL powers. The missing comparison is an actual ideal-germ algebra statement, not an invariant decrease claim. This chapter proves it before comparing any positive-mark intrinsic nonmonomial residual.

Lemma 97.1 (A true ambient principal denominator gives a genuine generator test and actual relative-colon normalization)**.** *An ACTUAL ambient principal denominator, with its real holomorphic equation and BOTH full ambient ideal directions, tests a relative colon by that SAME equation. Its real relative colon is already carrier-normalized in BOTH full germ directions. No principal, closure or quotient certificate is assumed beyond the actual independent denominator dictionary.*

Proof. The forward generator test uses the true equation's actual denominator membership. Conversely EVERY actual denominator germ has a genuine holomorphic cofactor times that equation. Multiplication by this actual cofactor and real germ congruence prove the required normalized numerator membership. For normalization, a true finite carrier-relative span of colon germs remains in the colon: multiply each ACTUAL generator relation by every denominator germ, retain the true coefficient functions, and flatten the resulting double relative span. The reverse uses the actual colon germ itself with coefficient one. □

Theorem 97.2 (Literal all-power relative saturation is unchanged by a real finite denominator-power colon)**.** *For the genuine ideal of an actual locally principal divisor and ANY actual numerator ideal, the independently defined all-power relative saturation of its finite-power relative colon equals its original all-power relative saturation in BOTH full germ directions. All finite exponents, including zero, are covered. No saturation-equality certificate is input.*

Proof. Choose the ACTUAL divisor equation. A literal saturation witness for the finite colon first yields a true colon germ after real normalization. Test its two genuine denominator powers in succession. Actual power addition gives a true numerator-colon witness at the SUM of those powers. Conversely, from a genuine numerator saturation witness, EVERY extra finite denominator power multiplies its actual normalized numerator relation into another such relation. The real generator test supplies the finite colon, and true relative normalization supplies its saturation witness. This proves the equality before any geometric or HS comparison. □

Theorem 97.3 (A REAL blowup internally identifies its actual positive-mark geometric-transform saturation with genuine raw-pullback relative STRICT)**.** *For ANY actual marked ideal and a REAL blowup, the WHOLE NEW-exceptional all-power saturation of its ACTUAL geometric transform ideal equals the independently defined next relative STRICT of its GENUINE raw pullback. The actual mark, carrier and EVERY ambient germ are retained. No desired transform/STRICT equality, coefficient presentation or invariant-history certificate is input.*

Proof. FIRST prove the real blowup centre-ideal pullback is the literal NEW exceptional vanishing ideal: actual local blowup equations give BOTH germ directions at incident points, while true off-divisor unit germs give BOTH directions elsewhere. Genuine blowup SNC geometry supplies an ACTUAL ambient divisor equation at EVERY point. Unfold the actual geometric transform: its numerator is the SAME raw pullback, its carrier the SAME strict carrier, and its denominator the ACTUAL centre-ideal pullback to the original mark. The independently proved real divisor-ideal dictionary and finite-colon invariance identify its genuine all-power saturation with the literal raw-pullback saturation. This is EXACTLY the independent next relative STRICT definition. No admissibility, residual or decrease result is inferred from this raw ideal comparison. Intrinsic ALL-boundary positive-mark residual comparison and sharp termination remain OPEN. □

98 A genuine decreasing ambient-contact budget above actual exceptional-difference points

The literal τ_D of [4, Lemma 12.8, p. 295] need not decrease at every blowup: a remaining exceptional power can require several steps. On a curve the genuine ambient contact orders give an independent natural-valued budget. This chapter proves its decrease for ACTUAL original-admissible steps; it does not supply a selected centre or replace the refined monomial driver by a desired decrease certificate.

Theorem 98.1 (The actual original-member ambient-contact budget loses exactly the actual incident-position count)**.** *Define the ACTUAL ambient-contact budget, independently of any decrease, by the literal sum over ORIGINAL member positions*

$$B_T(x) = \sum_{i\in \mathrm{Fin}\,|E|} \mathrm{ord}_{N_T,x}(\mathrm{Res}_{N_T}\,\mathcal{I}_{\mathrm{strictAlong}(T,E_i)}),$$

and define $A_T(x)$ as the actual finite set of ORIGINAL positions whose literal ambient strict member contains x. ONLY original closed smooth curve W, actual ambient SNC E, true whole-family density, ANY ORIGINAL mark-one product test T, an ACTUAL next original-admissible centre C and the SAME genuine blowup π internally give, at EVERY new strict-carrier point z above C,

$$B_{T'}(z) + |A_T(\pi z)| = B_T(\pi z), \qquad T' = T + (C,\pi).$$

This is the sum of TRUE ambient trace orders, not residual orders or a distinct-trace quotient count; all original positions and multiplicities are retained, including the empty-family case.

Proof. The preceding original-only genuine mixed-centre theorem internally supplies, for EACH original position, its actual new ambient trace order plus the true incidence indicator equal to its actual old normalized ambient trace order. Sum THESE genuine equations over the literal finite original index set. The actual step equations for the unchanged geometric template and EXISTING ambient strictAlong identify the summed new terms with the independently defined new budget. True finite sum distributivity and the genuine indicator/cardinality identity identify the remaining sum with the ACTUAL filtered original-position cardinality. Thus the displayed equation is proved from the real geometry, not supplied as an input invariant property. □

Theorem 98.2 (The genuine natural budget strictly decreases above every actual old exceptional-difference point)**.** *Under the SAME ONLY original hypotheses and ACTUAL original-admissible next step, if z is on its true new carrier, $\pi z \in C$ and $\pi z \in \Delta_T$ for the independently defined actual I/J exceptional difference, then*

$$B_{T'}(z) < B_T(\pi z).$$

No budget-decrease, nonempty-index or desired exceptional-difference support certificate is supplied. Membership in the genuinely defined old difference is the literal point condition, not an assumed transformation law.

Proof. The independently proved exact actual difference support identifies the old point with a point of the actual carrier, an actual accumulated exceptional member and at least one LITERAL original ambient strict member. The last condition supplies a genuine ORIGINAL position in $A_T(\pi z)$; hence that actual finite cardinality is positive. The first theorem independently gives the exact budget equation on this SAME blowup at this actual point. Natural-number arithmetic with this positive cardinality yields the strict inequality. No assertion that τ_D itself drops is made.

The actual choice of centres INSIDE Δ_T, their ORIGINAL positive record admissibility, the full old-exponent dictionary and a global refined driver/termination remain separate OPEN producers. In particular, mere containment in the current carrier does not establish original positive-record admissibility. FIRST Z, this actual difference and full sharp Σ remain distinct, and the real $I = DJ$ normalization is retained throughout. □

99 Actual original controlled order dominates the true curve contact budget

Retain the SAME smooth original curve W, ordered original ambient SNC family E, true density of $W \setminus \bigcup E$, and the ACTUAL original mark-one product record. For ANY genuine ORIGINAL test history T, write Q_T for its unchanged controlled record, N_T for its true carrier, B_T for the independently defined sum of actual ambient strict-member contact orders, and Δ_T for the independently defined actual ambient-trace/chronological-ideal difference locus. These are the source objects in [4, §12]; FIRST Z and the sharp Σ are not identified with Δ_T.

Lemma 99.1 (True original member incidence detects positive contact budget)**.** *ONLY the original curve/SNC/density and ANY genuine original test internally give, at EVERY actual current carrier point x, zero actual budget when no original strict member contains x, and positive actual budget when at least one original strict member contains x.*

Proof. The original bound-one domain and real test supply the literal template, genuine discrete centres and each actual current strict member's closed smoothness, trace coherence and nonzero. For a member absent at x, the actual normalized vanishing ideal contains the unit germ. Its true ordinary order is zero: a positive order would put that unit in the actual first point power, contradicting its value one at x. For an incident member, every genuine relative finite-span germ vanishes at x: evaluate the actual span on the true carrier at x, where each vanishing-ideal

generator is zero. The true mark-one cosupport dictionary therefore proves its actual ordinary order at least one. Summing the zero orders gives the first assertion. A genuine incident original position and nonnegative finite summands give the second assertion. No branch count, order value or support certificate is supplied. □

Theorem 99.2 (Every actual original curve test has controlled ordinary order at least its true ambient budget)**.** *ONLY the original curve/SNC/density and ANY genuine ORIGINAL mark-one product test internally prove, at EVERY actual current carrier point,*

$$B_T(x) \leq \mathrm{ord}_{N_T,x}(Q_T.\mathcal{I}).$$

The order on the right is the EXISTING ordinary order of the ACTUAL controlled ideal of the unchanged ORIGINAL record.

Proof. Induct on the ACTUAL original test history. Initially the genuine original product-order additivity theorem gives exactly the sum of the original member contact orders. The independently proved ordinary-order normalization dictionary identifies every actual normalized trace term, so the two sides are equal.

For an actual next step, original well formedness/nonzero and the real strict-carrier dictionary supply all current objects and the induction point. Off centre, the independently proved actual controlled ordinary order equality and true whole ambient budget equality transport the induction inequality. Above centre, let k be the actual number of incident ORIGINAL strict positions. The already proved real equations are $a' + 1 = a$ for the controlled orders and $B' + k = B$ for the true budgets. If $k > 0$, these equations and $B \leq a$ imply $B' \leq a'$ by natural arithmetic. If $k = 0$, the first lemma gives $B = 0$, and the genuine budget equation gives $B' = 0$, so the same inequality follows. Thus every actual centre, empty step and original position is retained; no canonicality, rebasing, selected-centre or desired comparison is assumed. □

Theorem 99.3 (The whole literal ambient difference locus lies in the SAME original positive cosupport)**.** *ONLY the same original inputs and ANY genuine ORIGINAL test internally prove* $\Delta_T \subset \mathrm{cosupp}(Q_T)$*. Thus the whole actual ambient difference set, rather than merely its carrier, is compatible with the unchanged original positive mark.*

Proof. The independently proved entire difference-locus dictionary gives a TRUE incident original strict member at each point of Δ_T. The first lemma supplies actual budget at least one, and the second supplies actual controlled ordinary order at least that budget. Genuine current domain inheritance supplies coherence/nonzero and the actual closed carrier before the ordinary-order/cosupport dictionary. The unchanged original mark remains exactly one along the whole test, so that dictionary puts the point in the SAME actual original cosupport. No desired support inclusion, unit-factor replacement or new marked record is supplied. Selection of the entire admissible difference centre, its finite erasure phase, refined higher-dimensional monomial scheduling and the full sharp Σ theorem remain subsequent OPEN producers. □

100 The true controlled exceptional factor and exact curve ordinary-order shift

Keep the SAME actual original marked record and genuine controlled transform. The ordinary order of its true normalized ideal, unlike its nonmonomial order, records remaining exceptional powers. The actual factor identity and curve shift below are genuine inputs for the source monomial phase of [4, Lemma 12.8, p. 295]; no desired factor, order change or driver certificate is supplied.

Theorem 100.1 (Actual coherence and admissibility construct the whole controlled mark-one factor)**.** *For ANY genuine coherent ideal its actual first ideal power is the SAME whole ideal.*

For ANY actual mark-one record with closed smooth carrier and coherent ideal, an ACTUAL admissible centre and the SAME genuine blowup π, put $N' = \mathrm{Strict}_\pi N$, $D = \pi^{-1}C$ and let Q be its ACTUAL controlled transform. Then as WHOLE actual ideals,

$$\mathrm{Res}_{N'}(\pi^* I) = \mathrm{Res}_{N'}(\mathcal{I}_D\, Q.I).$$

The outer carrier normalization is essential. No principality or nonzero certificate for the numerator or desired factorization is assumed.

Proof. For the first-power statement use the literal unit-times-ideal finite-span definition. True coherent membership is closed under the actual holomorphic coefficients, proving one whole direction. The actual unit times each original member and its genuine self-span prove the reverse direction.

Actual source admissibility and genuine Hadamard divisibility supply the real restricted inclusion of the raw pullback in the pulled-back centre ideal power. Mark one and the just proved true first-power equality, followed by the genuine whole pullback-centre/exceptional vanishing dictionary, make this precisely inclusion in $\mathcal{I}_D$. Real exceptional geometry makes this actual denominator invertible along the true strict carrier. The two independently proved relative-colon product directions therefore give BOTH actual restricted inclusions between the raw numerator and denominator times its actual colon. Genuine closed-carrier normalization turns them into the displayed WHOLE ideal equality. The same mark-one and first-power dictionaries identify that actual colon with the unchanged original controlled ideal. □

Theorem 100.2 (Actual curve raw-order transport and the genuine factor give exact controlled order drop)**.** *Assume the actual mark-one record is genuinely well formed, its carrier is a genuine pure smooth curve, its coherent ideal is genuinely nonzero along that carrier and its TRUE normalized source ideal has BOTH relative principal directions. Let C be an ACTUAL original-admissible discrete centre and π the SAME genuine blowup. At EVERY actual strict-carrier point z above C,*

$$\mathrm{ord}_{N',z}\, Q.I + 1 = \mathrm{ord}_{N,\pi z}\, I.$$

This is the EXISTING ordinary relative order of the ACTUAL controlled ideal, not nonmonomial order or a substituted trace ideal.

Proof. Actual well formedness and real admissibility supply new carrier/coherence, and genuine nonzero preservation supplies true new nonzero. The preceding normalized-source generator theorem constructs BOTH relative principal directions of the actual new controlled ideal. Actual NEW exceptional SNC and true carrier transversality independently construct its normalized vanishing ideal, genuine nonzero principal generators and actual order one at every incident carrier point.

The first theorem gives the WHOLE raw factor. True relative product normalization replaces its denominator by its actual carrier restriction in BOTH directions. Genuine relative principal product-order additivity, with the independently verified nonzero factors, therefore identifies the actual ordinary raw order with one plus actual controlled order. The essential outer normalization is removed only by the genuine ordinary-order normalization theorem. Independently the true curve strict-projection inverse, constructed from actual curve/centre geometry, preserves the raw relative ordinary order ABOVE the centre. Equating these two genuine expressions gives exactly the displayed drop. □

Theorem 100.3 (ONLY original curve product tests internally give every actual controlled ordinary-order shift)**.** *ONLY original closed smooth curve W, actual ambient SNC E, true whole-family density, ANY ORIGINAL mark-one product test T, an ACTUAL next ORIGINAL-admissible centre and its SAME genuine blowup internally supply the preceding exact ordinary-order drop for the ACTUAL original controlled record. No current principal/nonzero/pure-curve/discreteness or desired order-change certificate is supplied.*

Proof. Original smooth curve/SNC/coherence and genuine density internally construct the original mark-one dimension-one domain. Its genuine TEST preserves actual current well formedness/nonzero. The SAME actual original geometric template and genuine strict-carrier dimension preservation give the actual pure current curve, using the independently proved same-carrier dictionary before substitution. The actual next original TEST internally gives real discreteness of its centre. The preceding actual original ANY-mark/ANY-TEST normalized principal-history theorem constructs BOTH normalized current principal directions on this SAME unchanged record. All genuine premises of the second theorem are therefore supplied from the original inputs; apply it to the actual next centre and map.

The remaining comparison with the summed TRUE ambient strict-member budget and the positive-record admissibility of a selected whole τ_D centre are separate OPEN producers, as are global refined monomial scheduling and full sharp Σ termination. This theorem does not reset the original invariant or assume it is canonical. □

101 The actual original-member exceptional difference on a curve

We now use the genuine finite-history factor dictionary $I_{i,T} = \mathrm{Res}_{N_T}(D_{i,T}J_{i,T})$ for the EXISTING ambient strict member and independently chronological relative STRICT ideal. Following [4, Lemma 12.8, p. 295], count ORIGINAL member positions at which these actual ideal germs differ. This is the literal τ_D comparison; it is not the FIRST distinct-trace defect and no identification with the whole sharp Σ is asserted.

Lemma 101.1 (Actual normalized residual and unit tests)**.** *For ANY carrier N, ideal A, finite actual family F, and point outside EVERY member of F, the actual normalized residual of* $\mathrm{Res}_N A$ *has exactly the same whole germ membership as* $\mathrm{Res}_N A$*. Independently, if H is closed and $x \in N$, the actual normalized vanishing ideal* $\mathrm{Res}_N \mathcal{I}_H$ *contains the unit germ at x exactly when $x \notin H$.*

Proof. Outside the whole family, the actual unit germ satisfies every literal monomial-part condition vacuously. Testing the defining colon with this unit gives membership in the twice-normalized ideal; genuine finite-span transitivity removes the second normalization. Conversely, multiplication by any actual monomial germ preserves the normalized ideal, and the genuine self-span supplies the extra carrier restriction required by the colon. For the unit test, membership is represented by an ACTUAL finite relative span. If $x \in N \cap H$, every representing vanishing germ evaluates to zero at x, so the real span equation would give $1 = 0$. If $x \notin H$, the actual open complement of the closed set makes the unit a vanishing germ there; its genuine self-span supplies normalized membership. These proofs concern full ideal membership, not only zero sets or orders. □

Theorem 101.2 (Original curve histories internally identify every true ideal-germ difference)**.** *Let W be an original closed smooth curve, E its genuine ambient ordered SNC family, with true density of W away the WHOLE family. Let T be ANY genuine ORIGINAL mark-one product test and let F_T be its actual accumulated exceptional list. Define the ACTUAL finite set of original positions*

$$B_T(x) = \{i \in \mathrm{Fin}\,|E| : (I_{i,T})_x \neq (J_{i,T})_x\}, \qquad \tau_D(x) = |B_T(x)|,$$

where inequality means failure of equality of ALL genuine germ-membership predicates, and put $\Delta_T = \{x \in N_T : \tau_D(x) > 0\}$. For EVERY $x \in N_T$ and original position i,

$$i \in B_T(x) \quad \Longleftrightarrow \quad x \in \bigcup_{D \in F_T} D \ \ \textit{and} \ \ x \in \mathrm{strictAlong}(T, E_i).$$

No comparison, unit or exceptional-factor certificate is supplied.

Proof. Original smooth-curve/SNC/density data internally construct the original dimension-one domain and the SAME geometric-template history. The previous finite-history theorem gives actual trace coherence and its WHOLE residual equality with the independent chronological $J_{i,T}$, and genuine chronological geometry gives coherence of $J_{i,T}$. Off the actual whole exceptional union, the preceding residual test proves the whole equality of these two germs. At an actual exceptional point, the independently proved curve no-factor theorem internally makes EVERY chronological member a true unit. The preceding actual vanishing-unit test says the true trace is a unit precisely off its literal ambient strict member. Coherence turns a unit into membership of EVERY holomorphic germ in each ideal, so the two whole germ predicates agree off that member. On the member the actual unit belongs to $J_{i,T}$ and fails to belong to $I_{i,T}$, proving true inequality. This covers every original position, retaining repeated original positions and every empty history step. □

Theorem 101.3 (The true original-position count and its exact whole exceptional-difference locus)**.** *Under those ONLY original hypotheses, at EVERY carrier point x, the actual set $B_T(x)$ is empty off the exceptional union and otherwise is exactly the original positions whose literal ambient strict member contains x. Consequently, as ENTIRE actual ambient sets,*

$$\Delta_T = N_T \cap \Big(\bigcup_{D \in F_T} D \Big) \cap \Big(\bigcup_{i \in \mathrm{Fin}\,|E|} \mathrm{strictAlong}(T, E_i) \Big).$$

The count is over original positions, not distinct trace classes or the new exceptional positions.

Proof. Apply the preceding independently proved membership equivalence to each position of the actual finite set. The true exceptional-membership case distinction gives exactly the stated filtered original-position set or the empty set. Genuine finite-cardinality positivity is equivalent to an actual member position. This is exactly membership in the final finite union, and the definition of Δ_T retains actual carrier membership in both directions. Empty original families and empty histories are included by the same literal finite-set argument. □

Theorem 101.4 (The exact actual difference is analytic and genuinely discrete)**.** *Under the SAME ONLY original hypotheses, Δ_T is a genuine global complex analytic subset and EVERY one of its points has an actual open neighbourhood isolating it, so the set is genuinely discrete. No compactness, supplied analytic difference set, desired closedness or local-constancy certificate is input.*

Proof. Genuine original-template well formedness gives the actual closed smooth current carrier and actual accumulated SNC family. The finite-history ambient trace geometry internally makes EVERY literal original strict member a genuine smooth ambient hypersurface. Thus all three sets in the proved exact formula are genuinely analytic: use actual finite unions of the independently verified smooth sets and genuine analytic intersections. At a point of this exact difference, choose an ACTUAL incident accumulated exceptional member. True current pure-curve dimension and actual carrier/exceptional transversality give a real open isolating its carrier intersection. The exact difference is a subset of the exceptional union; shrink this open to avoid every nonincident member using the finite closed family, and intersect the finitely many isolating opens for incident members. Then the actual difference in that open is the SAME singleton as the chosen point. This real open-neighbourhood equality proves genuine discreteness of the literal difference at all its points.

This supplies an actual source-facing τ_D support producer on curves. It does not assert that τ_D strictly drops at each blowup: even one remaining power may require several admissible steps. The real remaining exponent evolution, the refined monomial invariant of Lemma 12.8, its admissible centres and termination remain OPEN, as do the general sharp Σ theorem and any unsupported trace/residual support equality. □

102 The actual remaining exceptional factor on a curve has the computed contact exponent

The factor in $I' = D'J'$ is an actual greatest exceptional factor, not a replacement residual. Compute its genuine exponent and every relative member germ from the old contact using the real first-step geometry. This supplies a numerical input for the separate τ_D phase of [4, pp. 293–295].

Theorem 102.1 (A genuine transverse curve member makes nearby divisibility exactly the actual point order)**.** *For an actual closed smooth carrier, actual SNC boundary transverse to it, ANY coherent ideal nonzero along the whole carrier, and an actual incident point where the carrier is a genuine curve, the TRUE exponent along that actual member equals the EXISTING ordinary trace order at that point. No isolating neighborhood, order or exponent certificate is supplied.*

Proof. Genuine simultaneous SNC coordinates, actual curve dimension uniqueness and the real free member coordinate supply an actual neighborhood where the member's trace on the carrier is the singleton point. For EVERY integer q, the literal nearby-divisibility predicate implies point cosupport by evaluation at that actual point. Conversely point cosupport implies nearby divisibility: every carrier/member point in that actual isolating neighborhood is precisely the same point. Thus BOTH actual nearby-divisibility and point-cosupport predicates agree.

A real carrier retraction and the actual nonzero germ make its true exceptional exponent finite. The true maximal-divisibility equivalence and EXISTING finite ordinary-order/cosupport dictionary now apply to these independently constructed objects. Apply the two predicate directions at the actual exponent and at the actual ordinary order. The resulting two integer bounds prove equality, without an assumed discrete divisor model. □

Theorem 102.2 (The true first remaining exceptional factor has old curve contact minus one)**.** *For ONLY original closed smooth curve N, singleton SNC H, genuine density away from H, actual discrete $C \subset N \cap H$ and the SAME actual blowup, put $D = \pi^{-1}C$, $I' = \mathrm{Res}_{N'} \mathcal{I}_{H'}$ and let M' be its ACTUAL greatest monomial factor along $[D]$. At EVERY actual $z \in N' \cap D$, writing the actual old contact as $r = \nu_{\mathcal{I}_H,N}(\pi z)$, one has*

$$e_D^{N'}(I', z) + 1 = r.$$

Moreover EVERY ambient germ g belongs to the TRUE M'_z iff it is holomorphic and its restriction to the actual carrier lies in the span of the actual power $(\mathcal{I}_D^{r-1})_z$. This is a BOTH-direction full relative germ dictionary, retaining carrier-normal germs.

Proof. The original-input first-step geometry independently supplies actual strict-carrier smoothness, singleton exceptional SNC/transversality and true strict-trace coherence/nonzero data. The genuine strict-carrier dimension theorem internally preserves curve dimension at the actual point. Apply the first theorem to THESE actual ideals and member; this identifies the actual remaining exceptional exponent with the actual strict-trace point order. The independently proved exact strict-trace contact drop then gives the displayed numerical formula. In particular its true exponent is $r - 1$.

Now unfold the independently defined actual greatest monomial factor. Its singleton list has exactly the true member D; both actual carrier and incident memberships hold at z. Its defining full relative span condition is therefore precisely the actual D power at its true exponent. Substitute the already proved numerical value in BOTH germ membership directions. No power, monomial or desired generator model is assumed. This gives the genuine remaining factor even when the separate all-power residual J' is unit. Mixed local centres and chronological ambient factors, the global τ_D driver, full sharp Σ and termination are not inferred here. □

103 Actual chronological curve residuals are units at EVERY accumulated exceptional point

Remove the FIRST-good hypothesis from the exceptional-point assertion. On a real smooth curve transverse to the actual accumulated SNC boundary, each incident member cuts out an isolated point. A genuine no-factor witness therefore cannot vanish at that point. Apply this to the independently constructed chronological ideals; no order-one, reduced trace, previously-good point or unit certificate is assumed.

Lemma 103.1 (A true transverse member cuts an actual pure curve in an isolated point)**.** *For actual regular dimension one of N at x, an actual SNC boundary transverse to N, and a true incident boundary member H, internally construct a genuine ambient open neighborhood U with $(N \cap H) \cap U \subset \{x\}$. No intersection-dimension or isolated trace certificate is supplied.*

Proof. The real simultaneous carrier/SNC normal form supplies the ACTUAL carrier plane with cutoff k and a free incident coordinate of index less than k. Its actual regular dimension is $\min(k, n)$, by real flattening. Uniqueness of actual regular dimension compares it to the given original curve dimension, forcing $\min(k, n) = 1$. The incident free index is then zero. At any true chart point of $N \cap H$, coordinate zero vanishes by membership in H, while every other actual coordinate vanishes by the carrier-plane equation. Its actual flattened point is therefore zero, as is the original point's. The genuine flattening and original-chart inverse laws on their TRUE domains identify it with x. Their real common flat-domain open is the required isolating neighborhood. □

Lemma 103.2 (A genuine no-factor witness on the curve forces a real unit at an incident transverse member)**.** *Let I be any actual coherent ideal on a genuine smooth carrier N regular of dimension one at an incident boundary point. An actual germ of I outside the TRUE relative member-ideal span internally gives $1 \in I_x$. No nonzero value, order bound, reduced support, FIRST-good hypothesis or unit certificate is input.*

Proof. Choose the genuine no-factor witness $a \in I_x$. If $a(x) = 0$, the real isolating neighborhood of the preceding lemma makes a vanish on the literal trace $N \cap H$ near x. Actual holomorphicity of that SAME member and the independently proved BOTH-direction transverse member trace-ideal dictionary then place a in the actual relative span of the boundary ideal, contrary to its genuine witness property. Hence $a(x)$ is nonzero. Its true holomorphic inverse and real coherent ideal-span closure give $1 \in I_x$. No relative principality or assumed residual unit is used in this argument. □

Theorem 103.3 (ONLY original curve inputs make every chronological member a unit over the WHOLE accumulated exceptional family)**.** *ONLY original smooth pure curve, original ambient SNC boundary, whole-original-boundary density and ANY actual finite template history internally give this at EVERY actual current carrier point of the ENTIRE literal accumulated exceptional family: EVERY original chronological member contains the genuine unit germ, the actual full chronological product has ordinary trace order zero, and the point lies outside literal current FIRST Z. No previously-FIRST-good, member-unit, factor, current curve or selected history certificate is supplied. Empty original and exceptional lists and empty steps are retained.*

Proof. Actual finite template wellformedness gives the SAME real upper carrier, its actual transverse accumulated SNC list, and its closedness. True original curve purity transports through the SAME actual history. For each original position, independent chronological geometry supplies its actual coherence, and true chronological continuation gives a genuine no-common-factor witness for EVERY incident actual accumulated position. The preceding producer therefore constructs its actual unit at every such point, including points not assumed FIRST-good.

Induct over the actual full chronological member list: units in both actual factors give the true unit in their ideal product, and actual carrier normalization preserves it. The empty product is the original unit ideal. The real full-product ordinary-order dictionary now excludes order at least one by testing its actual unit germ, so its ordinary order is zero. Since the true distinct proper-germ count is nonnegative, it cannot be strictly smaller than this zero order; the literal FIRST-defect inequality therefore fails. FIRST Z is consequently disjoint from the ENTIRE actual accumulated exceptional family in this true curve scope. This does not erase Z elsewhere, identify it with Sigma, prove off-centre FIRST-defect transport, a full sharp driver or higher-dimensional termination. □

104 A genuine finite whole-ambient-difference phase on the original curve record

Keep the unchanged original smooth curve product record, true ordered ambient SNC family and whole-family density. Begin at ANY genuine ORIGINAL test prefix P. The phase below computes its actual centres from the literal full-germ $I_i \neq J_i$ difference, retaining all original positions. This is a scoped curve phase in [4, §12], not a declaration of the general sharp driver or a substitution of FIRST Z or sharp Σ for the actual ambient difference Δ.

Theorem 104.1 (Literal whole-difference histories are genuine original TESTs and satisfy the real length/order bound)**.** *Define a literal tail recipe recursively: the empty tail has no condition; at an actual next step its centre is EXACTLY the independently computed Δ_{P*S} of the current appended ORIGINAL prefix. Such a tail is internally a TEST of the SAME ORIGINAL record. If the actual controlled ordinary order at every point of Δ_P is at most b, then at EVERY actual endpoint $z \in \Delta_{P*S}$ one has*

$$\operatorname{ord}(Q_{P*S}, z) + \operatorname{length}(S) \leq b.$$

The bound is an explicit ordinary numerical induction hypothesis for this generic theorem; no terminality, selected-history or desired decrease certificate occurs in the literal recipe.

Proof. Induct on the actual tail. The empty tail is the given original prefix, and its displayed inequality is the stated numerical starting bound. At a next step, the induction gives the actual original prefix TEST. The independently proved entire-Δ admissibility theorem makes the recipe's literal actual centre original-admissible, hence constructs the actual appended TEST before any order comparison. A point of the literal new difference maps into the literal old difference by the separately proved geometric support localization. The actual SAME original controlled-order shift above this centre is $a' + 1 = a$. Apply the induction inequality at that actual old difference point, and use the literal tail-length successor equation. Natural arithmetic gives the displayed new inequality. Every genuine empty centre and step remains in this induction. No original invariant is rebased and canonicality is not asserted. □

Theorem 104.2 (True compact current cosupport internally supplies the actual ordinary-order bound)**.** *ONLY the original curve inputs, ANY genuine original prefix P and true compactness of its ACTUAL current positive cosupport internally produce an ordinary natural bound on EVERY actual point of Δ_P. No ambient or carrier compactness, caller maximum or caller ordinary-order bound is assumed.*

Proof. Actual original domain inheritance gives the unchanged current record's closed smooth carrier, coherent ideal and nonzero at every carrier point. The independently proved EXISTING ordinary relative order is genuinely upper semicontinuous there, hence on the actual current cosupport. If that compact cosupport is empty choose zero; otherwise its genuine finite neighbourhood-cover maximum supplies a natural bound. The previously proved whole-Δ positive cosupport inclusion restricts this bound to every actual starting difference point. All maxima and orders refer to the same actual controlled ideal. □

Theorem 104.3 (ONLY original data and compact CURRENT cosupport construct a finite actual whole-difference erasure tail)**.** *ONLY the original curve/SNC/density, ANY genuine original prefix and true compactness of its actual CURRENT positive cosupport construct a genuine finite geometric tail whose EVERY centre is the whole independently computed current Δ, whose appended history is a TEST of the SAME ORIGINAL record, and whose final ACTUAL difference is empty. The actual tail projection is proper; the actual final domain and precise ordered SNC boundary are internally inherited. The initial ambient, carrier and initial original cosupport may be noncompact.*

Proof. First obtain the genuine natural bound b from the preceding theorem. Independently construct a literal recipe tail of EVERY finite desired length by induction: at each stage its internally proved original TEST supplies the genuine global blowup of the ENTIRE actual current Δ, and hence the next actual manifold, map and step. No terminal tower is supplied to this construction. Take the real constructed tail of length $b+1$. If its actual final Δ contained a point, the first theorem would give a nonnegative ordinary order plus $b+1$ at most b, a contradiction. Thus the actual final full-germ difference is empty. Real original TEST inheritance gives the actual final domain and ordered SNC boundary; genuine finite blowup projection geometry gives properness. This is a constructed finite curve phase with exact original centres and a real termination proof, not a renamed endpoint certificate. It does not assert general noncompact unbounded phase termination, preservation or erasure of FIRST Z, the whole carrier/boundary transversality conclusion, general refined monomial scheduling, or completion of the sharp Σ theorem. □

105 Genuine one-step erasure of the entire curve FIRST defect

Continue the SAME ORIGINAL mark-one product test history. Prove actual proper-position and quotient-class counts, rather than inferring a distinct count from order transport alone. The one-dimensional lower geometry allows a direct FIRST-phase erasure: whole-family off-centre germ transport in arbitrary dimension remains a separate API. This is the curve instance of the FIRST phase at printed pages 293–295 of [4]; it does not identify FIRST Z with the full sharp obstruction Σ.

Lemma 105.1 (Small actual product order internally determines every proper position and the true distinct count)**.** *For a genuine finite coherent, nonzero, relatively principal family on a real closed smooth carrier, if its ACTUAL normalized product order at a carrier point is at most one, the cardinality of its ACTUAL proper positions and its TRUE quotient-image distinct proper-germ count BOTH equal that product order. In particular the point is outside the literal FIRST defect. No count, factor order, unit/nonunit or quotient-injectivity certificate is supplied.*

Proof. True relative product-order additivity expresses the actual product order as the sum of ALL actual factor orders. The real order-one/cosupport and coherent properness dictionaries identify positive factor order with membership in the actual proper-position set. Each factor order is at most the actual sum, hence at most one. A proper position therefore has order exactly one, while a nonproper position has order zero. Summing these actual indicator values shows that the actual proper-position cardinality equals the product order and is at most one. Any two actual proper positions are therefore equal. The true full ideal-germ quotient map is consequently injective on this ACTUAL set; its image cardinality equals the actual cardinality. This is the required genuine distinct count dictionary, including zero, unit and empty-family cases. Equality of count and order excludes the literal strict FIRST inequality. □

Lemma 105.2 (Original curve FIRST-good points have actual full product order at most one)**.** *ONLY original smooth pure curve/SNC/density/template-history inputs give actual product order at most one at EVERY current carrier point outside the computed FIRST Z. No already-transported count or order bound is input.*

Proof. The genuine computed FIRST-good open and actual reduced curve-trace geometry make distinct indexed traces pairwise disjoint. At a true carrier point, real coherent properness identifies EACH proper position with membership in its ACTUAL chronological trace. Two different proper positions would therefore put the SAME actual open-subtype point in two disjoint actual traces. Thus the ACTUAL proper-position cardinality is at most one, and the TRUE quotient-image cardinality is at most one. Outside the literal FIRST defect its defining strict inequality fails; product order is consequently at most this actual distinct count, and hence at most one. No assertion that indexed and distinct counts agree at bad points is made. □

Theorem 105.3 (Every genuine SAME ORIGINAL-record blowup of the entire curve FIRST Z has empty NEW FIRST Z)**.** *ONLY the original curve/SNC/density inputs, a genuine ORIGINAL mark-one test prefix and ANY actual blowup with centre EXACTLY the entire literal current FIRST Z imply that the NEW literal chronological FIRST defect is empty. The SAME ORIGINAL history is genuinely extended internally; no new defect, count, unit, order-transport or desired history certificate is assumed.*

Proof. The independently constructed whole-Z original admissibility appends THIS real centre and blowup to the SAME ORIGINAL test prefix. Actual original wellformedness internally gives the genuine new template history, with every original and empty position retained.

Consider any actual new defect point. If its image lies in the centre, the point belongs to the NEW exceptional member of the actual new ordered accumulated boundary. Genuine new-history curve residual-unit geometry gives actual full product order zero there, and therefore directly excludes the NEW FIRST inequality.

Otherwise the real strict-carrier/preimage dictionary places its image in the old actual carrier and outside old Z. The preceding original curve theorem bounds its old product order by one. The genuine SAME ORIGINAL controlled residual/order dictionary identifies old and new chronological product orders with their corresponding original current Nm orders. The independently proved off-centre actual blowup order theorem equates those original orders, so the NEW product order is also at most one. Now apply the FIRST lemma to the ACTUAL NEW coherent, nonzero, relatively principal family, internally supplied by genuine new chronological geometry. Its ACTUAL proper-position cardinality and TRUE quotient-image count equal this new order, excluding the NEW defect. Thus neither case contains any new FIRST-defect point. Order equality alone is not used to infer a count equality. □

Theorem 105.4 (ONLY original curve inputs construct an actual global FIRST-Z-erasing original test step)**.** *The SAME original-only inputs construct a genuine global proper holomorphic blowup of EXACTLY the ENTIRE literal current FIRST Z, preserving its ENTIRE complement in both isomorphism directions, appending the SAME ORIGINAL mark-one test prefix with actual transformed domain and full ordered accumulated SNC boundary, and with NEW literal FIRST Z empty. Compactness and canonicality are not required.*

Proof. Use the independently constructed actual whole-Z blowup, whose centre, proper projection, complement isomorphism, genuine original test append, dimension-one domain and literal ordered SNC boundary are already proved from the original inputs. Apply the preceding theorem to THIS SAME actual projection; it internally verifies the needed new template geometry and proves empty new FIRST defect. No replacement projection or selected history is introduced. The result closes this curve FIRST-phase erasure only: full ambient strict ORIGINAL-family SNC, full sharp Σ preservation, higher ORIGINAL invariant comparison, the general-dimensional sharp driver, exceptional τ_D and termination remain independent frontiers. □

106 The full ORIGINAL ambient boundary along a genuine curve test history

Distinguish the actual accumulated-exceptional template, whose initial boundary is empty, from the WHOLE original ambient SNC family transported through the SAME maps. Genuine dimension-one positive cosupports make all actual original test centres zero-dimensional. Their actual singleton geometry supplies normal crossings with ANY independently tracked ambient SNC family, even when the actual curve is not yet transverse to that family. This advances the full original-boundary geometry needed in the sharp preparation at printed pages 293–295 of [4].

Theorem 106.1 (Every genuine dimension-one original test history is an actual ambient-boundary template history)**.** *For ANY genuine original marked-domain input with dimension bound one, ANY actual ORIGINAL test history T, and ANY independent original ambient SNC ordered family B, the SAME maps and centres form an actual geometric test history for the genuine ambient template $(X, B, 1, 0)$. Its actual transformed carrier is the WHOLE actual source manifold, and its ENTIRE ordered boundary is SNC: every original strict member and every actual exceptional position is retained. No centre-dimension, additional boundary admissibility, current SNC or ambient history certificate is supplied. The original carrier need not be transverse to B.*

Proof. Induct on the ACTUAL original test history, retaining all maps and empty steps. The empty ambient template is genuinely well formed: the whole ambient manifold is smooth, the ideal is the true unit, and the original ambient SNC family is transverse to the whole space. At the next original step its actual domain is still of dimension bound one. Coherent nonzero thinness and analytic dimension drop make its genuine positive cosupport zero-dimensional. The actual original centre is a real subset of this cosupport. Its true singleton neighborhoods therefore give normal crossings with the independently tracked WHOLE current ambient SNC family from the induction hypothesis. This is proved from real geometry before claiming ambient-template admissibility.

The ambient-template mark remains zero, so its actual cosupport is its actual carrier. The strict transform of the WHOLE ambient space is the whole new space by genuine exceptional nowhere density; hence that carrier is the whole space at every stage. The original smooth centre is therefore in the actual ambient-template cosupport. Its actual closed smoothness and the just-proved real normal crossings make it admissible for this SAME ambient template. Append the SAME actual centre and map. Genuine transformed wellformedness supplies the literal full ordered strict family followed by the new exceptional member SNC. No curve transversality to this independently tracked family has been used or inferred. □

Theorem 106.2 (ONLY original curve inputs retain the entire ORIGINAL ambient boundary on their SAME original product prefix)**.** *ONLY original smooth pure curve/SNC/whole-boundary density and ANY actual ORIGINAL mark-one product test prefix supply an actual ambient template history beginning with that SAME full ORIGINAL SNC boundary. Its carrier is the entire actual ambient source and its whole ordered boundary is SNC. No current full-family SNC or history is a premise.*

Proof. Actual original product coherence and genuine original-density nonzero germs, together with real pure curve geometry, directly construct a true dimension-one original mark-one product domain. Apply the preceding producer to the GIVEN actual original test prefix and the SAME original ambient SNC family. It constructs actual additional-boundary admissibility at every original centre internally, retaining the SAME projection and positions. The resulting family is the true ambient strict-transform recursion from the ORIGINAL family, not the different template that begins with an empty boundary. □

Theorem 106.3 (The genuine curve FIRST-Z-erasing step also retains the WHOLE original ambient family SNC)**.** *The original-only actual global FIRST-Z-erasing blowup retains all its*

proved SAME ORIGINAL-record properties and additionally has the WHOLE actual ambient strict ORIGINAL family plus ALL accumulated exceptional positions SNC on the SAME final manifold. Compactness and canonicality remain unnecessary. It does not yet identify the chronological trace supports with intersections of the ambient strict members and curve.

Proof. Construct the actual original-only FIRST-Z-erasing projection, exact centre, whole-complement isomorphism, genuine original test append, transformed domain and empty NEW FIRST defect. Apply the preceding full-boundary history producer to THIS SAME actual extended original prefix. Its genuine new centres already satisfy every dimension-one domain condition internally, so it gives the actual complete ambient family SNC without changing the projection or deleting any position. The next actual comparison must retain the exceptional factor $I_{ij} = D_{ij}J_{ij}$; equality of chronological support with the ambient strict-member trace is generally false. For example, start with the curve $y = x^2$ and original member $y = 0$ in the plane, then blow up the origin. In the genuine chart $y = xv$, the strict curve is $v = x$ and the strict original member is $v = 0$, meeting on the new exceptional divisor. Yet all-power exceptional saturation of the restricted total x^2 is the unit ideal. The actual remaining exceptional factor must therefore be tracked before carrier transversality or the full sharp obstruction can be controlled. Actual $I_{ij} = D_{ij}J_{ij}$, its exponent evolution, the τ_D phase, whole final carrier transversality, full sharp Σ and the general sharp driver remain separate producers. □

107 The actual curve endpoint preserves the ENTIRE initial normal-crossings good region

The preceding SAME-projection isomorphism is over the initial FIRST-defect complement. Its extension to the literal original normal-crossings good open requires independent local recognition. We now prove that recognition from actual transverse coordinates and true finite-product orders. FIRST Z, the ambient/residual difference Δ and the complete sharp Σ remain distinct throughout.

Lemma 107.1 (A genuine transverse curve has at most one actual incident indexed boundary position)**.** *Given a genuine original SNC family transverse to an actual carrier, and a point where that carrier is genuinely regular of dimension one, the ACTUAL indexed boundary count there is at most one. No distinct-germ count substitution or supplied coordinate rank/count certificate is used.*

Proof. The independently constructed actual simultaneous member/product normal form supplies the real flattening, true free coordinate indices for every incident original position, and actual injectivity of those indices on incident positions. Its carrier plane has actual dimension $\min(k, n)$, by genuine flattening regularity, and hence dimension one by uniqueness of real regular dimension. Every incident free coordinate is therefore strictly below one. Two incident indices have the SAME actual coordinate, so actual injectivity forces equality of their ORIGINAL positions. The literal finite incident set has cardinality at most one, which is exactly the EXISTING indexed boundary count. □

Theorem 107.2 (Actual initial normal-crossings recognition precedes genuine FIRST-defect avoidance)**.** *ONLY genuine original smooth-curve/SNC/whole-family-density data prove that every original NC-good carrier point has ACTUAL original product trace order at most one. The ENTIRE literal original good set, including points outside the original carrier, is therefore contained in the complement of the ACTUAL initial chronological FIRST Z. No good-set certificate, desired order identity or branch-count equality is supplied.*

Proof. At a true original normal-crossings point, actual closed incidence and whole-family density internally construct a real open neighbourhood where the WHOLE pulled-back SNC family is

genuinely transverse. Actual open pullback preserves the real carrier regular dimension one. The preceding lemma bounds its literal indexed incident count by one. The proved actual product order/cosupport theorem on true transverse coordinates identifies EVERY local product threshold with its genuine distinct trace count, and the separate true transverse count dictionary identifies that count with the literal indexed count. In particular the actual local order-two cosupport is absent. The ENTIRE original product marked record has genuine whole open pullback equality, so BOTH actual cosupport directions exclude the original order-two cosupport too. Original coherence/nonzero and the real finite order dictionary now prove actual original product order at most one.

At the REFLEXIVE original history the independently proved original controlled/nonmonomial dictionary and genuine whole relative product normalization give this same bound for the ACTUAL chronological family product. Every actual family member is independently coherent, nonzero and relative principal from original SNC/density geometry. Apply the proved small-order proper-position/quotient-image theorem to these REAL ideals: it constructs BOTH the actual proper-position cardinality and its genuine quotient-image distinct count before proving FIRST Z absence. Thus no numerical-order-to-distinct-count shortcut is taken. Outside the original carrier, FIRST Z is absent by its literal carrier field. This proves the ENTIRE original-good-set inclusion. □

Theorem 107.3 (ONLY original curve data and compact actual cosupport produce a whole–good-preserving transverse endpoint)**.** *ONLY an original smooth curve, genuine original SNC family, actual whole-family density and compact true initial product cosupport internally construct ONE real finite original TEST. Its SAME actual projection is proper and an EXISTING isomorphism over the ENTIRE literal initial NC-good open. Both independent actual terminal defects are empty. The literal full original-strict-plus-exceptional ordered family is genuinely SNC and transverse to the actual final carrier, and the true final original record is still a genuine bound-one domain. No selected history, centre, inverse, support equality, coordinate or terminal-geometry certificate is supplied.*

Proof. The original-only two-phase producer independently constructs the actual finite history, true proper projection, both real empty defects, and whole ordered SNC/transverse endpoint. It also proves the EXISTING isomorphism over the ENTIRE initial FIRST-defect complement by actual centre/differential-exceptional support, without inferring that locus from endpoint geometry. The preceding theorem internally supplies the literal initial-good inclusion. Restrict actual bijectivity to the real preimage of this subset: injectivity follows from the larger preimage, and for every true good point the genuine larger surjection supplies an actual preimage which automatically remains good. Actual nonexceptionality restricts by the same inclusion. This is exactly BOTH fields of the EXISTING isomorphism predicate for the SAME original composite projection, not a replacement comparison map.

The compactness hypothesis concerns the actual initial cosupport, not ambient compactness. This proves a genuine sharp-preserving curve endpoint in this scope. General unbounded/noncompact curve phases, higher-dimensional sharp Σ, general scheduling, functoriality and the complete Hironaka theorem remain separate OPEN producers. □

108 Actual local sharp endpoints for arbitrary original smooth curves

Compact ambient manifolds are not needed for the local geometric producer. The real product cosupport on a smooth curve is itself genuinely analytic of dimension at most zero. Actual isolated-point neighbourhoods therefore supply compact restricted cosupports internally. This chapter constructs such neighbourhoods and actual local sharp endpoints at EVERY basepoint. It does not assume a chosen finite local history or a compact presentation.

Lemma 108.1 (A true bound-one domain internally supplies a neighbourhood with compact actual pulled-back cosupport)**.** *For ANY genuine original bound-one domain M and ANY ambient point x, internally construct a real open U containing x on which the ENTIRE actual pulled-back marked record has compact actual cosupport. No ambient, carrier or original-cosupport compactness is assumed, and no isolating or compact-window certificate is supplied.*

Proof. Actual coherence, smooth carrier, nonzero-along and positive mark make the actual original cosupport analytic, closed and of dimension at most zero. If x is outside it, choose its genuine open complement. The true whole cosupport pullback dictionary then makes the actual restricted cosupport empty, hence compact.

If x belongs to it, independently proved analytic zero-dimensional regularity makes it genuinely regular of dimension zero there. Its actual flattening supplies a real isolating open U containing x. The actual restricted cosupport is closed in U by the original closedness and true continuous open inclusion, and its EVERY point equals the actual subtype point representing x by the isolating dictionary. It is therefore a closed subset of that actual compact singleton. The same BOTH-direction whole record/cosupport pullback dictionary identifies this true compact set with the EXISTING pulled-back marked cosupport. No compactness is postulated for an ambient neighbourhood closure. □

Theorem 108.2 (ONLY arbitrary original curve inputs construct actual sharp local endpoints at EVERY point)**.** *ONLY an original smooth curve W, genuine original SNC list E, actual whole-family density, and ANY original ambient point x internally construct a genuine neighbourhood U of x and an ACTUAL finite TEST of its literal original restricted product marked record. Its SAME projection is proper and an EXISTING isomorphism over the ENTIRE true NC-good set of this restricted original carrier/family. Both independent actual endpoint defects are empty. The WHOLE original-strict-plus-new exceptional ordered family is genuinely SNC and transverse to the real final carrier, whose original marked record is still a bound-one domain. No compactness hypothesis, chosen open, history, centre or final-geometry certificate is supplied.*

Proof. Original smooth-curve regularity, true SNC product coherence and actual whole-family density independently construct the genuine original bound-one product domain with its actual empty auxiliary boundary. Apply the first lemma at the actual point x, obtaining a real open U and actual compact pulled-back cosupport internally.

The true whole product pullback identity identifies ALL four fields of the literal original restricted product record with that actual open pullback. Actual coordinate pullback preserves the genuine original SNC list and real open-inclusion regularity preserves dimension one. Since the open inclusion is a genuine open map, original whole-family closure density pulls back to the literal entire restricted family, including EVERY original ordered position. None of these facts assumes a desired restricted geometric domain or selected history.

Apply the preceding chapter's original-only compact-actual-cosupport producer to precisely these real restricted data. It internally chooses the actual FIRST-defect blowup and finite literal whole-difference phase, and proves the proper SAME projection, entire original-local good-set isomorphism and actual complete ordered SNC/transverse endpoint. The compactness needed by that producer has been independently constructed here from zero-dimensional cosupport geometry.

This is a genuine local sharp curve producer on arbitrary noncompact ambient manifolds and arbitrary noncompact original cosupports. Actual comparison of the resulting finite histories on overlaps, gluing their terminal manifolds and projections, a globally locally finite sharp schedule, higher-dimensional sharp Σ and complete termination remain separate OPEN producers. □

109 Actual ordinary controlled and ambient-contact orders away from the real centre

The real exceptional-factor and curve order-drop theorems handle points above the centre. Complete the independent off-centre part needed by a genuine contact-budget comparison. All ideals, centres and maps remain the actual original objects in [4, §12]; ordinary order and nonmonomial order are not substituted for one another.

Theorem 109.1 (Actual denominator units give every controlled germ and every ordinary-order threshold off centre)**.** *For ANY actual coherent marked record and SAME genuine blowup, at a point whose actual base image is outside the centre, the ACTUAL controlled ideal has exactly the whole germ-membership predicate of the genuine raw pullback restricted to its actual strict carrier. This is independent of the actual mark. With actual closed carrier, EVERY ordinary-order threshold of the actual controlled ideal is the inverse-image threshold of the original ideal. If the original record is genuinely well formed, nonzero along its carrier and the actual centre is original-admissible, the EXISTING ordinary relative orders therefore agree at EVERY true new carrier point off centre. No curve or desired order transport is assumed.*

Proof. The actual pulled-back centre ideal contains the unit germ at an actual off-centre point, hence so does EVERY genuine denominator power. Testing the actual defining relative colon with that unit gives one whole membership direction. Conversely multiplication by each genuinely holomorphic denominator germ preserves the actual normalized raw ideal, giving the other colon direction. Actual denominator coherence supplies the required holomorphic germs; no supplied ideal equality is used.

For an arbitrary threshold q, form the genuine same-carrier, same-ideal marked record with mark q, without asserting it is admissible. The preceding independently proved mark-independent whole germ dictionary identifies its actual transformed membership with the original actual controlled ideal at this point. The genuine off-centre cosupport theorem applies to this actual mark-q record without admissibility, giving precisely BOTH desired threshold directions. Finally true original/new coherence and nonzero follow from actual well formedness/admissibility. Apply the genuine ordinary-order/cosupport dictionary at each of the two actual finite orders; the two threshold directions give both inequalities and hence exact existing ordinary-order equality. □

Theorem 109.2 (Actual mixed discrete strict-member contact preserves ordinary order off centre)**.** *ONLY an actual closed smooth carrier N, singleton ambient SNC member H, true density of $N \setminus H$, ANY actual discrete centre $C \subset N$ and the SAME genuine blowup internally give, at EVERY true strict-carrier point z off centre, equality of the EXISTING ordinary order of its TRUE normalized new ambient strict-member trace with the old ambient contact order. Neither global $C \subset H$ nor a desired strict/raw comparison, nonzero, principal or order certificate is supplied.*

Proof. The independently proved mixed-centre theorem constructs actual new trace coherence/nonzero. Apply the true whole accumulated trace/raw residual comparison to the EMPTY initial exceptional family; its actual new family is the SAME singleton NEW exceptional divisor. At this genuine off-centre point, the actual unit test in the monomial part removes that singleton residual on BOTH sides, proving equality of the entire normalized trace and raw germ predicates. Their real cosupport thresholds therefore agree.

The genuine zero-mark record with original ideal $\mathcal{I}_H$ is well formed and its actual centre is admissible because its cosupport is exactly N. Original true density supplies actual nonzero. Its actual controlled ideal is exactly the independently normalized raw pullback. The first theorem gives its actual ordinary order equal to old ambient contact. The true new trace and raw nonzero/coherence make their ordinary orders finite and the full threshold dictionary identifies them, giving the displayed trace order equality. No residual support substitution has been made. □

Theorem 109.3 (ONLY original curve tests preserve the whole true ambient budget off centre)**.** *ONLY original smooth curve/SNC/whole-family density, ANY genuine ORIGINAL mark-one product test and an ACTUAL next original-admissible centre with its SAME genuine blowup internally give $B_{T'}(z) = B_T(\pi z)$ at EVERY actual new carrier point off centre. All original positions and true ambient contact orders are retained.*

Proof. The original dimension-one domain and SAME original test internally give the real template history, all genuine discrete centres and actual next centre containment in the SAME current carrier. The genuine finite ambient trace producer supplies each original literal current strict member's actual singleton SNC and current density. Apply the second theorem to each of these actual members, on the SAME centre and map. Actual ordinary-order normalization identifies the old normalized trace order with the old unnormalized contact. The unchanged literal template and strictAlong step equations identify the new terms. Sum these proved equalities over the genuine finite ORIGINAL positions to obtain exactly the independently defined whole ambient budget equality.

This completes true off-centre order inputs; actual positive-record budget domination, selected τ_D centre admissibility and a refined global sharp driver remain separate OPEN producers. The full sharp Σ theorem and general termination are not asserted. □

110 A genuine local inverse of the actual strict projection on a smooth curve

The first-step exponent comparison requires actual raw pullback order preservation on the curve even above the centre. An off-centre ambient biholomorphism alone cannot supply this. Construct the genuine local inverse along the two actual carriers first. This uses the real monomial blowup charts behind [4, Proposition 3.13 and Remark 12.6]; it does not assume the desired curve isomorphism or an order certificate.

Lemma 110.1 (The genuine zero-centre monomial chart is the identity on the actual curve plane)**.** *For the genuine monomial blowup map with centre dimension zero, if the chosen actual chart index j satisfies $j < 1$ and w belongs to the actual coordinate curve plane $\mathbb{C}^1 \times 0$, then $\sigma_j(w) = w$. This is a literal equality of the actual coordinate vectors, including $w = 0$ and ambient dimension one.*

Proof. The actual inequality forces the index to be zero. At that coordinate the genuine monomial map keeps w_j. Every other coordinate has index at least one and is zero by the true curve-plane membership; its product with w_j remains zero. Coordinate extensionality proves the equality. □

Theorem 110.2 (Actual discrete-centre smooth-curve strict projection has a real local inverse along its carriers)**.** *Let $\pi : Y \to X$ be an ACTUAL blowup with $C \subset N$, where N is genuinely closed smooth and regular of dimension one at all its points, and C is genuinely regular of dimension zero at all its points. Let N' be the ACTUAL geometric strict transform, and let $z \in N'$ lie above C. These original geometric inputs internally construct an ambient map $s : X \to Y$, holomorphic near πz, with $s(\pi z) = z$, such that near πz EVERY actual $x \in N$ satisfies $s(x) \in N'$ and $\pi s(x) = x$, and near z EVERY actual $y \in N'$ satisfies $s(\pi y) = y$. No selected chart, local inverse, curve model, pullback ideal identity or order-preservation certificate is input.*

Proof. Actual nested flattening at πz simultaneously gives the true centre plane of dimension zero and the true carrier plane of dimension one. The genuine blowup local monomial inverse has some actual chart index j, a true open neighborhood of z, and holomorphic coordinate map W with its genuine monomial lift ℓ. Actual strict-carrier membership forces $j < 1$ and $W(z)$ into the real curve plane. The first lemma therefore makes the monomial map the identity on this plane. The genuine equality $\ell(W(y)) = y$ and $\pi\ell(w) = \Phi^{-1}(\sigma_j(w))$ give, for every nearby actual strict-carrier point, the literal identity $W(y) = \Phi(\pi y)$. At z, nested centering yields $W(z) = 0$.

The actual monomial domain is open and contains this real zero vector. Use its genuine lifted-open model to extend its lift to an ambient map near zero; outside that domain choose the fixed actual point z. Its coordinate map is the identity near zero, so the real lifted-open holomorphic criterion and the actual lift prove holomorphicity there. Compose this genuine map with the actual flattening coordinates to obtain s. The already proved equality at z gives $s(\pi z) = z$. For nearby actual carrier points, the first lemma and BOTH true inverse flattening laws imply $\pi s(x) = x$.

True centre regularity of dimension zero supplies an actual neighborhood on which C is the singleton πz. A nearby carrier point outside C lifts into N' by the genuine off-centre strict-transform membership dictionary. A point inside C is precisely πz, whose lift is the already known $z \in N'$. Thus the entire actual carrier neighborhood is mapped into N'. Finally the independently proved nearby identity $W(y) = \Phi(\pi y)$ and the genuine monomial lift inverse law give $s(\pi y) = y$ for EVERY nearby strict-carrier point, including points over C. These are both actual inverse equations along the actual carriers, not a replacement ambient inverse asserted at exceptional points.

Relative germ/order transport, actual contact-order decrease, mixed incident-member chronological factors, the separate τ_D phase, full sharp Σ and termination remain subsequent consumers. □

111 Actual raw curve order preservation and first strict-trace contact decrease

The actual first ambient strict trace is not its all-power residual. Use the independently constructed true curve inverse over the centre before deriving contact decrease for that true geometric trace, as required by [4, Remark 12.6 and pp. 293–295].

Theorem 111.1 (The genuine raw relative TOTAL preserves every actual curve cosupport order above the centre)**.** *Let N be an actual closed smooth curve, let the genuine blowup centre $C \subset N$ have dimension zero, and let I be ANY actual coherent holomorphic ideal on the original ambient manifold. Put $N' = \mathrm{Strict}_\pi N$ and $P = \mathrm{Res}_{N'}(\pi^* I)$ independently. At EVERY actual $z \in N'$ above C and EVERY $q \geq 0$,*

$$z \in \mathrm{cosupp}_{N'}(P, q) \quad \Longleftrightarrow \quad \pi z \in \mathrm{cosupp}_N(I, q).$$

No nonzero hypothesis is needed for this order-predicate dictionary, and no local inverse, pullback equality or order certificate is input.

Proof. The preceding geometric producer internally constructs the genuine ambient holomorphic s near πz with $s(\pi z) = z$, mapping the actual nearby carrier into N', and satisfying BOTH carrier inverse laws. True first derivatives on a neighborhood follow from its actual holomorphicity. Composition of actual point-ideal powers with a genuine holomorphic map preserves power membership. Hence an actual relative vanishing witness composes with π or s whenever that map carries the corresponding real nearby carrier into the other.

Actual relative normalization preserves BOTH cosupport directions. In the forward-pullback direction each genuine generator $a \circ \pi$ vanishes to the given order on N', and true finite holomorphic span substitution proves it for every actual pulled-back germ. In the reverse direction each original germ $a \in I_{\pi z}$ supplies its genuine pulled-back germ; compose its real relative vanishing witness with the internally constructed s. The true carrier identity $\pi s(x) = x$ identifies the resulting function with a on the entire nearby original carrier. This proves the reverse relative order predicate for every original germ. Carrier membership is independently supplied by the true strict-transform projection dictionary. Both implications include $q = 0$. □

Theorem 111.2 (Actual finite raw curve order is preserved above the centre)**.** *If the preceding actual I is nonzero along EVERY point of N, then for EVERY actual $z \in N'$ above C, $\nu_{P,N'}(z) = \nu_{I,N}(\pi z)$ for the EXISTING actual ordinary trace orders, including nonprincipal ideals and units.*

Proof. Genuine zero-mark geometry internally supplies the actual transformed carrier, raw ideal coherence and nonzero restriction on its entire carrier; zero-mark controlled transformation is the genuine restricted TOTAL. Thus both EXISTING finite ordinary-order/cosupport dictionaries apply. Use the preceding BOTH order-predicate directions at each of these two actual ordinary orders. The resulting two integer inequalities prove equality, without a supplied finite bound or desired order identity. □

Theorem 111.3 (The true first ambient strict trace contact drops by exactly one above the curve centre)**.** *Let the actual original singleton H be SNC, let the genuine original smooth curve N be dense away from H, and let the actual dimension-zero centre satisfy $C \subset N \cap H$. For the SAME actual blowup and its independently geometric strict carrier and ambient strict member, at EVERY $z \in N' \cap \pi^{-1}C$,*

$$\nu_{\mathrm{Res}_{N'}\mathcal{I}_{H'},N'}(z) + 1 = \nu_{\mathcal{I}_H,N}(\pi z).$$

In particular this TRUE ambient strict-trace contact order strictly decreases, including above the exceptional centre. No order/decrease, inverse, principal model, or desired factor certificate is supplied.

Proof. Original SNC and actual density internally supply coherence and nonzero restriction of $\mathcal{I}_H$ along EVERY original curve point. The preceding actual curve theorem identifies its genuine raw restricted TOTAL order with the old actual order at πz, using the independently constructed inverse even at the centre. Actual zero-mark normalization identifies THIS raw ideal with the independently defined member TOTAL. The preceding genuine first-step product theorem independently gives its actual order as one plus the TRUE ambient strict-trace order, from real exceptional transversality and relative product additivity. Substituting both actual dictionaries proves the displayed equality and its strict inequality. The chronological all-power residual J' is never substituted for the actual ambient strict trace. Mixed incident/nonincident centres, chronological ambient factors, the global τ_D algorithm, full sharp Σ and general termination remain independent subsequent producers. □

112 Preserving FIRST Z through the actual curve ambient-difference phase

Retain the unchanged original smooth curve product record, true original ordered ambient SNC family and whole-family density. The actual first phase uses the genuine quotient-image distinct count in FIRST Z; the second uses the independent full-germ ambient difference Δ. These are separate source quantities in [4, §12]. The phase theorem below constructs two actual successive histories, rather than assuming that erasing one locus automatically erases or preserves the other.

Theorem 112.1 (Every actual original-admissible curve step preserves emptiness of FIRST Z)**.** *ONLY the original curve/SNC/density, ANY genuine original test, actual emptiness of its FIRST Z, and ANY actual next ORIGINAL-admissible centre with its SAME genuine blowup give emptiness of the NEW FIRST Z. No new proper-position count, quotient-image equality, family transport or desired erasure certificate is supplied.*

Proof. The actual next original admissibility appends the genuine original TEST before constructing the next template, carrier and member geometry. Above centre, the point lies on the NEW actual exceptional divisor. The independently proved real curve no-common-factor theorem makes EVERY chronological residual member a genuine unit there, so NEW FIRST Z is absent. Off centre, project the actual strict-carrier point into the old carrier. Old FIRST Z

is empty, and genuine first-good curve trace separation therefore gives actual old chronological product order at most one. The independently proved whole original controlled/nonmonomial-product dictionaries and actual off-centre nonmonomial-order equality give actual new chronological product order at most one. Independently construct EVERY new chronological member's true coherence/nonzero/principal data. Genuine sum/properness rigidity at order at most one now constructs BOTH the actual proper-position count AND the genuine quotient-image distinct count, identifying them with actual new product order. Hence the NEW FIRST Z is absent. Ordinary order equality alone is never used to infer a distinct count. □

Theorem 112.2 (FIRST Z emptiness survives EVERY genuine original finite continuation)**.** *If FIRST Z is empty at an actual original prefix, then it is empty after EVERY finite tail whose appended history is a genuine TEST of the SAME ORIGINAL record. No canonicality or computed centre recipe is required for this preservation assertion.*

Proof. Induct on the actual finite tail. The empty tail is the stated old emptiness. At a step the actual appended TEST supplies the genuine preceding appended TEST and actual next original-admissible centre. Apply the induction to that preceding actual history, then the first theorem on the SAME actual centre and projection. Literal tower append step equations identify the resulting history, retaining every empty step and original position. □

Theorem 112.3 (ONLY original data and compact CURRENT cosupport construct successive actual FIRST-Z and whole-Δ erasure phases)**.** *ONLY the original curve/SNC/density, ANY genuine original prefix and true compactness of its ACTUAL current positive cosupport construct a real whole-FIRST-Z blowup followed by a genuine finite literal whole-Δ phase. The SAME appended ORIGINAL history is a TEST, its literal final FIRST Z and final full-germ Δ are BOTH empty, and its actual original projection is proper with genuine final domain and ordered accumulated SNC boundary.*

Proof. First invoke the genuine original-input whole-FIRST-Z erasing step. It constructs the actual original-admissible centre, SAME global blowup, appended original TEST and actual new FIRST-Z emptiness. Genuine original controlled cosupport inheritance places the new actual cosupport in the proper inverse image of the given compact current cosupport. Actual coherence/closed-carrier cosupport closedness therefore makes that new current cosupport compact; no ambient compactness is needed. Apply the genuine finite whole-Δ phase theorem to THIS actual new original prefix and its internally proved current compactness. It constructs the real finite tail, EVERY literal computed current centre, actual appended original TEST and final Δ emptiness. Apply the second theorem to the same actual tail and the independently proved FIRST-Z emptiness at its beginning. Thus the TWO independent literal defects are empty on the SAME final history. Genuine finite blowup projection geometry and actual original domain inheritance supply properness, final coherence/nonzero/carrier and ordered SNC. This does not yet assert simultaneous carrier transversality with the original ambient strict family, the full sharp Σ conclusion, general noncompact unbounded scheduling or higher-dimensional refined monomial termination. Those geometric consumers remain explicit OPEN obligations. □

113 The same actual two-phase curve projection is an isomorphism off the initial FIRST defect

The preceding two phases produce actual entire-family transversality. A separate mathematical issue is the isomorphism locus of their SAME composite projection. Empty final defects alone do not imply preservation of an initial open region. We prove the required support statement for EVERY actual centre and hence for the actual composite differential. Throughout, FIRST Z remains the distinct-ideal-germ order defect; it is not the full sharp simultaneous-normal-crossings bad locus.

Lemma 113.1 (Actual whole-difference tails localize both their new difference and their actual exceptional differential locus)**.** *Take ONLY genuine original smooth-curve/SNC/density data, ANY actual original TEST prefix P, and a finite real tail S whose EVERY centre is the independently computed whole difference of the actual appended prefix. Then the true new whole difference is contained in the inverse image under the actual tail projection of the original current whole difference. The ACTUAL exceptional differential locus of that same tail projection is contained in this same inverse image. No supplied centre-support, exceptional-locus or isomorphism certificate is assumed; every empty position is retained.*

Proof. Induct on the literal finite tower. At a reflexive tail the first inclusion is equality and the actual identity differential has empty exceptional locus. At a step the literal recipe identifies its actual centre with the entire independently computed current difference. The genuine previous producer internally constructs the actual original TEST before the real new-difference-over-old theorem applies. Combining its true support inclusion with the inductive inclusion gives the first assertion for the actual composite projection.

For the second assertion use the genuine equal-dimension differential composition formula: the composite exceptional locus is exactly the union of the actual new blowup exceptional locus and the inverse image of the actual previous exceptional locus. The actual new blowup exceptional locus lies over its literal centre by the true off-centre inverse theorem. That centre is over the initial difference by the first inductive assertion. The other term is over it by the second inductive assertion. Thus both terms are contained in the required inverse image. No implication from numerical order to a distinct quotient count or exceptional locus is used. □

Theorem 113.2 (The actual FIRST-defect step followed by any actual difference tail preserves its entire initial complement)**.** *Start with the genuine original curve record at its REFLEXIVE history, blow up its literal FIRST Z, and continue by ANY actual finite whole-difference recipe. The SAME entire composite projection has actual exceptional image contained in the initial FIRST Z and is an* existing *isomorphism over its ENTIRE complement. No compactness, final-defect emptiness or desired inverse-map certificate is required for this support/isomorphism theorem.*

Proof. The independent actual full-germ difference/support dictionary is used at the REAL first original TEST. Its ordered exceptional list has exactly the NEW entry $\pi^{-1}Z$. Consequently its entire actual difference is contained in $\pi^{-1}Z$, without replacing it by a residual support or the ambient strict trace. The preceding lemma therefore places the actual tail exceptional locus over this initial Z. The genuine first blowup exceptional locus also lies over Z. Apply the true differential composition equality to the SAME two actual maps, giving the claimed whole exceptional-image inclusion.

The genuine finite blowup sequence is surjective, and off its actual exceptional locus the established blowup-sequence inverse uniqueness proves true injectivity. The proved general blowup-sequence isomorphism theorem therefore constructs the EXISTING IsIsomorphismOver over Z^c, including both true bijectivity on its entire preimage and actual nonexceptionality. This uses no ambient compactness and is not a claimed isomorphism merely on the final carrier. □

Theorem 113.3 (ONLY original compact-cosupport curve inputs give the same transverse endpoint and the whole complement isomorphism)**.** *ONLY an original smooth curve, actual original SNC family, real whole-family density and compact actual initial product cosupport internally construct ONE actual finite original TEST whose same projection is proper and an isomorphism over the ENTIRE initial FIRST-defect complement. Its actual two independent defects are empty; the literal complete original-strict-plus-exceptional family is SNC and transverse to the actual final carrier. The current original marked record is still a genuine bound-one domain. No selected history, centre, final geometry, support equality or isomorphism certificate is supplied.*

Proof. Apply the genuine original-only two-phase construction to the REFLEXIVE original prefix and the actual compact initial cosupport. It internally constructs the actual FIRST Z blowup

and the finite literal whole-difference tail, including every original TEST, properness/domain and both actual empty-defect conclusions. Apply the preceding theorem to these SAME maps and the retained literal recipe. The independent whole-family endpoint theorem constructs the real ordered SNC and transverse coordinate dictionaries from the two empty defects of this SAME actual original history. Collect these genuinely proved conclusions, retaining the true actual projection throughout.

The further claim that the literal initial NC-good open lies in this FIRST-defect complement still requires its actual local ideal/count recognition, and remains a separate next consumer. General unbounded noncompact phases, higher-dimensional sharp Σ, the complete sharp driver and termination are not asserted. □

114 The entire literal curve ambient difference is an actual original centre

Let M be the unchanged ORIGINAL mark-one product record of the smooth curve W and ordered original ambient SNC family E, with true whole-family density. After ANY genuine ORIGINAL test T, keep the independently defined full-germ difference locus Δ_T and true ambient contact budget B_T. The locus is the actual $I_i \neq J_i$ exceptional difference of [4, §12]; it is neither FIRST Z nor the final sharp Σ. Its positive-record support inclusion has already been proved from actual finite-history ordinary-order domination.

Theorem 114.1 (The ENTIRE actual ambient difference is admissible for the SAME original record)**.** *ONLY the original curve/SNC/density and ANY genuine ORIGINAL test internally construct true original admissibility of the ENTIRE literal Δ_T. In particular it is a genuine closed smooth zero-dimensional centre, contained in the SAME actual original positive cosupport and with normal crossings with the ENTIRE actual accumulated ordered exceptional family. No compactness, canonical maximum, desired centre or admissibility certificate is supplied.*

Proof. Original curve geometry gives the actual bound-one initial domain, and real original test inheritance gives the SAME current bound-one domain. The preceding genuine budget-domination producer has already proved the WHOLE actual Δ_T lies in its positive cosupport. The original bound-one cosupport is analytic of dimension at most zero. Every subset of such a locus is closed smooth zero-dimensional in the ambient manifold; its genuine coordinates also give normal crossings with the actual ordered SNC boundary. Apply this established geometric subset theorem to THIS actual Δ_T and its independently proved support inclusion. Thus all three actual original admissibility fields are constructed, without changing the original mark or replacing Δ_T by its carrier. □

Theorem 114.2 (The SAME global blowup of the entire actual difference appends an original TEST and gives genuine budget drop)**.** *ONLY the same original inputs construct a genuine global blowup with centre EXACTLY Δ_T. The actual map is proper and holomorphic, is an isomorphism over the ENTIRE complement of Δ_T, appends a TEST of the SAME ORIGINAL record, and has the actual next bound-one domain and precise ordered transformed SNC boundary. At EVERY actual new carrier point above this centre, the true ambient contact budget strictly decreases and the ACTUAL original controlled ordinary order satisfies $a_{T'} + 1 = a_T$.*

Proof. The first theorem supplies actual original admissibility. Actual domain nonzero/coherence proves the whole current positive cosupport has empty interior; the genuine centre inclusion forces Δ_T to have empty interior as well. Construct its actual global blowup from this true closed smooth thin centre. The genuine blowup theorem gives properness, holomorphicity and the whole-complement inverse. Its original admissible centre appends exactly the original test, so actual original domain inheritance supplies next coherence/nonzero/carrier and ordered SNC. Above the centre, its point is by definition in the literal old actual difference. Apply the independently

proved actual original-position contact-budget recurrence and positive-incidence dictionary to obtain strict budget drop. The separately proved SAME original controlled mark-one order shift supplies the second real equation. No desired drop, new invariant, rebasing or canonicality is input. □

Theorem 114.3 (After the actual whole-difference step every new actual difference lies above the old difference)**.** *For ANY genuine blowup of the whole actual centre Δ_T, the literal new actual difference obeys $\Delta_{T'} \subset \pi^{-1}(\Delta_T)$. This is a geometric assertion about the independently defined true ambient-trace/chronological full-ideal differences at both stages.*

Proof. The first theorem makes this actual step an original test. The true entire difference-support dictionary therefore applies independently at both stages. A new difference point belongs to the actual strict carrier, a true original ambient strict member and an actual accumulated exceptional member. The new original strict member lies over the true old original strict member. If the exceptional member is NEW, its base image belongs to the actual centre, which is exactly the old difference. If it is an old member's genuine strict transform, its base image belongs to that actual old member, and actual strict-carrier membership gives the old carrier. The independently proved old support dictionary now puts that actual base point in the literal old difference. These exhaust the precise ordered old-strict-plus-NEW boundary list, including empty lists and empty centres. No transported family-germ or desired locus equality is assumed. Finite erasure, locally finite global sharp scheduling, the full sharp Σ statement and higher-dimensional refined monomial termination remain separate OPEN producers. □

115 Actual simultaneous transversality after both genuine curve phases

Keep the SAME original smooth curve W, ordered original ambient SNC family E, whole-family density, and ANY genuine ORIGINAL mark-one product TEST T. Write N_T and F_T for the actual template carrier and ordered accumulated exceptional family, $H_{i,T}$ for the literal original ambient strict members, $I_{i,T} = \mathcal{I}_{H_{i,T}}|_{N_T}$ for their actual normalized ambient ideals, and $J_{i,T}$ for the independently constructed chronological residuals. We prove actual common coordinates, not an identification of FIRST Z, ambient difference Δ or sharp Σ; compare [4, §12].

Lemma 115.1 (The whole actual ordered ambient boundary is literally the original strict list followed by actual exceptionals)**.** *For EVERY actual finite tower, the whole template starting with original ambient family E has ordered boundary $(E.\mathrm{map}(\mathrm{strictAlong}\, T)) +\!+ F_T$. For the stated original curve TEST this EXACT ordered list is genuinely SNC. No reordered, shortened or replacement boundary is used.*

Proof. Induct on the literal finite tower. Initially the accumulated family is empty, giving the exact list identity. At a step, BOTH genuine boundary recursions take the actual strict transform of EACH preceding member and append the SAME NEW exceptional divisor. Actual list-map composition, map over append and append associativity, together with the literal strictAlong step equation, prove the identity preserving all positions. The original bound-one domain and ANY real original test independently supply genuine SNC for the actual whole ambient template, as already proved using the true zero-dimensional centres. Substitute the actual ordered identity to obtain SNC for precisely the displayed list. □

Theorem 115.2 (Both actual empty defects internally supply true ambient germ identity and incidence/order-one rigidity)**.** *Assume the literal actual FIRST Z and full-germ ambient difference Δ_T are empty. At EVERY actual carrier point internally prove: EVERY original-position WHOLE ambient ideal germ $I_{i,T}$ equals the actual independent $J_{i,T}$ germ; each incident TRUE ambient strict member has EXISTING ordinary relative order one; at most one ORIGINAL*

strict member is incident; and no original strict member is incident at any actual accumulated-exceptional point. No desired family equality, order, distinct count or separation certificate is supplied.

Proof. Literal Δ_T emptiness at a true carrier point forces its actual finite difference-position cardinality zero. Therefore EVERY original position's full ideal-germ predicates agree, directly from the true independent definition. The actual normalized vanishing-ideal unit test makes each incident ambient member a proper germ; this WHOLE germ identity makes its actual chronological germ proper as well.

The original true FIRST-good rigidity independently gives chronological order one at EVERY such proper position. Both complete germ membership directions identify EVERY ordinary-order threshold of the normalized ambient and chronological ideals; genuine ordinary normalization then gives actual ambient contact order one. The actual FIRST-good open has pairwise disjoint chronological zero traces on the pure curve. An incident ambient position gives its actual chronological zero trace by real coherent proper-germ dictionary. Thus two incident ORIGINAL positions coincide, proving genuine indexed incidence uniqueness without replacing distinct counts at bad points. Finally the independent entire Δ support dictionary says exactly that an actual exceptional carrier point on an original strict member lies in Δ_T. Its emptiness proves the asserted whole-family avoidance. □

Theorem 115.3 (Both actual empty defects force genuine simultaneous transversality with the entire actual ambient boundary)**.** *ONLY the same original inputs, actual original TEST and literal emptiness of BOTH independent defects internally prove*

$$\mathrm{IsTransverse}\big(N_T, E.\,\mathrm{map}(\mathrm{strictAlong}\,T) +\!\!+ F_T\big).$$

Thus this SAME true ordered family is SNC and has actual simultaneous normal crossings with the true strict curve. No desired coordinate or transversality hypothesis is supplied.

Proof. Original domain/template inheritance supplies the actual smooth pure curve and genuine transversality with accumulated F_T. At a carrier point incident with F_T, the preceding theorem excludes EVERY original strict member. Retain the actual simultaneous curve/F_T coordinates and extend the family to the displayed full list: each incident member belongs to the actual old family, and all other member conditions are vacuous.

At a point outside F_T, either no original strict member is incident, when the true curve flattening already supplies the entire family, or there is a unique actual original incident position. Its independently constructed singleton SNC geometry, true current density and actual order-one contact supply the genuine common curve/member coordinates from the preceding contact-coordinate inverse theorem. Incidence uniqueness and absence of F_T then extend these SAME coordinates to EVERY member of the literal full ordered family. In each extension the actual coordinate assigned to an incident full-list member is that of its equal actual base member; absent members need no assertion. The real hyperplane cardinalities and disjointness from the true normal-coordinate set are retained. This gives BOTH actual ambient memberships in one chart at EVERY carrier point, proving the existing transversality predicate itself. The first lemma separately supplies full-list SNC. □

Theorem 115.4 (ONLY original data and compact CURRENT cosupport construct an actual full-boundary transverse curve history)**.** *ONLY the original curve/SNC/density, ANY genuine original prefix and true compactness of its ACTUAL CURRENT positive cosupport construct a finite actual continuation whose appended history is a TEST of the SAME ORIGINAL record, with BOTH actual defects empty, genuine final domain, proper original projection, and actual simultaneous transversality of the strict curve with the WHOLE original ambient strict list followed by ALL accumulated exceptionals. The ambient and carrier may be noncompact. No caller terminal history, desired geometry or comparison certificate is input.*

Proof. The real two-phase producer independently constructs the SAME original FIRST-Z step and literal finite whole-Δ tail, with actual appended original TEST and BOTH computed final emptiness conclusions. Apply the third theorem to precisely THAT actual appended history. It internally produces all original-member geometry, genuine order-one joint coordinate charts and actual full-family transversality; the first lemma gives SNC of the same ordered list. Retain its independently proved actual final domain and proper original projection. This is a source-backed curve geometric endpoint, not a replacement output record. Preservation of the ENTIRE INITIAL already normal-crossings open by the composite projection, general unbounded noncompact scheduling and the higher-dimensional sharp Σ driver remain explicit subsequent obligations. □

116 The entire literal curve FIRST defect is an actual original-record centre

Retain the ORIGINAL product marked input with mark one and its SAME actual test prefix. The FIRST defect $Z = \{\nu > t\}$ uses the TRUE quotient-image count of distinct proper chronological ideal germs. In pure curve dimension the positive cosupport itself is zero-dimensional. This makes the ENTIRE literal Z a genuine global centre, without selecting a maximum or assuming compactness. This is first-phase geometry in the setting of [4, pp. 293–295]; FIRST Z is still distinct from the full sharp obstruction Σ.

Lemma 116.1 (The actual current curve product has a true dimension-one domain and contains FIRST Z in its positive cosupport)**.** *ONLY original smooth pure curve, original ambient SNC boundary, whole-boundary density and ANY genuine finite template history internally give a genuine dimension-one marked domain for the actual chronological product with the ENTIRE ordered accumulated exceptional family and mark one. The whole literal FIRST Z is contained in this SAME actual positive cosupport. Neither a current domain, current purity nor a defect/support certificate is supplied.*

Proof. Actual template wellformedness gives the real current closed smooth carrier, transverse accumulated SNC boundary and actual ordered fields. True original curve purity transports through every actual strict-carrier step, including empty positions. Independently proved chronological finite-product geometry supplies actual coherence and nonzero restriction at EVERY current carrier point. These are precisely the true dimension-one domain conditions, with positive mark one. At a literal FIRST-defect point the genuine distinct proper count is nonnegative, so the true product order is at least one. The independently proved SAME chronological product dictionary and actual ordinary-order/cosupport equivalence place the point in this actual positive cosupport. □

Theorem 116.2 (The ENTIRE computed FIRST Z is an actual zero-dimensional centre for the SAME ORIGINAL curve record)**.** *ONLY the original curve/SNC/density inputs and ANY actual ORIGINAL mark-one product test prefix internally prove: the ENTIRE literal current FIRST Z is an admissible centre for the SAME ORIGINAL transformed record; every one of its actual points is regular of dimension zero; and Z is disjoint from the ENTIRE actual accumulated exceptional family. No compactness, original canonicality, selected centre, current domain, residual or admissibility certificate is input. Empty Z is allowed.*

Proof. Construct the true original mark-one dimension-one domain directly from the original smooth curve, actual original product coherence and genuine whole-boundary-density nonzero germs. Its SAME actual test prefix gives the real transformed dimension-one domain. Original test admissibility also internally supplies the SAME template history and literal current carrier/boundary equalities. Apply the preceding producer to this history.

The independently proved WHOLE original controlled residual equality identifies its actual Nm ideal with the chronological product. The original coherent ideal is contained in its genuine

Nm ideal; actual relative-span inclusion therefore carries the product's positive cosupport into the SAME ORIGINAL positive cosupport. FIRST Z is thus an actual subset of the latter, not a supplied support certificate.

Coherent nonzero germs make the true positive cosupport thin in the smooth pure curve. Actual analytic dimension drop makes it zero-dimensional. Every subset of this genuine closed discrete analytic set is closed and smooth, and true singleton SNC coordinates supply its normal crossings with the ENTIRE actual current boundary. This proves original admissibility of the WHOLE literal Z, with regular dimension zero at all its points. Independently, true chronological no-factor geometry on the same curve gives EVERY member a unit over EVERY actual accumulated exceptional point. The real product order there is zero, which excludes the literal FIRST inequality. Hence the two actual sets are disjoint. No already-good assumption is used. □

Theorem 116.3 (The whole computed curve FIRST defect has a genuine SAME ORIGINAL-record global blowup)**.** *The SAME original-only inputs internally construct a genuine global blowup with centre EXACTLY the ENTIRE computed FIRST Z. Its projection is proper and holomorphic and an isomorphism over the ENTIRE Z complement. It appends the SAME ORIGINAL mark-one test prefix, whose actual transformed record remains a true dimension-one domain with the WHOLE ordered strict accumulated boundary followed by the NEW exceptional member SNC. No compactness or supplied blowup/centre is required.*

Proof. The preceding original admissibility makes Z closed smooth and contained in the true original positive cosupport. That cosupport has empty ambient interior by genuine coherent nonzero domain geometry, so the actual centre also has empty interior. The global blowup-existence theorem therefore constructs its real source manifold and projection. The actual blowup theorem gives properness, holomorphicity and BOTH isomorphism directions on the whole centre complement. Append this precise centre and map to the given ORIGINAL prefix; its previously proved genuine admissibility proves the new original test history. True domain inheritance then supplies the transformed dimension-one domain, including the literal full ordered strict-boundary list followed by the new exceptional member. The construction also covers empty centres and noncompact zero-dimensional centres. It does not yet prove that the NEW FIRST Z is empty: actual off-centre WHOLE chronological-family transport is a separate next producer. No full Σ equality, higher original invariant decrease, sharp iteration or termination follows here. □

117 Genuine local member branches for an actual discrete blowup centre

An actual global discrete centre can contain both points on an original member and points away from it. It is incorrect to assume that the entire centre lies in every original member. Resolve these genuine local branches and retain the true ambient strict traces of [4, Remark 12.6, pp. 293–295].

Lemma 117.1 (Actual zero-dimensional centre geometry constructs both real local incidence branches)**.** *Let C be genuinely regular of dimension zero at each of its actual points and let H be actually closed. For EVERY $p \in C$, an actual source open $U \ni p$ is internally constructed such that its centre trace is precisely the singleton p. If $p \in H$, this true centre trace is contained in the actual H trace. If $p \notin H$, that entire H trace is empty. No selected open, local branch or centre-containment certificate is input.*

Proof. True zero-dimensional regularity gives an actual isolating neighborhood of p in C. If $p \in H$, retain that neighborhood; its actual centre trace is the point p and hence lies in H. If $p \notin H$, intersect it with the genuine open complement of H. It still contains p, still isolates the actual centre, and now misses the entire member. Both singleton directions follow from genuine point membership. □

Lemma 117.2 (True normalized vanishing traces have BOTH full germ directions on actual source opens)**.** *For ANY actual carrier N, a genuine closed smooth H and an actual open U, a germ of $\mathrm{Res}_{N\cap U}\mathcal{I}_{H\cap U}$ is precisely a germ whose actual extension by zero belongs to $\mathrm{Res}_N\mathcal{I}_H$. These are BOTH full relative germ directions, retaining normal germs and off-carrier units.*

Proof. Genuine smooth vanishing-ideal coherence applies to H. The actual open vanishing-germ dictionary and genuine coherent ideal pullback membership first identify the ENTIRE ideal of the actual H trace with the actual open pullback of $\mathcal{I}_H$. Independently, both true restricted finite-span substitution directions commute with the real open extension/restriction maps. Substitute the first whole ideal identity in that normalized open span dictionary. No zero-set-only identification or desired ideal dictionary is input. □

Theorem 117.3 (An actual incident discrete-centre branch produces the true restricted strict-trace factor and the original whole germ dictionary)**.** *ONLY actual smooth N, singleton SNC H, genuine density of $N\setminus H$ in N, actual discrete $C\subset N$, its actual blowup π and an actual $p\in C\cap H$ internally construct $U\ni p$, its real preimage open V, and the SAME restricted blowup $\rho: V\to U$. On these actual opens the genuine centre lies in BOTH actual carriers. The actual local ambient strict trace I'_U is coherent and nonzero along its actual strict carrier, its actual non-monomial part is the independently defined local all-power STRICT J'_U, and its actual greatest factor D'_U satisfies the WHOLE true identity $I'_U=\mathrm{Res}_{N'_U}(D'_UJ'_U)$.*

Moreover both local geometric strict transforms are EXACTLY the inverse images of the original geometric strict transforms under $V\hookrightarrow Y$. At EVERY actual point of V, EVERY local germ belongs to I'_U iff its real extension belongs to the TRUE original ambient strict-trace ideal $\mathrm{Res}_{N'}\mathcal{I}_{H'}$. The entire global centre is NOT assumed contained in H; no local map, factor or germ certificate is input.

Proof. The first lemma internally isolates the actual centre at p inside H. The actual inverse-image open and genuine blowup restriction theorem give the SAME restricted map and its genuine blowup proof. True open pullback gives smooth N_U and singleton SNC H_U. Because the inclusion of an actual open is continuous and open, its inverse image commutes with true closure; hence the original density gives genuine density of $N_U\setminus H_U$ in N_U. Actual centre containment supplies both required inclusions. Apply the independently proved genuine first-step factor producer to THESE actual restricted inputs, obtaining all local coherence, nonzero, actual residual and whole factor conclusions internally.

True geometric strict transforms commute with restriction to this SAME source open, proving both geometric set identities. To use the normalized full germ dictionary, original zero-dimensional centre isolation at each centre point transfers genuine singleton normal crossings to the actual centre. The real ambient SNC blowup theorem therefore makes the true original ambient strict member smooth, independently of any assumed carrier transversality. Apply the preceding actual normalized vanishing open dictionary to THIS true member, and substitute the two actual strict transform set identities. This proves both whole germ directions on the entire actual preimage open, not merely the selected point. Chronological gluing, exponent transport and the separate τ_D driver remain subsequent consumers of these real local data. □

Theorem 117.4 (The genuine nonincident branch makes every true ambient strict-trace germ a unit germ)**.** *For an actual discrete centre point $p\notin H$, the internally chosen $U\ni p$ makes the TRUE original ambient strict trace have exactly ALL holomorphic germs at EVERY point of $\pi^{-1}U$, along ANY actual strict carrier. This is a full relative unit-germ dictionary, not an asserted nonincident factor or order model.*

Proof. The first lemma chooses the actual source open disjoint from the entire closed H. The genuine strict transform of H lies in $\pi^{-1}H$, so it misses the entire preimage open. This strict transform is actually closed, by its geometric closure definition. At each real point of this open its actual complement therefore gives a neighborhood on which every holomorphic germ vanishes

vacuously along the true strict member. Thus that germ belongs to the genuine strict-member vanishing ideal, and its self relative span puts it in the true normalized trace. The converse is the literal holomorphic component of normalized membership. This covers every actual germ on the entire preimage open without coherence or a supplied unit certificate. Together with the incident producer this closes the real local incidence gap; whole chronological ambient factor transport, τ_D, full sharp Σ and termination are still separate. □

118 An actual dynamic FIRST DISTINCT maximum constructs a genuine next geometric blowup

The current ideals and product are the already defined ACTUAL iterated NEW-exceptional chronological objects. The centre below is constructed by maximizing a TRUE current-product invariant on literal dynamic FIRST Z. Its normal crossings are with the ENTIRE ACTUAL accumulated exceptional boundary. Comparison with strict transforms of the different original ambient member ideals and composite raw-TOTAL factor transport remain OPEN. No original-canonical-history or termination claim is made for this separately selected modified centre.

Lemma 118.1 (Actual coherent quotient classes have a true nonincrease neighborhood)**.** *For a genuine finite coherent current ideal family, its TRUE distinct proper ideal-germ quotient count cannot increase near ANY actual point. Consequently the SAME actual product-order stratum near a point of literal FIRST Z remains inside that SAME dynamic Z.*

Proof. Finite coherent presentations make EACH actual whole germ relation persist near the point. Closed literal coherent zero supports exclude new proper positions nearby. There are finitely many actual positions/relations, so all true persistence statements hold simultaneously. Actual quotient-map functoriality sends every old class to its genuine nearby class, and the true nearby proper-class image is contained in the image of the old proper classes. Real finite image cardinality therefore proves the stated actual count bound. Combine it with the actual strict deficit and genuine SAME product order, retaining real carrier membership, to get true stratum saturation. No indexed/DISTINCT replacement or difference USC is used. □

Theorem 118.2 (ONLY original history geometry supplies the genuine current product domain and its true first invariant coordinate)**.** *ONLY original closed smooth pure-dimensional compact carrier, ambient SNC, whole-boundary density and an ACTUAL geometric finite history supply a compact pure current carrier and the SAME accumulated SNC/transverse list. The ACTUAL full chronological product with mark one on THESE actual fields is a genuine original invariant-domain input. Its true first invariant coordinate at the reflexive current history equals the actual ordinary product order. SAME full invariant strata are genuinely saturated in Z.*

Proof. Induct through REAL geometric steps. Actual proper blowup preimages of the previous compact carrier contain the new closed strict carrier, proving its true compactness. Genuine centre containment and actual strict-transform regularity preserve its fixed pure dimension. Independently proved true finite product coherence/nonzero and actual template SNC/transversality supply the genuine marked domain. Actual full-product exceptional-factor exclusion gives its WHOLE nonmonomial=carrier-normalization dictionary. The independently defined empty-boundary ordinary order has the SAME true normalization; actual order-predicate congruence identifies the real first coordinate. Combine this actual dictionary with the first lemma and the literal first-coordinate equality extracted from the actual full invariant. No current-domain, constant-order or desired stratum certificate is input. □

Theorem 118.3 (Literal dynamic FIRST Z internally selects a genuine nonempty smooth maximum centre NC with ALL accumulated members)**.** *If that ACTUAL dynamic FIRST Z is nonempty, ONLY the preceding original inputs construct its TRUE full current-product invariant*

maximum centre. It is nonempty, closed smooth, lies in literal Z, and is genuinely admissible for the actual current-product marked input, hence NC with the ENTIRE SAME accumulated ordered boundary. It is also admissible for the ACTUAL transformed geometric template, not a caller-chosen centre.

Proof. Literal dynamic Z is independently analytic/closed and lies in the true compact current carrier, so it is compact. Its strict product deficit puts it in the actual mark-one cosupport. Genuine invariant USC attains a real maximum there. Its actual maximum locus is closed and nonempty. At each of its real points, original USC supplies a true neighborhood on which the full invariant is bounded by that point's value. Independently constructed stratum saturation puts the WHOLE local full level inside literal Z; therefore its true open preimage equals the open trace of the ACTUAL restricted-Z maximum centre. The PROVED original admissible-level theorem constructs genuine smoothness and NC with EVERY actual boundary member on that open. True open regularity and coordinate dictionaries transport both facts to each actual centre point. Closedness then gives the genuine global smooth centre and its ENTIRE boundary NC. Actual template mark zero has carrier cosupport, so true centre containment and these independently proved NC coordinates give actual template admissibility as well. □

Theorem 118.4 (The SAME dynamic centre has a genuine proper next blowup and extends the ACTUAL geometric history)**.** *The internally selected SAME centre has a genuine proper holomorphic blowup and an isomorphism over the ENTIRE complement of literal current FIRST Z. Appending this actual step gives a REAL geometric test history of the original template, with precise ordered strict-boundary-plus-new exceptional SNC geometry. Thus the genuine chronological ideal recursion really continues along it. No original canonical or decrease claim is made.*

Proof. The actual positive-domain cosupport is thin, and the independently proved centre containment gives empty ambient interior. Genuine smooth-centre blowup existence supplies the actual next manifold/map; true blowup properness and holomorphicity concern this SAME map. The actual centre lies in literal Z, so the true complement isomorphism restricts to its ENTIRE complement. Actual template admissibility supplies the REAL appended test-tower proof, and its genuine wellformed transform gives the precise whole ordered boundary update. The already independently defined finite chronological recursion is therefore applicable at EVERY original member position on this ACTUAL extended history. This constructs a genuine next modified step; canonicality/decrease relative to an earlier original invariant, composite original raw-TOTAL factor transport, recursive intersections/exceptional cleanup and sharp termination remain OPEN. □

119 Actual residual ideals admit cancellation of arbitrary exceptional powers

We prove a genuine arbitrary-power saturation statement for the actual residual ideal, before identifying saturation of the actual TOTAL ideal with relative STRICT. No residual is renamed strict. The real first-phase Z and the whole sharp bad locus remain distinct.

Lemma 119.1 (Genuine transverse SNC geometry supplies a real member generator and true nearby carrier nonzero points)**.** *For a true carrier point incident with an actual member of a transverse SNC family, internally construct a genuine holomorphic ambient member generator a with BOTH entire vanishing-ideal germ directions. Near the point, a vanishes identically on no nonempty relatively open subset of the actual carrier.*

Proof. Real simultaneous adapted coordinates give the SAME true member as one coordinate and make that coordinate free on the true carrier. Genuine coordinate Hadamard proves BOTH complete member-generator directions. Vary that actual free coordinate in the true coordinate plane, keeping carrier normal coordinates zero, then use the genuine inverse chart. These actual

nearby carrier points have nonzero chosen coordinate and accumulate at EVERY nearby carrier point. Restrict to the actual flat open; each nonempty relatively open carrier neighborhood therefore contains a point where the genuine generator is nonzero. No generator/nonzero field is a caller premise. □

Lemma 119.2 (True relative prime cancellation and actual nonzero geometry cancel a real divisor factor from a residual principal relation)**.** *For a genuine relative principal generator c not in the actual restricted member ideal, if ag is a true relative multiple of c, then g is a true relative multiple of c. Here a is the actual internally constructed member generator, and all functions are genuine holomorphic germs.*

Proof. Write the actual premise as $ag = bc$ on the true carrier. The left side is in the actual restricted member ideal. Genuine carrier germ congruence makes bc an actual member too. The independently proved relative member prime law and the actual exclusion of c give true membership of b. Substitute the genuine member-generator dictionary into that restricted span to obtain $b = ad$ on the carrier. The real equation becomes $ag = a(dc)$ there. Only now apply genuine restricted cancellation, justified by the first producer's actual nonzero points in every carrier neighborhood. The result is the true full relative principal relation $g = dc$. □

Theorem 119.3 (The actual greatest-factor residual is saturated by every power of each actual exceptional member)**.** *For a genuine coherent nonzero relatively principal total ideal on a smooth carrier transverse to an actual SNC family, its TRUE intrinsic greatest-factor residual J obeys, for EVERY actual member H and EVERY integer $k \geq 0$,*

$$(\mathrm{Res}_N\, J : \mathcal{I}_H^k)_x = J_x \qquad (x \in N).$$

This is equality of BOTH entire holomorphic germ predicates; normal carrier-vanishing germs are retained.

Proof. The independently verified genuine greatest-factor division constructs a true relative residual generator c. The actual no-common-member theorem supplies a genuine residual germ outside each incident restricted member ideal. If c belonged to that ideal, its actual relative principality and true span substitution would put every residual germ there, a contradiction. Thus c really avoids the actual member ideal, without a supplied exclusion certificate. The first two producers and induction on k cancel every power a^k from true residual membership.

A true colon germ multiplies EVERY actual member-power ideal germ into the restricted residual. The genuine generator power a^k is one of these actual germs; cancel its powers as just proved and substitute the actual residual-principal dictionary to obtain whole residual membership. Conversely actual residual membership multiplies by every holomorphic ideal-power germ inside that same genuine ideal, hence in its true carrier restriction. At nonincident members the actual vanishing ideal is unit, including every power, so evaluation at the genuine unit germ gives the reverse direction. The zero power is retained literally. No saturation, generator, prime or relative STRICT dictionary is supplied. □

Theorem 119.4 (Only original inputs give actual all-power residual saturation on the SAME first blowup)**.** *ONLY original compact smooth fixed-positive-dimensional carrier, original SNC, actual whole-boundary density and nonempty literal FIRST Z construct the SAME actual first blowup on which EVERY actual original member residual, at EVERY actual carrier point, is saturated by EVERY power of EACH actual exceptional position.*

Proof. The original-only first total-trace producer internally constructs the SAME real centre/projection, genuine per-member wellformed total geometry and true nonzero along every actual carrier point. Genuine original source SNC and the SAME actual projection independently construct the true relatively principal total generator. Apply the preceding actual all-power producer to these real objects. This supplies no selected centre, projection,

generator, no-common-factor or saturation certificate. Identifying arbitrary exceptional saturation of the TOTAL trace with this residual, chronological relative STRICT compatibility and modified-history termination remain OPEN. □

120 Actual finite histories of iterated NEW-exceptional relative saturation

Every chronological ideal in this chapter is LITERALLY constructed by actual old-ideal pullback followed by ALL powers of the NEW exceptional ideal at each genuine step. The history retains every original position and empty step. We do not identify it with stripping ALL accumulated exceptionals from the composite original raw TOTAL; the old-factor comparison is OPEN. No selected next centre or termination certificate is supplied.

Definition 120.1 (True geometric template and actual chronological ideal history)**.** The true geometric template has the SAME initial carrier and ordered exceptional family, actual unit ideal and mark zero. Starting with a genuine marked input, at each real tower step retain its true geometric strict carrier, precise old strict boundary followed by the new exceptional member, and put the LITERAL actual nextRelativeStrict in its ideal field with mark zero. The identity stage is the actual initial input.

Proof. These are actual existing MarkedIdeal records and real finite-tower recursion, not a new certificate. At each successor actual nextRelativeStrict is defined by true pullback and arbitrary NEW-exceptional-colon saturation before any coherence or history theorem. Literal tower recursion keeps every empty centre and each original ordered boundary position. □

Lemma 120.2 (The actual chronological history has exactly the real geometric template carrier and entire ordered boundary)**.** *At every actual finite tower, the true chronological record's carrier and ENTIRE ordered boundary equal those of the original geometric template's actual transformed record. After each genuine successor its mark is zero. At the identity stage its mark is the original actual mark.*

Proof. Induct on the literal tower. Identity fields agree by definition. At a real successor both sides use the SAME actual geometric strict transform and the SAME mapped ordered list followed by exactly the same new exceptional member. Substitute the two real previous field equalities. The independently changed ideal field cannot affect those true geometric operations. The literal new mark is zero; no deletion or reindexing of empty stages occurs. □

Theorem 120.3 (Genuine actual geometric history internally preserves coherent, nonzero and relatively principal chronological ideals)**.** *For a genuine initial wellformed mark-zero input with a true coherent ideal, nonzero along the real carrier and genuinely relatively principal there, EVERY real geometric test tower of its ACTUAL geometric template has a wellformed actual chronological final record. Its actual ideal is coherent, nonzero along the true carrier and genuinely relatively principal there. Its carrier and ENTIRE ordered boundary are the actual geometric ones. These statements apply to EVERY actual prefix by the same theorem.*

Proof. Induct on the real finite history. At identity use the actual initial facts. At a genuine next step, the real template test history supplies its actual prefix and true current centre admissibility. The preceding geometric field equalities transfer actual centre containment and NC with the ENTIRE prior exceptional list to the independently constructed chronological record. The induction has ALREADY proved its genuine wellformedness, coherent ideal, nonzero and relative principal dictionaries. Apply the actual next STRICT producer to these true data and the genuine projection to construct the new coherent/nonzero/principal ideal. Separately use actual whole-list NC and mark-zero cosupport equal to the real carrier to prove marked admissibility for the genuine prior-boundary record; its true transform gives precisely the actual new strict

carrier and accumulated SNC/transversality. These facts prove the literal new record wellformed, retaining every empty step. No caller chronological ideal, principal dictionary or desired new geometry certificate is supplied. □

Theorem 120.4 (Only original member data construct actual all-position finite chronological ideal geometry)**.** *ONLY original closed smooth carrier, genuine original ambient SNC family and actual whole-boundary density construct, for EVERY original member, the true initial relative member ideal and all its actual finite chronological NEW-exceptional-saturation history records. For ANY genuine geometric test history starting with the same carrier and EMPTY exceptional list, every actual final ideal is coherent, nonzero and relatively principal on its real carrier, with the precise entire ordered accumulated exceptional family SNC and transverse to that carrier. No compactness, fixed dimension, selected centre, desired ideal history or termination witness is a premise.*

Proof. The initial relative member ideal is the genuine carrier restriction of the actual original vanishing ideal. Real original SNC proves its ambient ideal coherent and supplies BOTH principal germ directions. Genuine smooth-carrier restriction gives its true coherence and BOTH relative principal directions. Whole original density constructs an actual nonzero trace germ at EVERY carrier point; its genuine original membership embeds in the true restricted ideal, retaining the same real nonzero accumulation. These independent facts prove the literal initial empty-exceptional mark-zero record wellformed and supply every input to the preceding genuine history producer. Its template is literally the SAME unit/zero-mark carrier/empty-list record for every original position, so a single actual geometric history supplies all true position conclusions. This is proved iterated NEW-exceptional saturation; absence of reintroduced OLD exceptional factors, equality with composite raw-TOTAL ALL-exceptional residual, dynamic DISTINCT driver/decrease, recursive intersections, exceptional τ_D and full termination remain OPEN. □

121 Actual finite chronological families have genuine relative products and full order sums

The finite product is defined from the ACTUAL current chronological ideals, with actual carrier normalization at each real multiplication. This retains normal-vanishing germs and every original position. The true order sum is constructed from genuine product geometry, never assumed as a driver field.

Definition 121.1 (Literal finite carrier-relative ideal product)**.** For a genuine carrier N and actual finite ordered ideal list, define the empty product as the actual holomorphic unit ideal and a nonempty product as the actual head ideal multiplied by the recursively defined tail product, then restricted to the SAME carrier. Define the chronological family from EVERY original member's ACTUAL recursively saturated ideal on the SAME geometric history, and its actual product by this literal finite recursion.

Proof. These are independent actual ideal operations, with no desired nonzero, coherence, order or principal dictionary in their definitions. The actual normalization is essential: a relative ideal contains carrier-normal germs, so its product is not identified with an ambient principal product. Original member positions and empty chronological blowup positions are all retained. □

Theorem 121.2 (Genuine finite products internally preserve coherence, nonzero, relative generators and full multiplicity sums)**.** *For genuine coherent ideals, nonzero along a smooth closed carrier and truly relatively principal at every carrier point, their ACTUAL finite product is coherent, nonzero at EVERY actual carrier point and genuinely relatively principal there. Its actual ordinary carrier order is the SUM of ALL actual factor orders, retaining multiplicities including repetitions and actual unit/empty positions.*

Proof. First prove actual carrier normalization preserves ordinary order by BOTH real cosupport inclusions and finite-order dictionaries; real nonzero witnesses pass by actual ideal inclusion. Induct on the actual list. For the empty list, the true unit has generator one, an actual nonzero witness at each carrier point and true order zero. For a genuine head, the already produced tail is coherent/nonzero/principal BEFORE any product claim. The genuine two-factor product theorem supplies a real nonzero product witness and exact literal product order additivity, while actual normalized product algebra supplies its real relative generator. Coherent product and true carrier restriction prove coherence independently. Actual normalization preserves order; substitute the genuine tail SUM. Every assertion is thus an induction conclusion from actual ideal operations, not a supplied product or full-family certificate. □

Theorem 121.3 (Only original finite-history inputs give the actual full chronological product and true whole order sum)**.** *ONLY original closed smooth carrier, genuine ambient SNC, whole-boundary density and an ACTUAL finite geometric history internally produce the literal product of ALL actual chronological ideals. It is coherent, nonzero and relatively principal on the SAME actual carrier, and its ordinary order equals the complete actual factor-order SUM at EVERY carrier point. No product geometry or sum certificate is input.*

Proof. Genuine geometric-template transformation independently supplies the actual current closed smooth carrier, including the empty-original-list case. For EACH actual original member position, the original-only finite-history producer supplies true wellformedness, coherence, nonzero and genuine relative generators. Its independent actual carrier-field identity is used BEFORE transporting these facts to the SAME template carrier. Apply the preceding genuine finite-product producer to the literal ordered family. This is actual dynamic product geometry for a subsequent DISTINCT-germ formula/centre consumer. Equality with the different composite original raw TOTAL stripped along ALL exceptionals, modified driver/decrease and sharp termination remain OPEN. FIRST Z is distinct from full Σ. □

122 The genuine whole chronological product has no accumulated exceptional factor

Every actual chronological member has independently proved absence of ALL old and new exceptional common factors. This chapter proves that the TRUE full relative product inherits this property by genuine relative member primality, then identifies its actual intrinsic monomial/nonmonomial parts. It does not identify a different composite original raw-TOTAL residual.

Theorem 122.1 (Genuine relative member primality constructs an excluded product germ for EVERY actual accumulated member)**.** *For a genuine SNC boundary transverse to the actual carrier, if EVERY real factor ideal has a genuine germ outside each incident restricted boundary ideal, the ACTUAL full relative product has such a genuine germ at EVERY incident boundary position. No product-factor exclusion is a caller premise.*

Proof. Induct on the ACTUAL finite ideal list. For the true empty unit, one is an actual product germ and cannot lie in the incident relative member ideal, whose genuine trace-vanishing dictionary would force one to vanish at the real carrier/member point. At a real head, choose its actual excluded germ and the independently constructed true tail excluded germ. Their genuine product belongs to the literal head-times-tail ideal, hence its real carrier normalization. If it belonged to the actual restricted member ideal, the independently proved TRUE transverse-member relative prime law would put one of those SAME factors in it, contradicting their genuine exclusions. No desired prime/product-factor certificate or ambient-principal replacement is used; all original positions and actual normal germs are retained. □

Theorem 122.2 (ONLY original histories supply actual full-product zero exponents and WHOLE intrinsic residual dictionaries)**.** *ONLY original smooth closed carrier, actual ambient SNC, whole-boundary density and an ACTUAL finite geometric history internally supply the SAME current ordered exceptional boundary and full relative product. EVERY actual incident product exponent is zero, its WHOLE intrinsic monomial part is the true unit ideal, and its WHOLE intrinsic nonmonomial part is the actual carrier normalization of that SAME product.*

Proof. Actual template geometry independently supplies the genuine current SNC/transverse boundary and closed smooth carrier. EACH original member's true accumulated no-factor witnesses are independently supplied by the already proved chronological theorem, with actual carrier/list fields identified BEFORE transport. Apply the preceding real finite-product primality producer to THESE actual objects. Actual full-product coherence and genuine excluded germs then supply the independently proved maximal-divisibility exponent-zero theorem, followed by WHOLE monomial-unit and true coherent-colon-by-unit nonmonomial normalization. No exponent, unit or residual certificate is input. □

Theorem 122.3 (At every actual carrier point the genuine full-product nonmonomial ideal has BOTH original product germ directions)**.** *At EVERY actual current carrier point, the actual intrinsic nonmonomial part of the TRUE full chronological product has exactly the SAME full holomorphic ideal-germ predicate as that product, in BOTH directions. No ambient principal or normal-germ deletion is asserted.*

Proof. Use the preceding WHOLE actual nonmonomial=carrier-normalization identity. The independently constructed true relative product generator internally substitutes every genuine finite product-ideal span into the SAME relative principal span, proving normalized membership implies actual product membership. Conversely genuine coherent ideal inclusion supplies the true normalization member. Thus BOTH full germ directions hold at every carrier point. This is actual input for a dynamic restricted-FIRST-Z centre consumer on the SAME ordered exceptional geometry. Modified history decrease, composite original raw-TOTAL factor transport and full sharp termination remain OPEN; FIRST Z is not full Σ. □

123 Actual first ambient strict-trace exceptional exponent evolution

The actual ambient strict trace retains an exceptional factor even when its chronological relative STRICT is a unit. Compute its true first-step exponent from genuine geometric ideals. Use actual ordinary-order additivity on EVERY incident carrier point before applying the literal nearby-order definition of exceptional exponent. This is the numerical first-step input to $I_{ij} = D_{ij}J_{ij}$ in [4, pp. 293–295]; it is not the full chronological or τ_D algorithm.

Lemma 123.1 (Actual normalized transverse member geometry supplies true order one)**.** *For an actual closed smooth carrier N transverse to a genuine singleton SNC member H, its TRUE normalized member ideal $A = \mathrm{Res}_N \mathcal{I}_H$ is coherent, nonzero along EVERY carrier point and genuinely relatively principal there. At EVERY $x \in N \cap H$ its actual ordinary trace order is exactly one. No generator, nonzero, order or transversal model certificate is supplied.*

Proof. True member and carrier coherence give normalized coherence. Original ambient SNC coordinates supply a genuine local generator of $\mathcal{I}_H$; BOTH relative-span substitutions give the SAME normalized generator dictionary. At an incident point the actual free coordinate and real carrier retraction give a genuine member germ which does not vanish to second order on N. If it vanished near the carrier, the actual zero germ would witness second-order vanishing; thus it has genuine frequent nonzero carrier values and supplies nonzero restriction. At a nonincident point the true member-complement open supplies its unit germ and actual nonzero carrier value. The

real order-one cosupport contains every incident point, while the internally constructed member germ excludes the second-order cosupport. The genuine least-failing-order definition therefore gives actual order one, without a global bound premise or an assumed ordinary-order identity. □

Theorem 123.2 (A true transverse member increases an actual relative principal factor exponent by one)**.** *Let B be any genuine coherent nonzero relatively principal ideal along the same actual N. For the independently defined actual normalized member A, at EVERY $x \in N \cap H$,*

$$\nu_{AB}(x) = 1 + \nu_B(x), \qquad e_H^N(AB, x) = 1 + e_H^N(B, x).$$

No product-order, exponent-shift or desired factor identity is a premise.

Proof. The preceding producer internally supplies all actual normalized member geometry. Genuine relative product-order additivity and actual member order one give the first equality at EVERY incident carrier point; true relative product geometry also constructs product nonzero witnesses. For every integer q, apply the actual ordinary-order/cosupport dictionary at each nearby point of $N \cap H$. The first equality gives the genuine successor shift $\mathrm{IsDivNear}_N(AB, H, q+1, x)$ iff $\mathrm{IsDivNear}_N(B, H, q, x)$ in BOTH directions. Choose a real carrier retraction at x. Genuine nonzero witnesses make both actual exceptional exponents finite, and the independently proved nearby-divisibility/actual-exponent equivalence applies. Divisibility at the actual B exponent gives the lower bound on the actual product exponent. A product exponent larger than that exponent plus one would, by the same genuine successor shift, give divisibility one above the actual B exponent, contradicting its true least-failing definition. These are both numerical bounds; their equality proves the second formula. □

Lemma 123.3 (Real first-step ambient SNC geometry supplies the actual strict-trace generators)**.** *For the original actual singleton SNC H, smooth carrier N dense away from H, real $C \subset N \cap H$ and actual blowup, the TRUE normalized ambient strict trace $I' = \mathrm{Res}_{N'} \mathcal{I}_{H'}$ has genuine relative generators at EVERY actual strict-carrier point, internally along with its actual coherence and nonzero witnesses. The actual singleton exceptional member is SNC and transverse to N'. No strict-trace generator, additional centre admissibility or current geometry certificate is a premise.*

Proof. The true centre is closed smooth and contained in the genuine ambient member. Actual containing-member normal-crossings insertion gives its normal crossings with $[H]$, starting from the real empty-family normal crossings of the centre itself. The genuine SNC blowup theorem therefore constructs the WHOLE actual ambient family $[H', D]$ SNC. Its true ambient H' generator and BOTH actual normalized span substitutions give real relative strict-trace generators at every point. The preceding actual strict-trace factor producer independently supplies coherence and nonzero witnesses from original density before this generator conclusion. The real empty-boundary carrier template makes the SAME centre admissible using true containment in N. Actual transformed wellformedness gives the genuine singleton exceptional SNC/transverse geometry on N'. □

Theorem 123.4 (The actual raw TOTAL exponent exceeds the true ambient strict-trace exponent by exactly one)**.** *For ONLY the preceding genuine original geometric inputs, let $P' = \mathrm{Res}_{N'} \pi^* \mathcal{I}_H$ be the independently computed actual mark-zero raw TOTAL, and let $D = \pi^{-1} C$. At EVERY $x \in N' \cap D$,*

$$\nu_{P'}(x) = 1 + \nu_{I'}(x), \qquad e_D^{N'}(P', x) = 1 + e_D^{N'}(I', x).$$

This compares the actual raw TOTAL and the actual ambient strict trace, not the different all-power residual J' or an assumed exceptional factor. Empty strata are included without a nonemptiness premise.

Proof. The real ambient blowup product identity is $\pi^*\mathcal{I}_H = \mathcal{I}_D\mathcal{I}_{H'}$ BEFORE restricting to the actual carrier. Both independently proved relative normalization inclusions for each factor, multiplied by genuine finite-span substitution, identify its true normalized product with $\mathrm{Res}_{N'}(AI')$, where $A = \mathrm{Res}_{N'}\mathcal{I}_D$. This gives a WHOLE actual ideal identity, retaining normal germs and off-carrier units.

The preceding actual strict-trace producer supplies all genuine relative generators, coherence and nonzero data. Actual singleton exceptional transversality supplies the real normalized member geometry. Apply the true product exponent/order theorem to THESE independently constructed objects. Both actual restricted-span directions make their positive and zero cosupports equal to those of the genuine raw TOTAL, hence preserve the true ordinary order and literal nearby-order exceptional exponent. Substituting these independently proved dictionaries gives both displayed equalities. In the tangent parabola example the raw exponent is two and the strict-trace exponent one, while the relative STRICT is unit. General local incident/nonincident branches, chronological ambient-strict factor transport, the τ_D algorithm, full sharp Σ and termination remain separate. □

124 The actual ambient strict trace retains its genuine exceptional factor

Let N be the original smooth carrier, H a genuine original ambient smooth hypersurface, $C \subset N \cap H$ a real closed smooth centre, and π its actual blowup. Write N', H' for the ACTUAL ambient geometric strict transforms and $D = \pi^{-1}C$. The independently constructed chronological relative STRICT ideal is the all-power saturation of the raw total pullback on N'. It is not the ideal of $N' \cap H'$. Construct the correct first-step $I' = D'J'$ comparison from actual geometric ideals, as in [4, Remark 12.6 and proof of Theorem 12.4, pp. 293–294]. The remaining exceptional factor can be nontrivial even after FIRST-Z erasure.

Lemma 124.1 (Genuine relative invertible-product cancellation gives the whole normalized quotient)**.** *For actual coherent ideals A, B on a real closed smooth carrier N, with A genuinely invertible along N,*

$$(\mathrm{Res}_N(AB) : A) = \mathrm{Res}_N B$$

as WHOLE ambient germ ideals. No product quotient, desired equality or principal numerator certificate is input; all normal-vanishing germs and off-carrier units are retained.

Proof. The actual product is contained in A by genuine coherent ideal-span closure. Real quotient coherence and the two actual invertible-along product/colon inclusions give $A(AB : A) = AB$ after restriction. Commute the real products and cancel the actual nonzero-along generator of A in the genuine relative finite-span relation. This supplies $\mathrm{Res}_N(AB : A) \subseteq \mathrm{Res}_N B$. Conversely each true $b \in B$ multiplies every actual $a \in A$ into the actual product AB, giving $B \subseteq (AB : A)$ before relative-span closure. These give BOTH normalized ideal directions. The true colon normalization theorem removes its outer restriction, by actual finite relative-span substitution. Closed-carrier complements supply the real off-carrier units. Thus the displayed WHOLE equality, rather than only an on-carrier membership statement, follows. □

Theorem 124.2 (The actual mark-one relative controlled transform is the true ambient strict-t-member trace ideal)**.** *For genuine closed smooth N, H, actual centre $C \subset N \cap H$ and actual blowup π, the literal relative mark-one transform of $(N, \varnothing, \mathcal{I}_H, 1)$ has WHOLE ideal*

$$I' = \mathrm{Res}_{N'}\mathcal{I}_{H'}.$$

No desired strict-transform ideal or trace dictionary is supplied. The carrier need not be transverse to H.

Proof. Real ambient blowup coordinates independently prove the ENTIRE original pullback equality $\pi^*\mathcal{I}_H = \mathcal{I}_D\mathcal{I}_{H'}$ for the actual smooth ambient H containing C. Genuine blowup equations give the ENTIRE centre-ideal pullback $\pi^*\mathcal{I}_C = \mathcal{I}_D$; its first power is this same actual invertible ideal. The true exceptional generator is nonzero along the actual strict carrier N', independently of N–H transversality. Actual strict-submanifold geometry supplies closed smoothness of N' and H', hence genuine coherence. Substitute the independently proved ambient product identity into the literal relative controlled-transform colon definition. Apply the preceding real relative cancellation theorem. This yields BOTH WHOLE ideal directions with the essential outer carrier normalization. □

Theorem 124.3 (The independently computed actual strict trace factors by its genuine greatest exceptional part and actual relative STRICT)**.** *In positive ambient dimension, assume the actual original singleton boundary $[H]$ is SNC and the original carrier has genuine density away from H. The preceding real geometric inputs internally construct actual coherence/nonzero of $I' = \mathrm{Res}_{N'}\mathcal{I}_{H'}$, and for the independently defined all-power relative STRICT J' prove*

$$\mathcal{N}_{N'}(I';[D]) = J', \qquad I' = \mathrm{Res}_{N'}(D'J'), \qquad D' = \mathcal{M}_{N'}(I';[D]).$$

The actual greatest exceptional part D' is computed from the TRUE ambient strict trace before the factor dictionary. It is coherent, nonzero along N' and genuinely monomial there. No residual, actual trace, factor, exponent or domain certificate is a premise.

Proof. Original density internally supplies genuine nonzero restriction of the original ambient member ideal. Actual $C \subset N \cap H$ and real centre smoothness make it admissible for the actual mark-one input, by the true order-one cosupport dictionary and empty-boundary normal crossings. Actual transformed wellformedness and nonzero inheritance precede all desired residual comparisons. The preceding theorem identifies its WHOLE ideal with the actual ambient strict trace. Independently, genuine controlled-versus-TOTAL monomial-division invariance identifies its nonmonomial ideal with that of the real normalized raw TOTAL, using the actual singleton exceptional family. Actual original member pullback generators, true TOTAL coherence/nonzero and genuine singleton SNC geometry give the independently proved all-power saturation/residual equality. Its raw numerator is the literal original member pullback, so restriction idempotence identifies that saturation with the independently defined J' in BOTH whole directions.

Apply the genuine monomial/nonmonomial parts theorem to this actual strict-trace input. It constructs its coherent, nonzero, monomial greatest exceptional part D' independently, with BOTH actual relative factor inclusions. Substitute the just-proved WHOLE residual identity and use the true double-restriction identity to obtain the displayed WHOLE factorization with OUTER $\mathrm{Res}_{N'}$. The factor is not silently set to one: for $N : y = x^2$, $H : y = 0$ and the origin blowup, $N' : v = x$, $H' : v = 0$ in $y = xv$, so the actual trace still has exceptional factor x although J' is a unit. General local incident/nonincident centres, chronological ambient-strict factor transport, numerical exponent evolution, τ_D, carrier transversality and full sharp termination remain separate producers. □

125 The genuine zero-dimensional lower base of the FIRST-good curve geometry

When the original smooth upper carrier is a curve, the actual reduced FIRST-good lower traces have genuine dimension zero. Use their real isolated-point neighborhoods to finish their lower restriction problem: all other normalized lower members and their whole product are units, the reduced traces are pairwise disjoint, and they avoid the ENTIRE actual accumulated exceptional family. This is a geometric lower base case, not a FIRST-Z-erasing or full sharp-termination theorem.

Lemma 125.1 (Genuine nonzero on an actual zero-dimensional carrier forces the whole normalized ideal to be a unit)**.** *Let S be an actual closed smooth carrier of pure dimension zero and I an actual coherent ideal. If its genuine carrier normalization is nonzero along S at EVERY actual point, then that ENTIRE normalized ideal is the actual ambient unit ideal. Both full germ directions hold at EVERY ambient point; no actual unit, nonzero value or selected isolated neighborhood is assumed.*

Proof. Actual regularity of dimension zero gives a true open neighborhood whose intersection with S is the single original point. At a point of S, choose the genuine nonzero restricted member supplied by the hypothesis. Its frequent nonzero points on S must meet this true isolating neighborhood and hence equal the original point. Thus that SAME member has a nonzero actual value there. Its genuine holomorphic inverse makes the constant one a member of the actual coherent normalized ideal. Coherent multiplication then gives EVERY actual holomorphic germ. The reverse direction is its true holomorphicity clause. Outside the closed carrier the normalization is already a unit on a genuine open complement. These two real cases prove equality of the WHOLE ambient ideals, retaining all normal germs and empty carriers. □

Theorem 125.2 (ONLY original smooth-curve inputs construct genuine FIRST-good separated reduced traces avoiding every actual exceptional member)**.** *ONLY original smooth pure curve, original ambient SNC boundary, whole-original-boundary density and a genuine finite template history internally construct the computed FIRST-Z complement open. EACH actual reduced trace is zero-dimensional and is disjoint from EVERY OTHER original position's actual reduced trace and from the ENTIRE literal accumulated exceptional family there. No disjointness, nonintersection, trace transversality or chosen good-open certificate is input.*

Proof. The actual lower-dimension producer supplies ONE computed open and the true zero-dimensional reduced traces, together with their whole actual exceptional-boundary density. At any such trace point take its genuine isolating open neighborhood. Actual density supplies a point of the SAME trace outside the ENTIRE accumulated family in this neighborhood; it must equal the original point. Thus the original point itself lies outside that whole family.

For another original trace position, the independently constructed original noncontainment theorem gives actual frequent points on the chosen trace outside the OTHER ideal's true zero locus. Carry this frequency through the actual open-embedding neighborhood identity and intersect with the SAME kind of genuine isolating neighborhood. The resulting point again equals the original point, which therefore does not belong to the other's literal trace. This proves BOTH actual intersection exclusions for every remaining position, retaining all empty traces and without assuming their separation. □

Theorem 125.3 (ONLY original smooth-curve inputs make every actual remaining normalized lower ideal and its whole product units)**.** *ONLY the same original smooth-curve data construct ONE computed FIRST-good open on which, for EACH chosen actual trace, EVERY OTHER normalized original lower member is the ENTIRE actual unit ideal, and the literal ordered ALL-remaining normalized lower product is the actual unit ideal. No lower ideal/unit certificate is supplied. All positions, multiplicities, normal germs and the empty product are retained.*

Proof. The independently constructed member geometry and actual dimension producer give the SAME literal FIRST-good open. Identify those true opens before transporting any member or point. The actual raw open pullback of each original coherent member is coherent; its real normalization is nonzero at EVERY actual lower-carrier point by the original noncontainment producer. The first lemma therefore gives WHOLE equality with the unit ideal for every other actual member. Induct over the literal ordered remaining-position list. The empty product is the genuine unit ideal. For a true next position, its already constructed unit equality and the inductive whole unit equality make their actual ideal product the unit; normalizing this genuine unit on the SAME carrier remains the unit. This proves the entire lower-product identity. FIRST

Z remains distinct from full Sigma; neither a Z-erasing step nor higher original invariant descent, full strict-original ambient-boundary compatibility, exceptional tauD or sharp termination follows from this lower base alone. □

126 Actual reduced-trace density outside the whole accumulated exceptional family

A genuine FIRST-good trace is smooth but need not be transverse to the accumulated exceptional boundary. The sharp induction requires the weaker TRUE whole-boundary density first. Construct it from actual transverse boundary coordinates and the independently proved chronological no-common-factor witnesses. The boundary below is the literal accumulated exceptional list of the actual template history, not an invented trace family or the original untransformed boundary.

Lemma 126.1 (Actual transverse boundary coordinates genuinely exclude second-order vanishing on the same carrier)**.** *At each actual incident member of an SNC boundary transverse to an actual closed smooth carrier N, construct a genuine holomorphic function in THAT member's actual vanishing ideal whose restriction to N does not vanish to order two at the point. No derivative, order or coordinate certificate is supplied.*

Proof. The real simultaneous carrier/SNC flattening supplies an actual free coordinate for this incident member whose index lies within the true carrier plane. Actual hypersurface Hadamard division makes that coordinate a genuine member of the ORIGINAL boundary ideal. Its actual first point power and next-power exclusion follow from the genuine one-coordinate monomial order theorem. The actual flattened carrier retraction fixes this coordinate near the original point. If it vanished to order two on N, true retraction transport would put the same original coordinate in the second ambient point-ideal power, contradicting that next-power exclusion. All maps are used only on their true chart neighborhoods. □

Lemma 126.2 (Actual no-factor and reduced order one force genuine exceptional noncontainment)**.** *For a genuine coherent nonzero relatively principal ideal I on the same smooth carrier N, at a real point of its literal trace $S = N \cap V(I)$ with actual order one, suppose the actual no-common-member-factor witness is known for an incident member of the TRUE transverse SNC boundary. Then EVERY ambient neighborhood of the point contains a point of this SAME trace outside that actual member. Nonincident members satisfy the same conclusion directly. No noncontainment, trace transversality or density certificate is assumed.*

Proof. Construct BOTH actual reduced-ideal directions for I from genuine order-one contact. Suppose the literal trace lay in the boundary member on some true ambient neighborhood. The preceding actual boundary coordinate would then vanish on that literal trace and so belong to I. Express it as a holomorphic cofactor times the genuine relative generator of I on N. This generator vanishes at the point. If the cofactor vanished there too, the boundary coordinate would vanish to order two on N, contradicting its independently constructed exclusion. Thus the cofactor is a genuine unit. Its holomorphic inverse expresses the true relative generator of I as a multiple of an ACTUAL boundary-ideal member. Substitution puts EVERY germ of I in the actual boundary ideal's relative span, contradicting the genuine no-factor witness. Negating neighborhood containment gives the required actual frequent trace points. For a nonincident boundary member, the chosen trace point itself lies outside it and belongs to every neighborhood used for the frequency criterion. □

Theorem 126.3 (ONLY original inputs construct whole accumulated-boundary density on every actual FIRST-good lower trace)**.** *ONLY original smooth positive pure carrier, true original SNC boundary, whole-original-boundary density and a genuine finite template history internally*

construct the actual FIRST-Z complement open. Its actual accumulated exceptional list pulls back to a genuine SNC boundary there. EACH actual reduced trace is a closed smooth lower carrier in that SAME computed open and is contained in the closure of its complement of the ENTIRE actual accumulated boundary. No chosen open, lower smoothness, density, transversality or factor certificate is input.

Proof. The genuine original reduced-trace theorem constructs the actual open and each closed smooth lower carrier. Real finite template wellformedness supplies the true accumulated SNC boundary transverse to the actual upper carrier. Original chronological geometry independently gives each SAME member ideal's coherence, nonzero and relative generators. At every FIRST-good trace point true factor rigidity constructs order one, and real chronological continuation supplies ALL actual no-factor witnesses. Apply the preceding noncontainment theorem to EVERY actual accumulated position. The genuine open-embedding neighborhood filter identity carries these frequent actual trace points to the SAME open lower carrier. True pullback of SNC coordinate data proves the boundary remains SNC.

Finally finite closed-family avoidance is proved topologically. In any true open neighborhood of a lower-carrier point, first choose a point outside the first closed member by the constructed frequency. Its intersection with that member's open complement is still an actual open neighborhood of the chosen point. Apply induction to the remaining finite closed list inside this smaller neighborhood. The resulting point lies outside EVERY member while staying in the original lower carrier and original neighborhood. Hence the entire boundary complement is genuinely dense. Empty and empty-position boundaries are included. This does not assert lower-carrier transversality or simultaneous SNC, FIRST Z equals Sigma, higher original invariant descent, a sharp recursive driver, exceptional tauD or termination. □

127 A genuine dimension drop for the actual sharp lower inputs

The lower recursive input must have a smaller TRUE carrier dimension, not just an ambient dimension bound. Prove actual pure-dimension preservation through the same finite geometric history, then use uniqueness of real regular dimension to identify the genuine reduced traces with pure dimension one lower. No compactness or canonicality is needed for this dimension argument. The actual accumulated boundary is retained as an ambient SNC family with proved lower density; lower transversality remains a different assertion.

Theorem 127.1 (Actual geometric-template history preserves the genuine original pure carrier dimension)**.** *An original closed smooth carrier of pure regular dimension k retains THAT dimension at EVERY point of its actual strict carrier after ANY genuine finite geometric-template history. No compactness, postulated current purity, canonicality or history comparison certificate is input. Every actual position, including empty-centre steps, is retained.*

Proof. Induct on the literal actual tower. The empty tower has precisely the original carrier and original pointwise regularity. At a true last blowup, original template wellformedness supplies the actual previous carrier's closedness; its actual admissibility supplies centre containment in that carrier. The genuine strict-submanifold theorem transports the inductively obtained pure regular dimension through THIS same blowup, both over and away from its actual centre. This is exactly the next carrier field. No step is deleted or replaced, and the proof makes no compactness assumption. □

Theorem 127.2 (ONLY original geometry constructs actual pure dimension-one-lower FIRST–good traces)**.** *ONLY original smooth pure dimension $d+1$, true original ambient SNC boundary, whole-boundary density and a genuine finite template history internally construct the actual FIRST-Z complement open on which EACH actual reduced trace is a closed smooth carrier of pure dimension d. The same true carrier is used; no lower pure-dimension certificate is assumed. The conclusion includes $d = 0$.*

Proof. Construct the actual computed FIRST-good open and true closed smooth traces from the genuine reduced-trace theorem. At each actual trace point that producer supplies an actual carrier dimension k and trace dimension $k-1$. The preceding finite-history theorem independently proves that SAME actual upper carrier is regular of dimension $d+1$ at the SAME original point. Uniqueness of actual regular dimension therefore gives $k = d+1$. Substituting into its actual open trace regularity gives dimension d. This proves pointwise pure lower dimension on the whole computed open, including empty traces and the zero-dimensional lower case. □

Theorem 127.3 (The genuine sharp lower data retain actual boundary density and a strictly smaller real invariant input)**.** *ONLY the same original data construct ONE computed FIRST-good open such that its TRUE accumulated exceptional family is ambient SNC, EACH actual lower trace is pure dimension d and genuinely dense outside that WHOLE boundary, and the TRUE remaining normalized lower product with EMPTY auxiliary boundary and mark one is an actual invariant-domain input with bound d. Its bound has genuinely dropped from the original pure carrier dimension $d+1$. No selected open, lower ideal, coherence, nonzero, dimension, density or domain certificate is supplied.*

Proof. The preceding theorem, genuine remaining-product constructor and actual exceptional-density constructor each produce the literal open complement of the SAME independently defined FIRST Z. Equality of the actual open sets identifies their underlying opens and ALL their real point/ideal objects. Thus the actual ordered accumulated boundary, actual reduced trace, actual normalized lower product, proved nonzero and genuine pure dimension d are on ONE and the SAME domain. Keep the whole genuine product's wellformedness and nonzero. At every true lower-carrier point, choose its now-proved actual regular dimension d; the dimension bound $d \leq d$ then constructs the actual mark-one domain of bound d. The EMPTY auxiliary boundary remains explicit. Actual ambient-boundary SNC and lower density do not imply that the lower carrier is transverse to it. The full ambient strict ORIGINAL boundary family, joint sharp SNC, higher original invariant descent, FIRST-Z-erasing and recursive drivers, exceptional tauD and termination remain separate unmet producers. □

128 Actual lower coherent products on the genuine FIRST-good traces

Construct real recursive lower ideal inputs on the computed FIRST-good open. The lower carrier is a literal actual reduced trace. The remaining member ideals are the TRUE open pullbacks, normalized on this SAME lower carrier. Their nonzero restrictions come from the genuine original noncontainment producer, not from an assumed recursive domain. The auxiliary boundary is EMPTY; no SNC with the accumulated exceptional boundary or sharp recursive driver is inferred.

Lemma 128.1 (Actual normalization on a true subcarrier retains the genuine relative generator through open restriction)**.** *For a coherent actual ideal relatively generated by f along N at an actual point, an actual subcarrier $S \subset N$ and a true ambient open, the WHOLE carrier-normalized open pullback to S is relatively generated by the SAME actual restricted f. Both full germ directions hold, including all germs vanishing on S. No principal restriction or model certificate is supplied.*

Proof. First restrict each actual generator relation from N to S. A genuine finite span in the original ideal then substitutes these true relations to become a span of f on S. Conversely f is a genuine member of the original ideal, so every span of f on S belongs to its actual normalization. This proves BOTH normalized germ directions on the true ambient carrier. For the open pullback, use the proved BOTH-direction normalized-ideal extension dictionary. An actual ambient cofactor restricts to a holomorphic cofactor on the open. Conversely extend an open cofactor by zero; it is holomorphic at the actual point, and the true open neighborhood

filter identity carries its relation back. The extensions agree with the original functions on the real open, so BOTH directions retain the same restricted generator and all normal germs. □

Theorem 128.2 (ONLY original geometry constructs every actual remaining lower member)**.** *ONLY original smooth positive pure carrier, true original SNC boundary, whole-boundary density and a genuine finite template history internally construct the computed FIRST-Z complement open. On EACH actual smooth reduced trace in that open, EVERY OTHER original member has an actual coherent, nonzero, relatively principal normalized open pullback. Both whole germ directions for its genuine generator hold at EVERY point of that lower carrier. No chosen good open, smooth lower carrier, nonzero, relative generator, or recursive domain certificate is supplied.*

Proof. The actual reduced-trace theorem constructs the literal FIRST-good open and each genuine closed smooth lower carrier there. Original finite history independently gives each original chronological ideal's actual coherence and genuine relative generator on the current carrier. Holomorphic open pullback preserves coherence; normalization on the proved actual lower smooth carrier is coherent by its true coherent vanishing ideal. The original noncontainment theorem constructs genuine nonzero along the literal ambient trace at every FIRST-good point. The actual open neighborhood filter identity carries THIS nonzero witness to the true open pullback. Actual inclusion in its normalization preserves that witness. Finally the preceding subcarrier/open producer supplies BOTH whole relative-generator directions on the SAME lower carrier. All remaining original positions, including unit members, are retained. □

Theorem 128.3 (The actual finite remaining lower product is a genuine invariant input)**.** *ONLY the same original inputs construct, on the SAME computed open and EACH actual reduced trace, the TRUE carrier-normalized finite product of ALL other original member restrictions. It is coherent, nonzero along the whole lower carrier, and genuinely relatively principal there. Its true ordinary trace order is the SUM of the actual remaining-factor orders, with multiplicities and unit factors retained. With this SAME lower carrier, EMPTY auxiliary boundary and mark one it is an actual invariant-domain input of ambient dimension bound. Empty remaining families give the genuine unit ideal. No lower product/domain or desired coherence/nonzero certificate is a premise.*

Proof. Form the literal ordered list of ALL original positions other than the chosen trace position, and use precisely their proved normalized open pullbacks. The preceding theorem supplies actual coherence, nonzero and relative generators of EVERY factor independently. The genuine finite relative-product theorem inducts over this actual list: true analytic order-product exclusion supplies nonzero products, actual normalized multiplication gives BOTH generator directions, and the true ordinary order is additive. The empty list is handled by the actual unit ideal. This proves the whole product geometry and exact SUM before any domain claim. For the mark-one input, the actual lower carrier is closed smooth, the product is genuinely coherent and nonzero there, and the EMPTY auxiliary boundary is SNC and transverse. Each actual smooth-carrier point has its true regular dimension bounded by the original ambient dimension. These are the real original domain fields, not supplied target-shaped certificates. This does not prove accumulated exceptional transversality, simultaneous sharp SNC, higher original invariant comparison, a terminating recursive intersection driver or exceptional tauD. □

129 Actual reduced smooth trace carriers on the genuine FIRST-Z complement

The real dynamic step now extends the SAME ORIGINAL test history and controls its first coordinate. The next recursive boundary-intersection phase needs TRUE lower trace carriers.

Construct them on the computed FIRST-Z complement from actual factor rigidity and reduced-ideal contact geometry. This does not identify FIRST Z with the full sharp defect Sigma, nor assert SNC for all traces or exceptional members.

Lemma 129.1 (Outside the actual FIRST defect, genuine original factors have order one and distinct whole proper germs)**.** *ONLY original smooth/SNC/whole-boundary density inputs and a genuine geometric-template history internally give this at EACH actual carrier point outside the literal computed current FIRST Z. Every actual proper chronological member ideal has ordinary trace order exactly one, and the quotient map on ACTUAL proper ideal-germ positions is injective. Thus two distinct proper original positions have distinct WHOLE ideal germs; raw indexed count is not substituted for distinct count.*

Proof. Independently original finite-history geometry supplies coherent nonzero relative-principal actual member ideals on the SAME real carrier. The proved true product order is at least the genuine quotient-image class count. Nonmembership in the literal computed FIRST defect gives the reverse inequality, hence actual equality. Apply the genuine factor- multiplicity/quotient-cardinality rigidity theorem: all proper factors have order one and the true proper quotient-image map is injective. A hypothetical full germ equality of distinct proper positions would give equality of their quotient classes, contradicting this actual injection. No equality of raw indexed and distinct counts was assumed. □

Theorem 129.2 (The genuine computed FIRST-good open contains actual smooth reduced lower trace carriers)**.** *ONLY original smooth positive pure carrier, true original ambient SNC boundary, whole-boundary density and a genuine finite template history internally construct the ACTUAL open complement of the literal current FIRST Z. For EACH original position, the preimage in THAT real open of the SAME current carrier intersected with its actual chronological ideal zero locus is a genuine closed smooth submanifold. At EACH of its points, its actual dimension is one below that of the actual carrier. On this ENTIRE actual open, the WHOLE carrier-normalized chronological ideal has BOTH actual ambient germ directions equal to the genuine vanishing ideal of that literal trace. No smooth/reduced trace, chosen contact or model certificate is supplied.*

Proof. The independently proved actual FIRST-Z finite analytic equations make its literal complement open. The actual carrier and genuine coherent member zero loci are closed, so each literal trace is closed there. At a trace point outside the actual FIRST defect, the preceding producer internally gives order one for THIS actual proper member. Genuine relative principality and nonzero are independently supplied by original finite- history geometry. The real order-one contact theorem constructs an actual contact function and unit cofactor BEFORE both whole reduced-ideal directions and genuine trace regularity. Actual ambient/open regularity transport gives the claimed true smooth trace submanifold in the computed open and its real codimension-one dimension dictionary.

For the ENTIRE localized ideal statement, at a true trace point use those actual whole reduced ideal directions and its genuine relative generator to collapse an additional carrier restriction. At a carrier point outside the trace, actual nonmembership in the coherent zero locus supplies a TRUE unit germ in the original member ideal; coherent multiplication then gives every actual holomorphic germ. The real trace is closed, so its vanishing ideal is also a unit there. Outside the SAME closed carrier the actual normalized ideal is a genuine unit, and the literal trace is absent, so both whole germ directions again agree. This retains all normal germs and proves the dictionary on the entire computed good open. Empty traces are included. Whole Z-empty traces follow when that phase condition holds, but no Z-erasing driver, simultaneous trace/exceptional SNC, recursive intersection resolution, exceptional tauD or termination is inferred. □

130 Genuine remaining-member nonvanishing on the actual reduced traces

The recursive intersection step of the sharp argument needs actual noncontainment in the OTHER original members. Separate reduced smooth traces do not supply this by themselves. Here true order-one unit cofactors convert containment into equality of full ideal germs, which contradicts the internally constructed proper-germ quotient injection. The conclusion concerns the actual chronological members; it does not identify them with intersections with arbitrary old ambient strict boundaries. FIRST Z remains distinct from the full sharp defect Sigma.

Lemma 130.1 (Order-one containment of genuine relative principal germs forces equality)**.** *Let the actual closed smooth carrier be N, and let I, J be coherent, nonzero along N, and genuinely relatively principal at a point x of N. Suppose that both true trace orders are one and their WHOLE ideal germs at x are distinct. Then J is nonzero along the literal trace $S = N \cap V(I)$ at x. No reduced trace, selected contact, nonzero restriction, or cofactor unit is assumed.*

Proof. Choose actual relative generators f and g. The genuine order-one contact theorem constructs BOTH full germ directions between I and the vanishing ideal of the literal trace S. If g vanished on S near x, then g would belong to I, so $g = cf$ on N near x for a holomorphic cofactor c. Actual order one makes $f(x) = 0$ and excludes order at least two for g. If $c(x) = 0$, then the product of the two true order-one vanishing germs would make g vanish to order at least two on N, a contradiction. Thus c is a genuine unit near x; its actual holomorphic inverse expresses f as a multiple of g on N. Substituting both actual relative generators gives BOTH full ideal germ inclusions, contradicting their distinctness. Hence g does not vanish identically on S near x. Negating the neighborhood vanishing condition gives actual points of S with $g \neq 0$ in every neighborhood of x, and g is an actual member of J. This is precisely the true nonzero restriction. □

Theorem 130.2 (Original inputs internally produce genuine other-member noncontainment on every FIRST-good trace)**.** *ONLY original smooth carrier, ambient SNC boundary, whole-boundary density and a genuine finite template history internally supply the following. At EACH point of a chosen actual chronological trace outside literal current FIRST Z, EACH OTHER original member ideal is nonzero on that literal trace. Moreover, every actual ambient neighborhood of that point contains a point of the SAME trace outside the OTHER member's true zero locus. Neither a nonzero restriction nor a germ-distinctness certificate is an input.*

Proof. Original finite-history geometry independently gives actual coherence, nonzero and relative generators of every member on the SAME current carrier. The genuine FIRST-good factor rigidity supplies order one for every actual proper member and injectivity of the true full ideal-germ quotient map. The chosen trace point makes its member proper. If the other member is proper, the two different original positions have DIFFERENT full ideal germs by that quotient injection. Apply the preceding unit-cofactor producer. If the other member is a unit, choose its actual unit germ; it is nonzero at the chosen trace point and cannot vanish on that trace in a neighborhood. Both branches therefore produce a genuine member nonzero on the actual trace arbitrarily near the point.

For actual geometric noncontainment, coherence continues this genuine member to every point of a sufficiently small ambient neighborhood. Intersect that neighborhood with its actual frequent nonzero trace points. A point where this continued member is nonzero cannot lie in the OTHER ideal's literal zero locus. Such points therefore occur in EVERY ambient neighborhood. This is the needed source-backed noncontainment of the true lower trace; simultaneous trace/exceptional SNC, recursive intersection resolution, a FIRST-Z-erasing driver, higher original invariant decrease, exceptional tauD and termination remain separate. □

131 Whole true ambient strict-trace geometry under an arbitrary discrete centre

Retain the genuine ambient strict trace, even when its remaining exceptional factor is nonunit. Establish its real whole geometry after ANY discrete centre inside the carrier, allowing both local incidence branches, before comparing chronological factors in [4, Remark 12.6 and pp. 293–295].

Lemma 131.1 (The genuine ambient pulled-back vanishing ideal lies in the true strict-member vanishing ideal)**.** *For ANY actual blowup and ANY genuinely closed H, $\pi^{*}\mathcal{I}_{H} \subset \mathcal{I}_{\mathrm{Strict}_{\pi} H}$ as ENTIRE actual ambient holomorphic germ ideals. No smoothness of H or its strict transform, coherence, local incidence or desired ideal comparison is a premise.*

Proof. The actual strict transform lies in $\pi^{-1}H$, because its defining off-centre preimage has that property and H is actually closed. A genuine pulled-back ideal germ has a true finite holomorphic span of actual original germs composed with π. Continuity pulls back each original vanishing neighborhood; actual strict membership places each projection in H, so every real composed generator vanishes there. Intersect these finitely many real neighborhoods and the actual finite span equality. The whole pulled-back germ vanishes on the actual strict member there and is genuinely holomorphic. This gives literal membership in its actual vanishing ideal, without assuming that ideal coherent. □

Lemma 131.2 (Actual nonzero normalized vanishing traces internally preserve true density)**.** *For ANY actual sets N, H, if the TRUE normalized ideal $\mathrm{Res}_{N}\mathcal{I}_{H}$ is nonzero along every actual point of N, then $N \subset \overline{N \setminus H}$. No smoothness, coherence, density, zero-set or generator certificate is supplied.*

Proof. At an actual carrier point choose its genuine nonzero trace germ and frequent nonzero carrier values. The literal normalized membership has a finite relative span of genuine H-vanishing germs. True eventual vanishing of each such germ and the real relative span equality force the chosen germ to vanish on the entire nearby $N \cap H$. Its frequent nonzero carrier values therefore lie outside H. Every actual ambient neighborhood consequently meets $N \setminus H$, proving true closure membership point by point. □

Theorem 131.3 (ONLY original geometry constructs the whole true strict-trace nonzero principal geometry and density for mixed discrete centres)**.** *ONLY original closed smooth N, singleton SNC H, actual density of $N \setminus H$, genuine discrete $C \subset N$ and its SAME actual blowup internally supply the following true objects. The actual strict carrier N' is closed smooth, the WHOLE actual ambient family $[H', D]$ is SNC, and the TRUE trace $I' = \mathrm{Res}_{N'}\mathcal{I}_{H'}$ is coherent and nonzero along EVERY point of N'. At EVERY such point it has an internally constructed genuine relative principal generator with BOTH full relative germ directions. Moreover $N' \subset \overline{N' \setminus H'}$ genuinely holds. The global centre is NOT assumed contained in H; no desired next geometry, nonzero, generator or density certificate is supplied.*

Proof. At each actual centre point zero-dimensional regularity gives a real isolating neighborhood. Transfer actual singleton-point normal crossings with the original SNC member along that genuine neighborhood; thus the actual whole centre has normal crossings with H. The real ambient blowup SNC theorem constructs the WHOLE family $[H', D]$, and true smooth-carrier strict-transform geometry independently makes N' closed smooth. Actual vanishing coherence and genuine normalization therefore give coherence of THIS true strict trace.

Original actual density supplies nonzero restriction of $\mathcal{I}_{H}$ along every original carrier point. Genuine zero-mark transformed geometry independently carries this to the ACTUAL raw restricted TOTAL on every point of N', with no centre containment in H required. The first lemma precedes any comparison with the new trace: actual relative span substitution makes the

raw restricted TOTAL a subideal of the TRUE normalized strict trace. Each real raw nonzero witness therefore gives a genuine new trace witness on the entire strict carrier. The second lemma gives the asserted new density from these internally constructed witnesses. This is an actual producer of density, not an assumed persistence clause.

Finally the already constructed whole ambient SNC family supplies a true local generator of $\mathcal{I}_{H'}$ at every ambient point. Both actual relative span substitutions give the full normalized principal dictionary on N', retaining normal germs and off-carrier units. These are the real next-step geometry inputs needed before whole nonmonomial/residual open comparison and chronological $I_i = D_i J_i$ gluing. Those comparisons, the separate τ_D algorithm, full sharp Σ and general termination remain subsequent producers. □

132 Actual mixed-centre curve contact and new exceptional exponent evolution

The literal τ_D counts genuine differences of original ambient traces and independently chronological residuals. Its refined monomial phase in [4, Lemma 12.8, pp. 294–295] must track ACTUAL remaining exponents. Prove the source-facing next-step recurrence for mixed discrete centres. There is no global assumption that the entire centre lies in a given original ambient member.

Lemma 132.1 (Actual ordinary relative order survives genuine open restriction)**.** *For any actual closed smooth carrier N and actual coherent ideal I nonzero at EVERY carrier point, restriction to ANY genuine ambient open preserves the EXISTING ordinary relative order of the actual pulled-back ideal at EVERY actual point. No order, coordinate or ideal-comparison certificate is supplied.*

Proof. Use the genuine ordinary marked input $(N, \varnothing, I, 1)$; actual smoothness and coherence supply its well formedness. Its actual marked open pullback is exactly the ordinary marked input of the literal pulled-back carrier and ideal, because the actual ordered boundary is empty. Apply genuine nonmonomial-order open invariance to this actual marked input, with the supplied true nonzero hypothesis. By the EXISTING definition, these orders are exactly the asserted ordinary relative orders. Thus neither a new order model nor a redefined scalar action is used. □

Theorem 132.2 (Actual mixed discrete-centre ambient strict contact loses the true incidence indicator)**.** *Let N be a genuine closed smooth curve, H an actual singleton ambient SNC member with true density of $N \setminus H$, $C \subset N$ ANY actual discrete centre, and π the SAME genuine blowup. Put $N' = \mathrm{Strict}_\pi N$ and $I' = \mathrm{Res}_{N'} \mathcal{I}_{\mathrm{Strict}_\pi H}$. At EVERY $z \in N'$ above C,*

$$\mathrm{ord}_{N',z} I' + \mathbf{1}_{\{\pi z \in H\}} = \mathrm{ord}_{N,\pi z} \mathcal{I}_H.$$

Actual current curve geometry and discreteness, not an order/drop certificate, are the only additional geometric inputs.

Proof. Independently construct actual whole new trace coherence and nonzero from the mixed-centre geometric producer. At an incident base point, genuine discrete-centre isolation constructs an actual source open on which the SAME restricted centre is contained in the SAME restricted ambient member. The real restricted projection is a genuine blowup, and both literal strict sets are exactly the true preimages of the global strict sets. Original curve regularity and centre discreteness restrict to this real open; true density restricts using the genuine open embedding. The previously proved first-incident contact theorem therefore applies to THIS actual local blowup, including the true point above the centre.

The actual whole normalized trace open-pullback equality is proved BEFORE comparing its orders. The preceding actual ordinary-order theorem on both source and target opens, together with genuine normalization invariance, transports the local drop equation back to the original

global objects. At a nonincident base point, the actual closed-member complement is a true open on which that member is absent. The literal strict member is contained in the inverse image of the original member, so the new point is absent from it too. True vanishing unit germs on both sides force their ACTUAL orders to zero. These two exhaustive real incidence branches give exactly the displayed indicator recurrence without any global $C \subset H$ premise. □

Theorem 132.3 (The actual NEW greatest exponent and every original-history member satisfy the genuine recurrence)**.** *In the preceding actual mixed situation let $D = \pi^{-1}C$. At EVERY $z \in N' \cap D$, the ACTUAL greatest exponent satisfies*

$$e_{D,z}(I') + \mathbf{1}_{\{\pi z \in H\}} = \operatorname{ord}_{N,\pi z} \mathcal{I}_H.$$

Finally, ONLY original smooth curve W, actual ambient SNC family E, true whole-family density, ANY ORIGINAL mark-one product test T, an ACTUAL next original-admissible centre and its SAME blowup internally supply this recurrence and the preceding contact recurrence for EVERY literal original strict member at that genuine next step. The actual old ordinary order can equally be taken on its true normalized ambient trace. No selected centre, desired new exponent, current-member geometry or history-comparison certificate is supplied.

Proof. The independently proved mixed-centre geometric theorem supplies the actual new closed smooth carrier, actual singleton NEW exceptional SNC family, actual carrier transversality and true trace coherence/nonzero. Genuine strict-carrier dimension preservation makes the actual new point a regular curve point. The real curve/member isolating-neighbourhood theorem therefore identifies the ACTUAL greatest exceptional exponent, defined using literal nearby divisibility, with the EXISTING ordinary relative point order. Substitute this proved equality into the preceding genuine indicator recurrence.

For the original-history consumer, original smoothness, actual SNC and true density internally construct the ORIGINAL dimension-one product domain. Its genuine TEST supplies the SAME geometric-template history and all actual discrete prefix centres. The actual next original-admissible centre lies in the true current cosupport; dimension-one domain preservation and actual cosupport geometry internally prove this next centre discrete. The genuine finite ambient trace theorem supplies EACH literal original strict member's true singleton SNC and current density. Genuine template well formedness gives the SAME current smooth carrier, and real pure-curve dimension preservation gives its actual dimension. Apply the two preceding producers to this genuine next step, retaining the SAME centre and map. Finally true normalization invariance identifies the old normalized trace order with the old unnormalized vanishing-ideal order, giving the asserted original-input recurrence.

This is actual finite-history NEW-exceptional exponent evolution. It does not assert transport of every OLD exponent, automatic original canonicality, strict decrease of τ_D alone, the refined monomial driver or full sharp Σ preservation/termination. Those remain separate genuine consumers. □

133 The genuine whole mixed-discrete-centre ambient trace factor dictionary

For the actual ambient trace I' retain its genuine greatest exceptional factor: I' need not equal relative STRICT J'. Prove the actual whole residual comparison for arbitrary mixed discrete centres in [4, Remark 12.6, pp. 293–295]. All local models and open dictionaries below concern the SAME actual map.

Lemma 133.1 (Literal local raw TOTAL has the SAME global residual germ dictionary)**.** *For actual smooth N, ANY actual coherent I nonzero along EVERY point of N, an actual blowup $C \subset N$, and ANY actual base open U, the literal restricted-map raw TOTAL residual on its*

actual preimage V contains EVERY local function g exactly when the SAME global raw TOTAL residual contains its actual extension. The local ideal is the actual pullback of the actual original open ideal. No local residual or coherence certificate is supplied.

Proof. The genuine empty-boundary zero-mark record is well formed; actual centre containment makes it admissible. Actual transformed geometry gives smooth strict carrier, coherent raw TOTAL, nonzero at EVERY strict-carrier point, and real SNC new exceptional member transverse to that carrier. The proved WHOLE raw open identity and literal strict-carrier restriction precede the whole nonmonomial open dictionary. Substitute them and the actual equality of restricted-map and global exceptional preimages to get BOTH full germ directions. No principal-ideal hypothesis is used here. □

Lemma 133.2 (A genuine incident source branch compares the SAME two global residuals)**.** *ONLY original smooth N, singleton SNC H, actual density away H, ANY genuine discrete centre $C \subset N$, its SAME blowup and an actual $p \in C \cap H$ internally give a true source open U containing p. On its ENTIRE actual preimage, EVERY local function's actual extended germ belongs to the global true trace residual exactly when it belongs to the global raw TOTAL residual. Global $C \subset H$ is absent.*

Proof. True zero-dimensional centre isolation supplies the genuine incident open, SAME restricted blowup, actual local smooth/SNC/density and real local centre containment in the member. The independently proved incident trace factor dictionary identifies its literal local trace residual with actual local all-power STRICT. Independently apply the ANY-member raw STRICT comparison to the actual restricted geometry: its raw residual equals that SAME independently defined all-power saturation. The genuine original open vanishing equality identifies the actual source open member ideal with the original whole ideal's pullback. Both preceding full open germ dictionaries therefore transport this independently established local comparison to the SAME two original global residuals at EVERY preimage point and for EVERY local function. No desired global comparison is input. □

Theorem 133.3 (ONLY original mixed discrete-centre data prove WHOLE actual trace residual equals independent raw STRICT)**.** *For ONLY original smooth N, singleton ambient SNC H, genuine density of $N \setminus H$, ANY actual discrete $C \subset N$ and its SAME actual blowup, put $N' = \operatorname{Strict} N$, $I' = \operatorname{Res}_{N'} \mathcal{I}_{\operatorname{Strict} H}$ and $D = \pi^{-1}C$. Then as WHOLE actual germ ideals*

$$\mathcal{N}_{N'}(I', [D]) = \operatorname{Sat}^{\infty}_{\mathcal{I}_D}\left(\operatorname{Res}_{N'} \pi^{*}\mathcal{I}_H\right) = J'.$$

The right side is the independently defined original raw all-power STRICT, including off-carrier units. No centre-in-member, residual, support, coherence or desired comparison certificate is input.

Proof. First independently produce the true whole trace geometry and the actual zero-mark raw geometry, including coherence of both residuals. At a point above $C \cap H$, use the preceding internally chosen incident branch. Restrict any ORIGINAL global function to its actual preimage open. Its actual extension agrees with the SAME original function on a genuine neighborhood, and actual coherent germ invariance transports BOTH residual memberships. At a point projecting outside H, a real closed-complement neighborhood makes the original member vanishing ideal contain 1; actual pullback and normalization give the raw unit germ. True strict membership gives the trace unit germ independently. Actual coherent ideal closure and the genuine residual inclusion make both residuals exactly all holomorphic germs there.

At every point outside the exceptional member, the actual singleton monomial ideal contains 1. Both actual colon directions identify its residual with its genuine normalized input. Off-centre strict membership is exactly preimage membership on a genuine open neighborhood, and the actual holomorphic inverse gives BOTH raw vanishing pullback directions. Finite relative

span substitution therefore identifies the two normalized inputs at that actual point. These cases cover EVERY ambient point; ideal extensionality gives whole residual equality. Finally the independent ANY-member raw STRICT comparison identifies the raw residual with the literal all-power saturation, rather than defining STRICT by the conclusion. □

Theorem 133.4 (The true mixed-centre ambient trace has its actual greatest factor times independent STRICT)**.** *Under the preceding ONLY original hypotheses, the independently defined greatest exceptional monomial part D' of the TRUE I' is coherent, nonzero and monomial along N'. With the independently defined raw STRICT J' above, the WHOLE actual ideal identity is*

$$I' = \mathrm{Res}_{N'}(D'\,J').$$

The OUTER carrier restriction is essential. The centre may have both incident and nonincident components; the actual exponent may vary.

Proof. The actual empty-old-boundary blowup supplies genuine new exceptional SNC and transversality, while the independent whole trace producer supplies smooth carrier, coherence and nonzero at EVERY actual carrier point. The genuine greatest-factor theorem internally constructs all asserted geometry of the ACTUAL monomial part BEFORE its true two factor directions. The preceding whole residual equality substitutes independent STRICT into those actual factor directions. Genuine restriction extensionality and normalization idempotence give the WHOLE ideal equality with the displayed outer restriction. This does not identify the trace with STRICT or erase its exceptional factor. Chronological accumulated factor transport and exponent evolution, the separate τ_D phase, full sharp Σ and termination remain OPEN; FIRST Z alone does not settle them. □

134 Genuine monomial division preserves the WHOLE intrinsic nonmonomial ideal

The composite mark-zero ORIGINAL dictionary is proved. Positive controlled transforms additionally divide by an actual exceptional power. Establish the genuine quotient geometry and factor uniqueness independently, before comparing controlled and TOTAL residuals. No full-invariant history or modified decrease is inferred.

Theorem 134.1 (Actual coherent monomial invertible-along division preserves the whole nonmonomial part)**.** *Take a genuine coherent ideal nonzero on a true smooth carrier transverse to an actual SNC boundary, and a coherent nonzero monomial denominator invertible along that carrier. If the TRUE ideal restricts into THAT actual denominator, then the intrinsic nonmonomial parts of the ideal and its independently defined genuine relative colon by this denominator are equal as WHOLE ambient germ ideals. Relative principality of the numerator is not assumed, and no factor or residual equality is input.*

Proof. Actual invertible-along division and the genuine restricted inclusion construct coherence of the REAL colon. Every original ideal germ belongs to that colon by actual holomorphic ideal multiplication, so original nonzero witnesses independently give colon nonzero. Construct the TRUE parts of this colon, including actual residual every-member excluded germs, both factor directions and its true coherent nonzero monomial part. Multiply that ACTUAL monomial part by the independently given actual denominator; genuine monomial product closure gives its real coherent/nonzero/monomial geometry BEFORE any uniqueness assertion.

Actual relative-colon inversion gives BOTH original ideal directions with the denominator times its true quotient. Multiply the true quotient parts directions by that denominator, and use genuine coherent commutativity and associativity to construct BOTH factor directions with the quotient residual and this actual combined monomial factor. Only then intrinsic factor uniqueness

identifies the original monomial denominator. Its real factor inclusion gives one whole residual direction; genuine nonzero cancellation gives the other restricted direction. The quotient residual is itself an actual normalized colon, so true finite-span colon normalization upgrades this to the FULL carrier germ direction. Outside the true closed carrier BOTH actual colons are units. No saturation=Nm shortcut for an arbitrary ideal is used. □

Lemma 134.2 (The real new exceptional denominator has genuine whole-boundary monomial geometry)**.** *For ONLY genuine smooth/SNC/transverse old geometry and a REAL contained whole-boundary NC blowup, EVERY actual power of the pulled-back centre ideal is coherent, nonzero and monomial along the SAME new strict carrier with respect to the ENTIRE ordered strict-old-plus-new boundary. No model, monomial denominator or exceptional exponent certificate is supplied.*

Proof. The genuine holomorphic unit ideal is nonzero and monomial along the true old carrier. Its actual mark-zero TOTAL is the whole unit ideal: zero-power division gives its true normalized pullback, and the actual pulled-back unit germ generates every holomorphic germ on and off the strict carrier. Apply the proved genuine monomial-factor chart theorem to THIS original unit ideal with order parameter equal to the desired denominator power and controlled mark zero. True carrier containment supplies its actual zero-mark cosupport condition. Its real monomial factor is precisely the actual exceptional power times the independently unit TOTAL, hence equals the true exceptional power. Genuine whole-boundary new SNC/transversality, coherence and the real principal-monomial theorem supply its actual nonzero/monomial geometry. All powers, including zero, and empty centres are included. □

Theorem 134.3 (Actual controlled transform and genuine TOTAL have equal whole accumulated nonmonomial parts)**.** *For ONLY a genuine original well-formed nonzero marked ideal, a REAL admissible centre and its ACTUAL blowup, the WHOLE intrinsic nonmonomial part of the actual positive-mark controlled transform equals that of the same numerator's genuine zero-mark TOTAL, computed on the SAME actual new carrier and ENTIRE actual accumulated boundary. This includes every mark, normal-vanishing germ and off-carrier unit. No principal numerator, residual comparison, desired factor or transformed history is supplied.*

Proof. Genuine admissibility and actual Hadamard geometry independently give the old restricted centre-power inclusion, and real strict-carrier pullback and power functoriality supply actual TOTAL divisibility by the true new exceptional power. Real TOTAL coherence/nonzero and new whole ordered geometry precede its factor computation. The preceding producer internally gives the denominator's actual monomial geometry; actual blowup geometry gives invertibility along the SAME strict carrier. Apply the genuine monomial-division theorem to THESE real inputs. The actual TOTAL is the carrier-normalized raw pullback. Real restriction idempotence identifies its true relative colon with the actual original controlled ideal, in BOTH whole germ directions. Thus the SAME-map ALL-boundary residuals agree. Earlier ORIGINAL full-value/history comparison, recursive sharp phases, exceptional tauD, modified decrease and termination remain OPEN. □

135 Actual normalized source generators give the same original controlled principal history

The ambient original product ideal need not contain all germs vanishing on its carrier. Consequently an ambient principal dictionary cannot be substituted for a whole RELATIVE principal dictionary. Keep the actual original marked record unchanged and prove the real normalized-source bridge needed by its controlled transforms. This is part of the literal ideal/factor mathematics in [4, §12, pp. 293–295].

Theorem 135.1 (Genuine normalized source spans construct the actual controlled quotient generator)**.** *Take ANY actual marked record with closed smooth carrier and coherent ideal, an ACTUAL admissible centre and the SAME genuine blowup. Suppose its actual normalized source ideal has BOTH relative principal germ directions at every carrier point. Then at EVERY true new carrier point, internally construct a genuine generator of the ACTUAL controlled ideal with BOTH whole relative germ directions. The source ideal itself is not assumed normalized and the actual original marked record is unchanged. No desired controlled generator or controlled-ideal comparison is supplied.*

Proof. Apply genuine raw relative generator transport to the actual normalized source ideal. Its defining real finite spans construct a generator of the true pullback restricted to the actual strict carrier. The actual whole pullback/normalization theorem, proved independently from the real strict-carrier map, identifies this raw normalized pullback with the normalized pullback of the ORIGINAL ambient ideal. This gives the true raw principal dictionary for the SAME controlled quotient numerator.

Original actual admissibility and genuine Hadamard divisibility give the real restricted inclusion in the actual exceptional denominator power. True exceptional geometry makes that denominator invertible along the actual strict carrier. The actual relative-colon producer therefore constructs its holomorphic cofactor and proves BOTH full quotient germ directions by genuine nonzero cancellation. That literal colon is exactly the unchanged original controlled ideal. No replacement marked history or desired factorization has been used. □

Lemma 135.2 (Original ambient SNC constructs genuine normalized product generators on any carrier)**.** *ONLY actual original ambient SNC and positive ambient dimension internally construct BOTH relative principal directions for the true normalized original product ideal on ANY carrier at EVERY carrier point. No carrier transversality, good-locus premise, chosen product generator or ideal comparison is supplied.*

Proof. Use the actual original SNC normal form on the ENTIRE ambient manifold, which is genuinely transverse to its SNC family. This internally gives a true holomorphic finite product generator and BOTH ambient germ-span directions for the literal original product, including all actual units and empty positions. The real restricted-span theorem then gives BOTH relative span directions on the independently supplied carrier, regardless of its incidence with the original family. Add the actual holomorphic condition from the defining normalization to obtain precisely the desired whole normalized ideal dictionary, retaining all normal-vanishing germs. □

Theorem 135.3 (ONLY original product geometry and ANY genuine original test internally supply every normalized principal stage)**.** *ONLY original closed smooth carrier W, actual ambient SNC family E, positive ambient dimension, ANY original product mark and ANY genuine ORIGINAL test T internally construct, at EVERY actual current carrier point, a real generator with BOTH full relative germ directions for the normalized ACTUAL current controlled ideal. Neither source/current principal generators nor a desired controlled record/history is input. True density, curve dimension, canonicality and carrier/original-boundary transversality are unnecessary for this principal-span conclusion.*

Proof. Induct on the SAME actual original test. At the empty history, the preceding original SNC theorem constructs the true normalized generator internally. At a genuine next original step, true original product coherence and actual original well formedness give the current closed smooth carrier and coherent ideal. The actual original test supplies that SAME centre's real original admissibility. The induction hypothesis supplies the true normalized current principal dictionary, so the first theorem constructs a generator of the ACTUAL next controlled ideal with BOTH whole relative directions. Normalizing this ideal again preserves that dictionary: real finite-span transitivity substitutes the actual generator in the forward direction, and the actual

generator's membership plus its true self-span gives the reverse direction. This retains every original map, centre, mark, ordered boundary entry and empty step.

This is a genuine missing principal-history producer on the unchanged original record. The subsequent real controlled ordinary-order shift, ambient-trace budget domination, positive-record admissibility of a selected τ_D centre and refined sharp driver/termination remain OPEN. It does not identify ambient ideals with their relative normalizations or remove the essential outer restriction in $I = DJ$. □

136 Genuine simultaneous curve and order-one member coordinates

The empty FIRST-Z and ambient-difference conclusions are not themselves simultaneous normal-crossings geometry. The missing local producer is a TRUE common coordinate system for the actual smooth curve and an actual smooth ambient member of ordinary contact order one. We construct it from real tangent derivatives and the holomorphic inverse theorem, as required in the geometric phase of [4, §12].

Lemma 136.1 (Actual order-one contact gives an AMBIENT unit cofactor)**.** *ONLY an actual closed smooth carrier N, true singleton ambient SNC member H, actual density $N \setminus H$, an actual carrier point on H and EXISTING ordinary relative order one internally give a genuine tangent contact u and actual member coordinate f, with $u = cf$ as AMBIENT germs and actual $c(x) \neq 0$. The equality is not restricted to the carrier and no common-coordinate or unit certificate is supplied.*

Proof. Actual order one and genuine original contact construction supply u, its true tangent field with nonzero derivative, and actual membership of u in the original ambient vanishing ideal of H. Choose the true singleton SNC coordinate f for H. Its independently proved BOTH ambient principal germ directions express $u = cf$ as an actual ambient germ with holomorphic c. If $c(x) = 0$, true point-power multiplication would force restricted order at least two for u, contradicting the nonzero genuine tangent derivative. Thus c is a genuine ambient unit near x. On a real open neighbourhood inside the actual SNC chart, H is therefore EXACTLY $u = 0$ in the whole ambient space. □

Theorem 136.2 (True order-one contact constructs an actual common curve/member coordinate family)**.** *Under the preceding hypotheses, assume the ACTUAL carrier is regular of dimension one at the point. Internally construct a genuine centered holomorphic coordinate system in which N is the first coordinate line and H is the hyperplane of its first coordinate. In particular the actual member coordinate is disjoint from the actual normal coordinates cutting out N.*

Proof. Take an actual dimension-one flattening Φ of N. The genuine nonzero tangent derivative of u forces its actual first flat partial to be nonzero; there is only one tangent coordinate. Replace this first flat coordinate by u, retaining EVERY normal coordinate. The proved actual contact-coordinate derivative calculation gives an injective complex differential. Holomorphicity on the actual common open and the real inverse theorem construct genuine centered coordinates Θ, with their actual image and inverse laws.

At the base point the entire new coordinate vector is zero: the old flattening is centered and $u(x) = 0$. On the actual inverse-theorem neighbourhood the normal coordinates remain EXACTLY the original normal coordinates, so membership in N is equivalent to all new coordinates except the first vanishing. The first new coordinate is EXACTLY u. The first lemma's actual ambient open equality identifies H with that coordinate's zero hyperplane. These are BOTH directions of actual ambient memberships, not merely a reduced intersection or dimension assertion. The first-coordinate singleton has cardinality one and is genuinely disjoint from the entire normal-coordinate set. This constructs the actual coordinate-family witness internally. □

Theorem 136.3 (Actual singleton transversality follows from ordinary order one along a smooth curve)**.** *ONLY the actual closed smooth pure curve, singleton ambient SNC member, true density, and actual EXISTING relative ordinary order one at EVERY incident carrier point internally prove* IsTransverse$(N, [H])$*.*

Proof. At a carrier point on H, the preceding theorem supplies the true simultaneous coordinate family, hyperplane cardinality and disjointness. At a carrier point outside H, use its genuine regular curve flattening and the coordinate-family empty-incidence construction; no condition on the absent member is required. The actual two cases give every field of the existing global singleton-transversality predicate. No transversality certificate is input. Combining whole original ambient families and accumulated exceptional families after the two actual phases remains a separate consumer; the full sharp Σ theorem is not asserted. □

137 WHOLE residual comparison on the SAME ORIGINAL positive-mark test history

Actual controlled-versus-TOTAL nonmonomial comparison is now proved for one genuine original step. Use it on the SAME original product marked input at every REAL original test prefix, retaining its positive mark and actual ordered history. This is a genuine original-history ideal comparison, without rebasing to a current reflexive invariant.

Lemma 137.1 (Every actual original test history supplies its true geometric-template history)**.** *ONLY a true original well-formed marked record and its ACTUAL original test tower internally give a genuine test tower for the independently defined unit-ideal zero-mark template with the SAME original carrier and ordered boundary. Every actual map, empty step and ordered exceptional position is unchanged; no canonicality or chosen comparison history is assumed.*

Proof. Induct on the actual original tower. The reflexive tower is a true geometric-template test tower. At a real next step its actual original centre is closed smooth, contained in the actual original carrier and NC with its ENTIRE ordered boundary. Independently proved carrier/list field equalities transfer these two actual geometric facts to the previous template position. Its true preceding test history and genuine well-formed initial unit template give real coherence there. The actual mark is zero, so the true template cosupport is its whole carrier. Thus that SAME centre is genuinely admissible for the actual template record. The same actual map and centre append its true next template step. No original cosupport is enlarged in the original history itself. □

Theorem 137.2 (SAME ORIGINAL full-product controlled history has the genuine chronological and composite-TOTAL whole residual dictionaries)**.** *Take ONLY an original smooth carrier, true original ambient SNC boundary and genuine density off its ENTIRE boundary. Fix ANY original mark and the original full-product marked record, with its actual original ambient product ideal and initially empty exceptional list. Along EVERY ACTUAL test tower for THIS SAME original marked record, its WHOLE intrinsic nonmonomial ideal is exactly the independently chronological full member product. It is also exactly the intrinsic ALL-accumulated-exceptional residual of the literal ORIGINAL ambient product pulled back by the SAME actual composite projection, computed on the SAME final strict carrier and ENTIRE ordered exceptional list. Every original mark, multiplicity, normal germ and empty step is retained. Neither the tower nor its next centres are assumed original canonical, and no residual or history comparison certificate is supplied.*

Proof. Original SNC and true density independently give actual original product coherence/nonzero and its well-formed marked record. The preceding theorem internally supplies the genuine SAME-map geometric-template test history. For the reflexive tower, the actual

empty-list nonmonomial colon is the true restricted original product, and independently proved whole finite product normalization identifies it with the chronological initial product.

Induct along the REAL ORIGINAL test tower. True original-domain geometry supplies well-formedness and genuine nonzero of its actual current controlled ideal BEFORE the next comparison. Its REAL original centre admissibility gives actual contained whole-boundary NC geometry. The proved genuine controlled-versus-TOTAL theorem FIRST identifies this actual next controlled residual with its SAME-map TOTAL residual. Actual one-step TOTAL nonmonomial transport replaces the genuine old numerator by its real old intrinsic residual. Only then the induction equality identifies it with the independently defined chronological full product. The independently proved whole current full-product geometry and genuine next all-boundary residual theorem give the real next chronological full product. Exact carrier/list field equalities retain the SAME original history throughout.

Finally the independently literal composite ORIGINAL full-product TOTAL residual theorem applies to the internally obtained actual template history. Its real carrier and ENTIRE ordered boundary coincide with those of the original controlled record. Compose the two WHOLE ideal equalities to obtain the literal same-projection TOTAL comparison. This compares real original-history nonmonomial ideals and supplies a true first-coordinate consumer; it does not identify the other ORIGINAL full invariant/history coordinates, infer canonicality, prove modified decrease or establish sharp termination. Recursive intersection and exceptional tauD phases and the FIRST Z versus full sharp Sigma distinction remain separate. □

138 A true SAME-ORIGINAL dynamic sharp step and genuine first-coordinate comparison

The chronological full product is now identified with the WHOLE nonmonomial part on the SAME ORIGINAL positive-mark test history. Use that actual dictionary to make the independently selected dynamic FIRST-Z centre genuinely admissible for this ORIGINAL record, and compare its TRUE original first coordinate. This is a real modified step, not an assertion of earlier-original canonicality or higher-history decrease.

Lemma 138.1 (The SAME ORIGINAL product history has the true current product first order)**.** *ONLY original smooth/SNC/density inputs and an ACTUAL test prefix for the SAME original full-product marked record internally identify its genuine nonmonomial order, at EVERY actual ambient point, with the order of the independently chronological product and with the true first order of its current product record. The original mark need not be one for this identity. No order/value or current-invariant comparison is supplied.*

Proof. The actual whole residual dictionary gives the independently chronological ideal as the nonmonomial ideal of the SAME original transformed record. Both real carrier/list dictionaries give its true carrier. The ordinary trace order of the actual chronological ideal uses its empty-boundary nonmonomial part, which is exactly its actual carrier restriction. Coherence supplies both restricted-span directions, hence BOTH actual cosupport threshold directions before equality of the genuine order functions. The independent current-product normalization theorem gives the same order for its actual current product record, establishing the three-way dictionary without using a selected full-value equality. □

Theorem 138.2 (The independently selected dynamic FIRST-Z centre is genuinely ORIGINAL-admissible and has constant ORIGINAL first order)**.** *ONLY original compact positive pure smooth carrier, true original ambient SNC boundary, whole-boundary density and a genuine SAME ORIGINAL mark-one product test prefix internally supply a nonempty actual dynamic FIRST-Z maximum centre, whenever its TRUE current distinct defect is nonempty. THAT SAME centre is admissible for the actual transformed ORIGINAL record, NC with its ENTIRE*

ordered boundary, and the ORIGINAL nonmonomial order is constant on the whole centre. No selected centre, constancy, original admissibility, count or defect certificate is supplied.

Proof. The preceding original-history theorem internally supplies the actual same-map template test history. The independently proved dynamic restricted-Z maximum producer constructs the true nonempty centre with its actual maximum-locus identity, genuine product-cosupport containment, smoothness and NC with the ENTIRE actual accumulated list. The true original controlled residual dictionary identifies that chronological product with the actual original nonmonomial ideal. Actual original ideal inclusion into its real nonmonomial part and coherent restricted-span transport therefore give genuine ORIGINAL mark-one cosupport containment. The actual geometric field dictionary transfers whole-boundary NC to the same original record, proving its actual marked admissibility.

Choose an actual point of this nonempty centre. BOTH real restricted maximum inequalities give equality of the actual auxiliary FULL values at every centre point. Taking their genuine first coordinates gives constant true current-product order. The preceding SAME ORIGINAL order dictionary converts this to actual ORIGINAL order constancy. The centre was independently constructed BEFORE either original admissibility or order constancy was asserted. □

Theorem 138.3 (The SAME actual dynamic sharp blowup extends the ORIGINAL test history and has true ORIGINAL first-coordinate nonincrease)**.** *Under ONLY the preceding real original inputs, the actual nonempty restricted dynamic FIRST-Z centre admits a genuine proper holomorphic blowup preserving the ENTIRE current FIRST-Z complement and APPENDING the SAME ORIGINAL mark-one product test history, retaining all old positions. At EVERY new ORIGINAL cosupport point the actual original image is in its old ORIGINAL cosupport and its genuine first nonmonomial coordinate does not increase; off the actual centre that coordinate is EQUAL. The whole actual new ordered boundary is SNC. Neither original canonicality nor higher full-coordinate decrease, exceptional tauD or termination is claimed.*

Proof. Use the actual independently selected centre and real map provided by the proved dynamic geometry theorem; its properness, holomorphicity, WHOLE current-defect-complement inverse and exact strict-old-plus-new SNC already concern THIS same map and centre. Apply the preceding actual original-admissibility/constancy construction to this actual maximum- locus identity. Original admissibility appends the SAME original test prefix, without rebasing. Original pure geometry supplies its genuine marked invariant domain and true transport along that original prefix. At a new cosupport point over the centre, use the PROVED genuine constant-order admissible-centre theorem with the independently established actual original centre order. This gives actual first-order nonincrease; original centre containment gives old original cosupport membership. Off the centre, the genuine original blowup-complement order theorem gives equality and actual cosupport transport gives old membership. This is a true SAME ORIGINAL comparison even at exceptional new points, but controls only its first coordinate. FIRST Z remains distinct from full sharp Sigma. □

139 Actual original mark-zero TOTAL histories and whole accumulated residuals

Retain the genuine original relative member and its actual mark-zero geometric transforms along the supplied REAL geometric-template history. These are independently defined from the chronological NEW-exceptional saturation ideals. Identify their whole intrinsic accumulated nonmonomial parts by real finite induction. A literal raw-pullback formula along the composite projection is a subsequent separate consumer; no invariant rebase or modified decrease follows here.

Lemma 139.1 (True zero-mark original TOTAL history geometry)**.** *ONLY an actual original well-formed coherent ideal, nonzero on its smooth carrier, and a genuine geometric-template test*

history internally supply well-formedness and nonzero of its ACTUAL geometric transform at EVERY real prefix. No desired raw-history domain, centre admissibility, coherence or nonzero certificate is supplied.

Proof. The existing actual entire-tower constructor first supplies real raw coherence and nonzero from actual contained-centre geometry, without raw cosupport admissibility. The independently constructed unit template is well-formed and has a genuine test history. Its actual well-formed transformed carrier, ENTIRE boundary and transversality transfer to the raw record by the previously proved true carrier/list field dictionaries. Thus the whole raw record is well-formed and nonzero at every real strict-carrier point. This holds for any original mark; the consumer below uses the actual original zero mark. No empty step is erased. □

Theorem 139.2 (Whole original-member TOTAL-history residual is the actual chronological ideal)**.** *ONLY original smooth carrier, genuine original ambient SNC boundary and true density outside that entire original boundary, internally give the following for EACH actual original boundary position and EVERY genuine finite geometric-template history. The WHOLE intrinsic nonmonomial part of its actual original mark-zero TOTAL record, computed using the SAME strict carrier and ENTIRE ordered accumulated exceptional list, is exactly the independently defined chronological relative member ideal. Both full ambient germ directions, carrier-normal germs and every empty step are retained. No desired history equality, principal model or no-factor certificate is input.*

Proof. Use the actual original restricted member ideal with empty exceptional list and zero mark. Genuine original SNC equations and whole-boundary density already construct its true initial well-formedness, nonzero and relative principality. For the reflexive position the genuine empty-list monomial ideal is the unit ideal. Testing the actual colon on the unit gives the first full germ direction, and genuine multiplication of restricted spans gives its reverse. Real restriction idempotence recovers the SAME actual initial member ideal.

For a true last blowup, the preceding raw geometry theorem independently supplies raw coherence/nonzero and ALL-boundary smooth/SNC/transverse geometry. The genuine template field dictionaries supply real centre containment and NC on both raw and chronological carriers. Actual one-step nonmonomial transport FIRST replaces the raw current ideal by its genuine old intrinsic nonmonomial part, without a principal premise. The induction hypothesis identifies THAT actual part with the separately constructed chronological ideal. Its independently proved finite-history geometry and whole old-member excluded germs then identify the ALL-new- boundary residual of its genuine next TOTAL with the literal next NEW- exceptional saturation. This is exactly the independent chronological recursion. The same zero mark and carrier/list field equalities give the stated WHOLE original raw-history dictionary. A literal composite projection pullback formula, earlier ORIGINAL full-invariant/history comparison, modified decrease and sharp termination remain OPEN. FIRST Z is distinct from the full sharp defect Sigma. □

140 Actual coherent relative-principal order-one ideals have genuine reduced smooth trace germs

Dynamic DISTINCT-defect recognition needs the ACTUAL chronological ideals, which need not be vanishing ideals of original ambient SNC hypersurfaces. This chapter constructs their real contact/unit, literal zero-support germ, reduced IDEAL and regularity from genuine coherent relative principality. No desired smooth/reduced support certificate is supplied.

Theorem 140.1 (Actual relative-principal order one internally constructs a true unit contact generator)**.** *For a genuine coherent nonzero ideal on a smooth carrier, with a TRUE relative principal generator at the actual point and ordinary carrier order one, internally construct a real tangent contact u on a true neighborhood. It belongs to the actual ideal, vanishes at the real point and generates that SAME ideal in BOTH full RELATIVE germ directions.*

Proof. Actual finite-order and true cosupport dictionaries give real order at least one and exclude order two. Genuine tangent geometry constructs a real contact u. Actual principal membership expresses $u = cf$ ON the true carrier. The original generator f belongs to the actual ideal, hence vanishes at the point by real order one. Thus u vanishes there. The proved actual restricted derivative/contact argument excludes order two for u, so the real cofactor c is a unit at that point. Its genuine holomorphic reciprocal supplies the reverse carrier principal relation. Actual finite span substitution gives BOTH full ideal directions with the SAME real contact; no ambient principal or original-hypersurface assumption is made. □

Theorem 140.2 (Real coherence and contact Hadamard prove the literal zero support smooth and the actual ideal reduced)**.** *At the actual order-one point, the LITERAL carrier intersected with the actual coherent ideal's zero locus has the same real germ as the genuine contact hypersurface. It is regular of carrier codimension one, and its TRUE vanishing-ideal germ equals the SAME actual relative ideal in BOTH complete holomorphic germ directions.*

Proof. Choose the actual finite coherent generators near the point. Their genuine membership and the first theorem's relative principal dictionary express EVERY such real generator as a true carrier multiple of u. Their finite actual germ equations hold simultaneously near the point. Thus every true contact point annihilates every actual coherent generator, hence EVERY actual ideal germ there by the real local presentation. Conversely actual coherent continuation keeps the SAME contact in the real ideal nearby, so any actual zero-support point has $u = 0$. This proves BOTH entire set-germ directions for the LITERAL support before smoothness is claimed.

Genuine contact Hadamard gives actual ambient generators consisting of u and true carrier-normal coordinates. The latter vanish on the real carrier and hence belong to the actual carrier-normalized ideal by its genuine relative principal dictionary. The contact also genuinely belongs. Actual coherent finite span closure therefore proves vanishing-ideal membership implies actual ideal membership; the reverse uses the TRUE relative contact multiple. Both whole germ directions hold. Transport through the just-proved literal support germ identity gives the actual reduced-ideal dictionary. Finally genuine contact multiplicity one gives real regular codimension-one geometry, transported to that SAME literal support. No equality of zero sets is substituted for the independently proved ideal dictionary. □

Theorem 140.3 (Only original data recognize every actual chronological order-one member by genuine reduced smooth trace geometry)**.** *ONLY original smooth closed carrier, genuine ambient SNC, whole-boundary density and an ACTUAL finite geometric history internally supply every chronological position's real coherent/nonzero/principal ideal. At each actual carrier point where its TRUE ordinary order is one, the preceding literal support regularity and BOTH actual reduced-ideal germ directions hold, through the internally constructed genuine contact, without a selected contact/model or desired dictionary.*

Proof. Apply the independently proved original-only finite-history producer to the SAME actual geometric tower and original position. It internally constructs true wellformedness, coherence, nonzero along the whole actual carrier and a real relative principal generator at this actual point. Apply the first two genuine geometric producers to THESE actual objects and the literal order-one assumption. These are real recognition conclusions for arbitrary genuine chronological ideals, not reuse of initial ambient SNC generators. The full dynamic finite-family DISTINCT-defect analytic formula/centre driver, composite original raw-TOTAL factor transport and sharp termination remain OPEN. □

141 Actual current relative-principal branches are recognized by genuine pair-intersection dimension

The actual chronological member ideals are coherent and relatively principal, including their carrier-normal germs. They are not replaced by original ambient hypersurface ideals. This chapter recognizes equality of TWO actual order-one ideal germs by the dimension of their LITERAL support intersection. It is a geometric producer for the next dynamic DISTINCT-defect formula, not a supplied branch-equality or analytic-centre certificate.

Theorem 141.1 (Large true pair dimension forces equality of actual order-one ideal germs)**.** *Let N be a genuine smooth closed carrier, regular of dimension $d+1$ at x. Let I, J be genuine coherent nonzero relatively principal ideals on N with actual ordinary carrier order one at x. If the LITERAL analytic pair $N \cap V(I) \cap V(J)$ has local dimension at least d there, then the SAME actual ideals have identical full holomorphic germ predicates at x.*

Proof. Actual coherence gives true analytic zero loci, hence the literal pair is analytic before any branch claim. For EACH real ideal, actual order-one geometry constructs its genuine contact generator. The pair lies in its true mark-one cosupport. The actual analytic codimension-one support theorem and true pair dimension give equality of the pair germ with the genuine contact hypersurface. The independently proved literal zero-support/contact germ identity therefore gives equality with EACH real support germ. Transport true vanishing-ideal membership through these BOTH set-germ directions, then use the independently proved actual reduced-ideal dictionaries in BOTH directions. Thus the actual ideal predicates agree; no support equality is substituted for an unproved reduced-ideal equality. □

Theorem 141.2 (Actual coherent ideal equality forces real pair dimension, with both recognition directions)**.** *Under the preceding genuine geometric assumptions, equality of the actual full ideal germs is EQUIVALENT to local dimension at least d of that SAME literal pair support. No caller-chosen reduced support or persistent branch relation is supplied.*

Proof. Actual finite coherent presentations make a true whole ideal-germ equality persist near x. Hence the two literal zero predicates agree there, by applying the actual equality to EVERY holomorphic germ. The actual pair germ therefore equals the first literal support germ. Independently established order-one contact geometry makes that support regular of carrier codimension one. Uniqueness of the real carrier regular dimension identifies this as d; transport regularity through the proved pair germ identity, then use true regular-point local dimension. The first theorem supplies the reverse. □

Theorem 141.3 (Only original finite-history inputs recognize actual chronological shared branches)**.** *ONLY original closed smooth carrier, genuine ambient SNC, whole-boundary density and an ACTUAL finite geometric history internally provide every chronological position's true coherent/nonzero/relative-principal ideal and common actual carrier. At any real carrier point of dimension $d+1$ where TWO actual orders are one, equality of their full ideal germs is EQUIVALENT to the genuine local dimension condition on their literal pair.*

Proof. Apply the original-only finite-history producer separately to the TWO real original positions on the SAME actual tower. Its independently proved whole geometric-field dictionary identifies their carriers with the SAME true template transform before transporting nonzero and relative generators. Apply the preceding geometric equivalence to these actual ideals and the actual carrier. No desired ideal relation, selected contact or replacement hypersurface is input. The dynamic finite DISTINCT-defect formula/driver, composite original raw-TOTAL exceptional-factor transport and sharp termination remain separate OPEN consumers; FIRST Z is not full Σ. □

142 Genuine single-step TOTAL product factorization uses the SUM of actual NEW exceptional exponents

The genuine WHOLE relative-product TOTAL transport is already proved. This chapter combines the independently proved actual individual exceptional factorizations on the SAME new strict carrier. The resulting exponent is the SUM of the actual individual normalized intrinsic exponents. Identifying this sum with the intrinsic greatest exponent of the full product, and iterating to composite original raw TOTAL, remain OPEN.

Lemma 142.1 (True ambient exceptional powers and actual relative residual generators multiply with their real exponent sum)**.** *A TRUE ambient principal divisor germ with actual equation a has genuine power ideal generator a^e at every exponent including zero. If actual factor ideals have independently established relative dictionaries $F_i = \mathrm{restrict}_N(D^{e_i} J_i)$ and each true J_i is relatively principal, then their actual relative finite product has BOTH full germ directions equal to $\mathrm{restrict}_N(D^{\sum_i e_i} \prod_i J_i)$. The outer restriction and actual residual product are retained.*

Proof. Induct on genuine ideal powers using the actual unit ideal and true ambient principal-product span algebra. Independently, actual normalized product finite spans allow a true ambient left generator and actual relative right generator to multiply in BOTH full germ directions. Induct on the actual ordered factor list: the genuine tail residual product has a real relative generator from the ACTUAL factors. The independently established head and tail factorizations give real generators $a^{e_i} c_i$ and $a^s c_{\mathrm{tail}}$. Their true product equals $a^{e_i+s}(c_i c_{\mathrm{tail}})$ by actual power addition and commutative multiplication. Genuine full relative principal dictionaries on BOTH sides therefore agree. The true empty unit supplies exponent zero. No desired full-product factorization or greatest-exponent certificate is input. □

Theorem 142.2 (Actual individual strict residuals give the genuine WHOLE next-product exceptional factorization)**.** *For actual coherent nonzero relatively principal old ideals along a smooth closed carrier and a REAL blowup with centre contained in that carrier, the WHOLE mark-zero TOTAL of their actual relative product has EXACTLY BOTH germ directions of the actual next residual product times the NEW exceptional ideal to the SUM of the actual individual normalized exponents, with essential outer new-carrier restriction. This holds at EVERY ambient point, not only at the carrier or at ideal zero sets.*

Proof. Actual empty-boundary geometry gives a genuine SNC NEW exceptional divisor and a transverse new strict carrier. Its true ambient equation, including a unit equation at nonincident points, is internally supplied from that actual SNC geometry. EACH actual old ideal independently yields its real new TOTAL, true NEW-exceptional all-power saturated strict residual, actual relative residual generators, and BOTH full normalized intrinsic factor directions. At a real carrier point apply the first genuine finite power-sum producer to THESE actual individual dictionaries and their REAL normalized exponents. The already proved WHOLE TOTAL/product commutation identifies the left side with the SAME actual old full-product TOTAL. Outside the SAME closed new strict carrier, BOTH actual ideals are genuine unit germs by their real outer restrictions. This upgrades the result to EVERY ambient germ predicate, retaining carrier-normal germs. □

Theorem 142.3 (ONLY original finite histories identify the actual next chronological full product in that true factorization)**.** *ONLY original smooth closed carrier, ambient SNC, whole-boundary density, an ACTUAL finite geometric history and a REAL next blowup whose centre lies in its actual current carrier internally give the preceding WHOLE factorization. Its residual product is EXACTLY the independently defined next chronological full product, with every original position and every actual empty step retained. No desired next ideal/product or exponent-sum dictionary is a caller premise.*

Proof. Original-only chronological geometry supplies EVERY actual current member ideal's coherence, nonzero and true relative generators. Its independent geometric-field identity puts all factors on the SAME actual current carrier BEFORE applying the second producer. Unfold the genuine chronological recursion at the actual next step: EACH ideal is literally old pullback then NEW-exceptional all-power saturation. Its actual carrier field is the SAME strict carrier, so finite ordered map/product equality identifies the constructed residual product with the ACTUAL next chronological full product. The displayed exponent retains the actual individual intrinsic normalized exponents and their true SUM. Greatest-full-product exponent identification, composite original RAW TOTAL transport, comparison/decrease of the modified original invariant/history, recursive intersections/exceptional cleanup and sharp termination remain independent OPEN mathematical producers. □

143 The ACTUAL next chronological product equals the genuine full-product relative STRICT

The independently constructed chronological family saturates EACH true member TOTAL by ALL powers of the NEW exceptional ideal. The previous actual exponent-SUM factorization does not by itself identify the full-product saturation. This chapter proves that missing equality using real transverse member primality and actual no-common-factor witnesses. Composite ORIGINAL raw TOTAL and the modified original-invariant decrease remain OPEN.

Lemma 143.1 (A genuine relatively principal ideal without an actual transverse-member factor has all-power colon cancellation)**.** *For an actual transverse SNC member and a true relatively principal ideal, a genuine ideal germ outside that actual restricted member ideal internally proves cancellation of EVERY member-ideal power, including exponent zero. BOTH full colon/ideal germ directions hold. No desired cancellation, primality or residual certificate is input.*

Proof. Choose a true relative generator. The actual excluded ideal germ forces THIS generator outside the member ideal by its genuine full span dictionary. At an incident member point the actual free SNC coordinate supplies a true ambient divisor equation and nonzero points in EVERY relative carrier neighborhood. Independently proved relative member primality forces the cofactor of a relation involving that equation to contain the equation. Real nonzero cancellation on the carrier yields the relation with one fewer factor. Actual induction cancels EVERY power. At a nonincident point the actual divisor ideal contains one, so testing its actual power at one supplies the result. The reverse direction follows real relative ideal multiplication. Carrier-normal germs are retained throughout. □

Theorem 143.2 (True normalized divisor-power factorization and independently proved cancellation identify the literal all-power saturation)**.** *If actual full germ dictionaries independently give* $I = \mathrm{restrict}_N(D^e S)$ *and true relatively principal* S *has a genuine excluded-member germ, then the independently defined literal all-power relative saturation of* I *by* D *has BOTH full germ directions of* S*. The exponent need not be asserted greatest.*

Proof. For saturation-to-residual, every actual I germ is in the true normalized relative span of S, by the established factor dictionary and real principal-product algebra. Thus a genuine saturation witness gives the corresponding power-colon relation for S; the preceding independent all-power cancellation removes that power. Conversely, for a true S germ, choose the ACTUAL factor exponent e. EVERY actual D^e germ times that S germ lies in the true product ideal, then in the normalized factor and hence in I by the independently supplied BOTH-direction dictionary. Actual restriction idempotence gives the required literal colon witness. No full-product saturation equality is assumed. □

Theorem 143.3 (ONLY actual old member geometry constructs the WHOLE genuine next STRICT/product equality)**.** *For real coherent nonzero relatively principal old ideals along the same smooth closed carrier and a REAL blowup with centre contained in it, the WHOLE independently defined next relative STRICT of their ACTUAL relative product equals the actual relative product of EACH independently constructed next STRICT. Consequently ONLY original smooth/SNC/density and an actual finite geometric history internally identify the next chronological full product with that genuine full-product STRICT. All ordered positions and actual empty steps remain. No desired product/saturation dictionary is a caller premise.*

Proof. Actual blowup geometry gives genuine singleton-NEW-exceptional SNC and true new-carrier transversality. Independently construct EACH actual next STRICT, its coherence and real relative generators, and its genuine NEW-member excluded germ by actual maximal-factor exclusion. Real transverse member primality constructs an excluded germ for their TRUE finite product. Only then apply the genuine exponent-SUM factorization of the ACTUAL old full-product TOTAL and the preceding saturation dictionary. The actual TOTAL is the true normalized raw pullback, and the already proved literal restriction/saturation idempotence identifies its saturation with the independent next STRICT definition. Outside the true closed new carrier, BOTH sides are actual units; thus the ideal equality is WHOLE. The original finite-history consumer independently supplies all actual current member coherence/nonzero/principal data and uses the literal next-family recursion. This proves NEW-exceptional chronological STRICT/product commutation, not composite original raw-TOTAL factor transport or sharp terminating history. □

144 Whole actual raw member open pullback and independent all-power STRICT

Keep literal original TOTAL pullback and independently defined all-power exceptional saturation distinct from the genuine ambient strict trace. The following supplies actual open gluing data for [4, Remark 12.6, pp. 293–295], including mixed centre incidence. No global containment of the centre in a member is presumed.

Lemma 144.1 (Genuine normalization commutes with the actual whole open pullback)**.** *For actual closed smooth N, ANY actual coherent germ ideal I, and ANY actual open U, the literal whole pullback of* $\mathrm{Res}_N I$ *is* $\mathrm{Res}_{N\cap U}(\iota_U^* I)$*. No desired normalization or germ comparison certificate is supplied.*

Proof. Actual coherence of I and the genuine vanishing ideal of smooth N constructs coherence of its actual relative normalization. The real restricted-span open theorem compares EVERY local germ to its actual extension, in BOTH directions, with literal restricted pullback generators. The coherent pullback theorem for the SAME normalized global ideal compares the SAME extended germ. Ideal extensionality gives the whole equality. □

Lemma 144.2 (The SAME restricted blowup has the genuine whole raw TOTAL open pullback)**.** *For actual smooth N, ANY coherent I, an actual blowup with $C \subset N$ and ANY actual base open U, form the ACTUAL preimage V and SAME restricted map $\rho : V \to U$. The literal raw restricted TOTAL* $\mathrm{Res}_{\mathrm{Strict}_\rho(N\cap U)}\, \rho^*(\iota_U^* I)$ *equals the WHOLE actual pullback to V of* $\mathrm{Res}_{\mathrm{Strict}_\pi N}\, \pi^* I$*.*

Proof. The actual restricted projection is holomorphic because composing it with the genuine open inclusion is the SAME original projection on the actual preimage open. Whole coherent pullback composition on both sides and the literal equality of the two composed maps give the genuine whole raw ideal equality BEFORE normalization. The real strict-transform restriction theorem identifies the literal local carrier with the preimage of the original actual strict carrier.

Genuine strict smoothness and pullback coherence then allow the preceding normalization theorem, yielding the asserted whole ideal equality. Neither a new map nor an ideal equality is an input. □

Theorem 144.3 (ANY actual member blowup has raw residual equal to independently defined all-power STRICT)**.** *ONLY actual smooth N, singleton ambient SNC H, genuine density of $N \setminus H$, ANY actual blowup with $C \subset N$ supply the following. The genuine residual of the WHOLE normalized raw TOTAL, with respect to the actual new exceptional member, equals the INDEPENDENT all-power relative STRICT ideal of the literal original member pullback. The centre need not lie in H, and no discreteness, selected-centre certificate, principal total model or desired saturation equality is supplied.*

Proof. Original actual SNC supplies vanishing coherence; true density supplies nonzero along EVERY carrier point. The real empty-boundary zero-mark record is well formed, and its genuine cosupport is exactly N, so the actual centre is genuinely admissible for this record. Real transformed well-formedness independently supplies actual smooth carrier, SNC new exceptional member and transversality. Genuine zero-mark geometry supplies coherence and nonzero on EVERY new carrier point. Actual ambient SNC principal generators composed with the SAME real projection supply BOTH relative principal directions for its actual raw TOTAL.

Only after these real inputs are constructed does the genuine arbitrary- power singleton cancellation theorem apply. It identifies the actual normalized raw saturation with its genuine greatest-factor residual as WHOLE ideals, including off-carrier unit germs. The independently proved normalization-idempotence directions identify that literal saturation with the independent definition using the ORIGINAL unnormalized raw pullback. Thus the result is not obtained by naming the residual STRICT. It is the raw member comparison needed before the mixed-incidence trace comparison; chronological all-member factors and τ_D remain open. □

145 True relative smooth member trace ideals have actual prime cancellation

The arbitrary exceptional-colon STRICT dictionary needs actual divisor prime cancellation. We construct it from genuine coordinate primeness and real member-preserving carrier retractions, using no desired-prime certificate and no Hilbert–Samuel import cone.

Lemma 145.1 (The actual coordinate hyperplane vanishing germ predicate is prime)**.** *For real holomorphic germs f, g at the origin of $\mathbb{C}^n$, if their actual product vanishes on a true coordinate hyperplane near the origin, then one actual factor vanishes there near the origin.*

Proof. Genuine coordinate Hadamard division identifies BOTH complete vanishing-ideal germ directions with the principal coordinate span. Actual holomorphic germ multiplication turns product membership into divisibility by that SAME coordinate germ. Its independently proved genuine primeness gives a divisor of one actual factor. A true germ representative of the quotient supplies a real holomorphic cofactor; Hadamard's reverse membership direction then gives the displayed actual factor vanishing. No prime or quotient witness is input. □

Theorem 145.2 (Original SNC member geometry internally gives actual ambient prime vanishing)**.** *At every real point of a genuine SNC family, the actual holomorphic-germ vanishing predicate of each SAME original member obeys prime cancellation: actual product membership forces membership of one actual factor.*

Proof. Outside the true closed member every holomorphic germ is an actual member of its vanishing ideal, so the assertion holds. At an incident point the genuine original SNC theorem internally supplies the SAME actual flattening, true zero centre and real member coordinate.

Pull an arbitrary original function through its genuine chart-and-flattening inverse. Actual inverse holomorphicity, continuity, the two real inverse equations and the literal original member coordinate dictionary prove BOTH entire vanishing-germ transport directions to the true coordinate hyperplane at the origin. Apply the first real prime producer to the actual flattened factors. Pull its actual factor membership back through that same whole dictionary. No original member-prime or chosen local generator certificate is supplied. □

Theorem 145.3 (Genuine transverse-carrier geometry internally constructs actual relative member prime cancellation)**.** *For a real SNC family transverse to the genuine carrier, at every true carrier point, membership of a holomorphic product in the SAME actual restricted member ideal forces actual restricted membership of one factor.*

Proof. The actual SNC/transverse theorem constructs a true holomorphic carrier retraction preserving every incident original member. The genuine restricted span/trace-vanishing dictionary turns the product premise into actual vanishing on the literal carrier/member intersection. Pull both actual factors through that real retraction. Its true nearby image lies in the carrier and in the SAME member when the original point lies there, so the pulled-back product genuinely vanishes on the entire ambient member germ. The second real ambient prime theorem supplies actual vanishing of one pulled-back factor. On the true carrier the actual retraction fixes every nearby point; the original factor therefore vanishes on the literal trace. The true trace vanishing/restricted-span dictionary gives its entire relative ideal membership. Outside the member, genuine closedness gives the unit ideal case. This proves actual relative prime cancellation from real geometry, without assuming a domain, prime ideal or desired divisibility certificate. □

Theorem 145.4 (Only original inputs construct the actual exceptional relative prime law on the SAME first blowup)**.** *For ONLY original compact smooth fixed-positive-dimensional carrier, original SNC, whole-boundary density and nonempty FIRST Z, the SAME actual first blowup has the genuine relative prime cancellation law for EVERY true exceptional position of each actual original member total carrier, at EVERY actual carrier point. No selected blowup, transverse/prime model or desired STRICT transform equality is an input.*

Proof. The independently verified original-only total-trace producer constructs the SAME real first blowup and every actual member total marked record. Its independent genuine wellformedness supplies the true exceptional SNC list and actual carrier transversality before the third prime theorem is applied. Thus the displayed law holds for all actual functions and original positions on those SAME real objects. Higher exceptional powers, arbitrary-colon saturation equals relative STRICT, chronological histories and modified termination remain subsequent producers. □

146 Genuine products of actual relative monomial factors on the SAME carrier and boundary

Composite ORIGINAL raw-TOTAL comparison needs genuine transported old monomial factors, not just a renamed residual. Two such factors must first be combined on the SAME actual strict carrier and accumulated boundary. This chapter proves real relative monomial-product closure from common SNC coordinates before any composite comparison.

Lemma 146.1 (True common adapted SNC coordinates construct actual principal-monomial geometry of the real product)**.** *For two ACTUAL coherent nonzero relative monomial ideals along the SAME smooth carrier and SNC boundary, internally construct real principal-monomial geometry of their ACTUAL product at EVERY carrier point. Its true common-coordinate exponent at every original member position is the SUM of the two actual intrinsic factor exponents. No product chart, principal model or desired product generator is input.*

Proof. Choose ONE actual simultaneous carrier/SNC adapted chart from genuine original geometry. The EXISTING actual monomial-generator theorem gives BOTH factor relative spans in THESE SAME coordinates, with EACH factor's ACTUAL intrinsic exponents; this old producer is reused, not assumed as a new target field. Genuine relative-product finite-span algebra multiplies THOSE actual generators. Real power addition and finite-product distribution identify the result with the coordinate monomial having the literal SUM exponent at EVERY actual original position. The same actual chart, carrier plane, full member equations and free-coordinate injection supply the complete real principal-monomial witness for the ACTUAL product. □

Theorem 146.2 (Real coherent nonzero monomial factors multiply on the SAME carrier and whole ordered SNC boundary)**.** *The ACTUAL product of two genuine coherent nonzero relative monomial ideals is coherent, nonzero along the SAME carrier and monomial for the SAME entire ordered boundary. This includes empty incidence and real unit factors. In particular independently constructed actual monomial parts of coherent nonzero original ideals supply such genuine factors internally. No desired product-monomial/principal or nonzero certificate is input.*

Proof. Internally choose ONE actual coordinate system adapted simultaneously to the real carrier and entire SNC list. Apply the preceding theorem to BOTH actual factors in THESE SAME coordinates. Genuine finite relative-product span algebra multiplies the actual two generator equations. True coordinate power addition and finite product distribution identify their real product with the coordinate monomial whose exponent at EVERY original position is the SUM of those two actual intrinsic exponents. This constructs genuine principal-monomial geometry at EVERY carrier point; it is not assumed. Actual product coherence precedes the proved principal-monomial theorem which supplies true nonzero and monomiality. For actual original monomial parts, the genuine existing parts constructor independently supplies each factor's coherence/nonzero/monomiality BEFORE this product producer applies. Composite raw-TOTAL transport/uniqueness and original modified-history decrease remain independent OPEN consumers. □

147 Actual relative-principal products retain genuine generators, finite order and nonzero germs

Actual chronological ideals include the carrier-normal germs. The initial ambient-principal product theorem cannot be substituted for their relative product geometry. This chapter proves genuine product algebra and true carrier order additivity from the actual relative generators. No desired product order or nonzero certificate is supplied.

Lemma 147.1 (The actual carrier-normalized product has the true relative product generator)**.** *If actual ideals I, J have true relative generators f, g at a carrier point, the actual product normalized along that carrier has generator fg in BOTH full relative germ directions. The literal unnormalized product's carrier cosupport is also exactly the true carrier vanishing-order condition on fg, at EVERY integer threshold including zero.*

Proof. Actual product membership is a genuine finite span of products of true factor germs. Substitute each real relative factor cofactor, multiply the cofactors and use actual finite restricted-span substitution. Conversely f and g themselves belong to their true factors, so their actual product belongs to the genuine product; restricted span gives the reverse entire normalized ideal direction. The outer carrier normalization retains all normal-vanishing germs and is not discarded. Each real relative multiple preserves every actual carrier vanishing order. True normalization/cosupport identities then give BOTH stated order predicates for the literal product. □

Theorem 147.2 (True carrier retractions prove exact relative-product order additivity and an actual nonzero product germ)**.** *For genuine coherent factor ideals nonzero along a smooth carrier*

and true relative principal generators at the point, the LITERAL product's actual ordinary carrier order equals the SUM of the two actual factor orders. Internally construct a true product germ nonzero arbitrarily near that point on the real carrier. No ambient principal or product-nonzero premise is used.

Proof. Actual finite-order/cosupport dictionaries give each true generator its real factor order and exclude its next order. A genuine local carrier retraction takes these to actual ambient point-ideal powers. True power multiplication proves the product's SUM lower bound; actual associated graded noncancellation proves exclusion of the next power. Retraction transports BOTH bounds back to the SAME carrier. The first producer then identifies these with actual product cosupports; genuine finite ordinary order gives exact equality. If this actual product generator vanished on the carrier in some true neighborhood, its equality there to the actual zero germ in EVERY point-ideal power would contradict the independently proved next-order exclusion. Thus it is genuinely nonzero arbitrarily near the point, and its actual product membership supplies the required witness. □

Theorem 147.3 (Only original finite-history data supply true two-position chronological product orders)**.** *ONLY original smooth closed carrier, genuine ambient SNC, whole-boundary density and an ACTUAL geometric finite history internally supply BOTH chronological positions on the SAME current carrier. Their literal product has actual carrier order equal to their true SUM and is nonzero at EVERY actual carrier point. Carrier normalization has the true relative product generator there. No desired product/domain/order certificate is supplied.*

Proof. The original-only finite chronological producer gives real wellformedness, coherence, nonzero and genuine relative generators separately at both actual positions. Its independent geometric-field dictionary identifies both carriers with the SAME actual template transform before these facts are transported. Apply the first two genuine producers to these actual objects at every actual carrier point. The result supplies real input for the next full finite-family product and dynamic DISTINCT-defect formula; that formula/driver, composite raw-TOTAL factor comparison and sharp termination remain OPEN. FIRST Z remains distinct from full Σ. □

148 The WHOLE genuine mark-zero TOTAL transform commutes with actual relative finite products

Before comparing the new chronological residual product with a transformed old product invariant, actual pullback/product normalization must genuinely commute. This chapter proves that WHOLE ideal equality from real relative generators and the ACTUAL strict-carrier projection. No desired transform or product comparison is supplied. New exceptional exponent sums and the composite original raw-TOTAL comparison remain separate OPEN consumers.

Lemma 148.1 (Actual paired relative generators supply a determined true finite-product equation and its real TOTAL pullback)**.** *For an actual finite ordered list of ideals with true relative generators f_i at a real carrier point, the ACTUAL relative product has the TRUE generator $\prod_i f_i$ in BOTH full relative germ directions. At any real point of the actual strict carrier of a genuine blowup, an actual old relative generator pulls back to THAT SAME generator composed with the actual projection, in BOTH true mark-zero TOTAL ideal directions.*

Proof. Induct on the real paired list, using the actual unit generator and the independently proved two-factor normalized-product dictionary. This fixes the actual product generator, rather than choosing an unrelated principal representative. Genuine strict-carrier containment in the real old carrier preimage supplies the old generator's point. Compose its actual holomorphic cofactors with the true blowup projection. Actual raw pullback finite spans, followed by the genuine outer strict-carrier normalization, supply the forward true relative generator direction. The actual old

generator itself belongs to its true ideal, so its real composition belongs to raw pullback; restricted span proves the reverse. Carrier-normal germs are retained. □

Theorem 148.2 (WHOLE mark-zero TOTAL of the actual relative product equals the real relative product of individual TOTALs)**.** *For genuine coherent relatively principal ideals along a smooth closed carrier and an ACTUAL blowup, the WHOLE mark-zero TOTAL transform of their ACTUAL finite relative product is EXACTLY the actual relative product, on the SAME new strict carrier, of EVERY individual real mark-zero TOTAL ideal. This is equality of full ideal predicates at EVERY ambient point, not only equality of zero loci or ordinary orders.*

Proof. At a real new-carrier point, genuine strict-carrier projection supplies the actual old-carrier point. Choose the actual relative generator for EVERY old position there. The first lemma supplies the determined old product generator and its true TOTAL pullback; separately it supplies each actual individual TOTAL generator. Genuine paired-list multiplication on the SAME strict carrier gives their actual product generator. The real finite product of the composed functions equals the composition of the SAME original finite function product, so BOTH full ideal predicates agree. Outside the real closed strict carrier, actual mark-zero TOTAL is its normalized raw pullback and hence a genuine unit germ. The true relative product is likewise unit outside that SAME carrier (including its empty unit list). These actual BOTH off-carrier directions upgrade the equality to EVERY ambient point. No ambient-principal or cancellation shortcut is used. □

Theorem 148.3 (ONLY original chronological inputs supply the genuine WHOLE product-TOTAL transport)**.** *ONLY original smooth closed carrier, ambient SNC, whole-boundary density and an ACTUAL finite geometric history internally supply the SAME current family and carrier. For EVERY actual next blowup, the WHOLE mark-zero TOTAL of the true full chronological product equals the true finite relative product of ALL individual actual member TOTALs, on the SAME new strict carrier. No desired product/transform dictionary is a caller premise.*

Proof. The original-only geometric-template history supplies the real current closed smooth carrier. EACH actual chronological member's coherence and relative generators are independently constructed and their carrier fields are identified with this SAME template before use. Their true relative finite product is the actual already defined chronological full product. Apply the preceding WHOLE actual TOTAL commutation theorem to these actual objects and the given REAL blowup. This closes actual product-pullback normalization transport, before individual exceptional factors are combined. True new-exceptional exponent-sum/greatest-factor comparison, iteration to the different composite original raw TOTAL, original modified-invariant comparison/decrease and sharp termination remain OPEN. □

149 Genuine whole open dictionaries for the actual ambient strict-trace residual

The local incidence branches must compare the SAME global true trace and its greatest-factor residual. Establish genuine whole open pullback and BOTH full germ directions before gluing in [4, Remark 12.6, pp. 293–295]. The ambient trace may retain its exceptional factor; it is not renamed as the relative STRICT ideal.

Lemma 149.1 (Actual normalized vanishing trace has the genuine whole open pullback ideal)**.** *For genuinely closed smooth N, H and ANY actual ambient open U,*

$$\mathrm{Res}_{N\cap U}\,\mathcal{I}_{H\cap U} = \iota_U^*(\mathrm{Res}_N\,\mathcal{I}_H)$$

as ENTIRE actual germ ideals on the open manifold. No desired open ideal equality or restricted generator certificate is supplied.

Proof. Actual vanishing coherence for N, H and genuine relative normalization first gives coherence of the SAME global normalized trace. The independently proved normalized vanishing open dictionary compares EVERY local germ with its actual extension to the original ambient manifold, in BOTH directions. True coherent pullback along the real open inclusion compares the very same local germ with the very same extended original germ. Compose these genuine equivalences and use ideal extensionality. The result is an ENTIRE ideal equality, not only support or order equality. □

Lemma 149.2 (Genuine greatest-factor residual keeps both full germ directions on every actual open)**.** *For actual closed smooth N, actual SNC ordered E transverse to N, and actual coherent I nonzero along EVERY point of N, let I_U be its genuine whole pullback to ANY actual open U. For EVERY local point x and EVERY local holomorphic germ representative g,*

$$\mathcal{N}_{N\cap U}(I_U, E\cap U)_x(g) \quad\Longleftrightarrow\quad \mathcal{N}_N(I,E)_{\iota_U x}(\mathrm{extend}_U\, g).$$

The displayed memberships themselves impose holomorphicity, so the statement is valid for EVERY function representative g. No local residual comparison, rank, exponent or coherence of the parts is an input.

Proof. From the actual original smooth/SNC/transverse/coherent/nonzero geometry, the genuine greatest-factor theorem internally constructs coherence of BOTH the actual monomial part and the actual nonmonomial part. The real open inclusion has genuine holomorphic local inverses with both actual inverse identities. Apply the previously proved whole nonmonomial pullback theorem to these independently constructed parts. Its whole ideal equality identifies the literal local residual with the actual pullback of the global residual. The coherent open pullback membership dictionary then gives BOTH full germ directions with the actual extension. No scalar action, ideal, exponent or germ representative is replaced. □

Theorem 149.3 (ONLY original mixed-discrete-centre inputs produce the true strict-trace residual dictionary on every actual source open)**.** *ONLY original smooth N, singleton ambient SNC H, actual density of $N\setminus H$, ANY genuine discrete centre $C\subset N$ and its SAME actual blowup produce the following. Set $N'=\mathrm{Strict}\,N$, $H'=\mathrm{Strict}\,H$, $D=\pi^{-1}C$ and $I'=\mathrm{Res}_{N'}\mathcal{I}_{H'}$. For EVERY actual source open V, EVERY $z\in V$ and EVERY actual function g on V, the literal normalized local trace residual with respect to $D\cap V$ contains g exactly when the SAME global residual $\mathcal{N}_{N'}(I',[D])$ contains its actual extension at the original point. Global $C\subset H$, desired next geometry/coherence and a local residual certificate are absent.*

Proof. The independently proved whole mixed-centre producer internally supplies the genuine smooth strict carrier, actual ambient SNC pair, trace coherence and nonzero witnesses at EVERY actual carrier point. Independently apply the genuine accumulated exceptional geometry to the original empty ordered list and the SAME actual blowup; true containment $C\subset N$ and empty-list normal crossings give actual strict-carrier transversality to the real new exceptional divisor. These are proved geometry conditions, not presumed transversality to H'. The first lemma identifies the literal normalized local trace as the actual whole open pullback of I'. The second lemma then applies to the internally verified new carrier, ordered singleton exceptional list and SAME actual trace. Substitute the proved whole local ideal equality to obtain BOTH asserted directions for EVERY original extended germ.

This supplies a genuine local-to-global residual dictionary. Comparing the local incident/nonincident residuals with the raw all-power STRICT ideal, chronological accumulated $I_i = D_iJ_i$, actual exponent evolution and the separate τ_D phase remain further producers. FIRST Z is still distinct from full sharp Σ; no false support equality is used. □

150 Actual ALL-accumulated-boundary residual of the genuine chronological product TOTAL

The actual next full-product relative STRICT is now independently identified. For the SAME mark-zero TOTAL, one must still check that stripping ALL accumulated exceptional members agrees with stripping only the NEW member. This requires genuine continuation along each OLD strict trace. No claim about a controlled positive-mark transform, composite ORIGINAL raw TOTAL or modified-invariant decrease is made here.

Lemma 150.1 (A genuine excluded germ along one fixed transverse member forces its actual incident exponent zero)**.** *For one FIXED actual member of a genuine transverse SNC list, actual coherent ideal germs outside that restricted member ideal at EVERY incident carrier point force its TRUE intrinsic incident exponent zero. No assertion of exclusion along the other members, especially the new exceptional member, is assumed.*

Proof. If the actual greatest exponent were positive, its true maximal-divisibility neighborhood would force first-order vanishing along the whole actual trace. Choose the genuine excluded germ at the reference point. Actual coherence continues that SAME germ nearby, so the true first-order cosupport dictionary makes it vanish on the actual carrier/member trace. Genuine transverse trace-ideal equality places it in the actual restricted member ideal, contradicting its independently supplied exclusion. This uses an actual neighborhood argument, not an exponent-zero certificate. □

Theorem 150.2 (Genuine OLD strict-trace continuation proves WHOLE accumulated monomial part equals the NEW singleton monomial part)**.** *For a true coherent old ideal with genuine excluded germs along ALL actual old boundary members, a REAL blowup whose centre is contained in the true carrier and has NC with the ENTIRE old ordered boundary gives: the ACTUAL TOTAL has exponent zero along EVERY OLD strict boundary member, and its WHOLE monomial part for the FULL new accumulated ordered list equals its monomial part for the NEW singleton member. No NEW-member exclusion is assumed or claimed for TOTAL.*

Proof. Actual marked admissibility at mark zero uses carrier containment and the REAL whole-boundary NC condition before wellformed transformed geometry and TOTAL coherence are supplied. Genuine old strict-intersection geometry and coherent continuation independently produce actual TOTAL excluded germs along EACH old strict trace. The first lemma forces EACH actual old incident exponent zero. Unfold the true monomial membership predicate at EVERY ambient point. Every old condition is now its real zero power unit condition; the final NEW condition is EXACTLY the singleton NEW condition. Use the actual ordered-list insertion dictionary in BOTH directions. Outside the carrier both genuine predicates contain every holomorphic germ. □

Theorem 150.3 (ONLY original finite histories give the WHOLE ALL-boundary TOTAL residual as the ACTUAL next chronological full product)**.** *ONLY original smooth closed carrier/SNC/whole-boundary density, an ACTUAL finite geometric history and a REAL next blowup with centre contained in its actual current carrier and NC with ALL accumulated members internally prove that the WHOLE nonmonomial part of the SAME current full-product mark-zero TOTAL, computed with the ENTIRE actual next accumulated list, is EXACTLY the independently constructed next chronological full product. No residual, old-factor or desired product dictionary is input.*

Proof. First use the preceding genuine WHOLE monomial equality in the actual nonmonomial colon definition. This makes the full-list residual the actual NEW-singleton residual. Independently constructed actual singleton saturation geometry identifies that true singleton residual with the genuine next relative STRICT. For the actual current full product, original-only finite

history geometry independently supplies its coherence/nonzero/relative principality and its genuine EVERY-old-member excluded germs. Only then apply the actual full-list producer and the proved genuine WHOLE STRICT/product commutation. The literal next-family recursion identifies the result with the ACTUAL next chronological full product, retaining all ordered members and empty steps. Positive-mark controlled-transform comparison, composite ORIGINAL raw-TOTAL factor transport, original modified-history decrease and sharp termination remain OPEN. □

151 Genuine old monomial-factor TOTAL transport before composite raw-TOTAL uniqueness

To compare chronological STRICT with composite ORIGINAL raw TOTAL, one must really transport the old exceptional factor. Here actual old nonmonomial/monomial parts independently supply a true factorization; its SAME real blowup TOTAL is proved to have the transported genuine monomial factor. Composite residual identification remains a subsequent uniqueness consumer.

Lemma 151.1 (A genuine old relative monomial factor has a true monomial TOTAL on the entire new accumulated boundary)**.** *For an ACTUAL coherent nonzero monomial ideal along a smooth closed carrier and its SNC/transverse ordered boundary, a REAL blowup with centre contained in the carrier and NC with ALL old members gives a genuine coherent nonzero monomial TOTAL on the SAME strict carrier and ENTIRE ordered strict-old-plus- NEW boundary. No transported monomial factor or model certificate is input.*

Proof. At mark zero the actual centre is genuinely admissible for the old factor: its real cosupport is the entire carrier, and the WHOLE old-boundary NC condition is explicitly used. The proved actual monomial-factor transform constructor at $o = d = 0$ supplies genuine principal-monomial geometry at EVERY new carrier point. Its actual zero-power multiplier is the TRUE unit, so the constructed factor is EXACTLY the independently defined old-factor TOTAL, in BOTH ambient ideal directions. Actual transform coherence and true new-carrier/SNC geometry precede the principal-monomial theorem giving real nonzero and monomiality. No maximum-centre condition is used or claimed. □

Theorem 151.2 (ONLY actual old ideal geometry internally proves its WHOLE next TOTAL factorization by the genuine transported old monomial part)**.** *For an ACTUAL coherent nonzero old ideal and genuine smooth/SNC/transverse carrier/boundary geometry, a REAL carrier-contained WHOLE-boundary NC blowup internally proves the WHOLE actual TOTAL equals the carrier-normalized product of the genuine TOTAL of its independently defined old nonmonomial part and the genuine TOTAL of its old monomial part. The latter factor is independently coherent, nonzero and monomial for the ENTIRE actual new accumulated list. No factorization or transported-factor certificate is input.*

Proof. FIRST the genuine actual parts theorem constructs BOTH coherent old parts, their true nonzero properties, actual old monomiality and the real restricted equivalence between the original marked ideal and their ACTUAL product. Use genuine restricted-equivalence transform equality on the SAME actual map BEFORE making any next factor claim. All three actual mark-zero records have REAL admissible centres from carrier containment and WHOLE boundary NC. The actual one-step product factor theorem at $o = d = 0$ gives BOTH real restricted TOTAL factor directions. Its independently constructed old monomial-factor transform is EXACTLY the true old monomial TOTAL by actual zero-power/unit algebra. The original TOTAL already has its true outer strict-carrier restriction, and genuine relative span idempotence upgrades those TWO restricted directions to the WHOLE ideal equality, retaining normal-vanishing germs and

off-carrier units. The preceding producer proves the transported monomial factor's actual coherence/nonzero/monomiality. This supplies REAL factor transport before combining it with the new nonmonomial-part TOTAL and applying uniqueness. Composite ORIGINAL raw-TOTAL residual identification, earlier original-history comparison/decrease and sharp termination remain OPEN. □

152 Genuine one-step TOTAL nonmonomial transport by actual factor uniqueness

The old monomial TOTAL factor has now been genuinely transported. Combine it with the TRUE new monomial part of the old nonmonomial TOTAL, before applying actual factor uniqueness. This supplies the one-step intrinsic nonmonomial comparison needed for composite ORIGINAL raw-TOTAL induction. No original invariant rebase or decrease is inferred.

Lemma 152.1 (Actual transported old and newly formed monomial factors give a true new factorization)**.** *ONLY genuine old coherent nonzero ideal and smooth/SNC/transverse geometry, and a REAL carrier-contained whole-boundary NC blowup, internally construct this actual new monomial factor: multiply the NEW monomial part of the TOTAL of the actual old nonmonomial part by the ACTUAL TOTAL of the old monomial part. It is genuinely coherent, nonzero and monomial on the SAME strict carrier and ENTIRE new ordered boundary. The TOTAL of the original ideal has BOTH real restricted factor directions equal to the actual new nonmonomial part times THIS actual factor. No desired factor/model or restricted equality is input.*

Proof. Independently construct actual old parts and real TOTAL geometry. The preceding transport producer FIRST gives the WHOLE original-TOTAL factor equality and true transported-old-monomial coherence/nonzero/monomiality. Actual transformed nonzero geometry permits the genuine parts theorem for the TOTAL of the old nonmonomial part. It constructs its TRUE new nonmonomial/monomial parts, actual every-member exclusion and BOTH factor directions. The actual new monomial part and the independently transported old monomial factor are on the SAME carrier and ENTIRE ordered boundary; real common-coordinate monomial-product closure supplies their true product geometry. Multiply the genuine new parts factor directions by the ACTUAL transported old factor, use real coherent product associativity, and combine with the independently established original-TOTAL factor directions. This proves the new factorization BEFORE uniqueness is invoked. □

Theorem 152.2 (Actual factor uniqueness gives the WHOLE one-step nonmonomial TOTAL transport)**.** *For ONLY the preceding real original inputs, the WHOLE nonmonomial part of the ACTUAL TOTAL of the original ideal equals the WHOLE nonmonomial part of the ACTUAL TOTAL of its independently defined old nonmonomial part, computed using the SAME actual new carrier and ENTIRE accumulated boundary. This holds for EVERY ambient ideal germ, not only zero sets or orders, and does not assume relative principality of the old ideal.*

Proof. FIRST construct the preceding actual factorization and its real coherent nonzero monomial factor. The residual in that factorization is the TRUE nonmonomial part of the old nonmonomial TOTAL; its genuine parts constructor independently supplies actual every-member excluded germs. Apply the proved factor-uniqueness theorem to THESE actual factors to identify the intrinsic monomial denominator of the original TOTAL. The genuine factor inclusion then gives this actual residual inside the original TOTAL nonmonomial part; real nonzero monomial cancellation supplies the reverse restricted span. The residual is itself an ACTUAL relative colon and is carrier-normalized by the independently proved true finite-span colon normalization, giving the reverse FULL germ direction on the carrier. Outside the SAME closed strict carrier BOTH actual nonmonomial colons are genuine units. Thus BOTH whole ideal directions agree everywhere. Composite original-history induction and its literal composite raw-pullback dictionary

remain subsequent consumers; full original invariant/history comparison, modified decrease and sharp termination remain OPEN. □

153 Actual nested smooth-carrier normal crossings inheritance: Bierstone–Milman Lemma 12.7

The true recursive boundary-intersection argument of printed page 294 needs a geometric bridge from an actual smaller smooth carrier to the actual containing smooth carrier. Here real original coordinates construct that bridge. No normal-crossings or transversality of the target carrier is input. The actual recursive history and first-phase analytic driver remain separate producers.

Lemma 153.1 (Actual noncontained smaller-carrier germs force free incident boundary coordinates)**.** *At a true point of the actual smaller carrier, its genuine pointwise NC family and absence of any contained incident boundary germ internally give real common coordinates in which every actual incident member coordinate is free on that smaller carrier.*

Proof. Use the SAME true NC coordinate family. Each incident actual member has a one-coordinate zero equation. If that coordinate were a smaller-carrier constraint, every actual nearby smaller-carrier point would lie in that member. The real preferred-chart neighborhood and literal coordinate membership laws would give precisely the forbidden contained germ. Thus the actual original coordinates are free. This proves the required genuine coordinate disjointness, rather than assuming target transversality. □

Theorem 153.2 (Genuine smaller-carrier NC and noncontained germs construct actual containing-carrier NC)**.** *In positive ambient dimension, let actual $W \subset Y$ in the original ambient manifold, with Y genuinely closed smooth. At a point of W, if W has actual NC with the original boundary family and no incident member contains the genuine W germ, then the SAME Y has actual NC with that SAME boundary family at the SAME point.*

Proof. Use the preceding true smaller-carrier coordinates and independently supplied smoothness coordinates of the actual containing carrier. Their genuine local biholomorphic transition maps the smaller coordinate plane into the containing coordinate plane, because actual $W \subset Y$. The original simultaneous-model coordinate theorem, applied with an empty list of extra member replacements, constructs new containing-carrier coordinates by changing only coordinates among the smaller-carrier constraints. Consequently it preserves every real free incident boundary coordinate. The containing carrier is now a true coordinate plane, and each SAME incident original member remains a genuine coordinate hyperplane. The actual model's centre image is zero: its constrained coordinates vanish on the containing plane and its remaining coordinates are preserved at zero. Compose the TRUE local biholomorphic maps with the SAME preferred original chart. Both actual membership directions, true centering and one-coordinate boundary cardinalities give literal containing-carrier NC. No desired target chart, derivative-rank, NC or transversality certificate is input. □

Theorem 153.3 (True original smaller-carrier density internally supplies the Lemma 12.7 geometric bridge)**.** *In positive ambient dimension, for actual $W \subset Y$, true smoothness of Y, genuine density of W off ALL original members and actual smaller-carrier pointwise NC at a point of W, the SAME containing carrier has actual NC with the SAME original members at that point. No contained-germ exclusion or target NC witness is given.*

Proof. If an incident original member contained the actual W germ, the true neighborhood containment and genuine smaller-carrier density off the whole family would supply an actual nearby W point both inside that member and outside every original member, a contradiction. Thus original density internally supplies each real noncontainment condition. Apply the preceding genuine coordinate inheritance producer to the SAME actual nested carriers and boundary. This supplies

the geometric Lemma 12.7 consumer; it does not construct the missing recursive resolving history, global first-phase Z analyticity/restricted maximum driver or exceptional τ_D termination. □

154 Actual pointwise normal crossings give genuine original good neighborhoods

The preceding genuine trace centre avoids the union of actual original NC neighborhoods. We now prove the missing pointwise dictionary, from the literal original adapted-coordinate definition. No neighborhood or preservation certificate is given by the caller. This proves openness in the actual complex topology, not global analyticity of the whole bad locus or termination of the modified driver.

Lemma 154.1 (Actual finite closed coordinate families propagate to genuine nearby charts)**.** *For a finite family of closed original subsets A_i, an original simultaneous adapted-coordinate witness at x gives a genuine open neighborhood where the same coordinate-index sets give adapted coordinates at every actual point.*

Proof. Intersect the actual chart-and-flattening source with the complements of the finitely many closed members which miss x. No unseen member can appear in this neighborhood. Fix a true nearby point y. Its preferred chart transitions holomorphically to the original chart. Recenter the original flattening at the actual chart value of y and subtract its actual flattened value. Compose with that genuine chart transition. Both directions are holomorphic and have the true inverse equations; the new source lies in the actual preferred-chart target and its true center maps to zero. For each member containing y, its old membership dictionary applies because it also contains x. That same dictionary puts the actual flattened value of y in its coordinate subspace. Translation by a vector in that subspace preserves it. Thus every true membership equation is retained by the produced centered coordinates at y. This constructs the whole nearby family, with the same actual finite coordinate sets. □

Lemma 154.2 (Every original pointwise normal-crossings witness gives an actual NC open)**.** *Let the original smooth carrier W and original boundary member H be closed. At any point of W, the literal original pointwise adapted coordinate witness for W and $[H]$ is equivalent to membership in the previously defined union of genuine original NC neighborhoods.*

Proof. Apply the preceding lemma to the actual two-member family indexed by the carrier and boundary positions. Its finite closed-member condition holds for both true original sets. At nearby carrier points the boundary cardinality-one condition is inherited from the original witness: every incident member already passed through the original point. Thus the whole actual neighborhood has the genuine pointwise NC property. Restrict its coordinates through the true open inclusion; the original coordinate-family open-restriction dictionary gives normal crossings of the literal restricted carrier and boundary on this true open. This is a witness for the original NC-open region. Conversely an actual NC-open witness supplies coordinates at the true subtype point over the original carrier point. The genuine open-inclusion coordinate dictionary identifies them with literal original adapted coordinates; its boundary positions and cardinalities are the same. □

Theorem 154.3 (The actual original good locus is exactly the genuine original NC-open region)**.** *The complement of W, together with its genuine pointwise normal-crossing points relative to $[H]$, equals the preceding genuine original NC-open region. In particular the whole original good locus is open.*

Proof. At points of W use the preceding actual dictionary in both directions. Outside closed W, its open complement has empty restricted carrier, so its true normal-crossings condition is vacuous and supplies a genuine NC-open witness. These cases prove the literal whole-set equality. The already proved openness of the genuine NC-open region gives openness of this actual original good locus. No new analytic bad-locus claim is substituted for this topological conclusion. □

Theorem 154.4 (The same genuine modified centre and blowup preserve every original good NC point)**.** *Under the genuine original hypotheses of the preceding trace-centre producer, the internally selected same original centre avoids every original smooth normal-crossing point. The same actually produced proper holomorphic blowup is isomorphic over the entire original good locus, with the same actual smooth strict carrier and literal transformed SNC boundary. No good-neighborhood certificate is an original input.*

Proof. Construct the actual centre and blowup from the preceding original-only trace producer, retaining its genuine compact original carrier and nonempty actual order-two defect hypotheses. Apply the just proved literal whole good-locus identity to its actual disjointness and its genuine whole NC-open isomorphism. All manifolds, maps, centre, projections, strict transforms and divisor members are those of the same actual producer. This advances the first single-original-boundary step to every true pointwise good NC point; the whole modified driver and its history termination remain separate producers. □

155 The actual product maximum centre has normal crossings with ALL original boundary members

These are genuine FIRST-centre constructions from the original smooth fixed-dimensional carrier, original SNC boundary, whole-boundary density, compact carrier and nonempty actual product cosupport. Actual factor multiplicities remain present. This does not assert the full dynamic DISTINCT-branch defect driver or termination.

Lemma 155.1 (True local containment in incident SNC members gives actual centre normal crossings)**.** *For a genuine closed smooth centre and actual SNC divisor, if near each centre point the centre is contained in every divisor member through that point, then the SAME centre has normal crossings with the WHOLE original divisor.*

Proof. Take a true open neighborhood simultaneously enforcing the finitely many incident-member containments. Its actual centre trace is closed and smooth, and the true pulled-back divisor is SNC. Start with real smooth-centre coordinates and insert each actual incident member by the original SNC coordinate insertion theorem: actual containment supplies precisely the required hypothesis. Finite insertion yields genuine common coordinates for the centre and all actual incident members. Nonincident positions impose no pointwise condition. The original open-coordinate equivalence transports these actual coordinates to the same original point. No centre normal-crossings certificate is assumed. □

Lemma 155.2 (True original factor USC and product-order sum force actual individual order rigidity)**.** *On any actual subset of the original carrier with constant actual original product trace order, each genuine original factor trace order is locally constant along that subset. Every incident original member then locally contains that subset.*

Proof. At a reference point intersect the genuine upper-semicontinuity neighborhoods of all actual factor orders. At another subset point each factor order is at most its reference value. The genuine whole original product-order sum and its actual constancy identify the two finite sums; if one factor were strictly smaller then the entire sum would be smaller. Thus every actual factor order equals its reference value. At a reference point in a member, actual order-one cosupport of the same original vanishing ideal is precisely carrier intersection with that member. Order equality therefore forces actual nearby subset points to stay in that member. This produces local containment internally from original ideal geometry, without a desired containment premise. □

Lemma 155.3 (The actual constructed product invariant maximum centre has constant genuine product first order)**.** *For the literal original product marked input with any positive mark, the actual maximum locus of the original constructed invariant at the identity history has constant SAME ordinary product trace order.*

Proof. At any two actual maximum points the two genuine maximum inequalities force equality of their original full invariant values. Take the first coordinate of that same actual value. The actual successor invariant definition identifies this first coordinate with the original nonmonomial order at the true identity transform. The genuine empty-boundary colon identity shows the SAME product order is independent of the positive mark. This yields real first-order constancy, not an assumed constant-order field. □

Theorem 155.4 (Only original inputs internally construct a genuine ALL-original-boundary admissible first product centre)**.** *For actual original smooth fixed-dimensional carrier, SNC boundary, genuine density off ALL members, compact carrier, positive product mark and nonempty literal product cosupport, internally select its SAME original invariant maximum centre. It is nonempty, closed and smooth, admissible for the true auxiliary product marked input, and has actual normal crossings with ALL original boundary members. It is also a genuine admissible centre for the literal original embedded carrier ideal with that whole boundary. No selected centre, constant-order, local-containment or original-boundary admissibility certificate is input.*

Proof. Original smooth geometry, whole product coherence and density supply the genuine original product marked domain, now at successor dimension so its actual first coordinate is explicit. The actual closed product cosupport is compact as a closed subset of the true compact carrier. Real original invariant upper semicontinuity attains its maximum on the nonempty true cosupport; the genuine original maximum-centre theorem supplies its closed smoothness, containment and auxiliary admissibility. The previous lemma supplies its actual constant product first order. Real factor USC and true sum rigidity internally give containment near every point in each actual incident original boundary member. The first lemma then proves actual centre NC with the WHOLE original SNC divisor. Actual centre containment in the carrier identifies its membership in the genuine mark-one embedded carrier cosupport, giving the actual whole-boundary embedded admissibility. Sharp good-region avoidance at a suitable deep mark, the actual blowup, dynamic DISTINCT-defect selection and the full modified history remain later consumers. □

156 The actual deep-product first blowup preserves the entire original good NC region

Fix the actual original product mark $\#E+1$. This is a genuine first centre for deep product-order defect, built from the original invariant and ideal. It need not select every milder DISTINCT-branch defect. The full sharp modified history remains open.

Lemma 156.1 (The literal original member product has its exact actual open restriction)**.** *On any genuine original open subset, the actual pullback of the LITERAL original member ideal product equals the LITERAL product of the actual pulled-back original member ideals. The whole product marked input and cosupport retain precisely their actual original open restrictions.*

Proof. The actual empty product is the genuine unit ideal and its true pullback is the same unit. Each real vanishing ideal restricts to the true vanishing ideal of its original member trace. Actual original SNC gives genuine member and prefix-product coherence; the real coherent ideal pullback-product theorem then proves the same equality by finite original-list induction. Carrier and mark are literal pullbacks and the auxiliary boundary remains empty. Actual original holomorphic ideal membership supplies the genuine cosupport open restriction theorem. No desired local product or marked-record equality is input. □

Theorem 156.2 (True deep product cosupport avoids EVERY original good NC point)**.** *For the genuine original smooth carrier, SNC boundary and density off ALL original members, the SAME original product cosupport at mark $\#E+1$ is disjoint from the ENTIRE actual original good locus (outside the carrier or with actual pointwise NC). It is contained in the literal original sharp bad set and original boundary support.*

Proof. Cosupport already lies in the actual original carrier. At a genuine original NC point, the independently proved true coordinate/density producer constructs an actual transverse open neighborhood. The preceding exact original product pullback identifies the SAME cosupport there. The actual good-geometry normal form says its order is the genuine DISTINCT proper trace-class count, which there equals the true finite incident-member count and is at most the whole original list length. This contradicts mark $\#E+1$. Thus no actual good point belongs. Actual smooth-carrier coordinates away from the original boundary give NC, so the remaining locus lies in the actual boundary support. The indexed-count identity is used ONLY on independently constructed good transverse geometry, never at bad points. □

Theorem 156.3 (Only original inputs select a true ALL-boundary first centre avoiding the whole good region)**.** *For original smooth fixed-dimensional carrier, actual SNC boundary, genuine whole-boundary density, compact carrier and nonempty literal deep product cosupport, the SAME internally selected original product maximum centre is nonempty, admissible for the actual product and literal embedded carrier with ALL original boundary, and avoids the WHOLE actual original good region. It lies in the genuine sharp bad locus and original boundary support. No selected centre, avoidance, original-boundary admissibility or modified-driver certificate is input.*

Proof. Use the independently constructed actual original maximum centre at mark $\#E+1$. Its true factor-order rigidity and real incident-member coordinate insertion give the already proved actual ALL-original-boundary admissibility. Its literal product cosupport containment and the preceding independently proved deep-cosupport avoidance supply the claimed whole-good avoidance and genuine bad/support containment for that SAME centre. □

Theorem 156.4 (The SAME actual deep product centre has a real proper holomorphic blowup preserving EVERY good NC point)**.** *The preceding original inputs internally construct the SAME actual centre and a genuine blowup along it, proper and holomorphic. Its true projection is an isomorphism over the ENTIRE original good region. The actual strict carrier transform is smooth, and the whole literal ordered family of strict transforms of ALL original boundary members followed by the real exceptional member is SNC.*

Proof. Actual auxiliary product domain and genuine cosupport containment make the actual centre have empty interior. Real blowup existence uses its independently constructed closed smoothness and this true thinness. Genuine blowup-off-centre geometry gives both inverse directions over the entire actual centre complement; true whole-good avoidance restricts those SAME directions to the original good region. True original whole-boundary density and actual centre containment in its support prove carrier density off the centre, so the genuine geometric strict carrier equals the literal strict transform. Actual smooth-carrier blowup geometry gives its smoothness. Finally the genuine original embedded marked input is wellformed with the WHOLE original SNC family; its independently proved actual ALL-boundary admissibility gives SNC of precisely the actual transformed ordered boundary, including the real exceptional divisor. Dynamic DISTINCT-defect selection, recursive boundary-intersection resolution, exceptional monomial cleanup and the full sharp terminating history remain separate open mathematical producers. □

157 The literal original FIRST-phase distinct-trace defect has genuine global analytic equations

For smooth W of fixed positive dimension $d+1$, original SNC members and density off their entire union, retain the literal FIRST-phase set of Bierstone–Milman, printed page 294:

$$Z = \{x \in W : \nu(x) > t(x)\},$$

where ν is the order of the SAME original member-product ideal on W and t counts actual DISTINCT proper restricted trace-ideal germs. We now prove actual finite analytic equations for this literal Z. It is not the whole sharp simultaneous-normal-crossings bad set Σ.

Theorem 157.1 (The literal FIRST-phase defect has actual higher-factor or shared-trace witnesses)**.** *At EVERY genuine original carrier point, literal membership in Z is equivalent to either some actual member trace having order at least two, or two DIFFERENT actual proper indexed positions having equal SAME entire restricted trace ideals. No pointwise carrier/boundary NC is assumed.*

Proof. The already proved actual product order equals the sum of ALL actual factor orders, retaining multiplicities. Its equality with true DISTINCT count holds exactly when every proper factor has order one and the actual quotient map is injective on the true proper positions. A factor of order at least two contradicts that equality, and actual DISTINCT count is always at most true product order, giving strict defect. Otherwise every incident factor has order one and each nonincident factor is truly zero. Failure of equality then forces a genuine quotient collision between DIFFERENT proper positions; actual quotient equality is precisely the SAME true trace ideal setoid relation. Conversely such an actual collision violates injectivity. Real order/proper-incidence and true quotient classes supply both directions; raw indexed count is never substituted for DISTINCT count at bad points. □

Theorem 157.2 (Actual FIRST-phase witnesses give BOTH directions of the genuine finite geometric equations)**.** *Only original geometry and true fixed positive dimension of W give the ENTIRE set equality*

$$Z = \bigcup_i \operatorname{cosupp}(\mathcal{I}_{E_i}|_W, 2) \ \cup \bigcup_{i \neq j} \{x : \operatorname{LocDimGE}(W \cap E_i \cap E_j, x, d)\}.$$

All ideals, intersections, original positions and the actual DISTINCT count remain the SAME original objects, at good and bad points alike.

Proof. A genuine high-factor witness is exactly the original relative order-two cosupport by the actual finite-order dictionary. If no factor has order at least two, both actual proper colliding factors have order one. The independently proved true shared-branch dimension theorem then turns that SAME trace ideal collision into the actual pair local dimension condition. Conversely a pair local dimension point belongs to the actual closed pair intersection, hence to the carrier and both genuine members. If either order is at least two, there is already a high-factor witness; otherwise both orders are truly one. The independent geometric converse supplies equality of their SAME entire restricted trace ideals, giving the actual DISTINCT collision because the positions are different. Each high-factor or pair equation lies in W, so both ENTIRE set directions also account for points outside the carrier. No desired branch, analytic Z, closedness or maximum-centre certificate is used, and no difference-of-semicontinuous-functions shortcut is taken. □

Theorem 157.3 (The literal original FIRST-phase defect is globally analytic, closed and compact on compact carrier)**.** *For ONLY those SAME original inputs, literal FIRST-phase Z is a genuine global complex analytic subset, hence closed. If the actual carrier is compact then this SAME Z is compact. There is no caller analytic/closed or restricted driver certificate.*

Proof. Actual original SNC supplies genuine coherence of each actual member ideal. True smooth carrier and positive mark two give real analytic relative cosupport equations. Every actual pair intersection is analytic because its SAME original carrier and members are closed smooth. The independently proved actual local dimension locus theorem gives genuine whole analytic equations for its dimension lower locus through true closure of regular dimension strata. There are only finitely many original positions and distinct pairs; real finite union analyticity and the preceding ENTIRE set identity prove literal Z analyticity. True analytic closedness and actual $Z \subset W$ then give compactness on a compact carrier. This discharges the initial FIRST-phase analytic bad-locus producer, not full Σ analyticity, restricted-Z maximum centre, dynamic relative strict-trace histories, recursive intersections, exceptional τ_D cleanup or termination. □

158 Genuine original first-phase DISTINCT defect recognition at ALL carrier points

Bierstone–Milman printed page 294 defines the FIRST-phase locus $Z = \{\nu(J) > t\}$, with t the number of DISTINCT proper trace germ ideals. It is not the whole sharp non-NC locus Σ^*. Even distinct order-one traces can require the subsequent simultaneous-crossings recursion of Lemma 12.7; exceptional monomial defects require Lemma 12.8. Here initial $J_i = I_i$ are the actual original member traces and all original multiplicities are retained.

Lemma 158.1 (Actual original member trace order detects precisely proper incidence)**.** *At every actual original carrier point, including bad points, a genuine original member trace has positive order exactly when that point lies in that original member; off the member its actual trace order is zero. The true proper-germ positions are exactly those incident positions.*

Proof. Actual original SNC gives the closed smooth member and coherent vanishing ideal. Whole-boundary density independently constructs its genuine nonzero restriction. The real ordinary-order cosupport dictionary and actual mark-one vanishing-ideal cosupport identify positive order with actual carrier/member intersection. Finiteness gives zero precisely off the member. Independently, the genuine whole restricted-span unit dictionary identifies absence of the unit germ with that SAME incidence. No good-NC hypothesis is used. □

Theorem 158.2 (The actual original whole product order dominates the true DISTINCT proper trace count everywhere)**.** *For only original smooth carrier, actual SNC boundary and whole-boundary density, at EVERY actual carrier point the SAME literal product trace order ν is at least the true quotient-class count t of DISTINCT proper trace germ ideals. The true count is at most the original list length, with no identification of DISTINCT count and indexed incidence at bad points.*

Proof. The actual quotient-class image of proper positions has cardinality at most the number of those positions. Every proper actual original factor has genuine order at least one and every other factor has order zero. Thus their true finite sum is at least the number of proper positions. The independently proved SAME whole original product-order sum identifies this sum with ν. The number of proper positions is at most the true finite original list length. This proves the inequalities retaining actual multiplicities, without any bad-point quotient injection premise. □

Theorem 158.3 (Equality recognizes exactly DISTINCT proper original traces of true order one)**.** *At any actual carrier point, the SAME $\nu = t$ if and only if EVERY actual proper original factor has true trace order one and the real quotient map on all proper positions is injective. Thus absence of the actual FIRST-phase defect means precisely order-one and distinct proper trace germs, including the true empty-family case; it does not assert simultaneous NC of all their crossings.*

Proof. The preceding inequalities form $t \le \#\text{proper} \le \nu$. If $t = \nu$, both inequalities are equalities. True finite-sum rigidity comparing each actual factor order with its actual zero/one incidence value forces each proper factor to have order one. Equality of the actual finite quotient-image cardinality and its actual source cardinality gives genuine quotient injectivity on proper positions. Conversely order one on proper factors and zero elsewhere makes the true original sum equal to the number of proper positions, while actual quotient injectivity makes that number exactly t. Both directions keep the SAME actual original ideals, orders and germ classes. □

Theorem 158.4 (The actual deep original maximum centre lies in the genuine first-phase DISTINCT defect)**.** *Define the actual initial first-phase set $Z = \{x \in W : t(x) < \nu(x)\}$ using the genuine whole product and actual quotient classes. The literal original deep cosupport at mark $\#E + 1$ lies in Z. Only the preceding original inputs therefore internally select the SAME actual*

ALL-boundary admissible maximum centre inside this true Z, while preserving the already proved entire good-region avoidance. No Z-analyticity, selected centre, defect membership or restricted-Z maximum certificate is input.

Proof. Actual product cosupport gives true product order at least $\#E + 1$ by its independently proved finite ordinary-order dictionary. The true DISTINCT count is at most $\#E$, so $t < \nu$. Actual maximum-centre containment in that SAME deep cosupport proves its real Z membership. The genuine original maximum centre, whole-boundary admissibility and good avoidance are constructed by the independently verified preceding producer from ONLY original inputs. Actual analytic closedness and saturated invariant strata for general Z, maximum centres selected on the whole Z, the actual transformed strict trace family, Lemma 12.7 recursion, exceptional τ_D cleanup and full sharp terminating history remain separate unmet mathematical producers. □

159 Actual coherent trace-germ persistence and genuine DISTINCT-count upper semicontinuity

The first-phase $Z = \{\nu > t\}$ of Bierstone–Milman printed page 294 has invariant strata whose germs remain in Z. We construct the required true DISTINCT-count upper semicontinuity from actual coherent original restricted ideals, rather than assuming a branch-count or stratum-saturation certificate. This alone does not prove global analytic closedness of Z.

Lemma 159.1 (Real coherent restricted germ equality persists on a true neighborhood)**.** *The SAME whole germ-membership inclusion between two genuine coherent ideals at an actual point persists near that point; both directions give genuine nearby germ equality. For original smooth carrier and SNC boundary, equality of two actual restricted holomorphic member trace-ideal predicates therefore persists near the SAME original point.*

Proof. Take a true finite coherent generator family for the source ideal. Its actual germs at the reference point belong to the target ideal by the given true pointwise inclusion. Genuine coherent germ continuation keeps every actual finite generator in the target near the point. The source's BOTH nearby finite-span directions and the target's actual span closure then transport membership of EVERY source germ near that point, not merely the displayed generators. Applying this in both directions proves the genuine germ equality. Actual original smooth carrier and SNC members internally give coherence of their genuine restricted extensions; those members are literally holomorphic germs with the SAME original whole restricted-span predicate. Thus actual trace equivalence gives precisely the true coherent point-germ equality and its internally constructed near-point persistence. □

Theorem 159.2 (The actual DISTINCT proper original trace-germ count is upper semicontinuous at every carrier point)**.** *For ONLY genuine original smooth carrier and SNC divisor, the true count of DISTINCT proper original trace germ-ideal classes is upper semicontinuous along the carrier at EVERY point, including bad points. No density or desired quotient-injection/count witness is input.*

Proof. Intersect the true persistence neighborhoods for all finitely many original pairs of equivalent trace germ ideals. Thus every actual old equivalence also holds at a nearby point. Closedness of every original member simultaneously ensures no nonincident original position becomes incident nearby; by the true restricted unit dictionary this means the actual nearby proper positions are a subset of the original proper positions. The identity on original positions now induces a GENUINE map from old quotient classes to nearby quotient classes, since the real old relation is preserved. Every nearby proper class is in the image of an old proper class. Actual finite-image cardinality therefore gives the SAME true nearby DISTINCT count at most its reference value. The proof retains true quotient classes and makes no raw-count substitution at bad points. □

Theorem 159.3 (The true original first-phase defect contains the germ of its actual order and invariant stratum)**.** *At any point of the actual original FIRST-phase Z, nearby original carrier points with the SAME true product trace order remain in Z. In particular the germ of the SAME original constructed full product-invariant value stratum is contained in Z. No order/invariant-constancy or saturated-stratum certificate is input.*

Proof. The preceding actual count theorem gives $t(y) \leq t(x) < \nu(x)$ near the actual reference point. At a true same-order carrier point $\nu(y) = \nu(x)$ this forces $t(y) < \nu(y)$, namely membership in the literal SAME Z. Taking the first coordinate of a SAME actual successor original full invariant-value equality gives precisely true original product-order equality, by the actual identity history and empty auxiliary boundary. Thus the entire genuine full-value stratum germ likewise remains in Z. These are actual internally constructed stratum neighborhoods; they do not prove global Z analyticity, admissibility of maxima restricted to the whole Z, dynamic strict-trace transforms, recursive-intersection completion, exceptional monomial cleanup or termination. □

160 Actual first member total traces have genuine greatest-exceptional-factor geometry

Bierstone–Milman printed pages 293–294 first pull each original member trace back to the real smooth blowup carrier, then remove its greatest exceptional monomial factor. We construct these actual intermediate objects from ONLY the original data. The BOTH-direction equality with arbitrary exceptional-colon saturation, and the chronological relative STRICT J_i history, remain explicit further producers; simply naming the nonmonomial part does not prove them.

Lemma 160.1 (A real mark-zero transform is exactly the entire carrier-restricted total pullback)**.** *For a genuine marked record with mark zero, its SAME geometric transformed ideal is literally the entire restricted extension of the true total ideal pullback to its actual geometric strict carrier. No positive-mark controlled or strict ideal is substituted for that total pullback.*

Proof. Unfold the actual geometric transform. Its exceptional denominator is the zeroth genuine ideal power, namely the whole unit ideal. Membership in the true colon therefore implies restricted-total membership by applying it to the actual constant one. Conversely genuine restricted-total membership multiplies by every holomorphic unit-ideal germ by true restricted ideal closure. These prove BOTH complete holomorphic germ directions and literal whole ideal equality. No desired total/controlled comparison is input. □

Theorem 160.2 (Only original inputs construct actual member total traces and their greatest exceptional-factor residual geometry)**.** *For ONLY original compact smooth fixed-positive-dimensional W, original SNC E, density off its whole union and nonempty literal FIRST Z, internally construct the SAME actual restricted-Z blowup. For EVERY original indexed member, its actual mark-zero transformed record has genuine wellformed geometry and actual nonzero along its true carrier. Its ideal is the SAME entire carrier-restricted total pullback of that original member ideal.*

For each actual member total ideal, division by its true greatest exceptional monomial gives a genuinely coherent residual ideal nonzero along the carrier. Every real total germ factors on the actual carrier into a genuine intrinsic exceptional monomial germ and a true residual germ. At every actual exceptional point there is a true residual germ not divisible along the carrier by that exceptional member, so a further COMMON exceptional factor cannot remain. No member admissibility, coherence, nonzero, chosen blowup or factorization certificate is supplied.

Proof. The independent original-only first-blowup producer constructs its actual centre, projection, genuine closed smoothness and centre containment in W. For each original position, form the literal initial member record with the SAME carrier, empty auxiliary boundary, actual member

ideal and mark zero. Original SNC supplies member coherence; its actual zero-mark cosupport is the whole carrier. Thus this SAME real centre is admissible for each such genuine zero-mark record, even where it is not in that member. Real geometric wellformedness transport supplies actual smooth strict carrier, true coherent total ideal and genuine exceptional SNC/transversality. Original whole-boundary density internally gives initial nonzero for EVERY actual member ideal. Actual coherent strict-carrier accumulation off the centre preserves that nonzero without assuming a positive member marked-domain or canonical history. The first lemma gives the entire genuine total-pullback dictionary.

Apply the independently proved actual monomial-part geometry to these SAME real coherent nonzero carrier and exceptional objects. Its exponents are the intrinsic actual divisor exponents, not chosen weights. Genuine local monomial generator and real restricted division construct actual coherent residual ideals and actual nonzero along the carrier. True original germ factorization supplies every displayed total-germ product. The actual maximal divisor-exponent theorem supplies at every incident exceptional point a true residual germ outside its genuine restricted member ideal, proving absence of any further common exceptional factor. This is actual total/residual geometry; BOTH arbitrary-colon saturation directions and chronological relative STRICT transform equality, product/history compatibility, modified decrease and termination are not inferred from these intermediate facts. □

161 The actual first member residual equals the literal relative STRICT saturation

This chapter proves BOTH arbitrary exceptional-colon directions for the SAME actual first TOTAL member trace. The result uses the original BM97 relative STRICT construction: restrict the genuine total pullback to the actual smooth strict carrier and saturate by ALL powers of the actual exceptional ideal. It is not defined by calling the residual strict. Chronological transformed-member histories are a later obligation.

Definition 161.1 (Literal arbitrary-exceptional-power saturation on the actual carrier)**.** For the actual carrier N, total ambient ideal I and exceptional set D, define its relative exceptional saturation by the ACTUAL germ predicate

$$S_x = \bigcup_{k\geq 0} (\mathrm{Res}_N\, I : \mathcal{I}_D^k)_x.$$

Thus each genuine holomorphic germ must multiply EVERY germ of the true ideal power into the actual carrier restriction. Passing first to the whole ambient extension of $\mathrm{Res}_N\, I$ does not change S.

Proof. Use the genuine colonAlong and actual idealPow, retaining $k = 0$ and its real unit ideal. A restricted span of restricted spans is a genuine restricted span of the original ideal by actual finite-span substitution. Conversely each true restricted ideal germ lies in its own actual restricted span. These BOTH whole germ directions prove idempotence of carrier restriction before applying it to every actual denominator germ and every power. No saturation equality is a defining input and no analytic/coherence premise is needed for this restriction identity. □

Theorem 161.2 (For a genuine single exceptional member, BOTH arbitrary-colon saturation directions give the actual greatest-factor residual)**.** *Let a real coherent nonzero total ideal be genuinely relatively principal on a smooth carrier transverse to the ACTUAL singleton exceptional family $[D]$. At EVERY actual carrier point, its literal saturation S_x equals its independently defined actual intrinsic greatest-exceptional-factor residual J_x, in BOTH complete holomorphic germ directions.*

Proof. A true coherent ideal lies in its actual residual. Thus a genuine total colon germ also multiplies every actual denominator germ into the restricted residual. The independently proved arbitrary-power residual cancellation therefore gives actual residual membership. No desired prime or saturation certificate is supplied.

Conversely, at a point incident with D, take the REAL intrinsic maximal exponent $e = \mathrm{divExpAlong}(I, D, x)$. Every actual germ of $\mathcal{I}_D^e$ belongs to the true monomial part: there is exactly the SAME one original position, and its actual power germ itself supplies the required restricted span. The genuine residual colon definition therefore multiplies EVERY such germ into the actual restricted total ideal. This constructs the real saturation witness e, not merely a divisibility of one chosen generator. Outside D, the genuine unit germ belongs to the monomial part; evaluating the actual residual colon at it gives true total restricted membership and hence the literal zero-power saturation witness. Normal carrier-vanishing germs remain in every dictionary. □

Theorem 161.3 (Only original inputs identify the true first relative STRICT member ideal with the actual residual on the SAME blowup)**.** *ONLY original compact smooth fixed-positive-dimensional carrier, true original SNC, actual whole-boundary density and nonempty literal FIRST Z construct the SAME first centre and blowup. For EVERY original member, its actual relative STRICT ideal, defined by saturating the GENUINE TOTAL pullback along the actual strict carrier by ALL powers of the real exceptional ideal, equals its independently defined greatest-factor residual at EVERY actual carrier point and for EVERY holomorphic germ.*

Proof. The independent original-only producer constructs the SAME real first projection, each actual member total marked record and its genuine wellformed/nonzero geometry. The mark-zero dictionary first identifies its ENTIRE ideal with the genuine carrier restriction of the ACTUAL total pullback. Its true original empty initial exceptional list transforms to precisely the singleton actual exceptional member. Genuine source SNC and actual projection composition independently construct total relative principality. Apply the preceding singleton saturation theorem, then the first producer's proved restriction idempotence to return to the actual RAW pullback saturation. This proves the literal BM97 relative STRICT equality rather than naming a residual strict. Multi-step chronological compatibility, modified-history canonical comparison/decrease, recursive traces, exceptional τ_D and termination remain OPEN. □

162 Actual first member residuals have genuine principal germs along the real carrier

Greatest-exceptional-factor division must preserve the actual principal trace ideals before their relative STRICT saturation dictionary is proved. We construct true BOTH-direction relative principal dictionaries, from actual original member generators and real nonzero exceptional-factor cancellation. No desired residual presentation is supplied by the original-input consumer.

Lemma 162.1 (Genuine relatively principal total ideals give true relatively principal residual ideals)**.** *For actual closed smooth carrier, genuine transverse SNC exceptional family, coherent nonzero total ideal and a real relative principal generator at the reference point, its actual greatest-exceptional-factor residual has a true relative principal generator, with BOTH entire holomorphic germ membership directions and an actual total-generator factorization on the carrier.*

Proof. Actual intrinsic monomial-part geometry supplies one genuine monomial germ, its BOTH nearby relative generator directions and true nonzero on every nearby carrier neighborhood. The actual total generator belongs to the total ideal, hence to that true monomial-part ideal, and its principal dictionary gives a real holomorphic quotient. Using the actual relative generator laws of the monomial part, multiplication by every actual monomial germ puts this quotient back in the

true total ideal on the carrier. Thus it is a genuine residual germ before any desired presentation is used.

For an arbitrary actual residual germ, multiply by the true monomial generator. Actual total principality writes that product as a multiple of the original total generator. Substitute the internally proved total-generator factorization. Genuine nonzero-on-carrier-neighborhood geometry permits real restricted cancellation, proving the residual is a true multiple of the constructed quotient on the carrier. Conversely any actual holomorphic germ equal to a multiple of that quotient on the carrier lies in the residual colon, by true restricted multiplication and germ congruence. These are BOTH entire germ membership directions, retaining all normal vanishing germs. They do not assert ambient principality of an ideal containing the carrier ideal. □

Lemma 162.2 (Actual original member generators give principal total traces on the SAME real blowup carrier)**.** *Original SNC internally supplies a real principal generator of each SAME original member ideal at the actual projected point. Its true composition with the real projection is a genuine principal generator, along the actual carrier, of that SAME mark-zero total trace ideal, in BOTH whole directions.*

Proof. Use true original member coordinates, or its actual unit germ outside that member, to obtain BOTH entire original ideal/span directions. Every genuine pullback ideal generator is an actual composition of an original member germ. Its real principal cofactor pulls back holomorphically and its true germ equality pulls back by continuity. Actual finite-span substitution therefore gives one entire pullback principal direction. The original generator itself is an actual member germ, so its true pullback belongs to the genuine coherent pullback ideal and gives the reverse direction by coherent span closure. The independently proved entire mark-zero/actual restricted-total equality then restricts both directions to the SAME true carrier. The real holomorphic germ condition is retained explicitly. □

Theorem 162.3 (Only original inputs construct BOTH relative principal directions for EVERY actual first member residual)**.** *For ONLY the original compact smooth fixed-positive-dimensional carrier, original SNC family, whole-boundary density and nonempty FIRST Z, the SAME internally constructed actual first blowup has true relatively principal residual ideals for EVERY original position at EVERY point of its genuine carrier. Their whole ideal memberships are exactly holomorphic germs equal on the carrier to true multiples of the internally constructed generators. No selected centre/projection or total/residual principal presentation is input.*

Proof. The preceding original-only producer supplies the SAME real blowup and actual per-member total geometry: closed smooth carrier, genuine exceptional SNC and transversality, real total coherence and independently proved nonzero along the carrier. The second lemma constructs the actual total principal generator from each SAME original member at the real projected point. Apply the first real division/cancellation producer to these independently obtained true objects. This constructs the residual generator and BOTH entire relative principal directions internally at every actual carrier point. Actual arbitrary-exceptional-colon saturation equals relative STRICT transform, chronological product/history comparison and full modified termination remain further mathematical obligations; they do not follow by naming this residual. □

163 Whole actual relative STRICT ideals and their genuine exceptional-power total factor

The relative STRICT construction is the independently defined arbitrary exceptional-colon saturation of the GENUINE raw total pullback. We upgrade its proved carrier dictionary to a WHOLE ambient ideal equality and actual coherence, then prove the genuinely normalized total factor $\mathrm{Res}_{N'}(\mathcal{I}_D^{e} J)$. Normal carrier-vanishing germs remain throughout; no ambient principal product equality is asserted.

Lemma 163.1 (Actual off-carrier unit geometry upgrades the true STRICT dictionary to whole ideal equality)**.** *For a genuine coherent nonzero total ideal on a closed smooth carrier transverse to the singleton actual exceptional family, with true relative total principality at every carrier point, its literal all-power saturation is ENTIRELY equal to its independent intrinsic residual ideal. It is a real coherent ideal and remains nonzero along the actual carrier.*

Proof. On the actual carrier use the preceding BOTH-direction saturation theorem with the independently supplied true relative principal generator. Outside the closed carrier its complement is a genuine neighborhood. Every genuine holomorphic germ and every denominator product has vacuous true restricted span there; both the literal saturation and actual residual therefore have EXACTLY all holomorphic germs. This proves BOTH whole ideal directions at EVERY ambient point before any coherence transfer. Actual intrinsic residual coherence and nonzero geometry then give those real properties of the independently defined saturation, without a desired ideal/analytic certificate. □

Lemma 163.2 (The genuine singleton monomial part has the actual intrinsic exceptional-power dictionary)**.** *At each actual carrier point set $e(x)$ to the true intrinsic* $\mathrm{divExpAlong}(I, D, x)$ *when incident with D, and to zero otherwise. The ENTIRE actual monomial-part germ predicate equals that of* $\mathrm{Res}_N \mathcal{I}_D^{e(x)}$*. No exponent or factor dictionary is input.*

Proof. At an incident point the SAME actual singleton position is the only defining monomial constraint. Its true restricted power span, together with actual holomorphicity, is exactly the complete restricted-power germ predicate in BOTH directions. Outside the member there are no incident constraints; the real exponent is zero, its actual power is the genuine unit ideal, and its carrier restriction has precisely all holomorphic germs. The zero exponent and actual normal-vanishing germs are retained literally. □

Theorem 163.3 (Genuine total factorization retains the necessary whole carrier restriction)**.** *For a real coherent nonzero total ideal I on the actual smooth transverse carrier and every actual ambient point x, the complete germ predicates of*

$$\mathrm{Res}_N I = \mathrm{Res}_N\big(\mathcal{I}_D^{e(x)} J\big)$$

agree, with J the independently defined actual residual.

Proof. The genuine intrinsic factorization proves BOTH carrier-span directions between the total ideal and monomial-part times residual. The preceding actual monomial dictionary identifies its carrier image with the true power. To substitute this INSIDE a real product, expand a genuine ambient product span, replace each actual left-factor germ by its true finite restricted power span, and multiply every real coefficient/generator by the unchanged actual residual germ. This produces genuine product generators and actual holomorphic coefficients; finite sums and true carrier equations give the whole forward span. Reverse substitution uses the actual power-to-monomial germ inclusion. Thus the two actual normalized whole germ predicates agree. Outside the carrier both normalized sides are true unit ideals by its real closed complement. The surrounding restriction is essential: relative ideals retain all carrier-normal vanishing germs and are not asserted to be ambient principal products. □

Theorem 163.4 (Only original inputs construct WHOLE coherent first relative STRICT ideals and actual normalized total exceptional factors)**.** *ONLY original compact smooth fixed-positive-dimensional carrier, true original SNC, whole-boundary density and nonempty FIRST Z construct the SAME first blowup. For EVERY original member, its independently defined actual relative STRICT is WHOLE equal to the actual residual, coherent and nonzero along the actual strict carrier. At EVERY ambient point the actual restricted TOTAL pullback equals the true restricted product of its intrinsic exceptional ideal power with that SAME actual STRICT ideal, in BOTH complete germ directions.*

Proof. Internally construct the SAME genuine first centre/projection and actual member well-formed/nonzero total geometry. Source SNC and true projection composition independently provide every real total relative principal generator. The actual empty initial exceptional list gives exactly the true singleton exceptional member. Apply the first producer before coherence is asserted, then the third genuine normalized factor theorem. The proven true mark-zero total identity identifies the actual total record with the carrier restriction of RAW pullback; genuine restriction idempotence removes only the repeated normalization, preserving the necessary restriction on the product. Replace the independently defined residual by the proved WHOLE actual STRICT equality. No chosen projection, exponent, factor, coherence or strict dictionary is a premise. Chronological transformed member/product histories, modified canonical comparison/decrease, recursive intersections, exceptional τ_D and full termination remain OPEN. □

164 The genuine original member product initializes its actual marked domain at every carrier point

This construction includes arbitrary original BAD carrier points. It uses original ambient SNC and density of the true carrier away from the whole original boundary, not normal crossings of that carrier. The product is the literal independently constructed original ideal product of Bierstone–Milman[4], Section 12, printed page 293 (PDF page 87). The auxiliary product marked ideal has empty auxiliary boundary; NC of its actual selected centre with ALL original boundary members is a separate required geometric producer.

Lemma 164.1 (The literal original finite member product is genuinely coherent)**.** *In positive ambient dimension, the actual product of the original SNC member ideals is coherent on the ENTIRE original manifold.*

Proof. Each genuine original SNC member is a closed smooth submanifold, by its actual member-intersection coordinates. Its actual vanishing ideal is therefore coherent. The independently defined literal finite ideal product starts with the true holomorphic unit ideal. Induct on that actual original factor list: genuine product coherence multiplies the two actual finite presentations. This constructs coherence of the same original whole product, without supplied equations or coherence field. □

Theorem 164.2 (True original density gives a nonzero product trace germ at EVERY carrier point)**.** *For original SNC E and any true original carrier $W \subset \overline{W \setminus \bigcup_{H\in E} H}$, the literal original member product is nonzero along W at every actual point of W. No carrier normal-crossings, nonzero-trace or local-order witness is input. Each literal original member ideal likewise has nonzero restriction everywhere on the original carrier.*

Proof. Original SNC is genuinely transverse to the whole ambient manifold. Apply the independently constructed actual product normal form there, at the true original reference point. It supplies a holomorphic product generator in the LITERAL original product ideal, with actual membership equations for every incident original member. Given any actual reference neighborhood, intersect with its true chart/flattening open and use the genuine original carrier density. This supplies a true nearby carrier point outside ALL original boundary members. Each incident product coordinate is nonzero there by its actual member dictionary, so the true product generator is nonzero there. Thus the actual ideal contains a holomorphic germ nonzero at arbitrarily near true carrier points, which is exactly genuine nonzero restriction along the original carrier. This argument includes empty products and arbitrary original bad points. For each original member, whole-boundary density implies density off that actual member. Its genuine original SNC singleton and the original single-member trace-germ construction supply the same nonzero statement. □

Theorem 164.3 (Only original geometry constructs the actual auxiliary product marked domain and global fixed-order equations)**.** *For a closed smooth original carrier of fixed dimension at most the ambient dimension, original SNC boundary and genuine whole-boundary density, every positive fixed mark gives the true marked-domain input* $(W, \varnothing, \prod_{H\in E} \mathcal{I}_H, q)$*. All genuine fixed-order loci of that SAME literal product along* W*, including order zero, are globally complex analytic.*

Proof. The original smooth carrier, genuinely constructed whole product coherence, empty auxiliary SNC/transversality and true positive mark give actual well-formedness. The true original fixed carrier dimension gives the domain dimension bound; the preceding independent true nonzero producer gives its required nonzero restriction at every actual carrier point. These constructs prove the actual original marked domain. For each positive fixed order, the genuine coherent ideal and smooth carrier supply actual relative cosupport equations. At zero, the actual holomorphic germ dictionary makes the order locus the entire original smooth carrier. These equations present the literal original product order sets globally, including bad points. No dynamic distinct-count defect analyticity, original-boundary admissible centre or completed modified sharp driver is certified by these auxiliary domain outputs. □

165 The true original member product has an actual restricted normal form

The literal product of the original member ideals is constructed before any normal form or sharp defect is claimed. This is the local product appearing in Bierstone–Milman[4], Section 12, printed page 293 (PDF page 87). All branch counts below use the independently defined DISTINCT proper restricted germ-ideal classes. No modified max-centre admissibility or history termination is claimed.

Lemma 165.1 (Actual products of original germ ideals retain whole principal membership dictionaries)**.** *The literal finite list product of germ ideals has the unit ideal as empty product. If each actual factor is generated at a point by a true holomorphic function, in both whole germ-membership directions, the actual product is generated by the true product of those functions, again with both whole membership directions. The actual restriction to the original carrier retains the same principal span dictionary.*

Proof. For two true factors, actual product membership is the finite holomorphic span of products of genuine member germs. Substitute both original principal membership dictionaries. Every such true product is a holomorphic multiple of the genuine product generator, by multiplying the actual coefficients and their two germ equations. Conversely the actual factor generators belong to their true factors, so their genuine product belongs to the literal product ideal; substituting that true generator in its finite spans proves the reverse direction. The unit case is the literal holomorphic unit ideal generated by one. Induction on the actual finite factor list gives the whole stated dictionary. Restricting actual ambient relations and substituting actual generators in both finite restricted spans gives the last assertion. □

Theorem 165.2 (The true restricted product order is the actual DISTINCT proper trace-germ count on good geometry)**.** *For genuine original transverse SNC geometry, at an actual carrier point the order of the LITERAL original member product restricted to that carrier is exactly the number of DISTINCT proper restricted member germ ideals. Equivalently, its true order-q cosupport condition holds there exactly when q is at most that actual distinct-germ count. The genuine whole neighborhood producer constructs this dictionary at every actual carrier point, without a supplied product-order certificate.*

Proof. The produced original product generator is the product of its real incident free coordinates. Its actual ambient order is the number of those factors, by the independently proved point-ideal

monomial order theorem: it belongs to that power and not to the next. The actual flat carrier retraction keeps each genuine free coordinate, hence keeps the whole product as a germ. Its actual restricted order therefore equals that actual ambient order. The true product generator belongs to its literal product ideal; any larger claimed ideal order would make that genuine generator vanish to higher restricted order and contradict the actual retraction/point-ideal dictionary. Conversely every real product ideal germ is a holomorphic multiple of that genuine generator, by the already proved full membership dictionary. Its ambient order at least the factor count therefore gives every required lower restricted order. The independently constructed actual quotient injection identifies this factor count with the true DISTINCT proper restricted germ-ideal count on this real transverse geometry. Apply these actual statements on the independently produced original whole transverse neighborhood. The count at arbitrary bad points remains the genuine quotient count; this proof does not replace it by indexed incidence there. □

Theorem 165.3 (Actual original transverse coordinates construct the true member-product normal form)**.** *On genuine original transverse SNC geometry, at every actual carrier point the literal product of ALL original member ideals has a genuine principal normal form equal to the product of the distinct free carrier coordinates belonging to the truly incident members. Both whole ideal membership directions and both original-carrier restricted-span directions are constructed. No product normal form is supplied.*

Proof. Produce actual simultaneous coordinates for the original carrier and whole original SNC list, with distinct free coordinates for incident members. Genuine Hadamard division gives the whole original ideal dictionary for each incident member coordinate. Every nonincident closed member has its real open complement, where its actual ideal is the unit ideal. Apply the actual finite product dictionary to these internally constructed original factor functions. The nonincident unit factors disappear; the remaining literal product is exactly the product of the actual incident free coordinates. Actual finite-list and indexed-product dictionaries identify the same original factors and no replacements. This proves the ambient and restricted normal forms for the true original product ideal. □

Theorem 165.4 (Only original good-point geometry supplies actual whole product neighborhoods)**.** *Original pointwise normal crossings, original SNC and true density away from the whole original boundary internally construct a genuine open neighborhood on which the actual restricted original member product has the preceding true principal normal form at every actual carrier point. These are the original member ideals on that SAME produced open. The full product defect at arbitrary bad points, its modified invariant and admissible maximum-centre driver remain separate constructions.*

Proof. Construct the genuine whole transverse SNC neighborhood from the independently verified original-only pointwise/density producer. At each actual carrier point there apply the preceding genuine product normal-form theorem to its literal restricted carrier and literal restricted boundary list. Every ideal, coordinate and factor belongs to those same actual original objects. This constructs the asserted whole neighborhood without a local normal form or preservation input. □

166 The actual original multi-boundary sharp defect is closed and compact

Here the original boundary is any finite closed list. The actual sharp defect is defined by failure of the literal original simultaneous adapted-coordinate predicate, rather than by every original boundary intersection. This supplies the topological part of Bierstone–Milman[4], Section 12, Lemma 12.5 (printed page 293, PDF page 87). Its analytic-presentation part and the genuine modified centre driver remain further work.

Lemma 166.1 (The literal original pointwise NC set is open for the whole finite boundary)**.** *For closed original carrier W and every actual member of a finite closed boundary list E, the set of points with the literal original pointwise normal-crossings witness for W, E is open. The whole original good set $W^c \cup \{x \in W : \mathrm{NC}(W, E, x)\}$ is consequently open.*

Proof. At a true pointwise NC point use the independently constructed genuine finite-closed-family neighborhood. Its real nearby preferred-chart coordinates retain every actual membership equation. No new incident boundary member appears, so every required cardinality-one condition is inherited. Thus a genuine neighborhood lies in the literal original NC set, proving its openness. The whole good set equals the union of this open set and the open complement of the closed original carrier, by splitting on original carrier membership. This proves whole-good-set openness without a caller local normal form or preservation witness. □

Theorem 166.2 (Original pointwise NC and density construct actual whole transverse boundary neighborhoods)**.** *At an actual original pointwise NC carrier point, for an original SNC boundary and true carrier density away from the whole boundary, original inputs construct an actual open neighborhood on which the literal restricted boundary is SNC and transverse to the literal restricted carrier. No transverse-neighborhood witness is supplied.*

Proof. Use the independently constructed original NC open set as the actual neighborhood. For each real carrier point, pull back its actual original adapted-coordinate witness through the genuine open inclusion, retaining the real finite boundary positions by the true list-map index dictionary. The original boundary SNC pulls back through that same actual open inclusion. True carrier density pulls back by its genuine open-map closure dictionary; the original boundary union pulls back to the union of the same actual list of restricted members. The genuine original density-to-transversality theorem excludes any incident restricted boundary coordinate from the carrier-defining coordinates. Thus the whole produced original open has precisely the asserted actual geometry. □

Lemma 166.3 (Actual transverse original member traces are proper and genuinely distinct germ ideals)**.** *For a genuine original transverse SNC carrier/boundary neighborhood and an actual carrier point, the proper restricted member germ ideals are exactly the incident original members. Two different incident positions have different whole restricted germ ideals. Consequently the number of DISTINCT proper restricted germ ideals there equals the actual indexed incidence count, by an internally proved injection; this equality is not asserted at arbitrary bad points.*

Proof. At an incident member, every finite restricted ideal span vanishes on the carrier-member intersection, so it cannot contain the unit germ at the actual point. At a nonincident closed member, its true open complement gives the actual unit germ in its ideal, hence in its restricted ideal. This proves the exact proper-member dictionary.

Use actual simultaneous transverse coordinates for the original carrier and whole original SNC list. Each incident member has its own distinct free carrier coordinate. For two different incident positions, the first member's actual coordinate belongs to its true ideal and hence its restricted germ ideal. If it belonged to the second restricted ideal, the whole genuine member-trace dictionary would make it vanish on the carrier-second-member intersection near the actual point. Actual points with the first free coordinate nonzero and all other coordinates fixed at their true center values accumulate at that point. They remain on the actual carrier and second member, contradicting that vanishing. Thus the whole two restricted ideal membership predicates differ.

Define equivalence by equality of these whole holomorphic-germ restricted span predicates, and count the true quotient classes of the proper member positions. The preceding separator proves that the actual quotient map is injective on that proper finite set. Its image cardinality is therefore the cardinality of the incident indexed set. No raw indexed count is substituted for the distinct-germ count at bad points, and this does not supply modified max-centre admissibility or termination. □

Theorem 166.4 (The actual original sharp defect is closed and compact on a compact carrier)**.** *The genuine original sharp defect* $\Sigma^* = \{x \in W : \neg \mathrm{NC}(W, E, x)\}$ *is precisely the complement of the whole original good set. It is closed, and when the original carrier is compact it is compact.*

Proof. The complement dictionary is a literal pointwise propositional identity: an original point is bad exactly when it belongs to the carrier and fails its genuine adapted-coordinate property. The preceding genuine whole-good-set openness therefore gives closedness. The actual defect is a closed subset of the true compact original carrier, so it is compact. No analytical bad-locus certificate or modified invariant is supplied or claimed. □

Lemma 166.5 (For an original smooth carrier the true sharp defect lies over the actual boundary)**.** *If the original carrier is a closed smooth submanifold, then* $\Sigma^* \subset W \cap \bigcup_{H \in E} H$*. This remains the literal true sharp defect, with every actual already normal-crossing intersection point excluded.*

Proof. At a true carrier point outside all original boundary members, original smoothness supplies genuine centered carrier-flattening coordinates. Use those same actual coordinates for the carrier position in the full family; every boundary position is nonincident, so its membership and cardinality requirements are vacuous. This constructs the literal pointwise NC witness for the whole original boundary, contradicting actual defect membership. Carrier membership already belongs to the defect's genuine definition. Thus only the stated boundary-supported original points can be bad. This is a topological and support producer, not analyticity or admissibility of a newly selected modified centre. □

167 Actual original order-one member traces have true smooth contact geometry

For the true first-phase $Z = \{\nu > t\}$ of Bierstone–Milman printed page 294, its complement has distinct proper order-one member traces. Actual smooth relative hypersurface geometry for those SAME ideals is needed to identify shared branches and prove genuine whole Z analyticity. We construct that geometry from real original member coordinates and actual tangent directions; no smooth-trace, unit-cofactor or target geometry certificate is input.

Lemma 167.1 (Actual tangent fields lower individual genuine restricted jet order)**.** *A genuine derivative tangent to the actual carrier lowers the true restricted jet order of an individual holomorphic germ by at most one. In particular the real multiplicity-one derivative of an actual tangent contact function rules out restricted order two for that SAME function at its zero.*

Proof. Use the actual point-ideal-power representative of the original restricted germ. Its difference from the original function vanishes on the true carrier near the point. The genuine tangent field preserves its actual vanishing ideal, while real point-ideal derivative powers lower order by one. Thus the actual derivative of the representative supplies the genuine lower-order point-power representative of the SAME original tangent derivative. At order two, its order-one derivative representative vanishes at the actual carrier point. This contradicts the genuine contact derivative nonzero value. No derivative rank or desired finite-order certificate is assumed. □

Theorem 167.2 (Only original order-one trace data construct genuine tangent contact and a true unit cofactor)**.** *At a true original carrier point with actual member trace order one, original smooth carrier, actual SNC family and whole-boundary density internally construct a genuine tangent contact open/function for that SAME member ideal. They also produce its true original principal generator and a holomorphic cofactor, nonzero at the point, with the SAME contact function equal to that unit times the SAME generator on the carrier near the point.*

Proof. The independently proved original order dictionary gives actual order-one cosupport and absence of order-two cosupport. Actual original coherence and smooth carrier therefore construct

the genuine tangent direction. Actual original SNC coordinates independently give the SAME member's real principal germ generator with BOTH actual whole-ideal membership directions, and its actual zero-set member dictionary. At mark one the tangent function belongs to the literal original member ideal itself, hence also its actual restriction. The true whole principal germ dictionary gives its real holomorphic generator cofactor on the entire ambient germ, and therefore on the same carrier. At the original point the generator vanishes and the contact function likewise vanishes. By the first lemma the real contact function is not of restricted order two. Actual order-one product arithmetic therefore forces the cofactor to be nonzero: a vanishing cofactor would give order at least two. This constructs the true unit internally without a supplied principal/contact normalization. □

Theorem 167.3 (The SAME original order-one member trace has genuine smooth codimension-one support)**.** *For ONLY the preceding original inputs, the literal original intersection of carrier with that SAME member has the true germ of the actual contact carrier, and is regular of dimension one less than the actual carrier dimension at the reference point. The genuine original member ideal, principal generator, contact function and unit cofactor all remain the SAME actual objects.*

Proof. The actual cofactor is nonzero on a true neighborhood by genuine continuity. The SAME contact/generator equation on the carrier and the actual original SNC member zero-set dictionary therefore identify BOTH nearby directions of member-trace membership with real contact-carrier membership, retaining its actual open. Real tangent-contact geometry independently makes that contact carrier a smooth submanifold of the carrier of codimension one. The true nearby set-germ equality transports its genuine regularity to the SAME literal original carrier/member intersection. This supplies real smooth trace geometry; shared-branch dimension recognition, whole Z analytic equations, restricted maximum centres, actual strict-trace histories, recursive resolutions and exceptional τ_D termination remain subsequent mathematical producers. □

168 The SAME genuine order-one original restricted trace ideal is the actual reduced trace ideal

Actual smooth order-one support alone is insufficient for the DISTINCT germ criterion. Bierstone–Milman printed page 294 uses equality of genuine trace IDEALS. We construct BOTH whole membership directions of the SAME original restricted extension and the actual vanishing ideal of its literal trace, using true tangent-contact Hadamard geometry before any shared-branch criterion.

Theorem 168.1 (Only original order-one data give a genuine principal contact generator for the SAME entire member ideal)**.** *The preceding original-only order-one member inputs internally construct an actual tangent contact function which is a true principal generator of the SAME entire original member ideal germ, with BOTH membership directions and the SAME actual trace/contact support germ.*

Proof. Use the actual original contact function and principal member generator already constructed. At mark one that contact function is a true member of the original ideal itself. Its full principal member dictionary therefore gives an actual ambient germ cofactor. The genuine contact nonmembership in restricted order two forces this true ambient cofactor to be a unit at the point. Holomorphic inverse of the actual unit then writes the same original generator as a true holomorphic multiple of the SAME contact function. Real finite-span substitution proves every actual member germ lies in its contact span. Conversely the contact function is an actual ideal germ and genuine coherent span closure gives the reverse membership direction. No principal-contact certificate is supplied. □

Lemma 168.2 (Actual tangent contact Hadamard geometry identifies its whole ideal with the genuine restricted extension)**.** *For true smooth carrier and actual tangent-contact function that genuinely generates the SAME coherent ideal at the reference point, its actual contact carrier vanishing ideal germ is exactly the genuine whole restricted ideal extension to the carrier, in BOTH membership directions.*

Proof. Actual carrier flattening and the genuine nonzero tangent derivative give the real contact Hadamard ideal dictionary: contact function plus the actual normal coordinates of the carrier generate its entire vanishing ideal germ. On the actual carrier those real normal coordinates vanish, so true finite restricted span substitution reduces precisely to the contact generator span. Conversely every genuine original restricted ideal germ is a true contact multiple on the carrier near the point, by the whole principal dictionary. It vanishes on the actual contact carrier, giving its actual vanishing-ideal membership. The same holomorphic germ condition is retained in both directions. No reduced-ideal, contact-carrier or desired dictionary certificate is input. □

Theorem 168.3 (Only original order-one trace inputs give the actual reduced ideal dictionary of the SAME literal member trace)**.** *At every actual original carrier point of genuine member trace order one, ONLY original smooth carrier, SNC family and whole-boundary density construct BOTH whole germ-membership directions*

$$\mathrm{restrictOn}_W(\mathcal{I}_{H_i})_x = \mathcal{I}_{W\cap H_i,x}.$$

These are the SAME true holomorphic germ predicates, not merely zero loci, orders or formal quotient fields.

Proof. The first original-only producer supplies the SAME actual contact function, true whole member principal dictionary and exact literal trace/contact carrier germ equality. The genuine contact Hadamard theorem then identifies its real vanishing ideal with the actual original restricted extension. A holomorphic germ vanishes on either actual carrier precisely when it vanishes on the other, by their genuine neighborhood set equality. Transport BOTH actual vanishing-ideal membership directions through that equality. This gives the literal original reduced trace ideal dictionary. True shared-branch dimension recognition and whole FIRST-phase Z analytic equations are subsequent consumers; restricted maximum driver, actual recursive history and exceptional termination remain open. □

169 Actual individual original trace orders and true product-order additivity

The original auxiliary marked inputs have EMPTY auxiliary boundary; their literal constructed first nonmonomial order must first be identified with actual ordinary restricted ideal order. This is proved from the genuine colon/restricted-span definitions, before using original factor-order upper semicontinuity or maximum-centre rigidity. The actual individual factors and product include BAD carrier points.

Lemma 169.1 (Empty auxiliary boundary has the same actual restricted ideal and ordinary order)**.** *For an actual original coherent ideal I, the true empty-boundary nonmonomial part is its literal restricted ideal extension to the original carrier. Every original restricted order cosupport is exactly the same one as for I. The literal constructed ordinary first order therefore has precisely the genuine actual restricted-order dictionary.*

Proof. With empty auxiliary boundary the genuine monomial part is the actual unit ideal: there are no divisor positions. The true nonmonomial part is the original colon of the restricted ideal extension by this unit. Taking the actual unit germ gives membership in that same restricted extension; conversely multiplication by every holomorphic unit-ideal member preserves it through

real restricted finite-span multiplication. This proves the actual ideal identity, not a replacement order premise. The restricted extension and original ideal have identical whole carrier-span predicates in both directions. Genuine restricted-order span substitution therefore gives the same order cosupports. The literal original nonmonomial-order definition and finite-order dictionary identify precisely that actual ordinary order. □

Theorem 169.2 (Actual original member and product trace orders are finite and upper semicontinuous)**.** *For original smooth carrier, actual SNC boundary and true whole-boundary density, all individual actual original member trace orders and the true original product trace order are finite and upper semicontinuous at every original carrier point, including bad points.*

Proof. Original SNC supplies each actual coherent member ideal and actual whole product coherence. Genuine original density constructs nonzero restriction for every actual factor and product everywhere on the carrier. The original smooth carrier, empty auxiliary boundary and those independently proved ideal facts construct each genuine marked input. Original finite-order and order upper-semicontinuity theorems apply to those SAME real inputs. The preceding actual ordinary-order dictionary identifies their literal values with actual original restricted ideal orders. No factor-order bound or finite-order witness is supplied by the caller. □

Lemma 169.3 (True principal restricted product orders add through an actual carrier retraction)**.** *For true principal ideal germs nonzero along the original smooth carrier, the actual restricted order of their literal product is the sum of their actual restricted orders.*

Proof. Produce a genuine local holomorphic carrier retraction fixing the actual point. A real ideal generator's restricted order is exactly its actual composed germ's ambient point-ideal order, by the actual retraction and whole principal membership dictionaries. Genuine nonzero restriction makes that order finite. Each composed germ belongs to its true order power and not the next. Actual point-ideal product multiplication gives the sum lower bound, while genuine holomorphic germ nonmembership multiplication gives nonmembership in the next sum power. Thus the actual product germ's order is exactly the sum. The genuine whole product-generator dictionary identifies it with the same literal ideal product, proving the claimed restricted order equality. □

Theorem 169.4 (The actual original member product trace order is the sum of its actual original factor orders everywhere)**.** *For only genuine original smooth-carrier, SNC and whole-boundary-density inputs, the LITERAL original whole product order along that carrier equals the sum of the LITERAL original individual member trace orders at every actual carrier point, including bad points. No product-order or normalized-factor certificate is input.*

Proof. At the actual original point, genuine ambient SNC internally constructs each member's real holomorphic principal germ dictionary, with true nonincident units. Original density supplies actual nonzero restriction of every factor. Apply the preceding true principal restricted-product additivity along the genuine finite original factor list, retaining the same literal ideals and their true generators. The unit product has actual order zero. Genuine finite-list/indexed-product identities give the stated whole original sum, and the first lemma identifies every literal marked first order with that actual restricted order. This supplies real sum rigidity for an internally constructed constant-first- order maximum centre; NC/admissibility of that actual centre with ALL original members remains its separate geometric consumer. □

170 The genuine restricted FIRST-Z centre has an actual whole-good-preserving blowup

Theorem 170.1 (Only original input geometry constructs the SAME true restricted-Z first blowup)**.** *For ONLY the preceding original compact smooth fixed positive-dimensional carrier, original SNC list, genuine whole-boundary density and nonempty literal FIRST Z, internally*

construct its true restricted full-invariant maximum centre and the actual blowup along that SAME centre. The real projection is proper holomorphic and an isomorphism over the ENTIRE original good NC locus. Its literal strict carrier is smooth, and the SAME whole ordered list of geometric strict original members followed by the actual exceptional member is SNC. The centre lies in literal Z and in the genuine sharp bad locus; good boundary intersections are preserved.

Proof. The independent preceding original-only theorem constructs the actual nonempty restricted-Z centre, real whole smoothness and genuine embedded admissibility with ALL original members. It is in true product cosupport. Actual positive marked-domain nonzero/coherence makes that cosupport have empty ambient interior, hence the centre has empty interior. The genuine blowup existence theorem therefore constructs the actual original blowup, its manifold and SAME projection, with actual properness and holomorphicity.

The true centre avoids EVERY original good point. The actual blowup isomorphism off its centre restricts to that ENTIRE good locus: both genuine injectivity and actual preimages are retained, together with actual local inverse holomorphicity. The centre lies in original sharp bad locus and hence in the true original boundary union. Whole-boundary density therefore gives actual carrier density off the centre. Genuine geometric strict transform then equals the literal strict carrier, whose real smoothness follows from the actual contained smooth centre. Independently true original embedded marked-domain wellformedness and real ALL-boundary admissibility transport along the SAME actual blowup; its actual boundary is precisely the whole ordered geometric strict original list followed by the actual exceptional member, and is SNC. These are independently proved geometric consequences, not supplied fields of a desired driver. This covers all nonempty initial FIRST Z, including mild defects missed by the deep order-length threshold. Actual dynamic relative strict-trace histories, recursive intersection resolution, exceptional τ_D cleanup and termination remain OPEN. □

171 The actual original maximum on FIRST Z gives a genuine all-boundary sharp first centre

The preceding actual closed analytic Z now supports the restricted maximum selection of Bierstone–Milman printed page 294. It is the literal original invariant on the actual original product, restricted to the true Z; no caller-selected centre, modified invariant certificate or formal driver is supplied. The producer concerns the FIRST step, before genuine relative strict-trace histories and exceptional phases.

Lemma 171.1 (Real original trace ideals transport to actual open restrictions)**.** *For genuine original SNC members, an actual equality of the SAME entire restricted trace ideals at an original point transports to the true open restriction, retaining each original indexed position.*

Proof. Original SNC gives actual coherent member ideals. The unchanged genuine open restricted-span theorem compares their whole holomorphic germ membership with actual extension by zero. Real vanishing-ideal open pullback identifies the true restricted members; extend an arbitrary actual open holomorphic germ, apply the SAME original trace relation, and pull back both germ directions. Actual length casts retain every original position. No desired open relation or quotient comparison certificate is input. □

Theorem 171.2 (The literal FIRST-phase Z avoids EVERY original good NC point)**.** *For ONLY original smooth carrier, SNC family and whole-boundary density, the SAME literal FIRST Z is disjoint from the ENTIRE original good NC locus, including good intersections of several original members. It is contained in the actual sharp bad locus, without identifying that larger locus with Z.*

Proof. At a true good point, the original producer internally constructs a genuine open where the actual carrier is transverse to the SAME restricted SNC family. Actual product order equations

on this open and true product pullback give original product order at most its actual indexed incidence count. The independently proved original sum bound gives the reverse inequality. Real free-coordinate separator geometry on that actual open separates every two different proper trace ideals. By the preceding genuine open ideal transport, a supposed original quotient collision contradicts that actual separator. Thus the TRUE original quotient is injective on proper positions, and its actual class count equals original incidence count ONLY at this good point. The original product order therefore equals actual DISTINCT count, excluding literal Z. Points outside W are already excluded by its literal definition. □

Theorem 171.3 (Only original input geometry constructs the genuine nonempty restricted-Z maximum centre)**.** *For actual compact smooth W of fixed positive dimension, original SNC and density, and nonempty literal Z, the SAME original product marked input with mark one and empty auxiliary boundary internally selects its genuine full-invariant maximum on Z. This true nonempty centre is closed smooth, is contained in Z, has NC with ALL original ordered E, and is genuinely admissible for the original embedded pair. It avoids the ENTIRE good NC locus.*

Proof. Original product coherence and actual nonzero construct its true marked domain. Every actual Z point has positive true product order and belongs to its real mark-one cosupport. Original upper semicontinuity restricts to the actual compact Z and internally selects a real maximum value. Its actual maximum level on closed Z is closed by true semicontinuity.

At each actual centre point, original full-invariant semicontinuity gives a true open on which every cosupport value is at most its value. Independently proved original invariant-stratum saturation gives a true neighborhood on which every actual same-value carrier point lies in Z. Intersect these real neighborhoods. On this actual open the restricted-Z maximum level is therefore EXACTLY the original whole cosupport level set. The unchanged true original local admissibility theorem makes that actual level set closed smooth. Its genuine open regularity transports to the SAME global restricted centre, proving smoothness everywhere without a supplied centre certificate.

True maximum-value equality gives real product first-order constancy on the entire centre. Actual individual factor USC and genuine product SUM rigidity force every incident factor's order to stay the same along it nearby. Real order-one incidence therefore gives actual local containment in EVERY incident original member. The preceding true all-boundary coordinate insertion theorem constructs NC with ALL original ordered E, before original embedded admissibility is formed. The actual centre lies in literal Z, so the genuine whole-good avoidance theorem supplies sharp preservation. This is FIRST-centre selection; actual relative strict-trace history comparison, recursive intersections, exceptional τ_D and termination remain open. □

172 Genuine shared original trace branches are recognized by actual local dimension

Bierstone–Milman printed page 294 defines the FIRST phase by actual DISTINCT proper restricted trace ideals. At genuine order-one points, the preceding BOTH-direction reduced-ideal theorem makes true local geometry recognize those SAME ideals. The actual pair intersection has a shared branch exactly when its local dimension is one less than the smooth carrier dimension. This concerns the FIRST phase, and does not identify it with the full sharp failure of simultaneous normal crossings.

Lemma 172.1 (Actual local dimension lower loci are genuine analytic dimension strata)**.** *For any genuine analytic set A and integer $q \geq 0$, the ENTIRE locus $\{x : \mathrm{LocDimGE}(A, x, q)\}$ is analytic. It is literally the closure of the actual regular points of A whose dimensions are at least q.*

Proof. The genuine local dimension definition says that every true neighborhood contains a regular point of dimension at least q. The neighborhood characterization of actual closure therefore identifies this ENTIRE set with the closure of the real regular dimension stratum, in both directions. The independently proved actual local analytic decomposition theorem gives analyticity of that closure for any dimension predicate. No desired dimension locus analytic, finite component presentation or upper-semicontinuity premise is supplied. □

Theorem 172.2 (Actual high-dimensional pair intersections internally give equality of the SAME order-one trace ideals)**.** *Only genuine original smooth carrier W, original SNC list E, whole-boundary density and actual order one of members i, j at x are given. If W is truly regular of dimension $d+1$ at x and the literal actual intersection $W \cap E_i \cap E_j$ has local dimension at least d, then the SAME entire restricted trace germ ideals of i and j are equal.*

Proof. Actual closed smooth original sets make the literal pair intersection analytic. It is a genuine subset of each actual order-one member cosupport on the carrier. The original-only contact producer gives a true tangent-contact open/function for each member. The actual codimension-one analytic support theorem, applied to the real pair intersection and each true contact function, identifies its whole set germ with each actual contact hypersurface. This uses the supplied actual local dimension lower bound, not a global maximal order premise. The true original trace/contact support dictionaries then give BOTH set-germ directions between the two literal member traces. Genuine vanishing-ideal membership transports through that set-germ equality, and the preceding actual reduced-ideal theorem identifies each vanishing ideal with the SAME entire original restricted ideal. Hence the actual trace setoid relation holds, without a desired branch or ideal-equality certificate. □

Theorem 172.3 (Equality of the SAME true order-one trace ideals forces actual shared-branch dimension)**.** *For those SAME genuine original inputs, equality of the actual whole restricted trace ideals at x internally gives local dimension at least d of the literal pair intersection, with carrier dimension $d+1$.*

Proof. True original restricted ideal coherence first continues the actual germ relation to all nearby points. The actual proper/unit dictionary on the carrier makes the SAME two member incidences equal nearby: the constant one belongs to the restricted span exactly outside that member. Thus the pair intersection has the same actual set germ as one literal member trace. The independently constructed genuine order-one trace regularity and true uniqueness of the carrier dimension make this SAME trace regular of dimension d. Transport its real regularity through the actual pair/trace germ equality and apply the genuine regular-point local dimension lower bound. No nearby trace-equality, reducedness or target pair-regularity certificate is input. □

Theorem 172.4 (The literal shared original order-one trace branch criterion has BOTH actual directions)**.** *Under the preceding original hypotheses, actual equality of the SAME entire restricted trace ideals is equivalent to local dimension at least d of $W \cap E_i \cap E_j$. This is the TRUE germ equivalence used in the actual DISTINCT proper class count, including bad points.*

Proof. Compose the two independent genuine geometric producers on the SAME original sets, ideals, members and point. Neither direction uses indexed count as the distinct count or simultaneous normal crossings of the carrier with the original family. Actual FIRST-phase analytic equations, restricted maximum centres, strict-trace histories, recursive intersections and exceptional termination remain subsequent producers. □

173 A genuine modified trace-order centre preserves the original smooth normal-crossings region

The original embedded marked ideal uses the ambient carrier and can blow up already resolved boundary intersections. Bierstone–Milman 1997, Section 12, pp. 292–294, instead uses the ideals

of the original boundary restricted to the smooth current variety. For one original smooth boundary member H, its first actual defect is order at least two of $\mathcal{I}_H\mathcal{O}_W$. We implement this genuine first centre-selection phase. The whole multi-member product/distinct-branch defect, recursive intersection phases, exceptional monomial cleanup and termination remain separate mathematical producers.

Lemma 173.1 (A true transverse boundary coordinate has restricted order one)**.** *Let W be a closed smooth submanifold of a positive-dimensional complex manifold, let $[H]$ be SNC, and suppose W has normal crossings with $[H]$ and is dense off H. Then the actual restricted marked ideal $(N = W, E = [], \mathcal{I}_H, 2)$ has empty cosupport.*

Proof. Density excludes a boundary coordinate containing W, so the actual normal-crossings family is transverse to W. At a point of $W \cap H$, use the genuine simultaneous adapted coordinates: W is a coordinate plane, and H is one of its free coordinate hyperplanes. The original ambient coordinate function belongs to the actual original vanishing ideal of H. Its restriction to W has order one: the true local retraction onto the coordinate plane preserves this free coordinate, while the original ambient flat-coordinate order theorem says it is not in the square of the actual point ideal. Thus it cannot have restricted order two, which contradicts cosupport membership. At a point of $W \setminus H$, mark-one cosupport is the actual zero locus and is already empty. Mark-two cosupport is contained in mark-one cosupport. This proves emptiness at every point without assuming a desired order or cosupport certificate. □

Lemma 173.2 (The actual original trace marked ideal lies in the original driver domain)**.** *Let W be closed and smooth of fixed dimension $d \leq n$, and let H be a smooth original boundary with $[H]$ SNC. Suppose $W \subset \overline{W \setminus H}$. The actual marked ideal $(N = W, E = [], \mathcal{I}_H, 2)$ is in the original canonical invariant domain. Its actual cosupport is closed and contained in $W \cap H$.*

Proof. The original boundary vanishing ideal is coherent because H is a closed smooth submanifold. Its actual restriction to W is nonzero at every point. Outside H use the original unit germ. At a point of H, the genuine SNC chart gives a local boundary coordinate generating a germ of $\mathcal{I}_H$. Density of $W \setminus H$ gives arbitrarily nearby points of W where this original coordinate is nonzero. Hence it does not vanish identically on W near the point. The true smooth carrier, empty divisor, positive mark two and fixed dimension give every original domain field. Actual coherent restricted cosupport is closed. Its containment in $W \cap H$ follows from monotonicity in the mark and the original mark-one zero-locus identity. □

Theorem 173.3 (Actual trace defect avoids every genuine original good normal-crossings neighborhood)**.** *Define the original good region as the union of the genuine original open neighborhoods on which the actual restriction of W has normal crossings with the actual restriction of $[H]$. The original trace marked ideal's mark-two cosupport is disjoint from this whole good region.*

Proof. Fix a witnessing actual original open. Actual open restriction preserves the smooth carrier, SNC divisor and the density away from H, by the open-map closure identity. The first lemma gives empty cosupport of the literal restricted marked ideal. The genuine marked open-restriction cosupport dictionary is the inverse image of the original cosupport. Thus no point of that open is in the original cosupport. This holds for every genuine witnessing open, giving disjointness from their whole union. The good region is open by its actual open-cover definition; no desired preservation or comparison certificate is part of this definition. □

Theorem 173.4 (Only original data produce a true modified smooth centre)**.** *Under the preceding original hypotheses, suppose W is compact and the actual mark-two trace cosupport is nonempty. The genuine original canonical invariant of the actual trace marked ideal produces a nonempty closed smooth centre C in that actual cosupport. In particular $C \subset W \cap H$, C avoids the entire original good normal-crossings region, and C has normal crossings with the true original boundary $[H]$. It is genuinely admissible for both the actual trace marked ideal and the original embedded marked ideal $(X, X, [H], \mathcal{I}_W, 1)$.*

Proof. The true trace cosupport is closed inside compact W. The original canonical invariant is upper semicontinuous there and attains a maximum. Its actual maximum locus is nonempty and is a closed smooth admissible centre by the original centre-production theorem, applied to the true trace domain and literal zero-length tower. This is an internally produced centre, not a supplied maximum or desired smooth-centre field. Its actual cosupport containment supplies $C \subset W \cap H$ and disjointness from the original good region. Normal crossings with the empty trace divisor follow from admissibility. Since the true original member H contains the whole centre, the proved containing-member normal-crossings theorem adds that member and proves normal crossings with $[H]$. Mark-one cosupport of the original embedded vanishing ideal is exactly W. Consequently the same genuine smooth centre is also admissible for the original embedded marked ideal, with its actual original boundary. □

Theorem 173.5 (The produced actual blowup preserves every original good normal-crossings point)**.** *The preceding original data produce an actual blowup along the internally selected centre, with proper holomorphic original projection and genuine isomorphism over the entire original good normal-crossings region. Its actual strict transform of W is smooth, and the actual transformed original boundary consists of the genuine geometric strict transform $\overline{\pi^{-1}(H \setminus C)}$ of H followed by the true exceptional divisor and is SNC.*

Proof. The centre is contained in the true restricted cosupport, which has empty interior by original domain membership. The existing genuine blowup construction therefore constructs the actual map and manifold. The projection is proper and holomorphic. Centre disjointness from the original good region and the true off-centre blowup isomorphism prove the asserted preservation over that whole region. Since $C \subset H$ and W is dense away from H, it is dense away from C. The genuine coordinate blowup theorem for a smooth submanifold containing a smooth centre makes its geometric strict transform smooth, and the actual density dictionary identifies that transform with the true strict transform. Original embedded admissibility and genuine transformed well-formedness give SNC of the literal original transformed boundary, retaining its original member and the actual exceptional member. No termination of a freshly initialized or whole modified driver is claimed. □

174 Calculus of blowups along smooth centres

This section collects the facts about a single blowup along a smooth centre, and about finite sequences of them, that the canonical resolution of marked ideals (Wlodarczyk's algorithm, in the analytic setting) uses: the exceptional divisor, the controlled transform of an ideal, and the restriction of a blowup to an open subset. The sources are written for smooth algebraic varieties (Wlodarczyk) or for manifolds and analytic spaces (Bierstone–Milman); here everything is stated for complex manifolds and for the blowup $\pi\colon Y \to X$ along a nowhere dense closed submanifold C in the sense of Definition **??**, i.e. by its universal property (its existence is Lemma **??**). The route differs from the sources as follows.

1. Ideals are ideal sheaves of $\mathcal{O}_X$ recorded by their germs (Definition **??**), and the controlled transform of a marked ideal $(\mathfrak{a}, \mu)$ is *defined* as the colon ideal $(\pi^*\mathfrak{a} : \mathcal{I}_D^{\mu})$, which needs no hypothesis; it is the quotient $\mathcal{I}_D^{-\mu}\pi^*\mathfrak{a}$ of the sources as soon as $\mathfrak{a} \subset \mathcal{I}_C^{\mu}$.

2. The hypothesis of the sources on the centre, $C \subset \operatorname{supp}(\mathfrak{a}, \mu) = \{x : \operatorname{ord}_x \mathfrak{a} \geq \mu\}$, enters only through its consequence $\mathfrak{a} \subset \mathcal{I}_C^{\mu}$ (Wlodarczyk, §2.2, Lemma, part (1)); the statements below take the inclusion as their hypothesis.

3. Open embeddings take the place of the smooth (étale) morphisms of the algebraic sources.

Lemma 174.1 (The exceptional divisor)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\pi\colon Y \to X$ be the blowup of X along C. The ideal $\pi^*\mathcal{I}_C = \mathcal{I}_C\mathcal{O}_Y$ is invertible; it is the ideal $\mathcal{O}_Y(-D)$ of an effective divisor D, the exceptional divisor*, and*

1. $\operatorname{supp} D = \pi^{-1}(C)$*, which is nowhere dense in Y;*
2. $\operatorname{Exc}\pi \subset \pi^{-1}(C)$*, and $\operatorname{Exc}\pi$ is nowhere dense;*
3. $\pi^*(\mathcal{I}_C^{\mu}) = \mathcal{O}_Y(-\mu D)$ *for every $\mu \geq 0$, and $\mathcal{O}_Y(-\mu D)$ is invertible;*
4. *for an effective divisor H on X the total transform π^*H, with ideal $\pi^*\mathcal{O}_X(-H)$, is an effective divisor with* $\operatorname{supp}\pi^*H = \pi^{-1}(\operatorname{supp} H)$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Invertibility of $\pi^*\mathcal{I}_C$ is part of Definition **??**. For a coherent ideal $\mathfrak{a}$ and a holomorphic map f, the zero locus of $f^*\mathfrak{a}$ is f^{-1} of the zero locus of $\mathfrak{a}$: a germ of $\mathfrak{a}$ not vanishing at $f(y)$ pulls back to a germ of $f^*\mathfrak{a}$ not vanishing at y, and conversely every germ of $f^*\mathfrak{a}$ at y is a combination of pullbacks of germs of $\mathfrak{a}$ at $f(y)$. The zero locus of $\mathcal{I}_A$ is A for a closed set A: off A the constant 1 belongs to $\mathcal{I}_A$. Since $\mathcal{I}_C$ is coherent, this gives $\operatorname{supp} D = \pi^{-1}(C)$, and likewise (4), the ideal of an effective divisor being coherent and total transforms of invertible ideals along maps whose exceptional locus is nowhere dense being invertible. The preimage $\pi^{-1}(C)$ is nowhere dense because a local generator of the invertible ideal $\pi^*\mathcal{I}_C$ vanishes on it and vanishes identically on no nonempty open set (Lemma **??**). Off $\pi^{-1}(C)$ the map π is a biholomorphism onto $X \setminus C$ (the universal property applied to the inclusion of $X \setminus C$, where $\mathcal{I}_C$ is the unit ideal), so $\operatorname{Exc}\pi \subset \pi^{-1}(C)$ and (2) follows from (1). Finally $\pi^*(\mathcal{I}_C^{\mu}) = (\pi^*\mathcal{I}_C)^{\mu}$ because pullback commutes with products of coherent ideals and sends the unit ideal to the unit ideal, and powers of invertible ideals are invertible. □

Lemma 174.2 (Cancelling an invertible ideal)**.** *Let $\mathfrak{l}$ be an invertible ideal and $\mathfrak{c}$ a coherent ideal. Then $(\mathfrak{l}\mathfrak{c} : \mathfrak{l}) = \mathfrak{c}$; in particular $(\mathfrak{c} : \mathcal{O}) = \mathfrak{c}$. Moreover, for coherent ideals,* $\mathrm{id}^*\mathfrak{c} = \mathfrak{c}$*, $\mathcal{O}^{\mu} = \mathcal{O}$, $\mathcal{O}\mathfrak{c} = \mathfrak{c}$, $(\mathfrak{a}\mathfrak{b})(\mathfrak{c}\mathfrak{d}) = (\mathfrak{a}\mathfrak{c})(\mathfrak{b}\mathfrak{d})$ and $(\mathfrak{a}\mathfrak{b})^{\mu} = \mathfrak{a}^{\mu}\mathfrak{b}^{\mu}$.*

Proof. If $f \in \mathfrak{c}_x$ then $gf \in \mathfrak{l}\mathfrak{c}$ for every $g \in \mathfrak{l}_x$. Conversely let f be holomorphic at x with $\mathfrak{l}_x f \subset (\mathfrak{l}\mathfrak{c})_x$, and let h generate $\mathfrak{l}$ near x. Then $hf = \sum_j c_j p_j q_j$ near x with $p_j \in \mathfrak{l}_x$, $q_j \in \mathfrak{c}_x$, and $p_j = e_j h$; so $h\big(f - \sum_j c_j e_j q_j\big) = 0$ near x. As h vanishes identically on no nonempty open set, the holomorphic function $f - \sum_j c_j e_j q_j$ vanishes on a dense subset of a neighbourhood of x, hence near x, and $f \in \mathfrak{c}_x$ since $\mathfrak{c}$ is coherent. The case $\mathfrak{l} = \mathcal{O}$ is the second claim ($\mathcal{O}\mathfrak{c} = \mathfrak{c}$). The remaining identities follow from commutativity and associativity of the product of coherent ideals (Lemma **??** and its context) by induction on μ. □

Definition 174.3 (Controlled transform)**.** Let $\pi\colon Y \to X$ be the blowup along C with exceptional divisor D, $\mathfrak{a}$ an ideal on X and $\mu \geq 0$. The *controlled transform* of the marked ideal $(\mathfrak{a}, \mu)$ is the colon ideal

$$\sigma^c(\mathfrak{a}, \mu) := \big(\pi^*\mathfrak{a} : \mathcal{O}_Y(-\mu D)\big).$$

Lemma 174.4 (The controlled transform is the quotient by $\mathcal{I}_D^{\mu}$)**.** *Let $\mathfrak{a}$ be coherent with $\mathfrak{a} \subset \mathcal{I}_C^{\mu}$. Then $\pi^*\mathfrak{a} \subset \mathcal{O}_Y(-\mu D)$,*

$$\pi^*\mathfrak{a} = \mathcal{O}_Y(-\mu D) \cdot \sigma^c(\mathfrak{a}, \mu),$$

$\sigma^c(\mathfrak{a}, \mu)$ is coherent, and it is the only coherent ideal $\mathfrak{c}$ with $\pi^*\mathfrak{a} = \mathcal{O}_Y(-\mu D)\,\mathfrak{c}$. For $\mu = 0$ (and $\mathfrak{a}$ coherent) $\sigma^c(\mathfrak{a}, 0) = \pi^*\mathfrak{a}$.

Proof. By Lemma 174.1(3), $\pi^*\mathfrak{a} \subset \pi^*(\mathcal{I}_C^{\mu}) = \mathcal{O}_Y(-\mu D)$ (Wlodarczyk, §2.2, Lemma, part (2)). The ideal $\pi^*\mathfrak{a}$ is coherent and $\mathcal{O}_Y(-\mu D)$ is invertible, so Lemma **??** gives the factorization with a coherent colon ideal, which is $\sigma^c(\mathfrak{a}, \mu)$ by definition. If also $\pi^*\mathfrak{a} = \mathcal{O}_Y(-\mu D)\mathfrak{c}$ with $\mathfrak{c}$ coherent, then $\sigma^c(\mathfrak{a}, \mu) = (\mathcal{O}_Y(-\mu D)\mathfrak{c} : \mathcal{O}_Y(-\mu D)) = \mathfrak{c}$ by Lemma 174.2. For $\mu = 0$, $\mathcal{O}_Y(-0 \cdot D) = \mathcal{O}_Y$ and $(\pi^*\mathfrak{a} : \mathcal{O}_Y) = \pi^*\mathfrak{a}$, again by Lemma 174.2. □

Lemma 174.5 (Restriction of a blowup to an open subset)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\pi\colon Y \to X$ be the blowup along C and $V \subset X$ open. Then the restriction $\pi^{-1}(V) \to V$ of π is the blowup of the complex manifold V along $C \cap V$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

$C \cap V$ is a closed submanifold of V (regularity is local, Lemma 443.2) with empty interior (an open subset of V inside C is open in X), and the restriction is holomorphic. A holomorphic map $g\colon Z \to V$ into the open set V is the same as a holomorphic map $\iota \circ g\colon Z \to X$ with values in V, and germs on V are germs on X at points of V (extend by 0); in particular $g^*\mathcal{I}_{C\cap V} = (\iota \circ g)^*\mathcal{I}_C$. For $g =$ the restriction of π to $\pi^{-1}(V)$ this is the restriction of the invertible ideal $\pi^*\mathcal{I}_C$ to the open set $\pi^{-1}(V)$, hence invertible. If $g^*\mathcal{I}_{C\cap V}$ is invertible, the universal property of π gives a unique holomorphic $h\colon Z \to Y$ with $\pi \circ h = \iota \circ g$; it takes values in $\pi^{-1}(V)$, so it is a holomorphic lift of g through the restriction, and any such lift, composed with the inclusion of $\pi^{-1}(V)$, is a lift of $\iota \circ g$ through π, hence equal to h. □

Towers. A *tower* over X is a finite sequence $X = X_0 \leftarrow X_1 \leftarrow \cdots \leftarrow X_r$ of blowups $\pi_i\colon X_i \to X_{i-1}$ along nowhere dense closed submanifolds $C_{i-1} \subset X_{i-1}$, recorded with all its spaces, centres and maps; its projection is $\pi_1 \circ \cdots \circ \pi_r\colon X_r \to X$. The transform of a marked ideal along a tower, multiple test blow-ups and resolutions are those of Bierstone–Milman, in the ambient form $(M, N, E, \mathcal{I}, d)$ used by the chapter on the resolution of marked ideals.

Lemma 174.6 (Restriction of a tower to an open subset)**.** *Let T be a tower over X with projection $p\colon X_r \to X$, and $U \subset X$ open. The open sets $U_i = (\pi_1 \circ \cdots \circ \pi_i)^{-1}(U) \subset X_i$, with the restrictions of the π_i and the centres $C_{i-1} \cap U_{i-1}$, form a tower over U of the same length, whose projection is the restriction of p to $p^{-1}(U) \to U$.*

Proof. Induction on the length, using Lemma 174.5 at each step: $\pi_i^{-1}(U_{i-1}) = U_i$, and the restriction of π_i to $U_i \to U_{i-1}$ is the blowup along $C_{i-1} \cap U_{i-1}$. □

Definition 174.7 (The exceptional divisor of a tower)**.** The exceptional divisor E_T of a tower T is 0 for the empty tower, and $\pi_r^*E_{T'} + D_r$ if T is T' followed by the blowup π_r with exceptional divisor D_r (Lemma 174.1); thus $E_T = \sum_i (\pi_{i+1} \circ \cdots \circ \pi_r)^* D_i$.

Lemma 174.8 (Transforms with $N = X$)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\pi\colon Y \to X$ be the blowup along C. The strict transform of X is Y. For a marked ideal $(X, X, E, \mathfrak{a}, d)$ with $\mathfrak{a}$ coherent, its transform by π is $(Y, Y, E', \sigma^c(\mathfrak{a}, d), d)$: the ideal of the transform, $\mathcal{I}_D^{-d}\pi^*\mathfrak{a}$ read on the strict transform Y, is the controlled transform of Definition 174.3.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

The strict transform of X is the closure of $Y \setminus \operatorname{Exc}\pi$, which is Y since $\operatorname{Exc}\pi$ is nowhere dense (Lemma 174.1(2)). Reading an ideal on the whole space changes nothing for a coherent ideal (a germ agreeing everywhere near the point with a combination of germs of the ideal is that combination), so the colon along Y of $\pi^*\mathfrak{a}$ by $\mathcal{O}_Y(-dD) = (\pi^*\mathcal{I}_C)^d$ is the colon $\sigma^c(\mathfrak{a}, d)$. □

Proposition 174.9 (Resolution gives principalization)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\mathfrak{a}$ be a coherent ideal on X and T a multiple test blow-up of the marked ideal $(X, X, E, \mathfrak{a}, 1)$, with projection p and transform $(X_r, X_r, E_r, \mathfrak{a}_r, 1)$. Then $\mathfrak{a}_r$ is coherent and $p^*\mathfrak{a} = \mathcal{O}(-E_T)\,\mathfrak{a}_r$. If T is a resolution (empty final cosupport), then $\mathfrak{a}_r = \mathcal{O}$ and $p^*\mathfrak{a} = \mathcal{O}(-E_T)$ is invertible.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Induction on the length. With the mark 1 the cosupport of a marked ideal $(X_i, X_i, E_i, \mathfrak{a}_i, 1)$ is the zero locus of $\mathfrak{a}_i$. An admissible centre C_i lies in it, so every germ of the coherent ideal $\mathfrak{a}_i$ at a point x vanishes on C_i near x (the germ belongs to $\mathfrak{a}_i$ at every point near x, by coherence), i.e. $\mathfrak{a}_i \subset \mathcal{I}_{C_i}$. By Lemmas 174.8 and 174.4, $\mathfrak{a}_{i+1} = \sigma^c(\mathfrak{a}_i, 1)$ is coherent and $\pi_{i+1}^*\mathfrak{a}_i = \mathcal{O}(-D_{i+1})\,\mathfrak{a}_{i+1}$. If $(\pi_1 \circ \cdots \circ \pi_i)^*\mathfrak{a} = \mathcal{O}(-E_i)\,\mathfrak{a}_i$, then pulling back by π_{i+1}, which commutes with products of coherent ideals, gives $(\pi_1 \circ \cdots \circ \pi_{i+1})^*\mathfrak{a} = \mathcal{O}(-\pi_{i+1}^*E_i - D_{i+1})\,\mathfrak{a}_{i+1}$. If the final cosupport is empty, the coherent ideal $\mathfrak{a}_r$ has no zero: near each point it contains a germ b with $b(x) \neq 0$, and then every holomorphic germ $g = (g/b)\,b$ belongs to it, so $\mathfrak{a}_r = \mathcal{O}$ and $p^*\mathfrak{a} = \mathcal{O}(-E_T)$ (Lemma 174.2). □

Submanifolds through the centre. In the remaining statements $\pi\colon Y \to X$ is the blowup along C and $N \supset C$ is a closed subset of X, regular of dimension k_N at each of its points. Coordinates are those of Definition **??**: a set is *flattened* by a local biholomorphism Φ (after the preferred chart) onto $\mathbf{C}^k \times 0$, the *last* $n-k$ coordinates vanishing.

Lemma 174.10 (Nested flattening coordinates)**.** *Let $C \subset N$ be regular at x of dimensions k_C and k_N. Then $k_C \leq k_N$, and one local biholomorphism Φ at x (with $\Phi(x) = 0$) flattens C onto $\mathbf{C}^{k_C} \times 0$ and N onto $\mathbf{C}^{k_N} \times 0$ simultaneously.*

Proof. Work in coordinates u flattening N, so $N = \mathbf{C}^{k_N} \times 0$ near 0, and let Θ flatten C there. The map $\varphi(v) = \Theta^{-1}(v, 0)$, $v \in \mathbf{C}^{k_C}$ near 0, parametrizes C and takes values in $\mathbf{C}^{k_N} \times 0$; write $\varphi = \iota \circ \varphi'$ with $\varphi'\colon \mathbf{C}^{k_C} \to \mathbf{C}^{k_N}$. Since $\Theta \circ \iota \circ \varphi' = (\cdot, 0)$ near 0, the differential $D = d\varphi'(0)$ is injective; in particular $k_C \leq k_N$. Let $J\colon \mathbf{C}^{k_N - k_C} \to \mathbf{C}^{k_N}$ be linear and injective onto a complement of $D(\mathbf{C}^{k_C})$, and write $w = (v, s, t) \in \mathbf{C}^{k_C} \times \mathbf{C}^{k_N - k_C} \times \mathbf{C}^{n - k_N}$. The map $F(v, s, t) = (\varphi'(v) + Js,\ t)$ has the invertible differential $(v, s, t) \mapsto (Dv + Js, t)$ at 0, so it has a local holomorphic inverse Φ (Lemma **??**). Now $F(w) \in \mathbf{C}^{k_N} \times 0$ iff $t = 0$, and $F(v, 0, 0) = \varphi(v) \in C$; conversely a point of C near 0 is $\varphi(v)$ with v small (it is $\Theta^{-1}(\Theta(\cdot))$ and $\Theta(\cdot) \in \mathbf{C}^{k_C} \times 0$), so Φ maps it to $(v, 0, 0)$. Hence near 0, $\Phi(C) = \mathbf{C}^{k_C} \times 0$ and $\Phi(N) = \mathbf{C}^{k_N} \times 0$; compose with the coordinates u. □

Lemma 174.11 (Monomial charts of an abstract blowup)**.** *Let C be flattened near y_0 by Φ onto $\mathbf{C}^k \times 0$, and for $j \geq k$ let $\sigma_j(w)_i = w_i w_j$ ($k \leq i \neq j$), $\sigma_j(w)_i = w_i$ otherwise, and $\gamma_j = e^{-1} \circ \Phi^{-1} \circ \sigma_j$ on $\sigma_j^{-1}(\Phi(\mathit{target}))$. Then $\gamma_j^*\mathcal{I}_C$ is generated by w_j, so γ_j lifts uniquely to a holomorphic ℓ_j with $\pi \circ \ell_j = \gamma_j$. Every point z of Y over the flattening domain has a neighbourhood O, an index $j \geq k$ and a holomorphic $W\colon O \to \mathbf{C}^n$ with $\sigma_j \circ W = \Phi \circ e \circ \pi$ and $\ell_j \circ W = \mathrm{id}$ on O; the differential of W (in a chart) is injective. Moreover $\sigma_j(w) \in \mathbf{C}^k \times 0 \iff w_j = 0$; for $k \leq j < m$, $\sigma_j(w) \in \mathbf{C}^m \times 0 \iff w_j = 0$ or $w \in \mathbf{C}^m \times 0$; for $j \geq m$, $\sigma_j(w) \in \mathbf{C}^m \times 0$ forces $w_j = 0$.*

Proof. The pullback of $\mathcal{I}_C$, generated by the flat coordinates τ_i ($i \geq k$, Lemma **??**), is generated by the $w_i w_j$ and w_j, i.e. by w_j, which vanishes on no open set; the universal property gives ℓ_j. Near z the pulled back coordinates are $\tau_i \circ \pi = \alpha_i h$ with h a local generator of $\pi^*\mathcal{I}_C$; writing h back in terms of the τ_i shows that some $\alpha_i(z) \neq 0$; for j with $|\alpha_j(z)|$ maximal, $W = (\alpha_i/\alpha_j\ (i \geq k,\ i \neq j);\ \tau_i \circ \pi$ otherwise) satisfies $\sigma_j \circ W = \Phi \circ e \circ \pi$, and $\ell_j \circ W$ and the inclusion are two lifts of π near z, hence equal; then $d\ell_j \circ dW = \mathrm{id}$ shows dW injective. The statements about σ_j are read off its formula. □

Theorem 174.12 (The strict transform of a submanifold through the centre)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

The strict transform $N' = \overline{\pi^{-1}(N \setminus C)}$ is a closed submanifold of Y, regular of dimension k_N at each of its points. In the monomial chart of Lemma 174.11 at a point z over C, built on nested coordinates (Lemma 174.10), N' near z is $W^{-1}(\mathbf{C}^{k_N} \times 0)$ if $j < k_N$, and N' does not meet the chart if $j \geq k_N$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Off $\pi^{-1}(C)$ the map π is a local biholomorphism (Lemma 174.1(2)) and $N' = \pi^{-1}(N)$ there ($\pi^{-1}(N)$ is closed, and agrees with $\pi^{-1}(N \setminus C)$ off $\pi^{-1}(C)$), so N' is regular of dimension k_N there (Lemma **??**). Let $\pi(z) \in C$; take nested coordinates for $C \subset N$ at $\pi(z)$ and the chart j, O, W at z. For $u \in O$: $\pi(u) \in C \iff W(u)_j = 0$, and $\pi(u) \in N \iff \sigma_j(W(u)) \in \mathbf{C}^{k_N} \times 0$. If $j \geq k_N$ this forces $W(u)_j = 0$, so $\pi^{-1}(N \setminus C) \cap O = \emptyset$ and $N' \cap O = \emptyset$. If $j < k_N$, $\pi^{-1}(N \setminus C) \cap O = \{u \in O : W(u) \in \mathbf{C}^{k_N} \times 0,\ W(u)_j \neq 0\}$. Its closure in O lies in the closed set $\{W \in \mathbf{C}^{k_N} \times 0\}$; conversely if $W(u) \in \mathbf{C}^{k_N} \times 0$ and $W(u)_j = 0$, then for $t \neq 0$ small $w_t = W(u) + te_j \in \mathbf{C}^{k_N} \times 0$ (as $j < k_N$) with $(w_t)_j = t \neq 0$, so $\ell_j(w_t) \in \pi^{-1}(N \setminus C)$, and $\ell_j(w_t) \to \ell_j(W(u)) = u$. So $N' \cap O = W^{-1}(\mathbf{C}^{k_N} \times 0) \cap O$. Since $W \circ e_z^{-1}$ has injective differential at $e_z(z)$, it is a local biholomorphism there (Lemma **??**); recentred by $W(z) \in \mathbf{C}^{k_N} \times 0$, it flattens N' onto $\mathbf{C}^{k_N} \times 0$ at z. □

Theorem 174.13 (The ideal of the strict transform)**.** *Let $N \supset C$ be a closed submanifold of X (of any, possibly varying, dimension) and $N' = \overline{\pi^{-1}(N \setminus C)}$. Then N' is a closed submanifold of Y,*

$$\pi^* \mathcal{I}_N = \mathcal{O}_Y(-D) \cdot \mathcal{I}_{N'}, \qquad \textit{equivalently} \qquad \sigma^c(\mathcal{I}_N, 1) = \mathcal{I}_{N'}.$$

Proof. N' is a closed submanifold by Theorem 174.12, applied at each point with the dimension of N there. Both sides of the identity are computed stalk by stalk at $z \in Y$. In general $f^* \mathcal{I}_A \subset \mathcal{I}_{f^{-1}(A)}$ for a continuous f: the generators $a \circ f$ vanish on $f^{-1}(A)$.

If $\pi(z) \notin C$, then π has a holomorphic section s over $X \setminus C$ with $s(\pi(z')) = z'$ for $\pi(z') \notin C$ (the universal property applied to the inclusion of $X \setminus C$, where $\mathcal{I}_C$ is the unit ideal). A germ g vanishing on $\pi^{-1}(N)$ near z is $a \circ \pi$ with $a = g \circ s$, and a vanishes on N near $\pi(z)$; so $\pi^* \mathcal{I}_N = \mathcal{I}_{\pi^{-1}(N)}$ at z. Near z, $N' = \pi^{-1}(N)$, and $\mathcal{O}_Y(-D)$ contains $1 = 1 \circ \pi$, so $\mathcal{O}_Y(-D) \cdot \mathcal{I}_{N'} = \mathcal{I}_{N'}$ at z.

Let $\pi(z) \in C$. Take nested coordinates Φ at $\pi(z)$ with $C = \mathbf{C}^{k_C} \times 0$ and $N = \mathbf{C}^{k_N} \times 0$ (Lemma 174.10) and the monomial chart $j \geq k_C$, O, W at z (Lemma 174.11), so $\sigma_j \circ W = \Phi \circ e \circ \pi$ on O. For $A = C$ or $A = N$, flattened onto $\mathbf{C}^m \times 0$, every germ of $\mathcal{I}_A$ at $\pi(z')$ is $\sum_{i \geq m} b_i \Phi_i$ (Lemma **??**); hence on O the ideal $\pi^* \mathcal{I}_A$ is generated by the functions $\sigma_j(W)_i$, $i \geq m$, each of which lies in it. For $A = C$ these are $W_i W_j$ ($i \neq j$) and W_j, so $\mathcal{O}_Y(-D) = (W_j)$ on O. For $A = N$:

- if $j < k_N$, all $i \geq k_N$ differ from j, so $\pi^* \mathcal{I}_N = (W_j W_i : i \geq k_N)$. By Theorem 174.12, $N' \cap O = W^{-1}(\mathbf{C}^{k_N} \times 0) \cap O$. If $z \in N'$, W has injective differential at z, so it is a local biholomorphism there (Lemma **??**); recentred by $W(z) \in \mathbf{C}^{k_N} \times 0$, it flattens N', and Hadamard's lemma in these coordinates gives $\mathcal{I}_{N'} = (W_i : i \geq k_N)$ near z; the product of the two presentations is $(W_j W_i : i \geq k_N) = \pi^* \mathcal{I}_N$. If $z \notin N'$, then $\mathcal{I}_{N'} = \mathcal{O}_Y$ near z, and some W_{i_0}, $i_0 \geq k_N$, does not vanish at z, so $W_j = W_{i_0}^{-1} \cdot W_j W_{i_0} \in \pi^* \mathcal{I}_N$ and $\pi^* \mathcal{I}_N = (W_j) = \mathcal{O}_Y(-D) \cdot \mathcal{O}_Y$.
- if $j \geq k_N$, then N' misses O, $\mathcal{I}_{N'} = \mathcal{O}_Y$ on O, and the generators of $\pi^* \mathcal{I}_N$ are W_j ($i = j$) and multiples $W_i W_j$ of it: $\pi^* \mathcal{I}_N = (W_j) = \mathcal{O}_Y(-D) \cdot \mathcal{O}_Y$.

Finally $\mathcal{I}_N \subset \mathcal{I}_C$, $\mathcal{I}_{N'}$ is coherent (it is the ideal of a closed submanifold) and $\mathcal{O}_Y(-D)^1 = \mathcal{O}_Y(-D)$, so the uniqueness part of Lemma 174.4 turns the identity into $\sigma^c(\mathcal{I}_N, 1) = \mathcal{I}_{N'}$. □

Lemma 174.14 (Geometric and analytic strict transforms)**.** *For $A \subset X$, $\overline{\pi^{-1}(A \setminus C)} \subset \overline{\pi^{-1}(A) \setminus \operatorname{Exc} \pi}$, with equality when C is nowhere dense in A ($A \subset \overline{A \setminus C}$). The first set is empty for $A = C$; the second is C itself when C is a smooth hypersurface (then π is a biholomorphism and $\operatorname{Exc} \pi = \emptyset$).*

Proof. $\operatorname{Exc} \pi \subset \pi^{-1}(C)$ (Lemma 174.1) gives the inclusion. Conversely let $\pi(u) \in A$, $u \notin \operatorname{Exc} \pi$. Then π maps every neighbourhood V of u onto a neighbourhood of $\pi(u)$ (Lemma **??**), which meets $A \setminus C$ because $\pi(u) \in \overline{A \setminus C}$; so V meets $\pi^{-1}(A \setminus C)$. The closure of $\pi^{-1}(A) \setminus \operatorname{Exc} \pi$ is therefore in $\overline{\pi^{-1}(A \setminus C)}$. □

The second clause is why the transform of a marked ideal must use $\overline{\pi^{-1}(A \setminus C)}$ for N and for the members of E: along a centre of codimension one that is a member H of E, the other set would keep H, and E' would list it twice.

Lemma 174.15 (The controlled transform of a function)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

A local equation *of D on an open set $U \subset Y$ is a holomorphic h on U, vanishing identically on no nonempty open subset, with $\mathcal{O}_Y(-D) = (h)$ on U; local equations exist around every point. On U, $\mathcal{O}_Y(-\mu D) = (h^\mu)$, and for $z \in U$:*

1. *if $u \in \mathcal{I}_C^\mu$ at $\pi(z)$, then $u \circ \pi = h^\mu u'$ near z for a germ u' holomorphic at z, unique as a germ (the* controlled transform *$u' = h^{-\mu}\pi^* u$ of u);*
2. *if moreover $u \in \mathfrak{a}$ at $\pi(z)$, then $u' \in \sigma^c(\mathfrak{a}, \mu)$ at z;*
3. *if $\mathfrak{a} \subset \mathcal{I}_C^\mu$ is coherent, $\sigma^c(\mathfrak{a}, \mu)$ is generated at z by the controlled transforms of the germs of $\mathfrak{a}$ at $\pi(z)$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Local equations are the local generators of the invertible ideal $\mathcal{O}_Y(-D)$. By induction on μ, the product of the presentations (h^μ) and (h) is $(h^{\mu+1})$. (1) $u \circ \pi \in \pi^*(\mathcal{I}_C^\mu) = \mathcal{O}_Y(-\mu D) = (h^\mu)$ (Lemma 174.1); uniqueness: if $h^\mu v = h^\mu w$ near z, then $v = w$ on the dense set $\{h \neq 0\}$, hence near z. (2) Every $\ell \in \mathcal{O}_Y(-\mu D)$ at z is ch^μ, and $\ell u' = c\,(u \circ \pi) \in \pi^*\mathfrak{a}$; so u' lies in the colon ideal $\sigma^c(\mathfrak{a}, \mu)$. (3) If $g \in \sigma^c(\mathfrak{a}, \mu)$, then $h^\mu g \in \pi^*\mathfrak{a}$, i.e. $h^\mu g = \sum_i a_i\,(u_i \circ \pi)$ with $u_i \in \mathfrak{a}$; writing $u_i \circ \pi = h^\mu u_i'$ by (1) and cancelling h^μ gives $g = \sum_i a_i u_i'$. Conversely the u_i' lie in the coherent ideal $\sigma^c(\mathfrak{a}, \mu)$ by (2) (Lemma 174.4), which absorbs their combinations. □

Corollary 174.16 (Strict transforms of hypersurfaces through the centre)**.** *Verified upstream; not exported by this standalone library.*

Let $N \supset C$ be a closed submanifold, $z \in U$, and suppose $\mathcal{I}_N = (u)$ at $\pi(z)$ with u holomorphic, and $u \circ \pi = h\,u'$ near z. Then $\mathcal{I}_{N'} = (u')$ at z, where $N' = \overline{\pi^{-1}(N \setminus C)}$; in particular $z \in N'$ if and only if $u'(z) = 0$. This is the statement "$V(u')$ is the strict transform of $V(u)$" of Włodarczyk's Lemma on tangent directions.

Proof. Verified upstream; not exported by this standalone library.

By Theorem 174.13, $\mathcal{I}_{N'} = \sigma^c(\mathcal{I}_N, 1)$, which by Lemma 174.15(3) is generated at z by the v with $w \circ \pi = hv$, $w \in \mathcal{I}_N$ at $\pi(z)$ (note $\mathcal{I}_N \subset \mathcal{I}_C$ as $C \subset N$). Such a w is cu, so $hv = (c \circ \pi)hu'$ and $v = (c \circ \pi)u'$; and u' itself is one of them. For the pointwise statement: if $z \in N'$, the generator u' vanishes at z; if $z \notin N'$, the constant 1 lies in $\mathcal{I}_{N'} = (u')$ at z, so $u'(z) \neq 0$. □

Lemma 174.17 (Zero sets of functions of multiplicity one)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let u be holomorphic on an open set $U \subset X$, of multiplicity one at its zeros in U (at each such zero some partial derivative of u in some chart does not vanish). Then $V(u) \cap U$ is closed in U and regular of dimension $n - 1$ at each of its points, and at every point of U the ideal of $V(u) \cap U$ is generated by u.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Closedness in U is the continuity of u. Let $x \in U$ with $u(x) = 0$, and let e be the preferred chart at x. The partial derivatives of u in any chart are holomorphic combinations of those in e (chain rule for the transition), so some $\partial_j(u \circ e^{-1})(e(x)) \neq 0$; in particular $n > 0$. Let τ be the transposition of j and $n-1$ and $W = (e_{\tau(0)}, \dots, e_{\tau(n-2)}, u)$. The differential of W at x is injective: a vector killed by it has all coordinates but the j-th equal to 0, and then $du(x)$ kills it only if that coordinate is 0. Hence $W - W(x)$ is a coordinate system at x (Lemma 174.121; inverse function

theorem, Lemma **??**) in which $V(u)\cap U = \mathbf{C}^{n-1}\times 0$: this is the regularity. By Hadamard's lemma in the coordinates W (as in the proof of Theorem 174.13) the ideal of $V(u)\cap U$ is generated near x by the coordinate $W_{n-1} = u$. At a point $x \in U$ with $u(x) \neq 0$, u is a unit near x and $V(u)$ misses a neighbourhood of x: both the ideal of $V(u) \cap U$ and (u) consist of all holomorphic germs. □

Theorem 174.18 (Włodarczyk's lemma on tangent directions on an open set)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let u be holomorphic on an open set $U \subset X$, of multiplicity one at its zeros in U, with $C \cap U \subset V(u)$, and put $N = V(u) \cap U$, $N' = \overline{\pi^{-1}(N \setminus C)}$. Let h be a local equation of the exceptional divisor on $U' \subset Y$, $z \in U' \cap \pi^{-1}(U)$, and $u \circ \pi = h\,u'$ near z with u' holomorphic at z. Then:

1. *the ideal of N' at z is (u'); hence N' and $V(u')$ agree near z (Włodarczyk's Lemma on tangent directions, part (3): $V(u')$ is the strict transform of $V(u)$ over U);*
2. *if $z \in N'$, some partial derivative of u' at z in some chart does not vanish (part (2): u' has multiplicity one).*

*The hypotheses on u hold for a tangent direction u of a marked ideal $(\mathfrak{a}, \mu)$, $\mu \geq 1$, on U (Definition **??**) and a centre C in the cosupport of $(\mathfrak{a}, \mu)$ (Lemma **??**).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

By Lemma 174.5 the restriction $\pi^{-1}(U) \to U$ of π is the blowup of the manifold U along $C \cap U$. In U, N is a closed submanifold containing $C \cap U$, of dimension $n-1$ at each of its points, with ideal (u) at every point (Lemma 174.17). The restriction of h to $U' \cap \pi^{-1}(U)$ is a local equation of the exceptional divisor of the restricted blowup: germs on an open subset are germs on the ambient manifold at its points (extension by zero), and the exceptional ideal of the restriction is the restriction of $\pi^*\mathcal{I}_C$. Strict transforms commute with the restriction: the closure in the open set $\pi^{-1}(U)$ of a subset of it is the trace of its closure in Y. Corollary 174.16 for the restricted blowup gives $\mathcal{I}_{N'} = (u')$ at z, read back on Y through the same identifications; as N' is closed, $z \in N'$ if and only if $u'(z) = 0$, and the hypotheses persist at the points near z, which gives the agreement near z. For (2): if $z \in N'$, then $u(\pi(z)) = 0$, so $n > 0$ by Lemma 174.17; N' is regular of dimension $n-1$ at z (Theorem 174.12 in $\pi^{-1}(U)$, regularity being local), and a generator of the ideal of a submanifold of dimension $n-1 < n$ has a nonvanishing partial derivative (Lemma 174.129). Finally, a tangent direction u is a germ of $\mathcal{D}^{\mu-1}(\mathfrak{a})$ at every point of U, hence holomorphic there, of multiplicity one by definition, and $C\cap U \subset \operatorname{cosupp}(\mathfrak{a}, \mu)\cap U \subset V(u)$ (Lemma **??**). □

Lemma 174.19 (A generator modulo a submanifold has a nonvanishing tangential derivative)**.** *Let $A \subset B$ be regular at z of dimensions $a < b$, and let f be holomorphic at z with $f(z) = 0$, such that every germ of $\mathcal{I}_A$ at z is a combination of germs of $\mathcal{I}_B$ and of f. Then some holomorphic vector field ξ tangent to B at z (Definition **??**) has $\xi(f)(z) \neq 0$. Moreover: a combination of germs vanishing at z vanishes at z; and if h is holomorphic at z and vanishes identically near no point, g is holomorphic at z, and hg is a combination of products ht with t in a family of holomorphic germs, then g is a combination of the t.*

Proof. Take coordinates ψ centred at z in which $A = \mathbf{C}^a \times 0$ and $B = \mathbf{C}^b \times 0$ (Lemma 174.10). The field $\xi = \partial/\partial\psi_a$ is tangent to B, as $a < b$ (Lemma **??**(2)), and $\psi_a \in \mathcal{I}_A$. Write $\psi_a = \sum_k c_k s_k$ near z with each s_k a germ of $\mathcal{I}_B$ or f. By the product rule $1 = \xi(\psi_a)(z) = \sum_k \xi(c_k)(z)s_k(z) + c_k(z)\xi(s_k)(z)$. Every s_k vanishes at $z \in A \subset B$; if $s_k \in \mathcal{I}_B$, then $\xi(s_k) \in \mathcal{I}_B$ (tangency) vanishes at z. So if $\xi(f)(z) = 0$, the right side vanishes, a contradiction. (For $B = X$ this is the argument of Lemma 174.129.) The second claim is evaluation at z. For the third, $hg = \sum c_k\, ht_k$ near z gives $h\,(g - \sum c_k t_k) = 0$ near z, and a holomorphic germ killed by h vanishes near z, since h vanishes identically near no point. □

Theorem 174.20 (Włodarczyk's lemma on tangent directions along a submanifold)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $N \supset C$ be a closed submanifold, u holomorphic on an open set $U \subset X$ with multiplicity one along N at its zeros on N (at each such point some holomorphic field tangent to N has $\xi(u) \neq 0$), and $C \cap U \subset V(u)$. Put $W = N \cap V(u) \cap U$, and let W', N' be the strict transforms. Let h be a local equation of the exceptional divisor on $U' \subset Y$, $z \in U' \cap \pi^{-1}(U)$, and $u \circ \pi = h\,u'$ near z with u' holomorphic at z. Then:

1. *if $\pi(z) \in W$, then $\mathcal{I}_{W'} = \mathcal{I}_{N'} + (u')$ at z;*
2. *near z, a point of N' lies on W' if and only if u' vanishes there (part (3) of Włodarczyk's lemma along N);*
3. *if $z \in W'$, some holomorphic field tangent to N' at z has $\xi(u')(z) \neq 0$: u' has multiplicity one along N' (part (2)).*

*The hypotheses on u hold for a tangent direction u along N of $(\mathfrak{a}, \mu)$, $\mu \geq 1$, on U (Definition **??**) and a centre C in $\operatorname{cosupp}(\mathfrak{a}|_N, \mu)$ (Lemma **??**). For $N = X$ this is Theorem 174.18.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Two facts on h: it vanishes nowhere on $U' \setminus \pi^{-1}(C)$ (there $\mathcal{O}(-D) \ni 1$, Lemma 174.1), and it vanishes identically near no point (Lemma 174.15). Hence u' vanishes on W' near z: on $\pi^{-1}(W \setminus C)$ we have $h\,u' = u \circ \pi = 0$ and $h \neq 0$, and u' is continuous.

(1) By Lemma **??**, W is a submanifold of U, closed in U, of dimension one less than N at each of its points, and near $\pi(z)$ its ideal is generated by u and the normal coordinates of N, so $\mathcal{I}_W \subset \mathcal{I}_N + (u)$ at $\pi(z)$. By Lemma 174.5 the restriction $\pi^{-1}(U) \to U$ is the blowup of U along $C \cap U \subset W$, and Theorem 174.13 for it, read back on Y as in Theorem 174.18, gives $h\,g \in \pi^*\mathcal{I}_W$ for every germ g of $\mathcal{I}_{W'}$ at z. Each germ of $\pi^*\mathcal{I}_W$ is a combination of $a \circ \pi$, $a \in \mathcal{I}_N$, and of $u \circ \pi = h\,u'$; and $\pi^*\mathcal{I}_N = \mathcal{O}(-D)\,\mathcal{I}_{N'}$ (Theorem 174.13 for N), so $a \circ \pi$ is a combination of products $h\,t$, $t \in \mathcal{I}_{N'}$. Thus $h\,g$ is a combination of the $h\,t$ ($t \in \mathcal{I}_{N'}$ or $t = u'$), and g is a combination of the t (Lemma 174.19). Conversely $\mathcal{I}_{N'} \subset \mathcal{I}_{W'}$ as $W' \subset N'$, and $u' \in \mathcal{I}_{W'}$ at z by the first paragraph.

(2) Let $y \in N'$ near z; the hypotheses hold at y. If $y \in W'$, then $u'(y) = 0$ by the first paragraph. Let $u'(y) = 0$. If $\pi(y) \notin C$, then $\pi(y) \in N$ and $u(\pi(y)) = h(y)u'(y) = 0$, so $y \in \pi^{-1}(W \setminus C) \subset W'$. If $\pi(y) \in C$, then $\pi(y) \in W$; were $y \notin W'$, the constant 1 would lie in $\mathcal{I}_{W'}$ at y, hence by (1) be a combination of germs vanishing at $y \in N'$, which is absurd.

(3) As W is closed in U, $\pi(z) \in W$. Let N have dimension k at $\pi(z)$; then $k \geq 1$ and W has dimension $k-1$ there (Lemma **??**). By Theorem 174.12 (for W in the restricted blowup, and for N), W' and N' are regular at z of dimensions $k - 1 < k$, and $W' \subset N'$. By (1) and $u'(z) = 0$, Lemma 174.19 applies with $A = W'$, $B = N'$, $f = u'$. Finally, a tangent direction u along N is a germ of $\mathcal{D}_N^{\mu-1}(\mathfrak{a})$ at every point of U, hence holomorphic there, of multiplicity one along N by definition, and $C \cap U \subset \operatorname{cosupp}(\mathfrak{a}|_N, \mu) \cap U \subset V(u)$ (Lemma **??**). □

Lemma 174.21 (Blowing up a codimension-one part of the support)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\mathfrak{a}$ be coherent, $\mu \geq 1$, and let the centre C of the blowup π lie in $\operatorname{cosupp}(\mathfrak{a}, \mu)$ (orders on the whole manifold). If $z \in \pi^{-1}(C)$, C has dimension $n-1$ at $x = \pi(z)$, and $\operatorname{ord}_x \mathfrak{a} \leq \mu$, then $z \notin \operatorname{cosupp}(\sigma^c(\mathfrak{a}, \mu), \mu)$. Consequently, if $(\mathfrak{a}, \mu)$ is of maximal order and C has codimension one at each of its points, the cosupport of $\sigma^c(\mathfrak{a}, \mu)$ does not meet the exceptional divisor $\pi^{-1}(C)$ (the transform half of Włodarczyk's Lemma "le:codim1", after Villamayor); the same holds for the corrected transform of a marked ideal with $N = M$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Near x, a flattening coordinate u generates $\mathcal{I}_C$ (Hadamard's lemma in flattening coordinates), so $\mathcal{I}_C^\mu = (u^\mu)$ (Lemma **??**). By the Hadamard bridge (Lemma **??**) $\mathfrak{a} \subset \mathcal{I}_C^\mu$. As $\operatorname{ord}_x \mathfrak{a} \le \mu$, some $f \in \mathfrak{a}_x$ does not vanish to order $\mu+1$ at x; write $f = c\,u^\mu$. If $c(x) = 0$, then c and u vanish at x and f vanishes to order $1+\mu$ (Lemma **??**); so $c(x) \neq 0$. Let h be a local equation of the exceptional divisor at z (Lemma 174.1). Both $u \circ \pi$ and h generate $\pi^*\mathcal{I}_C$ at z (the germs of $\pi^*\mathcal{I}_C$ are combinations of $a \circ \pi$, $a \in (u)$), so $u \circ \pi = d\,h$ and $h = d'\,(u \circ \pi)$, whence $h\,(d'd - 1) = 0$ and $d'd = 1$ near z (cancellation of h), and $d(z) \neq 0$. Then $f \circ \pi = h^\mu\,(c \circ \pi)\,d^\mu$, so $(c \circ \pi)\,d^\mu$ lies in $\sigma^c(\mathfrak{a}, \mu)$ at z (Lemma 174.15); it does not vanish at z, hence z is not in the cosupport of $(\sigma^c(\mathfrak{a}, \mu), \mu)$ for $\mu \ge 1$ (Definition **??**). If $(\mathfrak{a}, \mu)$ is of maximal order, $\mathcal{D}^\mu\mathfrak{a}$ contains 1, so $\operatorname{cosupp}(\mathfrak{a}, \mu+1) \subset \operatorname{cosupp}(\mathcal{D}^\mu\mathfrak{a}, 1) = \emptyset$ (Lemma **??**, Definition **??**): $\operatorname{ord}_x \mathfrak{a} \le \mu$ everywhere. For a marked ideal with $N = M$ the corrected transform has $N' = M'$ and ideal $\sigma^c(\mathfrak{a}, d)$ (Lemma **??**). □

Codimension-one components of the support. Włodarczyk's Lemma "le:codim1" has a second half: for a marked ideal of maximal order, the codimension-one components of the support are smooth and isolated. In the analytic category the components are the irreducible analytic components (Definition **??**); "isolated" is made precise as being a connected component of the support and meeting no other irreducible component; and Włodarczyk's argument "the support lies in the smooth hypersurface $V(u)$ of a tangent direction" is completed by an identity theorem on $V(u)$, Lemma 174.22. The analytic version of the lemma (Włodarczyk 2009, Lemma 5.3.7, printed p. 48) has the same statements for germs M_Z, under the additional hypothesis that the support has codimension one, which is not used here; its proof writes $\mathcal{I} = (u^\mu)$ locally, while Lemma 174.21 only uses $\mathfrak{a} \subset \mathcal{I}_C^\mu$ and $\operatorname{ord}\mathfrak{a} \le \mu$, and so applies to any centre of dimension $n-1$ inside the support (as in Step 1aa for $s = 1$, where the centre is a member).

Lemma 174.22 (Filling a regular set by an analytic set of the same local dimension)**.** *Let A be an analytic subset of a complex manifold X, let $H \subset X$ be regular of dimension k at x, and suppose $A \subset H$ near x and $\dim_x A \ge k$ (every neighbourhood of x contains a regular point of A of dimension $\ge k$, Remark 443.1). Then $A = H$ near x; in particular A is regular of dimension k at x.*

Proof. As A is closed and its regular points accumulate at x, $x \in A$. Let $g_1, \dots, g_m$ be holomorphic local equations of A at x in the chart e at x, defined on an open set $D \ni e(x)$, and let N_0 be an open neighbourhood of x on which $A \subset H$. Let Φ flatten H at x: $\Phi(e(x)) = 0$, and for y in the chart domain with $e(y)$ in the domain of Φ, $y \in H \iff \Phi(e(y)) \in \mathbf{C}^k \times 0$. Choose a ball B of radius r about 0 in the target of Φ with $\Phi^{-1}(B) \subset D \cap e(N_0)$, and put $W = e^{-1}\Phi^{-1}(B)$, an open neighbourhood of x inside N_0. Then $H \cap W = \psi(B \cap (\mathbf{C}^k \times 0))$ with $\psi = e^{-1} \circ \Phi^{-1}$, every point of $H \cap W$ is regular of dimension k for H, and $H \cap W$ is closed in W (Lemma **??**).

First, $A \cap W$ contains a nonempty relatively open part of $H \cap W$. Otherwise, regard W as a complex manifold. In it, $T = H \cap W$ is a closed submanifold, hence analytic (Lemma **??**), and all its regular points have dimension k (the dimension at a regular point is unique, Lemma **??**); $T' = A \cap W \subset T$ has empty interior relative to T. By the thin-dimension drop (Remark 443.8) every regular point of T' has dimension $< k$. But $\dim_x A \ge k$ gives a point $y \in W$ at which A is regular of some dimension $j \ge k$, and T' is regular of dimension j at y in W (regularity is local, Remark 443.2): a contradiction.

Second, the identity theorem on the flattened ball. Let $\iota\colon \mathbf{C}^k \to \mathbf{C}^n$ be the inclusion $w \mapsto (w, 0)$; the set $\iota^{-1}(B)$ is open and convex, and each $g_i \circ \Phi^{-1} \circ \iota$ is holomorphic on it. By the first step it vanishes on the nonempty open set of parameters w with $\psi(\iota(w))$ in the open part found there; hence it vanishes on all of $\iota^{-1}(B)$ (Lemma **??**), i.e. $H \cap W = \psi(\iota(\iota^{-1}B)) \subset A$. With $A \cap W \subset H$ this gives $A \cap W = H \cap W$, and regularity at x only depends on the germ (Lemma **??**). □

Lemma 174.23 (Irreducible analytic sets: connectedness and local dimension)**.** *Verified upstream; not exported by this standalone library.*

An analytically irreducible subset V of a complex manifold is connected. If V is a closed irreducible analytic subvariety, then $\dim_x V \geq \dim V$ at every point $x \in V$.

Proof. Verified upstream; not exported by this standalone library.

Let P, Q be closed sets with $V \subset P \cup Q$ and $V \cap P \cap Q = \emptyset$. Near each point of $V \cap P$ the set $V \cap P$ agrees with V (on the open set $X \setminus Q$), and $V \cap P$ is closed, so it is analytic (Lemma **??**); likewise $V \cap Q$. As $V = (V \cap P) \cup (V \cap Q)$, irreducibility (Definition **??**) gives $V = V \cap P$ or $V = V \cap Q$, i.e. $V \subset P$ or $V \subset Q$. For the second claim, every neighbourhood of $x \in V$ contains a regular point of V (Lemma **??**), and every regular point of V has dimension $\dim V$ (Lemma **??**). □

Lemma 174.24 (The support of a coherent marked ideal is analytic)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

If $\mathfrak{a}$ is coherent and $\mu \geq 1$, then $\operatorname{cosupp}(\mathfrak{a}, \mu)$ (orders on the whole manifold) is an analytic subset.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

By Lemma **??**, $\operatorname{cosupp}(\mathfrak{a}, \mu) = V(\mathcal{D}^{\mu-1}\mathfrak{a})$, and $\mathcal{D}^{\mu-1}\mathfrak{a}$ is coherent (Lemma **??**). The zero locus of a coherent ideal is analytic: near each point it is the common zero set of finitely many local generators (Lemma **??**). □

Lemma 174.25 (Near a point of local dimension $n-1$ the support is the hypersurface of a tangent direction)**.** *Verified upstream; not exported by this standalone library.*

*Let $\mathfrak{a}$ be coherent, $\mu \geq 1$, and $A \subset \operatorname{cosupp}(\mathfrak{a}, \mu)$ analytic with $\dim_x A \geq n-1$. If u is a tangent direction of $(\mathfrak{a}, \mu)$ on an open set $U \ni x$ (Definition **??**), then $A = V(u)$ near x. If moreover $(\mathfrak{a}, \mu)$ is of maximal order, A is regular of dimension $n-1$ at x.*

Proof. Verified upstream; not exported by this standalone library.

A is closed and its regular points accumulate at x, so $x \in A$; as $A \cap U \subset \operatorname{cosupp}(\mathfrak{a}, \mu) \cap U \subset V(u)$ (Lemma **??**), $u(x) = 0$. The tangent direction u is holomorphic on U and of multiplicity one at its zeros, so $H = V(u) \cap U$ is regular of dimension $n-1$ at x (Lemma 174.17), and $A \subset H$ on U. By Lemma 174.22, $A = H$ near x, i.e. $A = V(u)$ near x. If $(\mathfrak{a}, \mu)$ is of maximal order (Definition **??**), a tangent direction exists on a neighbourhood of every point of the support (Lemma **??**), and then A is regular of dimension $n-1$ at x because H is. □

Definition 174.26 (The codimension-one part of the support)**.** *Verified upstream; not exported by this standalone library.*

The *codimension-one part* $R(\mathfrak{a}, \mu)$ of the support $S = \operatorname{cosupp}(\mathfrak{a}, \mu)$ is the set of points at which S is regular of dimension $n-1$.

Theorem 174.27 (The codimension-one part of the support is a smooth isolated hypersurface)**.** *Verified upstream; not exported by this standalone library.*

Let $(\mathfrak{a}, \mu)$ be coherent, of maximal order, $\mu \geq 1$, with support S and codimension-one part R. Then:

1. *R is the set of points x with $\dim_x S \geq n-1$;*
2. *R is closed, open in S, and regular of dimension $n-1$ at each of its points; in particular R is a closed submanifold;*
3. *near each $x \in R$, $R = S = V(u)$ for every tangent direction u on a neighbourhood of x;*
4. *$S \setminus R$ is closed, and every regular point of $S \setminus R$ has dimension $< n-1$: $\dim(S \setminus R) \leq n-2$.*

Proof. Verified upstream; not exported by this standalone library.

(1) A regular point of dimension $n-1$ has local dimension $n-1$ (Remark 443.1); conversely S is analytic (Lemma 174.24), and Lemma 174.25 with $A = S$ shows that S is regular of dimension $n-1$ at a point of local dimension $\geq n-1$. (2) A flattening of S at $x \in R$ flattens S at all nearby points of S, with the same dimension (Theorem **??**); so $R = S$ near x. Hence R is open in S (the union of these neighbourhoods meets S in R), and R is regular of dimension $n-1$ at x because S is and the two agree near x. A limit of points of R has local dimension $\geq n-1$ for S, so it lies in R by (1): R is closed. (3) $R = S$ near x, and $S = V(u)$ near x by Lemma 174.25. (4) If O is open with $S \cap O = R$, then $S \setminus R = S \setminus O$, closed since S is (Lemma **??**). Let $S \setminus R$ be regular of dimension j at y. As R is closed, $S \setminus R$ and S agree on the open neighbourhood $X \setminus R$ of y, so S is regular of dimension j at y. With a tangent direction u on $U \ni y$ (Lemma **??**), $S \cap U \subset V(u) \cap U$ (Lemma **??**), which is regular of dimension $n-1$ at y (Lemma 174.17); a submanifold of a submanifold has smaller dimension (Lemma 174.10), so $j \leq n-1$; and $j \neq n-1$ because $y \notin R$. □

Theorem 174.28 (Codimension-one components of the support are smooth and isolated)**.** *Verified upstream; not exported by this standalone library.*

Let $(\mathfrak{a}, \mu)$ be coherent, of maximal order, $\mu \geq 1$, with support S, and let $V \subset S$ be a closed irreducible analytic subvariety with $\dim V = n-1$ (for instance an irreducible component of S of codimension one). Then $V \subset R(\mathfrak{a}, \mu)$, and:

1. *V is regular of dimension $n-1$ at each of its points;*
2. *near each $x \in V$, $V = V(u)$ for every tangent direction u on a neighbourhood of x, and $V = S$;*
3. *V is open in S, and it is the connected component of S through each of its points;*
4. *V meets no irreducible component $V' \neq V$ of S;*
5. *after the blowup π along V, the cosupport of $\sigma^c(\mathfrak{a}, \mu)$ does not meet the exceptional divisor $\pi^{-1}(V)$.*

Items (1)–(4) are the first half of Włodarczyk's Lemma "le:codim1" (after Villamayor), item (5) its second half for one component.

Proof. Verified upstream; not exported by this standalone library.

At every $x \in V$, $\dim_x V \geq \dim V = n-1$ (Lemma 174.23); hence $\dim_x S \geq n-1$ (local dimension is monotone, Remark 443.7; S is analytic by Lemma 174.24) and $x \in R$ by Theorem 174.27. Lemma 174.25 with $A = V$ gives (1) and the first half of (2). For the second half, take a tangent direction u on $U \ni x$ (Lemma **??**): near x, $S \cap U \subset V(u) = V \subset S$ (Lemma **??**). (3) The union O of the neighbourhoods of the points of V on which $V = S$ is open, contains V, and $S \cap O = V$. By Lemma 174.23, V is connected, so it lies in the connected component K of S through $x \in V$. Conversely $K \subset S \subset O \cup (X \setminus V)$ and K does not meet $O \setminus V = (S \cap O) \setminus V$ inside S; so the connected set K lies in O or in $X \setminus V$, and $x \in K \cap V$ excludes the second: $K \subset S \cap O = V$. (4) If an irreducible component V' of S (Definition **??**) contains $x \in V$, then V' is connected (Lemma 174.23) and contained in S, hence in the connected component of S through x, which is V by (3); since $V \subset S$, maximality of V' gives $V' = V$. (5) By (1), V is a centre of dimension $n-1$ at each of its points inside the support; apply Lemma 174.21. □

Corollary 174.29 (Blowing up the codimension-one part of the support)**.** *Verified upstream; not exported by this standalone library.*

Let $(\mathfrak{a}, \mu)$ be coherent, of maximal order, $\mu \geq 1$, and let π be the blowup along its codimension-one part R. Then the cosupport of $\sigma^c(\mathfrak{a}, \mu)$ does not meet the exceptional divisor $\pi^{-1}(R)$; the same holds for the corrected transform of a marked ideal with $N = M$. This is Step 1ba of Włodarczyk's algorithm, the elimination of the codimension-one part of the support.

Proof. Verified upstream; not exported by this standalone library.

R lies in the support and has dimension $n-1$ at each of its points (Theorem 174.27); apply Lemma 174.21. □

The transform away from the centre. Off the exceptional divisor a blowup is biholomorphic and the transform of a marked ideal is its pullback; in particular the cosupport does not change there.

Lemma 174.30 (A blowup is an isomorphism off its centre)**.** *Let $\pi\colon Y \to X$ be the blowup along C. There is a holomorphic map $s\colon X \setminus C \to Y$ with $\pi \circ s = \mathrm{id}$ and $s(\pi(z)) = z$ whenever $\pi(z) \notin C$. Hence π is an isomorphism over $X \setminus C$, and $\mathrm{Exc}(\pi) \subset \pi^{-1}(C)$.*

Proof. On the open manifold $X \setminus C$ the pullback of $\mathcal{I}_C$ by the inclusion is the unit ideal, which is invertible; by the universal property of the blowup (Definition **??**) the inclusion factors holomorphically as $\pi \circ s$. On the open set $\pi^{-1}(X \setminus C)$ both the inclusion and $s \circ \pi$ are holomorphic lifts of the restriction of π, whose ideal $\pi^*\mathcal{I}_C$ is again the unit ideal; by the uniqueness clause they agree, i.e. $s(\pi(z)) = z$. So π maps $\pi^{-1}(X \setminus C)$ bijectively onto $X \setminus C$, and the chain rule for $\pi \circ s = \mathrm{id}$ makes $d\pi$ surjective, hence bijective (equal dimensions), at every point over $X \setminus C$. □

Theorem 174.31 (The transform away from the centre)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be a marked ideal with N closed and $\mathfrak{a}$ an ideal of holomorphic germs, π the blowup along C, and $\underline{\mathcal{I}}' = (M', N', E', \mathfrak{a}', d)$ the transform (Definition **??**). If $\pi(z) \notin C$, then $N' = \pi^{-1}(N)$ near z, and $z \in \operatorname{cosupp}\underline{\mathcal{I}}'$ if and only if $\pi(z) \in \operatorname{cosupp}\underline{\mathcal{I}}$. Consequently $\operatorname{cosupp}\underline{\mathcal{I}}' \setminus \pi^{-1}(C) = \pi^{-1}(\operatorname{cosupp}\underline{\mathcal{I}} \setminus C)$, and $\operatorname{cosupp}\underline{\mathcal{I}}' \subset \pi^{-1}(\operatorname{cosupp}\underline{\mathcal{I}})$ when $C \subset \operatorname{cosupp}\underline{\mathcal{I}}$. For $N = M$ and $\mathfrak{a}$ coherent: $\operatorname{cosupp}(\sigma^c(\mathfrak{a}, \mu), \mu) \setminus \pi^{-1}(C) = \pi^{-1}(\operatorname{cosupp}(\mathfrak{a}, \mu) \setminus C)$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

$N' = \overline{\pi^{-1}(N \setminus C)} \subset \pi^{-1}(N)$ as N is closed, and on the open set $\pi^{-1}(X \setminus C) \ni z$ the set $\pi^{-1}(N)$ equals $\pi^{-1}(N \setminus C) \subset N'$. At z the constant 1 lies in $\pi^*\mathcal{I}_C$, hence in its d-th power (Lemma **??**), so a germ f of $\mathfrak{a}' = (\mathfrak{a}\mathcal{O}_{N'} : \pi^*\mathcal{I}_C^d)$ satisfies $f = 1 \cdot f \in \mathfrak{a}\mathcal{O}_{N'}$: on N' near z, f is a holomorphic combination of germs of $\pi^*\mathfrak{a}$, themselves combinations of $a \circ \pi$, $a \in \mathfrak{a}_{\pi(z)}$; conversely every $a \circ \pi$ lies in $\mathfrak{a}'$ at z, because $\mathfrak{a}\mathcal{O}_{N'}$ is closed under holomorphic multiples.

If $\pi(z) \in \operatorname{cosupp}\underline{\mathcal{I}}$, each $a \in \mathfrak{a}_{\pi(z)}$ agrees on N near $\pi(z)$ with some $h \in \mathfrak{m}^d_{\pi(z)}$ (Definition **??**, Lemma **??**); then $a \circ \pi = h \circ \pi$ on $N' \subset \pi^{-1}(N)$ near z, and $h \circ \pi \in \mathfrak{m}^d_z$ (Lemma **??**). Germs vanishing to order d along N' are closed under holomorphic combinations (Lemma **??**), so every germ of $\mathfrak{a}'$ at z vanishes to order d along N': $z \in \operatorname{cosupp}\underline{\mathcal{I}}'$.

Conversely let $z \in \operatorname{cosupp}\underline{\mathcal{I}}'$ and $a \in \mathfrak{a}_{\pi(z)}$. As $a \circ \pi \in \mathfrak{a}'_z$, there is $h \in \mathfrak{m}^d_z$ with $a \circ \pi = h$ on N' near z. Let s be the section of Lemma 174.30 on the open manifold $X \setminus C$, with $s(\pi(z)) = z$. Then $h \circ s \in \mathfrak{m}^d_{\pi(z)}$ on $X \setminus C$ (Lemma **??**), and its extension by zero is a germ of $\mathfrak{m}^d_{\pi(z)}$ on X (Lemma **??**). For $y \in N$ near $\pi(z)$, $s(y)$ is near z and lies in $\pi^{-1}(N) = N'$ there, so $h(s(y)) = a(\pi(s(y))) = a(y)$: a vanishes to order d along N at $\pi(z)$, and $\pi(z) \in N$. So $\pi(z) \in \operatorname{cosupp}\underline{\mathcal{I}}$. The set identities follow pointwise; for $N = M$ the transform has $N' = M'$ and ideal $\sigma^c(\mathfrak{a}, \mu)$ (Lemma **??**). □

Corollary 174.32 (Cosupports along multiple test blow-ups)**.** *Let $\underline{\mathcal{I}}$ have N a closed submanifold and $\mathfrak{a}$ coherent. Along every multiple test blow-up T of $\underline{\mathcal{I}}$, the cosupport of the transform $\underline{\mathcal{I}}_T$ lies over* $\operatorname{cosupp}\underline{\mathcal{I}}$.

Proof. Induction along T. Every stage has N a closed submanifold and a coherent ideal (Lemma **??**), and the next centre is admissible, so it lies in the cosupport of that stage; Theorem 174.31 puts the cosupport of the next stage over the cosupport of that stage, which lies over $\operatorname{cosupp}\underline{\mathcal{I}}$ by induction. □

Corollary 174.33 (After blowing up the codimension-one part the support has codimension at least two)**.** *Verified upstream; not exported by this standalone library.*

Let $(\mathfrak{a},\mu)$ be coherent, of maximal order, $\mu\geq 1$, with support S and codimension-one part R, and let π be the blowup along R. Then $\operatorname{cosupp}(\sigma^c(\mathfrak{a},\mu),\mu)\subset\pi^{-1}(S\setminus R)$*, where $S\setminus R$ is closed of dimension $\leq n-2$ (Theorem 174.27).*

Proof. Verified upstream; not exported by this standalone library.

Points over R are excluded by Corollary 174.29; at the other points Theorem 174.31 gives $\operatorname{cosupp}(\sigma^c(\mathfrak{a},\mu),\mu)\setminus\pi^{-1}(R)=\pi^{-1}(S\setminus R)$. □

Codimension one inside a submanifold. In the induction of the algorithm the marked ideals are $(M,N,E,\mathfrak{a},d)$ with N a submanifold of dimension $k+1$, and Step 1ba eliminates the part of the cosupport of codimension one in N. The transform half of Lemma 174.21 holds inside N with the same proof, read on N and on the strict transform N'.

Lemma 174.34 (Blowing up a centre of codimension one in N inside the support along N)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}}=(M,N,E,\mathfrak{a},d)$ have N a closed submanifold, $\mathfrak{a}$ coherent and $d\geq 1$, and let the centre C of the blowup π lie in $\operatorname{cosupp}\underline{\mathcal{I}}$*. Let $z\in\pi^{-1}(C)$, $x=\pi(z)$, and suppose that at x the centre is regular of dimension k and N is regular of dimension $k+1$, and that* $\operatorname{ord}_x(\mathfrak{a}|_N)\leq d$*. Then $z\notin$* $\operatorname{cosupp}\underline{\mathcal{I}}'$*. Consequently, if $\underline{\mathcal{I}}$ is of maximal order along N (Definition* **??***:* $\operatorname{ord}(\mathfrak{a}|_N)\leq d$ *everywhere) and C has codimension one in N at each of its points, the cosupport of the transform does not meet $\pi^{-1}(C)$. For $N=M$ this is Lemma 174.21.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Take nested flattening coordinates Φ at x (Lemma 174.10): $C=\mathbf{C}^k\times 0$, $N=\mathbf{C}^{k+1}\times 0$, and put $u=\Phi_k$, $H=\{u=0\}$; $\mathcal{I}_H=(u)$ near x (Lemma 174.39). At a point y of N near x a germ $g\in\mathcal{I}_C$ is $\sum_{i\geq k}b_i\Phi_i$ (Hadamard's lemma in flattening coordinates, Lemma **??**), and $\Phi_i=0$ on N for $i\geq k+1$; so on N near y, $g=b_ku$: $\operatorname{Res}_N\mathcal{I}_C\subset\operatorname{Res}_N\mathcal{I}_H$ near x, hence $\operatorname{Res}_N\mathcal{I}_C^d\subset\operatorname{Res}_N\mathcal{I}_H^d$ (restricted inclusions pass to powers, Lemma **??**), and $\mathcal{I}_H^d=(u^d)$ (Lemma **??**). As $C\subset\operatorname{cosupp}\underline{\mathcal{I}}$, $\operatorname{Res}_N\mathfrak{a}\subset\operatorname{Res}_N\mathcal{I}_C^d$ (Lemma **??**). Since $\operatorname{ord}_x(\mathfrak{a}|_N)\leq d$, some $f\in\mathfrak{a}_x$ does not vanish to order $d+1$ along N at x; by the above $f=c\,u^d$ on N near x, and $c(x)\neq 0$: otherwise $c\,u^d$ vanishes to order $d+1$ at x (Lemma **??**) and agrees with f on N. Suppose $z\in\operatorname{cosupp}\underline{\mathcal{I}}'$; then $z\in N'$. Take a monomial chart $j\geq k$, W around z with $\sigma_j\circ W=\Phi\circ\pi$ (Lemma 174.11). As $z\in N'$, $j<k+1$ (Lemma 174.121), so $j=k$ and $u\circ\pi=W_k$, which generates $\mathcal{O}(-D)$ near z (Lemmas 174.13 and 174.1); so $\mathcal{O}(-dD)=(W_k^d)$ near z. On $N'\subset\pi^{-1}(N)$ near z, $f\circ\pi=(c\circ\pi)\,W_k^d$; hence for every $l=b\,W_k^d\in\mathcal{O}(-dD)_z$, $l\,(c\circ\pi)=b\,(f\circ\pi)$ on N' near z, that is, $c\circ\pi$ lies in $\mathfrak{a}'=(\operatorname{Res}_{N'}\pi^*\mathfrak{a}:\mathcal{O}(-dD))$ at z (Definition **??**). It does not vanish at $z\in N'$, so $\mathfrak{a}'$ does not vanish to order $d\geq 1$ along N' at z (Definition **??**): a contradiction. The consequence is the pointwise statement at every $z\in\pi^{-1}(C)$. □

Lemma 174.35 (Near a point of local dimension $\dim N-1$ the support along N is a hypersurface of maximal contact)**.** *Let N be a closed submanifold, $\mathfrak{a}$ coherent and $\mu\geq 1$. Then $S=\operatorname{cosupp}(\mathfrak{a}|_N,\mu)$ is analytic. Let $A\subset S$ be analytic, $x\in M$, N regular of dimension $k+1$ at x, and $\dim_x A\geq k$. If u is a tangent direction of $(\mathfrak{a}|_N,\mu)$ along N on an open set $U\ni x$ (Definition* **??***), then $A=N\cap V(u)$ near x. If moreover $(\mathfrak{a}|_N,\mu)$ is of maximal order along N (Definition* **??***), such a u exists near every point of S, and A is regular of dimension k at x. The first conclusion holds for any w holomorphic on an open $O\ni x$ that agrees on N, near each point of $N\cap O$, with a combination of germs of $\mathcal{D}_N^{\mu-1}\mathfrak{a}$ and has multiplicity one along N at its zeros on $N\cap O$ (such w are the tangent directions of the transforms along a tower, which are controlled only on the strict transforms of N). For $N=M$ this is Lemma 174.25.*

Proof. $S = N \cap V(\mathcal{D}_N^{\mu-1}\mathfrak{a})$ (Lemma **??**); N is analytic (Lemma **??**), the zero locus of the coherent ideal $\mathcal{D}_N^{\mu-1}\mathfrak{a}$ (Lemma **??**) is analytic (Lemma 174.24), and so is the intersection (Definition **??**). Let A, x, u be as stated. A is closed and its regular points accumulate at x, so $x \in A$; $S \cap U \subset V(u)$ (Lemma **??**), so $u(x) = 0$, and $H = N \cap V(u) \cap U$ is regular at x of dimension one less than N (Lemma **??**), that is of dimension k (the dimension of a regular point is unique, Lemma **??**). Near x, $A \subset S \cap U \subset H$; by the filling lemma (Lemma 174.22), $A = H$ near x, i.e. $A = N \cap V(u)$ near x. For w as in the last sentence of the statement the same argument applies: w vanishes at the points $y \in S \cap O$, being on N near y a combination of germs of $\mathcal{D}_N^{\mu-1}\mathfrak{a}$, which vanish at y; and $N \cap V(w) \cap O$ is regular of dimension k at x by Lemma **??**, which uses only the holomorphy of w and its multiplicity one along N at x. Under maximal order along N, $\operatorname{ord}_x(\mathfrak{a}|_N) \leq \mu$ at every $x \in S$, so a tangent direction along N exists on a neighbourhood of x (Lemma **??**), and A is regular of dimension k at x because H is (Remark 443.1: regularity depends only on the germ). □

Definition 174.36 (The codimension-one part of the support along N)**.** Let $N \subset M$ be a submanifold. The *codimension-one part along* N of $S = \operatorname{cosupp}(\mathfrak{a}|_N, \mu)$ is the set $R_N(\mathfrak{a}, \mu)$ of points x at which N is regular of some dimension $k+1$ and S is regular of dimension k. For $N = M$ it is the codimension-one part (Definition 174.26).

Theorem 174.37 (The codimension-one part of the support along N is a smooth isolated hypersurface of N)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let N be a closed submanifold and $(\mathfrak{a}|_N, \mu)$ coherent, of maximal order along N, $\mu \geq 1$, with support S and codimension-one part $R = R_N(\mathfrak{a}, \mu)$ along N. Then:

1. *R is the set of points x at which N has some dimension $k+1$ and $\dim_x S \geq k$;*
2. *R is closed, open in S, and at each of its points regular of dimension one less than N; in particular R is a closed submanifold;*
3. *near each $x \in R$, $R = S = N \cap V(u)$ for every tangent direction u along N on a neighbourhood of x, and for every w controlled only on N as in Lemma 174.35;*
4. *$S \setminus R$ is closed, and at a point where N has dimension $k+1$ every regular point of $S \setminus R$ has dimension $< k$.*

For $N = M$ this is Theorem 174.27.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) A regular point of dimension k has local dimension $\geq k$ (Remark 443.1); conversely S is analytic and Lemma 174.35 with $A = S$ makes S regular of dimension k at a point of local dimension $\geq k$. (2) Near $x \in R$, with N of dimension $k+1$ and S of dimension k at x, every point of N is regular of dimension $k+1$ and every point of S regular of dimension k (Theorem **??**); as $S \subset N$, $R = S$ near x. Hence R is open in S and regular of dimension k at x. Let x be a limit of points of R; $x \in S$ since S is closed (Lemma **??**), N is regular of some dimension k_x at x, hence of dimension k_x at the points of N near x; at the points of R near x it also has dimension $k+1$ with S of dimension k there, so $k_x = k+1$ (Lemma **??**) and $\dim_x S \geq k$: $x \in R$ by (1). (3) $R = S$ near x, and $S = N \cap V(u)$ near x by Lemma 174.35. (4) If O is open with $S \cap O = R$, then $S \setminus R = S \setminus O$ is closed. Let $S \setminus R$ be regular of dimension j at y, where N has dimension $k+1$. As R is closed, S is regular of dimension j at y. A tangent direction u along N on $U \ni y$ exists (Lemma **??**); $S \cap U \subset N \cap V(u) \cap U$ (Lemma **??**), which is regular of dimension k at y (Lemma **??**); a submanifold of a submanifold has smaller dimension (Lemma 174.10), so $j \leq k$, and $j \neq k$ because $y \notin R$. □

Corollary 174.38 (Blowing up the codimension-one part of the support along N)**.** *Verified upstream; not exported by this standalone library.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N a closed submanifold, $\mathfrak{a}$ coherent, $d \geq 1$, and be of maximal order along N; let π be the blowup along the codimension-one part R of its support along N. Then the cosupport of the transform does not meet the exceptional divisor $\pi^{-1}(R)$, and it lies over $\operatorname{cosupp}\underline{\mathcal{I}} \setminus R$*, a closed set whose regular points have dimension at most* $\dim N - 2$*. This is Step 1ba of Włodarczyk's algorithm inside N. For $N = M$ it is Corollaries 174.29 and 174.33.*

Proof. Verified upstream; not exported by this standalone library.

R lies in the support and has codimension one in N at each of its points (Theorem 174.37(2)); Lemma 174.34 excludes the points over R. At the other points the transform has the same cosupport (Theorem 174.31: $\operatorname{cosupp}\underline{\mathcal{I}}' \setminus \pi^{-1}(R) = \pi^{-1}(\operatorname{cosupp}\underline{\mathcal{I}} \setminus R)$); the dimension bound is Theorem 174.37(4). □

Members of the exceptional divisor under a blowup. Let E be a simple normal crossings divisor (Definition **??**) such that the centre C has normal crossings with E. The ideals of the members of E transform monomially; this is the chart computation behind the monomial marked ideals of the algorithm.

Lemma 174.39 (Ideals of coordinate hyperplanes and principal ideals)**.** *(1) If Φ is a holomorphic coordinate system on an open set U and $A \cap U = \{\Phi_t = 0\}$, then at every point of U the ideal of A is generated by Φ_t; the same holds for a coordinate W_t of a holomorphic map W with injective differential. (2) If an ideal $\mathfrak{a}$ is generated by h at $f(z)$, then $f^*\mathfrak{a}$ is generated by $h \circ f$ at z; if $\mathfrak{a} = (p)$ and $\mathfrak{b} = (q)$ at x, then $\mathfrak{a}\mathfrak{b} = (pq)$ at x.*

Proof. (1) A biholomorphism has invertible differential (chain rule for $\Phi^{-1} \circ \Phi = \mathrm{id}$), so the row $d\Phi_t$ is nonzero at every point; the same holds for W, an injective endomorphism of $\mathbf{C}^n$ being surjective. Hence Φ_t (resp. W_t) has multiplicity one on U, and the ideal of its zero set is generated by it at every point of U (Lemma 174.17); A agrees with that zero set on U. (2) A germ of $f^*\mathfrak{a}$ is a holomorphic combination of the $a \circ f$, $a \in \mathfrak{a}_{f(z)}$, and $a = c\,h$ near $f(z)$ gives $a \circ f = (c \circ f)(h \circ f)$; conversely $h \circ f$ is one of the generators. Likewise $ab = (cp)(dq) = (cd)\,pq$, and pq is a product of generators. □

Theorem 174.40 (The pullback of the ideal of a member)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let π be the blowup along C, $D = \pi^{-1}(C)$, E a simple normal crossings divisor with which C has normal crossings, H a member of E and $H' = \overline{\pi^{-1}(H \setminus C)}$. At a point z of the blowup:

1. *if $C \subset H$ near $\pi(z)$, then $\pi^*\mathcal{I}_H = \mathcal{O}(-D) \cdot \mathcal{I}_{H'}$ at z;*
2. *otherwise $\pi^*\mathcal{I}_H = \mathcal{I}_{H'}$ at z.*

Statement (2) uses that H is a hypersurface: for $A = \{z_1 = z_3 = 0\}$ and $C = \{z_1 = z_2 = 0\}$, in the chart $z_1 = w_1$ the pullback of $\mathcal{I}_A$ is (w_1, w_3), while the strict transform of A misses that chart.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

If $\pi(z) \notin C$, π is biholomorphic near z, $H' = \pi^{-1}(H)$ near z and $\mathcal{O}(-D)$ contains 1 at z, so both sides agree (Theorem 174.13); in case (2) this does not occur, since C is closed and so $C \subset H$ near points off C. If $\pi(z) \in C \setminus H$, both ideals are the unit ideal at z (H is closed and $H' \subset \pi^{-1}(H)$), and $C \subset H$ near $\pi(z)$ fails. Let $x = \pi(z) \in C \cap H$. Take coordinates Φ at x in which $C = \mathbf{C}^k \times 0$ and $H = \{w_t = 0\}$ (Lemma 174.122 with $N = X$). Then $C \subset H$ near x if and only if $t \geq k$: if $t \geq k$ the coordinate w_t vanishes on $\mathbf{C}^k \times 0$; if $t < k$, the points $\Phi^{-1}(\Phi(x) + \varepsilon e_t)$, $\varepsilon \neq 0$ small, lie in C but not in H. Let $j \geq k$ and W be a monomial chart around z with $\sigma_j \circ W = \Phi \circ \pi$ (Lemma 174.11); W has injective differential (Lemma 174.121). By

Lemma 174.39, $\mathcal{I}_H = (\Phi_t)$ near x, so $\pi^*\mathcal{I}_H$ is generated by $\Phi_t \circ \pi = \sigma_j(W)_t$, which is W_t if $t < k$, W_tW_j if $t \geq k$, $t \neq j$, and W_j if $t = j$; $\mathcal{O}(-D)$ is generated by W_j (Theorem 174.13). On the chart, $H' = \{W_t = 0\}$ if $t \neq j$ and H' misses the chart if $t = j$ (Lemma 174.121); so $\mathcal{I}_{H'} = (W_t)$, resp. the unit ideal, at z (Lemma 174.39). In case (1), $t \geq k$ and $\pi^*\mathcal{I}_H = (W_jW_t) = (W_j)(W_t)$, resp. $(W_j) = (W_j) \cdot \mathcal{O}$; in case (2), $t < k \leq j$ and $\pi^*\mathcal{I}_H = (W_t) = \mathcal{I}_{H'}$. □

Lemma 174.41 (The monomial chart over a point of the centre)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let π be the blowup along C, $D = \pi^{-1}(C)$, $E = (H_1, \dots, H_s)$ a simple normal crossings divisor with which C has only normal crossings, $H_i' = \overline{\pi^{-1}(H_i \setminus C)}$, and z a point with $x = \pi(z) \in C$. There are an open $U \ni z$, a holomorphic map $W\colon U \to \mathbf{C}^n$ with injective differential at z, an index j and indices s_i ($x \in H_i$), pairwise distinct, such that:

1. *$W_j(z) = 0$ and $\mathcal{O}(-D) = (W_j)$ at every point of U; W_j vanishes identically near no point of U;*
2. *if $z \in H_i'$, then $x \in H_i$, $s_i \neq j$, $W_{s_i}(z) = 0$ and $\mathcal{I}_{H_i'} = (W_{s_i})$ at every point of U;*
3. *if u_i generates $\mathcal{I}_{H_i}$ at x ($x \in H_i$) and $a_i \in \mathbb{N}$, then near z*

$$\Big(\prod_{x \in H_i} u_i^{a_i}\Big) \circ \pi = c\, W_j^{\,e} \prod_{z \in H_i'} W_{s_i}^{a_i}, \qquad c(z) \neq 0, \qquad e = \sum_{x \in H_i,\ C \subset H_i \text{ near } x} a_i.$$

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Take coordinates Φ at x in which $C = \mathbf{C}^k \times 0$ and $H_i = \{w_{s_i} = 0\}$ for the members through x, the s_i distinct (Lemma 174.122 with $N = X$); as in the proof of Theorem 174.40, $C \subset H_i$ near x if and only if $s_i \geq k$. Let $j \geq k$ and W be a monomial chart around z with $\sigma_j \circ W = \Phi \circ \pi$ (Lemma 174.11); its differential is injective and $H_i' = \{W_{s_i} = 0\}$ on the chart if $s_i \neq j$, while H_i' misses the chart if $s_i = j$ (Lemma 174.121). Hence $\mathcal{O}(-D) = (W_j)$ and $\mathcal{I}_{H_i'} = (W_{s_i})$ on the chart (Theorem 174.13, Lemma 174.39), and $z \in H_i'$ forces $x \in H_i$ (H_i is closed), $s_i \neq j$ and $W_{s_i}(z) = 0$. The zero set of W_j in U is $D \cap U$, which has empty interior (Lemma **??**). (3): $\Phi_{s_i} \circ \pi = \sigma_j(W)_{s_i}$ is $W_{s_i}W_j$ if $k \leq s_i \neq j$, W_j if $s_i = j$ and W_{s_i} if $s_i < k$. Two generators u, φ of one principal ideal, φ vanishing identically near no point, differ by a unit: $u = a\varphi$ and $\varphi = bu$ give $(1 - ba)\varphi = 0$, so $ba = 1$ near the point (Lemma **??**). So $u_i = c_i\Phi_{s_i}$ near x with $c_i(x) \neq 0$ (Lemma 174.39), and the factors W_{s_i} with $W_{s_i}(z) \neq 0$, i.e. $z \notin H_i'$, $s_i \neq j$, are units at z. The exponent of W_j is the sum of the a_i with $s_i \geq k$, that is, with $C \subset H_i$ near x. □

Theorem 174.42 (Monomial ideals under the blowup of a centre with normal crossings)**.** *Verified upstream; not exported by this standalone library.*

Let π, D, z and $\mathfrak{a}$ be as follows.

1. *If $\mathfrak{a} = (m)$ at $\pi(z)$ and $m \circ \pi = c\, h^e q$ near z, with $\mathcal{O}(-D) = (h)$ at z, h vanishing identically near no point, $c(z) \neq 0$ and $\mu \leq e$, then $\sigma^c(\mathfrak{a}, \mu) = (h^{e-\mu}q)$ at z; for a marked ideal $(X, X, E, \mathfrak{a}, d)$ with $\mathfrak{a}$ coherent and $d \leq e$, the transformed ideal is $(h^{e-d}q)$ at z.*
2. *(Ambient case of Lemma **??**, generator form.) In the situation of Lemma 174.41, if $\mathfrak{a} = (\prod_{x \in H_i} u_i^{a_i})$ at $x = \pi(z)$ and $\mu \leq e = \sum_{x \in H_i,\ C \subset H_i \text{ near } x} a_i$, then*

$$\sigma^c(\mathfrak{a}, \mu) = \Big(W_j^{e-\mu} \prod_{z \in H_i'} W_{s_i}^{a_i}\Big) \quad \text{at } z:$$

the transform is monomial, with exponent $e - \mu$ on $D = \{W_j = 0\}$ and a_i on the strict transforms $H_i' = \{W_{s_i} = 0\}$ through z.

Proof. Verified upstream; not exported by this standalone library.

(1) $\pi^*\mathfrak{a} = (m \circ \pi) = (h^e q) = (h^\mu) \cdot (h^{e-\mu} q)$ at z, and $\mathcal{O}(-\mu D) = (h^\mu)$ (Lemma 174.1); since h^μ is a non-zero-divisor (Lemma **??**), the germs g with $h^\mu g \in \pi^*\mathfrak{a}$ are the multiples of $h^{e-\mu} q$ (Definition 174.3). For a marked ideal with $N = X$ and a coherent ideal, the transformed ideal is the controlled transform (Lemma **??**). (2) is (1) with $h = W_j$, $q = \prod_{z \in H_i'} W_{s_i}^{a_i}$ and the factorization of Lemma 174.41(3). □

Corollary 174.43 (The transform of a monomial marked ideal in adapted coordinates)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $(X, X, E, \mathfrak{a}, d)$ be a marked ideal with $\mathfrak{a}$ coherent, $E = (H_1, \dots, H_s)$ a simple normal crossings divisor, C a centre with only normal crossings with E, π the blowup along C and z a point with $x = \pi(z) \in C$. Suppose $\mathfrak{a} = (\prod_{x \in H_i} u_i^{a_i})$ at x, with u_i generating $\mathcal{I}_{H_i}$ at x (for instance the coordinates $\Phi_{\sigma_0(i)}$ of a flattening Φ at x in which the members through x are the hyperplanes $\{\Phi_{\sigma_0(i)} = 0\}$), and $d \le e = \sum_{x \in H_i,\ C \subset H_i \text{ near } x} a_i$. Let $E' = (H_1', \dots, H_s', D)$. Then there are coordinates Θ centred at z, indices $\sigma(i)$ for the members of E' through z, pairwise distinct, and exponents $a_i' = a_i$ $(i \le s)$, $a_D' = e - d$, such that every member H of E' through z is the hyperplane $\{\Theta_{\sigma(H)} = 0\}$ near z and the transformed ideal is

$$\mathfrak{a}' = \Big(\prod_{z \in H \in E'} \Theta_{\sigma(H)}^{a_H'} \Big) \quad \text{at } z.$$

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Take the monomial chart W of Lemma 174.41 and $\Theta = W - W(z)$, a coordinate system centred at z (Lemma 174.121); $\sigma(H_i') = s_i$, $\sigma(D) = j$. The coordinates W_j, W_{s_i} $(z \in H_i')$ vanish at z, so they agree with Θ_j, Θ_{s_i}. Where $\mathcal{O}(-D) = (W_j)$, the zero set of W_j is the support $\pi^{-1}(C)$ of D (Lemma 174.1); where $\mathcal{I}_{H'} = (W_{s_i})$ and H' is closed, the zero set of W_{s_i} is H' (off H' the ideal contains 1). The s_i are distinct and differ from j (Lemma 174.41(2)). The transformed ideal is $(W_j^{e-d} \prod_{z \in H_i'} W_{s_i}^{a_i})$ at z (Theorem 174.42), and the product over the members of E' through z splits into the strict transforms and the last member D, which passes through z (the indexing of E', Theorem 174.123; Definition **??**). Flat coordinates generate the ideals of their hyperplanes (Lemma 174.39), which gives the variant with $u_i = \Phi_{\sigma_0(i)}$. □

Theorem 174.44 (Monomial ideals under the blowup, along a submanifold)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $(M, N, E, \mathfrak{a}, d)$ be a marked ideal with N a closed submanifold and $\mathfrak{a}$ coherent, C an admissible centre, π the blowup along C, $N' = \overline{\pi^{-1}(N \setminus C)}$, $\mathfrak{a}'$ the transformed ideal, and $z \in N'$ with $\pi(z) \in C$. Write $\operatorname{Res}_W \mathfrak{b} = \operatorname{Res}_W \mathfrak{c}$ at a point when the same functions agree on W near the point with combinations of germs of $\mathfrak{b}$, resp. $\mathfrak{c}$.

1. *If $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N(m)$ at $\pi(z)$ and $m \circ \pi = c\, h^e q$ near z, with $\mathcal{O}(-D) = (h)$ at z, $c(z) \neq 0$ and $d \le e$, then $\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}(h^{e-d} q)$ at z.*

2. *(Lemma **??** along N', generator form.) In the monomial chart of Lemma 174.41, if $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N(\prod_{x \in H_i} u_i^{a_i})$ at $x = \pi(z)$, with u_i generating $\mathcal{I}_{H_i}$ at x, and $d \le e = \sum_{x \in H_i,\ C \subset H_i \text{ near } x} a_i$, then $\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}(W_j^{e-d} \prod_{z \in H_i'} W_{s_i}^{a_i})$ at z.*

No transversality of E to N enters: the statement concerns restrictions of ambient generators. Reading $W_j|_{N'}$ and $W_{s_i}|_{N'}$ as coordinates of N' (the monomial form of $E'|_{N'}$) needs a chart adapted to N' as well, which is not part of this statement; that form is Corollary 174.46, under transversality of E to N.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Restricted generators and restricted memberships pull back from N to N', since $\pi(N') \subset N$ (Lemma 174.12). By Lemma **??**, $\operatorname{Res}_{N'} \pi^*\mathfrak{a} = \operatorname{Res}_{N'} \mathcal{O}(-dD)\mathfrak{a}'$; near z, $\mathcal{O}(-dD) = (h_0)$ with h_0

vanishing identically on no relatively open piece of N' (Lemma **??**), and $h^d = \kappa h_0$ with $\kappa(z) \neq 0$, two generators of one principal ideal differing by a unit (Lemma 174.41; $h^d \in \mathcal{O}(-dD)$ and the germs of $\mathcal{O}(-dD)$ are multiples of h^d, Lemma **??**). So $m \circ \pi = h_0 \cdot c\kappa h^{e-d} q$. (1) A germ t of $\mathfrak{a}'$ gives $h_0 t \in \mathcal{O}(-dD)\mathfrak{a}'$, which on N' is a multiple $A\,(m\circ\pi) = h_0 \cdot Ac\kappa h^{e-d} q$; cancelling h_0 along N', $t = Ac\kappa h^{e-d} q$ on N'. Conversely $m\circ\pi$ is on N' a combination of germs of $\mathcal{O}(-dD)\mathfrak{a}'$ ($\mathfrak{a}'$ is coherent, Lemma **??**), each h_0 times a germ of $\mathfrak{a}'$; cancelling h_0, $c\kappa h^{e-d} q$ is on N' a combination of germs of $\mathfrak{a}'$, and $c\kappa$ is a unit at z. (2) is (1) with $h = W_j$, $q = \prod_{z \in H_i'} W_{s_i}^{a_i}$ and Lemma 174.41(3). □

Lemma 174.45 (The monomial chart adapted to a submanifold through the centre)**.** *Let π be the blowup along C, $C \subset N$, $E = (H_1, \dots, H_s)$ a divisor with closed members, and z a point with $x = \pi(z) \in C$. Suppose that $C \subset N$ and E admit adapted coordinates at x (Theorem 174.123): $C = \mathbf{C}^{k_C} \times 0$, $N = \mathbf{C}^{k_N} \times 0$, and the members through x are the hyperplanes $\{w_{s_i} = 0\}$, $s_i < k_N$, the s_i distinct. Then there are U, W, j and the s_i with all the properties (1)–(3) of Lemma 174.41, and moreover, with $N' = \overline{\pi^{-1}(N \setminus C)}$,*

$$N' \cap U = \{y \in U :\ W_t(y) = 0 \text{ for all } t \geq k_N\} \quad \text{if } j < k_N, \qquad N' \cap U = \emptyset \quad \text{if } j \geq k_N.$$

In particular $j < k_N$ if $z \in N'$.

Proof. Run the proof of Lemma 174.41 with the given adapted coordinates Φ at x in place of coordinates adapted to C and E alone: it uses only that $C = \mathbf{C}^{k_C} \times 0$ and that the members through x are the hyperplanes $\{w_{s_i} = 0\}$ with distinct s_i, and that the members are closed. The monomial chart W has index $j \geq k_C$ (Lemma 174.11). Since $N = L_T$ with $T = \{t : t \geq k_N\}$ in the coordinates Φ, Lemma 174.121 gives, for $y \in U$, $y \in N' \iff j \notin T$ and $W(y) \in L_T$, which is the description of $N' \cap U$; N' is a strict transform of a closed set (Lemma 174.12). □

Corollary 174.46 (Monomial ideals under the blowup, along a submanifold, in coordinates adapted to the strict transform)**.** *Let $(M, N, E, \mathfrak{a}, d)$ be a marked ideal with N a closed submanifold, $\mathfrak{a}$ coherent, $E = (H_1, \dots, H_s)$ a simple normal crossings divisor transverse to N, C an admissible centre, π the blowup along C, $N' = \overline{\pi^{-1}(N \setminus C)}$, $\mathfrak{a}'$ the transformed ideal, $E' = (H_1', \dots, H_s', D)$, and $z \in N'$ with $x = \pi(z) \in C$. Suppose $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N(\prod_{x \in H_i} u_i^{a_i})$ at x (notation of Theorem 174.44), with u_i generating $\mathcal{I}_{H_i}$ at x (for instance the coordinates $\Phi_{\sigma_0(i)}$ of a flattening Φ at x in which the members through x are the hyperplanes $\{\Phi_{\sigma_0(i)} = 0\}$; Φ need not be adapted to N), and $d \leq e = \sum_{x \in H_i,\ C \subset H_i \text{ near } x} a_i$. Then there are k, coordinates Θ centred at z, indices $\sigma(H) < k$ for the members H of E' through z, pairwise distinct, and exponents $a'_{H_i'} = a_i$, $a'_D = e - d$, such that near z*

$$N' = \{\Theta_t = 0,\ t \geq k\}, \qquad H = \{\Theta_{\sigma(H)} = 0\}\ (z \in H \in E'), \qquad \operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}\Big(\prod_{z \in H \in E'} \Theta_{\sigma(H)}^{a'_H} \Big) \text{ at } z.$$

*So $\Theta_0, \dots, \Theta_{k-1}$ restrict to coordinates of N' at z, the traces $H \cap N'$ of the members through z are coordinate hyperplanes of N', and $\mathfrak{a}'|_{N'}$ is the monomial $\prod \Theta_{\sigma(H)}^{a'_H}$ in these coordinates (the form of Lemma **??** along N').*

Proof. As $C \subset \operatorname{cosupp} \subset N$, $C \neq \emptyset$ and E is transverse to N, $C \subset N$ and E admit adapted coordinates at x (Lemma 174.122). Take the chart W of Lemma 174.45; $z \in N'$ gives $j < k_N$ and $W_t(z) = 0$ for $t \geq k_N$. Put $k = k_N$ and $\Theta = W - W(z)$, a coordinate system centred at z (Lemma 174.121). Near z, $y \in N' \iff W_t(y) = 0$ for $t \geq k \iff \Theta_t(y) = 0$ for $t \geq k$. As in Corollary 174.43, $\sigma(H_i') = s_i$ and $\sigma(D) = j$ are distinct, the members of E' through z are the hyperplanes $\{\Theta_{\sigma(H)} = 0\}$, and the coordinates W_j, W_{s_i} ($z \in H_i'$) agree with Θ_j, Θ_{s_i}; moreover $s_i < k$ ($x \in H_i$) and $j < k$. By Theorem 174.44(2), $\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}(W_j^{e-d} \prod_{z \in H_i'} W_{s_i}^{a_i})$ at z; this generator agrees near z with $\prod_{z \in H \in E'} \Theta_{\sigma(H)}^{a'_H}$ (the product over the members of E' through z splits into the strict transforms and D, which passes through z; Definition **??**, Theorem 174.123),

and whether a function is on N' near z a multiple of a generator depends only on the germ of the generator. Flat coordinates generate the ideals of their hyperplanes (Lemma 174.39), which gives the variant with $u_i = \Phi_{\sigma_0(i)}$. □

Lemma 174.47 (Monomial ideals along N' off the centre)**.** *Let $(M, N, E, \mathfrak{a}, d)$ be a marked ideal with N closed and $\mathfrak{a}$ coherent, $E = (H_1, \dots, H_s)$ a simple normal crossings divisor, π the blowup along C, N', $\mathfrak{a}'$, $E' = (H'_1, \dots, H'_s, D)$ as in Corollary 174.46, and z a point with $x = \pi(z) \notin C$.*

1. *$\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'} \pi^*\mathfrak{a}$ at z.*
2. *Let $n > 0$, and let Φ be coordinates centred at x in which $N = \{\Phi_t = 0,\ t \geq k\}$ and the members through x are the hyperplanes $\{\Phi_{\sigma_0(i)} = 0\}$, $\sigma_0(i) < k$ pairwise distinct, and suppose $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N(\prod_{x \in H_i} \Phi_{\sigma_0(i)}^{a_i})$ at x. Then $\Theta = \Phi \circ \pi$ are coordinates centred at z in which, near z, $N' = \{\Theta_t = 0,\ t \geq k\}$, the members of E' through z are the hyperplanes $\{\Theta_{\sigma_0(i)} = 0\}$ of the strict transforms H'_i, $x \in H_i$ (D does not pass through z), and $\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}(\prod_{z \in H'_i} \Theta_{\sigma_0(i)}^{a_i})$ at z.*

With Corollary 174.46 this gives the form of Lemma **??** *along N' at every point of N'.*

Proof. (1) Near z the ideal $\pi^*\mathcal{I}_C$ contains 1, and so does its d-th power; the transformed ideal is the colon along N' of $\pi^*\mathfrak{a}$ by this power (Definition **??**), so its germs are, on N' near z, combinations of germs of $\pi^*\mathfrak{a}$ (multiply by 1), and every germ t of $\pi^*\mathfrak{a}$ lies in it ($ht = h \cdot t$). (2) Off $\pi^{-1}(C)$ the blowup is a local biholomorphism, so $\Phi \circ \pi$, read in charts, is a coordinate system centred at z (Lemmas 174.30 and **??**); its domain lies off $\pi^{-1}(C)$. There $N' = \pi^{-1}(N)$ and $H'_i = \pi^{-1}(H_i)$, as N and the members are closed (Lemmas 174.12 and 174.31; the indexing of E', Theorem 174.123), which gives the coordinate descriptions, and the members of E' through z are the H'_i with $x \in H_i$. By (1) and the pullback of restricted generators along π (Theorem 174.44, $\pi(N') \subset N$), $\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}(m \circ \pi)$ with $m = \prod_{x \in H_i} \Phi_{\sigma_0(i)}^{a_i}$, and $m \circ \pi = \prod_{z \in H'_i} \Theta_{\sigma_0(i)}^{a_i}$ near z. □

The boundary count. Let $E = (H_1, \dots, H_s)$ be a simple normal crossings divisor (Definition **??**). For a set S of indices let $H_S = \bigcap_{i \in S} H_i$, and let $s(x) = \#\{i : x \in H_i\}$ be the number of members through x. Along a blowup π the count $s'(z)$ is that of the strict transforms $H'_i = \overline{\pi^{-1}(H_i \setminus C)}$ of the members; the new exceptional divisor is not counted (Włodarczyk identifies the strict transforms of E with E). Step 1a of the algorithm blows up maximal intersections H_S, which lowers s. Włodarczyk intersects irreducible components of the members; here the members themselves are intersected, and the maximality of s becomes the hypothesis that no member outside S meets H_S.

Lemma 174.48 (Intersections of members)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E be a simple normal crossings divisor on X and S a set of indices of members. Then H_S is closed and has only normal crossings with E. If $n > 0$, H_S is a closed submanifold. If E is the divisor of a marked ideal $\underline{\mathcal{I}}$ and $H_S \subset \operatorname{cosupp} \underline{\mathcal{I}}$, then H_S is an admissible centre for $\underline{\mathcal{I}}$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

H_S is an intersection of closed sets. Let $x \in H_S$ and take coordinates at x in which each member H_i through x is L_{T_i} (Definition **??**). Every member with index in S passes through x, so near x $H_S = \bigcap_{i \in S} L_{T_i} = L_{\bigcup_{i \in S} T_i}$ in the same coordinates; together with the L_{T_i} this is the normal crossings of H_S with E. Let $n > 0$. The divisor E is transverse to $N = X$ (Corollary 174.124), so Lemma 174.122 gives coordinates at x in which $H_S = \mathbf{C}^k \times 0$ near x; replacing k by $\min(k, n)$ does not change this plane. Hence H_S is a submanifold of dimension $\min(k, n)$ at every point near x (Theorem **??**). The last claim is Definition **??**. □

Lemma 174.49 (Intersections of members with a transverse submanifold)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E be a simple normal crossings divisor on X transverse to a closed set N, and S a set of indices of members. Then $N \cap H_S$ is closed and has only normal crossings with E. If $n > 0$, $N \cap H_S$ is a closed submanifold. If N and E are those of a marked ideal $\underline{\mathcal{I}}$ (for instance a well formed one) and $N \cap H_S \subset \operatorname{cosupp} \underline{\mathcal{I}}$, then $N \cap H_S$ is an admissible centre for $\underline{\mathcal{I}}$. (For $N = X$ this is Lemma 174.48; along N it gives the admissibility of intersections of the traces $H \cap N$ of members, as in the monomial case along N.)

Proof. Local and upstream declarations; individual reference status is recorded in the source.

$N \cap H_S$ is an intersection of closed sets. Let $x \in N \cap H_S$ and take coordinates at x in which $N = L_{S_0}$ and each member H_i through x is L_{T_i}, T_i a single index not in S_0 (transversality, Definition **??**). Every member with index in S passes through x, so near x $N \cap H_S = L_{S_0} \cap \bigcap_{i \in S} L_{T_i} = L_{S_0 \cup \bigcup_{i \in S} T_i}$ in the same coordinates; together with the L_{T_i} this is the normal crossings of $N \cap H_S$ with E. Let $n > 0$. As in Lemma 174.48, E is transverse to X (Corollary 174.124), so Lemma 174.122 gives coordinates at x in which $N \cap H_S = \mathbf{C}^k \times 0$ near x, and $N \cap H_S$ is a submanifold of dimension $\min(k, n)$ near x (Theorem **??**). The last claim is Definition **??**; a well formed marked ideal has N closed and E simple normal crossings and transverse to N (Definition **??**). □

Lemma 174.50 (The boundary count)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let E be a finite sequence of closed subsets of X. (1) s is upper semicontinuous: $s(y) \leq s(x)$ for all y near x. (2) For a continuous map $\pi \colon Y \to X$ and any C, $s'(z) \leq s(\pi(z))$ for every $z \in Y$, where $s'(z)$ counts the i with $z \in \overline{\pi^{-1}(H_i \setminus C)}$. (3) Along a tower of blowups T with projection p (either transform of Definitions **??** and **??**), a point z of the last space lies on at most $s(p(z))$ of the first $\#E$ members of the transformed divisor, which are the iterated strict transforms of the original members.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The complements of the finitely many members not through x are open, so near x every member through y passes through x. (2) $\overline{\pi^{-1}(H_i \setminus C)} \subset \pi^{-1}(H_i)$, the latter being closed (Theorem 174.12); so every i counted at z is counted at $\pi(z)$. The count along the sequence $(H'_1, \ldots, H'_s)$ is re-indexed by the length of the sequence. (3) Induction on the tower. At each stage the divisor is $E_{i+1} = (E_i$ strict-transformed) followed by the new exceptional divisor and has $\#E + i + 1$ members (Lemma **??**, Lemma **??**), so its first $\#E$ members are the strict transforms of the first $\#E$ members of E_i. These are closed (the original members by hypothesis, later ones as closures), and (2) at each stage composes with the induction hypothesis. □

Theorem 174.51 (Blowing up a maximal intersection separates the members)**.** *Verified upstream; not exported by this standalone library.*

Let E be a simple normal crossings divisor on X, S a set of indices such that no member H_i with $i \notin S$ meets $C = H_S$, and π the blowup along C. Then $s'(z) < \#S$ for every $z \in \pi^{-1}(C)$. (Włodarczyk, Step 1aa: "by blowing up $C = H^s_\alpha$ we separate divisors contributing to H^s_α, thus creating new points all with $s(x) < s$".) No hypothesis $S \neq \emptyset$ is needed: $H_\emptyset = X$, and a blowup centre has empty interior.

Local form. Let C be any blowup centre with only normal crossings with E, and $z \in \pi^{-1}(C)$. If $C \supset H_S$ near $\pi(z)$, then z does not lie on all the H'_i, $i \in S$. In particular, if near $x = \pi(z)$ the centre contains the intersection of the members through x, then $s'(z) < s(x)$; no global maximality is needed. This is the form used in Step 1aa: near each of its points the centre H^s_α is the intersection of the s members through that point.

Proof. Verified upstream; not exported by this standalone library.

Let $x = \pi(z) \in C$. If $z \in H_i'$ then $x \in H_i$ (Lemma 174.50), so $i \in S$; hence $s'(z) \leq \#S$, and equality means $z \in H_i'$ for every $i \in S$. Suppose this. Then $n > 0$ (Lemma 174.122), and C has only normal crossings with E (Lemma 174.48). Take coordinates Φ at x in which $C = \mathbf{C}^k \times 0$ and $H_i = \{w_{s_i} = 0\}$ for the members through x (Lemma 174.122 with $N = X$), and a monomial chart $j \geq k$, W around z with $\sigma_j \circ W = \Phi \circ \pi$ (Lemma 174.11). By Lemma 174.121, $z \in H_i'$ gives $s_i \neq j$ and $W_{s_i}(z) = 0$ for $i \in S$; and $W_j(z) = 0$ since $\pi(z) \in C$. For small $\varepsilon \neq 0$ the point $w_\varepsilon = \sigma_j(W(z) + \varepsilon e_j)$ lies in the target of Φ. For $i \in S$ its coordinate s_i is $W_{s_i}(z)$ or $W_{s_i}(z)\,\varepsilon$, hence 0, so $\Phi^{-1}(w_\varepsilon) \in H_i$ for all $i \in S$, that is $\Phi^{-1}(w_\varepsilon) \in C$. But the j-th coordinate of w_ε is $\varepsilon \neq 0$ with $j \geq k$, so $w_\varepsilon \notin \mathbf{C}^k \times 0$: a contradiction.

Local form: the same argument, with C adapted directly (it has normal crossings with E) and the members of S passing through x because z lies on their strict transforms. As $\varepsilon \to 0$, $\Phi^{-1}(w_\varepsilon) \to \Phi^{-1}(\sigma_j(W(z))) = x$, so for small ε the point lies in the neighbourhood of x where $H_S \subset C$. For the count, let S be the set of members through x: every H_i' through z has $i \in S$, and not all of them pass through z. □

Theorem 174.52 (Blowing up an intersection of members along a transverse submanifold separates them on the strict transform)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E be a simple normal crossings divisor on X transverse to a set N, $C \subset N$ the centre of the blowup π, with only normal crossings with E, $N' = \overline{\pi^{-1}(N \setminus C)}$ the strict transform, and S a set of indices of members. Let $z \in N'$ with $x = \pi(z) \in C$.

1. *If $C \supset N \cap H_S$ near x, then z does not lie on all the H_i', $i \in S$.*
2. *If near x the centre contains $N \cap \bigcap_{x \in H_i} H_i$, then $s'(z) < s(x)$.*
3. *If $C = N \cap H_S$ and no member H_i with $i \notin S$ meets C, then $s'(z) < \#S$.*

*This is Theorem 174.51 inside N: Step 1aa of Włodarczyk's algorithm for a marked ideal along N, and the drop of the monomial invariant along N (Lemma **??**(2)), where the centre is $N \cap H_I$ and the cosupport lies in N'. For $N = X$ it is the local form of Theorem 174.51; for $N \neq X$ the hypothesis $C \supset H_S$ of that theorem fails, and the point $z \in N'$ is what replaces it.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Suppose $z \in H_i'$ for every $i \in S$; then $x \in H_i$ for $i \in S$ (Lemma 174.50), and $n > 0$ since C is a nonempty centre. Take coordinates Φ at x adapted to $C \subset N$ and E (Lemma 174.122): $C = \mathbf{C}^{k_C} \times 0$, $N = \mathbf{C}^{k_N} \times 0$, $H_i = \{w_{s_i} = 0\}$ for the members through x; and a monomial chart $j \geq k_C$, W around z with $\sigma_j \circ W = \Phi \circ \pi$ (Lemma 174.11). By Lemma 174.121: $z \in N'$ gives $j < k_N$ and $W_t(z) = 0$ for $t \geq k_N$; $z \in H_i'$ gives $s_i \neq j$ and $W_{s_i}(z) = 0$; and $W_j(z) = 0$ since $x \in C$. For small $\varepsilon \neq 0$ put $w_\varepsilon = \sigma_j(W(z) + \varepsilon e_j)$. Its coordinate $t \geq k_N$ is $W_t(z)(W_j(z) + \varepsilon) = 0$ (as $t \neq j$), so $\Phi^{-1}(w_\varepsilon) \in N$; its coordinate s_i, $i \in S$, is $W_{s_i}(z)$ or $W_{s_i}(z)\varepsilon$, hence 0, so $\Phi^{-1}(w_\varepsilon) \in H_S$. As $\varepsilon \to 0$, $\Phi^{-1}(w_\varepsilon) \to \Phi^{-1}(\sigma_j(W(z))) = x$, so for small ε the point lies in the neighbourhood of x where $N \cap H_S \subset C$. But its j-th coordinate is $\varepsilon \neq 0$ with $j \geq k_C$, so it is not in $\mathbf{C}^{k_C} \times 0$: a contradiction. (2) Let S be the set of members through x. Every H_i' through z has $i \in S$ (Lemma 174.50), and by (1) not all of them pass through z. (3) C is closed with normal crossings with E (Lemma 174.49); the members through $x \in C$ are those of S, so near x the centre is $N \cap \bigcap_{x \in H_i} H_i$, and (2) gives $s'(z) < s(x) \leq \#S$. □

Lemma 174.53 (The members outside an intersection, restricted to a transverse submanifold)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E be a simple normal crossings divisor on X transverse to N, S a set of indices of members, and E_S the subsequence of the members with index not in S. Then E_S is a simple normal crossings divisor, transverse to $N \cap H_S$ (more generally, any finite sequence of members with indices not in

*S is transverse to $N \cap H_S$), and every member of E missing from E_S contains $N \cap H_S$. Hence, if $(M, N, E, \mathfrak{a}, d)$ is well formed, $n > 0$, and $\mathfrak{b}$ is coherent, then $(M, N \cap H_S, E_S, \mathfrak{b}, \mu)$ is well formed. This is the restriction of Step 1a of Włodarczyk's algorithm inside N: the marked ideal on $S = N \cap H^s_\alpha$ with the divisor without the s members containing it (Lemmas **??** and **??**).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

E_S is a subsequence of E (Lemma 174.126(2)). A member of E missing from E_S is H_i with $i \in S$, which contains H_S. Transversality: let $x \in N \cap H_S$ and take coordinates at x in which $N = L_{S_0}$ and every member H_i through x is L_{T_i}, T_i a single index not in S_0 (transversality of E to N, Definition **??**). Every member with index in S passes through x, so near x $N \cap H_S = L_{S_0 \cup \bigcup_{i \in S} T_i}$ (Lemma 174.49). Let H_j, $j \notin S$, pass through x. Its index T_j is not in S_0; and $T_j \neq T_i$ for $i \in S$: otherwise $H_j = L_{T_j} = L_{T_i} = H_i$ near x, and two members of a simple normal crossings divisor through x that agree near x are the same member (Lemma 174.122, first paragraph of the proof), so $j = i \in S$. Hence T_j is disjoint from $S_0 \cup \bigcup_{i \in S} T_i$, which is the transversality of E_S to $N \cap H_S$ at x. For the marked ideal, $N \cap H_S$ is a closed submanifold (Lemma 174.49, N closed and $n > 0$), and the conditions of Definition **??** are those just proved. □

Lemma 174.54 (Strict transforms of intersections of members)**.** *Let π be the blowup along C, $N \supset C$ closed, E a simple normal crossings divisor transverse to N with which C has only normal crossings, and S a set of indices of members. Then $\overline{\pi^{-1}((N \cap H_S) \setminus C)} = N' \cap \bigcap_{i \in S} H'_i$.*

Proof. As for one member (Lemma 174.117). $\subset$: strict transforms are monotone. $\supset$: let $z \in N' \cap \bigcap_{i \in S} H'_i$, so $\pi(z) \in N \cap H_S$. If $\pi(z) \notin C$, $z \in \pi^{-1}((N \cap H_S) \setminus C)$. If $\pi(z) \in C$, in coordinates adapted to $C \subset N$ and E (Lemma 174.122) $N = L_{T_N}$ and $H_i = L_{\{s_i\}}$, so $N \cap H_S = L_{T_N \cup \{s_i : i \in S\}}$. In a monomial chart j through z (Lemma 174.11), Lemma 174.121 turns $z \in N'$ into $j \notin T_N$, $W(z) \in L_{T_N}$, and $z \in H'_i$ into $j \neq s_i$, $W(z) \in L_{\{s_i\}}$; so $j \notin T_N \cup \{s_i : i \in S\}$ and $W(z) \in L_{T_N \cup \{s_i : i \in S\}}$, that is $z \in (N \cap H_S)'$ by the same lemma. □

Lemma 174.55 (Intersections of members along multiple test blow-ups)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let T be a tower over X, and write A_T for the iterated strict transform of $A \subset X$ (Definition **??**). Then $A \mapsto A_T$ is monotone; for A closed, A_T is closed and lies over A, so the strict transforms of disjoint closed sets are disjoint. For a marked ideal $(M, N, E, \mathfrak{a}, d)$, the first $\#E$ members of the transformed divisor E_T are the strict transforms $(H_i)_T$ of the members of E, in order (the exceptional divisors follow them). If the marked ideal is well formed and T is a multiple test blow-up, then for every set S of indices*

$$(N \cap H_S)_T = N_T \cap \bigcap_{i \in S} (H_i)_T.$$

In particular a point $z \in N_T$ lies on the strict transform of $N \cap H_{S(z)}$, $S(z)$ the set of original members whose strict transforms pass through z. This justifies, inside N, the identification of the strict transforms of E with E in Step 1a of Włodarczyk's algorithm: a point with $s(z) = s$ (Lemma 174.50) lies on the strict transform of the intersection it lies on at the start of Step 1.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Monotonicity, closedness and $A_T \subset T^{-1}(A)$ by induction on the length, each step being a closure of a preimage (Definition **??**). The members: E after one more blowup π along C is $E'_T = ((H_1)'_T, \dots, (H_{s+r})'_T, \pi^{-1}(C))$ (Definition **??**), whose first $\#E$ entries are the strict transforms of the first $\#E$ entries of E_T. The intersection by induction on the length: for the empty tower it is the definition of H_S. If T' is T followed by the blowup along C, the transform along T is well formed (Lemma **??**), so N_T is closed and E_T is simple normal crossings and transverse to N_T, and $C \subset N_T$ has only normal crossings with E_T (admissibility, Definition **??**). By induction $(N \cap H_S)_T = N_T \cap \bigcap_{i \in S} (H_i)_T$, an intersection of N_T with members of E_T, and

Lemma 174.54 applied to N_T, E_T, C gives $(N \cap H_S)_{T'} = N'_T \cap \bigcap_{i \in S} (H_i)'_T$, with $N_{T'} = N'_T$ and $(H_i)_{T'} = (H_i)'_T$. □

Lemma 174.56 (Localising the centres of a tower over an open set)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let T be a tower over X and $U \subset X$ open, and write $T|_U$ for its restriction (Lemma **??**). Then $(A \cup B)_T = A_T \cup B_T$. (1) If the centres of T lie in the strict transforms of $A \cup B$, with B closed and disjoint from U, then the centres of $T|_U$ lie in the strict transforms of $A \cap U$. (2) If every admissible centre C of every multiple test blow-up T of a marked ideal satisfies $C \cap T^{-1}(U) \subset A_T$, then the centres of the restriction to U of every multiple test blow-up lie in the strict transforms of $A \cap U$. In Step 1a inside N the centres lie in the strict transforms of the union of the intersections $S_\alpha = N \cap H^s_\alpha$, whose traces on the cosupport are disjoint; (1) with $A = S_\alpha$, $B = \bigcup_{\alpha' \neq \alpha} (S_{\alpha'} \cap \mathrm{cosupp})$ and $U = X \setminus B \supset S_\alpha \cap \mathrm{cosupp}$ puts the restricted tower in the setting of the restricted coefficient ideal with E_{S_α} (Corollary **??**) on U. In Step 1b, (2) applies to the strict transforms of a hypersurface of maximal contact $N \cap V(u) \cap U$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

The union: by induction on the length, as $\overline{\pi^{-1}((A \cup B) \setminus C)} = \overline{\pi^{-1}(A \setminus C)} \cup \overline{\pi^{-1}(B \setminus C)}$. The centres of $T|_U$ are the traces of those of T on the open preimages of U, and the strict transform of $A \cap U$ along $T|_U$ is the trace of A_T (Definition **??**). (1) A point p of a centre over U lies in $A_T \cup B_T$; it does not lie in B_T, which lies over B (Lemma 174.55), while p lies over U. (2) Each centre of T is an admissible centre of a multiple test blow-up (Definition **??**). □

Lemma 174.57 (The exceptional divisors are transverse to the strict transform of a submanifold containing the centres)**.** *Unresolved historical declaration reference; not counted as verified.*

Includes a preserved unresolved historical reference; compiled declarations are identified separately.

Let $(M, N, E, \mathfrak{a}, d)$ be well formed, W a closed submanifold and T a multiple test blow-up whose centres lie in the strict transforms of W. Then W_T is a closed submanifold, and the members of E_T after the first $\#E$ (the exceptional divisors of T) form a simple normal crossings divisor transverse to W_T; hence $(M_T, W_T, E_T \setminus E, \mathfrak{b}, \mu)$ is well formed for every coherent $\mathfrak{b}$. If E is transverse to W, all of E_T is transverse to W_T. This is Włodarczyk's observation that the new divisors, born from centres in the hypersurface of maximal contact, are transversal to it: with $W = N \cap V(u) \cap U$ and Lemma 174.56(2) it makes Step 1b's marked ideal on the strict transform of the hypersurface of maximal contact, with the old members dropped in Step 1a, well formed.

Proof. Includes a preserved unresolved historical reference; compiled declarations are identified separately.

Induction on the length. For the empty tower there are no exceptional members ($\emptyset$ is transverse to every closed submanifold, Lemma **??**). Let T' be T followed by the blowup π along $C \subset W_T$. The exceptional members of T' are the strict transforms of those of T followed by $\pi^{-1}(C)$ (Definition **??**). C has only normal crossings with E_T (admissibility), hence with its sub-sequence of exceptional members (Lemma 174.126), which is simple normal crossings and, by induction, transverse to the closed submanifold W_T. Theorem 174.123 (with $W_T \supset C$ in place of N) shows that the transformed sequence is simple normal crossings and transverse to $W'_T = W_{T'}$, a closed submanifold by Theorem 174.12. For all of E_T the same induction runs with E_T, simple normal crossings since the transform along T is well formed (Lemma **??**). □

Lemma 174.58 (Normal crossings and transversality are local)**.** *Adapted coordinates for a family at x (Definition **??**) are adapted coordinates for every family agreeing with it near x. Hence if near each of its points a set R agrees with a set W transverse to E (resp. having only normal crossings with E), then R is transverse to E (resp. has only normal crossings with E). If O is open and the trace of W on O is transverse, in O, to the traces of the members of E, then $W \cap O$ is transverse to E.*

Proof. For the first sentence apply Lemma **??** to the identity, whose differential is bijective, on an open set on which the families agree. The second follows at each point $x \in R$ from the coordinates of W at x. For the last, adapted coordinates at $z \in W \cap O$ on O are adapted coordinates on the ambient manifold for W and the members (Lemma **??**, open inclusions), and $W \cap O$ agrees with W near z. □

Lemma 174.59 (The exceptional divisors over an open set)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $(M, N, E, \mathfrak{a}, d)$ be well formed, $U \subset M$ open and A a set whose trace on U is a closed submanifold of U, and assume that every admissible centre C of every multiple test blow-up T satisfies $C \cap T^{-1}(U) \subset A_T$ (for a tangent direction u along N on U and $A = N \cap V(u) \cap U$ this is Theorem **??**). Then along every multiple test blow-up T the exceptional members of E_T are transverse to $A_T \cap T^{-1}(U)$, and so to every set R which near each of its points lies over U and agrees with A_T.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Restrict to U (Lemma **??**): the restricted tower is a multiple test blow-up of the well-formed restriction (Lemma **??**), its centres lie in the strict transforms of $A \cap U$ (Lemma 174.56(2)), so its exceptional members are transverse to the strict transform of $A \cap U$ (Lemma 174.57). The transform along the restricted tower is the restriction of the transform, and the strict transform of $A \cap U$ the trace of A_T; Lemma 174.58 returns to M_T, and gives the statement for R. □

Corollary 174.60 (Step 1ba's centre inside N is admissible)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $(M, N, E, \mathfrak{a}, d)$ have N a closed submanifold and $\mathfrak{a}$ coherent, be of maximal order along N, $d \geq 1$. If the codimension-one part along N of its cosupport (Definition 174.36) is transverse to E, it is an admissible centre. If $(M, N, E, \mathfrak{a}, d)$ is well formed, of maximal order along N, $d \geq 1$, then at every stage T of a multiple test blow-up the codimension-one part along N_T of the cosupport of the transform is transverse to the exceptional members of E_T (the old ones dropped in Step 1a); hence, if the transform is of maximal order along N_T, it is an admissible centre of $(M_T, N_T, E_T \setminus E, \mathfrak{a}_T, d)$. Indeed near each of its points the codimension-one part is the strict transform of the hypersurface of maximal contact $N \cap V(u) \cap U$ (Theorem 174.37 for the tangent direction of the transform, Theorem **??**), to which the exceptional members are transverse (Lemma 174.59). This is Włodarczyk's statement in Step 1ba (2009, p. 57) that the divisors are transversal to the codimension-one components.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

The codimension-one part is a closed submanifold contained in the cosupport (Theorem 174.37), and transversality implies normal crossings (Definition **??**). At a stage T: let x lie in the codimension-one part R_T. Then $T(x)$ lies in the cosupport of the original marked ideal (Lemma 174.32), where by maximal order there is a tangent direction u along N on an open $U \ni T(x)$ (Lemma **??**). Theorem **??** gives, on an open $O \ni x$ over U, a function w controlled on N_T with $N_T \cap V(w) \cap O = A_T \cap O$, $A = N \cap V(u) \cap U$, and with every admissible centre over U inside A_T. By Theorem 174.37 (its form for w controlled on N_T), $R_T = N_T \cap V(w)$ near x, so R_T agrees near x with $A_T \cap T^{-1}(U)$, to which the exceptional members are transverse (Lemma 174.59; $A \cap U$ is a closed submanifold of U, Lemma 174.20). Lemma 174.58 concludes. □

Canonicity of the codimension-one part and of the boundary count. The invariant of the maximal-order step (Proposition **??**) gives the points of the codimension-one part of the support the value $(0, \infty)$ and counts the original members through a point; its properties (3)–(7) (Definition **??**) need these two ingredients to commute with restriction to open sets and with

biholomorphisms, to be unchanged off the centre of a blowup and by blowups along $\emptyset$, and to depend only on the cosupports of equivalent marked ideals.

Lemma 174.61 (Regular points under biholomorphisms and local sections)**.** *Let $\Phi\colon M' \to M$ be a biholomorphism. Then x is a regular point of $\Phi^{-1}(A)$ of dimension k if and only if $\Phi(x)$ is a regular point of A of dimension k. More generally, let $\pi\colon Y \to X$ be holomorphic, $W \subset X$ open, $s\colon W \to Y$ holomorphic with $\pi \circ s = \mathrm{id}_W$, and $z \in Y$ with $\pi(z) \in W$ and $s(\pi(z)) = z$ (for instance a blowup along C, $W = X \setminus C$ and $\pi(z) \notin C$, Lemma 174.30). Then z is a regular point of $\pi^{-1}(A)$ of dimension k if and only if $\pi(z)$ is a regular point of A of dimension k.*

Proof. A biholomorphism has bijective differential everywhere (Lemma **??**), so Lemma **??** applied to Φ and A, and to Φ^{-1} and $\Phi^{-1}(A)$, gives the two directions. In the second statement, the chain rule for $\pi \circ s = \mathrm{id}_W$ at $x = \pi(z)$ gives $d\pi_z \circ ds_x = \mathrm{id}$ ($s(x) = z$); in equal finite dimensions both differentials are then bijective. Lemma **??** for π at z gives one direction; for s at x, with $s^{-1}(\pi^{-1}(A)) = A \cap W$, it gives that x is a regular point of $A \cap W$ in W, that is of A (Remark 443.2). □

Proposition 174.62 (The codimension-one part along N is canonical)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Write $R_N(\mathfrak{a}, \mu)$ for the codimension-one part along N of $\mathrm{cosupp}(\mathfrak{a}|_N, \mu)$ (Definition 174.36), and $R(\underline{\mathcal{I}}) = R_N(\mathfrak{a}, d)$ for $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$. Let $\mathfrak{a}$ consist of holomorphic germs.

1. *(Locality) Whether $x \in R_N(\mathfrak{a}, \mu)$ depends only on the germs at x of N and of $\mathrm{cosupp}(\mathfrak{a}|_N, \mu)$; equal cosupports along the same N give equal codimension-one parts.*
2. *(Biholomorphisms) For a biholomorphism $\Phi\colon M' \to M$, $\mathrm{cosupp}((\Phi^*\mathfrak{a})|_{\Phi^{-1}N}, \mu) = \Phi^{-1}\,\mathrm{cosupp}(\mathfrak{a}|_N, \mu)$ and $R_{\Phi^{-1}N}(\Phi^*\mathfrak{a}, \mu) = \Phi^{-1}R_N(\mathfrak{a}, \mu)$.*
3. *(Open sets) For $U \subset M$ open, $R_{N\cap U}(\mathfrak{a}|_U, \mu) = R_N(\mathfrak{a}, \mu) \cap U$.*
4. *(Transported towers) If $\mathfrak{a}$ is coherent, T a tower over M and Φ^*T its transport with last-stage biholomorphism Φ_T (Definition **??**), then $R(\Phi^*\underline{\mathcal{I}}_{\Phi^*T}) = \Phi_T^{-1}R(\underline{\mathcal{I}}_T)$.*
5. *(Restricted towers) If N is a closed submanifold, $\mathfrak{a}$ coherent, T a multiple test blow-up of $\underline{\mathcal{I}}$ and $T|_U$ its restriction (Lemma **??**), then $R((\underline{\mathcal{I}}|_U)_{T|_U})$ is the trace of $R(\underline{\mathcal{I}}_T)$ on the open preimage of U.*
6. *(Representatives) If $\underline{\mathcal{I}}$ and $\underline{\mathcal{J}}$ are restricted-equivalent (Definition **??**; N closed, coherent ideals), then $R(\underline{\mathcal{I}}_T) = R(\underline{\mathcal{J}}_T)$ for every tower T; if they are equivalent (Definition **??**) with the same N, then $R(\underline{\mathcal{I}}_T) = R(\underline{\mathcal{J}}_T)$ for every multiple test blow-up T of $\underline{\mathcal{I}}$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Regularity of a given dimension at x depends only on the germ of the set at x (Remark 443.1). (2) The cosupport identity is that of Lemma **??** for the marked ideal $(M, N, \emptyset, \mathfrak{a}, \mu)$; then apply Lemma 174.61 to N and to the cosupport. (3) The cosupport restricts (Lemma **??**), and regular points of $A \cap U$ in U are those of A in U (Remark 443.2). (4) For the empty tower this is (2); from the first blowup on the two transforms are equal and Φ_T is the identity (Lemma **??**). (5) Along a multiple test blow-up N_T stays a closed submanifold and $\mathfrak{a}_T$ coherent, and the transform of $\underline{\mathcal{I}}|_U$ along $T|_U$ is the restriction of $\underline{\mathcal{I}}_T$ to the open preimage of U (Lemma **??**); apply (3). (6) Restricted-equivalent marked ideals have the same N and cosupport, and equal transforms from the first blowup on (Lemmas **??** and **??**); equivalent marked ideals have equal cosupports along their multiple test blow-ups (Definition **??**), and N_T is the iterated strict transform of N, the same for both (Lemma **??**); apply (1). □

Lemma 174.63 (The codimension-one part off the centre)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N closed and $\mathfrak{a}$ consisting of holomorphic germs, π the blowup along C and $\underline{\mathcal{I}}'$ the transform. If $\pi(z) \notin C$, then $z \in R(\underline{\mathcal{I}}')$ if and only if $\pi(z) \in R(\underline{\mathcal{I}})$. In particular a blowup along $\emptyset$ gives $R(\underline{\mathcal{I}}') = \pi^{-1}R(\underline{\mathcal{I}})$, and along a multiple test blow-up T of $\underline{\mathcal{I}}$ (N a closed submanifold, $\mathfrak{a}$ coherent) followed by the blowup π along C, $z \in R(\underline{\mathcal{I}}_{T\pi})$ if and only if $\pi(z) \in R(\underline{\mathcal{I}}_T)$ whenever $\pi(z) \notin C$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Near z (on the open set $\pi^{-1}(M \setminus C)$) the strict transform N' is $\pi^{-1}(N)$ and the cosupport of $\underline{\mathcal{I}}'$ is the preimage of that of $\underline{\mathcal{I}}$ (Theorem 174.31). By Proposition 174.62(1), $z \in R(\underline{\mathcal{I}}')$ if and only if for some k the sets $\pi^{-1}(N)$ and $\pi^{-1}(\operatorname{cosupp}\underline{\mathcal{I}})$ are regular at z of dimensions $k+1$ and k. The section of π over $M \setminus C$ through z (Lemma 174.30) and Lemma 174.61 turn this into the same condition for N and $\operatorname{cosupp}\underline{\mathcal{I}}$ at $\pi(z)$. The tower form applies this to $\underline{\mathcal{I}}_T$, whose N_T is closed and ideal coherent (Lemma **??**). □

Lemma 174.64 (The boundary count is canonical)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

For a map f and a sequence E of sets, the number of members of $(f^{-1}H)_{H\in E}$ through z is the number of members of E through $f(z)$. Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have closed members, and write $s_T(z)$ for the number of the first $\#E$ members of E_T (the strict transforms of the members of E) through z (Definition 174.50). Then:

1. *for a blowup π along a closed C and $\pi(z) \notin C$, the number of strict transforms of members through z is the number of members through $\pi(z)$; in particular $s_{T\pi}(z) = s_T(\pi(z))$ for a tower T followed by the blowup π along C;*
2. *if $\mathfrak{a}$ is coherent, along the transport Φ^*T of a tower by a biholomorphism, $s_{\Phi^*T} = s_T \circ \Phi_T$;*
3. *if N is a closed submanifold and $\mathfrak{a}$ coherent, along the restriction $T|_U$ of a multiple test blow-up, $s_{T|_U}$ is the restriction of s_T to the open preimage of U.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

The first statement re-indexes the members. (1) Off the centre the strict transform of a closed H is $\pi^{-1}(H)$ near z (Theorem 174.31); the first $\#E$ members of the transform of $\underline{\mathcal{I}}_T$ are the strict transforms of the first $\#E$ members of E_T, which are closed (Lemma 174.50). (2) For the empty tower the members are $\Phi^{-1}H$; from the first blowup on the transforms are equal and Φ_T is the identity (Lemma **??**). (3) The transform along $T|_U$ is the restriction of the transform along T (Lemma **??**), whose members are the traces of the members. □

The codimension-one part along multiple test blow-ups. Step 1ba of the maximal-order step (Proposition **??**) blows up the codimension-one part of the support along N; Step 1bb evaluates the invariant of the restriction of the coefficient ideal to a hypersurface of maximal contact, and the inductive invariant (Definition **??**) is only defined for marked ideals whose ideal vanishes identically on no nonempty open subset of their submanifold. Włodarczyk states that the blowup of the codimension-one components "reduces the situation to the case when supp is of codimension ≥ 2" and then works at the points off codim(1)(supp); the three statements below make this precise inside N: the codimension-one part never reappears along the segment, and off it the restriction to a hypersurface of maximal contact is nonzero. As in Proposition 174.62, $R(\underline{\mathcal{I}})$ is the codimension-one part along N of the cosupport of $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$.

Theorem 174.65 (The codimension-one part along a multiple test blow-up lies over the original one)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed, of maximal order along N, with $d \geq 1$, and let T be a multiple test blow-up of $\underline{\mathcal{I}}$ with projection p. Then $R(\underline{\mathcal{I}}_T) \subset p^{-1}(R(\underline{\mathcal{I}}))$. In particular, if $R(\underline{\mathcal{I}}) = \emptyset$, then $R(\underline{\mathcal{I}}_T) = \emptyset$ for every multiple test blow-up T.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Induction on T; for the empty tower there is nothing to prove. Let T' be T followed by the blowup π along C, with projection $p' = p \circ \pi$, and $x \in R(\underline{\mathcal{I}}_{T'})$. Then x lies in the cosupport of $\underline{\mathcal{I}}_{T'}$, so $p'(x)$ lies in that of $\underline{\mathcal{I}}$ (Corollary 174.32), and by maximal order there is a tangent direction u along N on an open $U \ni p'(x)$ (Proposition **??**). Put $A = N \cap V(u) \cap U$. By Theorem **??** there are an open $O \ni x$ and a function w on O, controlled on $N_{T'}$ as in the last sentence of Lemma 174.35, whose zero set in $N_{T'} \cap O$ is the strict transform $A_{T'}$ (Definition **??**). By Theorem 174.37(3), $R(\underline{\mathcal{I}}_{T'}) = N_{T'} \cap V(w)$ near x; hence near x the points of $A_{T'} \cap O$ lie in $R(\underline{\mathcal{I}}_{T'})$, and $x \in A_{T'} = \overline{\pi^{-1}(A_T \setminus C)}$. So every neighbourhood of x contains a point $y \in \pi^{-1}(A_T \setminus C)$ so close to x that $y \in A_{T'} \cap O$ lies in $R(\underline{\mathcal{I}}_{T'})$. As $\pi(y) \notin C$, $\pi(y) \in R(\underline{\mathcal{I}}_T)$ (Lemma 174.63), and $p'(y) = p(\pi(y)) \in R(\underline{\mathcal{I}})$ by induction. Since $R(\underline{\mathcal{I}})$ is closed (Theorem 174.37(2)) and p' continuous, $x \in p'^{-1}(R(\underline{\mathcal{I}}))$. The transforms along multiple test blow-ups stay well formed (Lemma **??**), which the cited results need at each stage. □

Corollary 174.66 (After Step 1ba the codimension-one part is empty)**.** *Verified upstream; not exported by this standalone library.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N a closed submanifold, $\mathfrak{a}$ coherent, $d \geq 1$, and be of maximal order along N; let π be the blowup along $R(\underline{\mathcal{I}})$ and $\underline{\mathcal{I}}'$ the transform. Then $R(\underline{\mathcal{I}}') = \emptyset$. If $\underline{\mathcal{I}}$ is well formed, then $R(\underline{\mathcal{I}}_T) = \emptyset$ for every multiple test blow-up T of $\underline{\mathcal{I}}$ whose first blowup is π.

Proof. Verified upstream; not exported by this standalone library.

The cosupport of $\underline{\mathcal{I}}'$ does not meet $\pi^{-1}(R(\underline{\mathcal{I}}))$ (Corollary 174.38), and at a point z with $\pi(z) \notin R(\underline{\mathcal{I}})$, $z \in R(\underline{\mathcal{I}}')$ would give $\pi(z) \in R(\underline{\mathcal{I}})$ (Lemma 174.63). A multiple test blow-up whose first blowup is π is π followed by a multiple test blow-up of $\underline{\mathcal{I}}'$ (Lemma **??**), and $\underline{\mathcal{I}}'$ is well formed (Lemma **??**), of maximal order along N' (Lemma **??**), with the same mark; apply Theorem 174.65 to $\underline{\mathcal{I}}'$. □

Proposition 174.67 (Off the codimension-one part the restriction to a hypersurface of maximal contact is nonzero)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

1. *Let N be regular of dimension $k+1$ and A regular of dimension k at y, $\mathfrak{b}$ coherent, and suppose that near y $\operatorname{cosupp}(\mathfrak{a}|_N, \mu) \subset A$ and $\operatorname{cosupp}(\mathfrak{b}|_A, \nu) \subset \operatorname{cosupp}(\mathfrak{a}|_N, \mu)$. If $y \notin R_N(\mathfrak{a}, \mu)$, then some germ of $\mathfrak{b}$ at y does not vanish identically on A near y (Lemma **??**).*

2. *Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N a closed submanifold, $\mathfrak{a}$ coherent, $d \geq 1$, let u be a tangent direction along N on an open U (Definition **??**), and let $\underline{\mathcal{S}}_u = (U, N \cap V(u), E|_U, \mathcal{C}_N(\mathfrak{a})|_U, \mu_{\mathcal{C}})$ be the restricted marked ideal with $\underline{\mathcal{I}}|_U \simeq \underline{\mathcal{S}}_u$ (Lemma **??**). At every point y of $N \cap V(u) \cap U$, N is regular of some dimension $k+1$ and $N \cap V(u) \cap U$ of dimension k; if $y \notin R(\underline{\mathcal{I}})$, some germ of the ideal of $\underline{\mathcal{S}}_u$ at y does not vanish identically on $N \cap V(u)$ near y.*

3. *If $\underline{\mathcal{I}}$ is moreover well formed and of maximal order along N with $R(\underline{\mathcal{I}}) = \emptyset$ (for instance the transform after Step 1ba, Corollary 174.66), then for every multiple test blow-up T and every tangent direction w of $\underline{\mathcal{I}}_T$ along N_T on an open U, the conclusion of (2) holds for $\underline{\mathcal{S}}_w$ at every point of $N_T \cap V(w) \cap U$. In particular the ideal of $\underline{\mathcal{S}}_w$ vanishes identically on no nonempty open subset of $N_T \cap V(w) \cap U$, which is the hypothesis under which the inductive invariant of Definition **??** is defined, as Step 1bb of Proposition **??** uses it.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Suppose every germ of $\mathfrak{b}$ at y vanishes identically on A near y. Take generators $s_1, \dots, s_r$ of $\mathfrak{b}$ on a neighbourhood of y (Definition **??**); each s_i is a germ of $\mathfrak{b}$ at y, so all of them vanish on A near y, and then at every point z near y every germ of $\mathfrak{b}_z$, a combination of the s_i, vanishes on A near z. Such a germ agrees on A with 0, which lies in every power of the maximal ideal; so every point of A near y lies in $\operatorname{cosupp}(\mathfrak{b}|_A, \nu)$ (Definition **??**), hence in $\operatorname{cosupp}(\mathfrak{a}|_N, \mu) \subset A$. Thus $\operatorname{cosupp}(\mathfrak{a}|_N, \mu) = A$ near y, regular of dimension k at y, where N has dimension $k+1$: $y \in R_N(\mathfrak{a}, \mu)$ (Proposition 174.62(1)). (2) By Lemma **??**, N is regular at y of some dimension k' and $N \cap V(u) \cap U$ of dimension $k'-1$; and $k' \geq 1$, since a field tangent to N at y with $\xi(u)(y) \neq 0$ gives a nonzero partial derivative of u in a tangential coordinate of index $< k'$ (proof of that lemma). The equivalence $\underline{\mathcal{I}}|_U \simeq \underline{\mathcal{S}}_u$ gives equal cosupports (Definition **??**, the empty tower), and the cosupport of $\underline{\mathcal{S}}_u$ lies in its submanifold. The ideal of $\underline{\mathcal{S}}_u$ is coherent (Lemma **??** and Lemma **??**), and $R(\underline{\mathcal{I}}|_U) = R(\underline{\mathcal{I}}) \cap U$ (Proposition 174.62(3)). Apply (1) on U. (3) Along T, N_T is a closed submanifold, $\mathfrak{a}_T$ coherent and the mark unchanged (Lemma **??**), and $R(\underline{\mathcal{I}}_T) = \emptyset$ (Theorem 174.65); apply (2) to $\underline{\mathcal{I}}_T$. If the ideal vanished identically on a nonempty open subset of $N_T \cap V(w) \cap U$, all its germs at a point of that subset would vanish there. □

The member locus: Step 1aa inside N. In Step 1a of the maximal-order step (Proposition **??**) the points of an intersection $N \cap H_\alpha^s$ of s members contained in the cosupport get the value (s, ∞), and Włodarczyk notes that "all $H_\alpha^s \subset$ supp will be blown up first and we reduce the situation to the case where no H_α^s is contained in supp". Włodarczyk intersects irreducible components of the members; the invariant has to be local (Definition **??**(5)), and an intersection $N \cap H_S$ of members transverse to N is a submanifold, whose connected component through x lies in an analytic set as soon as a neighbourhood of x in it does. So the value set is defined pointwise, by the germ at x of the intersection of N with the members through x.

Definition 174.68 (The member locus)**.** Let $N, S \subset M$ and let E be a finite sequence of subsets of M. The *member locus* $L_N(E, S)$ is the set of points $x \in S$ near which every point of N lying on all the members of E through x lies in S: near x, $N \cap H_{S(x)} \subset S$, where $S(x)$ is the set of members through x. For a marked ideal $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ and a tower T we write $L(\underline{\mathcal{I}}_T)$ for $L_{N_T}(E_T^{\leq \#E}, \operatorname{cosupp} \underline{\mathcal{I}}_T)$, where $E_T^{\leq \#E}$ is the sequence of the first $\#E$ members of E_T, the strict transforms of the members of E (Lemma 174.50); its points with $s(x) = s$ are the points of value (s, ∞) of Step 1aa.

Proposition 174.69 (The member locus is canonical)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

1. *(Locality) Whether $x \in L_N(E, S)$ depends only on the germs at x of N, of S and of the members of E; members not through x play no role.*

2. *(Preimages) If $f\colon M' \to M$ maps the neighbourhoods of z onto a neighbourhood basis of $f(z)$ ($f_*\mathcal{N}_z = \mathcal{N}_{f(z)}$: a biholomorphism, the inclusion of an open set, a blowup at a point off its centre), then $z \in L_{f^{-1}N}((f^{-1}H)_{H \in E}, f^{-1}S)$ if and only if $f(z) \in L_N(E, S)$.*

3. *(Off the centre) Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N closed and $\mathfrak{a}$ consist of holomorphic germs, E_0 a sequence of closed sets, π the blowup along C and $\pi(z) \notin C$. Then z lies in the member locus of $\operatorname{cosupp} \underline{\mathcal{I}}'$ along N' and the strict transforms of E_0 if and only if $\pi(z) \in L_N(E_0, \operatorname{cosupp} \underline{\mathcal{I}})$. In particular a blowup along $\emptyset$ pulls L back, and along a multiple test blow-up T followed by π, $z \in L(\underline{\mathcal{I}}_{T\pi})$ if and only if $\pi(z) \in L(\underline{\mathcal{I}}_T)$ whenever $\pi(z) \notin C$ (N a closed submanifold, $\mathfrak{a}$ coherent, closed members).*

4. *(Transported and restricted towers) If $\mathfrak{a}$ is coherent, $L(\Phi^* \underline{\mathcal{I}}_{\Phi^* T}) = \Phi_T^{-1} L(\underline{\mathcal{I}}_T)$ for every tower T and biholomorphism Φ (Definition **??**); if moreover N is a closed submanifold and T a multiple test blow-up, $L((\underline{\mathcal{I}}|_U)_{T|_U})$ is the trace of $L(\underline{\mathcal{I}}_T)$ on the open preimage of U.*

5. *(Representatives) Restricted-equivalent marked ideals (N closed, coherent ideals) have the same L along every tower; equivalent marked ideals with the same N have the same L along every multiple test blow-up.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Quantifying over the indices of the members through x is quantifying over these members; if near x the sets agree, the defining condition at x reads the same for both, and a member not through x never enters it. (2) Membership of z in the preimages is membership of $f(z)$, and a property holds near z for the points $f(y)$ if and only if it holds near $f(z)$, since $f_*\mathcal{N}_z = \mathcal{N}_{f(z)}$. A biholomorphism and an open inclusion have this property; so does a blowup at z with $\pi(z) \notin C$: it is continuous, and its holomorphic section s over $M \setminus C$ with $s(\pi(z)) = z$ (Lemma 174.30) maps a neighbourhood of $\pi(z)$ into any given neighbourhood V of z, so $\pi(V)$ is a neighbourhood of $\pi(z)$. (3) Near z the strict transforms of N and of the members are their preimages, and the cosupport of $\underline{\mathcal{I}}'$ is the preimage of that of $\underline{\mathcal{I}}$ (Theorem 174.31); apply (1) and (2). The first $\#E$ members of the transform along $T\pi$ are the strict transforms of those along T (Lemma 174.50, Lemma **??**). (4) For the empty tower this is (2) with the cosupport identity of Lemma **??** (as in Proposition 174.62(2)); from the first blowup on the transforms are equal and Φ_T is the identity (Lemma **??**). The transform along $T|_U$ is the restriction of the transform along T to the open preimage of U (Lemmas **??** and **??**), with the traces of the members; apply (2). (5) Restricted-equivalent marked ideals have the same N, E and cosupport and equal transforms from the first blowup on (Lemmas **??** and **??**); equivalent ones have the same E, equal cosupports along their multiple test blow-ups (Definition **??**), and the same N_T and E_T (Definition **??**). □

Theorem 174.70 (Blowing up the level sets of the member locus)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed, E_0 a sequence of members of E (in the algorithm $E_0 = E$, or at a later stage the strict transforms of the original members), $s_0 \in \mathbf{N}$, and $C = \{x \in L_N(E_0, \operatorname{cosupp}\underline{\mathcal{I}}) : s(x) = s_0\}$, s counting the members of E_0.

1. *Near each $x \in C$, every point of N on all the members of E_0 through x lies in C: $C \supset N \cap H_{S(x)}$ near x.*

2. *If E_0 is a subsequence of E, C is an admissible centre and π the blowup along C, then no point of the transform with $s' = s_0$ lies in the member locus of $\operatorname{cosupp}\underline{\mathcal{I}}'$ along N' and the strict transforms of E_0.*

3. *Along every multiple test blow-up T with projection p, the points of the member locus of $\operatorname{cosupp}\underline{\mathcal{I}}_T$ along N_T and the strict transforms of E_0 with $s_T = s_0$ lie in $p^{-1}(\overline{C})$. Hence if $C = \emptyset$ this set is empty along every multiple test blow-up, and after the blowup of (2) it is empty along every multiple test blow-up that starts with that blowup.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The members of E_0 not through x are closed, so near x a point on all the members through x lies on exactly these members; its count is s_0, and near it $N \cap H_{S(x)} \subset \operatorname{cosupp}\underline{\mathcal{I}}$ still holds, since this holds on a neighbourhood of x. (2) Let z be such a point. If $\pi(z) \notin C$, $\pi(z)$ lies in $L_N(E_0, \operatorname{cosupp}\underline{\mathcal{I}})$ with $s(\pi(z)) = s'(z) = s_0$ (Proposition 174.69(3), Lemma 174.64(1)), that is $\pi(z) \in C$, a contradiction. If $x = \pi(z) \in C$, then $z \in N'$ and near x the centre contains $N \cap \bigcap_{x \in H} H$ by (1); a subsequence of a simple normal crossings divisor transverse to N is one, with which C has normal crossings (Lemma 174.126), so $s'(z) < s(x) = s_0$ (Theorem 174.52(2)), a contradiction. (3) Induction on T; for the empty tower the set is C. Let T' be T followed by the blowup π along C_T, and z a point of the set at stage T'. The intersection of $N_{T'}$ with the strict transforms of the members of E_0 through z is the strict transform of the intersection A of N with these members (Lemma 174.55), that is the closure of $\pi^{-1}(A_T \setminus C_T)$ (Definition **??**),

and it contains z. Take a point $y \in \pi^{-1}(A_T \setminus C_T)$ in a given neighbourhood of z, so close to z that $s_{T'}(y) \leq s_{T'}(z)$ (Lemma 174.50) and that the defining condition of the member locus at z holds near y. Then y lies on all the members through z and on no more, so y is in the member locus with $s_{T'}(y) = s_0$; as $\pi(y) \notin C_T$, $\pi(y)$ is a point of the set at stage T (Proposition 174.69(3), Lemma 174.64(1)), so $p'(y) \in \overline{C}$ by induction. As $p'^{-1}(\overline{C})$ is closed, $z \in p'^{-1}(\overline{C})$. For the last claim, a multiple test blow-up starting with π is π followed by a multiple test blow-up of the well-formed transform $\underline{\mathcal{I}}'$ (Lemmas **??** and **??**), the strict transforms of the members of E_0 are members of E', strict transforms along concatenated towers compose, and the set of (2) is empty. □

Proposition 174.71 (The Step 1aa centre at the maximal level is admissible)**.** *Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed with $d \geq 1$ and $n > 0$, E_0 a subsequence of E, and s_0 an upper bound of the number s of members of E_0 through the points of the cosupport. Then $C = \{x \in L_N(E_0, \operatorname{cosupp}\underline{\mathcal{I}}) : s(x) = s_0\}$ is an admissible centre: near each of its points x it is $N \cap H_{S(x)}$, the intersection of N with the members of E_0 through x. (For an open W on which $s \leq s_0$ on the cosupport, applied to $\underline{\mathcal{I}}|_W$, this is property (1) of Definition **??** for the value (s_0, ∞), and it discharges the admissibility hypothesis of Theorem 174.70(2).)*

Proof. Near $x \in C$, every point of $N \cap H_{S(x)}$ is in C (Theorem 174.70(1)); conversely a point $y \in C$ near x lies on no member missing x (the members are closed) and has $s(y) = s_0 = s(x)$, so it lies on exactly the members through x and $y \in N \cap H_{S(x)}$. The set $N \cap H_{S(x)}$ is a closed submanifold with normal crossings with E (Lemma 174.49, with E_0 simple normal crossings and transverse to N, Lemma 174.126); both properties are local (Lemma 174.58, Remark 443.1), so they pass to C once C is closed. $C \subset \operatorname{cosupp}\underline{\mathcal{I}}$ by definition. Closedness: let x be a limit of points of C. Then x lies in the cosupport, which is closed (Lemma **??**); s is upper semicontinuous (Lemma 174.50), so $s(x) \geq s_0$, and $s(x) \leq s_0$ by hypothesis. As above, the points $y \in C$ near x lie on exactly the members through x, and near each of them $N \cap H_{S(x)} = N \cap H_{S(y)} \subset \operatorname{cosupp}\underline{\mathcal{I}}$. Let $H = N \cap H_{S(x)}$, regular of some dimension k at x, hence at its points near x (Theorem **??**). The set $A = \operatorname{cosupp}\underline{\mathcal{I}} \cap H$ is analytic (Lemma 174.35, Lemma **??**, Definition **??**), contained in H, and agrees with H near the points y, which accumulate at x; so $\dim_x A \geq k$, and the filling lemma (Lemma 174.22) gives $A = H$ near x: $N \cap H_{S(x)} \subset \operatorname{cosupp}\underline{\mathcal{I}}$ near x, that is $x \in C$. The admissible centre is Definition **??**. □

Proposition 174.72 (Off the member locus the restriction to the intersection of the members is nonzero)**.** *Let $y \in N$, E a finite sequence of sets, $A = N \cap H_{S(y)}$ the intersection of N with the members through y, $\mathfrak{b}$ coherent, and suppose that near y $\operatorname{cosupp}(\mathfrak{b}|_A, \nu) \subset S$. If $y \notin L_N(E, S)$, some germ of $\mathfrak{b}$ at y does not vanish identically on A near y. In particular, for $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ with N a closed submanifold, $\mathfrak{a}$ coherent and $d \geq 1$, and $y \in N$ off $L_N(E_0, \operatorname{cosupp}\underline{\mathcal{I}})$, the coefficient ideal along N restricted to A (Lemma **??**, the restricted marked ideal of Step 1ab with any sequence of members) is nonzero along A near y: together with Theorem 174.70(3), at every stage after Step 1aa the restricted marked ideals of Step 1ab at the points with $s = s_0$ satisfy the hypothesis of Definition **??**.*

Proof. If every germ of $\mathfrak{b}$ at y vanished on A near y, then, $\mathfrak{b}$ being coherent, at every point z near y every germ of $\mathfrak{b}$ would vanish on A near z (proof of Proposition 174.67(1)), so every point of A near y would lie in $\operatorname{cosupp}(\mathfrak{b}|_A, \nu) \subset S$ (Definition **??**); with $y \in A$ this is $y \in L_N(E, S)$. For the coefficient ideal, its cosupport on $A \subset N$ is $\operatorname{cosupp}\underline{\mathcal{I}} \cap A$ (Lemma **??**) and it is coherent (Lemma **??**). □

Proposition 174.73 (The loci of Step 1 along isomorphic towers)**.** *Verified upstream; not exported by this standalone library.*

*Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N a closed submanifold and $\mathfrak{a}$ coherent, T a multiple test blow-up of $\underline{\mathcal{I}}$, $\Phi\colon M' \to M$ a biholomorphism and $\chi\colon T' \to T$ an isomorphism of towers over Φ with last-stage biholomorphism χ_r (Definition **??**). For $k \in \mathbf{N}$ write s_T for the number of the first*

k members of E_T through a point and $L_k(\underline{\mathcal{I}}_T)$ for the member locus of $\operatorname{cosupp}\underline{\mathcal{I}}_T$ *along N_T and these members (Definition 174.68; in the algorithm $k = \#E$). Then, with $\underline{\mathcal{I}}' = \Phi^*\underline{\mathcal{I}}$,*

1. *$R(\underline{\mathcal{I}}'_{T'}) = \chi_r^{-1}R(\underline{\mathcal{I}}_T)$ for the codimension-one parts along $N'_{T'}$ and N_T (Proposition 174.62);*
2. *$s_{T'} = s_T \circ \chi_r$;*
3. *$L_k(\underline{\mathcal{I}}'_{T'}) = \chi_r^{-1}L_k(\underline{\mathcal{I}}_T)$, and for every s_0 the level-s_0 parts (the centres of Step 1aa) correspond: $L_k(\underline{\mathcal{I}}'_{T'}) \cap \{s_{T'} = s_0\} = \chi_r^{-1}(L_k(\underline{\mathcal{I}}_T) \cap \{s_T = s_0\})$;*
4. *for a subset C of the last stage of T, $\chi_r^{-1}C$ is an admissible centre of $\underline{\mathcal{I}}'_{T'}$ if and only if C is one of $\underline{\mathcal{I}}_T$.*

These are the blowup-calculus halves of property (6) of Definition **??** *for the values $(0,\infty)$ and (s,∞) of Steps 1ba and 1aa and for the count s of Proposition* **??***; the transport Φ^*T is the case $\chi_i = \mathrm{id}$ for $i \geq 1$ (Propositions 174.62(4) and 174.69(4), Lemma 174.64(2), which hold even without the hypothesis that T be a multiple test blow-up). Applied to the transform at a later stage T_0 of the algorithm, the count k stays the number of the original members.*

Proof. Verified upstream; not exported by this standalone library.

By Lemma **??**, $\underline{\mathcal{I}}'_{T'} = \chi_r^*(\underline{\mathcal{I}}_T)$: its submanifold is $\chi_r^{-1}N_T$, its members are the preimages $\chi_r^{-1}H$ of the members of E_T in the same order, and its ideal is $\chi_r^*\mathfrak{a}_T$, coherent since $\mathfrak{a}_T$ is (Lemma **??**). (1) is Proposition 174.62(2) for χ_r. (2) is the re-indexing of Lemma 174.64: the first k members of $\chi_r^{-1}E_T$ are the preimages of the first k members of E_T. (3) The cosupport of $\chi_r^*(\underline{\mathcal{I}}_T)$ is $\chi_r^{-1}\operatorname{cosupp}\underline{\mathcal{I}}_T$ (Lemma **??**), and a biholomorphism maps neighbourhood filters onto neighbourhood filters, so Proposition 174.69(2) applies; the level sets follow with (2). (4) is the invariance of admissible centres under biholomorphisms (Lemma **??**) for χ_r and $\underline{\mathcal{I}}_T$. □

The domain of a resolution invariant. The inductive invariant of Definition **??** is defined on the well-formed marked ideals $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ with $\dim N \leq k$, $d \geq 1$ and $\mathfrak{a}|_N$ vanishing identically on no nonempty open subset of N. The general step forms the companion ideal from the transform at a later stage of a canonical segment (Włodarczyk: "Then we repeat the procedure for $(\mathcal{I}_{r_1}, \mu)$"), Step 1 forms its restricted marked ideals there, and the canonicity properties (5) and (6) are applied to restrictions and pullbacks. The two statements below show that the domain is stable under these operations; the point is that a transform stays nonzero along the strict transform over the centre, for every admissible centre.

Lemma 174.74 (A transform stays nonzero along the strict transform)**.** *Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have N a closed submanifold and $\mathfrak{a}$ coherent and nonzero along N at every point of N (some germ of $\mathfrak{a}$ at the point does not vanish identically on N near it, Definition* **??***). Let π be the blowup along an admissible centre C and $\underline{\mathcal{I}}' = (M', N', E', \mathfrak{a}', d)$ the transform. Then $\mathfrak{a}'$ is nonzero along N' at every point of N'; at a point z with $\pi(z) \notin C$ this needs only that $\mathfrak{a}$ be nonzero along N at $\pi(z)$. Consequently, along every multiple test blow-up T of $\underline{\mathcal{I}}$, $\mathfrak{a}_T$ is nonzero along N_T at every point of N_T.*

Proof. Let $\pi(z) \notin C$ and let a be a germ of $\mathfrak{a}$ at $\pi(z)$ not vanishing identically on N near $\pi(z)$. Then $a \circ \pi$ is a germ of $\pi^*\mathfrak{a}$ at z (Lemma **??**), and off the centre $\operatorname{Res}_{N'}\mathfrak{a}' = \operatorname{Res}_{N'}\pi^*\mathfrak{a}$ at z (Lemma 174.47): on N' near z, $a \circ \pi$ is a combination of germs of $\mathfrak{a}'$ at z. The holomorphic section s of π over $M \setminus C$ with $s(\pi(z)) = z$ (Lemma 174.30) maps the points of $N \setminus C$ near $\pi(z)$ where $a \neq 0$, which accumulate at $\pi(z)$, to points of $\pi^{-1}(N \setminus C) \subset N'$ near z where $a \circ \pi \neq 0$. If every germ of $\mathfrak{a}'$ at z vanished on N' near z, so would the combination $a \circ \pi$; hence $\mathfrak{a}'$ is nonzero along N' at z (Lemma **??**, whose proof uses the same step). Let now $\pi(z) \in C$, and suppose every germ of $\mathfrak{a}'$ at z vanishes identically on N' near z. The ideal $\mathfrak{a}'$ is coherent (Lemma **??**); taking finitely many sections generating it near z, every germ of $\mathfrak{a}'$ at every point z' near z vanishes on N' near z' (proof of Proposition 174.67(1)). But $N' = \overline{\pi^{-1}(N \setminus C)}$ (Theorem 174.12), so there is

such a z' with $\pi(z') \in N \setminus C$, at which $\mathfrak{a}'$ is nonzero along N' by the first case: a contradiction. For a multiple test blow-up, induction on its length: along it N_T stays a closed submanifold and $\mathfrak{a}_T$ coherent (Lemma **??**), and each centre is admissible (Definition **??**). □

Proposition 174.75 (The domain of a resolution invariant is stable)**.** *Say that $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ lies in the* domain in dimension $\leq k$ *if it lies in the domain of a resolution invariant in dimension $\leq k$ (Definition **??**): well formed, N regular of dimension $\leq k$ at each of its points, $d \geq 1$, and $\mathfrak{a}$ nonzero along N at every point of N.*

1. *If N is closed, $C \subset N$, π is the blowup along C and N is regular of dimension $\leq k$ at each of its points, so is the strict transform N': at $z \in N'$ of the dimension of N at $\pi(z)$.*
2. *If $\underline{\mathcal{I}}$ lies in the domain in dimension $\leq k$, so does $\underline{\mathcal{I}}_T$ for every multiple test blow-up T of $\underline{\mathcal{I}}$.*
3. *If $\underline{\mathcal{I}}$ lies in the domain in dimension $\leq k$, so do $\Phi^*\underline{\mathcal{I}}$ for a biholomorphism $\Phi\colon M' \to M$ and $\underline{\mathcal{I}}|_U$ for an open $U \subset M$.*

Proof. (1) Let N be regular of dimension $j \leq k$ at $\pi(z) \in N$ ($N' \subset \pi^{-1}(N)$ as N is closed). If $\pi(z) \notin C$, π is a local biholomorphism at z and N' is $\pi^{-1}(N)$ near z; if $\pi(z) \in C$, the centre is regular at $\pi(z)$ of a dimension $k_C \leq j$ (a submanifold of a submanifold), and N' is regular of dimension j at z (Theorem 174.12, in its pointwise form). (2) Induction on T: well-formedness passes along multiple test blow-ups (Lemma **??**), the mark does not change (Lemma **??**), N_T stays a closed submanifold (Lemma **??**) and the next centre lies in the cosupport, hence in N_T (Definition **??**), so (1) applies; nonzero along N_T is Lemma 174.74. (3) Well-formedness is invariant under biholomorphisms and restricts to open sets (Lemmas **??** and **??**); regular points and their dimensions correspond (Lemma 174.61, Remark 443.2); and a germ a of $\mathfrak{a}$ at $\Phi(p)$ not vanishing identically on N near $\Phi(p)$ pulls back to the germ $a \circ \Phi$ of $\Phi^*\mathfrak{a}$ (Lemma **??**), which does not vanish identically on $\Phi^{-1}(N)$ near p since Φ is a homeomorphism; likewise for the inclusion of U, an open embedding (Lemma **??**). □

These are the inputs that let the general step (Proposition **??**) and Step 1 (Proposition **??**) apply an invariant to marked ideals built at a later stage of a canonical segment.

Proposition 174.76 (The ideal of a smooth centre inside a submanifold)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $Z \subset N$ be closed submanifolds of M and $\underline{\mathcal{I}}_Z = (M, N, \emptyset, \mathcal{I}_Z, 1)$, with $\mathcal{I}_Z$ the ideal of germs of holomorphic functions vanishing on Z.

1. $\operatorname{cosupp}(\mathcal{I}_Z|_N, 1) = N \cap Z = Z$*, and Z is an admissible centre of $\underline{\mathcal{I}}_Z$.*
2. *A point $x \in Z$ where Z and N are regular of dimensions j and j' lies in the codimension-one part Z_1 of* $\operatorname{cosupp}(\mathcal{I}_Z|_N, 1)$ *along N (Definition 174.36) if and only if $j' = j + 1$. Hence $Z_1 = Z$ if Z has codimension one in N at each of its points, and $Z_1 = \emptyset$ if it has codimension at least two at each of its points.*
3. *If $\dim_x Z < \dim_x N$ at every $x \in Z$, then $\mathcal{I}_Z$ is nonzero along N at every point of N; if moreover $\dim N \leq k$, then $\underline{\mathcal{I}}_Z$ lies in the domain of a resolution invariant in dimension $\leq k$ (Definition **??**, Proposition 174.75).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The germs of $\mathcal{I}_Z$ are holomorphic, so the cosupport with mark 1 is the zero locus on N (Lemma **??**). Every germ vanishes at the points of Z; at $x \notin Z$ the constant 1 is a germ of $\mathcal{I}_Z$, Z being closed. So the zero locus is Z. Z is a closed submanifold contained in the cosupport, and has normal crossings with the empty divisor (Lemma **??**): an admissible centre (Definition **??**). (2) By (1), $x \in Z_1$ if and only if for some k the set N is regular of dimension $k + 1$ and Z of

dimension k at x; a set is regular of at most one dimension at a point (Lemma **??**), so this means $j' = j+1$. Points of Z_1 lie in the cosupport Z (Theorem 174.37). (3) At $y \in N \setminus Z$ the constant 1 is a germ of $\mathcal{I}_Z$ not vanishing on N. Let $x \in Z$. If $N \subset Z$ near x, then N and Z agree near x and are regular of the same dimension there (Lemma 174.22, regularity depends only on the germ), which is excluded; so points of $N \setminus Z$ accumulate at x. The ideal $\mathcal{I}_Z$ is coherent (Theorem **??**; for a submanifold, in flattening coordinates it is generated by the normal coordinates). If every germ of $\mathcal{I}_Z$ at x vanished on N near x, then every germ at every point near x would vanish on N near that point (proof of Proposition 174.67(1)), in particular 1 at a nearby point of $N \setminus Z$: a contradiction (Definition **??**). $\underline{\mathcal{I}}_Z$ is well formed (closed submanifold N, coherent ideal, the empty divisor is simple normal crossings and transverse to N, Lemma **??**), its mark is 1, and $\dim N \leq k$. □

In Lemma **??** this is applied first with $N = U$ and then, along the induction on the codimension, with $N = V(u) \cap U$; Włodarczyk's remark on the weak embedded desingularization ("Its support $\operatorname{supp}(\mathcal{I}_Y, 1)$ is equal to Y", and at a smooth point Steps 2a and 1b pass to a hypersurface until the algorithm ends in Step 1ba) is the same computation.

The domain of Step 1ab's restricted marked ideals. In Step 1a of the maximal-order step (Proposition **??**), at a point y of the cosupport on $s_0 \geq 1$ of the original members, the value is $(s_0, \operatorname{inv}(x))$ with the invariant of the restriction of the coefficient ideal to the intersection of N with the members through y. Włodarczyk forms these restrictions after Step 1aa: "After the blow-ups in Step 1aa we arrive at X_p for which no H^s_α is contained in $\operatorname{supp}(\overline{\mathcal{J}}_p, \overline{\mu})$ [...] Construct the canonical resolutions of $\overline{\mathcal{J}}_{p|H^s_\alpha}$." The two statements below show that these restrictions lie in the domain of the inductive invariant (Definition **??**) in dimension $\leq k-1$.

Lemma 174.77 (An intersection of members drops the dimension)**.** *Let E be a simple normal crossings divisor on M transverse to the closed set N ($n > 0$), S a nonempty set of indices of members and $x \in N \cap H_S$. If N is regular of dimension j at x, then $N \cap H_S$ is regular at x of some dimension $j' < j$. Hence if N is regular of dimension $\leq k$ at each of its points, $N \cap H_S$ is regular of dimension $\leq k-1$ at each of its points.*

Proof. Take coordinates at x adapted to $N \cap H_S \subset N$ and E (Lemma 174.122; $N \cap H_S$ has normal crossings with E, Lemma 174.49): near x, $N \cap H_S = \mathbf{C}^{k_C} \times 0$, $N = \mathbf{C}^{k_N} \times 0$ and each member H_i through x is $\{w_{s_i} = 0\}$ with $s_i < k_N$. So $N \cap H_S$ and N are regular at x of dimensions $\min(k_C, n)$ and $\min(k_N, n)$, and $j = \min(k_N, n)$ since a set is regular of at most one dimension at a point (Lemma **??**). If $\min(k_C, n) \geq \min(k_N, n)$, let c be the coordinate vector of x, $i \in S$ and $v = c + \varepsilon e_{s_i}$ with $\varepsilon > 0$ so small that v lies in the coordinate domain. As $c \in \mathbf{C}^{k_N} \times 0$, $c_{s_i} = 0$ and $s_i < k_N$, the vector v lies in $\mathbf{C}^{k_N} \times 0 \subset \mathbf{C}^{k_C} \times 0$, so the point with coordinates v lies in $N \cap H_S \subset H_i$; but $v_{s_i} = \varepsilon \neq 0$, so it does not lie on H_i: a contradiction. The second statement follows at each point. □

Proposition 174.78 (Step 1ab's restricted marked ideals lie in the domain)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed with $d \geq 1$, $n > 0$ and N regular of dimension $\leq k$ at each of its points. For a set S of indices of members let $\underline{\mathcal{S}}_S = (M, N \cap H_S, E_S, \mathcal{C}_N(\mathfrak{a}), \mu_{\mathcal{C}})$ be the restriction of the coefficient ideal along N to $N \cap H_S$, the members of S deleted from E (Corollary **??**).*

1. *If $\underline{\mathcal{I}}$ is nonzero along N at the points of N in an open set U, then $\underline{\mathcal{I}}|_U$ lies in the domain in dimension $\leq k$.*
2. *$\underline{\mathcal{S}}_S$ is well formed, $\mu_{\mathcal{C}} \geq 1$, $\operatorname{cosupp} \underline{\mathcal{S}}_S = \operatorname{cosupp} \underline{\mathcal{I}} \cap H_S$, and if $S \neq \emptyset$ then $N \cap H_S$ is regular of dimension $\leq k-1$ at each of its points. If E_0 is a finite sequence of sets and $z \in N$ is a point near which H_S is the intersection of the members of E_0 through z, and z is not in the member locus $L_N(E_0, \operatorname{cosupp} \underline{\mathcal{I}})$, then the ideal of $\underline{\mathcal{S}}_S$ is nonzero along $N \cap H_S$ at z.*

3. *Let E_0 be a sequence of members of E, y a point on $s_0 \geq 1$ members of E_0, $S(y)$ the set of members of E_0 through y, and assume that no point of $L_N(E_0, \operatorname{cosupp}\underline{\mathcal{I}})$ lies on exactly s_0 members of E_0. Then for every open U whose points lie on no member of E_0 missing y (for instance the complement of the union of these members, an open neighbourhood of y), $\underline{\mathcal{S}}_{S(y)|U}$ lies in the domain in dimension $\leq k-1$.*

4. *The same holds at every stage of a multiple test blow-up T of $\underline{\mathcal{I}}$ along which no point of the member locus of the cosupport along N_T and the strict transforms of E_0 lies on exactly s_0 of these strict transforms, for $\underline{\mathcal{S}}$ formed from $\underline{\mathcal{I}}_T$, at a point on $s_0 \geq 1$ of them; in particular at every stage after the blowup of Step 1aa at the level s_0 (Theorem 174.70(3)).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.
(1) As in Proposition 174.75(3): well-formedness restricts to open sets (Lemma **??**), so do the dimensions, and a germ g of $\mathfrak{a}$ at a point of U not vanishing identically on N near it restricts to a germ of the restricted ideal with the same property (Lemma **??**). (2) Well formed: Lemma 174.53, $\mathcal{C}_N(\mathfrak{a})$ being coherent (Lemma **??**); $\mu_{\mathcal{C}} = \prod_{j<d}(d-j) \geq 1$; the cosupport is Lemma **??**; the dimension is Lemma 174.77. Being nonzero along a set at z depends only on the germ of the set at z (Definition **??**), so the last claim is Proposition 174.72. (3) Let $z \in U \cap N \cap H_{S(y)}$. It lies on every member of E_0 through y and, being in U, on no other; so $H_{S(y)}$ is the intersection of the members of E_0 through z, and z lies on exactly s_0 members of E_0 (Lemma 174.50), hence not in the member locus. Now (2) and (1) apply, with $S(y) \neq \emptyset$. (4) Along T, $\underline{\mathcal{I}}_T$ is well formed (Lemma **??**) with the same mark (Lemma **??**) and $\dim N_T \leq k$ (Proposition 174.75(2)), and the strict transforms of the members of E_0 are members of E_T (Lemma 174.55), closed; apply (3) to $\underline{\mathcal{I}}_T$. □

These are the domain hypotheses of the inductive invariant for Step 1ab, the analogue for Step 1a of Proposition 174.67 for Step 1bb.

The domain of the companion ideal. In the general step (Proposition **??**), at a point where the order o of the nonmonomial part is positive, the value is computed from the invariant of the maximal-order step for the companion ideal (Definition **??**), formed from the transform at the stage where o was first reached and restricted to an open set (Włodarczyk, Step 2a: "By Step 1 we can resolve the marked ideal of maximal order $(\mathcal{J}, \mu_{\mathcal{J}}) := O(\mathcal{I}, \mu)$ [...] Then we repeat the procedure for $(\mathcal{I}_{r_1}, \mu)$"). The statement below shows that the companion ideal lies in the domain of that invariant.

Proposition 174.79 (The companion ideal lies in the domain)**.** *Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ lie in the domain of a resolution invariant in dimension $\leq k$ (Definition **??**), $n > 0$, and $o \geq 1$. Write $\mathcal{N}_N(\mathfrak{a})$ and $\mathcal{M}_N(\mathfrak{a})$ for the nonmonomial and monomial parts along N (Proposition **??**).*

1. *If $\mathfrak{b}$ is nonzero along N at y, so are $\mathfrak{b}^j$ and $\mathfrak{b}+\mathfrak{c}$; and $\mathcal{N}_N(\mathfrak{a})$ is nonzero along N at every point of N.*

2. *The marked ideal $\underline{\mathcal{O}} = (M, N, E, \mathcal{N}_N(\mathfrak{a}), o) + (M, N, E, \mathcal{M}_N(\mathfrak{a}), d-o)$ if $o < d$, and $\underline{\mathcal{O}} = (M, N, E, \mathcal{N}_N(\mathfrak{a}), o)$ if $o \geq d$ (the companion ideal of Lemma **??** with $\mathfrak{b} = \mathcal{N}_N(\mathfrak{a})$, $\mathfrak{c} = \mathcal{M}_N(\mathfrak{a})$), lies in the domain in dimension $\leq k$.*

3. *The same holds for the companion ideal of $\underline{\mathcal{I}}_T$ along every multiple test blow-up T, for the companion ideal of $\underline{\mathcal{I}}|_W$ and for $\underline{\mathcal{O}}|_W$, $W \subset M$ open.*

Proof. (1) If $g \in \mathfrak{b}_y$ does not vanish identically on N near y, neither does $g^j \in (\mathfrak{b}^j)_y$, and $g \in (\mathfrak{b}+\mathfrak{c})_y$ (Definition **??**, Lemma **??**). Let $y \in N$ and $g \in \mathfrak{a}_y$ not vanishing identically on N near y. On N near y, $g = m\,c$ with m the local generator of $\mathcal{M}_N(\mathfrak{a})$ and $c \in \mathcal{N}_N(\mathfrak{a})_y$ (Proposition **??**); at the points of N near y where $g \neq 0$, also $c \neq 0$, so c does not vanish identically on N near y. (2) $\underline{\mathcal{O}}$ has the submanifold N and the divisor E of $\underline{\mathcal{I}}$, so it is well formed

as soon as its ideal is coherent: $\mathcal{N}_N(\mathfrak{a})$ and $\mathcal{M}_N(\mathfrak{a})$ are coherent (Proposition **??**), and so are powers and sums of coherent ideals (Definition **??**). Its mark is $o(d-o) \geq 1$ (Definition **??**), resp. $o \geq 1$, and $\dim N \leq k$. Its ideal is $\mathcal{N}_N(\mathfrak{a})^{d-o} + \mathcal{M}_N(\mathfrak{a})^o$, resp. $\mathcal{N}_N(\mathfrak{a})$, nonzero along N at every point of N by (1). (3) $\underline{\mathcal{I}}_T$ and $\underline{\mathcal{I}}|_W$ lie in the domain (Proposition 174.75); apply (2), resp. restrict the conclusion of (2) to W (Proposition 174.75(3)). □

The maximal order of $\underline{\mathcal{O}}$ along N and its cosupport are Lemmas **??** and **??**; together with the proposition above this is what the general step needs of $\underline{\mathcal{O}}$ to apply the invariant of Proposition **??** to it.

Restricted towers with empty centres. The values of the invariant at a point are computed along the part of a canonical segment after the stage at which the relevant order (the boundary count, the codimension-one part, the order of the nonmonomial part) was reached near the point; before that stage the centres lie over a closed set missing a neighbourhood of the point, and over that neighbourhood the blowups are blowups along the empty set, which the re-basing property of the invariant (Definition **??**, Lemma **??**) removes.

Lemma 174.80 (A restricted tower whose centres miss the open set is empty)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let T be a tower over X with centres $C_i \subset X_i$ and projections $p_i \colon X_i \to X$, and $U \subset X$ open. The restriction $T|_U$ (Lemma **??**) has only empty centres if and only if $C_i \cap p_i^{-1}(U) = \emptyset$ for every i. In particular, if every C_i lies in $p_i^{-1}(F)$ for a set F with $U \cap F = \emptyset$, then $T|_U$ has only empty centres. Both conditions pass to concatenations $T = T_0T_1$ (for T_1, with U and F replaced by their preimages in the last stage of T_0), hence to the head T_0 of a split of T (Lemma **??**).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

The centres of $T|_U$ are the traces $C_i \cap p_i^{-1}(U)$ of the centres of T on the open preimages of U (Lemma **??**); a trace on an open subset is empty iff the set misses it. If $C_i \subset p_i^{-1}(F)$ and $U \cap F = \emptyset$, a point of $C_i \cap p_i^{-1}(U)$ would map into $U \cap F$. For T_0T_1 the centres are those of T_0 followed by those of T_1, and the projections of the stages of T_1 to X are those to the last stage of T_0 followed by p_{T_0} (Definition **??**). The head of a split followed by its tail is T (Lemma **??**). □

Re-basing Step 1a's restricted marked ideal. In Step 1a of the maximal-order step (Proposition **??**) the restricted marked ideal of Lemma 174.78 is formed at a stage depending on the point and restricted to a neighbourhood W of the image of the point. A nearby point whose stage is later is compared by re-basing across the steps between the two stages, which are empty over W (Lemma 174.80). The marked ideal formed at the earlier stage and transformed along these steps is then the one formed at the later stage, up to restricted equivalence (Definition **??**), which the invariant does not see (Definition **??**(7)). This is the lemma of the source on the commutativity of Step 1 with étale morphisms (the blow-ups whose centres "do not hit the images" pull back to isomorphisms, and $\phi^*(\overline{\mathcal{J}}_{i_l|H^s_{\alpha i_l}}) = \overline{\mathcal{J}}'_{j_{l'}|(H')^s_{\alpha j_{l'}}}$), with steps of empty centre in place of isomorphisms.

Lemma 174.81 (Re-basing Step 1a's restricted marked ideal)**.** *Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed with $d \geq 1$, $n > 0$, and E_0 a list of members of E. For a point y let $S(y)$ be the set of members of E_0 through y and $\underline{\mathcal{S}}(\underline{\mathcal{I}}, E_0, y) = (M, N \cap H_{S(y)}, E_{S(y)}, \mathcal{C}_N(\mathfrak{a}), \mu_{\mathcal{C}})$ the restricted marked ideal of Lemma 174.78.*

1. *$H_{S(y)}$ is the intersection of the members of E which belong to E_0 and pass through y, and $E_{S(y)}$ is E without these members, in its order.*
2. *For $U \subset M$ open and $y \in U$, $\underline{\mathcal{S}}(\underline{\mathcal{I}}, E_0, y)|_U = \underline{\mathcal{S}}(\underline{\mathcal{I}}|_U, E_0|_U, y)$, where $E_0|_U$ is the list of the traces of the members of E_0 on U.*

3. *For the blowup $\pi\colon M' \to M$ along the empty set and $z \in M'$, $\underline{\mathcal{S}}(\underline{\mathcal{I}}', E_0', z)$ is restricted-equivalent to $\underline{\mathcal{S}}(\underline{\mathcal{I}}, E_0, \pi(z))'$, where primes denote transforms and E_0' is the list of the strict transforms of the members of E_0. The same holds along a tower T with only empty centres: $\underline{\mathcal{S}}(\underline{\mathcal{I}}_T, E_{0,T}, z)$ is restricted-equivalent to $\underline{\mathcal{S}}(\underline{\mathcal{I}}, E_0, p_T(z))_T$.*

4. *Let T be a multiple test blow-up of $\underline{\mathcal{I}}$ whose centres miss the preimages of an open set $W \subset M$, and p a point of the last stage over W. Then $\underline{\mathcal{S}}(\underline{\mathcal{I}}_T, E_{0,T}, p)|_{p_T^{-1}(W)}$ is restricted-equivalent to $(\underline{\mathcal{S}}(\underline{\mathcal{I}}, E_0, p_T(p))|_W)_{T|_W}$.*

Proof. (1) A member H_i has $i \in S(y)$ exactly when H_i is a member of E_0 and $y \in H_i$; $E_{S(y)}$ keeps the other members in their order. (2) The coefficient ideal along N commutes with restriction to open sets (Lemma **??**), and so do intersections with members and the deletion of members, provided a member through y is deleted on U exactly when it is deleted on M. Let $H \in E$ pass through y and have the trace on U of a member H_0 of E_0. Then H and H_0 agree on the neighbourhood U of y and both pass through y, so they are the same member (Lemma 174.122, first paragraph of the proof), and $H \in E_0$. (3) A blowup along the empty set is a biholomorphism (Lemma **??**); it is surjective, by its section over the complement of the centre (Lemma 174.30). The strict transform of a closed set is its preimage, and $E' = (\pi^{-1}H_1, \dots, \pi^{-1}H_s, \emptyset)$ (Lemma **??**). Preimages under a surjection determine the set, so $\pi^{-1}H \in E_0'$ iff $H \in E_0$; and $z \in \pi^{-1}H$ iff $\pi(z) \in H$, while the last member misses z. By (1), $N' \cap H'_{S(z)} = \pi^{-1}(N \cap H_{S(\pi(z))})$, the strict transform of $N \cap H_{S(\pi(z))}$, and $E'_{S(z)} = (E_{S(\pi(z))})'$. The coefficient ideal of $\underline{\mathcal{I}}'$ restricted to this strict transform is restricted-equivalent to the transform of the restricted coefficient ideal (Lemma **??**). Along T argue by induction on its length: the stages are well formed (Lemma **??**; a tower with only empty centres is a multiple test blow-up, Lemma **??**), the strict transforms of the members of E_0 are members of E_T (Lemma 174.55), and restricted-equivalent marked ideals have the same transform (Lemma **??**), so the restricted equivalence of the previous stage passes to the transforms. (4) $T|_W$ has only empty centres (Lemma 174.80). Apply (3) to $\underline{\mathcal{I}}|_W$, $E_0|_W$ and $T|_W$ at p. Here $(\underline{\mathcal{I}}|_W)_{T|_W} = \underline{\mathcal{I}}_T|_{p_T^{-1}(W)}$ (Lemma **??**), the strict transforms of traces are the traces of strict transforms (Lemma **??**), and (2) at both ends (with $\underline{\mathcal{I}}$ and $\underline{\mathcal{I}}_T$ well formed, Lemma **??**) takes the restriction outside. □

Lemma 174.82 (Re-basing the values of a resolution invariant)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let inv *be a resolution invariant in dimension $\leq k$ (Definition **??**).*

1. *If $\underline{\mathcal{I}}$ lies in the domain and $\underline{\mathcal{J}}$, with a coherent ideal, is restricted-equivalent to it, then $\underline{\mathcal{J}}$ lies in the domain. Along a tower with only empty centres the transform of a marked ideal of the domain lies in the domain.*

2. *If $\underline{\mathcal{I}}$ and $\underline{\mathcal{J}}$ lie in the domain and are restricted-equivalent, every canonical segment of $\underline{\mathcal{I}}$ is a canonical segment of $\underline{\mathcal{J}}$.*

3. *Let $\underline{\mathcal{I}}$ lie in the domain, T_0 be a tower with only empty centres, and $\underline{\mathcal{J}}$, with a coherent ideal, be restricted-equivalent to $\underline{\mathcal{I}}_{T_0}$. If T_0T_1 is a canonical segment of $\underline{\mathcal{I}}$, then $\underline{\mathcal{J}}$ lies in the domain, T_1 is a canonical segment of $\underline{\mathcal{J}}$, and $\operatorname{inv}_{\underline{\mathcal{I}}}(T_0T_1) = \operatorname{inv}_{\underline{\mathcal{J}}}(T_1)$ on $\operatorname{cosupp}\underline{\mathcal{I}}_{T_0T_1}$.*

4. *(Step 1a) In the situation of Lemma 174.81(4), let $\underline{\mathcal{S}}_0 = \underline{\mathcal{S}}(\underline{\mathcal{I}}, E_0, p_T(p))|_W$ lie in the domain and $T|_WT_1$ be a canonical segment of it, and let $\underline{\mathcal{S}}_1 = \underline{\mathcal{S}}(\underline{\mathcal{I}}_T, E_{0,T}, p)|_{p_T^{-1}(W)}$. Then $\underline{\mathcal{S}}_1$ lies in the domain, T_1 is a canonical segment of it, its transforms along T_1 lie in the domain, and $\operatorname{inv}_{\underline{\mathcal{S}}_0}(T|_WT_1) = \operatorname{inv}_{\underline{\mathcal{S}}_1}(T_1)$ on the cosupport.*

5. *(The general step) Let $\underline{\mathcal{I}}$ lie in the domain, $n > 0$, $o \geq 1$, $\underline{\mathcal{O}}$ its companion ideal (Proposition 174.79), W open and T a tower whose centres miss the preimages of W. Then $(\underline{\mathcal{O}}|_W)_{T|_W}$ lies in the domain; and if $T|_WT_1$ is a canonical segment of $\underline{\mathcal{O}}|_W$, then T_1 is*

one of $(\underline{\mathcal{Q}}|_W)_{T|_W}$*, the transforms along* $T|_W T_1$ *lie in the domain, and* $\mathrm{inv}_{\underline{\mathcal{Q}}|_W}(T|_W T_1) = \mathrm{inv}_{(\underline{\mathcal{Q}}|_W)_{T|_W}}(T_1)$ *on the cosupport.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) $\underline{\mathcal{J}}$ has the submanifold, divisor and mark of $\underline{\mathcal{I}}$, so it is well formed with $\dim N \leq k$ and $d \geq 1$. At $y \in N$ let $g \in \mathfrak{a}_y$ not vanish identically on N near y; on N near y, $g = \sum_i a_i b_i$ with b_i in the stalk of the ideal of $\underline{\mathcal{J}}$, and if every b_i vanished identically on N near y, so would g (Lemma **??**). A tower with only empty centres is a multiple test blow-up (Lemma **??**(4)), along which the domain is stable (Proposition 174.75). (2) Induction along the segment. At every stage the transforms of $\underline{\mathcal{I}}$ and $\underline{\mathcal{J}}$ have the same cosupport (Lemma **??**), and on it their values agree by Definition **??**(7) along the canonical prefix; so their maximum loci agree, and a centre which is empty or the maximum locus for $\underline{\mathcal{I}}$ is so for $\underline{\mathcal{J}}$. (3) $\underline{\mathcal{I}}_{T_0}$ lies in the domain by (1), and T_1 is a canonical segment of it (Lemma **??**(4)). Restricted equivalence is symmetric, so (1) puts $\underline{\mathcal{J}}$ in the domain and (2) makes T_1 a canonical segment of $\underline{\mathcal{J}}$. For the values, re-base by Definition **??**(4') and then apply (7) along T_1. (4) Lemma 174.81(4) gives the restricted equivalence of $\underline{\mathcal{S}}_1$ with $(\underline{\mathcal{S}}_0)_{T|_W}$; $T|_W$ has only empty centres (Lemma 174.80), and $\underline{\mathcal{S}}_1$ has a coherent ideal (Lemma 174.78). Apply (3); the transforms along a canonical segment of a marked ideal of the domain lie in the domain (Theorem **??**). (5) $\underline{\mathcal{Q}}|_W$ lies in the domain (Proposition 174.79(3)) and $T|_W$ has only empty centres (Lemma 174.80); apply (1), Lemma **??**(4), Theorem **??** and Definition **??**(4'). □

The restricted equivalence of the companion ideals formed at two stages, the analogue of Lemma 174.81(4) for the general step, is not part of these statements.

Centres from local data. The invariant built in the inductive step (Propositions **??** and **??**) is given near each point by local data: on a neighbourhood of the point its values are (s_0, b), with b the invariant of a lower-dimensional marked ideal on the cosupport of that marked ideal, while the remaining points of the cosupport there have smaller values. The centre is the locus where the global invariant attains its maximum (Theorem **??**; Włodarczyk: "The blow-ups will be performed at the centers C [...] for which the invariant [...] attains its maximum"). The two statements below compute such a centre from the local data, and record how centres behave under restriction to open sets and under isomorphisms of towers.

Lemma 174.83 (Maximum loci from local data)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let Λ, Λ' *be totally ordered,* $f\colon Y \to \Lambda$, $s \subset Y$, $\iota\colon Y' \to Y$ *a map,* $s' \subset \iota^{-1}(s)$, $g\colon Y' \to \Lambda'$ *and* $\varphi\colon \Lambda' \to \Lambda$ *strictly increasing, with* $f(\iota(p)) = \varphi(g(p))$ *for* $p \in s'$*. Write* $\max_s f$ *for the set of points of* s *at which* f *attains its maximum on* s*.*

1. $\max_t(\varphi \circ g) = \max_t g$ *for every* t*.*
2. *If every point of* $\iota^{-1}(s) \setminus s'$ *has a value of* f *strictly below the value at some point of* s'*, then* $\iota^{-1}(\max_s f)$ *is empty or equal to* $\max_{s'} g$*. In particular, if* $f(\iota(p)) = g(p)$ *on* $\iota^{-1}(s)$*, then* $\iota^{-1}(\max_s f)$ *is empty or* $\max_{\iota^{-1}(s)} g$*.*
3. *If no point of* $\iota^{-1}(s) \setminus s'$ *has the value* l*, then* $\iota^{-1}\{y \in s : f(y) = l\}$ *is* $\{p \in s' : g(p) = \mu\}$ *if* $l = \varphi(\mu)$*, and empty if* l *is not a value of* φ*; and* $f \circ \iota \leq \varphi(\mu)$ *on* s' *gives* $g \leq \mu$ *on* s'*.*
4. *For a totally ordered* A *and* $a \in A$, $b \mapsto (a, b)$ *is strictly increasing from* Λ' *to* $A \times (\Lambda' \cup \{\infty\})$*, ordered lexicographically.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) $\varphi(b) \leq \varphi(b')$ iff $b \leq b'$. (2) A point p of $\iota^{-1}(\max_s f)$ outside s' would lie strictly below a point $\iota(q)$, $q \in s' \subset \iota^{-1}(s)$, against the maximality of $f(\iota(p))$; so $\iota^{-1}(\max_s f) \subset s'$. Let it contain p_0. If $p \in \iota^{-1}(\max_s f)$, then $p \in s'$ and $\varphi(g(q)) = f(\iota(q)) \leq f(\iota(p)) = \varphi(g(p))$ for $q \in s'$, so $p \in$

$\max_{s'} g$. Conversely, for $p \in \max_{s'} g$ and $y \in s$, $f(y) \le f(\iota(p_0)) = \varphi(g(p_0)) \le \varphi(g(p)) = f(\iota(p))$. The special case is $\varphi = \mathrm{id}$, $s' = \iota^{-1}(s)$. (3) On s', $f(\iota(p)) = l$ iff $\varphi(g(p)) = \varphi(\mu)$ iff $g(p) = \mu$ (φ is injective); off s' the value l does not occur. (4) The second components are compared when the first agree, and $\Lambda' \to \Lambda' \cup \{\infty\}$ is strictly increasing. □

Lemma 174.84 (Centres on open sets and along isomorphisms of towers)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ have a coherent ideal.

1. *For open sets $V \subset W \subset M$, the open subset $W \cap V$ of W is biholomorphic to V by the identity on points, and under this biholomorphism $(\underline{\mathcal{I}}|_W)|_{W\cap V}$ corresponds to $\underline{\mathcal{I}}|_V$.*

2. *Let $W \subset M$ be open and $C \subset M$. If every point of W has an open neighbourhood $V \subset W$ such that $C \cap V$ is an admissible centre of $\underline{\mathcal{I}}|_V$, then $C \cap W$ is an admissible centre of $\underline{\mathcal{I}}|_W$.*

3. *Let* inv *be a resolution invariant (Definition* **??***), $\underline{\mathcal{I}}$ in its domain and T a canonical segment. For $U \subset M$ open, the trace on the preimage of U of the maximum locus of* $\mathrm{inv}_{\underline{\mathcal{I}}}(T)$ *on* $\mathrm{cosupp}\,\mathcal{I}_T$ *is empty or the maximum locus of* $\mathrm{inv}_{\underline{\mathcal{I}}|_U}(T|_U)$ *on* $\mathrm{cosupp}(\underline{\mathcal{I}}|_U)_{T|_U}$*. For an isomorphism $\chi\colon T' \to T$ of towers over a biholomorphism Φ (Definition* **??***), the maximum locus of* $\mathrm{inv}_{\Phi^*\underline{\mathcal{I}}}(T')$ *on its cosupport is χ_r^{-1} of that of* $\mathrm{inv}_{\underline{\mathcal{I}}}(T)$*.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The identity on points and its inverse are holomorphic (Lemma 174.5). Restricting the members, N and the ideal to W and then to $W \cap V$ is restricting them to V along the composite inclusion, which is the inclusion of V after the biholomorphism; pullbacks of coherent ideals compose (Lemma **??**). (2) Admissible centres are local (Theorem **??**): apply this to $\underline{\mathcal{I}}|_W$ with the open sets $W \cap V$ of W; by (1) and the invariance of admissible centres under biholomorphisms (Lemma **??**), $C \cap W \cap V$ is admissible for $(\underline{\mathcal{I}}|_W)|_{W\cap V}$. (3) The canonical segment is a multiple test blow-up (Theorem **??**), so the cosupport along $T|_U$ is the trace of the cosupport along T (Lemma **??**) and, along T', its preimage under χ_r (Lemma **??**); the values agree there by Definition **??**(5), resp. (6). Apply Lemma 174.83(2) with $\varphi = \mathrm{id}$, resp. the pullback of a maximum locus along the surjection χ_r (Lemma **??**). □

Lemma 174.85 (A resolution invariant on a smaller open set)**.** *Let T be a tower over X, $V \subset W \subset X$ open, and $\underline{\mathcal{I}}$ a marked ideal on X with a coherent ideal.*

1. *Restricting T to W and then to the open subset $W \cap V$ of W gives a tower isomorphic to $T|_V$ over the biholomorphism $W \cap V \cong V$ of Lemma 174.84(1), with the identity on points at every stage (Definition* **??***).*

2. *Let* inv *be a resolution invariant (Definition* **??***), $\underline{\mathcal{I}}|_W$ in its domain and $T|_W$ a canonical segment of it. Then $\underline{\mathcal{I}}|_V$ lies in the domain, $T|_V$ is a canonical segment of it, and at the points of* $\mathrm{cosupp}(\underline{\mathcal{I}}|_V)_{T|_V}$ *the values* $\mathrm{inv}_{\underline{\mathcal{I}}|_V}(T|_V)$ *and* $\mathrm{inv}_{\underline{\mathcal{I}}|_W}(T|_W)$ *agree.*

Proof. (1) Induction along T: the stages of both towers are the open preimages of V, with the same points, and the identity on points between open sets with the same points is biholomorphic (Lemma 174.5); the centres of both are the traces of the centres of T (Lemma **??**), and the blowup maps commute with the identity on points. (2) $(\underline{\mathcal{I}}|_W)|_{W\cap V}$ lies in the domain (Proposition 174.75) and $(T|_W)|_{W\cap V}$ is a canonical segment of it (Lemma **??**(1)); by Lemma 174.84(1) this marked ideal corresponds to $\underline{\mathcal{I}}|_V$ under the biholomorphism, and by (1) and Lemma **??**(2) $T|_V$ is a canonical segment of $\underline{\mathcal{I}}|_V$, which lies in the domain (Proposition 174.75). The cosupports correspond (Lemma **??**), and the values agree by Definition **??**(6) along the isomorphism of (1) and (5) for the restriction from W to $W \cap V$. □

The monomial and nonmonomial parts under local biholomorphisms. In the general step (Proposition **??**) the companion ideal (Definition **??**) is formed from the monomial and nonmonomial parts along N of the transform at the stage where the order of the nonmonomial part was first reached near a point, restricted to an open set; nearby points are compared through restrictions to open sets, biholomorphisms and steps with empty centre (Definition **??**(4'), (5), (6)). All three are *local biholomorphisms*: holomorphic maps $\varphi\colon M' \to M$ with a holomorphic local inverse at every point (Lemma **??**): the inclusion of an open set, a biholomorphism, a blowup along $\emptyset$ (Lemma **??**). The lemma below shows that both parts commute with local biholomorphisms, depend only on the restriction of the ideal to N, and ignore empty members of E; this is how Włodarczyk's statement that the resolution procedure commutes with étale morphisms reaches the companion ideal.

Lemma 174.86 (Monomial and nonmonomial parts under local biholomorphisms)**.** *Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be as in Lemma **??**, with $\mathfrak{a}$ an ideal of holomorphic germs, and write $\mathcal{M}_N(\mathfrak{a})$, $\mathcal{N}_N(\mathfrak{a})$ for its monomial and nonmonomial parts along N (Definition **??**).*

1. *Let $\varphi\colon M' \to M$ be holomorphic with a holomorphic local inverse at every point. Then $\operatorname{cosupp}((\varphi^*\mathfrak{a})|_{\varphi^{-1}N}, e) = \varphi^{-1}\operatorname{cosupp}(\mathfrak{a}|_N, e)$ for every e, and $\varphi^*\mathcal{M}_N(\mathfrak{a}) = \mathcal{M}_{\varphi^{-1}N}(\varphi^*\mathfrak{a})$, the right-hand side formed with the members $\varphi^{-1}H_i$. For coherent $\mathfrak{b}$, $\mathfrak{l}$ with $(\widetilde{\mathfrak{b}}^W : \mathfrak{l})$ coherent (Lemma **??**), $\varphi^*(\widetilde{\mathfrak{b}}^W : \mathfrak{l}) = (\widetilde{\varphi^*\mathfrak{b}}^{\varphi^{-1}W} : \varphi^*\mathfrak{l})$; hence, if $\mathfrak{a}$, $\mathcal{M}_N(\mathfrak{a})$ and $\mathcal{N}_N(\mathfrak{a})$ are coherent, $\varphi^*\mathcal{N}_N(\mathfrak{a}) = \mathcal{N}_{\varphi^{-1}N}(\varphi^*\mathfrak{a})$.*
2. *If N is closed and $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N \mathfrak{a}'$, then $\mathcal{M}_N(\mathfrak{a}') = \mathcal{M}_N(\mathfrak{a})$ and $\mathcal{N}_N(\mathfrak{a}') = \mathcal{N}_N(\mathfrak{a})$.*
3. *Appending the empty set to E as a last member changes neither part.*

Proof. (1) Let $y \in M'$ and ψ a local inverse of φ at y. Composition with φ maps germs vanishing on H at $\varphi(y)$ to germs vanishing on $\varphi^{-1}H$ at y, and composition with ψ maps germs vanishing on $\varphi^{-1}H$ at y to germs vanishing on H at $\varphi(y)$ (Lemma **??**); by induction on e the same holds for the powers $\mathcal{I}_H^e$, whose germs are combinations of products of a germ of $\mathcal{I}_H^{e-1}$ and one of $\mathcal{I}_H$. Orders along N pull back along maps holomorphic near the point (Lemma **??**), and combinations of germs vanishing to order e along N vanish to order e (Lemma **??**): a germ of $\varphi^*\mathfrak{a}$ at y is a combination of germs $a \circ \varphi$, $a \in \mathfrak{a}_{\varphi(y)}$; conversely, if $a \circ \varphi$ vanishes to order e along $\varphi^{-1}N$ at y, then $a = (a \circ \varphi) \circ \psi$ near $\varphi(y)$ vanishes to order e along N, as $\varphi \circ \psi = \mathrm{id}$ near $\varphi(y)$. So the cosupports pull back; since φ maps the neighbourhoods of y onto those of $\varphi(y)$, the exponents pull back too, $e_{\varphi^{-1}H}(\varphi^*\mathfrak{a})(y) = e_H(\mathfrak{a})(\varphi(y))$ (Lemma **??**). Membership in $\mathcal{M}_N(\mathfrak{a})_y$ is a condition on each member through y and is closed under holomorphic combinations. A generator $b \circ \varphi$, $b \in \mathcal{M}_N(\mathfrak{a})_{\varphi(y)}$, satisfies it at y by the transfer of the powers along φ; conversely, for $g \in \mathcal{M}_{\varphi^{-1}N}(\varphi^*\mathfrak{a})_y$ the transfer along ψ gives $g \circ \psi \in \mathcal{M}_N(\mathfrak{a})_{\varphi(y)}$, and $g = (g \circ \psi) \circ \varphi$ near y. For a coherent $\mathfrak{c}$, $f \in (\varphi^*\mathfrak{c})_y$ if and only if $f \circ \psi \in \mathfrak{c}_{\varphi(y)}$: a coherent ideal is closed under holomorphic combinations and its membership only depends on the germ (Definition **??**), and $(b \circ \varphi) \circ \psi = b$ near $\varphi(y)$. The same holds for the restricted germs of $\widetilde{\mathfrak{b}}^W$ on W and of $\widetilde{\varphi^*\mathfrak{b}}^{\varphi^{-1}W}$ on $\varphi^{-1}W$, since ψ maps the points of W near $\varphi(y)$ into $\varphi^{-1}W$; so f lies in the quotient at y if and only if $f \circ \psi$ does at $\varphi(y)$. The nonmonomial part is the quotient $(\widetilde{\mathfrak{a}}^N : \mathcal{M}_N(\mathfrak{a}))$ (Definition **??**), and $\varphi^*\mathcal{M}_N(\mathfrak{a}) = \mathcal{M}_{\varphi^{-1}N}(\varphi^*\mathfrak{a})$. (2) The exponents only depend on the cosupports along N, and $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N \mathfrak{a}'$ gives the same cosupports and the same extensions $\widetilde{\mathfrak{a}}^N = \widetilde{\mathfrak{a}'}^N$ (Lemma **??**). (3) No point lies on the empty member, so its condition in the definition of $\mathcal{M}_N$ is vacuous, and the nonmonomial part is the quotient by the same monomial part. □

Re-basing the companion ideal. In the general step (Proposition **??**) the value at a point where the order o of the nonmonomial part is positive comes from the companion ideal formed at the stage where o was first reached near the point, restricted to an open set, and nearby points whose stages differ are compared by re-basing across the steps between the two stages,

which are empty near them (Definition **??**(4'), Lemma 174.80). This is the general-step twin of Lemma 174.81. For $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ and $o \geq 1$ write $\mathcal{O}(\underline{\mathcal{I}}, o) = (M, N, E, \mathcal{N}_N(\mathfrak{a}), o) + (M, N, E, \mathcal{M}_N(\mathfrak{a}), d - o)$ if $o < d$ and $(M, N, E, \mathcal{N}_N(\mathfrak{a}), o)$ if $o \geq d$, the companion ideal of Lemma **??** at the monomial and nonmonomial parts, as in Proposition 174.79.

Lemma 174.87 (The companion ideal under local biholomorphisms, empty steps and re-basing)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n > 0$ and $o \geq 1$.

1. *For coherent $\mathfrak{b}$, $\mathfrak{c}$ and a holomorphic $\varphi\colon M' \to M$, the pullback of $(M, N, E, \mathfrak{b}, o) + (M, N, E, \mathfrak{c}, d - o)$ is $(M', \varphi^{-1}N, \varphi^{-1}E, \varphi^*\mathfrak{b}, o) + (M', \varphi^{-1}N, \varphi^{-1}E, \varphi^*\mathfrak{c}, d - o)$, and likewise for the single term if $o \geq d$.*

2. *Let $\underline{\mathcal{I}}$ be well formed with $\mathfrak{a}$ nonzero along N at every point of N. For a holomorphic φ with a holomorphic local inverse at every point, $\varphi^*\mathcal{O}(\underline{\mathcal{I}}, o) = \mathcal{O}(\varphi^*\underline{\mathcal{I}}, o)$; in particular $\mathcal{O}(\underline{\mathcal{I}}, o)|_U = \mathcal{O}(\underline{\mathcal{I}}|_U, o)$ for $U \subset M$ open and $\Phi^*\mathcal{O}(\underline{\mathcal{I}}, o) = \mathcal{O}(\Phi^*\underline{\mathcal{I}}, o)$ for a biholomorphism Φ. For the blowup π along $\emptyset$, $\mathcal{O}(\underline{\mathcal{I}}', o) \simeq \mathcal{O}(\underline{\mathcal{I}}, o)'$ (restricted equivalence, Definition **??**).*

3. *If $\underline{\mathcal{I}} \simeq \underline{\mathcal{J}}$ and N is closed, then $\mathcal{O}(\underline{\mathcal{I}}, o) = \mathcal{O}(\underline{\mathcal{J}}, o)$.*

4. *Let $\underline{\mathcal{I}}$ lie in the domain of resolution invariants in dimension $\leq k$ (Definition **??**). Along a tower T with only empty centres, $\mathcal{O}(\underline{\mathcal{I}}_T, o) \simeq \mathcal{O}(\underline{\mathcal{I}}, o)_T$. If T is a multiple test blow-up of $\underline{\mathcal{I}}$ whose centres miss the open $W \subset M$, then $\mathcal{O}(\underline{\mathcal{I}}_T, o)|_{T^{-1}W} \simeq (\mathcal{O}(\underline{\mathcal{I}}, o)|_W)_{T|_W}$.*

5. *In the situation of (4), let* inv *be a resolution invariant in dimension $\leq k$ and $T|_W\, T_1$ a canonical segment of $\mathcal{O}(\underline{\mathcal{I}}, o)|_W$. Then $\mathcal{O}(\underline{\mathcal{I}}_T, o)|_{T^{-1}W}$ lies in the domain, T_1 is a canonical segment of it, its transforms along T_1 lie in the domain, and at the points of the cosupport along $T|_W\, T_1$*

$$\mathrm{inv}_{\mathcal{O}(\underline{\mathcal{I}}, o)|_W}(T|_W\, T_1) = \mathrm{inv}_{\mathcal{O}(\underline{\mathcal{I}}_T, o)|_{T^{-1}W}}(T_1).$$

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Pullback along a holomorphic map commutes with sums and powers of ideals of holomorphic germs (Lemmas **??** and 174.1), and the operations of Definition **??** only combine the ideals and the marks. (2) The parts are coherent (Proposition 174.79), so by (1) and Lemma 174.86(1) $\varphi^*\mathcal{O}(\underline{\mathcal{I}}, o)$ is the companion ideal of $\varphi^*\underline{\mathcal{I}}$; the inclusion of an open set and a biholomorphism have holomorphic local inverses. For π along $\emptyset$: both sides have the submanifold $\pi^{-1}N$ and the divisor $\pi^{-1}E$ followed by the empty exceptional member, and the same mark (Lemma **??**). On $\pi^{-1}N$ the transform $\mathfrak{a}'$ has the restriction of $\pi^*\mathfrak{a}$ (Lemma **??**), so $\mathcal{M}_{\pi^{-1}N}(\mathfrak{a}') = \mathcal{M}_{\pi^{-1}N}(\pi^*\mathfrak{a}) = \pi^*\mathcal{M}_N(\mathfrak{a})$ and likewise for $\mathcal{N}$ (Lemma 174.86(1)–(3), π having holomorphic local inverses); hence the ideal of $\mathcal{O}(\underline{\mathcal{I}}', o)$ is π^* of the ideal of $\mathcal{O}(\underline{\mathcal{I}}, o)$ by (1), which has on $\pi^{-1}N$ the restriction of the ideal of $\mathcal{O}(\underline{\mathcal{I}}, o)'$ (Lemma **??** again). (3) The parts only depend on $\mathrm{Res}_N\, \mathfrak{a}$ (Lemma 174.86(2)), and N, E, d agree. (4) Induction along T with (2): restricted-equivalent marked ideals have the same transform (Lemma **??**; the stages and their companion ideals lie in the domain, Propositions 174.75 and 174.79). For the second statement, $T|_W$ has only empty centres (Lemma 174.80), $(\underline{\mathcal{I}}|_W)_{T|_W}$ is $\underline{\mathcal{I}}_T|_{T^{-1}W}$ (Lemma **??**), and the companion ideal commutes with restriction to open sets by (2), at both stages. (5) Lemma 174.82 with $T_0 = T|_W$ and the restricted equivalence of (4); the transforms along a canonical segment lie in the domain (Theorem **??**). □

The companion ideal on an open set. The general step (Proposition **??**) takes, at a point where the order o of the nonmonomial part is positive, an open set W on which this order is at most o at the points of the cosupport, and the companion ideal of $\underline{\mathcal{I}}|_W$. Lemmas **??** and **??** are stated for an abstract factorization $(M, N, E, \mathfrak{bc}, d)$ with the bound on all of M; the statement below is their form for the marked ideal itself, on an open set, which is the open-set statement Lemma **??** records as missing. We write $\mathcal{O}(\underline{\mathcal{I}}, o)$ as in Lemma 174.87.

Lemma 174.88 (The companion ideal on an open set)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n > 0$ and $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed with $\mathfrak{a}$ nonzero along N at every point of N; write $\mathcal{N} = \mathcal{N}_N(\mathfrak{a})$, $\mathcal{M} = \mathcal{M}_N(\mathfrak{a})$.

1. *$\underline{\mathcal{I}} \simeq (M, N, E, \mathcal{N}\mathcal{M}, d)$ (restricted equivalence); in particular both have the cosupport of $\underline{\mathcal{I}}$. For every q the set of points of N where $\operatorname{ord}(\mathcal{N}|_N) \geq q$ is closed.*

2. *Let $o \geq 1$ with $\operatorname{ord}_x(\mathcal{N}|_N) \leq o$ at every $x \in \operatorname{cosupp}\underline{\mathcal{I}}$. Then $\mathcal{O}(\underline{\mathcal{I}}, o)$ is of maximal order along N (Definition **??**); every multiple test blow-up T of $\mathcal{O}(\underline{\mathcal{I}}, o)$ is one of $\underline{\mathcal{I}}$ and of $(M, N, E, \mathcal{N}, o)$; along it $\operatorname{ord}_z(\mathcal{N}_T|_{N_T}) \leq o$ at every $z \in \operatorname{cosupp}\underline{\mathcal{I}}_T$, and $\operatorname{cosupp}\mathcal{O}(\underline{\mathcal{I}}, o)_T = \operatorname{cosupp}(\mathcal{N}_T, o) \cap \operatorname{cosupp}\underline{\mathcal{I}}_T$.*

3. *Let $W \subset M$ be open, $o \geq 1$, and $\operatorname{ord}_x(\mathcal{N}|_N) \leq o$ only at the points $x \in \operatorname{cosupp}\underline{\mathcal{I}} \cap W$. Then the conclusions of (2) hold for $\underline{\mathcal{I}}|_W$ and $\mathcal{O}(\underline{\mathcal{I}}|_W, o) = \mathcal{O}(\underline{\mathcal{I}}, o)|_W$ (if $\underline{\mathcal{I}}$ lies in the domain of resolution invariants, Definition **??**).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) On N, $\mathfrak{a} = \mathcal{M}\mathcal{N}$ (Proposition **??**), so the restrictions agree; restricted-equivalent marked ideals have the same cosupport (Lemma **??**). $\mathcal{N}$ is coherent (Proposition **??**) and the cosupport along a closed submanifold of a coherent ideal is closed (Lemma **??**). (2) By (1) the bound holds on $\operatorname{cosupp}((\mathcal{N}\mathcal{M})|_N, d)$, and Lemmas **??** and **??** apply with $\mathfrak{b} = \mathcal{N}$, $\mathfrak{c} = \mathcal{M}$. They give the statements for $(M, N, E, \mathcal{N}\mathcal{M}, d)$, which is equivalent to $\underline{\mathcal{I}}$ by (1) (Lemma **??**: the same multiple test blow-ups and the same cosupports along them). (3) $\underline{\mathcal{I}}|_W$ is well formed and nonzero along $N \cap W$ (Proposition 174.75); its cosupport is $\operatorname{cosupp}\underline{\mathcal{I}} \cap W$ (Lemma **??**), its nonmonomial part is $\mathcal{N}|_W$ and the cosupports along $N \cap W$ of $\mathcal{N}|_W$ are the traces of those of $\mathcal{N}$ (Lemma 174.86(1), the inclusion of W having holomorphic local inverses, Proposition **??**). So the bound holds on $\operatorname{cosupp}(\underline{\mathcal{I}}|_W)$, and (2) applies to $\underline{\mathcal{I}}|_W$; the last equality is Lemma 174.87(2). □

Order zero of the nonmonomial part. In the general step (Proposition **??**), at a point where the order of the nonmonomial part is 0 the marked ideal is monomial near the point and the invariant is that of the monomial case (Lemma **??**). The lemma below gives the open set on which this holds.

Lemma 174.89 (Where the nonmonomial part has order zero)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ with $\mathfrak{a}$ coherent, and write $\mathcal{M} = \mathcal{M}_N(\mathfrak{a})$, $\mathcal{N} = \mathcal{N}_N(\mathfrak{a})$.

1. *If $y \in N$ and $\operatorname{ord}_y(\mathcal{N}|_N) = 0$, then $\operatorname{Res}_N \mathcal{M} \subset \operatorname{Res}_N \mathfrak{a}$ at y.*

2. *If $W \subset M$ is open and $\operatorname{ord}_y(\mathcal{N}|_N) = 0$ at every $y \in N \cap W$, then $\underline{\mathcal{I}}|_W$ is monomial along $N \cap W$ (Definition **??**).*

3. *If $n > 0$ and $\underline{\mathcal{I}}$ is well formed with $\mathfrak{a}$ nonzero along N at every point of N, the set W of points that are off N or where $\operatorname{ord}(\mathcal{N}|_N) = 0$ is open, and $\underline{\mathcal{I}}|_W$ is monomial along $N \cap W$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The order is 0, so some $c \in \mathcal{N}_y$ does not vanish at y (Lemma **??**). As $\mathcal{N}$ is the quotient $(\widetilde{\mathfrak{a}}^N : \mathcal{M})$ (Definition **??**, Lemma **??**), for $g \in \mathcal{M}_y$ the restriction of gc to N lies in $\operatorname{Res}_N \mathfrak{a}$ near y; c is holomorphic and nonzero near y, so it can be cancelled from this relation. (2) By (1) the restricted inclusion holds at the points of $N \cap W$; it pulls back along the inclusion of W (Lemma **??**), and $\mathcal{M}_{N\cap W}(\mathfrak{a}|_W) = \mathcal{M}|_W$ (Lemma 174.86(1), the inclusion having holomorphic local inverses, Proposition **??**). (3) The points of N where $\operatorname{ord}(\mathcal{N}|_N) \geq 1$ form a closed set (Lemma 174.88(1)); its complement is W; apply (2). □

The order of the nonmonomial part along a tower. In the general step (Proposition **??**) the value at a point y of the cosupport along a tower T begins with the order at y of the nonmonomial part of the transform $\underline{\mathcal{I}}_T$, taken with its whole divisor E_T. The lemma below collects the properties of this order that do not depend on the invariant.

Lemma 174.90 (The order of the nonmonomial part)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $n \geq 1$. For a well-formed marked ideal $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ with $\mathfrak{a}$ nonzero along N at every point of N, write $\nu_{\underline{\mathcal{I}}}(y) = \operatorname{ord}_y(\mathcal{N}_N(\mathfrak{a})|_N)$ for $y \in N$ (Proposition **??**).*

1. *$\nu_{\underline{\mathcal{I}}}(y)$ is finite for every $y \in N$, the sets $\{\nu_{\underline{\mathcal{I}}} \geq q\}$ are closed, and $\nu_{\underline{\mathcal{I}}}$ is upper semicontinuous on N.*

2. *If $\varphi\colon M' \to M$ is a local biholomorphism (for instance the inclusion of an open set, or a biholomorphism), then $\nu_{\varphi^*\underline{\mathcal{I}}} = \nu_{\underline{\mathcal{I}}} \circ \varphi$. If π is the blowup along $\emptyset$ and $\underline{\mathcal{I}}'$ the transform, then $\nu_{\underline{\mathcal{I}}'} = \nu_{\underline{\mathcal{I}}} \circ \pi$.*

3. *Let $\underline{\mathcal{I}}$ moreover lie in the domain of Definition **??**, and let T be a multiple test blow-up of it. If the centres of T miss an open set $W \subset M$, then $\nu_{\underline{\mathcal{I}}_T} = \nu_{\underline{\mathcal{I}}} \circ p_T$ at the points over W; in particular, for a test blowup σ along C, $\nu_{\underline{\mathcal{I}}_\sigma}(z) = \nu_{\underline{\mathcal{I}}}(\sigma(z))$ whenever $\sigma(z) \notin C$. If $\chi\colon T' \to T$ is an isomorphism of towers over a biholomorphism Φ (Definition **??**), then $\nu_{(\Phi^*\underline{\mathcal{I}})_{T'}} = \nu_{\underline{\mathcal{I}}_T} \circ \chi$.*

4. *Let $o \geq 1$ with $\nu_{\underline{\mathcal{I}}} \leq o$ on $\operatorname{cosupp}\underline{\mathcal{I}}$, and let T be a multiple test blow-up of the companion ideal $\mathcal{O}(\underline{\mathcal{I}}, o)$ (Lemma **??**). Then T is a multiple test blow-up of $\underline{\mathcal{I}}$, $\nu_{\underline{\mathcal{I}}_T} \leq o$ on $\operatorname{cosupp}\underline{\mathcal{I}}_T$, and $\operatorname{cosupp}\mathcal{O}(\underline{\mathcal{I}}, o)_T \subset \{\nu_{\underline{\mathcal{I}}_T} \geq o\} \cap \operatorname{cosupp}\underline{\mathcal{I}}_T$.*

5. *Let $\underline{\mathcal{I}}$ lie in the domain and σ be a test blowup of it along C with $\nu_{\underline{\mathcal{I}}} = q \geq 1$ on C. Then $\nu_{\underline{\mathcal{I}}_\sigma} \leq q$ at the points of $\operatorname{cosupp}\underline{\mathcal{I}}_\sigma$ over C.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) $\mathfrak{a} \subset \mathcal{N}_N(\mathfrak{a})$ (Theorem **??**), and a germ of $\mathfrak{a}_y$ not vanishing on N near y has finite order along N: through a local retraction onto N a germ of infinite order along N vanishes on N near y (Lemma **??**). The sets $\{\nu \geq q\}$ are closed by Lemma 174.88(1); hence near $y \in N$ the closed set $\{\nu \geq \nu(y) + 1\}$ is avoided, and $\nu \leq \nu(y)$ there. The order is the least q with $y \notin \{\nu \geq q + 1\}$, so it is determined by the sets $\{\nu \geq q\}$ (and $\{\nu \geq 0\} = N$, Lemma **??**).

(2) $\varphi^*\mathcal{N}_N(\mathfrak{a}) = \mathcal{N}_{\varphi^{-1}N}(\varphi^*\mathfrak{a})$, and cosupports along N pull back along local biholomorphisms (Lemma 174.86; inclusions of open sets and biholomorphisms have holomorphic local inverses, Proposition **??**, Lemma 174.87). Across a blowup along $\emptyset$, $N' = \pi^{-1}N$, the divisor is $\pi^{-1}E$ followed by an empty member, $\operatorname{Res}_{N'}\mathfrak{a}' = \operatorname{Res}_{N'}\pi^*\mathfrak{a}$, and π is a local biholomorphism (Lemma **??**); the nonmonomial part only depends on the restriction to N' and ignores an appended empty member (Lemma 174.86), so it is $\pi^*\mathcal{N}_N(\mathfrak{a})$, and the first assertion applies.

(3) Over W the tower restricts to one with only empty centres (Lemma 174.80), along which (2) applies stage by stage, the stages staying in the domain (Lemma 174.82); the transform of $\underline{\mathcal{I}}|_W$ along $T|_W$ is the restriction of $\underline{\mathcal{I}}_T$ to the preimage of W (Lemma **??**), restriction to open sets does not change the order by (2), and $p_{T|_W} = p_T$ on points (Lemma 174.6). For one blowup take $W = M \setminus C$. For isomorphisms of towers, $(\Phi^*\underline{\mathcal{I}})_{T'} = \chi^*\underline{\mathcal{I}}_T$ (Lemma **??**) with $\underline{\mathcal{I}}_T$ in the domain (Lemma 174.75); apply (2) to χ.

(4) Theorem **??** gives the first two assertions, and $\operatorname{Res}_{N_T}\mathcal{N}_{N_T}(\mathfrak{a}_T) \subset \operatorname{Res}_{N_T}\mathcal{N}_T$ for the transform $\mathcal{N}_T$ of $(M, N, E, \mathcal{N}_N(\mathfrak{a}), o)$, which lives on N_T. By Lemma 174.88(2), $\operatorname{cosupp}\mathcal{O}(\underline{\mathcal{I}}, o)_T = \operatorname{cosupp}(\mathcal{N}_T|_{N_T}, o) \cap \operatorname{cosupp}\underline{\mathcal{I}}_T$, and by the restricted inclusion $\operatorname{cosupp}(\mathcal{N}_T|_{N_T}, o) \subset \{\nu_{\underline{\mathcal{I}}_T} \geq o\}$ (Lemma **??**).

(5) The set $W = \{\nu_{\underline{\mathcal{I}}} \leq q\}$ is open by (1) and contains C; on W the order is at most q. By (2) and Lemma 174.88(2) (for the empty tower), C lies in $\{\nu \geq q\} \cap \operatorname{cosupp}\underline{\mathcal{I}}|_W = \operatorname{cosupp}\mathcal{O}(\underline{\mathcal{I}}|_W, q)$;

the companion ideal has the submanifold and the divisor of $\underline{\mathcal{I}}|_W$, and C is admissible for $\underline{\mathcal{I}}|_W$ (Lemma **??**), so the restriction of σ to $\sigma^{-1}W$ is a test blowup of $\mathcal{O}(\underline{\mathcal{I}}|_W, q)$ (Definition **??**). By (4), $\nu \le q$ on the cosupport of its transform, which is the restriction of $\underline{\mathcal{I}}_\sigma$ to $\sigma^{-1}W$ and contains the cosupport points over C; conclude with (2) for the inclusion of $\sigma^{-1}W$. □

The stage at which a point reached its order. In the general step the companion ideal at a point y of order $o \ge 1$ is formed at the stage at which the images of y first had order o, on a neighbourhood on which the order is at most o; nearby points are compared across the steps between their stages. The lemma below is the analogue for the general step of Lemmas **??** and **??** for Step 1.

Lemma 174.91 (The regime of the order of the nonmonomial part)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $n \ge 1$, $\underline{\mathcal{I}}$ a marked ideal in the domain of Definition **??**, ν the order of the nonmonomial part of Lemma 174.90, and $o \in \mathbf{N}$. A point y of the last stage of a tower T is* in the regime o *if $y \in \operatorname{cosupp} \underline{\mathcal{I}}_T$ and $\nu_{\underline{\mathcal{I}}_T}(y) = o$; write $r_o(T, y)$ for its run (Lemma **??**).*

1. *The runs of Lemma **??**(2), (3) under restriction to open sets and along isomorphisms of towers hold as soon as the regimes correspond along the towers with a property inherited by prefixes. Along multiple test blow-ups of $\underline{\mathcal{I}}$, the regime o and its run correspond under restriction to an open $U \subset X$ (for $\underline{\mathcal{I}}|_U$ and $T|_U$) and along an isomorphism of towers over a biholomorphism Φ (for $\Phi^*\underline{\mathcal{I}}$); off the centre of a test blowup σ, z is in the regime o iff $\sigma(z)$ is.*

2. *(The mask) Let $o \ge 1$, T_0T_1 a multiple test blow-up of $\underline{\mathcal{I}}$, and W an open set of the last stage of T_0 with $\nu_{\underline{\mathcal{I}}_{T_0}} \le o$ on $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0} \cap W$, such that $T_1|_W$ is a multiple test blow-up of the companion ideal $\mathcal{O}(\underline{\mathcal{I}}_{T_0}|_W, o)$. Then every point z of $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0T_1}$ over W with $\nu(z) \ge o$ is in the regime o and $r_o(T_0T_1, z) \ge \operatorname{length} T_1$.*

3. *Let $W \subset X$ be open with $\nu_{\underline{\mathcal{I}}} \le o$ on $\operatorname{cosupp} \underline{\mathcal{I}} \cap W$, and T a multiple test blow-up of $\underline{\mathcal{I}}$ each of whose centres lies in $\{\nu \ge o+1\}$ at its stage. Then the centres of T miss the preimages of W, and $\nu_{\underline{\mathcal{I}}_T} \le o$ at the points of $\operatorname{cosupp} \underline{\mathcal{I}}_T$ over W. Consequently, for $o \ge 1$, the companion ideal $\mathcal{O}(\underline{\mathcal{I}}_T, o)|_{p_T^{-1}W}$ re-bases to $(\mathcal{O}(\underline{\mathcal{I}}, o)|_W)_{T|_W}$ with the conclusions of Lemma 174.87(5).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Induction along the tower as in Lemma **??**, using the property only for the prefixes. Under restriction, the transform of $\underline{\mathcal{I}}|_U$ along $T|_U$ is the restriction of $\underline{\mathcal{I}}_T$ (Lemma **??**), cosupports restrict (Lemma **??**) and so does ν (Lemma 174.90(2)); along an isomorphism χ, cosupports correspond (Lemma **??**) and so does ν (Lemma 174.90(3)); off the centre the cosupport (Lemma 174.31) and ν (Lemma 174.90(3)) are those of the image.

(2) Write $\underline{\mathcal{I}}_0 = \underline{\mathcal{I}}_{T_0}$, in the domain (Lemma 174.75), and $\underline{\mathcal{I}}_{T_0T_1} = (\underline{\mathcal{I}}_0)_{T_1}$ (Lemma **??**). For every prefix S of T_1, $S|_W$ is a multiple test blow-up of $\mathcal{O}(\underline{\mathcal{I}}_0|_W, o)$, so by Lemma 174.90(4) on W (with (2) of that lemma for the restrictions) $\nu \le o$ at the cosupport points over W, and the centre of the next step, over W, lies in the cosupport of the transform of the companion ideal, hence in $\{\nu \ge o\}$. Induction along T_1: z is in the regime o ($\nu(z) \ge o$ and $\le o$); its image $w = \sigma(z)$ lies in the cosupport (in the centre, or off it by Lemma 174.31) and has $\nu(w) \ge o$ (in the centre by the above, off it by Lemma 174.90(3)); so w is in the regime, $r_o(z) = r_o(w) + 1$, and by induction $r_o(w) \ge \operatorname{length} S$.

(3) Induction along T. A centre C lies in the cosupport, where over W the order is $\le o$ by induction, and in $\{\nu \ge o + 1\}$; so it misses the preimage of W. Then a cosupport point over W is off the centre, its image lies in the cosupport (Lemma 174.31) and has the same order (Lemma 174.90(3)), at most o. The re-basing is Lemma 174.87(5), whose hypothesis is that the centres miss W (Lemma 174.80). □

The order of the nonmonomial part along canonical segments. In the general step (Proposition **??**) the value at a point y of the cosupport begins with the order $\nu(y)$ of the nonmonomial part, and along a canonical segment every centre is empty or the maximum locus of the value. The lemma below collects what follows from this first entry alone, for every value function whose order dominates ν; it supplies the hypotheses of Lemma 174.91(3) at the start of the run of a point.

Lemma 174.92 (The order of the nonmonomial part along canonical segments)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n \geq 1$, $\underline{\mathcal{I}}$ a marked ideal in the domain of Definition **??** *and ν the order of the nonmonomial part of Lemma 174.90. Let f assign to every tower T over X a function $f(T)$ on its last stage with values in a totally ordered set, such that along every multiple test blow-up T of $\underline{\mathcal{I}}$, $f(T)(y) \leq f(T)(y')$ implies $\nu_{\underline{\mathcal{I}}_T}(y) \leq \nu_{\underline{\mathcal{I}}_T}(y')$ for $y, y' \in \operatorname{cosupp} \underline{\mathcal{I}}_T$; for instance $f(T)(y) = (\nu_{\underline{\mathcal{I}}_T}(y), g(T)(y))$ ordered lexicographically, for any g.*

1. *Let σ be a test blowup of $\underline{\mathcal{I}}$ along C with $\nu_{\underline{\mathcal{I}}} = q$ on C, for any $q \geq 0$. Then $\nu_{\underline{\mathcal{I}}_\sigma} \leq q$ at the points of $\operatorname{cosupp} \underline{\mathcal{I}}_\sigma$ over C. If $q = 0$, $\nu_{\underline{\mathcal{I}}_\sigma} = 0$ at every point of the strict transform N' over C.*

2. *Let T be a multiple test blow-up of $\underline{\mathcal{I}}$. The maximum locus of $f(T)$ on $\operatorname{cosupp} \underline{\mathcal{I}}_T$ lies in one level set of $\nu_{\underline{\mathcal{I}}_T}$, on which ν takes its largest value on the cosupport; hence if C is empty or this maximum locus and w is a point of the cosupport with $\nu(w) \geq q$, then $C \subset \{\nu \geq q\}$. If $T_0 M$ is a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f (Definition* **??***), every point z of $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0 M}$ lies over $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0}$ and $\nu(z) \leq \nu(p_M(z))$: the order never goes up along a canonical segment.*

3. *Let T be a multiple test blow-up of $\underline{\mathcal{I}}$. If near every point y of $\operatorname{cosupp} \underline{\mathcal{I}}_T$ the points y' of the cosupport with $\nu(y') = \nu(y)$ have $f(T)(y') \leq f(T)(y)$, then $f(T)$ is upper semicontinuous on the cosupport (Definition* **??***).*

4. *Let T be a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f, and $y \in \operatorname{cosupp} \underline{\mathcal{I}}_T$ with $\nu(y) = o$ and run $r = r_o(T, y) < \operatorname{length} T$ in the regime o (Lemma 174.91). Then the image of y at the stage $\operatorname{length} T - r - 1$ lies in the cosupport and has order $\geq o + 1$.*

5. *Let $T_0 M\sigma$ (σ a blowup along C) be a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f, and w a point of $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0 M}$ with $\nu(w) \geq o + 1$. Then every centre of $M\sigma$ lies in $\{\nu \geq o + 1\}$ at its stage. Consequently, for an open set W of the last stage of T_0 with $\nu \leq o$ on $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0} \cap W$, the centres of $M\sigma$ miss the preimages of W, $\nu \leq o$ at the cosupport points over W, and for $o \geq 1$ $\mathcal{O}(\underline{\mathcal{I}}_{T_0 M\sigma}, o)|_{(M\sigma)^{-1}W} \simeq (\mathcal{O}(\underline{\mathcal{I}}_{T_0}, o)|_W)_{(M\sigma)|_W}$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) For $q \geq 1$ this is Lemma 174.90(5). Let $q = 0$, and write $\mathcal{N} = \mathcal{N}_N(\mathfrak{a})$, $\mathcal{M} = \mathcal{M}_N(\mathfrak{a})$. By Theorem **??**, $\operatorname{Res}_N \mathfrak{a} = \operatorname{Res}_N(\mathcal{N}\mathcal{M})$, both parts are coherent, $\mathcal{M}$ is nonzero and monomial along N, and $\mathcal{N}$ is nonzero along N and, restricted to N, contained in the ideal of no member at the points of N; restricted-equivalent marked ideals have the same cosupport and the same transform (Lemma **??**). Now $C \subset \operatorname{cosupp}(\mathcal{N}\mathcal{M}|_N, d)$, $C \subset N = \operatorname{cosupp}(\mathcal{N}|_N, 0)$ (Lemma **??**), and $\operatorname{ord}(\mathcal{N}|_N) = 0$ on C. So Lemma **??** with $o = 0$ gives $\operatorname{Res}_{N'} \mathfrak{a}' = \operatorname{Res}_{N'}(\mathfrak{b}'\mathcal{M}')$, where $\mathcal{M}'$ is principal monomial along N' at every point of N', hence nonzero and monomial along N' (Lemma **??**), and $\mathfrak{b}' = (\pi^*\mathcal{N} : \mathcal{O})$ is the ideal of the transform of $(M, N, E, \mathcal{N}, 0)$, coherent (Lemma **??**) and, restricted to N', in the ideal of no member of E'. The transform is well formed (Lemma 174.125), so by Lemma **??** $\mathcal{M}_{N'}(\mathfrak{a}') = \mathcal{M}_{N'}(\mathcal{M}')$ and $\mathfrak{b}' \subset \mathcal{N}_{N'}(\mathfrak{a}')$. Let $z \in N'$ with $\sigma(z) \in C$. Since $\nu(\sigma(z)) = 0$, some germ $c \in \mathcal{N}_{\sigma(z)}$ has $c(\sigma(z)) \neq 0$; then $c \circ \sigma$ is a germ of $\pi^*\mathcal{N} \subset \mathfrak{b}' \subset \mathcal{N}_{N'}(\mathfrak{a}')$ at z which does not vanish at z, so $\nu(z) = 0$.

(2) For c in the maximum locus and y in the cosupport, $f(T)(y) \leq f(T)(c)$, so $\nu(y) \leq \nu(c)$; two points of the maximum locus have the same value, hence the same order. If $\nu(w) \geq q$, a

point c of the maximum locus has $\nu(c) \geq q$, and $c \in N_T$, so $c \in \{\nu \geq q\}$ (Lemma 174.90(1); the stages lie in the domain, Proposition 174.75). For one canonical step σ along C and z in the new cosupport, $\sigma(z)$ lies in the old cosupport (in C, or off C by Lemma 174.31). If $\sigma(z) \notin C$, $\nu(z) = \nu(\sigma(z))$ (Lemma 174.90(3)); if $\sigma(z) \in C$, then C is the maximum locus, which lies in the level set of $\nu(\sigma(z))$, and (1) applies. Induction along M.

(3) Near y, $\nu \leq \nu(y)$ (Lemma 174.90(1): the set $\{\nu \geq \nu(y)+1\}$ is closed). At a nearby point y' of the cosupport with $\nu(y') < \nu(y)$, $f(T)(y) \leq f(T)(y')$ would give $\nu(y) \leq \nu(y')$; so $f(T)(y') < f(T)(y)$. At the points with $\nu(y') = \nu(y)$ the hypothesis applies.

(4) Split $T = T_0 M$ after its first length $T - r - 1$ blowups (Lemma **??**) and let $w = p_M(y)$. By the maximality of the run (Lemma **??**) w is not in the regime o. By (2) for $T_0 M$, w lies in the cosupport and $\nu(w) \geq \nu(y) = o$; hence $\nu(w) \geq o+1$.

(5) Induction along M. The centre C is empty or the maximum locus, so $C \subset \{\nu \geq o+1\}$ by (2). If $M = M'\tau$, the image of w under τ lies in $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0 M'}$ and has order $\geq \nu(w) \geq o+1$ by (2), and the induction hypothesis gives the centres of $M'\tau$. The tower $M\sigma$ is a multiple test blow-up of $\underline{\mathcal{I}}_{T_0}$ (Lemma **??**), with transforms $(\underline{\mathcal{I}}_{T_0})_{M'} = \underline{\mathcal{I}}_{T_0 M'}$; the rest is Lemma 174.91(3) and Lemma 174.87(4) for $\underline{\mathcal{I}}_{T_0}$. □

The companion phase after the start of a run. In the general step the value at a point y of order $o \geq 1$ is read off the companion ideal formed at the start of the run of y in the regime o, on the open set where $\nu \leq o$, along the rest of the segment. The lemma below shows that this rest, restricted to that open set, is a multiple test blow-up of the companion ideal (the hypothesis of Lemma 174.91(2)), and that nearby points of the same order started no later.

Lemma 174.93 (The restricted tail of the companion phase)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n \geq 1$, $\underline{\mathcal{I}}$ a marked ideal in the domain of Definition **??**, *ν the order of the nonmonomial part, $o \geq 1$, and f a value function whose order dominates ν as in Lemma 174.92. Write $W_o(\underline{\mathcal{J}})$ for the open set of the points outside the closure of $\{\nu_{\underline{\mathcal{J}}} \geq o+1\}$; on the domain it is $\{\nu_{\underline{\mathcal{J}}} \leq o\}$.*

1. *Along a multiple test blow-up T of a companion ideal $\mathcal{O}(\underline{\mathcal{J}}, o)$ ($\underline{\mathcal{J}}$ in the domain, $\nu \leq o$ on its cosupport),* $\operatorname{cosupp} \mathcal{O}(\underline{\mathcal{J}}, o)_T = \{\nu_{\underline{\mathcal{J}}_T} \geq o\} \cap \operatorname{cosupp} \underline{\mathcal{J}}_T$. *Transforms of marked ideals with the same divisor have the same divisor.*

2. *Let $T_0 T_1$ be a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f, W an open set of the last stage of T_0 with $\nu \leq o$ on* $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0} \cap W$, *and suppose some point has run* $\geq \operatorname{length} T_1$ *in the regime o (Lemma 174.91). Then $T_1|_W$ is a multiple test blow-up of $\mathcal{O}(\underline{\mathcal{I}}_{T_0}|_W, o)$.*

3. *Let T be a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f, $y \in$* $\operatorname{cosupp} \underline{\mathcal{I}}_T$ *with $\nu(y) = o$, r its run, and $T = T_0 T_1$ the split with* $\operatorname{length} T_1 = r$ *(Lemma* **??***). The image of y at the last stage of T_0 is in the regime o and in $W = W_o(\underline{\mathcal{I}}_{T_0})$, and $T_1|_W$ is a multiple test blow-up of $\mathcal{O}(\underline{\mathcal{I}}_{T_0}|_W, o)$.*

4. *In the situation of (3), every point y' of* $\operatorname{cosupp} \underline{\mathcal{I}}_T$ *near y with $\nu(y') \geq o$ is in the regime o, and its run is at least r.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The inclusion $\subset$ is Lemma 174.90(4). Conversely, by Lemma 174.88(2), $\operatorname{cosupp} \mathcal{O}(\underline{\mathcal{J}}, o)_T = \operatorname{cosupp}(\mathcal{N}_T|_{N_T}, o) \cap \operatorname{cosupp} \underline{\mathcal{J}}_T$, where $\mathcal{N}_T$ is the transform of $(M, N, E, \mathcal{N}_N(\mathfrak{a}), o)$, which lives on N_T and is contained in $\mathcal{N}_{N_T}(\mathfrak{a}_T)$ (Theorem **??**); a point where every germ of $\mathcal{N}_{N_T}(\mathfrak{a}_T)$ vanishes to order o along N_T is therefore in $\operatorname{cosupp}(\mathcal{N}_T|_{N_T}, o)$. The divisor of a transform is the list of strict transforms of the members followed by the exceptional member (Lemma **??**); induction along the tower.

(2) Induction along $T_1 = S\sigma$, σ along C. The run of the point y at the last stage is at least $\operatorname{length} S + 1 \geq 1$, so its image w at the last stage of $T_0 S$ is in the regime o and has run $\geq \operatorname{length} S$

(Lemma **??**); by induction $S|_W$ is a multiple test blow-up of $\mathcal{O} = \mathcal{O}(\underline{\mathcal{I}}_{T_0}|_W, o)$. The centre C is empty or the maximum locus of f; as w lies in the cosupport with $\nu(w) = o$, $C \subset \{\nu \geq o\}$ (Lemma 174.92(2)). The trace of C over W is an admissible centre of the transform of $\underline{\mathcal{I}}_{T_0}|_W$ along $S|_W$, which is $\underline{\mathcal{I}}_{T_0 S}$ restricted to the preimage of W (Lemma **??**, the tower S being a multiple test blow-up of $\underline{\mathcal{I}}_{T_0}$, Lemma **??**). The stages lie in the domain (Proposition 174.75) and the bound holds on W (Lemma 174.90(2)), so by (1) the cosupport of $\mathcal{O}_{S|_W}$ is $\{\nu \geq o\} \cap$ cosupp of that transform, which contains the trace of C (restriction to open sets does not change ν or the cosupport, Lemmas 174.90(2) and **??**); the divisors agree by (1), since $\mathcal{O}$ has the divisor of $\underline{\mathcal{I}}_{T_0}|_W$ (Lemma **??**). So the trace of C is an admissible centre of $\mathcal{O}_{S|_W}$ (Definition **??**).

(3) If $r = 0$ the image is y itself; otherwise the image at the start of the run is in the regime by Lemma **??**. On the domain the set $\{\nu \geq o + 1\}$ is closed (Lemma 174.90(1)), so a point in the regime o lies in W and the bound holds on W. Apply (2) with the point y, whose run is $r = \operatorname{length} T_1$.

(4) By (3) and Lemma 174.91(2), every point y' of the cosupport over W with $\nu(y') \geq o$ is in the regime o with run $\geq \operatorname{length} T_1 = r$; the points of the last stage over W form an open neighbourhood of y by (3). □

The order, its regime and its mask under the operations of the general step. Properties (4'), (5), (6) and (7) of the general step's value (Definition **??**) compare values across an empty prefix, a restriction to an open set, an isomorphism of towers and a change of representative. The lemma below gives these comparisons for the first entry of the value and for the data it determines: the regime o, its run, and the mask W_o of Lemma 174.93.

Lemma 174.94 (The order of the nonmonomial part under the operations of the general step)**.** *Let $n \geq 1$, ν the order of the nonmonomial part, $o \in \mathbf{N}$, r_o the run in the regime o (Lemma 174.91) and W_o the mask of Lemma 174.93.*

1. *(Representatives) If $\underline{\mathcal{I}} \simeq \underline{\mathcal{J}}$ (Definition **??**) with N closed and both ideals coherent, then along every tower T the sets $\{\nu \geq q\}$, the orders, the regimes o, their runs and the masks W_o of $\underline{\mathcal{I}}_T$ and $\underline{\mathcal{J}}_T$ agree.*

2. *(Open sets) Let $\underline{\mathcal{I}}$ lie in the domain of Definition **??**, T a multiple test blow-up of it and $U \subset X$ open. A point of the preimage of U lies in W_o of the transform of $\underline{\mathcal{I}}|_U$ along $T|_U$ iff it lies in $W_o(\underline{\mathcal{I}}_T)$.*

3. *(Isomorphisms) Let $\underline{\mathcal{I}}$ lie in the domain, T a multiple test blow-up of it, and $\chi\colon T' \to T$ an isomorphism of towers over a biholomorphism Φ (Definition **??**). Then $p \in W_o((\Phi^* \underline{\mathcal{I}})_{T'})$ iff $\chi(p) \in W_o(\underline{\mathcal{I}}_T)$.*

4. *(Empty prefixes) For every tower P, a point is in the regime o along PS iff it is in the regime o of $\underline{\mathcal{I}}_P$ along S. If $\underline{\mathcal{I}}$ lies in the domain and P has only empty centres, a point z of the last stage of P is in the regime o iff its image at the base is, its run is* length P *if so and* 0 *otherwise, and $W_o(\underline{\mathcal{I}}_P) = p_P^{-1} W_o(\underline{\mathcal{I}})$.*

Proof. (1) The nonmonomial part only depends on $\operatorname{Res}_N \mathfrak{a}$ (Lemma 174.86), and N, E agree; after one blowup restricted-equivalent marked ideals have equal transforms, and along every tower equal cosupports (Lemma **??**). So the sets $\{\nu \geq q\}$, which determine the order, agree at every stage, and so do the cosupports; the regimes agree pointwise, the runs by induction along the tower, and the masks, which are determined by $\{\nu \geq o + 1\}$.

(2) The transform of $\underline{\mathcal{I}}|_U$ along $T|_U$ is the restriction of $\underline{\mathcal{I}}_T$ to the preimage of U (Lemma **??**), whose set $\{\nu \geq o+1\}$ is the trace of that of $\underline{\mathcal{I}}_T$ (Lemma 174.90(2), the stages lying in the domain, Proposition 174.75). On an open subset the closure of a trace is the trace of the closure, since the inclusion is continuous and open.

(3) By Lemma 174.90(3) (with Lemma **??**) the set $\{\nu \geq o+1\}$ of $(\Phi^*\underline{\mathcal{I}})_{T'}$ is the preimage under χ of that of $\underline{\mathcal{I}}_T$; χ is a homeomorphism, so it maps closures to closures.

(4) $\underline{\mathcal{I}}_{PS} = (\underline{\mathcal{I}}_P)_S$ (Lemma **??**). Along P with only empty centres, the cosupport at the last stage is the preimage of the cosupport at the base (Lemma **??**, a blowup along $\emptyset$ creating no cosupport) and ν is that of the image (Lemma 174.90(2)); so the regime at every stage of P is that of the image at the base, and the run (Lemma **??**) is the whole tower or 0. The stages lie in the domain, where $W_o = \{\nu \leq o\}$, and the last assertion follows. □

Admissible centres and finitely many orders in the companion phase. Property (1) of the general step's value reads the level sets of the maximal-order invariant of the companion ideal as centres of the marked ideal itself, and property (8) needs that only finitely many orders occur over a compact set. The lemma below supplies both on the side of the order of the nonmonomial part.

Lemma 174.95 (The companion phase: admissible centres and a bound on the order)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n \geq 1$ and ν the order of the nonmonomial part.

1. *Let $\underline{\mathcal{J}}$ lie in the domain of Definition* **??**, *$o \geq 1$ with $\nu \leq o$ on* cosupp $\underline{\mathcal{J}}$, *and T a multiple test blow-up of $\mathcal{O}(\underline{\mathcal{J}}, o)$. A set C is an admissible centre of $\mathcal{O}(\underline{\mathcal{J}}, o)_T$ iff it is an admissible centre of $\underline{\mathcal{J}}_T$ contained in $\{\nu_{\underline{\mathcal{J}}_T} \geq o\}$; the same holds for the restrictions of both transforms to any open subset of the last stage.*

2. *Let $\underline{\mathcal{I}}$ be well formed with $\mathfrak{a}$ nonzero along N. On every compact $K \subset N$, ν is bounded.*

3. *Let $\underline{\mathcal{I}}$ lie in the domain, f a value function whose order dominates ν (Lemma 174.92), and K a compact subset of the base. There is q such that along every multiple test blow-up T of $\underline{\mathcal{I}}$ which is a canonical segment of f, $\nu \leq q$ at every point of* cosupp $\underline{\mathcal{I}}_T$ *over K.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The companion ideal has the divisor of $\underline{\mathcal{J}}$ (Proposition 174.79), so the transforms have the same divisor (Lemma 174.93(1)), and cosupp $\mathcal{O}(\underline{\mathcal{J}}, o)_T = \{\nu \geq o\} \cap$ cosupp $\underline{\mathcal{J}}_T$ (Lemma 174.93(1)). An admissible centre is a closed submanifold in the cosupport with normal crossings with the divisor (Definition **??**); both conditions on the divisor and on the cosupport therefore match. On an open set, restriction takes cosupports to their traces (Lemma **??**; the transforms lie in the domain, Proposition 174.75, and T is a multiple test blow-up of $\underline{\mathcal{J}}$, Lemma 174.88(2)) and divisors to their traces.

(2) By Lemma 174.90(1) every $y \in K$ has a neighbourhood on which $\nu \leq \nu(y)$; finitely many of them cover K, and the largest of the finitely many values bounds ν on K.

(3) N is closed, so $K \cap N$ is compact; let q bound ν on it by (2). A point y of cosupp $\underline{\mathcal{I}}_T$ over K lies over cosupp $\underline{\mathcal{I}} \cap K \subset N \cap K$, and $\nu(y) \leq \nu(p_T(y)) \leq q$ since the order never goes up along a canonical segment (Lemma 174.92(2), with T_0 the empty tower). □

The monomial phase of the general step. At a point where the nonmonomial part has order 0, the value of the general step is read off the invariant γ_N of the current marked ideal (Lemma **??**). The monomial case along N (Lemma **??**) needs a marked ideal which is monomial along all of N; the lemma below obtains its conclusions where only points of order 0 are involved.

Lemma 174.96 (The monomial phase: admissible level sets, and the invariant off and over the centre)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n \geq 1$, $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ lie in the domain of Definition **??**, *ν be the order of the nonmonomial part (Lemma 174.90), $\gamma = \gamma_N$ the invariant of Lemma* **??** *(with the mark d), and W_0 the open set of the points off N or of order $\nu = 0$ (Lemma 174.89(3); the mask of order 0 of Lemma 174.93).*

1. *$\underline{\mathcal{I}}|_{W_0}$ lies in the domain and is monomial along $N \cap W_0$; a point of order 0 lies in W_0.*

2. *(level sets) Let $W \subset M$ be open and v a value of γ or $\perp$, such that $\nu = 0$ and $\gamma \leq v$ at every point of $\operatorname{cosupp}\underline{\mathcal{I}} \cap W$. Then $\{y \in \operatorname{cosupp}\underline{\mathcal{I}} \cap W : \gamma(y) = v\}$ is an admissible centre of $\underline{\mathcal{I}}|_W$. In particular, if $\nu = 0$ on $\operatorname{cosupp}\underline{\mathcal{I}}$, the maximum locus of γ on $\operatorname{cosupp}\underline{\mathcal{I}}$ is an admissible centre.*

3. *(off the centre) Let C be an admissible centre (possibly empty), $\sigma\colon M' \to M$ the blowup along C, $\underline{\mathcal{I}}'$ the transform and γ' its invariant along N'. If $z \in N'$, $\sigma(z) \notin C$ and $\nu(\sigma(z)) = 0$, then $\gamma'(z) = \gamma(\sigma(z))$.*

4. *(over the centre) If $\nu = 0$ on $\operatorname{cosupp}\underline{\mathcal{I}}$ and C is the maximum locus of γ on $\operatorname{cosupp}\underline{\mathcal{I}}$, then $\gamma'(z) < \gamma(\sigma(z))$ at every $z \in N'$ with $\sigma(z) \in C$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The restriction to an open set stays in the domain (Proposition 174.75); it is monomial by Lemma 174.89(2), since W_0 misses $\{\nu \geq 1\}$. On the domain $\{\nu \geq 1\}$ is closed (Lemma 174.90), so W_0 is exactly the set of points off $\{\nu \geq 1\}$ (Lemma 174.93).

Throughout, restriction to an open set U is read back as follows: $\operatorname{cosupp}(\underline{\mathcal{I}}|_U) = \operatorname{cosupp}\underline{\mathcal{I}} \cap U$ (Lemma **??**); γ of $\underline{\mathcal{I}}|_U$ at p is $\gamma(p)$ (Lemma **??**(2)); the blowup restricts to the blowup of U along $C \cap U$ on $\sigma^{-1}U$ (Lemma 174.5), and the transform of $\underline{\mathcal{I}}|_U$ along it is $\underline{\mathcal{I}}'|_{\sigma^{-1}U}$ (Lemma **??**; the transform along an admissible centre is well formed, Corollary 174.125). Hence γ' of the transform of $\underline{\mathcal{I}}|_U$ at p is $\gamma'(p)$.

(2) By Lemma 174.84(2) it suffices that every $x \in W$ has an open neighbourhood $V \subset W$ on which the trace of the set is an admissible centre of $\underline{\mathcal{I}}|_V$. If $\nu(x) \geq 1$, x is not in the closed set $\operatorname{cosupp}\underline{\mathcal{I}}$ (Lemma **??**); on $V = W \setminus \operatorname{cosupp}\underline{\mathcal{I}}$ the trace is empty. If $\nu(x) = 0$, take $V = W \cap W_0$: $\underline{\mathcal{I}}|_V$ is monomial along $N \cap V$ by (1), so $\gamma = \perp$ exactly at the points of $N \cap V$ off the cosupport (Lemma **??**(2)). For $v = \perp$ the trace is empty. For $v \neq \perp$, $\gamma \leq v$ on all of $N \cap V$, and the trace is $\{y \in N \cap V : \gamma(y) = v\}$, an admissible centre by Lemma **??**(3). For the maximum locus: it is empty, or the level set of the value v of one of its points, on which $\gamma \leq v$ holds on the whole cosupport; take $W = M$.

(3) Take $U = W_0 \ni \sigma(z)$. By (1) and the reading back above, Lemma **??**(4) for $\underline{\mathcal{I}}|_{W_0}$ and the restricted blowup gives $\gamma'(z) = \gamma(\sigma(z))$.

(4) Again $U = W_0$, which now contains $\operatorname{cosupp}\underline{\mathcal{I}}$. The trace of C on W_0 is the maximum locus of γ of $\underline{\mathcal{I}}|_{W_0}$ on its cosupport: both are read off the same values at the same points, because every point of the cosupport lies in W_0. By (2), C is admissible, and Lemma **??**(3) for $\underline{\mathcal{I}}|_{W_0}$ gives $\gamma'(z) < \gamma(\sigma(z))$. □

The monomial phase along canonical segments. The next lemma applies Lemma 174.96 along canonical segments of any value function whose first entry is the order of the nonmonomial part and which, at the points of order 0, is a strictly monotone function of γ; the value of the general step is one (it reads $(0, \gamma)$ at the points of order 0).

Lemma 174.97 (The monomial phase along canonical segments)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n \geq 1$, $\underline{\mathcal{I}}$ lie in the domain of Definition **??**, *ν be the order of the nonmonomial part and, for a tower T, γ_T the invariant of Lemma* **??** *of $\underline{\mathcal{I}}_T$ (with its mark). Let f be a value function whose order dominates ν (Lemma 174.92), and ψ a strictly monotone map with $f(T)(y) = \psi(\gamma_T(y))$ at every point y of $\operatorname{cosupp}\underline{\mathcal{I}}_T$ with $\nu(y) = 0$, for every multiple test blow-up T of $\underline{\mathcal{I}}$. (For $f = (\nu, g)$ ordered lexicographically with $g = \gamma$ in the first summand of a lexicographic sum at the points of order 0, $\psi(w) = (0, w)$.) Let T be a multiple test blow-up of $\underline{\mathcal{I}}$.*

1. *If a point of order 0 lies on the maximum locus of $f(T)$ on $\operatorname{cosupp}\underline{\mathcal{I}}_T$, every point of the cosupport has order 0, and that maximum locus is the maximum locus of γ_T on the cosupport.*

2. *(level sets) Let W be an open subset of the last stage and l a value with $f(T) \leq l$ on* $\operatorname{cosupp}\underline{\mathcal{I}}_T \cap W$*, attained at a point of the cosupport of order* 0*. Then* $\{y \in \operatorname{cosupp}\underline{\mathcal{I}}_T \cap W : f(T)(y) = l\}$ *is an admissible centre of* $\underline{\mathcal{I}}_T|_W$*.*

3. *(upper semicontinuity) Near a point y of* $\operatorname{cosupp}\underline{\mathcal{I}}_T$ *with $\nu(y) = 0$, every point y' of the cosupport with $\nu(y') = 0$ has $f(T)(y') \leq f(T)(y)$.*

4. *(one step) Let $T\sigma$, σ the blowup along C, be a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f, and z a point of* $\operatorname{cosupp}\underline{\mathcal{I}}_{T\sigma}$ *with $\nu(\sigma(z)) = 0$. Then $\sigma(z) \in \operatorname{cosupp}\underline{\mathcal{I}}_T$, $\nu(z) = 0$, and $\gamma_{T\sigma}(z) < \gamma_T(\sigma(z))$, $f(T\sigma)(z) < f(T)(\sigma(z))$ if $\sigma(z) \in C$, while $\gamma_{T\sigma}(z) = \gamma_T(\sigma(z))$, $f(T\sigma)(z) = f(T)(\sigma(z))$ if $\sigma(z) \notin C$.*

5. *(segments) Let T_0M be a multiple test blow-up of $\underline{\mathcal{I}}$ and a canonical segment of f, and $z \in \operatorname{cosupp}\underline{\mathcal{I}}_{T_0M}$ with $\nu(p_M(z)) = 0$. Then $p_M(z) \in \operatorname{cosupp}\underline{\mathcal{I}}_{T_0}$, $\nu(z) = 0$, $\gamma_{T_0M}(z) \leq \gamma_{T_0}(p_M(z))$ and $f(T_0M)(z) \leq f(T_0)(p_M(z))$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

The transforms along multiple test blow-ups stay in the domain (Proposition 174.75), so Lemma 174.96 applies at every stage; on the domain a point of N has $\nu = 0$ iff it is off $\{\nu \geq 1\}$ (Lemma 174.90).

(1) A point y of the cosupport has $f(T)(y) \leq f(T)(c)$ for c on the maximum locus, so $\nu(y) \leq \nu(c) = 0$. Hence $f(T) = \psi \circ \gamma_T$ on the cosupport, and a strictly monotone change of values does not change maximum loci (Lemma 174.83; maximum loci only depend on the values on the set, Lemma **??**).

(2) Let y_0 be the point of order 0 with $f(T)(y_0) = l$. Every $y \in \operatorname{cosupp}\underline{\mathcal{I}}_T \cap W$ has $f(T)(y) \leq f(T)(y_0)$, hence $\nu(y) = 0$, $f(T)(y) = \psi(\gamma_T(y))$ and $\gamma_T(y) \leq \gamma_T(y_0)$; and $f(T)(y) = l$ iff $\gamma_T(y) = \gamma_T(y_0)$, ψ being injective. The set is therefore the level set of γ_T at $\gamma_T(y_0)$ of Lemma 174.96(2).

(3) γ_T is upper semicontinuous on N_T (Lemma **??**(1), which only uses that the ideal is nonzero along N_T), and $f(T) = \psi \circ \gamma_T$ at the points of the cosupport of order 0, ψ being monotone.

(4) By Lemma 174.92(2), $\sigma(z)$ lies in the old cosupport and $\nu(z) \leq \nu(\sigma(z)) = 0$. The centre C is admissible for $\underline{\mathcal{I}}_T$. If $\sigma(z) \in C$, then C is the maximum locus of $f(T)$ (it is not empty), and it contains $\sigma(z)$, of order 0; by (1) the cosupport has order 0 and C is the maximum locus of γ_T, so Lemma 174.96(4) gives $\gamma_{T\sigma}(z) < \gamma_T(\sigma(z))$. If $\sigma(z) \notin C$, Lemma 174.96(3) gives equality. Both points have order 0, so the values are ψ of the invariants.

(5) Induction on M. For $M = M'\sigma$, the image $\sigma(z)$ lies in the cosupport at the last stage of T_0M', with $\nu(\sigma(z)) \leq \nu(p_{M'}(\sigma(z))) = \nu(p_M(z)) = 0$ (Lemma 174.92(2)); (4) gives $\gamma_{T_0M}(z) \leq \gamma_{T_0M'}(\sigma(z))$ and the induction hypothesis $\gamma_{T_0M'}(\sigma(z)) \leq \gamma_{T_0}(p_M(z))$. The values follow as in (4), T_0 being a multiple test blow-up (Lemma **??**). □

The invariant of the monomial phase under the operations of the general step. Properties (4)–(7) of the general step's value need, at the points of order 0, that the invariant γ of the current transform behaves under the operations of the general step as the order ν does (Lemma 174.94); restriction to open sets is Lemma **??**.

Lemma 174.98 (The invariant of the monomial phase under re-basing, isomorphisms of towers and representatives)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

For a marked ideal $\underline{\mathcal{I}}$ and a tower T over its base let γ_T be the invariant of Lemma **??** *of $\underline{\mathcal{I}}_T$ (along N_T, with the members of E_T and the mark).*

1. *$\gamma_{N,\mathfrak{a},E,d}$ only depends on N, E, d and the cosupports* $\operatorname{cosupp}(\mathfrak{a}|_N, e)$*, $e \in \mathbb{N}$. In particular restricted-equivalent marked ideals (Lemma* **??***) have the same invariant.*

2. *(representatives) If $\underline{\mathcal{I}}$ and $\underline{\mathcal{I}}'$ are restricted-equivalent, with N closed and coherent ideals, then $\gamma_T(\underline{\mathcal{I}}) = \gamma_T(\underline{\mathcal{I}}')$ along every tower T.*

3. *(isomorphisms of towers) Let $\chi\colon T' \to T$ be an isomorphism of towers over a biholomorphism Φ (Definition **??**), T a multiple test blow-up of $\underline{\mathcal{I}}$, N a closed submanifold and the ideal coherent. Then $\gamma_{T'}(\Phi^*\underline{\mathcal{I}})(p) = \gamma_T(\underline{\mathcal{I}})(\chi(p))$.*

4. *(re-basing) $\gamma_{PT_1}(\underline{\mathcal{I}}) = \gamma_{T_1}(\underline{\mathcal{I}}_P)$ for all towers P, T_1.*

5. *(a trailing empty step at order 0) Let $n \geq 1$, $\underline{\mathcal{I}}$ lie in the domain of Definition **??**, T be a multiple test blow-up and π the blowup along $\emptyset$. At every point z of the strict transform $N_{T\pi}$ with $\nu(\pi(z)) = 0$, $\gamma_{T\pi}(z) = \gamma_T(\pi(z))$.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The exponent $e_H^N(\mathfrak{a})(x)$ is defined by the condition that $N \cap H$ lies in $\operatorname{cosupp}(\mathfrak{a}|_N, e)$ near x (Lemma **??**), and γ is read off the members through the point and these exponents. A restricted inclusion $\operatorname{Res}_N \mathfrak{a} \subset \operatorname{Res}_N \mathfrak{b}$ gives $\operatorname{cosupp}(\mathfrak{b}|_N, e) \subset \operatorname{cosupp}(\mathfrak{a}|_N, e)$ for every e (Lemma **??**); restricted-equivalent marked ideals have both inclusions and the same N, E, d.

(2) For the empty tower this is (1). After one blowup the transforms of restricted-equivalent marked ideals are equal (Lemma **??**), hence so are all later transforms.

(3) The transform of $\Phi^*\underline{\mathcal{I}}$ along T' is $\chi^*\underline{\mathcal{I}}_T$ (Lemma **??**; the transforms keep N a closed submanifold and the ideal coherent, Lemma **??**), and γ is invariant under biholomorphisms (Lemma **??**(2)).

(4) $\underline{\mathcal{I}}_{PT_1} = (\underline{\mathcal{I}}_P)_{T_1}$ (Lemma **??**).

(5) $\underline{\mathcal{I}}_T$ lies in the domain (Proposition 174.75), $\emptyset$ is an admissible centre, and a point of order 0 is off $\{\nu \geq 1\}$ (Lemma 174.97); Lemma 174.96(3) with $C = \emptyset$ applies. □

The general value at the points of order zero. At a point where the nonmonomial part has order 0 the general value (Definition **??**) is $(0, \gamma)$, with γ the invariant of the current transform. Its properties (4'), (5), (6) and (7) of Definition **??** at such points therefore follow from those of the order ν (Lemmas 174.90 and 174.94) and of γ (Lemmas 174.98 and **??**); the value function M of the companion phase plays no role there.

Lemma 174.99 (The general value at the points of order zero)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let M be a value function (Definition **??**) and* inv *the general value of Definition **??**. For a tower T let ν_T be the order of the nonmonomial part of $\underline{\mathcal{I}}_T$ and γ_T the invariant of $\underline{\mathcal{I}}_T$ of Lemma **??**.*

1. *If $\nu_T(y) = 0$, then $\operatorname{inv}_{\underline{\mathcal{I}}}(T)(y) = (0, \gamma_T(y))$; two points of order 0, for any two marked ideals and towers, with the same γ have the same value.*

2. *(Re-basing, every prefix) For all towers P, T_1 and every point z of the last stage of PT_1 with $\nu_{PT_1}(z) = 0$, $\operatorname{inv}_{\underline{\mathcal{I}}}(PT_1)(z) = \operatorname{inv}_{\underline{\mathcal{I}}_P}(T_1)(z)$. No emptiness of P is needed.*

3. *(Open sets) Let $n \geq 1$, $\underline{\mathcal{I}}$ lie in the domain of Definition **??**, $U \subset X$ be open and T a multiple test blow-up. At every point p over U with $\nu_T(p) = 0$, $\operatorname{inv}_{\underline{\mathcal{I}}|_U}(T|_U)(p) = \operatorname{inv}_{\underline{\mathcal{I}}}(T)(p)$.*

4. *(Isomorphisms of towers) Under the same hypotheses, let $\chi\colon T' \to T$ be an isomorphism of towers over a biholomorphism Φ (Definition **??**). At every point p with $\nu_T(\chi(p)) = 0$, $\operatorname{inv}_{\Phi^*\underline{\mathcal{I}}}(T')(p) = \operatorname{inv}_{\underline{\mathcal{I}}}(T)(\chi(p))$.*

5. *(Representatives) If $\underline{\mathcal{I}} \simeq \underline{\mathcal{J}}$ with N closed and coherent ideals, then along every tower T, at every point y with $\nu_T(y) = 0$ for $\underline{\mathcal{I}}$, $\operatorname{inv}_{\underline{\mathcal{I}}}(T)(y) = \operatorname{inv}_{\underline{\mathcal{J}}}(T)(y)$.*

6. *In particular, if T is a canonical segment of $\operatorname{inv}_{\underline{\mathcal{I}}}$ (M a value function for the marked ideals of maximal order, $n \geq 1$, $\underline{\mathcal{I}}$ in the domain), properties (4'), (5), (6) and (7) of Definition **??** hold for* inv *at the points of the cosupport of order 0.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) is the definition of the general value at order 0. In (2)–(5) the orders on both sides agree: in (2) because $\underline{\mathcal{I}}_{PT_1} = (\underline{\mathcal{I}}_P)_{T_1}$ (Lemma **??**); in (3) because the transform of $\underline{\mathcal{I}}|_U$ along $T|_U$ is the restriction of $\underline{\mathcal{I}}_T$ (Lemma **??**) and ν is local (Lemma 174.90(2)), the transforms staying in the domain (Proposition 174.75); in (4) by Lemma 174.90(3); in (5) by Lemma 174.94(1). So both points have order 0 and by (1) it suffices to compare γ: in (2) by Lemma 174.98(4), in (3) by Lemma **??** applied to the restriction of $\underline{\mathcal{I}}_T$, in (4) by Lemma 174.98(3) and in (5) by Lemma 174.98(2). For (6), canonical segments of the general value are multiple test blow-ups (Theorem **??**), so (3) and (4) apply; (2) and (5) need no hypothesis on T. □

An empty step in the middle of a sequence. Włodarczyk's extensions of a resolution insert isomorphisms anywhere in a sequence of blowups. Here a blowup along $\emptyset$ appends the empty set to the divisor E (Lemma **??**), so an empty step in the middle of a sequence moves every later member one position further, while the invariant γ of the monomial case records the members through a point by their positions in E (Lemma **??**). The lemma below computes the effect.

Lemma 174.100 (The monomial invariant under an empty member inserted in the divisor)**.** *For $j \in \mathbf{N}$ let $s_j(i) = i$ if $i < j$ and $s_j(i) = i+1$ if $i \geq j$, and let σ_j act on the values $(n-|K|, \sum_K a, K)$ of γ by $K \mapsto s_j(K)$ on the third entry (and fix $\perp$). For a marked ideal $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ and $j \leq \#E$ let $\underline{\mathcal{I}}^{(j)}$ be $\underline{\mathcal{I}}$ with $\emptyset$ inserted into E at position j.*

1. *σ_j is strictly increasing; it moves every value whose set K has an element $\geq j$. In particular maximum loci of $\sigma_j \circ f$ and of f agree.*

2. *$\gamma(\underline{\mathcal{I}}^{(j)}) = \sigma_j \circ \gamma(\underline{\mathcal{I}})$; the monomial and nonmonomial parts, the order ν of the nonmonomial part and the cosupport of $\underline{\mathcal{I}}^{(j)}$ are those of $\underline{\mathcal{I}}$.*

3. *Inserting $\emptyset$ commutes with the transforms: $(\underline{\mathcal{I}}^{(j)})_T = (\underline{\mathcal{I}}_T)^{(j)}$ along every tower T. Hence $\gamma_T(\underline{\mathcal{I}}^{(j)}) = \sigma_j \circ \gamma_T(\underline{\mathcal{I}})$, and ν, the cosupports and the maximum loci of γ along T are unchanged.*

4. *(An empty step in the middle) Let T_0 be a tower over the base of $\underline{\mathcal{I}}$, σ a blowup along $\emptyset$ of its last stage, S a tower over the source of σ, and $\underline{\mathcal{I}}'$ the transform of $\underline{\mathcal{I}}_{T_0}$ by σ with its last (empty) member removed. Then $\underline{\mathcal{I}}_{T_0\sigma} = (\underline{\mathcal{I}}')^{(m)}$ with $m = \#E + \operatorname{length} T_0$, and along $T_0\sigma S$: $\gamma = \sigma_m \circ \gamma_S(\underline{\mathcal{I}}')$, $\nu = \nu_S(\underline{\mathcal{I}}')$, the cosupport is that of $\underline{\mathcal{I}}'_S$ and the maximum loci of γ agree. If N and the members of $\underline{\mathcal{I}}_{T_0}$ are closed, $\underline{\mathcal{I}}'$ is restricted-equivalent to $\sigma^*\underline{\mathcal{I}}_{T_0}$, and (for coherent ideals) $\gamma_S(\underline{\mathcal{I}}') = \gamma_S(\sigma^*\underline{\mathcal{I}}_{T_0})$.*

5. *(The general value) At a point of order 0, the general value of Definition **??** of $\underline{\mathcal{I}}^{(j)}$ along T is $(0, \sigma_j(\gamma_T))$, and after an empty step in the middle as in (4) it is $(0, \sigma_m(\gamma_S(\underline{\mathcal{I}}')))$ while that of $\underline{\mathcal{I}}'$ along S is $(0, \gamma_S(\underline{\mathcal{I}}'))$. These two values differ whenever the set of $\gamma_S(\underline{\mathcal{I}}')$ at the point contains a position $\geq m$.*

So the values of the general step are not unchanged when an empty step is inserted in the middle of a sequence; they change by the strictly increasing relabelling σ_m, independent of the point, which keeps the maximum loci.

Proof. (1) The first two entries are kept and $K \mapsto s_j(K)$ preserves the colexicographic order since s_j is strictly increasing; if K has an element $\geq j$, its largest element k is sent to $k+1 \notin K$. Maximum loci of $g \circ f$ for strictly increasing g are those of f (Lemma 174.83). (2) The members through a point of $E^{(j)}$ are the images under s_j of those of E (the inserted member contains no point), with the same exponents (Lemma **??**). The feasible sets for $s_j(S)$ are the images of the feasible sets for S, with the same cardinalities and sums, so the largest key is the image of the largest key (Lemma **??**). The monomial part only involves the members through the point (Proposition **??**), so it does not change, nor do the nonmonomial part, ν (Lemma 174.90) and

the cosupport, which does not involve E. (3) The transform replaces each member by its strict transform and appends $\pi^{-1}C$ at the end (Lemma **??**); the strict transform of $\emptyset$ is $\emptyset$, so inserting $\emptyset$ at position $j \le \#E$ commutes with one transform, and by induction with the transforms along every tower (the length of E only grows). The rest is (2) and (1). (4) The transform by σ is $\underline{\mathcal{I}}'$ with $\sigma^{-1}\emptyset = \emptyset$ appended at position $\#E(\underline{\mathcal{I}}_{T_0}) = m$; apply (3) along S. For closed N and members the strict transforms along $\emptyset$ are the preimages, and the ideals of $\underline{\mathcal{I}}'$ and of $\sigma^*\underline{\mathcal{I}}_{T_0}$ agree on N' (Lemma **??**); restricted-equivalent marked ideals have the same γ along every tower (Lemma 174.98(2)). (5) By Lemma 174.99(1) and (3), (4) (the orders agree, so both points have order 0); the values differ by (1). □

Compact sets along sequences of blowups. Local finiteness of the resolution invariant (property (8) of Definition **??**, and its form along growing sequences of canonical segments) is proved phase by phase: at a stage T one counts the later nonempty centres meeting the preimage of a compact set K of the base against a compact subset of the stage T, inside an open set on which a lower-dimensional invariant lives. The lemma below gives the topological part.

Lemma 174.101 (Sequences of blowups are proper; the compact part of the cosupport)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let K be a compact subset of the base X.

1. *A blowup $\pi\colon Z \to M$ along a smooth centre is proper: $\pi^{-1}(K)$ is compact for every compact $K \subset M$. Hence $p_T^{-1}(K)$ is compact for the projection p_T of every tower T.*

2. *Let $\underline{\mathcal{I}}$ lie in the domain of Definition **??** and T be a multiple test blow-up of it. Then $K_T = p_T^{-1}(K) \cap \operatorname{cosupp}\underline{\mathcal{I}}_T$ is compact, and along every multiple test blow-up TS the points of $\operatorname{cosupp}\underline{\mathcal{I}}_{TS}$ lie over $\operatorname{cosupp}\underline{\mathcal{I}}_T$.*

3. *For towers T, S, the number of nonempty centres of TS meeting the preimage of K is that of T plus the number of nonempty centres of S meeting the preimage of $p_T^{-1}(K)$; if TS is a multiple test blow-up of $\underline{\mathcal{I}}$ (closed submanifold N, coherent ideal), the latter are the nonempty centres of S meeting the preimage of K_T.*

4. *If moreover W is an open set of the last stage of T containing K_T, then K_T is a compact subset of W and the nonempty centres of TS meeting the preimage of K are those of T plus those of the restricted tail $S|_W$ meeting the preimage of K_T. The restriction to W of S followed by a blowup is $S|_W$ followed by the restricted blowup, so restrictions of growing sequences are growing sequences.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Every point of M has a neighbourhood with compact preimage (the local part of Lemma **??**: on the centre a closed ball of a flattening chart, off it a compact neighbourhood missing the centre, over which π has a holomorphic section). Finitely many of them cover K, and $\pi^{-1}(K)$ is a closed subset of the union of their preimages (M is Hausdorff). For towers, induction on the length. (2) $\underline{\mathcal{I}}_T$ lies in the domain (Proposition 174.75), so its cosupport is closed (Lemma **??**) and K_T is a closed subset of the compact $p_T^{-1}(K)$. For one test blowup a point of the new cosupport lies over the old cosupport: over the centre, which is admissible, and off it, where the blowup is an isomorphism (Theorem **??**); induction along S. (3) The projection of TS is $p_T \circ p_S$ (Lemma **??**); induction along S. A centre of S is admissible, so it lies in the cosupport of its stage, which lies over $\operatorname{cosupp}\underline{\mathcal{I}}_T$ by (2); so it meets the preimage of $p_T^{-1}(K)$ iff it meets that of K_T. (4) K_T is compact in W since W carries the subspace topology. The count along $S|_W$ of a subset of W is the count along S (Lemma **??**), and the restriction of a tower is defined step by step (Lemma 174.6). □

The phases of a canonical segment. Along a canonical segment of the general value (Definition **??**) the nonempty centres are maximum loci, and their orders ν of the nonmonomial part form the sequence of phases of the segment. Local finiteness is proved phase by phase: the lemma below collects what the order contributes, for every value function whose order dominates ν (Lemma 174.92).

Lemma 174.102 (The phases of a canonical segment: the order of the nonmonomial part)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $n \geq 1$, $\underline{\mathcal{I}}$ lie in the domain of Definition **??***, and f be a value function whose order dominates ν (Lemma 174.92). Let T_0 be a multiple test blow-up with $\nu \leq o$ on* $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0}$.

1. *(The phase is absorbing) Along every canonical segment $T_0 S$ which is a multiple test blow-up, every point z of the cosupport lies over the cosupport at T_0, and $\nu(z) \leq \nu(p_S(z)) \leq o$; if $\nu(z) = o$, then $\nu(p_S(z)) = o$.*

2. *(Start of the run) Such a point z of order o is in the regime o (Lemma 174.91) and its run is at least the length of S: it reached the order o at the last stage of T_0 or before. Moreover* $\operatorname{cosupp} \underline{\mathcal{I}}_{T_0} \subset W_o(\underline{\mathcal{I}}_{T_0})$*, the mask of Lemma 174.93.*

3. *(Phase entry; the phases never go up) If T followed by a blowup along C is a canonical segment, every point of C bounds the order on* $\operatorname{cosupp} \underline{\mathcal{I}}_T$*. Along a canonical segment which is a multiple test blow-up the orders of the points of the nonempty centres, in the order of the steps, never increase, and every point of the last cosupport has order at most each of them.*

4. *(Over a compact set) For every compact $K \subset X$ there is q such that along every canonical segment which is a multiple test blow-up, every point of a nonempty centre meeting the preimage of K has order $\leq q$.*

5. *(The monomial phase) A value of the general step with first entry* 0 *is $(0, \gamma)$; no sequence of such values decreases strictly forever, since the values of γ are well-founded (Lemma* **??***). If the values of the maximal-order invariant are well-founded, so are the values of the general step.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The order never goes up along a canonical segment (Lemma 174.92), and the image of a point of the cosupport lies in the cosupport. (2) Induction along S: after one more step a point z of order o lies over a point of the cosupport of order exactly o by (1), which is in the regime with run at least the length so far (Lemma **??**). A point of the cosupport at T_0 has order $\leq o$, so it is off the closed set $\{\nu \geq o + 1\}$, i.e. in W_o (Lemma 174.93). (3) A nonempty centre of a canonical step is the maximum locus, and a point of the maximum locus has the largest order on the cosupport (Lemma 174.92). Induction along the segment: the new entry is the order at a point of the new centre, which lies in the cosupport and so has order at most every earlier entry; after the step the orders are at most those of the images, which are at most every entry, including the new one by phase entry. (4) Lemma 174.95 bounds ν at the points of the cosupport over K along all such segments; a nonempty centre meeting the preimage of K is the maximum locus, which lies in one level set of ν and contains a point of the cosupport over K. (5) Definition **??** and Lemma 174.99(1). The pairs $(0, \gamma)$ are ordered as the γ; lexicographic products and ordered sums of well-founded orders are well-founded (as for the tuples of Lemma **??**). □

Growing sequences: restriction and tails. Property (8) in its form along growing sequences (Definition **??**) bounds the numbers of nonempty centres meeting a compact set along a growing sequence of towers (each extending the previous one by one blowup). The phase argument passes, at a stage, to the rest of the sequence restricted to open sets of that stage.

Lemma 174.103 (Restriction and tails of growing sequences)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $s = (T_i)_{i \geq 0}$ be a growing sequence of towers over X (T_0 empty, T_{i+1} is T_i followed by one blowup).

1. *For an open $U \subset X$, the restrictions $T_i|_U$ form a growing sequence over U; for $K \subset U$ the numbers of nonempty centres meeting the preimage of K along it and along s agree.*

2. *For a stage a, write $T_{a+i} = T_a S_i$; the towers S_i over the last stage of T_a form a growing sequence (the tail of s from a), and the number of nonempty centres of T_{a+i} meeting the preimage of K is that of T_a plus the number of nonempty centres of S_i meeting the preimage of $p_{T_a}^{-1}(K)$. These numbers never decrease along s.*

3. *Let the T_i be multiple test blow-ups of a marked ideal $\underline{\mathcal{I}}$ of the domain of Definition **??**, a a stage, and $L_j \subset W_j$ (j in a finite set, W_j open in the last stage of T_a) with $p_{T_a}^{-1}(K) \cap \operatorname{cosupp} \underline{\mathcal{I}}_{T_a} \subset \bigcup_j L_j$. If for each j the numbers of nonempty centres meeting the preimage of L_j along the tail from a restricted to W_j are bounded, then the numbers of nonempty centres meeting the preimage of K along s are bounded.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The restriction of a tower followed by a blowup is the restriction followed by the restricted blowup (Lemma 174.101(4), Lemma 174.6), and the restriction of the empty tower is empty; the counts agree by Lemma **??**. (2) Choose the blowups of s after the stage a; by induction $T_{a+i} = T_a S_i$ (Lemma **??**), and the count splits by Lemma 174.101(3). A blowup adds one or no centre. (3) For $i \leq a$ the count is at most that of T_a by (2). For $i = a + d$, the count of T_a plus the count of S_d for $p_{T_a}^{-1}(K)$, which is the count for $p_{T_a}^{-1}(K) \cap \operatorname{cosupp} \underline{\mathcal{I}}_{T_a}$ (Lemma 174.101(3)), at most the sum over j of the counts for L_j (counts are monotone and subadditive in the set, Lemma **??**), which are those along the restricted tails by (1). □

The count of centres by phases. Along a canonical segment of the general value the values of the nonempty centres, in the order of the steps, strictly decrease (Theorem **??**, Theorem **??**); their first entries are the phases. The lemma below splits the count of property (8) along growing sequences (Definition **??**) by phases and settles the phase of order 0.

Lemma 174.104 (The count of centres by phases; the monomial phase is finite)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $n \geq 1$, $\underline{\mathcal{I}}$ lie in the domain of Definition **??**, M be a value function for the marked ideals of maximal order (Definition **??**) and* inv *the general value. For a tower T let $v(T)$ be the list of the values of the nonempty centres (Theorem **??**).*

1. *Prefix-growing lists, each strictly decreasing, with entries in a set on which no sequence strictly decreases forever, have bounded lengths.*

2. *Let $K \subset X$ and q be such that along every canonical segment every point of a nonempty centre meeting the preimage of K has order $\leq q$ (for compact K, Lemma 174.102(4)). Along a canonical segment T, the number of nonempty centres meeting the preimage of K is at most the number of entries of $v(T)$ with first entry $\leq q$, which is the sum over $o = 0, \ldots, q$ of the numbers $\ell_o(T)$ of entries with first entry o.*

3. *Along a growing sequence (T_i) of canonical segments, $\ell_0(T_i)$ is bounded: the monomial phase is finite.*

4. *Hence, if $\ell_o(T_i)$ is bounded for each $o = 1, \ldots, q$, the numbers of nonempty centres of T_i meeting the preimage of K are bounded.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) Otherwise, for each k choose a list of length $> k$ and take its k-th entry; by the prefix property all these entries are entries of one later list at the positions $0, 1, 2, \ldots$, so they strictly decrease. (2) Induction along T: a step along a nonempty centre adds the value at a point of the centre, whose first entry is the order of that point (Definition **??**); if the centre meets the preimage of K, it is $\leq q$. Partition the entries $\leq q$ by their first entry. (3) The lists $v(T_i)$ grow by prefixes and are strictly decreasing (Theorem **??** with the drop property of Theorem **??**); so are their entries of first entry 0, which are $(0, \gamma)$ (Lemma 174.102(5)); γ has well-founded values (Lemma **??**), so (1) applies. (4) By (2), the count is at most $\sum_{o \leq q} \ell_o(T_i)$, bounded by (3) and the hypothesis. □

The companion phase. While the order of the nonmonomial part stays $o \geq 1$, Włodarczyk's Step 2a "repeats the procedure" for one companion ideal. The lemma below makes this precise along canonical segments of the general value and bounds the phase of order o along growing sequences (Definition **??**) from property (8) of the value function for the marked ideals of maximal order.

Lemma 174.105 (One companion ideal for a phase; the phases of positive order are finite)**.** *Verified upstream; not exported by this standalone library.*

*Let $n \geq 1$, $\underline{\mathcal{I}}$ lie in the domain of Definition **??**, M be a value function for the marked ideals of maximal order (Definition **??**),* inv *the general value and ν the order of the nonmonomial part. Let $o \geq 1$, and let T_0 be a multiple test blow-up with $\nu \leq o$ on* cosupp $\underline{\mathcal{I}}_{T_0}$ *and every centre in $\{\nu \geq o+1\}$; let W be the open set $\{\nu \leq o\}$ of the last stage of T_0 and $\underline{\mathcal{O}}$ the companion ideal of order o formed there on W (Definition **??**).*

1. *If $T_0 S$ is a canonical segment of* inv *and every point of every centre of S has order $\geq o$, then $S|_W$ is a canonical segment of $\underline{\mathcal{O}}$ for M, and at every point q of the cosupport of the transform of $\underline{\mathcal{O}}$ along $S|_W$ the general value is $(o, M_{\underline{\mathcal{O}}}(S|_W)(q))$.*

2. *For $K \subset X$ let $c_o(K, T)$ be the number of steps of T whose centre has a point over K of order o. Then $c_o(K, T)$ is at most the number of centres meeting the preimage of K, it is additive along concatenations, and it vanishes when every centre lies in $\{\nu \geq o+1\}$. Along a canonical segment, with q as in Lemma 174.104(2), the number of centres meeting the preimage of K is at most $\sum_{o \leq q} c_o(K, T)$, and $c_o(K, T) \leq \ell_o(T)$.*

3. *Suppose that for every marked ideal of the domain of maximal order along its submanifold, every compact set and every growing sequence of canonical segments for M, the numbers of centres meeting the compact set are bounded (property (8) of M, per sequence). Then for K compact and $o \geq 1$, along every growing sequence (T_i) of canonical segments of* inv*, $c_o(K, T_i)$ is bounded.*

Proof. Verified upstream; not exported by this standalone library.

(1) By induction along S. A nonempty centre is the maximum locus of inv; its points have order $\leq o$ (the phase is absorbing, Lemma 174.102(1)) and $\geq o$, hence o, and lie over W. Over W the cosupport of the companion's transform is the set of points of order o of the cosupport (Theorem **??**). At such a point the run of order o started at or before T_0 (Lemma 174.102(1)); splitting $T_0 S$ at the start of the run (Lemma **??**), the steps from there to T_0 have centres in $\{\nu \geq o+1\}$, which miss the mask of the point (Lemma 174.93), so the companion value re-bases to T_0 (Theorem **??**): it is $(o, M_{\underline{\mathcal{O}}}(S|_W))$. Every other point of the cosupport has smaller order, hence smaller value, and the centre has a point of order o over W; so the trace of the maximum locus of inv on the preimage of W is the maximum locus of M (Lemma 174.83). (2) Induction along T. For the sum: a centre meeting the preimage of K at a point c has $\nu(c) \leq q$ and is counted in $c_{\nu(c)}$. For ℓ_o: the points of a canonical centre have one value, whose first entry is

their order (Lemma **??**, Lemma 174.92). (3) If at some stage $\nu < o$ on the whole cosupport, this persists (Lemma 174.92) and c_o is constant from there on. Otherwise every canonical centre has order $\geq o$. If at every stage some point of the cosupport has order $> o$, no centre has order o and $c_o = 0$. Otherwise let a be the first stage with $\nu \leq o$ on the cosupport: every centre before a lies in $\{\nu \geq o+1\}$ (it is a maximum locus and some point has order $> o$; Lemma 174.90), and every centre after a has order o. By (1) the restricted tails of the sequence after a (Lemma 174.103) are canonical segments of the one companion ideal $\underline{\mathcal{O}}$ at T_a, which lies in the domain and has maximal order (Theorem **??**). The part of the cosupport at T_a over K is compact and lies in W, and the centres after a meeting the preimage of K are those of the restricted tails meeting it (Lemma 174.101); the hypothesis bounds them, and c_o is at most the number of centres meeting the preimage of K. □

Admissible centres of the restricted marked ideal of Step 1ab. In Step 1ab of Proposition **??** the value at a point of level $s \geq 1$ is that of the lower-dimensional invariant of the restricted marked ideal of Lemma 174.78 on a mask. Admissibility of its maximum locus (Definition **??**(1)) is inherited from the inductive invariant through the lemma below, Włodarczyk's Lemma "le:S" for a single centre.

Lemma 174.106 (Admissible centres of the restricted marked ideal)**.** *Verified upstream; not exported by this standalone library.*

Let $n \geq 1$, $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be well formed with $d \geq 1$, S a set of members of E, and $\underline{\mathcal{S}}$ the restricted marked ideal $(M, N \cap H_S, E_S, \mathcal{C}_N(\mathfrak{a}), \mu_{\mathcal{C}})$ of Lemma 174.78.

1. *Along a multiple test blow-up T of $\underline{\mathcal{S}}$, a set C of the last stage is an admissible centre of $\underline{\mathcal{S}}_T$ if and only if it is an admissible centre of $\underline{\mathcal{I}}_T$ contained in the strict transform $(N \cap H_S)_T$, equivalently in* $\operatorname{cosupp} \underline{\mathcal{S}}_T$.

2. *In Step 1ab, let T_0T_1 be a multiple test blow-up of $\overline{\underline{\mathcal{J}}}$, y_0 a point of the last stage of T_0 lying in its own mask W, and $T_1|_W$ a multiple test blow-up of the lower marked ideal $\underline{\mathcal{S}}(y_0)|_W$ (Definition **??**). Then a set C of the preimage of W is an admissible centre of the transform of $\underline{\mathcal{S}}(y_0)|_W$ along $T_1|_W$ if and only if it is an admissible centre of $\overline{\underline{\mathcal{J}}}_{T_0T_1}$ restricted to the preimage of W and lies on the strict transforms along T_1 of the original members through y_0.*

Proof. Verified upstream; not exported by this standalone library.

(1) By Lemma **??**, T is a multiple test blow-up of $\underline{\mathcal{I}}$ with centres in the strict transforms of $N \cap H_S$, and $\operatorname{cosupp} \underline{\mathcal{S}}_T = \operatorname{cosupp} \underline{\mathcal{I}}_T \cap (N \cap H_S)_T$; the same holds for the restricted coefficient ideal with the full divisor (Lemma **??**), whose transform has the divisor of $\underline{\mathcal{I}}_T$ (Lemma **??**), simple normal crossings (Lemma **??**). The members of the transformed divisor that are not members of the transform of E_S contain $(N \cap H_S)_T$ (Lemma **??**), so for $C \subset (N \cap H_S)_T$ normal crossings with the two divisors are equivalent (Lemma 174.127); admissibility for $\underline{\mathcal{S}}_T$ is admissibility for the transform with the divisor replaced (Lemma **??**). Both conditions put C in $\operatorname{cosupp} \underline{\mathcal{I}}_T \cap (N \cap H_S)_T$. (2) On the mask the lower marked ideal is the restricted marked ideal of $\overline{\underline{\mathcal{J}}}_{T_0}|_W$ at the members through y_0 (Lemma 174.81), and $\overline{\underline{\mathcal{J}}}_{T_0}$ is well formed with mark ≥ 1 (Lemma **??**, Lemma **??**); apply (1) on W. The transform of $\overline{\underline{\mathcal{J}}}_{T_0}|_W$ along $T_1|_W$ is the restriction of $\overline{\underline{\mathcal{J}}}_{T_0T_1}$ (Lemmas **??** and **??**), and the cosupport of the lower transform consists of the cosupport points of $\overline{\underline{\mathcal{J}}}_{T_0T_1}$ on the strict transforms of the original members through y_0 (Lemma **??**; Lemma **??** for the restricted cosupport). □

Corollary 174.107 (Admissible centres of the restricted marked ideal on open sets)**.** *Let $\underline{\mathcal{A}}$, $\underline{\mathcal{B}}$ be marked ideals on one manifold, $\underline{\mathcal{A}}$ with coherent ideal and $\underline{\mathcal{B}}$ well formed, such that* $\operatorname{cosupp} \underline{\mathcal{A}} \subset \operatorname{cosupp} \underline{\mathcal{B}}$*, every member of the divisor of $\underline{\mathcal{A}}$ is a member of that of $\underline{\mathcal{B}}$, and every member of $\underline{\mathcal{B}}$ missing from $\underline{\mathcal{A}}$ contains* $\operatorname{cosupp} \underline{\mathcal{A}}$*. Then for every open set W'', every admissible centre of $\underline{\mathcal{A}}|_{W''}$*

is an admissible centre of $\underline{\mathcal{B}}|_{W''}$. The three relations hold for $\underline{\mathcal{A}} = \underline{\mathcal{S}}_T$, $\underline{\mathcal{B}} = \underline{\mathcal{I}}_T$ in the situation of Lemma 174.106(1), and for the lower marked ideal of Step 1ab and the restriction of $\overline{\underline{\mathcal{J}}}_{T_0T_1}$ to the preimage of its mask in the situation of Lemma 174.106(2); so the transfers of that lemma hold on every open set of the last stage.

Proof. The restricted centre is a closed submanifold of W'' for both; restriction to an open set takes cosupports to their traces (Lemma **??**), so the trace of $\operatorname{cosupp}\underline{\mathcal{A}} \subset \operatorname{cosupp}\underline{\mathcal{B}}$ gives the cosupport condition; the restricted divisors are the traces of the divisors, that of $\underline{\mathcal{B}}$ simple normal crossings, and the traces of the members of $\underline{\mathcal{B}}$ missing from $\underline{\mathcal{A}}$ contain the centre, so normal crossings pass (Lemma 174.127). The relations along T: the cosupport of $\underline{\mathcal{S}}_T$ is $\operatorname{cosupp}\underline{\mathcal{I}}_T \cap (N \cap H_S)_T$ (Lemma **??**), the members of the transform of E_S are members of E_T, and the others contain $(N \cap H_S)_T$ (Lemma **??**, with Lemmas **??**, **??** and **??**). For Step 1ab, the lower marked ideal on its mask is the restricted marked ideal of $\overline{\underline{\mathcal{J}}}_{T_0}|_W$ (Lemma 174.81), and the transform of $\overline{\underline{\mathcal{J}}}_{T_0}|_W$ along $T_1|_W$ is the restriction of $\overline{\underline{\mathcal{J}}}_{T_0T_1}$ (Lemmas **??** and **??**). □

The companion ideal along isomorphisms of towers. Property (6) of the general step (Proposition **??**) compares the companion ideal formed at the start stage of a point, on its mask, with the same construction for $\Phi^*\underline{\mathcal{I}}$ along an isomorphic tower. An isomorphism $T_0'T_1' \to T_0T_1$ over Φ splits into $\chi_0\colon T_0' \to T_0$ over Φ and $T_1' \to T_1$ over χ_0 (Lemma **??**(2)).

Lemma 174.108 (The companion ideal along isomorphisms of towers)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let $n \geq 1$, $\underline{\mathcal{I}}$ in the domain of Definition **??**, $o \geq 1$, $\chi_0\colon T_0' \to T_0$ an isomorphism of towers over a biholomorphism $\Phi\colon X' \to X$ with T_0 a multiple test blow-up of $\underline{\mathcal{I}}$, and W', W open sets of the last stages with $q \in W' \iff \chi_0(q) \in W$. Write $\mathcal{O}_0 = \mathcal{O}(\underline{\mathcal{I}}_{T_0}, o)|_W$ and $\mathcal{O}_0' = \mathcal{O}((\Phi^*\underline{\mathcal{I}})_{T_0'}, o)|_{W'}$.*

1. *$\mathcal{O}_0' = \psi^*\mathcal{O}_0$ for every biholomorphism $\psi\colon W' \to W$ agreeing with χ_0.*
2. *If $\nu \leq o$ on $\operatorname{cosupp}\underline{\mathcal{I}}_{T_0} \cap W$ (Lemma 174.90), then $\nu \leq o$ on $\operatorname{cosupp}(\Phi^*\underline{\mathcal{I}})_{T_0'} \cap W'$.*
3. *Let $T_1' \to T_1$ be an isomorphism of towers over χ_0. If $T_1|_W$ is a multiple test blow-up of $\mathcal{O}_0$, then $T_1'|_{W'}$ is one of $\mathcal{O}_0'$; if moreover* inv *is a resolution invariant and $T_1|_W$ a canonical segment of $\mathcal{O}_0$, then $\mathcal{O}_0'$ lies in the domain, $T_1'|_{W'}$ is a canonical segment of it, and $\operatorname{inv}_{\mathcal{O}_0'}(T_1'|_{W'})(p) = \operatorname{inv}_{\mathcal{O}_0}(T_1|_W)(q)$ at corresponding points of the cosupports.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) $(\Phi^*\underline{\mathcal{I}})_{T_0'} = \chi_0^*\underline{\mathcal{I}}_{T_0}$ (Lemma **??**), the companion ideal commutes with biholomorphisms (Lemma 174.87(2)), and restricting $\chi_0^*\mathcal{O}$ to W' is pulling back $\mathcal{O}|_W$ along ψ, as χ_0 and ψ agree on W' (Theorem **??**); $\mathcal{O}(\underline{\mathcal{I}}_{T_0}, o)$ has a coherent ideal (Proposition 174.79, the stage in the domain by Lemma 174.75). (2) Cosupports correspond along χ_0 (Lemma **??**) and so does ν (Lemma 174.90(3)). (3) The isomorphism restricts to $T_1'|_{W'} \to T_1|_W$ over the restriction ψ of χ_0 (Lemma **??**); multiple test blow-ups correspond along isomorphisms of towers (Lemma **??**) and so do canonical segments (Lemma **??**(2)); $\psi^*\mathcal{O}_0$ lies in the domain (Lemma 174.75); the values are those of Definition **??**(6). □

Step 1ab's lower values along isomorphisms of towers. Properties (5)–(7) of the value function of Step 1 (Proposition **??**, Definition **??**) compare lower values computed from different but corresponding data. At the levels $s \geq 1$ the lower marked ideal is the restricted marked ideal of Lemma 174.78; for level 0 see Theorem **??**.

Lemma 174.109 (Step 1ab's lower values along isomorphisms of towers)**.** *Let $n \geq 1$ and $\underline{\mathcal{J}}$ be well formed with $d \geq 1$.*

1. *For a biholomorphism $\chi\colon M' \to M$, a well-formed marked ideal $\underline{\mathcal{I}}$ on M with $d \geq 1$, a list E_0 of sets and $p \in M'$, the pullback along χ of the restricted marked ideal of $\underline{\mathcal{I}}$ at the members of E_0 through $\chi(p)$ is the restricted marked ideal of $\chi^*\underline{\mathcal{I}}$ at the members of $\chi^{-1}E_0$ through p.*

2. *(Anchor) Let T_0T_1 be a multiple test blow-up of $\overline{\underline{\mathcal{J}}}$ and p a point with lower data at level $s+1$ at the split T_0T_1 (Definition **??**), $\mathcal{M}$ the restricted marked ideal of $\overline{\underline{\mathcal{J}}}_{T_0}$ at the original members through the image y_0 of p. For every open $W \ni y_0$ on which $\mathcal{M}|_W$ lies in the domain of the inductive invariant R, with $T_1|_W$ one of its canonical segments, the lower value of p is $R_{\mathcal{M}|_W}(T_1|_W)(p)$.*

3. *(Property (6) at the levels $s+1$) Let $\chi_0\colon T_0' \to T_0$ be an isomorphism of towers over a biholomorphism Φ, $\chi\colon T_1' \to T_1$ one over χ_0, and T_0T_1 a multiple test blow-up of $\overline{\underline{\mathcal{J}}}$. If p' has lower data at level $s+1$ for $\Phi^*\underline{\mathcal{J}}$ at the split $T_0'T_1'$ and $\chi(p')$ has lower data at level $s+1$ for $\underline{\mathcal{J}}$ at the split T_0T_1, the two lower values agree.*

Proof. (1) The coefficient ideal commutes with biholomorphisms (Lemma **??**); the submanifold and the divisor of the restricted marked ideal are described by the members through the point (Lemma 174.81), and preimages under a bijection distinguish sets, so the members of $\chi^{-1}E_0$ through p are the preimages of those of E_0 through $\chi(p)$. (2) The lower value is R of $\mathcal{M}$ restricted to the mask along T_1 restricted to it; on the two open sets the values agree at the points of the cosupport over both (Lemma **??**, on the restricted marked ideal, well formed by Lemma 174.78, the stage $\overline{\underline{\mathcal{J}}}_{T_0}$ being well formed with mark ≥ 1, Lemmas **??** and **??**). (3) $\overline{(\Phi^*\underline{\mathcal{J}})} = \Phi^*\overline{\underline{\mathcal{J}}}$ (Lemma **??**), so its transform along T_0' is χ_0^* of that along T_0 (Lemma **??**) and its original members are the preimages (Theorem **??**); by (1) its restricted marked ideal at the image y_0' of p' is $\chi_0^*\mathcal{M}$, with $\chi_0(y_0') = y_0$. On $W' = \chi_0^{-1}(\text{mask of } y_0)$, $\chi_0^*\mathcal{M}|_{W'}$ is the pullback of $\mathcal{M}$ on the mask along the restriction ψ of χ_0 (Theorem **??**), it lies in the domain (Lemma 174.75), and $T_1'|_{W'}$ is isomorphic over ψ to the restriction of T_1 to the mask (Lemma **??**), hence a canonical segment with corresponding values (Lemma **??**(2), Definition **??**(6)). The concatenations are isomorphic (Lemma **??**), so $T_0'T_1'$ is a multiple test blow-up of $\overline{(\Phi^*\underline{\mathcal{J}})}$; conclude with (2) for $\Phi^*\underline{\mathcal{J}}$ on W'. □

Lemma 174.110 (Step 1ab's lower values under restriction and change of representative)**.** *Let $n \geq 1$ and $\underline{\mathcal{J}}$ be well formed with $d \geq 1$, and T_0T_1 a multiple test blow-up of $\overline{\underline{\mathcal{J}}}$.*

1. *(Transport) If $\mathcal{A}$ lies in the domain of the inductive invariant R with S one of its canonical segments, and $\xi\colon S' \to S$ is an isomorphism of towers over a biholomorphism ψ, then $\psi^*\mathcal{A}$ lies in the domain, S' is one of its canonical segments, and at every p with $\xi(p)$ in the cosupport along S, p lies in the cosupport along S' and $R_{\psi^*\mathcal{A}}(S')(p) = R_{\mathcal{A}}(S)(\xi(p))$.*

2. *(Property (5) at the levels $s+1$) Let U be open in the base, $\chi_0\colon T_0' \to T_0|_U$ an isomorphism of towers over the identity of U and $\chi\colon T_1' \to T_1|_{T_0^{-1}U}$ one over χ_0. If p' has lower data at level $s+1$ for $\underline{\mathcal{J}}|_U$ at the split $T_0'T_1'$ and $\chi(p')$ has lower data at level $s+1$ for $\underline{\mathcal{J}}$ at the split T_0T_1, the two lower values agree.*

3. *For $\underline{\mathcal{J}} \simeq \underline{\mathcal{J}}'$ restricted-equivalent with $\underline{\mathcal{J}}'$ coherent, every tower T_0 and every point y_0 of its last stage, the restricted marked ideals of $\overline{\underline{\mathcal{J}}}_{T_0}$ and $\overline{\underline{\mathcal{J}}'}_{T_0}$ at the original members through y_0 are restricted-equivalent.*

4. *(Property (7) at the levels $s+1$) For $\underline{\mathcal{J}} \simeq \underline{\mathcal{J}}'$ as in (3), if p has lower data at level $s+1$ at the split T_0T_1 both for $\underline{\mathcal{J}}$ and for $\underline{\mathcal{J}}'$, the two lower values agree.*

Proof. (1) Domain (Proposition 174.75), canonical segments (Lemma **??**(2)) and values (Definition **??**(6)) pass along isomorphisms of towers; cosupports correspond (Lemma **??**), S being a multiple test blow-up (Theorem **??**). (2) $\overline{(\underline{\mathcal{J}}|_U)} = \overline{\underline{\mathcal{J}}}|_U$ (Theorem **??**); multiple test blow-ups

and transforms restrict to open sets (Lemma **??**) and pass along the isomorphisms (Lemmas **??** and **??**), and the original members are the traces (Theorem **??**). By Lemma 174.109(1) and Lemma 174.81, the restricted marked ideal of $\underline{\mathcal{J}}|_U$ at the image of p' is $\chi_0^*(\mathcal{M}|_{T_0^{-1}U})$, with $\mathcal{M}$ that of $\underline{\mathcal{J}}$ at the image y_0 of $\chi(p')$. Shrink the mask A of y_0 to $W_1 = A \cap T_0^{-1}U$ (Lemma 174.85, values by Lemma **??**), view W_1 inside $T_0^{-1}U$ (Lemmas 174.84 and 174.85), transport by (1) along the restriction of χ_0 to $W' = \chi_0^{-1}W_1$ (Lemma **??**, Theorem **??**), and conclude with the anchor (Lemma 174.109(2)) for $\underline{\mathcal{J}}|_U$ on W'. (3) $\overline{\mathcal{J}} \simeq \overline{\mathcal{J}'}$ (Theorem **??**), so the coefficient ideals of their transforms along T_0 are restricted-equivalent (Lemma **??**), the divisors of the transforms agree (Lemma **??**) and so do the original members (Theorem **??**); the submanifolds and divisors of the two restricted marked ideals are described by the members through y_0 (Lemma 174.81), and the restricted inclusions pass to the smaller submanifold (Lemma **??**). (4) T_0T_1 is a multiple test blow-up of $\overline{\mathcal{J}'}$ as well (Lemma **??**). On the intersection W of the two masks both restricted marked ideals lie in the domain with $T_1|_W$ a canonical segment (Lemma 174.85) and p in the cosupport (Lemma **??**); their restrictions to W are restricted-equivalent by (3) (Theorem **??**), so the values agree (Definition **??**(7)); conclude with the anchor on both sides. □

The pushforward of $\mathcal{O}(-KD)$. For the blowup π along a smooth centre C with exceptional divisor D, a function whose pullback vanishes to order K along D vanishes to order K along C: $\pi_*\mathcal{O}(-KD) = \mathcal{I}_C^K$. The statement below is local on the blowup: the hypothesis is at a single point of D. It is the blowup half of the step "the nonmonomial part stays nonmonomial" in the proof of Lemma **??**.

Theorem 174.111 (The pushforward of $\mathcal{O}(-KD)$)**.** *Let $\pi\colon Y \to X$ be the blowup along C, $D = \pi^{-1}(C)$, $z \in D$ and f holomorphic at $\pi(z)$. If $f \circ \pi \in \mathcal{O}(-D)^K$ at z, then $f \in \mathcal{I}_C^K$ at $\pi(z)$.*

Proof. Let $x = \pi(z)$. Take flattening coordinates Φ at x, centred at 0, with $C = \mathbf{C}^k \times 0$, put $F = f \circ \Phi^{-1}$, and take a monomial chart $j \geq k$, W around z with its lift ℓ, $\pi \circ \ell = \Phi^{-1} \circ \sigma_j$ (Lemma 174.11); ℓ is read on $\mathbf{C}^n$ as in Lemma **??**. Since $\Phi(x) = 0$, the point $W(z)$ has $W_j(z) = 0$ and vanishing tangential coordinates.

(1) The germs of $\mathcal{O}(-D) = \pi^*\mathcal{I}_C$ at z vanish on D near z (Theorem 174.13), so a germ of $\mathcal{O}(-D)^K$ at z lies in $\mathfrak{m}_u^K$ at every $u \in D$ near z: induction on K, a germ of $\mathfrak{a}^{K+1}$ being a combination of products of a germ of $\mathfrak{a}^K$ and a germ of $\mathfrak{a}$. Pulling back along ℓ (Lemma **??**), $G = F \circ \sigma_j \in \mathfrak{m}_\lambda^K$ at every λ near $W(z)$ with $\lambda_j = 0$, so $D^iG(\lambda) = 0$ for $i < K$ (Lemma **??**).

(2) For $w \in \mathbf{C}^k \times 0$ and a direction h with $h_j \neq 0$, let $\lambda(w,h)$ have the tangential coordinates of w, the coordinates h_i/h_j for the normal $i \neq j$, and $\lambda_j = 0$, and let $v(h)$ have the tangential coordinates of h, $v_j = h_j$, and 0 otherwise. Then $\sigma_j(\lambda + tv) = w + th$ for every t. With $h_0 = W(z) + e_j$ one has $\lambda(0,h_0) = W(z)$, so for w near 0 and h in an open neighbourhood V of h_0, $\lambda(w,h)$ lies where (1) applies and

$$D^aF(w)(h,\dots,h) = \tfrac{d^a}{dt^a}\Big|_{t=0} F(w+th) = \tfrac{d^a}{dt^a}\Big|_{t=0} G(\lambda+tv) = D^aG(\lambda)(v,\dots,v) = 0 \qquad (a < K).$$

(3) The diagonal $h \mapsto D^aF(w)(h,\dots,h)$ is a polynomial vanishing on the open set V, hence everywhere (identity principle), and the symmetric form $D^aF(w)$ is zero by polarization (Lemma **??**). So every derivative of order $< K$ of F vanishes on $\mathbf{C}^k \times 0$ near 0; F lies in the K-th power of the ideal of the normal coordinates near 0 (Lemma **??**), and pulling back along Φ gives $f \in \mathcal{I}_C^K$ at x (Lemma **??**), as in the Hadamard bridge (Lemma **??**). □

Corollary 174.112 (A controlled transform inside $\mathcal{O}(-D)$ raises the order)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\mathfrak{a}$ be coherent, $C \subset \operatorname{cosupp}(\mathfrak{a},\mu)$, π the blowup along C and $z \in D$. If $\sigma^c(\mathfrak{a},\mu) \subset \mathcal{O}(-D)$ at z, then $\operatorname{ord}_{\pi(z)}\mathfrak{a} \geq \mu+1$. Hence, at every point of D over a point of C where $\operatorname{ord}\mathfrak{a} = \mu$, some germ of $\sigma^c(\mathfrak{a},\mu)$ is not in $\mathcal{O}(-D)$. Likewise for a closed set H with strict transform $H' = \overline{\pi^{-1}(H \setminus C)}$:

if $\sigma^c(\mathfrak{a},\mu)\subset\mathcal{I}_{H'}$ at a point $z\in H'$, then $\mathfrak{a}\subset\mathcal{I}_H$ at points of $H\setminus C$ arbitrarily close to $\pi(z)$; so if $\mathfrak{a}\not\subset\mathcal{I}_H$ at the points of $H\setminus C$ near $\pi(z)$, some germ of $\sigma^c(\mathfrak{a},\mu)$ at z is not in $\mathcal{I}_{H'}$. These are the two clauses used in the proof of Lemma **??**.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

By Corollary **??**, $\pi^*\mathfrak{a}=\mathcal{O}(-\mu D)\,\sigma^c(\mathfrak{a},\mu)$. For $a\in\mathfrak{a}_{\pi(z)}$, $a\circ\pi\in\pi^*\mathfrak{a}$ at z is a combination of products pq with $p\in\mathcal{O}(-\mu D)$ and $q\in\sigma^c(\mathfrak{a},\mu)\subset\mathcal{O}(-D)$ at z, so $a\circ\pi\in\mathcal{O}(-D)^{\mu+1}$ at z. By Theorem 174.111, $a\in\mathcal{I}_C^{\mu+1}\subset\mathfrak{m}_{\pi(z)}^{\mu+1}$ at $\pi(z)$ (as $\pi(z)\in C$), so $\operatorname{ord}_{\pi(z)}\mathfrak{a}\geq\mu+1$ (Definition **??**). The second statement is the contrapositive.

For H': $\sigma^c(\mathfrak{a},\mu)$ is coherent (Corollary **??**), so the containment in $\mathcal{I}_{H'}$ at z holds at every point near z (each of finitely many generators vanishes on H' near z). Points $z'\in\pi^{-1}(H\setminus C)$ accumulate at $z\in H'$. At such a z', for $a\in\mathfrak{a}_{\pi(z')}$, $a\circ\pi$ is a combination of products pq with $q\in\sigma^c(\mathfrak{a},\mu)\subset\mathcal{I}_{H'}$, so it vanishes on H' near z'. The section s of π over $X\setminus C$ with $s(\pi(z'))=z'$ (Lemma 174.30) maps the points $y\in H\setminus C$ near $\pi(z')$ into $\pi^{-1}(H\setminus C)\subset H'$ near z', so $a(y)=(a\circ\pi)(s(y))=0$: $a\in\mathcal{I}_H$ at $\pi(z')$, and $\pi(z')\to\pi(z)$. □

Along a submanifold. For a marked ideal $(M,N,E,\mathfrak{a},d)$ with $N\neq M$ only the restrictions of germs to N, and of the transformed germs to the strict transform $N'=\overline{\pi^{-1}(N\setminus C)}$, matter (Definition **??**). A germ g at z *lies in* $\mathfrak{b}$ *on* N' if near z it agrees on N' with a holomorphic combination of germs of $\mathfrak{b}$ at z. The retraction of Lemma 174.113 reduces the statements of this subsection along N' to their versions on the whole space; it is the blowup half of the normal form $\mathfrak{a}^\flat=r^*(\mathfrak{a}|_N)$ of Lemma **??**.

Lemma 174.113 (The chart retraction)**.** *Let $\pi\colon Y\to X$ be the blowup along C, $N\supset C$ a closed submanifold and $z\in N'$ with $\pi(z)\in C$. There are a holomorphic map r from a neighbourhood of $\pi(z)$ to N with $r=\mathrm{id}$ on N, and a holomorphic map r' from a neighbourhood of z to N' with $r'(z)=z$ and $\pi\circ r'=r\circ\pi$.*

Proof. Take nested flattening coordinates Φ at $\pi(z)$ (Lemma 174.10): with $\psi=\Phi\circ e$ and $\varphi=\psi^{-1}$, near $\pi(z)$ we have $\psi(\pi(z))=0$, $C=\{\psi\in\mathbf{C}^{k_C}\times 0\}$ and $N=\{\psi\in\mathbf{C}^{k_N}\times 0\}$. Let P be the linear retraction of $\mathbf{C}^n$ onto $\mathbf{C}^{k_N}\times 0$ and $r=\varphi\circ P\circ\psi$ (Lemma **??**); it is holomorphic near $\pi(z)$, maps into N, and fixes the points of N. Let $j\geq k_C$, O, W be a monomial chart through z with lift ℓ_j (Lemma 174.11): $\sigma_j(W)=\psi\circ\pi$ on O and $\ell_j(W(u))=u$. By Lemma 174.121 applied to N, $z\in N'$ gives $j<k_N$ and $W(z)\in\mathbf{C}^{k_N}\times 0$. For $j<k_N$, $\sigma_j\circ P=P\circ\sigma_j$: σ_j multiplies the coordinates w_t, $t\geq k_C$, $t\neq j$, by w_j, which P does not change. Put $r'=\ell_j\circ P\circ W$ near z (Lemma **??** reads ℓ_j on $\mathbf{C}^n$). Then $r'(z)=\ell_j(W(z))=z$, and $\pi\circ r'=\varphi\circ\sigma_j\circ P\circ W=\varphi\circ P\circ\sigma_j\circ W=\varphi\circ P\circ\psi\circ\pi=r\circ\pi$. Finally ℓ_j maps every point $q\in\mathbf{C}^{k_N}\times 0$ of its domain into N': if $q_j\neq 0$, then $\sigma_j(q)\in\mathbf{C}^{k_N}\times 0$ and $\sigma_j(q)\notin\mathbf{C}^{k_C}\times 0$, so $\pi(\ell_j(q))\in N\setminus C$; if $q_j=0$, $\ell_j(q)$ is the limit of $\ell_j(q+te_j)$, $t\to 0$, $t\neq 0$, and $q+te_j\in\mathbf{C}^{k_N}\times 0$ as $j<k_N$. As $P(W(u))\in\mathbf{C}^{k_N}\times 0$, r' takes values in N'. □

Theorem 174.114 (The pushforward of $\mathcal{O}(-KD)$ along the strict transform of a submanifold)**.** *Let π be the blowup along C, $N\supset C$ a closed submanifold, $z\in N'$ with $\pi(z)\in C$, and f holomorphic at $\pi(z)$. If $f\circ\pi$ lies in $\mathcal{O}(-D)^K$ on N' at z, then f lies in $\mathcal{I}_C^K$ on N at $\pi(z)$.*

Proof. Take r,r' of Lemma 174.113 and put $f^\flat=f\circ r$, holomorphic at $\pi(z)=r(\pi(z))$ and equal to f on N. A germ q of $\mathcal{O}(-D)=\pi^*\mathcal{I}_C$ at z is a combination $\sum b_i\,(c_i\circ\pi)$ with $c_i\in\mathcal{I}_C$ at $\pi(z)$; so $q\circ r'=\sum(b_i\circ r')(c_i\circ r\circ\pi)$, and $c_i\circ r\in\mathcal{I}_C$ because r fixes the points of $C\subset N$. Hence $q\circ r'\in\mathcal{O}(-D)$ at z, and by induction on K (using $r'(z)=z$) $g\circ r'\in\mathcal{O}(-D)^K$ at z for $g\in\mathcal{O}(-D)^K$ at z. If $f\circ\pi=\sum a_ig_i$ on N' near z with $g_i\in\mathcal{O}(-D)^K$, then near z, since r' takes values in N', $f^\flat\circ\pi=f\circ\pi\circ r'=\sum(a_i\circ r')(g_i\circ r')\in\mathcal{O}(-D)^K$ at z. By Theorem 174.111, $f^\flat\in\mathcal{I}_C^K$ at $\pi(z)$, and $f=f^\flat$ on N near $\pi(z)$. □

Corollary 174.115 (The transform along N' at the points of D and of strict transforms)**.** *Let $\underline{\mathcal{I}}=(M,N,E,\mathfrak{a},d)$ with N a closed submanifold and $\mathfrak{a}$ coherent, $C\subset\operatorname{cosupp}\underline{\mathcal{I}}$, π the blowup*

along C, and $\mathfrak{a}'$ the transformed ideal (Definition ??). Then: (0) every germ of $\pi^*\mathfrak{a}$ at a point of N' lies in $\mathcal{O}(-dD)\,\mathfrak{a}'$ on N'; (1) if at $z \in N'$ over C every germ of $\mathfrak{a}'$ lies in $\mathcal{O}(-D)$ on N', then $\operatorname{ord}_{\pi(z)}(\mathfrak{a}|_N) \geq d+1$; hence where $\operatorname{ord}(\mathfrak{a}|_N) = d$, some germ of $\mathfrak{a}'$ at every point of $D \cap N'$ over it is not in $\mathcal{O}(-D)$ on N'; (2) for a closed set $A \subset N$ (for instance $A = H \cap N$ for a member H of E) and $z \in A' = \overline{\pi^{-1}(A \setminus C)}$: if $\mathfrak{a}' \subset \mathcal{I}_{A'}$ at z, then $\mathfrak{a} \subset \mathcal{I}_A$ at points of $A \setminus C$ arbitrarily close to $\pi(z)$; so if $\mathfrak{a} \not\subset \mathcal{I}_A$ at the points of $A \setminus C$ near $\pi(z)$, some germ of $\mathfrak{a}'$ at z does not vanish on A' near z. These are the two clauses of Corollary 174.112 along N', as used in Lemma ?? for $N \neq M$.

Proof. (0) By the Hadamard bridge along N (Lemma ??), $\mathfrak{a} \subset \mathcal{I}_C^d$ on N; pulling back (Lemma ??) and restricting to $N' \subset \pi^{-1}(N)$, $\pi^*\mathfrak{a} \subset \pi^*\mathcal{I}_C^d = \mathcal{O}(-dD)$ on N' (Lemma 174.1). As $\mathcal{O}(-dD)$ is invertible along N' (Lemma ??) and $\mathfrak{a}'$ is the colon of $\pi^*\mathfrak{a}$ by it along N', Lemma ?? gives (0). (1) Let $a \in \mathfrak{a}$ at $\pi(z)$. By (0), $a \circ \pi$ is, on N', a combination of products pq with $p \in \mathcal{O}(-dD)$ and $q \in \mathfrak{a}'$ at z, and each q is on N' a combination of germs of $\mathcal{O}(-D)$; so $a \circ \pi$ lies in $\mathcal{O}(-D)^{d+1}$ on N'. Theorem 174.114 gives $a \in \mathcal{I}_C^{d+1} \subset \mathfrak{m}_{\pi(z)}^{d+1}$ on N (as $\pi(z) \in C$), that is $\operatorname{ord}_{\pi(z)}(a|_N) \geq d+1$ (Definition ??). (2) The ideal $\mathfrak{a}'$ is coherent (Lemma ??), so the containment in $\mathcal{I}_{A'}$ at z holds at every point near z; points $z' \in \pi^{-1}(A \setminus C) \subset N'$ accumulate at z. At such z', by (0), $a \circ \pi$ agrees on N' with a combination of germs of $\mathcal{O}(-dD)\,\mathfrak{a}'$, which vanish on $A' \subset N'$; so $a \circ \pi$ vanishes on A' near z'. The section of π over $X \setminus C$ (Lemma 174.30) maps the points of $A \setminus C$ near $\pi(z')$ into $\pi^{-1}(A \setminus C) \subset A'$, so a vanishes on A near $\pi(z')$, and $\pi(z') \to \pi(z)$. □

Lemma 174.116 (The transform of an ideal with an exceptional factor, along N')**.** *Verified upstream; not exported by this standalone library.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ with N a closed submanifold, $\mathfrak{a}$ coherent, C an admissible centre and $\mathfrak{a}'$ the transformed ideal. If $\pi^*\mathfrak{a}$ and $\mathcal{O}(-eD)\,\mathfrak{b}$ have the same restriction to N', for a coherent $\mathfrak{b}$ and $e \geq d$, then $\mathfrak{a}'$ and $\mathcal{O}(-(e-d)D)\,\mathfrak{b}$ have the same restriction to N'. With Lemma 174.40 ($\pi^*\mathcal{I}_H = \mathcal{O}(-D)\,\mathcal{I}_{H'}$ where $C \subset H$, $= \mathcal{I}_{H'}$ elsewhere) this computes the transform of a monomial marked ideal along N' (Lemma ??).

Proof. Verified upstream; not exported by this standalone library.

By Lemma ??, $\pi^*\mathfrak{a}$ and $\mathcal{O}(-dD)\,\mathfrak{a}'$ have the same restriction to N'. As $\mathcal{O}(-eD)\,\mathfrak{b} = (\mathcal{O}(-(e-d)D)\,\mathfrak{b})\,\mathcal{O}(-dD)$ (powers add, products are associative and commutative for coherent ideals), $\mathfrak{a}'\,\mathcal{O}(-dD)$ and $(\mathcal{O}(-(e-d)D)\,\mathfrak{b})\,\mathcal{O}(-dD)$ have the same restriction to N'. The ideal $\mathcal{O}(-dD)$ is invertible along N' (Lemma ??), so it cancels (Lemma ??); both ideals are coherent (Lemma ?? for $\mathfrak{a}'$). □

Lemma 174.117 (Strict transforms of the traces of the members)**.** *Let π be the blowup along C, $N \supset C$ closed, E a simple normal crossings divisor transverse to N with which C has only normal crossings, and H a member of E. Then $\overline{\pi^{-1}((H \cap N) \setminus C)} = H' \cap N'$. In particular, in Corollary 174.115 (2) with $A = H \cap N$, A' is the trace on N' of the member H' of the transformed divisor.*

Proof. $\subset$: strict transforms are monotone. $\supset$: let $z \in H' \cap N'$; then $\pi(z) \in H \cap N$, as $H' \subset \pi^{-1}(H)$ and $N' \subset \pi^{-1}(N)$. If $\pi(z) \notin C$, $z \in \pi^{-1}((H \cap N) \setminus C)$. If $\pi(z) \in C$, take coordinates at $\pi(z)$ adapted to $C \subset N$ and E (Lemma 174.122): $C = \mathbf{C}^{k_C} \times 0$, $N = L_{T_N}$, $T_N = \{t \geq k_N\}$, $H = L_{\{s\}}$, so $H \cap N = L_{\{s\} \cup T_N}$, and a monomial chart j through z (Lemma 174.11). By Lemma 174.121, $z \in H'$ gives $j \neq s$ and $W_s(z) = 0$, $z \in N'$ gives $j \notin T_N$ and $W(z) \in L_{T_N}$; so $j \notin \{s\} \cup T_N$ and $W(z) \in L_{\{s\} \cup T_N}$, that is $z \in (H \cap N)'$ by the same lemma. □

The trace of a member on a submanifold. In the cases of the algorithm along $N \neq M$ a member H of E enters through its trace $H \cap N$: the exponents along N are defined with $\mathcal{I}_{H \cap N}$ (Lemma ??), while the transform and the monomial part are written with the ideals $\mathcal{I}_H$ restricted to N. For E transverse to N the two agree on N, also for the exceptional divisor on the strict transform; a retraction onto N preserving the members compares them.

Lemma 174.118 (A retraction onto N preserving the members)**.** *Let E be a simple normal crossings divisor transverse to N (Definition **??**) and $x \in N$. There is a map ρ, holomorphic on a neighbourhood of x, with $\rho(x) = x$, $\rho(y) \in N$ and $\rho(y) = y$ for $y \in N$ near x, and $\rho(y) \in H$ for every member H of E and every $y \in H$ near x.*

Proof. If $n = 0$, the coordinates of the transversality at x make N a coordinate subspace of $\mathbf{C}^0$, so N is a neighbourhood of x, and $\rho = \mathrm{id}$. If $n \geq 1$, take coordinates $\psi = \Phi \circ e$ at x adapted to $N \subset N$ and E (Lemma 174.122, with $C = N$): near x, $N = \{\psi \in \mathbf{C}^k \times 0\}$ and each member H_i through x is $\{\psi_{s_i} = 0\}$ with $s_i < k$. Let P be the linear retraction of $\mathbf{C}^n$ onto $\mathbf{C}^k \times 0$, which does not change the coordinates w_t, $t < k$ (Lemma **??**), and $\rho = \psi^{-1} \circ P \circ \psi$, defined and holomorphic near x since $P(\psi(x)) = \psi(x)$ (Lemma **??**; the differentiability of ψ and ψ^{-1} as in Lemma 174.113). It maps into N, fixes N, and for $y \in H_i$ near x, $\psi_{s_i}(\rho(y)) = (P\psi(y))_{s_i} = \psi_{s_i}(y) = 0$, so $\rho(y) \in H_i$. A member not through x is closed, so no point near x lies on it. □

Lemma 174.119 (The ideal of the trace of a member)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

*Let E be a simple normal crossings divisor transverse to N, H a member of E, $x \in N$ and $K \geq 0$. Then $\mathcal{I}_H^K$ and $\mathcal{I}_{H\cap N}^K$ have the same restriction to N at x: a function agrees on N near x with a holomorphic combination of germs of the one iff it does so with germs of the other. In particular a germ g holomorphic at x lies in $\mathcal{I}_H$ on N iff it vanishes on $H \cap N$ near x: $\mathcal{I}_{H\cap N} = \mathcal{I}_H + \mathcal{I}_N$ at the points of N. This holds at every stage of a multiple test blow-up of a well-formed marked ideal, whose transforms are well formed (Lemma **??**).*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

$\mathcal{I}_H \subset \mathcal{I}_{H\cap N}$, and powers are monotone at x (by induction on K, as membership in $\mathfrak{l}^{K+1}$ at x only involves germs at x of $\mathfrak{l}^K$ and $\mathfrak{l}$); this gives one direction. Conversely take ρ of Lemma 174.118. For $q \in \mathcal{I}_{H\cap N}$ at x, $q \circ \rho$ is holomorphic at $x = \rho(x)$ and vanishes on H near x, since ρ maps the points of H near x into $H \cap N$ and tends to x; so $q \circ \rho \in \mathcal{I}_H$ at x. By induction on K (writing a germ of $\mathfrak{l}^{K+1}$ as $\sum a_i\, p_i q_i$ and composing each factor with ρ), $s \circ \rho \in \mathcal{I}_H^K$ at x for $s \in \mathcal{I}_{H\cap N}^K$ at x. If $g = \sum a_i s_i$ on N near x with $s_i \in \mathcal{I}_{H\cap N}^K$, then, ρ being the identity on N, $g = \sum a_i\,(s_i \circ \rho)$ on N near x. For the last sentence: if g is holomorphic and $g = \sum a_i s_i$ on N with s_i vanishing on $H \cap N$, then g vanishes on $H \cap N$; conversely g itself is a generator of $\mathcal{I}_{H\cap N}$ at x. □

Corollary 174.120 (The exceptional divisor on the strict transform)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let π be the blowup along C, $N \supset C$ a closed submanifold, $D = \pi^{-1}(C)$ and $z \in N'$ with $\pi(z) \in C$. Then $\mathcal{O}(-D) = \mathcal{I}_D$ at z, and $\mathcal{O}(-KD)$ and $\mathcal{I}_{N'\cap D}^K$ have the same restriction to N' at z. In particular a germ vanishing on $N' \cap D$ near z lies in $\mathcal{O}(-D)$ on N': $\mathcal{I}_{D\cap N'} \subset \mathcal{O}(-D) + \mathcal{I}_{N'}$ at z, and a holomorphic germ lies in $\mathcal{O}(-D)$ on N' iff it vanishes on $N' \cap D$ near z.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

$\mathcal{O}(-D) = \pi^*\mathcal{I}_C \subset \mathcal{I}_D$ (Lemma 174.13); conversely $\mathcal{I}_D \subset \mathcal{O}(-D)$ at the points of D (Lemma **??**, its first step, with $E = \emptyset$, with which every closed submanifold has normal crossings, Lemma **??**). So $\mathcal{O}(-D)^K = \mathcal{I}_D^K$ at z by monotonicity of powers at a point. The one-member divisor (D) is simple normal crossings and transverse to N' (Theorem 174.123 with $E = \emptyset$), and Lemma 174.119 applies to it at $z \in N'$. □

Normal crossings divisors. Let $E = (H_1, \ldots, H_s)$ be an ordered simple normal crossings divisor on X (Definition **??**) and $N \supset C$ a closed submanifold. The *transformed divisor* is $E' = (H_1', \ldots, H_s', \pi^{-1}(C))$ with $H' = \overline{\pi^{-1}(H \setminus C)}$, as in the transform of a marked ideal (Definition **??**). Coordinates Φ at $x \in C$ are *adapted* to $C \subset N$ and E if $C = \mathbf{C}^{k_C} \times 0$, $N = \mathbf{C}^{k_N} \times 0$ and each member of E through x is a coordinate hyperplane $\{w_{s_i} = 0\}$ with $s_i < k_N$, the indices s_i being distinct. This is the simultaneous normal crossings of N, C and E at x (for $E = \emptyset$ it is

Lemma 174.10; in general it follows from the normal crossings of C with E and the transversality of E to N, Lemma 174.122).

Lemma 174.121 (Strict transforms of coordinate subspaces)**.** *Let Φ flatten C onto $\mathbf{C}^k \times 0$ and a set A onto a coordinate subspace $L_T = \{w : w_t = 0,\ t \in T\}$, and let $j \geq k$, O, W be a monomial chart of the blowup (Lemma 174.11). For $u \in O$,*

$$u \in \overline{\pi^{-1}(A \setminus C)} \iff j \notin T \text{ and } W(u) \in L_T.$$

Moreover $W - W(z)$, read in a chart, is a coordinate system centred at any $z \in O$; and coordinates adapted to a family of sets at $g(z)$ pull back to coordinates adapted to the preimages at z along a holomorphic map g with invertible differential at z.

Proof. On O, $\pi(u) \in C \iff W_j(u) = 0$ and $\pi(u) \in A \iff \sigma_j(W(u)) \in L_T$, where $\sigma_j(w)_t = w_t w_j$ for $t \geq k$, $t \neq j$, and $\sigma_j(w)_t = w_t$ otherwise. Hence σ_j maps L_T into itself, and for $w_j \neq 0$: $\sigma_j(w) \in L_T \iff j \notin T$ and $w \in L_T$. So $\pi^{-1}(A \setminus C) \cap O = \{u : W(u) \in L_T,\ W_j(u) \neq 0\}$ if $j \notin T$, and $= \emptyset$ if $j \in T$. In the first case the closure in O lies in the closed set $\{W \in L_T\}$, and contains it: if $W(u) \in L_T$ and $W_j(u) = 0$, the points $\ell_j(W(u) + te_j)$, $t \neq 0$ small, lie in $\pi^{-1}(A \setminus C)$ (as $j \notin T$) and tend to u. The differential of W is injective (Lemma 174.11), so $W - W(z)$ is a local biholomorphism at z (Lemma **??**). For the last claim compose the adapted coordinates at $g(z)$ with a local holomorphic inverse of g read in charts, as in Lemma **??**. □

Lemma 174.122 (Simultaneous normal crossings coordinates)**.** *Let E be a simple normal crossings divisor transverse to N, let $C \subset N$, and suppose that C has only normal crossings with E. If $n > 0$, then $C \subset N$ and E admit adapted coordinates at every point of C. The hypothesis $n > 0$ holds whenever C is the centre of a blowup and $C \neq \emptyset$: in dimension 0 every point is open, while a centre has empty interior. (In dimension 0 no member of E passes through a point, but the index family (s_i) of adapted coordinates is required for every member.)*

Proof. Two members H_i, H_j of E through x that agree near x are equal: in the coordinates of E at x they are the hyperplanes $\{w_a = 0\}$, $\{w_b = 0\}$, with $a \neq b$ if $i \neq j$, and the points with coordinates τe_b, $\tau \neq 0$ small, lie in the first and not in the second.

Let $x \in C$. Take coordinates u at x, centred at 0, in which $C = L_Y = \{u_t = 0,\ t \in Y\}$ and each member H_i through x is $\{u_{t'_i} = 0\}$ (normal crossings of C and E), and coordinates v in which $N = L_{S_0}$ and $H_i = \{v_{t_i} = 0\}$ with $t_i \notin S_0$ (transversality). By the first paragraph $i \mapsto t'_i$ is injective on the members through x. Let $\xi = v \circ u^{-1}$ near 0 and $L = d\xi(0)$, an isomorphism. As $\xi(L_Y) \subset L_{S_0}$ (since $C \subset N$), the rows L_s, $s \in S_0$, vanish on L_Y. For a member with $t'_i \in Y$ (that is, $H_i \supset C$ near x), ξ_{t_i} vanishes on $\{u_{t'_i} = 0\}$, so $L_{t_i} = c_i\, du_{t'_i}$, and $c_i \neq 0$ as $L_{t_i} \neq 0$. Let $Y' = Y \setminus \{t'_i : t'_i \in Y\}$. Every $b \in \mathbf{C}^{S_0}$ is $(L_s w)_{s \in S_0}$ for some w supported on Y': choose w with $(Lw)_s = b_s$ for $s \in S_0$ and $(Lw)_r = 0$ otherwise; then $c_i w_{t'_i} = (Lw)_{t_i} = 0$ for $t'_i \in Y$ (as $t_i \notin S_0$), and replacing w by its projection to the coordinates in Y changes w by a vector of L_Y, hence does not change the $L_s w$. So the vectors $(L_s e_y)_{s \in S_0}$, $y \in Y'$, span $\mathbf{C}^{S_0}$; choose $X \subset Y'$ such that those with $y \in X$ form a basis, so $|X| = |S_0|$, and a bijection $\beta : X \to S_0$. Put $F(u)_t = \xi_{\beta(t)}(u)$ for $t \in X$ and $F(u)_t = u_t$ otherwise. If $dF(0)w = 0$, then w is supported on X and $L_s w = 0$ for all $s \in S_0$, so $w = 0$ by the choice of X. By the inverse function theorem (Lemma **??**) F is a coordinate system on a ball B around 0, and on B:

- $\xi(u) \in L_{S_0}$ if and only if $F_t(u) = 0$ for all $t \in X$, as β is onto S_0;
- $u \in L_Y$ if and only if $F_t(u) = 0$ for all $t \in Y$: if $u \in L_Y$ then $\xi(u) \in L_{S_0}$; conversely, let u' be u with the coordinates in Y put to 0; then $u' \in B \cap L_Y$, so $\xi(u') \in L_{S_0}$, and $F(u') = F(u)$, whence $u = u' \in L_Y$ by injectivity of F on B;
- each member through x is $\{F_{t'_i} = 0\}$, as $t'_i \notin X$ (either $t'_i \notin Y \supset X$ or $t'_i \in Y \setminus Y'$).

Finally a permutation σ of the coordinates with $\sigma(t) \geq n - |X| \iff t \in X$ and $\sigma(t) \geq n - |Y| \iff t \in Y$ (it exists as $X \subset Y$) gives coordinates, composed with u (Lemma **??**), in which $C = \mathbf{C}^{n-|Y|} \times 0$, $N = \mathbf{C}^{n-|S_0|} \times 0$, and each member through x is the hyperplane with index $s_i = \sigma(t'_i) < n - |S_0|$, these indices being distinct. □

Theorem 174.123 (The transformed divisor)**.** *Suppose that E is a simple normal crossings divisor and that $C \subset N$ and E admit adapted coordinates at every point of C. Then E' is a simple normal crossings divisor; if moreover E is transverse to N, then E' is transverse to $N' = \overline{\pi^{-1}(N \setminus C)}$. In particular $(\pi^{-1}(C))$ is a simple normal crossings divisor transverse to N'.*

Proof. Let $y \in Y$. If $\pi(y) \notin C$, then π is a local biholomorphism at y, near which $H' = \pi^{-1}(H)$, $N' = \pi^{-1}(N)$ and $\pi^{-1}(C) = \emptyset$; the coordinates of E (and N) at $\pi(y)$ pull back (Lemma 174.121). If $\pi(y) \in C$, take adapted coordinates Φ at $\pi(y)$ and the monomial chart $j \geq k_C$, O, W at y. By Lemma 174.121, near y: $\pi^{-1}(C) = \{W_j = 0\}$; for a member H through $\pi(y)$, $H' = \{W_{s_i} = 0\}$ if $s_i \neq j$ and H' misses O if $s_i = j$ (then $H \supset C$ has the centre as its trace in codimension one); $N' = \{W_t = 0,\ t \geq k_N\}$ if $j < k_N$, and N' misses O otherwise. A member not through $\pi(y)$ has a strict transform not through y ($H' \subset \pi^{-1}(H)$, H closed). In the coordinates $W - W(y)$ the members of E' through y are therefore the hyperplanes with the indices s_i (distinct, and $\neq j$) and j: E' is simple normal crossings at y. If $y \in N'$, then $j < k_N$, all these indices are $< k_N$, and N' is cut out by the coordinates $t \geq k_N$: E' is transverse to N' at y. The last statement is the case $E = \emptyset$, where adapted coordinates are nested flattening coordinates (Lemma 174.10). □

Corollary 174.124 (The transformed divisor of an admissible blowup)**.** *Let E be a simple normal crossings divisor and let the centre C of the blowup π have only normal crossings with E. Then $E' = (H'_1, \dots, H'_s, \pi^{-1}(C))$ is a simple normal crossings divisor. If moreover E is transverse to a closed set $N \supset C$, then E' is transverse to $N' = \overline{\pi^{-1}(N \setminus C)}$.*

Proof. A simple normal crossings divisor is transverse to the whole manifold (cut out by no coordinate). Lemma 174.122, applied to $C \subset X$, respectively to $C \subset N$, gives adapted coordinates at every point of C ($n > 0$ as C is a centre, and there is nothing to prove if $C = \emptyset$); conclude by Theorem 174.123. □

Corollary 174.125 (The transform of a marked ideal is well formed)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ be a well-formed marked ideal and C an admissible centre. Then the transform of $\underline{\mathcal{I}}$ by the blowup along C is well formed.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Since $C \subset \operatorname{cosupp} \underline{\mathcal{I}}$, $\mathfrak{a}|_N \subset \mathcal{I}_C^d|_N$ (Lemma **??**). Then N' is a closed submanifold and the transformed ideal is coherent (Lemma **??**). As $\underline{\mathcal{I}}$ is well formed and C is admissible, E is a simple normal crossings divisor transverse to $N \supset C$ with which C has only normal crossings, so $C \subset N$ and E admit adapted coordinates at every point of C (Lemma 174.122; $n > 0$ as C is a nonempty centre, and there is nothing to prove if $C = \emptyset$), and E' is a simple normal crossings divisor transverse to N' by Theorem 174.123. □

Dropping members that contain the centre. Step 1a of the algorithm replaces E by E_S, the divisor without the members that contain the submanifold S (Włodarczyk 2005, Step 1a; Lemma **??**). The submanifold, the ideal and the cosupport of a marked ideal and of its transforms do not depend on E; admissibility depends on E only through the normal crossings of the centre with E, and the members containing the centre do not change it.

Lemma 174.126 (Sub-divisors)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E be a finite sequence of closed subsets of X and E' a finite sequence each of whose members is a member of E. (1) If C has only normal crossings with E, it has only normal crossings with E'; if E is transverse to N, so is E'. (2) If E is a simple normal crossings divisor and E' is a subsequence of E, then E' is a simple normal crossings divisor. (3) If $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$ is well formed and E' is a subsequence of E, then $(M, N, E', \mathfrak{a}, d)$ is well formed.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

(1) The coordinates at a point x of C (of N) given by the hypothesis serve for E': every member of E' through x is a member of E through x, hence a coordinate hyperplane (whose coordinate is not among those cutting out N). (2) A subsequence is indexed by a strictly increasing, hence injective, map of the indices; composing the hyperplane indices of E at x with it keeps them injective on the members through x. For an arbitrary sequence of members this fails: a member through x repeated twice would need two distinct hyperplanes. (3) N and $\mathfrak{a}$ are unchanged; apply (1) and (2) to $E' \subset E$. □

Theorem 174.127 (Normal crossings and the members containing the centre)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E be a simple normal crossings divisor on X, $C \subset X$, and E' a finite sequence of members of E such that every member of E that is not a member of E' contains C. Then C has only normal crossings with E' if and only if it has only normal crossings with E. In particular, if A is a set of indices with $C \subset H_i$ for $i \in A$, then C has only normal crossings with E if and only if it has only normal crossings with the subsequence $E \setminus A$ of the members with index not in A; and if E is the divisor of a marked ideal $\underline{\mathcal{I}} = (M, N, E, \mathfrak{a}, d)$, then C is an admissible centre for $\underline{\mathcal{I}}$ if and only if it is one for $(M, N, E', \mathfrak{a}, d)$.

Proof. Local and upstream declarations; individual reference status is recorded in the source.

"If" is Lemma 174.126 (1). Conversely, let $x \in C$. Call holomorphic coordinates u at x, centred at 0, *adapted to a set I of members* if C is a coordinate subspace $L_Y = \{u_t = 0,\ t \in Y\}$ near x and every member of I through x is a coordinate hyperplane $\{u_{p_i} = 0\}$. Coordinates of the normal crossings of C with E' are adapted to the set of members of E that are members of E'. Let u be adapted to I and let $H_0 \notin I$ be a member containing C, so $x \in H_0$. In simple normal crossings coordinates v of E at x the members through x are $\{v_{t_i} = 0\}$ with distinct indices, so $t_i \neq t_0$ for the members of I through x. Read $\xi = v \circ u^{-1}$ near 0; $\xi(0) = 0$. Since $C \subset H_0$, ξ maps L_Y into $\{v_{t_0} = 0\}$, and ξ_{t_i} vanishes on $\{u_{p_i} = 0\}$ for the members of I through x. This is the situation of the proof of Lemma 174.122, with $S_0 = \{t_0\}$ (the hypersurface H_0 in the role of N) and the members of I in the role of the members through x: the construction there gives $X \subset Y$ with one element y_0, different from every p_i, and the coordinates F obtained from u by replacing u_{y_0} by ξ_{t_0} (a local biholomorphism by Lemma **??**, followed by a permutation of the coordinates), in which near 0 L_Y is a coordinate subspace, $\xi^{-1}(\{v_{t_0} = 0\})$, that is H_0, is a coordinate hyperplane, and the coordinates u_{p_i} are kept. As $\xi(0) = 0$, $F(0) = 0$. So $F \circ u$ is adapted to $I \cup \{H_0\}$. After finitely many steps, one for each member of E that is not a member of E', the coordinates are adapted to all members of E: C has only normal crossings with E.

For the index form take $E' = E \setminus A$: a member H_i of E that is not a member of $E \setminus A$ has $i \in A$, so it contains C. The cosupport of a marked ideal does not depend on E, so admissibility (Definition **??**) is compared through the normal crossings alone. □

Lemma 174.128 (Dropping members along one blowup)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let E and E' be finite sequences of subsets of X such that every member of E' is a member of E and every member of E that is not a member of E' contains S. Let $\pi\colon Y \to X$ be a map, $C \subset X$, and let $E^\pi = (H'_1, \dots, H'_s, \pi^{-1}(C))$, with $H' = \overline{\pi^{-1}(H \setminus C)}$, be the transformed sequence, E'^π that of E'. Then every member of E'^π is a member of E^π (and E'^π is a subsequence of E^π if E' is

*one of E), and every member of E^π that is not a member of E'^π contains $S' = \overline{\pi^{-1}(S \setminus C)}$. The transform of $(M, N, E', \mathfrak{a}, d)$ (Definition **??**) is the transform of $(M, N, E, \mathfrak{a}, d)$ with E^π replaced by E'^π.*

Proof. Local and upstream declarations; individual reference status is recorded in the source.

$A \mapsto \overline{\pi^{-1}(A \setminus C)}$ is monotone. A member of E^π that is not a member of E'^π is not $\pi^{-1}(C)$, which belongs to both, so it is H' for a member H of E; H is not a member of E' (otherwise H' would be a member of E'^π), so $S \subset H$ and $S' \subset H'$. The submanifold, the ideal and the mark of the transform do not involve E. □

Lemma 174.129 (Generators of the ideal of a submanifold have multiplicity one)**.** *Local and upstream declarations; individual reference status is recorded in the source.*

Let N be regular of dimension $k < n$ at y, and suppose $\mathcal{I}_N = (g)$ at y with g holomorphic. Then $dg(y) \neq 0$: some partial derivative of g in a chart does not vanish at y. Consequently, in the situation of Corollary 174.16 with N of dimension $k < n$ at each of its points, the controlled transform u' satisfies $du'(z) \neq 0$ at every $z \in N'$ (Włodarczyk's Lemma on tangent directions, part (2)).

Proof. Local and upstream declarations; individual reference status is recorded in the source.

Let Φ flatten N onto $\mathbf{C}^k \times 0$ at y. The coordinate Φ_k vanishes on N, so $\Phi_k = a\, g$ near y with a holomorphic; and $g(y) = 0$ since g itself lies in $\mathcal{I}_N$ and $y \in N$. Hence $d\Phi_k(y) = a(y)\, dg(y)$. The differential of Φ is invertible at y (it has a holomorphic inverse, Lemma **??**), so $d\Phi_k(y) \neq 0$ and therefore $dg(y) \neq 0$. For the consequence, N' is regular of dimension k at z (Theorem 174.12) and $\mathcal{I}_{N'} = (u')$ at z (Corollary 174.16). □

175 Literal finite generators of the actual pulled-back ideal

These supplementary bridges use the actual function-germ definition of ideal pullback. They retain the literal original finite family under composition with a holomorphic map. They do not substitute a zero-set relation for generation and need no assumed principalization or target finite presentation. The second bridge obtains the original family from genuine coherence.

Theorem 175.1 (The complete finite stalk pulls back to its literal family)**.** *Let $f : Y \to X$ be holomorphic near y, and let $\mathfrak{a}$ be an actual ideal of function germs on X. Suppose G_i are actual holomorphic germs at $f(y)$ and present its complete stalk:*

$$\forall a, \quad a \in \mathfrak{a}_{f(y)} \quad \Longleftrightarrow \quad [a] \in ([G_i])_{f(y)}.$$

Then for every literal $F : Y \to \mathbb{C}$,

$$F \in (f^*\mathfrak{a})_y \quad \Longleftrightarrow \quad [F] \in ([G_i \circ f])_y.$$

The source and target dimensions may differ, and may be zero; empty finite generator families use the actual empty span.

Proof. The actual pullback stalk consists of finite holomorphic combinations of $a \circ f$ with $a \in \mathfrak{a}_{f(y)}$. For such a generator, unpack the given complete finite generation into actual coefficients b_i holomorphic near $f(y)$ and a genuine neighborhood identity $a = \sum_i b_i G_i$. Holomorphy of f supplies its continuity and the actual holomorphic pullbacks of a and each b_i. Pull the neighborhood equality back under f. This gives $a \circ f = \sum_i (b_i \circ f)(G_i \circ f)$ near y, so every pullback generator is in the finite pulled-back span. Substitution of these actual relations in a finite span gives the forward inclusion.

Each G_i belongs to its own original finite span by its actual holomorphy. The reverse original generation implication therefore places G_i in $\mathfrak{a}_{f(y)}$. Consequently $G_i \circ f$ is an actual generator of the pullback stalk and is holomorphic near y. Substitution in every finite combination gives the reverse inclusion. No coherence, measure or comparison certificate is used in this first bridge. □

Theorem 175.2 (Coherence internally supplies the literal pulled-back finite presentation)**.** *For an actual coherent ideal $\mathfrak{a}$, a map f holomorphic near y, and arbitrary y, there are an actual open $U \ni f(y)$ and literal finite G_i holomorphic on U that present every original stalk on U, with the complete additional equivalence*

$$\forall F, \quad F \in (f^*\mathfrak{a})_y \quad \Longleftrightarrow \quad [F] \in ([G_i \circ f])_y.$$

No target presentation is an input.

Proof. Apply the actual coherence property at $f(y)$. It supplies U, all literal holomorphic generators G_i, and their complete generation relation at every point of U. Holomorphy on the genuine open U implies holomorphy of each generator germ at its center. Apply the preceding pullback theorem to the supplied relation at $f(y)$. Return the same original U and family with their original all-stalk relation and the new full pullback relation. Thus later genuine principalization consumers receive the literal original generators, not an independently assumed coordinate presentation. □

176 Actual finite ordered monomial procedures with arbitrary genuine Cartier blowup choices

Only an actual indexed SNC carrier-power list and a family choosing genuine blowups of actual SNC members are inputs. Every maximum, controlled datum, finite bound, tower, stage comparison and coherence conclusion is internally produced. No equality with the prior general separation or general invariant driver is inferred.

Lemma 176.1 (Whole indexed products and unique selected indices under actual biholomorphisms)**.** *For an actual biholomorphism Phi between genuine dimension-n manifolds and an actual SNC indexed list F, its entire literal preimage list is SNC. Its whole indexed product and complete marking-one record are the actual biholomorphic pullbacks. Positive nonempty indices, powers and old colex labels agree through the same-index equivalence. The independent actual selected indices therefore agree, and their whole maximum carriers correspond.*

Proof. Compose the original actual joint SNC coordinates with Phi; the genuine holomorphic inverse gives a true centred coordinate chart and its inverse. Every literal carrier is exactly the corresponding coordinate zero set. Actual stalkwise biholomorphic ideal transport identifies its whole vanishing ideal, including unit stalks outside the closed carrier. Coherent powers and finite products give whole indexed ideal equality and hence all four marked fields. Genuine surjectivity and exact inverse-image membership preserve nonempty positive indices. The singleton keys use the same dimension, coefficient and literal colex index, so they are equal. The internally attained unique selected maximum is consequently the same index. Its actual carrier is the whole old-gamma maximum by the actual indexed-product dictionary. Empty manifolds and dimension zero take the actual inactive branch. □

Lemma 176.2 (An actually chosen Cartier blowup has its true unique global inverse)**.** *For any actual SNC member C and any genuine IsBlowupAlong(C,pi), the actual projection pi is a global biholomorphism. Its comparison with the internally proved genuine identity Cartier blowup is unique. For every selected positive factor its entire real controlled marked successor is the literal pi-preimage of F.set(j,(empty,0)) appended with (C,k-1), with the genuine whole boundary. This successor is SNC and its true coefficient sum plus one equals sum(F).*

Proof. The true Cartier vanishing ideal of an actual SNC member internally makes id a genuine blowup. Compare pi to this actual identity blowup over the identity base by the true universal property; the actual comparison and its inverse are holomorphic and commute with projections. Since the identity projection is id, the comparison is literally pi, so pi has its genuine global

inverse. Uniqueness follows from that same universal property, including empty members. The real chosen positive factor is internally admissible for the whole marking-one product. The two genuine one-step towers are actually isomorphic. Entire coherent controlled marked transforms commute with this actual tower isomorphism. The identity-Cartier successor is the internally proved full normalized list, so its whole pi-pullback is the entire actual chosen successor, including old empty entries and the newly appended carrier. The true pulled SNC list follows from the actual biholomorphic dictionary. Literal preimage keeps all powers; the actual selected power k¿0 and replacing k by k-1 prove successor sum plus one equals original sum. □

Definition 176.3 (Genuine choice family and internally generated finite procedure)**.** A choice family returns a genuine manifold B, an actual map pi:B-¿A and IsBlowupAlong for each actual SNC member of an indexed list on A. Its default is the internally proved identity Cartier blowup. Define the actual bounded procedure recursively on q. Zero fuel or no actual active index returns the reflexive tower. Otherwise internally select the unique old-key maximum, choose its genuine Cartier blowup, internally form the full real pulled controlled successor and recursively run q-1 on this actual SNC successor. Append the true first blowup. The actual completed procedure uses its OWN sum(F). Every stored terminal manifold, actual tower, literal state and projection biholomorphism is produced by this recursion.

Theorem 176.4 (Every bounded chosen run has a true comparison with the independent identity procedure)**.** *For every actual SNC input and finite q, the independently generated chosen procedure has an internally produced actual tower isomorphism to the genuine identity procedure. Its endpoint biholomorphism is exactly the actual chosen whole projection. Every entire terminal marked record, literal transformed state, selected index and whole centre is transported by the actual stage biholomorphisms. Lengths are equal and at most the original true coefficient sum.*

Proof. Prove the stronger comparison simultaneously for every actual initial biholomorphism and every two genuine choice families, by induction on q. The inactive or zero-fuel case is the actual reflexive tower isomorphism. Otherwise the whole biholomorphic dictionary identifies the two independent selected indices and actual entire centres. The genuine universal property produces their unique commuting lift to the two actual chosen first blowups. The complete one-step marked dictionary identifies the real new indexed products under this SAME lift, with all historical entries retained. Apply induction to these actual successor SNC inputs and lift; append the genuine first-step and recursive tower isomorphisms. This constructs the comparison from original data, rather than assuming aligned towers or histories. Its commuting projection equation and actual global inverse of every Cartier step identify the endpoint map with the actual total projection in the identity-reference case. Whole marked transport through actual test towers and the real successor dictionaries give the complete actual terminal record; the same induction gives every stage/index/centre dictionary. Actual tower isomorphisms preserve lengths, and the independent reference procedure internally proves the true coefficient-sum bound. □

Theorem 176.5 (Own coefficient sum internally terminates every true chosen ordered procedure)**.** *From the original SNC input and genuine choice family alone, its OWN completed procedure is a real old-gamma canonical segment and geometric marking-one resolution with whole unit endpoint. For every bounded real run its entire recorded coefficient sum plus its actual performed-step length equals the original whole coefficient sum, including inactive empty-carrier entries. Every true literal prefix has the full actual marked domain and entire SNC boundary; all empty or repeated historical entries remain literal. No compactness or caller termination bound is assumed.*

Proof. Each true positive selected Cartier step is internally admissible and has actual entire old-gamma maximum carrier. Its real successor decreases the true coefficient sum by exactly one. Induct on the actual finite fuel and compose the genuine canonical/test steps using the complete marked successor identity. The actual invariant is transported under each genuine Cartier biholomorphism by the whole coherent ideal and unchanged indexed-exponent dictionary,

so it is the SAME old-gamma invariant of the real transformed original marked datum. At fuel sum(F), repeated actual drops give inactive terminal state. Alternatively, the already internally constructed reference comparison transports the genuine reference unit endpoint. Whole unit ideals, not only empty supports, are invariant under the actual holomorphic inverse. The same induction accounts exactly for each true one-unit coefficient drop: zero/inactive runs retain the whole sum, and each actual appended step adds one to the length and subtracts one from the true recorded sum. Inactive empty-carrier powers need not vanish and are retained literally. The actual test-tower prefix theorem supplies every literal prefix's smooth carrier, coherent ideal, SNC boundary, positive mark and monomial domain. Every bound used in this induction follows from the actual original list and its real successor; none is a public termination assumption. □

Theorem 176.6 (Unique real comparisons for arbitrary choices satisfy the actual cocycle)**.** *Two independently generated procedures from actual biholomorphically related SNC inputs and any two genuine Cartier choice families have an internally produced unique actual tower isomorphism. It transports the entire real marked record, index word and whole stage centres. For every triple of genuine choice families its actual endpoint comparisons satisfy identity, inverse and composition/cocycle exactly.*

Proof. The simultaneous genuine comparison induction above produces existence. Every true chosen Cartier step is a global biholomorphism, hence so is the actual total projection produced by the recursion. Any commuting endpoint map is therefore uniquely determined as the actual source projection, the genuine initial biholomorphism, and the inverse actual target projection. This proves uniqueness of the real comparison without a caller naturality certificate. For three actual procedures the adjacent target inverse and source projection cancel literally, giving the actual composite comparison. Reflexive and inverse comparisons follow from the SAME actual inverse identities. Their entire marked, centre and index dictionaries follow from the real tower comparison and whole controlled transform transport. No label permutation or prior general-separation identification is asserted. □

The general higher invariant, old general driver identification and arbitrary unrelated ideal admissibility remain OPEN. Nonsurjective maps require a separate actual trace/compression argument.

177 Actual open trace compression for the finite ordered marking-one monomial procedure

Primitive data are an actual dimension-n complex manifold A, an actual open U in A and a genuine entire finite indexed SNC carrier-power list F on A. Both procedures are the previously internally defined genuine identity-Cartier old-gamma marking-one procedure, completed with their OWN original coefficient sums. Global maximum centres need not meet U. No global-maximum pullback, compression, aligned history, reindexing, ideal equality, canonicality or termination certificate is an input. General higher-invariant resolution and prior general-separation identification remain OPEN.

Definition 177.1 (Actual retained-history mask and internally produced source index embedding)**.** Let L be the actual original F length. Run the actual global bounded procedure while simultaneously tracking the number p of already performed real source steps. Keep ALL L original source history entries, even if empty on U. For each new actual global appended exceptional history entry, record the genuine boolean saying whether its centre met U AT THE MOMENT OF THAT ACTUAL STEP. Keep precisely those new entries with true boolean. This actual provenance mask is not recomputed from current carrier nonemptiness: a formerly nonempty kept carrier can subsequently become an empty consumed historical entry. The source history index embedding into the global history is the increasing enumeration of this actual mask; it has

L+p entries and keeps all original indices literally. Compare the true source terminal state after p actual steps with the actual global current state. Its entire carrier list is the literal retained sublist of the global preimage carrier list. Actual coefficients agree at EVERY retained carrier which is nonempty on U; differing coefficients at empty retained carriers are genuine unit factors. No literal whole-power list equality is asserted.

Lemma 177.2 (Actual order of old singleton keys is preserved by the generated increasing history embedding)**.** *At any actual tracked stage, source positive nonempty indices correspond bijectively to the global indices whose current positive carriers meet U. The generated embedding preserves the actual coefficient and index order at these indices. It therefore preserves comparisons of their genuine old-gamma singleton keys. If the actual global selected maximum carrier meets U, its corresponding source index is the independently attained actual source maximum. No equality of numerical colex values after compression is claimed.*

Proof. Original indices are retained in their original order, and every retained new history index is appended after all prior retained indices. Thus the actual embedding is strictly increasing, including kept historical entries which have since become empty or have power zero. The tracked state dictionary identifies the positive nonempty source carrier-power factors exactly with the positive global factors meeting U. Old marking-one singleton keys compare first their actual power and then their literal colex singleton index. Equal actual powers and a strictly increasing embedding preserve those comparisons. The global selected index maximizes its true key against ALL global active indices; if its carrier meets U, it in particular maximizes against the genuine source-corresponding indices. The internally proved source singleton uniqueness then identifies this independent source selection. If there is an active source carrier, its genuine SNC coordinates internally force positive ambient dimension. The inactive and empty-manifold cases use their actual none branch. □

Lemma 177.3 (A genuinely empty open trace preserves the whole original controlled ideal and all retained intrinsic exponents)**.** *If an actual global selected Cartier carrier has empty trace on U, its actual restricted controlled step has an empty centre, a genuine identity Cartier projection, and the ENTIRE same controlled ideal and carrier on U. Its newly appended empty exceptional entry is the only newly deleted history entry. The tracked independent source procedure performs no step. The whole retained carrier list and all actual intrinsic exponents remain correct.*

Proof. The whole preimage of the actual selected closed SNC member is empty. Its ENTIRE vanishing ideal is the unit ideal at every stalk, not merely an ideal with empty support. Dividing the actual pulled whole marking-one ideal by this true unit ideal leaves it literally unchanged. Empty-centre strict transform leaves every actual closed carrier unchanged. The actual new boundary entry is empty and its positive or zero power contributes a true whole unit factor. The actual global coefficient update can zero an old carrier that was already empty on U, while the source independent procedure may retain its positive metadata power there. Both are genuine whole unit factors; every intrinsic exponent along an empty carrier is zero by its literal stalk dictionary. The source retained boundary entries remain literal. The new provenance bit is false and its actual appended empty entry is deleted. All retained indices remain ordered and all nonempty coefficients stay equal. This proves the actual tracked relation without a false whole metadata-power equality. □

Theorem 177.4 (The actual global procedure internally produces the entire independent source trace and boundary dictionary)**.** *Track the actual global procedure for its OWN finite fuel. Delete precisely those actual global steps whose literal centres have empty trace on U. The resulting actual nonempty centre word is exactly the centre word of the independently generated source procedure. At every tracked stage p, actual source controlled ideal equals the WHOLE pulled global controlled ideal; its entire actual boundary is the exact retained provenance sublist. Every genuine nonempty restricted step is the source's independently selected old-gamma maximum step. Every empty step is internally proved genuinely trivial.*

Proof. Induct on the actual global recursion fuel while tracking the real source performed count and generated provenance embedding. Initially both entire carrier-power states are the same literal preimage input and the embedding is the original identity. An inactive global stage forces source inactivity by the genuine positive-nonempty factor dictionary. In an active step split on the actual trace nonemptiness, decided from its genuine selected carrier. If empty, the previous whole unit/control lemma preserves the real source state and all actual retained history, while the global state gains a false provenance bit. If nonempty, the actual key-order lemma identifies the genuine independent source selected index and whole centre. Both actual Cartier steps are admissible for their true whole marked data. Their genuine controlled successors set that selected old carrier to empty with power zero and append the SAME true carrier at power k-1. Append a true provenance bit and its new increasing source index. All other whole carrier and nonempty-power entries agree literally. Genuine SNC preimage/sublist coordinates produce the full source and retained SNC boundary, including consumed empty historical entries. The exact whole component ideals, powers and finite products identify the entire marked ideals; every omitted or coefficient-discrepant factor has a true empty carrier and hence unit ideal. Thus the tracked relation and whole controlled dictionary persist at EVERY real stage. Every step of the claimed trace equality is derived from the genuine independently generated source selection, not supplied as an aligned-history hypothesis. □

Theorem 177.5 (OWN global and source finite fuels internally complete the actual compressed procedure)**.** *At global OWN completion, the actual tracked source run is inactive. The number p of its genuine steps is exactly its internally produced performed length and is at most its own original whole coefficient sum. Continuing that source run to its OWN completion adds no step. The internally compressed actual restricted identity-Cartier tower is consequently the genuine source completed tower, with exact retained boundary, whole ideal and true unit endpoint. No compactness, finite-value assumption or caller bound is required.*

Proof. The actual global coefficient-sum termination theorem gives its true inactive terminal state. The tracked positive nonempty index dictionary therefore forces the actual source stage to be inactive. Its performed p was generated by the preceding actual simultaneous recursion. Each genuine performed source step decreases its true whole coefficient sum by one, while an omitted step performs no source update. Hence p is internally bounded by the true original source sum. The source actual procedure append-factorization theorem shows that extending this genuine bounded run to its own sum is its actual tower followed by a procedure starting on the inactive actual terminal state. That latter procedure is literally reflexive, so the source completed tower and word are the produced compressed tower and word. Every retained boundary entry is still the exact provenance sublist; no list is replaced merely because its union is empty. Its whole controlled ideal is the actual global terminal unit ideal pulled to U, hence the whole source unit ideal. The actual genuine local blowup tower is constructed from the produced nonempty restricted centre word and internally supplied SNC member Cartier blowup predicates. A separate actual empty-step compression relation identifies it with the real restricted global tower; it has a genuine skip constructor for true empty-centre Cartier biholomorphisms and cannot be confused with an equal-length tower isomorphism. All source maximum, admissibility, canonicality and geometric resolution data were internally produced from the original source SNC input. □

Theorem 177.6 (Nested genuine open restrictions and actual compressed centre histories agree)**.** *For genuine nested opens V subset U subset A, direct actual trace compression to V agrees with compressing the actual U trace and then its actual V trace. Their actual nonempty centre words, retained original/new history provenance, independently generated source procedures, whole marked ideal dictionaries and performed finite lengths agree. Only actual order-preserving index identifications are asserted, not numerical equality of changed colex labels.*

Proof. An actual global centre meets V precisely when its actual U trace meets V. The two internally generated retention decisions therefore keep exactly the same original entries and the

same newly appended genuine history events. Increasing enumeration of these actual retained event indices composes exactly. The preceding genuine trace theorem identifies both produced nonempty words with the SAME independently generated procedure on the literal V preimage input. Entire preimages, coherent whole powers/products and the actual provenance sublist dictionaries compose literally. Thus the actual compressed towers, whole ideals and finite lengths agree under the genuine nested-open biholomorphism. No separate compression, colex-value, tower or naturality certificate is an input. □

Exact scope of the nested comparison. The genuine globally labelled creation-event mask filters by whether each literal centre met the actual open AT CREATION, and its direct/iterated preimage filters compose exactly. The generated increasing embeddings compose with their literal index-word lookup dictionary. For actual nested opens the independent source inputs coincide as entire carrier-power lists on the SAME source manifold, so their independently generated completed towers are literally equal. Each individual genuine open restriction has the produced unequal-length stage compression and all-prefix whole marked dictionary. Numerical colex values after historical compression are not claimed equal. The generic retained-relation hypotheses of helper lemmas are internally produced in every original open/SNC consumer; they are not original naturality, maximum, termination or compression certificates.

Formal status. This is OWN OFFLIB mathematics with a faithful offlib kernel checkpoint. All six nodes remain notready pending root registration, native explicit module target compilation and fresh native inventories. General higher-driver functoriality, prior general separation identification, unrelated arbitrary ideals and higher-invariant nonsurjective smooth base change remain OPEN.

178 Original inputs internally construct the actual smooth nonzero lower diagram

Construct genuine restricted submersions between the actual selected-component complements, and compare ENTIRE lower marked records. Then consume this construction on an actual original test history and its internally produced smooth pullback. The source contact function is explicitly the transported ORIGINAL tangent function, as in [5, Section 7]; no equality with an independently selected source tangent is asserted.

Theorem 178.1 (The actual selected complements carry a genuine restricted submersion and whole lower marked dictionary)**.** *For ONLY a genuine holomorphic submersion f, actual closed smooth marked carrier $L.N$, coherent ideal and positive mark, internally construct true smooth geometry for the literal pulled-back record P. Its actual selected complement G' is exactly $f^{-1}G$, where G is the original selected complement. The actual restricted map $q : G' \to G$ is a genuine holomorphic submersion and has its true ambient coordinate equation. The ENTIRE marked pullback by q of the actual lower remainder restricted to G equals the actual remainder of P restricted to G'. Every historical empty entry and actual ordered suffix is retained. No map, source geometry, open equality, component or whole-record certificate is a premise.*

Proof. Genuine submersion preimage geometry internally supplies the actual source closed smooth carrier. The preceding WHOLE selected-union theorem makes its true complement exactly the actual inverse image of G. Restrict f between these two real opens; actual submersion restriction supplies its holomorphic submersion property and its ambient coordinate equation. The previous full remainder theorem identifies the independently defined source remainder with the actual marked pullback of the original remainder. The latter's literal ideal is coherent because it is the original L ideal. Apply genuine mixed-dimensional marked pullback composition to the two actual inclusion/restricted-map squares. Their coordinate maps are literally the same. This

proves the ENTIRE four-field lower dictionary, rather than only its germs, carrier or cosupport. No statement about component bijections or arbitrary local test extension is needed. □

Theorem 178.2 (ONLY original maximal-order domain and ANY original test internally produce a real smooth lower diagram)**.** *ONLY the original maximal-order domain of dimension bound $k+2$, ANY genuine ORIGINAL test prefix, an actual point of its current cosupport, and a genuine smooth base map internally construct an actual source test tower, true stage submersion and ENTIRE current original marked pullback dictionary. The same original inputs internally choose the actual tangent open and function and nonzero lower model A in domain $k+1$.*

On the actual stage inverse-image open, internally construct the genuine restricted stage submersion p. Its literal marked pullback of the full contact model is the ACTUAL source contact model with the transported ORIGINAL tangent function. Internally construct the actual selected-complement restricted submersion q and actual source lower record B. One has the ENTIRE marked equality $q^*A = B$, and B is genuinely in domain $k+1+(m-n)$. All source tests, opens, geometry, maps, lower domains and whole-record comparisons are produced internally; canonicality is absent.

Proof. The actual smooth original tower producer constructs the source tower and real stage map from the genuine base submersion and the given ORIGINAL test. It supplies the true Cartesian tower relation, actual source test and whole current marked dictionary, preserving all positions. Independently the original nonzero contact producer supplies actual tangent existence, its fixed original choice, closed smooth coefficient model, actual selected complement and true lower domain $k+1$.

Genuine homogenized coefficient equivalence internally turns the original prefix into the actual coefficient test. The existing whole-contact open diagram theorem applies to the ACTUAL source tower just constructed. It supplies the real source inverse-image open, genuine restricted stage submersion p, its actual ambient coordinate equation and ENTIRE contact record equality with the transported original tangent. Apply the first theorem to this p and the independently constructed target contact model. It internally produces actual source smooth geometry, the precise selected complement, real restricted q and ENTIRE lower marked equality.

Finally genuine smooth pullback of the actual target lower domain through q internally gives domain $k+1+(m-n)$ on its true marked pullback. Substitute the already proved ENTIRE lower record equality to obtain the asserted actual source lower domain. Literal pullback preserves boundary length, so the same historical number is also the true original source number. No source-domain or rank certificate is introduced.

The source model uses the transported ORIGINAL tangent choice explicitly. Comparison with the independently chosen source tangent, earlier frozen FULL lower histories, future lower-tower identification, exceptional or infinitesimal semicoherence, higher decrease and general termination remain separate consumers. □

179 Literal actual smooth lower histories from ONLY original noncanonical inputs

Retain the actual original upper history and its genuine smooth pullback. A lower-history comparison must use the LITERAL nested restrictions of these maps, not an unrelated source resolution or a desired comparison certificate. Use real Cartesian restriction twice, then true controlled marked pullback, for the recursive locality in [5, Section 7]. The source contact function is explicitly the transported ORIGINAL choice.

Theorem 179.1 (ONLY original inputs construct actual smooth lower tests and whole controlled-history dictionaries on the same tails)**.** *ONLY the original maximal-order domain of dimension bound $k+2$, ANY genuine ORIGINAL full test tower T, an actual point of its current*

cosupport, ANY actual splitting position j, and a genuine smooth base map internally construct an actual smooth source upper tower Q with the same original Cartesian geometry. The actual head cosupport point internally fixes ONE original tangent choice and the genuine target contact model, selected complement and lower domain A of dimension bound $k+1$.

The internally constructed source contact model uses the transported ORIGINAL function. The actual selected-complement map q is a genuine submersion, and its whole marked pullback of A is the actual source lower record B of dimension bound $k+1+(m-n)$. The LITERAL nested restrictions of $\mathrm{tail}(Q,j)$ *and* $\mathrm{tail}(T,j)$ *to the actual contact opens and actual selected complements form a genuine Cartesian tower with base q and internally constructed true endpoint submersion. BOTH are genuine lower test towers. Their actual future lower domains are proved, and their ENTIRE controlled marked records satisfy the genuine endpoint pullback identity, with BOTH whole cosupport directions. No source history, open, selected map, lower test, future domain or desired record/cosupport comparison is supplied; canonicality is absent.*

Proof. The original smooth-tower producer internally constructs the actual source upper tower and actual Cartesian relation from the genuine base map and the ORIGINAL test. Split the ORIGINAL test at the actual integer j. Genuine head/tail test splitting supplies both actual tests. True domain preservation supplies closed smooth/coherent head geometry, and the original tail's projection takes the actual final cosupport point into the true head cosupport. Thus the original-input uniform nonzero contact producer applies to this actual head, with its actual projected point. It internally supplies ONE tangent choice, contact geometry, selected complement, initial lower domain and a genuine target lower test for the actual restricted future tail. The original head appended to this tail is the SAME given tower; no rebasing or separately supplied canonical lower history is used.

Actual homogenized coefficient equivalence internally gives the genuine coefficient upper test. Apply the real simultaneous split/contact diagram theorem to the ACTUAL upper Cartesian relation already constructed. It produces the actual head source map and source inverse-image contact open, a genuine restricted head submersion, ENTIRE contact marked identity with the transported original function, and the real Cartesian relation between the LITERAL once-restricted upper tails. The actual selected-complement lower diagram theorem now constructs the second pair of actual opens, their true preimage equality, genuine base submersion q and ENTIRE initial marked equality $q^*A = B$.

Apply genuine Cartesian open restriction to the just-proved once-restricted tail relation and THESE actual selected opens. It internally constructs the real endpoint map and an actual twice-restricted Cartesian relation. The constructed base map and q have the SAME ambient coordinate equation; subtype extensionality therefore identifies them. Thus the source tower is precisely the actual nested restriction of the original Q tail, preserving all actual maps, empty positions and centres.

The independently produced target lower-domain record is well formed, and the target nested tail is already a genuine test by the original nonzero presentation. Apply the genuine relative controlled marked tower dictionary to THIS actual twice-restricted Cartesian relation. It internally supplies the actual source lower test and ENTIRE future record identity for q^*A. Substitute the independently proved ENTIRE initial equality $q^*A = B$. Its genuine cosupport dictionary likewise supplies BOTH full future cosupport directions, not only equality on a centre or at the chosen point. Finally true initial lower-domain pullback gives the correct source fibre dimension shift; genuine domain preservation on both actual lower tests gives both future lower domains. Every asserted input condition was derived before its use, from the SAME original records and actual maps.

The source contact choice is transported explicitly. Comparison with the independently selected source contact and earlier frozen FULL original lower histories, exceptional or infinitesimal semicoherence, higher invariant decrease and general termination remain separate producers. No arbitrary local test is assumed to extend to an original global test. □

180 The actually computed original zero masks under smooth base change

Compare the actual original masked lower records on their true regime-run splits, including the independently chosen SOURCE tangent. This is the source-facing recursive locality step in [5, Sections 5 and 7]. Only genuine TESTS and WHOLE records are claimed; the current abstract invariant interface cannot supply noncanonical value equality.

Lemma 180.1 (ONLY original ANY-test zero/nonspecial data construct the actual computed mask domain, lower test and both cosupport directions)**.** *ONLY original maximal-order domain of bound $k+2$, ANY original TEST T and actual current original cosupport point y of level zero which is not special internally give the genuine computed run r, actual head point, its OWN original tangent mask U, and literal lower record J used in the original Step One definition. The head point belongs to U, J is a genuine domain of bound $k+1$, the literal restricted tail is a genuine lower TEST, and its WHOLE cosupport is exactly the actual trace of the ORIGINAL final cosupport, in BOTH directions. The actual final point lies in it and the future lower domain is proved. Canonicality and desired lower domain/test/cosupport data are absent.*

Proof. Original coefficient equivalence internally supplies the SAME coefficient- homogenized TEST and the genuine original cosupport point. The real regime- run recursion, not a canonical-tail certificate, supplies the actual head regime and actual head cosupport membership. On the zero/nonspecial branch, these give the actual chosen tangent mask membership. The proved actual masked lower-domain theorem supplies its domain with the true one-dimension drop. Genuine contact-tail TEST construction applies to THIS SAME actual head and tail, because the real zero mask lies inside the actual tangent preimage. Its full cosupport trace theorem and the independently proved original coefficient cosupport dictionary give BOTH asserted directions. The actual original point therefore belongs to the true lower cosupport; genuine domain preservation along THIS TEST supplies the future domain. No lower canonicality conclusion is used from a canonical-tail theorem. □

Theorem 180.2 (ONLY original smooth data construct actual independent-source zero-mask lower histories and all-prefix whole comparisons)**.** *ONLY original maximal-order domain M, ANY original TEST T and genuine smooth base map internally construct the actual source upper TEST Q and true Cartesian stage map. At EVERY actual source point mapping into the original current zero/nonspecial cosupport, the SAME true run supplies the actual target computed zero mask, the genuine common inverse-image source mask and the independently selected SOURCE OWN computed zero mask. Their actual point memberships, lower domains, lower TESTS, maps and whole common marked pullback are internally constructed.*

Inside BOTH actual source masks there are real subopens containing the actual head point. A genuine stage biholomorphism and endpoint biholomorphism make the literal restricted source tails isomorphic and fix the original final source point. The local contact parents are EXACTLY the genuine open restrictions of the actual common and own computed zero records. Their ENTIRE parents, endpoint controlled records and records at EVERY head are actual pullbacks, with compatible head/tail isomorphisms. No source history, independent choice, lower mask/test/domain, desired record or value comparison certificate is an input. Canonicality is absent.

Proof. The genuine original smooth tower producer constructs Q and its actual Cartesian relation before the source point is chosen. Original coefficient equivalence supplies the target and source coefficient TESTS; the true whole cosupport dictionary supplies the actual source point. Actual boundary-level, special-point and regime-run dictionaries identify the SAME target and source numeric run, including every empty position. Apply the first lemma separately to the ORIGINAL target and ACTUAL source records. It internally supplies both OWN computed zero masks, real point membership, actual domains with the true fibre-dimension shift, lower TESTS and whole cosupport dictionaries.

The genuine common-contact-mask theorem on THIS actual Cartesian relation constructs the literal inverse-image source mask, its true submersion maps, whole transported ORIGINAL contact record and actual restricted Cartesian tail. Actual target lower domain and TEST pull back through this real map to give the common source domain and TEST. Thus both actual source masks and parent records are constructed before comparing their choices.

At the actual original source base point, true tangent existence and smooth preservation independently verify the transported target and SOURCE OWN chosen directions. The actual whole projection square identifies the transported target base point; no independently chosen function is renamed. Both masks are inside the respective tangent preimages and avoid EVERY original source member: one by the common-mask theorem, the other directly from its genuine original zero-mask definition. Apply the actual ANY-test zero-mask tower comparison to THESE real inputs, the SAME split source tower and actual original cosupport point. It internally constructs the two smaller opens, real tower isomorphism, point fixation, genuine lower TESTS and ENTIRE parent, endpoint and EVERY-head marked comparisons.

Finally actual coherent pullback composition along the genuine nested open inclusions identifies both smaller contact parents with the actual open restrictions of their common and OWN computed zero-mask records. The whole projection identity identifies the source chosen function in the own record with the independently selected original source function. This proves comparison of the actual algorithmic zero records, with every historical empty entry, actual exceptional member and normal germ retained. Higher original noncanonical invariant values, the positive member branch's centre condition, exceptional or infinitesimal semicoherence and the general preparation driver remain OPEN. □

181 Original full curve values on ANY TEST under actual equal-dimension smooth basechange

The dimension induction of [5, Section 5] requires genuine smooth naturality of actual lower values after a preparatory original TEST. The preceding unchanged full curve-value formula applies on EVERY original TEST. Combining it with genuine whole controlled-record and relative monomial dictionaries gives the following equal-ambient-dimension case without canonicality. Positive relative dimension is deliberately not asserted here.

Theorem 181.1 (Actual equal-dimension Cartesian TESTs preserve every positive-depth unchanged full curve value)**.** *Let M be ONLY an original bound-one domain, T ANY genuine original TEST and f an actual smooth submersion between ambient manifolds of the SAME dimension. Given ANY genuine actual Cartesian tower pullback Q with actual endpoint map g, internally prove the actual source bound-one domain, source TEST and BOTH whole cosupport directions. At EVERY actual source endpoint mapping into the actual target cosupport, EVERY positive recursive-depth full original invariant of the actual pulled-back record along Q equals the unchanged full original target value along T. No canonicality, source-domain/TEST/cosupport/order, monomial or desired full-value certificate is supplied.*

Proof. Genuine smooth inverse image constructs the source domain; because the ambient dimensions agree its dimension bound is still one. The real Cartesian controlled-record theorem constructs the actual source TEST and ENTIRE current marked-record pullback identity, preserving carrier, EVERY ordered boundary position, ideal and mark. Its genuine cosupport dictionary gives BOTH membership directions before comparing values. Original-domain stability on the genuine target TEST independently gives actual current well formedness and nonzero.

The genuine whole nonmonomial-part pullback theorem therefore identifies EVERY nonmonomial-order threshold and hence the actual current nonmonomial orders. The independently proved whole relative exponent dictionary identifies every original indexed exponent and boundary incidence; equality of the ambient dimensions then identifies the actual whole

monomial value. Apply the unchanged ANY-test full curve-value formula on BOTH actual bound-one records and genuine TESTs, at every positive recursive depth. The order equality identifies the actual zero and positive cases, the whole monomial dictionary identifies the zero case, and both positive cases have the same actual computed special leaf $(0, \infty)$. Thus the original full values themselves agree, without invoking the canonical-only invariant naturality interface. □

Theorem 181.2 (ONLY original equal-dimension smooth data construct the same actual source history and all full values)**.** *ONLY an original bound-one domain, ANY genuine original TEST and actual equal-dimension smooth submersion internally construct an actual source tower, genuine smooth endpoint map and actual Cartesian square. On the SAME constructed history, internally prove the source domain, TEST, ENTIRE current marked record and BOTH cosupport directions, and the unchanged full original value equality at EVERY source cosupport point and EVERY positive recursive depth. All empty original positions remain in the actual source history. No chosen source history or naturality certificate is supplied.*

Proof. Apply the genuine original-input smooth tower construction to the given actual well-formed original record and TEST. It constructs the real source tower and smooth endpoint map, their actual Cartesian geometry, real source TEST, ENTIRE controlled-record dictionary and whole cosupport identity. Apply the first theorem to THIS actual constructed square, retaining its genuine current-record identity. Its internally constructed source-domain/TEST and both cosupport directions apply at every source endpoint, and its full-value conclusion applies at every positive depth. All objects precede the resulting value theorem.

This is an equal-dimension, carrier-bound-one ANY-test producer. It does not establish positive-relative-dimension noncanonical value transport, higher-dimensional full values, transformed exceptional or infinitesimal semicoherence, or general global scheduling. □

182 The ACTUAL complete original curve invariant at EVERY positive recursive depth

Compute the original GENERAL value in dimension bound one, including the literal companion formed at its ORIGINAL order-run year. The source [5, Section 5, Step II.B] uses actual companion marked ideals. No canonical-history value interface, globally companion-admissible tail or rebased invariant is substituted here.

Theorem 182.1 (ONLY original ANY-test curve data internally evaluate the actual original run-companion leaf)**.** *ONLY an ORIGINAL invariant-domain marked ideal M of bound one, ANY genuine ORIGINAL test T and an actual current ORIGINAL cosupport point y of positive ACTUAL nonmonomial order q internally supply its ACTUAL run r, ORIGINAL head H, ORIGINAL tail S, actual order-mask W at that head, and actual point $p \in S^{-1}W$. For EVERY original recursive depth k, the unchanged actual Step One value, using the original bound-k inductive invariant, of the ACTUAL original companion* $\mathrm{compAt}(M, q, H)$ *along the SAME literal restricted tail at p is exactly $(0, \top)$. No canonicality, companion TEST, candidate special property, chosen run, local replacement or desired value certificate is input.*

Proof. The genuine original order-run theorem, which requires only the ORIGINAL TEST, proves that the actual head image belongs to the actual order-q regime and mask W. Its order is exactly q and it belongs to the actual head cosupport. At THIS same head use the REFLEXIVE tail in the genuine companion-cosupport dictionary: reflexive source and companion tests are literal truths. This internally gives actual INITIAL membership of the head-image point in the original masked companion cosupport, without requiring that the original nonempty tail be its TEST.

The genuine original head/tail TEST dictionary and domain preservation supply the true head domain. The independently proved companion domain and maximal-order theorems construct

the actual masked companion's bound one and maximal order. Its actual initial carrier and ENTIRE ordered boundary equal those of the true masked head record. The SAME restricted original tail is a genuine TEST of this latter source record by actual open restriction of the original tail TEST.

The actual current original positive-order point is outside the ENTIRE accumulated boundary, by the genuine curve nonmonomial-unit theorem. Whole original head/tail and true open record equalities therefore put the SAME actual point outside the entire current boundary of the masked source TEST. Apply the independently proved same-carrier candidate leaf producer to THESE actual records, SAME restricted tail and actual point, using its internally proved initial companion-cosupport membership. That producer constructs its own true empty window and candidate TEST before actual special-point/zero-level evaluation. It gives the genuine special property and zero level on the ORIGINAL restricted tail. Both are independent of the inductive invariant. For EVERY original recursive depth k, the unchanged Step One definition at this PROVED special point ignores the lower invariant and returns exactly $(0, \top)$ with the actual bound-k lower target. Thus the assertion holds at EVERY recursive depth, without comparing any lower value. □

Theorem 182.2 (ONLY original curve-domain ANY-test data give the full unchanged original invariant formula)**.** *For ONLY the same original bound-one domain, ANY ORIGINAL TEST, EVERY actual current original cosupport point and EVERY $k \in \mathbb{N}$, the ACTUAL unchanged original bound-$(k+1)$ invariant is exactly*

$$R_{k+1}(M,T,y) = \begin{cases} (0, \operatorname{inl}\gamma(M_T, y)), & \nu(M_T, y) = 0, \\ (\nu(M_T, y), \operatorname{inr}(\uparrow (0, \top))), & \nu(M_T, y) > 0. \end{cases}$$

Here ν and γ are the EXISTING original nonmonomial order and literal monomial invariant, and the displayed upward inclusion is the actual finite companion value into its genuine with-top target. The lower target in $(0, \top)$ is the genuine original bound-k value type. No original canonicality, simplified substitute invariant, companion history, special-point or full-value certificate is supplied.

Proof. Unfold only the EXISTING original bound-$(k+1)$ general-step definition. If its actual nonmonomial order is zero, its literal defining conditional selects the monomial branch, giving precisely the first formula. Otherwise that natural order is positive. The first theorem, with the same actual run and unchanged original companion, supplies the actual restricted companion Step One value $(0, \top)$. The genuine original order-run mask membership selects the finite branch of the EXISTING companion-value definition. Substituting the first theorem in this SAME raw definition gives its actual finite inclusion $\uparrow (0, \top)$, and then the original GENERAL definition gives precisely the second formula. Both branches compute the original function itself, not a comparison to a newly defined order or an assumed canonical value.

This closes the complete original curve-value calculation at EVERY positive recursive depth on arbitrary ORIGINAL tests. It does not give a higher-dimensional noncanonical full value comparison, exceptional or infinitesimal semicoherence, an original canonical history after sharp preparation, or general driver termination. □

183 The genuine one-dimensional Step One value on every ORIGINAL test

At dimension one the original maximal-order Step One algorithm has no nonempty nonspecial lower branch. Prove this directly from actual masked geometry, without borrowing the canonical-history value interface, for the dimension induction in [5, Section 5]. This is an actual value theorem on every original TEST, not a statement that arbitrary lower TESTS are canonical.

Theorem 183.1 (ONLY original one-dimensional maximal-order ANY-test data force the actual special leaf and exact original Step One value)**.** *ONLY an ORIGINAL maximal-order invariant-domain marked ideal M of bound one, ANY genuine ORIGINAL test T and an actual point y of its current ORIGINAL cosupport imply that y is genuinely special. Consequently the ACTUAL unchanged masked Step One value using the original bound-zero inductive invariant is exactly* $(\mathrm{level}(y), \top)$*. No canonicality, lower canonical TEST, special-point certificate or desired value equality is an input.*

Proof. Original coefficient equivalence internally constructs the SAME actual coefficient-homogenized TEST and actual cosupport membership. Suppose the point were not special. At THIS current stage it is in the regime of its actual level by definition. The true masked lower-domain theorem therefore constructs its actual OWN mask membership and its actual lower marked ideal in dimension bound zero. This statement uses actual TEST geometry only.

Now use the REFLEXIVE tail at this SAME stage. It is genuinely a lower TEST without ANY centre condition or canonicality hypothesis. The actual masked lower cosupport dictionary, applied to this reflexive tail and the actual original level, puts the SAME current point in its true lower cosupport. But the independently constructed dimension-zero domain has EMPTY cosupport along every actual TEST, in particular the reflexive one. This is a contradiction. The point is therefore genuinely special. Unfold the ACTUAL original masked Step One definition at this proved special point. It gives exactly the asserted pair; no property of higher noncanonical lower values is invoked. □

Theorem 183.2 (ONLY original smooth inputs produce the actual Step One value comparison on every original one-dimensional maximal-order test)**.** *For a genuine actual Cartesian smooth source history of ANY ORIGINAL test of the preceding one-dimensional maximal-order input, EVERY actual source point mapping into the target ORIGINAL cosupport has the genuine original source Step One value equal to the actual dimension embedding of the original target Step One value. Both original ordered levels are equal and both actual second entries are* $\top$*.*

The final producer takes ONLY the original domain/maximal order, ANY original target TEST and the genuine smooth base map. It internally constructs the SAME actual source Cartesian TEST before providing this value equality at EVERY relevant actual source point. No source history, source canonicality, lower invariant comparison or special certificate is supplied. Positive original ambient/source dimensions are explicit.

Proof. The first theorem proves the actual target special-point property directly from the ORIGINAL test. Genuine coefficient equivalence internally supplies the SAME target coefficient TEST. The real all-branch special-point dictionary on the actual smooth Cartesian history supplies the ACTUAL source special-point property, before evaluating its function. The true whole original boundary dictionary supplies equality of the actual original levels, retaining every original and empty position.

Unfold the unchanged actual source and target masked Step One definitions. Both now evaluate in their PROVED special branches. The genuine recursive dimension map preserves the level and maps the actual top lower entry to top, so these actual values satisfy the asserted equality. This proof uses neither canonical value invariance nor a desired lower comparison. For the original-only final producer, the genuine original smooth-tower constructor first supplies the actual source tower, real Cartesian square and ORIGINAL source TEST. Apply the just-proved actual-history comparison to it at each actual source point. This keeps every actual map, centre and empty step and completes the original-input construction.

The full GENERAL invariant also uses a companion chosen at the original order-run year. Its genuine companion TEST/domain/cosupport must still be constructed internally on a non-canonical history before this leaf theorem can be applied there. General full-value nonincrease, arbitrary positive member-centre containment, exceptional or infinitesimal semicoherence and general termination remain OPEN. □

184 Genuine full original curve-value locality on arbitrary original tests

The open and isomorphism functoriality in [5, Theorem 1.3 and Section 5] must be applied to the ORIGINAL history after preparation, not silently replaced by a canonical history. In carrier dimension at most one the actual full formula now gives genuine value locality at EVERY positive recursive depth on ANY original test. This uses actual original order and monomial geometry, rather than the canonical-only full-value interface.

Theorem 184.1 (ONLY original curve ANY-test data give exact full-value open restriction at every recursive depth)**.** *ONLY an original bound-one domain, ANY genuine ORIGINAL test, ANY true base open and an actual endpoint-open point mapping into the original current cosupport internally give the SAME restricted original domain and TEST before exact equality of the unchanged original FULL values at EVERY positive recursive depth $k+1$. No canonicality, desired value comparison, replacement history or source-domain certificate is supplied.*

Proof. Genuine original open restriction internally gives the bound-one source domain and the SAME restricted TEST. Actual current whole-record open equality and real coherent-cosupport pullback give source cosupport membership. Apply the genuine full original curve formula to both actual histories at the SAME recursive depth.

The existing true nonmonomial-order open dictionary identifies their actual first entries, without canonicality. If that entry is positive, both actual full formulas have the SAME finite companion entry $\uparrow (0, \top)$ in the actual bound-k value type, so they are equal. If it is zero, the actual ordered monomial invariant pulls back under the genuine endpoint open inclusion, using coherence of the actual current ideal. Whole original record restriction identifies the actual mark and ENTIRE ordered family before this comparison. Its actual γ values therefore agree, giving equality in the zero branch. These two actual branches prove the full original value equality. □

Theorem 184.2 (ONLY original curve ANY-test data give exact full values on genuine isomorphic and internally transported towers)**.** *Let Φ be a genuine biholomorphism of original ambient manifolds and let two actual towers have a genuine tower isomorphism over Φ with actual endpoint biholomorphism χ. ONLY an original bound-one domain and ANY original target TEST internally supply the genuine source pullback domain and actual source TEST before exact equality of the unchanged full original values at EVERY positive recursive depth and EVERY point whose actual target image is in its current cosupport.*

In particular, the original-only final consumer takes just Φ and the original domain/ANY original target TEST. It internally uses the ACTUAL transported tower and ACTUAL back biholomorphism to give this full equality; no chosen source history or tower comparison is supplied.

Proof. True biholomorphic domain pullback internally supplies the original bound-one source domain. The genuine tower-isomorphism TEST dictionary gives its actual source TEST before any value comparison. Whole actual current-record transport and real coherent-cosupport pullback give the required source cosupport point. Apply the already proved actual full curve formula to both histories.

The genuine existing nonmonomial-order tower-isomorphism theorem identifies their actual first entries. In the positive branch both full values consequently have the SAME actual finite companion entry $\uparrow (0, \top)$. In the zero branch, whole current-record transport first identifies the actual source record with the TRUE endpoint biholomorphic pullback of the actual target record. Actual target coherence makes its every member germ holomorphic, so the real ordered monomial-invariant pullback theorem identifies the actual γ values and marks as well. Thus both actual full formulas agree in both branches, without using any canonical value theorem.

For the final original-only consumer, the genuine geometric transport constructor supplies the actual source tower, actual back biholomorphism and genuine tower isomorphism internally,

preserving every original map, centre and empty step. Apply the just-proved theorem to these actual objects. The result concerns carrier dimension at most one, even though the recursive depth is arbitrary. Higher-dimensional noncanonical full-value locality, exceptional or infinitesimal semicoherence and canonical continuation after arbitrary preparation remain OPEN. □

185 Original positive companion boundary thresholds are globally analytic

The actual companion starts at a point's literal nonmonomial-order history year. Its "original" boundary therefore includes precisely those genuine original and exceptional members already present at that year. The historical boundary list is not fixed over the whole final manifold. Its actual point-dependent indexed count nevertheless has global analytic thresholds, because it is an internally selected prefix of the same actual chronological final boundary and the real history-year bounds are analytic.

This identifies a family of genuine positive thresholds of the SAME constructed canonical invariant, rather than assigning a different invariant to the original input. These are the full companion-boundary-count thresholds with the true minimum lower value. Arbitrary later lower thresholds, special/contact presentations and the auxiliary Hilbert–Samuel chain remain separate producers. The historical convention agrees with Bierstone–Milman[5], Section 7, Step II.B, and the recursive historical boundary counts in Bierstone–Milman[4], Section 6.

Lemma 185.1 (The true minimum lower value makes the whole Step One threshold exactly its actual level)**.** *Let b_k be the actual minimum of the literal lower value type: 0 at induction dimension zero and $(0, \operatorname{inl} \bot)$ at a successor. Every actual masked Step One value satisfies*

$$(s, b_k) \le \overline{\operatorname{inv}}(y) \quad \Longleftrightarrow \quad s \le \operatorname{level}(y).$$

The statement includes special points and every later lower/contact branch.

Proof. At dimension zero, 0 is the minimum natural number. At a successor, either the first natural entry is positive, giving strict lexicographic inequality, or it is zero, where the left tagged bottom is at most every actual second entry. Thus b_k is the actual minimum, and its finite lift is at most every finite or top lower value. The literal Step One first entry is exactly its actual original-boundary level. Lexicographic comparison with (s, b_k) is therefore true precisely when that level is at least s, since the second comparison is automatic at equality. The special top value and all nonspecial finite values obey the same minimum comparison. □

Lemma 185.2 (The actual frozen companion boundary is the genuine chronological final prefix)**.** *For an original admissible tower T of length t with original boundary length e, and any actual split with r final stages, put*

$$E_T^{[r]} = \operatorname{take}_{e+(t-r)} E_T.$$

The original-boundary list of the actual companion formed on its true order mask at this split, along its actual restricted tail, is precisely the pullback of $E_T^{[r]}$. Consequently its actual Step One level at a point of the real tail mask equals the indexed boundary count of $E_T^{[r]}$ at that point's same underlying original final point.

Proof. Every genuine blowup appends its actual exceptional member after the strict transforms of the preceding members; empty members are retained at their true chronological positions. Therefore the prefix boundary length is exactly $e + (t - r)$, including literal empty steps. The original members at the final stage of the tail are the first this many members of its actual final boundary. The entire original transform dictionary identifies that boundary with E_T.

The actual companion construction preserves the entire prefix boundary, as does its actual homogenized/coefficient marked ideal used by Step One. Transforming the boundary only depends

on that same original boundary list. The original whole open-restriction transform dictionary therefore identifies its actual final boundary with the pullback of E_T on the true tail mask. Taking the actual original-length prefix commutes with that list pullback. This proves the entire list identity. The indexed boundary count is unchanged by pullback: an index passes through the restricted point precisely when its same original member passes through the underlying point. This gives the stated actual level identity without a caller boundary or value dictionary. □

Lemma 185.3 (The whole actual companion history count has a finite threshold dictionary)**.** *For the SAME original actual history, put* $c_q(y) = \operatorname{count}(E_T^{[R_q(y)]}, y)$ *and*

$$C_{q,s} = \{y \in \operatorname{cosupp} M_T : q \le \nu_T(y),\ s \le c_q(y)\}.$$

Then the exact entire identity is

$$C_{q,s} = \bigcup_{0 \le r \le t} \big(L_{q,r}(T) \cap \{y : s \le \operatorname{count}(E_T^{[r]}, y)\}\big).$$

No history year or selected prefix is supplied as additional data.

Proof. The literal run is at most the genuine finite tower length. A point of the left set therefore supplies its actual $r = R_q(y)$ in the true finite union; the selected prefix and count are exactly its own. Conversely membership in the r summand says $R_q(y) \le r$. Hence $t - r \le t - R_q(y)$ and $E_T^{[r]}$ is an actual indexed prefix of $E_T^{[R_q(y)]}$. The injection of the smaller prefix's indices into the larger one preserves each member and membership through the same point. The smaller indexed count is at most the larger indexed count, so its lower threshold gives the actual point-dependent count threshold. Both directions retain the same genuine current cosupport and current-order lower bound. This proves exactly the entire displayed identity, including zero or excessive count thresholds and all real empty chronological members. □

Theorem 185.4 (Actual chronological companion-count thresholds are genuinely global analytic sets)**.** *In positive ambient dimension, for only an original marked-domain input and its literal constructed canonical finite history, each* $C_{q,s}$ *is globally complex analytic, in arbitrary induction dimension without compactness or a curve envelope.*

Proof. Original canonical-domain transport internally supplies the actual final SNC boundary E_T. Every true indexed prefix $E_T^{[r]}$ inherits that same SNC structure, so its finite intersection dictionary supplies actual holomorphic equations for each indexed-count threshold. The actual history-year producer supplies genuine global equations for $L_{q,r}(T)$. Intersect those actual finite presentations in every true summand of the preceding dictionary and take their real finite union. The resulting holomorphic equations present exactly $C_{q,s}$ on the entire original final manifold. No global finite component family or analytic target certificate is assumed. □

Theorem 185.5 (Literal positive canonical companion minimum-tail thresholds have the exact whole original dictionary)**.** *For original marked-domain input and its SAME literal canonical finite history, and every positive* q *and indexed count threshold* s*, let*

$$\alpha_{q,s} = (q, \operatorname{inr}(s, b_k))$$

be the genuine tagged threshold in the literal canonical value type at induction dimension $k+1$*. Then*

$$\{y \in \operatorname{cosupp} M_T : \alpha_{q,s} \le \operatorname{inv}_{k+1}(M, T, y)\} = \big(\operatorname{cosupp} M_T \cap \{\nu_T \ge q+1\}\big) \cup C_{q,s}.$$

The left side is the original full value, with its actual original companion and point-dependent historical split; it is not a redefined count invariant.

Proof. At a current cosupport point of order below q, the original first entry makes the comparison false, and both right terms fail by their genuine order bounds. At order above q, the first entry gives strict inequality; the actual order-$q+1$ baseline includes the point regardless of its history count. At order exactly q, positivity and the genuine original canonical history internally supply the actual companion order mask, whole companion marked domain, true maximal order and canonical restricted tail. Its lower value is finite and equals the literal masked Step One value at the true restricted point. Therefore tagged lexicographic comparison is exactly comparison of that actual Step One value with (s, b_k). The first lemma turns this into the actual companion level threshold; the whole frozen boundary dictionary identifies that level with $c_q(y)$. This is precisely the right count term. All cases use actual original data; no companion value, mask-membership or chronological dictionary is an input. □

Theorem 185.6 (Only original inputs construct global analytic positive companion-count superlevels)**.** *Every literal original positive threshold $\alpha_{q,s}$ above has a genuine global analytic superlevel in arbitrary induction dimension. No compactness, curve, presentation, threshold-analyticity or selected-history certificate is supplied by the caller.*

Proof. The actual coherent nonmonomial ideal supplies the whole order-$q+1$ baseline and original coherence supplies the whole current cosupport. The chronological companion-count producer supplies genuine analytic equations for $C_{q,s}$. Take their finite union through the preceding exact whole original value dictionary. Its actual holomorphic presentations define precisely the literal canonical superlevel. This proves the positive companion boundary layer, while every later arbitrary recursive lower threshold still requires its own genuine semicoherent marked/contact presentation construction. □

186 Actual companion history-year thresholds have global analytic equations

The historical companion construction uses the first year at which a point has its current positive nonmonomial order. Its chronological year is genuine geometry of the original finite blowup tower, not an extra value certificate. The analytic threshold below includes the actual higher-order baseline, exactly as a lexicographic companion threshold does. On the actual fixed-order stratum it is the literal lower bound on that first year.

Bierstone–Milman[5], Section 7, Step II.B, defines the companion value at this first year and retains it under earlier isomorphisms. The present construction gives analytic equations for this actual history-selection layer. It is one input for the genuine semicoherent companion presentations required by Bierstone–Milman[4], Remark 6.14 and Lemma 13.7. It does not identify a whole recursive lower companion superlevel with this history-year threshold, and does not construct the simultaneous Hilbert–Samuel auxiliary-chain presentation.

Lemma 186.1 (Internally coherent nonmonomial ideals produce every actual prefix-order threshold)**.** *In positive ambient dimension, for an original invariant-domain marked ideal and an actual admissible finite tower, let P_j be its actual prefix in the literal split with the last j stages and let ρ_j be its actual tail projection. The nonmonomial ideal of the actual prefix transform is coherent. For every natural threshold a, the actual locus*

$$H_{j,a} = \rho_j^{-1}\{x \in N_j : a \le \nu_j(x)\}$$

is globally analytic, without compactness or a dimension restriction on its carrier. Its finite local equations are produced from the actual prefix nonmonomial ideal and composed with the true holomorphic tail projection.

Proof. Admissibility passes to every actual prefix. The original domain transport therefore gives the prefix's coherent ideal, closed smooth carrier, SNC boundary, genuine transversality and

nonvanishing along the carrier. Nonmonomial division on its actual SNC charts makes its actual nonmonomial ideal coherent. At threshold zero the order locus is the entire actual prefix carrier. At threshold $a > 0$ it is the intersection of that carrier with the zero locus of the coherent derivative ideal of order $a - 1$ along it. Those derivative generators and the carrier's actual local defining functions give finite holomorphic equations. The actual tail projection is a finite composition of genuine holomorphic blowup maps. Composing the finite defining functions with this projection presents exactly $H_{j,a}$ on its inverse-image neighborhood. Thus every such whole prefix threshold is genuinely analytic; no closedness-to-analyticity implication is used. □

Definition 186.2 (Literal enhanced companion history-year threshold)**.** For a genuine finite tower T of length t, let $R_q(y)$ be the literal number of trailing blowups whose previous images lie in the actual regime "current cosupport and nonmonomial order q". Define

$$L_{q,r}(T) = \{y \in \operatorname{cosupp} M_T : q \le \nu_T(y),\ R_q(y) \le r\}.$$

On the actual order-q stratum, this is the lower bound $t - r$ on the chronological companion starting year $t - R_q(y)$. No new invariant is substituted for the original value or its actual history.

Proof. The original trailing-run recursion computes R_q from the real successive points and original transformed ideals. The displayed set is defined by that same literal function and actual current order. If the current order is q, the original definition starts the companion precisely at the prefix preceding these $R_q(y)$ trailing blowups. Its year is $t - R_q(y)$. For $r \le t$, natural subtraction makes $R_q(y) \le r$ equivalent to $t - r \le t - R_q(y)$; the bound $R_q(y) \le t$ follows from the genuine finite tower. Thus the claimed history dictionary uses the original actual split. □

Lemma 186.3 (The entire actual history-year threshold has a finite analytic recursion)**.** *Write $T' = T\sigma$ for an actual canonical successor blowup of a value which dominates the actual nonmonomial order. Put $B_q(T') = \operatorname{cosupp} M_{T'} \cap \{\nu_{T'} \ge q\}$ and $A_{q+1}(T) = \{x \in N_T : \nu_T(x) \ge q+1\}$. Then the exact entire identities are*

$$L_{q,0}(T') = B_q(T') \cap \sigma^{-1} A_{q+1}(T),$$

$$L_{q,r+1}(T') = B_q(T') \cap \sigma^{-1}\big(A_{q+1}(T) \cup L_{q,r}(T)\big).$$

These identities include higher-order current points and all actual special or nonspecial points, not merely a selected contact open.

Proof. At a current cosupport point of order at least q, genuine canonical order monotonicity puts its previous image in the actual previous cosupport with order at least q. Its previous regime holds exactly when that previous order equals q. If it holds, the literal trailing-run recursion adds one to the previous run. If it does not, the run is zero; the previous order is then at least $q+1$, which is exactly the displayed actual order locus by the genuine nonmonomial-order dictionary. Hence run zero is exactly that first case. For the bound $r+1$, either that zero-run case holds, or the previous regime holds and the previous run is at most r. In the latter case the previous image belongs to $L_{q,r}(T)$. Conversely previous membership in $L_{q,r}(T)$ either already has order larger than q, giving zero current run, or has order exactly q, giving a current run at most $r+1$. This proves both directions of both entire identities. The order monotonicity and cosupport inheritance are derived from the actual domain, canonical centre, and blowup; they are not supplied as history dictionaries. □

Theorem 186.4 (Only original canonical inputs produce genuine global analytic companion history-year thresholds)**.** *For only an original invariant-domain marked ideal, its literal constructed canonical invariant and an actual canonical finite history, every $L_{q,r}(T)$ is globally analytic. This holds in every induction dimension in positive ambient dimension, without compactness, a curve envelope, finite value range, or any caller analyticity or presentation hypothesis.*

Proof. The literal constructed canonical value has first entry the genuine current nonmonomial order. It therefore supplies order dominance internally, and its actual canonical history is admissible by the proved original canonical-domain transport. Induct on the genuine finite tower. For the reflexive tower every run is zero, so the desired locus is the intersection of its actual cosupport and its actual order-q threshold; both have the actual coherent equations of the first lemma. At a successor, apply the preceding entire step dictionaries. The new actual cosupport and order threshold are analytic by internally transported original coherence. The old order-$q+1$ locus and old history-year locus are analytic by the first lemma and induction. Finite union and intersection of their actual finite equations, followed by composition with the genuine holomorphic successor blowup, give the exact equations in either dictionary. This constructs the entire analytic locus at each step. It uses upper semicontinuity nowhere as a substitute for analytic equations.

The next companion-presentation step must use these actual chronological strata to freeze the positive member/contact lower marked presentations and compare their true restricted histories. The arbitrary recursive lower thresholds and auxiliary-chain Hilbert–Samuel presentations are still separate mathematical producers; this theorem does not assume them. □

187 Actual computed zero-mask full curve values on nested original opens

The contact-history comparison in the dimension induction of [5, Section 5, Step I.B] must compare the unchanged full value of the actual computed lower record with its genuine smaller contact window. The computed large mask is fixed by the original regime run; it cannot be replaced by a convenient chosen mask. We first prove the required noncanonical nested-open transport from real tower geometry and the already proved ANY-test curve-value formula.

Theorem 187.1 (Genuine nested open restriction preserves every positive-depth full curve value on ANY original TEST)**.** *Let P be an actual marked ideal with coherent ideal on the original ambient manifold, $V \subset W$ actual opens, and T an actual tower. Assume only that the actual restriction $P|_W$ is a genuine bound-one domain and $T|_W$ is its genuine TEST. Internally construct the genuine bound-one domain and TEST on V. At EVERY endpoint of $T|_V$ its actual cosupport membership is equivalent to membership of the SAME underlying endpoint in the actual cosupport of $T|_W$. At such a point EVERY positive recursive-depth unchanged original full value on V equals the value on W. No canonicality, nested comparison, smaller domain/TEST/cosupport or full-value certificate is supplied.*

Proof. Let $B = W \cap V$, regarded as an open of W. First restrict the genuine domain and TEST on W to B. The actual identity-on-points biholomorphism $B \simeq V$ and actual tower isomorphism identify the doubly restricted tower with $T|_V$, preserving every map and empty position. True coherent pullback composition identifies the ENTIRE transported marked record with $P|_V$, including its ordered boundary. Transport of the genuine domain and TEST through this actual isomorphism supplies the asserted domain and TEST on V before any value comparison.

The genuine tower-isomorphism cosupport dictionary followed by the genuine open-restriction cosupport dictionary gives BOTH membership directions at the same underlying original endpoint. At such a point apply the proved ANY-test full curve-value isomorphism theorem to the actual identity-on-points isomorphism, and then apply the proved ANY-test full curve-value open-restriction theorem from W to B. Their composition gives the asserted equality at EVERY positive recursive depth. Only genuine TESTs are used; the canonical-only invariant restriction interface is never invoked. □

Theorem 187.2 (ONLY original zero-regime data supply the actual computed large mask and all full values on every genuine smaller mask)**.** *ONLY an original bound-two maximal-order domain, ANY genuine original TEST and an actual current cosupport point in the zero nonspecial*

regime internally supply the ACTUAL regime-run split, ACTUAL computed large zero mask W, unchanged computed lower record J and genuine restricted original tail. Its bound-one domain, TEST and actual cosupport point are internally proved. For EVERY actual open $V \subset W$ containing the actual head image, internally construct the true smaller kept contact domain, genuine restricted tail TEST and actual smaller cosupport point. At the SAME original endpoint, its unchanged full original value equals that of the ACTUAL computed large lower record for EVERY positive recursive depth. No caller lower record/domain/TEST, canonicality, smaller cosupport or desired value equality is input.

Proof. Apply the original-input zero-mask-history producer to the GIVEN original TEST, actual current cosupport and actual zero nonspecial regime. This internally constructs the true regime-run split, actual computed large mask membership, actual lower bound-one domain, real restricted tail TEST and BOTH whole cosupport directions. In particular the same original endpoint belongs to the actual computed lower cosupport.

The actual coefficient TEST head is supplied internally by genuine coefficient equivalence and the real head-tail decomposition. The original coherence theorem therefore proves coherence of the actual global contact-parent ideal on the head-stage manifold. By the original zero-mask definitions, its actual restriction to the computed mask is EXACTLY the computed lower record, including all four marked fields. For any actual smaller open as in the statement, apply the first theorem to THIS actual global contact parent, actual large mask and same original tail, using the internally proved large domain and TEST. Its BOTH cosupport dictionary and the internally constructed large cosupport point supply the smaller point. The all-depth full-value equality is then the first theorem's actual conclusion, with the unchanged computed lower parent identified by its defining equality.

This closes a genuine local-to-computed-record value bridge in the zero-contact branch with original carrier bound two. It does not assert the whole upper GENERAL comparison, a positive-member history bridge, higher-dimensional noncanonical full-value locality, exceptional or infinitesimal semicoherence, or the full global scheduling theorem. □

188 Actual contact-choice lower histories for every ORIGINAL test

Retain ONE true contact gluing before the ORIGINAL upper history is chosen. Use actual tower isomorphisms and the true selected-complement opens to compare the literal lower histories, as in [5, Sections 5 and 7]. The comparison retains every historical EMPTY position and every actual new exceptional member. Original nonempty boundary preservation is not presumed.

Lemma 188.1 (Actual original kept remainder is the remainder of the genuinely empty-position padded contact record)**.** *For ANY actual original K, function w, genuine tower prefix T_0 and actual stage open, let A_0 be its true contact record with empty INITIAL boundary. Pad A_0 by $\#K.E$ empty entries, preserving its WHOLE actual new exceptional list, to get A. Then the ACTUAL kept-position remainder of K is exactly* $\mathrm{remainder}_{\#K.E}(A)$ *as an ENTIRE marked record. No geometry, test or desired record equality is an input.*

Proof. The previously proved literal dictionary identifies the original kept remainder with the true remainder of A_0 padded by $\#K.E$ empty entries. For the padded parent A, its carrier, ideal and mark are those of A_0, so its actual selected components and remaining carrier are identical. Dropping its first $\#K.E$ boundary entries drops EXACTLY the just-added empty list and leaves the SAME whole exceptional list. The remainder replaces that initial block by the very same empty block. Thus all four marked fields agree, including the full ordered boundary with multiplicities. □

Theorem 188.2 (ONE actual contact gluing internally produces both genuine lower histories and whole records at every prefix of every ORIGINAL upper test)**.** *For actual well formed positive-mark original K, an ORIGINAL cosupport point and two genuine tangent directions on opens containing it, ONE actual gluing G is constructed BEFORE EVERY original split upper test $T_0\,T_1$. The SAME actual once-restricted upper tails have a real tower isomorphism. The true empty-initial-boundary contact records and their actual gluing are internally constructed, with smooth carrier, coherent ideal, positive mark and a genuine contact tail test.*

Let A be the true contact parent padded with exactly $\#K.E$ empty positions. Both original kept contact remainders are identified with the actual remainders of A and its genuine pullback. On the ACTUAL selected complement opens, the LITERAL twice-restricted upper tails are genuinely isomorphic lower TESTS. Their ENTIRE controlled marked records are genuine pullbacks at the endpoint and at EVERY actual head, with compatible real head/tail tower isomorphisms. The actual endpoint gluing fixes EVERY actual ORIGINAL transformed cosupport point. Canonicality, independent desired lower tests, padded-test certificates and record comparisons are absent.

Proof. Apply genuine contact gluing to the original smooth carrier, coherent ideal, positive mark, original cosupport point and actual tangent directions. It constructs G before the split tower. Coefficient-homogenized equivalence turns the SAME ORIGINAL appended test into the required actual empty-initial- boundary coefficient test. The genuine contact-tail theorem then constructs the actual lifted stage and endpoint biholomorphisms, a real tower isomorphism of the once-restricted tails, whole contact transport and actual smooth/coherent/positive geometry and TEST of its contact parent A_0.

Form the actual padded parent A. Genuine pullback commutes with padding by empty entries, hence the whole parent transport for A_0 gives that for A. The first lemma on both sides identifies the original kept remainders with the actual remainders of these padded parents. The marking, ideal and carrier of A have not changed. Its true boundary has length at least $\#K.E$, and actual padding preserves test admissibility: the original centres have normal crossings with every retained exceptional member and every new empty member is disjoint. Thus the actual padded parent tail is a genuine TEST, proved internally rather than assumed.

Apply the genuine selected-complement transport theorem to THESE actual parent and tower isomorphisms. The true remaining carrier is transported by the actual biholomorphism, so the complement opens and their endpoint inverse-image opens have real inverse maps. Restrict the SAME towers twice; this constructs the asserted lower tower isomorphism and both actual lower TESTS. Genuine controlled-transform transport gives the ENTIRE endpoint marked equality. The real split-isomorphism theorem applied to this actual lower isomorphism gives the compatible head and tail isomorphisms for EVERY numeric position, and whole controlled transport gives the ENTIRE marked equality at each such head. No centre, map or empty step is changed.

Finally the genuine coefficient and empty-boundary cosupport dictionaries identify the actual transformed ORIGINAL cosupport with the one fixed by the constructed endpoint map. Therefore the endpoint fixation applies to every original cosupport point in its actual domain. This constructs actual lower histories, rather than inferring them from a pointwise comparison. Earlier frozen FULL original lower invariants, comparison of independently chosen source values and exceptional or infinitesimal semicoherence remain separate obligations. □

189 The actual zero/nonzero contact split and whole remainder commute with smooth base change

Use the literal pulled-back holomorphic ideal and the actual inverse-image carrier under a genuine smooth holomorphic submersion, of possibly different ambient dimensions. The selected zero and remaining nonzero component unions are genuine geometric sets, not new component certificates. Their equality under base change supplies a recursive presentation ingredient for [5, Section 7]; it does not supply exceptional or infinitesimal stability.

Theorem 189.1 (Actual pullback nonzero reflects, and true submersion pullback preserves it)**.** *For an ANY actual carrier N, an actual holomorphic germ ideal I, a map continuous at y, and the independently defined true ideal pullback, nonzero along the actual inverse-image carrier at y implies nonzero along N at $f(y)$. No coherence or smooth-carrier assumption is required for reflection. If f is a genuine holomorphic submersion and I is coherent, BOTH nonzero conditions are equivalent. Ambient dimensions may differ; zero dimensions and empty carriers are included.*

Proof. For reflection, suppose every original ideal germ is eventually zero along N at $f(y)$. Let a genuine pulled-back ideal germ be given. Its actual finite span expression has finitely many original ideal germs composed with f, multiplied by genuine holomorphic coefficients, with a real neighborhood equality to the given germ. Continuity at y pulls back each original eventual zero relation. Intersect these finitely many actual neighborhoods and the true span-equality neighborhood. Every summand is zero on the entire inverse-image carrier there, so the given pulled-back germ is eventually zero along it. This contradicts any actual frequent nonzero witness and proves reflection, without an ideal or nonzero certificate. For preservation, genuine submersion local sections give the actual neighborhood-filter image equality. Compose an actual original nonzero germ with f; true ideal pullback gives its membership and the filter equality transports frequent nonzero carrier values. Combine the two independently proved implications. □

Theorem 189.2 (WHOLE actual selected and remaining component unions pull back under genuine submersions)**.** *For a genuine closed smooth marked carrier, coherent ideal and positive mark, under ANY actual holomorphic submersion, the selected component union of the literal pulled-back record equals the inverse image of the original selected union. The remaining component union likewise equals the actual inverse image of the original remaining union. No global surjectivity, connected-fibre, component-bijection or desired set equality is input; no canonicality is used.*

Proof. Actual submersion geometry internally makes the true inverse-image carrier closed smooth; genuine ideal pullback supplies coherence, and the literal record retains its positive mark. At a point of this actual carrier, the proved zero-germ criterion identifies selected-component membership with failure of nonzero along the carrier. Apply that criterion to BOTH real records, and then use the preceding BOTH-direction nonzero dictionary. This proves selected membership equality at every carrier point. Outside the carrier, neither selected union can contain the point, because each actual component is contained in its carrier. Thus the equality holds on the entire ambient manifold, not just on a chosen component or open image.

The true remaining union equals the carrier minus the actual selected union for each genuinely closed smooth carrier. Substitute the already proved whole selected equality and the literal inverse-image carrier. Inverse image preserves set difference, giving the asserted remaining equality. No claim that global components themselves are in bijection is needed or inferred. □

Theorem 189.3 (ALL FOUR fields of the actual kept-position contact remainder commute with smooth pullback)**.** *Under ONLY the preceding original geometric hypotheses, the literal smooth pullback of the actual contact remainder equals the actual contact remainder of the literal smooth pullback, as an ENTIRE marked record. The same number of historical empty positions, every actual remaining boundary position, the genuine pulled-back ideal and unchanged mark are retained. No lower-domain, component, transformed-record or naturality certificate is supplied.*

Proof. The preceding theorem gives equality of the actual carriers. Actual inverse-image mapping of finite lists commutes with dropping the same number of positions; mapping an actual empty member gives an actual empty member, and mapping a replicated list preserves its exact length. Thus both literal boundary lists have the SAME historical empty entries and ordered suffix, without deleting or reindexing any position. The ideal on both sides is the SAME independently defined genuine ideal pullback, since contact remainder changes only carrier and boundary; the

mark on both sides is the unchanged actual integer. These are the four literal field equalities, hence equality of the whole marked record.

This is genuine smooth base-change locality for the actual geometric component split and remainder. It does not assert that arbitrary local tests extend globally, compare earlier frozen FULL original lower histories, infer exceptional or infinitesimal semicoherence, or prove higher invariant decrease or general termination. □

190 Actual empty original-history windows at positive curve order

The ordered boundary in [5, Section 5, Step II] contains every new exceptional member, including empty positions. On a carrier of dimension at most one the genuine nonmonomial part is a unit at every incident transverse boundary point. Thus a positive-order point has an actual base neighborhood where the ENTIRE given history has empty centres. These statements concern ANY ORIGINAL test, without a canonical history assumption or a supplied empty-history certificate.

Theorem 190.1 (An actual point outside the entire accumulated boundary internally supplies an empty original-history window)**.** *For ANY marked record with closed original boundary members, ANY actual finite blowup tower T, and an actual point y outside its ENTIRE accumulated ordered boundary, internally construct an actual base open U containing $T(y)$ such that the restriction of the SAME ENTIRE tower to U has only empty centres. No compactness, marked admissibility, base neighborhood, inverse or empty-history certificate is supplied.*

Proof. The finite union F of actual current boundary members is closed: old members are closed initially, strict transforms are closed, and new members are inverse images of closed centres. The actual tower is proper, so its literal image $T(F)$ is closed. The current boundary contains the actual exceptional locus. A fibre through the given point $y \notin F$ is therefore a singleton. Consequently $T(y) \notin T(F)$: a point of F in that fibre would equal y. Put $U = X \setminus T(F)$.

Prove that all actual centres avoid U by induction on the SAME tower. At a last actual blowup π, surjectivity supplies a lift of every point of the preceding boundary. Its lift lies in the new boundary, since the inverse image of the old boundary is contained in the new ordered boundary. Hence the image of the preceding boundary is contained in the final image $T(F)$. The induction hypothesis applies to the SAME open U. Surjectivity also lifts every point of the last centre to the new exceptional member; its image in the original base therefore belongs to $T(F)$. That last centre is disjoint from the actual inverse image of U. This proves avoidance for every centre, retaining the actual maps and every empty step. The genuine restricted-centre dictionary now says that the ENTIRE restricted tower has only empty centres. □

Theorem 190.2 (ONLY a genuine dimension-at-most-one domain makes the actual nonmonomial part a unit on the whole boundary)**.** *ONLY a genuine invariant domain of dimension bound one implies that at EVERY actual carrier point on its entire ordered boundary the ACTUAL nonmonomial ideal contains* 1 *and has order zero. In particular, ONLY an original dimension-bound-one domain and ANY ORIGINAL test internally give the ACTUAL full original general invariant at every such current point as its literal monomial branch* $(0, \gamma)$*. No pure-curve, relative-principal, nonfactor, unit, order or desired full value certificate is input.*

Proof. Construct the true parts of the original marked ideal first. Actual wellformedness and nonzero-along-carrier hypotheses give coherence of the nonmonomial ideal, its genuine nonzero germs along the carrier, and an actual germ outside the relative ideal of EACH incident boundary member. At the chosen point, the real dimension bound supplies regular dimension $j \leq 1$. If $j = 1$, the genuine transverse curve isolating neighborhood and the actual no-factor witness force a nonzero value of that witness; its holomorphic inverse gives 1 in the actual nonmonomial ideal. If $j = 0$, a genuine isolating neighborhood in the carrier and the actual frequent nonzero witness

force that witness to be nonzero at the point itself, and its true holomorphic inverse again supplies 1.

Testing order at least one on this SAME unit germ is impossible, since its value is one. The actual order is therefore zero. For ANY original test, genuine domain preservation supplies all preceding hypotheses on the ACTUAL current record and entire ordered boundary. Finally unfold the unchanged original dimension-one general invariant: the first entry is the ACTUAL nonmonomial order just proved zero, so its actual defining conditional selects precisely the literal monomial γ branch. No companion TEST or noncanonical higher-value interface is used. □

Theorem 190.3 (ONLY original curve-domain ANY-test data give an actual empty entire-history window at every positive first entry)**.** *ONLY an original dimension-bound-one domain, ANY genuine ORIGINAL test and an actual current cosupport point of positive actual nonmonomial order internally supply a base open containing its actual original image on which the SAME ENTIRE restricted ORIGINAL history has empty centres. Its actual restricted original record remains in dimension bound one and the SAME actual restricted tower is genuinely its TEST. No canonicality, selected window, pure current carrier, replacement history or empty-history certificate is input.*

Proof. The current ORIGINAL cosupport point lies in the actual current carrier. Genuine domain preservation supplies its dimension-bound-one domain. If the point belonged to any actual current boundary member, the preceding theorem would make its actual nonmonomial order zero, contradicting its given positivity. It is therefore outside the ENTIRE actual accumulated boundary. Original wellformedness gives closedness of all ORIGINAL boundary members, so the first theorem internally constructs the required actual base open and empty ENTIRE restricted tower. True open restriction of the original domain gives the restricted domain, and true open restriction of the ORIGINAL TEST gives the SAME actual restricted TEST. The chosen point belongs to the actual endpoint open because its original image belongs to the constructed base open.

This is a genuine local geometric input for the positive GENERAL branch. Comparing the companion formed at its actual original order-run year, its actual restricted tail and the unchanged full noncanonical GENERAL value remains a separate obligation. No higher full-value locality, exceptional or infinitesimal semicoherence, or general driver termination is asserted here. □

191 The original positive history supplies a single true coherent companion presentation

At a reference cosupport point x of positive nonmonomial order q, use its actual original order run r, actual prefix, actual order mask W, and actual restricted tail S. Put $K = \mathcal{O}(M_{\mathrm{head}}|_W, q)$ and $V = S^{-1}_{\mathrm{unrestricted}} W$. These are the genuine original constructions of Bierstone–Milman[5], Section 7, Step II.B. The reference point determines one presentation on the entire true open V. Other points need not have the same historical year: the real original rebase comparison supplies the common value dictionary. This construction does not assume any lower superlevel is analytic.

Lemma 191.1 (Original inputs construct the true fixed companion domain and entire cosupport dictionary)**.** *For positive ambient dimension, original marked-domain input in induction level $k+1$, its literal original canonical finite history, and an original cosupport point x with $q \geq 1$, the true K is in that same marked domain, has maximal order, and the actual S is canonical for its literal original masked Step One value. Moreover $x \in V$ and the ENTIRE actual identity is*

$$p \in \operatorname{cosupp} K_S \quad \Longleftrightarrow \quad p \in \operatorname{cosupp} M_T \text{ and } \nu_T(p) = q, \qquad p \in V.$$

Proof. The original canonical invariant internally supplies a genuine test tower. Its literal Step One lower invariant is a maximal-order value function. Apply the genuine original canonical-tail

theorem at x: it supplies mask membership, the actual companion domain, maximal order and canonical restricted tail. Original tower splitting supplies the genuine prefix and tail test histories. The true companion-tail cosupport dictionary, using the actual companion test history, identifies its cosupport with the original current cosupport and current nonmonomial order q. The whole original head–tail transform identity identifies the tail transform with the same M_T. This proves the stated identity on every point of V; no mask, lower-data or companion-domain certificate is an input. □

Lemma 191.2 (The same true open has no higher-order cosupport points)**.** *Every original current cosupport point in V has nonmonomial order at most q.*

Proof. The original prefix is in the marked domain, and its true order mask excludes nonmonomial order $q+1$. The genuine original companion-tail test history preserves this bound on original current cosupport over that mask: the companion cosupport and actual order transformation give nonmembership in the true order-$(q+1)$ locus. The original finite order dictionary gives order at most q. The whole head–tail identity identifies this with the literal original current order. □

Theorem 191.3 (Independently selected original histories have one entire fixed companion value dictionary)**.** *For every point $p \in \operatorname{cosupp} K_S$, the SAME original full value is $(q, \operatorname{inr} \overline{\operatorname{inv}}(K,S,p))$. Consequently, for EVERY finite or top lower threshold β,*

$$V^{-1}\{y \in \operatorname{cosupp} M_T : (q, \operatorname{inr}\beta) \le \operatorname{inv}(M,T,y)\} = \{p \in \operatorname{cosupp} K_S : \beta \le \overline{\operatorname{inv}}(K,S,p)\}.$$

The one actual companion and its actual coherent history serve the whole family, including points with different genuine original historical runs.

Proof. The original canonical program supplies lower data at every true positive cosupport point. On V, the genuine fixed-companion comparison compares the point's independently selected original starting year with that of x. The actual intermediate centres avoid the earlier order mask; genuine empty-history restriction, companion rebase, canonical maximal-order value transport and the real common-open comparisons identify the value with that of the single reference companion. This proves the displayed value at every point of its true cosupport.

For the superlevel equality, original tuple comparison implies current order at least q. The preceding genuine upper bound implies equality, so the entire cosupport dictionary gives membership in the actual fixed companion cosupport. The whole value equality then reduces tagged lexicographic comparison exactly to the displayed lower comparison. Conversely fixed companion cosupport supplies original cosupport; the same value equality and lower comparison give the original full threshold. This includes top thresholds and makes no equal-year premise. □

Theorem 191.4 (Actual coefficient-history geometry supplies honest local holomorphic generators)**.** *Let $B = (\mathcal{C}(\mathcal{H}(K)))_S$ be the actual coefficient marked ideal along the true canonical restricted tail. It is genuinely well formed and its actual cosupport equals* $\operatorname{cosupp} K_S$*. At every point of V, its LITERAL ideal has a finite family of actual holomorphic functions on an actual open neighborhood generating every actual nearby ideal germ, with both membership directions. These are true functions of the original coefficient ideal, not supplied defining equations or a fresh substitute ideal.*

Proof. The original Step One canonical-tail theorem supplies the true admissible history of the actual homogenized/coefficient marked ideal, together with its entire cosupport equality. Original well-formedness of K supplies well-formedness of its true coefficient marked ideal; genuine test-history transport gives well-formedness of B, including actual ideal coherence, closed smooth carrier and actual SNC divisor. The coherence of that same literal ideal gives, at any point, a real open neighborhood, finite holomorphic functions and the two-direction local span membership dictionary. The constructions retain the actual coefficient ideal and the SAME fixed companion history throughout. □

Theorem 191.5 (Only original data produce analytic special-tail thresholds on the entire actual reference open)**.** *For every indexed count s, the inverse image on the whole actual V of the literal original threshold $(q, \mathrm{inr}(s, \top))$ is complex analytic in arbitrary induction level. The reference companion, its special locus and its presentations are internally constructed from the original inputs.*

Proof. The fixed companion's genuine domain and maximal order and the actual coefficient test history supply the real masked Step One special-top analytic producer. Its equations use the true coefficient cosupport, actual original-member incidence intersections and the actual codimension-one coefficient support at count zero. This proves analytic superlevels for $(s, \top)$ of the SAME masked Step One value on the entire actual tail open. The full original threshold dictionary transfers these actual equations to precisely the literal original full superlevel there. No lower-analyticity, special-locus or presentation premise is added. The result is explicitly local on this genuine reference open; global arbitrary-lower companion thresholds and recursive contact presentations remain subsequent mathematical work. □

192 Actual member/contact history supplies a genuine recursive lower presentation

The positive companion reference presentation reduces the full original value to actual masked Step One on one true open. At a nonspecial reference point, we now construct the genuine next member/contact marked ideal, its actual canonical restricted history, and one ENTIRE lower-threshold dictionary. The historical choice is the original one of Bierstone–Milman[5], Section 7, Step I. This advances the recursive original-input presentation; it does not assume lower analyticity or claim whole global closure.

Lemma 192.1 (The original nonspecial masked history constructs its actual lower marked domain)**.** *For original maximal-order marked-domain input at induction level $k+1$, its literal original canonical masked Step One history, and an actual nonspecial cosupport point x of level s, the true level-regime run selects an actual lower mask U and actual lower marked ideal L at its start. The reference point is over U, L is in the actual level-k domain, its true restricted tail is canonical for the SAME original level-k invariant, and the ENTIRE identity on its actual final open is*

$$p \in \mathrm{cosupp}\, L_{S|_U} \iff p \in \mathrm{cosupp}\, J_T \text{ and } \mathrm{level}(p) = s.$$

The original construction includes both true member and zero-level contact branches; no lower-domain or mask certificate is given.

Proof. The original Step One bridge internally supplies the actual test history of the true homogenized/coefficient ideal and its entire cosupport identity, and identifies its original canonical segments. The real canonical-tail theorem gives the point's mask membership, actual level-k lower domain and its true canonical restricted history. The original nonspecial level-regime theorem identifies the true start level and its cosupport point. Splitting and the canonical lower test history let the actual lower-marked cosupport theorem identify the whole lower cosupport with same-level true coefficient cosupport. The original whole coefficient cosupport identity gives the displayed original one. All hypotheses come from the actual original input and actual nonspecial branch, for either lower construction. □

Theorem 192.2 (One actual lower marked presentation serves the entire literal threshold family)**.** *At every point of the genuine lower cosupport, the SAME original masked Step One value is $(s, \mathrm{inv}_k(L, S|_U, p))$. For every finite or top α, the entire inverse image of its original threshold (s, α) is exactly*

$$\{p \in \mathrm{cosupp}\, L_{S|_U} : \alpha \leq \mathrm{inv}_k(L, S|_U, p)\}.$$

Other points need not have the reference point's selected history.

Proof. Original canonical-tail data are internally available along every actual canonical prefix. Genuine Step One common-history comparison and empty intermediate-history rebase therefore identify each actual lower cosupport point's selected lower value with that of the single reference lower presentation. The real lower mask prevents new original members from passing through its points, so their entire true boundary level is at most s. Original threshold comparison forces level at least s, hence equality and membership in the actual lower cosupport by the preceding dictionary. The common lower-value equality reduces tuple comparison exactly to the stated lower comparison. Conversely actual lower cosupport supplies original cosupport and the same value equality gives the original threshold. Top thresholds are included; junk values outside the actual lower mask are never substituted for real values. □

Theorem 192.3 (Only original history constructs honest generators for the true recursive lower ideal)**.** *At every point of the actual final lower open, the literal ideal of $L_{S|_U}$ has a genuine finite family of holomorphic generators on a true open neighborhood, with both directions of the membership/span dictionary at every nearby point.*

Proof. The real lower domain and original lower canonical history internally supply a genuine admissible lower tower. Original domain transport gives well-formedness of its actual final transform, including coherence of that literal original lower ideal. Its actual local coherence constructs the stated true open, actual finite holomorphic functions and both germ membership directions. The ideal is the original member/contact lower construction along its genuine history throughout. □

Theorem 192.4 (The genuine recursive lower input supplies the next positive analytic threshold family)**.** *For input induction level $k+2$, on the ENTIRE actual nonspecial reference open the SAME original Step One thresholds*

$$(s, \alpha_{a,b}), \qquad a \geq 1, \qquad \alpha_{a,b} = (a, \mathrm{inr}(b, b_k))$$

are genuinely complex analytic. The level-$k{+}1$ marked-domain input and its true canonical history needed by the global positive-count producer are internally constructed actual lower objects.

Proof. Apply the proved entire lower-threshold dictionary with the genuine finite lift of $\alpha_{a,b}$. The actual lower domain is level $k+1$, and its real canonical tail is for the SAME constructed original lower invariant. The preceding global positive companion-count theorem applies to these real lower objects with no new analytic premise. Its actual holomorphic equations present exactly the right set of that dictionary; finite-lift comparison gives precisely the original Step One threshold on the whole reference open. This includes both member and contact lower branches and retains the arbitrary genuine original history. □

Theorem 192.5 (Only original full-history inputs construct the next recursive literal analytic reference neighborhood)**.** *For original level-$k+2$ marked-domain input, original canonical history and a positive original cosupport reference point, internally lift that point to the true fixed companion's reference open. If this actual companion point is nonspecial of level s, the actual recursive member/ contact lower open contains it and the inverse image of the literal original full threshold*

$$(q, \mathrm{inr}(s, \alpha_{a,b})), \qquad a \geq 1,$$

is complex analytic on that ENTIRE true recursive open. No companion, lower input, history or lower-analyticity certificate is a caller premise.

Proof. The original fixed-companion producer supplies true reference-point membership, actual maximal-order companion domain, canonical masked history and its entire original cosupport dictionary. The canonical reference point lies in its actual cosupport because its original order is q. Thus the genuine nonspecial lower producer internally constructs the actual member/contact lower open and history. Apply the preceding actual next-positive Step One analytic theorem to

that SAME companion and true reference point. The fixed-companion whole threshold dictionary identifies the entire original full threshold on its first open with this literal Step One threshold. Taking its true inverse image along the actual recursive open inclusion identifies the original full threshold along the composed genuine inclusions with the analytic set just constructed. All maps and marked ideals are the original actual objects. This is a real next recursive analytic producer on a genuine reference neighborhood; whole arbitrary-lower global closure and special/higher-order extension remain open. □

193 Independent actual smooth source contact choices on every original test

The smooth lower diagram uses the transported ORIGINAL target contact function. Compare it with the independently selected actual source function, as required by the local-to-global coefficient comparison in [5, Sections 5 and 7]. The homogenized method in that source's discussion of Włodarczyk applies with empty original boundary. Accordingly the comparison below concerns the ACTUAL kept-position remainders: original boundary positions become EMPTY, and every actual new exceptional member is retained. It does not assert that contact gluing preserves the original nonempty boundary members.

Theorem 193.1 (Two actual tangent directions compare whole kept-position remainders on every ORIGINAL test)**.** *Let K be an actual well formed marked ideal with positive mark, and let x belong to its ORIGINAL cosupport. For two genuine tangent directions u, v on actual opens U, V containing x, ONE true contact gluing G has both its domain and codomain inside $U \cap V$ and contains x in both. For EVERY genuine ORIGINAL test tower T of K, the real gluing lifts to an actual biholomorphism of the inverse-image opens. Its WHOLE marked pullback transports the actual kept-position contact remainder for u to that for v, and it fixes every actual transformed ORIGINAL cosupport point. No coefficient test, canonical history or desired remainder equality is an input.*

Proof. The genuine homogenized contact construction internally supplies one G and its two inverse neighborhoods before the tower is chosen. Original well formedness supplies closed smooth carrier and coherent ideal; positive mark supplies the remaining premise. For any ORIGINAL test T, the proved coefficient-homogenized equivalence internally turns THIS SAME actual tower into a coefficient-homogenized test. Dropping the original boundary members then gives the genuine empty-original-boundary contact test required by the actual contact lifting theorem. That theorem constructs the real lifted biholomorphism, all actual inverse identities, whole empty-boundary contact record transport and transport of the actual remainder padded by ANY number of empty historical positions.

Choose exactly the original number $\#K.E$ of positions. The independently proved literal kept-history dictionary identifies this padded record with the ACTUAL contact remainder of K in all four fields, on both real opens. Substitute these two whole identities into the just-produced padded equality. The real coefficient equivalence also identifies the transformed ORIGINAL cosupport with that of the empty-original-boundary homogenized test. Hence the actual point-fixing property applies to EVERY transformed original cosupport point. No equality of the undeleted original boundary is needed. □

Theorem 193.2 (ONLY original smooth inputs compare the independently selected source contact with the transported target contact)**.** *Let M be an ORIGINAL maximal-order invariant-domain marked ideal, $f : Y_m \to X_n$ a genuine holomorphic submersion, and $x \in Y$ an actual point with $f(x) \in \operatorname{cosupp} M$. Put $K = f^*M$ using the actual whole marked pullback. ONLY these original inputs internally supply the actual target chosen tangent open/function at $f(x)$, its true inverse-image source tangent open/function, and the independently selected source tangent open/function for the SAME K at x. Both actual source directions are genuine and contain x.*

ONE true gluing between these actual source choices is produced BEFORE EVERY genuine ORIGINAL source test T of K. On each such SAME tower it lifts to a real biholomorphism identifying the ENTIRE actual kept-position remainder records and fixing the actual transformed source cosupport. Neither a source domain, tangent certificate, contact open, source test construction nor a desired choice-comparison certificate is supplied. Canonicality and surjectivity of f are absent.

Proof. Actual domain pullback along the genuine submersion first constructs the source invariant domain, with the true fibre-dimension shift. Thus source well formedness, coherent ideal, smooth carrier and positive mark are internally available. The true whole smooth cosupport dictionary sends the assumed ORIGINAL target cosupport membership to actual source cosupport membership. True maximal-order preservation gives maximal order of the SAME actual source ideal and carrier.

Apply the proved original tangent existence theorem at the target point and, independently, to this ACTUAL source domain and source point. The specification of each original choice gives its real open neighborhood and genuine tangent direction. Genuine smooth tangent preservation makes the transported target function a genuine tangent direction on the actual inverse-image open. Both source opens contain the original source point. Apply the first theorem to these independently verified source directions and source original data. Its gluing is chosen before any source test; therefore its universal ORIGINAL-test statement applies with every actual map, centre and empty position retained. This proves the asserted whole record equality and fixation, without replacing the independent source choice by the transported one.

Actual future lower tower isomorphisms and all-prefix lower records are the next consumer. Equality of earlier frozen FULL original lower invariants, exceptional or infinitesimal transformed semicoherence and the general preparation driver remain OPEN; the whole raw old-boundary contact record is not asserted to be preserved by this gluing. □

194 Independent actual smooth contact choices compare whole original lower histories

Construct the real independent source choice and the transported ORIGINAL target choice before the future upper history. Then apply genuine tower and selected-complement transport, as required by the original smooth locality in [5, Sections 5 and 7].

Theorem 194.1 (ONLY original smooth inputs internally compare both independently chosen contact lower tests at every head of every ORIGINAL source history)**.** *ONLY an ORIGINAL maximal-order invariant domain M, genuine submersion $f : Y_m \to X_n$ and actual source point x mapping into its ORIGINAL cosupport internally construct the actual whole source record $K = f^*M$, the genuine transported target tangent and the independently selected source tangent. Both true source opens and their actual tangent geometry are internally verified. ONE contact gluing is chosen BEFORE EVERY genuine ORIGINAL split source upper test T_0T_1 of the SAME K.*

For EVERY such actual history, the actual once-restricted source tails have true stage/endpoint biholomorphisms and a genuine tower isomorphism. The original kept-position contact remainders are the actual remainders of the true empty-position padded parents, in ALL FOUR fields. On their actual selected-complement opens, the LITERAL twice-restricted source tails are genuine lower TESTS with real tower isomorphisms. The ENTIRE controlled lower records are genuine endpoint pullbacks and genuine pullbacks at EVERY head, with compatible actual head/tail isomorphisms. The endpoint map fixes every actual transformed ORIGINAL source cosupport point. Neither source choice, source domain, lower history, canonicality, desired record comparison nor boundary-preserving certificate is an input.

Proof. The original-input independent-choice producer first constructs the actual whole source pullback, its true domain and actual cosupport membership. Actual smooth maximal-order preservation and true original tangent existence independently verify BOTH choices. In particular the source function is its own original tangent choice, rather than a renaming of the transported target function. The actual inverse-image tangent open and source chosen open both contain x.

Independently re-use the genuine whole smooth cosupport dictionary to obtain actual source cosupport membership from the original target membership; actual domain pullback internally gives source well formedness and positive mark. Apply the all-ORIGINAL-test contact lower-history producer to THESE actual verified directions and actual source record. It constructs ONE gluing before its universal split-history quantifier. Its actual empty- boundary contact TEST and real empty-position padding internally give the padded TEST before the selected-complement restrictions. Thus its two lower TESTS use precisely the literal restrictions of the given source upper history, including every actual map, centre and empty position.

The same theorem internally gives the actual lower tower isomorphism, ENTIRE endpoint controlled-record equality and ENTIRE equality at EVERY head, with the compatible actual tail isomorphisms. The independently constructed original cosupport dictionary gives its endpoint fixation on the ORIGINAL source transformed cosupport. These are constructed lower histories, not hypotheses inferred from the desired whole comparison.

This removes independent source contact choice as an obstruction to the actual lower RECORD and TEST comparison under smooth pullback. It does not identify undeleted raw original boundary members, earlier frozen FULL original invariant histories or actual higher recursive invariant values. Exceptional and infinitesimal transformed semicoherence and the general noncanonical preparation driver remain OPEN. □

195 A genuine lower contact presentation on the actual selected-component complement

Keep an ORIGINAL maximal-order domain of dimension bound $k+2$, ANY genuine ORIGINAL test prefix, and an actual point of its current cosupport. The preceding uniform contact producer internally fixes the actual tangent open, tangent function and coefficient contact model L. The fixed open G is the complement of the genuine union of components of $L.N$ contained in its actual cosupport. This is the remaining-component part of [5, Step I, Case A], retaining literal empty original boundary positions and the true exceptional suffix. No assertion that a selected-component point lies in G is made.

Theorem 195.1 (Original inputs internally construct the true nonzero lower contact model)**.** *ONLY the original inputs above internally choose the actual U, w, L and its true closed smooth carrier proof. Let G be the actual selected-component complement and let A be the true open restriction to G of the remaining contact record, with exactly the original number of empty historical boundary positions and the entire actual exceptional suffix. Then A is genuinely in invariant domain $k+1$. For EVERY actual $p \in L.N$, $p \in G$ iff the actual coefficient ideal is nonzero along $L.N$ at p. No actual tangent/open/model, lower-domain, nonzero, boundary-transversality or component-selection certificate is supplied.*

Proof. Apply the original-input uniform contact producer. Its fixed tangent choice follows genuine tangent existence; its real contact geometry gives closed smooth carrier, coherent ideal, positive mark and one lower regular dimension. The actual smooth component decomposition separates the closed selected and remaining unions. A carrier point is in the selected union precisely when the coherent ideal vanishes along its carrier germ: in one direction finite generation gives actual local vanishing, and in the other analytic continuation on the connected smooth component places the entire component in the actual cosupport. Taking complements gives the asserted pointwise equivalence. It also shows why selected points cannot silently be retained in the nonzero model.

Apply the genuine contact remainder domain theorem to the actual prefix and the reflexive future tower. Its proof uses the selected/remaining closed partition to transport actual strict carrier germs, proves nonzero and coherence by true controlled-colon induction, and uses the actual exceptional suffix transverse to the contact carrier. Exactly the original number of historical entries is replaced by empty entries; none of the actual exceptional entries is removed. Thus the actual remainder is in domain $k+1$. Genuine restriction of all domain fields to the actual open G gives the asserted domain of A. No transversality of the contact carrier to the original retained members is inferred. □

Theorem 195.2 (ONE internally constructed nonzero model supplies EVERY genuine restricted future test and whole controlled record)**.** *The SAME U, w, L, G, A are chosen BEFORE every genuine future ORIGINAL extension T_0Q. For EVERY such extension, the SAME nested actual restriction $Q|_{T_0^{-1}U}|_G$ is a genuine test tower of A. Its actual transform remains in domain $k+1$ and equals, in ALL FOUR marked fields, the true open restriction of the transformed remainder. Its carrier, ideal and mark separately agree with the true restriction of the transformed full coefficient contact model L. Consequently its cosupport is EXACTLY the pullback of the SAME ORIGINAL transformed cosupport by the actual nested inclusion. No model selected separately for Q, desired test, whole-record or cosupport comparison is supplied; canonicality is absent.*

Proof. Keep the actual choices of the first producer. The preceding uniform contact theorem internally makes each genuine restricted ORIGINAL future a test of L, with BOTH cosupport directions. The number of old positions is no greater than the length of its actual boundary: genuine boundary recursion appends each true exceptional entry, and restriction preserves length. Apply the true selected-complement restricted-test theorem to THESE objects. On G the entire remaining carrier equals the entire contact carrier, because selected membership was independently removed. The ideal and mark are unchanged. Dropping the old boundary prefix weakens normal-crossings admissibility, and reinserting the same number of literal empty entries preserves it; strict transforms of these entries stay empty. The SAME actually restricted maps therefore construct a genuine test tower, not a renamed unrestricted or clipped map.

The real open controlled-transform theorem gives the whole four-field restriction identity with the transformed remainder. The independent boundary dropping and empty insertion identities leave the carrier, ideal and mark equal to those of the restricted full model. These three literal field equalities give equality of cosupports. Real open cosupport pullback, followed by the already proved ORIGINAL/contact cosupport equality, gives the displayed BOTH-direction dictionary under the actual composite of nested inclusions. Finally apply true invariant-domain preservation to the actual initial domain of A and the internally produced genuine test tower. This supplies all future lower domains without a caller certificate.

The actual original test extension is essential; no assertion is made that all local tests extend globally. The model covers precisely the nonzero contact-component open. Selected-component branches, comparison with the earlier frozen FULL original lower history, original lower canonicality, complete exceptional/infinitesimal semicoherence, decrease and termination remain separate. □

196 A fixed actual coefficient contact model for every genuine original test extension

Retain the SAME original maximal-order marked ideal and ANY of its genuine test prefixes. One actual point of its current cosupport internally fixes an original tangent open and function. Keep that choice while comparing EVERY genuine original extension. This is the geometric coefficient-model part of [5, Step I, Cases A and B], adapted to the actual globally extended test histories rather than a claim about all local histories which need not extend. It is a concrete presentation ingredient in the sense discussed at printed page 222 of [4].

The actual contact carrier is not assumed transverse to the entire retained boundary, and the coefficient ideal can vanish identically on a contact component. Consequently no lower invariant domain is asserted before the separate member/component selection and nonzero analysis. In particular this does not silently identify an arbitrary noncanonical prefix with an earlier frozen canonical lower history.

Theorem 196.1 (ONLY original maximal-order domain and test-prefix data construct the actual chosen contact geometry)**.** *Let M be an ORIGINAL maximal-order invariant-domain input of dimension bound $k+1$, let T_0 be ANY actual ORIGINAL test prefix and let y lie in its current cosupport. These original inputs internally construct the actual chosen original tangent open $U \ni T_0(y)$ and its chosen tangent function w. Set $V = T_0^{-1}U$ and let L be the actual kept-boundary coefficient model on the strict transform of $M.N \cap \{w = 0\}$, restricted to V. Its carrier is genuinely closed smooth of dimension bound k, its literal coefficient ideal is coherent, its mark is positive, and its ENTIRE ordered boundary equals the actual current ORIGINAL boundary restricted to V. Its cosupport is EXACTLY the trace on V of the SAME original current cosupport. No tangent/open/model/geometry/coherence/test or cosupport dictionary certificate is supplied. Canonicality is absent.*

Proof. Genuine homogenized coefficient equivalence internally carries the given ORIGINAL test prefix to the actual homogenized coefficient test prefix, with equal current cosupports. Thus the given actual point is a real point of that actual coefficient cosupport. Original maximal order and coherent smooth geometry produce a tangent direction at its actual original projected point. Apply the actual fixed tangent choice only AFTER that existence proof; it supplies the real open, point membership and genuine tangent function.

Actual maximal-contact strict-transform geometry gives the true closed smooth contact carrier on V. Its dimension bound drops from $k+1$ to k by the genuine contact hyperplane dimension theorem, not by an input rank certificate. Coherence of the actual coefficient ideal at the stage, followed by true open pullback, gives coherence of the literal model ideal. The actual coefficient mark is positive. Coefficient formation and carrier replacement keep the boundary; original and homogenized coefficient records started with the SAME list, so their actual strict-boundary recursions give equality of the ENTIRE ordered lists before restriction. No position or empty member is deleted.

Use the genuine stage contact-tail theorem with the reflexive future extension. Actual Giraud cosupport containment in the strict contact carrier is proved for all actual original extensions, and the real coefficient restriction theorem supplies BOTH cosupport directions. Transfer through the already proved original/homogenized coefficient cosupport equality. This gives precisely the asserted current trace identity, so the chosen original point really lies in the model trace. It does not supply nonzero restriction on every contact component or transversality to every retained member. □

Theorem 196.2 (ONE internally chosen original contact model works for EVERY genuine noncanonical original extension)**.** *For ONLY the preceding original inputs, ONE such actual U, w, L also has the following property: for EVERY genuine ORIGINAL extension T_0Q, the SAME actual restricted future tower $Q|_V$ is a genuine test history of this fixed coefficient model L, and its actual transformed cosupport is EXACTLY the trace of the SAME ORIGINAL cosupport after T_0Q. The original prefix and extension need not be canonical. No choice of a model depending on Q, desired model test, overlap or transformed cosupport certificate is supplied.*

Proof. Keep the actual open and tangent function already constructed from the original prefix point; they are chosen BEFORE the future tower is given. Let an arbitrary genuine ORIGINAL extension T_0Q be given. The actual homogenized coefficient equivalence internally supplies its genuine coefficient test history and equal cosupport, preserving the SAME maps and every original position. Apply the proved contact-stage tail theorem to THESE actual data, with the already fixed U, w and $V = T_0^{-1}U$. Its Giraud argument supplies actual cosupport containment

on the genuine strict contact carrier at every future position. The real relative coefficient restriction theorem therefore constructs model-centre admissibility and the SAME restricted tower test internally, and gives BOTH whole cosupport directions. Substitute the independently proved original/coefficient cosupport equality. All actual future maps, restrictions and ordered boundary positions remain unchanged.

The conclusion quantifies over genuinely extended ORIGINAL histories; it does not claim that every tower over the local open extends globally, or infer an unrestricted local test-equivalence from that false premise. Lower-domain nonzero and boundary-transversality selection, comparison with earlier frozen full ORIGINAL lower histories, complete noncanonical recursive presentation equivalence, exceptional/infinitesimal semicoherence, invariant decrease and general termination remain separate producers. □

197 Genuine curve Step One leaves for an independently formed same-carrier candidate

The companion in [5, Section 5, Step II.B] is a genuinely different marked ideal. An ORIGINAL test of the source record need not be its global TEST. Use actual same-carrier coherence, actual open geometric identities and a genuinely empty local history to compute its special leaf, without supplying candidate-testness or using canonical full-value locality.

Lemma 197.1 (Actual same-carrier coefficient geometry restricts without candidate-testness)**.** *Let T be ANY genuine TEST of a source marked record M. Let J be ANY wellformed marked ideal of positive mark with the SAME actual initial carrier as M. On ANY true base open, its ACTUAL coefficient transform cosupport, original ordered level and special-point predicate restrict exactly along the SAME restricted tower. No TEST of J or its coefficient record, canonicality or value comparison is an input.*

Proof. The original source TEST puts every centre in the true transformed source carrier. Actual equality of initial carriers and the genuine same-carrier coherence theorem give the candidate coefficient transform a closed smooth carrier and a coherent ideal at every actual prefix, without candidate admissibility. The genuine entire same-carrier open identity, preceded by the actual coefficient/open identity at the initial stage, therefore gives its WHOLE coefficient marked record as the pullback on the endpoint open. Its actual cosupport is consequently the true inverse image.

The original members are the first $|J.E|$ actual coefficient boundary positions. Whole-record equality, the literal unchanged initial list length and the true finite-list map/take identity give EXACTLY the preimage of that entire ordered prefix. The real indexed boundary count then gives equality of the levels. The actual codimension-one cosupport part restricts because every member germ of the independently coherent current ideal is truly holomorphic; the actual member-intersection locus restricts by the genuine open neighborhood equivalence. Substitution in BOTH defining special branches gives the asserted special-point biconditional. No comparison of noncanonical higher invariant values has been used. □

Lemma 197.2 (The genuine candidate cosupport transports off the source's entire accumulated boundary)**.** *For ANY actual source TEST and ANY coherent candidate on the SAME initial closed smooth carrier, membership in its ACTUAL current cosupport at a point outside the source's ENTIRE accumulated boundary is equivalent to membership of its actual original image in the initial candidate cosupport. No candidate TEST or desired cosupport comparison is supplied.*

Proof. Induct on the SAME actual tower. At a last blowup, exclusion from the whole new boundary excludes both the actual last centre and the entire preceding boundary. Actual same-carrier coherence supplies a closed smooth candidate carrier and coherent current candidate ideal at the preceding prefix. The true ONE-step off-centre controlled-cosupport dictionary therefore

applies to that actual record; it requires holomorphic member germs and the actual blowup, not candidate admissibility. Compose this biconditional with the induction hypothesis at the preceding image. The reflexive history is literally the identity. This proves both directions along every original position and empty step. □

Theorem 197.3 (ONLY original same-carrier curve data internally compute the actual non-test candidate Step One leaf)**.** *ONLY a genuine source wellformed record M, ANY original source TEST, an actual independently formed dimension-bound-one maximal-order domain J with the SAME actual initial carrier and ordered boundary, and an actual point outside the source's ENTIRE current boundary whose original image belongs to J's initial cosupport imply the following: the SAME point is in the actual current candidate cosupport, is genuinely special for the ACTUAL candidate history, and its unchanged ACTUAL Step One value is exactly $(0, \top)$. No global candidate TEST, chosen local window, special-point certificate, canonicality or desired value equality is input.*

Proof. Actual initial coefficient equivalence transfers the genuine initial candidate-cosupport membership to its ACTUAL coefficient record. The preceding off-boundary dictionary transports it to the actual current coefficient cosupport, without any candidate-testness premise. The true proper-boundary-image theorem internally constructs ONE original base open through this point's actual image where the SAME entire restricted tower has only empty centres.

Every empty-centre tower is genuinely a TEST of the actual restricted candidate. The real open domain and maximal-order restriction theorems supply its original dimension-one maximal-order hypotheses. The preceding entire open coefficient identity supplies the actual coefficient cosupport membership on this window; genuine coefficient equivalence on this actual empty TEST gives its original candidate cosupport membership. The already proved ANY-test dimension-one Step One theorem therefore makes it genuinely special on this window. The first lemma transports this actual geometric special property to the ORIGINAL candidate history, without transporting any higher value.

Finally the actual candidate coefficient boundary is the SAME source boundary, since it has the SAME initial ordered family and actual maps. The point meets none of it, so none of its original prefix members pass through it; the actual indexed level is zero. The true special property and this actual zero level evaluate the unchanged actual Step One definition as $(0, \top)$. The second lemma also supplies the asserted actual current candidate-cosupport membership. This computes a genuine non-test candidate leaf. Applying it to the original GENERAL order-run companion and deriving that companion's initial membership internally are separate downstream obligations; no complete higher noncanonical value, exceptional semicoherence or driver claim is made. □

198 The actual selected zero branches and the complete contact cosupport

The nonzero lower contact model must not silently discard actual components on which the coefficient ideal is identically zero. Retain their genuine strict transforms and reconstruct the WHOLE cosupport from those zero branches and the true nonzero model. This is the geometric zero/nonzero component split in [5, Step I, Case A], for actual ORIGINAL histories; no full higher-invariant comparison is inferred.

Theorem 198.1 (True strict transforms of an actual initial cosupport subset stay in the actual cosupport)**.** *For ANY genuinely closed smooth marked carrier, coherent ideal and actual initial subset S of its cosupport, EVERY genuine test tower sends the TRUE geometric strict transform of S into the actual transformed cosupport. No nonzero, principal, invariant-domain, selected-component or persistence certificate is a premise.*

Proof. Induct over the actual maps. Initially the assertion is the given subset inclusion. At a genuine next blowup the actual previous and next carriers are closed smooth and their true ideals coherent. The actual next cosupport is therefore closed. Off the actual centre, the proved BOTH cosupport membership dictionary carries every point lying over the previous strict transform of S into the new cosupport. The real new geometric strict transform is the closure of precisely that off-centre preimage. Closedness therefore contains its closure. This proves the induction step for the actual maps and every position, including empty centres. □

Theorem 198.2 (The WHOLE actual contact cosupport is its selected strict branch union the true nonzero model image)**.** *Let L have a genuine closed smooth carrier and coherent ideal, and let S be its ACTUAL selected component union. Let $G = S^{c}$, and let A be the TRUE restriction to G of its remaining contact record, with any permitted number of empty historical boundary entries. For EVERY genuine test tower P of L, the WHOLE actual transformed cosupport equals*

$$\operatorname{strict}_P(S) \ \cup \ \iota\big(\operatorname{cosupp}(A_{P|_G})\big),$$

where $\iota : P^{-1}G \hookrightarrow Y$ is the actual inclusion. Both sets on the right are independently defined actual objects. No desired cosupport decomposition, lower test or transformation identity is supplied.

Proof. Each actual selected component lies in the original cosupport by its literal definition, so the first theorem puts its WHOLE strict transform in the new cosupport. Independently, the genuine selected-complement restricted-test theorem internally constructs $P|_G$ as a test of A; its carrier, ideal and mark are the true restrictions of the full transformed L. Thus BOTH cosupport directions give exactly the trace of the latter cosupport under ι.

The selected and remaining unions are disjoint closed smooth sets and partition $L.N$. True strict-transform geometry of a closed partition says that the selected strict transform is precisely the strict transform of $L.N$ intersected with the actual inverse image of S. A point of the full cosupport lies in that strict carrier. If its actual projection lies in S, it is therefore in the selected strict branch; otherwise it is a real point of $P^{-1}G$, and the independently proved trace dictionary places it in the image of the lower cosupport. Conversely selected strict points belong by the first theorem and lower-image points by the true trace dictionary. This proves equality of the entire sets, not merely pointwise equality near an originally chosen point. □

Theorem 198.3 (ONLY original noncanonical inputs construct a fixed complete future contact cosupport presentation)**.** *ONLY the original maximal-order domain of dimension bound $k+2$, ANY genuine ORIGINAL test prefix and actual current cosupport point internally choose ONE actual U, w, L, G, A BEFORE all genuine ORIGINAL future extensions. The true nonzero model A has lower domain $k+1$ and its SAME actual nested restricted future is a genuine test with true future domain. On the entire inverse image of U, the SAME ORIGINAL transformed cosupport is EXACTLY the union of the actual selected strict branch of L and the inclusion image of this actual nonzero lower cosupport. Canonicality and desired lower-domain/test/whole-cosupport certificates are absent.*

Proof. Use the actual original-input nonzero presentation producer, keeping all choices BEFORE the future tower. For each genuine ORIGINAL extension, actual coefficient equivalence and the contact-tail theorem internally supply the full L test and BOTH ORIGINAL/contact cosupport directions. Genuine boundary recursion gives the old-prefix length inequality. Apply the WHOLE decomposition theorem to these independently constructed objects and substitute the already proved ORIGINAL/contact cosupport identity. The original-input producer independently supplies the actual lower test and its future domain. Thus both actual zero and nonzero branches are covered without pretending the chosen point lies in the nonzero open, and all original positions and actual maps are retained.

This is a full geometric coefficient-cosupport presentation over genuinely extended ORIGINAL tests. It does not identify earlier frozen full lower histories, infer original lower canonicality, cover unrestricted local tests which fail to extend, or close exceptional/infinitesimal semicoherence, higher invariant decrease or general termination. □

199 The true masked Step One lower starting-year bounds are globally analytic

The literal original Step One regime is boundary level s and nonspecialness. Its actual trailing run selects the genuine member/contact lower starting year at nonspecial level-s points. We retain that original regime and run throughout. This follows the historical Step I convention of Bierstone–Milman[5], Section 7. Its reset occurs at a strictly higher previous level or at a special previous point of the same level; the latter is a real analytic special-top threshold, so it must be retained in the history equations.

Write $A_s(T) = \operatorname{cosupp} J_T \cap \{\operatorname{level}_T \geq s\}$ and

$$Z_s(T) = \{y \in \operatorname{cosupp} J_T : s < \operatorname{level}_T(y) \text{ or } (s = \operatorname{level}_T(y) \text{ and } y \text{ is special})\}.$$

The enhanced history set is $L_{s,r}(T) = A_s(T) \cap \{\operatorname{regimeRun}(\operatorname{stepRegime}(J, s), T) \leq r\}$. Its starting-year interpretation applies on the actual nonspecial level-s stratum; the whole analytic set retains the stated higher-level and special current baseline without claiming a lower value is evaluated there.

Lemma 199.1 (Actual coefficient, boundary and special equations construct both whole analytic ingredients)**.** *For original maximal-order marked-domain input of arbitrary induction level and its literal original canonical masked Step One history, both $A_s(T)$ and $Z_s(T)$ are globally complex analytic on the entire actual final manifold.*

Proof. Original canonical-domain transport supplies the genuine original ideal coherence and closed smooth carrier, whose actual cosupport is analytic. The original Step One bridge supplies the actual coefficient test history. Its genuine SNC boundary and indexed original-member intersection equations make the whole actual level threshold analytic. Intersect these two actual sets to obtain $A_s(T)$.

The original Step One bridge also identifies the whole coefficient cosupport with the same actual J_T cosupport. The true masked Step One special-top dictionary identifies $(s, \top)$ comparison exactly with a strictly larger level or the actual special point at equal level. Thus $Z_s(T)$ is exactly the existing actual special-top superlevel. Its genuine coherent coefficient, member-incidence and codimension-one equations give analyticity. No special predicate or analytic certificate is supplied. □

Lemma 199.2 (Every genuine original canonical successor inherits cosupport and cannot raise boundary level)**.** *Along an actual original canonical successor $\pi : T' \to T$, every current cosupport point maps to old cosupport, and its actual boundary level is at most the old boundary level.*

Proof. The genuine original masked Step One drop theorem supplies inherited cosupport and strict full-value decrease over the true centre, and exact full-value equality off it, including real empty steps. Hence the actual full value never increases. Its first entry is literally the actual indexed boundary level. Lexicographic comparison therefore implies the stated level inequality. No count-monotonicity or stage certificate is assumed. □

Lemma 199.3 (The entire true lower history has exact zero and successor run equations)**.** *The ENTIRE actual successor identities are*

$$L_{s,0}(T') = A_s(T') \cap \pi^{-1} Z_s(T), \qquad L_{s,r+1}(T') = A_s(T') \cap \pi^{-1}(Z_s(T) \cup L_{s,r}(T)).$$

Proof. A current baseline point maps to old cosupport with old level at least s. If that old point belongs to the literal original nonspecial level-s regime, the original run grows by exactly one. Otherwise its old level is strictly larger than s, or its level equals s and it is special; these are exactly $Z_s(T)$, and the original run resets to zero.

For the zero identity, run at most zero excludes the increment alternative, so precisely the reset set is necessary and sufficient. For the successor identity, an increment has run at most $r+1$ exactly when its old run is at most r, retaining the inherited old baseline. A reset has zero run and satisfies every successor bound. Conversely an old Z_s point resets, while an old $L_{s,r}$ point either increments its bounded run or resets. These cases prove both directions of the entire identities and retain every actual special, nonspecial and empty chronological step. □

Theorem 199.4 (Only original inputs produce every genuine global lower-history year analytic threshold)**.** *Each literal $L_{s,r}(T)$ is globally complex analytic for original maximal-order marked-domain input and its true original canonical history, in every induction level without compactness or a curve envelope.*

Proof. At the zero-length original history the actual run is zero, so its entire set is the true analytic current baseline. Induct on the actual finite original tower. The old reset set is analytic by the genuine special-top producer, and the old history threshold is analytic by induction. The zero successor is the intersection of the current baseline with the actual holomorphic blowup inverse image of the old reset set. A positive successor is the same genuine pullback of the true finite union of reset and old history locus. The exact whole identities prove that these actual holomorphic equations present precisely the stated original run threshold, not merely an upper-semicontinuous closed locus. □

Corollary 199.5 (An original full positive history internally supplies the true companion lower-year producer)**.** *For only original full marked-domain input, its literal original canonical history and a positive original cosupport reference point, every actual lower-year threshold of its genuine fixed companion is analytic on the ENTIRE actual companion final open.*

Proof. The previously constructed true reference companion supplies its actual marked domain, maximal order and SAME original canonical masked Step One tail internally. Apply the preceding genuine global lower-year producer to those real objects. Its entire final manifold is the actual original reference open, so the resulting equations give the asserted entire companion lower-year locus without a chosen companion/domain/history or analyticity premise. Full arbitrary recursive lower thresholds, special/ higher-order extension and the simultaneous auxiliary Hilbert–Samuel chain remain subsequent producers. □

200 Genuine whole controlled smooth pullback without a lower TEST

Actual smooth base change of presentations in [5, Section 7] must retain the entire original controlled ideal, genuine restricted colon and original ordered boundary. A member lower reference can have real smooth/coherent current geometry along an original upper TEST without being a TEST itself. The existing admissible-step theorem uses candidate admissibility to produce the next smooth strict carrier. Here we prove the actual ideal identity from that genuine geometry directly, and then supply it internally from original member data. No lower TEST or current point membership is inferred.

Theorem 200.1 (Actual coherent ideal and genuine next strict carrier give the whole smooth controlled step)**.** *Let $f : Y_m \to X_n$ be a genuine holomorphic submersion, π an actual blowup at C, and ρ an actual blowup at $f^{-1}C$. For ANY actual marked record B, assume only that its real ideal is coherent and its ACTUAL target successor strict carrier is closed smooth. Then its*

ENTIRE actual source controlled transform is the genuine actual-lift pullback of its ENTIRE target controlled transform. No candidate admissibility, centre containment, cosupport membership, SNC or positive mark is required. Both ambient dimensions, every boundary position and mark zero are retained.

Proof. The genuine canonical blowup lift is internally holomorphic and a submersion, with the true whole projection square. Actual coherent pullback composition on that square identifies BOTH full numerator ideals, with their original complex coefficients. True whole centre vanishing-ideal pullback, followed by actual ideal-power pullback, identifies the full denominator including power zero.

The actual target strict carrier is smooth by the explicit genuine geometric hypothesis. Thus the established entire restricted-colon pullback theorem applies to the actual coherent numerator and actual coherent centre-power denominator; no divisibility or admissibility hypothesis is used. Its restriction carrier is identified with the source strict carrier by the genuine arbitrary-set strict-transform preimage theorem. These equalities prove the ENTIRE actual controlled ideal equality, in BOTH germ-membership directions, not merely equality of zero sets. The same strict preimage theorem gives carrier equality. The actual ordered boundary preimage dictionary transports each old strict entry and appends precisely the same new exceptional inverse image, retaining empty and repeated entries. The mark is literally unchanged. All four marked fields therefore agree. □

Theorem 200.2 (Real reference-stage geometry gives whole controlled and cosupport dictionaries along any original upper Cartesian history)**.** *The auxiliary transport theorem takes a genuine original upper record M, a reference record K, and an explicit real geometric family: K_R is genuinely well formed at EVERY original upper M TEST R. For ANY actual Cartesian source/target history of an original upper TEST, prove ENTIRE K controlled-record pullback equality and ENTIRE cosupport preimage equality. No TEST of K or its pullback is assumed or concluded. The final original-only consumers below supply this geometric family internally; it is not a desired naturality or history certificate.*

Proof. Induct on the actual Cartesian history constructor. At the reflexive history both whole records are the original literal pullback record. At a step, the real upper TEST supplies its actual shorter upper TEST, and induction gives the whole shorter reference-record identity. The independently given real reference geometry supplies coherence BEFORE the step and the actual next strict carrier smoothness AFTER the step. Apply the first theorem to the SAME real target/source blowups and canonical lift, then substitute the shorter whole-record identity. This proves all four endpoint fields without claiming source or target reference admissibility. The true coherent target endpoint and genuine endpoint submersion then give BOTH directions of the ENTIRE actual cosupport preimage dictionary, using the independently defined germ ideal pullback. No point is asserted to belong to either cosupport. □

Theorem 200.3 (ONLY original bound-two member geometry gives untested whole controlled smooth naturality)**.** *ONLY an original bound-two domain, ANY genuine nonempty subset of its actual ordered member positions, ANY upper original TEST and an actual Cartesian pullback history internally supply the whole controlled pullback and BOTH cosupport dictionaries for its ACTUAL member remainder. No lower TEST, later domain/coherence or desired record comparison is supplied. Source and target ambient dimensions may differ.*

Proof. The genuine member-remainder theorem internally constructs its ENTIRE bound-one domain at EVERY genuine original upper TEST, including smooth strict carrier, actual ordered SNC/transversality and coherent nonzero controlled ideal. Forget only the extra domain fields to supply precisely the real reference well-formedness family needed in the preceding transport theorem. Apply that theorem to the SAME actual Cartesian history and original upper TEST. Every smoothness and coherence assertion used by the ideal recursion is therefore derived from the original member geometry before the whole comparison is proved. Source curve dimension, canonicality and lower admissibility are not inferred. □

Theorem 200.4 (Actual through-point member remainders have all-four-field smooth initial naturality)**.** *For genuine original well-formed positive-mark data, its actual original member sublist and ANY actual source point, the WHOLE smooth pullback of the target through-point member remainder equals the independently constructed SOURCE through-point member remainder. Both actual ambient dimensions are positive. No global component bijection, global preimage injectivity, lower TEST or desired component identity is supplied.*

Proof. The proved actual member-remainder/component-containment identity first expresses the target remainder as the genuine zero-padding contact remainder of its ACTUAL coefficient member restriction. True member restriction smooth naturality internally proves the entire source coefficient-member record: through-point original-member reflection uses actual SNC germ separation, and ordered filtering retains every position. Genuine whole contact-remainder smooth naturality uses the actual nonzero-germ and remaining-component preimage dictionaries, not a false bijection of global components. Substitute the independent whole member-restriction identity, and apply the actual source member-remainder/component-containment theorem. Source well formedness is internally provided by genuine original smooth pullback. Thus all four independent source fields agree, including the mark and literal ordered complement boundary. □

Theorem 200.5 (ONLY original through-member inputs supply independent source records and every literal controlled head)**.** *ONLY original bound-two data, an actual original member sublist, a real source base point with positive target through-member count, ANY original upper TEST and its actual Cartesian history internally prove WHOLE controlled-record and BOTH cosupport dictionaries for the INDEPENDENTLY constructed source and target through-member remainders. EVERY literal prefix has these same dictionaries under its genuine actual smooth endpoint map, including empty and beyond-length prefixes. No reference TEST, caller record identity, later geometry or desired map/cosupport membership certificate is assumed.*

Proof. The true original positive indexed through-count internally makes the actual target through-member subset nonempty. Apply the original-only member history theorem to construct the whole controlled dictionary. The preceding initial theorem independently identifies the source member-remainder constructor with the literal initial pullback; substitution proves the displayed record and cosupport equalities for those EXACT actual records.

For each literal prefix use the genuine actual Cartesian head producer and the true original upper TEST head. Repeat the same original member argument on that actual pair of prefixes, retaining the SAME original source base point, same original member positions and all empty entries. Thus every prefix map and every current reference assertion is supplied internally before its whole controlled and cosupport comparison.

This is genuine whole smooth controlled naturality without a false lower TEST. It does NOT supply actual lower-current cosupport membership of an arbitrary upper endpoint. The missing positive-run point-membership bridge, whole original higher-value comparison, exceptional/infinitesimal semicoherence and general unbounded global scheduling remain OPEN. □

201 Actual original positive-member GENERAL values without assuming a lower TEST

The genuine member tail of [5, Section 5, Step I.A] has actual smooth curve geometry at every original stage, even when original centres are not contained in the smaller member carrier. The preceding chapter proved original special leaves at actual coefficient cosupport points. We now construct the ACTUAL run companion point and its current coefficient cosupport, then compute the unchanged original GENERAL invariant. No lower TEST is inferred.

Lemma 201.1 (Real original-stage geometry transports actual candidate domains and off-boundary cosupports)**.** *The auxiliary geometric transport lemma takes a genuine original record m, a*

real reference curve record K whose ENTIRE current domains along original m TESTs are known, and an actual initial curve domain B with the same carrier and ordered boundary as K. It constructs ENTIRE current domains for B along those SAME original tests, without requiring a B TEST. These explicit reference geometry hypotheses are internally supplied in the original-only final theorem.

Proof. Induct on the actual original tower. Carrier and ordered boundary transform literally from the initial carrier and boundary, so the real reference-domain geometry supplies the actual smooth strict carrier, ordered SNC/transversality and bound one. Actual coherent pullback and relative colon supply the whole controlled ideal coherence. Genuine strict accumulation and the previous nonzero-along witness supply nonzero at EVERY new carrier point. The positive mark is unchanged. All four geometric/ideal domain fields are thus constructed without any candidate-centre containment assertion. The same induction, using the genuine off-centre whole cosupport dictionary at a point outside the actual ordered boundary, transports BOTH cosupport directions to the original point. It applies independently to the actual companion and its coefficient record. □

Theorem 201.2 (The ACTUAL run companion has genuine current coefficient cosupport without a lower TEST)**.** *For an actual current positive nonmonomial-order cosupport point of the true reference curve K along ANY original m TEST, internally construct its ORIGINAL order-regime run, head, genuine order-mask membership and ACTUAL original* `compAt`*. On the SAME restricted original tail its ORIGINAL Step One value is $(0, \infty)$ at EVERY recursive depth. The only auxiliary input is the stated genuine reference-geometry family; no candidate TEST, companion cosupport, run or desired value certificate is supplied.*

Proof. The real current domain makes its nonmonomial ideal a unit at every actual boundary point. Positive current nonmonomial order therefore excludes the ENTIRE current boundary. The literal regime-run recurrence supplies true start-regime membership. Its independently supplied actual head domain gives the genuine open order-mask membership from the existing order threshold dictionary.

Form the ACTUAL original companion on this mask. True coherent companion operations give its initial curve domain, unchanged carrier/boundary and actual maximal order. The genuine REFLEXIVE companion-cosupport dictionary, applied to the actual head domain itself, gives initial companion membership at the true projected point; it requires no TEST of the earlier reference record. Actual initial coefficient equivalence gives initial coefficient membership.

Use the first lemma and the independently given reference-stage geometry to construct whole companion AND coefficient domains on every original prefix of the real tail. Actual coherent open pullback/relative colon supplies BOTH entire record restriction identities, even without candidate admissibility. Genuine off-boundary cosupport transport, retaining each NEW actual exceptional position, then gives current coefficient membership at the SAME endpoint. The preceding chapter's real current-curve theorem now proves ORIGINAL specialness. No original member is incident because the endpoint avoids the ENTIRE real current boundary, so its actual level is zero. Only now unfold the unchanged original Step One definition, obtaining $(0, \infty)$ at EVERY depth. □

Theorem 201.3 (The unchanged whole GENERAL value on any genuine modeled curve history)**.** *Under precisely the explicit genuine reference-geometry family above, the UNCHANGED ORIGINAL full invariant at EVERY positive recursive depth on actual cosupport has its actual monomial value when the true current nonmonomial order is zero, and otherwise its original positive-order pair with the finite $(0, \infty)$ Step One value. No reference-record TEST or canonicality is assumed.*

Proof. Split on the literal existing current nonmonomial order. At zero use the UNCHANGED defining monomial branch and actual gamma. At positive order the preceding theorem internally supplies true mask membership and the ACTUAL computed run-companion Step One value.

Unfold only the unchanged GENERAL positive branch and use this actual membership to choose the genuine finite lower branch. Substitute the proved Step One value through the original WithTop and lexicographic maps. Thus every datum in the actual definition is constructed before evaluation; no locally chosen replacement invariant is used. □

Theorem 201.4 (ONLY original positive-member geometry supplies whole untested GENERAL values)**.** *ONLY an original bound-two domain, a genuine nonempty subset of its actual ordered member positions and ANY genuine original TEST internally supply the ACTUAL member-remainder record and all genuine geometric hypotheses above. At EVERY real point of its current positive cosupport and EVERY positive recursive depth, compute its unchanged ORIGINAL full GENERAL value by the preceding formula. No lower TEST, canonicality, stage-domain family or desired value certificate is supplied in this original-only consumer.*

Proof. The actual member-remainder constructor and genuine initial SNC/transversality supply its initial bound-one domain. The previously proved original member-tail geometry theorem applies to EVERY genuine original TEST and internally supplies the ENTIRE actual reference-domain family, including coherence and nonzero of its real controlled ideals and the unchanged ordered boundary. Apply the third theorem with precisely this actual member record and SAME original tower. The actual run, companion, prefix domains, coefficient cosupport and special leaf are therefore all constructed internally from the original data.

This is a whole unmasked member GENERAL producer without a false lower TEST. Comparison with the independently computed literal open member masks, whole upper GENERAL naturality, positive relative ambient dimension, higher-dimensional recursion and exceptional/infinitesimal semicoherence or general global scheduling remain distinct subsequent consumers. □

202 Actual current curve geometry supplies Step One special leaves without a lower TEST

The positive-member dimension induction in [5, Section 5, Step I.A] has a genuine scope issue after an arbitrary original TEST: the actual lower member carrier is smooth and its whole controlled ideals are coherent/nonzero, but the original centres need not lie in that smaller carrier. A lower TEST must therefore not be assumed. The next producer instead proves the required special leaf from the actual CURRENT coefficient curve geometry, and supplies that geometry internally along the real original member history.

Lemma 202.1 (A genuine bound-one cosupport point has actual carrier dimension one and cosupport dimension zero)**.** *For ANY actual marked bound-one domain and actual point of its positive cosupport, the real carrier is regular of dimension one at that point, and the real cosupport is regular of dimension zero there. No maximal order, history or regularity certificate is supplied.*

Proof. Actual coherence/nonzero and the positive mark make the genuine cosupport analytic and of dimension at most zero, so the proved analytic zero-dimensional regularity theorem gives its actual dimension zero. The original domain gives a real carrier dimension at most one at the point. If it were zero, its genuine isolating neighbourhood and actual nonzero-along witness would force some actual ideal germ to have nonzero value at that point. Positive cosupport forces EVERY actual ideal germ to vanish there, a contradiction. Thus the carrier dimension is one. □

Theorem 202.2 (Actual current coefficient curve geometry forces the original special branch on ANY history)**.** *Let A be ANY original marked record and T ANY actual tower. If its ACTUAL current coefficient record $\overline{A}_T$ is a genuine bound-one domain, then EVERY point of its actual positive cosupport is special for the ORIGINAL A, T. No TEST of A or $\overline{A}$, no canonicality, maximal-order or desired special-locus certificate is supplied.*

Proof. At such a point the first lemma gives actual carrier dimension one and actual cosupport dimension zero. If its original level is zero, these two genuine regularity witnesses are EXACTLY the defining original codimension-one-part membership, so it is special.

If the level is positive, its actual original member list is a prefix of the CURRENT ordered SNC boundary. The true finite indexed through-point set is nonempty. Actual current transversality and the genuine SNC member-intersection dimension theorem make the intersection of the real carrier with ALL these actual through-point members regular of dimension strictly below one, hence zero. Choose its genuine isolating neighbourhood. At every point there lying in the carrier and all original members through the reference point, the actual through-index/member-intersection dictionary gives membership in this isolated intersection, hence equality with the reference point. It therefore lies in the actual cosupport. This is BOTH fields of the ORIGINAL member-locus definition, proving specialness at positive level. Every member is the actual original indexed entry, not a distinct-branch replacement. □

Lemma 202.3 (Genuine original member geometry supplies entire curve domains for arbitrary same-geometry lower ideals)**.** *Let m be an actual original bound-two domain, S a genuine nonempty set of its actual ordered member positions, and B ANY genuine initial bound-one domain whose carrier and ordered boundary are EXACTLY those of the actual member remainder record. Along EVERY genuine original TEST of m, the ENTIRE actual transform of B is internally a true bound-one domain. No TEST or admissible centres for B, later coherence/nonzero or later domain certificate is supplied.*

Proof. The independently proved member-remainder geometric theorem supplies the actual strict carrier smoothness, whole ordered SNC boundary, transversality and dimension bound one at EVERY true original stage. Carrier and boundary transform depend only on their literal initial sets and ordered lists, so the assumed initial equalities transport these actual geometric assertions to B without assuming its centres are admissible.

Induct on the ACTUAL original tower to construct coherence and nonzero of the ENTIRE actual B ideal. Initially they are genuine initial domain fields. At a step use the real original-stage and next-stage smooth carrier geometry already supplied above. Genuine coherent pullback and relative colon produce the actual controlled ideal coherence even without an admissibility inclusion. Actual strict accumulation from the off-centre carrier and genuine nonzero there produce nonzero at EVERY true next carrier point. This proves the induction with the real controlled ideal and mark. Combine these independently constructed ideal assertions with the actual geometric fields and unchanged positive mark, obtaining every current domain field. This is not an assertion that the lower tower is a TEST. □

Theorem 202.4 (ONLY actual original positive-member history gives every untested coefficient curve special leaf)**.** *ONLY an original bound-two domain, ANY genuine split original TEST and an actual positive-level point at its head internally construct the ACTUAL original member-lower extension. At EVERY genuine prefix of the same original tail, internally prove true bound-one domains for its ENTIRE actual lower transform AND the ENTIRE actual transform of its original coefficient record. At EVERY point of that actual current coefficient cosupport, the ORIGINAL lower record is special along that SAME prefix and its unchanged sequence-aware Step One value is* (level, ∞) *at EVERY recursive depth. No lower TEST, canonicality, coefficient model or desired special/value certificate is supplied.*

Proof. Genuine original coefficient equivalence supplies the actual coefficient TEST and its real head/tail TEST decomposition. Original domain stability supplies the genuine head coefficient bound-two domain. The actual original head level is positive, so the finite through-point member subset in the ORIGINAL definition is internally nonempty. The actual member-remainder domain theorem therefore constructs the actual member-lower extension in bound one. Its genuine original coefficient-of-homogenization has the same ENTIRE carrier/boundary and is independently a bound-one domain, by true coherent operations and nonzero original summand.

At EVERY real tail prefix, the genuine original coefficient tail TEST supplies the independently proved member geometry. The preceding lemma applied to the actual original coefficient record, with its defining carrier/boundary equalities, internally constructs its WHOLE transformed bound-one domain. The actual lower domain is independently supplied by the genuine member-remainder transform theorem. Apply the second theorem to this ACTUAL current coefficient geometry and each actual cosupport point: it is special for the unchanged ORIGINAL lower record and same prefix. Only then unfold the original sequence-aware Step One definition, whose true special branch is precisely (level, ∞), independently of the lower recursive depth.

This supplies an actual previously missing NON-TEST lower-history special-value consumer. The companion current-cosupport/run dictionary needed for full positive-member GENERAL values is still a separate producer. No whole upper GENERAL, positive-relative-dimension smooth comparison, exceptional/infinitesimal semicoherence or general global scheduler is asserted. □

203 The literal original positive-member open mask has genuine full GENERAL values without a lower TEST

The preceding whole member-curve result must be consumed on the independently computed ORIGINAL open mask, rather than replacing that mask by a whole reference record. We first prove exact actual member-remainder open naturality, including its selected-component carrier and entire ordered complement family. This is the genuine local-presentation step in [5, Section 5, Step I.A].

Lemma 203.1 (Actual member remainder is the true remaining-component operation on its actual coefficient restriction)**.** *For genuine original well-formed marked data of positive mark and ANY true finite subset of ordered member positions, the ENTIRE actual member remainder record equals the actual zero-padding contact remainder of the actual member coefficient restriction. No desired carrier/component dictionary is input.*

Proof. The genuine coefficient member restriction has carrier $H = N \cap H_S$ and its actual cosupport is the original cosupport intersected with H_S. Every genuine global component of this smooth analytic H is contained in H. Consequently it is contained in the original cosupport exactly when it is contained in the actual member-restriction cosupport. The true unselected component subfamilies therefore coincide BEFORE taking their unions, proving ENTIRE carrier equality. Both actual records retain the same member-deleted ordered family, the actual coefficient ideal and its actual positive mark. At padding zero the literal list operation is the identity, giving equality of ALL FOUR marked fields, including empty original entries. □

Theorem 203.2 (WHOLE actual member remainder commutes with the real original open restriction)**.** *For ANY genuine original open U, actual point $p \in U$, and actual original member sublist, the WHOLE pullback of the actual through-p member remainder is EXACTLY the independently constructed through-p member remainder of the original record restricted to U. Both full ideals, the real remaining carrier, entire ordered complement boundary and mark are retained.*

Proof. The genuine whole member-restriction open dictionary, proved with actual coefficient ideal pullback, both member-intersection directions and true ordered list filtering, supplies equality of the actual coefficient member records on U. Apply the first lemma on both original manifolds. Genuine remaining-component open naturality is supplied by the existing actual coherent nonzero-germ dictionary, not by a false bijection of global components under an arbitrary open restriction. Its true zero-padding contact-remainder identity therefore transports ALL FOUR fields. Substitute the independently proved actual member-restriction record equality. No candidate TEST is involved. □

Theorem 203.3 (The genuine open member record has unchanged whole GENERAL values along every restricted original TEST)**.** *ONLY an original bound-two domain, actual nonempty through-point member positions, ANY genuine original open containing that actual point and ANY original TEST internally supply the independently constructed actual open member remainder. At EVERY actual current cosupport point and EVERY positive recursive depth compute its unchanged full original GENERAL value: its literal monomial branch at current nonmonomial order zero, otherwise its actual positive order paired with the finite* $(0, \infty)$ *special companion value. No lower TEST, canonicality, current model-domain family or desired value certificate is input.*

Proof. Original domain pullback constructs the genuine bound-two domain of the actual original record on U. The actual through-point cardinality and its true open member incidence dictionary prove that its actual finite indexed subset is nonempty. Actual original TEST open restriction gives the real restricted upper TEST. The second theorem identifies the literal open reference with the independently constructed member remainder on this actual original open. The preceding chapter's original-only theorem now internally supplies EVERY reference-domain, actual run, companion initial and current coefficient cosupport and original special leaf. Substitute the whole record equality to obtain the stated formula at the SAME original open endpoint. □

Theorem 203.4 (ONLY original positive-member data compute the ACTUAL literal lowerMarked full values)**.** *ONLY an original bound-two domain, ANY genuine original coefficient split TEST, an actual positive-level head point lying in its ORIGINAL computed member mask, and actual current lower cosupport internally supply all data for the preceding formula on the ACTUAL ORIGINAL* `lowerMarked` *and SAME restricted tail, at EVERY positive recursive depth. No replacement mask, lower TEST or desired record/value certificate is supplied.*

Proof. Original coefficient domain stability gives the genuine bound-two head record and genuine original tail TEST. Its real positive level internally gives a nonempty actual through-point member subset in the actual original prefix list. Use exactly the given point of the ORIGINAL computed open mask, and apply the third theorem with that SAME mask and the genuine original coefficient tail. The independent original mask theorem identifies the WHOLE actual pullback of the genuine member remainder with the literal ORIGINAL `lowerMarked`. Only after this ALL FOUR-field identity substitute the real original record into its unchanged GENERAL formula. Every actual history and point is retained.

This closes the actual computed positive-member curve lower-value producer. It does not assert whole upper GENERAL comparison, positive relative ambient dimension, higher-dimensional recursion, or exceptional/infinitesimal semicoherence and global scheduling. □

204 The genuine chosen contact remainder has a whole lower domain in every induction level

For the actual original nonspecial zero-level Step One branch, the real run selects a head H, tail Q and point $x_0 = Q(x)$. The original tangent choice at $H(x_0)$ has a genuine open U. We retain its entire inverse image $V = H^{-1}U$, the literal contact coefficient ideal, and the original historical boundary positions. The carrier remainder is precisely the existing union of smooth global contact components not contained in its actual cosupport. This is the contact reduction of Bierstone–Milman[5], Section 5, Step I, Case A, with the genuine history of Section 7. The remaining component construction enforces the original domain's nonzero condition on the whole chosen open. Replacing the historical entries by empty entries retains the literal ordered positions rather than silently deleting them.

Theorem 204.1 (Original zero-level input internally selects the genuine whole contact model)**.** *For original maximal-order domain input* K *of induction level* $k+2$*, its literal canonical masked*

Step One tower R, and a nonspecial zero-level cosupport point x, the actual tangent choice internally supplies one true open U containing $H(x_0)$ and its actual tangent function w. On V, the literal original contact marked ideal L has closed smooth carrier, coherent ideal and regular dimension at most $k+1$. The true restricted Q is a test tower of L, its entire final cosupport is the pullback of the original K_R cosupport, and the entire original contact mask lies in V.

Proof. The genuine canonical bridge internally gives the coefficient/homogenized original test tower and its whole original cosupport equality. The nonspecial regime theorem places x_0 in its true head cosupport. Maximal order gives a true tangent direction at $H(x_0)$, and the original tangent-choice specification supplies its chosen open and actual function. Giraud contact geometry along the true head gives the smooth strict contact carrier, of one lower regular dimension. The actual restricted coefficient ideal is coherent, and its open pullback is the literal ideal of L. Coefficient equivalence and the genuine contact-tail theorem supply both the actual restricted test tower and its entire cosupport equality. Concatenating the real head and tail recovers R, giving the displayed original cosupport. Zero level identifies the original lower mask with the actual contact mask; its defining interior is contained in the tangent inverse image. All data are chosen from the same actual original inputs. □

Theorem 204.2 (The true remaining contact components produce nonzero geometry in every dimension bound)**.** *For any genuine smooth closed marked carrier $L.N$, coherent ideal, positive mark and regular dimension bound b, the SAME original contact remainder has closed smooth carrier of dimension at most b, coherent ideal nonzero along every carrier point, and cosupport exactly $L.$*cosupp *intersected with that remainder carrier. The boundary keeps a specified number of actual empty historical entries and the remaining original entries.*

Proof. The actual locally finite smooth global component family divides the carrier into disjoint closed selected and remaining unions. Near each remaining point, remaining membership is exactly original membership. Coherent relative order therefore gives the entire asserted cosupport equality. The true remaining union is closed smooth, and regular dimensions inherit the original bound by the regular-dimension comparison for inclusion. Its ideal is unchanged. Analytic continuation on a smooth connected component shows that a nonempty relatively open piece of a remaining component cannot be contained in the actual analytic cosupport, since otherwise that entire component would be selected. Thus this cosupport trace is thin. If the coherent ideal vanished along a remainder point, its actual finite local generators would vanish along a whole carrier neighborhood. A nonempty relatively open piece would lie in the cosupport, contradicting thinness. This proves nonzero at every actual remainder point. □

Theorem 204.3 (Actual contact test prefixes produce the entire lower remainder domain)**.** *For genuine level-$k+2$ original K, an actual tangent open and function, and a true concatenated coefficient test history HQ, the entire transform of the SAME original remainder of its contact model along $Q|_V$ is in induction-level-$k+1$ domain. No centre containment in that remainder carrier is required as a hypothesis.*

Proof. The contact theorem internally supplies the raw smooth closed contact carrier, coherent ideal, lower dimension and actual restricted test history. The selected and remaining carrier partition has disjoint closed strict transforms and union the entire contact strict transform. At a remaining strict point, a neighborhood misses the transformed selected union. Its exact carrier germ is therefore the whole smooth strict contact carrier's germ. This gives smoothness and the lower dimension bound at every true prefix. Initially nonzero is supplied by the preceding theorem. At each actual blowup, smooth strict-carrier controlled-colon geometry gives the entire coherent ideal and preserves nonzero even without centre containment in the candidate carrier. Induction gives both fields along the actual whole history.

The true contact boundary is the open pullback of the full original coefficient boundary along HQ. Its exceptional suffix is SNC. Empty historical entries have empty strict transforms, so the

actual list recursion shows that the remainder boundary is the same number of empty entries followed by exactly that suffix, including new tail exceptions. Prefixing empty entries preserves SNC by assigning them no incident coordinates. Giraud gives actual source-centre containment over the tangent open in the real contact carrier. Actual contact geometry then gives its transversality to the exceptional suffix. Exact remaining strict-carrier germs inherit this transversality; empty entries impose no condition. These fields, unchanged positive marking and the real lower dimension bound prove the asserted entire lower domain. □

Theorem 204.4 (Only original computed zero-level input supplies every genuine remainder prefix)**.** *Only the original maximal-order domain-$k+2$ input, its original canonical masked Step One tower and its true nonspecial zero-level cosupport point internally produce one chosen tangent open, containing the entire original contact mask, such that EVERY true original tail prefix restricted to its whole inverse image has the SAME actual remainder in level-$k+1$ domain.*

Proof. Use the internally chosen open and tangent direction of the first theorem. The genuine original canonical bridge gives its whole coefficient test tower. Split this at the original computed run. Every actual head of its true tail is a true test prefix. Append this prefix to the same original head and apply the preceding entire remainder-domain theorem. Thus all domain fields are obtained from the actual original source history at every prefix on the one chosen open. There is no caller remainder-domain or lower-canonical history premise. □

Theorem 204.5 (The actual entire contact remainder gives genuine global prefix order equations)**.** *On the internally chosen entire final contact open, each actual remaining-tail projection into the corresponding true restricted original prefix is holomorphic. The actual prefix nonmonomial ideal is coherent, and the inverse image of every natural nonmonomial order threshold, including zero, is globally complex analytic on this entire chosen final open.*

Proof. The preceding original-only construction gives the actual whole prefix domain. True SNC monomial division gives coherence of its literal nonmonomial ideal. For positive order, the coherent relative derivative cosupport supplies its actual analytic equations. For order zero, the holomorphic germ dictionary identifies the order locus with its entire smooth closed carrier. The actual remaining-tail projection maps the chosen final open into the genuine prefix open by the real head-tail projection equation. Its ambient projection is holomorphic, and the actual open-subtype lift is holomorphic. Pulling back the actual prefix equations therefore gives the asserted globally analytic locus. These are true whole contact-remainder order equations. Equality with the entire original contact masked marked record, arbitrary deeper lower thresholds, higher/special boundary extension and the simultaneous auxiliary Hilbert–Samuel scheduler are subsequent producers. □

205 The genuine original member branch has an entire coherent lower history in every induction level

The original positive-member lower mask removes special component pieces and members not incident at the reference point. The genuine unselected-component member marked ideal supplies an actual whole extension of that literal masked lower input. We use the already proved generic whole controlled-tail geometry; we do not assume the whole extension's history is canonical for its lower invariant. Original source centres are genuinely admissible for the original coefficient input. Their true transformed member geometry supplies the whole lower domain and its actual coherent order equations at every prefix.

This follows the real member-intersection Step I.B of Bierstone–Milman[5], Sections 5 and 7. The existing full original consumers and first-entry dictionary were stated only in induction level three. Here the constructed original inputs in every level produce the same actual whole-member equations. Neither a curve envelope nor a discrete exceptional set is used.

Lemma 205.1 (The actual original nonspecial positive-member history produces its genuine entire lower extension)**.** *For original maximal-order marked-domain input of induction level $k+2$, its literal original canonical masked Step One history, and a genuine nonspecial cosupport point of positive level, its actual original run selects a true head H, tail Q and point $x_0 = Q(x)$. The actual whole member marked ideal L formed at that same H, x_0 has, at EVERY actual tail prefix, genuine induction-level-$k+1$ domain. Its whole final transform pulls back on the real original member mask to precisely the SAME literal original lower marked ideal along its actual restricted tail.*

Proof. The original canonical Step One bridge supplies the true coefficient test history. The genuine nonspecial regime theorem puts its real start point in the actual cosupport at the same original level. Original lower-mask membership is internally supplied by the true canonical lower-data theorem. Positive level selects the original member branch. Its original members through x_0 form a nonempty actual finite set of coefficient-boundary positions, retaining their original indices.

Intersect their actual SNC members with the true coefficient carrier, and take the genuine unselected smooth components, removing precisely those whole components contained in the actual coefficient cosupport. This is the existing whole member marked ideal L. The real unselected-component geometry supplies coherence and nonzero restriction on its entire carrier. Along every true source-admissible prefix of Q, the proved controlled member-tail geometry supplies smooth transformed carrier, SNC remaining boundary, genuine transverse structure, coherent controlled ideal and actual nonvanishing, with the real dimension drop. Consequently its whole transformed marked ideal is in level $k+1$ at every prefix. The existing whole open-restriction identity identifies its actual final pullback with the literal original member lower input and restricted tail. The selected split, mask and marked objects are the original computed ones; no whole-lower domain/history or identity is an input. □

Theorem 205.2 (Actual whole prefix nonmonomial ideals give genuine global analytic order equations)**.** *For every genuine prefix Q_i of that actual whole member tail, its actual nonmonomial ideal is coherent. For every $q \geq 0$, the inverse image of its actual nonmonomial order-q locus along the SAME actual remaining tail is globally complex analytic on the entire original final manifold.*

Proof. The preceding original-only producer gives the actual whole prefix marked domain, including genuine coherent ideal, SNC boundary, smooth carrier, transversality and nonzero restriction. True SNC monomial division gives coherence of its actual nonmonomial ideal. Positive order thresholds are the actual coherent relative derivative-ideal cosupports. At order zero, the coherent holomorphic germ dictionary identifies the locus with the entire actual smooth carrier. Thus every prefix order locus is analytic. Its actual remaining tail projection is holomorphic, so its true inverse image is globally analytic on the entire original final manifold. No prefix canonical-lower certificate or supplied defining ideal is used. □

Theorem 205.3 (The literal original lower first entry is exactly the restriction of the whole member order)**.** *On the ENTIRE true original final member mask, the SAME level-$k+1$ constructed original lower invariant has first entry equal to the actual nonmonomial order of the whole final L_Q at the underlying original point. Its first-entry threshold on its actual carrier equals exactly the inverse image of the whole L_Q order threshold, including zero.*

Proof. The literal lower value's first entry is its actual current nonmonomial order, by the definition of the original constructed general invariant. The preceding whole mask identity replaces its marked transform by the actual open pullback of L_Q. Original open nonmonomial-order transport, using the internally produced whole domain and its genuine nonvanishing, identifies that literal order with the same underlying whole member order. The marked pullback carrier is exactly the inverse image of the whole carrier. On that carrier the genuine finite-order dictionary

says order at least q precisely when the point belongs to its actual order-q cosupport. These two identities give both directions of the entire first-threshold equality, with no numeric/order-value certificate. □

Corollary 205.4 (Only original inputs construct a genuine analytic whole extension of each literal lower first threshold)**.** *Every literal original member lower first threshold has a genuine globally analytic whole extension supplied by the SAME actual whole member order locus; its original masked threshold is exactly its actual pullback.*

Proof. Apply the actual whole final member domain and its genuine coherent nonmonomial derivative equations to the actual whole final order locus. Its true open inverse image is analytic, and the preceding entire original lower first dictionary identifies this precisely with the original threshold. This is an actual whole analytic extension and exact mask restriction. The claim does not make arbitrary deeper lower values globally analytic or fill original higher-level/special boundary pieces. □

Theorem 205.5 (The original full positive companion internally supplies every true whole member prefix equation)**.** *For only original full level-$k+2$ marked-domain input, its literal canonical history and a positive original cosupport reference point whose actual internally lifted fixed-companion point is nonspecial and of positive member level, every genuine whole member prefix-order locus constructed above is analytic on the ENTIRE actual companion final open. No selected companion, whole member domain, history or analytic presentation is supplied.*

Proof. The original full fixed-companion producer internally constructs its actual domain, maximal order and canonical masked tail. Its internally constructed reference lift belongs to actual companion cosupport by the whole original order-equality dictionary. Apply the genuine all-dimensional whole member prefix producer to that SAME actual companion, SAME history and actual nonspecial positive-member point. The resulting entire analytic equations are on its actual final manifold, the true original companion open. All original source data and all marked transformations are retained. This advances the unmasked member presentation in every induction level; whole arbitrary lower threshold boundary extension, zero-level contact extension and the simultaneous auxiliary Hilbert–Samuel scheduler remain further producers. □

206 Actual noncanonical computed common zero-mask full curve values

The zero-contact comparison of [5, Section 5, Step I.B] requires the genuine endpoint coordinate law before comparing values at the same original point. An existential lower square without that law would not justify replacing its endpoint by the original global endpoint map. We retain this law from the real common-mask construction and use the actual ANY-test equal-dimension curve-value theorem.

Theorem 206.1 (Original ANY-test zero-contact data internally produce the actual common mask, original endpoint identity and all full lower values)**.** *ONLY an original bound-two maximal-order domain, ANY original TEST, an actual same-dimension Cartesian smooth tower pullback and a true current cosupport point in the zero nonspecial regime internally construct the ACTUAL target regime-run split, computed LARGE zero mask, unchanged lower record and genuine restricted tail. Internally construct its bound-one domain, TEST and actual cosupport point, the real source common preimage mask, genuine smooth base map, actual restricted-tail Cartesian square, ENTIRE source contact-record equality and BOTH whole cosupport directions. The real endpoint map on the common mask sends the SAME original point to the target point via the GIVEN original global endpoint map. At that actual point EVERY positive-depth unchanged full source-common lower value equals the ACTUAL computed LARGE target lower value. No lower canonicality/domain/TEST/endpoint identity/value certificate is supplied.*

Proof. First apply the original-input computed zero-mask-history theorem to the given original TEST and actual zero nonspecial current cosupport point. It internally supplies the actual run, LARGE target mask membership, unchanged lower record, genuine bound-one domain, real restricted tail TEST and actual lower cosupport point. Genuine original coefficient equivalence independently supplies the actual whole coefficient TEST and coefficient cosupport at the original point.

Apply the real original common-mask geometry theorem to THIS actual Cartesian original square and coefficient data. It constructs the real head map, actual common source open as the ENTIRE inverse image of the computed target mask, true smooth restricted base map, ENTIRE marked contact pullback equality and genuine tail Cartesian square. Retain its independently proved underlying-point equation at EVERY endpoint. At the chosen actual endpoint, subtype extensionality turns this into the true equation sending that actual source point to the target lower point obtained from the GIVEN original endpoint map.

Apply the proved ANY-test equal-dimension full curve-value theorem to the actual target lower bound-one domain, genuine lower TEST, true smooth base map and actual common-tail square. Its internally proved source domain/TEST and BOTH cosupport directions, followed by the ENTIRE contact-record equality, give those assertions for the actual source COMMON contact record. The retained endpoint equation supplies the required true target cosupport membership BEFORE any value comparison. The same endpoint equation and ENTIRE record equality then identify its all-depth full-value conclusion with the displayed unchanged common/target computed values at the SAME original point.

The source record here is the genuine COMMON pulled-tangent contact record. Its equality of values with the independently computed OWN source zero-mask record remains the subsequent genuine local-choice and nested-open comparison; it is not assumed in this statement. □

Theorem 206.2 (ONLY original inputs construct one actual source history and every computed common zero-mask value)**.** *ONLY an original bound-two maximal-order domain, ANY original TEST and actual same-dimension smooth submersion internally construct ONE actual source tower, genuine smooth global endpoint map and actual original Cartesian geometry. On THIS same history, at EVERY actual current cosupport point in the zero nonspecial regime, the first theorem's ACTUAL computed target/common masks, all genuine lower geometry, true same-point endpoint equation and every positive-depth original full lower value equality hold. No source history or desired comparison certificate is supplied.*

Proof. The original actual smooth tower construction uses only the given original well-formed record and genuine TEST. It supplies one real source tower and endpoint map with actual Cartesian geometry and true source TEST before any point or lower mask is considered. Apply the first theorem to this SAME actual square at every genuine target cosupport point of its global endpoint map in the stated zero nonspecial regime. Each lower object is internally produced from that actual point, and the true endpoint law retains the same original global map throughout. Equal ambient dimensions preserve the genuine lower bound one, so the real ANY-test full-value theorem applies at EVERY positive depth. OWN/common comparison, whole upper GENERAL, positive-member/higher-dimensional noncanonical values, exceptional or infinitesimal semicoherence and the full global driver remain OPEN. □

207 Actual noncanonical full lower values for independent zero-contact choices

In the dimension induction of [5, Section 5, Step I.B], actual contact-choice history isomorphisms must feed an actual lower invariant theorem. For original carrier dimension at most two, the real zero-contact lower domains have dimension at most one. The preceding ANY-test full curve-

value locality therefore supplies the genuine missing value consumer, without assuming lower canonicality.

Lemma 207.1 (A true subopen of the actual zero mask internally supplies the genuine lower curve domain)**.** *ONLY an original invariant domain of bound two, an actual original coefficient TEST prefix, ANY genuine original tangent direction on its true open, and an actual stage open contained BOTH in its original-open preimage and in the ACTUAL computed zero mask internally give the actual kept contact marked ideal in genuine dimension bound one. No lower domain, nonzero, transversality or selected-component certificate is input.*

Proof. Membership in the genuine computed zero mask implies membership in its actual defining interior. The true interior inclusion gives avoidance of EVERY original ordered member and of the ACTUAL codimension-one cosupport part, at every point of the supplied actual subopen. Restrict these statements to the true strict contact carrier. The independently proved contact-stage domain theorem now applies to the actual original tangent direction, original domain dimension bound two and same genuine coefficient TEST. It internally constructs closed smooth contact carrier, genuine coherence/nonzero, transverse ENTIRE actual kept boundary and positive mark, with dimension bound $2 - 1 = 1$. This is precisely the ACTUAL kept contact record, not an auxiliary normalized replacement. □

Theorem 207.2 (ONLY original dimension-two ANY-test zero-regime data give actual full lower-value independence of two genuine contact choices)**.** *ONLY an original domain of bound two of maximal order, an actual initial cosupport point and two genuine original tangent directions through that point internally supply ONE original neighborhood with the following property for EVERY original split TEST $H + S$. At ANY actual current original cosupport point whose original image is in that neighborhood and whose actual head image is in the zero nonspecial regime, internally construct actual contact opens, a genuine contact-choice tower isomorphism and its actual endpoint biholomorphism fixing the original point. The two ACTUAL kept contact records are genuine bound-one domains and their SAME restricted original tails are genuine TESTS. At the same actual point the unchanged full original LOWER invariants agree for EVERY positive recursive depth. No lower-domain/TEST/canonicality, chosen mask, comparison isomorphism or desired lower-value certificate is supplied.*

Proof. The genuine contact-choice gluing theorem first constructs ONE original neighborhood before any history. Shrink it inside BOTH original tangent opens; it still contains the actual original reference point. Fix a subsequent genuine original split TEST and actual point as in the statement. Genuine coefficient equivalence supplies its actual coefficient TEST and cosupport. Projection along its genuine coefficient tail puts the actual head image in the true head coefficient cosupport. Its proved zero nonspecial regime therefore puts it in its ACTUAL computed zero mask, by the original mask-membership theorem.

Intersect THIS actual mask with the two actual original tangent-open preimages. Both real opens contain the head image, avoid EVERY original ordered member, and the actual original image lies in the ONE gluing neighborhood. The genuine whole-history choice producer therefore constructs smaller actual opens, their real biholomorphism, true restricted tower isomorphism, point-fixing endpoint map, BOTH actual lower TESTS and WHOLE record pullback equality. The first lemma internally constructs BOTH actual lower bound-one domains on those genuine smaller opens; it uses actual computed-mask containment and the real tangent directions, not a caller domain certificate.

The genuine contact-tail cosupport dictionary puts the SAME original endpoint point in the actual target lower cosupport. Apply the preceding ANY-test full curve-value tower-isomorphism theorem to these ACTUAL bound-one lower records and true tower isomorphism, at EVERY original positive recursive depth. Whole original parent equality and the actual point-fixing equation identify its actual source value with the other contact record at the SAME original point, giving the asserted equality. The argument never transports a canonical-only value field.

It computes real noncanonical lower values on the actual restricted original tails. This is a zero-contact, carrier-bound-two consumer; the positive-member branch, general higher-dimensional full-value locality, exceptional or infinitesimal semicoherence and the full global driver remain OPEN. □

208 Actual original zero-mask contact tests without canonicality

The actual zero masked lower record keeps original boundary positions and restricts them to EMPTY. Compare THESE records, rather than replacing them by unrelated remainder models, in the contact-independence argument of [5, Sections 5 and 7]. The current resolution-invariant interface asserts value invariance only for canonical histories; the result below proves actual geometric TEST and WHOLE record comparison without invoking that interface or assuming lower canonicality.

Theorem 208.1 (Real original-member-avoiding contact masks give genuine lower tests and entire controlled records at every prefix of every ORIGINAL upper test)**.** *Let K be actual well formed with positive mark, and let x belong to its ORIGINAL cosupport. Two genuine tangent directions on real neighborhoods of x internally give ONE neighborhood O of x. For EVERY genuine ORIGINAL appended test T_0T_1, and ANY actual stage opens W, W' inside the respective real tangent preimages and avoiding EVERY original member, consider an actual transformed ORIGINAL cosupport point z whose head image belongs to both opens and whose original base belongs to O.*

Then actual subopens $V_u \subset W$, $V_v \subset W'$ contain that actual head point. Genuine stage and endpoint biholomorphisms give a real tower isomorphism of the LITERAL restricted tails, fixing z. The ACTUAL whole boundary-kept contact parents A, B are smooth/coherent and satisfy $\psi^* A = B$ in ALL FOUR marked fields. BOTH literal restricted tails are genuine lower TESTS. Their ENTIRE endpoint controlled records and ENTIRE controlled records at EVERY head are genuine pullbacks through compatible real head/tail isomorphisms. No lower domain/canonicality, desired lower test, record or value-comparison certificate is an input.

Proof. The genuine contact gluing theorem constructs O before the tower and masks. True coefficient equivalence internally turns the SAME ORIGINAL appended test into the empty-original-boundary homogenized test required for contact-tail gluing. The genuine cosupport dictionary supplies its actual cosupport point. Actual point-fixing gluing, restricted to the given two opens and their genuine inverse images, constructs V_u, V_v, the actual stage and endpoint biholomorphisms, a genuine tower isomorphism of the literal tails, fixation of z, and whole EMPTY-original-boundary contact record transport.

Each smaller open still avoids EVERY original member, because it lies in its respective original mask. The literal kept-record dictionary therefore identifies its actual kept contact parent with the empty-original-boundary contact parent padded by exactly $\#K.E$ EMPTY positions. Whole pullback commutes with THIS actual padding, so the empty-boundary parent equality gives the ENTIRE actual kept-parent equality. This step retains every true new exceptional member and does not assume raw old boundary preservation.

The actual upper head is itself an ORIGINAL test. The genuine tangent direction and true tangent-preimage containment therefore supply the actual smooth closed strict contact carrier; actual coefficient coherence supplies its coherent ideal. The actual coefficient appended TEST and tangent open containment independently give BOTH genuine kept-parent contact-tail TESTS. These premises are constructed before applying true controlled-transform transport through the actual tower isomorphism. It gives the ENTIRE endpoint record equality, independently of invariant canonicality or values.

Finally split THIS actual lower tower isomorphism at any numeric position. The real head and tail isomorphisms are compatible with the actual endpoint map. Each target lower head is

a true test by the genuine lower TEST already proved. Whole controlled-transform transport along that actual head isomorphism gives the ENTIRE corresponding marked-record equality. Empty steps, ordered boundary positions and normal germs are retained. Comparison of actual higher invariant values on arbitrary noncanonical histories, the positive selected-member centre condition, exceptional or infinitesimal semicoherence and general preparation termination remain OPEN. □

209 Actual independently computed OWN zero-mask full values on the same original smooth history

The common pulled-tangent presentation and the independently selected source presentation in [5, Section 5, Step I.B] have different actual large masks. The common lower full-value theorem alone cannot identify their values. We use genuine contact-choice history isomorphisms, actual curve-value locality and true nested open restrictions to compare the unchanged computed OWN lower record itself.

Lemma 209.1 (Two actual curve presentation opens give the same full values at their true common cosupport point)**.** *Let P be an actual coherent marked parent on an ambient manifold and T an actual tower. Let A, B be ANY two actual opens such that BOTH actual restricted parents have genuine bound-one domains and BOTH actual restricted towers are genuine TESTs. At a SAME underlying original endpoint in BOTH opens and in the actual cosupport on the first, internally prove cosupport membership on the second. EVERY positive-depth unchanged original full value on A equals that on B. No canonicality, common comparison window or desired value certificate is supplied.*

Proof. Use the actual intersection $A \cap B$ containing the real endpoint. The proved genuine nested-open theorem internally supplies its domain, TEST and BOTH cosupport dictionaries, once from each actual large open. The real cosupport point on A supplies the genuine intersection point; the second BOTH-direction dictionary supplies its real cosupport on B. At EVERY positive depth its same actual value equals the large value on A and equals the large value on B. Their genuine composition gives the asserted equality. Every tower retains the original maps and empty positions. No canonical-only invariant field is used. □

Theorem 209.2 (ONLY original same-dimension smooth data give full values on the independently computed OWN source zero mask)**.** *ONLY an original bound-two maximal-order domain, ANY genuine original TEST and an actual same-dimension smooth submersion internally construct ONE actual source tower and genuine global endpoint map. At EVERY real current cosupport point in the target zero nonspecial regime, internally construct the ACTUAL independently computed OWN source regime run, LARGE zero mask, unchanged lower record, true bound-one domain, TEST and cosupport point, as well as the corresponding ACTUAL computed target objects. At the SAME original point, EVERY positive recursive-depth unchanged full OWN source lower value equals the computed target lower value. No source choice/history/lower canonicality, domain, TEST, comparison window/isomorphism or desired value certificate is supplied.*

Proof. Construct ONE actual source history and Cartesian geometry from the original smooth data before choosing an endpoint. At a genuine endpoint in the stated regime, the independently proved whole-record, level, special-locus and exact run dictionaries internally give the source domain, TEST, source zero nonspecial regime and equality of the ACTUAL source/target run indices. Apply the original-input zero-mask-history producer to this SAME source TEST. This gives the ACTUAL OWN source large mask, unchanged lower record, real bound-one domain, TEST and cosupport point. Separately the genuine common-mask full-value producer constructs its ACTUAL common source lower record, true domain/TEST and cosupport, true original-point

endpoint identity, and every full-value comparison with the ACTUAL target computed large lower record.

At the genuine original source base point, the original smooth tangent producer internally supplies BOTH the pulled target tangent and the independently selected source tangent on their actual opens. Their proved contact-choice comparison constructs ONE source neighborhood before the given history is used. The actual source point lies in that neighborhood because its original image is precisely that reference point. The actual source regime at its true run head is zero and nonspecial, by the proved original run theorem. Therefore the genuine ANY-test contact-choice full-curve-value theorem applies to THIS actual head/tail split. It constructs actual smaller opens in the computed OWN mask, a real tower isomorphism fixing the original endpoint, ENTIRE parent pullback equality, genuine bound-one domains and TESTs, and full value equality on the two actual small contact records.

For the independent OWN tangent, true coefficient coherence supplies the actual global contact parent. Genuine nested-open transport from the actual OWN large mask gives BOTH cosupport directions and full values on its smaller comparison open. The whole parent equality and actual point-fixing tower isomorphism then transport this real smaller OWN cosupport point to the actual smaller pulled-tangent cosupport. For that pulled tangent, genuine coefficient coherence supplies its actual global parent independently. Its common large open and smaller comparison open need not be nested: apply the first lemma to their ACTUAL intersection, using the independently constructed curve domains, TESTs and true cosupport points on BOTH opens. Thus the smaller pulled-tangent value equals the COMMON large value. Compose the genuine OWN-large/smaller, contact-choice, two-open and common/target full-value equalities. The retained endpoint identities make every term a value at the SAME original point, giving precisely the unchanged independently computed OWN/target lower values at EVERY positive depth.

Every lower geometric assertion and mask membership precedes its value use. This closes this same-dimension, original carrier-bound-two, zero-contact lower-value naturality. It does not assert the entire upper GENERAL comparison, positive-member histories, positive relative dimension or higher-dimensional noncanonical full values, exceptional or infinitesimal semicoherence, or general global scheduling. □

210 The unchanged original Step One value in the noncanonical zero-contact regime

The preceding computed OWN lower comparison now feeds the ORIGINAL Step One function in [5, Section 5, Step I.B]. This is an exact value theorem on the same original history, rather than a statement only about an auxiliary lower field.

Theorem 210.1 (ONLY original equal-dimension smooth ANY-test data preserve the actual Step One value in the zero nonspecial regime)**.** *ONLY an original bound-two maximal-order domain, ANY original TEST and an actual smooth submersion of equal ambient dimensions internally construct ONE actual Cartesian source tower and genuine endpoint map. At EVERY source endpoint mapping into the target current cosupport and target zero nonspecial regime, for EVERY positive recursive-depth lower invariant, the UNCHANGED original sequence-aware Step One value of the actual pulled-back record equals the UNCHANGED target Step One value at its actual original endpoint image. No lower canonicality, source mask/domain/TEST, local comparison or desired Step One equality certificate is supplied. The Step One function itself is not redefined.*

Proof. The genuine computed OWN lower full-value producer first constructs ONE actual source tower and original Cartesian geometry. At each stated endpoint it internally constructs the ACTUAL target and independently computed OWN source run, both LARGE zero masks and actual lower records, their true domain/TEST/cosupport data and EVERY unchanged full lower value equality at the SAME original point.

Independently obtain the actual target coefficient TEST from original coefficient equivalence. The genuine whole-history special-locus and level dictionaries on the actual Cartesian square give source nonspeciality and source level zero BEFORE evaluating Step One. Unfold the ACTUAL sequence-aware Step One definition on BOTH original records. The internally proved nonspecialities select their true lower-value branches; the genuine levels make both first entries zero and select their ACTUAL zero-regime runs. The internally constructed mask memberships select the finite lower-value branches. The genuine OWN lower full-value equality, followed by the actual inclusions into `WithTop` and the original lexicographic pair, identifies exactly these two original Step One values. No canonical-only naturality field or modified invariant is used.

The conclusion is restricted to the zero nonspecial branch, original carrier bound two and equal ambient dimensions. Positive-member branches, the whole upper GENERAL companion comparison, positive relative dimension or higher-dimensional noncanonical full values, exceptional/infinitesimal semicoherence and general global scheduling remain OPEN. □

211 Actual recursive family values at smooth monomial leaves

The original recursive value types change with dimension. Smooth pullback therefore compares the complete original family by its genuine fibre-dimension embedding, preserving the whole indexed colex boundary. We prove this actual comparison at the monomial branch of every positive recursive depth and at every original test stage. This is the zero-nonmonomial-order branch in Włodarczyk's Step One/general construction, with the smooth functorial interpretation of Bierstone–Milman [3, 5].

Theorem 211.1 (Actual original family comparison at a monomial leaf)**.** *Let the positive ambient dimensions be n, m, let $t = m - n$, and let $f : Y \to X$ and $g : D' \to D$ form the actual smooth pullback of whole histories Q, T. Let M lie in the original invariant domain of dimension $k+1$, and let T be an actual test history of M. For $z \in D'$ suppose the actual nonmonomial order of M_T at $g(z)$ is zero. Then*

$$(R^{(m)}_{t+k+1})(f^*M, Q, z) = D_{t,k+1}\big((R^{(n)}_{k+1})(M, T, g(z))\big),$$

where D is the established strict map on the unchanged original value types. The source invariant domain with the true added fibre dimension is supplied internally. No source canonical history or value comparison is assumed.

Proof. The original domain and true test history give a well-formed transformed marked ideal and its genuine nonzero-along property. The actual marked pullback dictionary and smooth nonmonomial-order theorem then give actual source order zero. The domain pullback theorem gives dimension bound $(k+1) + t = t + k + 1$ and every other source-domain field internally. Expand each actual successor-family invariant into its original general step. At the proved order-zero branch, that whole value is precisely $(0, \operatorname{inl}\gamma)$, independently of its deeper companion. The genuine whole stage gamma theorem gives source gamma equal to the dimension-t map of target gamma; its proof supplies the SNC cardinality and actual fibre bounds internally and preserves every feasible subset and its full colex index. The original definition of $D_{t,k+1}$ maps the entire left branch to exactly this value. Thus the equality concerns the complete actual original family value at both recursive depths. □

Theorem 211.2 (Actual order reflection at two monomial leaves)**.** *For two points z, w of the same actual source history whose target nonmonomial orders are zero, comparison of the two complete source family values by $<$, $\leq$ or equality is equivalent to the corresponding comparison of their two complete target family values.*

Proof. Apply the preceding actual equality at both points. Strictness of the same dimension-t map reflects and preserves strict order, weak order and equality, yielding the three asserted equivalences. □

Theorem 211.3 (Original input constructs all actual prefix comparisons)**.** *Given only the original smooth f, original domain M and original test history T, construct the actual whole source history Q and its final smooth map g. It has the true transformed whole marked record, actual testness, cosupport preimage and correct source invariant domain. For every literal prefix index j, construct its actual smooth stage map a_j internally. At every source prefix point with target nonmonomial order zero, the complete actual family value obeys the preceding dimension formula, retaining the literal original M and both entire prefixes.*

Proof. The existing actual tower constructor applied to f, T supplies Q, g and its whole actual Cartesian witness. The relative marked and cosupport dictionaries give testness, the entire transformed marked record and the stated cosupport preimage, using only original well-formedness. The original domain pullback theorem supplies the correct source domain. At index j, the genuine whole prefix constructor gives a_j and the actual Cartesian prefix witness. The original test history supplies testness of this literal target prefix. Apply the first theorem to that actual witness and every indicated point. Every history, source-domain field and prefix map is thereby produced internally from the original inputs; no desired family comparison is a premise. □

Positive nonmonomial-order companion values, independent contact choice, maximum existence and a full canonical driver remain further obligations. No global monomialness, source compactness or smooth surjectivity is asserted.

212 Actual original source and target blowup charts from internally adapted submersion products

The original input consists only of a genuine submersion f of dimensions $n + r$ and n at y, an actual original target centre flattening chart Φ, its genuine zero-centre and preferred-chart-source conditions, the original whole equations of C, and a genuine target normal pivot $j \geq k$. No source chart, monomial dictionary, matching-chart or blowup-lift output is input.

Lemma 212.1 (Whole original-dimension monomial split dictionary)**.** *For the genuine internally constructed fibre-first split of $\mathbb{C}^{n+r}$, the source monomial polynomial with tangent dimension $r+k$ and rotated original pivot $r + j$ retains the ENTIRE fibre block and drops to the actual target monomial polynomial with tangent dimension k and original pivot j. These are equalities of whole actual polynomial maps, including the entire exceptional hyperplane.*

Proof. The preceding true finite rotation gives the individual target coordinate $r + i$ and fibre coordinate $a < r$. On a source fibre coordinate, the normal-index condition $r+k \leq a$ is impossible, hence the source monomial polynomial leaves that coordinate unchanged. On a source target coordinate, $r + k \leq r + i$ is equivalent to $k \leq i$, and equality with the rotated pivot is equivalent to $i = j$ by genuine finite permutation injectivity. The source monomial formula thus agrees with the target polynomial on EVERY target coordinate. Both complete coordinate equalities give the entire genuine split equality. Zero pivot values are allowed throughout. □

Theorem 212.2 (Original geometry internally constructs genuine corresponding FlatCharts)**.** *Internally produce actual open neighbourhoods R, S and genuine source and target* `BlowupConstruction.FlatCharts` *of the original preimage centre and original target centre. The target chart is the original chart genuinely restricted to the internally produced S; the source chart is the internally produced adapted product chart. Their actual base points, tangent dimensions and pivot indices are respectively fy, k, j and $y, r + k, r + j$. Their actual monomial chart targets are EXACTLY*

$$V = \sigma_{k,j}^{-1}(S), \qquad V' = \mathrm{split}^{-1}(R \times V).$$

For EVERY $w \in V'$, the actual inverse source blowup-chart base point maps to the actual target blowup-chart base point of its genuine target projection: $f(c'.\mathrm{pt}(w)) = c.\mathrm{pt}(\mathrm{drop}(w))$.

Proof. Apply the original adapted-product producer; its real product topology internally gives the actual R, S, source chart Ψ with target exactly the whole product, whole original preimage-centre equations and complete inverse point square. Restrict the original target chart to S using its actual inclusion in the original target and its zero membership. Its entire original centre equations and preferred-chart-source condition restrict to this SAME actual chart. These facts construct the target `FlatChart`, with the original normal pivot j. The produced source chart and its whole original preimage-centre equation construct the actual source `FlatChart`; the finite rotation gives $r + k \leq r + j$, so its pivot is genuinely normal. No corresponding-chart certificate enters. The target chart target is its literal monomial inverse image of S. For the source chart target, apply the preceding entire fibre/target polynomial split dictionary to its literal product target: the actual fibre is in R and the actual target polynomial is in S, precisely the stated product of the target chart open. For every actual source chart parameter, its actual source monomial image belongs to Ψ's actual product target; apply the genuine whole inverse point square there and substitute the actual target polynomial identity. The actual target restricted inverse is literally the original inverse, yielding the displayed equality even at exceptional parameters. □

Theorem 212.3 (Genuine original charts internally give an actual smooth local blowup map)**.** *If the original map is a genuine global submersion and the original target centre is closed smooth, internally produce the preceding SAME matching actual charts and a map on the actual source blowup space, defined on its real chart domain by* $G(p) = c^{-1}(\mathrm{drop}(c'(p)))$. *At EVERY point of this actual source chart domain,* G *is a genuine complex submersion with the SAME finite fibre complement, lands in the actual target blowup chart, and satisfies* $f(p.\mathrm{base}) = G(p).\mathrm{base}$. *Entirely exceptional points are included.*

Proof. The original closed smooth centre gives the target blowup manifold. Its genuine preimage is closed smooth by the existing original submersion regularity producer, so the actual source blowup manifold is constructed internally too. At any actual source chart point p, the genuine source chart image lies in its literal target. The whole product-target dictionary places its actual projection in the genuine target chart target, and the genuine target chart inverse therefore lands in the actual target chart source. Both actual `FlatChart.homeo`s belong to the genuine blowup atlases and hence their maximal atlases. The SAME finite split followed by factor interchange identifies source model with target model times fibre complement. In these SAME actual charts the written map is exactly its first projection, by the actual source and target inverse identities at EVERY actual source chart coordinate. The original submersion definition therefore holds at p. The actual source inverse-chart base point is the original source base by its genuine inverse identity, and the preceding whole point-square equality gives the actual base square. No pivot nonvanishing condition is assumed; membership in the actual blowup chart supplies only its genuine homogeneous-direction pivot, and its base may lie entirely in the exceptional divisor. This is a genuine local map producer, not yet identification with the global universal lift. □

This produces genuine original `FlatChart` objects and exact whole polynomial/open/base-point dictionaries. Identifying the global universal blowup lift on these SAME whole chart domains, proving every original source blowup point is covered by a genuinely matching pair, and gluing all charts remain required for full arbitrary-choice smooth base change. Higher invariant/old driver identification, whole G3/G8 and noncompact termination remain OPEN. OWN OFFLIB only; formal registration/native/publication pending root recovery.

213 Internally selected genuine matching blowup charts cover every original exceptional point

Continue with only original genuine submersion, original closed smooth centre and an actual point p of its true canonical source blowup whose base lies in the original preimage centre. A pivot, matching chart, direction image or coverage conclusion is not supplied.

Lemma 213.1 (Every actual source normal index has a unique original target index)**.** *An actual source index a of value at least $r+k$ has a genuine target index $j<n$ of value $a-r$, with $j\geq k$ and whose actual fibre-first rotation is exactly a.*

Proof. Since $a<n+r$ and $a\geq r$, integer subtraction gives $a-r<n$ and $r+(a-r)=a$. Construct the actual finite target index using that strict bound. The genuine finite rotation formula gives its value $r+j=a$ and hence exact finite-index equality. The normal bound gives $k\leq j$; injectivity of the finite rotation and first-block inclusion gives uniqueness. This includes zero complement dimension. In target dimension zero the original bounds would be contradictory, so no fabricated index is produced. □

Theorem 213.2 (Original direction internally selects a genuine matching chart containing its actual point)**.** *Given the original genuine submersion at p.*base *and an actual target centre flattening there, internally produce its SAME actual product source chart first, then a genuine original target normal pivot j, and actual matching target/source* `FlatCharts` *with their entire exact product-target and point-square dictionaries, such that the original blowup point p belongs to the produced actual source chart. No direction, pivot or source-chart coverage certificate is input.*

Proof. Use the original adapted product producer to choose the genuine R,S,Ψ BEFORE selecting the pivot. Its actual source centre equations and actual preferred-chart-source condition identify the entire original source tangent space with the common kernel of all source normal coordinate differentials. The original exceptional blowup point has a genuine direction vector outside this actual tangent space, by the original blowup-space definition. Hence some actual source normal coordinate has genuinely nonzero differential on that direction. The preceding integer producer identifies it with the rotation of an actual target normal pivot j. Restrict the original target chart to the SAME internally produced S and form the two actual `FlatCharts` using Ψ and this pivot. Their full target and point-square dictionaries follow from the preceding whole polynomial split and inverse point square. At p, the actual base lies in Ψ's flat domain, and the actual extended pivot value is the real coordinate differential on its actual exceptional direction, which is nonzero. Thus p is genuinely in this SAME produced source chart. Both chart membership conditions are internally proved; no geometric output certificate enters. □

Theorem 213.3 (Only the original closed smooth centre supplies coverage of every exceptional source point)**.** *The original genuine global submersion and original closed smooth target centre internally supply the true source closed smooth centre needed for the genuine canonical blowup manifolds. EVERY actual original source blowup point whose base lies in that preimage centre is contained in a genuinely matching source chart from the preceding producer, with actual target chart, SAME product target and whole base-point square. A target flattening, pivot or matching-chart hypothesis is not supplied here.*

Proof. The actual source closed smooth preimage is supplied by the existing original submersion regularity producer. The actual target image of the source base belongs to the original centre. Its original closed smooth regularity internally supplies its actual centred flattening, preferred source inclusion and whole original centre equations. Apply the preceding genuine direction/pivot producer at the actual source base. Thus every actual exceptional source point obtains a real matching pair and its whole dictionaries from the original geometry alone. □

This closes the actual exceptional matching-chart coverage step, not identification with the global universal lift. Its whole local-chart equality, off-centre case, arbitrary-choice transport and full Cartesian gluing remain necessary. Higher invariant/old driver/G3/G8/noncompact termination remain OPEN. OWN OFFLIB only, formal registration/native/publication pending root recovery.

214 Actual fibre-first adaptation and product shrinking of original submersion charts

Let the original complex manifold dimensions be $n + r$ and n. At an actual source point y the original map f is a genuine complex submersion. Let Φ be an actual centred holomorphic flattening chart of the original target smooth centre C, with target normal coordinates $j \geq k$. These are primitive geometric data, not assumed lifted charts, commuting-coordinate or product-domain conclusions.

Lemma 214.1 (Actual finite block rotation and whole plane dictionary)**.** *The genuine permutation of* $\mathrm{Fin}(n+r)$ *obtained by finite-sum decomposition, factor interchange and the actual equality* $r + n = n + r$ *moves every original target coordinate* $j < n$ *to literal coordinate* $r + j$*, and every original complement coordinate* $n + i$ *to literal coordinate* $i < r$*. The SAME actual finite split reads the whole target block after this rotation. Its actual complex linear isometry* L *internally identifies the entire original target-normal coordinate plane with the contiguous source plane of tangent dimension* $r + k$*.*

Proof. Decompose each original finite index into the genuine disjoint blocks $\mathrm{Fin}\, n + \mathrm{Fin}\, r$. Interchange the two summands and join using finite addition. The actual final finite cast preserves integer values. Thus the displayed formulas follow from finite-sum inverse identities. For the whole plane equivalence, a rotated coordinate of value at least $r + k$ cannot arise from the complement block, whose values are below r. It uniquely arises from target coordinate j with $j \geq k$; its value is exactly the original target coordinate. Conversely every such target index moves to a source normal index. This proves both directions for every vector, including $n = 0$, $r = 0$, and $k \geq n$. □

Lemma 214.2 (Actual postcomposition retains the entire local holomorphic chart)**.** *Postcomposing any genuine local holomorphic equivalence* Ξ *with* L *internally constructs a genuine local holomorphic equivalence with the SAME original source, actual target* $L_{\mathrm{inv}}^{-1}(\Xi.\mathrm{target})$*, forward map* $L\Xi$ *and inverse* $\Xi^{-1}L^{-1}$*. Its source, inverse, all coordinates and centre have their literal dictionaries.*

Proof. The target is open by continuity of the genuine inverse isometry. The original open source and centre remain unchanged. The holomorphic forward map is the composition with the actual complex continuous linear map; the inverse composition is holomorphic on its actual target. The original inverse identities together with $L^{-1}L = 1$ and $LL^{-1} = 1$ prove both inverse identities and actual source/target membership. At a centred chart $L(0) = 0$. The finite permutation formula gives every individual shifted target and complement coordinate, not only a plane or cardinality statement. □

Theorem 214.3 (Original submersion internally produces the SAME whole centre adapted chart)**.** *The actual submersion and original target flattening internally construct a centred source chart* Ψ *with source contained in the preferred source chart target. On its entire genuine flat domain,* f *lands in the original target flat domain and every shifted coordinate* $r + j$ *is exactly* $(\Phi \circ e_{fy})_j \circ f$*. If the entire original target centre has the actual equations* $j \geq k$*, the entire original preimage* $f^{-1}C$ *has the contiguous equations of the source coordinate plane of tangent dimension* $r + k$ *on this SAME source chart.*

Proof. Apply the existing actual submersion chart construction liftedCoordinatesAdd to the original submersion, actual target chart and its genuine zero-centre equality. Its fixed target-first block dictionary holds at every original source flat-domain point and gives true target-domain membership. Postcompose with the preceding internally constructed finite block rotation. Its original source is unchanged, the centre stays zero, and every target coordinate moves to literal $r + j$. Apply the actual target whole-centre equation and the whole plane dictionary in both

directions. No chart, injection, centre equation for the preimage or adapted-chart conclusion is assumed. □

Theorem 214.4 (The actual chart internally supplies a true product-open shrink)**.** *The SAME actual centred source chart internally supplies genuine opens $R \subset \mathbb{C}^r$ and $S \subset \mathbb{C}^n$, each containing zero, whose fibre-first product lies inside its actual target and whose target factor lies inside the original target chart target. The actual inverse point map also commutes: $f(e_y^{-1}\Psi^{-1}w) = e_{fy}^{-1}\Phi^{-1}(\operatorname{drop} w)$ for EVERY w in the entire actual source product target. Restricting the source chart to the inverse image of this product gives an actual local holomorphic chart with literal target exactly this product. On this entire actual restricted source domain the original map written in the original target chart is exactly the actual target block projection, and the whole preimage-centre equations remain those of the SAME contiguous source normal indices.*

Proof. The actual target contains zero because the constructed source centre is zero and belongs to its genuine source. Its preimage under the genuine finite fibre-first join is therefore an open neighbourhood of $(0, 0)$ in the actual product topology. The product neighbourhood basis produces true open neighbourhood factors R_0, S_0 whose whole product is inside this preimage. Intersect S_0 with the actual original target chart target, also a genuine open neighbourhood of zero. The resulting R, S retain the entire product inclusion. Restrict the original source to the inverse image of this literal product; it is open by the original chart's continuous-on source. The restricted target is exactly the product, the centre remains included, and original inverse identities give both directions and membership. Every genuine product point has an actual original inverse source point. The already produced whole coordinate dictionary proves the entire written-map projection and centre equations there. The actual source inverse point lies in the genuine source flat domain and has source coordinates exactly w. Its genuine target flat-domain membership and coordinate equality, followed by BOTH original target inverse identities, prove the entire inverse point square. This shrink is produced internally from the actual chart and its topology, never assumed as a chart or product-domain certificate. □

These are original geometric chart producers. Genuine arbitrary-choice blowup charts must still be connected to these SAME products and glued to identify the actual universal blowup lift with the true smooth Cartesian polynomial model. General higher invariant, old colex driver identification, G3/G8 and general noncompact termination remain open. Formal registration and native publication are pending root recovery; this chapter is OWN OFFLIB mathematics.

215 All actual Step One special branches under smooth pullback

All comparisons use the actual original coefficient-homogenized stage and whole original indexed member history. The zero-boundary branch is supplied by genuine geometric regularity reflection; no desired special-point or regime comparison is assumed.

Lemma 215.1 (The actual codimension-one support preimage)**.** *For an actual smooth map $f : Y^m \to X^n$, any original coherent ideal, original carrier N and marking, the actual pulled codimension-one support is exactly the preimage of the original codimension-one support.*

Proof. Actual relative order preservation identifies the entire pulled cosupport with the original cosupport preimage. By definition, the codimension-one support consists of points having consecutive regular dimensions for the carrier and cosupport. The genuine regular codimension-one pair theorem reflects and pulls back precisely those two original regularity statements. It supplies the exact set equality, including the added fibre dimension. □

Lemma 215.2 (The original full stage codimension-one dictionary)**.** *For the actual Cartesian pullback of an original coefficient-homogenized test history, the codimension-one support of its whole actual current stage is the preimage of that of the original whole current stage.*

Proof. The original well-formed datum and original test history internally give coherence of the actual original stage. The full original stage dictionary identifies the entire pulled marked record with its genuine smooth pullback. Apply the preceding original coherent-support producer to that actual stage. Every carrier, coefficient ideal and marking is therefore the original one, with no stage/codimension-one certificate supplied by the caller. □

Theorem 215.3 (Every actual original special branch and regime)**.** *At every actual stage point, being special is equivalent in the original and actual pulled histories, at all actual levels. Every fixed-level regime, including level zero, is likewise the original regime preimage.*

Proof. The literal original level is preserved by the original indexed-member dictionary. At level zero, the special condition is membership in the actual codimension-one support; the preceding theorem identifies that whole set. At positive levels, it is membership in the actual member-intersection locus, whose complete smooth preimage dictionary is already proved. Substituting these actual dictionaries proves both branches of the original special predicate. A fixed-level regime is the actual level equality and failure to be special; substitution proves it for every fixed level. □

Theorem 215.4 (The entire original fixed-regime run)**.** *The actual pulled original history has the same run length as the complete original history for every fixed Step One regime, including zero.*

Proof. Induct on the actual Cartesian tower. The initial identity histories have the same zero run. At a successor, the actual Cartesian square identifies the projected point with the original projected point. The preceding all-level regime dictionary gives the same branch of the literal run recursion. In its continuing branch the induction hypothesis identifies the prior runs, so adding one agrees; in its stopping branch both runs are zero. This compares the full actual recursion without a supplied run map. □

Theorem 215.5 (The actual entire original member mask at every level)**.** *The actual member mask W^A at every original stage and every actual level is exactly the smooth preimage of the original W^A.*

Proof. Its literal defining set requires that every original indexed member through a candidate point also passes through the selected point, and excludes special points of the selected level. The original indexed-member, actual level and all-level special dictionaries identify this entire set with the original set preimage. The actual stage map is continuous and open by its genuine submersion property, so taking interiors commutes with that preimage. This yields equality of the actual open masks. □

These producers close the zero-boundary special/regime obstruction. They do not yet compare the chosen contact masks or complete lower histories, normalize ambient-dependent monomial values, or prove the full higher smooth invariant comparison or resolution driver.

216 The genuine smooth blowup square is Cartesian for whole holomorphic cones

Definition 216.1 (The internally unique actual cone point map)**.** For original arbitrary-dimensional submersion f, ANY primitive genuine blowup choices π, π' and their actual internally generated lift G, and ordinary whole compatible functions $u : P \to Y$, $v : P \to D$ with $\pi v = fu$, define $L(p)$ to be the actual unique point internally produced by the genuine compatible-pair theorem. Then $\pi' L = u$ and $GL = v$ as WHOLE functions. No pointwise lift, chart, ideal invertibility or Cartesian output certificate is an input, and u, v need not be holomorphic for this point map and its two identities.

Proof. At every actual p, the original whole square supplies the genuine compatible pair (up, vp). The already proved original unique-pair theorem internally gives its unique actual point. Choose this point and use the producer's two whole point identities at every p. Function extensionality gives both entire function equalities. Empty P is handled by its unique empty function, without a fabricated point or dimension bound. □

Theorem 216.2 (Every genuine holomorphic cone point map is globally holomorphic)**.** *If the original compatible u, v are genuinely holomorphic on their WHOLE actual test manifold, the SAME actual point map L is globally holomorphic, even when the ENTIRE cone maps into the exceptional divisor.*

Proof. At every actual p, the original local-cone producer internally supplies a true open W containing p and a genuine map g_W holomorphic on W, with both whole projection identities throughout W. These inputs are precisely original f, primitive choices and the original holomorphic cone; no invertibility of its pulled centre ideal is assumed. At EVERY $q \in W$, both Lq and $g_W q$ have equal actual source base and actual lift image. Genuine arbitrary-choice global joint-point uniqueness makes them equal. Thus L and g_W agree on a real neighbourhood of p. The true open-domain holomorphy of g_W gives holomorphy at p, and its actual eventual equality with L transfers this to L. Since p was arbitrary, L is globally holomorphic. This is real gluing via internally proved global point uniqueness, not a caller overlap certificate or a density claim off the exceptional divisor. □

Theorem 216.3 (The actual original square has its full holomorphic universal property)**.** *Only original submersion and genuine primitive target/source blowup choices are inputs. For EVERY actual finite-dimensional complex test manifold in the SAME ambient universe and EVERY pair of globally holomorphic whole functions u, v satisfying the genuine square, there exists exactly one global holomorphic map g with $\pi' g = u$ and $Gg = v$ as WHOLE functions. Thus the produced square is Cartesian in that category of complex manifolds. No ideal invertibility assumption on the test cone is imposed.*

Proof. Take the internally chosen actual point map L. Its whole projection identities hold by the first producer and its global holomorphy holds by the preceding theorem. For any competing globally holomorphic map with the two whole projection identities, evaluate both equations at every actual point. The genuine arbitrary-choice joint-point theorem forces equality with L pointwise, hence as whole functions. No surjectivity of f is required, and empty or zero-dimensional test/ambient cases remain literal. This completes the whole cone universal property; it does not identify higher resolution invariants or general canonical drivers. □

Theorem 216.4 (The actual cone lift respects every genuine test-map composition)**.** *For any actual whole test function $h : Q \to P$, the internally constructed point map for (uh, vh) equals Lh as WHOLE functions. In particular all holomorphic test-map precompositions give the same global holomorphic Cartesian lift when the original cone is holomorphic.*

Proof. The original square composed with h is genuine at every actual point. Both the newly internally chosen point and $L(hq)$ have source base $u(hq)$ and actual lift image $v(hq)$ by their two true whole dictionaries. Arbitrary-choice joint-point uniqueness forces equality for every q; function extensionality proves the entire equality. No test-map surjectivity or nonempty/dimension certificate is supplied. □

Theorem 216.5 (Original geometry alone internally constructs the smooth Cartesian blowup)**.** *Only original arbitrary-dimensional submersion f and a genuine primitive target blowup are inputs. Internally produce the genuine source blowup π' along the true pulled centre and its SAME actual lift G. This G is a genuine global submersion, its original square commutes, and EVERY holomorphic compatible cone has exactly one WHOLE global holomorphic lift.*

Proof. The genuine original source-centre producer internally proves it is closed, smooth and nowhere dense and constructs a primitive actual source blowup. For that SAME choice, the actual lift producer gives the original whole square and the true arbitrary-choice geometry theorem gives its global submersion property. Apply the preceding entire universal theorem to this SAME primitive choice, EVERY actual test manifold and every genuine whole cone. No source blowup, smooth lift, Cartier, local lift, global gluing, termination or canonical sequence certificate is assumed. General higher invariant and canonical driver identification/functoriality remain OPEN. □

217 Arbitrary genuine blowup choices have the actual smooth universal lift

Lemma 217.1 (The original whole lift is the internally produced comparison conjugate)**.** *Let $f : Y^{n+r} \to X^n$ be a genuine submersion and let $\pi : D \to X$, $\pi' : D' \to Y$ be genuine blowup choices of C and $f^{-1}C$. Internally produce actual biholomorphisms $a : D \to \mathrm{Bl}_C X$ and $b : D' \to \mathrm{Bl}_{f^{-1}C} Y$ over the identity original bases. The actual uniquely generated universal lift for the two original choices is exactly $a^{-1}G_0 b$, where G_0 is the actual internally generated canonical universal lift.*

Proof. The original target blowup supplies the closed smooth centre and its empty interior. The original genuine submersion produces the same properties for its literal preimage. Therefore both true canonical blowups exist internally. Apply genuine aligned blowup uniqueness over the identity base twice. It constructs the actual biholomorphisms a, b and their whole projection identities, without a supplied comparison. The map $a^{-1}G_0 b$ is holomorphic because all three actual factors are holomorphic. Its projection is exactly $f\pi'$: use the inverse projection identity of a, the true canonical universal square, and the original source comparison identity of b. Actual target blowup universal uniqueness then identifies this entire map with the already generated actual original-choice lift. No restriction to the complement or density passage loses exceptional points. □

Theorem 217.2 (Actual arbitrary-choice whole lift is a same-r global submersion)**.** *For these same original genuine data, the actual universal lift of ANY two primitive genuine target/source blowup choices is a global submersion, with the actual fixed r-dimensional complement at every point.*

Proof. The preceding lemma internally produces the true comparisons and proves the entire original-choice lift formula. The previously proved canonical global lift is a submersion with the fixed complement $\mathbb{C}^r$. Transport its actual maximal-atlas submersion charts through the two produced GLOBAL biholomorphisms, using the existing genuine conjugation theorem. This proves submersion of the entire original chosen lift. If the general conjugation theorem chooses an equivalent complement, its actual finite-dimensional product equivalence internally gives rank r; the true complement-coordinate isomorphism fixes the complement to $\mathbb{C}^r$. No comparison, local chart, pointwise smoothness or global smoothness conclusion is an input. □

Theorem 217.3 (Arbitrary original dimensions, including empty sources)**.** *For a genuine submersion $f : Y^m \to X^n$ of ANY original finite dimensions and ANY genuine blowup choices of C and its literal preimage, the internally generated entire universal lift is a genuine global submersion.*

Proof. If its actual source blowup is empty, the submersion-of-complement property with the zero complement has no points to check, and gives the claimed global submersion. Otherwise select one actual source blowup point and its original projected point in Y. The actual original submersion chart there contains a linear product equivalence with target $\mathbb{C}^n$. Its true finite-dimensional

projection is surjective, so internally $n \le m$. Choose the actual natural difference r with $m = n+r$, and apply the preceding theorem to these SAME original manifolds/maps/centres/blowup choices. Thus no dimension bound or nonempty-source assumption is supplied. In particular no $m \ge n$ assertion is made for a globally empty source. □

Theorem 217.4 (Original geometry produces the actual complete smooth lift output)**.** *ONLY a genuine global submersion $f : Y^m \to X^n$ and a primitive genuine target blowup $\pi : D \to X$ internally produce a genuine source blowup of the literal preimage centre and its true holomorphic GLOBAL SUBMERSION lift. It satisfies the entire original projection square, the exact pullback identity for the WHOLE original centre ideal, and the literal centre-preimage divisor identity.*

Proof. Use the previously proved actual source-centre producer and genuine analytic blowup existence to choose the source blowup internally. Actual target universal uniqueness gives its whole holomorphic lift. The preceding theorem proves this SAME produced map is a global submersion of the actual original dimensions. The existing genuine coherent whole-ideal pullback composition and original vanishing-ideal preimage dictionary identify its ENTIRE pulled-back target blowup ideal with the actual source blowup ideal. The whole projection identity gives the literal centre-preimage set identity. All these conclusions concern the SAME internally selected source blowup and SAME true unique lift. No source, ideal, comparison, smoothness, Cartesian or result certificate is supplied. The Cartesian universal property for arbitrary compatible cones, higher invariant/canonical-driver functoriality and general G2/G3/G8 remain separate OPEN targets. □

218 Actual compatible target pairs have genuine blowup preimages

Lemma 218.1 (Actual target direction chooses the matching pair of charts)**.** *Let $f : Y^{n+r} \to X^n$ be a genuine submersion at y, and let an actual flattening Φ at fy carry the whole local centre to the coordinate plane of dimension k, with $\Phi(fy) = 0$. For an actual target blowup point z with $z.\mathrm{base} = fy \in C$, internally produce matching actual blowup charts c, c', an actual product open $R \times S$, and an actual source blowup point p. They satisfy $z \in c.\mathrm{source}$, $p \in c'.\mathrm{source}$, $p.\mathrm{base} = y$, and $\mathrm{flatChartMap}(c, c')(p) = z$, with the exact product targets and whole original base projection square. No chart, pivot or compatible-point output certificate is supplied.*

Proof. The actual target direction is not tangent to the true centre. The genuine flattening tangent dictionary therefore supplies a normal index $j \ge k$ whose differential on this actual direction is nonzero. Internally adapt the actual submersion coordinates to this SAME Φ and shrink them to the actual product target $R \times S$, with $0 \in R$, $0 \in S$ and source centre flattened in fibre-first coordinates. Restricting Φ to S preserves its entire coordinate functions and the actual chosen direction differential. Hence the monomial target chart with this actual j contains z: its base is the actual central point, and its true extended pivot value is nonzero. The source chart uses the corresponding actual normal index after block rotation. The genuine monomial/block formulas prove its target is exactly $\mathrm{split}^{-1}(R \times c.\mathrm{target})$, with the whole inverse base projection square. Set

$$w = \mathrm{split}^{-1}(0, c.\mathrm{toFun}(z)), \qquad p = c'.\mathrm{invFun}(w).$$

The actual chart target and $0 \in R$ give $w \in c'.\mathrm{target}$ and $p \in c'.\mathrm{source}$. Since $z.\mathrm{base} = fy$, the monomial target base coordinates are exactly $\Phi(fy) = 0$. The genuine source monomial/block identities therefore give monomial source coordinates zero. The actual source flattening sends its centre y to zero and has its actual inverse there, so $p.\mathrm{base} = y$. The true right inverse of c' and the true left inverse of c give the displayed actual map identity, including points on the entire exceptional divisor. No parameter inversion occurs. □

Theorem 218.2 (The actual universal lift hits every compatible exceptional pair)**.** *For ONLY a genuine global submersion $f : Y^{n+r} \to X^n$, a closed smooth nowhere-dense centre C, and an actual pair y, z with $fy = z$.base $\in C$, the internally generated canonical source blowup and actual universal lift G_0 have an actual point p with p.base $= y$ and $G_0 p = z$.*

Proof. The original closed smooth centre supplies the genuine flattening and its actual central zero value internally. Apply the preceding producer to the actual target direction at z, with the true original submersion at y. The existing entire universal-chart equality identifies G_0 with the internally produced flat-chart map on the entire source chart. The actual point belongs to this source, so the preceding actual map equality gives $G_0 p = z$. The original submersion internally supplies the source closed smooth nowhere-dense centre and both genuine blowups. No exceptional surjectivity or compatible-point conclusion is an input. □

Theorem 218.3 (Every actual compatible pair has a source blowup point)**.** *For the same ONLY original data and EVERY actual pair y, z satisfying $fy = z$.*base*, an actual source blowup point p satisfies p.base $= y$ and $G_0 p = z$.*

Proof. If $fy \in C$, use the preceding exceptional producer. Otherwise the actual source point is the genuine off-centre point over y. The true universal projection square gives the base of $G_0 p$ equal to $fy = z$.base. Off the centre the actual blowup point is uniquely determined by its base, so $G_0 p = z$. The two cases cover every actual compatible pair without a chosen output point or density hypothesis. □

Theorem 218.4 (Compatible pairs for arbitrary genuine blowup choices)**.** *ONLY original f and ANY primitive genuine target/source blowup choices internally give, for every actual pair y, z with $fy = \pi z$, a source point p with $\pi' p = y$ and $Gp = z$, where G is THEIR actual universal lift.*

Proof. Internally construct the true identity-base comparison biholomorphisms a, b with the canonical blowups and the entire lift formula $G = a^{-1} G_0 b$. The actual point az has base $\pi z = fy$, so the preceding producer gives P with canonical source base y and $G_0 P = az$. Set $p = b^{-1} P$. The genuine whole comparison projection square gives $\pi' p = y$, and the entire conjugation formula gives $Gp = a^{-1}(az) = z$. Comparisons and the canonical compatible-point witness are internally produced, not hypotheses. □

Theorem 218.5 (Actual surjective original submersions give surjective lifts)**.** *If the original genuine submersion is surjective, THEIR actual universal lift for ANY genuine blowup choices is surjective.*

Proof. For any actual target blowup point z, original surjectivity supplies an actual y with $fy = \pi z$. The preceding producer supplies p with $Gp = z$. No surjectivity of a produced lift is supplied. Empty centres and empty source/target sets obey the same genuine quantified statement; no nonexistent exceptional chart is selected. Joint-point uniqueness, holomorphic lifting of arbitrary cones, and the full Cartesian universal property remain separate OPEN targets, as do the general higher invariant and driver. □

219 Genuine holomorphic blowup lifts from original submersions

Let $f : B^m \to A^n$ be an actual complex submersion, without surjectivity, and let $\pi : Z^n \to A$ be any genuine blowup along its actual original centre C. The centre's actual closed-submanifold and nowhere-dense properties are intrinsic parts of the genuine input blowup. Neither a pulled-centre geometry, ideal equality, source blowup, lift, invertibility conclusion, Cartesian conclusion, smoothness of the lift nor naturality certificate is original input. This foundation constructs genuine source blowups and unique actual holomorphic lifts; it deliberately does not claim smoothness of the lifted map or the full unrestricted holomorphic Cartesian universal property.

Definition 219.1 (Actual original-input holomorphic blowup lift output)**.** The produced output consists of an actual complex m-manifold Z', a genuine blowup $\pi' : Z' \to B$ along the literal centre $f^{-1}C$, and a genuine holomorphic map $g : Z' \to Z$ with actual square $\pi g = f\pi'$. It includes the entire coherent-ideal identity $g^*(\pi^*\mathcal{I}_C) = \pi'^*\mathcal{I}_{f^{-1}C}$ and the literal centre-preimage equality $g^{-1}(\pi^{-1}C) = \pi'^{-1}(f^{-1}C)$. These are produced conclusions. Centre-preimage divisors are not identified with the differential exceptional locus: a Cartier centre has a biholomorphic blowup and need not have a differential exceptional locus.

Theorem 219.2 (Actual original submersion produces its true nowhere-dense source centre)**.** *The literal inverse image $C' = f^{-1}C$ is a genuine closed complex submanifold with empty interior, internally produced from original f and the genuine target blowup.*

Proof. The actual target closed submanifold and the actual source submersion charts give the true pulled closed submanifold with the internally produced dimension shift $m - n$. For nowhere density, a nonempty source open contained in $f^{-1}C$ would have a nonempty actual open image under f. Genuine local holomorphic sections of the original submersion internally prove that f is open, so this true image is a nonempty ambient open contained in C, contradicting the genuine target blowup's $\operatorname{int} C = \varnothing$. Thus source interior is empty without surjectivity, connectedness, compactness or a supplied source dimension/nowhere-dense conclusion. Empty source, empty centre and zero dimension use the same actual geometric statement. □

Theorem 219.3 (A genuine source blowup is internally constructed)**.** *Only original f and the actual target blowup produce an actual source manifold and a genuine blowup along the literal C'.*

Proof. Use the preceding true closed/nowhere-dense source centre in the already proved genuine analytic blowup existence construction. Package its actual type, topology, complex charts and genuine complex manifold structure as the actual complex m-manifold Z'. The resulting actual projection has the genuine blowup property along EXACTLY $f^{-1}C$. Its pulled whole source vanishing ideal is invertible by that true blowup property itself. No source manifold, source blowup, nonzero generator or existence result is caller data. In particular empty source or centre does not require an artificial nonempty coordinate family. □

Theorem 219.4 (The entire original ideal supplies true target-universal invertibility)**.** *For every genuine actual source blowup π' along this literal source centre, $(f\pi')^*\mathcal{I}_C = \pi'^*\mathcal{I}_{C'}$ as ENTIRE ideal sheaves, and the former actual ideal is genuinely invertible.*

Proof. The genuine original closed submanifold and original submersion internally produce the entire stalkwise identity $f^*\mathcal{I}_C = \mathcal{I}_{f^{-1}C}$ from actual lifted coordinate generators, with the unit presentation off the original centre. The original centre ideal is genuinely coherent. Compose true whole holomorphic finite-span pullbacks along the actual f and actual π' to obtain the written whole ideal identity. The right side is invertible by the genuine source blowup property; transport that actual property across the ENTIRE ideal equality. This argument does not pull an arbitrary invertible ideal through an arbitrary map: only the source blowup's genuine original-centre invertibility is used. □

Theorem 219.5 (Original geometry produces its unique genuine holomorphic lift and whole ideal square)**.** *Every genuine source blowup along C' has a unique holomorphic lift g with the actual original projection square. In particular the internally constructed actual source blowup gives the full output above, with whole ideal and literal centre-preimage dictionaries.*

Proof. The true original composite $f\pi'$ is holomorphic. The preceding internally proved ENTIRE pulled-ideal invertibility supplies the genuine target blowup universal property; it produces an actual holomorphic g and proves uniqueness among ALL actual holomorphic maps with this SAME projection square. To obtain the whole ideal square, the actual holomorphic maps g, π, f, π' and original coherent $\mathcal{I}_C$ give true whole finite-span pullback composition. Rewrite the real

commutative square and the preceding actual whole ideal transport to identify $g^*(\pi^*\mathcal{I}_C)$ with the literal actual source blowup ideal. Its invertibility is therefore also internally proved. Taking literal inverse images in the true square gives the exact written centre-preimage identity. This last equality concerns the centre preimage; no false identification with differential exceptional locus is made. □

Theorem 219.6 (Actual primitive blowup choices give genuine comparison and lift coherence)**.** *For two genuine source blowup choices along the SAME actual C' and two genuine target choices along the SAME actual C, internally produced genuine biholomorphisms over their original bases compare them. The corresponding actual holomorphic lifts obey the true comparison square and its identity/inverse/composite cocycles.*

Proof. Apply genuine blowup uniqueness over the identical actual centre to obtain the actual holomorphic inverse comparison maps, their real projection equalities and the genuine biholomorphism. For either composite of an actual source comparison, actual produced lift and actual target comparison, the true projection squares give the SAME original base map $f\pi'$. The source blowup's actual original-centre pullback is invertible; the preceding full original ideal dictionary therefore supplies the SAME target universal uniqueness hypothesis internally. Both actual composites are genuinely holomorphic, so target uniqueness proves their true equality. Identity, inverse and composite comparison maps preserve the SAME original projections; genuine uniqueness gives their exact cocycles. No chosen lift, comparison, Cartesian or naturality conclusion is caller data. □

Theorem 219.7 (Two original submersions internally supply genuine composite lift coherence)**.** *For original actual submersions $f : B \to A$ and $h : P \to B$, generate genuine source blowups successively along $f^{-1}C$ and $h^{-1}(f^{-1}C)$. The true composite of the two actually produced holomorphic lifts agrees with the direct target-universal lift from that SAME actual final source blowup along the literal composite base map fh; different genuine source choices are related by the preceding actual comparison.*

Proof. Apply actual pulled closed-centre geometry and nowhere-density twice, with actual source opens and original submersions. This internally constructs the true successive source blowups. Whole vanishing-ideal transport for BOTH original submersions and true coherent finite-span composition prove $(fh)^*\mathcal{I}_C = \mathcal{I}_{h^{-1}(f^{-1}C)}$ without requiring a caller composite-submersion certificate. The actual final source blowup therefore gives genuine invertibility of the direct target pullback, and the target universal property produces a direct holomorphic lift. The actual successive lift squares compose to the SAME original map $fh\pi''$, and the composite lift is holomorphic. True target universal uniqueness identifies it with the direct lift. Literal inverse-image composition identifies the final source centre with $(fh)^{-1}C$. Genuine final-source choice comparison gives the alternative-model statement. No higher algorithm or lifted-submersion conclusion is assumed. □

Actual next geometric frontier. A holomorphic compatible cone entirely inside an actual centre-preimage divisor can pull its nonzero invertible ideal back to the zero ideal. Thus the weak ideal-invertibility blowup universal property alone does NOT prove the unrestricted holomorphic Cartesian square for arbitrary cones. Genuine local product blowup charts, their actual atlas transitions and genuine choice comparison are still needed to produce smoothness of the actual lifted map and the unrestricted Cartesian universal property. General higher-invariant functoriality, noncompact general termination, old-driver identification, unrelated ideal admissibility, whole G3/G8 and formal/native/publication remain OPEN.

220 Actual holomorphic cones lift locally even entirely in the exceptional divisor

Lemma 220.1 (The actual matching-chart lift of every whole compatible cone)**.** *Consider actual source/target blowup charts c', c, with true block normal-index and pivot identities and the whole source-to-target flattening coordinate identity on the source flat domain. Genuine matching adapted charts are a special case. On an actual manifold P, let $u : P \to Y$ and $v : P \to \mathrm{Bl}_C X$ be holomorphic, with $fu = \pi v$. On the genuine open where u lies in the source flat domain and v lies in the target monomial chart, internally construct a holomorphic map into the actual source blowup chart, whose projection is u and whose flat-chart map is v. If the actual parameter projection maps the source chart target into the target chart target, two points in that SAME source chart with equal base and flat-chart image are equal. Thus the lift is unique among maps with image in that SAME chart and these whole identities.*

Proof. Write $U = \Psi e_y u$ and $V = c.\mathrm{toFun}\, v$ on this true open. Both are holomorphic there, by their genuine chart and flattening domains. The actual whole source-coordinate dictionary and the original compatible square give $\mathrm{drop}\, U = \sigma_j V$ at every point. Form the literal parameter map $w = \mathrm{split}^{-1}(\mathrm{fibre}\, U, V)$. The actual block monomial identities and original square give $\sigma_j' w = U$. The original u lies in the source flat domain, so U belongs to its genuine flattening target; therefore the entire image of w lies in the actual source monomial-chart target. Genuine continuous-linear maps make w holomorphic. The actual block/monomial identities give $\sigma_j' w = U$ and $\mathrm{drop}\, w = V$. Compose with the genuine entire holomorphic inverse of the source blowup chart to obtain g. The actual flattening inverse identities prove $\pi' g = u$; the true target chart inverse gives $\mathrm{flatChartMap}\, g = v$. This uses no division by the exceptional parameter. For two points in the SAME chart, equal source projections give equal monomial parameters. The fibre projection commutes with this genuine block monomial, so their fibre parameters agree. The parameter-projection target containment and true target chart right-inverse turn equality of the flat-chart images into equality of the dropped parameters. The actual linear split is injective, so these two parameter vectors are equal. The genuine source-chart left-inverse gives equality of the actual points, hence uniqueness of maps with these whole identities. This is uniqueness inside an actual chosen chart, not yet global joint-point uniqueness. □

Theorem 220.2 (Original geometry internally lifts every cone near an exceptional point)**.** *ONLY original global submersion $f : Y^{n+r} \to X^n$, closed smooth nowhere-dense centre C, and genuine holomorphic compatible cone $u : P \to Y$, $v : P \to \mathrm{Bl}_C X$ internally give a local holomorphic lift near any actual $p_0 \in P$ with $f(up_0) \in C$. The neighbourhood is a true open containing p_0, its lift goes into the actual source blowup, and BOTH whole projection identities hold on that entire neighbourhood.*

Proof. Set $y = up_0$ and $z = vp_0$. The genuine original square identifies the actual base of z with fy. Internally choose the genuine centre flattening at fy. The actual non-tangent direction of z supplies its true target pivot. The actual submersion chart producer supplies the SAME matching source chart and product target; restricting the target flattening preserves the selected pivot coordinate, so the target monomial chart contains z. The source flattening contains y by its true central inverse identity. The intersection of the inverse images of these two actual open domains under the original holomorphic u, v is therefore a true open neighbourhood of p_0. Apply the preceding actual entire join construction on this SAME open. Its source projection is u. Entire universal-chart equality identifies the actual canonical lift G_0 with the flat-chart map on the entire source chart, hence $G_0 g = v$ throughout this SAME neighbourhood, including when the ENTIRE original cone maps into the exceptional divisor. No chosen lift, product-domain, chart-coverage, local-lift or Cartier output certificate is a caller hypothesis. □

Theorem 220.3 (The original off-centre cone has its actual local lift)**.** *For the same original data and an actual point with $f(up_0) \notin C$, a true open neighbourhood of that point has a holomorphic source blowup lift satisfying BOTH entire original cone projection identities.*

Proof. Use the actual open inverse image of the source centre complement under u. It contains p_0 by the original off-centre assumption. The source primitive blowup internally gives the true complement biholomorphism, holomorphic in both directions on its whole actual complement. Compose its genuine inverse with the corestricted original u to construct the actual holomorphic lift g. Its projection is u on the entire open. The true universal square and the original compatible cone square give equal bases for $G_0 g$ and v. Those bases avoid C, so actual off-centre point uniqueness gives $G_0 g = v$ on the entire open. Empty complements require no nonexistent witness: the specified actual point supplies the true nonempty neighbourhood when needed. □

Theorem 220.4 (Every actual cone is locally lifted for arbitrary genuine choices)**.** *ONLY original submersion f and ANY primitive genuine target/source blowup choices, together with genuine holomorphic compatible cone u, v, internally give a true local holomorphic source lift at EVERY actual point of P, with both whole original projection identities. The cone may lie entirely in the exceptional divisor; its pulled-back centre ideal is not required invertible.*

Proof. For arbitrary original dimensions m, n, the specified actual reference point gives $y = up_0$. The actual original submersion at y internally proves $n \leq m$, hence $m = n+r$ for a genuine natural r. No caller dimension bound is assumed, and an empty test manifold has no reference point to lift. Internally construct the actual identity-base comparisons a, b with the canonical blowups and their whole lift formula $G = a^{-1} G_0 b$. The actual canonical cone is u, av and is holomorphic; its whole compatible square follows from the original square and the actual comparison projection. At each original point, centre membership decides the preceding exceptional or off-centre producer. Let g_0 be its actual local canonical lift on its actual produced neighbourhood. Then $g = b^{-1} g_0$ is holomorphic there, and the genuine comparison projection and whole conjugation identities give $\pi' g = u$ and $Gg = v$ on that entire neighbourhood. Every chart, neighbourhood, comparison and lift is internally generated. This local existence theorem does NOT assume or claim global joint-point uniqueness, overlap equality for arbitrary chosen lifts, a glued global lift, or the full Cartesian universal property. Those are the next independent mathematical obligations; higher invariant/canonical-driver functoriality is also still OPEN. □

221 Actual pivot correspondence and global joint-point uniqueness

Lemma 221.1 (Repivoting the same actual chart preserves the whole coordinate geometry)**.** *Replace only the normal pivot in an actual flattening blowup chart. Its base domain and all unnormalized coordinate functions remain the same. Repivoting to the original pivot returns the identical entire chart record. For actual matching source and target charts with the genuine block normal index and pivot dictionary, let the source chart parameter project into the target chart target. At every actual source-chart point q, with actual flat-chart image z, EVERY normal coordinate i has nonzero extended value at z exactly when its corresponding rotated source coordinate has nonzero extended value at q. Thus repivoting to an actual target pivot containing z produces a true source pivot chart containing q.*

Proof. The repivoted chart is the same literal flattening record with only its pivot and its normal-index proof replaced; its base domain and every raw coordinate are definitionally unchanged. For the old pivot, both extended values are nonzero by actual chart membership. For another normal index, the genuine target right-inverse gives target parameter equal to the drop of the source parameter. Evaluating coordinate i gives equality of the target and rotated source parameter entries. The actual monomial chart formulas write these entries as extended-value numerator

divided by their respective old-pivot extended value. Both denominators are genuinely nonzero. Therefore the numerators are nonzero simultaneously. This uses only the ordinary entire chart formulas, including at every point of the exceptional divisor, not a density argument away from it. The source base remains in the identical domain, hence its new chart membership follows. □

Lemma 221.2 (Original geometry produces one complete matching pivot family)**.** *Only the original submersion at an actual point and a genuine centre flattening internally produce a SAME source flattening, SAME target flattening, and matching monomial charts for every genuine normal pivot. At a specified actual exceptional source-blowup point, a genuine member contains that point. Every member has the true product target, whole inverse-point square and genuine block-index dictionary.*

Proof. Produce the genuine product neighbourhood $R \times S$ and source flattening Ψ from the original submersion and centre flattening. Restrict the SAME target flattening to S; its whole coordinate functions are unchanged. Use the source point's actual non-tangent direction and the actual tangent flattening dictionary to find a nonzero source normal extended coordinate. Its index is genuinely $r + j$ for a unique target normal index j. This member is a true source chart containing the specified point. Repivot both SAME flattenings at any genuine target normal index. The actual block monomial identity gives the product target for each member. The original whole inverse-flattening square gives its entire point square after that SAME block monomial identity. No chart family, pivot, product coverage or comparison output certificate is supplied. □

Theorem 221.3 (Canonical original blowup points are uniquely determined by BOTH projections)**.** *Only original global submersion, genuine closed smooth nowhere-dense centre and two actual source-blowup points with equal source base and equal actual canonical lift image imply equality of those points, with no chart input.*

Proof. Off the source centre, the actual blowup has a unique point over its base. At an exceptional common base, internally choose the genuine target centre flattening and preceding SAME matching family containing the first point. The second point's actual non-tangent direction in this SAME source flattening produces its own member. Entire universal-chart equality puts both actual canonical images in their true corresponding target charts. Their images are equal. The preceding whole extended-coordinate support correspondence in the second point's own member therefore puts the second point in the first point's SAME source chart. Actual whole chart equality and the original two projection equalities now give equal bases and equal flat-chart images in one SAME chart. The true split-coordinate joint-point uniqueness from the local-cone producer forces equality. Empty centres and zero-dimensional cases use the actual off-centre branch; an exceptional case supplies the genuine normal pivot internally. □

Theorem 221.4 (Arbitrary genuine choices have the actual unique compatible point)**.** *Only original arbitrary-dimensional submersion and genuine primitive source/target blowup choices internally give joint injectivity of their source projection and actual lift. Every genuine compatible pair (y, z) has exactly one actual point p with $\pi' p = y$ and $Gp = z$.*

Proof. For actual source points, the original submersion at their base internally gives $n \leq m$ and a genuine r with $m = n + r$. Produce the actual identity base comparisons with the canonical choices; their whole projection and conjugation formulas turn equal original projections into equal canonical projections. Apply the preceding genuine canonical joint-point uniqueness and invert the source comparison. Existence for every genuine compatible pair is the existing actual internally produced compatible-point theorem; uniqueness follows from the just-proved joint injectivity. The actual compatible base point also supplies the dimension bound internally. Empty manifolds require no fabricated point. This is a pointwise universal property, not yet global holomorphic gluing of every cone, full Cartesian manifold universal property, or higher-invariant driver identification. □

222 Actual ambient dimension shifts of the original canonical value types

The original invariant value types and gamma formulas are unchanged. For an actual smooth map $Y^m \to X^n$, its fibre dimension is $t = m - n$. Old monomial keys have first coordinate $n - |K|$, so their actual pulled values have first coordinate $(n - |K|) + t$. Their sum and complete colex indexed set stay literal. Numerical equality across dimensions is therefore not the correct desired conclusion.

Lemma 222.1 (The actual ambient key and bottom-preserving value shift)**.** *Construct D_t on the original monomial value triples by adding t to their first coordinate and retaining the entire second coordinate and whole colex member set. Its extension to WithBot retains bottom. Both maps are strictly increasing. For every actual indexed K with $|K| \le n$, $D_t(\mathrm{key}_n(a, K)) = \mathrm{key}_{n+t}(a, K)$.*

Proof. Addition of t is strictly increasing and reflects equality on natural numbers. Split the literal lexicographic order into unequal first coordinates and equal first coordinates; the former is preserved by addition, while the latter retains precisely the entire original remaining lexicographic pair. Bottom remains the unique least WithBot element, so this extension is also strictly increasing. For an actual key, the bound $|K| \le n$ gives $(n + t) - |K| = (n - |K|) + t$. The other two original coordinates are identical, proving the entire key equality. □

Theorem 222.2 (The whole original feasible maximum and actual smooth gamma)**.** *If the actual through-member set S has $|S| \le n$, the entire original finite feasible maximum in dimension $n + t$ is WithBot.map(D_t) of the whole original maximum in dimension n. Consequently the literal gammaAlong of an actual smooth pullback of an original coherent marked record with SNC boundary is this same actual dimension shift of its original gammaAlong.*

Proof. The whole feasible family is unchanged because its subsets, exponent sums and mark are unchanged. Each actual feasible K is contained in S, so its true cardinality is at most n and the preceding complete key formula applies. If the feasible family is empty, both actual maxima are bottom. Otherwise choose an actual original top set. Strict monotonicity preserves domination of every feasible key, so that same actual indexed set gives the pulled maximum and its exact dimension shift. For a genuine smooth map, the previously proved original exponent and indexed-member dictionaries supply precisely the same entire feasible family. Original SNC bounds the actual through-member cardinality internally, and the actual submersion supplies $m = n + (m - n)$. Apply the finite maximum formula with these original data. No gamma/value comparison is a caller premise. □

Theorem 222.3 (Explicit strictly increasing maps of all original value types)**.** *For each actual t, k, construct a strictly increasing map $F_{t,k} : \mathrm{ValTy}(k) \to \mathrm{ValTy}(t + k)$. At $k = 0$ use the natural strict seed: identity when $t = 0$, and the general value with outer coordinate equal to its input and monomial branch bottom when $t > 0$. At $k + 1$, preserve the original nonmonomial-order coordinate, apply the bottom-preserving D_t to the monomial branch, and retain the original Step One level while applying WithTop.map($F_{t,k}$) to its lower value. All actual top, bottom, branch and indexed colex information is retained.*

Proof. The seed is strictly increasing in its literal outer coordinate. For the successor, first lift $F_{t,k}$ through the actual WithTop lower entry and retain the original Step One level; the two literal lexicographic cases prove strict monotonicity. Lift that map through the general branch's WithTop and use the preceding WithBot monomial map on the other branch. The two equal-branch cases preserve order by these proved maps; every monomial branch remains below every nonmonomial branch. Retain the outer nonmonomial order and use its two literal lexicographic cases. Induction constructs the actual function and proves it strictly increasing at all k. No recursive invariant comparison property is assumed. □

Corollary 222.4 (Actual value order reflection and genuine monomial-stage normalization)**.** *The constructed original value maps reflect strict order, weak order and equality. At every original actual Cartesian test-history stage, the whole actual gammaAlong is the explicit ambient shift of the original stage gammaAlong, with all original indexed members retained.*

Proof. Strict monotonicity between linear orders reflects each comparison. At an actual Cartesian test stage, original well-formedness and original testness internally supply coherent stage ideals and the entire SNC stage boundary. The genuine whole marked-stage dictionary and the preceding original smooth gamma producer give the exact normalized monomial value. These are actual record/finite maximum constructions, with no stage-value certificate. □

These explicit maps and actual monomial value formulas are prerequisites for the full higher smooth comparison. They do not assert that the entire actual invFamily values are already related by these maps, nor prove the independent chosen-contact comparison, canonical driver or sheaf flatness.

223 Internally produced smooth Cartesian pullback of every genuine finite blowup tower

Definition 223.1 (The actual non-surjective stage history)**.** Let $f : Y \to X$ be a genuine submersion of arbitrary complex dimensions. Define an OUTPUT history relation on actual source and target finite blowup towers and their actual last-stage map. Its empty case is the original submersion and the two literal empty towers. Its step case extends a previous actual history with a genuine target blowup along C, a genuine source blowup along EXACTLY the previous actual map's preimage of C, and the internally defined actual blowup lift of these primitive choices. There is no surjectivity field, no prescribed final map, no Cartier-only condition and no caller square or Cartesian certificate in the step. Every such actual history has a genuine smooth last-stage map, exactly equal source/target tower lengths, and its original whole projection square.

Proof. The empty map is precisely f. At a step the original arbitrary-dimensional smooth blowup theorem proves that the SAME internally defined actual lift is a submersion. Thus its predecessor smoothness is actual geometry and not a supplied final result. Both lengths increase by one, including every literal empty or repeated centre. For the whole square, the actual step identity $\pi G_1 = G\rho$ followed by the predecessor identity gives $T_{\mathrm{new}} G_1 = f Q_{\mathrm{new}}$. This proves all three dictionaries by literal induction. The relation is used as a produced output below, not as an assumption replacing the original-only existence theorem. □

Theorem 223.2 (Actual step squares paste into the whole tower Cartesian square)**.** *Every actual history has the WHOLE holomorphic square universal property for its entire composite projections. For every actual finite-dimensional test manifold in the same universe and every two genuine whole holomorphic maps satisfying the original projection square, there is exactly one whole holomorphic map to the produced source last stage with both projection identities. The existing whole-square notation is reused; its old surjectivity-dependent stage-history predicate is not assumed.*

Proof. In the empty case the unique map is the original source-side test map, and its target-side equality is the given whole square. At a genuine step, first apply the internally obtained predecessor universal property to the source test map and the target test map followed by the genuine target blowup projection. This gives a unique WHOLE holomorphic predecessor map. Its actual lift-image identity supplies the true cone equation for the new step. The genuine single-blowup Cartesian theorem internally gives a unique WHOLE holomorphic map to the new source blowup, even for a test cone wholly in the exceptional divisor. Its source projection identity composed with the predecessor projection gives the desired whole source identity. For any

competitor, its genuine source blowup projection is a competing predecessor lift by the actual step square. Predecessor uniqueness identifies this map, then actual single-step uniqueness identifies the competitor. No invertibility of the test ideal, source/target surjectivity, overlap certificate or final universal-property certificate is supplied. □

Theorem 223.3 (Every literal geometric head retains its actual smooth Cartesian history)**.** *For EVERY natural j, the literal heads obtained by cutting the same last j steps from the source and target towers have an internally obtained actual stage map and actual history. Consequently their complete lengths, whole original projection squares and entire holomorphic universal properties hold. Cutting beyond the true length gives the two original empty towers; all empty and repeated steps remain literal before cutting.*

Proof. Induct on the actual history while allowing arbitrary j. At $j = 0$ use the SAME actual whole last-stage history. For a nonempty step and $j+1$, the literal split definitions remove its final source and target step; the induction hypothesis at j is therefore exactly the required head history. An empty history remains empty for every j, with actual map f. Equal lengths were proved from the real step induction, not from a compression certificate. Apply the preceding dictionary and universal theorems to each internally produced head history. This does not identify any general canonical algorithm or delete a geometrically empty step. □

Theorem 223.4 (Original submersion and original finite target tower internally produce every source stage)**.** *The ONLY inputs are an original genuine arbitrary-dimensional submersion f and a genuine ordinary finite target blowup tower T. Internally produce an actual source last space, an actual source blowup tower Q, a whole actual map G and its actual history. In particular G is smooth,* $\operatorname{length} Q = \operatorname{length} T$*, the original whole square commutes, the entire square is holomorphically Cartesian, and EVERY literal head has its corresponding actual smooth Cartesian square and equal length. No source tower, boundedness, termination, chosen-stage or naturality certificate is an input. The target tower is genuine primitive finite geometry, not asserted to be a canonical resolution tower.*

Proof. Induct on the supplied ordinary finite target tower. For its empty case take Y, its literal empty source tower, and f. At a target step, the induction hypothesis has ALREADY internally constructed the whole previous source tower and actual smooth stage map G. The original source-centre producer applied to this actual G and the genuine target step internally proves the preimage centre is closed, smooth and nowhere dense and provides its actual primitive source blowup. Extend the previous tower by that SAME blowup and take the actual lift. This is exactly the step constructor of the output relation, whose smoothness and square are genuine proved geometry. The preceding three theorems give all stated whole and literal-head outputs. Finite recursion is on the supplied real finite tower, not a hidden termination hypothesis for a general resolution algorithm. Empty initial spaces, dimension zero, non-surjective maps, empty or repeated centres are covered by the same primitive geometry and literal induction. □

Theorem 223.5 (Actual surjectivity internally recovers the old surjective stage history)**.** *If the original f is genuinely surjective, EVERY actual history internally has a surjective actual last-stage map and gives the old* `IsSmoothTowerPullback` *on the SAME literal source and target towers and SAME actual map. This consumes no caller stage-surjectivity or Cartesian certificate. It adds no assertion about whole marked transforms, higher invariant equality or a general canonical driver's chosen centres.*

Proof. Induct on the actual history. Surjectivity of f is the empty case. Given any point in the target step, predecessor surjectivity supplies a genuine source point over its target base. The actual compatible-pair theorem internally gives the unique source-blowup point over this source point and target point, and its actual lift-image identity proves stage surjectivity. This argument works in arbitrary dimensions without imposing a dimension difference as input. For the old

history, use its empty constructor and, at each literal step, the internally recovered previous old history, the primitive blowup choices, the real actual smooth lift, the just-proved actual stage surjectivity, the actual projection-square identity, the literal preimage-centre reflexivity, and the genuine single-step Cartesian universal theorem. Thus every old field is internally proved from original geometry. The prior general separation/colex driver identification, general higher-invariant functoriality and G3/G8 remain OPEN. □

224 Genuine composite submersion charts from the two original maps

Theorem 224.1 (Existing pure holomorphic analytic ingredient, recorded explicitly)**.** *A complex differentiable map on an actual open subset of a finite-dimensional complex normed space, with values in a complex Banach space, is analytic at EVERY point of that open set. This is an existing proved pure ingredient; its Lean source is unchanged, and this new node records its actual use.*

Proof. Fix an actual point c in the actual open domain. Complex differentiability gives continuity. Choose $R > 0$ so that the whole radius-R ball about c lies in that domain and the actual map's norm is bounded there by $\|f(c)\|+1$. The proved complex Cauchy estimates, applied to the actual one-variable restrictions $t \mapsto f(c+ty)$, give continuous symmetric multilinear Taylor coefficients and their genuine geometric majorant on the radius-R/e ball. For any vector in this smaller ball, choose $T = 2R/(R + \|y\|) > 1$; its radius-T line restriction lies in the original ball. The actual one-variable Taylor sum at $t = 1$, with the genuine iterated-line derivative dictionary, is exactly the multilinear Taylor series at $c+y$. The Cauchy bound gives convergence of that SAME power series on the entire smaller ball. Thus the original function, not an assumed series representative, is analytic at c. Since c was any actual point, it is analytic on a neighbourhood of every point of the actual open domain. This is precisely the existing pure analytic proof consumed below; no general smoothness certificate or new Lean interface is inserted. □

Definition 224.2 (Normalized actual target coordinates)**.** At any actual point in a genuine complex manifold, internally construct a whole local holomorphic coordinate equivalence in its preferred model chart whose source is EXACTLY the preferred chart target and whose actual point is sent to zero. No caller coordinate-normalization certificate is supplied.

Proof. Let p be the point's actual preferred chart coordinate and V the actual preferred chart target, an open set containing p. Use $w \mapsto w - p$ on V, with inverse $w \mapsto w + p$ on $(\cdot + p)^{-1}V$. Both maps are genuinely holomorphic, each maps its entire domain into the other domain, and the two inverse equations follow from actual addition and subtraction. The source is literally V, and the actual coordinate p maps to zero. Every zero-dimensional case uses the same true definitions. □

Theorem 224.3 (The actual holomorphic coordinate equivalence gives genuine maximal-atlas charts)**.** *For a true local holomorphic coordinate equivalence whose source lies in the actual preferred chart target, compose its whole partial homeomorphism with that preferred chart. This is a genuine chart with source equal to the actual coordinate domain, containing the original point; BOTH genuine holomorphic directions internally make it a maximal-atlas chart.*

Proof. The actual coordinate equivalence is a true open partial homeomorphism. Its actual preferred-chart composite has the stated source by the literal partial-map composition law. On this entire open source the forward map is the coordinate equivalence composed with the genuine preferred chart. On the entire target the inverse is the preferred chart inverse composed with the coordinate equivalence inverse; the true source inclusion gives every required intermediate actual target-domain membership. Both local holomorphic maps are complex analytic on their actual open domains by the proved holomorphic analytic theorem, hence complex smooth to every order.

The two actual smooth partial-map conditions give true maximal-atlas membership. No caller smooth-chart or atlas certificate is required. □

Theorem 224.4 (Two genuine original submersions produce the actual composite projection chart)**.** *Let $f : Y^{n+r} \to X^n$ and $h : Z^{(n+r)+s} \to Y^{n+r}$ be genuine submersions at the actual compatible points. Their composite is a genuine submersion there, with actual complement $\mathbf{C}^r \times \mathbf{C}^s$. For arbitrary three dimensions the same genuine pointwise conclusion is internally produced from the two original pointwise inputs; no composite submersion or derivative certificate is assumed.*

Proof. Internally construct the preceding normalized actual coordinates Φ on X at $f(hz)$. Apply the already proved actual lifted-coordinate producer to the ORIGINAL f and this SAME Φ. It gives a true local holomorphic equivalence Ψ on Y, normalized at hz, with its actual whole source inside the preferred chart target and the ENTIRE dictionary that its first n coordinate entries equal $\Phi(fy)$. Apply the SAME producer to the ORIGINAL h and this SAME Ψ. It gives a true local holomorphic equivalence Ω on Z, normalized at z, whose first $n+r$ entries equal $\Psi(hw)$ on its ENTIRE actual domain, and whose domain maps into the actual Ψ domain. The latter maps into the actual Φ domain. Thus on the ENTIRE Ω chart domain, the first n entries in the two nested coordinate blocks equal $\Phi(f(hw))$. The actual continuous complex linear equivalence first splitting $\mathbf{C}^{(n+r)+s}$ into $\mathbf{C}^{n+r} \times \mathbf{C}^s$, then splitting the first block and reassociating, identifies the model with $\mathbf{C}^n \times (\mathbf{C}^r \times \mathbf{C}^s)$. Its projection is EXACTLY those genuine first n entries. Use the preceding genuine maximal-atlas charts for Ω and Φ; their actual domain and inverse dictionaries give the entire extended-chart projection equation. This is precisely the true submersion-at-point definition. For arbitrary dimensions, the actual two original points internally give $n \le m \le \ell$. Write $m = n + r$ and $\ell = m + s$ and apply the already produced point chart. No ambient global inequality or finite-dimensional derivative-output certificate is supplied. □

Theorem 224.5 (Only the two original submersions produce genuine global composite smoothness)**.** *For arbitrary three complex dimensions and possibly empty or non-surjective original maps, TWO genuine global submersions internally imply that their composite is a genuine global submersion. The original maps and their genuine submersion hypotheses are the ONLY inputs.*

Proof. If the original source is nonempty, choose one actual point. The genuine original submersions internally force the actual dimension inequalities, so fix $m = n + r$ and $\ell = m + s$. At EVERY actual source point apply the preceding true two-map coordinate construction with the SAME fixed complement $\mathbf{C}^r \times \mathbf{C}^s$. This gives genuine global submersion with that fixed complement, hence genuine global submersion in the ordinary definition. If the actual source is empty, the pointwise requirement with fixed zero-dimensional complement is vacuous; no false global dimension inequality is asserted. Nothing requires original-map surjectivity. This provides the actual missing composite smoothness ingredient for ordinary finite Cartesian tower composition; it does not assert a higher canonical driver, arbitrary restricted controlled quotient naturality, general noncompact termination, whole G3/G8 or native publication. □

225 Actual open submersions and whole original contact coefficient restriction

Lemma 225.1 (Actual open inverse and restricted maximal charts)**.** *For an actual open manifold U and a point witnessing its nonemptiness, the actual open partial inclusion inverse is holomorphic on its true target. Every genuine maximal ambient chart restricts by that actual open inclusion to a genuine maximal chart of U; both actual coordinate maps are retained.*

Proof. The composite of the actual inclusion with its inverse equals the identity on the true open target, by the two inverse identities. It is holomorphic there. The exact open-subtype holomorphicity dictionary therefore gives holomorphicity of the inverse itself. A restricted chart

is the actual open partial inclusion followed by the original chart. Its forward map is the original holomorphic chart composed with the holomorphic inclusion; its inverse is the original chart inverse followed by the actual open inverse. On the true source and target of the composed partial homeomorphism the respective maps land in the next true domain. Ordinary composition thus gives both genuine holomorphic maps, which are exactly the maximal-atlas criterion. No chart or desired-atlas certificate is supplied. There is no nonemptiness requirement for an empty-open theorem below: at each actual point the point itself supplies the temporary nonemptiness witness. □

Theorem 225.2 (True original submersion restricts with the SAME actual complement)**.** *For an original genuine submersion $f : Y^m \to X^n$ and ANY actual target open U, internally form the ENTIRE actual preimage open $V = f^{-1}U$ and the actual map $f_U : V \to U$. It is a genuine submersion with the SAME globally chosen original complement, also for unequal dimensions, nonsurjective maps and empty opens. No open-map submersion or Cartesian/chart certificate is an input.*

Proof. Continuity of f internally makes V open; actual point membership makes f_U well defined. At an actual point of V, take the genuine original submersion charts and actual linear product decomposition. Restrict the source chart to V and the target chart to U by node one. Their actual partial inverse identities retain the same linear decomposition and the same coordinate projection. The original source chart maps into the original target chart, and every point of V maps into U, so the new source maps into the new target. At every point of the actual restricted source chart target, its true inverse is the original source inverse, with its guaranteed membership in V; the original projection equation therefore gives the ENTIRE new written-in-charts equation. The genuine submersion chart constructor gives the same-complement witness at every actual source point. The original globally chosen complement is constant across these witnesses, so this is a true global submersion, without choosing incompatible pointwise complements. Empty V has no point at which fictitious charts are demanded. □

Theorem 225.3 (Whole literal marked records on actual preimage opens)**.** *For ANY original coherent marked datum M, the actual pullback of its literal open restriction to U along f_U equals the literal open restriction to V of its actual pullback along f, in ALL fields N, E, I, d. Empty and repeated boundary entries are retained; every mark is allowed.*

Proof. The actual map square commutes pointwise by subtype membership. Entire coherent ideal pullback is functorial in arbitrary dimensions: literal finite original function spans compose by true holomorphic composition, then the reverse span inclusion expands every actual intermediate generator. Apply this whole composition theorem on both sides of the actual open map square to obtain the ENTIRE ideal equality. The carrier preimages are literally equal by membership. Mapping the actual boundary list twice gives exactly mapping the composite once, including every empty or repeated entry; the marks stay unchanged. These are exactly the four marked-record fields, not an assumed record comparison certificate. □

Theorem 225.4 (Whole coefficient ideals reduce on the actual original contact)**.** *Given a true original closed smooth contact carrier INSIDE U, form the actual contact-reduced coefficient record by taking the entire relative coefficient ideal of the literal original marked datum on U, restricting its whole ideal to that contact by the original* restrictOn*, changing its carrier to that contact, and retaining its literal boundary and coefficient mark. Its actual pullback along f_U equals the corresponding source construction on V and the literal preimage contact, in ALL four fields. No source reduced-ideal or coefficient comparison is input.*

Proof. The actual open submersion and genuine original smooth carrier first give the whole relative iterated derivative identity and the whole coefficient marked identity, for ANY original mark including zero. The actual entire coefficient ideal is coherent, by coherence of every actual

derivative, power and finite sum. True closed smooth contact internally has a coherent vanishing ideal. The original restriction operation is precisely the entire coefficient ideal plus this genuine vanishing ideal. Whole coherent sums commute with actual pullback, and the true submersion producer identifies the ENTIRE original vanishing ideal pullback with the vanishing ideal of the whole preimage contact. This gives the entire restricted coefficient ideal equality. Literal preimage carriers, unchanged marks and mapped boundary lists give the other three fields. No ambient closedness is claimed for the contact outside its genuine open manifold. □

Theorem 225.5 (Actual coherent reduced records and whole relative cosupport preimages)**.** *The true contact-reduced coefficient ideal is coherent and its whole source cosupport is exactly the true preimage of its whole original cosupport. This is the literal reduced marked datum, not an asserted equality with the original unreduced cosupport or an unproved SNC/admissibility condition.*

Proof. The entire relative coefficient ideal is coherent by its true finite power/sum presentation. The genuine closed smooth contact has coherent vanishing ideal, hence their actual sum, equal to the literal restriction, is coherent. Node four gives the ENTIRE source reduced ideal and its true preimage carrier and identical mark. The genuine whole relative order threshold equivalence for coherent ideals applies to this actual reduced ideal and carrier, at every source point. Unfolding cosupport gives the whole set equality, including infinite order and mark zero. No caller order, support or desired-zero-set certificate appears. □

Theorem 225.6 (ONLY original maximal-order data internally choose simultaneous actual reductions)**.** *ONLY genuine original submersion, coherent original marked ideal, actual closed smooth original carrier, positive original mark, and an actual source cosupport point with genuinely maximal original image order internally produce the SAME original open and contact function, BOTH genuine entire contact directions and closed smooth contacts in the two actual opens, the actual open submersion, ENTIRE reduced coefficient marked equality and the whole reduced cosupport preimage. No selected direction/open/contact, source comparison, SNC, admissibility, canonical or termination certificate is primitive.*

Proof. The real source cosupport point maps into original cosupport by the true relative order dictionary. Original positive maximal-order contact existence internally chooses actual U, u; the genuine tangent-field lift gives the whole direction on $V = f^{-1}U$, including every zero outside the original carrier. True contact regularity and holomorphicity give closed smooth carriers inside the two actual opens, and their ENTIRE carriers satisfy the literal preimage dictionary. Node two supplies the true same-complement open submersion. Node three gives the literal original marked open square. Apply node four to this actual original chosen contact and whole coherent coefficient record, then node five gives the full reduced cosupport equality. All intermediate geometry and whole ideal comparisons are internal outputs. No boundary transversality, independent contact-choice comparison, general higher invariant or noncompact termination follows from these producers; those remain OPEN. □

226 One actual nested contact-coordinate flag and its genuine smooth lift

Lemma 226.1 (The genuine contact coordinate replacement preserves the whole original carrier)**.** *In genuine original coordinates flattening N onto the plane of dimension k, replace one tangential coordinate $j < k$ by a true function u, moving that coordinate to slot $l = k-1$. The actual contact-coordinate map lies in the plane of dimension k exactly on N, and its slot l is the entire original function u. These are whole equations on the actual flat domain, not prescribed chart-output certificates.*

Proof. The original coordinate replacement swaps only j and l, both strictly below k. Every normal index $i \geq k$ is therefore fixed by that actual permutation and is distinct from l. At each such index, the literal new coordinate equals the original normal coordinate. Conversely every original normal coordinate occurs at that same new normal index. Thus vanishing of all actual new normal coordinates is precisely vanishing of all original normal coordinates, equivalent to N by the original genuine flattening. The replaced slot itself is u by the literal coordinate-map definition. The already proved contact-coordinate equation gives plane dimension $k-1$ exactly on $N \cap V(u)$; this calculation retains EVERY other original slot. □

Theorem 226.2 (ONLY original direction geometry internally creates one nested whole coordinate flag)**.** *Given original coherent I, actual closed smooth N, a genuine original whole tangent direction u on its open U and an actual contact point, internally produce $k > 0$, $l = k - 1$ and ONE true centred holomorphic coordinate chart whose ENTIRE domain lies in U. On that domain the actual original carrier is exactly the plane of dimension k, the actual contact is exactly the plane of dimension $k-1$, and the literal coordinate l equals u. No caller chart, index, positive dimension, normal form or desired equality certificate is supplied.*

Proof. Actual original smoothness chooses true flattening coordinates at the contact point. Coherence of the genuine tangent iterate gives holomorphicity of u throughout its actual open. Its real tangent-direction witness at the zero gives a true nonzero tangent derivative. The actual field decomposition internally chooses a tangential partial $j < k$ with nonzero value, so $k > 0$ and $l = k - 1$ exists. Form the genuine contact-coordinate map of node one. Its real differential is injective by the original coordinate replacement calculation and that nonzero partial. It is genuinely holomorphic on the actual intersection U with the old flat domain. The actual holomorphic inverse-function producer therefore constructs a true local coordinate inverse and recentres its value, with domain contained in this intersection and an ENTIRE coordinate equality to the contact-coordinate map minus its value at the point. The original chart was centred and u vanishes at the contact point, so that value is genuinely zero. Node one's whole original carrier equation, its contact equation and its literal u coordinate therefore hold on the ENTIRE new domain. This is an internally constructed joint flag, rather than separate independently chosen regularity charts. □

Theorem 226.3 (The SAME nested flag lifts in one genuine fibre-first source chart)**.** *For a genuine submersion of ANY two dimensions and a true point over that original contact, internally lift the SAME original chart to ONE true source chart with ALL original coordinate functions retained at actual injected indices. Its ENTIRE original-carrier preimage is the plane of dimension $k + (m - n)$, its ENTIRE contact preimage is the plane of dimension $(k - 1) + (m - n)$, and its actual contact coordinate is the WHOLE function $u \circ f$. Source points in this chart map into the original chart domain and actual U. No caller source chart, source flag, projection or equality certificate is primitive.*

Proof. At the actual source point, genuine submersion geometry internally gives $m = n + t$. Construct the actual fibre-first adapted chart from the SAME true original chart of node two. Its proved entire dictionary retains every original coordinate at its literal shifted index $t + j$, while the t fibre slots remain free. Apply the true whole original-centre producer to the actual original plane of dimension k and, in the SAME source chart, to the actual contact plane of dimension $k - 1$. It gives exact contiguous planes $t + k$ and $t + k - 1$, hence the asserted dimensions. The literal shifted contact coordinate equals the actual original u coordinate evaluated at $f(z)$, which node two identifies with $u(f(z))$ on the whole domain. The genuine coordinate dictionary also gives image-domain containment and EVERY retained original coordinate, not merely a stalk assertion. The actual linear projection and inverse-point equality are the already proved true adapted-chart producers. No points in an empty source demand fictitious indices; an actual point itself supplies the only required dimension comparison. Dimension zero is excluded only when the true nonzero tangent derivative produces the genuine positive k. □

Theorem 226.4 (Original maximal-order data internally select the joint contact flag)**.** *ONLY genuine original submersion, original coherent marked ideal, actual closed smooth original carrier, positive mark and a real source cosupport point with maximal original image order internally choose the SAME original open and contact function and produce the preceding ENTIRE original/source nested coordinate flags. No selected direction, positive carrier dimension, contact chart, source form, canonical or termination certificate is supplied.*

Proof. The genuine original-only maximal-order contact producer internally chooses actual U, u and BOTH whole tangent directions. Its actual original local cosupport inclusion puts the image point on the true contact, hence $u(f(y)) = 0$. The source point is in the ENTIRE preimage contact. Apply node three, with that actual chosen original direction, to produce one original chart and its genuine coupled source chart. All coordinates, the positive original dimension, actual injected slot and entire two-carrier flag identities are internal outputs. Coherent original input and genuine submersion are the only analytic primitives. Original boundary transversality, independent contact choices, higher canonical driver and noncompact termination remain OPEN; this flag does not assume or imply them. □

227 Genuine lower-dimensional contact parameter charts and their smooth square

Lemma 227.1 (Entire actual fibre-first projection commutes with genuine plane inclusion)**.** *For ANY a, n, use the genuine continuous linear inclusion $\iota_{n,a}$ of the first a coordinates. For the actual fibre-first projections $P_{n,t}$ and $P_{a,t}$ one has the ENTIRE map identity*

$$P_{n,t}\iota_{n+t,a+t} = \iota_{n,a}P_{a,t}.$$

The actual smaller projection is a genuine submersion with fibre $\mathbf{C}^t$, including $a = 0$ or $t = 0$.

Proof. For each original coordinate $j < n$, both sides read the source coordinate $t + j$ if $j < a$ and zero otherwise. This follows directly from the actual shifted-index formula and the genuine inclusion's finite coordinate formula. Coordinate extensionality proves the entire linear equality. The actual fibre-first linear equivalence splits $\mathbf{C}^{a+t}$ into its first t fibre coordinates and its last a original coordinates. Compose this true linear equivalence with interchange of the two factors. Identity charts on the model spaces then give the genuine submersion chart equation to first projection, with complement $\mathbf{C}^t$. No rank, coordinate, complement or submersion conclusion is supplied as a hypothesis. □

Theorem 227.2 (Actual arbitrary open-domain restrictions preserve the same smooth complement)**.** *Given a genuine original submersion f, genuine source and target opens V, U and the primitive actual image containment $f(V) \subset U$, the actual restricted map $V \to U$ is a genuine submersion with the SAME original complement. The source open may be any actual open, rather than the whole preimage of the target open. Empty opens require no artificial chart data.*

Proof. At each actual source point of V, take the genuine original submersion charts and its actual original complement. Compose each chart with the true open inclusion partial homeomorphism for V and for U respectively. Both composed charts genuinely belong to the maximal holomorphic atlases: their inverses are the original inverse charts restricted to actual opens, and their forward maps are holomorphic there. The primitive actual image containment makes the restricted map total into U. On the ENTIRE new source chart target, the true open inclusion right inverse identifies its ambient point with the original chart inverse. The original whole projection equation therefore proves the same new chart equation. The real submersion constructor produces the desired map with that SAME complement. Actual source points supply chart nonemptiness; for an empty source the universally quantified statement is vacuous, without invented dimensions or coordinates. □

Theorem 227.3 (ONLY original direction geometry internally creates lower-dimensional actual contact charts)**.** *ONLY original coherent I, actual closed smooth N, genuine whole original direction u on U, genuine submersion f and an actual source point over $N \cap V(u) \cap U$ internally produce $k > 0$, $t = m - n$, one original coordinate chart Θ and its actual adapted source chart Γ. There are actual opens $W \subset \mathbf{C}^{k-1}$ and $V \subset \mathbf{C}^{(k-1)+t}$ containing zero, and actual holomorphic parameter maps into the original and preimage contacts. They parametrize the ENTIRE respective contact portions in the chosen ambient chart domains, with true coordinate projection as inverse. The ENTIRE original f square in these parameters is precisely the actual projection $P_{k-1,t} : V \to W$, and that actual map is a genuine submersion with complement $\mathbf{C}^t$. No caller lower-dimensional contact chart, normal form, square, dimension or submersion conclusion is input.*

Proof. The true contact-coordinate producer internally creates $k > 0$ and one centred Θ whose entire domain lies in U and whose exact whole contact is the plane of dimension $a = k - 1$. Genuine submersion geometry internally gives $m = n + t$; construct its actual fibre-first adapted chart Γ from this SAME Θ. Its true entire dictionary flattens the preimage contact onto the plane of dimension $a + t$ and sends every source chart point into the original chart domain. Define $W = \iota_{n,a}^{-1}(\Theta.target)$ and $V = \iota_{n+t,a+t}^{-1}(\Gamma.target)$. Genuine continuous linear inclusions make both actual sets open; true centring places zero in both.

Define the actual parameter maps by $\Theta^{-1}\iota_{n,a}$ and $\Gamma^{-1}\iota_{n+t,a+t}$ using the real ambient inverse charts. Their whole inverse-point dictionaries put every parameter point into the actual contact and ambient chart domain. Conversely, each true contact point in that domain has its flattened coordinate in the actual coordinate plane; genuine plane inclusion composed with projection is identity there. The actual local chart round trips therefore give both entire inverse parameterizations, not just pointwise regularity.

The real adapted inverse-point square identifies $f\Gamma^{-1}(w)$ with $\Theta^{-1}(P_{n,t}w)$ on the ENTIRE source chart target. Node one's entire linear plane square identifies $P_{n,t}\iota_{n+t,a+t}$ with $\iota_{n,a}P_{a,t}$. The adapted chart's true image-domain containment proves that every V parameter is sent into W. Thus the whole parameter square is the genuine smaller projection. Node one's actual linear submersion and node two's true arbitrary-open restriction produce the genuine $V \to W$ submersion with complement $\mathbf{C}^t$. The genuine coordinate inverses are holomorphic on their actual targets; complex differentiability there gives smoothness of every order. Composing with the smooth original chart inverse and genuine continuous linear inclusion gives actual holomorphic parameter maps on the entire parameter opens. All dimensions, sets, maps and round trips arise internally from the original geometry. This also treats a zero-dimensional contact or zero fibre without creating nonexistent indices. □

Theorem 227.4 (Original marked maximal-order data internally select the actual contact parameter square)**.** *ONLY a genuine submersion, original coherent marked datum with actual closed smooth carrier, positive mark and an actual source cosupport point of maximal original image order internally select the SAME true original U, u and the preceding actual lower-dimensional contact parameterizations and submersion.*

Proof. The genuine original maximal-order contact producer internally chooses U, u and both true whole tangent directions. Whole relative cosupport pullback places the image point in original cosupport. Its true original local contact inclusion and actual membership in U put the image point on the genuine contact. Apply the preceding internal joint construction. No selected contact or lower-dimensional chart is supplied to this original-only consumer. Boundary transversality of independent contacts, higher canonical driver and noncompact termination remain open. □

228 Entire intrinsic coefficient ideals in the genuine contact parameters

Lemma 228.1 (Actual parameter-open maps are holomorphic into the original ambient manifold)**.** *Restrict each genuine inverse-coordinate parameter map constructed from the original contact to its actual parameter open. This actual map is holomorphic at every point, in the genuine parameter dimension. The original and parameter carrier universes may differ; no lift of the original carrier or supplied smoothness certificate is required.*

Proof. The previous actual parameter construction proves holomorphicity of each ambient-valued inverse-coordinate map on its entire actual open. At a true point of that open, openness makes its domain a neighbourhood; that whole holomorphic-on statement therefore gives actual holomorphicity at its ambient model point. Compose with the genuine smooth open inclusion to obtain the actual everywhere holomorphic map from the open manifold. Actual source membership is automatic for a point of the subtype. No nonempty open is assumed: all maps and statements remain literal on an empty subtype. □

Lemma 228.2 (Entire true contact restriction disappears under its actual parameter pullback)**.** *For ANY original coherent ideal I, actual holomorphic map σ and primitive actual image containment $\sigma(Z) \subset C$, internally prove the ENTIRE ideal equality*

$$\sigma^*(\mathrm{restrictOn}_C\, I) = \sigma^* I.$$

No closedness, finite generation frame, quotient ideal or desired equality is supplied. This applies to the actual contact parameters produced above.

Proof. An actual pulled-back restricted-ideal generator is a true original function a with an actual finite restricted span relation near $\sigma(z)$: it agrees on C with a finite sum of true members s_i of the entire original ideal, with genuine holomorphic coefficient functions b_i. Continuity of the actual holomorphic map pulls back the neighbourhood of that relation. The primitive image containment places every nearby parameter point in C, so the actual pulled-back function is the entire germ sum $\sum_i (b_i \circ \sigma)(s_i \circ \sigma)$. Original coherence internally supplies holomorphicity of every s_i; genuine holomorphic composition gives holomorphicity of all coefficient and generator compositions. Each actual $s_i \circ \sigma$ belongs to the true pullback of I by its actual one-term span. Thus the finite relation gives membership in the whole pullback ideal. Transitivity of true germ spans handles every outer pullback combination. For the converse, original coherence gives the actual inclusion of I into its entire contact restriction; monotonicity of the literal pullback span then gives the reverse inclusion. Stalk/function extensionality proves the whole ideal equality at every point, including empty spaces and zero parameter dimension. No ambient closed contact assertion is used. □

Lemma 228.3 (Entire coherent pullback composition uses the actual carrier universes)**.** *For actual holomorphic maps of ANY ambient dimensions and ANY three carrier universes, the entire coherent original ideal pullback along the composite equals the actual iterated pullback. The ENTIRE marked record composition retains the exact carrier, every literal old boundary member and mark.*

Proof. For a genuine original ideal member a at the composite image, original coherence gives holomorphicity there. Its true first composition belongs to the first actual pullback ideal by a genuine one-term span, with coefficient one. Its further composition is therefore an actual generator of the iterated pullback. Conversely, an actual member of the iterated pullback has a finite span of first-pullback members composed with the second map. Each first-pullback member has its actual finite original span. Composing that finite span along the true holomorphic second map, then using actual span transitivity and the literal equality of function compositions,

expresses the entire member in the composite pullback. Stalk/function extensionality gives whole ideal equality. No argument identifies the three type universes. For marked records the carrier is literal iterated preimage, the boundary is the same full original list mapped by genuine preimage composition, the mark is unchanged and the ideal is the just-proved entire equality. Extensionality of all four fields gives the true complete marked identity. This supplies the actual different-universe parameter application of the existing whole composition mathematics, rather than assuming a caller composite ideal. □

Theorem 228.4 (The actual intrinsic contact coefficient model has its entire original trace ideal)**.** *For the true original coefficient datum and actual contact parameter map, construct the genuine intrinsic marked model with full parameter carrier, every literal coefficient-boundary trace, entire original coefficient ideal pulled back along the actual parameter map, and the exact coefficient mark. Internally prove it equals the whole actual pullback of the contact-restricted coefficient model, and that its ideal is coherent. Empty or repeated boundary members are retained; boundary transversality is not asserted.*

Proof. Define its carrier to be the entire actual parameter open, its boundary to be the literal original coefficient-boundary list mapped by actual parameter preimage, its ideal to be the whole pullback of the original coefficient ideal and its mark to be the original coefficient mark. Actual contact membership proves that preimage of the contact carrier is the entire parameter carrier. The contact coefficient record already has the exact same boundary and mark. Node two gives equality of its entire pulled-back restricted ideal with the true original coefficient trace ideal. Thus all four fields are internally equal. Original coherence and actual smooth carrier geometry give coherence of the ENTIRE original relative coefficient ideal, including the zero-mark empty finite sum case. Its pullback is coherent by the genuine finite local holomorphic-generator pullback theorem. The actual intrinsic carrier is therefore the full regular parameter manifold; no original boundary member is filtered, merged or relabelled. This does not infer SNC for its boundary or canonicality of this new model. □

Theorem 228.5 (Original-only contact parameters produce the true entire smaller-dimensional coefficient square)**.** *ONLY original coherent marked datum, closed smooth carrier, genuine whole original contact direction, actual point over that contact and genuine arbitrary-dimensional submersion internally produce the actual target and source intrinsic coefficient models and their genuine smaller-dimensional parameter submersion. The ENTIRE source intrinsic marked model equals the actual pullback of the entire target intrinsic model. Every relative order threshold, the order including infinity, and whole cosupport are pulled back exactly. No caller chart, square, ideal/model/coherence/submersion/value or canonical certificate is supplied.*

Proof. The true original contact parameter construction internally supplies positive k, $m = n + t$, the actual opens and parameter maps, both ENTIRE contact parameterizations, their actual projection square and the genuine smaller submersion with fibre $\mathbf{C}^t$. Node four internally constructs the two intrinsic coefficient models from the SAME original marked datum and its actual original submersion pullback. Genuine relative derivative/coefficient pullback gives equality of the ENTIRE source ambient coefficient marked record with the actual original pullback record, retaining all four fields. Pulling it along the actual source parameter map and using node three gives its entire record as the pullback along the composite original map. The internally produced whole parameter square identifies that composite with the actual target parameter map composed with the actual smaller projection. Node three on that composite therefore gives the desired ENTIRE source intrinsic record as the true smaller-dimensional pullback of the target intrinsic record. All target/source contact restrictions have already been identified by node two, not assumed as quotient certificates. Node four supplies actual target coherence. The genuine relative order and cosupport submersion producers now apply to this actual smaller submersion and entire intrinsic target ideal; the full carrier is genuine and the entire source record is the produced equality. They

give every threshold, order including infinity and whole cosupport preimage. No boundary SNC or higher invariant equality follows from these coefficient records alone. □

Theorem 228.6 (Original marked maximal-order data internally select the whole intrinsic coefficient square)**.** *ONLY original coherent marked datum, actual closed smooth original carrier, positive mark, genuine submersion, actual source cosupport point and maximal original image order internally supply the preceding intrinsic coefficient models and every entire dictionary. No contact or reduced model is input.*

Proof. The true original maximal-order producer chooses the actual original open and whole contact function and both actual tangent directions. The genuine whole cosupport pullback places the image in actual original cosupport. Original contact inclusion and actual open membership put that image on the true contact. Apply the preceding original-direction producer. It constructs all dimensions, actual opens/maps, intrinsic marked models, whole trace ideals, coherence, smaller submersion, complete marked square and genuine relative order/cosupport conclusions internally. Independent contact choice comparison, boundary SNC, complete higher invariant functoriality, canonical driver identification, general noncompact termination and whole G3/G8 remain OPEN. Formal registration and native publication await root recovery. □

229 Actual smooth blowup pullback of whole controlled ideals and marked successors

Theorem 229.1 (Actual centre-relative strict carriers and whole literal boundary)**.** *For an original arbitrary-dimensional genuine submersion f, primitive genuine target and source blowups along C and its literal preimage, let G be the internally defined actual lift. For EVERY original set A, $G^{-1}\,\mathrm{geoStrict}_{\pi,C}(A) = \mathrm{geoStrict}_{\rho,f^{-1}C}(f^{-1}A)$. For EVERY literal finite ordered original boundary list, its whole strict-transformed list followed by the new centre-preimage divisor pulls back to EXACTLY the same literal source list. Empty, repeated and removed members are retained literally; this uses centre-relative strict transform, not the derivative exceptional locus for a Cartier blowup.*

Proof. The genuine actual lift G is a submersion, hence genuinely continuous and open. Openness and continuity give $G^{-1}\overline{S} = \overline{G^{-1}S}$ for every actual set S. Apply this to $S = \pi^{-1}(A \setminus C)$. The proved whole square $\pi G = f\rho$ then identifies its whole preimage with $\rho^{-1}(f^{-1}A \setminus f^{-1}C)$. Taking closure gives exactly the actual source strict carrier. Apply this genuine equality to every literal boundary member, with list map composition and append unchanged. The whole new divisor dictionary $G^{-1}\pi^{-1}C = \rho^{-1}f^{-1}C$ is the actual projection square. No closedness or regularity of A, boundary coordinate certificate, strict dictionary certificate or surjectivity is supplied. □

Theorem 229.2 (Every genuine original admissible centre transports internally)**.** *For ANY original marked record with coherent whole ideal, an original genuine admissible centre has its actual preimage admissible for the WHOLE marked pullback under a genuine arbitrary-dimensional submersion. The original carrier need not be the whole ambient manifold in this theorem.*

Proof. Original admissibility gives precisely a genuine closed smooth centre, its actual inclusion in the original relative cosupport, and genuine normal crossings with the whole ordered original boundary. The existing original-coordinate smooth preimage theorem internally produces the closed smooth preimage centre. The actual whole relative cosupport theorem identifies the pulled record's cosupport with the genuine preimage of the original cosupport, including all marks and infinite order, so the original inclusion pulls back directly. The genuine whole-coordinate normal-crossings preimage producer transports the last admissibility field, retaining all literal boundary indices. No caller pulled-centre or admissibility certificate, full-ambient assumption or maximum/termination input is used. □

Theorem 229.3 (Whole controlled ideal, genuine coherence and every actual cosupport threshold)**.** *Only the original genuine submersion, primitive actual blowup choices, the original whole coherent ideal I, its actual marking d, and the genuine original inclusion $C \subset \mathrm{cosupp}(I, d)$ in the WHOLE ambient manifold are inputs. The actual source controlled transform of f^*I equals the pullback under the SAME actual G of the actual target controlled transform of I, as ENTIRE germ ideals. It is genuinely coherent, and EVERY natural order threshold of this whole controlled ideal has its actual whole cosupport exactly pulled back. No ideal-inclusion, colon, factorization, coherence or threshold output certificate is supplied.*

Proof. Apply the genuine original coherent Hadamard bridge to the actual closed smooth centre and original cosupport inclusion. It internally gives $I \subset I_C^d$, then the actual target blowup internally gives $\pi^*I \subset L^d$ for its genuine invertible exceptional ideal L. The already proved arbitrary-dimensional smooth powered-colon theorem, applied to the SAME actual smooth G, identifies the entire pullback of $(\pi^*I : L^d)$ with $(G^*\pi^*I : (G^*L)^d)$. The genuine coherent whole pullback-composition law and actual whole square identify the first numerator with ρ^*f^*I. The actual whole blowup-ideal dictionary identifies G^*L with the source exceptional ideal, using the original vanishing ideal of the genuine source preimage centre. Thus this colon is EXACTLY the actual source controlled ideal, on every stalk. The original target controlled transform is coherent by its genuine Hadamard-derived inclusion. Pullback of this same ideal under the actual holomorphic G is coherent; the whole equality yields actual source coherence. Apply the genuine smooth whole-ideal cosupport theorem to the SAME coherent target transform for ANY threshold q, and use the same whole ideal equality. Mark $d = 0$, threshold $q = 0$, empty centres and non-surjective maps use these same entire identities. □

Theorem 229.4 (The entire actual controlled marked successor dictionary)**.** *For an original coherent marked record with literal carrier $N = \mathrm{univ}$ and a genuine original admissible centre, the geo-transform of its whole marked pullback by the SAME actual source blowup equals the whole marked pullback under the SAME actual G of its original target geo-transform. This is equality of the ENTIRE marked record: carrier, ordered boundary, whole ideal and unchanged mark. Surjectivity and marking one are not required.*

Proof. The actual strict-carrier theorem gives the whole carrier equality, and its actual ordered-list theorem gives the WHOLE boundary equality, keeping every empty and repeated historical entry and the exact appended new centre-preimage divisor. Since the original carrier is literally the whole ambient manifold, both original marked records have whole ambient carrier, and their actual blowup strict transforms are whole ambient carriers by genuine nowhere-density. The existing full-ambient geo-transform theorem therefore identifies both ENTIRE successor ideals with their genuine controlled ideals. Original admissibility supplies the actual original cosupport inclusion, so the preceding internally proved whole controlled equality gives the full successor ideal equality. Both marks are the unchanged original d. Marked-record extensionality then gives the entire equality, with no caller record, coefficient, chosen-centre or transform equality certificate. General non-full-carrier `colonAlong` naturality still needs a genuine restricted quotient producer; this theorem does not replace its restricted inclusion by a false ambient inclusion. General higher invariant and canonical driver equality, whole G3/G8 and formal/native/publication remain OPEN. □

230 Actual whole finite ideal intersections in genuine smooth stalks

Theorem 230.1 (The entire actual projected ideal is characterized by every genuine finite normal coefficient)**.** *For the actual tangential projection $\eta : B = \mathcal{O}_{\mathbb{C}^d,0} \to R = \mathcal{O}_{\mathbb{C}^{d+t},0}$ and ANY whole ideal $I \subset B$, an arbitrary actual $g \in R$ belongs to $\eta(I)R$ if and only if EVERY actual normal*

Taylor coefficient of EVERY finite truncation of g belongs to I. No generator, fixed-degree stability, flatness or membership certificate is input.

Proof. For a member of the ENTIRE extended ideal, the earlier true strong span induction includes all arbitrary R multipliers; therefore each coefficient in each actual finite truncation belongs to I. Conversely, if all those actual coefficients belong to I, the genuine finite reconstruction gives $g \in \eta(I)R + J^q$ for EVERY q, where J is the true normal-coordinate ideal. This actual ideal lies in the actual local maximal ideal. The proved local-ring Krull intersection theorem, applied to the ENTIRE ideal $\eta(I)R$, therefore gives $g \in \eta(I)R$. No uniform truncation bound is assumed. Zero normal dimension and degree zero use the literal earlier normal-index and unit-ideal conventions. □

Theorem 230.2 (Any two whole original ideals have their actual intersection extended exactly)**.** *For ANY TWO actual whole ideals $I, L \subset B$, internally prove $\eta(I \cap L)R = \eta(I)R \cap \eta(L)R$. Neither inclusion of the ideals, finiteness of prescribed generators nor a flatness certificate is supplied.*

Proof. Membership of an actual ring germ on the right is membership in both whole extended ideals. By node one it is equivalent to EVERY true finite coefficient belonging to both I and L. This is literally membership in $I \cap L$ for every such coefficient, so node one returns membership in the left ENTIRE extended ideal. The equivalence for EVERY actual germ is whole ideal extensionality. In particular no coefficient conclusion was assumed, and arbitrary source multipliers were handled by the original whole extended-ideal producer. □

Theorem 230.3 (Whole finite intersections retain all literal original indices)**.** *For ANY literal indexed family of actual whole ideals $I_i \subset B$, $\eta(\bigcap_i I_i)R = \bigcap_i \eta(I_i)R$. In particular every genuine finite family retains empty and repeated entries literally. This concerns fixed-origin projection ideals; it supplies no ONE common nearby open or analytic-extension assertion for an arbitrary family.*

Proof. For an arbitrary genuine source germ, node one identifies membership in EACH extended ideal with membership of EVERY true finite coefficient in that corresponding original ideal. Quantifying over all actual indices and interchanging these literal universal quantifiers yields that EVERY coefficient belongs to the ENTIRE intersection of the original ideals. Node one, applied to that actual intersection ideal, returns membership in its whole extension. This is genuine whole ideal extensionality, valid also for the empty index family, since its intersection is the unit ideal. Repeated ideals are retained by literal quantification. The previous strong span induction and actual Krull theorem account for arbitrary source coefficients; no infinite sum, convergence interchange or claimed nearby uniformity is used. Finite original families are a direct special case and will be the actual relative monomial consumers. □

Theorem 230.4 (Original coherent functions internally give the entire finite smooth stalk intersection dictionary)**.** *ONLY an original actual complex submersion $f : Y^{d+t} \to X^d$, original finite family of coherent GermIdeals I_i and a true point y internally construct the real target/source coordinates and prove that the intersection of EVERY actual source pulled stalk equals the extension of the intersection of the SAME whole target stalks. No chart, finite frame, target stalk, pulled-ideal equality or intersection-result certificate is a caller input.*

Proof. Normalize the true preferred target coordinates at $f(y)$. The actual submersion lift-coordinate producer constructs the source chart and its whole inverse projection square. For EACH original I_i, original coherence internally selects a true finite presentation by original holomorphic functions. The genuine whole pulled-stalk theorem identifies the ENTIRE actual pulled ideal in that source chart with the actual projection extension of the whole target ideal in the normalized chart. Apply node three to these actual target stalk ideals. Thus their finite whole source intersection has the asserted exact dictionary, including the empty family. No chosen source stalk or frame is input. This is the actual algebraic finite-intersection foundation for the subsequent

genuine relative monomial part: that part still requires its actual finite restricted-power stalk membership dictionary, internally computed original exponents and full coherent carrier producer. It does NOT yet assert that the whole original GermIdeal intersection is coherent or that a prescribed whole monomial part pullback equality has been proved. Higher invariant/canonical driver, noncompact termination, whole G3/G8 and formal/native publication remain OPEN. □

231 Entire coherent ideal quotients under genuine arbitrary-dimensional submersions

Lemma 231.1 (Actual adapted germ representatives and the entire original stalk)**.** *For an actual complex manifold X, actual point x and genuine centred local holomorphic coordinate equivalence Φ, define $\gamma_\Phi(F)$ by the literal original function $F \circ e^{-1} \circ \Phi^{-1}$, where e is the actual preferred chart. For an original coherent germ ideal I define I_Φ as the ideal span of ALL actual membership germs $\gamma_\Phi(F)$. Internally prove that every genuine complex germ has an actual original function representative, that γ_Φ preserves multiplication, and that $I.mem(x, F)$ is exactly $\gamma_\Phi(F) \in I_\Phi$. No finite generator, stalk equivalence or whole ideal membership dictionary is supplied.*

Proof. Holomorphy of the actual inverse-chart composite is equivalent to holomorphy of F at x: use the true preferred chart inverse identities, and then the two holomorphic directions and true inverse identities of Φ. For an arbitrary germ choose only its actual holomorphic representative H and set $F = H \circ \Phi \circ e$. Near zero the actual inverse composite returns H, since $\Phi^{-1}(0) = e(x)$, its inverse stays in the actual preferred target near zero, and both genuine chart inverse identities hold there. This gives actual holomorphy and exact germ equality. Multiplication follows from the literal product of these original inverse composites. Coherence internally selects a genuine finite family G_i generating I at x. The full preferred/adapted finite-span transport and genuine complex germ span theorem identify $I.mem(x, F)$ with membership in the actual span of $\gamma_\Phi(G_i)$. This span equals the definition using ALL original members: each original member belongs to the finite span by that exact finite-span theorem, and each chosen original generator really belongs to I by the original coherent presentation. These two actual span inclusions prove the full dictionary. No coherence model or frame is a caller input. □

Theorem 231.2 (The actual entire stalk of the original coherent quotient)**.** *For ANY TWO original coherent ideals I, L on X, the actual adapted ideal of their genuine sheaf quotient equals the ENTIRE ring ideal quotient: $(I : L)_\Phi = I_\Phi : L_\Phi$. No inclusion $I \leq L$ or invertibility is assumed.*

Proof. Oka internally gives coherence of the actual original quotient. For an arbitrary actual ring germ use the first node's internally constructed original representative F. Its membership in the quotient stalk is the original definition: F is holomorphic and $GF \in I$ for EVERY original $G \in L$. The multiplication and exact original membership dictionaries translate these conditions into the ring quotient. Conversely every actual ring germ in L_Φ has an internally produced original representative; the exact dictionary puts that representative in the SAME entire original L. Thus the original universal multiplication condition quantifies over ALL actual denominator germs, rather than only prescribed generators. Both implications and actual representability prove the whole ideal identity. No caller quotient or comparison certificate occurs. □

Lemma 231.3 (Actual submersion chart produces the whole original projection square)**.** *ONLY an original genuine complex submersion $f : Y \to X$ and a point y internally choose $m = n + t$, actual normalized target coordinates Φ and actual lifted source coordinates Ψ. EVERY original holomorphic F satisfies the exact whole germ identity $\gamma_\Psi(F \circ f) = \eta(\gamma_\Phi(F))$, for the actual tangential projection pullback η. No chart, square or stalk-map equality is input.*

Proof. The actual submersion's genuine linear complement internally proves $n \leq m$ at the actual point, so choose t with $m = n + t$. Normalize the preferred actual target chart by its own centre. The true lifted coordinate producer constructs Ψ and its entire first-block dictionary. On the genuine source inverse chart near zero, write $z = e_Y^{-1}(\Psi^{-1}(w))$; actual inverse identities put z in the true lifted domain and give its source coordinates exactly w. The dictionary says that the target coordinates of $f(z)$ are precisely the first n coordinates of w. The target actual inverse identity therefore gives $f(z) = e_X^{-1}(\Phi^{-1}(\mathrm{pr}(w)))$ on this actual neighbourhood. Compose with EVERY original F and use exact germ equality. Holomorphy is produced from the genuine original smooth map and both actual charts. This includes fibre dimension zero and either ambient dimension zero. □

Theorem 231.4 (The whole actual pulled ideal in the true lifted stalk)**.** *For EVERY original coherent ideal I, ONLY the same original f and its internally chosen charts give $(f^*I)_\Psi = \eta(I_\Phi)R$. This is the ENTIRE actual pulled ideal, with no caller pullback equality or finite-generator presentation.*

Proof. Original coherence internally chooses actual finite generators G_i at $f(y)$. The true finite-generator pullback theorem proves membership in the actual pulled ideal exactly by the literal family $G_i \circ f$ at y. The first node's true finite-span dictionary therefore represents its whole source ideal by the actual germs of that family. Each germ is exactly the projection pullback of the original target generator by the third node. Actual ideal-map span equality identifies this whole span with the extended original target ideal. The finite families were internally selected from original coherence and their literal composites; no caller family, source ideal or ideal-map identity entered the construction. □

Theorem 231.5 (Every actual coherent ideal quotient commutes with genuine smooth pullback)**.** *For ANY original genuine arbitrary-dimensional possibly NON-surjective submersion f and ANY TWO original coherent ideals I, L, internally prove $f^*(I : L) = (f^*I : f^*L)$ as the WHOLE actual GermIdeal. There is no original ideal inclusion, denominator invertibility, flatness or equality certificate.*

Proof. At EACH actual source point internally construct the true charts of node three. Oka and genuine coherent pullback internally supply coherence of both original and pulled quotients. Nodes two and four identify the source stalk of $f^*(I : L)$ with $\eta(I_\Phi : L_\Phi)R$, and that of the source quotient with $(\eta(I_\Phi)R : \eta(L_\Phi)R)$. The previously actually proved projection quotient theorem equates these ENTIRE actual ideals without assuming $I_\Phi \subseteq L_\Phi$. The first node's exact dictionary returns this equality to EVERY literal original source function membership. A nonholomorphic function belongs to neither coherent ideal; this follows from the internally selected actual finite coherent presentations. Pointwise extensionality proves the whole GermIdeal equality. Empty source needs no false dimension inequality, since the proof is genuinely pointwise. This removes the actual algebraic obstruction to arbitrary-carrier restricted quotient naturality. The restricted carrier dictionary and complete marked transform consumer are separate subsequent producers; general higher invariant/canonical driver, noncompact termination and whole G3/G8 remain OPEN, as do formal native registration and publication. □

232 The actual universal blowup lift is a submersion everywhere

Lemma 232.1 (The genuine complement biholomorphism)**.** *Let $\pi : Z \to X$ be an actual analytic blowup along a closed smooth centre C. The restriction $\pi : O = \pi^{-1}(X \setminus C) \to N = X \setminus C$ is a biholomorphism of the actual open submanifolds. For any actual off-centre point p, it internally determines a holomorphic open partial homeomorphism $b : Z \dashrightarrow X$ with source O, target N, and $b(z) = \pi(z)$ on its entire source. Both directions are holomorphic on their entire actual domains.*

Proof. Apply the actual blowup universal property to the inclusion $N \hookrightarrow X$. The whole centre ideal pulls back to the unit ideal on N, so this produces a holomorphic section $s : N \to Z$ with $\pi s = \mathrm{incl}$. Apply uniqueness again on the actual open O: its two holomorphic lifts $s\pi$ and the inclusion have the same projection and the same whole unit pullback ideal. Thus $s\pi$ is the identity on O. The actual maps $z \mapsto \pi z$ and $x \mapsto sx$, corestricted to these two opens, are inverse holomorphic maps. Corestriction preserves holomorphy because the target is an open submanifold. Compose the inverse of the open inclusion $O \hookrightarrow Z$, this biholomorphism, and the inclusion $N \hookrightarrow X$. This is the required actual partial homeomorphism. The open inclusion is holomorphic; its inverse is holomorphic because its composition with the subtype value map is the identity on O. This argument also applies to empty complements: for chart constructions at a specified off-centre point the needed nonempty witnesses are that actual point and its projection. No compactness or chosen complement section is supplied. □

Lemma 232.2 (True chart transport through local biholomorphisms)**.** *Let $b : Y' \dashrightarrow Y$ and $a : X' \dashrightarrow X$ be holomorphic open partial homeomorphisms, holomorphic in both directions on their actual open domains. Let $f : Y \to X$ be continuous and a submersion at $b(p)$ with complement F, and let $G : Y' \to X'$ be continuous at p. Suppose $p \in b.\mathrm{source}$, $G(b.\mathrm{source}) \subseteq a.\mathrm{source}$, and $a(Gq) = f(bq)$ on this whole source. Then G is a submersion at p with the same F, proved from actual maximal-atlas charts.*

Proof. Take the actual submersion charts d, c of f and its actual linear product equivalence L. Replace d by its restriction to $f^{-1}(a.\mathrm{target})$, an actual open containing $b(p)$, because the original f is continuous. Compose this restricted chart with b, and compose c with a. Both charts lie in the actual maximal atlases: their forward and inverse maps are compositions of the given holomorphic maps and actual maximal-atlas charts. They contain p and $G(p)$ respectively. For every parameter in the transported domain chart target, the two inverse identities for b and a, and the given whole local square, reduce the transported expression to cfd^{-1}. That expression is exactly $\mathrm{pr}_1 L$, by the actual original submersion chart dictionary. The continuous-at version of the submersion chart criterion now yields the asserted same-complement submersion. All source/target restrictions are the literal actual opens; no differentiable-surjectivity criterion or caller submersion conclusion is used. □

Lemma 232.3 (Off-centre submersion of the actual universal lift)**.** *Let $f : Y \to X$ be a genuine global holomorphic submersion, with dimensions $n + r$ and n. Let $\pi : Z \to X$ and $\pi' : Z' \to Y$ be genuine analytic blowups of C and its actual preimage. Let $G : Z' \to Z$ be the internally constructed holomorphic universal lift. At every p with $\pi' p \notin f^{-1}C$, G is a submersion with complement $\mathbb{C}^r$.*

Proof. Produce both actual complement partial biholomorphisms from the first lemma. The actual universal projection identity $\pi G = f\pi'$ implies that G maps the entire source complement into the target complement and gives the entire local square of the second lemma. The actual universal lift is holomorphic, hence continuous at p. The second lemma gives a submersion with an actual complement of the original submersion at $\pi' p$. The already proved actual `complementCoordinates` constructs an isomorphism from this complement to $\mathbb{C}^{(n+r)-n} = \mathbb{C}^r$: projection onto the complement is a surjective linear image of $\mathbb{C}^{n+r}$, so the complement is finite dimensional, and the actual product equivalence gives its rank r. Transport the complement along this genuine internally produced isomorphism. □

Theorem 232.4 (Global same-r submersion of the canonical actual lift)**.** *For a genuine global submersion $f : Y^{n+r} \to X^n$ and a closed smooth centre C with empty interior, the internally generated canonical blowup lift $\mathrm{Bl}_{f^{-1}C} Y \to \mathrm{Bl}_C X$ is a genuine global submersion, with the fixed complement $\mathbb{C}^r$ at every point.*

Proof. The original genuine submersion internally produces that its actual centre preimage is a closed smooth submanifold with empty interior. Hence both actual canonical blowups and their actual universal lift are defined without a caller source-centre or lift certificate. At a point whose base belongs to the centre, use the previously proved original exceptional-point matching-chart coverage and entire universal chart equality: these internally produce the same-r projection charts at that point. At every other point use the third lemma. The two exhaustive cases both have the identical fixed complement $\mathbb{C}^r$, so they give the actual global submersion-of-complement property; the actual global submersion follows. There is no compactness hypothesis and no assumption of a local or global submersion output certificate. Empty centre, empty source, $n = 0$, and $r = 0$ are covered by the same case split and actual finite-dimensional identities. This theorem does not assert the Cartesian universal property for arbitrary cones, comparison with arbitrary blowup choices, or functoriality of the whole higher invariant/general driver. □

233 The true historical companion masks at the internally computed original stopping index

Theorem 233.1 (Actual simultaneous head and tail maps)**.** *For a genuine original smooth blowup pullback and EVERY literal split index, internally produce a SINGLE actual head map, a genuine smooth pullback relation on the head, and a genuine smooth pullback relation of the tail with that same head map and the SAME original endpoint map. Both universal squares and the actual tail projection square follow internally.*

Proof. Induct on the genuine simultaneous blowup construction. At split zero the head is the whole actual tower and its endpoint map is already genuine; the tail is the identity tower, with the same endpoint map as its base map. At the identity tower every split has identity head and tail with the original map. At a successor split index, apply the induction to the preceding actual pair. Retain the exact same head map and head relation. Append the last actual source and target blowups to the genuine tail relation: their centres are the true preimages under its actual endpoint map, and their universal lift is the SAME actual last lift by the literal constructor. This is a real simultaneous split, not unrelated existential head maps or a supplied projection-square/tail certificate. Smoothness and universal squares are consequences of the two true relations. Beyond-length splits are the literal initial head and whole tail, so remain covered. □

Theorem 233.2 (The entire actual order mask on each true historical head)**.** *For an original invariant-domain datum and genuine original test tower, EVERY actual head and EVERY natural o have the source order mask equal to the entire preimage of the SAME original head's order mask under its actual head submersion. The actual restricted head map between these genuine open sets is holomorphic and a submersion, with its true inclusion square.*

Proof. The actual original head test condition and domain stability internally produce its coherent ideal, smooth carrier, SNC boundary, transversality and nonzero germs. The genuine head relation gives the ENTIRE current marked dictionary. True nonmonomial order masks commute with the actual head submersion: their underlying sets are complements of closures of the actual order-$o + 1$ locus; the scalar and whole order dictionary gives its preimage, while true open-map and continuity give exact closure transport. Thus the actual source mask maps into the actual target mask. Form the literal restriction to these two opens. Choose the genuine complement already supplied internally by this actual submersion. Restrict both of its true local source and target charts to the actual opens, preserving the same complement and written-in-charts projection equation. This proves true submersion of the restricted map and hence its holomorphicity. Empty opens and $o = 0$ are literal; no mask equality or complement is input. □

Theorem 233.3 (The entire previously formed head-mask companion record)**.** *At each genuine head and EVERY o, the actual original compAt record on its true order mask pulls back along*

the actual restricted head map to the SAME source compAt record: carrier, full ordered boundary, ENTIRE original ideal and exact marking all agree.

Proof. The old compAt is the companion of the actual current marked datum AFTER its genuine restriction to the true order mask. Original head domain stability and actual open restriction give well formedness and nonzero germs of this precise original restricted datum. Whole companion pullback under the actual restricted head submersion therefore applies. Its entire pulled original restricted datum equals the source current datum restricted to the source mask: prove this by actual inclusion square, coherent whole marked pullback composition, and the genuine original head marked dictionary. The full carrier, each literal old/created boundary entry, entire ideal and mark commute in those true compositions. Substitute this whole equality into companion formation. This identifies the actual old compAt definition itself, rather than substituting a fresh endpoint companion for an earlier formed one. It does not yet assert equality of its controlled transform along the remaining restricted tail. □

Theorem 233.4 (The own current order selects the SAME true historical head and mask)**.** *At EVERY actual endpoint source point, the SAME original own orderAt and runAt select the simultaneous split from the first theorem. The true source maskAt is the whole preimage of the true target maskAt under its actual selected head map, and the actual companionAt is the genuine full marked pullback of the target companionAt along that restricted map.*

Proof. The preceding actual nonmonomial-history producer identifies both current orderAt and the full trailing OrdRegime count runAt internally, by the genuine simultaneous-constructor recursion. Use this actual common natural index in the first theorem and this common own order in the second and third. The old definitions of maskAt and companionAt are precisely the order mask and compAt on that genuine historical head. Rewriting the internally produced order and run equalities identifies the two literal definitions and gives their entire record/preimage dictionaries. No stopping index, current order, source canonicality or history certificate is assumed. □

Theorem 233.5 (The actual entire endpoint patch is the preimage of the original patch)**.** *The whole actual source patchAt at its own source reference point is exactly the endpoint-map preimage of the SAME original patchAt at its actual image, including every boundary and special point in this precise patch.*

Proof. The old patchAt is the preimage of maskAt under the true remaining-tail projection. The first theorem gives the genuine tail projection square for the SAME selected head map and original endpoint map. Combine it with the fourth theorem's actual entire mask preimage. For EVERY endpoint point, its source tail projection is in the source mask exactly when its image by the actual head map is in the target mask, exactly when the target tail projection of its actual endpoint image is in that mask. Extensionality gives the whole patch equality. No patch membership, centre inversion, surjectivity, canonical value or output set equality is supplied. □

Theorem 233.6 (Only genuine original inputs internally produce this historical masked model)**.** *Only a genuine arbitrary-dimensional original submersion, original invariant-domain marked datum in positive original ambient dimension and its actual original multiple test tower internally construct the actual source tower and endpoint map, and for EVERY actual reference point the true selected historical head/tail pair, restricted mask submersion, entire mask and patch preimages and whole companionAt marked dictionary. No caller source tower, head/tail geometry, mask, companion or value certificate is input.*

Proof. First construct the genuine actual source blowups and universal lifts from original data. The true original nonmonomial-history producer internally supplies all original current orders and trailing runs along this constructed output. Apply the first five theorems at every actual reference point. All domain, ideal coherence, geometry and open-mask inputs used there are produced by the true original test tower and actual simultaneous construction. Neither compactness nor

surjectivity is used; empty and repeated boundary history and all literal split indices are retained. The entire CONTROLLED transform of this previously formed companion along its remaining restricted tail, special-locus and tangent-choice comparison, general higher canonical driver and noncompact termination remain OPEN. □

234 Actual open restrictions and the earlier formed companion's controlled tail

The primitive inputs are genuine smooth submersions, actual original blowup towers, an original marked datum in its invariant domain, and the original target canonical segment. Target canonicality is a property of the original target history; source canonicality is not assumed. Every retained stage, including an empty centre, is preserved. The construction transports the literal earlier formed companion, not a fresh companion at the endpoint. This is the smooth pullback of the multiple test blow-ups of [3], Section 2.4, applied after the Step 2a companion formation; the statements retain all indexed boundary fields and do not infer higher invariant functoriality.

Lemma 234.1 (A genuine projection square identifies the actual chosen lift)**.** *Let $a : Y \to X$ be a genuine holomorphic submersion and let $\rho : Y' \to Y$ and $\pi : X' \to X$ be actual blowups along $a^{-1}C$ and C. Any actual holomorphic map $b : Y' \to X'$ satisfying $\pi b = a\rho$ equals the internally produced geometric lift. Consequently a genuine simultaneous history followed by these two actual blowups is again the same simultaneous history with endpoint b.*

Proof. The entire pullback of the true ideal of C under $a\rho$ is the actual source exceptional ideal, hence invertible. The genuine universal property of the actual target blowup therefore supplies a unique holomorphic map over $a\rho$. Both the internally constructed lift and b satisfy that universal property, so they agree as actual maps. Substitute this equality in the simultaneous history's one-step constructor. Equality of a source centre with its literal inverse image suffices to make this substitution; no source-lift identity or desired history property is input. □

Lemma 234.2 (True open preimages restrict every simultaneous blowup history)**.** *For a genuine simultaneous history Q over Y and T over X with true base map f and endpoint g, and an actual target base open U, let $V = f^{-1}U$. The literal restricted towers $Q|_V$ and $T|_U$ form a genuine simultaneous history. Its true base and endpoint maps are the actual restrictions of f and g, its length equals the original literal length, and its whole projection square is holomorphically universal. The endpoint source open is exactly the inverse image under g of the endpoint target open.*

Proof. At the empty tower, the true restricted map is a holomorphic submersion: open source and target inclusions identify both tangent spaces with the original ones, so the original split-surjective derivative remains split-surjective. The simultaneous empty-history constructor applies. For a last blowup π along C and its actual pulled blowup ρ along $g_0^{-1}C$, assume the restricted previous history has been produced. Its endpoint map is exactly g_0 on the actual source open. Therefore the trace of $g_0^{-1}C$ in that open is exactly the inverse image of the trace of C in the target open. Both restricted projections are actual blowups along those literal traces, by the genuine open-restriction universal property. The original last-step square implies that the actual endpoint lift takes every point over V into the target open over U; the true restricted endpoint map is holomorphic. Its square with the true restricted projections follows pointwise from the original square. By the preceding lift-uniqueness lemma this actual restricted map is the genuinely produced one-step lift, so the simultaneous-history constructor applies. This proves the claim by induction. Its projection square gives the endpoint-open equality pointwise: $T\,\mathrm{proj}(gz) = f(Q\,\mathrm{proj}(z))$. The existing genuine simultaneous-history length and whole holomorphic universal-property producers now apply to precisely these literal restricted towers. No stage is removed when its trace is empty. □

Lemma 234.3 (Controlled entire marked records on every literal restricted prefix)**.** *If the original target marked datum is well formed and the original target history is a multiple test blowup of it, the actual source restricted tower is a multiple test blowup of the whole marked pullback on the source open. At every literal prefix, the entire controlled transform is the whole marked pullback of the target restricted-prefix transform under the actual prefix map. The full carrier, ordered boundary, ideal and marking agree, and every actual cosupport is the literal inverse image.*

Proof. The genuine target open restriction is a test tower for the original marked record restricted to its open: every actual restricted centre is the trace of the actual original admissible centre; restriction preserves its carrier inclusion, SNC condition and cosupport inclusion. The original record's closed smooth carrier, entire coherent ideal and original indexed SNC and transversality restrict to that open. Apply the actual controlled-tower dictionary to the simultaneous restricted history produced by the preceding lemma. This constructs source admissibility at every step from target admissibility and the true smooth submersion, and transports the actual strict carrier, full indexed boundary, entire pullback numerator and full coherent colon along the strict carrier. Thus all four marked fields agree at its endpoint. For any literal prefix apply the actual simultaneous-prefix constructor to that same restricted history and the actual test-prefix property of the target tower, and repeat the same dictionary. Entire cosupport follows from the true order dictionary for the actual submersion. Every empty or repeated boundary entry remains in its literal list position because whole marked pullback maps the complete list. □

Lemma 234.4 (The real historical masks give a simultaneous restricted remaining tail)**.** *For an original invariant-domain marked record in positive original ambient dimension and a genuine simultaneous original test history, at any endpoint point use its actual order and actual regime-run index. The actual source order and run index equal the original target values. On precisely the two actual historical masks, their actual restricted remaining tails form a genuine simultaneous history. Its base map pulls back the original earlier formed target companion to the actual earlier formed source companion; its endpoint map is the actual restriction of the original endpoint map to the whole two endpoint patches.*

Proof. The genuine all-stage whole marked dictionary gives actual nonmonomial-order equality and literal fixed-order regime equivalence. Induction along the original simultaneous constructors then gives equality of the actual run index, and hence the source and target own indices agree. Split the same genuine simultaneous history at this actual index; the existing simultaneous-split producer yields one actual intermediate map which serves both the genuine head relation and the genuine remaining-tail relation. The actual whole head order-mask dictionary identifies the full source mask with the true inverse image of the target mask. The actual historical mask producer restricts this intermediate map to those two true masks, retains its genuine submersion property, and gives equality of the entire earlier formed companion records. Apply the open-history lemma to the genuine remaining-tail relation and this actual target mask; the source mask is precisely its true inverse image, so the resulting literal restricted towers are the original two historical companion tails. The endpoint map equals the original endpoint map at every patch point. The original tail square proves the exact patch preimage and endpoint membership; no patch comparison certificate is input. □

Theorem 234.5 (Only original target canonicality produces the controlled earlier companion tail)**.** *Suppose the original target test history is canonical for an original value whose order comparison dominates the actual nonmonomial order (in particular the actual original general value), and the own original target order at the reference endpoint is positive. The genuine historical companion-tail model of the preceding lemma internally has both actual target and source test-tail properties. Every literal source-tail prefix has an entire controlled companion record equal to the marked pullback of its actual target-tail prefix; all four marked fields and actual cosupports are retained. Source canonicality is not a hypothesis or conclusion.*

Proof. The original target value supplies the actual nonmonomial-order comparison. In the actual original general value this comparison is supplied internally because the order is its lexicographic first component. The genuine original canonical-tail theorem therefore makes the original remaining tail, restricted to its actual historical order mask, a multiple test blowup of the same earlier formed target companion. This uses original target canonicality and the positive own original order, with no source-canonical assumption. The target companion itself is in the original invariant domain: the head transform is in that domain by the original test-head property, its open restriction is in that domain, and its positive-order companion has the same closed smooth carrier and indexed boundary, coherent entire companion ideal, positive marking and actual nonzero restricted ideal. Apply the entire controlled-tower dictionary to the actual simultaneous historical-tail relation and this internally produced target companion/test tail. The already proved whole earlier-companion equality identifies its whole source marked pullback with the actual source historical companion, so the actual source test-tail property and all controlled endpoint records follow. For every literal prefix use the genuine simultaneous prefix relation and actual target test-prefix property. The same controlled dictionary gives every full prefix record and its entire cosupport inverse image. Nothing identifies the source higher value or makes this source tail canonical. □

Theorem 234.6 (Original input constructs its genuine historical controlled companion tails)**.** *Only an original genuine smooth submersion, original invariant-domain marked datum and original finite target test history canonical for the actual original general value construct a source blowup history and genuine endpoint submersion. At each point of positive own original order they internally construct the actual historical companion-mask map, genuine simultaneous remaining restricted tails, source and target companion test tails, whole controlled record and cosupport dictionaries at every literal inner prefix. No source history, canonicality, companion-test, selected-tail or whole-record certificate is input. No surjectivity or compactness is required.*

Proof. Induct the actual original target blowup tower and at each stage construct the actual source blowup along the true centre inverse image and its unique genuine smooth lift. This yields a source tower with the same literal length, genuine simultaneous relation and complete holomorphic universal square. For each actual source endpoint point, apply the preceding historical-tail and controlled-tail producers using its genuine target image. The original target general value has the required actual first order key internally. Every true source mask, own run index, earlier formed companion, actual restricted remaining tail and full prefix transform is thereby produced; none is chosen by the caller. The simultaneous relation additionally supplies actual smooth endpoint maps and full universal projection squares for these same literal tails and prefixes. All finitely recorded stages and complete indexed boundary lists remain. The construction does not supply higher invariant equality, source canonicality, a general canonical driver or noncompact termination. □

235 Actual Cartesian fibre-product geometry of the monomial blowup charts

Let n, r, k be natural numbers and $j \in \mathrm{Fin}(n)$. The fibre-first source coordinates are $\mathbb{C}^{r+n} = \mathbb{C}^r \times \mathbb{C}^n$. Write fib and drop for the two genuine finite continuous linear projections and join for their genuine continuous linear inverse. Let $j' = r + j$ be the literal shifted index. The target monomial map is the existing actual polynomial $\sigma_{k,j}$, which multiplies coordinates $i \geq k$, $i \neq j$ by the pivot coordinate w_j. The source map is the actual $\sigma_{r+k,j'}$. No chart-square, smoothness, lift or universality conclusion is an input.

Theorem 235.1 (Entire genuine fibre-first finite coordinate dictionary)**.** *The internally defined maps* $\mathrm{fib}, \mathrm{drop}, \mathrm{join}$ *are holomorphic, with* $\mathrm{fib}(\mathrm{join}(a, b)) = a$, $\mathrm{drop}(\mathrm{join}(a, b)) = b$ *and*

$\operatorname{join}(\operatorname{fib}(w), \operatorname{drop}(w)) = w$. *Each genuine fibre coordinate and each shifted target coordinate is read literally, including* $r = 0$ *and* $k = 0$.

Proof. Use the actual finite Euclidean continuous linear equivalence for the split of $\mathrm{Fin}(r+n)$ into $\mathrm{Fin}(r)$ and $\mathrm{Fin}(n)$. The two actual projections are continuous linear maps and join is the inverse actual continuous linear equivalence. Their inverse identities follow from its actual left and right inverse equalities. The coordinate formula is obtained by the true sum-index equivalence; hence no chosen coordinate-index certificate is needed. Continuous linear maps are holomorphic. If either block is empty, the genuine finite equivalence still gives these SAME identities. □

Theorem 235.2 (Actual source monomial map retains all fibres and drops to the true target map)**.** *For EVERY actual* w*, the polynomial source chart projection has* $\operatorname{fib}(\sigma_{r+k,j'}(w)) = \operatorname{fib}(w)$ *and* $\operatorname{drop}(\sigma_{r+k,j'}(w)) = \sigma_{k,j}(\operatorname{drop}(w))$*. Equivalently its entire polynomial is* $\operatorname{join}(\operatorname{fib}(w), \sigma_{k,j}(\operatorname{drop}(w)))$.

Proof. Every fibre index has value below r, hence below $r + k$, and is different from $j' = r + j$. The actual source monomial formula therefore leaves it unchanged. At a shifted target index $r + i$ the inequality $r + k \leq r + i$ is exactly $k \leq i$, and $r + i \neq r + j$ is exactly $i \neq j$. Read the shifted coordinate of the source polynomial using the preceding genuine finite dictionary. Both its multiplication branch and its unchanged branch are therefore exactly the target monomial coordinate. Equality of the entire finite vectors proves the written formulas; join's true inverse then proves the whole polynomial equality. □

Theorem 235.3 (The actual lifted monomial-chart map is a genuine complex submersion)**.** *The actual map* $\operatorname{drop} : \mathbb{C}^{r+n} \to \mathbb{C}^n$ *is a genuine complex submersion with the actual fibre complement* $\mathbb{C}^r$.

Proof. Use actual global complex vector-space charts on source and target. The true continuous linear split followed by the actual factor interchange is an equivalence from the source model to target model times the SAME fibre complement. In these genuine maximal-atlas charts the map is literally the first projection. Thus the definition of submersion with complement holds at every actual point. No derivative-rank or submersion conclusion is caller data. For $r = 0$ this is the genuine isomorphism case; an empty target coordinate index only means there is no chosen pivot chart. □

Theorem 235.4 (Every actual holomorphic cone has an internally constructed unique monomial-chart lift)**.** *For every space* P *genuinely charted on the complex* ℓ*-model, in particular every genuine complex* ℓ*-manifold, any holomorphic* $u : P \to \mathbb{C}^{r+n}$ *and* $v : P \to \mathbb{C}^n$ *with* $\operatorname{drop} u = \sigma_{k,j} v$ *have a unique actual holomorphic* $w : P \to \mathbb{C}^{r+n}$ *satisfying* $\sigma_{r+k,j'} w = u$ *and* $\operatorname{drop} w = v$*. No nonvanishing, invertibility, dense-image or exclusion of exceptional cones is required.*

Proof. Define the genuine lift $w = \operatorname{join}(\operatorname{fib} u, v)$. Its actual component functions are holomorphic; join is the sum of the genuine fibre and block continuous linear inclusions, so the lift is holomorphic. The whole source polynomial dictionary gives $\sigma' w = \operatorname{join}(\operatorname{fib} u, \sigma v)$, which is u by the true compatible square and join's actual inverse. Its drop is v by the real inverse identity. A competing lift w' has drop v and fibre $\operatorname{fib}(\sigma' w') = \operatorname{fib} u$, using actual fibre preservation; the entire coordinate split therefore forces $w' = w$. This proof also applies if both maps are entirely within the pivot-zero exceptional hyperplane, where ideal-invertibility alone would not supply a lift. □

Theorem 235.5 (Literal product-open chart domains retain the entire holomorphic Cartesian square)**.** *Let* $R \subseteq \mathbb{C}^r$ *and* $S \subseteq \mathbb{C}^n$ *be actual opens. The source base is the literal open* $\{a : \operatorname{fib} a \in R, \operatorname{drop} a \in S\}$*, the target base is* S*, the actual source monomial chart is* $\{w : \operatorname{fib} w \in R, \sigma \operatorname{drop} w \in S\}$ *and the target chart is* $\{v : \sigma v \in S\}$*. Their literal source/target polynomial projections and drop maps are holomorphic. Every actual compatible holomorphic cone into these SAME base and chart opens has a unique genuine holomorphic lift into the SAME source monomial-chart open.*

Proof. All displayed sets are open by genuine polynomial continuity and finite continuous linear projections. The entire monomial dictionary shows that the polynomial projection lands in the literal source base and drop lands in the literal target chart. Their subtype-valued maps are holomorphic by the actual open-subtype holomorphic equivalence. Given a genuine compatible cone, compose with the literal inclusions and apply the preceding explicit join lift. Its actual fibre is the original source fibre in R and its actual target monomial image is the target chart's given point in S, so its value lies in the written source chart open. Subtype-valued holomorphicity follows from the genuine included join map. The two square identities and uniqueness follow by inclusion injectivity from the entire model identities. Empty opens retain these literal statements. □

These are actual full monomial model and product-open chart producers. They do not assume a Cartesian certificate. To apply them to the original arbitrary blowup lift requires internally adapted original submersion/centre charts, genuine arbitrary-choice comparison, and all-chart gluing of the actual lift. Those next geometric conclusions, higher invariant functoriality, old general-driver identification and general noncompact termination remain open.

236 Actual nonmonomial and original-boundary histories under smooth blowup pullback

Theorem 236.1 (Literal original indices and levels at every actual stage)**.** *Let a genuine smooth submersion pull back an actual original multiple test blowup tower of an original well formed marked datum. At each actual pair of stages, the original-member list of StepOne is the literal preimage list of the original target list; its level at every source point equals the original target level at its actual image. Empty and repeated old entries and every created exceptional suffix remain literal.*

Proof. The genuine joint tower producer gives equality of the entire transformed marked datum with the actual marked pullback at every stage, hence equality of the entire boundary lists. The original members of StepOne are exactly the first original number of entries of that boundary: homogenization and coefficient formation preserve the entire boundary, and the true boundary transform appends the exceptional entry after strict-transforming every existing entry. Thus taking the original length commutes with the actual preimage list, whose length equals the original length. Each finite indexed entry contains a source point precisely when its original target contains the image. The literal indexed count is equal, proving equality of levels. No initial label, count, boundary equality or chosen value certificate is input. This statement asserts neither equality of special loci nor a higher invariant value. □

Theorem 236.2 (Actual current whole nonmonomial order)**.** *An original invariant-domain datum and its actual original multiple test tower have, under this genuine pulled tower, exactly equal current nonmonomial orders at every actual source/image pair, including points outside cosupport. The entire current cosupport is the actual preimage.*

Proof. Original domain stability along the genuine test tower internally produces the actual current smooth carrier, coherent ideal, entire SNC ordered boundary, transversality and nonzero germs along the whole carrier. The actual joint marked dictionary identifies the whole source current ideal with its actual submersion pullback. The true whole nonmonomial-part pullback theorem applies to these internally produced hypotheses. The existing entire ideal order threshold equivalence therefore identifies all current nonmonomial order thresholds, and their literal natural-order definition gives equality, even outside the carrier. Whole coherent cosupport pullback gives the entire current cosupport preimage. A positive original ambient dimension is assumed as required by the whole monomial part theorem; the source may be empty and no surjectivity is needed. □

Theorem 236.3 (Actual fixed-order regime and its complete trailing history)**.** *For every natural order o, the genuine original OrdRegime predicates correspond at actual stage/image pairs, and their complete trailing regimeRun counts are equal along the independently produced pulled tower.*

Proof. OrdRegime is the conjunction of membership in the whole current cosupport and current nonmonomial order equal to o. The preceding theorem identifies both actual predicates. Prove trailing-run equality by induction on the genuine simultaneous blowup construction. Both zero-stage runs are zero. At a successor, the actual universal lift projection square identifies the true original preceding-stage image of the source point. Prefix inheritance of the original test condition gives the actual preceding original domain, so the predicate dictionary applies there. If this actual preceding point is in the order-o regime, both literal recursions add one to their equal preceding runs. Otherwise both reset to zero. The argument uses the actual entire prefix history, without a supplied regime equivalence, run value, termination bound, canonicality or naturality certificate. □

Theorem 236.4 (The own-order historical stopping position is exactly preserved)**.** *The actual general-step orderAt and runAt of every source point equal the SAME original target functions at its actual image. Thus the literal historical stopping index selected by its own current order agrees.*

Proof. The old orderAt definition is precisely the current nonmonomial order, whose entire dictionary is the second theorem. The old runAt selects the OrdRegime run at that very orderAt. Substitute its true equality and apply the third theorem at the original image's actual order. No own order or stopping index is assumed. This does not yet identify the chosen tangent mask, special points, lower canonical driver or a complete higher invariant. □

Theorem 236.5 (Every literal head and its genuine whole fresh companion)**.** *At every literal head, including indices beyond the tower length, internally produce the actual head submersion and whole transformed marked dictionary, original-member list and level dictionary, all current nonmonomial orders, all fixed-order trailing runs and own runAt dictionary. For every o, the entire fresh companion of that actual current target datum pulls back to the entire fresh companion of the source current datum, with carrier, ordered boundary, ideal and marking all exact.*

Proof. The genuine simultaneous tower construction internally supplies its actual head relation at every literal index. Original test-tower prefix inheritance produces a true original head test tower, hence the actual head invariant domain. Apply the first four theorems to this genuine relation. The true whole companion pullback theorem applies to the actual internally produced current domain and head submersion; substituting the whole marked dictionary gives all four fresh companion fields at once. Heads beyond the length are the literal initial head and are handled by the same true constructor recursion. The statement concerns the fresh companion of each current datum; it does not silently replace an earlier formed, masked controlled companion tail by a newly formed one. □

Theorem 236.6 (Only original data internally supply the entire actual history)**.** *Only a genuine arbitrary-dimensional original submersion, an original invariant-domain marked datum in positive original ambient dimension, and its actual original multiple test tower internally construct a genuine source tower, endpoint submersion, same literal length and whole universal projection square, and all the preceding endpoint and every-head order, level, fixed-regime run, own-run and fresh companion dictionaries. No source tower, source canonicality or history/value/output certificate is input.*

Proof. Inductively construct the actual source blowup at the true preimage of each actual original centre and the genuine universal lift. The existing genuine tower producer supplies this construction, smoothness, universal square, length and complete marked/test dictionaries from

just original data. Apply the first four theorems to this constructed output; then the fifth produces every literal head and each entire fresh companion. All nonzero, coherent, SNC and order assumptions used in those applications are actual original inputs or internally produced original test-tower domain consequences. No compactness or surjectivity is used, and empty steps and repeated boundary labels are preserved. Historical masked companion tails, special-locus geometry, higher invariant comparison/canonical driver and noncompact termination remain OPEN. □

237 Actual full normal-crossings pullback under complex submersions

The sole geometric inputs are a genuine holomorphic submersion between the actual manifolds and the entire original finite SNC family. Surjectivity is used only for global nonempty-index and global-maximum statements. No pulled SNC, ideal, exponent, maximum, centre or naturality certificate is an input.

Lemma 237.1 (Actual normalized complement and full lifted coordinates)**.** *Let $f : Y^m \to X^n$ be a genuine complex submersion at an actual y. For every actual centred holomorphic coordinate change Phi at f(y), there is a genuine centred preferred-chart change Xi at y and an injection $rho : Fin(n) \to Fin(m)$ such that its true domain maps into Phi's true domain and, throughout that whole domain, Xi's rho(j) coordinate is Phi's j coordinate composed with f.*

Proof. Use the actual submersion witnesses: open maximal-atlas charts phi and psi, and an actual continuous complex-linear equivalence $L : C^m \simeq C^n \times F$, such that $psifphi^{-1}$ is first projection on the entire true submersion chart domain. The projection through L is surjective, so finite dimension gives n¡=m. Second projection through L is likewise surjective, so F is finite dimensional. The actual equivalence gives m=n+dim F; hence dim F=m-n. An actual complex basis of F and finite-dimensional continuity produce a continuous linear equivalence $F \simeq C^{(m-n)}$. The canonical first/last finite-index split identifies C^m with $C^n \times C^{(m-n)}$, with its first-coordinate injection rho.

Compose the target preferred chart, Phi, and the inverse of psi to form the actual local biholomorphism $Gamma = Phie_X psi^{-1}$. This is defined on the true open chart overlap and is holomorphic in both directions, by maximal atlas compatibility and the two genuine Phi holomorphic directions. Form Gamma times the fibre translation $v \mapsto v - v_0$, $v_0 = (Lphi(y))_2$. Compose the entire true partial-homeomorphism chain e_Y^{-1}, phi, L, Gamma times fibre translation, the actual finite-index split inverse (with the chosen actual F-coordinate equivalence). Every composition is restricted to its true open source and target, so this chain is an actual local biholomorphism. Its source contains $e_Y(y)$, as Gamma sends psi(f(y)) to zero; fibre translation sends v_0 to zero. It therefore gives a genuine centred preferred-chart Xi. On its entire true source, the original submersion chart equation gives first-coordinate identity. The last finite coordinates are unconstrained. Source restriction ensures the domain maps into the true original Phi domain. No inverse function or local-coordinate certificate is assumed; all charts, complements, restrictions and inverse functions are constructed from actual witnesses. This argument includes n=0, m=n, empty complement and zero-dimensional source; it only asserts pointwise existence where an actual y exists. □

Theorem 237.2 (The whole literal SNC family pulls back)**.** *If the entire original E is SNC, then the entire literal list $E.map(f^{-1})$ is SNC in dimension m, without surjectivity.*

Proof. Each member is closed, and f is genuinely continuous, so each entire preimage is closed. At an actual y apply original whole SNC at f(y) to obtain one genuine common centred Phi and its literal finite-index subsets. Lift that SAME whole coordinate system using the preceding lemma, sending all subsets by the actual injection rho. A through pulled member is exactly an original through member. The entire-domain coordinate equation therefore identifies each pulled member

with its true mapped coordinate subspace. Mapped through subsets retain singleton cardinality. Finite-subset mapping by an injection reflects equality; hence the actual original through-index injectivity gives actual pulled through-index injectivity. This is the full SNC definition for all literal indices, including empty members and zero source/target dimensions. No entry is deleted and no pulled-SNC assumption occurs. □

Theorem 237.3 (Every whole actual member ideal pulls back exactly)**.** *For every original literal SNC member C, the COMPLETE ideal f^*I_C equals the COMPLETE vanishing ideal $I_{(f^{-1}C)}$, at EVERY actual stalk of Y.*

Proof. If f(y) is outside the closed C, the actual original ideal is unit near f(y), and both pulled and actual preimage ideals are unit near y. If f(y) is on C, the SAME true original SNC chart gives C's actual singleton coordinate u. Actual Hadamard division gives its COMPLETE vanishing ideal generated by u at every nearby stalk. Lift the whole chart as above; the actual preimage is exactly the corresponding singleton coordinate u f. Hadamard division in the actual m-dimensional chart gives its COMPLETE vanishing ideal generated by u f, at every nearby stalk. By the definition of genuine ideal pullback, u f belongs to f^*I_C, and every pulled generator is locally an actual holomorphic multiple of u f. Finite holomorphic sums remain multiples, while every multiple of u f is in the pullback span. Both complete ideals therefore coincide at y. Actual holomorphicity follows from the genuine submersion and original chart domains. This covers every stalk, without any caller ideal equality or support-only substitution. For n=0 no original member can pass through an actual f(y), by original SNC; hence only the unit case occurs. For empty Y the full equality is extensional on its genuinely empty domain. □

Theorem 237.4 (Whole indexed marked product and intrinsic exponents)**.** *Let F be an actual finite carrier-power list whose entire original carrier list is SNC. Put $f^*F = F.map(C, a \mapsto (f^{-1}C, a))$. Then its entire actual carrier list is SNC and its COMPLETE component power product equals the genuine pullback of the original COMPLETE product. Its whole-ambient mark-one marked datum is exactly the original genuine marked pullback. Every true through-index intrinsic exponent equals the original actual power. All literal zero powers and empty/repeated historical entries remain.*

Proof. The second theorem internally supplies the entire pulled SNC family. Every original member ideal is coherent and invertible by actual SNC. The third theorem identifies its entire genuine pullback ideal with the actual pulled vanishing ideal. Genuine pullback preserves coherent whole ideal powers, multiplication and unit. Induction on the actual finite list therefore proves the WHOLE product equality, including the empty product. The carrier lists, full marked ambient and marking one agree definitionally, so whole ideal equality gives full marked equality. In positive source dimension the actual complete pulled SNC product dictionary computes every through exponent from the SAME whole ideal; alternatively the genuine submersion exponent dictionary reads the original exponent at its actual image point. In dimension zero there are no true through indices. No coefficient factorization, pulled-SNC or intrinsic exponent certificate is supplied. □

Theorem 237.5 (True global active indices and old singleton centre)**.** *If the genuine submersion is surjective, the actual nonempty positive literal index family of f^*F is exactly that of F. For each actual nonempty positive singleton maximum in F, the internally generated actual old-gamma maximum centre of the pulled WHOLE marked product is precisely the entire preimage of the internally generated original centre. The maximum literal index is the same, including every old colex tie break.*

Proof. Natural powers and literal index length are unchanged. Whole preimage nonemptiness reflects original nonemptiness by surjectivity and preserves it in the other direction for every actual member; hence active families agree exactly under the literal same-index equivalence. In

the nonempty case an actual source preimage point produces the genuine n¡=m inequality. All singleton cardinalities are one, so their dimension-first coordinates change uniformly from n-1 to m-1, with the same actual coefficient and old colex singleton. Actual key comparison is preserved and reflected; unique literal maximum indices, produced internally by the finite families, agree. The preceding whole marked equality and the actual gamma-order theorem identify the literal maximum locus on the actual cosupport with the whole preimage of the original locus. The original internally produced maximum locus is its entire selected nonempty Cartier carrier, so the pulled internally produced centre is exactly its entire preimage. The inactive case gives whole unit products on both sides and no canonical centre. For empty manifolds, surjectivity makes the source and target active families empty; there is no spurious ambient dimension inequality assertion. No global maximum-preimage claim is made for a nonsurjective submersion. □

Theorem 237.6 (Actual next Cartier control respects the entire pulled history)**.** *For every positive original literal factor, including the internally generated maximum index, the true identity Cartier blowups and their SAME complete mark-one controlled transforms commute with genuine pullback: every literal old strict carrier and its coefficient agrees, the selected old empty carrier is retained with zero power, and the actual new selected carrier is appended with power k-1. The whole new controlled marked datum and every true intrinsic through exponent agree by the actual pulled indexed dictionary. The actual full coefficient sum drops by one on both sides. This whole next-step dictionary does not require surjectivity; only the preceding global maximum comparison uses surjectivity.*

Proof. Both selected carriers are actual members of their internally proved entire SNC lists. True Cartier universal property and genuine positive-factor admissibility supply their actual identity blowups and controlled division. Actual indexed Cartier control replaces the selected old index by (empty,0) and appends its same actual carrier with exponent k-1, leaving every other literal entry. Entire preimage commutes with that actual finite-list set operation, empty carrier and append. Thus the two actual normalized lists are exact whole preimages with unchanged real coefficients. The preceding WHOLE indexed ideal pullback equality gives exact controlled marked equality. True intrinsic exponents use those SAME controlled ideals, and actual coefficient arithmetic gives new sum plus one equal to old on both sides. This proves the actual NEXT ordered monomial step, not naturality of a full higher-dimensional blowup driver. Full finite procedure comparison and its empty-step deletion for nonsurjective opens are separate remaining work; prior general separation's identification with the old driver is OPEN. □

238 Actual smooth base change of independent arbitrary-choice ordered monomial procedures

The primitive inputs are dimension-n and dimension-m complex manifolds A and B, an actual surjective complex submersion f:B-¿A, an entire literal SNC carrier-power list F on A, and two genuine primitive Cartier blowup-choice families in their respective dimensions. No caller comparison, stage map, Cartesian square, marked transport, maximum, termination or prefix certificate is an input. All whole finite lists retain their actual empty and repeated historical entries. The general higher-invariant driver and nonsurjective step compression remain open.

Lemma 238.1 (Genuine submersion charts are transported by actual biholomorphisms)**.** *Let a:A'-¿A and b:B'-¿B be actual same-dimensional biholomorphisms. Then h=a^{-1} o f o b is a genuine dimension-m-to-dimension-n complex submersion. If f is surjective, so is h. Both conclusions are internally produced.*

Proof. At z in B', take genuine submersion domain and codomain charts for f at b(z), with its actual complement and continuous linear equivalence to base times complement. Compose the domain chart with b and the codomain chart with a. Both compositions and their inverses are

holomorphic: this follows from the actual biholomorphisms and both holomorphic directions of the original maximal-atlas charts. Consequently these are actual maximal-atlas charts on B' and A'. Their sources contain z and h(z), and the whole new domain source maps into the whole new codomain source. In these charts the exact inverse identities of a and b cancel, so the SAME original split linear projection represents h on its entire chart target. This gives the actual submersion predicate with the genuine original complement, without surjectivity of a derivative supplied as an output certificate. The complement remains the same globally chosen complement when f is globally a submersion. For a target point t, lift a(t) through f, then lift the chosen source point through b. This constructs a preimage under h and proves surjectivity, including empty manifolds. □

Definition 238.2 (Real stage pullback predicate and actual endpoint lift)**.** For actual towers T' over B and T over A, the stage pullback predicate is inductive. At empty towers the actual map is f. At an actual simultaneous step it stores a genuinely produced surjective submersion between the new stage manifolds, exact commutation of the two genuine blowup projections, whole preimage equality of their actual centres, and the genuine holomorphic universal pullback property of the actual step square. Its strong marked version additionally records the ENTIRE real controlled marked identity at every actual geometric stage, proved by this SAME simultaneous recursion. These fields are conclusions proved about the newly generated procedures, never hypotheses required of their initial input. For independently generated bounded chosen procedures on pulled F and F, let s and t be their internally produced actual total projection biholomorphisms. Define the actual endpoint lift by t^{-1} o f o s.

Lemma 238.3 (The actual Cartier square has its internally produced holomorphic universal property)**.** *For genuine chosen Cartier blowups p':D'-¿B and p:D-¿A of the internally identified corresponding selected carriers, let b and a be the genuine projection biholomorphisms internally produced by the universal property. The lift h=a^{-1} o f o b is a surjective submersion and p o h=f o p'. This actual square is a holomorphic pullback: for any actual complex manifold P and genuine holomorphic maps u:P-¿B and v:P-¿D with f o u=p o v, there is a unique genuine holomorphic map w:P-¿D' with p' o w=u and h o w=v.*

Proof. The true Cartier inverse producer identifies b with p' and a with p. The preceding actual chart lemma gives the real submersion and surjectivity. The genuine inverse identities give exact commutation. Construct w=b^{-1} o u; it is actually holomorphic. Then p' o w=u. Applying the injective actual a to h o w and v and using the given genuine cone commutation proves h o w=v. Any other such map agrees with w after applying the injective actual b, so equals w. This proves a real holomorphic universal property, not just a commutative square or an assumed Cartesian certificate. The same argument with the internally produced actual total projections gives the universal property for the entire generated tower square. Equivalently, paste the genuine step universal properties: first lift the compatible cone to the previous source stage, then lift it through the actual last square. Uniqueness follows successively from the previous and last actual square properties. This proves the entire projection-square universal property for every literal geometric head. □

Theorem 238.4 (Two independent genuine choice procedures internally generate every actual smooth stage lift)**.** *For every bounded fuel q, the independent chosen procedure on the genuine whole pulled input $f^{-1}F$ and the independent chosen procedure on F have an internally produced stage pullback predicate. Every genuine stage map is a surjective complex submersion; the actual whole selected centres are literal inverse images, and every actual simultaneous Cartier step square has the full holomorphic universal property. The endpoint map is exactly t^{-1} o f o s. The performed tower lengths and entire old-key index words are equal.*

Proof. Induct simultaneously on q, with arbitrary genuine original manifolds, SNC data, surjective submersions and both primitive choice families. The actual submersion-chart SNC producer

internally proves the entire literal pulled SNC input. At zero fuel or inactive input both real towers are empty. Actual surjectivity preserves the nonempty positive index set and the original literal colex keys; the genuine internally attained unique selected indices therefore agree. In the active case choose the two genuine first Cartier blowups by their respective primitive families. Their actual global projection biholomorphisms are internally proved, so the preceding square lemma constructs the true first lift h and its genuine universal property. The entire source successor is exactly the h-preimage of the entire target successor: the literal preimages commute, the true selected index is the same, and the SAME carrier is appended at its actual power k-1. This includes all removed-index empty entries and all preceding history, without any boundary reindexing. The target successor is genuinely SNC and the source successor is genuinely its whole h-preimage. Apply induction to these real successors and actual h; append the genuinely produced first-step and recursive stage pullbacks. The endpoint is determined by projection commutation and the actual target projection inverse, and equals t^{-1} o f o s. Actual appended lengths and the recursively identical literal selected indices give equal lengths and entire old-key words. No aligned history or prefix assumption is used. Strong induction stores the entire actual controlled marked identity at each step: the true first Cartier controlled dictionaries give the two genuine successor products, and the inductive tail identities concatenate by the actual transform-append equation. Induction on this strong output then extracts the WHOLE marked dictionary for every literal split head, not merely for separate bounded terminal runs. □

Theorem 238.5 (Whole actual controlled marked records and internally completed procedures pull back)**.** *For every genuine bounded pair above, the complete actual controlled marked datum of the source procedure equals the whole marked pullback of the complete actual target datum under the actual endpoint lift. All four fields, each entire boundary entry, and every genuine stage controlled ideal agree. The whole computed coefficient sums agree. Thus the two procedures completed with their OWN original sums have the same genuine finite length, actual stage pullback, whole unit endpoints and real geometric marking-one resolutions; no compactness or external finite bound is assumed.*

Proof. The previous actual induction gives exact whole terminal indexed-list preimage equality under the SAME real endpoint lift. Its genuine submersion charts internally give the whole vanishing-ideal pullback dictionary for every actual terminal SNC member. True coherent powers and whole finite products then identify the ENTIRE terminal ideal, including unit stalks outside the carriers. Together with whole carrier, boundary and mark identities this proves all four fields of the actual controlled marked record. Apply this at each actual stage produced by the preceding induction, not at a caller-provided prefix. Literal preimage preserves every actual coefficient; hence original whole sums agree even when inactive empty entries have positive powers. Each procedure's own internally proved sum bound and true unit endpoint apply at these equal OWN fuels. Whole unit ideals pull back to whole unit ideals. The actual admissibility, canonical old-gamma segment and geometric resolution of each real chosen procedure were internally generated from its own original SNC input. General higher invariant or unrelated-ideal admissibility is not inferred. □

Theorem 238.6 (The actual base-change maps obey composition and choice cocycles)**.** *For a genuine composable pair of surjective complex submersions and three independently generated genuine choice procedures on the exact whole successive pulled inputs, the actual stage lifts compose to the lift of the genuine composite. Identity maps induce the true identity lift. For two changes of genuine choice families, the actual base-change maps commute with the internally produced unique real choice comparisons, at every actual stage and at the completed endpoint.*

Proof. All actual lifts are uniquely determined by the genuine source total projection, primitive base map and inverse genuine target projection. For each literal geometric head, its actual projection has a genuine holomorphic inverse: compose the actual tail projection with the internally

produced inverse full projection. The actual split projection equation proves its right-inverse identity; genuine noncompact surjectivity of the actual tail blowup sequence proves its left-inverse identity. Both maps are holomorphic. Thus the unique actual head lift is exactly the target head projection inverse, primitive base map and source head projection. True same-dimensional choice tower comparisons restrict to the actual heads; their genuine commuting projections identify their actual head comparison with this same inverse-projection formula. In a composite the adjacent actual target projection inverse and source projection cancel by their true inverse identities. The literal whole finite-list preimages of the genuine composite equal the successive whole preimages, so the independent composite procedure is the SAME input procedure, up to proof irrelevance of actual SNC proofs. Identity follows from the same inverse identities. The internally produced real choice comparison is the actual old source projection followed by the inverse new projection. Both routes around the resulting square therefore reduce to the same actual primitive base map between the same genuine projections. Their exact equality follows, including every stage's entire controlled marked dictionary from the internally generated stage pullback theorem. The composite-input SNC list is internally obtained by the two genuine successive submersion pullbacks and exact literal preimage composition. The independently defined actual composite endpoint formula therefore exists from these true SNC data without assuming a composite-map submersion certificate. The conclusion here is exact composition of these actual endpoint maps and dictionaries; no additional general composite-submersion API or higher-invariant functoriality is inferred. □

239 Actual nonsurjective traces of independently chosen marking-one Cartier procedures

Original inputs are a genuine holomorphic submersion $f : B^m \to A^n$, the genuine entire finite indexed SNC list F on A, and primitive genuine Cartier blowup choice families in dimensions m and n. The two procedures, actual prefix count, maxima, manifolds, chosen blowups, projection inverses, controlled ideals, history masks and finite completion are produced internally. There is no caller Cartesian square, pulled ideal, maximum, tower comparison, termination or canonicality output certificate. This new own-scratch route awaits root registration. It concerns actual ordered marking-one monomials; higher-invariant resolution, prior general separation identification, and unrelated original ideals remain open.

Definition 239.1 (Produced unequal-prefix lift and retained whole marked record)**.** For each original prefix q let p(q) be the actual number of nonempty f-preimages in the genuine original independent centre word. Let $T_g(q)$ be the actual independently chosen global procedure and $T_s(p)$ the actual independently chosen source procedure on the true entire pulled original F. Both actual Cartier procedures internally produce genuine biholomorphic projection inverses b_g,b_s. Define λ(q)=b_g^{-1}∘f∘b_s between their actual terminal manifolds. The retained boundary indices are ALL original indices followed by the genuine global creation labels whose centre had nonempty f-preimage at creation; retained carriers which subsequently become empty remain present. The output trace record has carrier and entire ideal pulled by λ from the actual global record, the original marking, and the actual retained sublist of the pulled global boundary. At completion each chosen procedure instead uses its OWN internally computed initial coefficient sum. None of these objects is original caller data.

Lemma 239.2 (Whole holomorphic germ-ideal pullback composes across arbitrary dimensions)**.** *For holomorphic maps $Z^\ell \to Y^m \to X^n$, the actual full pullback of any original germ ideal all of whose members are holomorphic is the iterated actual pullback. In particular this holds for every original coherent ideal, and for ENTIRE marked records with coherent ideals; carriers, ordered boundary lists and marking are transported literally.*

Proof. Fix an actual point z and an actual germ in the direct pullback. Its literal finite-span presentation uses original ideal members a composed with f∘g. Original holomorphy of a and genuine holomorphy of f internally put a∘f in the intermediate actual pullback, using the true one-term span and constant coefficient one. Thus the SAME finite-span presentation belongs to the iterated pullback. Conversely, unpack the outer actual finite span. Each intermediate member has its true inner finite-span presentation. Compose that entire presentation by the genuine holomorphic g, including each actual holomorphic coefficient and the true neighborhood equality. Substitute into the outer finite span. Finite-span transitivity and literal function composition then give the original direct pullback. This is a whole membership equivalence at EVERY stalk, with no selected frame or pullback certificate. Coherence supplies original member holomorphy internally. The full marked record follows carrier inverse-image composition, whole list-map composition, this entire ideal equality, and unchanged marking. □

Lemma 239.3 (The actual unequal-prefix lift is a genuine unique holomorphic projection-square lift)**.** *Only original geometry and primitive genuine choices internally produce submersion of λ(q), the actual projection square, and its genuine holomorphic universal property. A compatible actual holomorphic cone has its unique actual holomorphic lift. No equality of source and global lengths is claimed.*

Proof. Every actual chosen blowup is of a genuine SNC member and is therefore a genuine Cartier blowup. The already proved unique Cartier inverse producer, applied recursively to the actual independent choices, gives b_g and b_s with their literal projections. Conjugating the original genuine submersion by these actual biholomorphisms gives a genuine submersion λ(q). The projection square follows the true inverse identities. Given a compatible actual holomorphic cone into the lower two objects, compose its source component with b_s^{-1}; it is holomorphic and satisfies the true upper square by compatibility and the genuine inverse identities. Any other lift is equal because its actual source projection is the bijective b_s. The same argument makes λ unique among all maps commuting with the produced whole projections. Empty source and dimension zero require no nonempty witness or false surjectivity assumption. □

Theorem 239.4 (Actual independently chosen unequal prefixes have the complete retained controlled dictionary)**.** *For every original q, the actual independent source prefix p(q) has its ENTIRE carrier, controlled ideal, marking and literal retained boundary equal to the output trace record above. Its real tower length is exactly p(q), and it is a genuine selected-factor test tower and ordered canonical monomial segment.*

Proof. The already produced nonsurjective ImageTrace internally computes p(q), the independent reference source word, and the ENTIRE identity-Cartier controlled record at that prefix. The genuine chosen-reference comparisons are internally produced by the real Cartier recursion, not caller assumptions. Transport this full reference record by b_s. Transport the actual global reference record by b_g. The whole cross-dimensional pullback composition lemma and genuine inverse cancellations identify the source entire ideal and carrier with the actual λ-pullback of the global controlled ideal and carrier. The literal boundary transport is list-map composition with the SAME generated increasing creation mask. The chosen record/reference dictionaries preserve every original entry and genuine retained creation event. A genuine source prefix takes precisely p(q) real active steps because ImageTrace gives its centre word equal to the p(q)-element actual nonempty trace word. Chosen-reference comparison therefore gives real tower length p(q). Actual selected-factor controls and the internally unique old singleton maximum give the genuine test tower and canonical segment. No equality with a false unfiltered full global boundary is asserted. □

Theorem 239.5 (Actual completed chosen source procedures retain the true finite unit trace)**.** *Each actual independent chosen procedure completes at its OWN original coefficient sum, with its genuine unit ideal. The source real length is the internally retained number of nonempty original*

traces; its entire completed controlled record is the retained actual global completed record pulled by the produced completed lift.

Proof. Apply the already proved actual independent chosen completion and genuine ImageTrace OWN completion. The real chosen-reference comparisons transport the complete entire unit endpoint and exact centre-length dictionary. The true entire pullback of the unit ideal remains unit. Apply the same actual full ideal and literal creation-mask argument as at bounded prefixes, using the two OWN completed references. Empty carriers and zero coefficients may leave ineffective metadata, but their true whole ideals are unit factors and no literal boundary entry is removed except a new event proved actually empty at creation. Compactness, global externally supplied order bounds, and a termination certificate are never used. □

Theorem 239.6 (Actual chosen unequal-prefix lifts obey genuine choice cocycles)**.** *Changing either actual primitive choice family produces the internally generated genuine same-dimension choice comparisons; these comparisons commute with the actual unequal-prefix lifts. Their actual identity, inverse and composite cocycles, and the whole retained records, follow the genuine projection-square uniqueness.*

Proof. The existing real two-choice procedure recursion internally gives the actual comparison biholomorphisms over the SAME original source or global manifold at their respective true prefix counts. Compose a produced lift with either actual comparison. Both resulting maps commute with the SAME actual projections because the genuine comparisons preserve their whole projection squares. Apply the proved unique lift property to identify them. In the explicit formulas this is just the true cancellation of the intermediate projection and its genuine inverse. The existing actual comparison identity/inverse/cocycle dictionary then gives the corresponding actual lift coherence. Transport the exact entire ideal and retained ordered history by these same real maps; all creation labels remain literal. This does not assert higher-invariant functoriality, general-driver identification, or a caller-supplied compression certificate. □

240 Actual nonsurjective smooth image trace for ordered marking-one monomials

Only a genuine holomorphic submersion $f : B^m \to A^n$ and a true entire finite indexed SNC carrier-power family F on A are original inputs. No surjectivity, chosen image open, image submersion, pulled-SNC, ideal, maximum, compression, marked-record or termination result is caller data. The following is a new own-scratch route awaiting root registration. It concerns the genuine finite identity-Cartier ordered marking-one procedure only; general higher-invariant base change and the old prior general-separation driver identification remain OPEN.

Definition 240.1 (The actual open image factor)**.** O is the actual set-theoretic range of f with the open structure produced below. The genuine factor f_O:B→O sends b to the literal pair (f(b), range witness b). Its inclusion composite is exactly f, and its surjectivity follows from the true range definition. The original list restricted to O is the entire literal preimage list, retaining every original index and every original coefficient, including empty factors. For an original actual global prefix q, first take its genuine OpenTrace marked record on O, with ALL original entries plus the internally retained creation-time events; then pull this ENTIRE record by the literal factor f_O. Denote this actual record by $M_f(q)$. It is an output definition, with no marked equality or compressed history certificate as original input.

Lemma 240.2 (Genuine submersion charts internally make the actual image open)**.** *Every genuine submersion f is an open map. Its actual range is therefore a true open, including the empty-source case.*

Proof. For an open W in B and a point a=f(b) with b∈W, the actual submersion normal-form charts internally produce a local holomorphic section s through b, with s(a)=b and f∘s=id near a. This is the proved actual holomorphic local-section producer, without a caller section certificate. Its eventual differentiability includes a and gives actual continuity there. Thus s(z)∈W near a; intersect with the true near-a projection-square identity. Each such z is genuinely f(s(z)) with s(z)∈W, so f(W) is a neighborhood of a. The pointwise neighborhood criterion proves openness of f(W). For W=B this is the actual range. Empty B gives an empty range and requires no nonempty or chart witness. □

Lemma 240.3 (The actual range factor is a genuine surjective submersion)**.** *The literal f_O is a genuine holomorphic submersion with the SAME internally selected global model complement as f, and is genuinely surjective.*

Proof. At b keep the original domain chart and complement equivalence. Restrict the original codomain chart to the genuine open O. Its forward values are the original chart on the actual included point. Restricting a genuine maximal-atlas chart to an actual open gives a genuine maximal-atlas chart of O. The original domain source maps into this restricted codomain source because every f-value belongs to its actual range. The written-chart identity remains the original projection onto the first factor, with the same equivalence and complement. These actual charts produce the true local submersion predicate. The original globally chosen complement works at every b, hence gives the genuine global submersion predicate. If B is empty the quantified local proof is vacuous. At any actual b the needed nonempty O witness is the literal f(b), not a global nonempty assumption. Surjectivity follows by unpacking each actual range witness. The actual chart derivatives give holomorphy, and the inclusion composite equals f pointwise. □

Lemma 240.4 (The same whole original indexed ideal has its genuine image and source dictionaries)**.** *The actual restricted family F_O is SNC internally from actual open charts; the true source family $f_O^* F_O$ is the literal f^*F and is SNC internally from the genuine produced submersion. Their entire vanishing ideals, powers and finite products are the actual whole ideal pullbacks; empty metadata factors contribute genuine unit ideals. Every original index and coefficient is retained before actual history compression. More precisely, let M(q) be the actual entire original geometric marked transform, L the original list length, and $K_f(q)$ the chronological labels of actual new global steps whose centre had nonempty f-preimage AT CREATION. Then $M_f(q).N = f^{-1}M(q).N$, $M_f(q).I = f^*M(q).I$, $M_f(q).d = 1$ and its whole literal E is the list of f-preimages of M(q).E at the indices range(L) followed by $K_f(q)$. Previously kept events remain even when their current carrier later becomes empty.*

Proof. Consume the genuine open holomorphic embedding of O and the actual original SNC coordinates to construct the entire pulled coordinate system. The produced genuine submersion f_O then supplies source SNC coordinates of the actual further pulled family. Pointwise inverse-image composition identifies the entire list with f^*F, including indices and coefficients. Actual coherent vanishing-ideal pullback equalities, genuine ideal power and finite product pullback identify the whole original monomial ideal at every stalk, not merely support. Empty factors are true unit ideals by their literal vanishing dictionary. For every genuine original prefix, its whole marked dictionary expresses its ideal as the true SNC terminal indexed product. Apply the actual open product pullback and then the genuine submersion product pullback, and use the literal double-weight equality. This proves equality with the direct f-pullback of the ENTIRE original controlled ideal without a generic composition certificate. The record carrier follows literal inverse-image composition and its marking follows the actual marking-one dictionary. The true range witness makes meeting O equivalent to having nonempty f-preimage; induction over the actual original choices therefore gives exactly the SAME creation labels $K_f(q)$. Literal list-map composition gives the boundary formula. No pulled-SNC or whole ideal equality is an original hypothesis. □

Theorem 240.5 (Actual source centres are the genuine nonempty f-traces of the original completed word)**.** *The independently generated source procedure has exactly the original actual completed centre word pulled by f and filtered by true nonemptiness. Its full literal history is the internally produced creation-time retained mask; each kept original/new entry and every true controlled coefficient is calculated from the actual procedures. For every original prefix q, the source prefix number is internally p=the length of the nonempty f-traces of its actual centre word. The independent source centre word at p is this exact filtered word, and its whole marked record at p is exactly $M_f(q)$.*

Proof. First apply the internally produced OpenTrace procedure to O and F. It deletes precisely original actual steps whose literal centre misses O, gives the independent O procedure, all-prefix full marked dictionaries and genuine unequal-length empty-stage compression. Actual old singleton ordering is preserved by the produced increasing retained embedding, not by asserting numerical equality of compressed colex labels. Next consume the produced surjective submersion f_O. The genuine surjective ordered-procedure producer identifies every independent source index, centre, state, controlled ideal and full marked record with the exact pullback of the independent O procedure, across the actual dimensions m/n. A centre in O is nonempty exactly when its actual f_O-preimage is nonempty by genuine surjectivity. Thus filtering original centres by meeting O and pulling by f_O is precisely filtering their actual f-preimages by nonemptiness. Every history decision refers to its actual creation event. Already-kept entries which later become empty remain literal. All dictionaries are derived from original f,F; no global maximum preimage statement is made for nonsurjective f. □

Theorem 240.6 (Actual nonsurjective mark-one identity-Cartier procedures have whole finite trace completion)**.** *The true independently generated source identity-Cartier procedure completes at its OWN internally calculated coefficient sum, with its genuine full unit endpoint and actual finite centre word above. Its actual finite identity-Cartier blowups form the genuine source trace of the original geometric procedure after the produced empty-step compression on the actual image. All-prefix full marked dictionaries and literal creation history are internally produced.*

Proof. The actual image procedure has its genuine compressed Cartier tower and full unit endpoint by OpenTrace at its OWN sum. The produced surjective submersion lifts its genuine identity Cartier steps using the actual pulled SNC member coordinates and true controlled-transform dictionary; equivalently consume the previously proved real surjective procedure producer. Its actual identity-Cartier projections give the genuine dimensional projection squares: both vertical maps are the produced literal identity projections. No cross-dimensional tower isomorphism is asserted. Every genuine source step decreases its actual coefficient sum by one, while internally omitted image steps are genuinely empty. The true source OWN sum terminates the independent source procedure; its entire controlled ideal is unit by its actual SNC inactive dictionary. All original and kept exceptional histories are retained in the precise ordered lists. This is an actual marking-one monomial producer, without compactness, caller bound, maximum, tower or completion certificate. It does not identify a general higher-invariant driver, arbitrary unrelated ideal, or prior separation with that driver. □

241 Independent finite old-gamma monomial procedures under genuine surjective submersions

The original input is an actual finite carrier-power list F on a genuine complex manifold. Its entire carrier list is SNC. The target input is a genuine surjective holomorphic submersion and the entire literal preimage list internally produced from F. No tower, maximum, value, factorization, termination, prefix dictionary or naturality certificate is supplied.

Lemma 241.1 (Internally selected unique literal old-key index)**.** *For every finite F and ambient dimension n, define sel(n,F) to be none when the actual finite positive nonempty index family is empty, and otherwise its unique maximum singleton key, with the unchanged coefficient and old colex index. Then none occurs exactly for the empty active family; every selected index is genuinely active and dominates every genuinely active singleton key. For actual SNC F, a selected index has an entire genuine nonempty Cartier carrier equal to the whole old-gamma maximum locus on the actual cosupport. These conclusions are internally produced.*

Proof. The actual finite active set is either empty or has an attained maximum under the true singleton key. The old colex coordinate makes the key injective on literal indices. Antisymmetry therefore gives uniqueness, so the actual classical selection is independent of any existence proof. If an active index exists, an actual point on its original carrier and the true SNC singleton cardinality force n¿0. Apply the actual indexed-product order, through-exponent and singleton-max dictionary in that genuine positive dimension. Its internally attained maximum is exactly the entire selected carrier on the whole actual cosupport. If n=0 or the manifold is empty, no nonempty carrier can occur under actual SNC; none is selected. No caller maximum or centre is involved. □

Lemma 241.2 (Real controlled successor and internal finite bound)**.** *For a genuinely selected index j, put next(F)=F.set(j,(empty,0)) appended with the SAME selected carrier and power k-1. It is the genuine complete mark-one controlled datum for the true identity Cartier blowup, with the entire actual transformed boundary. It is SNC and its whole coefficient sum is exactly the original sum minus one. Inactive F has complete unit ideal and supplies the actual reflexive canonical segment.*

Proof. A selected index is active, so k¿0 and its carrier is nonempty. The genuine SNC Cartier universal property internally supplies IsBlowupAlong for id; the positive indexed factor internally supplies marked admissibility. Complete controlled division gives the true full marked successor exactly as stated, including the selected old empty entry and appended exceptional factor. Coherent whole ideal equality, rather than support equality, justifies its zero-power empty entry. Real transformed SNC follows from the genuine marked test-step theorem. Removing the old k from its actual literal coefficient sum and appending k-1 gives new sum plus one equal to old sum. If inactive, every actual carrier either is empty or has zero power, so its whole ideal factor is unit. Induction over the actual list makes the complete product unit and its actual mark-one cosupport empty. Hence the genuine reflexive tower is a canonical test resolution, with no empty canonical step inserted. □

Definition 241.3 (Actual bounded procedures from the original input)**.** Define a bounded recursion on a natural q. At q=0 take the original state and the reflexive tower. At q+1 take the reflexive output if sel(n,F)=none; otherwise record the original state, literal selected index and whole selected carrier, perform the genuine identity Cartier step, and recursively run q steps on next(F). The actual procedure uses q=sum(F), computed from its literal original coefficients, and the actual SNC input internally supplies every genuine step and successor.

Theorem 241.4 (The internally generated procedure is a true finite resolution)**.** *The actual procedure is a genuine BlowupTower on the same original manifold, with projection id, old-gamma IsCanonicalSegment and actual IsGeoResolution. Its terminal WHOLE ideal is unit and its length is at most the original actual coefficient sum. Every literal state, selected index, entire centre and appended historical boundary agrees with the genuine controlled marked prefix. Every actual prefix has the entire SNC boundary and marked domain. No compactness hypothesis is required.*

Proof. Prove the private recursion specification by induction on q for every actual SNC F with sum(F)¡=q. At q=0 all coefficients are zero and the active set is empty, so the actual reflexive case applies. In the inactive successor case the same unit/reflexive dictionary applies. Otherwise

the preceding genuine step gives admissibility, actual centre equality with the whole old-gamma maximum and nonempty centre. Its coefficient sum drops by exactly one, so the actual successor satisfies the induction bound. Append the genuine recursive tower. Complete geoTransform equality makes the recursive whole marked datum literally the current transformed datum; thus canonical-segment and genuine test-resolution composition apply to the SAME original ideal and invariant. The whole unit endpoint, projection identity and length inequality follow from the actual append equations. The recorded states and boundaries are exactly these real step equations, so every literal prefix has its complete marked/SNC dictionary. Specialize q to the internally computed sum(F). This is a genuine procedure producer; the private induction bound is proved from its real input and is not an external termination assumption. □

Theorem 241.5 (Two independent procedures have literal smooth pullback histories)**.** *Let $f : Y^m - > X^n$ be a genuine surjective holomorphic submersion. Independently run the actual procedure in dimension n from F and in dimension m from its internally constructed whole preimage list. Their actual selected natural index words agree, their complete state histories are literal whole preimages, and their centre histories are entire preimages, with exactly the same length. Every genuine controlled marked prefix on Y is precisely the genuine marked pullback of the corresponding original prefix on X. The actual whole terminal marked datum has unit ideal on both sides.*

Proof. The true submersion chart construction internally gives the whole pulled SNC family and the COMPLETE marked product pullback. Genuine surjectivity identifies the actual nonempty positive finite index sets. If inactive, both independent procedures take their actual reflexive branch. Otherwise the actual chart dimension inequality, positive through SNC dimension, unchanged powers and old colex singleton key give order preservation and reflection across n and m. Uniqueness of the internally selected maximum therefore identifies the two independently selected literal indices. The two genuine Cartier successors commute with literal whole preimage, including the retained empty old entry and appended new actual carrier; the actual complete controlled marked dictionary also commutes. Induct on the common q. At every successor use the internally proved transformed SNC to apply the same argument to the independently generated new states. This proves the entire state, index and centre words and all whole marked prefix identities, without assuming a comparison certificate. Whole coefficient lists are unchanged by preimage, so the two independently computed initial sums are equal. The actual complete procedures consequently use the SAME internally computed q. Their genuine geometric resolutions, canonical segments, equal lengths and whole unit endpoints were already produced from the actual input by the preceding theorem. Empty manifolds and zero dimensions take the true inactive branch. Empty or repeated old historical entries are retained literally and never silently compressed. □

Theorem 241.6 (Literal naturality under identities and composition)**.** *For two composable genuine surjective submersions, the actual full finite state and centre dictionaries for their composite equal the successive preimage dictionaries, and the selected natural index words are the SAME. Identity pullback leaves the actual procedure unchanged.*

Proof. Whole finite-list preimage is literally functorial because entire set preimage is functorial and powers are unchanged. Apply the preceding independent-procedure equality first to the original submersion and then to the second, whose SNC input is internally produced by the first; compose those actual history equalities. The independently defined procedure on the composite full preimage list is literally the one on the twice-pulled list, so its SAME states and indices give coherence without a caller certificate. Identity follows from the actual identity preimage equation and the same deterministic unique-index recursion. □

These conclusions compare actual identity-Cartier old-gamma procedures. They do not identify prior general NC separation with the old general invariant driver, prove higher-invariant

functoriality, or assert global maximum-preimage equality for a nonsurjective map. Arbitrary Cartier blowup choices require a further genuine universal-property comparison; nonsurjective open restriction requires actual empty-step traces and compression, separately.

242 The original chosen zero-contact model along actual smooth histories

The initial marked ideal M, original chosen tangent direction and actual original histories are retained. We first produce the common contact model given by transporting that literal original choice. The independent choice in the pulled datum need not have the same mask; comparison of its actual recursive values with this common model remains a subsequent producer. No equality of the two independently chosen directions or masks is claimed.

Lemma 242.1 (The entire strict carrier through an actual Cartesian history)**.** *For any actual Cartesian history (Q, T, g) over an original smooth map f and any original subset B, even one closed only on an open,*

$$\operatorname{strictAlong}(Q, f^{-1}B) = g^{-1} \operatorname{strictAlong}(T, B).$$

Proof. Induct on the actual Cartesian constructors. The initial formula is the literal preimage. At a last step the actual lift is smooth, hence open and continuous; its preimage commutes with closure. The actual square identifies the whole preimages outside the actual pulled centre. Thus its preimage of the actual target strict transform is the actual source strict transform. Apply that one-step theorem to the induction hypothesis. No global closedness, transversality or desired carrier equality is required. □

Theorem 242.2 (The whole original contact coefficient record)**.** *Fix original well-formed M, an actual test history T of its original bar record, its actual Cartesian pullback Q, and any original function w. The entire unmasked contact coefficient record at Q, using $w \circ f$, is the genuine pullback by g of the entire original contact record at T, using w.*

Proof. The actual original bar-stage internally supplies well-formedness and the whole bar-stage dictionary. True relative coefficient pullback identifies the entire coefficient ideal, mark and retained indexed boundary. The preceding strict-carrier producer applied to the actual original set $M.N \cap V(w)$ identifies the full contact carrier; its original smooth preimage is literally $f^{-1}M.N \cap V(w \circ f)$. Replacing the carrier in the two coefficient records therefore identifies all four fields. The function is original and no chosen-contact equality is an input. □

Theorem 242.3 (The actual original contact model on the entire preimage mask)**.** *For any actual original open U in the stage of T, construct its actual smooth preimage open V in the stage of Q and its actual restricted map $p : V \to U$. The whole original contactMarkedKept record for M, w, T, U pulls back to the whole contactMarkedKept record for $f^*M, w \circ f, Q, V$. Every stage coordinate of p is the original map g.*

Proof. The preimage is open by actual continuity and the restricted map is smooth. The preceding full contact record theorem identifies the actual unmasked record. Its actual coefficient ideal is coherent internally, since the original bar-stage is well formed. Genuine whole marked pullback composition along the two actual open inclusions identifies the entire masked records. No masked-record comparison or preimage equality is supplied by the caller. □

Theorem 242.4 (The actual original selected zero-contact model and whole tail)**.** *For an original maximal-order marked ideal with positive mark, an actual original bar test history, a cosupport point of actual level zero outside the actual special locus and its actual current regime run, the simultaneous split and actual original mask produce a common chosen-contact lower model and*

the whole actual Cartesian relation of the restricted remaining tails. The common direction is the literal original tangent choice pulled by f.

Proof. The actual all-level regime/run dictionary supplies the same current run; split the actual histories at this literal index. The original test history gives head testness internally. At the selected projected point, the actual original cosupport, level and nonspecial branch supply membership in the actual original zero mask by the genuine original mask theorem. Its defining condition and original maximal order supply validity of the literal chosen direction on its actual original open. Smooth tangent pullback supplies its true tangent-direction property on the entire pulled open. The previous record theorem constructs the actual common lower model on the actual preimage of the original mask. Actual open restriction of the tail relation constructs both maps and the whole Cartesian restricted tails. The original target lower record is exactly the original lowerMarkedM at zero; the source record is explicitly the original transported-choice contact model. It is not identified with the independently chosen source lowerMarkedM. That final choice comparison and actual recursive values are further obligations. □

243 Complete original curve-family smooth functoriality

At original dimension bound one, every nonspecial maximal-order point would produce a dimension-zero lower cosupport, which is empty. This closes the whole first positive recursion level, rather than only a monomial or special branch. Original histories, centres and ordered boundary entries are retained. Sources are Włodarczyk's dimension induction and Bierstone–Milman's smooth functoriality [3, 5].

Theorem 243.1 (Original curve maximal-order canonical cosupport is special)**.** *For $n > 0$, original M in the dimension-one domain, original maximal order, and original Step One $R_0^{(n)}$-canonical whole history T, every point of the actual final cosupport is special.*

Proof. The existing Step One bridge internally supplies the true bar-marked test history, its canonicality and the cosupport equality. If a given cosupport point were nonspecial, the original canonical-tail theorem would supply its genuine masked lower marked ideal, an original domain with dimension bound $1 - 1 = 0$, its original $R_0^{(n)}$-canonical restricted history, and membership of that same point in the lower transformed cosupport. The zero-domain theorem makes that cosupport empty, a contradiction. This handles both positive-member and zero-contact branches without comparing independently chosen contact functions or assuming a lower certificate. □

Theorem 243.2 (Complete actual original curve-family smooth values)**.** *Let $n, m > 0$, $t = m - n$, and let an actual smooth Cartesian pair (Q, T) pull back the original dimension-one domain M and its original whole $R_1^{(n)}$-canonical history. At every z whose image lies in the actual target cosupport, the complete actual original family values satisfy*

$$R_{t+1}^{(m)}(f^*M, Q, z) = D_{t,1}(R_1^{(n)}(M, T, gz)).$$

Only original target canonicality is assumed; source canonicality and a source value formula are not premises.

Proof. Target canonicality gives its true original test history. If its actual nonmonomial order at the image is zero, consume the established complete smooth monomial-leaf formula at depth one. Otherwise write the actual order as a positive integer o. The original general canonical-tail theorem, with the genuine Step One maximal-order value over $R_0^{(n)}$, internally supplies the selected order-mask membership, whole companion domain of dimension one, maximal order, its whole Step One canonical history and its actual cosupport point. The first theorem makes this companion point special. The existing actual historical-tail constructor gives the literal whole source

companion record, both restricted smooth maps, the actual whole Cartesian restricted histories and their point identity. Original specialness transports by the actual geometric dictionary, and its full Step One value is the literal pair (level, $\top$) on both sides, with the same actual level. Thus the established strict Step One dimension map compares both entire companion values. The actual order and run dictionaries give the same outer o and exact split index; expanding the original general-value definition then yields precisely $D_{t,1}$ of the whole target value. No positive-inner or independent-contact case is left at this true depth. □

Theorem 243.3 (Complete original curve-family canonical history pullback)**.** *Under precisely the preceding original inputs, the actual whole source history Q is canonical for the actual original $R_{t+1}^{(m)}$, retaining every empty step and every literal preimage centre.*

Proof. Induct on the actual Cartesian history witness. The trivial pair is canonical. The original last centre is either empty or the full maximum locus of the preceding target history. The induction hypothesis supplies source canonicality of the preceding history. An empty target centre or an empty source preimage yields an allowed empty source step. Otherwise choose a source point mapping to the actual full target maximum. The actual cosupport dictionary and the preceding complete value theorem, applied at every source cosupport point, show that the source maximum locus is exactly the preimage of that full target maximum. Indeed, a source maximum dominates the chosen visible target maximum by strict order reflection, which makes its image a full target maximum; the reverse implication follows from domination of all target points. The strict unchanged $D_{t,1}$ supplies both order implications. The literal source centre is therefore its own full maximum locus. No surjectivity, source compactness or global-maximum existence is assumed, and the required actual value formulas are proved internally at every inductive stage. □

Theorem 243.4 (Original compact curve input constructs both terminal histories)**.** *For compact original target X, original smooth $f : Y \to X$, $n, m > 0$ and original dimension-one domain M, internally construct a target whole canonical terminal history T for $R_1^{(n)}$ and its actual whole source Q, g. The source has the correct domain bound $t + 1$, a whole canonical terminal history for the actual original $R_{t+1}^{(m)}$, unchanged history length, and the literal whole marked and cosupport pullback dictionaries.*

Proof. The existing per-sequence invariant driver applied to the original $R_1^{(n)}$ and original M supplies target canonicality and terminal geometric resolution, using compactness only of X. The actual smooth tower constructor supplies Q, g and its entire Cartesian witness. Apply the preceding theorem for source canonicity. The original-domain pullback theorem supplies the true source bound $1 + t = t + 1$ and all other fields internally. The actual marked dictionary supplies source testness and whole transformed-record equality. The cosupport dictionary makes the final source cosupport the preimage of the target's empty final cosupport, giving source terminal resolution. The actual witness supplies unchanged length. This closes the whole original bound-one smooth family case; arbitrary higher original dimension remains an active obligation. □

244 Original general companions and complete smooth family values

The actual general step evaluates its companion at the beginning of its own positive-order run, on its original open mask. To compare full original family values one must construct that whole smooth companion diagram from original canonical input. We do so before invoking the actual Step One branch comparisons, retaining the literal original history and every boundary entry [3, 5].

Theorem 244.1 (Complete original special Step One comparison)**.** *Let M be in original dimension domain $k + 1$ and of maximal order, let T be its actual specific-family Step One canonical*

history, and let Q be its genuine smooth pullback. At any target special point $g(z)$, the complete source Step One value is the strict Step One map induced by $D_{m-n,k}$ of the complete target value. This includes original level zero.

Proof. The original bridge gives actual bar-testness internally. Its true smooth all-branch dictionary preserves the original special predicate and level. Both entire Step One values are consequently $(s, \top)$ at their identical level s. The genuine strict Step One map retains s and maps WithTop infinity to infinity, proving the asserted entire pair formula without any lower-domain, contact-choice or value certificate. □

Theorem 244.2 (Original general canonical input constructs the actual whole companion diagram)**.** *Let M be in original dimension domain $k+2$, let T be its genuine full R_{k+2} canonical history, and let Q be its actual smooth pullback. At a target transformed-cosupport point $g(z)$ of positive actual nonmonomial order o, set r to its true order-regime run. Retain the literal target mask U at* $\mathrm{head}(T, r)$*, source mask V at* $\mathrm{head}(Q, r)$*, target companion P and both whole restricted tails L, L'. The original inputs internally supply both mask members, the target companion domain/maximal order/whole Step One canonicity and cosupport, the entire source companion as a true marked pullback of P, and actual smooth mask/tail maps sending the true source point to the true target point. Source companion testness and cosupport are supplied internally. No source companion canonical-history certificate is assumed or asserted.*

Proof. The actual original family canonical value gives target testness. Its specific sequence-form Step One family is a genuine maximal-order value function. Apply the general canonical-tail theorem with this actual maximal-order value to obtain target mask membership, companion domain and maximal order, its whole Step One canonical restricted tail and real cosupport point. The established actual historical companion diagram at the same o, r constructs both smooth maps, the entire marked-record equality and the whole actual restricted Cartesian tails. The actual order-mask preimage dictionary and commuting projection square give source mask membership. The final coordinate formula gives the exact restricted point equality. Target companion canonicity gives testness; the true relative marked/cosupport dictionaries supply source testness and cosupport internally. All data belong to the original M, T, Q. □

Theorem 244.3 (Complete original family comparison on the actual special companion branch)**.** *For the preceding original inputs, if the internally supplied target companion point is special on L, the complete original source $R_{m-n+k+2}$ value equals $D_{m-n,k+2}$ of the complete target R_{k+2} value. The assertion includes a special companion of original level zero.*

Proof. Construct the full actual companion diagram above. The complete special Step One theorem applies to its internally supplied companion domain, maximal order, canonicity and genuine smooth diagram. The marked-record and final point identities identify its entire values with the two actual general companion values. The true original order and run dictionaries give the same positive outer order and its exact two own-run splits. Both internally supplied mask memberships make their general companion values finite. Expand the actual successor-family definitions on the positive general branch. Their common outer order is retained, and the true dimension map on the right WithTop branch is precisely the whole Step One comparison already proved. This gives the entire family formula. □

Theorem 244.4 (Complete original family comparison on the positive companion lower leaf)**.** *For these same original inputs, suppose the actual target companion point is nonspecial on L and has positive original level $s+1$. Its canonical input internally supplies its own lower marked ideal, entire canonical tail and cosupport point. At actual nonmonomial order zero of that lower transform, the entire original source $R_{m-n+k+2}$ value equals $D_{m-n,k+2}$ of the entire target R_{k+2} value. Neither desired source canonical data nor a value comparison is supplied as an input.*

Proof. Construct the true whole companion diagram from original data. Apply the original positive Step One lower-leaf theorem to its internally supplied target companion domain/maximal order/canonicity/cosupport, nonspecialness and actual positive level. This internally constructs the further lower mask member and gives the complete companion Step One comparison at the genuine lower order-zero branch. Use the same whole marked-record and point identities, actual outer order/run dictionaries and original mask members as in the special-companion proof. Expand both complete original family definitions on their positive outer branch. Their unchanged outer order and mapped finite companion values give exactly the recursive $D_{m-n,k+2}$ formula, retaining every literal original split and record. □

Positive lower nonmonomial order, independent nonspecial level-zero contact, global maximum existence and the full canonical driver remain further work.

245 Actual positive-member recursion through the complete curve family

The complete original curve-family theorem closes all inner orders of the positive-member branch at original dimension bound two. We retain every original boundary index, whole lower marked record and actual restricted tail. This is the original dimension induction in [3, 5].

Theorem 245.1 (Complete positive-member Step One value at original bound two)**.** *Let $n, m > 0$, $t = m - n$, and let an actual smooth Cartesian pair (Q, T) pull back original maximal-order M in dimension bound two. Assume only the original target Step One history is canonical for the lower original $R_1^{(n)}$. At a target cosupport image which is nonspecial and has actual positive level $s + 1$, the entire original Step One values obey*

$$\mathrm{Step}_{R_{t+1}^{(m)}}(f^*M, Q, z) = \mathrm{stepMap}(D_{t,1})(\mathrm{Step}_{R_1^{(n)}}(M, T, gz)).$$

The actual inner nonmonomial order is arbitrary, and no source canonical or value certificate is assumed.

Proof. The existing original positive canonical lower-diagram producer internally constructs both true mask memberships, literal lower marked records, their actual whole Cartesian restricted histories, final point identity, original lower domain of dimension one, its full original $R_1^{(n)}$ canonical history and cosupport point. Apply the complete curve-family formula to this internally constructed genuine lower diagram. It compares the entire actual lower values without restricting their inner order. Rewrite its source record and final point using the produced literal equalities. The actual original level, regime-run and specialness dictionaries give the same $s + 1$, same selected run and nonspecialness on both sides. Expanding each original Step One value and its proved mask membership gives its actual lower value as a finite WithTop member. The strict Step One map preserves the first coordinate and maps that entire finite lower value by $D_{t,1}$, which is exactly the claimed formula. □

Theorem 245.2 (Complete positive-companion recursive value at original bound two)**.** *Let the original bound-two M, T be canonical for the whole original $R_2^{(n)}$, with an actual smooth whole Cartesian pair (Q, T). At a target cosupport image of positive outer order o, internally obtain its actual selected companion mask point, dimension-two companion domain, maximal order, whole original Step One $R_1^{(n)}$ canonical history and cosupport point. If that actual companion point is nonspecial and has positive level $s + 1$, then the entire actual family values satisfy*

$$R_{t+2}^{(m)}(f^*M, Q, z) = D_{t,2}(R_2^{(n)}(M, T, gz)).$$

There is no restriction on the actual inner order and no supplied lower history, source canonicality or value comparison.

Proof. The existing original whole-family canonical companion producer constructs the stated target data and the exact whole source companion, both actual restricted smooth histories and their point identity internally. Apply the preceding theorem to this true companion diagram and its actual nonspecial positive-level point. Rewrite the whole source companion and final point by the produced equalities. The original whole-history order and regime-run dictionaries identify both positive outer orders with o and both actual selected split indices. Expanding the two original general values at their produced order-mask memberships reduces the whole values to the preceding complete Step One comparison. The definition of the unchanged $D_{t,2}$ is precisely the general map over that Step One map. This removes the monomial-inner-order restriction at this true recursive depth. Independent zero-contact comparison and higher-dimensional recursion remain active obligations. □

246 Actual positive Step One leaves under smooth pullback

We compare the entire actual Step One value at a positive original level whose internally constructed lower history reaches a monomial leaf. Only the original target canonical history is assumed; source lower geometry, domain, testness and cosupport are produced from its genuine smooth diagram. This is a recursive branch of Włodarczyk's original Step One, interpreted with actual smooth pullbacks [3, 5].

Theorem 246.1 (Original canonical positive input produces the genuine lower smooth diagram)**.** *Let M be in dimension domain $k+2$, of original maximal order, and let T be its specific-family Step One canonical history. Let Q be its actual smooth pullback, with final map g, and let z map to a nonspecial target transformed-cosupport point of positive level $s+1$. Set r to that point's actual current-level run; use the literal splits at r. The original inputs internally supply both actual lower-mask memberships, the entire target lower marked ideal in dimension domain $k+1$, its whole canonical restricted tail and true cosupport point, actual smooth maps of the two lower masks and the two whole restricted tails, the exact source lower marked record as pullback, and the source lower test history and cosupport point with correct added-fibre-dimension domain. No source lower canonical-history certificate is supplied or asserted.*

Proof. The actual original canonical-tail theorem supplies target lower mask membership, domain, whole canonicity and cosupport at its own true run. The original Step One bridge supplies target bar-testness and equality of transformed cosupports. The established positive lower diagram producer applies to this bar-test history and this same actual run, constructing the entire marked-record equality and actual restricted Cartesian tails. Its two stage maps are the restrictions of the genuine smooth stage maps. The target mask membership and exact mask preimage dictionary imply source mask membership, since the true tail projection square commutes. The final restricted map sends the source point to the target point, by its proved underlying-coordinate formula. The target lower canonical history is a genuine test history. Its actual relative marked/cosupport dictionaries therefore supply source testness and cosupport internally. Pullback of the actual target lower domain supplies the correct source domain $(k+1)+(m-n)$. All these witnesses belong to the same literal original M, T, Q, z and selected run. □

Theorem 246.2 (Complete actual positive Step One comparison at the lower monomial leaf)**.** *For these original inputs let J, L, p be the actual target lower marked ideal, whole restricted tail and cosupport point produced above. If the actual nonmonomial order of J_L at p is zero, the whole source Step One value is the strict Step One map induced by $D_{m-n,k+1}$ of the whole target Step One value. Both original positive levels remain $s+1$, and their actual runs agree. The lower leaf condition is a genuine geometric branch, with its mask member supplied internally; no desired source value or lower canonical history is a premise.*

Proof. Obtain the complete actual lower diagram of the first theorem. The actual all-branch dictionary preserves level, nonspecialness and the current-level run. Thus both whole original

Step One definitions use these precise literal lower records and restricted tails, with no special infinity branch. Both internally supplied mask memberships identify their lower values with the complete actual lower family values. The preceding actual-family monomial-leaf theorem applies to the genuine restricted lower diagram and the internally supplied target lower domain and test history. It gives precisely $D_{m-n,k+1}$ on their entire actual lower family values. The exact marked-record equality and final point formula identify these values with the original Step One expressions. Mapping the unchanged positive first coordinate and the finite WithTop lower value gives the asserted complete Step One pair formula. □

Positive lower nonmonomial order, chosen level-zero contact comparison, the full outer general companion and the canonical driver remain open.

247 Original zero-contact lower diagrams and the actual source mask

The genuine contact-value anchor already in the original Step One stack requires an actual tangent preimage mask and avoidance of the original indexed boundary. We construct precisely those data for the common source contact model using the actual transported target choice; we never equate that choice with the independent original source choice. This is the smooth contact-lowering step of [3, 5].

Theorem 247.1 (Original actual common-contact mask geometry)**.** *Let an actual smooth Cartesian pair (Q, T) pull back original well-formed maximal-order M with positive mark. Suppose T is an actual bar-marked test history and the image of z is in its actual bar cosupport, nonspecial at level zero. Internally obtain the actual run r, target's actual chosen tangent function w, both whole split Cartesian pairs, the full source preimage V of the actual target zero mask, its point membership, the restricted maps and exact whole common-contact source record. In addition V lies in the actual source head preimage of the transported target tangent open, and is disjoint from EVERY original indexed source member. The transported $w \circ f$ is a true source tangent direction there.*

Proof. Consume the existing original zero-contact selected-diagram producer; it constructs all split histories, actual open preimage and maps, whole source record and membership internally, with the actual transported direction. For a point of V, its image by the head map lies in the target's literal maskB. Interior membership implies its target head projection lies in the true chosen tangent open. Actual head projection commutation therefore puts its source head projection in the transported tangent preimage. For original-member avoidance, the genuine whole bar-original-member dictionary identifies the entire source original list as literal preimages of the target original list, retaining every index. An image point in the actual maskB lies on no such target member by its definition. Thus a source point cannot belong simultaneously to V and any original source member; this proves all required disjointness assertions, including empty or repeated inactive entries. No source canonical or value premise occurs. □

Theorem 247.2 (Original canonical zero branch produces its entire lower diagram)**.** *Let $n > 0$, let original M lie in dimension bound $k+2$ and have original maximal order, and let its original Step One history be canonical for the lower original $R^{(n)}_{k+1}$. For an actual Cartesian pair and a target cosupport image nonspecial at level zero, construct the target lower ideal of dimension $k+1$, its whole original canonical restricted history and cosupport point, the entire actual common source contact record using the transported target choice, the genuine whole restricted Cartesian diagram and point equality. Construct its source domain with true bound $t+k+1$, actual testness and cosupport membership, and the full source tangent-mask and original-member avoidance data above. All these are outputs from the original input. Source lower CANONICALITY and the*

complete lower VALUE formula are supplied by the subsequent depth-lowering induction, not by a premise of this producer.

Proof. The original Step One bridge supplies target bar-testness, bar canonicality and its actual cosupport equality. Apply the original canonical-tail theorem at the true nonspecial level-zero point. It internally gives the true target zero-mask membership, complete original lower record, domain $k+1$, original $R^{(n)}_{k+1}$-canonical restricted tail and its actual cosupport point. Apply the preceding original mask-geometry producer to the internally obtained bar data; it yields the genuine whole common source contact diagram. The coordinate equality of its restricted final map identifies the exact two cosupport points. The original-domain pullback theorem gives the common source lower domain with bound $(k+1)+t = t+k+1$, using its actual restricted base submersion. The original lower canonical history supplies actual lower testness, and the genuine whole marked and cosupport dictionaries then give actual common source testness and point membership. Rewrite by the proved literal whole source record equality. These true source data, together with the two mask conditions, are exactly those needed to apply the existing contact-value anchor AFTER the depth induction has internally constructed common source lower canonicality. The source's own lower canonicality is separately constructed from current source canonicity in that induction. No desired source-canonicity or value certificate is an endpoint input. □

248 Actual positive Step One lower histories under arbitrary smooth maps

Fix an original well-formed marked ideal M on X^n, its actual coefficient-homogenized test history T, and an actual smooth Cartesian history Q over $f : Y^m \to X^n$, with actual final map g. No surjectivity, positive source compactness, injectivity on global boundary sets, desired lower-record comparison or restricted-tail certificate is assumed. Every empty indexed boundary entry and original ordering is retained.

Lemma 248.1 (Reflection of original-member membership through a point)**.** *Let E be an actual SNC divisor and E_0 a list of its actual members. For a smooth map g, a member $H \in E$ through $g(z)$ whose whole preimage belongs to E_0's preimage list already belongs to E_0.*

Proof. Choose $H_0 \in E_0$ with $g^{-1}H_0 = g^{-1}H$ and retain their actual indices i, j in E. Smooth pullback supplies the SNC structure of the entire indexed list $g^{-1}E$. Its indexed members i, j pass through z and agree as whole sets, hence as germs. The actual SNC germ separation theorem gives $i = j$. Thus $H = H_0 \in E_0$. No reflection is claimed for members missing z; coincident empty preimages remain literal entries. □

Theorem 248.2 (The whole member-restricted record)**.** *For an original well-formed J, a list E_0 of its actual members and any smooth g, the smooth pullback of the whole member-restricted record selected by E_0 through gz equals the corresponding whole record for g^*J at z.*

Proof. The true smooth coefficient-ideal theorem identifies the entire coefficient record, thus its literal ideal and marking. For the carrier, write the intersection as the condition on members through the point. Pullback preserves membership at the point; the preceding lemma reflects membership in E_0 when it is needed. The same reflection in the existing filter-map theorem identifies the entire surviving indexed divisor in its original order, with multiplicities and empty entries. These four field equalities identify the entire original marked record. □

Theorem 248.3 (The actual positive masked lower record)**.** *For every positive lower index $s+1$ and every actual stage point z, the actual restricted smooth map between the original masks pulls back the whole original lowerMarkedM record to the pulled history's whole lowerMarkedM record.*

Proof. The actual original bar-stage gives its well-formedness internally. The original-member list is literally a prefix of its whole divisor, so membership inclusion is internal. The bar-stage and original-member dictionaries and the preceding producer identify the whole unmasked member-restricted record. The all-branch maskA theorem identifies the actual open mask with the smooth preimage. Restrict the actual smooth map to these actual opens. Coherence of the coefficient ideal, supplied internally by the original stage, makes whole marked pullback commute with both compositions of these actual maps. Substitution identifies all four fields of the actual positive masked lower record. The formula holds for every stage point; no positive-level or nonspecial-point certificate is needed to construct the record itself. □

Theorem 248.4 (The actual selected positive lower Cartesian tail)**.** *At every literal split index j, and in particular at the literal current fixed regime run r at a point of positive level $s+1$, the actual simultaneous split supplies both head and tail maps. There exist actual maps between the actual lower masks and their actual tail preimages, which compare the whole lower records and give an actual Cartesian relation of the restricted tail histories.*

Proof. First apply the actual simultaneous split at the arbitrary index j. For the actual selected diagram, the full original level/run dictionaries identify the current selected run in both original histories, so apply the arbitrary-split construction at precisely that literal index. The original full test history supplies head test data internally. The tail square identifies the selected projected points, and the preceding theorem gives the exact lower record at those points. The actual mask dictionary and actual open-restriction producer construct the maps and the true Cartesian restricted histories. Their stage coordinates remain the original stage coordinates. Neither the lower record nor the restricted-tail conclusion is a premise. □

The chosen level-zero contact comparison, the ambient dimension shift of higher values and the full recursive canonical invariant comparison remain further producers. This result constructs actual positive lower records and tails and does not assert equality of ambient-dependent resolution invariant values.

249 Actual positive member histories under smooth pullback

This extension supplies the positive-boundary branch needed in the smooth comparison of the original recursive invariant. It uses the literal marked ideal, the original indexed divisor, the actual coefficient-homogenized ideal $\overline{J}$, and actual Cartesian test histories. The zero-boundary codimension-one branch, chosen tangent-direction masks, and the complete higher recursive comparison remain separate obligations. No equality of chosen tangent directions or of the entire higher invariant is assumed. This is the smooth-map extension of the original Step One construction in Wlodarczyk, proof of the marked-ideal resolution proposition, Step 1aa/1ab.

Lemma 249.1 (The actual coefficient-homogenized record)**.** *For a smooth holomorphic map $f : Y \to X$ and an original well-formed marked ideal J, $f^*\overline{J} = \overline{f^*J}$. If $Q \to T$ is an actual Cartesian history and T is a geometric test history for $\overline{J}$, then Q is a test history for $\overline{f^*J}$ and their entire transformed records correspond under the actual last-stage smooth map g.*

Proof. Relative derivative ideals, homogenization and coefficient ideals commute with smooth pullback, including the actual carrier, marking and full divisor list. Apply those two identities successively to $\overline{J} = C_N(H_N(J))$. The established geometric Cartesian controlled-transform identity, applied to the well-formed $\overline{J}$, gives both the test-history assertion and the transformed record. The original $|E|$ strict transforms are the initial $|E|$ entries of that transformed record, so their lists and boundary counts are exactly pulled back. □

Lemma 249.2 (Actual member locus and positive special branch)**.** *At every stage, the original member-intersection locus of the cosupport of $\overline{f^*J}$ is the inverse image of the literal original*

member locus for $\overline{J}$. If the target boundary count at $g(y)$ is positive, y is special for the source Step One history if and only if $g(y)$ is special for the target history. For every fixed positive boundary count s, the actual Step One regime at level s corresponds at all points, and its complete recursive run index is unchanged.

Proof. A smooth map is continuous and open; thus it maps the neighbourhood filter at y onto the neighbourhood filter at $g(y)$. In the definition of the member locus, transport the cosupport, carrier, full original divisor list, and the quantified nearby member-intersection condition with this exact filter equality. The special branch at positive boundary count is precisely membership in this member locus; the level-zero alternative is false. For a fixed positive s, points whose level differs from s satisfy neither regime, and the remaining points use the preceding special-point comparison. Induction on the actual history compares the regime at every intermediate stage and therefore the recursive run index. □

Lemma 249.3 (The actual positive member mask)**.** *For any point of positive boundary count, the literal Step One member mask $A_{f^*J,Q}(y)$ is $g^{-1}A_{J,T}(g(y))$.*

Proof. The condition excluding every original member missing the base point corresponds by the original list dictionary. In the remaining condition, if a point has the same boundary count as the base point, this count is positive, and the preceding special-point comparison applies; if the counts differ, both exclusions are false. The raw sets in the two masks therefore correspond exactly. For a continuous open map, inverse image commutes with interior, proving the statement for the actual open masks. No chosen contact mask is substituted for these masks. □

250 Actual projection of holomorphic germs preserves whole ideal quotients

Lemma 250.1 (Internally produced entire normal-jet coefficients)**.** *Let $B = \mathcal{O}_{\mathbf{C}^d,0}$, $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$, and let $\eta : B \to R$ be the actual tangential projection pullback. Let J be the actual ideal of the last t coordinate germs. For EVERY q, internally define the whole coefficient map c_q of R/J^q by the inverse of its actually constructed finite normal-monomial B-linear equivalence. It is B-linear, has literal kernel J^q, and its actual reconstruction is $g \equiv \sum_{|\alpha|<q} \eta(c_q(g)_\alpha)y^\alpha \pmod{J^q}$. No finite frame, tuple inverse, kernel or representation certificate is input.*

Proof. Compose the true quotient map $R \to R/J^q$ with the inverse of the internally constructed normal-monomial linear equivalence. The latter is genuinely bijective by actual normal Taylor coefficient separation and the proved entire monomial representative theorem. Its linearity gives additivity and $c_q(\eta(a)g) = ac_q(g)$ for the ENTIRE original base germ a. Its inverse identity gives the exact whole finite reconstruction using the actual monomial evaluation dictionary. Its injectivity and the genuine quotient kernel give $c_q(g) = 0$ exactly when $g \in J^q$. This includes $q = 0$ (zero quotient and empty tuple), $t = 0$ and $d = 0$, without a prescribed basis input. □

Theorem 250.2 (Every original ideal has an exact actual coefficient membership dictionary)**.** *For ANY actual original ideal $I \subseteq B$, EVERY q, and EVERY actual germ $g \in R$, one has $g \in \eta(I)R + J^q$ if and only if EVERY actual coefficient $c_q(g)_\alpha$ is in the SAME entire original I. No finite generation or coefficient membership/output certificate is supplied.*

Proof. First prove the stronger generator-span assertion that for every actual $g \in \eta(I)R$ and every whole multiplier $r \in R$, all coefficients of rg belong to I. Induct on actual ideal-span membership. A true image generator is $\eta(a)$ with actual $a \in I$; B-linearity rewrites its coefficient tuple after ANY multiplier as $ac_q(r)$, whose entries belong to I. The zero and additive cases use genuine linearity. In the arbitrary R-multiple case, absorb the new multiplier into the universal whole r; no coefficient convolution or fake base-linearity over R is asserted. Thus the assertion holds for the ENTIRE extended original ideal. A member of $\eta(I)R + J^q$ is an actual sum of a member

of this ideal and an element of J^q; the second coefficient tuple is zero, proving one implication. Conversely, use the actual finite reconstruction of the first node. Each finite term $\eta(c_q(g)_\alpha)y^\alpha$ belongs to $\eta(I)R$ by its actual coefficient membership. Their whole finite sum therefore lies there. The genuine inverse identity says the difference from g lies in J^q, proving the reverse implication with no chosen polynomial model. □

Lemma 250.3 (Genuine normal powers detect the whole extended ideal)**.** *Every actual ideal $K \subseteq R$ satisfies $g \in K$ if and only if $g \in K + J^q$ for ALL q. This is internally proved for the SAME genuine normal coordinate ideal, without caller separatedness or Krull certificate.*

Proof. Every genuine normal coordinate germ vanishes at the actual origin. The actual local-germ maximal-ideal evaluation dictionary therefore puts every original generator in the genuine maximal ideal $\mathfrak{m}_R$. Actual ideal-span monotonicity gives $J \subseteq \mathfrak{m}_R$, and hence $J^q \subseteq \mathfrak{m}_R^q$. Membership in $K + J^q$ at ALL q thus implies membership in $K + \mathfrak{m}_R^q$ at ALL q. The already proved faithful point-truncation theorem internally installs genuine holomorphic-germ Noetherianity and local-ring Krull separation, giving $g \in K$. The forward implication is the genuine inclusion into each sum. Empty coordinate sets and $q = 0$ are literal and do not require a positive codimension. □

Theorem 250.4 (Actual projection preserves the ENTIRE colon of ANY two original ideals)**.** *For ANY actual original ideals $I, L \subseteq B$ the entire projection pullback satisfies $\eta(I : L)R = (\eta(I)R : \eta(L)R)$. There is no hypothesis $I \subseteq L$, denominator invertibility, caller flatness, finite presentation, finite generator, truncation bound or quotient identity. Both ideals and the projection are the actual original objects.*

Proof. For the easy inclusion fix an actual $a \in I : L$. For every original $b \in L$, one has $ab \in I$, so $\eta(a)\eta(b) = \eta(ab) \in \eta(I)R$. Genuine span induction over the ENTIRE extended denominator ideal extends this multiplication inclusion to all its members. Thus $\eta(a)$ lies in its true quotient; actual ideal-span closure extends to the ENTIRE $\eta(I : L)R$. Conversely take a true g in the right-hand colon. For EVERY actual $b \in L$, $\eta(b)g$ belongs to the ENTIRE extended I. For EVERY q its actual tuple entries therefore belong to I by the exact whole-ideal coefficient criterion. Genuine B-linearity identifies those entries with $bc_q(g)_\alpha$. Since this holds for EVERY original $b \in L$, every actual $c_q(g)_\alpha$ lies in the SAME original $I : L$. The real finite coefficient criterion then gives $g \in \eta(I : L)R + J^q$ for EVERY q. Actual normal-power separation gives $g \in \eta(I : L)R$, completing equality. No finiteness over all degrees, inverse denominator or ideal-inclusion certificate entered the argument. This proves a real algebraic prerequisite for non-full-carrier controlled transform naturality. The still necessary actual arbitrary submersion chart/stalk transport, whole sheaf-colon bridge, relative marked-transform consumer and general higher invariant/canonical driver/termination remain OPEN; this projection-germ theorem alone does not claim those conclusions or formal native publication. □

251 Regularity of actual saturated cylinders

This chapter proves the geometric reflection step for a genuine smooth submersion. Product coordinates are supplied by the original submersion; no reflected-regularity or chosen-normal-equation certificate is an input. The local product calculation is stated first and then consumed by those actual charts.

Lemma 251.1 (Actual vertical-preserving normal coordinates)**.** *Let Ξ be a local holomorphic equivalence at the origin of $\mathbb{C}^{n+t}$. Suppose its inverse image of a specified coordinate plane contains the vertical plane $\{0\} \times \mathbb{C}^t$ on its actual source. There are new local holomorphic coordinates at the origin, a normal cutting index set X contained in the first n coordinates, and an actual source contained in that of Ξ, in which the same normal zero set is the coordinate subspace cut by X. Every vertical coordinate is retained literally. The number of cutting indices is the original normal codimension.*

Proof. Apply the existing simultaneous model-coordinate construction with the centre cutting indices equal to the first n coordinates and with the divisor family empty. Containment of the vertical plane is exactly its centre-containment premise. That construction internally chooses independent normal differential columns from those first n coordinates, then applies the holomorphic inverse function theorem. It gives a normal index set X contained in this first block and a permutation σ. Undo that permutation in the resulting coordinate values. Its literal unchanged-coordinate formula then says that every coordinate outside X is retained, hence all vertical coordinates are retained. The normal plane dictionary becomes precisely vanishing on X. The permutation identifies X with the old last normal block, giving the asserted cardinal. □

Lemma 251.2 (An actual horizontal slice of vertical-preserving coordinates)**.** *A local holomorphic equivalence at the origin of $\mathbb{C}^{n+t}$ which retains every vertical coordinate restricts to a genuine local holomorphic equivalence at the origin of $\mathbb{C}^n$. Its source and target are the literal preimages of the original source and target under $u \mapsto (u, 0)$; its maps are the first-block projections of the original maps on that slice.*

Proof. The source and target are open by continuity of the linear slice inclusion. The forward map sends a horizontal input to a horizontal output because all vertical coordinates are retained. Conversely, the inverse of a horizontal output is horizontal: applying the forward map to that inverse shows that its retained vertical coordinates vanish. The original inverse identities therefore imply both slice inverse identities and both genuine source-to-target maps. Holomorphicity follows by composition with the linear inclusion and projection. The origin is retained, so the actual slice source contains it. □

Theorem 251.3 (Reflection of regularity of an actual cylinder)**.** *Let $A \subseteq \mathbb{C}^n$ contain the origin. If its actual cylinder $A \times \mathbb{C}^t$, read in the fixed finite product coordinates, is regular of dimension j at the origin, then $t \leq j$ and A is regular of dimension $j - t$ at the origin.*

Proof. Take the actual flattening supplied by cylinder regularity. The vertical plane lies in its normal zero set, since $0 \in A$. The first lemma supplies genuine vertical-preserving coordinates with normal cutting indices X in the first block, of cardinal $n + t - j$. Restrict those actual coordinates to the horizontal slice using the second lemma. The cylinder dictionary on that slice says exactly that membership in A is vanishing on those same first-block indices. Permute only this first block to put its $|X|$ cutting indices last. This is a genuine flattening of A of dimension $n - |X|$. Because $|X| \leq n$, we have $t \leq j$, and the displayed cardinal gives $n - |X| = j - t$. □

The subsequent original-manifold consumer composes these actual slice coordinates with the actual lifted submersion and original preferred charts. The cylinder calculation alone does not yet identify the zero-branch contact choices or complete higher invariant values under a smooth map.

Theorem 251.4 (Original regularity reflection under an arbitrary smooth map)**.** *Let $f : Y^m \to X^n$ be an actual holomorphic submersion at y, and let $A \subseteq X$ be any original subset. If the actual preimage $f^{-1}A$ is regular of dimension j at y, then $m - n \leq j$ and the original A is regular of dimension $j - (m - n)$ at $f(y)$.*

Proof. The true submersion supplies $n \leq m$ and its actual lifted holomorphic coordinates, centred at y, with first block equal to the actual centred preferred coordinates of f. Compose their genuine inverse with the actual flattening of the whole source preimage. Its source is the actual intersection where both coordinate maps and both original preferred charts apply. On that source the normal plane dictionary identifies membership with the literal cylinder over the original A in the centred target coordinates: the first-block equality and both chart inverse identities prove this directly. The source point belongs to the preimage, hence the original target point belongs to A. Apply the cylinder reflection producer on this actual local source. It supplies the fibre-dimension bound and an actual target flattening of dimension $j - (m - n)$ in the centred target coordinates. Compose

this flattening with the actual centred preferred target coordinates. The literal inverse identities give the original membership dictionary, and its source lies in the preferred target chart. This is precisely the original definition of target regularity. □

Corollary 251.5 (Actual regular codimension-one pairs)**.** *For any original subsets S, N and an actual holomorphic submersion at y, the existence of regular dimensions $k+1, k$ for $f^{-1}N, f^{-1}S$ at y is equivalent to the existence of such consecutive regular dimensions for N, S at $f(y)$.*

Proof. For reflection, apply the original reflection theorem to both subsets. The regular dimension k of the smaller source subset is at least the fibre dimension t, so $(k+1)-t = (k-t)+1$. This supplies the target pair. Conversely, the original forward regularity producer adds the same actual fibre dimension to both original dimensions, retaining their difference of one. No supplied choice of target regular dimension or coordinates is needed in either direction. □

252 Genuine restricted controlled transforms on arbitrary original carriers

Theorem 252.1 (The whole original restricted ideal commutes with smooth pullback)**.** *For an original genuine submersion f, original closed smooth carrier N and original coherent ideal I, internally prove $f^*(\mathrm{restrictOn}_N\, I) = \mathrm{restrictOn}_{f^{-1}N}(f^*I)$. The two dimensions may differ and neither $N = X$ nor surjectivity is assumed.*

Proof. The actual restriction definition equals the sum of the ENTIRE original I and the genuine whole vanishing ideal of N: take the remainder of a true finite restricted relation; conversely a vanishing remainder gives the literal restriction. The original closed smooth carrier internally gives coherence of that whole vanishing ideal. Actual pullback of the entire sum and the whole original vanishing-ideal pullback identity give $f^*I + \mathcal{I}_{f^{-1}N}$. Original smoothness also internally gives that $f^{-1}N$ is a closed smooth carrier and f^*I is coherent. Their actual restriction sum dictionary now identifies the ENTIRE source restriction. No restricted presentation, vanishing equality or source carrier certificate is supplied. These cited whole vanishing/sum ingredients are the exact previously kernel-compiled snapshot proof bodies, not an assumed output or a claim that the root's full unpublished chapter has been accepted. □

Theorem 252.2 (The actual arbitrary-carrier quotient commutes with smooth pullback)**.** *For the same original geometry and ANY original coherent denominator L, $f^*(\mathrm{colonAlong}_N(I, L)) = \mathrm{colonAlong}_{f^{-1}N}(f^*I, f^*L)$. Every genuine natural power of an original invertible denominator is included. No $I \le L$, restricted divisibility, invertibility or flatness certificate is needed for the quotient identity.*

Proof. The literal original definition is $\mathrm{colon}(\mathrm{restrictOn}_N\, I, L)$. Original coherence and the actual carrier vanishing ideal internally prove coherence of its ENTIRE numerator. The new genuine submersion whole GermIdeal colon theorem applies to this numerator and L without the old ambient inclusion hypothesis. Node one identifies the actual pulled numerator, giving the literal source quotient. For powers use the actually coherent denominator power and the actual whole ideal-power pullback identity. The zero power is the actual unit ideal, so mark zero is retained without an extra condition. □

Theorem 252.3 (Every actual admissible marked successor has the entire pulled record)**.** *ONLY an original genuine submersion, genuine target/source blowups at the literal pulled centre, original well formed marked M on ANY closed smooth carrier and its actual admissible centre internally prove the ENTIRE marked successor equality under the true canonical smooth lift. The whole N/E/I/d record agrees, including all old empty or repeated boundary entries.*

Proof. Original admissibility gives the actual centre inclusion in $M.N$. The true strict-transform carrier theorem internally proves that the target successor carrier is closed smooth. The actual lift is internally a submersion, so node two applies to the ENTIRE actual restricted quotient defining its controlled ideal. Genuine coherent pullback composition and the actual projection square identify its original numerator with the true source blowup pullback. The actual entire exceptional vanishing-ideal pullback and power dictionary identify its denominator. The true strict carrier preimage identity identifies the restriction carrier with the source strict carrier. These equalities give the literal whole successor ideal, without replacing relative cosupport admissibility by an unjustified ambient inclusion. The existing genuine strict-boundary dictionary transports each literal old list entry and appends the true exceptional preimage. N agrees by the true strict-carrier identity and d agrees literally. Thus marked-record extensionality gives the entire equality; source admissibility was genuinely produced from original relative order/cosupport and NC geometry. □

Theorem 252.4 (Joint actual geometric and whole marked recursion on every carrier)**.** *For a genuinely generated actual smooth pullback history of an original ordinary finite target test tower on original well formed M, internally produce the source test tower and the entire marked endpoint equality. No full-ambient carrier hypothesis or source test/equality certificate occurs. The ENTIRE actual source cosupport equals the actual endpoint map's preimage of the target cosupport.*

Proof. Induct on the ACTUAL geometric constructor history. At the empty tower both records are literal initial records. At a genuine last target/source blowup, apply induction to the actually recorded shorter target test tower. Original well formedness and the true shorter target test conditions internally give well formedness of the actual current target marked datum. The entire shorter source equality identifies its actual datum with the true current pullback. Genuine relative admissibility transport therefore proves source admissibility at the literal pulled centre. Node three identifies its whole actual successor record with the pulled target successor. These two results prove source test validity and the entire endpoint equality simultaneously. Every created boundary entry stays in the literal constructor history; no empty-step compression, removal of repeated entries or caller successor record is used. The actual history itself is produced in node five from the original ordinary target geometry. The genuine entire relative cosupport identity applies to the actual current coherent marked record; after the produced whole record equality it gives the whole source cosupport dictionary. □

Theorem 252.5 (Original-only production of all relative marked Cartesian heads)**.** *ONLY original f, original well formed M on arbitrary carrier, an original ordinary finite target tower and its genuine original test conditions internally produce an actual source tower, smooth endpoint map, whole Cartesian square, true same length, source test conditions and ENTIRE marked endpoint equality. EVERY literal head has the same genuine geometric, whole marked and cosupport dictionaries. No source tower, termination, canonical or naturality certificate is supplied.*

Proof. The genuine original finite-tower geometric producer internally constructs the actual source tower, every true pulled centre and canonical smooth lift. Its actual constructor history proves length equality, endpoint smoothness and the WHOLE holomorphic Cartesian universal property. Node four internally supplies source test conditions and the entire marked endpoint. True marked pullback well formedness and true test-stage preservation give source well formedness. The true entire relative cosupport identity applied to that actual endpoint gives the full source cosupport preimage equality. For EVERY literal head, including indices beyond the true length, use the actual geometric head producer and the genuine target test head conditions; repeat node four for that actually generated pair of heads. Thus all source head tests, entire marked records, well formedness and whole relative cosupports are produced internally. Beyond-length heads are the actual empty prefix under the existing split convention. This is a genuine ordinary-test-tower

functoriality producer and removes the earlier full-carrier restriction. It does NOT construct a canonical target algorithm, identify old separation/colex with the general driver, prove higher invariant equality or noncompact termination, or assume a target resolution certificate in order to declare Hironaka complete. Those goals, whole G3/G8 and formal native integration/publication remain OPEN. □

253 Actual tangent differential ideals along the entire original carrier under submersions

Lemma 253.1 (Actual fibre-first projection and every coordinate direction)**.** *The genuine existing fibre-first split internally gives a continuous complex linear map $P : \mathbb{C}^{n+t} \to \mathbb{C}^n$, whose original whole coordinate function is precisely the actual adaptedTargetProjection. For EVERY real source coordinate direction, $P(e_i) = 0$ for $i < t$, and $P(e_{t+j}) = e_j$ for EVERY original target index j. No desired matrix or direction dictionary is supplied. The formulas include empty coordinate blocks.*

Proof. Take the second projection of the actual continuous linear fibreFirstSplit, which was constructed from the real whole coordinate permutation and direct sum splitting. Its explicit original coordinate identity says $(Pw)_j = w_{\mathrm{rotate}(\mathrm{castAdd}(j))}$, and the true permutation producer identifies this original index with $t + j$. Apply it to each actual standard coordinate vector. A fibre index $i < t$ cannot equal $t + j$, so every coordinate is zero. At $i = t + j$ the coordinate is the actual Kronecker delta at j. Coordinate extensionality gives the WHOLE linear identities, rather than a claim restricted to chosen nonzero directions. □

Lemma 253.2 (Genuine common chart inverse produces the actual entire differential chain rule)**.** *ONLY an actual submersion at a real source point, ANY genuine original centred target holomorphic coordinates, and an original holomorphic function internally produce real source coordinates on which its ENTIRE pulled function is the original flattened function composed with P. The entire differential on the true open chart is $D(F \circ \phi^{-1})(Pw) \circ P$; in particular all actual fibre partials vanish and every lifted target partial is the true pulled original target partial. No inverse/function/differential comparison certificate is an original caller input.*

Proof. Use the genuine existing internally constructed fibre-first adapted chart, whose whole forward coordinate square holds on its actual flat domain. Every point of its actual target has a genuine flat inverse source point, whose coordinates are that whole point. Apply the forward coordinate square and both true target inverse identities to obtain the WHOLE inverse projection square on its entire actual target. Compose that square with the original function F. Around every actual point of this open target, the genuine target-chart inverse and the given holomorphic function are holomorphic, so the entire equality holds on a real neighbourhood there. Derivative congruence for equal actual germs and the ordinary complex chain rule identify its derivative with the derivative of the true flattened target function composed with the actual continuous linear P. Apply this equality to every source coordinate direction and use node one. The actual flatPartial definition is exactly this differential evaluated on that coordinate direction, by the proved field/flattening identity. Consequently the fibre rows are genuinely zero and the target rows are their actual pulled functions, at EVERY true source point of the common neighbourhood, not merely at a supplied sample point. □

Lemma 253.3 (Original smooth carrier supplies the same actual finite tangent generator rows)**.** *For ANY genuine original centred target chart, actual original finite functions holomorphic at the real image point, and genuine submersion, the ENTIRE generated stalk span of their pulled target derivative family equals the span of the true source derivative family in the internally constructed fibre-first chart of tangent dimension $t + k$. EVERY target normal and tangent row has its actual source row, and EVERY source row is a pulled target row or a genuine zero fibre row. These are*

actual function calculations, with no caller span or derivative equality certificate. In the whole original-ideal consumer of node four, original coherence and closed smoothness internally choose these genuine target charts and finite functions; no presentation or source frame is a primitive input to that consumer.

Proof. At an actual point over N, original smoothness internally chooses its true flattening coordinates of carrier dimension k. Original coherence internally chooses finite original functions on one true neighbourhood; intersect that neighbourhood with the target flat chart. The actual fibre-first chart producer then gives source fibre dimension t, a true preimage carrier cut by the last $n-k$ coordinates, and carrier dimension $t+k$. Restrict its actual open to the preimage of the chosen target presentation open. Whole coherent pullback presents the original pulled ideal by the same actual functions composed with f; no prescribed source frame is assumed.

Apply the actual flat derivative-along generator theorem to both real carriers and their internally obtained true presentations. On the target it uses the original functions, the actual normal coordinates $j \geq k$ and the true tangential partials $j < k$. On the source it uses the pulled functions, normal coordinates $t+j$ with $j \geq k$, fibre tangential partials $i < t$ and lifted target tangential partials $t+j$ with $j < k$. The actual original coordinate square identifies the normal rows, and node two identifies every derivative row; the additional fibre rows are zero. Every target row has its actual source row and every source row is a target pullback or an actual zero row. Finite span transitivity and the actual zero membership prove equality of the ENTIRE generated stalks. Empty generator families, zero tangent/fibre blocks and repeated functions remain literal. No arbitrary source tangent field is omitted: the original generator theorem proves that the whole intrinsic tangent-field derivative ideal is exactly this finite family. □

Theorem 253.4 (Entire original tangent derivative ideal commutes with genuine smooth pullback)**.** *ONLY original coherent I, original closed smooth N and genuine arbitrary possibly non-surjective submersion f internally prove the ENTIRE original identity $f^{*}\mathcal{D}_{N}(I) = \mathcal{D}_{f^{-1}N}(f^{*}I)$. There is no caller output certificate, positivity, compactness or ambient ideal inclusion assumption.*

Proof. For every actual source point over N, actual submersion geometry internally provides $m = n + t$ and node three identifies the entire stalk ideals, using the same genuine original finite functions. Target and source derivative ideals are coherent from the actual original carrier geometry. Use the genuine old derivative-family presentation in each actual flat chart. Compose every finite target span relation with the actual holomorphic map, and use every real target row membership for the reverse inclusion. This identifies the actual WHOLE pulled derivative ideal with the pulled target-family span. Node three and the true source family presentation then give membership equivalence for EVERY source function; the actual span definitions themselves exclude nonholomorphic members.

Off $f^{-1}N$, the source derivative-along ideal is the entire holomorphic unit ideal, because it contains the whole vanishing ideal of the closed carrier. The target derivative-along ideal contains the genuine unit function at the true image point for the same reason. Its actual pullback therefore contains every source holomorphic function and excludes every nonholomorphic one. This proves the entire membership equivalence there as well. Whole GermIdeal extensionality proves the identity. If the source is empty the same extensionality is vacuous and no false dimension inequality is asserted. All original ambient/carrier dimension-zero cases are legitimate; neither a centre cosupport nor a canonical driver has been claimed. □

Theorem 253.5 (Every actual iterated derivative and literal tangent ideal is transported)**.** *Under only the same original inputs, EVERY natural iterate has the true ENTIRE pullback identity, and therefore the literal tangent ideal with index $\mu - 1$ does too for EVERY natural μ. No finite-order witness or iteration/result certificate is supplied.*

Proof. Induct on the actual natural iteration index. At zero the statement is the literal original ideal pullback. At the successor, actual coherence of EVERY original iterate follows from the

closed original carrier and coherence of I. Apply the whole one-step identity to that real iterate, then use the induction equality inside the literal next derivative-along operation. This produces every whole iterate internally. The tangent ideal is literally the iterate at the natural truncated difference $\mu - 1$; substitute that actual index, including $\mu = 0$ where the difference is zero. There is no infinity or stabilization claim hidden in this natural induction. □

Theorem 253.6 (Actual relative homogenization and entire coefficient records at every mark)**.** *ONLY an original coherent marked ideal on a closed smooth carrier, ANY natural mark d and actual submersion internally produce ENTIRE relative homogenized ideal/marked record and relative coefficient marked record pullback identities, and their true whole relative cosupport preimages. ALL N/E/I/d fields use the original literal definitions. No higher constructed invariant or canonical driver is assumed as a substitute for these identities.*

Proof. Unfold the genuine relative homogenized ideal: its finitely many summands are the true iterated derivative ideals multiplied by actual powers of the literal tangent ideal. Their coherence is internally produced from the original coherent I and closed smooth N. Node five identifies every true iterate and tangent ideal. Actual whole product, power and finite-sum pullback identities therefore identify the entire homogenized ideal, including all original source multipliers. Marked extensionality preserves the real carrier preimage, the complete ordered boundary list-map, the whole actual ideal and the unchanged mark.

The coefficient marked ideal is the literal finite marked sum of derivative marked ideals with marks $d-i$ for $i < d$. Its old actual definition ALWAYS retains N/E from its term zero, including $d = 0$, and has actual ideal powers whose exponents are its literal common-mark construction. Node five and genuine whole powers/finite sums identify that ENTIRE ideal; numerical exponents and marking are literally the same. Four-field extensionality gives the whole record identity. Genuine relative whole-order/cosupport pullback applies to these original coherent whole ideals and their actual carriers, giving the whole preimage sets. At $d = 0$ its original sum ideal is the genuine empty finite-span ideal, its mark is the literal empty product 1, and its N/E fields still belong to term zero; actual empty finite-span pullback and the unchanged empty product give the same full record identity. Homogenization likewise works for every natural index, including its actual empty sum at zero. These are actual analytic calculus producers. Arbitrary unrelated-ideal centre admissibility, higher constructed invariant/ old canonical driver identification, noncompact termination, whole G3/G8 and formal/native/publication remain subsequent work. □

254 Whole original relative geometry under genuine smooth submersions

Theorem 254.1 (Every original simultaneous coordinate family)**.** *For a genuine holomorphic submersion $f : Y^m \to X^n$ and any original coordinate family at $f(y)$, there is an internally produced coordinate embedding $\rho : \{1, \dots, n\} \hookrightarrow \{1, \dots, m\}$ such that every entire preimage member is the coordinate subspace cut out by the image under ρ of precisely its original cutting-coordinate set.*

Proof. Apply the genuine lifted product-coordinate construction to the actual centered original chart. It supplies a centered chart at y, defined on a genuine ambient open neighborhood, whose coordinates indexed by ρ are exactly all original coordinates composed with f. Every point of its domain maps into the original chart domain. Substitute this complete dictionary into each original member equation. A vector vanishes on the mapped finite index set exactly when its original coordinates vanish on the original index set, since ρ is injective. Thus the same internally constructed chart simultaneously gives all original preimage members; no preimage-chart certificate is an input. □

Theorem 254.2 (Original regular points and closed submanifolds)**.** *If the original set N is regular of dimension j at $f(y)$, then $f^{-1}N$ is regular of dimension $j+m-n$ at y. Consequently a closed original submanifold has a closed submanifold preimage under a genuine global holomorphic submersion.*

Proof. The original regular chart cuts out N by exactly the last $n-j$ coordinates. The preceding simultaneous coordinate dictionary cuts out its preimage by the injective image of those same $n-j$ coordinates. An actual finite coordinate permutation puts precisely those cutting coordinates last. Its induced complex linear isometry is a global holomorphic equivalence, with holomorphic inverse; compose it with the actual lifted chart. This is a true regular flattening of dimension $m-(n-j)=j+(m-n)$. The actual submersion internally implies $n \leq m$, and the original regularity implies $j \leq n$, so the equality and dimension bound hold also when either dimension is zero. For the global statement, continuity gives closedness and apply this actual regular flattening at every preimage point to its original regular dimension. □

Theorem 254.3 (Entire original normal crossings and transversality)**.** *The whole preimage of an original centre with normal crossings with the original ordered boundary again has normal crossings with the entire pulled boundary. The entire pulled boundary is transverse to the whole preimage of an original transverse underlying set.*

Proof. At each actual preimage point apply the preceding family construction to the original simultaneous family consisting of the underlying set or centre and every indexed original boundary member. Identify the pulled list index with the same original index; list length and order are unchanged. Images under the single injective coordinate map retain the cardinality one of each actual through-member cutting set. For transversality, two mapped index sets intersect only when their original index sets intersect, by injectivity. Hence every original disjointness relation with the underlying cutting set is retained. These are exactly the original definitions on the entire preimage; empty and repeated historical list entries remain literal. □

Theorem 254.4 (Actual original nonvanishing along the whole preimage)**.** *A coherent original ideal which is nonzero along the original underlying set at $f(y)$ has an actual nonzero germ along the entire preimage at y. The original underlying set is arbitrary.*

Proof. Choose the actual original ideal germ nonzero at points of the original set arbitrarily close to $f(y)$. Its composition with the genuine smooth map belongs to the whole original pulled ideal. The genuine local holomorphic section through y gives exactly $f_*(\mathcal{N}_y)=\mathcal{N}_{f(y)}$: continuity gives one inclusion and composition with the section gives the other. Transport the original frequent nonzero predicate through this equality. It says precisely that the actual composed germ is nonzero at points of the entire original preimage set arbitrarily close to y. No caller witness on the preimage is used and no analyticity of that set is required. □

Theorem 254.5 (Whole original marked geometry and invariant domain)**.** *The entire actual marked preimage of an original well-formed marked ideal is well formed. An original invariant domain of underlying dimension at most k gives an actual preimage invariant domain of underlying dimension at most $k+m-n$, with the same original marking.*

Proof. The actual marked pullback retains the entire preimage underlying set, whole ideal pullback, original ordered preimage boundary and original mark. The preceding producers give closed submanifold geometry, transversality and nonvanishing. Genuine holomorphic pullback gives coherence and the genuine whole SNC producer gives the entire boundary condition. At each original point select its actual regular dimension $j \leq k$ from the original domain. The preceding regular producer gives the preimage dimension $j+(m-n) \leq k+(m-n)$. The original positive mark is unchanged. These are all fields of the true original invariant-domain definition; no preimage well-formedness, dimension or nonvanishing certificate is supplied. Higher invariant equality, genuine blowup base change and complete canonical smooth functoriality remain open. □

255 Whole relative monomial stages under smooth submersions

The original underlying set is arbitrary. In particular, none of the following statements substitutes the whole ambient space for that set. The ideal, complete ordered boundary and original marking are retained.

Theorem 255.1 (Every original divisibility predicate and exponent along the original set)**.** *For a genuine holomorphic submersion $f : Y^m \to X^n$, any original sets $N, H \subset X$ and coherent original ideal I, every original nearby divisibility predicate along N is exactly equivalent to its counterpart along $f^{-1}N$ and $f^{-1}H$. The actual original exponent is unchanged.*

Proof. The actual holomorphic local section through each y proves the exact neighbourhood-filter identity $f_*(\mathcal{N}_y) = \mathcal{N}_{f(y)}$: continuity gives one inclusion, and composition with the actual section gives the reverse inclusion. The entire relative cosupport in every natural degree is the actual preimage of its original counterpart. Substitute this equality into the original nearby divisibility predicate, and transport the eventual statement through the filter identity. The underlying conditions are exactly membership in N and H, so this identifies every degree predicate, without regularity of either set. The original exponent is defined by the first failed successor-degree predicate, with its actual zero fallback when no failure exists. Equality of the whole predicate family identifies both the first failure and the fallback, proving the literal exponent equality. □

Theorem 255.2 (Entire indexed original relative exponent and invariant dictionary)**.** *The whole actual indexed exponent function is preserved for every original ordered boundary entry. The original relative monomial invariant on the preimage is computed by those same exponents and original through-indices in the genuine source dimension. In equal ambient dimensions its whole value is literally equal to the original value.*

Proof. Preimage preserves the literal boundary-list length and every indexed entry. For an index in range apply the preceding actual exponent equality to precisely its original member. For an index outside the list, both original functions have their actual value zero. Thus the entire indexed functions agree, including historical empty and repeated entries. Membership in each pulled member is literally membership of its image in the same original entry; hence the whole through-index sets are equal. Substitute both full equalities into the original definition of the relative monomial invariant, retaining the genuine ambient dimension. When the dimensions agree, this is exactly the original entire value, including bottom and all colex ties. □

Theorem 255.3 (Whole original relative ordering across genuine ambient dimensions)**.** *If the entire original ordered boundary is SNC, every comparison and every equality between original relative monomial values at two actual image points is preserved and reflected by the smooth preimage.*

Proof. The actual submersion linear coordinates internally imply $n \leq m$. The entire original SNC coordinates internally bound the cardinality of each genuine through-index set by n. The preceding whole indexed dictionary supplies exactly the same feasible subsets in both dimensions. The actual dimension-comparison theorem for the original monomial invariant therefore identifies both weak comparisons, retaining empty feasible families, bottom values and original colex ordering. Equality is the conjunction of the two opposite comparisons. Neither dimension nor cardinality bounds, invariant values or ordering certificates are inputs. □

Theorem 255.4 (Whole original relative marked maximum centre)**.** *For genuine surjective smooth f, a coherent original marked ideal and entire original SNC boundary, the whole relative monomial maximum locus on its actual relative cosupport pulls back exactly. The original underlying set and marking are arbitrary.*

Proof. The complete original relative cosupport dictionary gives exactly the preimage set. A pulled point is maximal precisely when every other actual pulled cosupport point has value at most its value. Each original cosupport point has an actual preimage by genuine surjectivity; that preimage lies in the pulled cosupport by the same whole dictionary. Reflect every comparison to obtain maximality on the entire original cosupport. Conversely original maximality preserves every comparison at each actual pulled cosupport point. This proves equality of the entire ambient maximum sets. It retains empty cosupports and arbitrary markings, and uses no compactness, caller maximum or ambient-set certificate. Higher invariant equality, canonical blowup base change and complete canonical smooth functoriality remain open. □

256 Actual relative orders under smooth submersions

All statements concern the original germ ideals and the original order defined by nearby agreement on a set. The set need not be smooth or analytic. No caller order, relative ideal, section or value certificate is assumed.

Lemma 256.1 (Whole relative function-order dictionary)**.** *Let $f : Y^m \to X^n$ be a holomorphic submersion at y, $N \subset X$ any set, and a any actual function on X. For every natural q, $a \circ f$ vanishes to order at least q along $f^{-1}N$ at y if and only if a vanishes to order at least q along N at $f(y)$.*

Proof. Submersion charts internally produce a holomorphic local section s through y, with $s(f(y)) = y$ and $f \circ s = \mathrm{id}$ near $f(y)$. If $a \circ f$ agrees along $f^{-1}N$ with an actual germ $b \in \mathfrak{m}_y^q$, compose b with s. Holomorphic composition preserves powers of point ideals, so $b \circ s \in \mathfrak{m}_{f(y)}^q$. Continuity of s transports the entire nearby agreement. For every nearby $x \in N$, the section identity puts $s(x)$ in $f^{-1}N$ and gives $a(x) = a(f(s(x))) = b(s(x))$. This is precisely the original relative order. Conversely a genuine representative $b \in \mathfrak{m}_{f(y)}^q$ agreeing with a along N pulls back to $b \circ f \in \mathfrak{m}_y^q$; continuity transports the whole agreement to $f^{-1}N$. Both arguments use actual point-ideal representatives and hold also for $q = 0$, empty sets, dimension zero and points outside N. □

Theorem 256.2 (All thresholds of the whole relative ideal)**.** *For the same actual submersion and any original germ ideal I whose members are holomorphic at $f(y)$, every threshold of f^*I along $f^{-1}N$ equals the corresponding threshold of I along N.*

Proof. Each original $a \in I_{f(y)}$ gives the actual pullback generator $a \circ f \in (f^*I)_y$, using the holomorphic unit coefficient. A threshold for the whole pulled-back ideal therefore descends for every original member by the preceding equivalence. In the other direction, every original member pulls back with the same threshold. The actual pullback ideal is the whole germ span of these generators. Relative order is preserved by holomorphic multiplication and finite sums, so every genuine member of this entire germ span has the threshold. There is no choice of a finite presentation and no reduced-ideal substitution for the original ideal. □

Theorem 256.3 (Original extended-natural relative order)**.** *The original extended-natural order of f^*I along $f^{-1}N$ at y is exactly the original order of I along N at $f(y)$.*

Proof. The original order is the supremum of exactly those natural thresholds which hold. The preceding theorem identifies this entire indexed family, so its supremum is unchanged, including infinite order. No finite order or nonvanishing assumption is introduced. □

Theorem 256.4 (Whole relative marked cosupport)**.** *If f is a holomorphic submersion everywhere and I is coherent, then for every natural marking q the entire cosupport along $f^{-1}N$ of f^*I is the preimage of the entire cosupport along N of I.*

Proof. At every actual y, coherence makes all original ideal members holomorphic. Apply the whole relative threshold dictionary at y. Membership in either cosupport consists of membership in the actual underlying set and its original order threshold. The underlying sets are literally related by preimage, and the thresholds coincide, proving equality of the entire ambient sets. Surjectivity is unnecessary; empty preimages and marking zero are retained. Higher invariant equality, canonical blowup base change and complete smooth functoriality remain open. □

257 Genuine entire relative monomial parts, nonmonomial order and companions

Theorem 257.1 (Whole original monomial stalk is the actual finite restricted-power intersection)**.** *Let $n > 0$, N be an original closed smooth carrier, E its actual ordered SNC boundary transverse to N, and I an original coherent ideal genuinely nonzero along N. At a true point x in any genuine centred adapted chart, the ENTIRE actual stalk of $\mathcal{M}_N(I,E)$ is the intersection, over ALL literal indices i with $x \in N \cap E_i$, of the actual stalks of* $\mathrm{restrictOn}_N(\mathcal{I}_{E_i}^{e_i(x)})$ *where $e_i(x)$ is the literal original* divExpAlong*. Inactive indices contribute the actual unit ideal. No finite frame or desired intersection identity is supplied. Every original index remains represented.*

Proof. Original SNC coordinates internally prove each actual member a closed smooth hypersurface, including literal empty members. Original geometry and genuine nonzero-along input internally supply coherence of the ENTIRE monomial part. Each actual vanishing ideal, its true natural power and actual restriction are coherent. For EVERY ring germ, the genuine adapted-representative producer returns an original holomorphic function F. The whole original membership dictionary identifies membership of its germ in the monomial stalk with the literal definition of the monomial part: for each active original index, F belongs to the true restricted span of that actual vanishing-ideal power. This restricted-span predicate is exactly membership in the original restriction extension. The adapted membership dictionary therefore equates it to that ENTIRE restricted-power stalk. All conditions are actual finite original indices. Off N every condition is vacuous, and an empty active index family gives the true whole unit ideal. Repeated empty entries and zero exponents are retained by the literal predicate. Whole ring ideal extensionality gives the asserted dictionary, with no assumed stalk or generator comparison. □

Theorem 257.2 (Genuine whole restrictions of vanishing-ideal powers commute with submersion pullback)**.** *ONLY an original genuine possibly non-surjective submersion, original closed smooth carriers N, H and ANY actual natural e internally prove $f^*\,\mathrm{restrictOn}_N(\mathcal{I}_H^e) = \mathrm{restrictOn}_{f^{-1}N}(\mathcal{I}_{f^{-1}H}^e)$ as the ENTIRE original GermIdeal. No source ideal or desired ideal equality is supplied.*

Proof. The whole actual vanishing ideal of the genuine closed smooth H is coherent. Its actual natural power is coherent as well. The genuine arbitrary-carrier restriction theorem identifies the whole pulled extension with the actual restriction to $f^{-1}N$ of that pulled power. Genuine coherent ideal-power pullback, including exponent zero and the whole unit ideal, identifies that power with the power of the actual pulled vanishing ideal. Finally the real whole vanishing-ideal pullback theorem identifies it with $\mathcal{I}_{f^{-1}H}$. These are equalities of the ENTIRE actual original ideals, not only supports or generators. The later monomial consumer internally specializes H to EVERY actual indexed SNC member and e to its literal original exponent; its true exponent preimage identity is an independently proved ingredient. □

Theorem 257.3 (The entire genuine original relative monomial ideal commutes with smooth pullback)**.** *ONLY the same original geometry and actual submersion internally prove $f^*\mathcal{M}_N(I,E) = \mathcal{M}_{f^{-1}N}(f^*I, f^{-1}E)$ as the ENTIRE original GermIdeal, in arbitrary source/target dimensions. No caller source ideal, exponent function, monomial factorization or equality certificate occurs.*

Proof. At EACH actual source point the real submersion complement gives $n \leq m$ and hence internally $m = n + t$. Since $n > 0$, source positivity is genuinely implied at that point. Actual source coherence of the monomial part follows from the internally transported true closed carrier/SNC/transversality and nonzero-along data. The whole original nearby divisibility predicates internally give the literal original exponent preimage identity, and the genuine list-map indices retain EVERY member. Nodes one and two express its entire adapted stalk as the intersection of the same actual pulled restricted-power ideals as the whole pullback of the target monomial part. The genuine fixed-origin projection map preserves this whole intersection, so the two ENTIRE actual stalk ideals are equal. The adapted entire-membership dictionary returns equality to EVERY original source holomorphic function. A nonholomorphic function belongs to neither coherent ideal, by the original internally chosen coherent presentations. Whole GermIdeal extensionality therefore proves the full equality. If the source is empty, extensionality is vacuous and does not assert a false dimension inequality. Empty/disconnected carriers or boundary families, inactive and repeated empty entries and zero exponents use the literal same dictionaries. This asserts no higher invariant or old canonical-driver identity. □

Theorem 257.4 (The entire actual nonmonomial ideal is transported without ambient inclusion)**.** *The same ONLY original inputs internally prove* $f^*\mathcal{N}_N(I,E) = \mathcal{N}_{f^{-1}N}(f^*I, f^{-1}E)$ *as the ENTIRE GermIdeal. No false* $I \subseteq \mathcal{M}_N(I,E)$ *ambient divisibility or invertibility certificate is assumed.*

Proof. The literal original nonmonomial part is $\mathrm{colonAlong}_N(I, \mathcal{M}_N(I,E))$. Original geometry and nonzero-along input internally prove coherence of its actual whole monomial denominator. Apply genuine arbitrary-carrier colonAlong naturality, whose original coherent ideal quotient theorem assumes no ambient inclusion. Node three identifies the ENTIRE actual pulled monomial denominator. Literal definitions give the entire source nonmonomial part. All quotient, restriction and power ideals are the original actual ones; no claimed factorization or source equality is provided as input. □

Theorem 257.5 (Whole original nonmonomial order predicates and literal masks are actual preimages)**.** *For an original marked datum with the same genuine geometric inputs, EVERY actual nonmonomial order threshold is exactly pulled back. The literal numeric* nmOrd *agrees at EVERY true source point, and the ENTIRE literal* ordMask(q) *is its actual open preimage for EVERY natural* q*. Neither surjectivity nor a global maximum is supplied.*

Proof. Original geometry internally gives coherence of the genuine nonmonomial part. Its ENTIRE pullback equality from node four and the actual whole relative cosupport theorem identify EACH actual order predicate, including threshold zero. The numeric nonmonomial order is the literal infimum of the natural first-failure set; those actual predicate identities equate the sets exactly at each point, so the numbers agree without a supplied finite-order witness. The literal mask is the complement of the closure of the next order threshold. A genuine submersion is an actual open continuous map from its internally produced local sections. Therefore its whole preimage commutes with that actual closure, and the threshold identity gives the literal source mask. This is true open-preimage equality, not a false global maximum preimage statement for a non-surjective map. □

Theorem 257.6 (Same original whole companion record is transported exactly)**.** *For EVERY natural chosen order* o*, the actual original companion marked record pulls back to the literal companion of the actual original marked pullback. The ENTIRE N/E/I/d record agrees in arbitrary dimensions, retaining every original boundary index and both actual* $o < d$ *cases.*

Proof. Unfold the literal companion: it uses the true whole original nonmonomial and monomial parts, powers $d-o$ and o, their sum if $o < d$, and the true nonmonomial part otherwise; its marking is literally $o(d-o)$ or o. Nodes three and four identify the ENTIRE actual pulled ideals. Genuine

coherent ideal-power and sum pullback identities produce the whole source ideal in the same actual numerical case. The original carrier is its real preimage, the boundary is the complete literal list-map and the marking is unchanged, so whole marked-record extensionality gives the identity. This includes $o = 0$ and mark zero literally and does not claim they satisfy an invariant-domain or positive-order condition. Node five separately identifies the genuine original order at each point; no caller desired order, companion ideal or canonical tower is supplied to a higher-invariant completion claim. Historical regime-run comparisons and full constructed invariant/canonical driver under smooth maps remain subsequent producers; noncompact termination, whole G3/G8 and formal/native publication remain OPEN. □

258 Whole relative ideal restriction under genuine smooth submersions

Theorem 258.1 (The entire ideal of a simultaneous coordinate zero locus)**.** *Let A contain x in a complex manifold of dimension n. Let Φ be a genuine centered local holomorphic coordinate equivalence at x, and let $S \subset \{0, \dots, n-1\}$ be finite. Suppose that, on its actual chart domain, A is exactly the locus where all coordinates indexed by S vanish. Then the whole vanishing ideal at x is generated by those actual coordinate functions. This includes $S = \varnothing$, whose ideal is the zero ideal.*

Proof. Choose a genuine finite permutation taking S to the last $|S|$ coordinates. Its complex linear isometry and holomorphic inverse compose with Φ to give another genuine centered chart. The same original set A is now exactly the coordinate plane of dimension $n-|S|$. Hadamard's lemma in this chart expresses every holomorphic germ vanishing on A as the sum of the last $|S|$ coordinate functions with holomorphic coefficients; all earlier coefficients are literally zero. Compose back with the inverse permutation. Every surviving coordinate is one of the original coordinates indexed by S, so this is a finite holomorphic span of the claimed actual functions. Conversely each such coordinate is holomorphic and vanishes on A on the actual chart domain. Every finite holomorphic combination consequently vanishes on A near x. Empty S leaves the empty sum, so the same argument gives the literal zero germ without a nonemptiness assumption. □

Theorem 258.2 (Actual finite presentations pull back in arbitrary dimensions)**.** *Let $f : Y^m \to X^n$ be holomorphic. Suppose the entire germ ideal I at $f(y)$ is presented by a genuine finite family of holomorphic functions g_i. Then f^*I at y is exactly the holomorphic span of the actual functions $g_i \circ f$, with no equality of ambient dimensions required.*

Proof. By definition a member of f^*I is a holomorphic combination of compositions $a \circ f$, with $a \in I_{f(y)}$. Substitute the actual finite presentation of each a; continuity pulls its whole germ identity back to a neighborhood of y, and composition preserves the holomorphic coefficients. Transitivity of finite holomorphic spans gives the claimed span of $g_i \circ f$. Conversely every g_i belongs to $I_{f(y)}$ by its own finite presentation, so its composition belongs to the defining generating family of f^*I. Closure under finite holomorphic combinations gives the reverse inclusion. The empty presentation is handled by the same empty-span argument. □

Theorem 258.3 (The whole original submanifold ideal pulls back)**.** *For a genuine holomorphic submersion $f : Y^m \to X^n$ and an original closed complex submanifold $N \subset X$, the entire germ ideals satisfy $f^*\mathcal{I}_N = \mathcal{I}_{f^{-1}N}$.*

Proof. Fix y. If $f(y) \in N$, choose the actual original flattening chart supplied by regularity of N. Its last $n - j$ coordinates define the whole original set N locally. The genuine submersion chart construction internally lifts the entire coordinate family to one centered chart at y, with an actual embedding of all original coordinate indices. Thus $f^{-1}N$ is exactly the common zero locus of the lifted last coordinates on this actual chart. The first theorem gives the entire ideal

on both manifolds. The second theorem identifies the pullback with the span of the composed original coordinates. The actual chart dictionary identifies these functions as germs with the lifted coordinates, so the spans, and therefore the whole ideals, are equal. If $f(y) \notin N$, closedness gives an actual neighborhood disjoint from N and its inverse image disjoint from $f^{-1}N$. Both ideals are the unit ideal there; the genuine generator 1 pulls back to 1. Pointwise equality for every holomorphic function and every y gives equality of the entire germ-ideal records. No caller flattening or ideal-equality certificate is supplied. □

Theorem 258.4 (Whole sums pull back across ambient dimensions)**.** *For a holomorphic map $f : Y^m \to X^n$ and original coherent germ ideals I, J, $f^*(I + J) = f^*I + f^*J$ as entire germ-ideal records.*

Proof. A member of the left side is a holomorphic combination of compositions of members of $I+J$. Expand each such member as a finite holomorphic combination of members of I or J, pull the genuine germ identity back by continuity, and compose its holomorphic coefficients. Transitivity gives membership in the sum on the right. Conversely a composed member of either original ideal is a composed member of $I + J$, because coherence makes the original members holomorphic and the sum contains both ideals. Every generator of the sum on the right therefore belongs to the pullback on the left, and transitivity of the defining finite holomorphic spans proves the reverse inclusion. These constructions use the actual dimensions m and n throughout. □

Theorem 258.5 (The complete original restricted ideal dictionary)**.** *For the same genuine submersion, original closed submanifold N and coherent original ideal I, the ambient representative of its restriction satisfies $f^* \operatorname{Res}_N I = \operatorname{Res}_{f^{-1}N}(f^*I)$ as an entire germ-ideal record.*

Proof. The actual restricted-ideal construction is $\operatorname{Res}_N I = I + \mathcal{I}_N$: subtract the finite restricted-span identity to obtain a holomorphic remainder vanishing on N, and conversely add such a remainder to any original finite span. Apply the fourth theorem to this genuine sum, and the third theorem to the whole original vanishing ideal. The pullback of the coherent original ideal is coherent, so the same actual restricted-ideal identity on Y identifies the resulting sum with the right side. Every operation concerns the original whole records and the actual original set. □

Theorem 258.6 (Coherence and the entire restricted marked record)**.** *The whole pulled restricted ideal above is coherent. Replacing the ideal in an original marked record by its actual restriction commutes with the entire marked pullback, retaining the original marking and every original ordered boundary entry.*

Proof. The original submanifold ideal is coherent by its actual finite flattening coordinates, including the genuine unit presentation off the submanifold. Its sum with the coherent original ideal is coherent. Holomorphic pullback preserves coherence, and the fifth theorem identifies this actual coherent ideal with the claimed pulled restricted ideal. For the marked record, the underlying set on both sides is literally $f^{-1}N$, the ordered boundary is the same original list mapped entry by entry by inverse image, and the marking is unchanged. The fifth theorem equates the whole ideal field. Equality of all fields consequently gives equality of the entire marked records, including empty and repeated boundary entries. No higher invariant equality or general blowup base change is asserted here; those parts of G2 remain open. □

259 Genuine relative tangent directions and original maximal contact under submersions

Lemma 259.1 (Every actual holomorphic field is tangent off a closed carrier)**.** *For an original closed carrier N, a true point outside N, an actual chart containing that point and actual holomorphic field coefficients, the field is genuinely tangent to N there. No tangency or vanishing-output certificate is supplied.*

Proof. The complement of the actual closed carrier is a genuine neighbourhood of the point. Every holomorphic function germ, including every true field derivative, therefore belongs to the actual vanishing ideal of N: its required equality on nearby carrier points is vacuous on this neighbourhood. The ordinary field product/derivative calculus gives holomorphicity of the derivative of every member of the vanishing ideal, whose function is itself holomorphic. The original chart membership and coefficient holomorphicity are exactly the remaining two fields of the actual tangency definition. □

Lemma 259.2 (A true nonzero original tangent derivative has a genuine source tangent derivative)**.** *Only genuine submersion f, original closed smooth N, an original function holomorphic at fy, and a genuine original tangent field with nonzero function derivative at fy internally produce a genuine holomorphic field at y tangent to $f^{-1}N$ whose derivative of the entire pulled function is nonzero. Dimensions may differ and f need not be surjective. No lifted field, chart, tangent-space or derivative comparison certificate is supplied.*

Proof. At a point over N, internally choose actual flattening coordinates of N. The original whole field/flat-partial decomposition, together with tangency, makes all normal-coordinate coefficient values zero at that point. A nonzero whole derivative therefore has a genuinely nonzero tangential partial with index $j < k$. Construct the true fibre-first source chart from the actual submersion. Its whole preimage carrier is the contiguous plane of dimension $t + k$, and its actual coordinate field at index $t + j$ is tangent to this carrier. The proved genuine function chain rule identifies its derivative of $u \circ f$ with the nonzero original partial.

At a point outside N, use genuine centred ambient coordinates and regard the original field as tangent to the whole ambient manifold. The same whole flat decomposition yields a nonzero ambient partial. Its lifted coordinate field has the same nonzero derivative by the actual chain rule. Source field coefficients are holomorphic; node one makes it tangent to the true closed preimage carrier outside that carrier. Thus no off-carrier zeros are discarded. If the source is nonempty actual submersion geometry internally gives $m = n + t$; otherwise there is no point at which a fictitious field is demanded. The argument itself excludes a nonexistent nonzero derivative in dimension zero without introducing a false positive-dimension hypothesis. □

Theorem 259.3 (Entire original tangent directions pull back on the whole true preimage open)**.** *For original coherent I, original closed smooth N, actual submersion and ANY genuine original tangent direction u on its true open U, the actual function $u \circ f$ is a tangent direction of the true pulled ideal along $f^{-1}N$ on the ENTIRE actual preimage $f^{-1}U$. This is a primitive true direction as intermediate input; the original-only consumer below internally chooses it. No source-direction or desired-preimage conclusion is assumed.*

Proof. Continuity gives the true preimage open. At every actual source point there, original tangent-direction membership puts u in the actual original iterate $\mathcal{D}_N^{\mu-1} I$. Its coherence gives actual holomorphicity. Genuine function membership in actual ideal pullback, followed by the WHOLE iterated derivative identity, gives the literal source iterate membership. At EVERY zero of $u \circ f$ the original direction supplies a nonzero true tangent field derivative at the image. Apply node two, including its off-carrier case. These are exactly all fields of the original source tangent-direction structure, on the whole preimage open, not merely a chosen smaller chart. □

Theorem 259.4 (Original maximal order internally produces a simultaneous genuine contact direction)**.** *ONLY original coherent I, original closed smooth N, positive original mark and a genuine maximal-order original current point over a source cosupport point internally produce a true original open U and function u, with BOTH original and source whole tangent-direction witnesses on U and $f^{-1}U$. Global maximal order likewise pulls back from its true relative order thresholds. No caller selected direction/contact/canonical/result certificate is supplied.*

Proof. The true relative cosupport preimage identity gives the actual image point in the original cosupport. Its real original nonmembership in order threshold $\mu + 1$ is the original maximal-order point input. The existing original coherent tangent-direction existence theorem internally chooses actual U, u containing that point and gives the whole original direction. Apply node three to produce the source direction on its ENTIRE preimage, containing the original source point. If global maximal order is given, the same actual order-threshold equivalence at every source point proves the source global maximal-order property, without any finite-order or boundedness certificate. □

Theorem 259.5 (Whole actual contact carrier and its real dimension dictionary)**.** *For a genuine original direction u on U, the ENTIRE source contact carrier is exactly the preimage of $N \cap V(u) \cap U$. At each true contact source point, the actual original carrier has some internally produced positive dimension k, its actual contact has dimension $k - 1$, and the genuine source carrier and contact have dimensions $k + (m - n)$ and $(k - 1) + (m - n)$ respectively. No caller positive dimension, contact-regularity or desired-carrier certificate is supplied.*

Proof. Literal membership in the actual preimage, intersection and composed zero set gives the whole set equality. Original contact regularity internally chooses true original carrier coordinates and produces contact dimension $k - 1$. At a real contact point the original tangent-direction field has a nonzero derivative; its actual tangential flat-partial producer gives an index below k, hence $k > 0$. The two genuine original regular carriers then have actual smooth preimages by the already proved dimension-shift producer. Use the whole set equality for the source contact. Its positive tangent codimension one is now an actual consequence, including arbitrary fibre dimension, rather than a guessed arithmetic formula at $k = 0$. □

Theorem 259.6 (Actual selected contacts are closed smooth carriers in their whole original opens)**.** *The original-only chosen contact internally gives genuine closed smooth carriers in the actual original open U and the ENTIRE source preimage open. Their whole ambient contact sets are not falsely called closed in the full ambient manifolds. Their true local cosupports lie on these real carriers. Contact coefficient restriction and independent contact-choice/higher canonical comparison are subsequent work.*

Proof. Every actual direction function is holomorphic throughout its real open from membership in the true coherent tangent iterate. Its zero set in that open is therefore closed by continuity. Intersect with the actual closed original carrier to get a true closed contact in the open manifold. At every actual contact point node five gives real regularity; the exact open-subtype regularity dictionary transports it into that open manifold. The SAME argument applies to the whole source direction produced by node three, with true pulled coherence and closed preimage carrier, on the whole source open. The actual Giraud tangent-direction support inclusion gives the entire relative cosupport intersection contained in each selected contact. Apply these producers to the genuine original-only choices of node four; no open, closed-carrier or support certificate is primitive. No whole coefficient contact restriction, canonical contact choice, higher invariant or global noncompact termination is inferred from these genuine contact carriers. □

260 The actual global universal lift has the entire genuine matching-chart formula

Only the original submersion $f : Y^{n+r} \to X^n$, original closed smooth nowhere-dense target centre C, and genuine canonical blowups are original inputs. Ordinary actual charts are intermediate geometry internally supplied by the preceding adapted product and coverage producers. A global Cartesian or lift-chart equality certificate is never supplied.

Lemma 260.1 (The entire actual monomial inverse parameter map is holomorphic)**.** *For any genuine canonical blowup FlatChart c, its inverse on the universe-lifted actual open monomial parameter domain is holomorphic. Its entire projected map equals the genuine original monomialChart, including every parameter on the exceptional coordinate hyperplane.*

Proof. The actual FlatChart homeomorphism belongs to the genuine blowup maximal atlas. Its inverse is holomorphic at every actual point of its exact open target. The genuine lifted-open coordinate map is holomorphic and lands in that entire actual target; composition gives the claimed whole-domain map. The original inverse-chart base dictionary identifies its projection with the actual monomialChart at every parameter. No density argument discards exceptional parameters. □

Lemma 260.2 (The whole original target ideal is genuinely invertible on the entire source chart)**.** *Given the original genuine global submersion and actual source FlatChart c', the whole ideal of C pulled back by f composed with the original source monomialChart equals the whole original source-centre ideal pulled back by that monomialChart and is invertible on its entire true open domain.*

Proof. The original closed smooth target centre has coherent entire vanishing ideal. Genuine holomorphic coherent pullback composition and the original submersion whole vanishing-ideal preimage dictionary identify the two ENTIRE ideals, rather than their supports. The actual source FlatChart has genuine original source-centre equations and a true normal pivot. The existing whole monomialChart ideal producer gives an invertible principal ideal generated by that genuine pivot coordinate. Its coordinate is nonzero somewhere in every nonempty actual open parameter neighbourhood, so this remains invertible on the entire domain including exceptional parameters. The equal entire target pullback is therefore genuinely invertible without a caller ideal/Cartier certificate. □

Theorem 260.3 (The original global canonical universal lift equals the whole matching-chart formula)**.** *Let c, c' be genuine original target/source FlatCharts internally obtained from the preceding adapted product producer, with actual whole source target equal to the split preimage of R times the target chart target, and their actual whole inverse-parameter base square. The uniquely produced original global canonical blowup lift equals $c^{-1} \circ \mathrm{drop} \circ c'$ on the ENTIRE actual source chart source.*

Proof. Use the entire genuine source parameter domain of the first lemma. The global actual lift composed with its holomorphic inverse parameter map is a holomorphic lift of f composed with the actual source monomialChart. The explicit matching target inverse of the linear dropped source parameter is also holomorphic on this same entire domain: its genuine product-target dictionary supplies actual target membership. Its projection is the SAME whole base map by the original matching inverse-parameter square. The second lemma supplies genuine whole-ideal invertibility on this ENTIRE domain. Apply the actual target blowup universal uniqueness (equivalently its proved continuous lift uniqueness) to these two actual holomorphic lifts. This proves pointwise equality including every exceptional parameter; only then transport by the actual source inverse/right-inverse dictionary to every point in its entire actual source chart. No arbitrary exceptional-cone invertibility or merely off-centre equality is assumed. □

Theorem 260.4 (The actual global canonical lift is a genuine submersion at every original exceptional point)**.** *Only the original global submersion and original closed smooth nowhere-dense target centre internally imply that its genuine global canonical universal lift is a submersion at EVERY genuine source blowup point whose actual base lies in the original preimage centre.*

Proof. The original geometry produces the source closed smooth nowhere-dense centre, hence its canonical genuine blowup. Actual original exceptional matching-chart coverage internally chooses a genuine chart pair CONTAINING the given original source blowup point, together with its exact

product-target and whole base-square dictionaries. The real source/target FlatChart homeomorphisms belong to the actual maximal atlases. In these SAME charts the actual matching local map is the fixed continuous linear projection with the entire original r-dimensional complement; inverse identities prove its entire chart expression. Thus it is a genuine submersion with that complement at the actual original point. The preceding theorem proves equality with the actual global universal lift on this OPEN chart containing that point, and the local congruence theorem transports submersion. The pivot and chart coverage are internally produced, never assumptions of this original-geometry consumer. □

This proves the whole-chart equality and the exceptional-point submersion case only. The off-centre case, arbitrary actual blowup-choice smoothness, full arbitrary-cone Cartesian gluing and higher canonical driver remain OPEN. OWN OFFLIB only; formal registration/native target/publication await root recovery.

261 Entire ambient positive-first-entry superlevels of actual Step1

The actual Step1 datum has domain three, true maximal order and its genuine canonical masked Step1 tower for the constructed invFamily2. The final theorem produces those data from the original invFamily3 canonical input. All thresholds with positive first entry are treated, with arbitrary entire second threshold including top. Original first-entry-zero contact and whole General3 remain OPEN.

Lemma 261.1 (A real special point has the exact top-baseline dictionary)**.** *At an actual special point, comparison with any threshold (s,beta) is equivalent to comparison with (s,top) for the SAME literal masked Step1 value.*

Proof. The literal masked Step1 definition at a special point is the pair of its actual original-boundary level and top. Lexicographic comparison gives either a strictly smaller first entry or equality of first entries and comparison with top. Every beta is at most top, so the second comparison is automatic in both cases. This proves the exact equivalence without a value certificate. □

Theorem 261.2 (A true finite analytic exception freezes every positive threshold on the actual curve)**.** *For a genuine domain-three maximal-order datum and actual canonical masked Step1 tower, on any true analytic curve Z inside actual current cosupport, internally produce an analytic dimension-zero E subset Z. On each genuine global component of Z outside E, membership in EVERY positive-first-entry whole Step1 superlevel is constant. One E works for all such thresholds.*

Proof. The actual canonical bridge internally gives the true barMarked test tower and equality of its actual current cosupport with the original datum's. The proven HeadIncidence history produces its analytic dimension-zero exception, making actual current levels equal on each component outside it. The actual across-mask producer supplies its true exception, where entire positive nonspecial Step1 values agree across different actual basepoints.

Take the real finite family of current whole top-second-entry superlevels with first entries from zero through the ORIGINAL finite boundary length. Each predicate is genuinely analytic by the actual top-superlevel producer. The true finite curve predicate producer gives an analytic dimension-zero exception outside which all their memberships agree on each component. The literal indexed level is bounded by that true original list length, so the predicate at a point's OWN actual current level is in the real finite family. At real cosupport points the top dictionary says this own threshold is exactly actual specialness. Thus specialness agrees as well. Union these three true exceptions; the union is analytic of dimension zero inside Z.

For any two component points outside the union, if both are special their whole actual values are the same pair of their equal level and top. If they are nonspecial and of positive level, the actual across-mask dictionary makes their entire Step1 values equal, hence all thresholds agree. If their common level is zero, every positive-first-entry threshold fails at both points by the literal first-entry definition and lexicographic order, independently of their possibly different zero-contact lower values. Therefore EVERY positive threshold has the stated component dictionary on the SAME genuine exception. No finite lower-order range or global maximum is assumed. □

Theorem 261.3 (Whole ambient positive Step1 thresholds are genuinely analytic)**.** *Assume the true analytic curve envelope Z has dimension at most one, is inside actual current cosupport, and contains every actual nonspecial cosupport point. For every s positive and every beta, the ENTIRE ambient set of true current cosupport points with actual masked Step1 value at least (s,beta) is analytic. Actual closedness and all curve dictionaries are produced internally.*

Proof. The actual current marked domain internally gives the coherent positive-mark cosupport analytic and closed. Actual canonical Step1 upper semicontinuity, proved for the constructed lower invariant, makes its COMPLETE superlevel L closed. The preceding theorem produces the true analytic dimension-zero exception and exact component membership dictionary on the actual Z. Thus L intersect Z is closed, contained in the real analytic Z, and has constant membership on each genuine component outside that exception.

The already proved actual component-filling producer identifies this curve set exactly as the union of the genuine selected global components contained in L and its intersection with the actual exception. Real analytic density outside the exception and closedness fill each selected component completely. Local finiteness of the true global component family makes that selected subunion genuinely analytic. The remaining intersection is a true closed subset of an actual analytic dimension-zero set, hence a genuine closed smooth discrete analytic carrier. Therefore L intersect Z is analytic; closedness of an arbitrary curve subset has never been treated as analyticity.

The actual top-baseline superlevel B at (s,top) is analytic by the previous genuine producer and is contained in L because beta is at most top. Every point of L outside Z is truly special by the stated genuine nonspecial envelope property. The first lemma puts it in B. Consequently the ENTIRE ambient identity is L equals B union (L intersect Z). Both sets are actually analytic, so L is analytic. This includes beta=top, empty curve and empty cosupport, and uses neither compactness nor a caller closedness certificate. □

Theorem 261.4 (Original constructed General3 gives every entire positive companion Step1 threshold)**.** *From ONLY the original domain-three datum, its literal invFamily3 canonical tower, a real current cosupport point and positive original order, every positive-first-entry superlevel of its SAME actual companion's masked Step1 value on the ENTIRE literal companion patch is analytic, with arbitrary second threshold. No caller curve, predicate, value or closedness certificate is supplied.*

Proof. The actual original companion dictionary produces its genuine domain-three companion, true maximal order and literal canonical masked Step1 tail. The real curve-envelope producer gives Z analytic of dimension at most one, inside actual current companion cosupport and containing all its actual nonspecial cosupport points. Apply the preceding entire ambient theorem to these internally produced data and the arbitrary positive threshold. The ambient here is the ENTIRE actual original companion patch; the value is the literal Seq.StepOneValueMasked value for invFamily2. This does not assert a complete original invFamily3 superlevel or solve its missing first-entry-zero contact histories. Original zero-contact, whole General3 and general global G3 termination remain OPEN. □

262 Actual zero-contact stalks and admissible clipped original centres

Let L be the actual enlarged contact coefficient model and let k be the number of original historical boundary entries. Write A for the true union of components of $L.N$ not contained in its actual cosupport, B for the selected union, and L^{rem} for the already constructed remainder. The ordered remainder boundary retains exactly k empty historical entries. All towers below are the SAME literal original towers; their maps blow up their original centres. Admissibility of a clipped centre does not identify any of these maps with a blowup along that clipped centre.

Theorem 262.1 (Entire controlled ideals agree on the actual selected-component complement)**.** *For closed smooth $L.N$, coherent $L.\mathrm{ideal}$, every natural k, and every actual test tower T of L, put $G = M \setminus B$. The entire controlled ideals of L^{rem} and L restricted to $T^{-1}G$ are equal.*

Proof. Both component subunions are closed smooth and disjoint, with union $L.N$. Thus G is genuinely open and $A = L.N \cap G$. At each strict point of A, closedness of the strict selected union gives a neighborhood where strict remaining and full contact carrier membership agree. The actual parent test tower gives smooth full strict carriers; these identical local germs and closedness give smooth remaining strict carriers at every literal prefix. Induction on the original tower now proves coherence of the entire remaining controlled ideal: pull back the actual coherent ideal and take its coherent colon along the actual smooth strict carrier by the power of the actual exceptional ideal. This does not require the centre to lie in the remainder.

Initially on G, the remaining and full records have the same carrier, ideal and mark. Replacing the full record's boundary by the remaining boundary makes the whole restricted records equal. Changing only the boundary leaves the entire transformed carrier, ideal and mark unchanged by the literal recursion. Coherent controlled transformation commutes with genuine open restriction at every prefix, since both source and target ideals have just been proved coherent. Induct on the original tower and compare these two restrictions. This gives equality of entire ideals on the whole inverse image of G. □

Theorem 262.2 (All actual stalks and all ordinary-order traces agree on the remaining carrier)**.** *Under the preceding genuine inputs, at EVERY point of the actual remaining strict carrier, membership of EVERY ambient function in the two controlled ideal stalks is equivalent. At that point, membership in their along-carrier ordinary-order cosupports is equivalent for EVERY natural threshold.*

Proof. The strict transform of the closed remaining carrier lies over that carrier, so every such point belongs to $T^{-1}G$. A coherent ideal restricted to an open set has exactly its original stalk there: restrict an ambient function, then extend it back; the extension agrees with the original near the point, and coherent germ membership is invariant under this agreement. Apply this in both directions to the entire open ideal equality just proved. This gives the asserted equivalence for all functions, without finite-generator or coordinate-only truncation.

The closed strict partition also gives identical remaining and full carrier membership germs near the point. In the definition of order at least q, a function agrees on the carrier with a germ in the qth power of the point ideal. Intersect its witnessing neighborhood with the neighborhood of exact carrier membership to transfer this witness in each direction. Combine this all-order identity with all-function ideal-stalk equivalence and with actual point membership in both carriers. This proves the claimed cosupport trace dictionary for every q, including zero. □

Theorem 262.3 (An actual parent centre internally produces an admissible clipped centre)**.** *Assume in addition $k \le \#L.E$. For any actual parent test prefix T and actual admissible parent centre C, the set $C \cap (L^{\mathrm{rem}})_T.N$ is an admissible centre of the actual entire remaining controlled datum at that SAME prefix.*

Proof. The intersection is closed. At a point of it, actual parent admissibility puts the entire centre in the full current carrier. The exact remaining/full carrier membership germ therefore makes the clipped centre germ equal to the original centre germ. Its actual regular coordinate presentation transfers across this identity, proving closed smoothness. The all-order stalk dictionary at the unchanged marking transfers the original centre's true cosupport inclusion to the remaining controlled datum.

The boundary recursion gives precisely k empty entries followed by the full transformed boundary with its first k entries dropped. Indeed strict transform preserves an empty entry; mapping commutes with dropping, and each new exceptional entry is appended beyond the first k entries, whose length bound persists at every prefix. Normal-crossing coordinates of the original centre restrict to this actual dropped sublist. Transfer those same coordinates to the clipped centre by its exact membership germ, shrinking the chart source within the witnessing neighborhood. Adding the k empty entries imposes no additional incident coordinate condition. This internally proves all three fields of actual admissibility. No new blowup map is claimed. □

Theorem 262.4 (Every literal original centre has an admissible actual clipping)**.** *For the same genuine full contact test tower, EVERY actual recorded centre, evaluated at its SAME literal prefix, has an admissible intersection with the current true remaining carrier.*

Proof. Induct on the actual tower. A zero-length tower records no centre. At a literal last blowup the true parent test condition supplies both its actual earlier test tower and admissibility of its actual last centre. Apply the induction hypothesis to all earlier recorded centres, then the preceding theorem at the actual last prefix. Their conjunction is exactly the recursive property of all recorded centres. This is produced as output, never supplied as input. □

Theorem 262.5 (Original zero-level Step1 produces one remainder with the complete stalk and centre dictionary)**.** *Original domain-three maximal-order data, the literal constructed canonical Step1 tower and an actual nonspecial zero-level cosupport point internally produce ONE true chosen tangent open and the SAME actual remainder. On it, the joint output includes the existing whole original masked record dictionary, true domain two at every literal original tail prefix, all-function stalk and all-threshold ordinary-order dictionaries at every remaining point at every prefix, and admissible clippings of EVERY literal restricted original centre.*

Proof. Use the existing original-input producer to obtain one true chosen tangent open, its actual mask inclusion and the complete whole-record/domain dictionary. Actual Giraud and the original canonical bridge internally give the enlarged contact model's genuine test tower on its full tangent inverse image. Original closed smoothness and coherence internally give the enlarged model's true smooth carrier and coherent ideal; the literal ordered boundary has at least its original historical length. Apply the preceding stalk and order producer to every literal prefix of that SAME test tower, and the all-centres producer to the SAME whole restricted tail. Retain all original whole-mask equalities and lower domains already produced. All objects in this conjunction use the one chosen tangent open; no candidate admissibility, nonzero, canonicality, lower domain or identity is a caller certificate. □

Theorem 262.6 (Literal General3 internally supplies the actual companion and clipped-centre output)**.** *Original domain-three data and literal canonical invFamily3, at a true positive ordinary-order point and an actual nonspecial zero-level point in its true companion cosupport, internally produce the entire preceding joint output for that SAME actual companion and literal companion tail.*

Proof. The genuine original companion producer internally gives its domain-three datum, its maximal-order property and literal canonical Step1 tail, together with the exact original cosupport dictionary. Apply the preceding theorem to those actual data and the given real companion cosupport point. Its full conjunction is returned unchanged, with the actual original companion and tail substituted. No independent contact model or tower is supplied. □

Open frontier. The original maps still blow up their original centres. Construction of a clipped blowup tower and its comparison with the original maps, candidate canonicality, deeper zero-contact histories, cross-contact patch descent, whole original zero-level and General3 superlevels, and general noncompact global termination remain open.

263 Actual zero-contact coefficient models across the genuine mask boundary

The genuine nonspecial lower patch and its complete local value dictionary already exist. The new construction extends their SAME contact coefficient model to the entire inverse image of its genuine tangent neighbourhood, retaining the codimension-one and history boundaries excluded by maskB. It proves actual controlled-transform and ideal-order dictionaries there; it does not claim that this larger contact model is globally canonical, nonzero, or already in the lower invariant domain. Cross-contact history gluing, original whole zero-level analytic thresholds and whole G3 remain OPEN.

Theorem 263.1 (Original zero-level data produce the actual enlarged contact model)**.** *From a genuine domain-three maximal-order datum K, its literal constructed canonical masked Step1 tower R, and a real nonspecial current cosupport point p of actual level zero, internally produce its TRUE chosen tangent open U. At its actual run head H, put V equal to the whole inverse image of U, and put L equal to the SAME actual contact coefficient marked ideal on V, with the original boundary kept. Its carrier is closed smooth of dimension at most two and its ideal is coherent. The TRUE restricted tail is a test tower of L, and its entire cosupport is exactly the trace of the true original current cosupport. The actual maskB is contained in V.*

Proof. The actual canonical bridge produces the true barMarked test tower and its equality with the original current cosupport. The true regime-head producer puts the image of p in the head cosupport. Maximal order internally produces the chosen tangent direction at its original base point, with that point in its true open U. Its inverse image V is therefore a true open. The actual stage tangent construction identifies the strict contact carrier locally with a nonsingular tangent zero hypersurface in the actual transformed smooth carrier. Thus its trace on V is closed smooth; the actual regular ambient dimension and the genuine original dimension-three bound give its dimension at most two. The entire head coefficient ideal is coherent by actual coherent derivatives and finite coefficient sums, and remains coherent under the genuine open pullback.

Giraud places every actual later cosupport, and hence every actual centre, in the strict transform of this SAME contact carrier over U. The true coefficient restriction theorem therefore produces the actual restricted tail as a test tower of L and identifies its ENTIRE cosupport exactly with the trace of the original barMarked cosupport. The canonical bridge returns this to the actual original K cosupport. The literal maskB definition requires its points to map into this U; it further removes original members and the codimension-one part, so it is genuinely contained in V. Neither removal is imposed on V, and no nonzero or candidate canonicality conclusion is inferred there. □

Theorem 263.2 (Every literal enlarged-contact prefix has actual geometry, coherence and analytic ideal-order predicates)**.** *For EVERY literal prefix of the actual enlarged-contact restricted tail, its true controlled transform has closed smooth carrier, coherent ideal, carrier dimension at most two and positive mark. For ALL natural ideal-order thresholds q, its whole actual carrier ideal-order superlevel is analytic, including across the codimension-one/history boundaries retained in V.*

Proof. Each literal prefix of the genuinely produced test tail is itself a test tower. Smooth strict-transform geometry and actual coherent controlled colon therefore preserve the true closed smooth carrier and coherent ideal at every prefix. Actual geometric dimension cannot increase under those

strict transforms, so the original bound of two remains valid. The true coefficient mark is positive and the literal tower preserves it. For each natural q, the actual derivative ideal of order q minus one presents the ideal-order superlevel inside this SAME actual carrier. At q zero this is the entire closed smooth carrier; for positive q it is its intersection with the true coherent derivative ideal zero set. Hence every such whole superlevel is analytic. An ideal identically zero on a genuine carrier component is allowed; these are ordinary ideal-order predicates, and no nonmonomial finite-order or lower invariant domain claim is made on the enlarged model. □

Theorem 263.3 (The enlarged contact model has the SAME literal masked controlled transforms)**.** *The actual enlarged contact model restricted to the genuine original maskB is the SAME original masked zero lower record, with carrier, ideal, boundary and mark. For every literal prefix of the ORIGINAL tail, restrict it separately to V and maskB. The actual whole controlled marked transform agrees under the genuine identity biholomorphism of nested opens with the restriction of the SAME enlarged contact transform. In particular its ideal-order predicates on the original mask are restrictions of the SAME genuine analytic enlarged-model predicates.*

Proof. Both initial marked records are genuine open pullbacks of the same full head coefficient ideal with its same strict chosen-contact carrier, boundary and mark. The literal mask inclusion into V identifies the double open restriction with the single original mask restriction by the genuine nested open biholomorphism. Actual coherent pullback composition proves equality of entire ideals, not only supports, and the other three marked fields follow from exact inverse-image identities.

The genuine restriction of a test tower to nested opens is isomorphic to its single restriction, with the actual identity point maps at every stage. Whole marked controlled transforms commute with genuine open restriction, using the internally proved carrier smoothness and ideal coherence at each literal stage. Transport this actual entire marked equality through the nested-open tower identity; thus EVERY prefix has the SAME ideal, carrier, boundary and mark as the original literal masked controlled transform. Genuine open pullback preserves ordinary along-carrier ideal orders, giving its exact predicate restriction. No caller tower, ideal equality, domain or value certificate is introduced. This bridge ties the enlarged analytic predicates to the original lower construction rather than producing an unrelated coherent ideal. □

Theorem 263.4 (Original constructed General3 internally supplies the actual enlarged zero-contact model)**.** *Only the original genuine domain-three datum, literal invFamily3 canonical tower and positive actual original order internally supply its SAME true companion and canonical Step1 tail. For any real nonspecial zero-level cosupport point of this companion, the enlarged model, all-prefix analytic ideal-order predicates and whole original masked controlled-transform identity above are internally produced.*

Proof. The actual original companion dictionary produces domain three, genuine maximal order and the actual literal canonical masked Step1 tail. Apply the three preceding producers to the stated real companion zero-level point, using ONE and the SAME chosen open throughout. A private output proposition only abbreviates this full conclusion at the default elaboration budget; it is never a caller assumption. Every chosen tangent, contact carrier, test tail, coherent ideal and masked record is internally obtained from these same original data. The enlarged contact model is not asserted globally canonical or nonzero, and the result does not solve the remaining cross-contact/history gluing or whole original zero-level and General3 analytic thresholds. □

264 Actual zero-contact remainder, historical empty entries and whole original mask dictionary

The genuine enlarged contact coefficient model L is internally produced from original domain-three maximal-order and literal canonical Step1 data. Put $k = \#K.E$. Retain the actual ideal

and marking. The new carrier is the union of the TRUE smooth global components of $L.N$ not contained in its actual cosupport. Retain precisely k literal historical empty entries, followed by the actual existing exceptional boundary. These empty entries are essential to preserve the entire original ordered lower record on maskB. No enlarged-model nonzero, candidate canonicality, admissibility, lower domain, transversality or result identity is a caller hypothesis.

Definition 264.1 (The actual contact remainder with historical positions retained)**.** For a genuine marked L, set A equal to the true remaining global component union of $L.N$ relative to L.cosupp. The actual remainder has carrier A, the SAME entire ideal and mark, and ordered boundary $[\varnothing, \dots, \varnothing]_k +\!\!+ \operatorname{drop}_k(L.E)$.

Theorem 264.2 (Genuine coherent contact components internally produce nonzero remainder geometry)**.** *For closed smooth $L.N$, coherent actual ideal, positive marking and true regular dimension bound two, the actual remainder carrier is closed smooth of regular dimension at most two; its ideal is coherent and nonzero along EVERY point of this carrier. Its actual cosupport is exactly the intersection of the original contact cosupport with this genuine remaining union.*

Proof. The real locally finite global smooth component family partitions the carrier into the closed selected and remaining subfamilies. Their unions are disjoint. At each remaining point a true neighborhood misses the selected union, so remaining-carrier membership is exactly original-carrier membership there. The true remaining union is closed smooth, and its regular dimensions are inherited from this exact germ identity. Its ideal is literally unchanged. Along-carrier ordinary orders agree at each point by these genuine membership germs; the full cosupport is therefore the asserted actual intersection.

Analytic continuation on a real smooth connected component prevents a nonempty relatively open part of a remaining component from being contained in the true analytic cosupport: otherwise that entire component would have been selected. Thus the actual trace is thin in the genuine remaining union. If the entire coherent ideal were zero along this carrier at a point, its actual finite local generators would all vanish on a whole carrier neighborhood. A nonempty relatively open part of that same carrier would then lie in its actual cosupport, contradicting thinness. This internally proves nonzero everywhere, including arbitrary varying-dimensional and empty component families. There is no supplied nonzero or selection certificate. □

Theorem 264.3 (Every actual original prefix internally produces a full lower domain)**.** *Original genuine domain-three, maximal-order, literal canonical Step1, and real nonspecial zero-level cosupport point internally produce ONE true chosen tangent open and actual enlarged contact model. For EVERY literal prefix of the ORIGINAL tail restricted to its full tangent inverse image, the actual entire controlled transform of the remainder is in domain two. The actual original centres need not lie in its remaining carrier.*

Proof. Giraud and the actual original canonical bridge give each restricted original prefix as a true test tower of the entire enlarged contact model. The real closed selected/remaining partition has disjoint closed strict transforms. Their union is the entire strict contact carrier, and the remaining strict carrier is that carrier intersected with the original projection preimage of its initial remaining union. At a remaining strict point a neighborhood misses the transformed selected union. Thus its exact germs are those of the true contact test transform, giving closed smoothness and regular dimension at most two.

At every actual blowup, coherent pullback and the true smooth strict-carrier vanishing ideal produce the entire coherent controlled colon, without centre containment in the candidate. Away from the actual exceptional divisor, nonzero follows by the genuine blowup local isomorphism. At an exceptional point, failure of nonzero would make all actual coherent generators vanish on a whole strict-carrier neighborhood. Its defining strict accumulation supplies real off-centre points in that neighborhood, where the already proved nonzero gives a contradiction. Induction therefore produces actual coherence and nonzero at every prefix.

Literal empty historical entries have empty strict transforms. The boundary map/append recursion and the actual initial list-length bound show that every candidate boundary is precisely k empty entries followed by the true transformed exceptional block, including ALL new actual tail exceptions. The true original bar tower internally gives SNC; restricting its genuine exceptional sublist preserves SNC, and prefixing actual empty entries preserves it by assigning no incident coordinate to those entries. Giraud places every actual centre over the chosen tangent open in its real strict contact carrier. The genuine base smooth contact and this internally proved centre containment give actual transversality of the whole strict contact carrier to the entire exceptional block at each original concatenated prefix. Exact remaining-carrier germs inherit this transversality; empty historical entries impose no condition. Combine these internally proved fields, unchanged positive marking and regular dimension bound to obtain the full actual domain-two assertion. No candidate admissibility or canonical assertion is inferred from this domain construction. □

Theorem 264.4 (The ENTIRE original mask has the SAME actual marked remainder record)**.** *The actual remainder restricted to the ENTIRE SAME original maskB, under the true nested-open identity, equals the literal original zero-level lowerMarkedB as a WHOLE carrier, ideal, ordered boundary and mark.*

Proof. The original true canonical lower-domain construction internally proves the masked contact coefficient nonzero. If an actual original masked carrier point belonged to a selected full contact component, closedness of the remaining union would give a whole contact-carrier neighborhood in the true cosupport. Openness of the actual mask and the genuine finite coherent presentation would then make the restricted ideal zero along its true masked carrier near that point, contradicting its actual lower nonzero. Thus every original masked contact carrier point lies in the SAME remaining carrier. The whole initial carrier restriction agrees, not merely at the chosen reference point. The entire ideal and marking were retained.

Every true original historical member misses the literal maskB by its actual definition. Its pullback is therefore the empty set entry, in its SAME original index position. The historical block retains its exact original length. Hence its whole ordered list is precisely the k empty entries, followed by the genuine exceptional block; no empty list entry is silently removed. True coherent double-open pullback and the actual nested-open biholomorphism give equality of entire ideals. These four exact fields produce the literal whole original marked-record equality on the full mask. □

Theorem 264.5 (All original masked prefix transforms have the SAME actual remainder dictionary)**.** *On ONE true chosen tangent open, the output jointly includes the whole initial original-mask record equality, domain two at EVERY actual original tail prefix, and equality of the whole controlled marked transforms of the remainder on its true enlarged inverse image, restricted to the ENTIRE original mask inverse image, agree with the original literal masked lower transforms under the genuine nested-open tower identity.*

Proof. Actual open restriction commutes with the true controlled colon when its actual source and strict-target ideals are coherent. The preceding producer internally supplies smooth strict carriers and whole coherent controlled ideals at every original prefix. Induct on the actual original prefix, using coherent pullback composition and the genuine open strict-carrier identity at each blowup; the centre is not required to lie in the candidate carrier. Transport the true double restriction through its actual nested-open tower identity. Start with the entire original mask equality of the preceding theorem. Every literal map/append step retains exact carrier, ideal, mark and the SAME empty historical positions plus actual exceptionals. Thus every prefix has the claimed WHOLE marked-record equality, not only support or a smaller local neighborhood. Original lower canonicality remains available on that SAME original mask; it is not asserted on the enlarged remainder. □

Theorem 264.6 (Original literal General3 internally supplies the actual remainder and all dictionaries)**.** *Only the original domain-three marked datum, literal invFamily3 canonical tower, positive actual original order and a real nonspecial zero-level point of its SAME actual companion internally produce the actual chosen enlarged remainder, all-prefix domain-two geometry and whole original mask dictionaries.*

Proof. The actual original companion dictionary produces its genuine domain three, maximal order and actual literal canonical Step1 tail. Apply the preceding producers to these internally supplied data on ONE true chosen tangent open. Every real dead component, historical position, exceptional boundary, controlled ideal and complete mask dictionary is obtained from those SAME original data. No enlarged global domain, nonzero, canonicality, tower or identity certificate is an input. Candidate global canonicality, deeper zero-contact history, cross-contact patch descent, whole original zero-level and General3 thresholds and noncompact global G3 termination remain OPEN. □

265 Actual zero-contact remainders under contact glueing

Let $L = (N, E, I, d)$ be a marked ideal. Write $S(L)$ for the union of the actual global components of N contained in its actual cosupport, and $R(L) = N \setminus S(L)$. A positive marking is essential in the following zero-germ characterization. No bound on dimension, component count, or ideal orders is imposed. Put

$$L^{\mathrm{rem},k} = (R(L), [\varnothing]^k +\!\!+ E_{\geq k}, I, d).$$

Theorem 265.1 (The actual selected union is precisely the zero-germ locus)**.** *Suppose N is closed smooth, I is coherent, and $d \geq 1$. At each $x \in N$, membership in $S(L)$ is equivalent to the failure of nonzero along N at x. Equivalently every germ of I_x vanishes identically on the germ N_x.*

Proof. The selected and remaining component unions are disjoint closed smooth sets with union N. If x is selected, the complement of the remaining union is a neighborhood of x on which every point of N lies in the actual cosupport. A coherent ideal germ extends to a member at every nearby point. At a nearby point of N, order at least $d \geq 1$ makes the value of this member zero. Thus no ideal member is frequently nonzero along N near x.

Conversely let x be remaining. The actual cosupport is analytic, and analytic continuation on each genuine connected smooth component proves that its intersection with the remaining union contains no nonempty relatively open subset. Near every remaining point, closedness of the selected union makes the remaining and original carrier germs equal; their along-carrier order predicates therefore agree. If all ideal germs were zero along the remaining carrier at x, a finite coherent generating family would give a single neighborhood where all ideal members vanish along that carrier. This neighborhood would be contained in its cosupport, contradicting the preceding thinness. A germ frequently nonzero along the remaining carrier is also frequently nonzero along N.

Finally, the negation of frequent nonvanishing for every ideal member says that each member is eventually zero along N. Coherence makes each member holomorphic. These are exactly the two requirements for membership in the vanishing-ideal stalk of N. Conversely such stalk membership contradicts frequent nonvanishing. This proves the zero-germ equivalence. □

Theorem 265.2 (Actual nonzero and selected germs commute with open restriction)**.** *For a coherent ideal I, an arbitrary carrier N, an actual open U, and $p \in U$, nonzero along the restricted carrier for $I|_U$ at p is equivalent to nonzero along N at the same ambient point. If N is closed smooth and $d \geq 1$, the remaining carrier of $L|_U$ is precisely the restriction of $R(L)$ to U.*

Proof. Restrict an actual nonzero germ to U. Coherent ideal restriction preserves its membership, and the open embedding identifies the neighborhood filter at p with that at its ambient point. It therefore preserves frequent nonvanishing along the restricted carrier. In the reverse direction,

extend a restricted function back to the ambient manifold. Coherent restriction identifies its ideal membership, while its values on U are unchanged. The same equality of neighborhood filters transfers frequent nonvanishing back. Apply the preceding zero-germ characterization to the genuinely closed smooth restricted carrier and to N. Since remaining membership means carrier membership and nonzero along, this gives equality of the entire remaining carriers. No assertion about equality of the two global component index sets is needed. □

Theorem 265.3 (Actual nonzero and remaining germs commute with biholomorphisms)**.** *For a biholomorphism Φ, coherent I, arbitrary N, and p in its source, nonzero along $\Phi^{-1}N$ for the actual entire pullback ideal at p is equivalent to nonzero along N at Φp. For closed smooth N and positive marking, the remaining carrier of $\Phi^{*}L$ is $\Phi^{-1}R(L)$.*

Proof. Compose an actual ideal member with Φ; it belongs to the entire coherent pullback ideal. The homeomorphism's equality of neighborhood filters transfers frequent nonvanishing. For the reverse implication apply the same argument to Φ^{-1}. Genuine coherent pullback functoriality and the inverse identities identify the twice-pulled ideal with the original ideal and the twice-pulled carrier with N. Closed smoothness and coherence are preserved by this actual biholomorphism. The zero-germ characterization on both sides now identifies remaining membership at every point and hence the entire remaining carriers. □

Theorem 265.4 (The entire remainder record commutes with actual transport)**.** *For closed smooth N, coherent I, positive marking and every natural k, the entire record $L^{\mathrm{rem},k}$ commutes with genuine open restriction and with genuine biholomorphic pullback.*

Proof. The preceding theorems give the exact carrier identities. The ideal and mark are unchanged by the remainder construction, so their identities follow from the definitions of actual pullback. On the boundary, taking preimages commutes with dropping the first k entries and with list concatenation. The preimage of an empty set is empty, so the first k historical entries remain exactly k empty entries. These four field identities give equality of the entire records. In particular a genuine whole contact-record transport induces whole remainder-record transport. □

Theorem 265.5 (True contact lifts transport the actual remainders at every prefix)**.** *Suppose $K.N$ is closed smooth, $K.I$ is coherent, $K.d \geq 1$, and x is in its actual cosupport. Let u, v be genuine tangent directions on actual opens containing x. There is actual glueing data G, with both neighborhoods inside both tangent opens and containing x, such that every actual test prefix T of the coefficient-homogenized datum has an actual lifted biholomorphism transporting the entire coefficient contact remainder for u to that for v, retaining every actual exceptional entry and adding any prescribed number of empty historical entries. This lift fixes the actual transformed cosupport. The padding does not delete exceptional entries.*

Proof. The actual homogenized contact construction gives glueing maps trivial modulo the genuine glue ideal. Intersect its open neighborhood with the two genuine tangent opens. The inverse identities, ideal invariance, contact-coordinate identity, carrier invariance and trivial action restrict to this intersection. Both actual glueing neighborhoods are therefore contained in the two tangent opens and contain x, since the maps fix the original cosupport.

For every actual coefficient-homogenized test prefix, the genuine universal-property lifting construction produces a biholomorphism of the two inverse-image neighborhoods, whole transformed coefficient-ideal transport, whole contact-record transport, and pointwise fixation of the actual transformed cosupport. The true contact-stage coordinate theorem, applied inside the actual tangent opens, produces closed smooth strict contact carriers on both neighborhoods. The transformed coefficient ideal is coherent by the actual coherent coefficient construction and the actual test tower; its marking is positive by the genuine factorial coefficient marking. Thus all premises of whole remainder-record transport have been produced internally. Apply it to the actual whole contact-record equality for each k. This preserves the actual exceptional boundary and

exactly the padded historical positions. It imposes no equality on undeleted original boundary members. □

Theorem 265.6 (The padded datum is exactly the original boundary-kept contact remainder)**.** *At every tower prefix, restricting the contact datum of the coefficient homogenization with empty initial boundary, removing its actual selected components, and padding $\#K.E$ empty entries gives exactly the actual remainder of the original boundary-kept contact datum with its first $\#K.E$ entries replaced by empty entries.*

Proof. Initially the coefficient homogenization with empty initial boundary is the original bar datum with all original boundary entries dropped. The actual map-and-append transformation recursion commutes with dropping these entries: new exceptional entries are appended after the original block, whose length persists at every prefix. The coefficient operation and actual open restriction preserve the carrier, ideal and mark identities; taking preimages of boundary members commutes with dropping. The two actual contact records consequently have the same carrier, ideal, mark and cosupport, and their boundaries differ precisely by dropping the original block. They select precisely the same actual contact components. Padding the dropped contact datum by the original number of empty entries gives the defined original contact remainder in all four fields. No exceptional entry is dropped by this padding. □

Theorem 265.7 (Original chosen contact inputs produce the actual remainder glueing)**.** *Let K have its actual domain-three and maximal-order properties, and let R be a canonical sequence for its literal constructed Step One invariant. The genuine chosen contact directions at original cosupport points x, y, with x inside the actual chosen tangent open at y, internally produce whole remainder dictionaries for the SAME original boundary-kept contact records, with their historical block replaced by empty entries, along every literal prefix of R. The same actual contact maps and actual lifts fix the original transformed cosupport.*

Proof. The actual domain gives coherent ideal, closed smooth carrier, positive marking and well-formedness. Maximal order at the actual original cosupport points produces tangent directions, so the literal chosen directions are genuine and their opens contain their own points. The stated actual incidence puts x in the second chosen open as well. Hence the preceding construction applies to these two actual directions, without any caller-supplied contact-domain or glueing witness.

The genuine Step One canonical bridge internally makes the literal sequence a test sequence for its actual bar datum. Deleting only its initial historical boundary gives a true test sequence for the coefficient-homogenized datum, whose carrier, ideal and marking have the same literal transformation recursion. Every literal prefix inherits this genuine test property. Apply the preceding actual lifts at each prefix, and identify the bar and coefficient-homogenized transformed cosupports by the genuine weak-equivalence and Step One bridge. This gives precisely the stated joint whole record dictionaries and pointwise fixation on the actual original cosupport. No enlarged remainder canonicality has been used or asserted. □

Scope. The conclusions are actual local contact transport and entire remainder record dictionaries. Canonicality on enlarged contact neighborhoods, deeper zero-contact histories, descent of all zero-level thresholds across contact patches, whole General Three analytic superlevels and general noncompact global termination remain open.

266 Actual selected-complement remainder tails under contact glueing

The preceding zero-germ construction produces the actual remaining carrier and entire contact remainder, with no dimension or component-count bounds. The next construction uses the

SAME original test tail on the actual open complement of its selected contact components. It produces genuine isomorphisms of these actual restricted tails, their actual test properties and whole controlled-transform dictionaries. No tail, centre-containment, canonicality, coherence, nonzero or transport result is supplied as a caller certificate in the contact producer. The generic transport calculus is consumed internally.

Theorem 266.1 (Actual selected complements correspond)**.** *For a genuine biholomorphism $\Phi : X' \to X$, closed smooth N, coherent I and positive marking, the actual selected union of $\Phi^* L$ is $\Phi^{-1} S(L)$. Consequently its actual complement open G' is exactly $\Phi^{-1} G$, where $G = X \setminus S(L)$.*

Proof. Every selected component lies in the true carrier. At a point of the pulled carrier, the selected-union criterion is failure of nonzero along the carrier. The actual pointwise nonzero pullback equivalence therefore gives selected membership equivalence. At a point outside the carrier neither selected union contains the point. This proves equality at every point, without claiming equality of global component index sets. Taking complements gives the matching-point equivalence for the genuine complement opens. □

Theorem 266.2 (The actual restricted remainder records correspond)**.** *For every natural k, the actual restricted biholomorphism $\Phi_G : G' \to G$ pulls the entire actual remainder record $L^{\mathrm{rem},k}|_G$ back to the actual record $(\Phi^* L)^{\mathrm{rem},k}|_{G'}$.*

Proof. The preceding exact complement dictionary constructs the actual restricted biholomorphism with inverse. Genuine coherent marked-ideal pullback functoriality applies because the remainder ideal is the original coherent ideal. Both composites from G' to X have pointwise value $\Phi(q)$. Thus the pullback of $L^{\mathrm{rem},k}$ through either composite agrees. The preceding whole remainder pullback identity identifies the second composite with the actual remainder of $\Phi^* L$ restricted to G'. These are entire four-field equalities, including the dropped and padded historical boundary positions. □

Theorem 266.3 (Actual selected-complement tails and controlled records correspond)**.** *Suppose the genuine towers T', T are actually isomorphic over Φ and T is a test tower for L, with $k \le \#L.E$. On the actual selected complement opens, the SAME restricted towers are genuinely isomorphic over Φ_G. Both are actual test towers of the corresponding restricted remainder records. Their entire controlled-transform records correspond under the actual restricted last-stage biholomorphism.*

Proof. The actual complement membership equivalence, together with the original projection square, identifies the two inverse-image complement opens at the last stages. Restrict the original genuine tower isomorphism to these actual opens, retaining each original restricted map and centre trace. The true remainder test-tower producer on G internally supplies its test property, with no candidate centre-containment premise. The restricted remainder carrier is closed smooth by the genuine remaining-component geometry and open restriction; its ideal is coherent by open pullback. Whole marked-ideal transport along the now-produced genuine isomorphism therefore identifies all four controlled fields. The isomorphism test iff and the proved initial whole record equality internally yield the test property of the second actual remainder tower as well. The original restricted maps are genuine blowups of the true centre traces; no unrestricted ambient map is renamed as a clipped-centre blowup. □

Theorem 266.4 (Every literal prefix has the actual remainder dictionary)**.** *Under the preceding genuine geometric inputs, for every natural prefix index j the literal heads of the SAME two restricted remainder tails admit a produced genuine isomorphism. Their entire controlled records correspond, and their remaining tails correspond under the same head map. No bound on length or exclusion of empty steps is imposed.*

Proof. The actual isomorphism of the entire restricted tails admits a genuine split at each j. The tower-isomorphism split induction follows the original maps, produces the head map and the residual-tail isomorphism, and treats the terminal identity case literally. The genuine test property of the entire remainder tower implies that of its head. Apply full controlled-transform naturality to the produced head isomorphism, using the proved whole initial remainder equality. This gives all four fields at the exact literal head, rather than at a different truncated or concatenated presentation. The genuine split theorem supplies the remaining-tail isomorphism in the same output. □

Theorem 266.5 (Literal empty history entries preserve genuine tests and full transforms)**.** *Prepending any prescribed number k of literal empty boundary entries to a marked ideal commutes with its ENTIRE actual tower transform and preserves its genuine test-tower property along the SAME tower.*

Proof. Induct along the actual tower. The identity case is literal. At a step, carrier, controlled ideal and mark are independent of the boundary field. The actual boundary recursion maps every old boundary member to its strict transform and appends the new exceptional member. The strict transform of the empty set is empty, so the prefixed empty block survives and all actual exceptional members remain in their exact order. This proves the entire transform equality. For the test property, use the same induction. At each real centre the carrier smoothness and cosupport containment are unchanged. The normal-crossing predicate for the enlarged list follows from the actual original predicate: every new member is empty and hence disjoint from the centre, while all genuine old members retain their previous coordinate dictionary. Thus every literal centre is admissible, including empty centres, with no altered tower or additional premise. □

Theorem 266.6 (Original contact tests internally produce genuine remainder tail glueing)**.** *Let K have closed smooth carrier, coherent ideal and positive marking. At a point of its actual cosupport choose two genuine tangent directions on true opens containing the point. There is genuine contact glueing data inside both opens such that every actual concatenated test tower of the coefficient-homogenized datum with EMPTY initial boundary internally produces actual contact-stage models with their genuine parent-tail test properties, a whole model equality, and a genuine isomorphism of their SAME tails restricted to the actual selected-component complements. These are genuine remainder test tails, with entire controlled record transport at the endpoint and every literal head. The actual endpoint map fixes the trace of the original transformed cosupport wherever it lies in those complement neighborhoods.*

Proof. Genuine tangent contact glueing shrinks its domain and codomain inside both original tangent opens, produces the actual prefix lift, and simultaneously lifts through the actual entire tail. The lift gives a genuine tower isomorphism and whole coefficient-contact model equality. The real strict contact-coordinate theorem produces the closed smooth carrier of the first actual model. The genuine coherent coefficient construction and the actual prefix test produce its coherent ideal; the actual factorial coefficient marking is positive. The same contact tail theorem on a smaller actual open internally proves that the actual restricted original tail is a test tower of this contact model. No candidate tests are assumed.

Apply the proved selected-complement calculus to this actual model and produced isomorphism. The whole contact model equality identifies the pulled model with the second genuine model. Hence its selected complement, actual remainder and original restricted tail are exactly the second objects, not independent existence witnesses. The calculus now supplies both true remainder tests, the genuine restricted tower isomorphism and whole endpoint controlled equality. Its literal-prefix theorem gives all actual head identities. The original contact lift fixes the transformed parent cosupport at the endpoint. Restriction preserves that pointwise identity on the actual selected-complement trace. □

Theorem 266.7 (The original literal constructed Step One tower supplies every contact tail)**.** *For original domain-three and maximal-order K, a canonical sequence R for the SAME literal constructed Step One invariant, actual cosupport points x, y, and actual incidence of x in the chosen tangent open at y, ONE produced genuine chosen-direction glueing datum supplies the preceding actual remainder-tail construction at EVERY literal split $R = \mathrm{head}_j R \,\mathrm{tail}_j R$. The contact models here have the genuine empty initial boundary; their entire accumulated exceptional boundary is retained. The conclusions do not require caller test, model-domain, tail-isomorphism or canonical-remainder witnesses.*

Proof. The actual original domain gives closed smoothness, coherence, well-formedness and positive marking. Maximal order at each original cosupport point produces genuine tangent directions, hence verifies the two literal choices and their actual opens. The stated actual incidence places the first point in both chosen opens. Apply the true contact-tail producer once, obtaining ONE glueing datum with its joint universal conclusion for every actual test concatenation.

The actual Step One canonical bridge internally makes the SAME literal sequence R a true test tower of its actual bar datum. The coefficient homogenized datum with empty initial boundary has the identical carrier, ideal and mark; its genuine test property follows by deleting only the initial boundary block. For each natural j, the exact literal split identity reconstructs R from its head and tail. Its concatenated test property therefore follows from that of R. Apply the joint produced contact-tail construction to this exact head and tail. All complement opens, restricted maps, remainder tests, endpoint records and literal subprefix dictionaries are thus attached to the SAME original sequence. No conclusion about enlarged remainder canonicality is used. □

Theorem 266.8 (The SAME original historical contact remainders have genuine whole tail dictionaries)**.** *For the SAME original inputs in the preceding theorem, ONE genuine chosen contact glueing datum supplies actual remainder tests and whole endpoint and literal-prefix dictionaries for the original boundary-KEPT contact remainders with their first $\#K.E$ historical entries replaced by empty entries. The underlying tails are the SAME original maps restricted to the actual selected complement. Every genuine accumulated exceptional entry survives. Actual original record identities are part of the result.*

Proof. At a literal head, let A_0, B_0 be the two genuine coefficient contact models with empty initial boundary. The previous producer internally supplies the actual whole relation $\Phi^* A_0 = B_0$, smooth carrier, coherent ideal, positive mark, and genuine contact model tail test. Put $k = \#K.E$ and prepend exactly k empty entries to A_0, obtaining A_k. Its true carrier, ideal and mark agree with A_0, and the preceding literal empty-history theorem internally gives its SAME actual tail test. Its boundary length is at least k.

The original map-and-append/drop identity identifies the boundary-kept contact remainder EXACTLY with $A_k^{\mathrm{rem},k}$: replacing the old history first and dropping the first k entries of the padded model both leave precisely k empty entries followed by ALL actual exceptional entries. The two constructions have the same selected/remaining carrier, ideal and mark, since the boundary field does not enter cosupport. Pullback of empty history commutes with Φ, so the corresponding pulled padded model is EXACTLY the padded B_0. Thus the second original historical contact remainder is its actual remainder, too. These are proved record equalities, rather than caller model identifications.

Apply the genuine selected-complement tower calculus with A_k and this actual k. Its closed smoothness, coherence, positivity, length condition, parent test and tower isomorphism have ALL been produced internally. It gives the actual restricted remainder tests and entire endpoint controlled equality. The literal-prefix calculus gives the actual head and residual-tail maps and all four controlled fields at EVERY literal prefix. Consume the proved original record identities to identify these objects with the SAME original historical contact remainders. No original boundary is discarded outside these actual marked-record identities; no actual exceptional member is accidentally dropped as historical. □

Scope. The generic transport assertions explicitly take genuine geometric isomorphisms; the contact producer constructs them from actual tests and tangent directions. Canonicality of the enlarged remainder tails, deeper zero-contact histories, full zero-level patch descent, whole GeneralThree analytic superlevels and general noncompact global termination remain OPEN. The contact endpoint fixation assertion is on its genuine complement trace; a cosupport point lying on a selected dead component is not silently moved to the complement.

267 Actual remainder test towers on the selected-component complement

The original contact tail is a true test tower of the enlarged contact model. Its original centres can meet components removed from the remainder. Passing to the actual complement of the selected component union removes precisely this obstruction and retains the entire remaining carrier. The resulting maps are the actual open restrictions of the original maps. On this open, historical boundary members may remain nonempty outside the carrier. Entire ordered marked-record equality with the original contact model is therefore asserted only on the original mask, where these members really miss every point.

Definition 267.1 (The actual selected-component complement)**.** For a true closed smooth carrier $L.N$, let G be the complement of the actual union of its global components contained in $L.\mathrm{cosupp}$. This is a genuine open set, since that selected locally finite component union is closed. It contains the entire true remaining carrier.

Theorem 267.2 (The actual original restriction is a genuine remainder test tower)**.** *For actual closed smooth $L.N$, coherent $L.\mathrm{ideal}$, $k \le \#L.E$ and a genuine parent test tower T, the SAME original tower restricted to G is a genuine test tower of the actual remainder restricted to G. At every literal prefix, its entire carrier, ideal and mark equal those of the full parent transform restricted to the corresponding inverse image of G. Its boundary is its own actual ordered boundary, with k empty historical entries and real exceptions. Its WHOLE marked record equals the actual ambient remainder transform restricted to that inverse image; this last comparison is between the SAME remainder.*

Proof. Initially the carrier identity is $A = L.N \cap G$, the actual ideal and mark were retained, and the candidate boundary is precisely the parent boundary after dropping its first k entries and prefixing k empty entries. The genuine full parent test tower restricts to a genuine full parent test tower on G, by actual open blowup restriction and coherent controlled pullback.

For any genuine test tower, dropping an initial boundary block preserves admissibility: every remaining incident member is an original member in its original adapted coordinates. Prefixing empty entries also preserves admissibility, since they have no incident member at any point. The literal strict-transform and exceptional-append recursion preserves those empty entries and all new actual exceptions. Induct on the original restricted tower: its genuine last centre keeps the same closed smoothness and cosupport, and the restricted adapted coordinates prove normal crossings with the actual padded dropped boundary. Hence every actual restricted centre is admissible for the candidate and every actual restricted map is its true blowup.

Replacing only the boundary leaves all transformed carrier, ideal and mark fields unchanged. Apply this actual recursion to the initial restricted record identity and then the parent coherent open-transform identity. This gives the exact three fields at every literal prefix, without asserting whole boundary equality on G. The true closed strict partition supplies the same carrier equality for the ambient remainder restricted to that inverse image; the preceding entire open ideal producer supplies its ideal equality and the mark is unchanged. The empty-prefix/drop/map identities give equality of its own boundary with the restricted candidate boundary. These four fields prove the whole-record restriction identity for the SAME remainder, not for the full parent. □

Theorem 267.3 (The actual restricted map blows up the true clipped centre trace)**.** *At any actual parent test prefix followed by its true admissible centre and blowup, the centre's trace on the corresponding inverse image of G equals the trace of its intersection with the true remaining strict carrier. This actual traced centre is admissible for the current restricted remainder, and the SAME actual restricted original map is a blowup along this clipped trace.*

Proof. The original centre lies in the full current carrier. The true closed component partition gives the strict remaining carrier as the intersection of the full strict carrier with the projection preimage of the initial remaining union. A point over G cannot project into the selected union. If it lies in the full strict carrier, its projection lies in the initial full carrier, hence in the initial remaining union. Thus every traced original centre point lies in the actual remaining strict carrier, proving the two centre traces equal. The reverse inclusion is immediate from the intersection.

Apply the preceding genuine test-tower producer to the actual original tower with this last step appended. Its actual last test condition proves admissibility of the traced centre for the actual current restricted candidate. Actual open restriction of the original blowup satisfies the genuine blowup universal property along the traced original centre. Substitute the proved trace equality to obtain the SAME genuine restricted map as a blowup along the clipped trace. No statement is made about the unrestricted original map. □

Theorem 267.4 (Original zero-contact data produce one genuine remainder test tower and retain the whole mask)**.** *Original domain-three maximal-order, literal canonical Step1 and actual nonspecial zero-level cosupport data internally produce ONE chosen tangent open, the SAME actual remainder and its genuine selected-complement open G. The ENTIRE original mask lies in G. The actual restricted original tail is a true remainder test tower. All its actual prefix domains are in domain two, and the original whole-mask controlled marked-record dictionary is retained.*

Proof. The preceding original producer internally supplies one chosen tangent open, the genuine contact model, whole original masked records and every original prefix remainder domain. Original Giraud and the literal canonical bridge produce the actual full contact test tail. Actual contact geometry gives its closed smooth carrier, coherent ideal and historical boundary length bound. Apply the genuine selected-complement test-tower producer to that SAME model and SAME original tail; no candidate test condition is an input.

If a point of the original mask were selected, it would lie in the full contact carrier. The already proved entire original-mask carrier equality would put it also in the remaining carrier, contradicting disjointness of the true selected and remaining unions. Thus the entire mask lies in G. For each literal prefix, source and target remainder controlled ideals are coherent by their internally produced domains. Induct on that prefix using coherent controlled open restriction, which requires no unrestricted candidate-centre containment. The whole restricted candidate transform is the pullback of its actual full inverse-image transform. Actual open pullback preserves every domain-two field, so all actual restricted prefix domains are produced. Retain the entire original masked record equalities under the true nested-open identities. Historical empty positions remain literal; there is no whole boundary comparison with the full parent on G. □

Theorem 267.5 (The original literal General3 internally supplies its genuine remainder test tower)**.** *Original domain-three data and literal canonical invFamily3, a true positive ordinary-order point and an actual nonspecial zero-level point of its true companion cosupport internally produce the preceding entire joint output for that SAME actual companion and literal tail.*

Proof. Use the genuine original companion producer to obtain its actual domain-three record, maximal order and literal canonical Step1 tail. Apply the preceding theorem to these internally produced data and the actual companion cosupport point. Return its whole conjunction with this SAME companion and tail. No candidate domain, nonzero, test tower, admissibility or canonicality is supplied as a hypothesis. □

Open frontier. These are actual original maps restricted to an internally produced open containing the entire remaining carrier and original mask. Unrestricted ambient clipped-tower construction/comparison, canonicality on the entire selected complement, deeper zero-contact histories, cross-contact descent, whole original zero-level/General3 superlevels and noncompact global termination remain open.

268 Faithful point truncations of holomorphic germ ideals

The following local algebra bridge uses the actual ring of holomorphic germs. It is a supplementary form of Krull separation, distinct from the earlier torus-coefficient criterion. It identifies membership in an ideal from all of its point truncations; it does not assert a valuative membership criterion.

Theorem 268.1 (Membership detected by every point truncation)**.** *Let $\mathcal{O}_n$ be the ring of holomorphic germs at the origin of $\mathbf{C}^n$, for any $n \in \mathbf{N}$, and let $\mathfrak{m}$ be its actual maximal ideal. For every ideal $I \subseteq \mathcal{O}_n$ and every germ $f \in \mathcal{O}_n$,*

$$f \in I \quad \Longleftrightarrow \quad \forall N \in \mathbf{N}, \quad f \in I + \mathfrak{m}^N.$$

The assertion includes $I = 0$, $I = \mathcal{O}_n$, $N = 0$, and $n = 0$. No properness, finite presentation, nonzero germ, or ideal-membership certificate is a hypothesis.

Proof. Write $R = \mathcal{O}_n$. The genuine holomorphic germ ring is local: a germ is a unit precisely when its value at the origin is nonzero, and its nonunits form $\mathfrak{m}$. The already proved Weierstrass induction gives Noetherianity of R, including the zero-dimensional field case. The quotient $M = R/I$ is a finite R-module, generated by the class of 1. Since $\mathfrak{m}$ is a proper ideal of the Noetherian local ring R, the module form of Krull's intersection theorem gives

$$\bigcap_{N \geq 0} \mathfrak{m}^N M = 0.$$

Suppose $f \in I + \mathfrak{m}^N$ for every N. At a fixed N, write $f = a_N + b_N$ with $a_N \in I$ and $b_N \in \mathfrak{m}^N$. Its class in M equals the class of b_N, which is b_N times the class of 1 and hence belongs to $\mathfrak{m}^N M$. This holds for every N, so the class of f is zero by the displayed intersection identity. The kernel of $R \to R/I$ is exactly I, and thus $f \in I$. Conversely, inclusion of I into each sum $I + \mathfrak{m}^N$ immediately gives all truncation memberships. This argument also covers the zero quotient when $I = R$ and the maximal ideal 0 when $n = 0$. □

269 Actual formal extension contracts to the same entire original ideal

Let $R = \mathcal{O}_{\mathbb{C}^n,0}$, $S = \mathbb{C}[[X_1, \ldots, X_n]]$ and let $\mathcal{T} : R \hookrightarrow S$ be the independently constructed GENUINE original complex Taylor algebra homomorphism. Original coefficients, both actual ring products and original maximal powers are unchanged. This is a convergent-to-formal ideal bridge for the standard-basis route in [4, §7], not a formal division or general strict-transform theorem.

Lemma 269.1 (Every formal coefficient has an actual original finite polynomial approximation)**.** *For EVERY genuine formal series $A \in S$ and $k \geq 0$, evaluate its actual finite total-degree truncation below $k+1$ at the ORIGINAL coordinate germs, obtaining $a_k \in R$. Then $[X^\alpha]\mathcal{T}(a_k) = [X^\alpha]A$ whenever $|\alpha| < k+1$. Moreover for ANY $B \in S$ these same genuine coefficients give $[X^\alpha](\mathcal{T}(a_k)B) = [X^\alpha](AB)$ in this range.*

Proof. The genuine original polynomial evaluation followed by $\mathcal{T}$ is the usual actual polynomial-to-formal-series map: prove this on constants and original variables, then under polynomial addition and multiplication. The actual formal total-degree truncation retains exactly the displayed original coefficients, giving the first equality. In each actual product antidiagonal $\beta+\gamma=\alpha$ the true nonnegative total degree of β is at most $|\alpha|$; the genuine first equality therefore replaces every left coefficient in the actual convolution formula. This proves the product equality without any convergence assumption on the arbitrary original formal input A. □

Theorem 269.2 (Every finite actual formal ideal expression has true original jet representatives)**.** *Let $g, f_1, \dots, f_r \in R$ and $A_1, \dots, A_r \in S$ be actual original numerators and actual formal coefficients. If their literal formal identity is $\mathcal{T}(g) = \sum_i A_i \mathcal{T}(f_i)$, then for EVERY $k \geq 0$ the preceding internally constructed original approximants satisfy*

$$g - \sum_i a_{i,k} f_i \in \mathfrak{m}^{k+1}.$$

Proof. Apply the genuine algebra map to the stated original whole remainder. Its α coefficient, for $|\alpha| < k+1$, is the difference between the coefficient of the given literal formal identity and the sum of the coefficients of $\mathcal{T}(a_{i,k})\mathcal{T}(f_i)$. The first lemma identifies EACH of these actual product coefficients with $A_i\mathcal{T}(f_i)$, so the difference is zero. The independently proved actual ORIGINAL maximal-power/low-formal-coefficient dictionary then gives the displayed ENTIRE original power membership. Empty finite generator families are included. □

Theorem 269.3 (Every original whole ideal is recovered from its actual formal extension)**.** *For EVERY original ideal $I \subset R$ and EVERY original $g \in R$,*

$$g \in I \quad \Longleftrightarrow \quad \mathcal{T}(g) \in S\mathcal{T}(I).$$

Thus $\mathcal{T}^{-1}(S\mathcal{T}(I)) = I$ as ENTIRE original ideals, and actual formal extension is injective on the set of all original ideals.

Proof. The forward implication is actual original image-ideal membership. For the reverse, genuine original Noetherianity internally chooses finite original generators f_i of I. Actual image-of-span equality identifies the ENTIRE formal extension with the ideal generated by $\mathcal{T}(f_i)$. The assumed actual formal membership therefore provides a finite literal formal identity of precisely the type in the preceding theorem. At every k its genuine original polynomial combination lies in I and its genuine original remainder lies in $\mathfrak{m}^{k+1}$. Hence $g \in I + \mathfrak{m}^{k+1}$ at EVERY order. Actual whole original Krull separation forces $g \in I$. Both ideal directions give the contraction equality. Contract two equal actual formal extensions to recover the two original ideals, proving injectivity. There is no supplied ideal equality, faithful-flatness or formal-completion certificate. □

Theorem 269.4 (Original maximal thickenings extend to the same genuine variable-power thickenings)**.** *Write $\mathfrak{n} = (X_1, \dots, X_n) \subset S$. Actual formal extension sends the original $\mathfrak{m}$ to $\mathfrak{n}$, sends EVERY $\mathfrak{m}^q$ to $\mathfrak{n}^q$, and sends EVERY original $I + \mathfrak{m}^q$ to $S\mathcal{T}(I) + \mathfrak{n}^q$.*

Proof. The genuine original zero-tangent Hadamard dictionary identifies the ENTIRE original $\mathfrak{m}$ with the ideal spanned by the SAME original coordinate germs. Actual image-of-span and the independently proved coordinate formula $\mathcal{T}(x_i) = X_i$ give precisely $S\mathcal{T}(\mathfrak{m}) = \mathfrak{n}$. Actual algebra image preserves ideal products and powers, and preserves ideal sums; these give the other two equalities. All are equalities of the genuine ENTIRE ideals, including $q = 0$ and $n = 0$. No completed quotient dimension, flatness or division is inferred in this step. □

270 Actual formal initial diagrams and their finite monic vertex series

Let $S = \mathbb{C}[[X_1, \dots, X_n]]$ be the GENUINE formal series ring. Fix an additive, pointwise-monotone well monomial order $\preceq$ on the actual exponent set $\mathbb{N}^n$. These are actual coefficient supports and the actual finite-antidiagonal product. This is the initial-diagram construction used in [4, §3, §7, pp. 261–264], before proving formal division. The degree-lexicographic specialization uses a fixed ordering of the original coordinates; reversing the coordinate priority only permutes the convention permitted in the source.

Lemma 270.1 (Actual lowest nonzero coefficients and genuine convolution)**.** *For every nonzero actual $F \in S$, there is a unique lowest exponent $\exp_{\preceq}(F)$ in its actual nonzero coefficient support. Its actual coefficient is nonzero and every smaller coefficient vanishes. For actual nonzero F, G,*

$$\exp_{\preceq}(FG) = \exp_{\preceq}(F) + \exp_{\preceq}(G), \qquad [X^{\exp(F)+\exp(G)}](FG) = [X^{\exp(F)}]F\,[X^{\exp(G)}]G.$$

In particular the lowest exponent of the genuine monomial X^β is β, and multiplying F by it shifts its actual lowest exponent by β.

Proof. The actual nonzero coefficient support is nonempty, so its image in the genuine well ordered exponent type has a minimum. Pull it back to the same exponent set. Membership gives its nonzero coefficient; minimality gives vanishing below it. Conversely these two properties characterize that exponent by antisymmetry. Put $a = \exp(F)$ and $b = \exp(G)$. In the ACTUAL finite convolution for degree $a + b$, the pair (a, b) contributes the displayed product. For every other pair (u, v) with $u + v = a + b$, either $u \prec a$ or $v \prec b$: if both were at least their respective minima, ordered addition and cancellation force equality of both. Thus every other actual summand vanishes. The displayed product is nonzero over $\mathbb{C}$. If $c \prec a + b$, then for every pair $u + v = c$ one of $u \prec a$, $v \prec b$ again holds by ordered addition; hence that whole convolution vanishes. The characterization proves the product formula. The monomial has precisely its one actual nonzero coefficient, so the same characterization and product formula prove the final claim. □

Theorem 270.2 (The whole genuine ideal determines its actual upper diagram)**.** *For every actual ideal $I \subset S$, define its actual initial diagram*

$$\mathcal{N}_{\preceq}(I) = \{\exp_{\preceq}(F) : F \in I,\ F \neq 0\}.$$

Then $a \in \mathcal{N}_{\preceq}(I)$ implies $a + \beta \in \mathcal{N}_{\preceq}(I)$ for every actual exponent β. Equivalently, this actual diagram is upward closed for the original pointwise exponent order. The zero ideal has empty diagram.

Proof. Take the actual ideal member F witnessing a. The actual monomial X^β times F is a member of the SAME whole ideal, is nonzero, and has lowest exponent $\beta + a$ by the preceding result. If $a \leq c$ pointwise, write $c = (c - a) + a$ in the actual finitely supported exponent monoid and apply this assertion. The zero ideal has no actual nonzero member, proving its empty-diagram statement directly. □

Theorem 270.3 (Dickson vertices have internally constructed monic genuine series)**.** *For every actual $I \subset S$, its pointwise-minimal diagram elements form a finite set V. For EVERY actual exponent c,*

$$c \in \mathcal{N}_{\preceq}(I) \quad \Longleftrightarrow \quad \exists a \in V :\ a \leq c.$$

For each $a \in V$, one can internally choose a GENUINE $f_a \in I$ whose actual lowest exponent is a and whose actual coefficient there is 1. In particular all these f_a are nonzero. No vertex set or series with the required exponents is a premise.

Proof. Dickson's lemma applies to the SAME finite-coordinate pointwise exponent order. Minimal elements form an antichain, hence are finite. For each actual diagram member c, pointwise well foundedness gives a minimal diagram member $a \le c$; conversely upper closure of the actual diagram proves membership from any such a. For each a in this finite set, actual diagram membership supplies an actual nonzero $F_a \in I$ with lowest exponent a. Its coefficient $c_a = [X^a]F_a$ is a genuine nonzero complex number. Set $f_a = C(c_a^{-1})F_a$ in the ACTUAL formal series ring. The whole ideal is closed under this actual product. The true coefficient formula for a constant series product makes its coefficient at a equal to 1 and preserves vanishing of every lower coefficient. The preceding characterization therefore gives the SAME lowest exponent a. This constructs every claimed series internally. It does not yet claim that this family generates the whole ideal: that is a consequence of the subsequent genuine formal division theorem. □

Lemma 270.4 (Degree-compatible lowest exponents detect actual formal powers)**.** *Specialize to the genuine degree-lexicographic monomial order. For each actual nonzero $F \in S$ and each $q \ge 0$,*

$$F \in (X_1, \dots, X_n)^q \quad \Longleftrightarrow \quad |\exp_{\mathrm{deglex}}(F)| \ge q.$$

Proof. If F lies in the true variable power, the already proved actual power kernel kills every coefficient below total degree q. Its genuine lowest coefficient is nonzero, so its degree is at least q. Conversely suppose that degree is at least q. Every exponent of degree below q is strictly smaller in the genuine degree-first order, so its actual coefficient vanishes by the lowest-coefficient lemma. The BOTH-direction genuine power kernel places F in that same actual formal variable power. This includes $q = 0$ and the empty coordinate family; it assumes no Hilbert–Samuel or standard-basis certificate. □

271 Actual formal division by internally constructed vertex series

Keep the ACTUAL whole formal ideal $I \subset S = \mathbb{C}[[X_1, \dots, X_n]]$, its genuine initial diagram $\mathcal{N}$, its internally constructed finite vertex set V, and the internally chosen genuine monic series $f_a \in I$ with lowest exponent a, for $a \in V$. All products are the genuine finite-antidiagonal formal products. The construction below is a direct coefficient version of formal division in [4, Theorem 3.17, Lemma 7.3, pp. 261–264]. We allow any fixed internally chosen owner of each exponent in the diagram; this refines the same vertex cones into disjoint parts. The usual first-vertex partition is one such choice. This harmless partition convention does not assume or modify the whole ideal or the actual series ring.

Lemma 271.1 (Actual vertex owners and well founded genuine coefficient recursion)**.** *Internally choose for each $c \in \mathcal{N}$ one actual $o(c) \in V$ with $o(c) \le c$; for $c \notin \mathcal{N}$, put $o(c) =$ none. For every actual $F \in S$ there is a genuinely well defined complex coefficient function $t_F(c)$ given by*

$$t_F(c) = [X^c]F - \sum_{a \in V} \sum_{\substack{\beta+\gamma=c \\ \beta+a \prec c,\ o(\beta+a)=a}} t_F(\beta + a)[X^\gamma] f_a.$$

Every displayed inner sum is the actual finite exponent antidiagonal; every recursive call is strictly smaller in the genuine well monomial order. Neither the owner nor a coefficient solution is a caller premise.

Proof. The actual finite-vertex coverage theorem gives an actual dominating vertex for every diagram exponent, so choice constructs the stated owner, with the stated pointwise inequality. Outside the actual diagram there is no owner. Apply well founded recursion to the actual well monomial exponent order. At stage c, the formula uses only the finitely many actual antidiagonal pairs and only the already defined values at $\beta + a \prec c$. Thus it gives an actual complex number at every exponent, with the displayed defining equation. No infinite numerical sum or convergence assertion enters the construction. □

Theorem 271.2 (Actual formal quotients and remainder have the required disjoint supports)**.** *Define GENUINE actual formal series $Q_a(F), R(F) \in S$ by their coefficients:*

$$[X^\beta]Q_a(F) = \begin{cases} t_F(\beta+a) & o(\beta+a)=a, \\ 0 & \textit{otherwise}, \end{cases} \qquad [X^c]R(F) = \begin{cases} t_F(c) & c \notin \mathcal{N}, \\ 0 & c \in \mathcal{N}. \end{cases}$$

The shifted supports $a + \operatorname{supp} Q_a(F)$ lie in the actual disjoint owner parts of $\mathcal{N}$, and $\operatorname{supp} R(F) \cap \mathcal{N} = \varnothing$.

Proof. An actual formal series is an arbitrary complex coefficient function on the actual exponent set, with its existing genuine ring operations. The displayed functions therefore define actual series directly. A nonzero quotient coefficient forces its defining owner condition; that condition places the shifted exponent in the true diagram. Distinct owners cannot have the same shifted exponent because an actual owner is a single value. A remainder coefficient at a diagram exponent is literally zero by its definition, proving the last assertion. □

Theorem 271.3 (The actual finite sum of genuine formal products equals the original series)**.** *For EVERY actual $F \in S$, the internally constructed series satisfy the WHOLE genuine formal equality*

$$F = \sum_{a \in V} Q_a(F) f_a + R(F).$$

Proof. Fix an actual exponent c and expand the coefficient of the displayed finite sum using the genuine product antidiagonals. The terms with $\beta + a \prec c$ are exactly the correction sum in the defining recursion, by the actual quotient coefficient formula. For every remaining antidiagonal pair, a nonzero coefficient of f_a forces $a \preceq \gamma$. If $\beta + a \not\prec c = \beta + \gamma$, ordered cancellation then forces $\gamma = a$. Such a term exists only when $a \le c$ pointwise, has $\beta = c - a$, and contributes $t_F(c)$ exactly when $o(c) = a$, since f_a has actual coefficient 1 there. Thus if $c \in \mathcal{N}$, exactly its one genuine owner contributes that remaining $t_F(c)$ and the remainder coefficient is zero. If $c \notin \mathcal{N}$, no owner contributes and the actual remainder coefficient is $t_F(c)$. In both cases the coefficient of the whole right side is the correction plus $t_F(c)$, which is precisely $[X^c]F$ by the true recursion equation. Equality of EVERY actual coefficient proves the actual formal ring equality. □

Theorem 271.4 (Internally constructed actual vertex series generate the whole original formal ideal)**.** *For every actual $F \in I$, its internally constructed remainder is zero. Consequently I is generated by the internally constructed finite actual family $(f_a)_{a \in V}$, with the genuine whole-series coefficient decomposition of the preceding theorem. This includes the zero and unit ideals and requires no basis, quotient, or division certificate.*

Proof. Each actual f_a belongs to the SAME whole ideal I, so the actual finite sum $\sum Q_a(F) f_a$ belongs to I. The genuine identity therefore puts $R(F)$ in I. If this actual remainder were nonzero, its actual lowest nonzero coefficient exponent would belong to $\mathcal{N}$ by the definition of that diagram. But the actual remainder has zero coefficient at every diagram exponent, contradicting its genuine lowest nonzero coefficient. Thus $R(F) = 0$. The resulting identity puts every actual member of I in the span of the actual f_a; the converse inclusion follows from their actual membership in I. This proves BOTH whole ideal inclusions internally. □

272 Genuine whole original and formal jet quotients have the same HS values

Keep the ACTUAL original Taylor complex algebra embedding $\mathcal{T} : R = \mathcal{O}_{\mathbb{C}^n,0} \hookrightarrow S = \mathbb{C}[[X_1, \dots, X_n]]$. Write $\mathfrak{m}$ for the original maximal ideal and $\mathfrak{n} = (X_1, \dots, X_n)$ for the genuine formal variable ideal. This chapter proves the original-to-formal Hilbert–Samuel bridge used in the standard-basis setting of [4, §7, pp. 260–271]. Every complex scalar action below is the genuine existing action on its actual ring.

Lemma 272.1 (The genuine formal variable powers have exactly the actual low-coefficient kernel)**.** *For every genuine $F \in S$ and $q \geq 0$,*

$$F \in \mathfrak{n}^q \quad \Longleftrightarrow \quad [X^\alpha]F = 0 \text{ for every } |\alpha| < q.$$

Proof. Use the canonical, coefficient-defined algebra equivalence from the ACTUAL formal series ring to the adic completion of the genuine polynomial ring $P = \mathbb{C}[X]$ at its true variable ideal J. Its actual variable formula sends EACH X_i to the canonical completion image of the SAME polynomial variable. Actual image-of-span therefore sends the WHOLE $\mathfrak{n}$ to $J\widehat{P}$, and sends $\mathfrak{n}^q$ to $J^q\widehat{P}$. The original finite variable family makes J finitely generated. The genuine completion evaluation kernel theorem identifies $J^q\widehat{P}$ with the kernel of evaluation to P/J^q. The canonical series equivalence evaluates at order q to precisely the class of the actual total-degree truncation of F. In the genuine polynomial ring, membership in J^q means exactly vanishing of ALL coefficients of degree below q; the actual truncation retains precisely these coefficients. These equalities in BOTH directions give the claimed kernel. This applies also to the empty variable family and to $q = 0$. No equivalence or desired power-kernel premise is supplied by a caller. □

Theorem 272.2 (The actual original-to-formal whole quotient map is bijective at every order)**.** *For EVERY original ideal $I \subset R$ and EVERY $q \geq 0$, the genuine original Taylor map induces the canonical bijective ORIGINAL complex algebra map*

$$R/(I + \mathfrak{m}^q) \longrightarrow S/(S\mathcal{T}(I) + \mathfrak{n}^q), \qquad [g] \longmapsto [\mathcal{T}(g)].$$

Proof. Actual extension sends the ENTIRE original $I+\mathfrak{m}^q$ to $S\mathcal{T}(I)+\mathfrak{n}^q$, so the displayed original algebra map is well defined. If the images of two genuine original classes agree, $\mathcal{T}(g-h)$ belongs to that true extended ideal. Actual whole original extension-contraction forces $g-h \in I+\mathfrak{m}^q$, proving injectivity. For any genuine formal numerator F, evaluate its actual total-degree truncation below q at the SAME original coordinate germs to obtain $g_F \in R$. The already proved original polynomial/Taylor identity shows that $\mathcal{T}(g_F)$ has precisely the lower coefficients of F. The preceding genuine formal power kernel therefore gives $\mathcal{T}(g_F)-F \in \mathfrak{n}^q$. Their true formal quotient classes agree, proving surjectivity on EVERY entire formal class. The same construction handles $q = 0$, where both quotient rings are trivial. □

Theorem 272.3 (Canonical whole quotient algebra equivalences retain every actual original numerator)**.** *The preceding map constructs an actual ORIGINAL complex algebra equivalence for every I, q, whose value at EVERY original numerator class is its literal Taylor numerator class as displayed above.*

Proof. Apply the genuine algebra equivalence constructor to the ACTUAL bijective quotient algebra map, retaining that same forward map. Its numerator formula is therefore the defining actual quotient map formula. The scalar field acts through the original constant-germ map on the source, and the genuine constant formal series on the target; the true Taylor algebra commutation gives complex linearity of this equivalence. There is no changed or transported algebra action on either actual quotient. □

Theorem 272.4 (Every existing original HS value equals the genuine formal extension HS value)**.** *Every $S/(K + \mathfrak{n}^q)$, for arbitrary formal ideal $K \subset S$, is genuinely finite dimensional over the existing complex field. Define $H_K^{\mathrm{formal}}(\ell) = \dim_{\mathbb{C}} S/(K + \mathfrak{n}^{\ell+1})$. Then for EVERY original ideal I and EVERY $\ell \geq 0$,*

$$H_I(\ell) = H_{S\mathcal{T}(I)}^{\mathrm{formal}}(\ell).$$

Proof. The genuine original quotient $R/\mathfrak{m}^q$ is internally finite dimensional by actual holomorphic finite jets. The true complex-linear quotient equivalence at $I = 0$ therefore makes $S/\mathfrak{n}^q$ finite dimensional. Its actual surjective quotient map to $S/(K + \mathfrak{n}^q)$ gives the first assertion, with no formal Noetherian or finite-quotient certificate assumed. Finally take the preceding actual whole

quotient complex-linear equivalence for the SAME original I and $q = \ell + 1$. Its true complex dimension equality is precisely the EXISTING original HS value and the displayed genuine formal value, proving the assertion simultaneously at every order. □

273 The genuine original multivariate formal Taylor algebra embedding

Let $R = \mathcal{O}_{\mathbb{C}^n,0}$ with its ORIGINAL complex scalar structure, and let $c_\alpha(g)$ be the actual stabilized coefficients already constructed from genuine finite quotient polynomials. Define $\mathcal{T}(g) = \sum_\alpha c_\alpha(g) X^\alpha$ in the ACTUAL multivariate formal power-series ring $\mathbb{C}[[X_1, \ldots, X_n]]$, whose multiplication is its genuine coefficient convolution. The zero-tangential-block coordinate identification retains the same original ambient coordinates. This supplies a faithful convergent-to-formal producer for [4, §7]; completion surjectivity, faithful flatness, formal division and transformed standard bases remain separate obligations.

Lemma 273.1 (Actual constants and original coordinates have their genuine formal values)**.** *The actual formal series $\mathcal{T}(a)$ of an original complex constant is the genuine constant formal series a, and the actual formal series of EACH original coordinate germ is the corresponding formal variable X_i. All coefficients are the SAME original complex numbers.*

Proof. A genuine constant polynomial has the exact original constant germ as its evaluation and has no coefficient outside degree zero. The already proved finite Taylor uniqueness therefore gives $T_k(a) = a$ for every k. For an original coordinate, its genuine polynomial X_i has the exact same evaluation and degree one; uniqueness gives $T_k(x_i) = X_i$ whenever $k \geq 1$. To compute ANY original coefficient in either statement, use coefficient stabilization at an actual truncation containing that exponent and, for a coordinate, also degree one. These are precisely the genuine constant and variable coefficients in the actual formal-series ring. □

Theorem 273.2 (Actual whole original products give the genuine formal coefficient convolution)**.** *For ANY original numerators $g, h \in R$ and EVERY actual exponent α,*

$$c_\alpha(gh) = \sum_{\beta+\gamma=\alpha} c_\beta(g) c_\gamma(h).$$

Thus $\mathcal{T}(gh) = \mathcal{T}(g)\mathcal{T}(h)$ for the ACTUAL formal power-series multiplication, rather than pointwise coefficient products.

Proof. Set $k = |\alpha|$ and $A = T_k(g)$, $B = T_k(h)$, $C = T_k(gh)$. All three true original remainders lie in $\mathfrak{m}^{k+1}$. The literal original ring identity

$$\operatorname{ev}(C) - \operatorname{ev}(A)\operatorname{ev}(B) = -(gh - \operatorname{ev}(C)) + g(h - \operatorname{ev}(B)) + (g - \operatorname{ev}(A))\operatorname{ev}(B)$$

puts the actual evaluation of $C - AB$ in that SAME true maximal power. Genuine lower coefficient separation gives $[X^\alpha]C = [X^\alpha](AB)$. The genuine polynomial product formula expresses the right side as its finite exponent antidiagonal sum. At each actual pair $\beta + \gamma = \alpha$, nonnegative degree additivity gives $|\beta|, |\gamma| \leq k$; true coefficient stabilization therefore replaces both polynomial coefficients by $c_\beta(g)$ and $c_\gamma(h)$. This proves the displayed convolution. Actual formal coefficient extensionality and the genuine formal product formula give the stated multiplication identity. □

Theorem 273.3 (An actual injective algebra homomorphism retains every original constant and coordinate)**.** *The SAME $\mathcal{T}$ is an injective ORIGINAL complex algebra homomorphism from the ENTIRE original holomorphic germ ring to the ACTUAL multivariate formal power-series ring, retaining the displayed original constants and coordinates.*

Proof. Original coefficient linearity gives zero and addition for the genuine formal additive structure. The preceding actual constant computation gives the genuine unit and the original constant action; the preceding convolution gives the actual formal multiplication. These fields construct the complex algebra homomorphism on the SAME entire original numerator map. If two actual formal images agree, each actual original coefficient agrees; the already proved injectivity of the original complex coefficient function map forces the two ORIGINAL germs equal. No algebra structure is transported or substituted on either ring. □

Theorem 273.4 (Actual original maximal powers have precisely the genuine low-coefficient kernel)**.** *For every original $g \in R$ and $q \geq 0$,*

$$g \in \mathfrak{m}^q \quad \Longleftrightarrow \quad [X^\alpha]\mathcal{T}(g) = 0 \quad \textit{for all } |\alpha| < q.$$

This is the actual original filtration dictionary with the genuine formal coefficients, without a supplied coefficient or kernel certificate.

Proof. Suppose $g \in \mathfrak{m}^q$ and $|\alpha| < q$. Put $k = |\alpha|$. The actual power inclusion gives $g \in \mathfrak{m}^{k+1}$, and its true Taylor remainder lies there too. Subtracting yields $\mathrm{ev}(T_k(g)) \in \mathfrak{m}^{k+1}$. Actual lower coefficient separation kills its α coefficient, which is the actual formal coefficient by stabilization. Conversely $q = 0$ is the whole original ring. For $q = k + 1$, all lower coefficients of $T_k(g)$ vanish by the assumed actual formal coefficients and stabilization; all higher ones vanish by its true support bound. Thus $T_k(g) = 0$, and the actual whole remainder gives $g \in \mathfrak{m}^q$ as required. No identification with a completed-ideal quotient or flatness is inferred. □

274 Actual original complex Taylor polynomials and stable coefficients

This is the convergent original-coefficient bridge toward the formal standard-basis setting of [4, §7]. Every polynomial is evaluated at the ACTUAL original coordinate germs, with the ORIGINAL constant-germ complex action. The already proved zero-tangential-block identification $\mathbb{C}^{0+n} = \mathbb{C}^n$ retains these same coordinates. No formal-series multiplication or division is asserted in this chapter. Write $R = \mathcal{O}_{\mathbb{C}^n,0}$, $\mathfrak{m}$ for its actual maximal ideal, X^α for the actual coordinate monomial, and $|\alpha| = \sum_i \alpha_i$ for the true original exponent degree.

Lemma 274.1 (Actual original polynomial evaluation separates all lower coefficients)**.** *The actual evaluation* $\mathrm{ev} : \mathbb{C}[X_1, \ldots, X_n] \to R$ *is an algebra homomorphism with* $\mathrm{ev}(aX^\alpha) = a\,X^\alpha$ *in the original whole ring. If* $\mathrm{ev}(P) \in \mathfrak{m}^q$*, EVERY actual coefficient* $[X^\alpha]P$ *with* $|\alpha| < q$ *is zero.*

Proof. Use literal original coordinate germs in polynomial evaluation; its monomial formula is the original finite product formula. Map the same complex polynomial coefficients to their actual zero-dimensional constant germs. True zero-block pullback of each such constant is the SAME original ambient complex constant, so evaluating this coefficient lift by the existing whole normal polynomial map gives precisely $\mathrm{ev}(P)$. The actual zero-tangent normal ideal is $\mathfrak{m}$. Existing genuine whole normal coefficient separation therefore annihilates every coefficient below q as a zero-dimensional whole germ; actual evaluation at its unique point gives the displayed original complex coefficient zero. This includes zero dimension and $q = 0$, where the lower-coefficient condition is empty. □

Theorem 274.2 (Every original numerator has a canonical genuine finite Taylor polynomial)**.** *For every original $g \in R$ and $k \geq 0$, internally define $T_k(g) \in \mathbb{C}[X]$ using the actual distinct-monomial complex coordinates of the TRUE quotient $R/\mathfrak{m}^{k+1}$. Its coefficients vanish when $|\alpha| \geq k + 1$, and*

$$g - \mathrm{ev}(T_k(g)) \in \mathfrak{m}^{k+1}.$$

It is the UNIQUE polynomial with that coefficient support bound and that actual whole remainder. The construction is complex linear in the original numerator.

Proof. Apply the inverse of the already constructed original complex monomial quotient equivalence to the ACTUAL class of g. If its coordinate tuple is c_α, set $T_k(g) = \sum_{|\alpha|<k+1} c_\alpha X^\alpha$. The genuine finite original exponent family gives the asserted support bound. Expanding its coefficient tuple as the finite sum of standard basis vectors and using the actual monomial-class formula for that SAME quotient equivalence proves that the class of its evaluation is exactly the original class of g. This is precisely the stated whole remainder. If P has the same support bound and remainder, evaluation of $P - T_k(g)$ lies in $\mathfrak{m}^{k+1}$. The first lemma annihilates all its lower coefficients; both original support bounds annihilate the others. Polynomial coefficient extensionality gives uniqueness. All constituent quotient, inverse-basis and coefficient combination maps are linear for the original complex action, giving linearity. □

Theorem 274.3 (True original Taylor coefficients stabilize across all actual truncations)**.** *If $k \leq l$ and $|\alpha| < k+1$, then $[X^\alpha]T_l(g) = [X^\alpha]T_k(g)$. Thus the original complex coefficient $c_\alpha(g) := [X^\alpha]T_{|\alpha|}(g)$ equals $[X^\alpha]T_k(g)$ for EVERY k with $|\alpha| < k+1$.*

Proof. The true higher remainder belongs to $\mathfrak{m}^{l+1} \subseteq \mathfrak{m}^{k+1}$, and the true lower remainder belongs there too. Subtracting these genuine original whole remainders shows that $\mathrm{ev}(T_l(g) - T_k(g)) \in \mathfrak{m}^{k+1}$. Actual lower polynomial coefficient separation gives the first equality. Apply it with lower truncation $|\alpha|$ to obtain the stated equality for the actual stable coefficient. No coefficient compatibility premise is supplied by a caller. □

Theorem 274.4 (All genuine original coefficients give a faithful complex linear map)**.** *The actual original coefficient map $g \mapsto (c_\alpha(g))_\alpha$ is complex linear and injective into the actual coefficient function space $(\mathbb{N}^n \to \mathbb{C})$.*

Proof. Linearity follows from the true finite polynomial construction at the original degree $|\alpha|$. If all actual coefficients of g vanish, stabilization annihilates every lower coefficient of EVERY $T_k(g)$; its genuine support bound annihilates the others, so $T_k(g) = 0$. The true remainder then places the original g in EVERY $\mathfrak{m}^{k+1}$. Whole original ideal separation at $I = 0$ gives $g = 0$. Apply this kernel conclusion to a difference, using actual complex linearity, to obtain injectivity. This is a faithful original linear coefficient map; formal-series algebra multiplication remains the next explicit obligation. □

275 Canonical reduced standard bases of the actual formal ideal

Let $S = \mathbb{C}[[X_1, \ldots, X_n]]$, let $\prec$ be a monomial order, and let $\mathcal{N}(I)$ be the actual initial-exponent diagram of an arbitrary ideal $I \subset S$. Write R_I for the remainder already constructed by actual formal division. Its coefficients vanish on $\mathcal{N}(I)$. The vertex members chosen for that division are monic but need not be reduced. We now construct reduced members from the remainder itself. This is the normal-form construction used in the standard-basis argument in [1], Section 6.3; here existence, uniqueness and generation are established directly for the actual formal ideal, before any geometric hypothesis.

Theorem 275.1 (The actual remainder is the unique normal representative)**.** *For every $F \in S$, $F - R_I(F) \in I$. If $F - H \in I$ and every coefficient of H on $\mathcal{N}(I)$ is zero, then $H = R_I(F)$. Consequently $R_I(F) = 0$ if and only if $F \in I$, and R_I is a $\mathbb{C}$-linear idempotent with kernel I and image the space of series supported outside $\mathcal{N}(I)$.*

Proof. The division identity expresses $F - R_I(F)$ as a finite sum of multiples of actual members of I, proving membership. The difference $H - R_I(F) = (F - R_I(F)) - (F - H)$ belongs to I and has zero coefficients on its entire diagram. If it were nonzero, its actual initial exponent would belong to $\mathcal{N}(I)$, contradicting that vanishing. This proves uniqueness. A member of I has zero remainder by the same argument; conversely a zero remainder and the membership just proved put F in I.

For $F, G \in S$ and $c \in \mathbb{C}$, the series $R_I(F)+R_I(G)$ and $cR_I(F)$ vanish on the diagram and differ from $F+G$ and cF by members of I. Uniqueness proves additivity and homogeneity. Any series vanishing on the diagram is its own remainder, again by uniqueness, proving both idempotence and the asserted image. No decomposition is supplied by the caller. □

Lemma 275.2 (Actual division preserves every initial segment)**.** *If $[X^c]F = 0$ for every $c \prec a$, then $[X^c]R_I(F) = 0$ for every $c \prec a$.*

Proof. Use well-founded induction on c in the actual division recursion. For $c \prec a$, its input coefficient is zero. Every correction summand contains a division coefficient at an exponent $b \prec c$, and therefore $b \prec a$; the induction hypothesis makes that summand zero. The recursion thus makes the division coefficient at c zero. A remainder coefficient is either this division coefficient or zero according as c lies outside or inside the actual diagram, proving the claim for all initial segments, including monomial orders with infinitely many predecessors. □

Theorem 275.3 (Internally constructed canonical reduced vertex members)**.** *For each actual vertex a of $\mathcal{N}(I)$, define*

$$G_a = X^a - R_I(X^a).$$

Then $G_a \in I$, its actual initial exponent is a, its coefficient there is one, and its coefficient at every other exponent of $\mathcal{N}(I)$ is zero. It is the unique member of I with this entire diagram coefficient vector. In particular it depends only on the actual ideal and monomial order.

Proof. Membership follows from the first theorem. At a diagram exponent b, the remainder coefficient is zero, so $[X^b]G_a$ is the coefficient of X^a, namely one for $b = a$ and zero otherwise. All coefficients of X^a below a vanish; the initial-segment lemma makes the remainder coefficients there vanish as well. Thus the nonzero coefficient at a is the actual first one. If another member has the same coefficients on the whole diagram, its difference from G_a belongs to I and vanishes on the diagram, so the first proof makes that difference zero. This also proves independence from all choices made during division. □

Theorem 275.4 (The genuine reduced standard basis generates the whole ideal)**.** *The finite family (G_a) is a reduced standard basis of the SAME ideal:*

$$I = (G_a : a \text{ an actual vertex of } \mathcal{N}(I)).$$

This includes the zero ideal, the unit ideal and ambient dimension zero.

Proof. Let J be the displayed span. Its generators belong to I, hence $J \subset I$. Each actual vertex a of $\mathcal{N}(I)$ is the actual initial exponent of $G_a \in J$. Upward closure of the actual diagram and the vertex cover imply $\mathcal{N}(I) \subset \mathcal{N}(J)$; the reverse inclusion follows directly from $J \subset I$.

For any $F \in I$, division by the actual ideal J gives $R_J(F) = F - H$ with $H \in J \subset I$. Thus $R_J(F) \in I$. If it were nonzero, its actual initial exponent would lie in $\mathcal{N}(I) \subset \mathcal{N}(J)$, where all remainder coefficients vanish, a contradiction. Therefore $R_J(F) = 0$ and the first theorem gives $F \in J$. This proves the other whole-ideal inclusion. No geometric compatibility, normal-flatness, or strict-transform conclusion has been assumed or asserted. □

276 Actual whole-ideal detection by all original finite jets

This chapter supplies the convergent original-ring separation needed before formal initial-diagram or division arguments can identify an ENTIRE original ideal. It is an auxiliary bridge for the standard-basis route in [4, §7, pp. 260–263]; it does not prove formal division or a transformed-standard-basis theorem. Write $R = \mathcal{O}_{\mathbb{C}^n,0}$, with its ACTUAL original complex algebra structure, $\mathfrak{m}$ its true maximal ideal, and j_q the already constructed genuine finite derivative jet of every original germ. All ideals below are arbitrary original ideals, including zero and the unit ideal, and zero ambient dimension is allowed.

Theorem 276.1 (Every original whole ideal is separated in the actual maximal filtration)**.** *For EVERY original ideal $I \subset R$,*

$$\bigcap_{q\geq 0}(I+\mathfrak{m}^q) = I.$$

Consequently an original numerator belongs to I exactly when it belongs to all these genuine enlarged ideals, equivalently when it belongs to $I+\mathfrak{m}^{\ell+1}$ for every $\ell \geq 0$.

Proof. The original holomorphic germ ring is internally Noetherian and local. Its quotient $M = R/I$, regarded as an ORIGINAL R-module, is finite even when $I = R$. Module Krull intersection for the actual proper maximal ideal therefore gives $\bigcap_q \mathfrak{m}^q M = 0$. If g lies in every displayed enlarged ideal, write $g = a_q + z_q$ with $a_q \in I$ and $z_q \in \mathfrak{m}^q$. The actual quotient class of g is the class of $z_q = z_q \cdot 1$, hence belongs to every $\mathfrak{m}^q M$. It is zero, so $g \in I$. The other inclusion is literal original ideal inclusion. The order-zero enlarged ideal is the whole ring; removing that one condition gives precisely the positive-order version. No quotient nontriviality or proposed ideal equality is assumed. □

Theorem 276.2 (True finite derivative images determine entire original ideals)**.** *For every original $g \in R$, ideal I, and $q \geq 0$,*

$$j_q(g) \in j_q(I) \quad \Longleftrightarrow \quad g \in I + \mathfrak{m}^q.$$

Thus $g \in I$ iff $j_q(g) \in j_q(I)$ for EVERY q, and equivalently for every positive q. For any two original ideals I, J, actual inclusion $I \subseteq J$ is equivalent to $j_q(I) \subseteq j_q(J)$ for every q, and equality is equivalent to equality of ALL these actual complex subspaces.

Proof. A genuine image witness $a \in I$ with $j_q(a) = j_q(g)$ has $g - a \in \ker j_q = \mathfrak{m}^q$, giving the enlarged-ideal membership. Conversely the actual decomposition $g = a + z$, with $a \in I$ and $z \in \mathfrak{m}^q$, gives $j_q(g) = j_q(a)$ by the already proved WHOLE jet kernel. Intersect the resulting equivalences using the first theorem. Actual ideal inclusion implies every image inclusion directly. If all image inclusions hold, apply the membership equivalence to EACH original $g \in I$ to obtain $g \in J$. Both inclusion directions give the equality assertion. In particular this does not select or transport a finite surrogate ideal in place of the original one. □

Theorem 276.3 (A proper original ideal inclusion is detected by an existing HS value)**.** *If $I \subsetneq J$ are original ideals, there is an actual $\ell \geq 0$ with*

$$H_J(\ell) < H_I(\ell).$$

Consequently, for actual original $I \subseteq J$, equality of EVERY existing original HS value is equivalent to WHOLE ideal equality.

Proof. Choose a genuine original $g \in J \setminus I$. The preceding positive-order jet detection supplies some $q = \ell + 1$ with $j_q(g) \notin j_q(I)$. Since $j_q(g) \in j_q(J)$ and actual image inclusion holds, $j_q(I)$ is a PROPER complex subspace of $j_q(J)$. All are finite dimensional actual derivative subspaces. Their original complex dimensions therefore satisfy $\dim j_q(I) < \dim j_q(J) \leq \dim \operatorname{im} j_q$. The already proved ENTIRE original quotient dictionary gives $H_K(\ell) = \dim \operatorname{im} j_q - \dim j_q(K)$ for $K = I, J$; these genuine finite dimension inequalities give the displayed strict inequality. If $I \subseteq J$ and all original HS values agree, a proper inclusion would contradict it, so $I = J$. The converse follows from literal ideal equality. Without the inclusion premise, equal HS profiles are not claimed to identify ideals. □

277 Controlled restriction to a smaller strict carrier

Marked ideals on a smooth carrier represent ideals of its structure sheaf; their ambient representatives are compared by actual restricted holomorphic spans. The controlled transform of [5, Definition 2.2] uses the same exceptional power for two records with the same marking. The following comparison requires admissibility only for the full reference record. The smaller carrier need not contain the centre, and its history need not be a test history.

Theorem 277.1 (One-sided controlled restriction from a genuine admissible reference)**.** *Let $B = (X, N, E, I, d)$ and $K = (X, S, F, A, d)$ be actual marked records. Assume that N is closed smooth, I is coherent, S is closed, $S \subseteq N$, and $A|_S \subseteq I|_S$ in the literal restricted holomorphic-span sense. Let C be admissible for B, and let $\pi : Y \to X$ be an actual blowup of C. Put $N' = \overline{\pi^{-1}(N \setminus C)}$ and $S' = \overline{\pi^{-1}(S \setminus C)}$, and let I' and A' be the actual controlled ideals of these same records. Then*

$$A'|_{S'} \subseteq I'|_{S'}.$$

The marking may be zero. No admissibility for K, containment $C \subseteq S$, smoothness of S' or coherence of A' is assumed.

Proof. Set $L = (\pi^*\mathcal{I}_C)^d$. The definition of the restricted colon gives $LA'|_{S'} \subseteq (\pi^*A)|_{S'}$ without any coherence or divisibility hypothesis: each generating product uv, with $u \in L$ and $v \in A'$, is by the colon definition a restricted holomorphic span of π^*A; expand a finite sum of such products and compose its finite span relations.

Since S is closed, continuity gives $\pi(S') \subseteq S$. Pulling back the original restricted span relations, with their actual holomorphic coefficient functions, therefore gives $(\pi^*A)|_{S'} \subseteq (\pi^*I)|_{S'}$. The genuine Hadamard divisibility theorem for the admissible reference B gives $(\pi^*I)|_{N'} = LI'|_{N'}$. Monotonicity of closure gives $S' \subseteq N'$, so restriction and transitivity yield

$$LA'|_{S'} \subseteq LI'|_{S'}.$$

The reference controlled ideal I' is coherent, by its genuine admissibility and coherent numerator.

To cancel L, choose at $z \in S'$ a true holomorphic generator h of L on an open neighbourhood U. It vanishes on no nonempty relatively open subset of S'. Indeed every open set meeting S' meets the dense set $\pi^{-1}(S \setminus C)$; at such a point L is the unit ideal, so its generator cannot vanish. This argument also covers the zeroth power. For $f \in A'_z$, the actual colon makes f holomorphic. The displayed product inclusion writes hf on S' as a finite holomorphic span of germs of LI'. Expand each generating product and express its L factor as h times a genuine holomorphic cofactor. Coherence of I' then collects the other factors into germs $t_i \in I'_z$, giving $hf = h\sum_i a_i t_i$ on S' near z. The nonzero locus of h is relatively dense in $S' \cap U$; continuity of the two holomorphic functions extends their equality from that dense locus to all of S' near z. Thus $f = \sum_i a_i t_i$ on S' near z, precisely the required restricted span. This proves the inclusion at every point without a smaller-record test or a supplied quotient germ. □

Theorem 277.2 (Entire original reference histories preserve the one-sided restricted comparison)**.** *Let B be a genuine well-formed original marked record and K any actual marked record with closed carrier contained in B's carrier, the same marking, and $K.\mathrm{ideal}|_{K.N} \subseteq B.\mathrm{ideal}|_{K.N}$. For every genuine original B test history T, the actual current K carrier is closed and contained in the actual current B carrier, the markings remain equal, and*

$$(K_T.\mathrm{ideal})|_{K_T.N} \subseteq (B_T.\mathrm{ideal})|_{K_T.N}.$$

This concerns the literal original controlled history of K; it neither assumes nor constructs a K test history.

Proof. Induct on the genuine blowup-history constructor. For the reflexive history the four assertions are exactly the original hypotheses. For a successor, the original test predicate first supplies the genuine shorter B test history and admissibility of its next centre for the actual current B record. Original well-formedness along that shorter test supplies its actual smooth carrier and coherent ideal. The induction hypothesis supplies the closed smaller carrier, its inclusion, the same marking and the restricted ideal comparison. Apply the first theorem to these actual current records and that same actual blowup. The new smaller carrier is closed because it is a closure. Its inclusion in the new reference carrier follows by applying preimage and closure to the old carrier inclusion, after deleting the same centre. The controlled construction retains both old markings. These four facts are exactly the induction assertions at the successor. All original positions and empty steps remain part of the same history. □

Theorem 277.3 (Reference cosupport on the actual smaller current carrier)**.** *Under the hypotheses of the preceding theorem, at every genuine original reference test history T one has*

$$B_T.\mathrm{cosupp} \cap K_T.N \subseteq K_T.\mathrm{cosupp}.$$

The smaller-carrier membership is retained explicitly here. It will be constructed from the actual retained-positive regime of the original member run, rather than assumed in that original-input consumer.

Proof. Take a point in the displayed intersection. Every germ of the actual reference ideal has order at least the current reference marking along its current carrier. Restrict its genuine maximal-ideal-power witness to the contained current smaller carrier; the same witness still proves the same order. The history theorem expresses each germ of the current smaller ideal on that carrier as a finite holomorphic combination of these reference germs. Each product and finite sum retains the common order, so every smaller germ has that order. The markings are equal by the history theorem, and actual smaller-carrier membership was part of the given intersection. This is exactly membership in its cosupport. No converse inclusion or smaller test predicate is used. □

278 Count the genuine original complex base jets

For $B_d = \mathcal{O}_{\mathbb{C}^d,0}$ and its ACTUAL original maximal ideal $\mathfrak{m}_d$, the required tangent weight at order $k+1$ has complex dimension $\binom{d+k}{d}$. We count genuine distinct monomials, not the full multilinear derivative codomain, whose redundant nonsymmetric tensors would give an incorrect dimension.

Lemma 278.1 (The actual zero-dimensional germ algebra is the original complex field)**.** *Evaluation is a genuine complex-algebra isomorphism $B_0 \simeq_{\mathbb{C}} \mathbb{C}$. In particular its actual maximal ideal is zero and the actual normal ideal $J_{0,d}$ in B_d is precisely $\mathfrak{m}_d$.*

Proof. The actual model $\mathbb{C}^0$ has one point. Every genuine holomorphic germ representative therefore agrees everywhere with its original constant value. The original germ equality relation gives injectivity of evaluation. Every complex constant has its actual constant germ, which gives surjectivity and the original complex scalar compatibility. This constructs the stated algebra isomorphism. Membership in the actual zero-dimensional maximal ideal is exactly zero evaluation, hence that ideal is zero. Its whole image in B_d is zero. The original relative Hadamard identity $\mathfrak{m}_d = \mathfrak{m}_0 B_d + J_{0,d}$ proves the last assertion. □

Theorem 278.2 (The true original base-jet quotient has its actual distinct complex monomial coordinates)**.** *The actual quotient $B_d/\mathfrak{m}_d^{k+1}$ is genuinely complex-linearly isomorphic to complex coefficient families indexed by original normal exponents $\alpha : \mathrm{Fin}\, d \to_0 \mathbb{N}$ with $|\alpha| \leq k$. Each coordinate basis vector is the class of its same original monomial.*

Proof. Apply the previous genuine whole normal-monomial basis construction with tangent dimension zero and normal dimension d. Actual derivative coefficient separation proves independence and actual Hadamard remainder representatives prove surjectivity; neither is a premise here. Its actual base action is the original B_0 action. Use the preceding true evaluation isomorphism for each original coefficient. The actual tangent pullback of a zero-dimensional constant is the same original ambient constant, so these maps preserve the genuine complex action. Identify the same normal ideal with the genuine maximal ideal. The original basis-vector formula retains each whole monomial numerator and therefore gives the asserted genuine original complex coefficient dictionary. □

Lemma 278.3 (Actual bounded original exponent families have the binomial cardinality)**.** *The genuine distinct exponent family* $\{\alpha : |\alpha| \le k\}$ *has cardinality* $\binom{d+k}{d}$*, including* $d = 0$ *and* $k = 0$*.*

Proof. An exponent is its actual multiplicity multiset on $\operatorname{Fin} d$. Map that multiset into $\operatorname{Option}(\operatorname{Fin} d)$ using some and fill its missing $k - |\alpha|$ entries with none. This gives a multiset of exactly k entries on an alphabet of $d+1$ symbols. Conversely discard the none entries and retain the actual multiplicities of every some(i). The total retained cardinality is at most k, giving a valid original exponent. Counting each original symbol proves both inverse identities, including the slack symbol count $k - |\alpha|$. This is a true finite equivalence with $\operatorname{Sym}(\operatorname{Option}(\operatorname{Fin} d), k)$. The genuine symmetric-power cardinality is $\binom{d+k}{k}$, equal to $\binom{d+k}{d}$ by binomial symmetry. This proves the formula without counting repeated normal words as distinct monomials. □

Theorem 278.4 (The actual original base-jet dimension is the tangent binomial weight)**.** *For every* d, k*, the actual original complex dimension is* $\dim_{\mathbb{C}}(B_d/\mathfrak{m}_d^{k+1}) = \binom{d+k}{d}$*.*

Proof. The original complex coefficient isomorphism identifies the genuine jet quotient with a finite family of complex coefficients, one for each actual distinct exponent. That complex vector space has dimension the cardinality of its genuine coordinate index. Apply the preceding exact binomial cardinality and invariance of dimension under the constructed true complex-linear isomorphism. All finiteness and index witnesses were produced internally from the actual original monomials. □

279 Actual power lifts give the entire compatible normal filtration

For a commutative algebra $B \to R$ and actual original ideals I, J, write $L_q = \operatorname{im}(J^q \to R/(I + J^{q+1}))$ and $D = \bigoplus_{q \ge 0} L_q$. Suppose these genuine original layers are free over B. Use their internally constructed power lifts $\lambda_q : L_q \to J^q$. Put

$$\Lambda(v) = \sum_q \lambda_q(v_q), \quad T_s = R/(I + J^s), \quad \Gamma_s(v) = [\Lambda(v)].$$

Every sum has finite support. The original coherent-ideal consumer below obtains these free layers internally on the same actual nearby open.

Definition 279.1 (The true whole representative and every normal truncation map)**.** The preceding original power representatives define a genuine B-linear whole map $\Lambda : D \to R$ and, for every s, a genuine B-linear map $\Gamma_s : D \to T_s$. The degree-q image is the same original numerator $\lambda_q(x)$ followed by the actual quotient map.

Proof. The qth whole power lift followed by its original power-submodule inclusion is B-linear. The finite-support direct-sum universal property therefore constructs one whole linear map whose restriction in every degree is this composite. The actual whole quotient map to T_s is B-linear. Composing it with the same whole map gives Γ_s and the asserted original numerator formula in every degree. □

Lemma 279.2 (Every actual power representative has a preimage in the correct tail)**.** *For every r, s and every actual original $g \in J^r$, its class in T_s is $\Gamma_s(v)$ for some $v \in D$ with $v_q = 0$ for all $q < r$. In particular every Γ_s is surjective.*

Proof. If $r \geq s$, then $J^r \subseteq J^s$, so the actual class of g is zero and $v = 0$ works. For $r < s$, put $x_r = p_r(g)$ in the true original layer. The lift identity gives $g - \lambda_r(x_r) \in I + J^{r+1}$. Write this difference as $i_r + g_{r+1}$ with $i_r \in I$ and $g_{r+1} \in J^{r+1}$. Thus its actual class in T_s is the sum of the class of $\lambda_r(x_r)$ and the class of g_{r+1}. Repeat the construction at $r + 1$; induction on $s - r$ terminates at degree s, whose whole power is killed in T_s. The resulting finite sum of degree inclusions has all degrees below r zero and maps to the original class of g. For unrestricted surjectivity take $r = 0$, when the actual original power is all of R, and use surjectivity of the original quotient map. □

Theorem 279.3 (The full kernel is exactly the original high-degree tail)**.** *For every s,*

$$\Gamma_s(v) = 0 \quad \Longleftrightarrow \quad v_q = 0 \text{ for every } q < s.$$

Moreover $\Lambda(v) \in J^r$ whenever $v_q = 0$ for every $q < r$.

Proof. In the last assertion each nonzero summand of the finite representative sum has degree at least r and lies in its actual power $J^q \subseteq J^r$. The zero lower-degree summands also lie in J^r, so closure under finite sums proves the assertion. If every degree below s vanishes, apply it with $r = s$ to prove $\Gamma_s(v) = 0$.

Conversely, suppose $\Lambda(v) \in I + J^s$. Prove $v_q = 0$ for $q < s$ by induction on q. Assume every lower degree is zero and subtract the degree-q inclusion $\iota_q(v_q)$ from v. This remaining vector has every degree below $q + 1$ zero, so its true representative lies in J^{q+1}. Since $q < s$, $J^s \subseteq J^{q+1}$. Linearity therefore gives $\lambda_q(v_q) \in I + J^{q+1}$. Its true layer class is zero, while the power-lift identity identifies that same class with v_q. Hence $v_q = 0$, completing the induction and the full kernel identity. □

Theorem 279.4 (The actual maps preserve the entire normal filtration and every truncation comparison)**.** *Let $D^{\geq r}$ be the original direct-sum tail and let $F_r(s)$ be the actual image of J^r in T_s. Then $\Gamma_s(D^{\geq r}) = F_r(s)$ for every r, s. For $s \leq t$, the true original quotient map $T_t \to T_s$ composed with Γ_t is Γ_s. In particular $D/D^{\geq s} \simeq_B T_s$, retaining the whole original normal filtration.*

Proof. The preceding whole representative containment gives one inclusion in the image identity. The actual power-tail preimage theorem gives the reverse inclusion for every actual representative of $F_r(s)$. The comparison quotient sends the class of every original numerator to its same class. Thus it sends each degree image of Γ_t to the same degree image of Γ_s. Equality on every direct-sum degree proves equality of the whole maps. Their previously proved surjectivity and full kernel identity give the quotient isomorphism. The image identity proves that this same isomorphism preserves all original normal filtration levels. □

Theorem 279.5 (Only original coherence gives the entire actual compatible filtration on one nearby open)**.** *Original coherence of the ideal on $\mathbb{C}^{d+t}$ gives one genuine base open meeting every origin neighborhood, the original uniform degree ranks, and, at every actual plane point in that open, all the preceding compatible original truncation maps, whole kernels and filtration identities. No power lift, filtration splitting or desired truncation dictionary is a caller premise.*

Proof. The original-only all-power-lift producer supplies one true open and all actual original degree modules with their finite freeness and uniform ranks. Apply the preceding constructions at each original plane point to its same whole translated original ideal and actual normal-coordinate ideal. Their proofs apply simultaneously to all orders using the same internally selected degree lifts. Retain the previous original open and rank data. This gives every stated map and identity from the original coherent ideal alone. □

280 Original normal layers lift to genuine whole power representatives

The weighted ambient quotient requires compatible finite normal-filtration coordinates. The first construction lifts each free original graded layer back into its actual whole normal power. No splitting or basis is supplied to the original coherent-ideal consumer.

Lemma 280.1 (A free original layer has a true linear power lift)**.** *Let $B \to R$ be a commutative algebra, I, J actual whole ideals of R, and $L_q = \mathrm{im}(J^q \to R/(I + J^{q+1}))$ with its actual B action. If L_q is free over B, there is a genuine B-linear map $\lambda_q : L_q \to J^q$ for which the actual original power-layer map p_q satisfies $p_q\lambda_q = \mathrm{id}_{L_q}$.*

Proof. Choose a basis of the free module L_q internally. The actual map $p_q : J^q \to L_q$ is surjective by the definition of its image. For each chosen basis vector, choose a genuine element of J^q mapping to that same vector. Extend these actual representatives B-linearly using the basis universal property. The composite with p_q is the identity on every chosen basis vector. Two linear maps agreeing on a basis agree on every vector, so the composite is the full identity. This proves the assertion for the actual power module and actual quotient map, including degree zero and the zero module. □

Lemma 280.2 (Every lifted whole numerator has the same actual original class)**.** *For every $x \in L_q$, the actual whole numerator $g = (\lambda_q x) \in J^q$ has exactly the original class x in $R/(I + J^{q+1})$.*

Proof. Evaluate $p_q\lambda_q = \mathrm{id}$ at x. By the definition of the actual original power-layer map, $p_q(g)$ is precisely the class of the whole numerator g in the same original quotient. Forgetting the image-submodule membership from this equality gives the asserted original quotient-class identity. □

Theorem 280.3 (Only original coherence produces all actual nearby power lifts on one open)**.** *For an original coherent ideal sheaf on $\mathbb{C}^{d+t}$, original coherence supplies one genuine base open U meeting every neighborhood of the origin and one rank sequence $r(q)$ such that, at every original plane point $a \in U$ and every q, its true translated original layer is finite free of rank $r(q)$ and has a genuine linear lift to the whole original normal power. The same U works for all degrees.*

Proof. Apply the original-only all-degree uniform-rank producer to this same coherent ideal. It internally supplies U, its origin-neighborhood intersection property, and every actual original layer's finite freeness and rank, all on that same open. At each actual plane point and each degree, apply the first lemma to its same whole translated original ideal, same original normal-coordinate ideal, and actual tangent-base action. This produces the true power lift and its full right-inverse identity. Retain all previously supplied ranks, finite freeness and open properties. No further neighborhood, degree-dependent denominator, free-module witness, basis or desired power lift is an input to this original coherent-ideal theorem. □

281 The genuine mixed ambient quotient has all weighted original layer coordinates

Let $B \to R$ be the actual whole commutative algebra, I, J genuine original ideals and $\mathfrak{m}$ an actual base ideal. On the original coherent-ideal open the actual layers $L_j = \mathrm{im}(J^j \to R/(I + J^{j+1}))$ are free. Use the preceding single whole representative map Λ and $\Gamma_s : D = \bigoplus_j L_j \to T_s = R/(I+J^s)$. Put $M = \mathfrak{m}R$ and retain the literal mixed ideal $A_s = \sum_{q=0}^{s} M^{s-q}J^q$. Every action below is the original B action, including its restriction to the original complex scalars.

Lemma 281.1 (The actual mixed ideal image has precisely the weighted original power images)**.** *The genuine image of A_s in T_s, with its original B action, is $K_s = \sum_{q=0}^{s} \mathfrak{m}^{s-q}F_q(s)$, where $F_q(s)$ is the actual image of J^q in the same normal truncation.*

Proof. The algebra map sends ideal powers to powers of mapped ideals. Thus M^{s-q} is the image ideal of $\mathfrak{m}^{s-q}$. The product $M^{s-q}J^q$, restricted to B, is precisely the submodule generated by actual products bg with $b \in \mathfrak{m}^{s-q}$ and $g \in J^q$: restriction of the actual extension-ideal action equals the original base-ideal action. The true B-linear quotient map carries an ideal-generated submodule to the same ideal acting on its true image. Apply this identity to each original product. Both restriction of scalars and submodule image preserve finite joins, giving the stated equality for their whole sum. Surjectivity of the actual quotient map identifies this submodule image with the genuine image ideal, rather than only its generators. □

Lemma 281.2 (The whole weighted sum of true tails is exactly the coordinate-weight submodule)**.** *Let $D^{\geq q}$ be the original whole direct-sum tail. Then*

$$W_s := \sum_{q=0}^{s} \mathfrak{m}^{s-q} D^{\geq q} = \{v \in D : \ v_j \in \mathfrak{m}^{s-j} L_j \text{ for every } j\}.$$

For $j \geq s$ the exponent is zero and imposes no restriction.

Proof. Project each true tail to degree j. If $j < q$ its image is zero; if $j \geq q$ its image lies in all of L_j. Since $\mathfrak{m}^{s-q} \subseteq \mathfrak{m}^{s-j}$ for $j \geq q$, every projected summand has the displayed weight. Projection is B-linear, so the whole finite sum does too. Conversely write v as the finite sum of its actual degree inclusions. If $j < s$, the given coefficient membership and linearity of the degree inclusion place this summand in $\mathfrak{m}^{s-j} D^{\geq j}$. If $j \geq s$, this summand lies in $D^{\geq s}$, the $q = s$ summand with unit weight. Each term therefore lies in the true finite weighted tail sum. Closure under addition proves the reverse inclusion. □

Theorem 281.3 (The genuine weighted ideal has exactly the weighted coordinate preimage)**.** *For the true whole map, $\Gamma_s(W_s) = K_s$ and $\Gamma_s^{-1}(K_s) = W_s$. Hence the original whole map to the weighted truncation T_s/K_s is surjective with complete kernel W_s.*

Proof. The genuine whole map sends each true tail onto the actual original power image $F_q(s)$. Linear images commute with the original base ideal action and with finite sums. The preceding mixed-image identity therefore gives $\Gamma_s(W_s) = K_s$. For any linear map the preimage of the image of a submodule is its sum with the kernel. Here the previous full kernel is $D^{\geq s}$, which already lies in W_s by its unit-weight final summand. Thus the preimage is exactly W_s. The true quotient map to T_s/K_s is surjective, as is Γ_s; its kernel is the asserted preimage. This proves the complete statement. □

Theorem 281.4 (The original whole weighted quotient retains every actual numerator)**.** *The actual original map induces a true B-linear isomorphism $D/W_s \simeq_B T_s/K_s$ sending the class of every v to the class of the same original whole numerator $\Lambda(v)$. This same quotient is also genuinely B-linearly isomorphic to $R/(I + A_s)$, keeping every original numerator class.*

Proof. Apply the linear first isomorphism theorem to the genuine surjective whole map whose entire kernel was just computed. Identify that kernel with the literal coordinate-weight submodule W_s. On every original quotient representative the first-isomorphism map is the same whole map, so the image is exactly the class of $\Lambda(v)$. This fixes the original actions and the whole representative dictionary. The genuine image submodule is the actual quotient image ideal restricted to B. Identify the underlying additive quotients by the preceding whole image identity. For each original quotient representative, the genuine algebra quotient map is B-linear, so this same additive identification preserves the actual base action; this proves linearity without silently identifying potentially different instance paths. The true degree-s mixed term gives $J^s \subseteq A_s$, hence $(I + J^s) + A_s = I + A_s$. The actual double-ideal quotient isomorphism therefore identifies this same whole weighted quotient with $R/(I + A_s)$ and fixes every original numerator. □

Theorem 281.5 (The same quotient is the finite sum of the actual weighted original layers)**.** *The degree quotient maps induce a genuine B-linear isomorphism*

$$D/W_s \simeq_B \bigoplus_{0\le j<s} L_j/(\mathfrak{m}^{s-j}L_j).$$

Proof. Map each actual original degree below s by its true whole quotient map and place it in the corresponding finite direct-sum degree; map every higher degree to zero. This gives a B-linear map from all of D by its finite-support universal property. Every target has finitely many coordinates. For each of them choose an original representative using surjectivity of its quotient map and sum those actual degree inclusions. This proves surjectivity of the whole map. Its kernel consists precisely of vectors with each lower coefficient in the actual weighted submodule. For higher degrees the weight is $\mathfrak{m}^0 = B$, so the same condition is automatic. The preceding coordinate dictionary therefore makes the full kernel exactly W_s. Apply the same true linear first isomorphism theorem. □

Theorem 281.6 (Only original coherence produces the actual ambient weighted decomposition on one open)**.** *For an original coherent ideal on $\mathbb{C}^{d+t}$, one genuine nearby base open and the original degree-rank sequence work for all actual plane points and all s. The true ambient quotient $R/(I+\mathfrak{m}_R^s)$ at each point is genuinely complex-linearly isomorphic to the finite sum of its actual original weighted layers $\bigoplus_{j<s} L_j/(\mathfrak{m}_B^{s-j}L_j)$.*

Proof. Original coherence supplies all original layer freeness on one genuine open, retaining all original finite ranks. Apply the preceding actual constructions to the same whole translated original ideal, same normal coordinate ideal and actual base maximal ideal, for all s simultaneously. The actual original relative Hadamard identity gives $\mathfrak{m}_R = \mathfrak{m}_B R + J$, so its genuine mixed-power quotient is the existing full ambient quotient. Compose that true quotient isomorphism with the two preceding whole weighted isomorphisms. Their B actions are the actual tangent-base actions; restriction along the original complex algebra map therefore gives genuine complex linear maps and inverse maps. No transport of invented actions, caller splitting, desired dimension formula or quotient dictionary is a premise. Retain the same original open and rank data. □

This chapter constructs actual weighted coordinates. Evaluation of the finite free layer-quotient dimensions and the genuine base-jet binomial count, then ambient Hilbert–Samuel constancy and true blowup descent, remain subsequent mathematical producers.

282 The genuine nearby original deformation and entire normal-cone dictionary

All maps below act on actual holomorphic FUNCTIONS and their actual pullback sheaves. No map from arbitrary origin-germ classes to nearby germs is invented. The genuine original coherent ideal is the only ideal input of the final producer. Every selected exponent/open/family/initial ideal is internal.

Lemma 282.1 (Actual affine inverse gives whole ideal and holomorphic membership dictionaries)**.** *The genuine function maps $z \mapsto a+z$ and $z \mapsto -a+z$ are entire holomorphic inverse maps at every point. Actual whole coherent-ideal pullback membership is exactly original whole membership of the actual function composed with the true inverse, at the true image point. Holomorphic function germs satisfy the same equivalence.*

Proof. Addition and cancellation give both actual inverse identities globally; the true smooth translation maps give every local inverse defining field. Apply the genuine local-biholomorphism membership theorem to the original coherent ideal. The holomorphicity implication uses the true continuous image-neighbourhood map and complex chain rule. Composing with the actual inverse

proves the converse, since the two true functions compose to the literal original function. These dictionaries concern actual functions at actual points; there is no nearby image of an arbitrary fixed origin germ. □

Lemma 282.2 (ONE actual tangent-base shift preserves every original deformation equation)**.** *Define the actual enlarged translation by the central inclusion of the original tangent inclusion of a. Original projection commutes with this shift and the true original ambient translation. The actual deformation parameter and every fiber coordinate remain unchanged. Every original equation $uv_i - y_i$ is unchanged. The actual central inclusion commutes with the SAME actual enlarged shift and the true ambient translation.*

Proof. Each original linear coordinate applied to a sum is its actual sum. The true central inclusion's original tangent coordinates equal the original tangent coordinates, its original normal coordinates and parameter are zero, and its added fiber coordinates equal the original normal coordinates. At the tangent inclusion of a every one of those original normal coordinates is zero. Hence the enlarged translation changes only the original tangent base, leaves the parameter, every original normal and every added fiber coordinate unchanged, and retains all literal original equations. Original projection therefore gives exactly the true ambient translation. Linearity of central inclusion gives the central square on the whole function domains. Every coordinate identity remains literal if either dimension is zero. □

Lemma 282.3 (The entire genuine ORIGINAL equation ideal commutes with the actual base shift)**.** *For the real translated original coherent ideal, its ENTIRE genuine deformation equation ideal equals the actual pullback of the original entire deformation ideal under this SAME enlarged base shift.*

Proof. The original finite entire equation functions are genuinely holomorphic everywhere and give exactly their defining entire functional span. The true finite-generator pullback theorem says that their actual pullback is generated by their actual compositions with the enlarged shift. The preceding coordinate identities make each composed equation the literal same original equation, hence identify their entire equation ideal. Genuine coherent whole pullback composition identifies the projection pullback of the true translated original ideal with the enlarged-shift pullback of the original projection ideal, by the actual projection square. Whole finite coherent ideal-sum pullback distributes over these two actual factors. Their sum is precisely the original deformation definition on each side. No equation model, source ideal equality or finite presentation certificate is input. □

Lemma 282.4 (Every actual parameter colon and entire central restriction commute on the SAME shift)**.** *For EVERY natural q, the entire actual qth parameter-colon ideal of the true translated original ideal is the SAME enlarged-shift pullback of the original qth colon. Its actual central pullback is the SAME true ambient translation pullback of the original central ideal at index q.*

Proof. For a coherent ideal and the actual globally invertible shift, the first lemma gives whole membership in the pulled ideal by composition with the genuine inverse. Induct on the literal recursive parameter-colon definition. At zero the dictionary is exactly the third lemma. At a successor its two defining clauses are holomorphicity of the actual test function and membership of its product with the original parameter in the preceding colon. The first lemma identifies holomorphicity, the inductive whole membership dictionary identifies the product membership, and the second lemma leaves the actual parameter unchanged under both the true shift and inverse. Thus both full predicates agree at every point for every function. Each actual colon is internally coherent by the original Oka producer, so the true pullback membership theorem applies throughout. Compose this whole equality with the actual central inclusion. Genuine cross-dimension coherent pullback composition and the exact central square give the asserted whole central-ideal equality. □

Lemma 282.5 (Two genuine internally stabilized indices give the SAME entire origin central ideal)**.** *If the actual original colon chain stabilizes at two true indices at the actual origin point, their ENTIRE central origin ideals are equal; no global equality of the original sheaves or chosen common index is assumed.*

Proof. Order the two natural indices. The smaller stabilization identifies its entire functional colon predicate at the origin with the larger one by the actual nonnegative difference. Pullback membership at the origin of the genuine central inclusion uses only this entire original membership predicate at its actual image zero. Hence the two central pullbacks have exactly the same membership for EVERY actual function at origin. The origin ideals are the spans of their true whole holomorphic member germs, so equality follows by equality of these entire generating sets. This is an equality of complete ideals, not finite jets, radical/support equality or a generator certificate. □

Theorem 282.6 (Only original coherent data identify the ENTIRE nearby original normal cone with the SAME true coefficient-germ model)**.** *Only original coherent I internally supplies SAME N and a genuine base open B containing zero, and the SAME finite original homogeneous relation family with its actual nearby entire coefficient germs. At EVERY $a \in B$, the internally formed homogeneous polynomial ideal is exactly the ENTIRE genuine original normal initial ideal of the actual translated original coherent ideal. Its analytic image is the entire central origin ideal of the SAME selected original deformation model translated by the true ambient map. Every actual original colon stabilization dictionary remains on SAME B. The ENTIRE literal original associated-graded module is genuinely linearly equivalent to the quotient by this SAME nearby homogeneous polynomial ideal; no degree-wise or truncation freeness is inferred.*

Proof. The preceding true coefficient-germ producer supplies SAME N, B, whole translated central ideal and its entire polynomial inverse-image kernel. By the fourth lemma, this is the actual original translated ideal's central model at index N. The original all-colon stabilization dictionary at the true enlarged central image of a and the actual inverse membership dictionary give entire stabilization at the origin for the translated original ideal at SAME N. Apply the previously established ORIGINAL-input whole origin normal-cone theorem to this genuine translated coherent ideal. It internally selects another exponent and gives its entire origin central ideal as the analytic image of its genuine whole normal initial ideal. Its own actual stabilization and the fifth lemma identify that full central ideal with the SAME index-N full central ideal. Both the internally formed nearby coefficient ideal and the true original initial ideal are homogeneous, and the genuine whole homogeneous analytic contraction theorem identifies their entire polynomial inverse-images. Their whole ideals are therefore equal. The true entire original associated-graded quotient equivalence, composed with quotient transport through this literal whole ideal equality, gives the entire original associated-graded module as the SAME polynomial quotient. This uses the actual original symbol map and retains every actual whole quotient class. Every object and every comparison is produced internally from only original coherent data. A single common ALL-order finite-free/normal-flat open remains a subsequent algebraic/analytic producer; no such result is assumed or claimed here. □

283 Actual original finite normal relations and leading lifts have ONE genuine analytic coefficient neighborhood

Let $B = \mathcal{O}_{\mathbb{C}^d,0}$, $R = \mathcal{O}_{\mathbb{C}^{d+t},0}$, J the actual normal ideal, and $I \subset R$ an arbitrary ORIGINAL whole ideal. All finite relations, leading lifts, multipliers and the denominator below are internally produced from this actual I. An entire function representative ring is used only to express finite genuine identities; its functions are proved holomorphic on one actual open neighborhood. There is no canonical homomorphism from all germs at zero to all germs at an arbitrary nearby point.

The finite model produced here is not yet identified with the WHOLE original normal cone at nearby stalks.

Lemma 283.1 (Internally chosen whole germ representatives share one actual holomorphic open)**.** *Every actual entire $h \in B$ internally has a function representative $r_h : \mathbb{C}^d \to \mathbb{C}$ holomorphic near zero whose WHOLE germ is h. Choose $r_0 = 0$ literally. Every finite actual family of these coefficients has ONE genuine open neighborhood U of zero on which ALL its chosen representatives are holomorphic at EVERY point.*

Proof. Membership in the actual holomorphic-germ subring is precisely existence of an actual function differentiable throughout some neighborhood of zero, with that ENTIRE germ equal to the given element. Internally choose such an actual representative for each nonzero germ, and choose the literal zero function for zero. The subtype equality proves its whole germ identity, not merely an equality of values at the centre. For a finite family, take the finite intersection of their genuine neighborhood conditions. The resulting condition belongs to the neighborhood filter; choose an actual open neighborhood of zero contained in it. Every chosen coefficient is then genuinely differentiable, hence holomorphic, at EVERY point of this same open. The empty family permits the whole space and the zero-dimensional base uses the same neighborhood argument. No caller representative, open set or point-value replacement is supplied. □

Lemma 283.2 (Finite-support polynomial representatives retain their entire original polynomial germs and degrees)**.** *For each actual $P \in B[z_1, \ldots, z_t]$ form the genuine polynomial $\widetilde{P}$ with coefficient functions r_{P_α} on its ACTUAL finite support. Its coefficient at EVERY exponent is exactly r_{P_α}, including the literal zero coefficients outside that support. The genuine function-to-germ ring homomorphism maps this WHOLE polynomial to the original P under the injective holomorphic-germ subtype map. The actual specialization at any point has support contained in the SAME finite support; every actual homogeneous degree of P is retained.*

Proof. Define $\widetilde{P} = \sum_{\alpha \in \operatorname{supp} P} z^\alpha r_{P_\alpha}$ in the TRUE polynomial ring over the ring of actual functions. Coefficient extraction gives the stated function at an exponent in the actual finite support. Outside it the sum coefficient is zero, and the original coefficient and its chosen representative are literally zero, giving the SAME dictionary at every exponent. Apply the genuine function-to-germ ring map coefficientwise. Each coefficient then equals the corresponding entire original holomorphic germ by the first lemma; polynomial extensionality gives the complete polynomial identity. Specialization is the genuine evaluation ring map on this actual function ring, not an invented map on all zero-centred germs. Outside the SAME support it remains zero, so actual homogeneous original support stays in its true original total degree at every point, even when some coefficients vanish. A common coefficient open follows by applying the first lemma to the finite union of the actual polynomial supports. Empty supports and zero polynomials retain all these exact dictionaries. □

Lemma 283.3 (Whole finite polynomial germ identities spread to ONE common genuine open)**.** *For any ACTUAL finite polynomial family A_i, P_i, the chosen representative of the complete original polynomial $\sum_i A_i P_i$ agrees throughout ONE actual open neighborhood of zero with the sum of products of the chosen representatives. This is a WHOLE polynomial functional identity.*

Proof. Subtract these two actual sides in the genuine polynomial ring of coefficient functions. The actual function-to-germ ring map takes this difference to zero, since the complete original polynomial sum-product identity holds and the preceding full coefficient dictionary is exact. Each coefficient in its actual finite support therefore has zero ENTIRE germ. Equality to the zero germ means genuine equality of the function to zero on some neighborhood of zero. Intersect these actual neighborhoods for every coefficient of this actual difference polynomial. Choose an open neighborhood of zero contained in that finite intersection. Each entire difference polynomial specializes to zero at EVERY point there, by its actual finite coefficient support and polynomial

extensionality. Thus this original whole finite sum-product identity holds on ONE common actual open. For the internally produced FINITE family of original lift identities below, intersect these genuine opens and the actual coefficient holomorphic open; this retains every whole identity and holomorphicity on the SAME open. No compactness, finite-order truncation or caller functional identity is used. □

Theorem 283.4 (The original whole ideal internally supplies its actual finite homogeneous relations and leading-lift multipliers)**.** *The actual whole original initial ideal $H(I)$ internally has a finite subset S of its ACTUAL homogeneous normal relations with $H(I) = (S)$. The actual internally selected finite leading-lift family F consists of nonzero whole members of $H(I)$. For EVERY $g \in F$, internally produce actual polynomial multipliers $A_{g,s} \in B[z]$ satisfying the COMPLETE original polynomial identity $g = \sum_{s \in S} A_{g,s} s$. Internally obtain ONE genuine common open containing zero on which ALL chosen coefficients of S, F, A are holomorphic and ALL these WHOLE functional identities hold.*

Proof. The actual original normal associated-graded producer identifies the ENTIRE initial ideal with the span of the actual original homogeneous relations. True Noetherianity internally extracts a finite subset S of this SAME relation family generating the whole ideal, retaining each actual original homogeneous degree and its underlying original relation. The actual field leading-generator construction and genuine denominator clearing internally give the finite family F of nonzero original polynomials belonging to this SAME $H(I)$. Substitute $H(I) = (S)$ in each actual membership. The genuine finite-span membership characterization then yields finitely many actual polynomial multipliers, one per true finite generator, whose sum is the ENTIRE original member. Choose these actual witnesses internally. There are only finitely many g and s; all multipliers therefore have a finite total coefficient family. Apply the actual whole coefficient-representative producer to the finite union of their supports and all S, F supports. The third lemma spreads each TRUE internally supplied sum-product identity to an actual open. Finiteness of the original lift family permits their finite intersection with the coefficient holomorphic open, producing ONE common actual open containing zero that retains ALL coefficient and whole functional dictionaries. The empty leading family and empty relation family use genuine empty sums. No caller finite relation, leading-lift, Groebner or multiplier certificate is passed to this ORIGINAL ideal consumer. □

Theorem 283.5 (The SAME genuine leading-coefficient product has an actual nonvanishing open with exact leading exponents)**.** *The actual product $\beta = \prod_{g \in F} r_{\mathrm{LC}(g)}$ is holomorphic on ONE actual coefficient open $U \ni 0$, and its ENTIRE germ is the SAME actual nonzero uniform denominator $b(I)$, independent of normal order. The true set $U_\beta = \{y \in U : \beta(y) \neq 0\}$ is open and meets EVERY neighborhood of zero. At EVERY $y \in U_\beta$ ALL actual specialized $\widetilde{g}(y)$ have their SAME original leading exponents.*

Proof. Every original lift is nonzero, so its actual leading coefficient is a nonzero entire base germ. The actual uniform denominator is literally the product of these SAME coefficients, and true domain arithmetic makes it nonzero. The genuine whole representative product has that exact ENTIRE germ because function-to-germ is a ring homomorphism. The first lemma supplies one actual open on which all factors are holomorphic; the finite product is holomorphic there. Continuity at every actual point makes its nonvanishing set open relative to this actual open, hence actually open. If any neighborhood of zero missed U_β, the function β would vanish throughout its intersection with U, a genuine neighborhood of zero. Its ENTIRE germ would then be zero, contradicting the actual nonzero $b(I)$. Hence this real open meets EVERY origin neighborhood, including zero-dimensional and empty-product cases. At a point of this SAME open each genuine product factor is nonzero. The preceding polynomial dictionary gives that the original leading coefficient remains nonzero and every coefficient beyond the original leading exponent remains

literally zero. The actual monomial-order maximum of the finite support is consequently unchanged. This is an exact finite-polynomial leading dictionary, not a nearby whole normal-cone or normal-flatness conclusion. □

Theorem 283.6 (ONLY the original whole ideal produces the complete common analytic finite model)**.** *For the ORIGINAL arbitrary I, internally produce the actual S, F, A, ONE actual open $U \ni 0$, their whole holomorphic coefficient and polynomial germ dictionaries, true homogeneous relation degrees, and the SAME real nonzero denominator representative β. Throughout this SAME U, ALL complete identities $\widetilde{g}(y) = \sum_{s\in S} \widetilde{A}_{g,s}(y)\widetilde{s}(y)$ hold. Its SAME nonvanishing open meets EVERY origin neighborhood and preserves ALL true leading exponents.*

Proof. Apply the fourth theorem to the actual original I to obtain the TRUE internally generated homogeneous relations, leading lifts and multipliers. Take the finite union of ALL their genuine coefficient supports, including all actual leading coefficients. Apply the first two lemmas to construct actual whole representatives and a common actual holomorphic open. Apply the third lemma to the FINITE internally supplied complete membership identities. Shrink the SAME actual open to retain EVERY functional identity and EVERY coefficient holomorphic condition. Form the actual product β of the SAME real leading-coefficient representatives. The fifth theorem applies on this very open: its ENTIRE germ remains the SAME actual nonzero denominator, so its nonvanishing open still meets EVERY origin neighborhood and ALL actual leading exponents remain exact. Retain all original ideal, whole germ, degree, product and finite identity dictionaries. Every construction consumes only the actual original whole ideal. No nearby original normal-cone equality is asserted from this finite model. □

Remark 283.7. Actual coherent relative Rees/initial-ideal identification of this SAME finite analytic model with the WHOLE original normal cone at nearby stalks remains OPEN. Until that genuine producer exists, analytic normal-flat loci, nearby ALL-order finite freeness, higher HS and general G8 remain OPEN.

284 The actual analytic graph quotient and its saturated central model

Let $\mathfrak{a}$ be the original coherent ideal on $\mathbf{C}^{d+t}$. The only original inputs are $d, t, \mathfrak{a}$ and its genuine coherence. Put $D = d+t+1+t$, $G = d+1+t$, with coordinates (x, y, u, v) and (x, u, v). All rings below are the actual WHOLE convergent holomorphic germ rings at zero. This chapter proves a true graph quotient and the actual saturated central fiber dictionary. It does not identify it with the nearby ORIGINAL normal cone.

Theorem 284.1 (Actual graph maps and a whole split surjection)**.** *Let $\rho(x, y, u, v) = (x, u, v)$ and $\iota(x, u, v) = (x, uv, u, v)$. Both maps are genuinely holomorphic, preserve zero and satisfy $\rho\iota = 1$. The actual whole pullback $T = \iota^* : A \to C$ is surjective, with genuine right inverse $\rho^* : C \to A$. Their coordinate dictionaries include zero tangent/normal dimensions.*

Proof. Construct ρ as the actual continuous linear coordinate projection. Construct ι coordinate by coordinate: the original tangent coordinates are x, original normal coordinates are the actual products uv_i, and parameter and fiber coordinates are the literal u, v. Finite coordinate complex differentiability proves whole holomorphy, and each coordinate vanishes at zero. Direct substitution in every block proves $\rho\iota = 1$. On every actual germ represented by a holomorphic function, composition of these genuine maps gives the same WHOLE representative. Thus $T\rho^* = 1$ as actual algebra maps; every entire germ in C has the actual preimage $\rho^* c$, proving surjectivity internally. □

Theorem 284.2 (Whole analytic Hadamard gives the exact actual graph kernel)**.** *Let E be the actual entire origin ideal of the genuine equation sheaf generated by $uv_i - y_i$. For EVERY whole ambient germ $F \in A$, $(\iota\rho)^* F - F \in E$. Consequently* $\ker T = E$.

Proof. Use actual doubled coordinates (z, δ) in $\mathbf{C}^{D+D}$. The whole germ $F(z+\delta) - F(z)$ has actual zero plane trace. The genuine whole Hadamard decomposition writes it as the finite sum $\sum_j \delta_j H_j(z, \delta)$ with entire holomorphic coefficients H_j. Pull this WHOLE identity back by the actual holomorphic map $z \mapsto (z, \iota\rho(z) - z)$. Its tangent, parameter and fiber differences are literally zero; its original normal differences are exactly $uv_i - y_i$. Hence EVERY ambient entire difference is in the genuine equation ideal. Each actual equation vanishes under ι, so $E \subseteq \ker T$. If $TF = 0$, actual retraction gives $(\iota\rho)^* F = \rho^*(TF) = 0$; the whole difference identity then puts $-F$, and hence F, in E. The finite functional span/origin-germ dictionary identifies the entire true equation ideal with the ring span of these actual equation germs. No polynomial-only argument or caller kernel certificate is used. □

Theorem 284.3 (Actual coherent origin ideals commute with genuine pullback and sum)**.** *For a genuine zero-preserving continuous linear map η and an actual coherent ideal $\mathfrak{b}$, the WHOLE origin ideal of its actual sheaf pullback equals the genuine ring-ideal image under η^*. The WHOLE origin ideal of the actual sum of two coherent ideals is the sum of their whole origin ideals.*

Proof. One pullback inclusion follows by taking every genuine original stalk member to its actual composite. For the reverse, unfold actual sheaf pullback membership: it is a finite WHOLE holomorphic coefficient combination of actual functions $f_i \circ \eta$ with each f_i in the original stalk. Coherence supplies genuine holomorphy of every f_i. The full eventual functional identity passes to the entire germ identity, so every actual pullback member belongs to the ring-ideal image. For sums, actual sheaf membership is a finite functional span of the union of the two actual stalks. Coherence supplies every generator's whole holomorphy, and its entire germ lies in the corresponding original ideal. The same whole finite coefficient identity proves inclusion in the ring sum. Conversely the actual coherent sheaf sum contains both original stalks and is coherent; its whole origin ideal contains both entire ring ideals and therefore their sum. No flatness certificate is needed. □

Theorem 284.4 (The original enlarged equation quotient is the actual scaled graph quotient)**.** *Write I for the actual original whole origin ideal, K for the actual origin ideal of $p^*\mathfrak{a} + \mathfrak{e}$, and $L = (T \circ p^*)I$ for the genuine scaled ideal in C. Then $TK = L$, $K = T^{-1}L$, and the actual quotient rings satisfy $A/K \simeq C/L$. The actual scaled functions are $f(x, uv)$ for every original WHOLE f, including arbitrary convergent germs.*

Proof. The preceding actual pullback/sum identities give $K = p^* I + E$. Taking the genuine ideal image under T kills $E = \ker T$ and gives precisely the ideal image under the actual composite $T \circ p^*$. The actual split surjectivity and $\ker T \subseteq K$ imply $T^{-1}(TK) = K$. Compose T with the actual quotient map $C \to C/L$; it is surjective with kernel K. The genuine first isomorphism theorem therefore gives the claimed WHOLE quotient ring equivalence. Its value on every quotient representative is the actual trace followed by the actual quotient map. No caller presentation or quotient certificate occurs. □

Theorem 284.5 (Literal parameter colons and the actual central fiber dictionary)**.** *Let a be the actual parameter germ in A, $\bar{a} = Ta$ its actual graph parameter, and $K_q = K : a^q$, $L_q = L : \bar{a}^q$ the literal entire ideal colons. For ALL q, $K_q = T^{-1}L_q$ and $TK_q = L_q$. Let $\gamma(x, v) = (x, 0, v)$ in the graph model. The whole origin ideal of the actual central sheaf pullback of the qth original colon is exactly $\gamma^* L_q$ as a genuine ideal image, with the full coordinate dictionary.*

Proof. For every whole ambient g, true colon membership is $ga^q \in K$. Using $K = T^{-1}L$ and multiplicativity of the ACTUAL trace, this is exactly $(Tg)\bar{a}^q \in L$, or $Tg \in L_q$. Thus $K_q = T^{-1}L_q$. Surjectivity of T proves that the entire image of this preimage is L_q. The actual linear central

graph map γ satisfies $\iota\gamma = j$, where j is the genuine previously constructed central inclusion. Whole representative composition therefore gives $j^* = \gamma^* T$. Actual coherence of the original qth colon and the genuine linear pullback/origin-ideal equality identify its central origin ideal with $j^* K_q$. Genuine ideal-image composition and the preceding full image identity now give exactly $\gamma^* L_q$. No nearby-original-cone identity is inferred from this actual central model dictionary. □

Theorem 284.6 (Only the original coherent input supplies the true saturated graph model)**.** *Only $d, t, \mathfrak{a}$ and actual coherence internally supply SAME N, U, the actual whole graph quotient and central Nth graph-colon image identities above, and every original enlarged-equation all-power saturation and whole-union dictionary at EVERY point of the SAME true open U. At zero the whole graph saturation union equals the SAME L_N.*

Proof. Internally select the genuine coherent original-equation saturation index and ONE open from its proven actual Noetherian/coherent spreading producer. Retain every all-point, all-power and entire functional saturation-union identity on that SAME open. At zero, representing each WHOLE ambient germ by its genuine holomorphic function identifies these actual functional colon dictionaries with the literal ring colons. Their full entire images under the internally surjective trace are the corresponding graph colons, so all graph colons beyond N and their entire increasing union equal L_N. The preceding actual quotient and central-image identities apply to this internally selected N without external index, generator, kernel, Rees, central-fiber or flatness certificates. Return these actual joint results. □

Open frontier. The actual graph-saturated central model is not yet identified with the WHOLE nearby ORIGINAL Rees algebra/associated graded cone. The reverse whole analytic initial-ideal dictionary, analytic polynomial-extension flatness and convergence/spreading identification require genuine proofs. Nearby ALL-degree freeness, analytic normal-flatness, higher HS and G8 remain OPEN. Origin whole ring equalities do not invent maps into arbitrary nearby stalks.

285 One actual open for nearby original normal-module freeness

Theorem 285.1 (The same genuine open makes every nearby entire original associated module free)**.** *An original coherent ideal I supplies an actual open set U meeting every neighbourhood of zero, such that at every $a \in U$ the entire associated graded module of the actual translated original ideal is free over the actual base holomorphic germ ring.*

Proof. Take the internally supplied common domain D, genuine base neighbourhood W and finite polynomial ideal H presenting every nearby entire initial ideal. The uniform standard-monomial construction gives $b \in D$, $b \neq 0$. Choose a representative c of b in the actual coefficient-function subalgebra. Its actual function g has nonzero whole origin germ: otherwise c would belong to the entire origin kernel and its quotient class b would be zero. Put $U = \{a \in W : g(a) \neq 0\}$. Holomorphicity on W makes U open. If an origin neighbourhood missed U, g would vanish on its intersection with W, contradicting its nonzero whole origin germ. At every $a \in U$, the actual nearby image of b is the translated whole germ of g, whose value at zero is $g(a) \neq 0$; it is therefore a unit. The uniform free coefficient-image theorem gives the entire nearby polynomial quotient free over the actual base germ ring. Its whole ideal is the entire original initial ideal, and the genuine original symbol quotient equivalence transfers freeness to the entire actual associated module. □

Lemma 285.2 (Every original normal layer is finite)**.** *For every original germ ideal and every natural q, its actual normal degree-q layer is finite over the base holomorphic germ ring.*

Proof. The degree-q homogeneous polynomial submodule is generated by the finitely many monomials in t variables of total degree q. The true homogeneous symbol map surjects from this finite module onto the entire actual original normal layer. Its image is consequently finite. □

Theorem 285.3 (Every original normal degree is finite free on that same open)**.** *The same original-input open U makes every actual normal degree-q layer finite free over the actual base germ ring at every $a \in U$, for all natural q simultaneously.*

Proof. Each degree layer is a direct summand of the entire associated graded direct sum, by its genuine degree inclusion and component projection. A free entire associated module therefore makes each layer projective, hence flat. The preceding finite-layer lemma and the local nature of the holomorphic germ ring make each finite flat layer free. The same U supports the entire free module, so it supports this argument for every q simultaneously; no degree-dependent shrinking is used. □

Theorem 285.4 (Every original normal truncation is finite free on the same open)**.** *The same original-input open U makes every actual original quotient by the sum of the original ideal and the qth normal-ideal power finite free over the actual base germ ring, for every point of U and every natural q.*

Proof. For each nearby original ideal, induct on q. The zero truncation is zero, since the zeroth normal-ideal power is the entire ring. The genuine original filtration gives a short exact sequence with degree-q layer on the left, the $(q+1)$st truncation in the middle, and the qth truncation on the right. Both end terms are finite free, by the preceding all-degree theorem and the inductive hypothesis. The right term is projective, so this short exact sequence splits; equivalently the free short-exact-sequence theorem gives a free middle term. The finite-generation exact-sequence theorem gives a finite middle term. The induction is carried out at every point of the same U and makes no further shrinking. □

286 A common coefficient algebra for the nearby original normal cones

The coefficient algebra here consists of actual functions holomorphic near every point of a fixed open base set. It is distinct from the ring of germs at the origin: passage to each nearby germ is defined on the actual functions. This supplies a common algebraic presentation of the previously identified nearby normal cones. All-degree generic freeness for this presentation is established in the subsequent base-change and nearby-freeness sections. This coordinate normal-module calculation is an additional commutative-algebra step in the resolution route of Włodarczyk's marked-ideal construction. The common function algebra, whole-kernel shrinking, and uniform polynomial denominator below are the explicit analytic adaptation used here; they do not assert the geometric singular-locus-centre or higher Hilbert–Samuel driver.

Definition 286.1 (The algebra of holomorphic coefficient functions)**.** For a base set $B \subset \mathbb{C}^d$, let A_B be the complex subalgebra of functions $\mathbb{C}^d \to \mathbb{C}$ that are holomorphic in a neighbourhood of each point of B. Its addition and multiplication are pointwise operations.

Proof. Constant functions are holomorphic. The sum, product and negative of functions holomorphic near a point are holomorphic near that point, by their complex derivative formulas on the intersection of the two neighbourhoods. These closures give the asserted complex subalgebra. □

Lemma 286.2 (Nearby germ maps on the common coefficient algebra)**.** *For every $a \in B$ there is a complex algebra homomorphism $\rho_a : A_B \to \mathcal{O}_{\mathbb{C}^d,0}$ sending an actual function f to the whole germ of $z \mapsto f(a+z)$.*

Proof. Holomorphicity of f near a and the complex chain rule make its translated function holomorphic near zero. Translation commutes pointwise with constants, sums and products, and taking a whole germ preserves these operations. Consequently this formula defines the asserted algebra homomorphism. □

Lemma 286.3 (Whole polynomial lifts over the actual coefficient algebra)**.** *Suppose the chosen representative of every coefficient of P is holomorphic near every point of B. The finite sum with those same coefficient functions defines $\widetilde{P} \in A_B[Y_1, \ldots, Y_t]$. At every exponent its coefficient is the actual chosen coefficient function of P. Thus its support is contained in that of P, and its homogeneous degree is retained. For every $a \in B$, mapping $\widetilde{P}$ by ρ_a gives the entire nearby coefficient polynomial P_a. If $0 \in B$, mapping by ρ_0 gives P.*

Proof. Use the sum over the finite support of P. Coefficient extraction gives the asserted coefficient at a supported exponent. Outside that support the original coefficient is zero and its chosen function is identically zero, so the same formula holds there. Every nonzero coefficient of the lift therefore has an original supported exponent, which proves preservation of homogeneous degree. Extract coefficients after applying each algebra homomorphism: they are exactly the translated whole germs used to define P_a. At zero the translation is the identity and the chosen function has its prescribed original whole germ. Polynomial equality follows at every exponent. □

Theorem 286.4 (One actual algebra presents every entire nearby original initial ideal)**.** *An original coherent ideal I internally supplies one open base neighbourhood B of zero and lifts over A_B of its selected finite original homogeneous relation family. If K_B is the ideal generated by those lifts, its image under the polynomial map induced by ρ_0 is the whole original normal initial ideal. At every $a \in B$, its image under the polynomial map induced by ρ_a is the whole normal initial ideal of the actual translated original ideal. The generators over A_B are homogeneous.*

Proof. The entire nearby original normal-cone theorem supplies a single open B on which every selected coefficient function is holomorphic near every base point and on which the nearby finite relation polynomials generate precisely the whole original initial ideals. Form their lifts by the preceding lemma. The image of an ideal generated by a family is the ideal generated by the images of that family. At each a those images are exactly the nearby polynomials, so the existing entire nearby initial-ideal equality gives the desired equality. At zero each image is its original relation polynomial, and the selected original finite relations generate the whole original initial ideal. Each selected relation has its original homogeneous degree; the preceding lemma retains that degree over A_B. □

Lemma 286.5 (Finite type coefficient algebra for finitely many polynomials)**.** *For finitely many polynomials over A_B, the complex subalgebra C generated by their finitely many supported coefficients is of finite type and Noetherian. Each polynomial lifts to $C[Y_1, \ldots, Y_t]$ with exactly its original image under coefficient inclusion, retaining homogeneous degree.*

Proof. The union, over the finite family, of the coefficient sets indexed by each finite support is finite. Its complex algebraic span is therefore a finite type algebra and a quotient of a polynomial algebra in finitely many variables over the Noetherian field $\mathbb{C}$, hence Noetherian. Every supported coefficient lies in this subalgebra by construction; unsupported coefficients are zero and also lie there. Form the same finite sum over each support with these coefficients in C. Inclusion recovers every original coefficient, and injectivity of inclusion shows that the lifted support is exactly the original support. Homogeneous degree is consequently unchanged. □

Theorem 286.6 (Original nearby initial ideals over one finite type coefficient algebra)**.** *Only the original coherent ideal I supplies an open B containing zero, a finite type complex subalgebra $C \subset A_B$, and a finite homogeneous relation family over C. Its entire ideal image under the actual origin germ map is the original normal initial ideal. Its image under the actual nearby germ map at every $a \in B$ is the entire normal initial ideal of the actual translated original ideal.*

Proof. The original-input common function-algebra theorem gives B and its finite homogeneous relations over A_B. Take C to be the complex subalgebra generated by their finite supported coefficient set. The finite type lemma lifts every relation to C, with its original image under inclusion and its original homogeneous degree. Composing the origin and nearby maps with this inclusion gives exactly the previous images of every generator. The ideal map formula for generated ideals therefore retains each entire initial-ideal equality, both at origin and at every point of B. □

Lemma 286.7 (One neighbourhood kills the entire origin coefficient kernel)**.** *Let B be open and contain zero, and let $C \subset A_B$ be a Noetherian complex subalgebra. There is one open neighbourhood $W \subset B$ of zero such that $\ker(\rho_0|_C) \subset \ker(\rho_a|_C)$ for every $a \in W$.*

Proof. Noetherianity makes the origin kernel finitely generated. Each of its selected generators is an actual function with zero whole germ at the origin, so it vanishes on a neighbourhood of zero. Intersect these finitely many neighbourhoods and B, and take an open neighbourhood W of zero inside that intersection. At every $a \in W$, each selected actual function is identically zero near a. Its translated whole germ at zero is therefore zero. The nearby homomorphism kills the selected generators; since its kernel is an ideal, it kills their whole span, which is exactly the origin kernel. □

Theorem 286.8 (A common domain coefficient algebra with genuine nearby homomorphisms)**.** *The quotient $C_0 = C/\ker(\rho_0|_C)$ is a domain and, when C is of finite type over $\mathbb{C}$, remains of finite type. On one open neighbourhood W of zero every actual nearby homomorphism $\rho_a|_C$ factors as a complex algebra homomorphism through C_0, with its original actual-function formula preserved after composition with the quotient map.*

Proof. The origin homomorphism takes values in the domain of holomorphic germs, so its kernel is prime and the quotient is a domain. A quotient of a finite type algebra is of finite type. The preceding lemma gives one W on which this entire origin kernel lies in every nearby kernel. The universal property of the quotient therefore defines each asserted homomorphism, and its defining composition identity recovers the actual translated whole germ map. No map on arbitrary origin germs is required: only the finitely generated actual coefficient algebra is factored through its internally determined kernel. □

Theorem 286.9 (A single finite type domain presents all nearby original initial ideals)**.** *Only the original coherent ideal I supplies a finite type domain $C_0 = C/\ker(\rho_0|_C)$, an open base neighbourhood W of zero, and one finite homogeneous relation family over C_0. There are genuine complex algebra homomorphisms from C_0 to the translated base germ rings at every $a \in W$. Their composition with the quotient map equals the actual translated whole germ map on coefficient functions. At every $a \in W$, the image of the ideal generated by this same relation family is the entire normal initial ideal of the actual translated original ideal.*

Proof. Apply the preceding original-input theorem to obtain B, C and its finite homogeneous relations. Finite type over $\mathbb{C}$ makes C Noetherian. The common-domain theorem produces one $W \subset B$ on which every nearby actual coefficient map factors through C_0. Map each relation's coefficients through $C \to C_0$. Coefficient maps preserve homogeneous degree. At every $a \in W$, composition of polynomial coefficient maps and the defining factorisation identity give exactly the original nearby relation polynomial. Since ideal maps preserve generated ideals, the entire nearby initial-ideal equality follows for the same finite family. The origin kernel is prime and quotient finite type is preserved, giving the claimed domain and finite type properties. This presentation alone does not assert freeness of its degree pieces or analytic normal flatness. □

287 Whole coefficient extension of polynomial quotients

Lemma 287.1 (The entire coefficient extension quotient is the actual tensor base change)**.** *Let $R \to S$ be a homomorphism of commutative rings and let $H \subset R[Y_1, \dots, Y_t]$ be an ideal. There is an S-algebra isomorphism*

$$S \otimes_R (R[Y]/H) \cong S[Y]/H_S,$$

where H_S is the whole ideal generated by the coefficient images of all members of H.

Proof. Tensoring the whole quotient gives the quotient of $S \otimes_R R[Y]$ by the ideal generated by $1 \otimes H$. The polynomial tensor isomorphism sends $s \otimes P$ to sP_S, where P_S is coefficient extension. It therefore sends the whole ideal generated by $1 \otimes H$ exactly to H_S. Passing to quotients gives the asserted isomorphism; both maps preserve the scalar inclusion of S. □

Corollary 287.2 (Whole polynomial quotient freeness survives actual coefficient extension)**.** *If $R[Y]/H$ is free as an R-module, then $S[Y]/H_S$ is free as an S-module.*

Proof. Tensoring a free R-module with S gives a free S-module, with the same basis indices and extended basis vectors. Transfer this free module structure through the preceding whole quotient isomorphism. □

Theorem 287.3 (A single uniform denominator controls every coefficient-image quotient)**.** *For an ideal $H \subset D[Y]$ over a domain D, let $b \neq 0$ be its uniform standard-monomial denominator. For every homomorphism $f : D \to S$ such that $f(b)$ is a unit, the entire quotient $S[Y]/H_S$ is free as an S-module. The same b works for every such coefficient homomorphism.*

Proof. The unit condition extends f to $D[b^{-1}] \to S$. The entire coefficient-localized quotient is free by the uniform standard-monomial theorem. Extend its coefficients through this actual localisation map and apply the preceding free base-change theorem. The composition of the localisation map with the extended map is the original f, so its whole ideal image is exactly H_S. This proves the claimed free quotient using the same internally determined denominator for every f. □

288 One genuine analytic open for all powers of an actual coherent saturation

Let $X = \mathbb{C}^n$, let $\mathfrak{a}$ be an ORIGINAL coherent ideal of holomorphic germs on X, and let h be an actual function holomorphic near the origin. Membership is recorded by entire functions and their WHOLE germs; no map from all origin germs to arbitrary nearby stalks is postulated. Put $C_0 = \mathfrak{a}$ and $C_{q+1} = (C_q : h)$, with $(C_q : h)_y = \{g \in \mathcal{O}_{X,y} : gh \in (C_q)_y\}$. Thus C_q is the literal colon by h^q. The same argument applies to an ideal locally generated by an actual finite family of holomorphic functions. The stable exponent and neighborhood below are produced internally from these original data.

Lemma 288.1 (Entire finite analytic spans and genuine origin ideals)**.** *For a finite actual holomorphic family s_i, membership of an entire function g in its analytic span at the origin is equivalent to $[g]_0 \in ([s_i]_0)_i$ in the TRUE ring of holomorphic germs. If g has this membership at the origin, it has membership in the same actual analytic span at every point of one neighborhood of the origin. More generally, if each member of a finite family t_j is in the s_i span at the origin, then their entire analytic span is contained in the s_i span at every point of one actual neighborhood.*

Proof. An analytic span witness consists of finitely many whole holomorphic coefficient functions a_i and the eventual identity $g = \sum_i a_i s_i$. Taking WHOLE germs gives the literal ring ideal membership. Conversely, finite ideal membership gives an actual coefficient tuple of whole germs. Choose genuine holomorphic function representatives for those finitely many coefficients; the ring

identity becomes an eventual identity of the actual entire functions. For nearby membership, shrink to an open neighborhood where all coefficient functions and all s_i are holomorphic and the identity holds. At every point of this SAME open the identity holds as an entire germ, hence gives the required analytic span witness. There are only finitely many t_j, so intersect their actual neighborhoods. Substitution of any actual analytic combination of the t_j into these finite representations gives an actual analytic combination of the s_i. The coefficient product and finite sum are genuinely holomorphic at that nearby point. Empty families use the literal zero sum and the same eventual-identity argument. □

Lemma 288.2 (The literal original colon sheaves are coherent and retain their entire origin dictionaries)**.** *Every C_q is coherent near the origin. Internally select a finite actual family generating $\mathfrak{a}$ on a neighborhood and let A_0 be its TRUE origin-germ ideal. Define $A_{q+1} = (A_q : [h]_0)$ in $\mathcal{O}_{X,0}$. For every q and every actual holomorphic function g, original membership $g \in (C_q)_0$ is equivalent to $[g]_0 \in A_q$. At every point where h is holomorphic and the original ideal is coherent, $g \in (C_q)_y$ is equivalent to $g \in \mathcal{O}_{X,y}$ and $gh^q \in \mathfrak{a}_y$.*

Proof. The original coherence hypothesis supplies actual finite holomorphic generators on a genuine open containing zero; these are selected internally. The preceding entire-span dictionary identifies their literal origin ideal with original membership. Oka's theorem applied to the row $(h, s_1, \ldots, s_r)$ constructs actual finitely many holomorphic relations on an open neighborhood. Their first components generate the literal colon at EVERY point of that open. Iterating this genuine colon-coherence construction proves coherence of every C_q; no next-colon presentation is supplied by a caller. Taking WHOLE germs of the actual product gh gives $[gh]_0 = [g]_0[h]_0$, so induction gives the exact origin dictionary for every A_q. Finally repeated colon membership is exactly membership of gh^q in the original ideal: at each induction step the extra holomorphicity condition on gh follows from that of g and the actual h, and the identity $(gh)h^q = gh^{q+1}$ holds for the entire functions. The zero power uses original coherent membership, which already implies holomorphicity. □

Lemma 288.3 (The stable exponent is internally selected from the actual origin Noetherian chain)**.** *There exists an internally produced integer N such that $A_N = A_{N+1}$, and hence $A_{N+q} = A_N$ for every $q \geq 0$.*

Proof. Every ideal is contained in its colon by a single element: if $a \in A_q$, then $a[h]_0 \in A_q$. Consequently the actual A_q form an increasing chain of TRUE ideals. The holomorphic germ ring $\mathcal{O}_{\mathbb{C}^n,0}$ is Noetherian, so its genuine ascending chain condition produces an integer N after which this very chain is constant. In particular $A_N = A_{N+1}$. Applying the same colon operation to both sides and using induction proves all later equalities. The integer is selected from the original chain, rather than assumed as a stabilization or saturation certificate. The zero-dimensional case uses the same proved Noetherian ring. □

Theorem 288.4 (One actual analytic open has exact equality of the two actual coherent colon sheaves)**.** *There is a genuine open $U \ni 0$ on which h is holomorphic and $(C_N)_y = (C_{N+1})_y$ for EVERY $y \in U$ and every WHOLE holomorphic germ at y.*

Proof. Apply the genuine coherent-colon constructions to the actual internally selected N and $N+1$. They supply finite actual holomorphic families s_i and t_j generating C_N and C_{N+1} at EVERY point of respective neighborhoods of zero. Every t_j is in $(C_{N+1})_0$. The exact whole origin dictionary and the genuine ring equality $A_N = A_{N+1}$ place it in $(C_N)_0$. Thus each ACTUAL t_j is an actual analytic combination of the s_i at the origin. The finite-span lemma spreads these particular whole identities on ONE actual neighborhood. Intersect this open with both original coherent presentation opens and an actual holomorphic open for h. At every point of the resulting SAME U, every member of C_{N+1} is an actual analytic combination of the t_j, hence one of the s_i, and belongs to C_N. Conversely $C_N \subset C_{N+1}$ at every such point: in the literal

ideal C_N, multiplication by the actual holomorphic h preserves membership. This ideal closure follows directly from the actual finite analytic span presentation there. The two inclusions prove WHOLE stalk equality for EVERY point of this same U. An equality only at the origin would not suffice; the finite functional identities and the genuine coherent presentation at ALL points are essential. □

Theorem 288.5 (Only original coherence and an actual holomorphic function produce uniform all-power saturation)**.** *ONLY the original coherent ideal and actual holomorphic h supply an integer N and an actual open $U \ni 0$ such that for EVERY $y \in U$, every $q \geq N$, and every actual holomorphic g,*

$$gh^q \in \mathfrak{a}_y \quad \Longleftrightarrow \quad gh^N \in \mathfrak{a}_y.$$

Equivalently the literal saturation $\bigcup_q (\mathfrak{a}_y : h_y^q)$ is $\mathfrak{a}_y : h_y^N$ at EVERY point of this SAME U.

Proof. The preceding theorem supplies an internally constructed N and the genuine U. At a fixed arbitrary $y \in U$, exact equality $C_N = C_{N+1}$ as predicates of WHOLE holomorphic germs persists after taking a colon by the same actual h. Induction therefore gives $C_{N+r} = C_N$ at this y for every r; this is a pointwise algebraic induction using the SAME previously produced open, with no new shrinking as r varies. Write $q = N + r$ and use the literal power-colon dictionary to obtain the displayed equivalence. All earlier C_q are contained in C_N by actual ideal closure, and all later ones equal it, proving the saturation statement. Original finite generators and the finite generators of both colons, the true Noetherian exponent, all coefficient witnesses and all actual presentation opens are internally produced; no caller stable exponent, finite-colon relations, result or saturation certificate is an input. Zero or unit ideals and an identically zero h use the same ideal-chain and coherence proof. □

Remark 288.6. This is a genuine uniform analytic SATURATION producer. It does not yet identify a deformation-to-normal-cone presentation with the WHOLE original nearby graded cone. The actual deformation equations, their saturated ideal and the nearby original associated-graded dictionary still require proofs. Nearby all-order finite freeness, analytic normal-flat loci, higher HS and general G8 remain OPEN.

289 The same original coherent generators in actual nearby translated normal jets

Let $\mathfrak{a}$ be an original coherent ideal sheaf on $\mathbf{C}^{d+t}$. An actual translation is a genuine map of these manifolds and an actual pullback of the original SHEAF. No map from arbitrary origin germs to nearby germs is introduced. The literal translated normal ideal below is generated by the actual normal coordinates of the translated chart; identification with an unshifted original centre is asserted only after a genuine centre chart has been specified, and is not assumed here.

Lemma 289.1 (Actual affine translations and whole sheaf pullbacks)**.** *For every actual a, the map $\tau_a(z) = a + z$ is genuinely globally holomorphic and sends zero to a. Its actual original sheaf pullback $\tau_a^* \mathfrak{a}$ is coherent. Precomposition by τ_a is a genuine ring homomorphism of FUNCTION rings; it is not a homomorphism from arbitrary origin germ rings.*

Proof. Addition of the actual constant vector to the identity is an actual affine holomorphic diffeomorphism, with inverse translation by $-a$. Its genuine smooth complex derivative and continuity give the required holomorphic pullback hypotheses. Apply true coherent sheaf pullback stability. Function precomposition preserves the literal zero, one, sum and product pointwise; these elementary identities supply the actual function-ring homomorphism. No source germ is evaluated at a point to define a nearby germ map. □

Theorem 289.2 (One original finite family has true whole translated germs everywhere on its original open)**.** *Only original coherence internally selects ONE genuine finite family of functions f_i and ONE original open $V \ni 0$ on which they are holomorphic and generate the original sheaf at EVERY point. For EVERY $a \in V$, each actual function $z \mapsto f_i(a+z)$ is holomorphic near zero and supplies a true ENTIRE germ $g_{a,i}$. Its raw germ is literally the translated function.*

Proof. Choose the original coherence neighborhood at zero once, retaining its literal finite index and function family. At $a \in V$, openness and continuity of τ_a give an actual zero-neighborhood mapped into V. Original holomorphy at EVERY point of V composed with actual affine translation therefore proves differentiability of each translated function throughout this true zero-neighborhood. Form its genuine holomorphic germ. The raw whole-germ dictionary is definitional and preserves the chosen function; there is no arbitrary representative choice at the nearby point. □

Theorem 289.3 (The entire actual translated original ideal is generated by these same germs)**.** *For EVERY actual $a \in V$, the literal original origin ideal of the ACTUAL sheaf $\tau_a^* \mathfrak{a}$ is exactly* $\operatorname{span}\{g_{a,i}\}$*. Equivalently membership of every genuine holomorphic function at zero in that actual pulled sheaf is exactly its true finite whole-function germ span in $f_i \circ \tau_a$.*

Proof. The original finite generator dictionary holds at the ACTUAL source point $\tau_a(0) = a$. Actual ideal pullback is generated by all original stalk members composed with the actual map. Each such original member is a finite holomorphic combination of the original f_i at a. Compose those actual functions and the entire eventual equality with the genuine continuous τ_a to produce a true holomorphic combination at zero. Conversely every f_i is a true original stalk member at a, hence its actual composition belongs to the literal pulled sheaf. The transitivity of genuine finite holomorphic span proves equality of the two actual sheaf stalk spans. Finally the full original coherent ideal-germ dictionary and genuine whole finite sums/products identify the ENTIRE origin ideal with the actual finite span of these same translated germs. No inclusion or spanning result is an external certificate. □

Theorem 289.4 (Actual nearby original finite normal jets retain the same whole generator columns)**.** *At EVERY actual $a \in V$ and EVERY literal q, use the true translated original ideal I_a and the actual normal quotient R/J^q in its translated chart. Its entire actual ideal image is exactly the B-span of $e_\alpha \overline{g_{a,i}}$, $|\alpha| < q$, formed from THESE SAME original functions translated by τ_a. Every column has a real original ideal witness, and whole-image membership is exactly a genuine finite coefficient combination of these SAME columns.*

Proof. The preceding actual ideal dictionary supplies $I_a = \operatorname{span} g_{a,i}$ internally. Each normal monomial times a genuine translated generator belongs to this actual ideal and maps literally to the written column. For any true I_a element, take its actual finite generator expression and pass the whole identity to the normal quotient. The genuine entire normal monomial basis expands every actual coefficient jet over B; true finite sums and quotient multiplication give exactly the stated column combination. Conversely the actual original ideal image is a B-submodule containing every actual column. This proves the entire equality and actual finite coefficient dictionary at every genuine point, including empty families and literal order zero. □

Theorem 289.5 (Only original coherent input jointly produces the actual nearby sheaf and jet dictionaries)**.** *Only actual $d, t, \mathfrak{a}$ and coherence internally produce ONE original finite holomorphic family/open and ALL actual translated whole generator, original ideal and finite normal-jet image dictionaries above. For every $a \in V$ and every actual q the SAME original functions are used.*

Proof. Choose the original coherence data once. Retain its true open and full original sheaf membership dictionary. Apply the genuine affine function translation and actual sheaf-pullback producer at each point of that open, then the entire translated ideal and genuine finite normal-jet proofs. Return those same original functions with all actual point/order dictionaries. No

neighborhood, generator, translated chart, coefficient, image, basis or freeness conclusion is caller data. □

Open frontier. These actual nearby ORIGINAL jet columns have not yet been identified with the earlier same-denominator analytic projection functions. Genuine normal Taylor function/germ coefficient identities on a common open and actual finite cleared-denominator column identities are still required. Neither these pointwise whole original dictionaries nor the finite projector model alone implies one all-order open, actual nearby whole Rees/normal-cone identification, analytic normal-flatness, higher Hilbert–Samuel or G8. Formal registration, native build and publication remain pending root recovery.

290 Finite original whole initial relations spread into the actual central model

Only the original coherent ideal $\mathfrak{a}$ and d, t are original inputs. Write I for its internally constructed WHOLE origin ideal, H for its actual entire normal initial ideal, and S for the actual internally selected finite homogeneous original relation family. None of these is caller data. We prove a true one-sided whole analytic ideal inclusion on ONE neighborhood. This finite relation model is not identified with the whole nearby ORIGINAL normal initial ideal, and no reverse inclusion or flatness is asserted.

Theorem 290.1 (The actual finite original family gives the whole analytic extension ideal)**.** *The internally selected S consists of genuine original homogeneous relations and generates the ENTIRE original initial ideal H. Its actual analytic evaluations generate precisely the whole ideal image of H under normal polynomial evaluation.*

Proof. Apply the actual original Noetherian extraction to the genuine original origin ideal. Its selected finite family has the exact whole ideal-generation dictionary and each of its members is an actual original homogeneous symbol. The genuine ideal-image theorem carries the span of S to the span of its actual evaluated image. This is a whole analytic ideal image, closed under arbitrary whole holomorphic multipliers, not only polynomial coefficients. No analytic extension-flatness assertion is needed for this exact image identity. □

Theorem 290.2 (The actual finite evaluations have common whole holomorphic representatives)**.** *For every actual $P \in S$, select the genuine entire holomorphic representative f_P of its actual WHOLE normal-polynomial evaluation. The zero evaluation uses the literal zero representative. All f_P share ONE actual open neighborhood A of zero on which they are holomorphic, and their WHOLE origin germs equal those actual evaluations. In particular the entire origin ideal generated by their true germs is exactly the analytic extension ideal above.*

Proof. Use the already proved actual coefficient representative construction on each whole ambient holomorphic evaluation germ. Its full germ equality and genuine eventual complex derivative supply its true holomorphicity near zero. The finite image of S under actual normal evaluation is an actual finite family of ambient germs, so the common-coefficient-open producer gives ONE open containing zero with all their actual derivatives. No canonical map from arbitrary origin germs into other stalks is used: the functions f_P themselves are genuine selected holomorphic functions on this actual open. Their full origin germ identities identify the generated whole origin ideal with the preceding actual analytic-extension image ideal. □

Theorem 290.3 (The actual finite functional model is coherent on its genuine open)**.** *Let F_y be the actual finite holomorphic span of the SAME functions f_P at y. At every $y \in A$ this is a true finitely generated coherent analytic ideal predicate. At zero, for every actual holomorphic function g, $g \in F_0$ if and only if its whole germ belongs to the ENTIRE analytic extension image of the original H.*

Proof. Index the actual finite relation set by its genuine finite subtype. On the open A every function in this SAME finite family is holomorphic. It supplies a literal finite presentation of its span predicate at every point of A: the neighborhood is A, the generating functions are the actual f_P and the full span dictionary is the definition itself. Their whole holomorphy on A follows from its openness and actual derivatives at every point of A. The genuine entire finite analytic-span theorem identifies zero-stalk span membership with membership in the ring ideal generated by these whole germs. Use the preceding actual image-ideal equality. No finite-presentation or nearby result certificate is external input. □

Theorem 290.4 (Actual coherent membership spreads the ENTIRE finite model inclusion)**.** *For the genuinely internally selected central model, there is ONE true open V containing zero and contained in A, such that at EVERY $y \in V$, the ENTIRE actual finite functional model F_y lies in the SAME actual coherent central ideal stalk. The same original input jointly retains the actual enlarged saturation neighborhood and all of its literal all-power and whole-union dictionaries with this SAME selected index. This includes all genuine holomorphic coefficient combinations and not only the selected functions' values.*

Proof. Internally choose the actual saturation index and neighborhood from the original-input whole initial-inclusion producer. Its WHOLE origin image inclusion and the preceding finite functional dictionary give genuine origin membership inclusion between the entire finite model and the SAME central model. The finite model supplies its actual finite presentation at zero, and actual central coherence supplies the target presentation. The genuine coherent inclusion-spreading theorem spreads the finite whole membership witnesses on ONE actual neighborhood. It gives an inclusion for every entire analytic span member at every point of that neighborhood. Intersect with A to retain all actual representatives' holomorphy. Every stalk map used is evaluation of an actual whole functional witness; a map on all origin germs into arbitrary nearby stalks is not invented. □

Theorem 290.5 (The original coherent input jointly supplies the actual near-central finite model)**.** *ONLY the original coherent $\mathfrak{a}, d, t$ internally supply SAME N, V, the genuine coherent central model, all true finite relation functions and their holomorphy on V, the ENTIRE finite-model inclusion in every actual central stalk on V, and the actual all-power saturation dictionaries for the enlarged original equation ideal at every actual central image $j(V)$.*

Proof. Retain the internally chosen N, U and all-power stabilization and whole saturation-union dictionaries of the actual enlarged equation model. The previous producer gives a genuine neighborhood for the entire finite relation model inclusion in this SAME actual central ideal. The actual central linear inclusion j is continuous and sends zero to zero. Intersect this neighborhood with $j^{-1}(U)$. It is still open and contains zero. Restriction preserves the entire finite-model inclusion and all actual representative holomorphy. Every actual central image now belongs to the original SAME saturation open, so its exact all-power and whole union dictionaries are retained at every point. Return these literal internally constructed objects and joint output; no index, open, source relation family, target inclusion or flatness certificate is an original hypothesis. □

Open frontier. The finite actual origin-relation analytic model is NOT proved to equal the WHOLE nearby ORIGINAL normal initial ideal. The reverse whole analytic central ideal dictionary, nearby original Rees/central-fiber/graded identification, nearby ALL-degree freeness, analytic normal-flatness, higher HS and G8 remain OPEN. A true neighborhood finite-model inclusion does not close these gaps.

291 Entire original initial relations lie in the actual deformation central saturation

Only d, t and an ORIGINAL coherent ideal sheaf $\mathfrak{a}$ on $\mathbb{C}^{d+t}$ are original consumer inputs. The origin ideal I below is internally constructed from ENTIRE genuine stalk membership; the actual equation ideal $\mathcal{K}$, parameter u, selected N, U and central map $j(x, v) = (x, 0, 0, v)$ are the literal preceding original-input constructions. This chapter proves one genuine WHOLE-ideal inclusion at the origin. It does not claim the reverse inclusion or a nearby whole cone dictionary.

Theorem 291.1 (The actual original coherent stalk is a whole germ ideal)**.** *Internally define the actual origin germ ideal of any original coherent $\mathfrak{a}$ as the ideal generated by ALL whole germs $[f]$ with $f \in \mathfrak{a}_0$. For EVERY actual holomorphic representative f, $[f]$ belongs to this entire ideal if and only if $f \in \mathfrak{a}_0$. Its literal repeated holomorphic functional colons correspond to the literal repeated ring colons of this SAME origin ideal.*

Proof. Original coherence internally supplies a finite family s_i, a genuine open containing zero and its whole stalk span dictionary. The actual finite holomorphic span theorem identifies that dictionary with membership in the genuine whole germ ideal generated by $[s_i]$. Each s_i belongs to the original stalk, by its finite span witness with coefficient one. Conversely every original stalk member has a whole finite coefficient identity, hence its entire germ belongs to that finite germ ideal. The ideal generated by ALL original stalk members therefore equals this finite germ ideal, independently of the chosen coherent presentation. This proves the all-representative dictionary. At a successor colon the literal functional predicate is holomorphy of f and original membership of fh; the exact product germ dictionary makes this precisely the genuine ring colon by $[h]$. Induction gives every repeated colon. Each functional colon is already genuinely coherent internally, so the dictionary applies to its entire stalk without an additional presentation input. □

Theorem 291.2 (The actual polynomial normal scaling has the whole graph ideal dictionary)**.** *The actual internally constructed holomorphic map $\Theta(x, y, u, v) = (x, uv)$ fixes the tangent coordinates and multiplies each true normal coordinate by the actual parameter. It sends zero to zero and gives a genuine whole germ pullback. For EVERY original whole germ g, $\Theta^* g - p^* g$ belongs to the WHOLE ideal generated by the true equations $uv_i - y_i$, not merely their pointwise zero locus.*

Proof. Construct Θ using the actual finite coordinate functionals and their literal coordinate products; their genuine complex derivatives supply its holomorphy and actual zero identity. In the actual doubled ambient $(X, \delta) \in \mathbb{C}^{d+t} \times \mathbb{C}^{d+t}$ consider the entire germ $g(X + \delta) - g(X)$. The actual coordinate-plane trace is zero, because the literal second coordinate block is zero on that plane. Actual relative Hadamard decomposition gives a WHOLE finite identity $g(X+\delta)-g(X) = \sum_i \delta_i H_i(X, \delta)$ with genuine holomorphic coefficient germs. Pull this entire identity back along the actual holomorphic map $(p, \Theta - p)$. Its tangent difference coordinates are zero and each normal difference coordinate is exactly $uv_i - y_i$. Its literal sum coordinate is Θ. Thus the true finite coefficient identity places $\Theta^* g - p^* g$ in the actual equation germ ideal. No graph division or kernel certificate is supplied by the caller. □

Theorem 291.3 (Every actual original normal-power germ has its genuine divided scaling lift)**.** *For every $g \in J^q$ there exists a genuine whole enlarged germ F_g with $\Theta^* g = u^q F_g$. Its actual central pullback is EXACTLY the analytic normal polynomial evaluation of the internally produced degree q original initial symbol of g.*

Proof. The exact normal coordinate dictionary puts the image of J in the actual principal ideal (u). Genuine ideal-power multiplication therefore puts the image of J^q in $(u)^q = (u^q)$; whole principal ideal membership internally supplies a genuine germ F_g. Write P for the actual homogeneous

degree q representative of g modulo J^{q+1}, supplied by the proved whole normal Taylor producer. The literal scaling pullback of each monomial of P is u^q times its actual fiber monomial with its true tangent coefficient unchanged, since all supported monomials have exactly degree q. The whole remainder in J^{q+1} similarly pulls back to $u^{q+1}G$. Hence $u^q F_g = u^q \widetilde{P} + u^{q+1}G$ as an entire enlarged germ identity. The actual parameter is a nonzero whole coordinate germ, proved by its actual derivative, and the genuine holomorphic germ ring is an integral domain. Cancel u^q to obtain $F_g = \widetilde{P} + uG$. The actual central pullback sends u to zero, every added fiber coordinate to its literal central normal coordinate, and all tangent coefficients to themselves. It therefore gives $j^* F_g = \mathrm{eval}(P)$ as an ENTIRE germ identity. This is the internally produced initial symbol, with its actual whole Taylor uniqueness, not a supplied jet or pointvalue statement. □

Theorem 291.4 (Every original initial relation belongs to the actual selected central model)**.** *ONLY the original coherent ideal internally supplies an actual selected N with coherent colon and central models, such that EVERY original $g \in I \cap J^q$ has its entire degree q initial symbol evaluation in the actual origin ideal of that SAME coherent central restriction $j^*\mathcal{K}_N$. The internally selected N is obtained with the genuine saturation neighborhood, without an externally supplied index or membership dictionary.*

Proof. The whole original stalk dictionary identifies $g \in I$ with genuine original sheaf stalk membership of every chosen actual representative. Its actual projection pullback belongs to the genuine original pullback ideal and hence to $\mathcal{K}$. The entire graph equation dictionary puts $\Theta^* g - p^* g$ in the equation ideal and hence also in $\mathcal{K}$. True ideal addition gives $\Theta^* g = u^q F_g \in \mathcal{K}_0$. The whole functional/ring colon dictionary therefore puts F_g in the actual qth parameter colon. Since zero belongs to the SAME actual U, the entire saturation union dictionary puts this genuine germ in the internally selected Nth colon, regardless of q. Pull back its genuine stalk membership by the actual central map. The whole central dictionary from the preceding theorem identifies this image with the ENTIRE analytic evaluation of the actual original initial symbol. No N, colon membership, central relation or result certificate is original caller data. □

Theorem 291.5 (The entire original initial ideal has a genuine analytic-extension inclusion)**.** *ONLY the original coherent ideal internally supplies SAME N, U, genuine coherence of its literal colon and central model, all-point/all-power stabilization and whole saturation-union dictionaries, such that the ideal image under actual analytic normal polynomial evaluation of the ENTIRE internally constructed original homogeneous initial ideal lies in the actual origin ideal of this SAME selected coherent central model.*

Proof. Internally choose the genuine SAME N, U from the original saturated-model producer, retaining its actual coherence and all-point/all-power dictionaries. The whole initial-symbol argument above applies to this SAME actual selected index through its genuine origin saturation-union output; no index or relation membership is caller data. The original associated-graded producer proves that its WHOLE initial ideal is the genuine ideal generated by ALL actual homogeneous initial symbols of original germs in $I \cap J^q$, over all degrees. The preceding theorem places the analytic evaluation of each of these actual generators in the SAME selected central ideal. That genuine ideal is closed under arbitrary whole analytic germ multiplication and addition. The universal property of the genuine generated ideal, followed by the exact actual ring ideal image dictionary, gives the claimed ENTIRE inclusion. This does not assume that extension to holomorphic germs is flat and does not claim surjectivity or the reverse inclusion. □

Theorem 291.6 (The original coherent input jointly supplies its actual one-sided cone dictionary)**.** *ONLY original coherent $\mathfrak{a}, d, t$ internally produce an actual N with the SAME actual coherent colon and central models and the ENTIRE origin initial-ideal inclusion. This direct coherent-model consumer extracts these actual fields from the joint original-input producer.*

Proof. Apply the preceding joint original-input producer and retain its actual selected N, genuine coherence of the literal colon and central models, and ENTIRE analytic-extension ideal inclusion. Discard only its additional neighborhood dictionaries from this simpler conclusion; they remain part of the joint producer. Every retained object is the literal internally constructed object of the SAME original ideal, not an independently supplied cone presentation. □

Open frontier. The reverse whole analytic central relation dictionary, its extension to nearby ORIGINAL stalks, the genuine Rees/central-fiber equivalence and nearby ALL-degree freeness remain OPEN. The one-sided origin inclusion does not settle them. Analytic normal-flat neighborhoods, higher HS strata and G8 remain OPEN, and no canonical map of all origin germs to nearby stalks is used.

292 Actual deformation equations and one analytic saturation neighborhood

Fix natural d, t and an ORIGINAL coherent ideal sheaf $\mathfrak{a}$ on $\mathbb{C}^{d+t}$. No deformation model, equation presentation, stable index, saturation or normal-cone identity is an additional input. Put $D = d+t+1+t$ and use the literal coordinates (x, y, u, v) on $\mathbb{C}^D$, with $x \in \mathbb{C}^d$ and $y, v \in \mathbb{C}^t$. All memberships below are memberships of whole holomorphic germs.

Theorem 292.1 (The actual coordinate projection and deformation equations)**.** *The internally defined continuous linear projection is $p(x, y, u, v) = (x, y)$. The internally defined parameter and fiber coordinates are u and v_i. The actual finite equations are the literal entire functions $e_i(x, y, u, v) = uv_i - y_i$. The parameter and all equations are holomorphic on the whole enlarged ambient space.*

Proof. Project to the first $d + t$ coordinates using the actual first-coordinate continuous linear projection. Each specified scalar coordinate is its genuine continuous linear functional, not a chosen point value. The parameter is therefore everywhere complex differentiable. Multiplication of the two actual coordinate functions, followed by subtraction of the actual original normal coordinate, gives the literal equation and everywhere complex differentiability. The first-coordinate projection is an actual smooth holomorphic map. All constructions also apply to $d = 0$ and $t = 0$. □

Theorem 292.2 (The actual original deformation equation ideal is coherent)**.** *Internally define $\mathcal{K} = p^*\mathfrak{a} + (e_i : i \in \{1, \dots, t\})$ as the sum of the genuine original ideal pullback and the entire finite equation ideal. Original coherence alone makes this actual ideal coherent.*

Proof. The finite equation ideal records precisely the holomorphic finite span of the actual family e_i at each point. Its global entire generators give a coherent presentation on the entire enlarged space, including the empty family when $t = 0$. Actual smooth holomorphic pullback carries the original coherent ideal to a coherent ideal by the genuine coherent pullback theorem. The actual sheaf sum consists of the finite holomorphic span of the union of these two actual stalk predicates. The coherent sum theorem supplies a local finite presentation of this literal sum. Thus the equation model is built from the original ideal; no presentation or kernel conclusion is supplied. □

Theorem 292.3 (Every actual repeated parameter colon is a coherent ideal)**.** *Let $\mathcal{K}_0 = \mathcal{K}$ and let $\mathcal{K}_{q+1}$ consist of holomorphic germs g with $gu \in \mathcal{K}_q$. These are the literal repeated parameter colons. Each is a genuine coherent ideal on the whole enlarged space, internally produced for every natural q.*

Proof. At an arbitrary actual point, the original coherent presentation converts manifold holomorphic finite spans to ordinary complex differentiable finite spans, using the model-space derivative equivalence. Actual Euclidean Oka coherence then produces a finite presentation for the colon by the actual holomorphic parameter. Induct on q at this same arbitrary point. Reindex any finite generator type by its genuine equivalence with a finite ordinal; finite sums are unchanged under this equivalence, so conversion back gives the required manifold coherent presentation. No finite colon generators are caller data. The recorded predicate is exactly the recursive functional colon, including its holomorphic-germ condition, at every point. □

Theorem 292.4 (One genuine open stabilizes all powers and the entire actual saturation)**.** *Original coherence internally produces a natural N and ONE open neighborhood U of the origin such that, at EVERY $z \in U$ and for EVERY whole ambient function g, all $\mathcal{K}_{N+q,z}$ have the same membership as $\mathcal{K}_{N,z}$ for EVERY natural q. For every $q \geq N$ and every holomorphic g at z, $gu^q \in \mathcal{K}_z \iff gu^N \in \mathcal{K}_z$. Moreover g belongs to SOME $\mathcal{K}_{q,z}$ if and only if it belongs to the one actual coherent ideal $\mathcal{K}_{N,z}$.*

Proof. Apply the genuine original-input coherent saturation producer to the actual coherent equation ideal and the actual holomorphic parameter. Internally Noetherianity of the whole origin germ ring selects N. Actual Oka coherent presentations of the two consecutive colons and whole finite functional membership witnesses shrink to ONE neighborhood where the two entire stalks agree at EVERY point. Literal colon induction propagates this equality to ALL subsequent powers on that SAME open. The true entire-power dictionary and actual original ideal closure identify the entire increasing saturation union with the internally selected colon. Substitute the exact recursive GermIdeal predicates just defined. No stabilization index, neighborhood, saturation certificate or nearby cone identity is supplied by the caller. □

Theorem 292.5 (The actual effective central restriction)**.** *The internally constructed continuous linear map $j : \mathbb{C}^{d+t} \to \mathbb{C}^D$ is $j(x, v) = (x, 0, 0, v)$. It has exactly the displayed tangent, original normal, parameter and fiber coordinate dictionary; EVERY actual equation vanishes identically on its image. The actual central restriction $j^*\mathcal{K}_N$ is a genuine coherent ideal for every N.*

Proof. Use the actual finite product of coordinate continuous linear functionals, followed by the genuine product-to-Euclidean continuous linear equivalence. In the tangent block use the corresponding original tangent functional; in the added fiber block use the corresponding original fiber functional; in the original normal and parameter blocks use the zero functional. The index blocks are disjoint. Evaluating at each specified coordinate gives precisely its literal coordinate, or zero for an absent block. In the actual equation this yields $0 \cdot v_i - 0 = 0$ as an identity of entire functions. The map is a genuine continuous linear map, hence smooth holomorphic. Pull back the actual coherent selected colon along this map using the genuine coherent pullback theorem. This constructs the actual restriction; it does not identify it with an associated graded ideal before that separate whole-ideal dictionary is proved. □

Theorem 292.6 (The original ideal jointly produces the actual coherent saturation model)**.** *Only d, t and original coherent $\mathfrak{a}$ jointly produce the actual projection, parameter and equation ideal, the internally selected N, U, the SAME actual coherent $\mathcal{K}_N$ with the entire all-power and saturation dictionaries on U, and its actual coherent central restriction.*

Proof. Retain the actual N, U from the uniform original-input producer, and append coherence of the literal Nth colon and coherence of its literal central pullback, already proved for every index. The entire all-point/all-power dictionary is unchanged. Thus all output objects are constructed from the same original ideal and the same actual equations. The conclusion is a joint actual model, not an independently supplied result certificate. □

Open frontier. The actual equation saturation model and its actual central restriction have NOT yet been identified with the nearby entire original Rees algebra or whole original associated graded cone. These dictionaries are independent remaining obligations. The finite analytic coefficient model and the origin localized all-degree algebra are not promoted to nearby all-degree freeness. Analytic normal-flat neighborhoods, higher Hilbert–Samuel strata and G8 remain open. No canonical map of all origin germs into arbitrary nearby stalks is used.

293 The whole original fibre Hilbert–Samuel dictionary

Write $B = \mathcal{O}_{\mathbb{C}^d,0}$, $R = \mathcal{O}_{\mathbb{C}^{d+t},0}$, J for the actual normal-coordinate ideal, and M for the whole image of $\mathfrak{m}_B$ under the actual tangent pullback. The genuine original fibre ideal of an original ambient germ ideal I is $I + M$. The objective is the existing whole local-ring Hilbert–Samuel quotient in Bierstone–Milman's "Use of the Hilbert–Samuel function". The argument below identifies its entire sequence on the one original-input open from the uniform relative ranks; it does not assert descent after blowing up.

Definition 293.1 (The genuine whole original base-fibre ideal)**.** Let M be the whole ideal image of $\mathfrak{m}_B$ under the original tangent pullback. The fibre ideal of an arbitrary original I is $I + M$. All quotients use the original whole germ ring, its actual multiplication, and its original complex scalar structure.

Lemma 293.2 (The actual split-coordinate maximal ideal)**.** *The literal maximal ideal of R is exactly $M + J$.*

Proof. Every base germ in $\mathfrak{m}_B$ evaluates to zero at the original base origin, and its true tangent pullback has the same value at the ambient origin. Therefore $M \subseteq \mathfrak{m}_R$. Every normal coordinate also evaluates to zero, so $J \subseteq \mathfrak{m}_R$. Conversely, for $g \in \mathfrak{m}_R$, the actual relative holomorphic Hadamard decomposition writes $g = \text{tangentPullback}(\text{planeTrace}(g)) + \sum_i z_i h_i$, with the sum over the original normal coordinates. The actual plane trace has the same origin value as g, hence belongs to $\mathfrak{m}_B$. Its true tangent pullback therefore belongs to M. Every normal-coordinate summand belongs to J. Thus $g \in M + J$. □

Lemma 293.3 (All actual fibre maximal-ideal powers are the same normal-power quotients)**.** *For every original whole I and every $s \in \mathbb{N}$,*

$$I + M + \mathfrak{m}_R^s = I + M + J^s.$$

Thus the existing whole Hilbert–Samuel quotient of the genuine fibre ideal $I + M$ at order ℓ is literally the original quotient $R/(I + M + J^{\ell+1})$.

Proof. Let $\pi : R \to R/M$ be the genuine whole quotient map. The images of $M + J$ and J under π agree, because the image of M is zero. Ideal image commutes with taking powers, so the images of $(M + J)^s$ and J^s agree for every s, including zero. The inverse image of the image of any whole ideal K under this surjective quotient map is exactly $K + M$. Taking inverse images therefore gives $M + (M + J)^s = M + J^s$. Substitute the preceding actual identity $\mathfrak{m}_R = M + J$ and add the same whole original ideal I. This proves precisely the asserted equality, with M retained on both sides. For positive base dimension it does not identify the ambient quotient by $I + \mathfrak{m}_R^s$ with a normal-power or fibre quotient; that quotient retains mixed tangent degrees. □

Theorem 293.4 (Actual residue tensor is the whole original fibre-power algebra)**.** *Using the genuine base residue evaluation $B \to \mathbb{C}$, there is a $\mathbb{C}$-algebra isomorphism*

$$\mathbb{C} \otimes_B R/(I + J^s) \;\simeq_{\mathbb{C}}\; R/(I + M + J^s).$$

It maps every pure original numerator $c \otimes [g]$ to the actual class $[cg]$, where c is the original constant germ. Its inverse maps every original class $[g]$ to $1 \otimes [g]$.

Proof. The proposed forward map is induced by multiplication of the two actual maps into the whole original quotient: complex constant germs and the original quotient of R. It is balanced over B. Indeed, for $b \in B$, $b - b(0)$ belongs to $\mathfrak{m}_B$; its true tangent pullback therefore belongs to M and is zero in the target. Thus the base action in that quotient agrees with actual origin evaluation. The tensor universal property gives the forward unital algebra map. For the inverse, first map $g \in R$ to $1 \otimes [g]$. The map kills $I + J^s$ by the actual original quotient, and it kills every bg with $b \in \mathfrak{m}_B$ by tensor balancing and $b(0) = 0$. It therefore kills the whole generated image ideal M and descends to the asserted whole fibre-power quotient. Both maps preserve the actual complex constants. Their composition on every original quotient numerator is the identity. On a pure tensor $c \otimes [g]$, the other composition is $1 \otimes [cg] = c \otimes [g]$, because evaluation of the constant germ is c. Pure tensors generate the tensor algebra additively, so this second composition is also the identity. These two identities prove the stated algebra isomorphism and both numerator formulas. □

Theorem 293.5 (One original-input open gives the entire whole fibre Hilbert–Samuel sequence)**.** *Original coherence alone supplies one actual open U meeting every origin neighbourhood and one sequence $r(q)$, such that for every $a \in U$ and every ℓ, the EXISTING whole Hilbert–Samuel value of the genuine original fibre ideal $I_a + M$ equals*

$$H_{R/(I_a+M)}(\ell) = \sum_{q<\ell+1} r(q).$$

The same U works for the entire sequence. No geometric descent or singular-resolution driver is a conclusion of this theorem.

Proof. The original-input uniform-rank producer gives the true finite complex dimension of every residue tensor fibre at every normal order, all on its same actual open. Apply the preceding genuine fibre tensor algebra isomorphism at $s = \ell+1$. Its underlying complex linear isomorphism retains that dimension. The preceding whole ideal-power equality then identifies its target with the existing whole maximal-ideal quotient that defines the original fibre ideal's Hilbert–Samuel value. Taking complex dimensions proves the equality for every ℓ, with no new open, denominator, branch, or rank certificate. □

294 Internally split actual finite original normal jets

Let $B = \mathcal{O}_{\mathbf{C}^d,0}$ and $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$ be the genuine entire holomorphic germ rings, $J = (y_1, \dots, y_t)$ the actual normal ideal, and I an arbitrary actual original ideal. Write b for the SAME internally constructed nonzero base germ of the actual original all-order truncation producer. No caller provides b, a basis, section, kernel, matrix or flatness/result certificate.

Theorem 294.1 (The actual original finite-jet surjection and whole ideal kernel)**.** *For every literal q, the actual B-linear projection $\pi_q : R/J^q \to R/(I + J^q)$ is surjective and takes EVERY actual numerator to the same original numerator class. Its entire kernel is precisely the actual image of the original I in R/J^q. This is an exact sequence of whole actual modules, not pointwise values.*

Proof. The genuine inclusion $J^q \subseteq I + J^q$ defines the actual quotient factor algebra map. Every target quotient class has its actual original numerator, which provides a preimage. If a source class maps to zero, its actual numerator g lies in $I + J^q$. Write $g = a + c$ with actual $a \in I$, $c \in J^q$; the source class is exactly the actual image of a. Conversely every actual $a \in I$ maps to zero. These entire numerator identities identify the literal whole kernel with the range of the true original ideal inclusion. □

Theorem 294.2 (The same internally selected denominator preserves the entire exact sequence)**.** *Localize ALL these genuine modules and maps at powers of the SAME actual b. The actual finite-jet map remains surjective and the entire original ideal-image sequence remains exact. Both the ambient actual finite jet module and the actual original quotient are finite free over B_b. The ambient basis is the genuinely localized actual normal monomial basis. Every true numerator/denominator pair retains its literal original numerator and the SAME denominator; $q = 0$ and empty index sets are literal.*

Proof. Use the actual original uniform denominator and its proved genuine entire nonvanishing, not a point value. The actual localization functor maps each literal true quotient map and actual original ideal inclusion, retaining numerator and denominator. Genuine localization surjectivity and exactness apply to the actual whole exact sequence above. The actual normal monomial basis of R/J^q localizes to a true basis of the ambient module over B_b; its index is the actual finite family of exponents of degree below q. The proved actual SAME-denominator original truncation theorem supplies finite freeness of the target for EVERY q. Thus all hypotheses used here are internally produced from the actual original ideal. □

Theorem 294.3 (Actual projectivity internally supplies a whole section and complementary projections)**.** *For each actual q, genuine target projectivity internally supplies a B_b-linear section s_q of the actual surjection. Put $P_q = s_q\pi_q$ and $Q_q = 1 - P_q$. Then $P_q^2 = P_q$, $Q_q^2 = Q_q$, $P_q + Q_q = 1$, $\pi_q P_q = \pi_q$, $\pi_q Q_q = 0$, and $\ker P_q = \ker \pi_q = \operatorname{im} Q_q$.*

Proof. The actual finite free target is genuinely projective. Apply its true projective lifting property to the ACTUAL surjection and its literal identity map. This internally supplies s_q and the WHOLE linear-map identity $\pi_q s_q = 1$. Compose these actual maps to get P_q. Associativity and the section identity give $P_q^2 = P_q$. For every actual ambient vector x, $Q_q x = x - P_q x$; direct linear-map algebra gives the stated complementary identities. If $\pi_q x = 0$, then $P_q x = 0$ and $Q_q x = x$, so x lies in the actual range of Q_q. Conversely $\pi_q Q_q = 0$ puts its entire range in the actual kernel. Applying π_q to $P_q x$ proves equality of both whole kernels. No section or projection identity is caller data. □

Theorem 294.4 (The actual finite matrices retain the complete original ideal-image dictionary)**.** *In the actual localized normal monomial basis, the real finite coefficient matrices of P_q, Q_q have entries in the genuine B_b and satisfy $P_q P_q = P_q$, $Q_q Q_q = Q_q$, $P_q + Q_q = 1$. The ENTIRE image of Q_q is precisely the actual localized original ideal image supplied above. Every column of its true coefficient matrix therefore has an actual original-ideal-image witness, and every vector in this entire image has an actual finite combination of these SAME columns.*

Proof. Apply the true basis coordinate equivalence to the actual linear maps. Each actual entry is the coordinate of the whole image of one true basis vector. The genuine basis-to-matrix composition dictionary carries the preceding whole linear-map identities to actual matrix identities. The true localized exact sequence and whole projection-kernel equality identify the entire range of Q_q with the actual localized ideal image. Its true finite basis expansion writes $Q_q x$ as the finite combination of its actual basis-vector images with genuine coordinates of x. Conversely all such combinations lie in the actual whole range by linearity. This supplies real coefficient identities suitable for subsequent finite analytic representative spreading; that spreading has not yet been asserted. □

Theorem 294.5 (The original coherent equation model selects its genuine finite jet splitting)**.** *Only original coherent $\mathfrak{a}, d, t$ internally supply its actual whole origin ideal I, the SAME real nonzero b, and the internally selected actual saturation N, U. At the genuine finite order $N + 1$ the whole ambient jet projection, original quotient, actual internally produced section/projections and real finite matrices have the exact dictionaries above. The actual graph quotient and the original SAME-U all-point saturation identities are retained. No fixed index or finite matrix is supplied as a conclusion certificate.*

Proof. Construct the actual whole origin ideal from true original coherent stalk members. Apply the genuine actual graph-saturation producer to these SAME original inputs, internally retaining its N, U and full original functional saturation identities. Apply the actual finite-jet constructions at literal order $N+1$ to this actual original ideal and its SAME internally selected uniform denominator. Their true quotient freeness supplies projectivity, the true original surjection supplies the section, and the actual monomial basis gives real finite coefficient matrices. Return these actual joint constructions and identities, with no external index, basis, section, projection, kernel or flatness certificate. □

Open frontier. The actual finite localized coefficient identities still require true common holomorphic representatives and an actual denominator-nonzero open before nearby analytic assertions. Even a fixed-q analytic spread does not by itself give one all-q neighborhood. The whole nearby ORIGINAL Rees/graded cone dictionary, reverse analytic initial ideal, analytic extension flatness, nearby all-degree freeness, analytic normal-flatness, higher HS and G8 remain OPEN.

295 The intrinsic algebra of the original normal filtration

Write $R = \mathcal{O}_{\mathbb{C}^{d+t},0}$ and $B = \mathcal{O}_{\mathbb{C}^d,0}$, with the actual tangent pullback giving the B-algebra structure on R. Let J be the original normal ideal and let $I \subset R$ be any original whole germ ideal. Its existing literal degree-q module is the image of J^q in $R/(I + J^{q+1})$. The algebra below is constructed from multiplication of these genuine representatives; the polynomial quotient is a subsequent theorem about that algebra. This commutative-algebra construction is the explicit intrinsic algebra adaptation needed by the relative normal-flatness and Hilbert–Samuel route; the quotient never defines the multiplication.

The source-facing objective is the whole local-ring Hilbert–Samuel quotient used in Bierstone–Milman, "Use of the Hilbert–Samuel function". The following constructions supply its relative normal-filtration algebra ingredient directly from the original germ ideals. They add the intrinsic multiplication to the preceding literal-module dictionary; they do not assert the source's semicontinuity, admissible-centre theorem, or descent under blowing up.

Definition 295.1 (Actual multiplication of original normal layers)**.** For x in degree p and y in degree q, choose actual representatives $g \in J^p$ and $h \in J^q$ of their whole quotient classes and define $x \cdot y$ to be the class of gh in the image of J^{p+q} in $R/(I + J^{p+q+1})$.

Proof. The representatives exist by the literal image-module definition. Their product lies in $J^p J^q = J^{p+q}$, hence gives an element of the stated actual layer. Independence from these representatives is proved in the following lemma. □

Lemma 295.2 (The whole original remainder makes multiplication well-defined)**.** *The actual product class is independent of the chosen whole original representatives. In particular, the product of the degree-p class of $g \in J^p$ and the degree-q class of $h \in J^q$ is exactly the degree-$(p+q)$ class of their actual product gh.*

Proof. Suppose $g - g' \in I + J^{p+1}$ and $h - h' \in I + J^{q+1}$, with each representative in its asserted original power. For $v \in I + J^{p+1}$ and $w \in J^q$, write $v = i + u$ with $i \in I$ and $u \in J^{p+1}$. Then $iw \in I$ and $uw \in J^{p+1}J^q = J^{p+q+1}$, so $vw \in I + J^{p+q+1}$. Apply this first to $(g - g')h$, and then, with the two degrees interchanged, to $g'(h - h')$. Their sum is $gh - g'h'$, which therefore belongs to the whole next-power remainder $I + J^{p+q+1}$. The two product classes coincide in exactly the original quotient that defines the layer. □

Theorem 295.3 (The literal original direct sum is an intrinsic graded algebra)**.** *The existing literal direct sum of all original normal layers is a commutative B-algebra, with the product of its degree-p and degree-q summands given by the preceding actual multiplication and lying in degree*

$p+q$. *Its unit is the actual class of* 1 *in degree zero. Its* B*-scalar action is the existing tangent-pullback action on each layer.*

Proof. The family of actual powers $J^q \subset R$, with ordinary multiplication and degree-zero unit 1, is a graded commutative ring. The true quotient map from each J^q onto its literal layer is additive and B-linear, and the preceding lemma makes these maps multiplicative for the already defined actual layer products. Every layer class has an actual power representative. Thus associativity, commutativity, the two unit laws, and both distributive laws follow by lifting the classes to these genuine power representatives and applying the corresponding identities in R. The degree-zero scalar class is the class of the actual tangent pullback of the scalar; multiplying it by a degree-q representative gives exactly the existing scalar action on the whole quotient. Extend the homogeneous products to finite sums by distributivity. All sums involved are finite, and these proved homogeneous laws give the stated commutative algebra on the original direct sum. □

Lemma 295.4 (The actual entire polynomial symbol map is an algebra homomorphism)**.** *The existing true polynomial symbol map from* $B[Y_1, \ldots, Y_t]$ *to the literal original associated module is a unital* B*-algebra homomorphism for its intrinsic multiplication. Its complete kernel is the existing whole original normal initial ideal.*

Proof. For homogeneous P of degree p and Q of degree q, their symbol classes are the actual classes of their genuine normal evaluations in their respective layers. The intrinsic product lemma identifies their product with the class of the product of these evaluations in degree $p+q$. Actual normal polynomial evaluation is an algebra homomorphism, so this is exactly the symbol of the homogeneous polynomial PQ. Write arbitrary P and Q as their finite sums of monomials, each of which is homogeneous of its actual exponent degree. Induct through these two finite sums. Additivity of the existing symbol map and the two distributive laws reduce every induction step to the already proved homogeneous product identity. This gives the same multiplicative identity for the two whole polynomials. The symbol of 1 is its actual degree-zero class, which is the intrinsic unit. A constant coefficient has the actual degree-zero tangent pullback class, proving preservation of scalars. The underlying map is the existing entire symbol map, so its previously proved complete kernel dictionary gives exactly the existing original initial ideal. □

Theorem 295.5 (The polynomial quotient is the intrinsic original associated algebra)**.** *There is a* B*-algebra isomorphism from the whole polynomial quotient by the existing original normal initial ideal onto the intrinsic original associated algebra. It sends every actual polynomial numerator to its same existing entire original symbol, retaining every homogeneous degree.*

Proof. The original homogeneous symbol surjections give surjectivity of the entire symbol map onto the literal direct sum. Its intrinsic algebra homomorphism has exactly the existing whole original initial ideal as kernel. The algebraic first isomorphism theorem therefore gives the asserted quotient isomorphism. Its defining quotient map sends every polynomial numerator to that same actual symbol. For a homogeneous numerator the original component dictionary places its symbol in its actual same-degree summand, proving the degree assertion. Since every original quotient class has a polynomial numerator, the numerator identity also proves that the underlying linear isomorphism is exactly the previous literal quotient-module dictionary. The ring and scalar structures used here were constructed before this quotient comparison from genuine original representatives. □

296 Original analytic saturation internally produces genuine finite graph-jet relations

This OWN OFFLIB continuation connects the actual whole original coherent ideal and its true graph saturation to the previously proved multiplicative finite polynomial jets. All finite generators, whole analytic coefficient relations and polynomial representatives are selected internally.

Reverse central initial-ideal inclusion, its entire-ideal closure and nearby original cone/Rees/ALL-order analytic freeness remain OPEN.

Lemma 296.1 (The actual scaling is the whole graph pullback of the original ideal)**.** *The actual entire scaling pullback agrees on EVERY original whole germ with the genuine graph trace after original-coordinate pullback. Hence its actual ideal map is exactly the already constructed original graph-scaled ideal. The actual graph parameter germ is the SAME literal parameter normal coordinate used by actual polynomial evaluation.*

Proof. Choose an actual holomorphic representative for every original whole germ. The genuine pullback representative formulas compose the actual graph embedding with the true original-coordinate projection, precisely the actual scaling map. Thus the two whole germ algebra homomorphisms agree on every representative and are equal. Their genuine ideal maps are therefore equal. The actual finite parameter index gives the identical coordinate function on the genuine graph model, so its whole parameter germ and literal normal-coordinate germ agree. No ideal-map or parameter dictionary certificate is input. □

Lemma 296.2 (Every genuine saturated graph germ has its actual finite whole original-ideal relation)**.** *For a genuine graph germ in the actual Nth parameter colon of the original scaled ideal, internally select a finite list of actual original-ideal germs and actual entire graph coefficient germs such that parameter to power N times the given germ equals their genuine scaled finite linear combination. The original germs belong to the WHOLE actual original origin ideal.*

Proof. Unfold the true actual colon membership: multiplying by the genuine parameter power belongs to the actual scaled original ideal. The preceding whole ideal dictionary identifies this ideal with the genuine scaling map of the original origin ideal. A true ideal map is the ideal span of its actual original-ideal image. Genuine span membership supplies a finite sum with whole analytic coefficient germs. Each finite image element has an actual preimage in the original origin ideal; choose those actual preimages internally. Their whole image identities yield the required genuine finite relation. Commutativity fixes the parameter multiplication order. No finite relation, generator family or result certificate is assumed, and no Noetherian finite-generation certificate is needed for this elementwise finite relation. □

Lemma 296.3 (The genuine whole relation gives an internally selected multiplicative finite polynomial relation)**.** *For every true finite relation above and EVERY finite order L, internally choose actual original-normal polynomial representatives of each original germ, actual graph-normal polynomial representatives of every coefficient and of the given graph germ. Their true whole remainders belong to the corresponding SAME normal powers. The literal finite polynomial relation $\text{parameter}^N H - \sum A_i \sigma(P_i)$ belongs to the actual graph variable ideal to power L.*

Proof. Apply genuine internally produced finite monomial representatives separately to the actual whole given graph germ, every actual whole coefficient germ and every original-ideal germ. Actual graph scaling carries the latter original remainder to the graph normal power $2L$, hence to its SAME Lth power. For each actual product, subtract its genuine polynomial-evaluation product. It is exactly coefficient remainder times scaled original germ plus polynomial coefficient times scaled original remainder; ideal absorption and additivity put it in the graph Lth normal power. Sum the finite product remainders and subtract parameter^N times the given germ's remainder. The genuine whole relation cancels the remaining entire term. Actual polynomial evaluation's algebra laws identify this expression with the entire evaluation of the displayed literal polynomial relation. The true evaluation/power coefficient dictionary puts that actual polynomial in the SAME literal variable ideal power. This is a whole analytic and multiplicative assertion, not equality of values or chosen Taylor coefficients at a point. □

Theorem 296.4 (Only the original coherent ideal supplies SAME uniform saturation and finite relations for EVERY order)**.** *Only the actual original coherent ideal internally selects the genuine*

uniform saturation index and actual open neighbourhood supplied by its whole deformation saturation model. On that SAME index, EVERY actual graph saturation element obtains a finite whole relation with original-ideal germs, and those SAME actual finite germs and entire coefficients obtain the preceding internally produced finite polynomial relation for EVERY order. The actual all-point/all-power coherent colon stabilization on the SAME open is retained.

Proof. Apply the already proved original coherent graph saturated-model producer, retaining its internally selected N, U, actual origin saturation dictionary and true all-point coherent colon stabilization. For a given actual saturation element its genuine union-membership dictionary supplies membership in the SAME Nth graph colon. The second lemma internally selects its finite original-ideal germs and whole coefficient germs ONCE, giving an exact whole relation. Apply the third lemma to that SAME relation for every finite L. This returns the asserted joint output from original inputs; no saturation index, finite whole relation, polynomial remainder or nearby normal-cone dictionary is a caller certificate. Uniform coherent saturation and ALL finite polynomial relations are not yet the reverse initial-ideal or nearby ALL-order free conclusion. □

297 Actual whole analytic reverse inclusion by genuine local-ring Krull closure

This OWN OFFLIB continuation passes the internally produced original initial symbols to the WHOLE actual analytic central ideal. It uses actual finite polynomial representatives, entire coefficient uniqueness and the true Noetherian local-ring Krull theorem on the literal quotient module. No closure or nearby normal-cone certificate is assumed. Nearby whole normal-cone identification and one ALL-order analytic free open remain OPEN.

Lemma 297.1 (Every true ideal in a Noetherian local ring is closed for a genuine proper-ideal filtration)**.** *Let R be a genuine Noetherian local commutative ring, let $J \neq R$ be an actual proper ideal and let A be ANY actual ideal. If $g \in A + J^q$ for EVERY natural q, then $g \in A$. The actual original normal ideal is proper internally, so this applies to its genuine whole analytic germ ring without a caller closedness or properness result certificate.*

Proof. Take the literal quotient module R/A, finite over the actual R. Membership in each actual ideal sum internally gives $g = a + j$ with $a \in A$, $j \in J^q$. Its literal quotient class is the class of j, equal to j times the class of one, hence lies in the actual submodule $J^q(R/A)$. Genuine Noetherian local-ring Krull intersection for the proper actual J gives intersection of these submodules zero. Thus the quotient class is zero, exactly $g \in A$. This handles $A = R$ directly through the same zero quotient module and needs no local or nontrivial-ring assumption on R/A. For the original normal ideal, its already proved whole plane-trace kernel identifies it as proper: an ideal containing one would give genuine plane trace of one zero, contradicting the nontrivial entire base germ ring. □

Lemma 297.2 (Actual finite normal representatives have true bounded support and low-degree uniqueness)**.** *For EVERY genuine whole germ g and EVERY natural q, internally produce an actual polynomial Q with $g - \mathrm{eval}(Q) \in J^q$ and no genuine coefficient of degree at least q. If two actual polynomials represent the SAME whole germ modulo J^{k+1}, their genuine degree-k homogeneous parts are equal. No representative, support or coefficient dictionary is supplied as a hypothesis to the whole-germ producer.*

Proof. The genuine finite original monomial basis gives the actual finite sum indexed by exponents of degree below q, with whole coefficient germs. Its proved whole remainder gives the asserted jet. At every exponent of degree at least q, each true monomial coefficient is zero, since none of the actual finite indices equals it. Thus the whole support bound holds, including $q = 0$ with empty index set. For two true representatives, subtract their whole remainders: the entire evaluation of their difference lies in J^{k+1}. The genuine power/low-coefficient dictionary makes every degree-k

coefficient of their difference zero. The actual homogeneous extraction coefficient formula discards all other exponents, so its whole degree-k component is zero. Actual linearity gives equality of the two true homogeneous components. □

Theorem 297.3 (Every genuine saturated central germ is approximated inside the actual analytic original initial ideal)**.** *Only the actual original coherent ideal internally supplies the SAME N, U. For EVERY genuine h in its actual graph saturation, the actual whole central germ c^*h belongs, for EVERY q, to the sum of original J^q and the genuine analytic ideal generated by the WHOLE original initial ideal. Every polynomial and original initial relation used to prove this is internally selected.*

Proof. Choose the actual bounded polynomial Q representing c^*h modulo J^q. For every $k < q$, the original-input initial-symbol producer internally gives H_k, F_k with its actual central homogeneous component equal to the genuine original initial symbol of $F_k \in I \cap J^k$. The whole central pullback carries its retained graph remainder into original $J^{N+k+1} \subset J^{k+1}$, and evaluates the SAME parameter-zero polynomial. Compare it with Q, whose remainder lies in $J^q \subset J^{k+1}$. The preceding coefficient uniqueness identifies the degree-k part of Q with that internally produced original initial relation. Hence every such component belongs to the actual WHOLE polynomial initial ideal. Components of degree at least q vanish by the true support bound. Every actual polynomial is its genuinely finite sum of actual homogeneous components, so Q belongs to this SAME original initial ideal. Actual entire evaluation then belongs to its genuine analytic ideal map. Add the chosen original J^q remainder to obtain the asserted genuine whole approximation for all q. □

Theorem 297.4 (The whole actual saturated central ideal has the genuine reverse initial-ideal inclusion)**.** *Only original coherent I internally supplies the SAME genuine N, U and actual all-point/all-power colon stabilization. The entire origin ideal of its actual selected central deformation model is contained in the genuine analytic extension of the WHOLE original normal initial ideal. No caller closure, central-fibre, initial-ideal, flattened-frame or output certificate is supplied.*

Proof. The preceding original-input producer gives all genuine normal-power approximations of each actual central image of a true graph saturation element. The actual whole analytic germ ring is Noetherian and local internally, and its genuine normal ideal is proper by the first lemma. Apply that actual quotient-module Krull closure to the internally constructed analytic ideal map of the WHOLE original initial ideal. Thus every true graph saturation image belongs to that genuine analytic ideal. In particular every image of the selected Nth graph colon does. Genuine ideal-map span membership extends this whole elementwise assertion to the entire image ideal. The existing exact original central-model origin dictionary identifies this image ideal with the whole origin ideal of the SAME actual central deformation model. Retain the actual open and uniform colon outputs. This is the genuine origin reverse inclusion; pairing it with forward inclusion and spreading a finite whole original cone dictionary near the centre remain subsequent actual producers. □

298 An actual whole finite analytic model for the original central sheaf near the origin

This OWN OFFLIB continuation spreads the genuine whole origin central equality through actual finite coherent presentations. The finite original homogeneous relations and every whole analytic representative are internally selected. Arbitrary whole representatives are not declared literal fibre polynomials; the nearby original normal-cone dictionary and one all-order free neighbourhood remain subsequent mathematical producers.

Theorem 298.1 (Original coherent input supplies an exact whole finite model at every nearby central stalk)**.** *Only original coherent I internally supplies SAME genuine N, U and an actual open A containing the origin. The selected original central ideal is coherent, every actual finite original initial-relation evaluation representative is holomorphic on A, its whole functional span is actually finitely generated there, and at EVERY $a \in A$ that WHOLE span is equivalent to original central-ideal membership for EVERY function germ. All genuine original colon, parameter-power and saturation dictionaries on SAME U are retained.*

Proof. The preceding original-input whole ideal equality chooses genuine N, U, coherent selected colon and central ideals, all-point/all-power dictionaries, and whole equality between selected origin central ideal and analytic extension of the entire original polynomial initial ideal. The original finite homogeneous-relation producer selects a genuine finite set generating this whole initial ideal. Its actual whole analytic evaluation representatives have exactly their original evaluation germs. The true finite-span origin dictionary therefore identifies its entire functional span with membership in the analytic initial ideal. For central membership, original coherence internally supplies the germ's genuine holomorphicity through its actual finite presentation; apply the same origin dictionary in this direction as well. The whole ideal equality thus gives an actual equivalence of BOTH entire functional membership predicates at the origin, without any chosen representative, index matching or fibre-equality hypothesis.

The finite source model has a genuine common open of holomorphic generators and actual finite generation at the origin. Original central coherence supplies its genuine finite presentation there. The already proved actual coherent equality-spreading theorem uses the finitely many actual generator memberships in BOTH directions, and genuine common shrinking, to produce one open W containing the origin on which the two WHOLE membership predicates are equivalent at EVERY point for EVERY germ. Intersect W with the genuine common holomorphic-source open to obtain A. Retain the SAME internally chosen N, U, all original saturation dictionaries, generator holomorphicity and source finite generation. No all-order normalcone or free-frame certificate is introduced. □

Theorem 298.2 (The same exact whole finite model retains original all-power dictionaries at every actual central image)**.** *Only original coherent I internally supplies SAME N and one genuine V containing the origin, on which the entire selected original central ideal equals the actual finite original relation-evaluation span at EVERY point and for EVERY germ. At EVERY actual original central image of a point of SAME V, all genuine original colon indices $N + q$ stabilize, all original parameter-power membership dictionaries hold, and the WHOLE original saturation union is exactly the selected Nth colon.*

Proof. Internally choose N, U, A from the preceding original-input producer. Form the actual intersection $V = A \cap c^{-1}(U)$ using the actual original central inclusion c. Its genuine holomorphicity gives continuity, hence this is an actual open; its true value at the origin is zero, so the origin remains. Restrict every whole finite model, coefficient-holomorphicity and source finite-generation field to V. For every $a \in V$, its actual central image is genuinely in U, so specialize every all-point/all-power original colon, parameter-power and saturation-union field at this SAME image. All quantities are the SAME original model, SAME finite relation family and SAME internally chosen N; no nearby zero-origin-germ homomorphism, caller normalcone, saturation or conclusion certificate is assumed. These exact whole nearby central sheaf dictionaries still do not identify the nearby original graded normal cone with a literal homogeneous polynomial model. □

299 Whole actual original central ideal equality and its literal quotient

This OWN OFFLIB continuation consumes the genuine whole reverse inclusion, never a caller central-fibre or result certificate. It identifies the entire original central ideal at the origin with

the analytic extension of its entire polynomial initial ideal. It does not identify nearby original normal cones or produce one all-order free neighbourhood.

Lemma 299.1 (Every actual selected central colon has the genuine whole reverse inclusion)**.** *Only original coherent I internally supplies, for EVERY natural selection index M, the whole inclusion of the actual origin ideal of the original central Mth colon model into the genuine analytic extension of the WHOLE original normal initial ideal.*

Proof. The preceding original-input Krull producer internally supplies membership of the actual central pullback of EVERY genuine element of the union of all true graph colons in the same whole analytic initial-ideal extension. Every element of the actual Mth graph colon is in this union, using its literal index M. The exact original central origin dictionary identifies its entire ideal with the actual ideal image of that same graph colon. Genuine ideal-map membership, extended across finite ideal sums, gives the entire stated inclusion. No stabilization index is supplied as a hypothesis and no comparison between two independently chosen indices is needed. □

Theorem 299.2 (Only original coherent data produce one actual central ideal equal to the entire analytic original initial ideal)**.** *Only original coherent I internally supplies a SAME genuine N, U, with U open and containing the origin, actual coherent selected colon and central ideals, all-point/all-power colon stabilization on U, true parameter-power and saturation dictionaries there, and equality of the entire origin ideal of its selected central deformation model with the analytic extension of the WHOLE original normal initial ideal.*

Proof. The genuine original forward-inclusion producer internally selects its true N, U and supplies the entire analytic original initial ideal contained in the selected whole central origin ideal, along with coherent ideals and all the stated genuine nearby colon and parameter-power dictionaries. Apply the preceding reverse inclusion at this SAME internally selected N. It applies to every actual graph colon index, so none of its premises require caller index identification, saturation or fibre equality. The two true whole ideal inclusions give equality by antisymmetry. Retain the original producer's SAME N, U and all genuine all-point/all-power outputs. Equality here is at the actual origin and does not turn these saturation dictionaries into a nearby original normal-cone identification. □

Theorem 299.3 (The actual selected central quotient is the literal whole analytic original initial quotient)**.** *Only the original coherent ideal internally produces the SAME genuine N, U, coherent selected central model, actual whole origin equality, and a genuine ring equivalence between its entire analytic initial quotient and its selected central origin quotient. This equivalence sends the literal quotient class of EVERY whole analytic germ to the class of that SAME germ.*

Proof. Choose the actual N, U and the whole ideal equality internally from the preceding theorem. The genuine quotient equivalence of equal ideals in the actual same analytic ring has these two literal quotient rings as source and target. Its actual quotient-class formula sends every original germ's class to the class of exactly the same germ; no representative, basis or inverse certificate is supplied. Retain the SAME open, coherent central ideal and genuine origin equality. This is an actual analytic quotient equivalence at the origin, rather than a polynomial quotient, a nearby fibre identification or an all-order analytic free frame. □

300 Actual original initial symbols of genuine saturated central graph jets

This OWN OFFLIB continuation consumes only internally constructed finite analytic reconstruction and whole diagonal coefficients. It constructs genuine original-ideal initial symbols for every degree of a saturated graph germ. Whole analytic central-ideal equality, Krull closure, nearby original normal cone and one ALL-order analytic free open remain OPEN.

Lemma 300.1 (True whole analytic graph scaling reflects actual odd normal powers)**.** *For EVERY whole original analytic germ g, actual $\sigma(g)$ belongs to graph normal power $2k+1$ if and only if g belongs to original normal power $k+1$. This is an actual whole germ assertion, without a caller jet or coefficient certificate.*

Proof. The forward implication internally selects a genuine polynomial representative P of g modulo original J^{k+1}. Its true graph-scaled remainder belongs to graph J^{2k+2} and hence to graph J^{2k+1}. Subtract it from the given genuine scaled germ membership; the entire evaluation of $\sigma(P)$ belongs to graph J^{2k+1}. Genuine polynomial evaluation's power dictionary and the already proved odd diagonal coefficient dictionary give $P \in (Y)^{k+1}$. Its whole evaluation lies in original J^{k+1}; adding the internally selected remainder gives actual whole g membership. Conversely true graph scaling carries original J^{k+1} to graph J^{2k+2}, hence to the desired odd power. Every representative and remainder is produced internally from the original whole germ. □

Lemma 300.2 (The genuine central graph pullback evaluates the SAME whole central polynomial)**.** *The actual graph central pullback fixes every entire base germ, sends the SAME graph parameter to zero and every SAME genuine fibre germ to its original normal coordinate. On EVERY polynomial H it therefore satisfies $c^*\,\mathrm{eval}(H) = \mathrm{eval}(C(H))$, where C is the actual parameter-zero polynomial map. It carries the actual graph normal Lth power into the original normal Lth power. The internally defined degree-k central part is genuinely homogeneous of degree k for the original polynomial grading.*

Proof. Take true holomorphic representatives. The actual central inclusion has its proved whole tangent-coordinate identity, parameter-zero identity and every genuine fibre identity. Composition with representatives gives the three asserted whole germ dictionaries, including arbitrary entire base coefficients. Polynomial algebra extensionality on these genuine coefficients and variables gives the entire evaluation identity. The actual graph normal ideal is generated by these SAME coordinate germs: the parameter image is zero and fibre images are original generators. Its true ideal-map inclusion, genuine powers and elementwise ideal-map membership give preservation of every actual normal power. The actual central degree-k part is the weighted component for uniform weight one. The whole coefficient predicate says its nonzero exponents have degree k, giving genuine homogeneous membership in the original grading. No central pullback or homogeneous conclusion certificate is supplied. □

Theorem 300.3 (Every saturated central degree internally gives a true original-ideal initial relation)**.** *Only actual original coherent I internally supplies the SAME saturation N, U. For EVERY genuine saturated graph germ h and EVERY natural k, internally produce a true graph polynomial H representing h modulo graph J^{N+k+1} and a genuine $F \in I \cap J^k$ such that the actual degree-k parameter-zero part of H is EXACTLY the original initial symbol of F. Hence this entire base-coefficient homogeneous polynomial belongs to the actual WHOLE original initial ideal. All-point/all-power coherent saturation on the SAME U is retained.*

Proof. The genuine original-input analytic reconstruction internally gives H and whole $F \in I$ with actual $\sigma(F) - \mathrm{eval}(U^k \pi_{-k} H)$ in graph J^{2k+1}. The proved true diagonal central homogeneous congruence identifies the latter term modulo that SAME power with actual $\sigma(\mathrm{eval}(C_k H))$. Thus $\sigma(F - \mathrm{eval}(C_k H))$ belongs to graph J^{2k+1}. The first lemma gives the WHOLE original remainder in J^{k+1}. The preceding genuine homogeneity gives $\mathrm{eval}(C_k H) \in J^k$, hence $F \in J^k$. Its internally constructed original initial symbol is homogeneous of degree k and represents the SAME actual F modulo J^{k+1}. True homogeneous representative uniqueness makes it EXACTLY $C_k H$. Since $F \in I$, its genuine original relation belongs to the actual whole initial ideal. Retain H's actual remainder and the genuine uniform saturation outputs. This identifies every actual finite central degree; it does not yet pass to the entire analytic ideal by Krull closure or spread the whole initial ideal to nearby original stalks. □

301 Literal normal-polynomial functions for the whole original nearby central model

This OWN OFFLIB continuation replaces the arbitrary whole evaluation representatives by genuine finite normal-polynomial functions with the SAME genuine coefficient functions. All original data and finite families are selected internally. Nearby original normal-cone identification and one all-order free open remain OPEN.

Lemma 301.1 (Actual normal monomials are entire functions with their true whole germs)**.** *For an actual exponent α, the literal function on the original ambient model is the product of its actual normal coordinates raised to the entries of α. It is holomorphic everywhere and its whole origin germ is the SAME product of the actual original normal coordinate germs, including the empty product when there are no normal coordinates.*

Proof. Each actual coordinate is a genuine continuous complex linear functional, hence complex differentiable at every point. Its natural power remains differentiable, and the genuinely finite product is differentiable everywhere. The whole germ map preserves each finite product and natural power; the germ of each actual coordinate is exactly its defined original normal coordinate. Thus its true whole germ equals the SAME actual germ product. Empty products are literal one, so no dimension, nonzero or invertibility certificate enters. □

Lemma 301.2 (Literal finite polynomial functions have genuine whole evaluation and actual specialization dictionaries)**.** *For EVERY actual polynomial P over entire base holomorphic germs, define its ambient function as the finite sum of its actual coefficient representatives composed with original tangent projection, times the literal actual normal monomial functions. Its whole origin germ is exactly original normal-polynomial evaluation. At EVERY ambient point its value is evaluation at the actual normal coordinates of the genuine coefficient-function specialization at that point's true tangent projection. On every genuine coefficient-holomorphic open it is genuinely holomorphic over the actual projection preimage.*

Proof. The support is genuinely finite. Each actual coefficient representative has the SAME entire base germ, and true tangent-projection pullback produces its SAME original ambient coefficient germ. Multiply by the preceding genuine monomial germs and sum over actual finite support. The actual finite monomial expansion of P and the whole evaluation ring map identify this whole germ with its true original evaluation, rather than a point value or truncated jet. At every actual point, the genuine specialization coefficient formula gives exactly the same coefficient function values; the finite polynomial monomial expansion gives the pointwise equality without mapping all origin germs to a nearby stalk. Over an actual coefficient-holomorphic open, genuine complex chain rule, finite products and sums give actual differentiability throughout its projection preimage, hence genuine holomorphicity there. □

Lemma 301.3 (A true finite original relation family has ONE whole functional comparison open)**.** *The internally selected finite original homogeneous relation family shares ONE genuine open containing the origin, on which its literal normal-polynomial functions equal the SAME previously selected entire analytic evaluation representatives for EVERY relation and at EVERY point. Their entire finite spans are consequently equivalent at EVERY point of this SAME open and for EVERY germ.*

Proof. Both actual whole origin germs of each relation's functions equal the SAME original analytic evaluation by the preceding lemma and the existing exact original representative dictionary. Genuine whole germ equality gives an eventual equality of these functions near the origin. Because the internally selected relation family is truly finite, intersect these actual equality neighbourhoods to obtain ONE genuine open on which every relation equality holds throughout. At any point of this open, the SAME open is a true neighbourhood. Every finite coefficient relation witnessing

span membership on one family therefore witnesses membership in the other family: restrict its true eventual equality to this actual neighbourhood and replace each of the finitely many actual generator values by its equal value. Retain the whole holomorphic germ of the tested function and of every multiplier. This proves BOTH membership directions for ALL germs, without caller family identities or generator dictionaries. □

Theorem 301.4 (Only original coherent input gives its whole nearby central ideal by literal homogeneous coefficient polynomials)**.** *Only original coherent I internally supplies SAME N and ONE actual open containing the origin. At EVERY point of this open, original selected central membership for EVERY whole germ is equivalent to the finite span of the literal normal-polynomial functions of the SAME finite original homogeneous relations. Every true coefficient function is holomorphic at the actual tangent projection throughout this open, the original homogeneous degree and specialization dictionary are retained, and the SAME true all-power original colon/parameter/saturation dictionaries hold at every actual central image.*

Proof. Choose the SAME genuine selected central model and its whole finite-span equality open internally from the preceding original-input central finite-model producer. Intersect with the genuine finite functional comparison open just constructed and the actual tangent-projection preimage of the common coefficient-holomorphic open of the SAME internally selected original relation family. The origin remains because actual tangent projection sends zero to zero; continuity makes the intersection genuinely open. The preceding full span equivalence and the retained exact original finite model dictionary identify original selected central membership with the entire literal polynomial-functional span at EVERY point and for EVERY germ. Genuine coefficient holomorphicity, the actual pointwise specialization formula and original homogeneous relation degrees persist on this SAME open. Restrict every original all-power colon, parameter-power and saturation-union dictionary to its actual central images. No arbitrary original-germ-to-nearby-germ map or nearby normal-cone equality is assumed. Identifying this genuine homogeneous analytic model with the WHOLE nearby original graded ideal is the next independent producer. □

302 Actual diagonal coefficients and genuine central homogeneous jets

This OWN OFFLIB continuation identifies the actual coefficients of the literal substitution $Y_i \mapsto UV_i$. Every statement uses whole base germs. It supplies the missing genuine diagonal dictionary for the already constructed original-ideal analytic reconstruction. Whole analytic central-ideal equality, Krull closure, nearby normal cone and one ALL-order free open remain OPEN.

Lemma 302.1 (The genuine diagonal exponent is internally injective with exact degree)**.** *For an original normal exponent α, internally define $\Delta\alpha$ by parameter exponent $|\alpha|$ and fibre exponents α_i. These are the literal genuine graph exponents. The map is additive and injective, has actual total degree $2|\alpha|$, and weight zero.*

Proof. Form the actual finite sum of the parameter singleton with exponent $\sum_i \alpha_i$ and the genuine fibre singleton exponents. At the parameter, every genuine fibre singleton vanishes; at fibre i, the parameter singleton vanishes and the actual finite sum equals α_i. Fibre coordinate equality therefore proves injectivity. The true finite sum defining total degree splits into the unique parameter and the actual fibre indices, giving $|\alpha| + |\alpha| = 2|\alpha|$. Additivity follows from the genuine pointwise formulas. The previously proved integer-weight formula gives $|\alpha| - |\alpha| = 0$. No coefficient or injectivity certificate is input. □

Theorem 302.2 (The entire genuine substitution has exactly its diagonal coefficients)**.** *For EVERY actual polynomial P over the whole base germ ring, its true substitution has coefficient at $\Delta\alpha$ equal to the actual coefficient of P at α, and every coefficient outside the actual diagonal*

vanishes. Consequently $\sigma(P)$ belongs to graph variable power $2k$ exactly when P belongs to original variable power k, and belongs to graph power $2k+1$ exactly when P belongs to original power $k+1$.

Proof. On each genuine monomial, the actual substitution algebra map sends its coefficient to the same base coefficient and each variable to UV_i. The finite product monomial formula gives precisely the monomial with exponent $\Delta\alpha$ and the SAME whole coefficient. Every polynomial is its actual finite sum of monomials. The preceding true exponent injectivity makes exactly one coefficient contribute at a genuine diagonal exponent; no monomial contributes away from the diagonal. Thus the whole coefficient dictionary holds. Apply the genuine variable-power low-coefficient dictionary and the exact degree $2|\alpha|$. If the substituted polynomial lies in graph power $2k$, each original coefficient of degree below k vanishes. Conversely every graph exponent of degree below $2k$ is either off the diagonal with coefficient zero, or is $\Delta\alpha$ with original degree below k and therefore has coefficient zero. For odd power $2k+1$, the identical argument uses the precise natural-number equivalence $2|\alpha| < 2k+1$ if and only if $|\alpha| < k+1$; this internally gives the stated genuine odd-order dictionary. □

Theorem 302.3 (The selected negative weight is the actual central homogeneous jet)**.** *Internally set the parameter to zero in an actual graph polynomial H and take its genuine fibre homogeneous component of degree k, denoted $C_k(H)$. Then*

$$U^k \pi_{-k} H - \sigma(C_k(H)) \in (U,V)^{2k+1}.$$

The polynomial $C_k(H)$ is internally constructed from the SAME actual H, not a central homogeneous output hypothesis.

Proof. Expand the genuine finite monomials of H. Actual weight selection retains exactly exponents with parameter degree a and fibre degree $b = a+k$. Multiplication by U^k gives parameter degree $a+k=b$ and total graph degree $2b$. If $a=0$, the fibre degree is exactly k, and the monomial is precisely the genuine substitution of its actual parameter-zero degree-k fibre monomial. If $a>0$, its total degree is $2(a+k) > 2k$, so that entire monomial lies in the genuine variable power $2k+1$. Monomials of other weights are discarded by both actual selections; actual parameter-zero homogeneous extraction retains precisely the $a=0, b=k$ cases. Finite ideal additivity proves the displayed whole congruence, including $k=0$, no fibre variables and zero coefficients. Combining it with the actual original-ideal analytic reconstruction is a subsequent consumer; the whole analytic central-ideal and nearby cone claims are not assumed here. □

303 Actual original analytic reconstruction of every finite central-weight jet

This OWN OFFLIB continuation consumes internally selected finite original analytic relations and genuine weight cancellation. It constructs whole analytic elements of the original ideal whose actual scaled jets match the selected central-weight term. Whole analytic central-ideal equality, diagonal coefficient identification, Krull closure, nearby original cone and one ALL-order free open remain OPEN.

Lemma 303.1 (Reconstructed coefficients multiply true whole original-ideal germs)**.** *For the actual finite original-ideal germ family and actual internally reconstructed coefficient polynomials, the finite sum of their genuine entire evaluations times the original germs belongs to the WHOLE original ideal. Its genuine scaling is the literal finite scaled sum.*

Proof. Each actual coefficient polynomial evaluates to a whole original analytic germ. Genuine ideal absorption puts its product with each actual original-ideal germ in that SAME original ideal. Finite ideal additivity gives the whole sum. The true scaling algebra homomorphism preserves

every actual product and finite sum, giving the asserted entire scaling identity. No original-ideal membership or scaling output certificate is input. □

Theorem 303.2 (Every genuine finite analytic relation internally produces the original-ideal central-weight jet)**.** *From an exact genuine finite relation $u^N h = \sum a_i \sigma(g_i)$ with every g_i in the actual whole original ideal, for EVERY natural k internally construct a true graph polynomial H representing h modulo graph normal power $N + k + 1$ and an actual whole original-ideal germ F such that*

$$\sigma(F) - \operatorname{eval}\big(U^k \pi_{-k} H\big) \in J^{2k+1}_{\text{graph}}.$$

Proof. Choose the actual finite polynomial jet relation at the internally specified order $L = N + k + 1$. The previously proved genuine weight selection and literal parameter cancellation construct coefficient polynomials B_i and give $U^k \pi_{-k} H - \sigma(\sum B_i P_i)$ in the graph variable power $L+k-N = 2k+1$. The whole evaluation/power dictionary transfers this true congruence to actual analytic germs. Construct $F = \sum \operatorname{eval}(B_i) g_i$; the preceding lemma gives true whole original-ideal membership. Each original remainder $g_i - \operatorname{eval}(P_i)$ belongs to actual original J^L, so its true graph scaling belongs to graph J^{2L}. Multiply by the actual scaled coefficient germ and sum; the resulting entire difference between $\sigma(F)$ and the actual substituted polynomial evaluation is in graph J^{2L}, hence in graph J^{2k+1}. Combine with the genuine cancelled polynomial congruence. The same internally selected H retains its actual remainder statement for h. All objects are whole analytic germs and entire polynomial evaluations. □

Theorem 303.3 (Only the original coherent ideal supplies uniform saturation and EVERY reconstructed analytic jet)**.** *Only the actual original coherent ideal internally supplies the SAME genuine saturation index and open neighbourhood, actual all-origin and all-nearby coherent colon stabilization, and for EVERY actual graph saturation germ and EVERY natural k the preceding original-ideal entire analytic reconstruction. No finite relation, coefficient family, selected weight or analytic reconstructed output is a caller certificate.*

Proof. Apply the genuine original-input finite-relation producer. Retain its internally selected actual N, U and all-point/all-power colon dictionary. Every actual saturation element gives an exact whole finite relation with internally selected original-ideal germs and whole coefficient germs. Apply the preceding analytic reconstruction to that relation for each k. Every claimed output is constructed from original inputs and genuine saturation membership. The resulting finite analytic congruences do not by themselves identify every central homogeneous coefficient with an original initial form or close the whole analytic central ideal; those steps remain explicit subsequent producers. □

304 Actual parameter-power cancellation in true finite graph-jet relations

This OWN OFFLIB continuation applies actual integer-weight selection and natural-power reconstruction to genuine finite graph relations modulo actual normal powers. Parameter cancellation uses literal polynomial coefficient shifting, not a parameter inverse or a saturation certificate. Reverse whole analytic initial-ideal inclusion, Krull closure, nearby original cone/Rees and one ALL-order analytic free open remain OPEN.

Lemma 304.1 (Actual homogeneous multiplication shifts genuine weight extraction)**.** *For a genuine weighted homogeneous polynomial Q of weight m, actual finite extraction satisfies $\pi_w(AQ) = \pi_{w-m}(A)Q$. In particular the genuine parameter power has actual weight N, and $\pi_{N-k}(U^N H) = U^N \pi_{-k}(H)$ for every actual polynomial H and natural N, k.*

Proof. Compare all true coefficients using finite antidiagonal multiplication. When the actual coefficient of Q vanishes both corresponding terms vanish. Otherwise its exponent has genuine

weight m. For every actual pair summing to the target exponent, actual weight additivity gives $\mathrm{weight}(\alpha) = w - m$ exactly when the target has weight w. Thus the two genuine coefficient-selection predicates agree, giving the whole product identity. The actual parameter variable has weight one, and genuine graded powers give weight N for its Nth power. Apply the product identity, commutativity and the literal integer subtraction identity; all coefficients remain actual entire base germs. □

Lemma 304.2 (Actual polynomial parameter powers cancel with a true jet-order loss)**.** *If $U^N Q$ belongs to the actual graph variable ideal to power M and $N \le M$, then Q belongs to that SAME ideal to power $M - N$. Also multiplying any element of its Lth power by U^k gives an element of its genuine $(L+k)$th power. No domain or nonzero parameter hypothesis is needed.*

Proof. The true parameter power is the literal monomial with parameter exponent N and unit coefficient. Its genuine product coefficient at that exponent plus α is precisely the actual coefficient of Q at α. The actual total degree of that shifted exponent is $N + |\alpha|$. If $|\alpha| < M - N$, the genuine whole power coefficient dictionary makes this coefficient vanish. The same dictionary gives Q in the $(M-N)$th power. The parameter generator lies in the actual variable ideal, so its kth power lies in the genuine kth ideal power; actual product ideal membership and the true power-addition identity prove the second assertion. This includes zero coefficient rings, $N = 0$, $M = N$ and no fibre variables. □

Theorem 304.3 (The genuine finite relation internally produces its actual reconstructed diagonal congruence)**.** *For any genuine finite relation $U^N H - \sum A_i \sigma(P_i)$ in the actual graph variable ideal to power L, internally construct $B_i = \mathrm{forget}(\pi_{N-k} A_i)$ and $Q = \sum B_i P_i$. Whenever $N \le L + k$, the true polynomial*

$$U^k \pi_{-k} H - \sigma(Q)$$

lies in the actual graph variable ideal to power $L + k - N$.

Proof. The actual weighted component of the original relation belongs to its SAME Lth power, by genuine power preservation. Multiply by the genuine parameter power U^k, obtaining actual power $L + k$. Use the first lemma on its $U^N H$ term and the existing internally reconstructed finite-relation identity on its finite product sum. Actual linearity and ring distributivity identify this whole polynomial with $U^N(U^k \pi_{-k} H - \sigma(Q))$. Apply the preceding literal parameter cancellation to this actual polynomial membership. Every coefficient polynomial B_i is actually constructed from the given true finite coefficient polynomial; no reconstructed congruence or parameter-divisibility certificate is supplied. This polynomial statement will next be consumed with the internally selected original analytic finite relations; it alone does not yet identify the original analytic central ideal or nearby normal cone. □

305 Actual integer-weight components of whole graph jets

This OWN OFFLIB continuation uses the actual whole graph scaling $(x, u, v) \mapsto (x, uv)$ and actual polynomial/germ jet algebra. Integer weights assign 1 to the genuine parameter and -1 to every genuine fibre variable. They select actual finite polynomial coefficients, never a formal-series surrogate or a supplied diagonal-weight certificate. Reverse saturated-central initial-ideal inclusion, its entire-ideal passage, nearby original cone/Rees identification and one common ALL-order analytic free open remain OPEN.

Lemma 305.1 (Every actual graph-substituted polynomial has true integer weight zero)**.** *The actual integer-weight function has parameter weight 1 and every fibre weight -1. Every entire-base-coefficient polynomial under the literal actual substitution $Y_i \mapsto UV_i$ is genuinely weighted homogeneous of weight zero. In particular every coefficient at an exponent of nonzero integer weight vanishes.*

Proof. The actual parameter and fibre indices are distinct, by their literal finite values zero and $1+i$. Thus their actual weights are 1 and -1. Their product has genuine weight zero, using the actual graded polynomial multiplication. Entire base constants have weight zero. Polynomial induction on constants, sums and multiplication by original variables, followed by the actual algebra-homomorphism identities, proves weight zero for every actual substituted polynomial. The definition of weighted homogeneity then forces each coefficient of nonzero integer weight to vanish. No support, exponent or homogeneous-output certificate is input. □

Lemma 305.2 (Actual integer-weight extraction commutes with multiplication by the genuine graph-scaled polynomial)**.** *For every actual graph polynomial A, actual original polynomial P and integer w, the true finite weighted component satisfies*

$$\pi_w(A\,\sigma(P)) = \pi_w(A)\,\sigma(P).$$

All coefficients are genuine whole base germs. Finite sums of such actual products satisfy the corresponding literal extraction identity.

Proof. Compare every actual coefficient of the two polynomials. Genuine polynomial multiplication is the finite antidiagonal convolution over pairs $\alpha+\eta=\delta$. If the actual coefficient of $\sigma(P)$ at η vanishes, both terms vanish. Otherwise the preceding internal producer gives $\mathrm{weight}(\eta)=0$, so additivity of the actual integer-weight homomorphism gives $\mathrm{weight}(\alpha)=\mathrm{weight}(\delta)$. Hence the two genuine coefficient-selection predicates agree term by term. The finite antidiagonal sums agree, proving the whole polynomial identity. The actual weighted component is a base linear map, so finite sum extraction follows from its genuine additivity. This proof uses full coefficients and actual convolution, not a value at a point. □

Lemma 305.3 (Actual weight extraction preserves every literal normal-power jet congruence)**.** *Every actual finite integer-weight component of a polynomial in the literal parameter/fibre variable ideal to power L lies in that SAME power. Therefore whole-polynomial congruence modulo that true power is preserved by every integer-weight component.*

Proof. The proven entire coefficient-separation dictionary characterizes the actual variable ideal to power L by vanishing of every coefficient of total normal degree below L. Weighted component extraction either retains that same coefficient or replaces it by zero. Every required low coefficient therefore still vanishes, and the SAME genuine power dictionary gives membership. Apply this to the actual polynomial difference; the genuine linear component map preserves subtraction, proving congruence. This holds for every integer weight and $L=0$, including no fibre coordinates. □

Theorem 305.4 (The actual whole scaled germ has an internally produced finite weight-zero representative)**.** *For every actual original entire germ and finite graph-jet order, internally produce a genuine finite parameter/fibre polynomial of weight zero representing its actual whole graph-scaled jet. Every other genuine representative of that SAME whole jet has each nonzero-weight component in the actual jet ideal power.*

Proof. The actual whole graph-jet producer selects a genuine finite original polynomial and its actual diagonal substitution, with an entire analytic remainder in the literal graph normal power. The first lemma internally proves that substituted polynomial has integer weight zero. This gives the genuine finite whole-jet representative. For another true polynomial representative, subtract the two whole evaluated polynomials. Both actual whole remainders lie in the same normal power, so their difference does. The genuine entire polynomial-evaluation power equivalence gives membership of the polynomial difference in the literal variable ideal power. The preceding preservation lemma applies. The selected representative has zero component at every nonzero weight, so the other representative's whole component lies in that same power. This proves a genuine finite-jet diagonal-weight dictionary; no reverse initial-ideal or nearby ALL-order conclusion is inferred. □

306 Actual parameter/fibre scaling acts on entire multiplicative normal jets

This OWN OFFLIB producer supplies genuine finite polynomial parameter/fibre jets of the actual graph-scaled original germ. It is a prerequisite for the reverse saturated-central initial-ideal dictionary. That reverse dictionary, nearby original whole normal cone/Rees and one ALL-order analytic free open remain OPEN. No caller scaling, coefficient, jet, kernel or result certificate occurs.

Lemma 306.1 (Original graph geometry supplies the actual entire base-algebra scaling)**.** *The genuine original graph embedding followed by actual original-coordinate projection sends (x, u, v) to (x, uv). Its actual whole holomorphic germ pullback is a genuine base algebra homomorphism. It fixes every entire base germ and sends each original normal coordinate to the actual product of parameter and corresponding fibre coordinate.*

Proof. Use the previously constructed actual graph embedding, its true holomorphy and origin identity. Compose its actual whole germ pullback with the actual original-coordinate projection pullback. The exact pointwise tangent and normal graph equations give the stated identities on coordinate germs. On every entire base germ, use its actual holomorphic representative and the exact tangent-projection square to prove equality of entire germs; not merely equality of values. Thus the actual ring homomorphism commutes with the genuine entire base algebra maps, yielding the true base algebra homomorphism internally. □

Lemma 306.2 (Actual original polynomial scaling is the literal parameter/fibre substitution)**.** *The actual graph pullback of every whole original evaluated polynomial equals entire evaluation of the literal polynomial substitution $Y_i \mapsto UV_i$. All coefficients are the SAME entire base germs. Every literal monomial has parameter exponent equal to its original normal exponent sum and the corresponding actual fibre monomial.*

Proof. Construct the actual polynomial base algebra homomorphism by evaluating each variable at the literal product of parameter variable and corresponding fibre variable. The whole analytic homomorphism identity follows from genuine base-algebra polynomial extensionality, using its actual coefficient identity and every true coordinate-product identity from the preceding lemma. For a monomial the actual finite product substitution gives the product of $(UV_i)^{\alpha_i}$ times its whole base coefficient. Finite power-product laws separate the genuine parameter factor $U^{\sum \alpha_i}$ and the actual fibre monomial. No coefficient or diagonal-weight certificate is input. □

Lemma 306.3 (Actual scaling carries the whole original normal power into the doubled graph normal power)**.** *The actual graph scaling sends every entire original J^q germ into the genuine graph parameter/fibre normal ideal to power $2q$. Thus every genuine finite original normal-polynomial representative has its actual scaled whole remainder in that literal doubled power.*

Proof. Every actual original normal generator maps to the product of the genuine parameter generator and one fibre generator, both in the graph's literal normal ideal. Their product belongs to its actual square. Mapping the entire original ideal generated by those generators therefore lies inside that square. Genuine ideal-map/power identities carry its actual qth power into the actual $(J^2_{u,v})^q = J^{2q}_{u,v}$, including q=0 and empty coordinate blocks. Apply this to the genuine entire remainder of the internally produced finite original monomial representative; no point-value or formal-series replacement is made. □

Theorem 306.4 (Every entire original germ has its genuine finite diagonal graph jet)**.** *For every actual original whole germ and every real finite graph-jet order L, internally produce a genuine finite original normal polynomial representative and the exact literal polynomial model of its actual graph-scaled entire L-jet. Its true parameter/fibre polynomial is the preceding diagonal*

substitution, so all coefficient products are genuine operations in the actual truncated polynomial algebra.

Proof. Choose the actual finite original monomial representative modulo J^L. Its true scaled remainder lies in $J^{2L}_{u,v} \subseteq J^L_{u,v}$. The genuine graph truncated polynomial algebra equivalence therefore identifies the actual scaled whole jet with the actual polynomial-evaluation jet of its diagonal substitution. The proved exact polynomial evaluation formula gives precisely that literal quotient polynomial class. Since the whole jet is an actual algebra homomorphism, every finite sum/product/convolution identity used in later weight extraction refers to genuine entire germ multiplication and genuine literal polynomial quotient operations. All representatives and remainders are internally produced; the only input is the actual original whole germ and dimensions/order. Reverse central initial-ideal inclusion and uniform nearby cone/freeness are still separate unproved steps. Formal registration/native target/publication await root recovery. □

307 Actual finite reconstruction of integer-weight graph coefficients

This OWN OFFLIB producer reconstructs actual finite selected graph coefficients as genuine original normal polynomials, with whole base-germ coefficients. It uses the actual finite indices, integer-weight selection and literal graph substitution. It introduces no original-ideal result, weight, reconstruction or free-basis certificate. The reverse saturated-central initial ideal, entire-ideal closure, nearby original cone/Rees dictionary and ALL-order analytic free open remain OPEN.

Lemma 307.1 (Actual parameter forgetting is a genuine whole-base polynomial algebra map)**.** *Internally construct the actual whole-base algebra map sending the true parameter variable to one and each true fibre variable to its corresponding original normal variable. Its composition with the genuine diagonal graph substitution is the literal identity on EVERY original polynomial. Hence that genuine diagonal substitution is injective.*

Proof. At the literal parameter finite index set the value one. At every other finite index use its actual positive index minus one, whose original finite bound follows from the genuine ambient bound. The real parameter value and every fibre-index value give one and the corresponding original variable, respectively. Genuine polynomial algebra extensionality compares the composition with the identity: every original variable maps first to parameter times fibre, hence after forgetting to one times itself; every entire coefficient is fixed by both actual base algebra maps. This proves the entire identity and therefore injectivity, without a supplied retraction or coefficient-separation certificate. □

Lemma 307.2 (Actual integer weight and monomial forgetting have their literal finite-coordinate formulas)**.** *For an actual parameter/fibre exponent α, its integer weight is its genuine parameter exponent minus the sum of its genuine fibre exponents. Forgetting its literal monomial retains the entire coefficient times the original fibre-variable monomial. Substituting this forgotten monomial back gives the SAME fibre monomial and parameter exponent equal to that genuine fibre sum.*

Proof. The actual finite coordinate block is parameter first and the entire genuine fibre block after it. Split the full finite sum defining the actual exponent weight into that genuine parameter coordinate and the genuine fibre coordinates. Their already proved weights one and minus one give the stated actual integer difference. Split the full finite monomial evaluation product in the same literal way. Forgetting replaces the actual parameter factor by one and each fibre factor by its corresponding original variable. The actual diagonal substitution then replaces those genuine variables by parameter times fibre. Genuine finite product and power laws give the stated parameter power and SAME whole fibre monomial, with the actual entire coefficient unchanged. □

Lemma 307.3 (Two actual natural parameter powers reconstruct every selected whole polynomial)**.** *For ANY natural N, k and genuine finite graph polynomial A, internally let A_w be its actual component of integer weight $N - k$ and let B be its actual forgotten original polynomial. Then the genuine whole polynomial identity is*

$$U^k A_w = U^N \sigma(B).$$

It has only genuine natural powers, with no division or inversion of the parameter.

Proof. The actual component is genuinely weighted homogeneous of degree $N-k$. Expand it into its actual finite monomial sum. For each actual nonzero coefficient, true weighted homogeneity and the preceding literal weight formula give parameter-exponent plus k equal to fibre-exponent-sum plus N, as actual natural numbers. The preceding whole monomial dictionary therefore identifies the two entire monomial products after multiplying by the stated actual parameter powers. For a zero coefficient both sides vanish. Genuine finite sums and distributivity give the whole polynomial identity. This argument treats every N, k, including $k > N$, without negative parameter powers or a caller weight certificate. □

Theorem 307.4 (Actual finite graph relations produce genuine original coefficient polynomials)**.** *For ANY actual finite family of graph coefficient polynomials A_i and genuine original polynomials P_i, internally produce the original coefficient polynomial B_i by actual weight selection followed by actual parameter forgetting. Their genuine finite relation satisfies*

$$U^k \pi_{N-k}\left(\sum_i A_i \sigma(P_i)\right) = U^N \sigma\left(\sum_i B_i P_i\right).$$

If the actual original P_i belong to an actual original polynomial ideal, the internally produced sum belongs to THAT ideal.

Proof. Use the preceding actual construction independently for each finite coefficient polynomial. The actual product extraction identity moves integer-weight selection to its coefficient polynomial, and actual linearity moves it through the finite sum. Multiply by the genuine parameter power and apply the preceding natural-power reconstruction to each term. Genuine multiplication, distributivity and the actual diagonal algebra-homomorphism product/sum laws give the displayed whole identity. Every reconstructed coefficient is an actual original polynomial with entire base-germ coefficients. If every genuine P_i belongs to the stated actual ideal, true ideal absorption puts each actual $B_i P_i$ in it and genuine finite ideal additivity puts their actual sum in it. This is an ordinary reusable ideal-membership lemma; final analytic reverse-ideal consumers must internally obtain their original finite generators and relation, which remain separate unfinished steps. □

308 Genuine whole contraction for homogeneous polynomial ideals under analytic normal evaluation

This OWN OFFLIB continuation recovers entire homogeneous polynomial ideals from their genuine analytic extensions. It uses real finite multiplier jets and whole coefficient separation, without a flatness certificate. Arbitrary nonhomogeneous polynomial ideals need not have this property.

Theorem 308.1 (A genuine homogeneous polynomial ideal is exactly the contraction of its whole analytic extension)**.** *Let H be an actual homogeneous ideal in the normal polynomial ring over the ENTIRE base holomorphic germ ring. A true polynomial P belongs to H if and only if its whole analytic normal evaluation belongs to the genuine entire analytic ideal image of H. Thus the actual inverse image of that whole analytic ideal is precisely H. No caller flatness, representative, coefficient or output certificate is supplied.*

Proof. The forward direction is true ideal-map membership. Conversely, membership in the actual analytic image ideal gives a genuine finite sum of actual holomorphic germ multipliers times evaluations of actual polynomials $R_i \in H$. Choose a natural q strictly larger than the degree of EVERY actual exponent in the finite support of P, internally from that finite support. For each true analytic multiplier select its genuine bounded polynomial representative Q_i modulo original J^q by the actual normal monomial basis. The true analytic remainder of every product lies in J^q, hence whole evaluation of $P - Q$ lies in J^q for the SAME actual polynomial $Q = \sum_i Q_i R_i \in H$.

For every $k < q$, the whole original low-coefficient dictionary identifies the actual degree-k homogeneous parts of P and Q. Genuine homogeneity of the actual H puts this whole component of Q in H, so the SAME component of P belongs to H. Every component of P of degree at least q is literally zero by the internally chosen support bound. Thus EVERY actual homogeneous component of P belongs to H. The genuine finite homogeneous decomposition reconstructs the ENTIRE P as a finite sum of such actual components, proving $P \in H$. Ideal extensionality gives equality of the actual inverse-image ideal with H. This is whole polynomial contraction for genuine homogeneous ideals, without declaring all analytic extensions faithfully flat. □

Theorem 308.2 (Equality of whole analytic extensions identifies the genuine homogeneous polynomial ideals)**.** *For two actual homogeneous polynomial ideals H, K, equality of their genuine entire analytic normal-evaluation ideal images implies $H = K$.*

Proof. Apply the actual inverse-image operation of the SAME actual normal evaluation ring map to the whole analytic ideal equality. The preceding actual homogeneous contraction identifies its two sides exactly with the original whole H and K. No ideal, degree, generator or free-frame correspondence is externally supplied. This ordinary algebraic comparison will be consumed with nearby homogeneous ideals internally constructed from actual original coefficient germs. □

Theorem 308.3 (Only original coherent input internally gives the entire original polynomial kernel dictionary of its actual selected central model)**.** *Only original coherent I internally supplies SAME genuine N, U, coherent selected original central model and its true all-point/all-power saturation dictionaries, such that for EVERY actual normal polynomial P, whole evaluation belongs to the ENTIRE selected original central origin ideal if and only if P belongs to the WHOLE original normal initial ideal.*

Proof. The genuine original-input whole central equality producer internally selects N, U and identifies the entire selected central origin ideal with the actual analytic extension of the WHOLE original initial ideal. Its genuine homogeneity is internally proved by the original associated-graded producer from the actual original ideal, rather than assumed as a caller kernel or initial-ideal certificate. Apply the first theorem to this SAME actual homogeneous initial ideal and substitute the true whole central equality. Retain every genuine original N, U and actual colon, parameter-power, saturation and coherence output. This is the whole origin polynomial kernel dictionary. The actual nearby original normal-cone/Rees identity and single all-order free open remain subsequent independent producers. □

309 Literal original associated-module localization and every actual normal graded-piece basis

Use the SAME internally produced nonzero entire tangential base germ b of the preceding uniform localization chapter. Put $B_b = B[1/b]$ and let H be the SAME actual whole original homogeneous normal initial ideal. The actual original graded pieces are the genuine J^q images in $R/(I + J^{q+1})$; the original associated module is their literal direct sum. No caller localization, basis, homogeneous division, flatness or result certificate is supplied. All scalar extensions below are genuine module localizations.

Theorem 309.1 (Actual polynomial quotient coefficient maps are genuine module localizations)**.** *For an actual whole polynomial ideal H and an actual base element b, the canonical map $B[Z]/H \longrightarrow B_b[Z]/H_b$ is a genuine localization of the SAME original quotient module at the powers of b. The canonical localized-module equivalence is B_b-linear and sends each genuine original numerator class divided by b^n to its actual localized polynomial class.*

Proof. The actual polynomial coefficient ring $B_b[Z]$ is the localization of $B[Z]$ at the image under the actual constant-polynomial map of the powers of b. The quotient coefficient map is well-defined because every original whole H member maps into the actual coefficient extension H_b. Every image of a power of b is a unit in the genuine localized coefficient ring and hence in its actual quotient. Every whole localized polynomial has an original polynomial numerator with a power-of-b constant denominator; passing to its actual quotient class gives the genuine module localization surjectivity condition. If an original quotient class maps to zero, its actual original polynomial maps into H_b. The actual ideal membership localization theorem supplies a genuine constant power-of-b multiplier whose product with that SAME original polynomial belongs to H. Thus a power of b kills the SAME original quotient class. These three internally proved conditions give genuine module localization. Its universal linear equivalence extends to B_b scalars by true localization universality. □

Theorem 309.2 (The SAME literal original associated module has its actual localized whole quotient model)**.** *The coefficient-localized polynomial quotient by the SAME original normal initial ideal is canonically B_b-linearly equivalent to the genuine localization of the literal original associated module. This map is the localization of the actual original whole polynomial-symbol quotient equivalence and retains its actual original homogeneous symbols. Transporting the ENTIRE internally produced original standard monomial basis through this SAME equivalence supplies an actual whole basis of the literal localized original associated module and proves that SAME original module free over B_b.*

Proof. The actual original polynomial quotient by H is already internally identified with the literal direct sum of the actual original J-adic pieces through the actual whole normal symbol map and its COMPLETE kernel. Localize that SAME genuine original linear equivalence at the powers of the SAME b. Genuine localized-module functoriality gives a B_b-linear equivalence between the actual two localizations. Compose it with the canonical actual polynomial quotient localization equivalence of the first node. Naturality on original numerators proves the literal dictionary: every original normal polynomial class maps to its actual original associated symbol, localized with numerator denominator 1. Transport the ENTIRE actual original standard monomial basis of the preceding uniform localized whole polynomial quotient through this genuine equivalence. The actual basis vectors and their true finite coefficient representations retain this SAME original numerator dictionary. Genuine equivalence transport gives linear independence and spanning of the entire literal localized original associated module, so this actual internally produced whole basis proves that SAME original module free over B_b. This is whole-module freeness; its precise degree restrictions are separately proved in the subsequent nodes. □

Theorem 309.3 (Genuine localization commutes with the literal original direct sum)**.** *Localizing the literal original direct sum at the powers of the SAME b gives the literal direct sum of the genuine localizations of its actual original graded pieces. The genuine B_b-linear equivalence commutes with EVERY actual degree inclusion and projection on original numerators.*

Proof. Genuine module localization is canonically the tensor product with the actual localized coefficient ring B_b. The actual tensor/direct-sum linear equivalence identifies $B_b \otimes_B \bigoplus_q M_q$ with $\bigoplus_q (B_b \otimes_B M_q)$. Compose these true equivalences with the genuine piece localization tensor equivalences. Thus this is the literal direct sum of the SAME actual original localized pieces. The actual pure-tensor dictionary sends an original finite direct-sum numerator with denominator a power of b to its SAME original components, each with that SAME actual denominator. Applying

this dictionary to a true degree inclusion retains precisely its original one nonzero component. It proves EVERY actual component and degree-inclusion naturality directly. Finite support in the actual direct sum supplies all tensor/direct-sum arguments; there is no finite-degree bound and no compactness. □

Theorem 309.4 (The actual SAME-denominator monomial basis respects each literal original degree)**.** *Transport the entire actual original standard monomial basis through the preceding literal equivalences. Its vector of actual exponent α lies in the literal actual localized original degree $|\alpha|$ piece, and is the localization of the actual ORIGINAL normal monomial symbol in that piece. Thus the genuine whole localized basis has an actual, proved degree dictionary.*

Proof. Every actual localized polynomial quotient basis vector is its LITERAL original normal monomial class. The actual original monomial is genuinely homogeneous of total degree $|\alpha|$. In the actual original polynomial homogeneous decomposition all components except that degree vanish, while that component is the same entire monomial. The actual original normal symbol map therefore has exactly one actual direct-sum component, its original homogeneous normal monomial symbol in the degree $|\alpha|$ piece. The actual original monomial germ lies in the true $J^{|\alpha|}$ by the whole normal-coordinate ideal power proof. The literal localization bridge and direct-sum naturality retain that exact original component dictionary. No claim that whole ungraded freeness implies degree freeness is used. □

Theorem 309.5 (Every actual original graded piece has its internally produced finite original monomial basis on the SAME localization)**.** *For ALL actual degrees q simultaneously, the actual original qth J-adic piece localized at the powers of the SAME nonzero original b has its true original normal monomial basis indexed by precisely the ACTUAL standard exponents of total degree q. This true index is finite; thus every actual original localized graded piece is finite free on that same localization.*

Proof. Restrict the internally produced whole actual basis to its indices of degree q and use the actual degree-q vector dictionary of the previous node. Injectivity follows from the actual whole basis coefficient injectivity and the literal degree inclusion. For surjectivity, include a true localized piece vector into its actual direct sum and take its genuine whole basis coefficient function. Every coefficient outside degree q is zero: project to that coefficient's actual degree $k \neq q$, whose projection of the original vector is zero. The actual basis degree dictionary makes this projected vector the whole basis combination of exactly the degree-k coefficients. Whole basis injectivity therefore forces all those coefficients to vanish. The remaining finite coefficient function is supported exactly on the degree-q standard exponent subset and gives a true representative of the original localized piece vector. Hence the restricted actual monomial coefficient combination is a linear equivalence and supplies its actual single-vector transported basis. Each actual exponent coordinate is between 0 and q, since the nonnegative coordinate sum is q; finitely many normal coordinates therefore give a finite true degree-q exponent set, and its actual standard subset is finite too. The actual empty-piece, zero-variable, zero/unit-ideal and infinite-whole-index cases are included. □

Remark 309.6. This is literal localization of the actual ORIGINAL associated module and true degree-compatible bases of ALL actual ORIGINAL graded pieces at ONE internally produced nonzero entire base germ. Freeness of ALL actual original truncations $R/(I + J^q)$ requires their genuine localized exact filtration and internally split finite extensions; uniform analytic spread to original nearby stalks is a further distinct obligation. Neither is inferred from whole ungraded freeness. Higher HS, analytic normal flatness and general G8 remain OPEN until those actual producers and their consumers are proved.

310 Actual ambient powers retain every mixed tangent degree

Let $B = \mathcal{O}_{\mathbb{C}^d,0}$, $R = \mathcal{O}_{\mathbb{C}^{d+t},0}$, let $M = \mathfrak{m}_B R$ through the original tangent pullback, and let J be the actual normal-coordinate ideal. The genuine relative Hadamard decomposition gives $\mathfrak{m}_R = M + J$. The ambient Hilbert–Samuel quotient of R/I therefore retains mixed tangent and normal orders. This chapter makes that weighted quotient explicit, including every original numerator. It is an algebraic prerequisite for the ambient dimension calculation; no blowup descent or Hilbert–Samuel constancy is a conclusion here.

Lemma 310.1 (Every power of a whole ideal sum retains its mixed weights)**.** *For ideals M, J in a commutative ring and every $s \geq 0$,*

$$(M + J)^s = \sum_{q=0}^{s} M^{s-q} J^q.$$

Proof. The binomial theorem in the semiring of ideals writes this power as the sum of the mixed products multiplied by the binomial coefficients. Ideal addition is ideal sum and is idempotent. Hence multiplication of an ideal by a positive natural-number coefficient leaves that ideal unchanged. For each $0 \leq q \leq s$, the binomial coefficient is positive, so the corresponding term is exactly $M^{s-q}J^q$. There are no other terms in the binomial sum. This proves the equality, including $s = 0$, when both sides are the unit ideal. □

Theorem 310.2 (The genuine original ambient maximal power is the mixed ideal)**.** *For every actual original whole ideal $I \subset R$ and every s,*

$$I + \mathfrak{m}_R^s = I + \sum_{q=0}^{s} M^{s-q} J^q.$$

In particular the degree-q term retains its original tangent weight M^{s-q}, rather than replacing the ambient quotient by a fibre quotient.

Proof. Substitute the previously proved whole original identity $\mathfrak{m}_R = M + J$ into the first lemma and add the same original whole ideal I to both sides. This retains the actual original ring, all its original powers, and all mixed products. □

Theorem 310.3 (Every actual ambient jet is a weighted quotient of the normal truncation)**.** *Put $T_s = R/(I + J^s)$, let $\pi_s : R \to T_s$ be the actual quotient map, and let K_s be the image in T_s of $\sum_{q=0}^{s} M^{s-q}J^q$. There is a genuine complex algebra isomorphism*

$$T_s/K_s \simeq R/(I + \mathfrak{m}_R^s)$$

that sends the double class of every actual original numerator g to its original ambient class $[g]$.

Proof. The mixed sum contains J^s as its $q = s$ term. Thus the actual normal truncation quotient map is followed by a well-defined map to the ambient quotient, and the image ideal K_s lies in its kernel. The third isomorphism theorem for the actual quotient identifies T_s/K_s with $R/(I + J^s + \sum_q M^{s-q}J^q)$. Containment of J^s in the mixed sum removes precisely that redundant term, and the preceding identity identifies the remaining ideal with $I + \mathfrak{m}_R^s$. The quotient maps and the third-isomorphism map all send an original numerator to its same class and preserve complex constants. Their composite therefore has the stated complex algebra structure and the original numerator formula. □

Corollary 310.4 (The existing whole ambient Hilbert–Samuel value uses the same weighted quotient)**.** *For every ℓ, the existing whole ambient Hilbert–Samuel value of the original ideal I is the complex dimension of $T_{\ell+1}/K_{\ell+1}$. This quotient is finite dimensional for every original I, without a normal-flatness hypothesis.*

Proof. The preceding complex algebra isomorphism gives a complex linear isomorphism with the actual original quotient defining the existing Hilbert–Samuel value. That original quotient is finite dimensional by the genuine original finite-jet theorem. Transport finite dimension and equality of complex dimensions through this linear isomorphism. □

311 Actual nearby coefficient germs and the entire translated central polynomial model

This OWN OFFLIB continuation uses actual affine maps of FUNCTIONS, never a canonical map from all origin germs to nearby germs. The original coherent ideal internally selects the SAME finite homogeneous relation family, true neighbourhood and actual central model. Identifying this entire translated central model with the nearby ORIGINAL normal cone remains a subsequent producer.

Lemma 311.1 (Actual affine function translations produce genuine entire nearby coefficient germs)**.** *An actual coefficient function holomorphic at a genuine base point a has a true entire origin germ after composition with the genuine affine function map $z \mapsto a + z$. On the original base-preserving ambient affine map $w \mapsto \iota(a) + w$, actual tangent projection translates by a and EVERY actual normal coordinate is unchanged.*

Proof. The actual affine translation is a genuine entire holomorphic map of original manifolds, and its actual value at zero is a. The genuine local complex chain rule transfers coefficient holomorphicity at a to a true whole holomorphic germ of its translated function at zero. Tangent projection is genuinely linear and retracts actual tangent inclusion, so projection of $\iota(a) + w$ is exactly $a + \mathrm{pr}(w)$. Every actual normal coordinate of $\iota(a)$ is literally zero; linear evaluation therefore leaves its normal coordinate equal to that of w. These are actual whole function and coordinate identities, not a map on arbitrary origin-germ equivalence classes. □

Lemma 311.2 (Whole nearby polynomial coefficient germs retain literal finite support and true homogeneous degree)**.** *For a genuine original polynomial P whose actual coefficient functions are holomorphic at a, form its actual finite polynomial P_a over entire translated base coefficient germs. At EVERY exponent its coefficient is the WHOLE true translated germ of the SAME original coefficient function; its true support is contained in the original finite support, and EVERY genuine original homogeneous degree is retained.*

Proof. Define P_a by the finite sum over original support of monomials with actual translated coefficient germs. Actual coefficient extraction gives the asserted coefficient at every supported exponent. At an exponent outside original support, the original coefficient is zero and its actual representative is the literal zero function. Its actual affine translation remains zero, hence its entire coefficient germ is zero. Thus the same formula holds at EVERY exponent and the actual support is contained in the original finite support. Every exponent with nonzero actual new coefficient therefore belonged to the true original support; its original homogeneous degree gives the SAME actual degree. No point-value coefficient surrogate or nearby-germ ring map on arbitrary origin germs is used. □

Lemma 311.3 (Translated entire ambient polynomial-function germs are genuine evaluation of their nearby coefficient polynomials)**.** *The WHOLE germ of the original literal normal-polynomial function translated by the actual tangent inclusion of a equals normal evaluation of the genuine entire coefficient-germ polynomial P_a.*

Proof. The actual affine coordinate identities of the first lemma rewrite the entire translated function as the SAME true finite sum of actual coefficient functions translated on the base and composed with true tangent projection, times unchanged original normal monomial functions. The true whole germ map preserves every finite sum and product. The base coefficient germ

pulls back through the SAME genuine tangent projection; each true normal monomial germ is the actual original normal monomial. The actual finite monomial evaluation formula therefore identifies the WHOLE sum germ with actual normal evaluation of P_a. Nothing is inferred from equality of values only, and every coefficient germ and actual normal-coordinate identity is retained. □

Theorem 311.4 (Only original coherent input supplies the entire genuine translated nearby central ideal by actual homogeneous coefficient-germ polynomials)**.** *Only original coherent I internally supplies SAME N and a genuine base open B containing zero. At EVERY $a \in B$, the ENTIRE origin ideal of the actual affine pullback of the SAME selected original central ideal equals the analytic normal-evaluation ideal image of the span of the SAME finite original relations with their WHOLE actual translated coefficient germs. The latter is an actual homogeneous polynomial ideal, constructed internally from actual original homogeneous degrees, and is the ENTIRE polynomial inverse-image kernel of this actual translated central ideal. No original translated normal-cone equality or all-order free frame is supplied as an input or claimed.*

Proof. The preceding original-input literal polynomial model supplies SAME N, V, original selected central coherence, actual whole finite polynomial-functional span at EVERY point of V, true coefficient holomorphicity and original homogeneous relation degrees. Take $B = \iota^{-1}(V)$ using the genuine original tangent inclusion. It is an actual open containing zero. At any actual $a \in B$, every original finite coefficient function is genuinely holomorphic at a by the actual retraction identity. The preceding lemmas internally form every entire nearby coefficient-germ polynomial of the SAME finite original relation family and retain its homogeneous degree.

The selected original central coherence and genuine finite functional presentation at $\iota(a)$ internally supply the ENTIRE actual affine pullback's finite generators, equal to these same literal generator functions composed with the actual affine translation. The genuine original origin ideal of this coherent pullback is exactly the span of their actual whole origin germs. The preceding WHOLE evaluation dictionary identifies each such germ with actual normal evaluation of the SAME true coefficient-germ polynomial. Genuine ideal-map span equality therefore gives the asserted whole ideal image. Every actual nearby polynomial is homogeneous at its internally retained original degree, so the span of this SAME finite family is a genuine homogeneous ideal by the actual graded ideal constructor. The actual homogeneous analytic contraction theorem internally identifies its WHOLE polynomial inverse-image kernel as exactly this SAME actual homogeneous ideal. This entire translated central-model dictionary still requires separate identification with the nearby ORIGINAL deformation saturation and original normal initial ideal before all-degree flattening can be transported. □

312 Uniform original normal ranks on the genuine common open

Let $\mathfrak{I}$ be an original coherent whole ideal sheaf in the split ambient coordinates, and let I_a be its actual translated original whole germ ideal at the point $(a, 0)$. The common coefficient construction gives a finite type domain D, a finite family of true homogeneous relations, and their whole generated ideal $H \subset D[Y_1, \dots, Y_t]$, together with genuine coefficient homomorphisms ρ_a whose whole ideal images are exactly the original initial ideals of I_a. Let $b \in D \setminus \{0\}$ be the actual uniform leading denominator of this one whole ideal, and write $L = D[b^{-1}]$. These choices are independent of all degrees and truncation orders. The source-facing objective is the whole local-ring Hilbert–Samuel quotient in Bierstone–Milman's "Use of the Hilbert–Samuel function". The following argument supplies its relative-filtration rank and literal residue-tensor dimensions first; the geometric maximal-ideal quotient comparison and descent remain separate producers.

Lemma 312.1 (Actual coefficient-image standard monomial bases)**.** *Fix the actual field-standard exponent family E of H, computed over the fraction field of D. For every coefficient homomorphism $f : D \to T$ with $f(b)$ a unit, the whole polynomial quotient by the image of H has a T-basis*

indexed by this same E. Its vector indexed by α is the actual quotient class of Y^α with coefficient 1.

Proof. The unit condition gives a genuine homomorphism $L \to T$. The existing uniform standard monomial basis of $L[Y]/HL[Y]$ has its literal monomial numerator formula. Tensor that basis with T over L. The whole polynomial coefficient base-change algebra isomorphism sends $1 \otimes [Y^\alpha]$ to the actual class of Y^α in $T[Y]$ modulo the extended whole ideal. The composition of $D \to L \to T$ is f, so this extended ideal is exactly the whole image of H under f. Tensoring a basis and applying these genuine isomorphisms proves both the asserted basis and its actual numerator formula, without requiring f to be injective. □

Lemma 312.2 (Actual homogeneous degree fibres of a whole monomial basis)**.** *For any actual nearby original quotient dictionary, the preceding whole basis identifies each literal original degree-q layer with the finite free coefficient module indexed by $E_q = \{\alpha \in E : |\alpha| = q\}$. Thus its rank is $\#E_q$.*

Proof. The intrinsic original quotient algebra dictionary sends each whole basis vector to the actual degree-$|\alpha|$ original monomial symbol. For a coefficient function supported in E_q, include it in the whole exponent family, apply the whole basis isomorphism, and project to degree q. Its original degree inclusion recovers the whole vector, because each basis vector used in the sum already lies in degree q. This gives injectivity. Conversely, include any literal degree-q vector in the whole direct sum and take its unique whole basis coefficients. Projection to degree q kills every basis vector of other degree and retains every basis vector of degree q. Uniqueness of coefficients therefore forces the original coefficient function to be supported in E_q. This gives surjectivity and the claimed genuine degree isomorphism. Every exponent of degree q has all its entries at most q, so E_q injects into a finite set of bounded normal exponents. Its cardinal is therefore finite, and the degree isomorphism gives the asserted rank. □

Theorem 312.3 (One original-input open and one complete original rank sequence)**.** *Original coherence alone supplies one open set U in the actual base, meeting every origin neighbourhood, and one sequence $r(q) \in \mathbb{N}$, such that for every $a \in U$ and every q, the actual original layer $L_q(I_a)$ is finite free over the actual base germ ring and has rank $r(q)$. The same U works for all q; it need not contain the origin.*

Proof. Apply the original-input common-domain relation producer to obtain the actual D, H, ρ_a above on its common neighbourhood W. Represent the one nonzero denominator b by its true coefficient function g. Its original germ is nonzero, because D was the quotient by the complete origin kernel. Set $U = W \cap \{g \neq 0\}$. Holomorphicity makes U open. If some origin neighbourhood failed to meet U, g would vanish on its intersection with W, contradicting that nonzero original germ. For $a \in U$, the actual translated germ $\rho_a(b)$ evaluates to $g(a) \neq 0$, hence is a unit. Apply the actual coefficient-image monomial basis and the whole ideal equality supplied internally by the common-domain producer. Apply the preceding actual homogeneous degree dictionary to the original translated ideal I_a. Its degree-q index family is always the same finite set E_q, independently of a. Taking $r(q) = \#E_q$ proves all assertions with this single U. □

Theorem 312.4 (Every actual whole original truncation has the common rank sum)**.** *For the same original-input open and the same sequence r, every actual whole original power truncation $R/(I_a + J^s)$ is finite free over the base germ ring and has rank $\sum_{q<s} r(q)$, for all $a \in U$ and all $s \in \mathbb{N}$.*

Proof. At $s = 0$ the ideal contains $J^0 = R$, so the quotient is zero and its rank is the empty sum. At the successor step, the literal original power filtration gives the actual short exact sequence with left term $L_s(I_a)$, middle term $R/(I_a + J^{s+1})$, and right term $R/(I_a + J^s)$. The right term is finite free by induction and therefore projective, so this genuine sequence splits. The middle module is therefore the direct sum of the two finite free endpoint modules. Their ranks add,

giving $r(s) + \sum_{q<s} r(q) = \sum_{q<s+1} r(q)$. No new neighbourhood or denominator is chosen in this induction. □

Theorem 312.5 (Complete dimensions of the actual original algebraic fibre powers)**.** *Specialize the actual base germ ring at its origin residue homomorphism to $\mathbb{C}$. On the same U, the complex dimension of every literal fibre module $\mathbb{C} \otimes_B R/(I_a + J^s)$ is $\sum_{q<s} r(q)$. These are dimensions of the genuine algebraic fibres of the whole original normal-power quotients.*

Proof. Each actual truncation has a finite basis with the preceding common cardinal. Tensor this true basis with $\mathbb{C}$ using the actual base evaluation homomorphism. The resulting basis of the genuine tensor fibre has the same index family, so its complex dimension equals that same cardinal. The already proved common truncation rank therefore gives the asserted full sequence. Identification with the geometric fibre whole maximal-ideal Hilbert–Samuel quotient requires the separate literal base-maximal-ideal extension and ideal-power comparison; no geometric Hilbert–Samuel descent is asserted by this tensor dimension statement. □

313 Actual coherent generators and the entire original finite normal-jet image

Let $B = \mathcal{O}_{\mathbf{C}^d,0}$, $R = \mathcal{O}_{\mathbf{C}^{d+t},0}$, J the literal normal ideal, and $\mathfrak{a}$ an original coherent ideal sheaf. All generators, their common actual holomorphic open, the entire origin ideal and its finite normal-jet columns are produced internally from this input. No finite-generation, kernel, Taylor-coefficient, basis, image or result certificate is supplied to the original consumer.

Theorem 313.1 (Original coherence supplies its entire finite generator dictionary)**.** *Only actual $\mathfrak{a}$ and coherence internally supply ONE actual open $V \ni 0$, a genuine finite index μ, and actual functions f_i holomorphic on V. At EVERY point of V, membership in the original sheaf is exactly membership in the actual whole-germ span of these SAME functions. Their true entire holomorphic origin germs g_i satisfy $I = \operatorname{span}_R\{g_i : i \in \mu\}$, where I is the literal original origin ideal generated by all actual original stalk members.*

Proof. Apply original coherence at zero and retain its actual finite functions and entire common open. Holomorphy on that open internally constructs their whole origin germs. Every generator belongs to the actual original sheaf stalk because it is a genuine one-term holomorphic span member. Thus its germ is in I. Conversely every literal generator used to define I is the germ of an actual original stalk member. The SAME original coherence dictionary writes that actual function near zero as a finite holomorphic combination of the chosen functions. Genuine whole germ sums and products transfer that identity to a finite R-combination of the actual g_i. Taking true ideal spans proves both inclusions. The entire nearby sheaf dictionary is retained unchanged. □

Theorem 313.2 (Every true finite normal generator column has an original ideal witness)**.** *For each actual q, let $M_q = R/J^q$ with its genuine entire B-monomial basis e_α, $|\alpha| < q$. Define the actual finite columns $c_{i,\alpha} = e_\alpha \overline{g_i}$ in this WHOLE quotient. Every column belongs to the literal image of $I \to M_q$, with its true original witness $y^\alpha g_i \in I$.*

Proof. The genuine normal monomial basis dictionary identifies e_α with the actual quotient class of the literal original normal monomial y^α. The internally produced entire origin generator belongs to I; true ideal closure puts its actual product with this monomial in I. Quotient multiplication identifies its literal class with the written column. This proves genuine image membership, including all empty finite-index cases, without replacing a whole germ by its point value. □

Theorem 313.3 (These genuine finite columns generate the entire original jet image over the base)**.** *The ENTIRE literal image of $I \to R/J^q$ is exactly the B-submodule spanned by all actual*

$c_{i,\alpha}$. *Equivalently a genuine whole jet is in that original image precisely when it is a true finite B-linear combination of these SAME columns. No particular element or matrix is a caller image test.*

Proof. For any actual $a \in I$, the internally proved finite whole ideal span gives $a = \sum_i r_i g_i$ with actual coefficients $r_i \in R$. Project this ENTIRE identity into M_q. The genuine whole normal monomial basis writes each true class $\overline{r_i}$ as $\sum_\alpha b_{i,\alpha} e_\alpha$ with actual $b_{i,\alpha} \in B$. Distribute genuine finite quotient multiplication; the true scalar-tower dictionary makes the result $\sum_{i,\alpha} b_{i,\alpha} c_{i,\alpha}$. This gives image containment in the written finite base span. Conversely each actual column has the preceding original ideal witness and the true image of the original B-linear map is a B-submodule. It contains all genuine finite base combinations, proving whole equality and the literal finite coefficient membership dictionary. □

Theorem 313.4 (Only original coherent input jointly supplies its sheaf and entire finite jet dictionaries)**.** *Only actual $d, t, \mathfrak{a}$, coherence and the actual order q internally supply ONE genuine finite generator family and open, the entire origin ideal span dictionary, and the exact WHOLE original finite jet image and its finite base-column membership dictionary. The SAME generator functions generate the original sheaf at EVERY point of the produced original open.*

Proof. Consume the actual original coherence producer once. Construct its entire origin germs and columns internally and apply the whole origin ideal equality and true finite normal monomial basis expansion above. Return these SAME functions/open together with both actual entire dictionaries. No family, finite presentation, coefficient relation, basis or nearby image result is an external input; every finite index is literal, including empty families and order zero. □

Open frontier. The genuine nearby original sheaf generator dictionary and the entire origin finite-jet column dictionary have been distinguished explicitly. This chapter alone does not identify those origin columns with nearby ORIGINAL finite jets; that requires actual normal Taylor representative identities on a common open. Nearby projection-image identification, ONE all-order open, the whole nearby ORIGINAL Rees/graded normal cone dictionary, analytic normal-flatness, higher Hilbert–Samuel and G8 remain OPEN. Formal registration/native build/GitHub publication also remain pending root recovery.

314 Genuine multiplicative polynomial models of whole normal jets

This OWN OFFLIB producer supplies the actual multiplicative truncation model needed for true finite parameter/fibre coefficient extraction in the deformation model. It uses the literal entire base holomorphic germ ring and the actual entire normal ideal. No caller coefficient, kernel, Taylor, ideal-image, free or result certificate occurs. The nearby normal-cone/Rees dictionary and one ALL-order analytic free open remain OPEN.

Lemma 314.1 (Literal variable powers equal the genuine low-coefficient kernel)**.** *Let B be the genuine base holomorphic germ ring and X the actual ideal generated by all polynomial normal variables. A whole normal polynomial belongs to X^q exactly when every actual coefficient of total degree less than q vanishes. This is exactly the kernel of the genuine polynomial-evaluation map to the entire analytic R/J^q.*

Proof. Every actual polynomial monomial belongs to the variable ideal to the power of its literal exponent sum: each variable power belongs to its corresponding ideal power, and finite product membership sums those powers. A coefficient times such a monomial has the same membership. If every coefficient of degree less than q vanishes, the actual finite support contains only degrees at least q, so its genuine monomial sum belongs to X^q. Conversely the genuine normal polynomial

evaluation maps the actual variable ideal to the entire analytic normal ideal, hence maps its qth power to J^q. The previously proved actual analytic derivative/whole coefficient separation then forces every low entire base coefficient to vanish. This proves both implications and identifies the genuine kernel, including $q = 0$ and an empty normal block. □

Theorem 314.2 (Actual polynomial evaluation gives the whole truncated algebra)**.** *The actual original evaluation induces a genuine B-algebra equivalence $B[Y]/X^q \simeq R/J^q$. Every literal polynomial class goes to its entire analytic evaluation class.*

Proof. The actual evaluation followed by the true analytic quotient map is a genuine base algebra homomorphism. The actual whole finite normal monomial representative of every analytic germ proves surjectivity: its entire remainder is in J^q, and its actual polynomial evaluates to that representative. The preceding exact kernel identity and the genuine first algebra isomorphism theorem give the stated equivalence. The genuine quotient-of-equal-ideals equivalence retains each literal polynomial numerator. No finite coordinate linear equivalence is silently treated as a multiplicative one; this equivalence is induced by the actual algebra homomorphism itself. □

Theorem 314.3 (The actual analytic jet map is multiplicative and respects every whole coefficient truncation)**.** *The inverse equivalence composed with the literal analytic quotient map gives a genuine entire Taylor-jet algebra homomorphism. It maps every actual polynomial evaluation to its literal polynomial quotient class, so addition, multiplication and every coefficient computation modulo the genuine variable power are actual ring operations on the entire jets.*

Proof. Compose genuine algebra homomorphisms, rather than assigning coefficients by a caller dictionary. The preceding numerator equality and the actual inverse-equivalence identity prove the exact polynomial formula. Algebra-homomorphism laws then give the genuine sum/product identities. Every analytic germ has the actual finite monomial representative used in the surjectivity proof, so the entire jet can be computed from that actual representative in the literal quotient. For two representatives the proved evaluation kernel says their difference lies in exactly X^q, so the formula is independent of choices. This supplies true finite convolution in the truncated polynomial algebra without claiming an infinite formal series or a nearby all-order normal-cone model. □

Theorem 314.4 (Every entire original ideal has a genuine relative polynomial quotient model)**.** *For any genuine original entire ideal I, the true evaluation to $R/(I + J^q)$ is surjective and has its internally defined exact whole polynomial preimage ideal. Its polynomial quotient is genuinely equivalent as a base algebra to the literal original $R/(I + J^q)$. For original coherent sheaf input use its actual entire origin ideal internally.*

Proof. For every literal analytic numerator use the same actual finite whole normal polynomial representative; the remainder lies in J^q and hence in $I + J^q$. Thus actual evaluation to this larger literal quotient remains surjective. Define its exact whole preimage ideal by the actual original evaluation and the literal sum $I + J^q$, and prove it is the kernel by the genuine quotient zero criterion. The first algebra isomorphism theorem gives the equivalence and its literal numerator formula. In the coherent sheaf consumer the only input ideal is internally constructed as the span of all actual holomorphic origin germs belonging to that original sheaf; no finite generator, initial-ideal, coefficient or kernel certificate is presumed. No unsupported equality with the polynomial preimage of I plus X^q is claimed. A nearby ENTIRE initial ideal/Rees dictionary, uniform analytic freeness and G8 remain OPEN. Formal registration/native target/publication await root recovery. □

315 The SAME original denominator makes ALL actual original normal truncations finite free

Let R be the actual whole ambient holomorphic germ ring, B its actual whole tangential germ ring, J the original normal-coordinate ideal, and $I \subset R$ the arbitrary actual original ideal. Retain the SAME internally produced nonzero whole base germ $b \in B$ of the original normal initial ideal. No new denominator depends on a truncation order. Put $T_q = R/(I + J^q)$ and M_q equal to the literal actual image of J^q in T_{q+1}. Every localization here is at the actual powers of this SAME b. All original maps, exactness, localized maps, projective sections and freeness are produced internally from d, t, I; no caller exactness, splitting, basis, free, flatness or result certificate is supplied.

Theorem 315.1 (Actual original truncation maps and their complete original kernels)**.** *The original ideal inclusion $I + J^{q+1} \subset I + J^q$ gives the ACTUAL canonical B-linear original quotient map $p_q : T_{q+1} \to T_q$. Its kernel is precisely the SAME literal original graded piece $M_q \subset T_{q+1}$; the kernel dictionary holds for every whole original quotient numerator.*

Proof. The genuine ideal-power inequality $J^{q+1} \subset J^q$ gives the actual original sum-ideal inclusion and hence the original quotient map. It sends the class of each whole original germ x to the SAME germ class in T_q. A whole class lies in its kernel exactly when $x \in I + J^q$. Such membership has actual whole original witnesses $a \in I$ and $z \in J^q$ with $x = a + z$. The SAME actual class of a is zero already in T_{q+1}, so the original class of x is the class of the SAME whole $z \in J^q$, and belongs to the literal original image M_q. Conversely a whole original J^q image class maps to zero in T_q. The original quotient map and subtype inclusion are B-linear for the actual tangential algebra action, proving the complete original kernel equality, without a pointwise surrogate or chosen frame. □

Theorem 315.2 (The actual original truncation filtration is genuinely short exact)**.** *The TRUE original sequence $0 \to M_q \xrightarrow{i_q} T_{q+1} \xrightarrow{p_q} T_q \to 0$ is short exact over the actual whole base ring B. Both maps retain their literal original germ class dictionaries.*

Proof. The map i_q is the actual subtype inclusion of the literal original J^q image submodule and is injective. The complete kernel dictionary of the preceding theorem says precisely that its actual range is the kernel of p_q, and therefore gives the actual exactness equation at T_{q+1}. Every true class of T_q has an actual whole original germ numerator x. The SAME numerator class in T_{q+1} maps to it, proving actual quotient surjectivity. Each dictionary is an equality in the genuine whole original quotient module, including the zero/unit ideal and $q = 0$ cases. □

Theorem 315.3 (Localization preserves the SAME actual original short exact sequence)**.** *Localizing these SAME original modules and maps at powers of the SAME original b gives a TRUE short exact sequence over B_b. Its first term is the SAME actual localized original degree-q piece already internally proved finite free; no replacement associated module is used.*

Proof. The genuine localized map of each actual original linear map sends the original numerator-denominator representative (x, b^n) to the SAME original image numerator with the SAME denominator. Actual localization exactness therefore transports the exactness equation of the original sequence. Explicitly, if $p_q(x)/b^n = 0$, true localization supplies an actual multiplier b^r with $p_q(b^r x) = 0$. True original exactness supplies $z \in M_q$ with $i_q(z) = b^r x$, and the localized element z/b^{r+n} maps to x/b^n. Conversely the localized composite is zero. If an actual localized M_q element maps to zero, the true zero criterion gives an actual multiplier killing its original included numerator, and original inclusion injectivity kills that original numerator itself. This proves localized inclusion injectivity. Surjectivity follows by lifting each actual localized target numerator through the actual original surjection and retaining its actual denominator. Localization universality extends these actual maps to the B_b scalar action, retaining all dictionaries. □

Theorem 315.4 (ALL actual original localized normal truncations are finite free at ONE original denominator)**.** *For EVERY actual order q simultaneously, the literal localization of $R/(I + J^q)$ at the powers of the SAME original nonzero b is finite free over B_b. The proof takes only the original dimensions and arbitrary actual original ideal; it takes no caller finite-presentation or free certificate. After this internal induction, its TRUE projective-section construction also exposes the actual consecutive truncation product equivalence at EVERY order.*

Proof. A true internally produced free module is projective. Apply its genuine lifting property to the actual surjection in the preceding short exact sequence and the identity on its true target, obtaining an actual linear section s_q with $p_q s_q = 1$. This section is produced internally, rather than passed by the caller. The genuine kernel is linearly equivalent to the SAME first term through the actual injective range-kernel inclusion. For an actual next-truncation vector x, the vector $x - s_q(p_q x)$ lies in the ACTUAL kernel. Send x to this true kernel vector and $p_q x$. Send a true kernel vector k and preceding-truncation vector y back to $k + s_q y$. The actual section equality and the actual kernel zero equation prove both composites are identity; linearity follows from the genuine original maps. Transport the ACTUAL kernel through its actual first-term equivalence. The result is the asserted genuine product linear equivalence, for the literal localized original modules.

For $q = 0$, the genuine ideal J^0 is the whole ambient ring, so the ACTUAL original quotient T_0 is zero and its genuine localization is zero. The empty true basis makes it finite free. Suppose actual T_q localization is finite free by the internally established induction hypothesis. Its true free structure gives projectivity and hence the internal section and actual product equivalence of the internal splitting construction above. The SAME actual localized original piece M_q has its internally produced finite degree standard monomial basis from the earlier all-degree theorem. The genuine product of these two actual finite free modules is finite free; transport this structure through the actual product linear equivalence to the SAME localized original T_{q+1}. Thus induction proves the assertion for ALL orders. Throughout the induction the denominator is exactly the one whole original initial-ideal denominator; no additional inverse or q-dependent open set is introduced. Actual zero-variable and zero-dimensional-piece cases are included by their actual empty bases. The internally proved preceding-order free structure supplies the actual projective section and TRUE product equivalence as a genuine original-datum producer at every order. □

Theorem 315.5 (The original consumer exposes ONE nonzero entire base germ and ALL literal original truncation quotients)**.** *There exists ONE genuine nonzero entire original tangential base germ b such that ALL literal original $R/(I + J^q)$ localizations at powers of this SAME b are finite free. This is an actual uniform algebraic normal truncation producer on the original quotient, with no caller conclusion certificate.*

Proof. Choose the already internally constructed original normal initial-ideal leading-coefficient product. Its genuine whole original base nonvanishing was proved from the actual fraction-field leading lifts. Apply the preceding ALL-order theorem to this SAME actual denominator and retain its universal quantifier over actual original truncation orders. Both the denominator and the literal original quotient modules are exposed in the conclusion. No fixed-order denominator is substituted and no compactness argument is used. □

Remark 315.6. This proves uniform algebraic finite freeness of ALL literal original truncations at ONE entire original germ localization. Uniform analytic spread from this entire-germ localization to actual nearby analytic stalks, analytic normal-flat loci and their higher HS/G8 consumers remain distinct OPEN obligations. No analytic neighborhood conclusion is inferred from algebraic localized freeness alone.

316 Actual curve defects under genuine open restriction

Let W be a closed smooth pure curve in a positive-dimensional complex manifold X, let E be its actual finite ordered SNC family, and assume $W \subset \overline{W \setminus \bigcup E}$. All histories below are TEST histories for the original mark-one product record M. Write A_T for the actual transformed unit geometric template, J_T for the actual chronological product, and Q_T for the transformed original record. The FIRST defect Z_T uses distinct full ideal-germ classes; the independent exceptional difference Δ_T uses the literal original-member ideal-germ inequalities. Neither is identified with the whole sharp bad locus.

Theorem 316.1 (The actual FIRST defect is the chronological product's second-order cosupport)**.** *For EVERY genuine original TEST prefix T,*

$$Z_T = \operatorname{cosupp}_{A_T.N}(J_T, 2).$$

This is equality of entire ambient subsets, including points outside the actual carrier, and retains every original member position.

Proof. The actual original domain and TEST internally supply the true geometric template history. Each chronological member is coherent, nonzero along the actual carrier and locally principal there; the actual finite product has the same properties. In particular its actual relative order is finite at every carrier point, and its order-two cosupport is precisely where that order is at least two.

At a carrier point outside Z_T, true separation of the chronological traces and their actual proper positions bounds the product order by one. Conversely, product order at most one makes each proper factor have order one and gives at most one proper position. The quotient map from these positions to the genuine full ideal-germ classes is then injective; its actual image cardinal equals the product order. The defining strict inequality for Z_T cannot hold. Thus outside Z_T is equivalent to product order at most one, or equivalently membership in Z_T is product order at least two. Both Z_T and the actual relative cosupport are empty outside the actual carrier, proving the entire equality. This argument constructs the required distinct-class count; it does not substitute an indexed incidence count at bad points. □

Theorem 316.2 (The same defect is the original controlled record's second nonmonomial cut)**.** *For the SAME original M and SAME T, $Z_T = \operatorname{ordGe}(Q_T, 2)$. No replacement current record or order dictionary is supplied.*

Proof. The original TEST domain internally supplies the actual current record. Its carrier equals $A_T.N$. The independently proved whole controlled-residual dictionary identifies the actual nonmonomial IDEAL of Q_T with J_T. By definition $\operatorname{ordGe}(Q_T, 2)$ is the relative cosupport of precisely that actual ideal at mark two. Substitute the whole ideal equality and the carrier equality in this definition. The result is the preceding entire cosupport equality, including points outside the common carrier. □

Theorem 316.3 (The actual FIRST centre commutes with every genuine open restriction)**.** *For ANY actual open $U \subset X$, let W_U and E_U be the literal preimages, T_U the actual restricted tower and $j : T^{-1}(U) \longrightarrow Y_T$ its real open inclusion. Then*

$$Z_{T_U}(W_U, E_U) = j^{-1}(Z_T(W, E)).$$

The original restricted TEST, all four fields of its current record, curve geometry and whole-family density are constructed internally.

Proof. The actual original product pullback identity identifies all four fields of the original restricted record. Genuine TEST restriction therefore supplies its actual TEST history. Open

pullback preserves closed smooth carriers, dimension one at every carrier point and the entire ordered SNC list. The true open-map closure dictionary pulls back $W \subset \overline{W \setminus \bigcup E}$, supplying whole-family density for these literal restricted inputs.

The preceding theorem identifies both actual FIRST defects with their actual original-record second cuts. The whole transformed-record restriction identity identifies the restricted current record with the true open pullback of Q_T. Nonmonomial cuts pull back under this actual open inclusion, since Q_T is internally well formed and nonzero along its carrier. Substituting these three identities proves both membership directions for every point of $T^{-1}(U)$. Restriction retains every centre position, including those with empty trace; no step is deleted. □

Theorem 316.4 (The literal exceptional difference commutes with every genuine open restriction)**.** *For the SAME actual restricted data and inclusion,*

$$\Delta_{T_U}(W_U, E_U) = j^{-1}(\Delta_T(W, E)).$$

This transports the literal original-position ideal-germ difference, independently of FIRST Z, ordinary product order and distinct counts.

Proof. Use the internally constructed restricted original curve data and TEST from the previous proof. The independent full ideal-germ dictionary gives

$$\Delta_T = (A_T.N \cap \bigcup A_T.E) \cap \bigcup_i \mathrm{strictAlong}_T(H_i)$$

for EVERY genuine original TEST; apply it on both actual manifolds. The unit template pulls back to the literal restricted unit template, and its actual transformed record pulls back along j. Hence its carrier and EVERY ordered exceptional entry are the literal preimages. Actual strict-set restriction separately identifies each original member's strict transform with its preimage under j. The list mapping retains EVERY original position, and preimages commute with these finite unions and intersections. Substitution in the two independent difference formulas proves the claimed whole-set equality, including outside the carrier and all empty exceptional positions.

These are centre naturality producers. They do not compare arbitrary terminal witnesses, produce padded histories or establish a cocycle. Those require retaining the actual first-centre and subsequent literal Δ recipe and using genuine same-centre blowup uniqueness. □

317 Actual sharp-curve centre comparison at corresponding stages

Work over the SAME original base manifold and original curve product record. An actual stage isomorphism of two genuine histories records all previous projection squares and original centre matches. This chapter constructs the next comparison from the two independently computed curve centres, rather than assuming their equality or a desired new lift.

Theorem 317.1 (Both actual curve centres correspond through genuine stage isomorphisms)**.** *Let W, E satisfy the original smooth-curve, ordered SNC and whole-density hypotheses. Let T be ANY original product TEST, and let T' be a tower with an actual stagewise tower isomorphism over the identity of the SAME base, whose final genuine biholomorphism is $\chi : Y' \to Y$. Then the true original TEST of T' is supplied internally, and*

$$Z_{T'} = \chi^{-1}(Z_T), \qquad \Delta_{T'} = \chi^{-1}(\Delta_T).$$

No centre, residual, current-record or member-germ comparison is supplied.

Proof. The original product is coherent and its actual identity pullback is the same complete marked record. The genuine tower isomorphism transports TEST histories, hence internally makes T' an original TEST. The actual original curve domain supplies well-formedness and nonzero-along at every prefix. The preceding chapter identifies each FIRST defect with the second nonmonomial cut of the SAME original current record. Actual tower-record transport and genuine biholomorphic cut locality therefore give the first equality at every ambient point.

For the second equality, first use the independent literal exceptional difference formula on both actual TEST histories. The coherent unit geometric template is its own identity pullback. Transform transport identifies its ENTIRE current record through χ, hence identifies its carrier and every ordered exceptional member by true preimage. Actual tower-isomorphism transport separately identifies the strict transform of EVERY original member. Preimages preserve the displayed unions and intersections in the literal difference formula. This proves the second equality without any inference from FIRST Z, product order or an indexed substitute for distinct germ classes. □

Theorem 317.2 (Actual corresponding blowups have a unique holomorphic lift)**.** *Let $\chi : Y' \to Y$ be a genuine biholomorphism, let $C \subset Y$, and let $\pi' : Z' \to Y'$ and $\pi : Z \to Y$ be actual blowups of $\chi^{-1}(C)$ and C. Internally construct a genuine biholomorphism $e : Z' \to Z$ with*

$$\pi \circ e = \chi \circ \pi'.$$

EVERY holomorphic $k : Z' \to Z$ satisfying this actual square equals e. This includes an empty centre as a genuine length-one blowup.

Proof. Compose π with χ^{-1}. Actual base-biholomorphism transport makes this an actual blowup of the SAME centre $\chi^{-1}(C)$ as π'. The actual blowup universal property supplies a unique holomorphic map e from Z' to Z over Y'. Apply it in the reverse direction to obtain a holomorphic map f over Y'. Their composites commute with the real projections, so uniqueness in the two universal properties makes both composites the identity. Thus these maps are a genuine biholomorphism with both inverse laws. Applying χ to the original square gives the stated square over Y.

If k satisfies that square, composing with χ^{-1} gives the SAME square over Y' as for e. Uniqueness in the actual universal property therefore gives $k = e$. No arbitrary terminal-map uniqueness over a composite projection is inferred. □

Theorem 317.3 (Actual computed FIRST or difference steps extend the real stage comparison)**.** *Under the original hypotheses and actual stage isomorphism of the first theorem, take two actual next blowups, both of their own independently computed FIRST centres, or both of their own independently computed literal difference centres. Internally construct the next genuine biholomorphism and an actual isomorphism of the extended towers, with the true next projection square and uniqueness among all holomorphic maps satisfying that square. The centre match and desired new map are NOT hypotheses.*

Proof. The first theorem internally identifies the two actual centres by preimage under the already constructed stage biholomorphism. Substitute that equality into the source blowup predicate. The second theorem then constructs the real next biholomorphism, its projection square and its holomorphic uniqueness. The previous actual tower isomorphism, these two actual blowups, that square and the internally proved centre equality are exactly the constructor of an isomorphism of the extended towers. Apply this argument separately to the true FIRST centre and to the literal exceptional-difference centre.

No centre is omitted when it becomes empty: both actual next blowups remain length one and create their genuine ordered empty slots. These are one-step comparison producers; finite recipe retention, unequal terminal lengths, genuine padding independence and overlap cocycles remain separate constructions. □

318 Actual coefficient presentations supply formal centre powers

Write the original coordinates as (u, y, z), with u tangent to a smooth centre, y normal to it inside the presentation carrier, and z normal to that carrier. This chapter spells out the coefficient-ideal step used in the proof of Bierstone–Milman Theorem 9.4. We use literal power-series coefficients, and do not identify an unnormalized derivative with a coefficient: any factorial normalization must be included in the actual coefficient data. No Hilbert–Samuel equality or transformed presentation is concluded here.

Lemma 318.1 (Whole formal centre powers from actual coefficient series)**.** *Let $F \in \mathbb{C}[[w, z]]$, where $w \in \mathbb{C}^d$, $z \in \mathbb{C}^r$, and let $A \subset \{1, \dots, d\}$. Write $F = \sum_\gamma F_\gamma(w) z^\gamma$ using its literal coefficients. Let P_A be the actual ideal of the selected w variables and let $P_{A,z}$ also include every z variable. For every $q \geq 0$,*

$$F \in P_{A,z}^q \quad \Longleftrightarrow \quad F_\gamma \in P_A^{q-|\gamma|} \text{ for every } |\gamma| < q.$$

Proof. Split each exponent uniquely as (β, γ). Its weight for the whole centre is the selected w weight of β plus $|\gamma|$. The actual coordinate-power coefficient kernel proves the forward implication coefficient by coefficient. Conversely, if that sum is less than q, then $|\gamma| < q$ and the selected w weight is less than $q - |\gamma|$, so the assumed coefficient-series power makes the literal coefficient of F zero. The same actual kernel gives the reverse whole ideal inclusion. No convergence, chosen quotient or finite truncation is assumed. □

Lemma 318.2 (Actual functional powers give the original whole plane-germ powers)**.** *Let $C = \{y = 0\} \subset \mathbb{C}^{k+s}$. If a holomorphic representative f belongs at the origin to the genuine functional ideal $\mathcal{I}_C^q$, then its original holomorphic germ belongs to the q-th power of the actual ideal generated by the same y coordinate germs. Taylor expansion takes this ideal power into the actual formal ideal power of those same variables.*

Proof. For $q = 0$ the ring ideal is the unit ideal. For a successor, the functional power is the actual finite holomorphic span of products of a previous-power member and a member vanishing on C. Its latter germ belongs to the true coordinate-germ ideal by the already proved entire plane-kernel dictionary. Induction, the actual product identity for holomorphic germs, and their actual finite-sum identity place the full representative in the ring ideal power. Taylor expansion is the genuine original algebra homomorphism; its value on every original coordinate is that same formal variable, so ideal-map and ideal-power operations supply the final inclusion. □

Theorem 318.3 (A genuine coefficient-marked centre internally supplies the full formal bound)**.** *Let $F = \sum_\gamma \widehat{c_\gamma}(u, y) z^\gamma$ have actual original holomorphic coefficient representatives c_γ. For each $|\gamma| < q$, let $\mathfrak{a}_\gamma$ be an original coherent ideal containing c_γ at the origin. Assume the actual smooth coordinate centre $C = \{y = 0\}$ lies in the original coefficient cosupport*

$$C \subset \operatorname{cosupp}(\mathfrak{a}_\gamma, q - |\gamma|) \qquad (|\gamma| < q).$$

Then F belongs to the actual ambient centre ideal $(y, z)^q$. In particular this constructs, rather than assumes, the required centre power of every original vertex member represented by those coefficients.

Proof. For each low coefficient the original coherent Hadamard bridge applies to the genuine smooth centre and its actual cosupport containment. It puts the coefficient representative in the functional power $\mathcal{I}_C^{q-|\gamma|}$ at the origin. The previous lemma puts its original germ, and then its genuine Taylor series, in the corresponding coordinate ideal power. The coefficient identity in the original data and the first lemma now put the ENTIRE formal member in $(y, z)^q$. This argument does not assume that arbitrary monic representatives have the needed centre order, nor that the formal properties (7.2)(4),(5) or the full semicoherent presentation have already been constructed. □

319 True coordinate-ideal powers and genuine centre-order division memberships

In the ACTUAL formal ring $S = \mathbb{C}[[X_1, \ldots, X_n]]$, fix ANY actual subset A of the original finite coordinate family. Put $P_A = (X_i : i \in A)$, $w_i = 1$ for $i \in A$ and $w_i = 0$ otherwise, and $w(\alpha) = \sum_i w_i \alpha_i$. This chapter gives the actual coordinate-power meaning of the coefficient bounds in [4, Lemma 7.22 and Corollary 7.24, pp. 269–270]. The ideal, powers, formal products and scalar actions are the genuine existing ones. No centre compatibility or strict-transform certificate is manufactured by these formal algebra statements.

Lemma 319.1 (Genuine weight-vanishing ideals contain the true coordinate powers)**.** *For $q \geq 0$, the actual set $V_q = \{F \in S : [X^\alpha]F = 0$ whenever $w(\alpha) < q\}$ is a genuine ideal of S. One has $P_A^q \subset V_q$, and for EVERY actual exponent α, the actual monomial X^α belongs to $P_A^{w(\alpha)}$. In particular $w(\alpha) \geq q$ implies $X^\alpha \in P_A^q$.*

Proof. Zero and sums have the stated coefficient vanishing. For an ACTUAL product GF with $F \in V_q$, every genuine antidiagonal pair $\beta + \gamma = \alpha$ satisfies $w(\gamma) \leq w(\alpha)$. Thus when $w(\alpha) < q$, the F coefficient at γ is zero, and the entire genuine finite convolution vanishes. This proves ideal closure for the actual ring multiplication. Similarly, if $F \in V_p$ and $G \in V_q$, every pair with $w(\alpha) < p + q$ has $w(\beta) < p$ or $w(\gamma) < q$, so actual convolution gives $V_p V_q \subset V_{p+q}$. Each selected actual variable belongs to V_1, so actual span and power induction give $P_A^q \subset V_q$, including $q = 0$. Finally the genuine monomial is the actual finite product $X^\alpha = \prod_i X_i^{\alpha_i}$. For selected i the corresponding factor lies in $P_A^{\alpha_i}$; for unselected i it lies in the actual whole ideal $P_A^0 = S$. Multiplying these genuine ideal memberships gives membership in $P_A^{\sum_{i \in A} \alpha_i}$, exactly the claimed power. Monotonicity of genuine ideal powers gives the final assertion. No infinite series division is used in this part. □

Theorem 319.2 (The actual coordinate power has precisely the true coefficient kernel)**.** *For every genuine monomial order, the actual initial diagram of P_A^q is exactly $\{\alpha : w(\alpha) \geq q\}$. For EVERY actual $F \in S$,*

$$F \in P_A^q \quad \Longleftrightarrow \quad [X^\alpha]F = 0 \text{ for every } w(\alpha) < q.$$

These statements include empty and full coordinate subsets, $n = 0$ and $q = 0$; the actual unselected coordinates are unrestricted.

Proof. An actual nonzero member of P_A^q belongs to the preceding true V_q. Its actual lowest nonzero coefficient therefore has weight at least q. Conversely each exponent of weight at least q has its actual monomial in P_A^q by the preceding lemma, and that monomial's genuine lowest exponent is exactly the same exponent. This proves BOTH actual diagram inclusions. Power membership implies the displayed coefficient condition by $P_A^q \subset V_q$. Conversely suppose $F \in V_q$ and apply the already internally proved ACTUAL formal division for the WHOLE ideal P_A^q. The finite sum of its genuine vertex products belongs to $P_A^q \subset V_q$, so its actual remainder belongs to V_q as well. At every exponent of weight below q that remainder's actual coefficient therefore vanishes. At every exponent of weight at least q, the just proved true diagram identity puts the exponent in the actual diagram, so the genuine remainder support theorem again makes its coefficient zero. Thus EVERY actual remainder coefficient is zero. The WHOLE genuine division identity now places F in P_A^q. No coefficient kernel, quotient model or desired membership is a caller premise, and no formal completion identification is needed. □

Theorem 319.3 (Actual whole division quotients belong to the true centre-order ideal powers)**.** *Let $I \subset S$ be ANY actual ideal and let its actual internally constructed monic vertex series be*

f_a, for the fixed genuine monomial order. Suppose each SAME actual f_a belongs to $P_A^{w(a)}$. For EVERY actual $F \in I \cap P_A^d$, its SAME genuinely constructed division quotients satisfy

$$F = \sum_a Q_a(F) f_a, \qquad Q_a(F) \in P_A^{d-w(a)}.$$

The natural-number subtraction convention handles $d < w(a)$. The stated actual vertex power hypothesis is explicit; deriving it from genuine admissible-centre geometry remains the separate transformation step.

Proof. The BOTH-direction actual coordinate-power coefficient kernel turns each genuine vertex power membership into exactly the actual coefficient weight compatibility of the preceding division theorem. It similarly turns the genuine numerator power membership into vanishing below weight d. That theorem gives the actual WHOLE ideal member division identity with its SAME quotients, and shows that a nonzero actual coefficient of $Q_a(F)$ has weight at least $d - w(a)$. Equivalently every actual coefficient below that weight is zero. The SAME true coordinate-power kernel, now in the reverse direction, therefore places the entire actual quotient series in $P_A^{d-w(a)}$, as claimed. No quotient bound, division decomposition or ideal model is assumed. □

320 Genuine coefficient division preserves actual compatible weight filtrations

Keep the genuine internally constructed finite monic vertex series of an ACTUAL whole formal ideal I, the actual owner partition and actual coefficient-recursive division. Let $w_i \in \mathbb{N}$ be actual coordinate weights and put $w(\alpha) = \sum_i w_i \alpha_i$. This is the coefficient mechanism behind the centre-order quotient bounds of [4, Lemma 7.22, Corollary 7.24, pp. 269–270]. The following states its genuine weight compatibility assumption explicitly; it does not assert that arbitrary chosen centres automatically satisfy it.

Lemma 320.1 (True compatible vertex coefficients force genuine recursive weight vanishing)**.** *Suppose EACH actual internally constructed vertex series f_a has zero coefficient at EVERY exponent γ with $w(\gamma) < w(a)$. If an ACTUAL $F \in S$ has zero coefficient below weight d, then its actual internally constructed division coefficient $t_F(c)$ is zero for EVERY c with $w(c) < d$.*

Proof. Use well founded induction in the SAME actual monomial order that defines the genuine division recursion. At exponent c with $w(c) < d$, the actual numerator coefficient is zero. Consider an actual correction summand indexed by vertex a and genuine pair $\beta + \gamma = c$. If the actual coefficient of f_a at γ is zero, the genuine product is zero. Otherwise the stated weight compatibility forces $w(a) \le w(\gamma)$. Thus $w(\beta + a) = w(\beta) + w(a) \le w(\beta) + w(\gamma) = w(c) < d$. Every recursive correction uses only $\beta + a \prec c$, so induction annihilates its actual recursive coefficient. The finite whole correction sum is therefore zero, and the actual defining equation yields $t_F(c) = 0$. The compatibility concerns actual vertex coefficients and the conclusion concerns genuinely constructed quotients; no desired quotient bound or recursion solution is a premise. □

Theorem 320.2 (Actual whole formal division quotients retain their genuine weight bounds)**.** *Under the preceding actual coefficient hypotheses, EVERY coefficient of $Q_a(F)$ at β vanishes if $w(\beta) + w(a) < d$; EVERY actual remainder coefficient below weight d vanishes. For an actual $F \in I$, these are bounds on the genuine WHOLE series coefficients in the internally proved equality $F = \sum_a Q_a(F) f_a$. For 0–1 coordinate weights, they are precisely the coefficient centre-order bounds $w(\beta) \ge d - w(a)$ for the nonzero quotient terms.*

Proof. An actual quotient coefficient is either zero by the actual owner condition, or is precisely $t_F(\beta + a)$. Additivity of the actual weight and the preceding lemma give its vanishing under the displayed strict inequality. An actual remainder coefficient is likewise either literally zero or the

actual $t_F(c)$, giving its bound. For $F \in I$ the already proved WHOLE ideal member division identity has zero remainder; it uses exactly these SAME genuinely constructed quotients. For coordinate indicator weights the weight is the actual sum of the selected coordinate exponents, so the displayed inequality is the stated usual centre-order coefficient bound, with the nonnegative truncated subtraction convention. □

Theorem 320.3 (Degree-lex division internally preserves the genuine formal maximal filtration)**.** *For the ACTUAL degree-lex initial diagram of ANY actual formal ideal, weight compatibility for total degree holds internally. If $F \in \mathfrak{n}^d$, then the genuine remainder is in $\mathfrak{n}^d$, and its SAME actual quotient series satisfy*

$$[X^\beta]Q_a(F) = 0 \quad \textit{whenever} \quad |\beta| + |a| < d.$$

This conclusion needs only the ACTUAL whole ideal and ACTUAL formal power membership; no vertex compatibility certificate is supplied in this case.

Proof. Every actual vertex series has lowest degree-lex exponent exactly a. An exponent γ of degree smaller than $|a|$ is strictly smaller in the true degree-first order, so its actual coefficient is zero. This internally supplies the preceding weight compatibility for the constant weight family $w_i = 1$. The true maximal-power coefficient kernel supplies numerator vanishing below total degree d. Apply the preceding actual quotient and remainder bounds, and use the BOTH-direction true maximal-power kernel again to place the genuine remainder in $\mathfrak{n}^d$. No transformed standard basis or general strict Bennett inequality is asserted by this filtration result. □

321 Actual whole division by the canonical reduced standard basis

The earlier division constructs arbitrary monic vertex members of the actual whole ideal. Those members need not have the same selected-centre orders as the canonical reduced members. We therefore perform the actual recursion for a specified monic family, then instantiate it with the internally constructed canonical reduced basis. The construction follows the standard-basis division in [1], Section 6.3. This chapter does not infer centre compatibility from an arbitrary monic choice or construct a semicoherent presentation.

Theorem 321.1 (Whole division by an actual monic vertex family)**.** *Let $I \subset \mathbb{C}[[x]]$ have its genuine initial diagram and actual vertices. For each actual vertex a, let $B_a \in I$ have coefficient one at a and zero coefficients strictly below a. Every formal F has internally constructed series Q_a, R satisfying*

$$F = \sum_a Q_a B_a + R,$$

with the actual disjoint owner-part support for each Q_a and zero coefficients of R on the entire actual diagram. If $F \in I$, then $R = 0$, and the B_a generate the whole ideal.

Proof. Use the already constructed actual owner partition of the whole diagram. At exponent c, define the correction coefficient by well-founded recursion: subtract all finite antidiagonal products whose quotient exponent plus its owner vertex is strictly below c. Put the correction in that owner's quotient, or in the remainder if no owner exists. For each antidiagonal product, the monic coefficient supplies the owner term at c; the remaining terms are exactly the recursive tail. Terms whose source vertex would lie above their second exponent vanish by the assumed genuine lower coefficients. Summing all actual owners proves the identity coefficient by coefficient. If $F \in I$, the finite product sum lies in I, hence so does R. A nonzero such R would have its genuine initial exponent in the diagram, contradicting its vanishing there. Whole ideal generation follows in both directions. □

Theorem 321.2 (The same actual canonical normal representative)**.** *The actual remainder produced by any such family is exactly the previously constructed canonical normal representative of the same class modulo I.*

Proof. The whole division identity puts $F - R$ in I. Its actual remainder coefficients vanish on the diagram. Apply uniqueness of the canonical normal representative to these two already proved properties. No equality of arbitrary chosen monic representatives is required. □

Theorem 321.3 (Actual selected-centre filtration of the same quotients)**.** *Let P_A be the actual coordinate ideal of a selected subset and write $w_A(a)$ for its literal exponent weight. If $B_a \in P_A^{w_A(a)}$ for every actual vertex and $F \in I \cap P_A^q$, then the very same whole division quotients satisfy $Q_a \in P_A^{q-w_A(a)}$.*

Proof. The actual coordinate-power kernel translates the hypotheses into lower coefficient vanishing. Induct along the very same well-founded recursion. In a nonzero tail product, the second factor has weight at least the owner vertex weight. Its first exponent plus the owner vertex therefore has weight at most the total exponent. At a total weight below q the induction hypothesis kills its correction coefficient. Thus every correction coefficient below q vanishes. The quotient's actual owner formula shifts this bound by the vertex weight. Translate it back through the entire coordinate-power kernel. Power zero and empty coordinate blocks are covered by the same argument. □

Corollary 321.4 (Internally supplied canonical reduced family)**.** *Use the actual reduced members $G_a = X^a - R_I(X^a)$ of the same whole ideal. They internally supply every monic-family hypothesis above. Hence there is genuine whole division by these very reduced members, with the same canonical remainder and the asserted centre-power quotient bounds whenever their actual selected-centre powers hold.*

Proof. The previous construction proves actual ideal membership, coefficient one at the genuine vertex, and vanishing below it for every G_a. These are exactly the three inputs to the actual recursion, with no supplied quotient, normal form, generation certificate or filtered division. Substitute this internally constructed family in the preceding theorems. Actual original coefficient cosupports can supply the remaining centre-power hypotheses by the separate coefficient-presentation producer; general presentation construction and Bennett inequalities are still separate obligations. □

322 Literal open restriction of independently chosen monomial blowups

Let A be the actual complex n-manifold, $U \subseteq A$ any genuine open, and F the entire original finite indexed SNC weighted list, including empty, repeated, and zero-weight metadata. Original choices are only two genuine Cartier blowup choice families. Both actual bounded procedures, their true projection inverses, their whole marked controls and their finite completion are internally generated. Write $p(q)$ for the internally computed number of nonempty U-traces in the actual reference centre word of the original prefix q. This section constructs a literal all-head geometric compression from the actual restriction of the independently chosen global tower to the independently chosen local tower. Higher invariant functoriality and identification of prior general separation with the old driver remain open.

Lemma 322.1 (Genuine target isomorphisms transport every retained or silent stage)**.** *Suppose an actual geometric stage compression sends T' to T over a genuine base biholomorphism Φ, with actual terminal map χ, and an actual tower isomorphism sends T to S over Ψ, with terminal map θ. Then their genuine composite is a stage compression from T' to S over $\Psi\Phi$ with terminal map $\theta\chi$.*

Proof. Induct on the actual compression. At the empty compression the target tower is empty; invert the actual tower isomorphism to identify its target as empty and its terminal map with its base map. At a matched step, invert the actual target tower isomorphism at its genuine last blowup, obtaining its actual preceding isomorphism, centre equality and projection square. Apply the induction hypothesis to the true preceding compression and isomorphism. Compose the actual terminal maps and their projection squares; the centre inverse-image equality is genuine inverse-image composition. At a silent step only the source has an actual blowup along the empty centre. Apply induction to the preceding compression with the same actual target isomorphism and compose its true silent projection square. This retains every matched stage and omits only an actual empty-centre source blowup. No new step, centre or comparison is supplied as a certificate. □

Lemma 322.2 (The actual chosen-reference isomorphism restricts literally)**.** *The independently generated global chosen tower restricted to U is genuinely tower-isomorphic to the literal restriction of its generated identity-Cartier reference tower. Its actual terminal map is the restriction of the true chosen projection inverse. No restricted tower isomorphism is original input.*

Proof. Apply the previously produced chosen-reference comparison to the actual original F, q and primitive choice family. Its base map is the actual identity of A. The projection square internally identifies both actual open preimages of U. Restrict this genuine tower isomorphism stage by stage to these actual open subsets using the existing genuine restriction producer. Its actual base restriction is the identity of U and its terminal map is the restriction of the actual global projection inverse. This is an isomorphism of the literal restricted towers, preserving all original steps, even if their restricted centres are empty. □

Theorem 322.3 (Every actual bounded chosen open restriction has genuine all-head empty-step compression)**.** *For every original q there is an internally produced actual biholomorphism from the terminal manifold of the literal U-restriction of the actual global chosen procedure to the terminal manifold of the independently chosen local procedure at $p(q)$. Together these actual towers and map satisfy the genuine inductive all-head stage-compression relation over the identity of U. The local real length is exactly $p(q)$, and no literal global step is discarded unless its actual restricted centre is empty.*

Proof. The original SNC input internally makes the entire pulled weighted list SNC on U. The already produced literal reference restriction is genuinely isomorphic to the uncompressed trace tower. The actual centre-trace induction gives genuine silent compression to the compressed trace tower and identifies that actual compressed tower with the independently generated identity-Cartier local procedure at $p(q)$. Compose these actual source transports. The actual local chosen-reference comparison, inverted, is a genuine isomorphism from the local reference tower to the actual local chosen tower. Apply the target-transport lemma and the preceding actual restricted global comparison. The resulting composite terminal biholomorphism and inductive compression use no caller tower, trace, maximum, centre, termination or naturality certificate. The exact local length follows the previously produced independent trace-centre word and genuine chosen-reference length equality. The matched/silent constructors preserve every actual retained head and delete only true empty heads; formerly nonempty history components that later become empty stay retained. □

Theorem 322.4 (The same literal geometric comparison holds for both OWN finite completions)**.** *The actual global procedure completed at its own coefficient sum, literally restricted to U, compresses genuinely to the independently chosen local procedure completed at its own local initial coefficient sum. The actual local length is exactly the number of nonempty actual traces in the completed global centre word. Its genuine whole controlled ideal is unit and the same actual local tower resolves the actual original pulled ideal.*

Proof. Use the already produced completed reference open compression, whose source and target run at their OWN actual initial coefficient sums. Restrict the genuine actual global chosen-reference comparison and invert the genuine actual local chosen-reference comparison at these two independently computed bounds. Apply genuine source and target transport as before. The true internally generated chosen completions already have whole unit endpoint and genuine marked resolution, and the completed independent trace word gives exact local length. The bound uses the finite original coefficient sum; no compactness, externally supplied bound or termination certificate enters. An empty open, dimension zero, all zero coefficients and metadata with empty support require no fabricated positive-length or nonempty input. □

Theorem 322.5 (The actual compression preserves the entire controlled ideal and literal retained history)**.** *Let χ be the internally produced actual bounded or completed compression. The independently chosen local entire controlled carrier, ideal and marking equal the true χ^{-1}-pullback of the raw literal restricted global controlled record. Its ordered boundary is precisely the actual retained sublist of that pulled raw boundary: ALL original indices followed by exactly the genuine global creation indices whose centre was nonempty on U at creation. Empty entries retained in this literal sublist are not deleted.*

Proof. The compression's actual projection square and the internally produced true Cartier projection inverses identify the original cross-manifold lift with the inclusion of the terminal raw restricted manifold composed with χ^{-1}. The actual local chosen-reference whole-record identity and the already produced genuine reference open-trace dictionary identify the entire independent local record with the local chosen projection pullback of the true retained reference trace record. The corresponding global chosen-reference whole-record identity identifies the actual global record with the global chosen projection pullback of the SAME original reference record. Genuine marked-test restriction supplies the whole raw restricted record as the actual open inclusion pullback of this SAME global record. Its true stage ideal is coherent, internally produced by the actual marked tests. Whole holomorphic pullback composition therefore identifies the entire local carrier and ideal with the actual χ^{-1}-pullback of the raw controlled carrier and ideal. Marking stays literally one. The entire raw ordered boundary is the actual pullback of every global boundary entry. List-map composition and true getD inverse-image identities carry the SAME increasing creation mask into its actual retained sublist. Only new events actually empty at creation are omitted; all old indices and all genuine retained events stay literal. □

Theorem 322.6 (The produced actual terminal compression is unique and compatible with genuine choices)**.** *The actual terminal compression map is uniquely determined by its true projection square. Changing either primitive genuine choice family transports the actual compression by the internally produced genuine two-choice comparisons on the literal global restriction and local chosen tower. Their real endpoint maps obey the exact projection-square cocycle; no equal-length source/target comparison is claimed.*

Proof. The true local chosen projection is the internally produced biholomorphic inverse and is therefore injective. Any two actual compression endpoint maps have the SAME original projection square over U; apply this actual injectivity pointwise to identify them. For two actual pairs of primitive choices, the independently produced genuine two-choice comparisons preserve the true whole global and local projections. Restrict the actual global comparison to the true matching open preimages, then compose with either produced compression and the actual local comparison. Both resulting actual maps have the same true projection square. The same injectivity proves their equality. The true comparison identity, inverse and composite cocycles yield the corresponding genuine compression endpoint coherence. This is a theorem about the actual chosen monomial procedures, not a supplied canonical algorithm or a higher-invariant naturality certificate. □

Formal registration, native target build, leanok and GitHub publication remain pending root recovery. The reference trace theorem preserves the old singleton-key order under the actual

increasing retained creation-label embedding; it does not claim numerical equality of changed colex labels. Arbitrary unrelated ideals and the higher general driver remain OPEN.

323 One genuine source history compares every original StepOne branch

The ambient dimensions agree and are positive. The original target record has DOMAIN2 and actual maximal order. Its history is ANY genuine original TEST; canonicality is not a hypothesis. We retain one actual smooth Cartesian source history for all points, every branch and every depth.

Lemma 323.1 (The genuine zero-fibre recursive dimension maps are identity)**.** *For every original value depth k, the actual recursive dimension map with fibre dimension zero is the identity on the original value type, using only the actual equality $0+k=k$ to identify its codomain. Consequently its induced StepOne map is also the identity.*

Proof. Induct on the actual value depth. At depth zero, the defining seed map is identity. At a successor, the defining general map preserves the actual order coordinate and maps its actual monomial and StepOne alternatives. The zero-fibre monomial map adds zero to its dimension key, hence is identity on every original key and on bottom. The induction hypothesis makes the lower map identity, so its WithTop map preserves every finite value and top. The actual StepOne map therefore preserves both the level and lower value; the enclosing sum and general map preserve every original value. This proves identity on the unchanged recursive types, including all bottom/top alternatives, rather than a comparison only on attained values. □

Theorem 323.2 (One actual source TEST compares all original StepOne values)**.** *Let $f : Y \to X$ be an actual smooth submersion between the same positive ambient dimensions. For ONLY an original DOMAIN2 record M of actual maximal order and ANY genuine original TEST T, construct ONE actual smooth Cartesian source history Q, its endpoint map and the genuine original pulled-back TEST. At EVERY actual source endpoint whose image lies in the target current cosupport, the unchanged original sequence-aware StepOne values agree at EVERY positive lower GENERAL depth. No branch, source history, lower TEST, current lower membership or value comparison is supplied.*

Proof. The original-only zero-own-contact producer constructs one actual Cartesian source history and its genuine pulled-back upper TEST before any point or depth is selected. Its genuine zero-contact whole-value comparison holds on this SAME history at every nonspecial level-zero point; it is not an independently chosen source history for that branch.

Fix a current target cosupport point and depth. Genuine original coefficient-homogenization weak equivalence supplies the actual bar TEST. Actual smooth SPECIAL and indexed-level dictionaries hold for this constructed history. At a special point both unchanged original StepOne definitions give the same level and top. At a nonspecial point of level zero, use the already proved original zero-own StepOne comparison for this exact history. At a nonspecial positive level, write that actual level as a successor and apply the original ANY-TEST positive StepOne producer. It internally supplies the actual lower current point, both independent masks and full lower GENERAL comparison, without candidate admissibility. Since the dimensions agree, its genuine fibre shift is zero; the preceding identity reduces its actual map to identity. These exhaust all original branches at every point and depth on the ONE constructed history.

This statement concerns the original StepOne invariant in DOMAIN2. It does not identify arbitrary positive-relative-dimension zero-contact choices, assert arbitrary higher-dimensional noncanonical full GENERAL naturality, exceptional/infinitesimal semicoherence, or a general unbounded global scheduler. □

324 Actual current cosupport of a retained positive member lower

The positive member reduction of [5, Sections 5–6] must retain its original coefficient reference, actual indexed members and literal run. A centre admissible for the upper record can miss the smaller member carrier. The argument below proves point membership without declaring the lower tail admissible or canonical. It uses the controlled one-sided restriction theorem on the same original history.

Theorem 324.1 (The retained positive endpoint belongs to the actual remaining strict carrier)**.** *Let M be an original well-formed positive-mark record in positive ambient dimension. Let H followed by R be a genuine test history of its actual $\overline{M}$. Let y be an actual current cosupport point, $y_0 = R(y)$, and $s+1$ its current positive level. Assume that the head point y_0 is in that same positive nonspecial regime. Form the ACTUAL member remainder K from $\overline{M}_H$ and its original members through y_0. Then $y \in K_R.N$. No lower test or current lower cosupport membership is assumed.*

Proof. Write $m = \overline{M}_H$, S for the actual indexed member positions through y_0, and $V = \mathrm{maskA}(M, H, y_0)$. Genuine positive nonspecial head data internally gives $y_0 \in V$. Projecting the actual upper cosupport point along the genuine upper tail gives $y_0 \in m.N$. Every selected original member contains y_0 by its definition, so y_0 lies in the full member carrier $F = m.N \cap \bigcap_{i\in S} m.E_i$. The original lower record on V is the literal pullback of the member restriction. The already proved whole initial-record identity also identifies it with the pullback of the actual remaining-component record K. Reading their actual carrier fields at $y_0 \in V$ therefore gives $y_0 \in K.N$, before any transformed lower membership is asserted.

The exact indexed level dictionary on V applies at $R(y) = y_0$. Equality of current and head levels forces y to lie on the strict transform of EVERY original member through y_0. Actual upper carrier membership and the genuine strict-intersection theorem thus give $y \in \mathrm{strictAlong}_R(F)$. The remaining and selected components form a disjoint closed partition $F = K.N \cup F_{\mathrm{sel}}$: both are genuine unions of components of the closed smooth member carrier. Strict transform preserves finite unions, and the strict transforms of the two closed disjoint parts remain disjoint since their projections lie in the respective old parts. Consequently

$$\mathrm{strictAlong}_R(K.N) = \mathrm{strictAlong}_R(F) \cap R^{-1}(K.N).$$

The two independently proved endpoint facts, together with $R(y) = y_0 \in K.N$, give membership in this strict transform. It is literally the actual current carrier $K_R.N$. □

Theorem 324.2 (Actual computed positive-member current cosupport without a lower test)**.** *For an original DOMAIN b record M, with arbitrary natural bound b, ANY genuine original upper test history HR, an actual upper current cosupport point and an actual retained positive nonspecial head regime as above, construct the true endpoint subtype on $V = \mathrm{maskA}(M, H, R(y))$ and prove*

$$y_V \in \mathrm{cosupp}\left((\mathrm{lowerMarked}(M, s+1, H, R(y)))_{R|_V}\right).$$

The current member lower is the unchanged original computed record. There is no input lower test or desired endpoint membership.

Proof. Original homogenized-coefficient weak equivalence supplies the genuine $\overline{M}$ upper test history and its same whole current cosupport. Split that actual history into the head and tail. At the head write $m = \overline{M}_H$ and take the full reference $B = \mathcal{C}(m)$ formed ONCE there. The true coefficient theorem makes R a test history of this full B and identifies B_R's cosupport with m_R's cosupport. Original well-formedness internally supplies the coherent smooth full reference. The actual member remainder K has a closed carrier contained in $m.N = B.N$, with LITERALLY the same ideal and marking as B. Its initial restricted comparison is therefore reflexive.

The preceding theorem internally gives $y \in K_R.N$. The controlled reference history theorem now gives $B_R.\mathrm{cosupp} \cap K_R.N \subseteq K_R.\mathrm{cosupp}$, so the actual upper endpoint belongs to K_R's cosupport. Original DOMAIN b member geometry supplies its real coherent current ideal and whole actual open controlled-record identity. Pulling the actual cosupport back through the literal endpoint inclusion, then substituting that whole identity, gives the displayed current lower membership. This uses the original lower ideal and ordered boundary, including every empty step; it never changes the upper history into a lower test. □

Theorem 324.3 (The original run constructs its positive lower current cosupport internally)**.** *Let M be an original DOMAIN b record, with arbitrary natural bound b, and T ANY original upper test history, and y an actual current upper cosupport point that is nonspecial and has positive current level $s+1$. Use the UNCHANGED original regime-run counter to form its head H, tail R and base point $R(y)$. These inputs ALONE construct the actual mask endpoint and prove its membership in the current original positive member lower cosupport. No run, head, mask, reference geometry or lower membership certificate is supplied.*

Proof. The original run-head theorem, applied to the true upper test and cosupport point, internally proves that the projected head point lies in the same current-level nonspecial regime and in the true head cosupport. Positivity chooses the actual member mask. The genuine head–tail theorem retains the SAME original history, all positions and its literal endpoint, and the actual current level equals the head regime level by that theorem. Apply the preceding constructed-membership producer to these exact head and tail objects and that genuine retained-level equality. Its returned endpoint is precisely the subtype used in the original run's lower marked ideal, so its actual current membership is the asserted one. □

Theorem 324.4 (Whole original member-lower values with membership supplied by the actual run)**.** *For the SAME original inputs with the additional bound DOMAIN2, and the actual run, EVERY depth $k+1$ of the original lower GENERAL invariant has its genuine monomial formula when its actual current nonmonomial order is zero, and otherwise its genuine positive-order formula whose lower StepOne value is $(0, \top)$. The formula is the existing full lower-value formula for the exact computed marked record and restricted tail, now with actual current membership internally supplied. The membership constructions above permit arbitrary DOMAIN bounds; this value formula retains DOMAIN2. It is not a claim of arbitrary upper higher-value equality or lower canonicality.*

Proof. The previous theorem first constructs the actual mask endpoint and its membership in the current original lower cosupport. The true run-head regime supplies its positive head level and mask membership, and the literal head–tail history remains a genuine original upper test. Apply the already proved entire computed-positive-member lower-value theorem at each natural depth k. Its endpoint-cosupport hypothesis is exactly the internally constructed membership, not a new assumption. Substitute the unchanged original run, lower record and restricted tail. The theorem's two branches use that record's own actual nonmonomial order and monomial exponent data. Its positive branch follows from the proved actual curve-current domain and original companion specialness; no lower test or canonical-history value law is invoked. □

325 Original positive StepOne smooth values on every upper TEST

Both ambient dimensions are positive. The target original record has DOMAIN2. Its actual upper history is an arbitrary genuine TEST, with no canonicality assumption. We assemble the actual run, independently computed source member mask and true current lower point before evaluating the unchanged original StepOne invariant.

Theorem 325.1 (Actual open member references have whole smooth GENERAL values without a lower TEST)**.** *Let m be an original DOMAIN2 record, let E_0 be an actual boundary sublist, and choose an actual point in an original open U with positive indexed through-member count. On ANY genuine original upper TEST restricted to U, the literal open member remainder has its WHOLE positive-depth GENERAL values compared under any actual smooth Cartesian pullback by the genuine dimension map. Its actual current target cosupport point is retained. No reference TEST, matching source record or value comparison is supplied.*

Proof. Pull the actual original upper record to U. Its real DOMAIN2 and genuine restricted upper TEST follow from original coherent open restriction. Map the actual original member sublist positionwise. True indexed boundary-count preimage equality makes its actual through-member subset nonempty before applying any member theorem.

The genuine whole open member-remainder identity identifies the literal open pullback with the independently computed member remainder of this open original upper record. It retains its actual remaining components, ordered boundary, ideal and mark. Substitute this equality into the actual endpoint cosupport and both value arguments. The accepted original-only member smooth-value theorem applies to the SAME actual Cartesian history, internally supplying all candidate domains and source codimension-one geometry. It proves the whole displayed value comparison without inferring candidate admissibility. □

Theorem 325.2 (The actual original positive run constructs both lower points and their whole values)**.** *For ONLY the original DOMAIN2 record, ANY upper TEST, its actual smooth Cartesian pullback and a genuine target current cosupport point that is nonspecial of positive level, construct the two actual run-mask endpoints and the genuine restricted endpoint map. The unchanged independently computed SOURCE lower GENERAL value at every positive depth is the dimension map of the actual target lower value. No run, lower TEST, endpoint membership or source-record matching certificate is an input.*

Proof. Actual original homogenized-coefficient weak equivalence supplies the true bar TEST and identical current cosupport. The unchanged original run-head theorem supplies its actual retained positive regime and head cosupport. The accepted original-run membership theorem internally constructs the actual target mask endpoint AND current original lower cosupport point before any lower-value theorem is applied.

The actual Cartesian split and positive member-mask producer construct the independent source and target lower records and their WHOLE initial pullback identity. The true original mask preimage dictionary and actual tail projection square give source mask membership; the restricted endpoint coordinate dictionary identifies its image with the internally constructed target endpoint.

At the actual head take the real current bar upper record, whose DOMAIN2 is internally supplied by the genuine upper TEST. On its actual target mask the original memberLowerExtension identity identifies the lower record with the literal open pullback of the genuine member remainder. True positive indexed head level makes the through-member count nonempty. The preceding theorem now applies to these EXACT objects. Substitute the independently proved initial source lower identity and the actual endpoint equation. The resulting full-value identity concerns both unchanged original lower constructors, even though neither lower history is asserted admissible. □

Theorem 325.3 (Whole original positive StepOne values commute with smooth pullback without canonicality)**.** *For the SAME original inputs and actual positive nonspecial current point, EVERY unchanged original StepOne value whose target lower GENERAL depth is $k+1$ satisfies the actual smooth comparison with source lower depth $(m-n)+k+1$. The map is the genuine StepOne dimension map induced by the original family dimension map. No lower TEST, upper canonicality or value comparison premise is supplied.*

Proof. The genuine original bar TEST supplies BOTH actual upper SPECIAL directions and indexed level equality, so the actual source upper point is nonspecial of the same positive level. The literal entire fixed-regime run dictionary identifies the source's OWN original run with the target's actual run before either StepOne value is unfolded. Use the preceding theorem to obtain the actual two mask endpoints and the whole lower GENERAL value comparison for the independently computed original lower records.

Now unfold the unchanged sequence-aware StepOne definition on both sides. Actual nonspecialness selects the lower branch; actual level and actual run equality select the SAME real split and positive member case. The constructed mask memberships replace the conditional lower values by their literal original GENERAL values. Apply the already proved whole lower-value identity. The genuine StepOne dimension map retains the actual level and maps the true finite lower value, yielding the claimed entire original StepOne equality.

This closes the positive member branch in original DOMAIN2 on arbitrary upper TESTs. The zero-contact branch, arbitrary higher-dimensional noncanonical recursive GENERAL, exceptional/infinitesimal semicoherence and the general unbounded global scheduler remain separate obligations. □

326 Actual order companions on untested smooth member histories

The original GENERAL invariant selects its own order-regime run and companion on the actual order mask. The member lower need not be a TEST. We construct these original objects from genuine current domains along an upper TEST, before any comparison of recursive values. Throughout this chapter the target ambient dimension satisfies $0 < n$.

Theorem 326.1 (Actual modeled nonmonomial orders and fixed-order runs)**.** *Let M be the actual upper record and K a real candidate whose ENTIRE current DOMAIN b is supplied for every original M TEST. Along any actual smooth Cartesian pullback of such a TEST, the unchanged original nonmonomial order, every fixed-order cosupport regime and its literal run equal those at the actual target image. The candidate history is not assumed admissible or canonical. The final original member consumer below constructs the geometric family internally.*

Proof. The genuine modeled controlled theorem first constructs the ENTIRE current K record pullback from the real coherent ideals and smooth successor carriers. The independently given actual target domain supplies SNC, carrier transversality and nonzero-along witnesses. The actual nonmonomial-part smooth theorem therefore applies to the whole current ideals and proves equality of their original orders. The modeled whole cosupport preimage dictionary and this order equality prove both directions of the literal order regime.

Induct on the genuine Cartesian history for a fixed order. At a successor use its shorter upper TEST before applying the regime dictionary. The true lift projection square identifies the two actual points in the recursive run equations. In the regime both counters increment and the induction hypothesis applies; outside it both reset to zero. Reflexive and empty-centre steps are retained literally. □

Theorem 326.2 (Genuine modeled order masks and whole independently computed companions)**.** *Under the SAME genuine current-domain family, the actual source order mask is the whole preimage of the target order mask. Construct the actual restricted endpoint submersion between them and prove ENTIRE marked-record equality between the true source and target* `compAt` *constructors. Their actual underlying map is the given endpoint map. No mask map, companion identity or candidate TEST is supplied.*

Proof. Read the actual whole current controlled identity from the first theorem. The genuine nonmonomial order-threshold pullback identifies the closed next-order threshold and its closure.

The actual smooth map is continuous and open, so it preserves that closure in both directions. Taking complements gives the precise original open order masks.

Restrict the actual endpoint map to those real opens. The same true submersion normal forms prove it is a submersion and its underlying point map remains the original one. On the target order mask, the actual domain pulls back genuinely to the open subset. The two composites through the endpoint square are literally equal; whole coherent marked-ideal pullback composition proves equality of the ENTIRE restricted current records. Only after this whole equality use the genuine companion smooth theorem, with the true coherent nonzero target record, to identify all four fields of the independently computed companions. Ordered boundaries and markings are retained. □

Theorem 326.3 (True simultaneous heads and tails retain modeled companions)**.** *At every actual split index, construct ONE real endpoint map that is simultaneously the Cartesian head map and the base map of the Cartesian tail. On its independently computed candidate order masks, construct the genuine submersion and the WHOLE original companion record identity of the preceding theorem. No independently selected head, tail or companion matching certificate is supplied.*

Proof. The genuine simultaneous split producer constructs the actual head and tail Cartesian histories sharing the SAME map. The original upper TEST gives its literal head TEST. Apply the preceding companion producer to that actual pair of heads, with the original geometric family unchanged. It constructs the true order opens, actual restricted submersion and whole companion identity. All original positions remain in the heads and tails, including indices beyond the history length. No claim that the candidate tail is a TEST is used. □

Theorem 326.4 (Only original member inputs construct untested order-run companions)**.** *An ORIGINAL DOMAIN b record for ANY natural b, a nonempty set of its actual indexed member positions, ANY original upper TEST and its actual smooth Cartesian history ALONE construct the original candidate fixed-order run comparison and EVERY actual split's whole companion/order-mask diagram. All source relative dimensions are allowed. Neither source nor target member candidate is asserted to have a TEST history.*

Proof. The genuine original member theorem supplies the ENTIRE DOMAIN $(b-1)$ of the literal member remainder at EVERY original upper TEST: actual strict carrier smoothness, coherent nonzero ideal, ordered SNC and transversality are proved before invoking any transport theorem. Use this internally constructed family in the first and third theorems. They supply the actual original run, simultaneous split, order mask and full independently computed companion records. No desired later geometry, point membership, source run or companion equality appears among the original inputs.

This constructs actual source objects needed for later recursive comparison. It does not yet prove all companion-current cosupport transport or whole upper GENERAL equality on noncanonical histories. Exceptional and infinitesimal semicoherence and the general unbounded scheduler remain separate obligations. □

327 Whole smooth GENERAL values on genuine untested curve references

The actual member curve can be transformed on an original upper TEST without being admissible itself. Its smooth pullback has dimension increased by the relative dimension and need not be a curve. We prove the genuine current companion geometry before evaluating its unchanged original GENERAL invariant. Both ambient dimensions are positive.

Theorem 327.1 (An arbitrary actual source open retains the genuine Cartesian history)**.** *For ANY actual finite blowup history and any original source open, the literal restricted history is its genuine Cartesian pullback along the open inclusion. Thus any genuine smooth Cartesian history can be restricted ONLY on its source, retaining the ENTIRE target history, with its literal composite endpoint map. No candidate TEST or matching local history is assumed.*

Proof. For an open inclusion, induct on the actual blowup history. At the reflexive history it is a genuine submersion, including the empty open. At a successor the actual restricted map is the actual blowup of the true inverse-image centre. Its projection square with the endpoint open inclusion commutes literally, and that inclusion is holomorphic and a submersion. True blowup universal uniqueness identifies this actual square with the canonical lift, giving the genuine Cartesian successor constructor. Every empty centre stays an actual step. Finally compose this actual open-inclusion pullback with the given Cartesian history using the already proved whole actual composition theorem. Both base and endpoint maps are the literal composites. □

Theorem 327.2 (The actual curve order-run companion has real current coefficient cosupport)**.** *Let the real candidate K_R have DOMAIN1 on every genuine original upper M TEST. At an actual current K cosupport point of positive nonmonomial order, construct its UNCHANGED original order-run, head, tail and actual order-mask point. Its ACTUAL masked run companion has current coefficient DOMAIN1 and actual current coefficient cosupport. The actual whole coefficient record is the endpoint open pullback of the global head companion's coefficient transform. All its genuine current domains on the same upper tail tests are internally supplied. No companion TEST, current companion point or presentation equality is an input.*

Proof. The actual current curve domain makes the nonmonomial part a unit on every actual boundary member. Positive order therefore excludes the ENTIRE current boundary. The original regime-run equation supplies the true start-regime cosupport and order; the actual head domain gives its genuine open order-mask membership.

Form the actual companion and its coefficient record ONCE at that head. The true coherent companion operations give initial DOMAIN1 with the SAME carrier and ordered boundary as the actual head reference. The genuine REFLEXIVE companion-cosupport dictionary gives the initial companion point, and actual coefficient weak equivalence gives initial coefficient membership. Independently transport actual candidate domains on each upper tail TEST using the genuine same-carrier/E geometry and coherent restricted colons.

The coefficient boundary at the endpoint is literally the reference's current boundary, which the actual point avoids. The real off-centre cosupport recursion therefore transports BOTH coefficient cosupport directions from the initial point to the endpoint. The preceding open-inclusion Cartesian history and whole modeled controlled theorem prove the ENTIRE controlled open restriction identity without assuming candidate admissibility. This actual record identity proves the masked current coefficient DOMAIN1 and point membership. □

Theorem 327.3 (True smooth codimension-one geometry gives the source companion special leaf)**.** *For ANY genuine smooth Cartesian pullback of the same upper history, construct the actual SOURCE order-mask endpoint of its independently computed original run companion. At EVERY recursive StepOne depth its unchanged original value is $(0, \top)$. Source relative dimension is arbitrary. Source DOMAIN1 and candidate TEST are neither assumed nor concluded.*

Proof. Actual nonmonomial-order and literal order-run naturality identify the source's OWN order and run with the target's actual ones. The genuine simultaneous split and whole independent companion diagram give the source companion as the actual restricted smooth pullback of the target companion, with its actual mask point internally constructed. Actual initial coefficient smooth naturality and true coherent pullback composition identify that source coefficient record with the pullback of the global target head coefficient record.

Restrict only the actual source tail to its real mask and retain the global target tail by the first theorem. The preceding theorem's real coefficient-domain family supplies coherence and strict smoothness BEFORE whole controlled smooth naturality is applied. Thus the actual source current coefficient record is the genuine endpoint pullback of the global target current coefficient record.

That target current coefficient cosupport is genuinely regular of codimension one in its smooth curve carrier at the actual point, by the independently proved DOMAIN1 and cosupport facts. Genuine smooth codimension-one reflection therefore places the actual source point in its actual coefficient codimension-one support. It avoids the ENTIRE actual source coefficient boundary by the whole ordered record identity, so its original indexed level is zero. Only now infer the unchanged original zero-branch SPECIAL predicate and unfold StepOne, obtaining $(0, \top)$ at EVERY depth. No source curve-dimension claim is used. □

Theorem 327.4 (Whole original GENERAL values reflect true smooth pullback on modeled curve histories)**.** *For every k, every actual current candidate cosupport point and ANY actual smooth Cartesian pullback of the upper TEST, the unchanged source GENERAL value at depth $(m-n)+k+1$ equals the genuine original dimension map of the target GENERAL value at depth $k+1$. Only the explicit real current curve-domain family is used; there is no candidate TEST, canonicality, source-value or recursive comparison certificate.*

Proof. The true modeled whole controlled identity gives nonmonomial order equality. If it is zero, the unchanged GENERAL definition uses the actual monomial invariant. Its genuine whole smooth dimension-shift theorem retains each original indexed exponent and gives precisely the original family dimension map on this branch.

If the order is positive, the preceding theorem computes the actual source run companion StepOne value as $(0, \top)$ at every required source depth. The independently constructed target curve formula computes the same original target leaf. Substitute the true order/run and mask identities into the unchanged original GENERAL definition. The genuine dimension map preserves the actual order and level-zero top branch, proving the displayed whole value equality. The argument uses actual values of the original records and original histories; it does not transport a newly invented action or invariant. □

Theorem 327.5 (Only original bound-two member data supply the whole untested smooth values)**.** *For an ORIGINAL DOMAIN2 record, a nonempty actual indexed member subset, ANY original upper TEST and an actual current cosupport point of its genuine member remainder, EVERY whole positive-depth GENERAL value of the genuine smooth pullback of that member remainder is the actual dimension map of the original target value. All current candidate geometry is constructed internally from these original data. Neither candidate history is declared admissible or canonical.*

Proof. The genuine original member-remainder theorem supplies the ENTIRE DOMAIN1 of its actual controlled transform at EVERY original upper TEST, retaining the actual carrier, coherent nonzero ideal and ordered boundary. Use this internally constructed geometric family in the preceding theorem for the SAME actual Cartesian history and actual current target point. Thus no desired value comparison or companion geometry appears in the input.

This theorem concerns the actual smooth pullback of the original member remainder. The upper positive-branch assembly and general higher noncanonical histories still require their genuine original masks and independent source-record identifications. Exceptional/infinitesimal semicoherence and general unbounded scheduling are separate obligations. □

328 Actual smooth special regimes without a candidate TEST

The member lower can have genuine smooth coherent transforms on every original upper TEST while the upper centres miss its smaller carrier. Smooth naturality of the original special regimes

must therefore be proved from those actual records, rather than by claiming a lower TEST. The generic statements below expose only real reference geometry; the last original member consumer constructs it internally. Source and target ambient dimensions may differ.

Theorem 328.1 (Whole modeled coefficient records retain actual indexed original members)**.** *Let M be the original upper record and K an actual well-formed candidate. Assume the explicit real geometric family that $\overline{K}_R$ is well formed for EVERY genuine upper M TEST R. For any actual Cartesian pullback of such a TEST, construct the ENTIRE controlled $\overline{K}$ record identity, the exact preimage of its indexed original-member LIST, and equality of its actual indexed levels. No TEST of K, $\overline{K}$ or their source pullbacks is assumed.*

Proof. The actual initial smooth coefficient/homogenization identities give $f^*\overline{K} = \overline{f^*K}$, retaining all four marked fields. The genuine modeled controlled-history theorem applies to the candidate $\overline{K}$ because the stated family supplies actual current coherence and next strict-carrier smoothness BEFORE every step. This constructs its whole endpoint record and cosupport identities, without candidate admissibility.

The indexed original-member list is literally the first $\#K.E$ entries of the current coefficient record's ordered boundary. Read the boundary field of the whole record identity. The initial pullback list has the same length, and taking that many entries commutes with mapping each set to its true inverse image. This proves equality of the entire original-member list, including empty and repeated entries. Actual boundary-count preimage equality then proves the indexed level equality. No distinct-germ count replaces this count. □

Theorem 328.2 (Actual modeled zero and positive special branches reflect under smooth pullback)**.** *Under precisely the real modeled geometry above, the unchanged original* IsSpecial *predicate of f^*K along the actual source history holds if and only if the original predicate of K holds at its actual target image. Every actual fixed-level regime has the same two-sided dictionary. No desired specialness, regularity comparison or candidate cosupport is supplied.*

Proof. The preceding theorem first constructs the whole coefficient record and exact original-member list. The actual target coefficient ideal is coherent by the independent geometric family. True smooth cosupport reflection identifies both whole current cosupports. The established codimension-one support theorem uses genuine submersion local geometry to identify the entire zero-branch codimension-one support as a preimage. For the positive branch, the actual submersion maps the neighborhood filter at each source point onto the target neighborhood filter. Together with the whole carrier, member list and cosupport identities this gives both directions of the literal member-intersection germ locus.

Unfold the original special predicate only after these true geometric identities. Its level-zero/codimension-one and positive-level/member branches are the same at both actual points, so both implications follow. Combine this with the preceding indexed level equality to compare the actual conjunction defining each fixed-level nonspecial regime. □

Theorem 328.3 (The unchanged entire modeled special-regime run is natural)**.** *For every fixed natural level, the UNCHANGED original regime-run counter of the actual source candidate equals that of the target candidate at the actual image. This is the literal run of the SAME histories; no source run, matching head or canonical segment is assumed.*

Proof. Induct on the actual Cartesian history. In the reflexive case both run counters are zero. At a successor the real upper TEST supplies its shorter upper TEST. The preceding theorem therefore compares the actual fixed-level regime at the shorter source and target points. The genuine blowup lift projection square identifies the two target points used in the actual recursive run equations. If the regime holds, both counters increment and the induction hypothesis equates the old counters. If it does not hold, both literal equations reset to zero. Every actual empty centre remains a successor in this induction. □

Theorem 328.4 (Only original member geometry supplies all untested smooth special histories)**.** *In positive target ambient dimension n, let M satisfy the ORIGINAL DOMAIN b for ANY natural bound b, let S be a nonempty subset of its actual indexed boundary positions, and take ANY original upper TEST with an actual smooth Cartesian pullback history. For its ACTUAL member remainder K, construct the whole coefficient-record pullback, indexed level and both special-regime dictionaries at EVERY literal head, and equality of the actual fixed-level run on the whole history. These original inputs ALONE supply all reference geometry. Neither source nor target lower history is asserted to be a TEST.*

Proof. The genuine original member-remainder geometry supplies the actual DOMAIN $(b-1)$ of K_R at every upper M TEST. At the reflexive history it gives the real well-formed initial record K, hence that of $\overline{K}$. Independently induct on the SAME original TEST to construct well formedness for this actual coefficient record. Its carrier and ordered boundary transform literally from those of K, so the actual next smooth strict carrier, SNC and transversality follow from the true member domain. Genuine coherent pullback and relative colon supply its entire next ideal coherence from the previous coherent ideal; no candidate admissibility is used. These four facts construct actual current well formedness before the generic modeled theorem is applied. Thus no desired smooth naturality, candidate TEST or candidate geometry certificate is an input here.

Apply the preceding run theorem on the original history. For each literal head, the genuine Cartesian head producer constructs the real endpoint submersion, and the upper TEST head theorem supplies its genuine original TEST. Apply the first two theorems using the SAME internally constructed candidate-domain family. All whole records, indexed lists, levels and special branches concern the literal member remainder and retain every empty position.

This supplies the source-facing special-history bridge. It does not yet identify every candidate order-companion mask or prove the whole upper GENERAL invariant on arbitrary noncanonical histories. Exceptional and infinitesimal semicoherence and general unbounded scheduling remain open. □

329 Actual empty-blowup padding of terminal sharp-curve histories

An empty centre is a genuine position in an actual blowup history. We use actual identity blowups of the empty centre, retaining their exceptional slots and proving their precise geometric effect. No deletion of steps or canonical-history compression is used.

Definition 329.1 (Actual identity empty tails)**.** For a genuine complex manifold Y and $r \in \mathbb{N}$, construct an actual tower $B_r(Y) : Y \to Y$ by starting with the reflexive tower and appending r genuine identity blowups of the empty centre. Its length is exactly r, its actual projection is the identity and every actual centre is empty. Both inverse laws of its projection are literal identity laws; the tower itself has r positions.

Proof. The existing identity blowup theorem supplies an ACTUAL blowup predicate for id_Y at the empty centre. Use it at each successor of the recursion. The tower constructor increases the length by one, composes the previous projection with the identity and appends the actual empty centre. Induction from length zero proves each assertion. □

Theorem 329.2 (Actual empty padding preserves carrier and cosupport and records every empty boundary slot)**.** *Let M be an actual well-formed marked record and P a genuine original M TEST. Appending $B_r(Y_P)$ gives another genuine original TEST. Its actual current carrier, marking and cosupport equal those at P, and its ENTIRE ordered current exceptional list is*

$$E_P +\!\!+ \operatorname{replicate}(r, \varnothing).$$

No equality of the old and new ambient ideal representatives is asserted.

Proof. An empty centre is admissible for EVERY actual marked record, so induction constructs the actual appended TEST before current geometry is used. For one step, the strict carrier is the closure of its preimage under the identity after deleting the empty set. The old carrier is closed, so it is unchanged. Every old exceptional member is closed by actual well-formedness, hence the same computation fixes each of them. The new exceptional member is exactly the empty preimage. The marking is retained by the actual controlled transform definition.

At every point the actual projection is off its empty centre. The independently proved off-centre controlled cosupport equivalence gives both cosupport directions; its smooth-carrier and coherent-ideal requirements are supplied by actual current well-formedness. Iterating these one-step identities proves the stated carrier, marking, whole cosupport and ordered-list formulas. The controlled colon can normalize an ambient ideal along its carrier even in this empty step, which is why no literal whole-ideal equality follows from these geometric identities. □

Theorem 329.3 (A terminal literal difference remains empty along actual recipe padding)**.** *For positive ambient dimension n and the original curve/SNC/density data, ANY genuine original product TEST P with its ACTUAL literal difference $\Delta_P = \varnothing$ and ANY r, the genuine tail $B_r(Y_P)$ is a literal-Δ recipe tail on that SAME original prefix, and the actual difference at its endpoint is still empty. Every original position and every padded empty centre is retained.*

Proof. At $r = 0$ the recipe is reflexive and the difference is the stated actual terminal difference. At a successor, the genuine original TEST theorem for actual literal-difference recipes applies to the shorter recipe already constructed by induction, supplying the original shorter TEST internally. The induction hypothesis says its ACTUAL computed difference is empty; therefore the true next identity empty blowup is precisely a blowup of that computed difference. Actual whole-difference support localization gives $\Delta_{\text{next}} \subseteq \pi^{-1}(\Delta_{\text{old}})$. The right side is empty, proving the actual new difference empty. The shorter recipe plus equality of the REAL next empty centre with its computed old difference is exactly the next recipe condition. No new termination, order bound or canonicality assertion is used. □

Theorem 329.4 (Actual empty padding concatenates and transports through genuine biholomorphisms)**.** *The actual towers satisfy $B_{r+s}(Y) = B_r(Y)B_s(Y)$. For EVERY genuine biholomorphism $e : Y' \to Y$ and EVERY r, construct an actual stagewise isomorphism between $B_r(Y')$ and $B_r(Y)$ over e, whose terminal map is the SAME actual e.*

Proof. Induction on s gives the first equality: both successor histories are the same actual identity empty blowup appended to their shorter equal histories. For the second assertion, start from the reflexive tower isomorphism over e. At each successor both actual centres are empty, whose true preimages under e are empty; both next projections are the identity, so $\mathrm{id} \circ e = e \circ \mathrm{id}$ is the actual next square. The genuine tower-isomorphism constructor extends the shorter comparison with terminal map still e. This construction keeps all empty entries. Unequal stopping-length comparison, padding independence and actual overlap cocycles require the separate finite recipe comparison and remain subsequent consumers. □

330 Genuine finite comparisons of the actual sharp-curve recipe

The finite histories here retain EVERY actual centre, including empty centres. Equality of tail lengths is explicit. Unequal stopping lengths will require actual empty-blowup padding; no canonical-history compression is applied to these original TEST histories.

Theorem 330.1 (An actual stagewise tower comparison is unique)**.** *For two fixed actual finite towers T', T and a fixed genuine base biholomorphism Φ, two actual stagewise tower isomorphisms over Φ have the SAME terminal biholomorphism. This is uniqueness of stagewise comparisons, not of arbitrary maps over the composite base.*

Proof. Induct on the first actual stagewise isomorphism. At the reflexive towers the other isomorphism must also have terminal map Φ, by the reflexive constructor. At a successor, invert the other isomorphism's last constructor. It supplies an actual shorter stagewise isomorphism, a real next projection square and a true centre match. The induction hypothesis equates its shorter stage map with that of the first isomorphism. Both actual terminal maps are therefore holomorphic maps of the SAME two actual blowups satisfying the SAME next projection square. The preceding genuine same-centre uniqueness makes their underlying functions equal. Equality of the actual biholomorphisms follows pointwise. This also treats every empty-centre position. □

Theorem 330.2 (Literal difference recipe tails of the same length have an actual comparison)**.** *Let P', P be actual prefixes over the SAME original curve data, with a genuine stagewise isomorphism over the identity and actual terminal map χ_0. Assume ONLY that P is an original TEST; the other TEST is transported internally. Let S', S be actual tails of the SAME finite length such that EVERY real next centre is its own literal current Δ, computed on the SAME appended original prefix. Then construct a true tail isomorphism over χ_0 and a true full-history isomorphism of $P'S'$ with PS, with the SAME terminal biholomorphism. Every stage and ordered empty slot is retained.*

Proof. Induct simultaneously through the actual tail constructors, decreasing the first tail. Equal lengths force both tails to be reflexive at length zero, when the tail isomorphism is the actual reflexive one over χ_0 and the full comparison is the given prefix comparison. A successor on one side forces a successor on the other; their shorter tails have equal lengths and still satisfy the literal centre recipe. Apply the induction hypothesis to construct the real shorter tail and full-prefix comparisons. The genuine literal-recipe TEST theorem internally supplies the original TEST of the actual longer right-hand prefix. The two next centres are by the recipe their OWN independently computed actual differences. The preceding chapter therefore constructs the real next comparison with its actual projection square and centre equality internally. That square extends the tail isomorphism; append it to the original prefix comparison to obtain the full-history isomorphism. Empty centres remain actual successor steps and are compared by the same universal property. □

Theorem 330.3 (Two genuine first-then-difference recipes have a unique equal-length comparison)**.** *Under ONLY the original smooth curve/SNC/density hypotheses, take two actual first blowups of the actual initial FIRST defect and two actual subsequent literal-Δ recipe tails of the SAME length. There is a UNIQUE terminal biholomorphism which is an actual stagewise isomorphism of these full first-then-difference histories over the identity of the original base. No already matched centres, endpoint map, finite stopping or canonicality certificate is supplied.*

Proof. Start with the actual identity isomorphism of the original reflexive histories; the original mark-one TEST there holds directly. The preceding computed-FIRST-step comparison internally supplies the actual initial centre match and genuine first-stage biholomorphism. The FIRST-centre admissibility theorem, using the original geometry, internally supplies the original TEST of the actual right first step. The second theorem now constructs actual tail and full comparisons for the given literal equal-length recipes. Uniqueness follows from the first theorem applied to these SAME full towers. No FIRST-erasure or terminal-geometry equality is used to guess a recipe.

This is an actual finite comparison of original recipe histories. Restricting recipe histories to a common open, proving the real nested open identifications, retaining recipe outputs in an original-only local endpoint constructor, padding unequal terminal lengths and proving padding independence and overlap cocycles remain subsequent producers. □

331 Actual sharp-curve difference recipes restrict to true opens

Fix positive ambient dimension and the original smooth curve, ordered SNC and whole-family density data. An open restriction retains every actual centre position, including those whose true traces become empty. We restrict the real difference recipe, rather than recovering a centre rule from an arbitrary terminal geometry witness.

Theorem 331.1 (The actual concatenated restricted prefix has the true difference preimage)**.** *Let P, S be actual towers whose actual full concatenation is a genuine original product TEST, and let U be ANY original base open. Construct the genuine TEST of the actual concatenation*

$$(P|_U)(S|_{P^{-1}U})$$

for the literal original product pulled back to U. Its independently computed whole difference is exactly the inverse image of the actual whole difference on PS under the endpoint open inclusion. Both original input records and all ordered member positions are the actual ones. No TEST or difference identity of that concatenation is supplied.

Proof. The original open-input producer gives the actual closed smooth curve, ordered SNC family, whole-family density and regularity on U. The actual whole product pullback identity identifies the restricted initial record in all four fields. Genuine TEST restriction therefore gives the TEST of $(PS)|_U$.

The actual restriction-of-concatenation theorem constructs a stagewise isomorphism from $(PS)|_U$ to $(P|_U)(S|_{P^{-1}U})$. Its endpoint map is the genuine opens-congruence map induced by equality of the two actual open preimages, and its underlying map is the identity on the ambient endpoint points. The true coherent identity pullback and actual tower TEST transport give the concatenated restricted TEST. Actual difference naturality under this stagewise isomorphism and the independently proved original difference open-restriction theorem give the desired whole preimage identity. Surjectivity of the genuine opens-congruence map supplies the reverse set direction. No replacement ideal or difference certificate is used. □

Theorem 331.2 (Every actual literal difference recipe restricts with all its positions)**.** *Let P be any genuine original product TEST and S an actual literal difference recipe on it. For EVERY original base open U, the real restricted tail $S|_{P^{-1}U}$ is the actual literal difference recipe on the actual prefix $P|_U$ for the true pulled-back original data. Its length is EXACTLY the original length of S. Centres whose traces are empty remain actual successors.*

Proof. Induct on the actual tail constructors. The reflexive tail has the trivial recipe and length zero. At a successor, the original recipe supplies the actual shorter recipe and the literal equality of its next centre with the computed difference of the full shorter prefix. The genuine recipe TEST producer supplies that full original TEST before the first theorem is invoked.

Actual tower restriction uses exactly the trace of this real next centre and the actual restricted blowup map. The first theorem identifies that centre trace with the independently computed difference of the concatenated restricted shorter prefix. Together with the induction hypothesis this is precisely the successor recipe condition. The existing restriction length theorem retains every actual successor, including an empty trace; it proves the exact length equality. □

Theorem 331.3 (Actual terminal recipes stay terminal on every true base open)**.** *Under the preceding original inputs, suppose the ACTUAL difference at PS is empty. On every original base open, construct the genuine restricted literal recipe, its original TEST and its independently computed empty endpoint difference, with all actual positions retained. No restricted endpoint or terminality witness is supplied.*

Proof. The actual recipe theorem supplies the full original TEST. Apply the first theorem to obtain the true restricted concatenated TEST and its whole difference preimage identity, and the

second theorem to obtain the actual restricted recipe with exact length. The preimage of the given computed empty difference is empty. This proves actual restricted terminality for the same towers, maps and original ordered family. Comparisons on overlaps and their cocycles still require the genuine retained local two-phase recipes; no arbitrary existential endpoint is identified with such a recipe here. □

332 Original local sharp endpoints retain their actual two-phase recipes

An existential sharp endpoint does not remember its actual centre choices. We instead start with the proved actual first-defect blowup and its terminating literal-difference tail, and prove endpoint geometry for those SAME witnesses. Local compactness is constructed from the actual original cosupport, rather than supplied by a caller. Throughout the ambient dimension is positive.

Theorem 332.1 (A compact actual initial cosupport supplies a sharp endpoint with its real recipe)**.** *For the original smooth curve, ordered SNC and whole-family density data with true compact initial product cosupport, construct an ACTUAL blowup of the literal original FIRST-Z locus and an actual finite tail whose EVERY centre is its independently computed current whole difference. Retain both maps and both tower pieces. Their SAME full history is an original product TEST; its actual FIRST-Z and whole difference are empty; its map is proper; its current original record has DOMAIN1; the complete ordered original strict boundary plus accumulated exceptional family is SNC and transverse to the actual strict carrier; and the SAME projection is an isomorphism over the ENTIRE literal original good locus.*

Proof. Apply the genuine two-phase producer to the reflexive original prefix. It internally constructs the first blowup and the actual terminating literal-difference tail from compact original cosupport, retaining the literal recipe and the actual original TEST, both empty defects, properness and current domain. Read the first-prefix TEST from the actual full TEST by the genuine concatenation theorem.

Apply actual ambient-boundary SNC and the proved two-empty-defect transversality theorem to that SAME full original history. The true first-then-difference support theorem gives isomorphism of its actual projection over original FIRST-Z complement, using the retained real recipe. Actual original good-locus recognition places the entire good locus in that complement. Restrict both inverse directions and the actual differential-exceptional disjointness to this smaller open. Every conclusion concerns the original first blowup followed by the same actual tail. No arbitrary endpoint is selected and then assigned a centre rule. □

Theorem 332.2 (Every original point has an actual sharp local two-phase recipe)**.** *ONLY the original smooth curve, ordered SNC, whole-family density and ANY original ambient point supply an actual open neighborhood and a sharp local endpoint with all the real witnesses of the preceding theorem. The ENTIRE original product record on that open is the true pullback of the original record, and all ordered members are the true inverse images. The first actual centre is the locally computed original FIRST-Z and the actual finite tail retains its literal whole difference recipe. No compactness, open, history, centre or endpoint geometry certificate is supplied.*

Proof. The genuine original coherent nonzero product and smooth curve data construct the actual original DOMAIN1. Its true analytic zero-dimensional cosupport supplies an actual isolating open at a cosupport point and its open complement at a point outside cosupport. The proved original compact-cosupport-open producer therefore constructs a neighborhood of the given point with compact actual pulled-back cosupport internally.

The actual whole product pullback identity gives all four marked fields of the original record on this SAME open, including the literal ordered family. True open regularity and SNC pullback supply smoothness and the actual curve dimension; the open inclusion's genuine closure-preimage

identity supplies whole-family density. The compactness just constructed is therefore compactness of this exact original local product record. Apply the preceding retained-recipe producer to these internally constructed data. This proves sharp endpoint geometry and retains the first map, actual first centre and real terminating tail simultaneously. Comparisons of these recipes on nested overlap opens and their descent remain subsequent producers. □

333 Actual terminal sharp-curve recipes with unequal stopping lengths

The histories in this chapter retain their actual literal difference recipes. A terminal geometry witness alone does not determine such a recipe. Each empty padding position is a genuine blowup and remains in the ordered exceptional list. We compare those real histories after padding to a common length and prove that the resulting terminal map does not depend on that length.

Theorem 333.1 (Literal difference recipes concatenate exactly)**.** *For actual towers P, S, R, the appended tail SR is a literal difference recipe on P if and only if S is such a recipe on P and R is such a recipe on the actual full prefix PS.*

Proof. Induct on the actual constructors of R. For the reflexive tail, appending changes neither tower and the second recipe condition is true. At a successor, the recipe consists of the shorter recipe and the equality between its actual next centre and the independently computed difference on its full old prefix. The induction hypothesis splits the shorter recipe into the claimed two pieces. Actual tower associativity identifies $(PS)R$ with $P(SR)$, so the two next-centre equalities are literally the same. Reassemble the conjunction in both directions. No step, including an empty centre, is removed. □

Theorem 333.2 (Actual terminated recipes can be padded to every larger length)**.** *Take positive ambient dimension and the original smooth curve, ordered SNC and whole-family density data. Let P be ANY original product TEST, let S be a genuine literal-difference recipe on P, and assume its ACTUAL endpoint difference is empty. For every $L \geq \operatorname{length} S$, construct the actual tail*

$$S_L = S\, B_{L-\operatorname{length} S}(Y_S).$$

It has length exactly L, is a genuine literal-difference recipe on the SAME original prefix, and its actual endpoint difference is empty.

Proof. The real original recipe TEST theorem first supplies the genuine TEST of PS. The preceding actual identity-empty producer then applies to this true current prefix and its computed empty difference. It supplies the real padding recipe and the actual empty endpoint difference. The first theorem concatenates the original and padding recipes. Exact tower length additivity and the proved length of the identity-empty tail give length $S + (L - \operatorname{length} S) = L$. The endpoint manifold is the actual old endpoint because the padding projections are identities; all added positions are retained. □

Theorem 333.3 (Unequal terminated actual recipes have unique common-length comparisons)**.** *For the SAME original curve data, let P', P have an actual stagewise isomorphism over the identity, and let P be a genuine original TEST. Let S', S be actual literal-difference recipes on these respective prefixes with ACTUAL empty endpoint differences. Their lengths may differ. For every L at least both lengths, construct a UNIQUE genuine terminal biholomorphism that is the terminal map of an actual stagewise isomorphism of their full padded histories $P'S'_L$ and PS_L over the identity.*

Proof. Actual tower-isomorphism transport of the original TEST internally supplies the TEST of P'. Apply the preceding theorem independently to both real terminal recipes. Both resulting

recipes have exactly length L. The proved equal-length actual recipe comparison now constructs a genuine tail isomorphism over the supplied prefix map and the actual full-history isomorphism over the identity. The fixed-tower stagewise uniqueness theorem proves uniqueness among ALL terminal biholomorphisms that give such actual stagewise isomorphisms. No terminal-map or centre-matching certificate is supplied. This does not claim uniqueness among arbitrary holomorphic maps over a composite projection. □

Theorem 333.4 (The genuine terminal comparison is independent of the padding length)**.** *For fixed actual prefixes and tails, an actual padded stagewise comparison at any common length L extends to $L+r$ with the SAME terminal map. Consequently two genuine comparisons constructed at possibly different common lengths have exactly the same terminal biholomorphism. This extension fact itself only requires the actual stagewise comparisons and the true length bounds.*

Proof. Append to the given comparison the actual comparison between r identity-empty tails induced by its own genuine terminal map. Actual tower associativity and the exact concatenation law for identity-empty tails identify the resulting histories with the histories padded to $L+r$. The elementary subtraction equality $(L+r)-s=(L-s)+r$, valid for $s \leq L$, supplies the exact numbers of retained slots on each side. The terminal map remains unchanged.

For comparisons at L and L', extend the first by L' and the second by L. Both now compare the same actual full histories at length $L+L'$, using commutativity of addition for the second. Fixed-tower stagewise uniqueness equates their terminal maps. The argument does not delete the empty slots, identify the two differently sized tower objects, or assert canonical compression. Actual restriction of local recipes and their overlap cocycles remain subsequent consumers. □

334 Independent actual arbitrary-centre choices and entire finite Cartesian tower comparison

Definition 334.1 (Primitive genuine arbitrary smooth-centre choices)**.** A primitive choice family in complex dimension m assigns to EVERY actual closed smooth nowhere-dense centre in EVERY genuine m-dimensional manifold one genuine individual blowup, including the empty centre and Cartier centres. It supplies no tower, stage map, Cartesian square, marked identity, bound, termination or naturality certificate. Such a family exists from the actual geometric blowup-existence theorem.

Proof. For each genuine manifold and actual centre with its real closed-smooth and empty-interior hypotheses, apply actual blowup existence. Choose its underlying manifold, topology, charts, complex manifold structure and actual projection with the genuine blowup proof. Assemble exactly these individual geometric choices into the family. This is ordinary choice of actual objects; no downstream comparison or tower certificate is a field. □

Theorem 334.2 (Each primitive family internally produces its own literal actual finite tower)**.** *From ONLY an original arbitrary-dimensional possibly non-surjective genuine submersion f, an original genuine ordinary finite target tower T, and one primitive choice family, construct the family's OWN actual source tower and actual endpoint map. Every source centre is literally the true preimage of the SAME target centre under the previous internally produced actual map. Its actual stage history is produced internally, hence its map is genuinely smooth, its entire original square is Cartesian, its length is exactly the target length, and EVERY literal head has that actual history.*

Proof. Recurse on the genuine ordinary target tower. For the empty tower use the original source manifold, literal empty source tower and original map. For a genuine target step, the recursion has already produced the source predecessor and its actual smooth map. The actual geometric preimage-centre theorem supplies the current source centre's real closed smoothness and empty

interior from these original step inputs. Invoke the primitive family on this actual centre, then append its genuine chosen blowup and the internally defined actual lift. This is precisely the step constructor of the actual non-surjective history. The already proved geometric history theorems give smoothness, whole square, entire universal property, literal equal lengths and every true head. Every empty or repeated centre remains a literal step. The history is an output of this recursion, not an input. □

Theorem 334.3 (Entire universal properties produce a unique actual biholomorphic comparison)**.** *Two actual smooth Cartesian projection squares over the SAME original map and SAME target tower have exactly one genuine biholomorphism between their source last stages that preserves BOTH the original source projection and the entire target-stage map. This auxiliary universal-property theorem is applied below to squares actually generated from original geometry.*

Proof. Apply the second square's ENTIRE universal property to the first source projection and first actual stage map. Their genuine projection identity gives a compatible whole cone; this produces a whole holomorphic map a with both desired equalities. Reverse the squares to produce a whole holomorphic map b in the other direction. The composite ba and the identity map have the SAME two whole projections into the first square. Its ENTIRE uniqueness makes $ba = \mathrm{id}$. Similarly the second square's uniqueness gives $ab = \mathrm{id}$. Thus the two whole holomorphic maps form an actual biholomorphism. Any other biholomorphism with these two projection identities is a competing whole holomorphic lift in the second square, so that same uniqueness identifies it with a. Cones lying entirely in any exceptional divisor are included; no off-centre density or invertible test ideal is assumed. □

Theorem 334.4 (Original independent families internally produce entire choice comparison at every literal head)**.** *The ONLY inputs are an original genuine arbitrary-dimensional possibly non-surjective submersion, a genuine ordinary finite target tower and two independent primitive genuine blowup-choice families. Internally construct the two OWN actual source towers and their unique entire comparison. Both source lengths equal the SAME target length. For EVERY literal split/head index, internally produce the actual two head-stage maps and their unique biholomorphic comparison preserving BOTH head projections. There is no caller source tower, Cartesian, stage, comparison or prefix certificate.*

Proof. Run the independent actual finite recursion for each primitive family. Both genuine histories internally give smoothness, whole source projection identities and entire Cartesian universal properties. The previous theorem therefore produces the unique actual comparison. Its selected value is the comparison function, with its two whole projection identities and its uniqueness from exactly the same theorem. The actual length theorem gives the two equal lengths. For any literal head index apply each SAME generated history's head theorem, including indices beyond length. Those actual heads internally give their true smooth maps, squares and ENTIRE universal properties; the previous theorem produces their actual unique comparison. No head is replaced by a prescribed candidate tower, and no literal empty or repeated step is deleted. No higher-invariant canonical centre or canonical-driver identification is inferred from ordinary finite geometry. □

Theorem 334.5 (The unique geometric comparison preserves the whole controlled marked endpoint)**.** *For an original well-formed marked datum with literal whole ambient carrier, and a genuine ordinary finite target test tower, BOTH independently generated source towers are genuine test towers. Pullback by the SAME actual geometric choice comparison identifies their ENTIRE controlled marked endpoints, including the whole ideal, carrier, marking and every ordered old and newly created boundary member, even empty or repeated ones. No caller ideal, admissibility, endpoint or naturality identity is supplied.*

Proof. The genuine marked-history joint induction applied to each internally generated history supplies that source's actual test admissibility and entire endpoint equality with the target endpoint's marked pullback by its actual stage map. The target well-formed-stage theorem supplies

its true current coherent ideal. Pull back the second whole equality by the actual comparison. The genuine whole coherent marked-pullback composition theorem rewrites this as pullback by the second stage map composed with the comparison. Its actual target-stage equality identifies that map with the first stage map. The first internally generated whole marked identity now gives exactly the first source endpoint. This is an equality of ENTIRE records, so every literal ordered boundary entry and whole ideal is retained. It asserts naturality of these actual ordinary towers; no general resolution existence, higher canonical invariant equality or termination follows. □

Theorem 334.6 (Actual independent-choice comparisons satisfy the genuine cocycle)**.** *The actual comparison of one family with itself is the identity, and for THREE independent primitive genuine choice families the composite of the first-to-second and second-to-third comparisons is the first-to-third comparison as an ENTIRE actual biholomorphism.*

Proof. The identity biholomorphism preserves both actual whole projections, so unique comparison gives the reflexive identity. For the composite, compose the two genuine original-source projection identities and separately the two entire target-stage identities. The composed biholomorphism is thus a genuine competitor with BOTH required projections in the first-to-third entire Cartesian square. Its actual uniqueness proves the cocycle. There are no supplied choice-compatibility or cocycle fields. General composite-submersion chart production, non-full-carrier controlled quotient naturality, general higher invariant/driver, noncompact termination, whole G3/G8 and formal native registration/publication remain OPEN. □

335 Actual direct and iterated ordinary finite Cartesian pullbacks compose

Theorem 335.1 (Two original submersions give the actual single-blowup composite lift)**.** *For TWO genuine original submersions f, h, a genuine original target blowup along C, a genuine middle blowup along the true f-preimage of C, and a genuine source blowup along the true iterated preimage, the internally defined actual lift for the composite submersion equals the composite of the two SAME actual individual lifts as a WHOLE map. Composite submersion is produced internally from the two original maps, not supplied.*

Proof. The genuine two-map smoothness theorem internally gives the actual composite submersion. Its centre preimage is literally $h^{-1}(f^{-1}C)$. The two actual single-blowup lifts are whole holomorphic maps. Their composite is whole holomorphic and, by the two genuine whole projection identities, its target projection equals $f \circ h$ followed by the SAME source blowup projection. The actual composite lift has precisely that same projection identity. The genuine original smooth blowup-lift uniqueness theorem therefore identifies the two whole maps. No prescribed composite map, square or equality certificate is an input, and the argument includes the entire exceptional divisor. □

Theorem 335.2 (Genuine actual stage histories compose by their own literal recursion)**.** *Actual histories over f and then over h yield the actual history over $f \circ h$ of the SAME iterated source tower with the SAME composite stage map. From ONLY two original submersions, a genuine ordinary finite target tower and primitive genuine middle/source individual blowup choices, internally produce that entire iterated tower and its actual composite history. There is no caller iterated tower, history, Cartesian or comparison certificate.*

Proof. For the auxiliary history statement induct on the actual source history over h, simultaneously using the compatible actual history over f. At the literal empty case both compatible towers must be empty, and the actual composite original submersion gives the actual empty history. At a true step, the compatible middle history's literal last constructor identifies its target centre with the actual preceding f-stage map's preimage of the SAME target centre. The source

step is its actual h-stage preimage. Thus it is literally the composite stage map's preimage of the SAME original centre. The induction has already produced the genuine composite predecessor history. The genuine stage maps are smooth internally; the two-map theorem gives genuine composite stage smoothness. Append the actual source blowup and original target blowup, then use the preceding whole actual-lift identity to identify the produced step map with the SAME two individual lift maps composed. This is the true step constructor of the actual composite history. For original-only production, independently run the actual primitive-choice recursion first for f and the original target tower, then for h and that SAME internally produced middle tower. The just-proved induction supplies the entire composite history on that SAME actual source tower. All literal empty and repeated steps are retained. □

Theorem 335.3 (Internally produced direct and iterated towers have a unique entire comparison)**.** *From ONLY the same two original possibly non-surjective submersions, genuine ordinary finite target tower and THREE primitive genuine families for the middle, iterated-source and direct-source blowups, internally construct the iterated source tower and an INDEPENDENT direct source tower for the actual composite submersion. They have a UNIQUE actual whole biholomorphic comparison preserving BOTH the original source projection and the whole target-stage map. The direct composite smoothness, both towers and all naturality identities are produced internally.*

Proof. The preceding producer internally constructs the iterated source tower and its actual composite stage history. The genuine two-map theorem internally gives original composite smoothness; the independent direct primitive family then genuinely constructs its OWN actual tower for the SAME original target tower. Both actual histories have the SAME original composite map and SAME genuine target tower. Their internally proved ENTIRE Cartesian universal properties therefore give the genuinely unique entire comparison by the proved two-whole-square comparison theorem. It constructs two actual whole holomorphic maps and proves both inverse identities by ENTIRE uniqueness, including cones wholly in the exceptional divisor. Select this actual unique comparison. Its two real whole projection identities and its actual uniqueness are precisely the theorem's conclusions; none is assumed by either independent recursive construction. □

Theorem 335.4 (Original input internally produces composition geometry at every literal head)**.** *For the SAME original-only inputs, BOTH actually produced source lengths equal the SAME original target length, their actual last-stage maps are genuine submersions, and both whole projection squares are genuinely Cartesian. EVERY literal split/head index internally gives true actual stage maps and actual histories for the SAME original composite map and SAME target head, with their UNIQUE entire biholomorphic comparison preserving both whole head projections. Cutting beyond length gives genuine empty heads; empty or repeated steps are never silently compressed.*

Proof. Both actual histories are internally produced by the preceding constructions. Their real history theorems give genuine smoothness, exactly equal lengths, whole source/target squares and entire Cartesian universal properties. For any literal split/head index, apply the SAME generated histories' actual head theorem to both actual towers. This internally gives their actual head stage maps and genuine composite-map histories with the SAME target head. The real entire universal properties of these true heads give their actual unique comparison. No supplied prefix tower, empty-step suppression, stage-domain identity, map or comparison certificate enters this production. The construction is finite recursion on genuine supplied ordinary geometry, not an assertion of higher canonical-driver termination. □

Theorem 335.5 (The same direct-versus-iterated geometric comparison preserves the entire controlled marked record)**.** *For an original well-formed marked datum with literal full ambient carrier and a genuine original ordinary finite admissible target tower, both OWN source towers are genuine test towers. The SAME actual whole comparison pulls the ENTIRE direct controlled marked endpoint back to the ENTIRE iterated controlled endpoint starting from the actual iterated*

original marked pullback. This includes the whole ideal, carrier, marking and ALL literal ordered old/created boundary entries. No caller source test, coherence, marked or comparison identity is assumed.

Proof. Genuine original coherence and the actual original maps give the entire initial-record composition dictionary: direct pullback of the original record by $f \circ h$ equals its actual two iterated whole pullbacks. Apply the actual marked-history joint induction separately to the internally produced composite iterated history and independent direct history. It supplies both genuine source test conditions and ENTIRE endpoint identities with marked pullbacks of the SAME target endpoint by the SAME actual maps. The target stage is well formed internally, hence its entire ideal is actually coherent. Pull back the direct endpoint equality by the actual geometric comparison; genuine whole coherent marked composition rewrites it as pullback by the direct map composed with this comparison. The actual target-stage identity identifies that map with the actual iterated map. The iterated whole endpoint identity then gives precisely the iterated controlled marked record. Finally the genuine initial-record composition dictionary identifies the original input on that tower with the actual two iterated pullbacks, without deleting any literal boundary or history. General non-full-carrier controlled quotient naturality, higher canonical invariant/driver identification, noncompact termination, whole G3/G8 and formal native registration/publication remain OPEN. □

336 Actual whole controlled marked dictionaries along a produced finite smooth Cartesian tower

Theorem 336.1 (Actual history internally carries genuine admissibility and the whole marked record)**.** *Let an original arbitrary-dimensional submersion have its genuinely produced actual Cartesian stage history. Let the original marked record be well formed with literal whole ambient carrier, and let the original finite target tower be genuinely admissible for it. Then the SAME actual source tower is genuinely admissible for the whole original marked pullback, and its ENTIRE controlled marked endpoint equals the whole marked pullback of the target endpoint under the SAME actual produced stage map. In particular its whole relative cosupport equals the true preimage of the target cosupport. The actual history is an output of original geometry; the original-only producer below supplies it internally, without a caller marked, stage, coherence or admissibility output certificate.*

Proof. Induct on the true actual stage history. The empty towers have the original pullback marked record literally, and their test conditions are true. At a step, genuine target admissibility gives predecessor test admissibility and the current original centre's real admissibility. The induction hypothesis has already supplied source predecessor test admissibility and its ENTIRE marked equality. The genuine target well-formed-stage theorem internally gives its current coherence, closed smooth carrier and SNC/transversality. The existing whole-ambient test-tower theorem gives the actual current carrier still equal to univ. The actual predecessor map is smooth by the internally produced geometry. Apply the genuine admissibility preimage theorem to the actual target current record and current centre; use the previous WHOLE record equality to identify the actual source admissibility. Thus the literal new source step is a true admissible test step. Rewrite the previous full record equality into the source successor and apply the genuinely proved entire arbitrary-mark controlled successor theorem. This supplies its entire N/E/I/d equality, preserving every literal empty or repeated old/new boundary member. No source test-tower field is assumed. For cosupport, the internally obtained target well-formedness gives actual coherence, and the proved whole relative cosupport theorem applied to the SAME actual smooth endpoint map gives the dictionary after rewriting the whole marked equality. Non-surjectivity and marks other than one remain covered; no canonical centre identification is inferred. □

Theorem 336.2 (Original actual geometry and original test tower produce all literal marked Cartesian stages)**.** *The ONLY inputs are an original genuine arbitrary-dimensional possibly non-surjective submersion, an original well-formed marked record with carrier univ, and a genuine original ordinary finite admissible target tower. Internally produce its literal finite source tower and actual smooth endpoint map. The full original projection square is holomorphically Cartesian, lengths are exactly equal, and the source tower is genuinely admissible with the ENTIRE original controlled marked pullback equality, genuine endpoint well-formedness and actual whole cosupport dictionary. For EVERY literal split/head index, including indices beyond length, internally produce its actual smooth Cartesian stage map, its true source test head, WHOLE controlled marked dictionary, genuine well-formed endpoint and actual cosupport dictionary. No source tower, naturality, prefix, coefficient, bound, termination or Cartier certificate is supplied.*

Proof. Apply the original-only actual Cartesian-tower producer to the original submersion and genuine finite target tower. It internally supplies the actual source tower and actual stage history, with literal pulled centres at every true step. The preceding joint induction internally yields source test admissibility and the whole endpoint marked dictionary. The original marked pullback is well formed by genuine original smooth geometry; the internally proved source test condition then gives endpoint well-formedness. The actual geometry gives smoothness, equal lengths and the WHOLE Cartesian universal property; the preceding cosupport theorem gives the actual endpoint cosupport. For any literal head index, use the SAME generated history's real head theorem. The genuine target test-tower split law gives target head admissibility, without a caller prefix certificate. Apply the same joint induction and source well-formed propagation to this SAME actual head. Its genuine geometry supplies its smooth full Cartesian square; its entire original marked equality and cosupport dictionary follow. This is actual simultaneous production, not a record populated from caller prefix equalities. Finite recursion is on supplied genuine finite geometry, not a asserted general algorithm termination result. Every empty and repeated step and ordered boundary entry remains literal. □

Theorem 336.3 (A genuinely resolved target tower has its internally produced resolved source tower)**.** *For the same original submersion and original whole-ambient well-formed marked record, if the supplied genuine finite target tower actually has empty terminal cosupport, the source tower internally produced from that SAME ordinary target tower is a genuine resolution of the original marked pullback. Its ENTIRE marked dictionary and actual smooth Cartesian geometry hold with exact literal equal length. This transports a genuinely resolved target tower; it does not assert existence or termination of a general canonical resolution driver.*

Proof. The real target resolution condition gives its actual test-tower condition and literal terminal cosupport equality with the empty set. Apply the preceding original-only construction, which supplies the source's real test-tower condition and exact WHOLE terminal cosupport dictionary. Rewriting the actual target empty cosupport in this dictionary gives the genuine source empty cosupport, with no source terminal certificate assumed. Combine its true source test condition with this genuine empty cosupport. All actual Cartesian geometry and entire marked identities are from the SAME construction, without replacing the target tower or deleting empty steps. General non-full-carrier restricted quotient naturality, higher invariant/old canonical-driver equality, general noncompact termination, whole G3/G8 and formal/native/publication remain OPEN. □

337 Genuine cross-dimensional all-head pullback after empty image steps

Let $f : B^m \to A^n$ be an actual holomorphic submersion, with no surjectivity assumption, and let F be the entire finite indexed SNC weighted list on A. Its actual image $U = f(B)$ is open, its actual factor $\bar{f} : B \to U$ is a surjective submersion, and F_U is the literal whole open pullback. All

these objects and their geometry have already been produced from the original input. Primitive Cartier blowup choice families are ordinary choices of actual blowup models, without any tower, comparison, marked-control, centre, bound, trace, canonicality or termination certificate.

Definition 337.1 (An actual smooth tower trace retains every matched and every silent target head)**.** For a genuine smooth base map $g : B^m \to A^n$, define a cross-dimensional tower-trace relation inductively on actual blowup towers. At the empty towers the terminal map is g and the actual base map is a submersion. A matched constructor adjoins actual blowups to both towers, with an actual submersion lift, its genuine projection square, equality of the entire source centre with the inverse image of the entire target centre, and the true holomorphic universal property of that square. A silent constructor adjoins an actual target blowup but no source step. Its actual new submersion lift commutes with that target projection, the preceding lift's inverse image of the actual target centre is literally empty, and its square with the identity source projection has the true holomorphic universal property. These are conclusions of the actual producer below, not certificates requested from original inputs. No same-dimension tower isomorphism is asserted for different dimensions, and a silent target centre need not itself be empty.

Theorem 337.2 (The true relation carries all heads, real length and genuine universal squares)**.** *An actual smooth tower trace supplies a true terminal submersion and the entire original projection square. Its source length is at most its target length. Each literal target head has a genuine source head, possibly with a different index, and an actual trace between those real heads. Actual trace towers append, and their real holomorphic universal properties compose to the full original projection square.*

Proof. The last constructor supplies the terminal submersion. Induct on the trace to compose true projection squares. At a matched stage both lengths grow by one; at a silent stage only the target length grows. For a target split index zero use the entire produced trace. For a positive target split index, invert the final constructor: a matched final stage shifts both real head indices, while a silent final stage shifts only the target index. Induction supplies the actual preceding source head and real lift, preserving the literal target history. For append, induct on the appended actual trace. The empty trace does nothing; matched and silent steps use their corresponding actual constructors with the internally obtained preceding trace. For universal squares, start with the identity square. At a matched stage first lift the cone to the preceding true square, then to the actual blowup square, and use their genuine uniqueness in reverse order. At a silent stage the preceding source map is unchanged: first lift through the preceding square and then through the genuine target blowup square with identity source projection. Uniqueness follows from exactly these two actual universal properties. No endpoint relation substitutes for the inductive whole-history relation. □

Theorem 337.3 (True geometric compression and true smooth stage pullback produce a genuine cross-dimensional trace)**.** *Suppose internally produced same-dimensional target towers R, S have an actual stage compression $R \to S$ over an actual base biholomorphism Φ, with actual endpoint biholomorphism χ. Suppose the actual source tower T has a genuine surjective smooth stage pullback $T \to S$ over g, with actual endpoint h. Then these true data produce an actual smooth tower trace $T \to R$ over $\Phi^{-1}g$, with endpoint $\chi^{-1}h$. The original-input applications specialize Φ to the identity. All matched centre identities and all silent empty inverse images are proved from the real constructors.*

Proof. Induct on the actual compression, jointly with the real smooth stage pullback. At the empty compression its target is empty; the true stage pullback has empty source, and the genuine endpoint is the actual transported base map $\Phi^{-1}g$. At a matched target-compression stage invert the actual smooth pullback at the real target blowup. It has a real source blowup, preceding actual pullback, genuine full centre equality, and actual holomorphic square. Apply induction

to the preceding real compression and pullback. The actual preceding and current target biholomorphisms transport the new map to $\chi^{-1}h$. The original real commutative squares prove its new square. Inverse-image composition and the actual centre dictionary prove the entire transformed source-centre equality. Conjugation of the actual submersion gives the true new submersion. To prove the genuine transported universal property, map a compatible target cone by the current target biholomorphism into the true old square, use its unique actual lift, and apply the inverse target biholomorphism to its equality; actual injectivity proves uniqueness.

At a silent compression stage the uncompressed tower has a genuine empty-centre blowup and the compressed target tower is unchanged. Keep the real source tower. The actual endpoint maps identify this target projection with a genuine biholomorphism, so its real inverse uniquely lifts every compatible holomorphic cone. The true preceding and current projection squares identify the new endpoint with $\chi^{-1}h$, conjugation gives its actual submersion, and the inverse image of the literal empty target centre is empty. Apply the silent trace constructor. The compression and smooth pullback assumptions are generic algebra inputs only here; the original-input producer internally proves both, rather than requesting them as caller certificates. □

Theorem 337.4 (Only original nonsurjective geometry produces the actual bounded all-head image trace)**.** *For every original prefix q, let $p(q)$ be the internally produced number of nonempty image traces in its real centre word. Restrict the actual globally chosen tower literally to U. Independently generate the actual image procedure at $p(q)$ and the actual source procedure at $p(q)$ on the true whole pullback $\bar{f}^*F_U = f^*F$. The primitive genuine choice families internally produce an actual cross-dimensional smooth tower trace from this independently chosen source tower to that literal restricted global tower. Its true endpoint maps by the literal terminal inclusion to the already produced actual nonsurjective lift to the unrestricted global terminal manifold.*

Proof. The actual original submersion charts internally supply the genuine open image and actual surjective image factor. Original SNC coordinates internally supply the entire image and source weighted lists, including all empty and repeated old metadata. The true independent open-procedure induction supplies $p(q)$ and actual all-head stage compression from the literal global restriction to the independently chosen image procedure. The actual surjective smooth-choice producer supplies a genuine all-head smooth stage pullback from the independently generated source procedure to this SAME actual image procedure at $p(q)$. Apply the preceding real composition theorem. No new centre choice, smooth-pullback, comparison, finite trace, bound or termination certificate is original input.

The resulting terminal map is the inverse of the actual open compression followed by the true smooth source-to-image lift. Its projection square and the genuine Cartier inverse projections identify its image under literal terminal inclusion with the unique already produced actual nonsurjective global lift. True image-factor composition identifies the entire initial weighted input with f^*F, not merely its support. The inductive trace compares source and literal image-restricted global towers at every matched/silent head. Globalizing this all-head relation to the unrestricted global tower requires the separate genuine open-square producer and is not claimed here. □

Theorem 337.5 (Independent OWN completions retain genuine all-head geometry and whole marked controls)**.** *The same true all-head construction compares the actual global OWN-completed tower, literally restricted to U, with the independently generated actual source OWN completion. The source true length is the nonempty traced length of the completed global word and its actual whole controlled ideal is unit. The entire source marked record equals the pullback of the whole actual raw restricted global record by the true compressed smooth lift, with ordered boundary filtered by ALL original indices followed by exactly the actual nonempty creation events. Retained entries that later become empty remain literal.*

Proof. The genuine completed open-compression producer uses the global and image procedures at their independently computed OWN coefficient sums. The surjective source-to-image geometry

preserves the entire initial coefficient word, so those real source and image OWN bounds agree internally. Its completed genuine smooth stage pullback therefore applies at that exact bound. Compose these true all-head comparisons as above. Actual trace word identities and the genuine source-to-image length theorem give the exact source length. The independently generated actual completed source procedure has whole unit endpoint and genuine marked resolution, without compactness.

For the whole record, the genuine surjective stage pullback identifies the entire actual source controlled record with the actual image record pullback. The true open-compression dictionary identifies that entire image record with the inverse-compression pullback of the raw actual literal global restriction, keeping ALL old indices and its actual creation-event mask. Whole holomorphic ideal pullback composition is valid because all these actual ideals are internally coherent. Inverse images and ordered list-map composition give the SAME entire carrier and literal retained boundary, the entire coherent ideal equality follows stalkwise finite-span composition, and marking remains literally one. No supplied marked identity or coefficient certificate occurs. □

Theorem 337.6 (Actual whole controlled records retain every original and genuine creation entry)**.** *The entire bounded or independently completed source marked record is the true compressed-lift pullback of the whole actual raw restricted global controlled record, with boundary filtered by ALL original indices followed by precisely the nonempty-at-creation global events. Its carrier, ideal and marking are exact equalities; retained entries that subsequently become empty remain literal. The raw ideal is internally coherent.*

Proof. The actual source-to-image smooth pullback identifies the entire genuine source controlled record with the actual image record pullback. The genuine open-compression dictionary identifies that image record with the inverse-compression pullback of the raw literal global restriction, retaining ALL old indices and its actual creation-event mask. The actual globally chosen stage ideal is internally coherent by its genuine marked test and whole chosen-reference dictionary, and open pullback internally gives coherence of the entire raw restricted ideal. Whole holomorphic finite-span ideal pullback therefore composes for these SAME actual ideals. Inverse images and literal list-map composition prove equality of the entire carrier and the SAME retained ordered boundary. The finite-span composition identity gives the entire ideal equality, and both markings are literally one. Extensionality in these four actual fields gives the whole marked equality. Apply this argument both at the genuine bounded trace count and at the independently computed OWN source completion bound. No ideal, coherence, history or marked-identity conclusion is caller input. □

Theorem 337.7 (True projection uniqueness removes intermediate choices and gives the actual choice cocycle)**.** *The resulting actual compressed smooth lift is uniquely determined by its genuine original projection square. Changing the original primitive source/global Cartier choices transports it by the internally produced genuine source and restricted global choice comparisons. Changing the intermediate image choice leaves the actual resulting lift unchanged, and all actual endpoint maps obey the true choice cocycle.*

Proof. The actual raw restricted global tower has a genuine bijective projection because its original chosen Cartier projection is biholomorphic and its literal open restriction preserves the actual inverse. Apply this true injectivity to two candidate terminal maps with the SAME actual base projection square. The internally produced compressed lift satisfies that square by the genuine inductive trace. The choice comparisons on the source and literal global restriction preserve their actual whole projections. Their two compositions with the compressed lifts therefore have the SAME original square; true target projection injectivity proves the exact cocycle. The same argument with fixed source/global choices proves that arbitrary intermediate image choices give the identical actual lift. This is real universal uniqueness for actual produced maps, not an original naturality certificate. Higher invariant functoriality, general old-driver identification and unrelated ideal admissibility remain OPEN. □

Formal root registration, native target build, leanok and publication remain pending. The all-head target is the literal image restriction; full unrestricted-global all-head geometry remains a separate genuine producer. Higher invariants, prior general old-driver identification and unrelated ideal admissibility remain OPEN.

338 Actual unrestricted global smooth traces of the original monomial procedure

The original inputs remain a genuine possibly nonsurjective complex submersion $f : B^m \to A^n$, the entire original indexed SNC weighted list F, and primitive genuine source, image and global Cartier blowup choice families. The preceding complete producer gives a true inductive matched/silent source trace to the literal image-restricted global tower. We now supply the actual geometric open-square ingredient and globalize EVERY literal target head to the unrestricted original global tower. No Cartesian, centre, submersion-of-lift, marked identity, finite trace, bound or termination result is requested as original input.

Theorem 338.1 (Actual open inclusions preserve genuine submersion charts)**.** *For any actual open V of a complex n-manifold and any genuine submersion $g : P^m \to V$, the actual composite $\iota_V g : P \to A$ is a submersion with the SAME globally chosen complement. In particular the actual inclusion $V \to A$ is a submersion, without nonempty or surjective assumptions.*

Proof. At a genuine source point p, $g(p)$ internally supplies nonemptiness of V. The actual inclusion is an open partial homeomorphism with entire source and target V. Extend the actual codomain submersion chart on V to A by composing it with the inverse of this actual open inclusion. Its source is the image of the original chart source, which is open. It is a genuine maximal-atlas chart: its forward map is holomorphic because on V the inverse inclusion is holomorphic into the open manifold (the actual subtype-val holomorphic equivalence applied to the identity), and its inverse is the original genuine holomorphic chart inverse followed by the holomorphic inclusion. The original source chart and SAME actual continuous linear splitting now give exactly the original coordinate projection for $\iota_V g$. Actual inclusion/inverse identities on these true chart domains prove all source and target memberships and written-in-charts equality. This proves genuine submersion of the composite internally. For inclusion itself start with the genuine identity submersion on V. If V or the source is empty, the pointwise proof has no fabricated point or complement change. □

Theorem 338.2 (Every actual open restriction is a true holomorphic universal square)**.** *Let $\pi : Z \to D$ be any genuine blowup and $V \subseteq D$ an actual open. Its actual restricted projection $\pi_V : \pi^{-1}V \to V$, with genuine blowup property already internally produced, and the two literal inclusions form a true holomorphic pullback square. No Cartier or trivial-blowup assumption is needed.*

Proof. For any actual holomorphic compatible cone $u : P \to V$, $v : P \to Z$ with $\iota_V u = \pi v$, define the lift $w(p) = (v(p), h_p)$ where the true square identifies $\pi(v(p))$ with $u(p) \in V$. The actual subtype-val holomorphic equivalence proves that w is holomorphic because its literal inclusion composite is the original genuine v. The true square and subtype extensionality give $\pi_V w = u$; inclusion gives $\iota w = v$. Any competing actual cone lift equals w by actual inclusion injectivity. Thus the full real holomorphic universal property is produced, even for an empty open or an empty actual centre. □

Definition 338.3 (Actual matched open tower geometry)**.** A matched open tower pullback is genuine geometric output: at the empty stage its actual map is an actual open inclusion preceded by a genuine biholomorphism onto that open. Each matched actual blowup step carries its true projection square, full inverse-image centre equality and real holomorphic universal square, and

its actual new map is again a genuine open inclusion preceded by a genuine biholomorphism onto that true open. This is an intermediate inductive relation internally constructed for the literal restriction of ANY real original blowup tower, not a certificate requested by the original-input consumers.

Theorem 338.4 (Literal-restriction traces globalize every matched and every silent head)**.** *Let T be any actual blowup tower on A, U an actual base open, and $g : B \to U$. A genuine inductive smooth tower trace from an actual source tower S to the literal restriction $T|_U$, with actual endpoint h, internally produces a genuine smooth tower trace over $\iota_U g$ from S to the original unrestricted T, with endpoint $\iota_{T^{-1}U} h$.*

Proof. First induct on the original real target tower to internally produce matched open tower geometry for its literal restriction: every actual endpoint map is the literal inclusion with the genuine identity biholomorphism, the true restricted centre dictionary is definitional and the preceding open-square theorem supplies each actual universal square. Compose this genuine matched open geometry with the actual smooth trace by joint induction on the matched geometry and the real smooth trace. For an empty target both source and actual trace are empty; the preceding true chart theorem gives the genuine composite base submersion. At a matched final head the actual target open map is a genuine biholomorphism onto its actual open followed by literal inclusion. Its submersion composite is produced by genuine biholomorphic conjugation and the preceding actual open-chart theorem. The restricted actual target blowup is precisely the literal restriction of the original true global blowup. Induction globalizes the preceding real trace. Compose the actual new lift with the literal final inclusion; the actual chart theorem gives its genuine submersion. Composing the true restricted projection square with inclusion gives the global projection square. The literal inverse-image definition of the restricted centre gives EXACT full source-centre equality with the inverse image of the original global centre. Compose the genuine matched universal square with the preceding actual open-square universal property: first restrict any compatible global cone into the actual target open using its true projection equality, then use the original genuine matched universal lift; subtype extensionality gives all equalities and uniqueness. At a silent restricted head source is unchanged. Apply the SAME geometry; its genuine empty inverse-image restricted centre becomes the literal empty inverse image of the original global centre, although the original global centre itself may be nonempty. The actual identity-source universal square composes with the same genuine open restriction square. Apply the corresponding matched or silent original-target trace constructor. This induction retains every literal target head, rather than replacing the relation by an endpoint square. □

Theorem 338.5 (Original nonsurjective geometry internally produces unrestricted bounded all-head traces)**.** *For every original prefix q, let $p(q)$ be the actual internally generated nonempty image trace count. The independently generated genuine source procedure at $p(q)$ on the SAME actual entire pulled input f^*F admits an actual matched/silent all-head smooth trace to the globally chosen original tower at q, with actual terminal map equal to the genuine conjugate of f by the two true Cartier projections. Its ENTIRE actual source controlled carrier, ideal and marking equal the true unrestricted global controlled pullback; its literal boundary retains ALL old indices and exactly the nonempty-at-creation original events.*

Proof. The actual original submersion internally produces its actual open image and genuine surjective image factor. The previously proved original-input compressed image producer supplies the entire matched/silent trace to the actual literal image restriction, with no caller comparison or trace certificate. Apply the preceding genuine globalization theorem to this SAME produced actual relation. The actual image inclusion composed with its actual factor is literally f, so this is the trace over the original map. The genuine projection inverse uniqueness identifies the terminal map with the already produced conjugate. Genuine trace-head and full-universal theorems then give every literal global target head its true source head, actual submersion, exact matched/empty centre dictionary and genuine holomorphic universal square. The target and

source bounds remain q and the internally computed $p(q)$ and are not falsely identified. For the whole actual controlled record, true marked-test restriction identifies the raw literal image record with the whole global open-inclusion pullback. The genuine globally chosen ideal is coherent internally; compose whole finite-span holomorphic ideal pullbacks with the true source lift. The existing whole raw-record dictionary gives the entire carrier and ideal equality, literal marking one, and the SAME boundary mask by ordered list-map composition. All old entries and genuine retained creation events remain literal even if subsequently empty. □

Theorem 338.6 (OWN-completed global traces preserve the SAME whole actual controlled data)**.** *The independently generated actual source OWN completion admits the same true all-head relation to the globally chosen original OWN completion. Its genuine length is the internally computed nonempty image trace count of the completed global word, and its actual controlled ideal is unit. The whole actual source carrier, ideal and marking are the true pullback of the entire unrestricted actual global controlled record by the produced terminal map. Its boundary is precisely the pullback of ALL old indices followed by retained nonempty-at-creation global events; subsequent empty metadata remains literal.*

Proof. Use the previously produced true independent source OWN completion and literal-image all-head trace, and globalize it by the same real open-square induction. True source OWN length and unit/resolution were internally established from the actual finite original coefficient sum, with no compactness or caller bound. The whole literal raw restricted global record is the actual open-inclusion pullback of the SAME genuine global controlled record, by true marked-test restriction. Its entire ideal is coherent internally from the genuine global marked test. Compose whole holomorphic finite-span pullbacks for this actual coherent ideal and the actual compressed source lift. The already proved exact whole raw restricted dictionary then gives the actual global entire ideal and carrier pullback equalities and the literally unchanged marking. Ordered list-map composition proves the SAME literal ALL-old plus true retained creation-mask equality. No unrelated ideal or discarded retained event enters. □

Theorem 338.7 (Actual original projection uniqueness and primitive-choice cocycles)**.** *The genuine unrestricted terminal lift is uniquely determined by the actual original projection square. It is independent of the intermediate image choice and transforms under the internally generated genuine source/global Cartier choice comparisons, both at bounded prefixes and OWN completion.*

Proof. The actual original global Cartier projection is biholomorphic, so its real injectivity identifies any two compatible terminal maps pointwise. The preceding produced true trace gives the projection square. The actual source/global primitive-choice comparisons preserve the same real original projections; their two composites therefore satisfy the SAME square. True global projection injectivity proves the actual cocycle. Intermediate-image-choice independence follows either from this uniqueness or from the preceding literal-image independence followed by literal inclusion. These are actual mathematical results for the independently generated marking-one monomial procedures. Higher invariant/general driver identification, arbitrary unrelated ideals, noncompact general termination, all-order normal-flat G8 and whole G3 remain OPEN. □

Formal registration, native target/audits and publication remain pending root recovery. This actual monomial smooth trace includes the unrestricted original global tower and every literal target head. Higher general invariants, general old-driver identification, unrelated ideal admissibility and whole G3/G8 remain OPEN.

339 Actual coefficient cosupport and finite reduced strict generators

This chapter connects the actual original coefficient-centre producer to whole division by the canonical reduced basis. Its strict ideal is the existing genuine formal strict transform, generated

by all nonzero original members normalized at their actual maximal centre orders. This is the finite strict-generation component of the presentation argument in [4, Section 7, p. 268; Theorem 9.4, pp. 277–278]. Full semicoherent-presentation construction, transformed derivative/Jacobian properties, analytic strict-transform equality and Bennett are not asserted.

Lemma 339.1 (Actual centre orders of the same canonical reduced members)**.** *Let G_a be the actual canonical reduced vertex members of I and let P_A be a genuine selected-coordinate centre ideal. If $G_a \in P_A^{w_A(a)}$, its actual maximal centre order is exactly $w_A(a)$. For every real chart $e \in A$ and every direction b, actual exceptional division therefore constructs the same normalized member G'_a with $X_e^{w_A(a)} G'_a = \sigma_b(G_a)$.*

Proof. The genuine coefficient of G_a at a is one, giving the upper bound for its actual least nonzero coefficient weight. The proved full coordinate-power characterization and the assumed actual membership give the lower bound. Actual blowup divisibility constructs the quotient and formal-domain cancellation proves its uniqueness. No arbitrary original monic representative is identified with this reduced member. □

Theorem 339.2 (The entire genuine formal strict ideal from reduced generators)**.** *With actual reduced-centre compatibility as above, the entire genuine formal strict transform in every chart is $(G'_a : a \text{ an actual vertex})$.*

Proof. The genuine strict ideal is already identified with the ideal of all normalized original members of all actual centre powers. For an actual $F \in I \cap P_A^q$, true reduced-basis division gives $F = \sum_a Q_a G_a$ with $Q_a \in P_A^{q - w_A(a)}$. Normalize every actual quotient at that genuine power. Applying the actual chart homomorphism to the whole division identity, and factoring the true exceptional powers, gives

$$[F]_q = \sum_a X_e^{w_A(a)-q} [Q_a]_{q-w_A(a)} G'_a.$$

The equality follows by multiplying both sides by X_e^q and using formal-domain cancellation; the natural truncated-exponent identity $(q - w) + w = q + (w - q)$ handles both inequalities. Thus all normalized original members lie in the finite reduced span. Conversely each G_a is an actual original member with its proved centre power, so its normalization belongs to the genuine strict ideal. This proves both whole ideal inclusions, including the zero ideal and weight-zero cases. □

Theorem 339.3 (Original coefficient cosupport internally supplies finite strict generators)**.** *Write the actual coordinates as (u, y, z) and let $C = \{y = 0\}$ in the presentation carrier. For every actual reduced member G_a and every low normal multiindex γ with $|\gamma| < w_A(a)$, assume genuine original holomorphic coefficient representatives $c_{a\gamma}$ in actual original coherent ideals $\mathfrak{a}_{a\gamma}$, with the explicit literal formal coefficient identity for G_a. Assume the actual smooth coordinate centre lies in $\operatorname{cosupp}(\mathfrak{a}_{a\gamma}, w_A(a) - |\gamma|)$. Then in every genuine chart of the actual ambient centre (y, z) there is an internally constructed family G'_a generating the ENTIRE genuine formal strict ideal and satisfying all its literal exceptional-product identities. The actual centre powers of the reduced members are not additional inputs. Here the coefficient ideals are coherent on the whole coordinate model, and the cosupport hypothesis concerns the whole coordinate plane. Producing such data from arbitrary local presentation germs, without assuming global extendability, remains a separate local-domain construction.*

Proof. For each vertex the actual original coefficient-centre theorem applies to these real holomorphic representatives, actual coherent ideals, genuine cosupport containment and literal coefficient identities. It produces $G_a \in (y, z)^{w_A(a)}$ internally by Hadamard, actual functional and germ powers, genuine Taylor expansion and the whole coefficient kernel. The preceding theorem now constructs the actual normalized reduced members and proves both whole strict-ideal inclusions. Its actual product identities are the original chart identities. The coefficient data contain no quotient, centre-power, strict-generation or Hilbert–Samuel certificate. Existence of all source presentation data and subsequent transformed stability remain separate goals. □

340 Actual center shears and whole-power reflection in every formal blowup direction

Keep the genuine coordinate center $P = (X_i : i \in A)$, the actual $e \in A$, the literal blowup chart σ_b, and $B = A \setminus \{e\}$. The true centered linear shear is

$$L_b(X_i) = X_i + \mathbf{1}_B(i) b_i X_e.$$

The genuine formal substitutions, rather than a supplied change-of-model certificate, implement the changes of direction in [4, §7, pp. 267–272]. All complex scalar actions and whole ideals remain the existing ones.

Lemma 340.1 (The actual centered shear constructs a genuine formal complex algebra map)**.** *The literal finite coordinate family $L_b(X_i)$ has actual zero constant coefficients and internally defines a genuine formal complex algebra map $L_b : S \to S$ with precisely those variable images. It fixes X_e.*

Proof. Each actual variable and each actual scalar multiple of X_e has zero constant coefficient. The genuine coefficient homomorphism is additive, so every displayed coordinate has zero constant coefficient. The finite family therefore supplies a genuine formal substitution internally. Its literal variable formula is the displayed family, and it retains the original complex algebra action. Since $e \notin B$, its actual e-coordinate is exactly X_e. □

Theorem 340.2 (Actual inverse shears and both whole center-power image directions)**.** *The actual maps L_b and L_{-b} are inverse on EVERY formal series. They are genuine formal algebra automorphisms, and for every $d \in \mathbb{N}$,*

$$(L_b)_* P = P, \qquad (L_b)_*(P^d) = P^d, \qquad F \in P^d \iff L_b(F) \in P^d.$$

Proof. Genuine formal substitution composition identifies the composite with the substitution by the actual composite coordinate family. At e both maps fix X_e; at every $i \notin B$ both fix X_i; at $i \in B$ the true image is $(X_i - b_i X_e) + b_i X_e = X_i$. Thus the actual composite coordinate family is the genuine variable family, whose formal substitution is the identity on EVERY series. Replacing b by $-b$ proves the other composite identity and hence constructs the true algebra inverse. Every selected coordinate image belongs to the same actual P, so image of the actual selected-variable span is contained in P. For the converse, each X_i with $i \in A$ equals its actual image minus, when $i \in B$, the scalar multiple b_i of the SAME actual image of X_e. Both terms belong to the actual image ideal since $e \in A$, proving the reverse whole inclusion. True image-of-power gives the assertion for every d. Forward membership follows from that whole image formula; reverse membership applies the actual inverse shear and the SAME whole image formula for $-b$. □

Theorem 340.3 (The SAME actual arbitrary-direction blowup factors through the genuine zero chart)**.** *For EVERY actual formal series F,*

$$\sigma_b(F) = \sigma_0(L_b(F)).$$

Proof. Use the genuine formal substitution composition theorem for the two actual finite zero-constant coordinate families. For $i \notin A$ the actual composite image is X_i. For $i = e$ it is X_e. For $i \in B$ it is $X_e X_i + b_i X_e = X_e(b_i + X_i)$. These are precisely the literal coordinate images of the SAME actual σ_b. Equality of the actual composite coordinate families, and thus equality of their genuine formal substitutions, proves the identity on EVERY series, with no arbitrary algebra-map extensionality principle for discontinuous homomorphisms being assumed. □

Theorem 340.4 (Genuine injectivity and both whole center-power reflection directions in every chart)**.** *The SAME actual map σ_b is injective. For EVERY $F \in S$ and every $d \in \mathbb{N}$,*

$$F \in P^d \Longleftrightarrow X_e^d \mid \sigma_b(F), \qquad \sigma_b^{-1}((X_e^d)) = P^d.$$

Every internally normalized quotient of a nonzero actual $F \in P^d$ is nonzero, in every actual direction b.

Proof. The preceding factorization expresses σ_b as the composite of the actual invertible shear and the already proved injective genuine zero-direction substitution, so it is injective. The true zero-chart whole-power reflection applied to $L_b(F)$ says that exceptional power divisibility of $\sigma_b(F)$ is equivalent to $L_b(F) \in P^d$. The true shear inverse and whole-power preservation make the latter equivalent to $F \in P^d$. This proves both membership directions and hence equality of the actual whole ideal inverse images. Finally, a zero actual normalized quotient would make its true exceptional-product numerator $\sigma_b(F)$ zero. Genuine actual injectivity forces $F = 0$, contradicting the given nonvanishing. This constructs the valuation-reflection facts internally. Identification of normalized-member ideals with the independent all-power strict saturation, transformed standard-basis stability and general Bennett remain subsequent separate proofs. □

341 Actual formal blowup coordinates and generation of normalized original members

Work in the genuine formal complex algebra $S = \mathbb{C}[[X_1, \ldots, X_n]]$. Fix an actual coordinate subset $A \subset \{1, \ldots, n\}$, an index $e \in A$, and actual complex constants b_i. Put $P = (X_i : i \in A)$ and $x = X_e$. The actual formal chart substitution in [4, §7, pp. 267–272, especially Lemma 7.22 and Corollary 7.24] is

$$\sigma(X_i) = \begin{cases} X_e & i = e, \\ X_e(b_i + X_i) & i \in A \setminus \{e\}, \\ X_i & i \notin A. \end{cases}$$

We construct the genuine formal algebra map and the actual exceptional quotients, then use the already proved whole coordinate-power division. The endpoint below concerns the ideal generated by normalized images of EVERY original formal member. Identifying this ideal with the independently specified all-power saturation, and deriving geometric compatibility of standard vertices with an admissible center, are separate subsequent steps.

Lemma 341.1 (The actual formal chart sends every whole center power to its literal exceptional power)**.** *Each displayed coordinate image has actual zero constant coefficient, so it internally defines a genuine complex formal substitution $\sigma : S \to S$. Its images of the WHOLE original coordinate ideals satisfy*

$$\sigma_* P = (x), \qquad \sigma_*(P^d) = (x^d) \quad (d \in \mathbb{N}).$$

Proof. Every displayed image contains an actual variable factor, and thus has zero constant coefficient. The finite zero-constant family therefore admits the genuine formal substitution, retaining the existing complex scalar action. The variable formula is precisely the displayed formula. Every selected coordinate image is divisible by the SAME actual x; image of the true selected-variable span is therefore contained in (x). Conversely $e \in A$, and its actual image is x itself, so (x) is contained in that whole ideal image. Actual image-of-power and the principal-ideal power identity give the assertion for EVERY d, including $d = 0$. □

Lemma 341.2 (Every genuine center-power member has a unique internally constructed exceptional quotient)**.** *For every actual $F \in P^d$ there is a unique genuine formal series $[F]_d \in S$ satisfying*

$$x^d [F]_d = \sigma(F).$$

The quotient is constructed from this actual membership and the preceding whole-ideal image formula, with no exceptional-divisibility certificate supplied.

Proof. Actual membership supplies $\sigma(F) \in \sigma_*(P^d) = (x^d)$. The true principal-ideal membership identity supplies an actual quotient, which defines $[F]_d$. The actual variable x has nonzero coefficient at its unit exponent, hence is nonzero. The genuine formal power-series ring over $\mathbb{C}$ is a domain, so $x^d \neq 0$ and cancellation proves uniqueness of the actual quotient. □

Theorem 341.3 (Actual normalized original members are generated by the actual normalized vertices)**.** *Fix any actual ideal $I \subset S$ and any monomial order. Let f_a be the internally chosen actual monic vertex series, and put $w_a = \sum_{i \in A} a_i$. Assume explicitly the actual source conditions $f_a \in P^{w_a}$ for every actual vertex a. Then for EVERY actual $F \in I \cap P^d$, its genuine quotient belongs to the actual finite ideal*

$$[F]_d \in \big([f_a]_{w_a} : a \text{ an actual vertex of } I\big).$$

Proof. The actual whole coordinate-power division already constructs the SAME whole quotients Q_a with

$$F = \sum_a Q_a f_a, \qquad Q_a \in P^{d-w_a}.$$

Normalize these actual quotients using the preceding internally constructed division: $\sigma(Q_a) = x^{d-w_a}[Q_a]_{d-w_a}$, and independently $\sigma(f_a) = x^{w_a}[f_a]_{w_a}$. Since $(d - w_a) + w_a = d + (w_a - d)$ in the natural numbers, actual multiplicativity and the finite actual sum give

$$\sigma(F) = x^d \sum_a x^{w_a - d}[Q_a]_{d-w_a}[f_a]_{w_a}.$$

The SAME actual left side is $x^d[F]_d$. Genuine domain cancellation therefore identifies $[F]_d$ with the displayed finite sum, which belongs to the claimed actual generated ideal. This includes both $w_a \leq d$ and $w_a > d$, with the genuine truncated natural exponents, and includes the zero ideal and the empty vertex family. □

Theorem 341.4 (Both whole normalized-member ideals coincide)**.** *Under the SAME explicit source vertex-power hypotheses, the actual whole ideal spanned by $[F]_d$ for EVERY $d \in \mathbb{N}$ and EVERY actual $F \in I \cap P^d$ equals the actual finite ideal spanned by the $[f_a]_{w_a}$.*

Proof. Every generator on the first side belongs to the second side by the preceding genuine whole-member theorem; the actual ideal-span induction gives that whole inclusion. Conversely each actual f_a belongs to I and, by the explicit source condition, to P^{w_a}. Its SAME normalized series is therefore literally among the generators on the first side. The actual ideal-span property gives the reverse whole inclusion. No saturation equality or geometric admissibility hypothesis is silently supplied by either ideal definition or either inclusion. □

342 Genuine monomial blowup coefficients reflect the whole center-power filtration

Keep the actual coordinate center $P = (X_i : i \in A)$ and $e \in A$. For this chapter take the actual zero-direction chart σ_0 of the preceding genuine formal blowup construction. Put $B = A \setminus \{e\}$. Its literal exponent map is

$$T(\alpha) = \alpha + \big(\sum_{i \in B} \alpha_i\big)\mathbf{e}_e.$$

This is the monomial chart underlying the substitutions in [4, §7, pp. 267–272]. The conclusion is a genuine reflection theorem, rather than a supplied exceptional-valuation or saturation comparison. Other chart directions and the all-power strict-transform saturation bridge remain subsequent work.

Lemma 342.1 (The actual chart exponent map is injective and gives the true monomial product)**.** *The actual map T is additive and injective. For every genuine exponent α,*

$$T(\alpha)_e = \sum_{i \in A} \alpha_i, \qquad \prod_i \sigma_0(X_i)^{\alpha_i} = X^{T(\alpha)}.$$

Proof. The true B-weight and the actual single-coordinate insertion are additive, proving additivity. The actual B-weight of $T(\alpha)$ equals the B-weight of α, because $e \notin B$. If $T(\alpha) = T(\beta)$, apply this true weight equality, then evaluate at e to recover $\alpha_e = \beta_e$. At every other index the inserted single exponent is zero, so equality gives every remaining coordinate. Thus $\alpha = \beta$. Since $e \in A$, the true A-weight is the sum of the true B-weight and the exponent at e, proving the displayed coordinate formula. The literal zero-direction coordinate image is $X_i X_e^{\mathbf{1}_B(i)}$. Genuine power multiplicativity and the finite product over the actual support therefore give $X^\alpha X_e^{\sum_{i \in B} \alpha_i} = X^{T(\alpha)}$ in the true formal ring. □

Theorem 342.2 (Every original coefficient survives at its literal true blown-up exponent)**.** *For EVERY actual formal series F and EVERY exponent α,*

$$[X^{T(\alpha)}]\sigma_0(F) = [X^\alpha]F.$$

In particular the genuine formal algebra map σ_0 is injective.

Proof. Apply the genuine formal substitution coefficient formula to the finite zero-constant coordinate family. Its actual summand at a source exponent β is $[X^\beta]F$ times the coefficient at $T(\alpha)$ of the true product $\prod_i \sigma_0(X_i)^{\beta_i}$. By the preceding literal monomial formula, this second coefficient is one when $T(\beta) = T(\alpha)$ and zero otherwise. Actual injectivity of T makes the sole surviving exponent exactly $\beta = \alpha$. The genuine coefficient sum therefore equals the claimed original coefficient. If two actual image series agree, this identity recovers every original coefficient; actual coefficient extensionality proves the original series equal, establishing injectivity. □

Theorem 342.3 (Both whole actual center-power reflection directions and true normalized nonvanishing)**.** *For EVERY actual $F \in S$ and every $d \in \mathbb{N}$,*

$$F \in P^d \quad \Longleftrightarrow \quad X_e^d \mid \sigma_0(F).$$

Thus the genuine inverse image of the whole exceptional principal power (X_e^d) is exactly the actual whole P^d. For every nonzero actual $F \in P^d$, the internally constructed genuine normalized quotient $[F]_d$ in this chart is nonzero.

Proof. Forward divisibility is supplied internally by the already proved whole-power chart image formula. Conversely assume the actual exceptional power divides $\sigma_0(F)$. The true variable-power coefficient kernel kills every target coefficient whose exponent at e is less than d. For any actual source α of A-weight less than d, the preceding exponent formula gives $T(\alpha)_e < d$. The true coefficient-reflection identity therefore kills the actual source coefficient at α. The genuine whole coordinate-power kernel puts F in P^d. This proves both membership directions at every actual numerator and therefore equality of the actual whole ideal inverse images. Finally, if the actual normalized quotient of a nonzero F were zero, its defining literal exceptional-product identity would give $\sigma_0(F) = 0$. The independently proved actual substitution injectivity would force $F = 0$, a contradiction. No target-shaped valuation, normalization or saturation premise is used. □

343 The genuine source formal strict ideal and its finite normalized vertex generators

Keep the actual coordinate center $P = (X_i : i \in A)$, index $e \in A$, true chart σ_b, and the internally constructed quotients $[F]_d$. Here we use exactly the formal strict-transform definition in [4, §7,

p. 268, followed by Lemma 7.22 and Corollary 7.24]: for nonzero F its center order is the largest d with $F \in P^d$, and the formal strict ideal is generated by the SAME normalized $[F]_{\mu_P(F)}$ for all nonzero $F \in I$. Zero members contribute only zero. The definition uses the actual original series, true center powers and genuine existing complex algebra map. It assumes no strict-generation, Hilbert–Samuel, division or all-power-saturation certificate.

Lemma 343.1 (The actual coefficient support internally constructs the genuine finite center order)**.** *For every actual nonzero F, the genuine nonzero coefficient support has a least A-weight, denoted $\mu_P(F) \in \mathbb{N}$, and*

$$F \in P^d \Longleftrightarrow d \le \mu_P(F).$$

In particular $F \in P^{\mu_P(F)}$ and $F \notin P^{\mu_P(F)+1}$. For an actual monic vertex series f_a explicitly satisfying $f_a \in P^{w_a}$, where $w_a = \sum_{i \in A} a_i$, its actual center order is exactly w_a.

Proof. Since $F \neq 0$, actual coefficient extensionality supplies a nonzero coefficient. Its true A-weight makes the set of such natural weights nonempty; natural well ordering supplies a least weight and an actual nonzero coefficient at that weight. If $F \in P^d$, the genuine whole coordinate-power kernel forbids this least weight from being less than d. Conversely if d is at most the least weight, every actual coefficient of A-weight less than d is zero; the SAME true whole power kernel gives $F \in P^d$. Taking d equal to the least weight and its successor proves the asserted largest-order characterization. For a monic actual f_a, its actual coefficient at a is one, so the least weight is at most w_a. Its explicit genuine power membership and the just proved characterization give the opposite inequality. Thus its actual source center order is w_a, without assuming it. □

Theorem 343.2 (The exact source formal strict ideal equals the whole normalized-original-member ideal)**.** *For an actual ideal $I \subset S$, define its genuine source formal strict transform I'_b by the actual span of $[F]_{\mu_P(F)}$ for all actual nonzero $F \in I$. Then the actual WHOLE ideal I'_b equals the ideal spanned by $[F]_d$ for EVERY $d \in \mathbb{N}$ and EVERY actual $F \in I \cap P^d$.*

Proof. Each source generator belongs to that whole normalized-member ideal, because $F \in P^{\mu_P(F)}$; this proves one actual span inclusion. Conversely take ANY actual normalized original member $[F]_d$ with $F \in I \cap P^d$. If $F = 0$, its true exceptional-product identity and domain cancellation give $[F]_d = 0$, hence membership in I'_b. If $F \neq 0$, the preceding characterization gives $d \le \mu_P(F)$. The SAME actual exceptional-product identities give

$$X_e^d\big(X_e^{\mu_P(F)-d}[F]_{\mu_P(F)}\big) = \sigma_b(F) = X_e^d[F]_d.$$

True domain cancellation therefore gives $[F]_d = X_e^{\mu_P(F)-d}[F]_{\mu_P(F)}$. The second factor is literally one of the source generators of I'_b, so this product belongs to that actual whole ideal. The actual span property proves the reverse whole inclusion. No independent analytic saturation equality is part of this formal source definition or either inclusion. □

Theorem 343.3 (Genuine finite source formal strict generators)**.** *Let f_a be the internally chosen actual monic vertex series of an actual ideal I for any monomial order, and explicitly assume only the actual source power memberships $f_a \in P^{w_a}$. The genuine source formal strict transform is the actual WHOLE finite ideal*

$$I'_b = \big([f_a]_{w_a} : a \text{ an actual vertex of } I\big).$$

The exponent w_a is the internally proved actual center order of the SAME f_a. For degree-compatible vertices with $w_a = |a|$, these are precisely the normalized generators in Corollary 7.24.

Proof. The preceding actual source-definition identification proves that I'_b equals the WHOLE normalized-original-member ideal. Genuine whole coordinate-power division already proves that every actual member of that ideal is in the finite normalized-vertex span, and that every actual

normalized vertex belongs to the whole member ideal. Thus the already proved BOTH whole ideal inclusions give the displayed equality. The least-weight theorem identifies each vertex's genuine source center order with w_a, making these literally the source strict transforms of those actual vertices. This is a finite strict-generation producer in the actual formal ring; geometric admissibility supplying the source vertex memberships, transformed-standard-basis stability, the analytic all-power-saturation bridge and general Bennett remain subsequent work. □

344 Genuine local coefficient cosupport and whole reduced strict generation

The coefficient ideals are required coherent only on an actual open neighbourhood U of the origin. Cosupport containment is required only on $U \cap \{y = 0\}$. No global extension of local holomorphic functions or coherent ideals is assumed. The real flattening pullback already produces exactly this local coherence notion. This is the local coefficient-centre component of [4, Theorem 9.4, pp. 277–278]; full presentation construction, transformed derivative/Jacobian properties and Bennett remain separate goals.

Lemma 344.1 (Local coherent germs propagate on their real domain)**.** *Let $\mathfrak{a}$ be an original ideal coherent on an actual open domain U, and let $x \in U$. Every original membership $f \in \mathfrak{a}_x$ remains a membership at all sufficiently nearby points.*

Proof. Take the actual finite original generating family and its membership characterization near x, supplied by local coherence. Express f as a finite holomorphic linear combination of these actual generators. Shrink to a common open where the coefficients, f and this literal equality remain holomorphic and valid. At each point of that open the same finite relation, read in the original membership characterization, proves the claimed membership. No extension outside the real domain is needed. □

Theorem 344.2 (Local plane cosupport produces actual whole germ powers)**.** *Let $U \subset \mathbb{C}^{k+s}$ be open with $0 \in U$, and let $\mathfrak{a}$ be coherent only on U. Suppose $U \cap \{y = 0\} \subset \operatorname{cosupp}(\mathfrak{a}, q)$. For every actual original member $f \in \mathfrak{a}_0$ holomorphic at zero, its entire germ lies in the actual normal-coordinate ideal power $(y_1, \dots, y_s)^q$. Its genuine Taylor series consequently lies in the same whole formal coordinate ideal power.*

Proof. Propagate the original membership on a neighbourhood contained in U. At every nearby point of the coordinate plane the genuine cosupport condition puts f in the actual maximal-ideal power of degree q. The original chart-derivative dictionary therefore makes all iterated ordinary derivatives of length less than q vanish along that plane near zero. The proved multivariable Hadamard–Taylor decomposition gives the actual recursive normal-power expression of degree q. Its genuine whole-germ normal-power dictionary yields membership in $(y)^q$. Apply the actual Taylor ring homomorphism and its whole coordinate-ideal identity to obtain the formal power. The argument includes $q = 0$ and an empty normal block, and uses neither coherence away from U nor a globally smooth coordinate-centre certificate. □

Theorem 344.3 (True local coefficient data produce the whole formal centre power)**.** *Let $F(w, z)$ be an actual formal series. On a genuine open neighbourhood $U \ni 0$ in $w = (u, y)$ suppose its low literal normal coefficients are Taylor expansions of original holomorphic coefficient functions $c_\gamma \in \mathfrak{a}_{\gamma,0}$, where $\mathfrak{a}_\gamma$ is coherent only on U and $U \cap \{y = 0\}$ lies in its genuine cosupport of weight $q - |\gamma|$. Then $F \in (y, z)^q$.*

Proof. For every low normal exponent, the preceding theorem gives its whole formal tangential-centre power of degree $q - |\gamma|$. The literal coefficient identity and the proved entire formal coefficient-power criterion imply $F \in (y, z)^q$. That criterion treats every original coefficient, including the high normal exponents whose weight alone already exceeds the required bound. No factorial convention or unproved identification with raw derivatives is used. □

Theorem 344.4 (Local original coefficients internally produce entire reduced strict generators)**.** *Suppose the actual canonical reduced members G_a have the preceding literal local coefficient data, with weight equal to their actual selected vertex weight. In every genuine blowup chart of (y, z) there is an internally constructed finite family generating the ENTIRE existing formal strict transform and satisfying the literal exceptional-product identities for these same reduced members. Centre-power membership and strict generation are not inputs to the coefficient data.*

Proof. The preceding producer supplies every reduced member's required actual centre power from the original local coherence, local cosupport and literal coefficient identities. Genuine reduced filtered division now expands every original ideal member with the actual quotient powers. Normalize the same reduced members and those actual quotients in the true chart; formal-domain cancellation proves the whole strict-ideal equality and the actual exceptional-product identities. Thus the previously proved whole reduced strict-generation theorem applies without any global extension premise. Existence of all source coefficient presentations and the later analytic/formal and Hilbert–Samuel transformation arguments are not asserted. □

345 Deletion of a genuine empty blowup in an original recursive history

Keep the original marked record, its entire indexed boundary and original head $H : X \leftarrow A$. An actual empty blowup $\pi : B \to A$ may have a different source manifold and need not be an identity. Keep its entire original suffix $S : B \leftarrow Y$. The ordered-boundary route of [3, 5] is adapted by internally normalizing the actual suffix through the empty blowup's genuine biholomorphism, before deleting only that empty step. The original marked record is never rebased and all actual later blowups remain recorded.

Theorem 345.1 (The actual empty blowup supplies a whole suffix normalization)**.** *For an original head H, actual empty-centre blowup $\pi : B \to A$ and entire original suffix S, there are a genuine normalized suffix S' over A and an actual final biholomorphism $\chi : Y' \to Y$ such that $|S'| = |S|$ and the ENTIRE normalized identity-inserted history $H\epsilon S'$ is isomorphic to $H\pi S$ over the identity of the ORIGINAL base X, with final map χ. No biholomorphism, normalized history or tower comparison is a caller input.*

Proof. The actual empty blowup's universal uniqueness against the genuine identity empty blowup constructs a biholomorphism $\varphi : B \to A$ whose underlying map is exactly π. Transport the SAME entire S through φ^{-1}. At the empty suffix the normalized suffix is empty over A and the final map is φ^{-1}. At a next original centre and projection, pull back the centre through the preceding final biholomorphism and compose the projection with its inverse. The genuine blowup universal property gives the normalized blowup and a commuting next-stage square. Its new source is exactly the original source and its final biholomorphism is the identity. Consequently every source stage after the first, and every later original centre and projection, remain literal; the first centre and projection are conjugated by the actual empty blowup's biholomorphism. This constructs a whole suffix tower isomorphism and preserves its number of actual steps. The original head is isomorphic to itself over identities. Extend this by the identity empty step on the normalized side and the actual π on the original side, with intermediate map φ^{-1}. The square commutes because $\pi\varphi^{-1} = \mathrm{id}$ and both centres are empty. Concatenate with the constructed WHOLE suffix isomorphism. This proves the stated whole-history comparison over the unchanged original base, including empty suffixes and empty source manifolds. □

Theorem 345.2 (Original actual family canonicality computes genuine empty deletion)**.** *For EVERY actual family depth k, ONLY original bound-k marked input and canonicality of the actual entire $H\pi S$ internally give such a normalized S' and χ, with deleted HS' canonical for the*

SAME actual R_k. The entire original transformed cosupport is the χ^{-1}-preimage of the deleted transformed cosupport. At EVERY original cosupport point y,

$$R_k(M, H\pi S, y) = H_{k,j}\big(R_k(M, HS', \chi^{-1}y)\big), \qquad j = |M.E| + |H|.$$

The actual original history is a terminal geometric resolution if and only if the normalized deleted history is. No deleted or normalized canonicality, cosupport, branch, value or termination certificate is a caller premise.

Proof. Construct the entire actual normalization by the first theorem. Actual canonical tower-isomorphism transport supplies canonicality of $H\epsilon S'$ from the GIVEN original actual canonical history. Pullback by the base identity is literally the original marked record, using its internally supplied coherence. The completed actual identity empty deletion theorem then constructs deleted whole canonicality and its whole-value formula. The GIVEN original canonical history internally supplies real testness; tower-isomorphism transport gives normalized inserted testness and the whole transformed-cosupport preimage dictionary. The deleted canonical history internally supplies its real testness and original head testness. The actual identity insertion dictionary identifies normalized inserted cosupport with deleted cosupport. Applying the inverse final biholomorphism gives the stated entire original cosupport equality. For each original cosupport point y, the equality proves that $\chi^{-1}y$ is a genuine normalized inserted cosupport point. The actual specific family's canonical tower-isomorphism value theorem equates its normalized inserted value with the GIVEN original value at y. Compose this equality with the internally supplied entire identity deletion formula. No independent choice of a contact or assignment history is introduced. Actual whole tower-isomorphism resolution transport and actual identity deletion terminal equivalence give the stated terminal iff. All orders, true levels, contact-zero branches, empty suffixes and actual empty manifolds are included. □

Theorem 345.3 (The actual maximal-order Step One history has the same genuine deletion)**.** *For EVERY lower depth k, original bound-$(k+1)$ maximal-order input and canonicality of the actual entire Step One R_k history $H\pi S$ internally construct normalized S' and χ, with deleted HS' canonical for the SAME actual masked Step One value. Its entire original cosupport is the χ^{-1}-preimage of the deleted cosupport. At EVERY original cosupport point, its entire original Step One value is $S_{k,j}$ of the entire deleted Step One value at $\chi^{-1}y$, with the same $j = |M.E| + |H|$. Actual original and normalized deleted terminal resolutions are equivalent. All normalizing geometry, lower-order choices and deleted canonicality are supplied internally.*

Proof. Construct the real entire normalization from the first theorem. The actual masked Step One of the specific lower family is a genuine maximal-order value function. Apply its true canonical tower-isomorphism transport to the GIVEN original whole Step One canonical history and the constructed whole isomorphism. Actual coherent identity pullback keeps the original marked record. The completed Step One identity empty deletion constructs deleted whole canonicality, genuine testness and its whole-value formula. True Step One canonical testness on the original side supplies the actual entire cosupport preimage dictionary; deleted canonical testness and the head dictionary equate normalized inserted and deleted cosupport. Inverting the final biholomorphism gives the stated entire cosupport equality. At each actual original point the genuine maximal-order tower-isomorphism value theorem compares the normalized inserted value with the original value; compose with the actual deletion formula. Whole geometric resolution isomorphism transport and the internally supplied Step One deletion terminal iff prove the terminal equivalence. The proof uses the actual masked value including independently chosen contact-zero branches and every positive inner depth, not a monomial-only surrogate. Noncompact canonical scheduling and global resolution gluing remain separate source obligations. □

346 Actual ambient terminal choice comparison including dimension zero

Keep every actual recorded empty step and the original marked record. The finite canonical-choice route of [3, 5] is completed at ambient dimension zero by deriving genuine whole projection geometry from the original bound-zero canonical history. No nonvacuous value comparison on empty cosupport is asserted.

Theorem 346.1 (Original bound-zero canonical input supplies a real whole projection biholomorphism)**.** *On ANY ambient dimension, original bound-zero domain input and a GIVEN actual whole canonical history internally give a genuine biholomorphism from its ORIGINAL final manifold to its original base whose underlying map is EXACTLY the whole original projection. The value function may be any function into an ordered type; no empty-centre, projection-invertibility or terminal comparison certificate is a caller premise.*

Proof. Induct on the actual history. The empty history has identity projection. At a next original step, its actual preceding history is canonical. The original bound-zero canonical testness producer supplies its genuine test history, and the actual bound-zero cosupport producer proves its entire transformed cosupport empty. The actual centre alternative supplied by canonicality is empty or the entire maximum locus on this empty cosupport. The latter locus is literally empty. Thus THIS genuine original step has empty centre. Universal uniqueness against the genuine identity empty blowup internally constructs its actual projection biholomorphism. The induction constructs the actual preceding whole projection biholomorphism. Compose these two REAL biholomorphisms. Actual tower projection recursion identifies their composite with the entire original next projection at every point. This retains arbitrary actual empty projections and every recorded empty step, including empty manifolds and empty histories. Only original canonicality/domain are assumed; the output is genuine geometry rather than a vacuous invariant-value formula. □

Theorem 346.2 (Original ambient canonical terminal choices compare in every ambient dimension)**.** *For EVERY ambient dimension n, ONLY original bound-n marked input and TWO GIVEN whole canonical terminal histories for the ACTUAL ambient canonical invariant internally supply a genuine biholomorphism of their ORIGINAL final manifolds commuting with the entire original projections. Arbitrary and differing original empty-step counts, ambient dimension zero and empty manifolds are included. No empty/compressed-history, nonempty-centre, comparison or whole projection certificate is supplied.*

Proof. If n is positive, the actual ambient canonical invariant is its defining specific family at depth n. Consume the completed actual original terminal-choice producer on these GIVEN whole canonical terminal histories. It internally compresses their real empty steps before actual canonical terminal comparison and supplies precisely the original final-space map and whole projection formula. If $n = 0$, the original bound-n condition is actual bound-zero input. The preceding theorem supplies a genuine whole projection biholomorphism for EACH given original canonical history. Compose the first projection biholomorphism with the inverse second one. Its actual original projection formula and the inverse identity give the claimed equality at every original source point. No cosupport point or nonvacuous zero-depth value comparison is used. These actual final-space comparisons do not assert an isomorphism of differently sized recorded towers, arbitrary-map uniqueness/cocycles, noncompact scheduling or global resolution gluing. □

347 Actual initial-boundary compatibility of the whole history map

The indexed Step One histories retain the original boundary list, including empty or inactive repetitions. This supplies the compatibility between the initial-boundary insertion formula and

the true level-dependent history map needed for higher original recursive histories. The argument adapts the indexed recursive route of [3, 5] by deriving the necessary empty prefix from the actual original geometry.

Lemma 347.1 (The true lower empty prefix contains the clamped history index)**.** *Let M be an original well-formed marked record, H an actual test history for its original barred record, p an actual endpoint, and $j \le |M.E|$. Put $\ell = \mathrm{level}(M, H, p)$ and $d = j - \ell$, with subtraction clamped at zero. The ACTUAL selected lower record $L(M, \ell, H, p)$ has at least d boundary entries and its first d entries are empty. At positive level, d is bounded above by the actual retained-prefix insertion index. No lower record, empty-prefix or index comparison is assumed.*

Proof. Write E for the full actual barred boundary, $m = |M.E|$ and $E_0 = E|_m$ for its original prefix. Actual testness and original well-formedness give the true SNC boundary. The established indexed SNC count therefore says that the retained original prefix, filtered by $\neg(W \in E_0 \wedge p \in W)$, has length $m - \ell$. Also $\ell \le m$, since it counts a subset of the m original indices. Split the literal original prefix into its first j entries and its next $m - j$ entries. Filtering preserves concatenation; the retained suffix has length at most $m - j$. If c is the actual retained length of the first j entries, then $m - \ell \le c + (m - j)$, hence $j - \ell \le c$. The actual original lower-prefix producer makes the first c lower entries empty at positive level. Taking their first $d \le c$ entries proves the claim with its boundary-length bound. At level zero the actual chosen-contact lower prefix already has its first j entries empty; here $d = j$. No equality of independent contact choices is needed: this is the actual lower record selected by the unchanged original input. □

Theorem 347.2 (The entire actual Step One value has one compatible initial-boundary history map)**.** *For EVERY original well-formed M, entire barred test history T, actual point y, every actual original initial-boundary index $j \le |M.E|$, and EVERY family depth k, the entire actual Step One value v satisfies*

$$\mathcal{H}_{k,j}(v) = \mathrm{pairShift}(\mathrm{shiftFamily}(k, 0))(v), \qquad v = \mathrm{StepOne}(R_k; M, T, y).$$

The value is unrestricted in its level, chosen-contact branch and all inner nonmonomial orders. No canonicality, lower-value, branch or compatibility certificate is supplied.

Proof. At any actual assignment head H and suffix Q, take the level of the actual assignment point $p = Qy$. The preceding result gives a true empty prefix of length $j - \ell$ in the ACTUAL lower marked record. If p misses the actual lower mask, the masked lower value is infinite and both maps fix it. If it lies in the mask, restrict the actual whole suffix and use the specific-family empty-prefix theorem on this actual lower record. It gives $\mathrm{shiftFamily}(k, j - \ell)$ equal to $\mathrm{shiftFamily}(k, 0)$ on its ENTIRE R_k value, at every inner order. Coercion into the masked value gives the same equality. For the whole Step One value, a special point has infinite lower coordinate, so the two pair maps agree directly. At a nonspecial point put $\ell = \mathrm{level}(M, T, y)$, let r be the actual run of that regime, and take the literal full split $T = HQ$ with suffix length $\min(r, |T|)$. Current regime membership and this true suffix length imply regime membership at the original assignment head, by the established reaching-run induction. Thus its ACTUAL point level is exactly ℓ. Whole testness supplies actual barred testness of that head. Apply the masked comparison just proved and expand the unchanged Step One definition. Both first coordinates are ℓ; the second coordinates coincide by that actual lower comparison. This gives the displayed entire value identity, including empty suffixes and level zero. It supplies a compatibility producer; higher outer canonical-history transport and global scheduling remain separate obligations. □

348 Actual canonical normalization of the original crossing lower tail

The literal original crossing lower data are retained at every recursive dimension. Normalizing a restriction of a concatenation is a real tower isomorphism, whose complete original canoni-

cal history and values must be transported before the positive-order recursive branch can use them. The proof uses the genuine sequence-form canonical invariant; it does not replace its local finiteness by a uniform noncompact bound. This supplies explicit original-history bookkeeping in Włodarczyk's Step One and the trivial-isomorphism convention of Bierstone–Milman [3, 5].

Theorem 348.1 (Original canonical input supplies the normalized whole lower history)**.** *Let M be in the original dimension-$(k+1)$ domain and of actual maximal order. Let HS be its genuine specific-family Step One canonical history and y an original nonspecial transformed-cosupport point whose actual current-level run r reaches S. Retain the true crossing assignment $K = \mathrm{head}(H, r-|S|)$, $B = \mathrm{tail}(H, r-|S|)$, $p = (BS).\mathrm{proj}(y)$, $W = \mathrm{lowerMask}(M, \ell, K, p)$ and $J = \mathrm{lowerMarkedM}(M, \ell, K, p)$. These original data internally supply $p \in W$, the true dimension-k domain of J, BOTH the canonical whole old restricted history $(BS)|_W$ and the canonical whole normalized history $B|_W\, S|_{B^{-1}W}$, the corresponding actual cosupport points, and equality of their actual full R_k values at the same underlying y. This applies to every actual lower nonmonomial order, including its positive branch, and to an empty or nonempty S.*

Proof. The original canonical crossing lower-data producer internally supplies the genuine mask member, lower domain, entire old canonical restricted tail and real cosupport point. The true restriction-of-concatenation isomorphism identifies this old restricted tail with the stated normalized history. Apply the actual sequence-form canonical tower-isomorphism theorem to its inverse, whose base and final identifications are the actual identity maps on the corresponding equal opens. The genuine coherent identity-pullback dictionary identifies its initial marked ideal with the same actual J; thus it gives canonicity of the entire normalized history. The actual old canonical tail is a test history. Its genuine transform cosupport dictionary across this same inverse isomorphism supplies the normalized cosupport point. The established actual-family tower-isomorphism value property, applied to the true old canonical tail and that internally supplied cosupport point, gives the full value equality at y, without an order-zero branch assumption. All lower geometry, canonical histories, point identifications and values are produced from the original inputs. □

Canonicity of a history with an additional interior empty step, its positive-order recursive value transport, the outer general companion history and noncompact scheduling remain further obligations. Normalizing the existing actual canonical history does not by itself prove those claims.

349 The exact whole lower index of an original crossing run

The original crossing lower history retains the same marked datum and a true interior empty step. Its actual insertion index is its WHOLE original lower boundary length plus the true remaining head length. Actual SNC geometry computes that index from the original level; it is not a comparison premise. This refines the indexed Step One bookkeeping of [3, 5].

Lemma 349.1 (Whole original lower boundary length at the actual level)**.** *Let M be originally well formed, let K be an actual bar-test history, and let p be an actual stage point with level ℓ. The WHOLE actual lower marked record formed at its true level has boundary length*

$$|\,\mathrm{lowerMarkedM}(M, \ell, K, p).E| = |M.E| + |K| - \ell.$$

At level zero this is the genuine chosen-contact lower record; at every positive level it is the genuine member-restricted lower record. No carrier, choice, index or boundary-count certificate is assumed.

Proof. At level zero the actual coefficient/contact record retains the entire bar transform boundary, and open pullback preserves its length. The true bar transform length is $|M.E| + |K|$, and subtraction by zero leaves it unchanged. At positive level the actual lower boundary is the open

pullback of the entire bar boundary filtered by removal of the original members through p. Actual bar-testness makes this bar transform genuinely well formed and SNC. Its literal original member list is its prefix of length $|M.E|$. Apply the proved exact original filtered-prefix count with larger prefix equal to its WHOLE boundary. The boundary count of the original prefix is the true level ℓ. The map into the actual open lower carrier preserves the retained length, yielding the displayed identity. This counts every actual boundary index and permits inactive or empty repeated entries. □

Theorem 349.2 (Original crossing canonicity internally computes the exact lower index)**.** *Assume only an original dimension-$(k+1)$ maximal-order datum and an actual canonical Step One history HS for the specific original lower family R_k. Let y be an actual nonspecial point whose TRUE own-level run r reaches the insertion after H, so $r \geq |S|$. Put $K = \mathrm{head}(H, r - |S|)$, $B = \mathrm{tail}(H, r - |S|)$, $p = (BS)(y)$, $\ell = \mathrm{level}(M, HS, y)$, and $J = \mathrm{lowerMarkedM}(M, \ell, K, p)$. Then the exact actual lower insertion position is*

$$|J.E| + |B| = |M.E| + |H| - \ell.$$

Every actual lower order, including positive recursive orders, is allowed.

Proof. The genuine original canonical bridge supplies actual whole bar-testness and hence bar-testness of the literal selected prefix K. Its actual nonspecial current point lies in its own level regime. Since the true suffix length $|B| + |S|$ is at most r, actual reaching-run membership gives the SAME level ℓ at the assignment point p. Apply the preceding whole lower-length identity there. This also gives $\ell \leq |M.E| + |K|$, since the true level counts indices in an original prefix. Actual head/tail lengths sum to $|H|$. Consequently adding $|B|$ to the computed whole lower boundary length yields exactly $|M.E| + |H| - \ell$. Nothing about lower values or inserted canonical histories is assumed. □

Theorem 349.3 (The actual crossing leaf uses the same whole Step One history map)**.** *For the preceding original canonical inputs AT AN ACTUAL ORIGINAL COSUPPORT POINT, assume the suffix S is nonempty. The genuine original lower-data producer internally constructs the whole actual lower canonical tail, domain and cosupport point. On its actual lower nonmonomial-order-zero branch, the ENTIRE original Step One value on $H\epsilon S$ is $F_{k,|M.E|+|H|}$ of the original value on HS. The original point may have zero or positive Step One level.*

Proof. The genuine original canonical crossing-leaf producer supplies the actual lower witness, domain, full canonical tail and cosupport internally. On its stated actual order-zero branch it proves the complete Step One pair formula with actual family index $|J.E| + |B|$. Substitute the exact original index computed above. The first coordinate of the whole original Step One value is its actual level, so this pair formula is precisely the definition of the single level-indexed strict history map $F_{k,j}$. The lower order-zero condition is an explicit real branch on internally supplied original data; positive crossing recursion remains open. Together with the complete short Step One formula, this supplies the common map for actual short points and actual crossing leaf points, without claiming full higher crossing transport or noncompact scheduling. □

350 The actual selected lower history crossing an interior empty step

These producers retain the actual initial marked ideal, the whole original history and the run computed at the actual current Step One level. Here that run reaches the insertion. No desired assignment-stage, tail, lower record or invariant comparison is an input.

Lemma 350.1 (A run reaching the base of a concatenated tail)**.** *For an actual regime P, actual histories H, S and final point y, if $P(HS, y)$ holds and $|S| \leq \mathrm{run}_P(HS, y)$, then $P(H, S.\mathrm{proj}(y))$*

holds. Moreover, for every $r \geq |S|$, the entire actual split of HS at r has stage and head those of the split of H at $r - |S|$, and tail $\mathrm{tail}(H, r - |S|)S$.

Proof. For regime membership induct on S. The empty tail uses current membership. At a nonempty last step, a run reaching the positive length of S cannot have stopped at that step. The actual predecessor therefore satisfies P, and the run recursion gives a predecessor run reaching the predecessor tail length. Induction gives the original base membership. For the full split formula, remove the last actual steps of S recursively. Once the tail is exhausted, the remaining $r - |S|$ steps are removed from H. The actual split constructors retain precisely the stated original head and append S to its actual tail. This applies also to saturated split indices beyond the whole length. □

Theorem 350.2 (The original crossing run and its whole selected splits)**.** *Let M be originally well formed, H an actual test tower of $\overline{M}$, S any actual tail, and y an actual nonspecial final point of HS. Put $\ell = \mathrm{level}(M, HS, y)$ and $r = \mathrm{run}_{M,\ell}(HS, y)$, and assume $|S| \leq r$. The actual current-level run in $H\epsilon S$ is $r+1$. Write $a = r - |S|$, $D = \mathrm{stage}(H, a)$, $K = \mathrm{head}(H, a)$ and $B = \mathrm{tail}(H, a)$. The two actual selected full split records are precisely (D, K, BS) and $(D, K, B\epsilon S)$.*

Proof. The actual nonspecial point satisfies the regime of its own actual level. The first lemma therefore supplies membership at the original H point. The exact original run formula, with $r \geq |S|$, gives $r + 1$ after insertion; the original level dictionary also identifies the new actual current level. The preceding whole split formula gives the old split. Apply it to the new prefix $H\epsilon$ at $r + 1$. Because $r \geq |S|$, its remaining split count is $a + 1$. The last step of this prefix is the identity empty blowup, so its split at $a + 1$ has stage/head (D, K) and tail $B\epsilon$ by the actual split recursion. Appending S gives precisely the new whole split record. This includes level zero and an empty S. □

Lemma 350.3 (Actual restriction of an interior identity empty step)**.** *For any actual open W and histories B, S, the restriction of $B\epsilon S$ to W is genuinely isomorphic over the identity of W to the restriction of B followed by the genuine identity empty blowup and the restriction of S to the actual open preimage of W under B. The final identification is the natural equality of the actual open preimages, with the same points.*

Proof. The empty centre restricts to the empty centre, the identity projection restricts to the identity on the same actual open preimage, and the actual restricted blowup has the true empty blowup universal property. Thus the whole restricted prefix is exactly the restricted B followed by that identity empty step. The genuine restriction-of-concatenation dictionary then gives the asserted whole tower isomorphism and its natural final open identification. No restricted-tail isomorphism is assumed. □

Theorem 350.4 (The actual crossing lower assignment and restricted history)**.** *For the actual selected splits above, the images of y in D agree, both selected heads are literally K, and the actual lower masks and masked lower marked ideals at their actual level ℓ agree. The new whole restricted lower tail is the genuine interior identity empty insertion in the actual restricted old tail, as in the preceding restriction diagram. No positive-level restriction is required for this crossing branch.*

Proof. The two actual tails are BS and $B\epsilon S$, whose composite projections are equal by the actual identity-step projection dictionary. Their images $p \in D$ therefore agree. The full split calculation has already supplied the same original head K. The actual lower mask and lowerMarkedM depend on the same M, ℓ, K, p, so their equality follows by substitution, also for the original chosen contact direction at level zero. Apply the actual restriction producer to this literal common lower open. It supplies the new whole restricted tail diagram internally. □

Complete lower invariant value/order transport along these actual histories, the zero-contact carrier in the short-run branch, and full higher recursive invariant and noncompact maximum scheduling remain subsequent producers.

351 Original canonical crossing tails at their actual monomial leaves

This is a source-facing consumer of the genuine interior identity empty step. The initial marked ideal, the actual current level and run, and the whole selected original tails are retained. Lower domain, testness and cosupport are conclusions from the original canonical history. The value comparison is restricted explicitly to the actual order-zero branch of that lower history; it does not assert positive-order recursive transport. This adds the indexed-history bookkeeping needed in Włodarczyk's Step One and the trivial-blowup convention of Bierstone–Milman [3, 5].

Theorem 351.1 (Actual canonical lower data on the original crossing split)**.** *Let $n > 0$, let M belong to the original invariant domain in dimension $k+1$, and assume that M has maximal order. Let HS be a canonical history for its actual Step One value built from the specific invariant R_k, and let y be a nonspecial point in the actual transformed cosupport. Put $\ell = \mathrm{level}(M, HS, y)$, $r = \mathrm{run}_{M,\ell}(HS, y)$ and assume $|S| \le r$. Put $a = r - |S|$, $K = \mathrm{head}(H, a)$, $B = \mathrm{tail}(H, a)$, $p = (BS).\mathrm{proj}(y)$, $W = \mathrm{lowerMask}(M, \ell, K, p)$ and $J = \mathrm{lowerMarkedM}(M, \ell, K, p)$. Then $p \in W$, J belongs to the genuine invariant domain in dimension k, the entire restricted tail $(BS)|_W$ is canonical for $R_k(J)$, and the actual point y over W belongs to its transformed cosupport.*

Proof. The original canonical Step One history is a genuine multiple test history both of M and of $\overline{M}$, with identical transformed cosupports. Its canonicity is also the canonicity used for $\overline{M}$. The canonical-tail theorem therefore gives actual lower data at the run split of this nonspecial cosupport point. The true crossing split calculation identifies its entire split record with $(\mathrm{stage}(H, a), K, BS)$. Substitute this whole record in the canonical-tail conclusion. The lower mask, marked ideal, restricted tail and cosupport point are exactly the ones stated, rather than separately selected representatives. The dimension bound is $(k+1) - 1 = k$. □

Lemma 351.2 (Actual restricted interior tails at a monomial leaf)**.** *Let W be an actual open subset of a stage D, J an actual marked ideal on W in the dimension-k domain, and B, S actual consecutive histories from D. Suppose $|S| > 0$ and $(BS)|_W$ is a test history of J. At any actual point y over W whose nonmonomial order on this restricted old history is zero, the specific full invariant R_k on $(B\epsilon S)|_W$ equals its value on $(BS)|_W$ under the actual index map $\mathrm{shift}_{k,j}$, where $j = |J.E| + |B|$. Both points are the same underlying point y; membership in the new open preimage is supplied by the identity-step projection.*

Proof. The true restriction-of-concatenation isomorphism identifies the old restricted history with $B|_W$ followed by $S|_{B^{-1}W}$. Its base and final identifications are the identity on points. Testness transfers across this isomorphism, and the genuine order-isomorphism dictionary transfers the assumed actual order zero. The restriction of S has length $|S| > 0$. The proved same-base interior test equivalence supplies testness after inserting the identity empty step into this normalized history. The actual full-family monomial leaf theorem on this history gives the index $|J.E| + |B|_W = |J.E| + |B|$ and the stated shift, including $k = 0$. The genuine nonmonomial-order equality supplies order zero after insertion. Finally the whole restricted empty-history isomorphism transfers the new value back to $(B\epsilon S)|_W$. At order zero this uses only the actual testness and domain: the general value is the true monomial invariant, so no desired new canonical segment or lower-value equality is required. For $k > 0$ this is exactly the order-zero branch of R_k; for $k = 0$ its value is constant. All final identifications fix y. □

Theorem 351.3 (The actual original crossing Step One value at its lower leaf)**.** *Retain the original canonical hypotheses and literal ℓ, r, K, B, W, J, p above and assume $|S| > 0$. The actual lower data at p exist internally. If the nonmonomial order at their actual cosupport point on $(BS)|_W$ is zero, then the whole actual Step One value at y on $H\epsilon S$ is*

$$\mathrm{pairShift}(\mathrm{shift}_{k,|J.E|+|B|})\big(\mathrm{StepOne}(R_k, M, HS, y)\big).$$

The original initial M is unchanged and both Step One values use their own actual current-level runs. This holds at all actual levels, including the original chosen-contact level zero.

Proof. The first theorem supplies genuine mask membership, the actual lower domain and canonical old tail. The resolution invariant makes that canonical tail a test history. The original whole history also supplies testness of its original head H. The actual crossing-run theorem therefore gives run $r + 1$ and the two entire selected split records (D, K, BS) and $(D, K, B\epsilon S)$. Their projected point is literally the same p, hence their lower mask and whole lower marked ideal are the same W, J, also in the zero-contact branch. The level dictionary identifies both current levels with ℓ; the special-point dictionary makes the new point nonspecial. Unfolding the actual Step One values now selects these two genuine lower tails. The mask membership supplied by the first theorem selects the finite lower-value branch on both histories. The preceding restricted-leaf theorem gives the full R_k shift of this true lower value. Keeping the common first coordinate ℓ gives exactly the displayed pair map. No independently chosen lower ideal, contact direction, tail, new canonicity or value comparison is an input. □

Positive-order recursion inside this actual crossing lower tail, the full outer general-value transport, mixed spectra and noncompact scheduling remain further obligations.

352 Actual interior test histories and the recursive monomial leaf

An original crossing lower diagram has the same actual initial lower record but a different entire tail, containing an internal identity empty blowup. Its admissibility must therefore be supplied from the original data, not assumed as a lower-history comparison. We first establish that real test-history preservation. The actual monomial branch of the full recursive invariant is then transported along these original histories. This is additional explicit indexed-history bookkeeping for Wlodarczyk's Step One and Bierstone–Milman's trivial blowup convention.

Lemma 352.1 (Normal crossings ignore an inserted empty boundary entry)**.** *For every actual ordered boundary list E, actual subset C and index $j \le |E|$, C has normal crossings with $E.\mathrm{insertIdx}(j, \varnothing)$ if and only if it has normal crossings with E. No boundary entry is deleted from either history.*

Proof. At a point $x \in C$, retain the actual adapted chart and coordinate support of C. A member of the inserted list through x cannot be its empty entry, so it is an actual old member. Choose its original list index and use that member's actual coordinate support; for a member missing x choose empty support. Each required coordinate formula and one-coordinate cardinality is precisely the retained old member's formula. The inserted entry has no required formula or cardinality, because no point belongs to it. This constructs adapted coordinates for the entire inserted indexed list. Conversely every old member occurs in the actual inserted list. At x, choose its actual index there and retain that coordinate support, with the same chart and C support. Its formula and cardinality are the supplied inserted-list ones. These constructions give both directions pointwise, and hence for all $x \in C$. They use genuine membership equality of the nonempty members; no global index-preimage injectivity or new normal-crossings certificate is needed. □

Lemma 352.2 (Actual test towers under an initial empty boundary insertion)**.** *For any marked ideal J and $j \leq |J.E|$, any actual tower R is a test tower of* $\mathrm{insertEmpty}_j(J)$ *if and only if it is a test tower of J. The actual admissibility of every centre also agrees.*

Proof. Inserting the empty member changes neither the carrier, coefficient ideal, mark nor cosupport. A centre has the same closed submanifold and cosupport conditions, and its normal-crossings condition agrees by the previous lemma. Thus its entire admissibility condition agrees. Induct on the actual last steps of R. The empty tower is a test tower on both sides. At a last step, actual transformation commutes with the same indexed insertion; the index remains valid because the whole boundary has gained exactly one entry at each previous step. The induction hypothesis gives testness of the predecessor and the proved admissibility equivalence gives its actual last centre. This proves both directions for the entire tower, with all original blowup data kept. □

Theorem 352.3 (Genuine original testness across the internal identity empty step)**.** *Let M be originally well formed and H an actual test tower of M. For every actual tail S, $H\epsilon S$ is an actual test tower of M if and only if HS is, where ϵ is the genuine identity empty blowup. This includes an empty tail.*

Proof. The actual transformed original record $K = M_H$ is well formed. Its genuine identity empty step is admissible, so $H\epsilon$ is an actual test head. The actual transformed record after that step is $\mathrm{insertEmpty}_{|M.E|+|H|}(K_\epsilon^-)$, where K_ϵ^- is the actual empty-step base with only its newly created empty member removed. The actual identity empty-blowup calculus gives restricted equivalence of K_ϵ^- and K, genuine coherence of both ideals, closedness of their common carrier, and equality of their original whole boundary. Restricted-equivalent coherent records have the same actual admissible test towers. Removing the new empty boundary entry for testing is also legitimate by the previous lemma, with its index bounded by the real whole empty-step base boundary. Thus S is a test tail of the actual new head record exactly when it is a test tail of K. Testness of a concatenation is testness of its actual head and tail; the already proved test heads H and $H\epsilon$ now give precisely the asserted equivalence. No testness of the inserted lower tail is a caller conclusion certificate. □

Theorem 352.4 (The actual full-family monomial leaf in the same original history)**.** *Let $n > 0$, $k \geq 0$, M originally well formed, H an actual test history of M, S a nonempty actual tail and y an actual final point. Assume that the actual nonmonomial order of M_{HS} at y is zero. Then the actual nonmonomial order at $H\epsilon S, y$ is also zero and*

$$\mathrm{invFamily}(n,k)(M, H\epsilon S, y) = \mathrm{shiftFamily}(k, |M.E| + |H|)\big(\mathrm{invFamily}(n,k)(M, HS, y)\big).$$

For any two such actual zero-order points, strict order, weak order and equality of these actual full invariant values are preserved and reflected. The original initial ideal and entire history remain unre-based.

Proof. The nonempty-tail whole-record producer gives the actual transformed record at $H\epsilon S$ as the old transformed record with the empty entry inserted at the true position $j = |M.E| + |H|$, bounded by the old whole boundary. Actual nonmonomial order is unchanged by this insertion, so both points lie in the same actual zero-order branch. For $k = 0$ the actual family is constant and its shift is the identity. For $k > 0$ its definition in this branch is its unchanged first coordinate zero and the actual monomial invariant of its transformed original record. That monomial invariant is changed precisely by the proved indexed shift at j. The definition of shiftFamily on this branch is exactly that map, giving the entire full-family equality. The map is strictly increasing on the full value type at every k, so applying it to two genuine zero-order points preserves and reflects all three comparisons. This leaf computation uses no proposed arbitrary-invariant insertion axiom. It does not replace the still-needed positive-order recursive crossing transport or claim maximum existence or scheduling. □

353 Entire original curve-family histories across an interior empty step

At original dimension bound one every positive-order companion cosupport point is special for its actual Step One lower family. Together with actual whole short histories this closes the curve-family interior empty-step comparison on every suffix. Higher-dimensional crossing recursion remains separate. This is the original dimension induction and ordered-boundary trivial-blowup convention of [3, 5].

Lemma 353.1 (Actual whole monomial transport on every original suffix)**.** *Original well-formed M and actual original test head H internally supply, for EVERY suffix S, including the empty suffix,*

$$\gamma(M, H\epsilon S, y) = \mathrm{shiftVal}(|M.E| + |H|)(\gamma(M, HS, y)).$$

No ambient marked-record equality or nonempty-suffix premise is assumed.

Proof. Actual head testness makes its true marked transform $K = M_H$ well formed. The genuine identity empty-blowup geometry supplies restricted equivalence between its actual empty-step base and K, with the SAME original boundary, carrier and mark. Actual closedness of K and both actual coherent ideals permit the existing WHOLE monomial invariant restricted-equivalence theorem along EVERY suffix. This theorem includes the empty suffix, where the ideals need agree only along their literal carrier; after a genuine blowup their whole records agree. The actual empty transform is the empty-step base with an empty boundary inserted at $|K.E| = |M.E| + |H|$. The complete actual indexed monomial insertion identity supplies the displayed shift, and restricted equivalence supplies the whole old gamma value on the SAME S. Thus no false ambient equality at the immediate empty step is needed. □

Theorem 353.2 (Actual entire original monomial leaf on every suffix)**.** *At EVERY actual original nonmonomial-order-zero point, for EVERY family depth k and EVERY suffix S, the whole unrebased original family value on $H\epsilon S$ equals* $\mathrm{shiftFamily}(k, |M.E| + |H|)$ *of its whole value on HS. There is no nonempty-suffix or canonical input.*

Proof. At depth zero the family is the constant original seed and the shift is the identity. At successor depth the genuine all-tail order dictionary keeps both actual outer orders zero. The whole original general value is therefore its monomial sum branch on both histories. Apply the preceding true gamma identity to that complete branch; the successor family shift acts there by exactly the same indexed monomial shift. This gives the entire family equality, not merely equality of one projected coordinate. □

Theorem 353.3 (The complete original curve-family value and order dictionary)**.** *An original dimension-one datum and its actual original R_1 canonical history HS internally supply at EVERY original cosupport point, for EVERY suffix S,*

$$R_1(M, H\epsilon S, y) = \mathrm{shiftFamily}(1, |M.E| + |H|)(R_1(M, HS, y)).$$

Strict order, weak order and equality of the entire original curve-family values on the whole actual cosupport are consequently reflected and preserved. No caller short-run, special-companion, lower-order, source canonicity or value-equality certificate is supplied.

Proof. Original full canonicity gives actual testness, including head testness. At outer order zero apply the preceding all-tail actual monomial-leaf theorem. At positive order, inspect the TRUE computed original outer run. If short, the complete actual whole short-family theorem supplies the formula for all inner orders. If reaching, original full canonical companion data internally give the actual dimension-one companion, maximal order, whole Step One R_0 canonical restricted tail and its actual cosupport point. The existing original curve Step One producer makes that point

special: a nonspecial point would give an actual dimension-zero canonical lower history and a real cosupport point, contrary to its proved empty cosupport. The genuine original special-companion transport therefore gives equality of the two entire positive-order family values. At depth one the inner family shift is the identity seed, so the whole successor shift fixes every actual positive-order value. This equality is consequently the required shift formula. The same strict whole family map for both actual points gives all three comparisons. No maximum existence is inferred from order preservation. □

Theorem 353.4 (Actual original curve canonical and terminal history producer)**.** *Every original dimension-one actual R_1 canonical history HS remains canonical after inserting the literal identity empty blowup after H. If the original history is terminal, the actual inserted history is terminal as well. Every original centre and suffix step is retained; no additional canonical-history or terminality certificate is assumed.*

Proof. Induct on the SAME actual suffix S. For the empty suffix the original head is canonical and its new identity step has empty centre, hence is canonical. At a further original step the preceding original prefix is canonical, and its inserted prefix is canonical by induction. If the actual next original centre is empty, it remains empty. Otherwise it is the actual maximum-dominance locus on the old prefix cosupport. The genuine geometric dictionary identifies both whole cosupports; the preceding complete curve value/order formula identifies their entire dominance loci under a common strict map. Thus the SAME actual next centre is canonical for the inserted prefix. This requires no new maximum existence or compactness assumption. Genuine original testness transfers across the interior identity empty history, and actual whole cosupport equality transfers empty terminal cosupport. These facts give the actual inserted terminal resolution when the original is terminal. □

354 Internally removing all actual empty steps of a finite canonical history

Keep the ORIGINAL marked record and all of its indexed boundary members. The ordered-boundary route of [3, 5] is adapted by repeatedly deleting only an actual empty blowup and its internally normalized suffix. Each deletion strictly decreases the actual finite history length. The result concerns a GIVEN whole finite canonical history, not existence of an infinite or noncompact global resolution schedule.

Lemma 354.1 (A failed actual nonempty-centre condition locates a true empty step)**.** *If some actual recorded centre of a finite tower is empty, the ENTIRE tower is literally $H\pi S$, with a genuine actual empty-centre blowup π and its SAME original entire suffix S. Otherwise every recorded centre is nonempty. No chosen split or comparison history is supplied.*

Proof. Use the actual tower recursion. The empty tower has no failing centre. For a last step, if its actual centre is empty, take the actual preceding tower as H and the empty suffix. If its actual centre is nonempty but some earlier centre fails, the actual preceding tower fails the same condition. The recursion supplies its entire empty-step split $H\pi S$. Append the original last step to THIS S. The actual append recursion identifies the whole resulting tower with the given tower. Thus the split retains every original later step and records a genuine empty blowup. □

Theorem 354.2 (The original whole family history internally compresses all empty steps)**.** *At EVERY actual family depth k, ONLY the original bound-k marked record and the GIVEN entire actual R_k canonical tower T internally supply a whole canonical tower Q over the SAME original base, an actual final biholomorphism $\chi : Z \to Y$, and ONE strict order embedding F of the actual depth-k value type. Every actual recorded centre of Q is nonempty and $|Q| \le |T|$. Whole projections satisfy $p_T\chi = p_Q$ and ENTIRE original transformed cosupport satisfies*

$$\operatorname{cosupp} M_T = \chi_{\mathrm{inv}}^{-1}(\operatorname{cosupp} M_Q), \qquad R_k(M,T,y) = F\big(R_k(M,Q,\chi^{-1}y)\big)$$

for EVERY original cosupport point y. Here the displayed preimage uses the inverse map χ^{-1} : $Y \to Z$. The actual original and compressed histories are terminal geometric resolutions exactly together. The SAME strict F therefore reflects all mixed-point strict, weak and equality comparisons; no maximum existence or compressed history/value certificate is a caller input.

Proof. Use strong induction on the actual whole length, uniformly over every actual final manifold and every original canonical tower of that length. If every centre is nonempty, choose $Q = T$, identity final biholomorphism and identity order embedding. All dictionaries are literal identities. Otherwise the first lemma internally gives a true whole empty-step split. The genuine actual empty-deletion producer constructs the normalized suffix, deleted whole canonical tower D, final biholomorphism $\chi_D : D_{\text{final}} \to Y$, entire cosupport equality, full-value formula with strict chronological embedding $H_{k,j}$, and terminal iff. Its normalized suffix has the original suffix length, so $|D| = |H| + |S| < |H| + 1 + |S| = |T|$. Apply the length induction to this INTERNALLY supplied D and its internally derived whole canonicality. It gives Q, final χ_Q, and a single strict embedding F_Q with the entire dictionaries. Choose $\chi = \chi_Q$ followed by χ_D and $F = H_{k,j} \circ F_Q$. Composition of genuine strict embeddings is strict.

The actual whole normalization isomorphism commutes with original base projections; the identity empty insertion has literally the deleted projection. Compose these projection identities with the induction's actual projection identity to obtain $p_T\chi = p_Q$. Compose the two entire inverse-preimage cosupport identities. At each original cosupport point the first identity supplies a genuine D cosupport point, so the induction's WHOLE value formula applies there. Substitution into the single-deletion WHOLE formula gives precisely F. Compose actual terminal iff statements. The induction constructs a finite tower with every centre nonempty; it never postulates its existence or chooses a desired value comparison. The ONE strict F supplies all mixed-point order reflection because the actual values at both points use this same embedding. □

Theorem 354.3 (The actual maximal-order Step One history has the same full compression)**.** *For EVERY lower depth k, original bound-$(k+1)$ maximal-order input and its GIVEN entire actual masked Step One canonical history internally give such a whole nonempty-centre canonical history, shorter-or-equal actual length, true final biholomorphism, SAME original projection dictionary, ENTIRE cosupport equality, terminal iff and ONE strict embedding of the entire masked Step One value type computing all original cosupport values. Every true level and lower order, chosen-contact zero and empty histories are included. No compressed canonicality, lower history, value, branch or termination certificate is supplied.*

Proof. Use strong induction on the actual entire tower length, retaining the same original maximal-order record and all of its indexed members. If all actual centres are nonempty, take the original whole tower and both identities. Otherwise internally locate an actual empty step and its entire original suffix. The actual Step One empty-deletion producer gives deleted WHOLE Step One canonicality, the real normalized suffix of exactly the old suffix length, its final biholomorphism, entire cosupport/whole values and terminal iff. Thus its actual length is strictly smaller. The length induction on THIS internally supplied original canonical deleted history gives a whole nonempty-centre Q and one strict embedding F_Q. Compose actual final biholomorphisms and the single-deletion masked chronological embedding $S_{k,j}$ with F_Q. Whole projection identities compose using actual identity-insertion projection equality. Entire cosupport equalities compose via inverse maps. At every actual original cosupport point they internally give the intermediate cosupport membership, so substitution of the two genuine masked WHOLE formulas gives the stated one-map formula. True terminal iff statements compose. This uses the entire actual masked Step One values at every original branch and lower order, rather than a monomial or independently chosen contact surrogate. No noncompact scheduling or global gluing conclusion is asserted. □

355 The actual original General history crossing an interior empty step

The original general invariant keeps its actual nonmonomial order and the companion formed at the start of its original order regime. To compare an interior empty insertion, that order and its whole selected history must be computed from the original datum, including an empty tail. The following are genuine history and canonical-input producers for the positive General branch. They do not assume a companion value or new canonical-history comparison. The source route is Włodarczyk's General step and Bierstone–Milman's trivial-blowup convention [3, 5].

Theorem 355.1 (Original nonmonomial order on every interior empty history)**.** *Let M be originally well formed, H an actual test history of M, S any actual subsequent history, and y any actual final point. Then the nonmonomial orders of the SAME original M on $H\epsilon S$ and HS at y agree. No nonempty-tail, positive-order or desired equality condition is required.*

Proof. The actual transformed head $K = M_H$ is well formed. Its genuine identity empty blowup is admissible, hence has a coherent transformed ideal. Removing its new empty boundary entry leaves the actual empty-step base $K_\epsilon^{\mathrm{base}}$, coherently restricted-equivalent to K by the genuine identity pullback dictionary. Their carriers are the same closed original carrier, and both ideals are coherent. Restricted equivalence therefore gives equal nonmonomial orders after the entire actual suffix S. The actual new empty boundary entry changes no nonmonomial order, also after S; its index is internally valid because the base boundary has exactly $|K.E|$ members. The real transform-append and transform-step dictionaries identify these two computed transforms with $M_{H\epsilon S}$ and M_{HS}. This proves the stated original order equality for every S, including the immediate empty step. □

Theorem 355.2 (The actual original order-regime run across insertion)**.** *For these actual data and any fixed original order o, put $r = \mathrm{run}_{M,o}(HS, y)$. The actual run on $H\epsilon S$ is r if $r < |S|$, is $r + 1$ if $r \geq |S|$ and the original point $S.\mathrm{proj}(y)$ is in order regime o at H, and is $|S|$ otherwise. The regime is the literal original transformed-cosupport membership together with $\nu = o$.*

Proof. Induct on the actual last steps of S. At the empty tail, the run recursion on the genuine identity empty step reads the original regime at H: it adds one to the original run if that regime holds, and is zero otherwise. This is exactly the displayed formula when $|S| = 0$. At a nonempty last step, the preceding theorem identifies the original order regime at the two predecessor points. If the predecessor fails the regime, both runs are zero and the old run is shorter than the positive tail length. If it satisfies the regime, each actual run is the predecessor run plus one. Apply induction and compare the old predecessor run with the predecessor tail length. Each of the three inductive cases gives precisely the corresponding stated current case. The actual whole cosupport dictionary supplies the cosupport component of the two predecessor regimes; the order dictionary supplies their order component. The composite projections identify the original base point throughout. □

Theorem 355.3 (Both whole General splits retain the original assignment head)**.** *Assume y is in the actual original transformed cosupport, its actual current order is o, and the original run r reaches S. The actual new order-regime run is $r + 1$. Put $a = r - |S|$, $D = \mathrm{stage}(H, a)$, $K = \mathrm{head}(H, a)$ and $B = \mathrm{tail}(H, a)$. Both entire selected General split records have the SAME original stage/head (D, K), with tails BS and $B\epsilon S$. Hence the actual original order mask $U = \mathrm{ordMask}(M_K, o)$ and WHOLE companion J formed from M, o, K are identical at the two assignments. The new restricted companion tail has the genuine restricted-interior-empty diagram.*

Proof. Actual cosupport membership and current order supply current order-regime membership. A run reaching S internally supplies membership at the original H point by the true run-reaching induction. The preceding run formula therefore gives $r + 1$. The actual full split formula above

the tail length gives the old split (D, K, BS). Apply it to $H\epsilon, S$ at $r+1$: the remaining split count is $a+1$, whose first removed step is the identity empty blowup. The actual split recursion therefore gives $(D, K, B\epsilon S)$. Both lower constructions use literally the same M, o, K, so their order mask and entire companion agree without a chosen comparison hypothesis. Restrict the real empty-history diagram to this literal original open U; its identity base and actual final open identifications give the entire asserted tail isomorphism. □

Theorem 355.4 (Original full canonical input supplies the crossing companion data)**.** *Let $n > 0$, let M belong to the original dimension-$(k+1)$ invariant domain, and let HS be canonical for its actual full R_{k+1}. Let y be an original transformed-cosupport point with actual positive order o and actual order-regime run reaching S. Use the true original K, B, U, J just computed. Original data internally supply mask membership, the true dimension-$(k+1)$ domain and maximal order of J, the entire OLD restricted companion tail canonical for the actual Step One value formed from R_k, and its actual cosupport point. They also supply the WHOLE new restricted companion tail diagram and its genuine testness. No companion domain, maximal-order, test, canonical-history, mask or value comparison is a caller premise.*

Proof. The actual original family supplies testness of its canonical HS and hence of H. Its positive branch is the genuine General value whose inner maximal-order value is the actual Step One construction from R_k. That construction has its proved actual maximal-order value properties. The General canonical-tail theorem therefore gives actual companion lower data at the original order-regime split. Substitute the preceding WHOLE computed original split before unpacking the dependent mask member and cosupport point. This gives exactly the domain, maximal order, canonical restricted old tail and cosupport stated for the literal U, J, BS. The preceding new split retains that same original U, J and gives its genuine whole restricted empty-history diagram. The true old canonical companion tail is a test history. Normalize its restriction of concatenation by the actual tower isomorphism; testness transfers and the real identity-pullback gives the same coherent J. The original interior-empty test equivalence supplies testness of the normalized inserted companion tail. The true new restriction diagram transfers this testness back to the whole new restricted companion tail. These constructions use only the internally supplied original companion data and actual geometry. □

The added interior companion history is not asserted to be canonical or to have the required positive recursive value yet. Full positive inner insertion, outer General values, mixed spectra and noncompact scheduling remain further obligations.

356 The complete original General value at a special companion

The original positive General branch evaluates the actual Step One companion formed at its own order-regime head. An interior identity empty step preserves that head in the reaching branch. Its whole restricted tail must still be compared without assuming that the newly inserted history is canonical. The actual special companion branch permits this comparison directly, including its level-zero case. This is original indexed history bookkeeping for Włodarczyk's General step and Bierstone–Milman's trivial-blowup convention [3, 5].

Theorem 356.1 (Actual whole restricted special-companion dictionary)**.** *Let U be an actual original open in a stage D, let J be originally well formed on U with mark at least one, and let B, S be actual original histories after D. Suppose the WHOLE old restricted BS is an actual test history of $\overline{J}$. For every actual final point y whose projection lies in U, the whole restricted $B\epsilon S$ also contains y. The actual level and special predicate at that point agree for the two WHOLE restricted histories, including empty S and level zero.*

Proof. The original empty identity step leaves the composite projection unchanged, so old open membership gives new membership. Normalize the whole old restriction into $(B|_U)(S|_{B^{-1}U})$ by

the actual restriction-append tower isomorphism. Its base map is the identity of the original U, whose pullback of the coherent J is literally J. The genuine geometric tower isomorphism dictionary transfers bar-testness and identifies actual levels and special predicates at the point with the same underlying y. Its bar-test prefix supplies real geometry for the interior identity empty step. The actual original interior-empty dictionaries preserve level and specialness, with no contact-choice or canonical-history input, and the true interior-empty test equivalence gives bar-testness of the normalized inserted history. The whole restricted-empty-history isomorphism identifies that inserted normalization with the entire new restriction, again over the identity on the SAME original U and with the same underlying point. Its geometric level/special dictionary completes the two comparisons. All three point identifications follow from the literal underlying point; no newly inserted canonical segment or desired predicate equality is assumed. □

Theorem 356.2 (Entire original General value preserved at its special companion)**.** *Let $n > 0$ and let M belong to the original dimension-$(k+1)$ invariant domain. Let HS be canonical for its actual full R_{k+1}. Let y lie in its actual transformed cosupport, have actual positive order o, and have original order-regime run r reaching S. Put $K = \mathrm{head}(H, r - |S|)$, $B = \mathrm{tail}(H, r - |S|)$, $U = \mathrm{ordMask}(M_K, o)$ and J its actual whole companion. Original data internally give actual old mask membership, companion domain, maximal order, whole Step One canonical old tail and real cosupport. If that internally supplied companion point is special on the OLD whole restricted BS, then the ENTIRE original R_{k+1} value of M at y on $H\epsilon S$ equals its entire value on HS. No companion-domain, new canonicity, contact-choice or value comparison certificate is an input.*

Proof. Use the original full canonical-tail producer to construct the stated companion data. Its actual Step One bridge supplies bar-testness of the whole old companion restriction. The preceding dictionary therefore supplies new mask membership, equal actual companion levels and specialness at the new whole restricted point. The actual Step One definition at a special point is the entire pair $(\ell, \top)$, so both complete companion values agree, including when $\ell = 0$. Original full canonical testness supplies the actual head geometry. The true original all-tail order dictionary and reaching-run split computation give the same positive outer order and both WHOLE selected General records (K, BS) and $(K, B\epsilon S)$. Their companion construction uses literally the same original M, o, K. Both mask memberships make the actual General lower value the finite inclusion of its whole companion Step One value. Expanding the positive branch of the entire original successor family and substituting the two whole splits consequently gives equal outer order and equal finite right branch. This is equality of the complete actual original R_{k+1} values. No canonicity of the newly inserted companion or outer history is used. □

Nonspecial companions, their genuine recursive lower branches, mixed spectra, maximum existence and noncompact scheduling remain further producers. The special-companion branch is a true computed branch, not a surrogate conclusion certificate, and the full Hironaka goal remains open.

357 An actual interior empty step with the original history retained

The source-facing goal is the empty-step invariance used in Włodarczyk's resolution invariant: original boundary members remain original, whereas a new exceptional member belongs to the later history. This chapter supplies an unrebased geometric prerequisite. Throughout, the same initial marked ideal and the same initial space are retained. No value from the transformed initial marked ideal is substituted for a value from the original history.

The step treated here is the genuine blowup of the empty centre with identity projection. It is sufficient to isolate deletion of a redundant interior stage without a change of manifold. Arbitrary empty-centre biholomorphisms and the complete recursive invariant comparison still require further proofs.

Definition 357.1 (The original history with an interior identity empty step)**.** For actual blowup towers $H : X \leftarrow A$ and $S : A \leftarrow Y$, let

$$H\epsilon S = (H \xleftarrow{\mathrm{id}_A} A)S, \qquad C_\epsilon = \varnothing.$$

The identity is an actual blowup along the empty centre: the empty vanishing ideal is the unit ideal, and the blowup universal property has the identity as its unique lift. The initial space, final space and all stages of H and S are retained. The composite projection of $H\epsilon S$ is literally that of HS, and its length is $|H| + 1 + |S|$.

Lemma 357.2 (The actual geometric dictionary without rebasing)**.** *Let $J = (N, E, I, d)$ be well formed and let H be an actual test tower of J. For any actual tower S starting at the final stage of H, put $j = |E| + |H|$, $J^+ = (J)_{H\epsilon S}$ and $J^- = (J)_{HS}$. Then*

$$N^+ = N^-, \quad d^+ = d^-, \quad E^+ = (E^-).\,\mathrm{insert}_j(\varnothing), \quad \mathrm{cosupp}\, J^+ = \mathrm{cosupp}\, J^-.$$

Their codimension-one cosupport parts along $N^+ = N^-$ also agree. If S has at least one blowup, the entire literal record satisfies $J^+ = \mathrm{insertEmpty}_j(J^-)$. No admissibility hypothesis on the tail S is required for these comparisons.

Proof. Write $K = J_H$, and let B be the transform of K by the identity empty blowup with its appended empty boundary member removed. Closedness of $K.N$ and of every member of $K.E$ implies that their strict transforms by this identity map equal themselves. The empty-step ideal comparison on $K.N$, together with the identity pullback of its coherent ideal, gives restricted equivalence $B \sim_N K$. Both ideals are coherent: K is a stage of a true test tower, and the empty centre is admissible for K.

The actual transform definition, including its ordered boundary list, gives

$$J^+ = \mathrm{insertEmpty}_j(B_S).$$

Restricted equivalence preserves cosupports along every tail and, after the first blowup, the entire transformed records are equal. The carrier and boundary transforms along any tower depend only on their initial carrier and boundary, so $(B_S).N = (K_S).N$ and $(B_S).E = (K_S).E$, also for an empty tail. The mark is retained by every transform. Inserting an empty boundary member changes neither the carrier, ideal, mark nor cosupport. Finally the codimension-one part depends only on the carrier and cosupport, which have just been proved equal. If S is nonempty, apply equality of transforms of restricted-equivalent coherent marked ideals after its first blowup; this proves the claimed literal record equality. □

Lemma 357.3 (The original Step One members and all special branches)**.** *Let the original marked ideal M be well formed, and let H be an actual test tower of the original coefficient-homogenized marked ideal $\overline{M}$. For every actual tail S, the ordered list of original members, its indexed membership count at every point, both branches of the Step One special locus, and each fixed Step One regime coincide for $H\epsilon S$ and HS. This includes the zero-member codimension-one branch.*

Proof. The coefficient-homogenized marked ideal is well formed and has the same initial boundary list as M. Apply the geometric dictionary to $\overline{M}$. The first $|M.E|$ elements of its transformed boundary list are precisely the original members. Since $j = |M.E| + |H| \geq |M.E|$, insertion at j leaves this whole ordered prefix unchanged. Consequently its membership count agrees pointwise. In the zero-member branch, the dictionary identifies the actual carrier and codimension-one cosupport part. In the positive branch it identifies the actual carrier, ordered original-member list and cosupport; these are all arguments of the member-intersection locus. Both special predicates therefore coincide. A fixed regime is the conjunction of equality to its chosen level and failure of the special predicate, so it also coincides. □

Lemma 357.4 (The actual contact and member masks of the original history)**.** *Under the preceding hypotheses, at each final point p the actual open sets* $\mathrm{maskA}(M, H\epsilon S, p)$ *and* $\mathrm{maskB}(M, H\epsilon S, p)$ *equal their counterparts for HS. Hence the actual lower mask agrees at every level.*

Proof. The raw set whose interior defines the member mask uses only the ordered original members, their levels and the special predicate at each point. All of these agree by the preceding lemma, so its interior agrees.

For the zero-level contact mask, the initial marked ideal is literally the same M. Both composite projections equal that of HS. Thus the chosen tangent-direction pair is evaluated at exactly the same initial point; its choice is not replaced or identified using a transport certificate. The remaining raw-set arguments are absence from each original member and absence from the actual codimension-one cosupport part. The geometric and Step One dictionaries prove both equalities. The raw sets and their interiors therefore agree. The lower mask selects the contact mask at level zero and the member mask at every positive level, which completes the proof. □

Lemma 357.5 (The exact actual fixed-regime run across the interior insertion)**.** *Fix any Step One level s under the preceding hypotheses. At $y \in Y$ put $r = \mathrm{run}_{M,s}(HS, y)$, $t = |S|$, and $b = \mathrm{stepRegime}_{M,s}(H, S.\mathrm{proj}(y))$. Then the actual recursively computed run satisfies*

$$\mathrm{run}_{M,s}(H\epsilon S, y) = \begin{cases} r, & r < t, \\ r+1, & t \le r \text{ and } b, \\ t, & t \le r \text{ and } \neg b. \end{cases}$$

This is an equality of the literal original fixed-regime run, not a supplied comparison of split times. No final regime membership is assumed.

Proof. Induct on the actual tail S. For the empty tail, its length is zero and its projection is the identity. The new last step is the identity empty blowup, so the defining recursion gives $\mathrm{run}(H, y) + 1$ precisely when H, y satisfies the chosen regime, and zero otherwise. This is the stated formula, including the third case when a run of H itself is nonzero but its current point fails the regime.

For a tail ending in π, the special-branch and level dictionary makes the predecessor regime test identical in both histories. If that test fails, both runs are zero, which is shorter than the now positive tail length. If it holds, both recursively computed predecessor runs increase by one. Apply the induction hypothesis at πy. The inequality $r < t$ is preserved when both sides increase by one; in the other cases the old run, or the tail length respectively, increases by one. The base-point test b is unchanged because the tail projection composes with π. These three cases give the formula. □

These results give geometric producers and the actual fixed-regime run for retaining the original history when an interior empty step is removed. They do not yet identify the recursive invariant's lower marked ideals or restricted tails, prove the whole higher value unchanged, provide local regularity of its maximum locus, or construct global maximum scheduling on a noncompact space. The full G3 goal remains open. The argument is an adaptation of the empty-step part of the marked-ideal resolution invariant, using the existing literal controlled-transform and original-member definitions; it does not supply an independent resolution driver. Source routes: `wlodarczyk05-hironaka`, resolution-invariant empty-step property and original boundary histories; `bierstone-milman08-functoriality`, marked-ideal equivalence and blowup histories.

358 The actual positive lower record after an interior empty step

The initial marked ideal, original members and complete original head remain fixed. This chapter computes the actual lower marked ideal at a positive Step One level after insertion of an interior identity empty blowup. It uses actual full records, not an assumed equality of lower values.

Lemma 358.1 (Future empty insertion in a member restriction)**.** *Let J be any marked ideal, E_0 its specified original-member list, p a point, and $j \le |J.E|$. Write*

$$c = \big|(J.E.\,\mathrm{take}(j)).\,\mathrm{filter}(W \notin E_0 \text{ or } p \notin W)\big|.$$

Inserting an empty future member at j in J, while leaving E_0 unchanged, commutes with the actual member restriction through p, with insertion index c in the restricted boundary list.

Proof. The empty member never passes through p. It is therefore never among the selected members, even if E_0 already contains an empty member. Insertion changes neither the carrier intersection nor the coefficient ideal nor the mark of the member restriction. Its actual boundary is the original whole boundary filtered by failure to be an original member through p. The inserted empty member is retained by this filter. Induction on the first j boundary elements shows that filtering an insertion of a retained element inserts it after exactly the retained elements of that prefix. This is precisely c, and proves equality of all literal marked fields. □

Definition 358.2 (The actual lower index of the interior future member)**.** For original histories $H : X \leftarrow A$ and $S : A \leftarrow Y$, set $J = \overline{M}_{HS}$, $E_0 = \mathrm{origMembers}(M, HS)$, $j = |M.E| + |H|$ and let c be the preceding filtered-prefix cardinal. Then c is at most the length of the actual boundary of the original positive lower marked ideal, at every positive Step One level.

Proof. The actual positive lower marked ideal is the pullback of the member restriction to the actual member mask. Its boundary is the filtered whole boundary $J.E$, then mapped by the mask inclusion preimage. Mapping retains its length. The filtered first j elements form a sublist of this whole filtered boundary, so its length c is bounded by the actual lower boundary length. This argument does not require zero lower nonmonomial order or a bound inside an empty original prefix. □

Theorem 358.3 (The genuine original positive lower record)**.** *Let M be originally well formed and H an actual test tower of $\overline{M}$. For every nonempty actual tail S, every final point p, and every positive Step One level $s+1$, the actual masked lower marked ideal for M at $H\epsilon S, p$ equals the actual lower marked ideal for M at HS, p with an empty boundary member inserted at the computed index c. Equality is heterogeneous only because the equal mask opens are kept as their respective literal carrier types. All fields agree under the equality of those actual masks.*

Proof. The nonempty-tail whole-record producer gives $\overline{M}_{H\epsilon S} = \mathrm{insertEmpty}_j(\overline{M}_{HS})$. The original-member producer supplies the unchanged entire E_0 and $j \le |J.E|$ follows from the actual transformed boundary length. Apply the future member-restriction lemma to these actual records. The member mask producer gives equality of the two literal open masks. Pullback of an empty insertion maps its empty member to the empty set and commutes with insertion at the same computed index. Substituting the mask equality therefore proves the stated actual masked lower-record equality. The masked positive lower definition is exactly this member-mask pullback. □

The boundary index is computed from the actual retained prefix; it may extend past the empty original prefix. No assertion of a common recursive value shift, equality of lower values, or global order/scheduling follows here. The zero-contact lower carrier still needs its own local closedness comparison on the chosen contact open. Actual restricted lower tail histories and full higher recursive invariant comparisons remain open.

359 The original zero-contact lower record across an interior empty step

The initial marked ideal and its actual chosen tangent direction are retained. The argument is an explicit adaptation of the Step One contact construction and empty-history functoriality:

the contact zero set is closed only on the chosen tangent open, so it is first replaced there by a genuinely closed representative. No contact, lower-record, tail or value comparison is an input. An absent tangent choice is handled by its actual empty open.

Lemma 359.1 (Closed carriers across an interior identity empty blowup)**.** *For every closed subset $B \subseteq X$ and actual histories H, S, the entire strict transforms of B along $H\epsilon S$ and HS are equal, where ϵ is the genuine identity empty blowup. No testness of either history is required.*

Proof. The strict transform B_H is closed: at the empty history this is the hypothesis, and at every last blowup step the strict transform is a closure. The identity empty blowup takes B_H to the closure of B_H, hence to B_H itself. Strict transformation along a concatenation is transformation first along its head and then along its tail, by induction on the actual last steps of that tail. Applying the same tail S to these equal carriers proves the claim, including an empty H or S. □

Lemma 359.2 (A closed representative of the actual chosen contact carrier)**.** *Let M be well formed and $x \in X$. Put $(U, w) = \mathrm{tangentChoice}(M, x)$ and $A = M.N \cap V(w)$. The actual set U is open, and there exists a closed $B \subseteq X$ such that $A \cap U = B \cap U$.*

Proof. If the original tangent-direction existence condition fails, the actual choice is $(\varnothing, 0)$; take $B = \varnothing$. The open is empty and both intersections are empty. Otherwise the choice specification gives a tangent direction on its actual open U. Its derivative-ideal membership and coherence of the original ideal imply holomorphicity of w at every point of U, hence continuity there. Set $B = M.N \cap \overline{V(w) \cap U}$, which is closed because $M.N$ is closed. If $z \in A \cap U$, then $z \in B \cap U$ by inclusion into the closure. Conversely, if $z \in B \cap U$ and $w(z) \neq 0$, continuity at z supplies a neighbourhood on which w is nonzero. Since $z \in \overline{V(w) \cap U}$, that neighbourhood meets $V(w) \cap U$, a contradiction. Thus $w(z) = 0$ and $z \in A \cap U$. □

Theorem 359.3 (The whole actual masked contact carrier)**.** *Let M be originally well formed, H, S actual histories and p a final point. The strict transforms of the actual original chosen contact carrier in the two histories HS and $H\epsilon S$ have equal traces on the actual zero-contact mask for HS, p. The tangent direction in the second history is the same literal chosen direction.*

Proof. The composite projections of the two histories are equal, so their projections of p and hence their original tangent-choice input agree. Choose the closed representative B supplied by the preceding lemma for this actual input. Strict transforms are local over an open subset: sets agreeing on U have transforms agreeing over the projection preimage of U. To see this along a tower, induct on its actual last steps. At one step, intersecting a closure with the open projection preimage equals the closure relative to that open; the agreed preimages outside the same centre therefore give agreed strict transforms there. The induction gives the whole-tower statement. Both contact transforms thus agree, over this same projection preimage, with their respective transforms of B. Those entire transforms of B are equal by the closed-carrier lemma. Every point of the actual mask belongs to its defining set, since the mask is its interior; its projection therefore lies in U. Restricting the preceding equality to the entire actual mask proves equality of its contact carrier traces. This also covers the absent-choice branch, whose actual mask has no points. □

Theorem 359.4 (The entire original zero-contact lower marked record)**.** *Let M be originally well formed, H an actual test history of $\overline{M}$, S a nonempty actual tail, and p any final point. Put $j = |M.E| + |H|$. The actual original masked lower marked ideal at level zero for $H\epsilon S, p$ equals that for HS, p with one empty boundary entry inserted at j. The index j is bounded by the actual whole lower boundary. Equality is heterogeneous only because the two equal actual masks remain their respective literal carrier types.*

Proof. The nonempty-tail producer supplies the entire actual record

$$\overline{M}_{H\epsilon S} = \mathrm{insertEmpty}_j(\overline{M}_{HS}).$$

The coefficient construction along the carrier depends on its carrier, ideal and mark and retains the entire boundary, so it commutes with this future empty insertion. The zero lower record changes its carrier to the strict transform of the chosen original contact carrier. The preceding theorem identifies the full trace of that carrier on the actual old mask. The original mask dictionary identifies the two whole actual mask opens. Pullback along their inclusions therefore gives equal carrier, coefficient ideal and mark. Its boundary is the full transformed boundary mapped by preimage; mapping commutes with list insertion and takes an empty set to an empty set. This proves equality of all four marked fields with the stated insertion. The old lower boundary has length $|M.E| + |H| + |S|$, because coefficient formation retains the full boundary and mapping by inclusion preserves length. It therefore bounds j. □

Theorem 359.5 (The actual current-level short splits at every level)**.** *Let M, H, S, y be as above, with S any tail. Set $\ell = \mathrm{level}(M, HS, y)$ and $r = \mathrm{run}_{M,\ell}(HS, y)$, and suppose $r < |S|$. The actual current-level run in $H\epsilon S$ is precisely r. Writing $D = \mathrm{stage}(S, r)$, $P = \mathrm{head}(S, r)$ and $Q = \mathrm{tail}(S, r)$, the two entire actual split records are (D, HP, Q) and $(D, H\epsilon P, Q)$.*

Proof. The actual level dictionary identifies the new current level with ℓ. The original run formula uses its first branch because $r < |S|$, and gives the same r. Since $r \leq |S|$, the actual concatenation split is obtained by removing precisely the last r steps from S, leaving head HP and tail Q. Induction on those last steps supplies the full split record, not merely the lengths. Applying the same computation to the actual prefix $H\epsilon$ gives head $H\epsilon P$, with exactly the same stage D and tail Q. No positivity of ℓ is required. □

Theorem 359.6 (The genuine selected short zero-contact lower diagram)**.** *Let M be originally well formed, H an actual test history of $\overline{M}$, S an actual tail and y an actual final point with $\mathrm{level}(M, HS, y) = 0$. Assume that its actual zero-level run r satisfies $r < |S|$. Put $P = \mathrm{head}(S, r)$, $Q = \mathrm{tail}(S, r)$, $p = Q.\mathrm{proj}(y)$ and $j = |M.E| + |H|$. The new actual current-level run is r, its selected zero lower record is the old actual selected zero lower record with an empty boundary entry inserted at j, and j is a valid actual lower boundary index. The two whole actual restricted tails of the common literal Q are equal under equality of the respective actual masks.*

Proof. The preceding all-level short split theorem applies at the actual current level zero. It supplies both whole splits and the actual shared stage and tail Q. The selected prefix P has length $|S| - r > 0$, derived directly from the given strict inequality. Thus the original zero lower record theorem applies to H, P, p and supplies its full record comparison and the valid whole-boundary index. The original all-level mask dictionary at level zero identifies the two actual masks; replacing one mask by that proved equal open makes the entire restricted Q records reflexively equal, including all their centres, blowup data and projections. No contact, selected-prefix nonemptiness, lower-index, lower-record or restricted-tail comparison is supplied by the caller. □

These statements compare the actual zero-contact carrier, whole lower records and whole restricted tails in the short-run branch. They do not assert transport of complete higher recursive invariant values, a global maximum, noncompact scheduling or completion of the G3 parent.

360 Empty original entries in the entire recursive canonical value

The ordered boundary is part of the actual marked ideal. Inserting an empty original entry changes its indices, even though it changes neither its ideal nor its cosupport. The comparison below retains this change throughout the entire recursive value. It uses the actual masks and

histories of the masked Step-One construction in Włodarczyk's marked-ideal algorithm. The explicit recursive index map describes how the retained indexed boundary adapts the insertion of isomorphisms in the original resolution history.

Definition 360.1 (The recursive empty-entry value map)**.** Let $F_{0,j}$ be the identity on the zero-dimensional value type. For the successor value type, $F_{k+1,j}$ retains its first, nonmonomial-order entry. On its monomial branch it retains bottom and applies the increasing index insertion map at j to the finite index component of the monomial value. On its positive-order branch it retains infinity; on a finite Step-One value it retains the original level and applies $F_{k,0}$ to the recursive lower value, retaining infinity there as well. Thus all recursive monomial index components after the first one use the insertion map at zero.

Lemma 360.2 (The recursive map is a strict order embedding)**.** *Each $F_{k,j}$ is strictly increasing and consequently reflects strict order, weak order, and equality on the entire recursive value type.*

Proof. Induct on k. The identity has the assertion at zero. At a successor, the monomial index insertion preserves and reflects the colexicographic order of finite sets, and leaves the other monomial entries unchanged. Extending a strictly increasing map by a fixed bottom or top preserves strict order. The induction therefore applies to the lower-value map, its top extension, and the Step-One lexicographic pair with unchanged level. Extend that map again by a fixed top. The sum order puts every monomial branch value below every positive branch value, so applying these two increasing branch maps preserves strict order on the sum. Finally the outer lexicographic pair has an unchanged first entry. If two first entries differ their comparison is unchanged; if they agree their second-entry comparison is preserved. This proves the successor assertion. A strictly increasing map between linear orders reflects both orders and is injective, giving all three reflections. □

Theorem 360.3 (The entire actual invariant commutes with empty insertion)**.** *For every k, actual marked ideal m, valid original insertion position j, genuine blowup tower T, and endpoint y, the actual canonical value satisfies*

$$\mathrm{inv}_k(\mathrm{insertEmpty}_j\, m, T, y) = F_{k,j}(\mathrm{inv}_k(m, T, y)).$$

Proof. Induct on k. At zero the invariant is constant. At a successor the actual nonmonomial orders agree at every stage. If the endpoint has order zero, the whole transformed boundary has the same insertion at j, so its monomial value is precisely the monomial branch of $F_{k+1,j}$. For positive order, the actual order regimes agree along every stage, hence their computed runs and genuine head-tail splits agree. The actual order masks agree and their actual companion marked ideals have the same insertion at j. The true Step-One original levels, specialness, regimes, and runs also agree. Special endpoints give identical infinite Step-One values. Otherwise the actual lower masks agree. At level zero, the masked contact record has the same insertion at j; its entire original prefix has empty traces on this actual mask because the zero mask requires avoidance of every actual original member. Pulling each of these members back to the mask therefore gives the empty set. At positive level the genuine member restriction computes the new insertion index c as the number of retained entries before j. Its whole retained original prefix is empty on the actual mask, and c lies inside that prefix. Apply the induction to the real lower record and genuine restricted tail. At either level the positions selected in the current monomial branch lie beyond the empty prefix; insertion at j, respectively c, therefore acts on those positions exactly as insertion at zero. The positive branch of $F_{k,c}$ is already independent of c. Thus the actual recursive lower value is $F_{k,0}$ of its old value. Extending through both actual infinity cases gives the stated positive branch of $F_{k+1,j}$. All masks and runs here are those computed by the original value; no comparison, selected-index, or history alignment is assumed. □

Corollary 360.4 (All actual value comparisons and maximum loci are retained)**.** *At fixed k and insertion position j, all comparisons of actual values before and after insertion agree, including*

comparisons between different genuine histories. In particular every maximum locus on any set of endpoints of a fixed history agrees.

Proof. Rewrite each actual inserted value by the preceding theorem and apply the three order reflections of the same strictly increasing map $F_{k,j}$. A point belongs to a maximum locus exactly when it belongs to its comparison set and dominates every value on that set. Weak-order reflection identifies every one of these comparisons, proving equality of the whole maximum loci. □

Corollary 360.5 (Entire actual canonical histories are retained)**.** *For each actual marked ideal and valid empty original insertion, a genuine history is canonical for the inserted datum if and only if it is canonical for the old datum. This holds both when empty centres are permitted and when every centre must be nonempty.*

Proof. Induct on the genuine history. At length zero both canonical predicates hold. At a successor, their preceding histories agree by induction. The whole transformed cosupports agree because insertion retains the carrier, ideal, and marking at every stage. The preceding maximum-locus corollary therefore identifies the actual maximum centres at the preceding stage. The successor centre is the same set for both inputs, so its emptiness or nonemptiness is also the same. Thus every clause of either recursive canonical-history definition agrees, giving both equivalences. □

361 The actual positive member lower boundary has an empty original prefix

At a positive Step-One head, the original member restriction selects the actual original members through its reference point and keeps all other indexed members in order. Its actual mask excludes every original member missing that reference point. The retained original boundary prefix is therefore literally a list of empty sets on this SAME mask. Future members are retained with their original order. The numerical position of the end of this empty prefix may depend on the reference point, but any monomial top set has every position after it. Inserting an empty member at its end consequently shifts every position used by the literal monomial value by one, independently of the reference point. This is an order embedding, and does not assert equality of the old literal values. It supplies the zero-order branch of the recursive member comparison; positive recursive lower orders and the complete middle-empty-step comparison remain subsequent producers.

Lemma 361.1 (The literal positive lower boundary splits after its retained original prefix)**.** *Let J be the actual coefficient-homogenized transform at any original head H, let E_0 be its original prefix, and let p be the actual reference point. Set A to the ordered sublist of E_0 missing p, and set F to the ordered sublist of the actual future boundary not selected by the original members-through-p restriction. On the actual original positive lower mask, the entire boundary of the old lower marked record is exactly $[\varnothing]^{|A|}$ followed by the full masked preimage of F. All actual entries, including repeated and empty entries, retain their order.*

Proof. The original members-through restriction keeps exactly those indexed entries W for which it is false that $W \in E_0$ and $p \in W$. Split the actual original boundary into its initial original prefix and the remaining future list, and apply this SAME filter to both ordered lists. On the original prefix this filter is exactly the predicate $p \notin W$, because each entry belongs to that original prefix. For every retained original entry W, the defining predicate of the actual lower mask says that any masked point of W would force $p \in W$. The retained entry has $p \notin W$, so its entire masked preimage is empty. Thus the entire masked original sublist is precisely $|A|$ empty entries. Map distributes over the ordered concatenation, giving the stated literal whole boundary identity. No incidence, chosen member subset, mask, lower model or boundary decomposition is supplied by the caller. □

Lemma 361.2 (Every actual lower tail preserves the same empty original prefix)**.** *Along every genuine blowup tail on this SAME original positive lower mask, the first $|A|$ members of the whole actual transformed lower record are empty. This prefix length is at most the actual transformed boundary length.*

Proof. The preceding literal identity establishes the statement at the base. At every genuine blowup, the actual transformed boundary is the ordered strict-transform map of the preceding entire boundary followed by its true exceptional member. A strict transform of an empty set is empty. The new member is appended after the preceding list, whose length is at least $|A|$, so the first $|A|$ members remain exactly the same empty list. Induction on the actual tail gives the literal prefix identity at every stage. The true transformed boundary length is the original lower boundary length plus the tail length, which gives the stated bound. This applies to every actual tail, without canonicality or ideal hypotheses because it is a structural boundary statement. □

Lemma 361.3 (The actual monomial value uses only positions after an empty prefix)**.** *For an actual entire indexed boundary whose first j entries are empty, every member position through any point is at least j. Whenever its literal monomial value is a finite key, every position of that key is at least j.*

Proof. A member position smaller than j indexes the empty prefix and hence cannot contain the point. Every feasible subset is a subset of the true set of member positions through that point. If the monomial value is finite, its actual maximum is achieved by one of those feasible subsets, so every position of its selected key is in that true set and is at least j. If there is no feasible subset, the literal value is bottom, which is retained by every insertion relabelling. No top-set, feasible-set or value-profile certificate is input. □

Lemma 361.4 (The future-only insertion relabelling is uniform)**.** *For every literal monomial value computed from a boundary with j empty initial members, the insertion relabelling at j equals the SAME relabelling at 0. In particular the actual transformed original positive lower marked record with an empty member inserted at its internally computed retained-original-prefix length has its entire monomial value equal to the uniform relabelling at 0 of the SAME actual original lower monomial value, at every point of every actual lower tail.*

Proof. On a position at least j, insertion at j and insertion at 0 both send that position to its successor. Therefore they give identical images of every selected feasible subset, preserving its cardinality, exponent sum and other key entries. The preceding actual maximum description gives equality on every finite literal value; bottom is unchanged by definition. Whole transformation commutes with explicit empty insertion, retaining the same original insertion position. The preceding actual empty-prefix producer and length bound apply at every stage, so the original whole insertion dictionary gives this SAME uniform relabelling for every actual transformed lower monomial value. The value definition and its absolute member indices remain literal. □

Theorem 361.5 (Original zero-order lower values have the same strict order after the actual insertion)**.** *At points of actual nonmonomial order zero in any two such original positive lower marked records and actual lower tails, insertion at EACH internally computed retained-original-prefix length preserves equality, non-strict order and strict order of the entire old general invariant values, even when these computed lengths or reference masks differ.*

Proof. Whole empty insertion preserves the actual nonmonomial order, so both old and inserted general values lie in the literal zero-order monomial branch. Their first component is zero and their second is the monomial value in the left branch of the original lexicographic sum. The preceding original-input producer gives the SAME uniform insertion relabelling at 0 for both monomial values. That relabelling is strictly increasing, hence injective and order-reflecting on the actual linear order, and consequently preserves and reflects equality, non-strict order and strict order. The common lexicographic outer components and left injections preserve those three comparisons. This uses no recursively supplied invariant relation or canonicality certificate. □

Corollary 361.6 (Original zero-order exact maximum loci and centres agree)**.** *On any actual subset consisting of nonmonomial-order-zero points of one such genuine original lower tail, the exact maximum locus of its whole old general value equals that after empty insertion at the internally computed retained-original-prefix length.*

Proof. A point belongs to the exact maximum locus precisely when it belongs to the stated actual subset and every value at another point of that subset is at most its value. Apply the preceding original comparison equivalence to each such pair of actual points. This proves equality of the entire exact loci with all actual points retained. Since these are equal subsets of the SAME actual ambient mask, any admissibility property or geometric centre already proved for that exact original locus concerns precisely the SAME inserted locus. No existence of a global maximum on a noncompact space, recursive positive comparison or scheduling statement follows from this locus identity. □

362 The whole original family at its actual short outer run

An empty step after H lies before the actual assignment head of every outer run shorter than the remaining suffix S. The selected companion therefore receives an INITIAL boundary entry, rather than a future entry of its own lower history. The entire specific-family initial-boundary producer handles its genuine positive and zero-contact lower records internally. This is the ordered-boundary adaptation of the general step in [3, 5].

Lemma 362.1 (Both actual original short order splits)**.** *Let M be originally well formed, let H be its actual test head, and put $r = \mathrm{regimeRun}(\mathrm{OrdRegime}(M, o), HS, y)$. If $r < |S|$, the corresponding original run on $H\epsilon S$ is exactly r. Both entire computed splits have the SAME actual stage and tail $Q = \mathrm{tail}(S, r)$, with heads $H\,\mathrm{head}(S, r)$ and $H\epsilon\,\mathrm{head}(S, r)$. The selected prefix has positive length.*

Proof. The genuine whole original order-regime run dictionary is obtained from the actual geometric cosupport and order dictionaries at every prefix. Its short branch is literally r, so the new computed run is r. The actual split-of-append identity at $r \le |S|$ identifies both entire dependent split records with the stated heads and common tail. The true selected prefix length is $|S| - r > 0$, so no prefix nonemptiness hypothesis is supplied. □

Lemma 362.2 (Whole selected original short companion and actual tail)**.** *Under the preceding original inputs, put $j = |M.E| + |H|$, $P = \mathrm{head}(S, r)$, $U = \mathrm{ordMask}(M_{HP}, o)$ and $J = \mathrm{compAt}(M, o, HP)$. The actual new assignment mask equals U, the whole actual new companion is J with an INITIAL empty boundary entry inserted at j, and both entire selected restricted tails are the same restriction of Q along this mask. The insertion index satisfies $j \le |J.E|$ internally.*

Proof. The selected prefix is nonempty by the preceding lemma. Genuine head testness and original well-formedness therefore give literal whole transformed-record equality $M_{H\epsilon P} = \mathrm{insertEmpty}(j, M_{HP})$. The true empty-boundary order dictionary identifies all order superlevel sets, hence their closures and the actual open order masks. Pullback along this same mask carries the inserted empty set to the empty set and preserves the entire old marked record. The actual monomial and nonmonomial part dictionaries imply that companion formation commutes with this genuine initial-boundary insertion; the ideal, carrier and mark remain literal. Consequently the whole new companion has precisely the asserted record. Companion formation and open pullback preserve the actual boundary length, which is $|M.E| + |H| + |P|$, proving the required index bound internally. Equality of the actual masks identifies both entire restrictions of the SAME original Q. No mask, record, lower-history or value equality is a caller premise. □

Theorem 362.3 (Entire actual original family value on the short outer branch)**.** *For every original successor depth $k+1$, original well-formed M, actual test head H and actual point y with TRUE own-order outer run shorter than $|S|$,*

$$R_{k+1}(M, H\epsilon S, y) = \text{shiftFamily}(k+1, |M.E| + |H|)(R_{k+1}(M, HS, y)).$$

No canonical new-history, arbitrary invariant transport, lower-order or positive-companion restriction is supplied.

Proof. The short-run condition implies S is nonempty. At actual nonmonomial order zero, the genuine whole original monomial-leaf producer supplies the stated family shift directly, with the actual insertion index. At positive outer order, both actual whole splits from the preceding lemmas retain the SAME original tail, and the whole new companion is the true initial-boundary insertion in the old companion. The complete specific-family initial-boundary identity for Step One follows from its genuine actual level, specialness and run dictionaries. On the nonspecial branch, each true lower mask is equal and each whole lower record receives its internally computed initial insertion index. The complete actual lower family insertion theorem applies at every recursive order. Its actual empty-prefix dictionary identifies this index with zero on these true lower family values. This treats both genuine contact choice at level zero and positive-member lower data, including missing-mask infinity; no desired lower comparison is assumed. Thus the ENTIRE new companion Step One value is the old one under pairShift(shiftFamily$(k, 0)$). The whole order mask equality handles both the present and absent mask cases of the general value. Its actual outer order is unchanged. Expanding the two entire original general values therefore gives exactly the defining successor family shift at j, which has the just-proved inner map. The proof keeps the actual original M, H, S rather than rebasing the statement. □

Corollary 362.4 (The entire actual short outer spectrum)**.** *At any TWO actual original points satisfying their own true short outer run, strict order, weak order and equality of the entire original successor-family values are preserved and reflected across the interior empty step, including zero and positive outer orders and all recursive inner orders.*

Proof. Both entire actual values are sent by the SAME genuine strict family embedding with the fixed original insertion index $|M.E| + |H|$. Strict monotonicity of this whole family map preserves and reflects strict and weak order and, by injectivity, equality. No comparison of distinct computed indices is assumed. Crossing positive recursive histories, full mixed short/crossing spectra, maximum existence and noncompact global scheduling remain open. □

363 Original positive companions and actual histories under inserted empty boundary members

This independent OWN OFFLIB fragment supplies the original companion and regime dictionaries needed before a middle-empty-step comparison. It keeps all original boundary indices. It proves whole companion equality up to the explicit inserted empty member, not equality of the entire recursively constructed general invariant. Whole strict-order alignment of lower StepOne values and noncompact gluing remain OPEN.

Lemma 363.1 (The whole actual companion commutes with the explicit empty insertion)**.** *For any actual marked ideal m, index $j \leq |E_m|$ and natural order o, the companion of m with an empty member inserted at j equals the companion of m with the SAME empty member inserted at j. Equality is of the ENTIRE marked ideal, including its actual ideal and mark; the indexed boundary list is retained literally.*

Proof. The actual monomial part is unchanged by the inserted empty member, since the new member contains no point. The actual nonmonomial part is also unchanged by the whole original

monomial-part identity. The true companion is the marked sum of these original parts with orders $o, d-o$ if $o<d$, and the nonmonomial part with order o otherwise. In each of these two actual cases its ideal, carrier and mark are independent of the empty member, while its boundary is exactly the original boundary with that SAME insertion. Thus every field of the entire original companion agrees. □

Lemma 363.2 (The whole original companion dictionary holds at every tower stage)**.** *Along ANY actual blowup tower T, companion formation after transforming the inserted marked ideal equals explicit insertion at the SAME original j into the companion of the actual original transform m_T.*

Proof. The original geometric transform commutes with explicit empty insertion along the entire tower, including every exceptional member appended after the original list. Its boundary length is $|E_m|+|T|$, so the original insertion position is still valid. Apply the preceding whole companion equality to this SAME actual transform. No independently supplied transformed or companion model is used. □

Theorem 363.3 (Original coherent input gives the entire companion ideal dictionary at every order and point)**.** *Only actual original invariant-domain data and a genuine geometric test tower internally give, for EVERY positive natural order o, coherence of the entire companion ideal of the transformed inserted datum and equality of its membership with the companion of the SAME original transform at EVERY ambient point and EVERY function germ. Points of higher original nonmonomial order are included; no order-bound or maximal-order assumption is supplied.*

Proof. Original domain stability along the genuine geometric test tower gives the actual transformed original datum in the SAME invariant domain. For each true $o \geq 1$, the original companion-domain producer proves that its whole companion is coherent; it needs no bound on the original nonmonomial order. The preceding ENTIRE marked-ideal equality identifies the inserted companion ideal with this SAME coherent original ideal, since inserting an empty member changes only the boundary field. Substitute this whole ideal equality for every ambient point and every germ. This proves real coherent extension of the original companion as an ideal across higher-order points; it does not extend its maximal-order hypothesis or its full StepOne canonical-value dictionary across those points. □

Lemma 363.4 (Every actual order mask agrees on the entire stage)**.** *At every tower stage and every natural o, the genuine original open order mask of the inserted transformed datum equals the SAME open mask of the original transformed datum.*

Proof. Whole transformation commutes with insertion as above. The entire original nonmonomial part, hence every actual order-at-least locus, is unchanged by insertion. The actual open mask is defined as the complement of the closure of the order-at-least-$o+1$ locus. Substituting the entire locus equality proves equality of the WHOLE open mask, without a pointwise chosen neighbourhood or an order-bound assumption. □

Lemma 363.5 (Actual regimes at every original stage agree)**.** *At EVERY actual tower stage and EVERY point, the original regime of order o for the inserted datum is equivalent to that for the original datum.*

Proof. The actual regime is the conjunction of original current cosupport membership and equality of the current nonmonomial order with o. Whole cosupport and actual nonmonomial order are unchanged by empty insertion along the tower, so both actual conjuncts agree. □

Lemma 363.6 (The original entire run length agrees)**.** *For EVERY order, actual tower and endpoint, the original run length of the inserted datum equals that of the original datum. Thus the actual tower is split at the SAME integer.*

Proof. Induct on the actual tower. Both runs are zero at the base. At each actual final blowup the run tests the actual preceding endpoint's regime, either continuing by one or resetting to zero. The preceding regime equivalence identifies these two actual branch predicates; the induction identifies the preceding runs in the continuing case. Both resetting cases are literally zero. This proves equality of the entire point-dependent run, without a caller run or split certificate. □

Lemma 363.7 (The actual regime across a middle empty step)**.** *For actual T_0, a genuine blowup σ along the empty centre, and ANY actual tail S, the regime along $T_0\sigma S$ equals the regime along S of the actual transported base with its new empty member removed. This uses the original explicit empty-step base, retaining its genuine transformed ideal and every original boundary member.*

Proof. The original middle-empty-step transform formula identifies the ENTIRE actual transform with explicit insertion at $|E_m| + |T_0|$ into the tail transform of that SAME original empty-step base. The original middle-empty identities give identical whole cosupports and nonmonomial orders. Substitute them into the two actual regime conjuncts. □

Theorem 363.8 (The whole actual run across a middle empty step has an internally determined cap)**.** *For EVERY actual order and endpoint let r be the actual run in S for the original transported empty-step base. The run in $T_0\sigma S$ is r if $r < |S|$, and otherwise $|S|$ plus the actual run in $T_0\sigma$ at the image of the endpoint. The split is determined by these SAME original computed integers. No history, run cap or chosen alignment is supplied.*

Proof. Actual runs are bounded by the true tail length. The preceding actual regime equivalence holds on EVERY prefix of S. Apply the original concatenation-run identities: in the strict case the run stays wholly inside S and equals r; in the remaining case the bound forces $r = |S|$, so the true original run continues from its actual preceding image into $T_0\sigma$. These two internally exhausted cases give the stated conditional formula. This is an actual original-history producer; comparison of lower StepOne invariant orders after deleting the empty step remains a subsequent producer. □

364 The actual positive original lower record and its computed insertion index

The empty original member is never selected by the actual members-through-reference-point restriction. After that genuine restriction, its position is the number of retained entries before its original index. This rank is computed from the actual original list and actual reference point. It can vary between masks, but it remains within the retained original empty prefix already proved for the literal positive lower record. Thus the entire actual positive lower records, rather than caller-supplied models, have a precise insertion dictionary needed for the recursive order comparison.

Lemma 364.1 (An ordered filter computes the exact new position of a retained inserted entry)**.** *For an actual finite ordered list, a genuine insertion position within it, and an ordinary predicate keeping the inserted entry, filtering after insertion equals insertion into the filtered list at the number of kept entries of the actual prefix before that position. The entire filtered list retains its order, including repeated entries.*

Proof. Induct on the original finite ordered list and its insertion position. At position zero, the retained inserted entry is the new first entry, and the prefix has zero kept entries. At a positive position the original list has a first entry. If the predicate keeps that entry, it appears first on both sides and contributes one to the kept-prefix length; apply the induction to the remaining list and the preceding insertion position. If the predicate omits that entry, it appears on neither side and contributes zero to the kept-prefix length; the SAME induction applies to the remaining list. The genuine insertion bound excludes any missing-list case at a positive position. These

cases prove the literal ordered list identity, without a supplied rank, relabelling or permutation certificate. □

Lemma 364.2 (The whole true member restriction commutes with the actual computed empty insertion)**.** *Let J be the actual head marked record and E_0 its actual original member list, with a valid insertion position in both. Insert an empty member at that SAME position into the whole boundary and original list. Apply the true original members-through-p restriction. The resulting entire marked record equals empty insertion into the SAME old actual restricted record at the number of genuinely retained entries of the original boundary prefix before that position. Its carrier, ideal, mark and whole ordered boundary are retained literally.*

Proof. The actual member selection keeps an entry precisely when it is false that it belongs to the actual original list and contains p. Adding an empty entry to the original list leaves this predicate unchanged on EVERY boundary entry, since an empty entry never contains p. The new empty entry is genuinely kept. Apply the preceding ordered-filter identity to this SAME actual predicate: it computes the retained-boundary insertion position internally. The original member-restricted carrier is the intersection of the actual carrier with every selected member through p. The new empty member contributes no selected condition, so the entire selected intersection is unchanged. The actual coefficient ideal and marking depend on carrier, ideal and old mark and retain the boundary; inserting the empty member therefore does not change those fields. These four literal field identities prove equality of the entire actual member-restricted marked record, without an input selected set or lower-record certificate. □

Theorem 364.3 (The entire actual original positive lower record has an internal insertion dictionary)**.** *For ONLY an actual original marked record, its valid original empty insertion, a genuine head and reference point, the literal positive lowerMarked record for the inserted original input is heterogeneously equal, after identifying its internally computed original mask, to insertion at the internally computed retained-prefix rank into the literal positive lowerMarked record of the original input.*

Proof. The original Step-One whole coefficient-homogenized transform dictionary identifies the inserted actual head marked record with the SAME insertion into the actual old head record. Its true original member list has the SAME insertion, and its actual positive mask equals the actual old positive mask. Apply the preceding whole member restriction dictionary to those SAME actual head records and true original lists. A mask pullback acts entrywise on the whole ordered boundary, takes the inserted empty entry to empty, and commutes with insertion at the internally computed index. Substitute the internally proved equal masks. The original ideal, mark and entire carrier are preserved as actual fields; all original and future entries retain their actual order. No lower model, mask or desired equality is a hypothesis. □

Theorem 364.4 (The computed index is within the actual retained original empty prefix)**.** *The actual retained-prefix insertion rank is at most the true number of original members missing the reference point and at most the actual old positive lower boundary length.*

Proof. The actual insertion position lies within the original prefix. Therefore its original boundary prefix is exactly the corresponding prefix of the true original member list. On that prefix the genuine retention predicate is precisely failure to contain the reference point. Its filtered list is a sublist of the actual ordered list of ALL original members missing the reference point; its length is at most that whole retained-original length. The true positive lower boundary contains exactly that whole retained-original list, restricted to its actual mask, followed by its actual future entries, so its length is at least the same retained-original length. The preceding original lower empty-prefix producer proves every entry of this whole original prefix is empty on the SAME mask. Thus the internally computed insertion lies within the actual empty prefix, the input needed for the already proved uniform future-only monomial order map. No upper bound, selected count or history certificate is supplied. □

365 Actual canonical initial-boundary compatibility at every recursive depth

Keep every original indexed member and entire original canonical history. The recursive chronological history maps differ from the uniform metadata insertion maps on arbitrary symbols. Their equality on ACTUAL canonical values is proved from the true lower canonical data and original empty prefixes. This is the ordered-boundary recursive route of [3, 5], adapted by constructing each lower input internally from the original history.

Lemma 365.1 (Recursive chronological maps on unchanged whole value types)**.** *On the unchanged original types define*

$$H_{0,j} = \mathrm{id}, \qquad S_{k,j}(\ell, v) = (\ell, \mathrm{map}(H_{k,j-\ell})(v)), \qquad H_{k+1,j} = \mathrm{generalShift}(j, S_{k,j}),$$

where subtraction is clamped at zero and the masked map fixes infinity. Every $H_{k,j}$ and $S_{k,j}$ is strict and therefore an order embedding. For every depth, $H_{k,0} = \mathrm{shiftFamily}(k, 0)$; also $H_{1,j} = \mathrm{shiftFamily}(1, j)$. No assertion on actual history values follows from these algebraic facts alone.

Proof. Induct on k, simultaneously for all natural indices. At zero the family map is the identity. For the Step One map, strict first-coordinate inequalities are retained. If both first coordinates equal ℓ, the same index $j - \ell$ occurs on both lower coordinates and strictness of $H_{k,j-\ell}$, extended by infinity, reflects the lower inequality. The established strict general map therefore makes $H_{k+1,j}$ strict. At index zero, $0 - \ell = 0$ for every level, so $S_{k,0}$ is the pair map of $H_{k,0}$. The induction identity at zero gives precisely the defining recursion of the original uniform zero-index family shift. At depth one, $S_{0,j}$ is the identity pair map for every level; the defining original first-depth shift has that same pair map. These are identities on the unchanged recursive types; the zero-index monomial map is not replaced by an identity. □

Theorem 365.2 (Actual original canonical Step One values have initial-boundary compatibility)**.** *For EVERY family depth k, original maximal-order record M in the actual bound-$(k+1)$ domain, its OLD entire Step One R_k-canonical history T, actual transformed cosupport point y, and true ORIGINAL index $j \le |M.E|$, internally prove*

$$S_{k,j}(\mathrm{StepOne}(R_k; M, T, y)) = \mathrm{pairShift}(\mathrm{shiftFamily}(k, 0))(\mathrm{StepOne}(R_k; M, T, y)).$$

No lower-domain, lower-canonicality, empty-prefix, branch or value certificate is a caller premise. The family depth induction is discharged internally.

Proof. The proof is simultaneous with the following entire-family theorem by natural-number depth induction. At lower depth zero the map is the identity. Assume the actual entire-family initial-boundary comparison at depth k. Original Step One canonicality internally gives barred testness, actual barred cosupport and genuine canonical lower data at every nonspecial cosupport point. A special point has infinite lower value, fixed by both maps. At a nonspecial point put $\ell = \mathrm{level}(M, T, y)$ and split at its actual own-level run, obtaining the true head K, suffix Q and actual point $p = Qy$. Current regime membership and the true suffix length imply that the head point has this SAME level ℓ; whole barred testness supplies actual head testness. Original canonical-tail data give the actual lower marked record L, its bound-k domain, OLD entire restricted R_k-canonical history, mask membership and genuine cosupport. The proved original initial-boundary prefix theorem internally supplies $d = j - \ell \le |L.E|$ and the actual first d entries empty, including the true chosen-contact branch at zero level. Apply the depth induction hypothesis to THIS actual lower record and history at index d. It identifies $H_{k,d}$ on its whole original value with the original uniform shift at d. The actual specific-family empty-prefix theorem makes that uniform shift equal the zero-index shift on THIS value. Expanding the true masked value and retaining first coordinate ℓ gives the displayed whole comparison. Every lower property used here is constructed from original canonical input, rather than supplied as a desired conclusion. □

Theorem 365.3 (Every actual original canonical family has recursive initial-boundary compatibility)**.** *For EVERY depth k, ONLY original bound-k record M, its OLD whole R_k-canonical history T, actual transformed cosupport point y and true ORIGINAL index $j \leq |M.E|$, internally prove*

$$H_{k,j}(R_k(M,T,y)) = \text{shiftFamily}(k,j)(R_k(M,T,y)).$$

Consequently a genuine first-j empty original boundary prefix gives $H_{k,j}(R_k(M,T,y)) = H_{k,0}(R_k(M,T,y))$. No caller comparison, lower canonicality, branch or arbitrary-invariant certificate is assumed.

Proof. At depth zero both maps are the identity. Given the entire depth-k result, the preceding argument proves Step One initial-boundary compatibility for its actual lower family. Now consider original depth $k+1$ and inspect its actual outer order. At order zero, both maps apply exactly the same monomial chronological insertion at index j, so they agree on the ENTIRE actual family leaf without a lower premise. At positive order original full canonical-tail data internally supply the actual assignment head, order mask, true companion J, bound-$(k+1)$ domain, maximal order, OLD whole Step One R_k-canonical restricted tail, mask membership and cosupport. The entire companion boundary retains $|M.E| + |K|$ entries, so the ORIGINAL bound $j \leq |M.E|$ gives $j \leq |J.E|$. Apply the Step One result just proved to THIS actual companion. Thus its true recursive inner map agrees with the original uniform zero-index pair map. The outer general map expansion is exactly the original uniform depth-$(k+1)$ shift at index j. This completes the genuine specific-family depth induction, uniformly over all original records, real restricted carriers and whole histories. Finally, if the first j original members are empty, the established actual specific-family empty-prefix theorem equates the uniform shifts at j and zero on the actual original value; the first lemma identifies that zero shift with $H_{k,0}$. The comparison is at a valid INITIAL boundary index; it supplies the actual compatibility needed for future lower records, whose initial lists include earlier chronological exceptional members. Arbitrary higher interior history transport and noncompact scheduling remain further obligations. □

366 Actual deletion of an interior identity empty step at every depth

Keep the original marked record and every original indexed member. Delete only the actual identity empty blowup between the original head and suffix; the entire original suffix and all its centres remain literal. The ordered-boundary route of [3, 5] is adapted by deriving original prefix canonicality during deletion, before using the established whole-value and dominance transport.

Theorem 366.1 (Whole original canonical histories reflect interior identity empty deletion)**.** *For EVERY original family depth, original bound-k record M, original head H and entire suffix S, the literal inserted history $H\epsilon S$ is canonical for the actual R_k if and only if HS is canonical. The same statement holds for the actual Step One R_k at original bound-$(k+1)$ maximal-order inputs. In the deletion direction ONLY the actual inserted whole canonical history is assumed; no deleted canonicality or value comparison certificate is supplied.*

Proof. Insertion follows the established original-input whole canonical-history producer. For deletion induct on the SAME suffix S. If it is empty, canonicality of the last identity empty step contains original head canonicality directly. At a next actual step, inserted whole canonicality supplies inserted prefix canonicality and the actual centre condition. Apply the induction hypothesis FIRST to construct deleted OLD prefix canonicality. That genuine prefix now satisfies the hypotheses of the established actual dominance dictionary. If the centre is empty it remains empty. Otherwise it equals the entire inserted prefix dominance locus; the proved dictionary identifies that locus with the entire deleted prefix dominance locus. Thus the literal OLD next step is canonical. This constructs deleted whole canonicality internally and proves both equivalences. No maximum existence, branch, level or lower-order condition is required. □

Theorem 366.2 (Actual inserted canonical input internally computes the deleted whole values)**.** *ONLY original domain data and actual INSERTED whole canonicality internally supply the deleted whole canonical history. At every actual inserted cosupport point, the entire inserted family value equals $H_{k,j}$ of the literal deleted whole value, and the inserted Step One value equals $S_{k,j}$ of the deleted value, where $j = |M.E| + |H|$. All true levels and orders and empty suffixes are included; no OLD canonicality, cosupport or value certificate is a caller premise.*

Proof. The first theorem constructs deleted canonicality from the actual inserted canonical history. The original invariant then supplies genuine testness of the deleted whole history and original head. The actual geometric dictionary gives equality of inserted and deleted transformed cosupport. An inserted cosupport point is therefore internally a genuine deleted cosupport point. Apply the established ENTIRE original value formula to this internally supplied OLD canonical history and point. Its SAME strict chronological map gives the stated complete value without selecting a new assignment history or rebasing the original marked record. □

Theorem 366.3 (Whole canonical terminal histories reflect the same actual deletion)**.** *At EVERY family and Step One depth, the literal inserted history is both canonical and a terminal geometric resolution if and only if the literal deleted history has these properties. No terminal or canonicality certificate for the deleted history is supplied in the deletion direction.*

Proof. For insertion use the established whole canonical terminal producer. For deletion the first theorem internally gives deleted canonicality. The genuine original invariant supplies actual testness of that deleted history and head. Original transformed cosupport equality then transfers inserted terminal emptiness to the deleted history. Together these give the true deleted geometric resolution and canonicality. This transports an actual given history; arbitrary empty biholomorphic maps, noncompact canonical scheduling and global resolution gluing remain distinct source obligations. □

367 Whole original recursive histories across an interior identity empty step

Retain the SAME original marked record, every indexed original and exceptional member, the original assignment heads and the entire original suffix. The chronological maps $H_{k,j}$ and $S_{k,j}$ are those already constructed on the unchanged original value types. The ordered-boundary route of [3, 5] is adapted by a simultaneous natural-depth construction of inserted canonical histories: inserted canonicality is proved before it is used by canonical tower-isomorphism value transport. No arbitrary-invariant or desired-value certificate is an original caller premise.

Theorem 367.1 (Canonical short Step One values use the true recursive chronological map)**.** *For EVERY depth k, original bound-$(k+1)$ maximal-order M, OLD whole Step One R_k-canonical HS, true cosupport point y and its actual own-level short run $r < |S|$, put $j = |M.E| + |H|$. Then the ENTIRE inserted Step One value is $S_{k,j}$ of its entire OLD value. This includes every level, chosen-contact zero, every positive inner order and the actual masked domains supplied by original canonical data. No lower-domain, lower-canonicality, lower-index or lower-value certificate is assumed.*

Proof. The established actual short-history formula gives the uniform indexed map on the whole OLD value. A special point has infinite lower coordinate, fixed by either map. Otherwise original Step One canonicality supplies barred whole testness, barred cosupport and genuine lower data at the actual own-level split $T = KQ$: actual mask membership, original lower record L, its bound-k domain, OLD whole restricted R_k-canonical tail and cosupport point. Put $\ell = \mathrm{level}(M, T, y)$ and $p = Qy$. Current nonspecial regime membership and $|Q| \leq r$ supply the SAME head level ℓ at p. The genuine lower-boundary count gives $|L.E| = |M.E| + |K| - \ell$. Since the run is short,

$|K| \geq |H|$, hence $d = j - \ell \leq |L.E|$. Apply the already internally discharged all-depth initial-boundary theorem to THIS canonical lower record at this valid initial index. It equates $H_{k,d}$ with the uniform shift at d on THIS actual whole lower value. The original first coordinate is exactly ℓ; expanding the true finite masked value makes the two whole Step One maps equal. Combine this with the actual insertion formula. No assertion that the first d entries of the lower record are empty is used: earlier chronological exceptional members remain present in that initial list. □

Theorem 367.2 (Original recursive canonical crossing data construct both entire lower histories)**.** *At EVERY depth, original canonical inputs in a genuine reaching nonspecial Step One branch internally construct BOTH whole OLD and inserted restricted lower R_k-canonical histories, their true cosupport points and entire values. Their actual lower insertion index is $|L.E|+|B| = |M.E|+|H| - \ell$. Likewise a reaching positive outer branch internally constructs BOTH whole OLD and inserted restricted Step One R_k-canonical companion histories; their true index is $|J.E| + |B| = |M.E| + |H|$. The lower records, canonicality, values and index identities are outputs.*

Proof. This construction is made inside the following natural-depth induction. For a Step One crossing at depth k, original canonical input supplies the actual lower domain, whole OLD restricted canonical history, whole OLD normalized restricted-append history and genuine cosupport. The actual restriction-append tower isomorphism retains the same real point and original record by coherent identity pullback. Apply the already proved preceding family-depth history producer to THAT normalized lower history. It constructs its inserted whole canonical history and its entire value first. The genuine restriction-empty-history isomorphism now transfers this NEW canonicality to the literal inserted restricted tail. Actual geometric cosupport transports the SAME point. Only after NEW canonicality has been constructed apply the specific-family canonical tower-isomorphism value comparison. Compose with the OLD normalized comparison and the preceding depth's actual value formula. Actual restricted lengths and the original lower-boundary count give the stated exact index, including the independently chosen original zero contact. For the outer crossing at depth $k+1$, original full family canonicality internally supplies the actual maximal-order companion and OLD whole Step One R_k-canonical restricted tail. Normalize through the real restriction-append isomorphism using the genuine maximal-order Step One invariant. Apply the Step One history producer just constructed at depth k to obtain NEW normalized Step One canonicality and the entire value. The genuine restriction-empty isomorphism then supplies inserted restricted canonicality and cosupport; canonical maximal-order value transport is applied only at this point. The companion's actual boundary retains $|M.E| + |K|$ entries, and the split lengths give its exact index. Empty suffixes are included throughout. □

Theorem 367.3 (Whole Step One recursive values and canonical histories survive interior insertion)**.** *For EVERY k, ONLY original bound-$(k+1)$ maximal-order M and OLD whole Step One R_k-canonical HS internally produce the inserted whole Step One canonical history. At EVERY actual OLD transformed cosupport point its ENTIRE value is $S_{k,j}$ of the OLD value, $j = |M.E|+|H|$. No branch, level, lower order, inserted canonicality or comparison certificate is a caller premise.*

Proof. Use the preceding family-depth result in the simultaneous induction. At a special point the actual level/special dictionaries preserve the same level and infinite lower value; the map fixes infinity. At a nonspecial short run the first theorem supplies the whole recursive value, including zero contact. At a reaching run the second theorem internally supplies BOTH whole restricted lower canonical histories and their entire value. The two true own-level split records have the SAME original assignment head; actual whole projection equality retains the SAME lower record and mask. Expand the actual masked value and use the internally exact lower index $j-\ell$ to obtain the whole $S_{k,j}$ formula. Thus all true points, without any common-level restriction, use ONE strict whole-value map. The actual geometric cosupport dictionary and its weak-order reflection identify the entire maximum-DOMINANCE loci; existence of a maximum is not required. Induct through the SAME original suffix S. At the empty suffix the inserted centre is empty, allowed

by the original canonical-history definition. At a next step the OLD canonical prefix gives an inserted canonical prefix by induction; an original empty centre remains empty, and an original full dominance centre remains the full inserted dominance centre by the proved dictionary. This constructs NEW whole Step One canonicality internally. □

Theorem 367.4 (Every original family depth has whole recursive interior history transport)**.** *For EVERY depth k, ONLY original bound-k record M and OLD whole R_k-canonical HS internally construct the inserted whole R_k-canonical history. At EVERY actual OLD transformed cosupport point its ENTIRE value is $H_{k,j}$ of the OLD value, $j = |M.E| + |H|$. All original lower records and histories remain literal. No inserted canonicality, lower comparison, maximum-existence or recursive-family certificate is assumed.*

Proof. Induct on the natural family depth, uniformly over ALL original records, real carriers and whole histories. At depth zero the actual entire value has its defining constant value and $H_{0,j}$ is the identity. Genuine cosupport equality and this whole value equality give the dominance dictionary; the same suffix induction constructs inserted canonicality. Given the entire family result at depth k, the preceding two constructions prove whole Step One R_k value and canonicality transport. Now consider family depth $k+1$. At actual outer order zero, the genuine all-tail monomial leaf formula applies, including an immediate empty suffix; both outer maps have the SAME monomial chronological insertion and do not use inner values. At a short positive-order run, the true whole splits select a nonempty prefix and the SAME actual tail. The established actual whole-family short insertion formula gives the uniform family shift. Original full canonical data applied to the true full split internally supply the actual maximal-order companion, its bound-$(k+1)$ domain, OLD whole Step One canonical restricted tail and cosupport. The original insertion index j is valid in that companion's initial boundary, because the selected prefix retains the whole H. The proved ALL-depth canonical Step One initial-boundary comparison identifies $S_{k,j}$ on THIS companion value with the uniform zero-index inner pair map. Expanding the actual masked companion value therefore identifies the uniform short family formula with $H_{k+1,j}$. At a reaching positive-order run the second theorem supplies BOTH actual whole companion canonical histories and the entire Step One value. The two actual full order splits retain the SAME original assignment head and mask. The internally exact companion index is j, so expanding the true positive-order family value gives precisely $H_{k+1,j}$ of the OLD value. These cases exhaust every original cosupport point. Strictness and genuine cosupport equality give the full dominance dictionary. The SAME suffix induction, retaining original empty centres and original full dominance centres, constructs NEW whole family canonicality. This completes the simultaneous natural-depth induction; its preceding-depth premise is discharged internally and is never an original caller assumption. □

Corollary 367.5 (Entire mixed spectra and terminal histories are preserved)**.** *The entire Step One and family spectra on actual original cosupport reflect strict order, weak order and equality under interior identity empty insertion, with no common branch, level or lower-order condition. Original terminal canonical resolutions produce inserted terminal canonical resolutions. Neither maximum existence nor noncompact scheduling or global gluing is claimed.*

Proof. The whole original value formulas use the SAME strict map for all points of one original history. Applying its order reflection and injectivity gives the three comparisons, including pairs in different branches or levels. Combine constructed inserted canonicality with the genuine geometric test transport and cosupport equality. An original empty terminal cosupport thus remains empty, supplying the actual inserted terminal resolution. All centres, assignment histories and original indexed members are retained. These results transport a GIVEN whole original canonical history; existence and gluing of arbitrary noncompact canonical schedules remain distinct obligations. □

368 Actual order-run companions and their SPECIAL leaves

All records and histories below are the original constructors. A candidate record on a genuine reference TEST need not itself have a TEST. Actual raw controlled geometry supplies the comparisons instead. Ambient dimensions are positive when used in original companion and invariant constructions.

Theorem 368.1 (Independent order masks have a genuine Cartesian companion diagram)**.** *Let M be a reference record and suppose the actual raw transforms of K on every genuine M TEST have DOMAINb. Take a genuine M TEST and an actual smooth Cartesian source history. For every head/tail index and every order, construct the actual maps between both independently computed order opens and their actual tail inverse images. The two masked tail histories are genuinely Cartesian, the endpoint map retains the literal original endpoint map, and ALL FOUR independently computed initial compAt fields are related by the genuine initial smooth pullback. No companion TEST is inferred.*

Proof. Simultaneously split the actual Cartesian tower at the given index, retaining every successor and empty centre. The genuine controlled and nonmonomial order dictionaries prove that the source order mask is the whole inverse image of the target mask. Apply actual open Cartesian restriction to the two true tail histories and these exact opens. Independently, actual smooth companion naturality and coherent pullback composition construct the whole compAt initial pullback identity on the independently defined masks. Both initial restricted maps have the same underlying split map at every point; subtype extensionality identifies them. Substitute this actual map equality in the whole initial record identity. The endpoint remains the actual restricted original endpoint map. No matching history is postulated. □

Theorem 368.2 (Raw masked companions have actual SPECIAL and level dictionaries)**.** *For the same genuine reference geometry and a positive order, take an actual source endpoint whose target tail projection is in the true target order open. Internally construct its source order-open membership. The two actual compAt records on the corresponding masked tails have equivalent original SPECIAL predicates and equal true indexed levels at these actual points. Only the stated genuine reference-domain family is assumed; no candidate TEST or desired SPECIAL, level or value certificate is an input.*

Proof. The genuine reference TEST splits into actual head and tail TESTs. Appending any genuine tail TEST to that head supplies the full candidate domain family at the head. Positive-companion and bar initial domains are actual original constructions. Their carrier and whole ordered boundary equal the current candidate's. The arbitrary-bound same-carrier theorem therefore constructs every raw bar-companion domain on the genuine reference tail TESTs.

Compare each masked companion separately with that SAME whole, unmasked positive companion. The target open inclusion gives a genuine Cartesian history. The source tail followed by its real source-open restriction gives a second genuine Cartesian history over the same target tail. Coherent pullback composition and the preceding whole compAt identity identify their initial pullback records with the independently computed target and source compAt constructors. Apply the genuine modeled SPECIAL and indexed-prefix level dictionaries, using the independently constructed entire bar family. The two equivalences meet at the same actual target endpoint; their composition gives SPECIAL in both directions and exact level equality. This does not use an admissible history for either companion. □

Theorem 368.3 (Only original DOMAIN data supply the actual positive-run dictionary)**.** *Take ONLY an original DOMAINb record, ANY genuine original TEST, an actual smooth Cartesian pullback and a source endpoint whose image lies in the actual target current cosupport with positive nonmonomial order. Compute the original attained order and the original fixed-order run. Construct both actual mask memberships and the SPECIAL/level dictionary for the two independently computed compAt records on those exact run tails.*

Proof. Genuine original TEST inheritance supplies the complete candidate-domain family internally, taking the candidate to be the unchanged original record. The literal original regime-start equation, applied to actual current cosupport membership and the attained order, gives genuine head cosupport membership and exact order there. The original order-mask dictionary then excludes order one higher and constructs the actual target mask membership. Apply the preceding theorem at precisely the computed original run and attained positive order. Its true mask-preimage equation constructs the source membership, and its genuine comparison supplies both SPECIAL and level assertions. No run, lower TEST or current lower-membership certificate has been supplied. □

Theorem 368.4 (Genuine curve run geometry gives source SPECIAL leaves in every relative dimension)**.** *For ONLY the preceding original inputs with target DOMAIN*1*, the actual source run compAt is SPECIAL with true level zero. Consequently its unchanged original sequence-aware StepOne value is* $(0, \infty)$ *at EVERY recursive depth. The actual source carrier may have any smooth relative dimension; it is not assumed to be a curve. No source or target companion TEST, desired SPECIAL or value certificate is supplied.*

Proof. Independently construct the actual target run bar-companion current domain, its current cosupport point, full ordered boundary avoidance and the whole order-open pullback dictionary from the genuine original curve model. Actual nonzero-along curve geometry makes the cosupport zero-dimensional inside its smooth one-dimensional carrier. This gives the original target SPECIAL predicate. Boundary avoidance and the literal prefix count make its level zero; every original position is retained. Apply the preceding original-only run dictionary to transport that actual SPECIAL predicate and level to the independently computed source compAt. Finally unfold the unchanged original sequence-aware StepOne definition in its proved SPECIAL branch. This is actual leaf geometry, not a source-curve dimension shortcut or general higher-dimensional noncanonical GENERAL naturality. □

369 Actual same-carrier candidate geometry in every carrier bound

Candidates below are not declared admissible histories. Their current coherence, nonzero germs and entire smooth carrier/boundary geometry are constructed from a genuine reference TEST. The carrier bound is arbitrary; in particular a smooth pullback of a curve is not called a curve.

Theorem 369.1 (Same actual carrier and ordered boundary supply every candidate domain)**.** *Let* M *be a reference record, let* K, B *have the SAME actual initial carrier and ordered boundary, and let* B *have genuine DOMAIN*b*. Suppose the actual* K *transforms on every genuine* M *TEST have DOMAIN*b*. Construct DOMAIN*b *of the actual raw controlled transforms of* B *on every such TEST. No* B *TEST or candidate admissibility is assumed, and the ideal and positive mark of* B *may differ from* K*.*

Proof. Induct on actual tower constructors. At a successor, read the real reference next carrier and ordered boundary from the independently known K domain. Actual strict carrier and ordered-boundary transform formulas depend only on their initial carrier and ordered boundary; thus they are exactly the candidate's next carrier and boundary. The genuine smooth-strict-carrier controlled transform theorem gives coherence of the candidate ideal without candidate admissibility. The genuine nonzero-along transform theorem gives its nonzero germs. Reference geometry supplies smoothness, SNC, transversality and the same actual dimension bound b; the candidate's positive mark is unchanged. This constructs every field of its real domain. The argument does not use $b = 1$. □

Theorem 369.2 (Original positive companions have every actual bar domain on any original TEST)**.** *For ONLY an original DOMAINb record M in positive ambient dimension n and any positive integer q, construct DOMAINb of the actual bar of its actual positive companion on EVERY genuine original M TEST. No companion or bar TEST is supplied or inferred.*

Proof. The actual original companion construction gives its initial domain; real homogenization/coefficient geometry gives the bar domain. The actual carrier and entire ordered boundary of the bar companion are exactly those of M. Genuine original TEST stability independently gives every current DOMAINb of M. Apply the preceding theorem with $K = M$ to obtain the actual bar-companion domains on all original tests. All geometry inputs are internal constructions from M, q. □

Theorem 369.3 (Independent raw bar companions have whole smooth current records)**.** *For ONLY such original data, ANY genuine original upper TEST and a real smooth Cartesian pullback, the independently computed source bar positive-companion raw controlled record is the ENTIRE pullback of the actual target bar-companion record at the current stage. Their whole cosupports have both preimage directions. The target candidate has real DOMAINb, and the source candidate has the actual shifted DOMAIN$b + (m - n)$. No candidate TEST or current-member certificate is an input.*

Proof. First construct all target candidate bar domains by the preceding theorem. These actual coherence and smooth-strict-carrier facts supply the genuine raw controlled smooth history dictionary. Actual smooth companion pullback and actual bar pullback, independently justified by the original coherent nonzero record, identify ALL FOUR initial fields with the independently computed source constructor. Substitute that initial identity into the whole controlled and cosupport preimage dictionaries. The target domain was already constructed; true smooth domain pullback and the proved whole current equality supply the actual shifted source domain. No candidate admissibility has been used to obtain these whole records. □

Theorem 369.4 (Actual companion SPECIAL and level are transported without a candidate TEST)**.** *For the SAME original inputs, the two independently constructed positive companions have equivalent actual SPECIAL predicates and equal true indexed levels at every actual source endpoint of the Cartesian history. No companion TEST, desired SPECIAL or level certificate is supplied.*

Proof. Actual companion initial domain supplies its well-formed record. The second theorem supplies every current bar well-formed record along the genuine original reference TEST family. Apply the genuine modeled SPECIAL and indexed-prefix level dictionaries to this reference and companion. Real initial smooth companion pullback identifies the source's independently computed companion before substitution in both dictionaries. They preserve both actual SPECIAL branches and every indexed ordered boundary position. This supplies the transport needed for small-to-large curve-model leaves; no arbitrary higher-dimensional noncanonical GENERAL naturality or source OWN LARGE value equality is asserted here. □

370 The actual selected positive lower tail after an interior empty step

The original Step One value uses the actual run of the current level to choose its assignment stage. The following producer computes its actual lower-history diagram when that run stays inside the final tail; it does not assume a desired comparison of assignment stages or lower values.

Lemma 370.1 (The genuine selected short positive run and its full splits)**.** *Let M be originally well formed, H an actual test tower of $\overline{M}$, S any actual tail and y its final point. Suppose the actual original level at (HS, y) equals $s + 1$. Put $r = \mathrm{regimeRun}(\mathrm{stepRegime}(M, s + 1), HS, y)$,*

and suppose $r < |S|$. The actual run at the actual current level of $H\epsilon S, y$ is also r. Put $D = (\mathrm{split}(S,r)).\mathrm{stage}$, $P = \mathrm{head}(S,r)$ and $Q = \mathrm{tail}(S,r)$. Then the full original splits selected by those literal runs are precisely (D, HP, Q) and $(D, H\epsilon P, Q)$, respectively. In particular they have the same actual stage and the same whole actual tail Q.

Proof. The original level dictionary supplies the same actual current level in both histories. The proved exact run formula gives equality to r because the old run is shorter than $|S|$. The split of a concatenation inside its final tail is obtained by recursively removing the same last r blowups from S; thus its full stage/head/tail record is (D, HP, Q). Apply that actual split formula also to the prefix $H\epsilon$, and substitute the internally computed new run r. This gives the two claimed literal full splits. No comparison of heads or tails was an input. □

Theorem 370.2 (The actual masked lower-record and restricted-tail diagram)**.** *Under the preceding hypotheses, write $p = Q.\mathrm{proj}(y)$. The actual lower masks at the two selected original heads agree. The actual masked lower marked ideal at $H\epsilon P, p$ is the true empty insertion at the internally computed filtered-prefix index c into the actual lower marked ideal at HP, p, at level $s+1$. This c is within the actual old lower boundary. The whole restricted actual tails $Q|_{W^+}$ and $Q|_{W^-}$ agree under the equality of the actual masks, with heterogeneous equality expressing only their retained dependent endpoint carrier types.*

Proof. Since $r < |S|$, the actual prefix P has length $|S| - r > 0$. Apply the actual positive lower-record producer to M, H, P, p, using this internally proved nonempty prefix. It gives the asserted whole lower record and its valid computed insertion index without a bound inside an empty original prefix. The original lower-mask dictionary gives equality of W^+ and W^-. Both restricted histories are restrictions of the same actual tail Q from the preceding full-split computation. Substitution of that actual open-set equality proves equality of their whole restricted tower records. No supplied lower record, mask, assignment-stage or restricted-tail equality is needed. □

This is the actual short-run positive lower-history producer. The case where the selected run crosses the interior empty step, the zero-contact carrier, the complete lower-value transport/order comparison, higher recursive values and noncompact maximum scheduling remain open. The index c can extend past an empty original prefix, so a uniform recursive value map is not inferred.

371 Actual recursively constructed lower values along original short histories

The invariant is the actual recursively constructed family $\mathrm{invFamily}(n,k)$. No property of an arbitrary lower invariant, desired value equality or boundary-insertion certificate is assumed. The initial marked ideal and entire actual original histories remain fixed. The proved original short positive, short zero-contact and crossing diagrams provide the selected records; here the original short diagrams are consumed by actual values. The crossing diagram has a different actual lower tail and still requires recursive transport through that history.

Theorem 371.1 (The whole actual zero lower value, in every recursive dimension)**.** *Let $n > 0$, $k \geq 0$, M originally well formed, H an actual test history of $\overline{M}$, P a nonempty actual tail and Q any subsequent actual history. Set $j = |M.E| + |H|$. The actual zero lower value of M formed at $H\epsilon P$ and computed along Q is*

$$\mathrm{WithTop}.\,\mathrm{map}(\mathrm{shiftFamily}(k,j))\big(\mathrm{lowerValue}_k(M, 0, HP, Q, y)\big).$$

Both lower values use the unchanged original family and the unchanged original lowerMarkedM definition, including the actual missing-mask top value.

Proof. The actual whole zero lower record theorem supplies its future empty insertion at j, the whole mask equality and the actual lower-boundary index bound. Identify the equal actual opens, keeping their genuine dependent carrier types. If the projection of y misses the old mask, it misses the new equal mask: both values are top, which the map preserves. Otherwise the real lower value is the actual invariant on the real restricted Q at the corresponding actual subtype point. The inserted lower record and the proved actual all-k invariant insertion theorem give precisely $\mathrm{shiftFamily}(k, j)$ of the old value on that same whole restricted history. Coercing into WithTop gives the stated formula. The needed insertion theorem is proved for this specific recursive family and supplied internally; it is not a hypothesis about a proposed invariant. □

Theorem 371.2 (The positive lower value at its actual index)**.** *With the same original inputs, at every positive lower index $s + 1$ put*

$$p = Q.\,\mathrm{proj}(y), \qquad c = \mathrm{interiorPositiveLowerIndex}(M, H, P, p).$$

The actual lower value formed at $H\epsilon P$ is WithTop.map$(\mathrm{shiftFamily}(k, c))$ *of the actual lower value formed at HP, both computed along the same literal Q.*

Proof. The true member filter computes c in the actual whole lower boundary, and the actual positive lower producer supplies the whole record insertion there and equality of the actual masks. The previous argument now applies with this actual c: the missing-mask branch is top on both sides, while the genuine masked branch invokes the proved specific all-k insertion theorem on the actual lower record and the entire actual restriction of Q. No empty-prefix hypothesis is needed. The index may lie beyond the empty original prefix, so no uniform replacement of c by zero is asserted. □

Theorem 371.3 (The literal own-level short zero Step One value)**.** *Let S be an actual tail and y an actual point whose original current level is zero and whose actual zero-level run r satisfies $r < |S|$. For every k, the actual Step One value over invFamily(n, k) along $H\epsilon S$ equals*

$$\mathrm{pairShift}(\mathrm{shiftFamily}(k, j))\big(\mathrm{stepOneInvM}_k(M, HS, y)\big), \qquad j = |M.E| + |H|.$$

No current nonspecialness or canonical-lower-value certificate is required.

Proof. The actual level and special-point dictionaries identify the corresponding original branches. At a special point both actual values are $(0, \top)$, and the map preserves this value. Otherwise the actual short split theorem computes the new current-level run as r and supplies both complete split records. Their actual heads are HP and $H\epsilon P$, with the same literal Q, where $P = \mathrm{head}(S, r)$ and $Q = \mathrm{tail}(S, r)$. The strict run inequality gives $|P| = |S| - r > 0$ internally. Apply the first theorem to these actual inputs; the lower value is changed by exactly WithTop.map$(\mathrm{shiftFamily}(k, j))$. The unchanged outer level zero gives the asserted whole lexicographic pair formula. □

Theorem 371.4 (The literal own-level short positive Step One value)**.** *If instead the actual current level is $s + 1$ and its actual run r is shorter than $|S|$, put $P = \mathrm{head}(S, r)$, $Q = \mathrm{tail}(S, r)$, $p = Q.\,\mathrm{proj}(y)$ and let c be the actual filtered lower insertion index. The actual new Step One value is pairShift*$(\mathrm{shiftFamily}(k, c))$ *of the actual old value.*

Proof. Both actual levels and special branches agree. At special points the lower entry is top and is preserved. At nonspecial points the proved positive short splits supply the identical actual run, selected stage and tail Q. The computed selected prefix has positive length by the strict run inequality. The second theorem gives the actual lower-value map at its genuine computed index c, and retaining the outer actual level $s + 1$ gives the pair formula. No comparison of different computed indices or both-zero monomial spectra is inferred from this pointwise equation. □

Theorem 371.5 (Order and dominance on the entire actual short-zero spectrum)**.** *For a fixed original M, H, S and every k, let Z be any set of actual points, each with actual level zero and actual zero-level run shorter than $|S|$. The entire old and new actual Step One spectra on Z have the same strict order, weak order and equality comparisons. Their maximum-dominance loci on Z are equal.*

Proof. The third theorem maps every actual value on Z by the same map

$$\text{pairShift}(\text{shiftFamily}(k, j)), \qquad j = |M.E| + |H|.$$

This index does not depend on the selected point or run. The proved shiftFamily is strictly increasing, as is its WithTop extension and therefore the whole lexicographic pair map. Strictly increasing maps on linear orders reflect and preserve strict order, weak order and equality. Apply these facts to any two actual points of Z. Membership in the maximum-dominance locus is membership in Z together with domination of every actual value on Z; the weak-order equivalence proves equality of those predicates. This does not assert maximum existence and makes no noncompact scheduling claim. □

The crossing original diagrams remain real inputs for a subsequent recursive history transport theorem. This chapter closes the actual short-branch whole lower-value producer at every k and its uniform zero-spectrum order comparison. It does not identify complete outer general invariants across arbitrary unrebased middle deletions or close G3.

372 The exact original short-history index and whole Step One spectrum

A removed original member is an actual boundary index, not merely a set. The existing actual SNC germ-index uniqueness prevents two boundary entries through the same point from being the same set. This supplies the missing count needed to compare original positive-level short histories by a single map, rather than assuming equality of their point-dependent filtered indices. This indexed deletion argument is a local adaptation of Włodarczyk's Step One and Bierstone–Milman's ordered-boundary trivial-blowup convention [3, 5].

Lemma 372.1 (The actual filtered original boundary prefix has its exact count)**.** *Let E be an actual SNC boundary and let $m \le j \le |E|$. Select the members of its original prefix $E[0, m)$ through x, as indices in all of E. The prefix $E[0, j)$ with these selected indices removed has length $j - \text{boundaryCount}(E[0, m), x)$.*

Proof. Every selected index is below m. If an index at or above m were selected, its through-point member would equal some actual member in the prefix below m; the genuine SNC germ-index uniqueness would identify these two indices, a contradiction. For indices below m, selection is exactly membership of x in that indexed boundary entry, since the entry belongs to the original prefix. Therefore the selected indices inside the larger prefix are in bijection with the through-point indices of the original prefix. Their number is its actual boundary count. The j prefix indices partition into the selected and retained ones, yielding the stated natural-number subtraction. The proof counts every original index and uses no supplied count certificate. □

Theorem 372.2 (The true original positive lower insertion index)**.** *Let M be originally well formed and let HP be an actual test history of $\overline{M}$. At any actual stage point x, the true positive lower index for an empty insertion after H is*

$$c = |M.E| + |H| - \text{level}(M, HP, x).$$

There is no supplied filtered-index or count equality premise.

Proof. Actual bar-testness makes the whole original transformed bar datum well formed, so its entire boundary is genuinely SNC. The true transform dictionary gives its length $|M.E|+|H|+|P|$. Its actual original member list is literally the prefix of length $|M.E|$. Its level at x is the boundary count of precisely that original prefix. The actual lower insertion index is the retained length of the whole prefix of length $|M.E| + |H|$. Both required prefix bounds hold internally from these actual lengths. Apply the preceding exact filtered-prefix count to get the asserted original formula. □

Definition 372.3 (The actual level-indexed history map)**.** For original lower dimension k and original insertion position j, define on the entire Step One value order

$$F_{k,j}(\ell, v) = (\ell, \mathrm{WithTop.\,map}(\mathrm{shiftFamily}(k, j-\ell))(v)).$$

This is a strict order embedding, including infinite lower values.

Proof. For distinct first coordinates the strict comparison is preserved because that coordinate is unchanged. At a common first coordinate ℓ, both lower entries use the SAME genuine strict family map with index $j-\ell$. Its extension to WithTop is strictly increasing, including infinity. The lexicographic comparison is therefore preserved and reflected, giving the strict order embedding. No inequality between different family insertion indices is assumed. □

Theorem 372.4 (The entire original short-history Step One spectrum)**.** *Let M be originally well formed and let HS be an actual test history of $\overline{M}$. Put $j = |M.E| + |H|$. At every actual point y whose true own-level Step One run on HS is shorter than $|S|$, the ENTIRE actual specific-family Step One value on $H\epsilon S$ is $F_{k,j}$ of its entire value on HS, for every lower dimension k. Consequently strict order, weak order and equality of actual values at any two such original points are reflected and preserved, without a common-level or nonzero-lower-order restriction.*

Proof. Original whole bar-testness internally supplies the head bar-test dictionary. If y is special, the true special/level dictionaries preserve its whole pair $(\ell, \top)$, and $F_{k,j}$ fixes that pair. If it is nonspecial at level zero, the genuine original short-zero formula supplies the entire pair map with family index j, which is $j - 0$ as required. Otherwise write its true level as $s + 1$. The genuine positive-short formula computes the lower insertion index at the whole selected assignment head $H\,\mathrm{head}(S, r)$ and the actual image point, where $r < |S|$ is its true own-level run. This head is an actual prefix of the original bar-test history. True run-reaching membership on the selected suffix identifies its assignment level with the original current level $s + 1$. The preceding exact original index formula therefore gives $c = j - (s + 1)$. Substitute this internally computed equality in the complete positive-short value formula to obtain $F_{k,j}$ of the whole original value. The single strict order embedding just proved now compares any two true short-run points; unequal levels are handled by its unchanged first coordinate, and equal levels by its common internally computed index. No lower-value equality, variable-index comparison or canonical new-history certificate is assumed. Dominance predicates on this actual short-run subset are consequently preserved, but no maximum existence or global scheduling is asserted. □

Crossing runs and their positive recursive companions, full outer transport, maximum existence and noncompact scheduling remain open. These local actual SNC/count producers do not assert global distinctness of all boundary entries.

373 Actual source choice isomorphisms give genuine Cartesian histories

The next source-own contact comparison needs actual smooth Cartesian maps, not only a commuting square. The geometric inputs here are real stagewise tower isomorphisms with their actual centre laws. No lower-value comparison or lower-canonicality certificate is assumed.

Theorem 373.1 (Every actual stagewise tower isomorphism is the genuine smooth pullback)**.** *For a genuine stagewise isomorphism of finite original towers over a true base biholomorphism, construct the actual smooth Cartesian tower history whose base and endpoint maps are exactly the supplied biholomorphisms. Every actual successor, including every empty centre, is retained.*

Proof. Induct on the actual isomorphism constructors. The reflexive case is the genuine smooth submersion given by the base biholomorphism. At a successor, the actual earlier stage biholomorphism is a submersion and its inverse image of the actual target centre is exactly the actual source centre by the isomorphism's centre law. Both given arrows are genuine blowups. The actual next biholomorphism is holomorphic and satisfies the genuine blowup square. Actual blowup-lift uniqueness identifies it with the canonical smooth lift, proving the actual Cartesian successor. This constructs the full actual history, rather than inferring Cartesianity from an unsupported commuting square. □

Theorem 373.2 (A true source choice composes with the actual smooth history)**.** *Let an actual smooth Cartesian source history over a target tower and a genuine stagewise isomorphism into that source history be given. Construct the actual Cartesian history for the source-choice tower; its base and endpoint maps are the literal composites of the original maps with the actual choice biholomorphisms. No Cartesianity of this new history is supplied.*

Proof. The preceding theorem constructs the true actual Cartesian history for the source-choice isomorphism. Actual Cartesian composition constructs the composite history on the same finite towers and actual blowups. Its constructor-level proof retains each centre as the true composite preimage and uses actual blowup-lift uniqueness at every successor; therefore the maps are the literal composites, including every empty position. No canonicality or marked-ideal assumption is needed for this geometric construction. □

Theorem 373.3 (Actual source choice models have entire curve-reference smooth GENERAL values)**.** *Assume both ambient dimensions positive. For ONLY a genuine DOMAIN1 reference record and ANY genuine reference TEST, an actual smooth Cartesian history and any genuine source-choice tower isomorphism, the literal original reference pullback through the composite base map has its WHOLE original GENERAL value at EVERY positive depth equal to the genuine dimension map of the target value. The point is any actual source endpoint whose composite image is in the reference current cosupport. No source-curve hypothesis, source value, lower canonicality or desired Cartesian history is supplied.*

Proof. First construct the actual composite Cartesian history by the preceding theorem; its maps are the literal composites of the real geometric inputs. The original reference DOMAIN1 and genuine TEST internally give DOMAIN1 at every actual reference TEST prefix, supplying the genuine model family needed for the full curve-reference smooth-value producer. Apply that producer to the reference itself as model, on the actual composite Cartesian history and its given true target current point. It independently constructs all source codimension-one geometry and recursive companion values, with no source-curve or lower TEST premise. The result is the unchanged whole original GENERAL comparison for the literal composite pullback. Transport to a separately constructed source-own marked record must still use its genuine whole initial record identity. This node does not assert equality of a SMALL source model with its source OWN LARGE presentation or complete the latter's open-locality obligation. □

374 Whole original Step One histories over the complete curve family

Original canonical lower histories now carry a complete original curve-family interior empty-step producer. It constructs the inserted lower canonicity before any canonical isomorphism value

interface is used. This gives the first full positive-inner crossing consumer of the indexed Step One route in [3, 5].

Theorem 374.1 (Actual entire restricted curve lower history from original crossing data)**.** *Assume ONLY original dimension-two maximal-order M, its actual whole canonical Step One history HS over original R_1, and an actual nonspecial cosupport point whose own-level run reaches the insertion after H. Internally obtain the actual selected lower marked datum J, real mask W, true split K, BS, lower dimension-one domain, both actual whole old/new restricted canonical lower histories and cosupport points. At EVERY actual lower nonmonomial order,*

$$R_1(J,(B\epsilon S)|_W,y') = \mathrm{shiftFamily}(1,|J.E|+|B|)(R_1(J,(BS)|_W,y)).$$

Empty suffixes are included. No inserted lower CAN, equality, lower domain or lower value certificate is supplied by the caller.

Proof. The genuine original canonical normalized-lower producer constructs actual J, W, its dimension-one domain, OLD entire restricted R_1 canonicity and cosupport, and OLD normalized history $B'S'$ with its canonicity, cosupport and complete value, where $B' = B|_W$ and $S' = S|_{B^{-1}W}$. Apply the COMPLETE original curve-history producer to this internally supplied normalized history. It constructs actual NEW normalized R_1 canonicity and the entire new normalized value at the same actual underlying point, under index $|J.E|+|B'| = |J.E|+|B|$. Genuine identity pullback of the actual coherent J is literally J. The actual restriction/interior-empty tower isomorphism then transfers this newly PROVED normalized canonicity to the WHOLE actual new restricted history. Its geometric whole cosupport equality gives the true new cosupport point. Only NOW apply the original specific-family tower isomorphism value interface, using that internally proved new normalized canonical premise. The literal endpoint point identity and the old normalized value identity give exactly the stated ENTIRE new/old lower formula. No canonical interface is applied to an unproved inserted history. □

Theorem 374.2 (Entire actual positive-inner crossing Step One value)**.** *For the preceding original inputs, with $j = |M.E| + |H|$, the ENTIRE original Step One value over R_1 on $H\epsilon S$ is $F_{1,j}$ of its value on HS. Both zero-contact and positive-member levels, EVERY actual lower order and the empty suffix are included. No monomial-inner restriction remains.*

Proof. Both genuine original own-level reaching-run splits have the SAME assignment head K and tails BS and $B\epsilon S$. Actual level/specialness dictionaries keep the level and nonspecial branch. The new tail projection is literally the old one, so the actual original lower mask and entire marked record at K are the SAME on both sides, including the chosen tangent function at level zero. The preceding producer supplies the entire lower formula on the two genuine restricted histories, with actual lower domain/canonicity/ cosupport and mask membership internally obtained. The exact original crossing index producer computes its true $|J.E| + |B|$ as $j - \ell$. Expand both entire Step One values along their true splits and memberships. Their first coordinate is the actual level ℓ, and the entire finite lower value receives precisely $\mathrm{shiftFamily}(1, j - \ell)$. This is the whole map $F_{1,j}$. No independently chosen smooth-source contact function is compared here; these histories have the SAME original point and tangent choice. □

Theorem 374.3 (Complete whole Step One curve-history spectrum)**.** *An original dimension-two maximal-order datum and its actual original Step One R_1 canonical history HS internally give the complete $F_{1,|M.E|+|H|}$ value formula at EVERY actual cosupport point, and preserve and reflect strict order, weak order and equality of its ENTIRE spectrum. There is no short/reaching, special/nonspecial, level or lower-order premise.*

Proof. Original canonicity gives actual WHOLE bar-testness internally. At a special point both actual values have the same level and infinite lower component, which the whole map fixes. Otherwise inspect the TRUE own-level run. A short run is covered by the complete actual short

Step One theorem at every lower order. A reaching run is covered by the preceding COMPLETE positive-inner crossing producer, with its original lower data internally supplied. Thus every real point has the same complete formula under one strict map. Apply its genuine strict order embedding to two actual cosupport values for all three comparisons, including mixed short/reaching points and unequal levels. No separate branch-comparison certificate is needed. □

Theorem 374.4 (Actual complete Step One curve canonical history)**.** *Every original dimension-two maximal-order actual Step One R_1 canonical history remains canonical after the literal interior identity empty step. The actual entire original suffix and all its centres are retained, including an empty suffix. Terminality transfers when the original history is terminal.*

Proof. Induct on the SAME actual suffix. At the empty suffix, append the literal empty canonical step to the actual original canonical head. At a next step, use the induction hypothesis at its true prefix. Empty centres remain empty. The other actual old centre is its true whole maximum-dominance locus; genuine cosupport equality and the preceding complete value/order producer identify it with the new whole maximum-dominance locus under the SAME strict map. Thus this SAME actual centre is canonical for the new prefix. Original geometric testness transfers and empty final cosupport transfers, giving terminality when present. No global maximum existence, compactness, higher outer recursion or full noncompact scheduling is inferred. □

375 The entire original Step-One histories under an inserted empty original member

An inserted empty original boundary entry leaves the actual coefficient-homogenized ideal unchanged. It shifts the actual original indexed boundary list, including every tower stage, but contributes neither to the original boundary count nor to the real member special locus. Consequently all actual Step-One regime histories and both actual lower masks agree. The literal zero-level contact-lower marked records agree after identifying those SAME computed masks. The remaining positive selected-index comparison and full recursive invariant-order embedding are separate producers.

Lemma 375.1 (The actual entire coefficient-homogenized transform and original members keep the same insertion)**.** *For every actual marked record, actual insertion position within its original boundary, and genuine blowup tower, the whole coefficient-homogenized transform equals the SAME insertion into the old actual transformed record. The actual original list equals the SAME insertion into the old original transformed list.*

Proof. The original homogenization and coefficient ideal use only the actual carrier, ideal and marking, which insertion does not change; their boundary is retained literally. Thus coefficient-homogenization commutes with the SAME insertion field by field. Whole geometric transformation commutes with insertion at every actual stage, since strict transformation sends the empty entry to an empty entry and each exceptional member is appended after the original list. The actual original list is the ordered strict-transform map of the base original list. That map commutes with insertion and sends the empty set to the empty set, giving the stated literal list identity. □

Lemma 375.2 (The actual original level and entire special locus agree)**.** *At every actual tower stage and every point, original levels agree and original Step-One specialness is equivalent for the inserted and old original datum.*

Proof. The true indexed set of original members through a point is carried by the insertion's increasing index map, because the new entry contains no point. This map is injective, so its finite set has the SAME cardinality; the real original boundary count agrees. The whole actual transformed carrier, ideal and mark are unchanged by the preceding entire transform dictionary, so the original codimension-one special locus agrees. In the positive-level branch, the genuine

member special locus tests whether the carrier intersection of the original members through the point is contained in the actual cosupport near that point. The newly inserted empty member contains no point, hence imposes no condition in that intersection at any nearby point. The entire eventual implication defining this real special locus is therefore equivalent. Both actual branches of original specialness have identical levels and predicates. □

Lemma 375.3 (Both entire original lower masks and the true tangent choice agree)**.** *At every actual head/reference point, both actual original positive and zero-level lower masks agree. The actual original tangent-choice record also agrees, because all its genuine input fields are unchanged.*

Proof. The true tangent choice uses only the original carrier, ideal, marking and point, which insertion keeps literally. The positive mask requires that an original member containing a candidate point also contain the reference point, and excludes the actual level-specific special points. The new empty entry satisfies this implication vacuously; the preceding level and special dictionaries identify the other predicate. The whole positive interiors agree. The zero mask uses the SAME tangent choice, requires avoidance of every actual original member, and excludes the codimension-one locus of the true whole transformed coefficient-homogenized ideal. Avoidance of the inserted empty member is automatic, and every other input was identified above. Thus the whole zero interiors agree. Neither a chosen mask nor a tangent compatibility certificate is supplied. □

Theorem 375.4 (The entire literal original zero-level lower record agrees)**.** *For every original actual head and reference point, the literal old zero-level lowerMarked record of the inserted original datum is heterogeneously equal to the literal old zero-level lowerMarked record of the original datum, after identifying their internally computed equal masks. This includes its genuine ideal, carrier, mark and entire ordered retained exceptional boundary.*

Proof. The original zero-level contact construction uses the original carrier, ideal, marking, true tangent choice, actual tower and the actual zero mask. Its coefficient-homogenized base has no original boundary: the old contact construction omitted the entire original prefix before transforming. All these genuine inputs agree by the preceding dictionaries. Substitute the actual mask equality; every field of the SAME contact marked record is then literal equality, including every created exceptional entry. No ideal is replaced by its support, no original value definition is changed, and no recursively supplied lower record is assumed. □

Theorem 375.5 (The complete actual Step-One regime run agrees)**.** *For every actual original level, tower and endpoint, Step-One regimes are equivalent and their whole internally computed run lengths agree under the inserted empty original member. Therefore the actual tower splits at the SAME integer.*

Proof. The actual Step-One regime is equality of the true original level with the given level and failure of original specialness. The preceding whole dictionaries identify both conjuncts at every actual stage. Induct on the genuine tower to compare the actual runs. At the base each is zero. At every genuine final blowup the run tests the preceding endpoint's true regime, either incrementing the preceding run or resetting to zero. The branch tests agree by the regime equivalence; in the incrementing case the induction identifies the preceding runs, and in the reset case both are zero. This proves the whole point-dependent run and its SAME actual split, without a caller history/run/split/alignment certificate. It does not yet compare positive recursively computed lower invariant values. □

Theorem 375.6 (The actual masked zero lower record keeps the SAME inserted empty index)**.** *At every actual original head and reference point, the whole zero-level masked lowerMarkedB record used by the constructed recursive Step-One invariant for the inserted original datum is heterogeneously equal, after identifying the internally computed equal masks, to insertion at the SAME original index into the whole old actual lowerMarkedB record. The old contact-lower record*

omits originals; the masked record retains their empty traces and every created exceptional. These distinct literal definitions are kept.

Proof. The actual masked lower record is the genuine mask pullback of the coefficient ideal of the actual coefficient-homogenized head transform, with its carrier replaced by the true strict transform of the original chosen contact hypersurface. The preceding whole transform identity gives the SAME empty insertion in the entire head marked record. Coefficient ideal formation depends only on carrier, ideal and marking and retains the entire boundary, so it commutes with that SAME insertion. The original carrier and true tangent choice agree, giving equality of the actual chosen strict contact carrier. Replacing the carrier field therefore still commutes with insertion. The actual masks agree by the preceding mask producer. A mask pullback takes the indexed boundary entrywise to its full preimage; preimage of the inserted empty entry is empty, and the ordered map commutes with insertion at the SAME index. Substitution of the actual mask equality gives heterogeneous equality of the whole literal masked record, including its genuine ideal, mark and full indexed boundary. No ideal equivalence, caller lower record or choice-independence assertion is assumed. □

376 Original-input local terminal atlases and actual topological gluing

The local resolution and canonical overlap route of [3, 5] is applied from ONLY original marked ambient-domain input. Local histories are constructed by the genuine existing per-sequence LOCAL driver on actual compact singletons, rather than supplied as a desired conclusion. The result below is actual topological gluing. A complex atlas, proper resolution geometry and a noncompact canonical scheduled sequence require subsequent genuine producers and are not certificates assumed here.

Theorem 376.1 (Original marked-domain input internally constructs a covering local canonical terminal atlas)**.** *On ANY original complex manifold, ONLY the original marked ambient-domain input internally constructs a family indexed by ORIGINAL base points. At each point it supplies an actual open neighbourhood containing that point, an actual finite tower over that open, its actual final complex manifold, WHOLE canonicality for the actual ambient invariant and actual terminality for the original restricted marked record. These actual opens cover the original base. No caller local history, compact neighbourhood, local canonicality or scheduled sequence is an input.*

Proof. The actual ambient invariant is the established per-sequence invariant with its true well-ordered value type. For each ORIGINAL base point, its actual singleton is compact. Apply the genuine local canonical resolution driver to this original marked-domain input and THIS compact singleton. It internally produces an actual open neighbourhood of the singleton and a genuine finite canonical terminal history of the actual original restricted record. The singleton inclusion gives point membership of that actual neighbourhood. Select these internally proved witnesses for each original point, retaining all their actual tower and canonical terminal fields. The resulting family covers the base because each original point belongs to its OWN actual chosen neighbourhood. Positive dimensional neighbourhoods need not be compact; only the original singleton compactness was used. No global maximum or noncompact growing history was presumed. Dimension zero and empty original bases are included. □

Definition 376.2 (Internally supplied original local histories give actual topological gluing data)**.** The actual final spaces of the INTERNALLY constructed atlas, their literal final projection preimages of pairwise original-open intersections and the genuine internally constructed overlap maps form ACTUAL topological gluing data. Every self-domain is the entire actual final space, every self-transition is identity, every triple-domain condition and every pointwise cocycle is a proved original consequence. No glue-data, transition or cocycle witness is a caller input.

Proof. At two original indices use the literal preimage of their actual original open intersection under the actual first local tower projection. It is open by the true projection continuity. The true independent-local comparison producer applies to the internally supplied marked restrictions and genuine canonical terminal histories, giving an ACTUAL biholomorphism of these literal overlap domains. Its genuine continuous forward map is the topological transition. The self-domain is whole because every actual local projection lands in its original local base open. Actual comparison self-uniqueness gives identity there. For a point belonging to two such actual pairwise domains, the proved original pairwise triple-domain producer supplies the true next membership and the ENTIRE pointwise transition cocycle. Thus every field of the standard pointwise topological gluing constructor is internally supplied by actual geometry. Apply the real constructor, retaining the original actual spaces and transitions. No common global finite tower or equal uncompressed histories is assumed. □

Theorem 376.3 (Original inputs supply an actual glued topological space with genuine open chart inclusions)**.** *ONLY original marked-domain input internally determines an ACTUAL glued topological space of these original local terminal spaces. Their actual gluing inclusions are open embeddings and jointly cover this entire space. No glued-space, embedding or cover certificate is supplied.*

Proof. Take the actual topological multicoequalizer of the preceding INTERNALLY constructed true overlap gluing data. Its actual local inclusions are the multicoequalizer structure maps. The genuine topological gluing theorem for open overlap data makes every such inclusion an open embedding and proves these inclusions jointly surjective: every actual quotient point is represented in one of the original local final spaces. All requisite open-domain and cocycle facts were proved before this construction. This is an actual topological space and actual open gluing. It does not yet install a complex manifold atlas or assert Hausdorffness, second countability, proper resolution geometry or an existing noncompact canonical schedule without their subsequent proofs. □

Theorem 376.4 (The actual original local whole projections descend to one continuous whole base map)**.** *The ENTIRE actual local tower projections, followed by their genuine original-open inclusions, internally descend to one continuous map from this actual glued topological space to the ORIGINAL base. Its composite with EVERY actual local inclusion is exactly that original local whole projection, at EVERY original local final-space point. No desired global projection, compatible family or descent certificate is a caller input.*

Proof. Each actual local projection is holomorphic, hence continuous, and its true original-open inclusion is continuous. Their ENTIRE composites are therefore actual continuous maps to the SAME original base. The internally constructed actual pairwise comparison's WHOLE original-base formula proves these maps agree on each REAL overlap transition. Thus they give an actual compatible cocone for the real topological gluing diagram. The actual multicoequalizer universal property constructs its continuous descent. The true structure-map/descent identities retain EVERY original local whole projection formula. Any second continuous map with those same whole formulas agrees on every actual local inclusion; the actual inclusions jointly cover the glued space, so it is equal. All family/projection compatibility was constructed from original inputs. Complex holomorphic structure, actual proper resolution and noncompact scheduled-history existence remain subsequent source obligations. □

Theorem 376.5 (Each actual local inclusion range is the entire original-open projection preimage)**.** *For EVERY original index, the range of its ACTUAL local final-space inclusion in the glued space is exactly the ENTIRE preimage of its original chosen base open under the actual glued whole projection. Only original marked-domain input is used. No range equality, final projection surjectivity, Hausdorff or countable-cover certificate is assumed.*

Proof. An actual local included point has glued projection equal to the actual local whole projection followed by its original-open inclusion. That projection genuinely lands in that chosen base open, giving one inclusion. Conversely, represent an arbitrary actual glued point over the chosen open by a genuine local final point at some index, using actual joint surjectivity. Its whole original-base projection lies in the chosen open by the proved global projection formula. It already lies in its own original base open. Thus it belongs to the REAL overlap domain between these two local histories. Their internally constructed true transition supplies a real point of the desired local final space. The genuine actual gluing condition identifies its inclusion with the original local representative, hence with the arbitrary glued point. This gives the entire reverse inclusion from original actual overlap geometry. The proof does not choose a point above an arbitrary base point or presume surjectivity of the glued projection. It supplies the true range dictionary needed by later Hausdorff, countable-cover and complex-atlas producers. □

377 Actual terminal choice comparison after internally omitting empty steps

Keep the ORIGINAL marked record and every original indexed boundary member. Two actual finite canonical histories may use different genuine blowup choices and different numbers of actual empty steps. The route of [3, 5] is adapted by internally compressing their actual empty steps BEFORE the genuine canonical terminal-history comparison. The original histories need not have equal recorded length and are not asserted to be isomorphic as uncompressed recorded towers.

Lemma 377.1 (Actual nonempty canonical histories satisfy the true canonical segment condition)**.** *A whole canonical history whose EVERY actual recorded centre is nonempty is a genuine canonical segment with no empty steps. This supplies the entire segment condition from true whole canonicality and the existing every-centre predicate, rather than a chosen list of active centres.*

Proof. Induct on the actual tower. The empty tower satisfies both whole conditions. At a next genuine blowup, whole canonicality supplies canonicality of its actual prefix and the alternative that its actual centre is empty or the entire original maximum locus. The actual every-centre condition supplies prefix nonemptiness and actual centre nonemptiness. The empty alternative is therefore impossible. The induction gives the prefix canonical segment, and this genuine nonempty maximum centre gives the entire next canonical segment. No existence of a maximum is inferred beyond the given true canonical centre. □

Theorem 377.2 (Original actual terminal histories compare after internal empty compression)**.** *At EVERY actual specific family depth, ONLY original bound-k marked input and TWO GIVEN entire actual canonical terminal histories internally supply an actual biholomorphism between their ORIGINAL final manifolds commuting with their WHOLE original projections to the SAME original base. Their recorded lengths and actual empty-step choices may differ. No nonempty-centre, compressed history, canonical segment or final comparison certificate is supplied by a caller, and no compactness is required.*

Proof. Use the following genuine original base-biholomorphic comparison with the identity of the SAME original base. Original domain coherence makes its actual identity pullback literally the original marked record. The TWO GIVEN original canonical terminal histories therefore supply its actual original and pulled-back inputs. That producer internally constructs both nonempty canonical terminal histories before whole canonical terminal comparison, and the composed true final biholomorphism retains the original whole projection formula. Since the base map is identity, its exact formula is the asserted equality of original projections. No compressed condition or comparison witness is supplied by the caller. The original histories may have different empty-step counts; only their actual final spaces are compared, without an uncompressed tower or arbitrary-map uniqueness claim. □

Theorem 377.3 (Actual terminal choices compare over a genuine original base biholomorphism)**.** *Let the original base map be a genuine biholomorphism $\Phi : X' \to X$. At EVERY actual family depth, original bound-k M over X and TWO GIVEN actual canonical terminal histories of M and its ACTUAL Φ pullback internally supply a genuine original-final-space biholomorphism $\eta : Y' \to Y$ satisfying $p_T\eta = \Phi p_{T'}$. No pulled-back domain, nonempty/compressed history, equal length or terminal comparison certificate is a caller input.*

Proof. The original domain is stable under the genuine base biholomorphism; its actual holomorphic pullback internally supplies the correct original pulled-back marked record and domain. Apply finite actual empty compression to both GIVEN original whole canonical histories. Each result retains its original record, internally constructs all actual centres nonempty and whole canonicality, and gives the true final biholomorphism, whole projection identity and terminal iff. Given terminality then internally gives both compressed terminal histories. The first lemma gives the true canonical segments. The established per-sequence canonical terminal comparison over Φ aligns these ACTUAL original and pulled-back records and gives their whole tower isomorphism, internally with equal compressed length. Compose its final map with the inverse source compression final map and target compression final map. Their true whole projection identities give precisely $p_T\eta = \Phi p_{T'}$ at EVERY original source point. Empty histories/manifolds and differing original empty-step counts are included. Noncompact global scheduling, intrinsic uniqueness/cocycles and global resolution gluing remain distinct obligations. □

378 Finite canonical histories identify the original global terminal manifold

The local-to-global comparison in [5, 3] can be made on the already constructed original terminal manifold whenever an actual finite canonical terminal history is available. The finite history is genuine geometric input; no local comparison or compatibility datum is assumed. For compact original bases the canonical driver supplies this history internally. The construction below does not assert existence of a finite history on an arbitrary noncompact base.

Definition 378.1 (Actual local comparison with a finite terminal history)**.** Let M be an original marked ideal in the invariant domain on X, and let $T : D \to X$ be an actual finite canonical terminal history. For every original atlas open U_i, construct a biholomorphism $c_i : D_i \to p_T^{-1}U_i$ from the original local final space. It satisfies $p_T(c_i(y)) = p_i(y)$ at every point.

Proof. Restrict the actual history T to the entire open X. Its base is the literal whole-open subtype and its final space is the literal whole projection preimage. Both are canonically biholomorphic to the original spaces by subtype inclusion. The original restriction theorem supplies canonicality and terminality. Compare this actual whole-open history with the original history on U_i, over the common open U_i. The common-open comparison theorem supplies a genuine biholomorphism, including both holomorphic directions and the whole projection equality. The local history's preimage of its entire base is canonically its whole final space; the whole-open history's preimage of U_i is canonically the literal $p_T^{-1}U_i$ by forgetting the redundant whole-open subtype. Composing these actual subtype identifications gives c_i and the stated equality. □

Lemma 378.2 (Finite local comparisons obey the original gluing relation)**.** *On every actual original overlap, the finite local comparison after the original transition equals the first finite local comparison, as an equality of points of D.*

Proof. Use the original triple-transition theorem on the histories over U_i, U_j, and the entire open X, with the third history the actual restriction of T to X. Every point of the first overlap also lies over X, so the required third-open membership is automatic. That theorem identifies the first-to-third comparison with the composite first-to-second-to-third comparison. The smaller-open

comparison theorem identifies these actual comparisons with the restrictions of the comparisons defining c_i and c_j. Forget the redundant literal whole-open subtypes. The resulting equality is precisely the claimed identity in D. □

Theorem 378.3 (The original global terminal manifold is biholomorphic to a finite terminal history)**.** *For the same original input and actual finite canonical terminal history, there is a genuine biholomorphism $C : \widetilde{X}_M \to D$ from the original global terminal manifold. It equals c_i on every original local inclusion, and its entire projection is the same original glued projection: $p_T C = P_M$.*

Proof. The preceding compatibility gives a continuous map from the actual topological gluing to D. Its equality with each c_i proves the whole projection identity, since the original inclusions are jointly surjective. For surjectivity choose, for a point $d \in D$, the original atlas index at $p_T(d)$; the corresponding open contains that base point, so d belongs to $p_T^{-1}U_i$ and the inverse of c_i supplies a preimage. For injectivity, equal images have equal base projections. The saturation identity therefore puts both original global points in the same original local range. Use that range's actual local inverse representatives. Their images under the same injective c_i coincide, so the representatives and then the global points coincide.

To prove forward holomorphicity at a global point, use an original saturated range containing it. On that genuine open neighborhood the glued map is c_i composed with the actual local range inverse. Both are holomorphic. For inverse holomorphicity at d, use the true open neighborhood $p_T^{-1}U_i$ selected above. There the inverse equals the original local inclusion composed with c_i^{-1}. Both are holomorphic. Equality on these actual open neighborhoods transfers the regularity to the whole maps. Bundle bijectivity and the two proved holomorphic directions. These arguments also include empty manifolds and dimension zero. □

379 Original local terminal charts are genuine biholomorphisms onto their saturated ranges

The original local-to-global construction of [5, 3] needs actual biholomorphic local identifications before transporting ideal/divisor geometry or smooth terminal arrows. Only original marked ambient-domain input is used. Inverse holomorphicity is PROVED from the actual lifted charts; it is not inferred from an unspecified regularity or inverse certificate. All actual original point and projection formulas are retained.

Definition 379.1 (Original data construct the literal saturated local range and its real homeomorphism)**.** For EVERY original local atlas index, form the TRUE open range of its internally constructed local inclusion into the actual global complex terminal manifold. Its entire range is the actual original-open preimage under the whole projection. The actual open embedding internally gives a genuine homeomorphism from that local final space onto THIS literal range. Its forward underlying value is the SAME original inclusion. No open range, homeomorphism or inverse witness is supplied.

Proof. The original topological producer supplies an actual open embedding. Thus its literal range is genuinely open; form this actual open subtype inside the constructed global complex manifold. Genuine embedding injectivity and induced topology give its real homeomorphism onto the ENTIRE literal range, with its actual inverse. The original saturated range theorem identifies this true range with the whole original-open projection preimage. Genuine homeomorphism inverse equations retain each local inclusion value; all this is constructed from original inputs. □

Theorem 379.2 (The internally constructed original local range identification is genuinely biholomorphic)**.** *For ONLY original marked-domain input, EVERY actual local final space is genuinely biholomorphic to its actual saturated open range inside the constructed global complex terminal*

manifold. The forward underlying map is the SAME original inclusion. BOTH holomorphic directions and whole inverse equations are internally proved; no desired inverse regularity or local geometric isomorphism certificate is a caller premise.

Proof. The true original inclusion is holomorphic by the completed producer. Holomorphicity into the actual range-open subtype follows from its independently proved true range membership. For inverse regularity, fix an actual range point and take its genuine local inverse representative from the first node's true homeomorphism. Choose its original preferred local chart and that chart's true global lift. The actual lifted chart belongs to the genuine global maximal holomorphic atlas. Its composite with the true range-open inclusion is holomorphic at the actual point. The original local inverse chart is holomorphic at the genuine coordinate value; the true lifted forward formula identifies this value with the original local chart coordinate of the chosen inverse representative. Thus their actual composite is holomorphic at the original range point.

On the true inverse image of the lifted chart source, an arbitrary actual range point is the original local inclusion of a true local chart-source representative. Real embedding injectivity and genuine homeomorphism inverse equations identify that representative with the actual inverse value. The true lifted coordinate formula and original chart left inverse identify the holomorphic composite with that entire actual inverse map on this real open neighborhood. Holomorphicity therefore transfers to the actual inverse at every range point. Bundle the genuine homeomorphism and both PROVED holomorphic directions into the actual biholomorphism. No inverse of a general holomorphic embedding is silently asserted regular; the true lifted charts prove this actual inverse. □

Theorem 379.3 (The actual local range projection is the original local whole terminal projection)**.** *From ONLY original marked-domain input, restrict the genuine global whole projection to the actual local saturated range and its true original base open. Its entire composite with the internally constructed local biholomorphism equals the ORIGINAL local finite tower projection; on an arbitrary actual range point it equals that original tower projection applied to the actual inverse representative. This range projection is holomorphic. No caller projection dictionary, local holomorphicity or transported ideal geometry certificate is assumed.*

Proof. Every actual range point's global projection belongs to the original chosen open by the true saturated range equality; this internally gives the target subtype membership for the actual range projection. The original local projection formula and actual forward inclusion value identify the whole composite with the original finite tower projection, using true subtype injectivity. At an arbitrary actual range point, apply this identity to its real inverse representative and cancel the genuine forward/inverse maps to get the ENTIRE inverse formula. The original finite tower projection is genuinely holomorphic and the second node proves the actual inverse is holomorphic. Their entire composite is therefore holomorphic and is THIS actual range projection by the proved whole formula. This supplies the genuine local dictionary needed by global principalization and smooth-map gluing without assuming either desired downstream global conclusion. □

380 Original marked input internally constructs a genuine global complex terminal manifold

The original noncompact resolution atlas of [3, 5] now acquires its ACTUAL holomorphic manifold structure and whole projection. Only the original marked record and actual ambient domain are inputs. The genuine local driver, canonical comparisons and topological gluing are internally constructed before any global geometric property is consumed. No local histories, global charts, separation, countability, biholomorphic overlap or projection holomorphicity certificate is a caller premise. Properness and genuine globally scheduled locally finite histories are separate producers.

Lemma 380.1 (Original canonical transitions identify the actual embedding overlap domains)**.** *From ONLY original marked-domain input, the literal preimage under each actual local gluing inclusion of another inclusion's ENTIRE range equals the internally constructed original canonical overlap domain. Internally transport the actual canonical transition through THESE true open equalities. It is a genuine biholomorphism between the literal embedding overlap domains and preserves their WHOLE actual gluing inclusions. No domain equality, comparison or inclusion-compatibility certificate is supplied.*

Proof. The actual second inclusion range is the ENTIRE original-open preimage of the internally constructed glued projection. Pull it back through the actual first inclusion. The true whole local projection identity identifies this with the literal first tower projection preimage of that original second open, exactly the actual original canonical overlap domain. Apply this equality in both directions. Genuine open-congruence biholomorphisms transport the already internally constructed canonical transition to the literal embedding overlap domains, preserving the SAME underlying original final points. The REAL topological gluing condition for these two original local objects proves the whole inclusion equation at every actual overlap point. All data came from the original local driver and original canonical comparison producer, not from assumed global holomorphic structure. □

Theorem 380.2 (Original marked-domain input internally constructs a Hausdorff second-countable global holomorphic manifold)**.** *On ANY original complex manifold, ONLY the original marked-domain input internally constructs an ACTUAL global terminal complex manifold on the actual original local-resolution glued space. Its genuine charted space is built from ALL true local atlas lifts. It is Hausdorff and second countable and its whole atlas transitions are genuinely holomorphic. No original base compactness, countable atlas index or global complex manifold certificate is assumed. Empty bases and ambient dimension zero are included.*

Proof. The genuine singleton local canonical driver supplies original covering opens and actual local final complex manifolds; the actual topological gluing producer supplies true joint open embeddings, continuous whole projection and ENTIRE saturated range identities. Original Hausdorffness, true local Hausdorffness and these actual projection data apply to the proved saturated-atlas separation theorem. Original second countability and true local second countability apply to its genuine internally selected countable-subcover theorem; no countable original point index is presumed. Actual joint coverage supplies preferred-chart representatives and ALL true local charts lift to the actual global charted-space structure. The first lemma internally supplies every genuine pairwise embedding-overlap biholomorphism and its whole inclusion identity. The proved holomorphic atlas constructor now establishes true coordinate compatibility of ALL lifted atlas charts. These internally supplied Hausdorff, second-countable and genuine holomorphic atlas structures construct the actual bundled complex manifold. No separation, chart or overlap conclusion is a premise. □

Theorem 380.3 (Every internally constructed original local terminal inclusion is genuinely holomorphic)**.** *ONLY original marked-domain input internally gives holomorphicity of EVERY actual original local terminal inclusion into the genuine constructed global complex terminal manifold. Its WHOLE original local final-space values and actual topological open-embedding properties are retained. No map regularity or chosen global chart certificate is supplied.*

Proof. Use the actual constructed global charted-space structure. All actual pairwise embedding-overlap comparisons and whole inclusion equations were supplied internally in the first node. The genuine local-inclusion holomorphicity theorem applies to these true original inputs. It uses the original local preferred chart, its true global lift and the actual inverse-chart holomorphicity to derive local regularity at EVERY original local point, before returning the entire actual inclusion's holomorphicity. No change is made to the literal original gluing map or its point values. □

Theorem 380.4 (The internally constructed whole original global projection is holomorphic)**.** *ONLY original marked-domain input internally gives holomorphicity of the ENTIRE actual original global glued projection to the original base, for the genuine constructed complex manifold structure. Every original local whole projection formula is retained. No desired projection or local/global holomorphicity certificate is an input.*

Proof. At an arbitrary actual global final point, true joint coverage internally supplies an original local final representative. Choose its genuine local preferred chart and true lift into the global atlas. The actual global lifted chart is in the genuine maximal holomorphic atlas and is holomorphic at that global point. The original local inverse chart is holomorphic at its true coordinate value. The actual local whole tower projection followed by its actual original-open inclusion is genuinely holomorphic, since both real component maps are. Compose these three actual holomorphic maps at the original global point. On the TRUE lifted chart source, every point is the actual local inclusion of an original local chart-source point. The real lifted forward chart formula and original chart inverse equation identify the composite with that actual local whole projection. The proved original global projection formula on EVERY local included point therefore identifies it with the actual global projection on this ENTIRE true source. The lifted chart source is an actual open neighbourhood of the point, so holomorphicity transfers through the genuine local equality. Since the actual global point was arbitrary, the entire whole projection is holomorphic. Properness and locally finite scheduled-history statements still require their own original producers; they are not assumed here. □

381 Literal open restrictions of original terminal comparison maps

The finite comparison route of [3, 5] is kept on the ORIGINAL recorded towers and their literal final-space open preimages. All canonicality, terminality and both genuine open-map domains are supplied from original inputs. These provide actual local comparison dictionaries; no global noncompact scheduling or glued resolution is presumed.

Lemma 381.1 (Original terminal histories genuinely restrict to their literal open preimages)**.** *ONLY original marked ambient-domain input, a GIVEN actual whole canonical terminal history T and a true original open U internally supply the actual restricted marked domain, whole restricted canonicality and terminality. The entire transformed marked record is the true restriction of the original transformed record to $p_T^{-1}U$, with every original indexed member retained.*

Proof. The genuine domain restriction producer preserves original regularity, coherence, nonzero-along germs and marking. The established PER-SEQUENCE canonical restriction producer applies to the actual ambient invariant and THIS given whole original canonical history. At each true restricted step the original empty centre stays empty; a nonempty original maximum trace is the restricted maximum, and an empty trace is an allowed genuine empty step. The genuine whole test-tower restriction dictionary follows from actual coherence, closed submanifold data and original terminal testness. Its literal transform recursion proves equality of the ENTIRE marked record, including the original mapped indexed boundary list. The actual transformed ideal remains coherent. Its cosupport restriction is exactly the preimage of the original empty terminal cosupport, and therefore empty. Hence the actual restricted history is terminal, with no caller restricted-domain, canonicality or transform certificate. □

Definition 381.2 (An original terminal comparison restricts as an actual biholomorphism)**.** From ONLY original marked-domain input and TWO given actual canonical terminal histories S, T, the internally constructed unique final comparison η restricts to a genuine biholomorphism $\eta_U : p_S^{-1}U \to p_T^{-1}U$ for EVERY true original open U. Its actual forward and inverse maps are exactly η and η^{-1} on the literal subtypes. Its WHOLE restricted projection formula holds at every original point. No open-domain preservation or lift certificate is a caller premise.

Proof. The completed original producer constructs the actual η and its whole projection identity $p_T\eta = p_S$ internally. If a point lies in $p_S^{-1}U$, that identity puts its genuine image in $p_T^{-1}U$. For a point of $p_T^{-1}U$, apply the SAME identity at its actual inverse point and use $\eta\eta^{-1} = \mathrm{id}$ to derive membership of its inverse in $p_S^{-1}U$. Thus both entire actual maps have the required true subtype domains. Their genuine inverse equations hold as subtype equalities. The forward and inverse ambient maps are holomorphic; compose each with the true open-subtype inclusion and apply actual holomorphicity into an open subset. This constructs a real biholomorphism, with its literal ambient value formulas. The true restricted tower projection dictionary and the retained whole projection identity imply the entire restricted projection identity as an equality in U. Empty opens, empty manifolds and dimension zero are included. □

Theorem 381.3 (The literal restricted comparison is the internally constructed open canonical comparison)**.** *At EVERY original open U, the actual literal restriction η_U is EXACTLY the unique original canonical terminal comparison constructed on U between the two entire restricted histories. Both histories' required canonicality, terminality and original restricted marked-domain inputs are supplied INTERNALLY. No restricted comparison or open-canonicality certificate is a caller input.*

Proof. The first lemma internally supplies the common original restricted marked record/domain and both actual whole restricted canonical terminal histories. The completed all-ambient canonical terminal producer therefore constructs a genuine comparison between these actual restricted final spaces, with its entire projection identity on U. The second lemma constructs the literal restricted original comparison with the SAME entire identity. Apply genuine projection uniqueness to the actual target restricted tower and these two genuine biholomorphisms. They are equal as entire maps, rather than merely on cosupport or an assumed dense locus. This retains every true empty step and permits different original empty-step counts. The original ambient and local comparisons are not input witnesses. □

Theorem 381.4 (Actual open terminal comparisons satisfy identity, inverse and triple cocycles)**.** *For GIVEN original canonical terminal histories and EVERY true original open, the actual self-comparison on the literal final open preimage is identity, the actual reverse comparison is the genuine inverse, and THREE original histories' actual restricted comparisons obey the entire triple cocycle. All required open histories and maps are internally constructed from original inputs, with differing actual empty-step counts.*

Proof. Each actual open comparison retains its WHOLE projection formula into U. The actual identity has the same self-formula. The actual inverse formula follows by substituting the genuine inverse point and applying the real right inverse. For three histories their two genuine comparison formulas compose to the direct first-to-third whole formula. Projection uniqueness on the relevant ACTUAL restricted target tower therefore identifies the self-comparison, inverse and triple composite with the internally constructed direct maps. The preceding theorem also identifies them with the true open canonical comparisons, so these are genuine literal open naturality/cocycle dictionaries, not independent local choices assumed compatible. Existence of a noncompact canonical schedule and actual global manifold/resolution gluing remain subsequent obligations. □

382 Unique true overlap comparisons of independent original local terminal histories

Keep original marked data on the original base and independently GIVEN finite canonical terminal histories on different original opens. The finite comparison route of [3, 5] is applied on true common opens only AFTER constructing their actual nested-base biholomorphism and whole marked dictionary. No overlap-map or restricted canonicality certificate is assumed.

Theorem 382.1 (Actual original ambient terminal histories compare uniquely over a genuine base biholomorphism)**.** *At EVERY ambient dimension, ONLY original marked-domain input, a genuine base biholomorphism and TWO GIVEN original ambient canonical terminal histories, one for the true marked pullback, internally give EXACTLY ONE actual final biholomorphism covering that base map on the WHOLE original projections. Actual empty steps and differing history lengths are allowed.*

Proof. For positive ambient dimension the actual ambient invariant is its defining specific family. The completed original finite base-comparison producer internally compresses both given histories, supplies their genuine canonical segments and terminality, then returns the real original-final map and whole projection identity. At ambient dimension zero, the true domain pulls back through the given genuine base biholomorphism. The actual bound-zero whole canonical projection producer gives a genuine entire projection biholomorphism for EACH original history. Compose the source projection, the given base biholomorphism and the inverse target projection. Their true pointwise formulas prove the required whole projection identity. Any two actual maps satisfying it have equal target whole projections at every source point. The actual projection uniqueness producer identifies them. No supplied compressed history, inverse projection or desired map is used. □

Lemma 382.2 (True common-open nested bases retain the ENTIRE original marked pullback)**.** *For original opens U, V and a true common open $W \subseteq U \cap V$, the actual traces of W inside U and V are genuinely biholomorphic through their real identification with W. This actual map is identity on the underlying ORIGINAL base points. Its pullback of the ENTIRE nested restriction of the original marked record through V is exactly the nested restriction through U, including every indexed boundary entry, marking and whole coherent germ ideal.*

Proof. The true open containments construct actual nested-open biholomorphisms from each trace to W, with true inverses and both holomorphic directions. Compose the first identification with the inverse second one. Its real underlying point formula is identity. Original marked-domain data supplies original coherence, so genuine holomorphic pullback composition gives the entire nested record through each open as the pullback of the SAME original record on W. Under the constructed nested-base map the second identification composes literally to the first. The genuine coherent commuting-square pullback dictionary therefore identifies the entire original ideals; preimage composition identifies original carrier and marking, and actual list-map composition identifies EVERY original indexed boundary member. True identity pullback removes the intermediate identity. No desired marked equality, boundary filtering or scalar/germ comparison is a premise. □

Theorem 382.3 (Independent original local terminal histories internally supply one genuine unique overlap map)**.** *ONLY original marked-domain input and independently GIVEN actual canonical terminal histories over U and V internally construct a unique genuine biholomorphism between their literal final open preimages over EVERY true common open W. Its WHOLE original-base projection formula holds at every original point. In particular their actual overlap $W = U \cap V$ requires no additional containment, restricted-history, comparison or lift certificate.*

Proof. True original domain restriction supplies the marked inputs over U and V. Actual whole terminal restriction supplies BOTH genuine canonical terminal histories on their actual traces of W, retaining all their recorded steps. The preceding lemma internally constructs their genuine nested-base biholomorphism and identifies the entire source marked record with its true target pullback. Transport the internally supplied source canonicality and terminality through THIS proved literal record equality. The first theorem applies to these actual restricted histories and internally supplies a real final biholomorphism. Its whole nested-base projection identity becomes the claimed original-base identity by the constructed actual base point dictionary and true restricted projection formulas. Conversely equality over the original base is equality of

nested target-base projections, since both actual subtype inclusions are injective. Genuine target restricted-tower projection uniqueness supplies uniqueness. Select the actual map from this proved existence and uniqueness. For $W = U \cap V$ both required containments are internally the true intersection projections. The original local histories need no common uncompressed length or common boundary list after different empty steps. □

Theorem 382.4 (Independent original local terminal comparisons satisfy actual common-open identity, inverse and cocycle)**.** *For independently GIVEN local canonical terminal histories, the internally constructed actual comparison on a true common open is identity for a history against itself, reverses by its genuine inverse and satisfies the entire triple cocycle for THREE histories on a true common original open. The common open may be their actual triple intersection. These compare the independently given ORIGINAL local final spaces and their genuine restricted maps, rather than restrictions of a caller common global tower.*

Proof. Every actual constructed comparison has the same entire original-base projection identity on the true common open. The identity, reversed inverse and true triple composite satisfy their respective identities by the real inverse equations and composition of whole formulas. For any two such genuine maps, their target nested projections have equal original-base values, so actual subtype injectivity gives equality in the target nested base. Actual target restricted-tower projection uniqueness therefore proves the entire map equalities. Original canonicality and terminality on each trace were internally supplied before choosing any map. This is the genuine finite local overlap/cocycle prerequisite; actual global manifold/resolution gluing and noncompact canonical scheduling/existence are still subsequent source obligations. □

383 True further restriction and triple transition laws for independent local terminal histories

All local histories remain independently GIVEN over their original opens. The finite canonical overlap route of [3, 5] now keeps the literal original final-space points under a further common-open restriction. No pairwise transition compatibility or triple-overlap certificate is supplied.

Lemma 383.1 (Actual final common-open inclusions are genuine holomorphic maps)**.** *For ANY actual local tower on an original open U and true opens $W' \subseteq W$, its literal final open preimage over W' includes holomorphically in its literal final open preimage over W. The actual map preserves the SAME original final-space point. No history or map certificate is needed.*

Proof. Membership of the smaller final preimage says its whole original-base projection lies in W'. True inclusion puts that SAME projection in W. Thus the map sends a point to the SAME underlying point with the actual larger membership proof. Its composite with the larger final-open inclusion is the true smaller final-open inclusion, which is holomorphic. Genuine holomorphicity into an actual open target therefore proves the map is holomorphic. This retains every recorded step and actual whole projection. □

Theorem 383.2 (A constructed independent-local overlap comparison genuinely restricts to the constructed smaller-open comparison)**.** *ONLY original marked-domain input and TWO independently GIVEN original local canonical terminal histories internally construct the genuine restriction of their actual overlap comparison over W to any true $W' \subseteq W$. Its forward and inverse maps retain EXACTLY the larger comparison's original final-space values, and this real restricted biholomorphism is the internally constructed unique comparison over W'. No desired restricted map, domain-preservation or equality certificate is a caller input.*

Proof. The preceding genuine inclusion puts each smaller final point into its larger overlap domain. The actual larger comparison preserves the WHOLE original-base projection. Thus its image belongs to the smaller target domain exactly when the original smaller point's projection does;

the actual inverse satisfies the reverse identity by its genuine right inverse. Construct both literal smaller subtype maps with THESE internally supplied memberships. Their actual inverse equations are the original larger inverse equations as subtype equalities. Their ambient maps are genuine composites of the true common-open inclusion, larger biholomorphism and larger open inclusion; actual holomorphicity into the smaller targets proves both holomorphic directions. The resulting actual biholomorphism retains the SAME whole original-base projection formula. The completed original smaller-open producer constructs another actual biholomorphism with that formula, with its local canonical histories supplied internally. True projection uniqueness on the ACTUAL target restricted tower identifies them. The formula holds at every smaller original point, not only on cosupport or an assumed dense locus. □

Theorem 383.3 (Original pairwise local transitions have genuine triple-domain preservation and pointwise cocycles)**.** *For THREE independently GIVEN original local canonical terminal histories on U, V, L, their internally constructed pairwise actual final overlap transitions preserve the genuine triple domains. At EVERY original first final point over both V and L, the actual first-to-second transition lies in the second-to-third overlap domain, and the actual two transitions have the SAME underlying original third final-space value as the direct first-to-third transition. No supplied triple point, intermediate domain membership or pairwise cocycle certificate is required.*

Proof. The first-to-second WHOLE projection formula retains the original first point's base value, so its membership over L supplies the actual second point's membership over L. Actual projections already land in their respective original base opens, giving its full second-to-third pair domain. The original first point lies over the true triple intersection $Q = U \cap V \cap L$. Restrict each genuine pairwise comparison to Q by the preceding theorem. Each becomes the internally constructed actual comparison on THIS SAME true common Q, retaining every original ambient final-space point. The proved actual common-open triple cocycle identifies the two restricted transitions with the direct restricted transition. Their retained original point formulas therefore give the claimed ENTIRE pairwise transition identity at the original point. These are the genuine pointwise domain and cocycle equations required by topological gluing data; no global glued manifold/resolution or noncompact schedule is yet asserted. □

384 True uniqueness and cocycles of original terminal comparison maps

Keep the actual recorded finite blowup tower and every original marked input. The finite canonical-choice route of [3, 5] is extended from existence of true original-final comparisons to their intrinsic uniqueness over the WHOLE original projection. The true first-centre ideal pullback is supplied internally from the actual blowup and its entire tail; no stage lift, invertibility or dense-isomorphism certificate is a caller input.

Lemma 384.1 (An actual positive-length tower has a genuine first-step factorization)**.** *Every actual positive-length recorded tower has a literal first genuine blowup followed by its ENTIRE actual remaining tail. The tail has strictly smaller actual length. This includes true empty first centres and actual nonidentity empty projections.*

Proof. Use actual tower recursion. An empty tower contradicts positive length. For a next step, if its preceding tower is empty, this step is the first and its tail is empty. Otherwise the actual preceding tower has positive length and its genuine first step and entire tail are internally supplied by recursion. Append the given actual last step to THIS tail; the actual append recursion identifies the whole result with the original tower. True length append gives one plus tail length, hence strict decrease. □

Theorem 384.2 (Actual final biholomorphisms over a whole blowup projection are intrinsically unique)**.** *For ANY actual finite tower $T : X \leftarrow Y$ and two genuine biholomorphisms $a, b : Z \to Y$,*

equality $p_T a = p_T b$ INTERNALLY implies $a = b$. No source canonical history, chosen stage lift, first-centre pullback invertibility or generic-density certificate is required.

Proof. Use strong induction on ACTUAL target-tower length, uniformly over all original source manifolds and actual final biholomorphisms. For the empty tower its projection is identity, so the given equality is exactly $a = b$. Otherwise the first lemma internally supplies a genuine first blowup $\pi : A \to X$ and its entire remaining tower $S : A \leftarrow Y$ with $p_T = \pi p_S$. The actual first blowup's universal property includes invertibility of the ENTIRE actual pullback of its actual centre ideal J on A. Genuine pullback along an ACTUAL recorded tower preserves an invertible ideal: each true blowup has exceptional locus of empty interior, which preserves nonzero local holomorphic generators, and actual coherent ideal pullback composition gives the whole tower assertion. Apply this established intrinsic producer to THIS entire S and THIS internally supplied $\pi^* J$. Pull back further along the genuine biholomorphism a; its inverse differential makes its exceptional locus empty. Thus the ENTIRE original pullback of J along $\pi p_S a$ is internally invertible. Actual coherent ideal pullback composition supplies exactly this original map's ideal, not a substitute denominator or generation certificate.

The genuine universal property of π, applied to THIS actual holomorphic map $\pi p_S a$, gives uniqueness of its actual holomorphic lift into A. Both $p_S a$ and $p_S b$ are real lifts: the former by definition and the latter by the GIVEN entire projection equality. Hence $p_S a = p_S b$. The target-tail length is strictly smaller, so the induction on THIS actual tail and SAME genuine final biholomorphisms gives $a = b$. All invertibility and lift conditions were supplied from the actual original blowups; no arbitrary holomorphic pullback preservation or maximum existence was assumed. Empty manifolds, empty steps and dimension zero are included. □

Theorem 384.3 (Original ambient canonical terminal comparisons are internally unique)**.** *At EVERY ambient dimension, ONLY the original marked-domain input and TWO GIVEN whole actual canonical terminal histories internally supply EXACTLY ONE original-final biholomorphism commuting with their WHOLE original projections. Let their canonical comparison be this INTERNALLY constructed unique actual map. All original empty steps/counts remain allowed; no compressed, nonempty, invertibility, comparison or cocycle certificate is supplied by a caller.*

Proof. The completed genuine all-ambient original terminal-choice producer internally gives an actual original-final biholomorphism with the whole projection formula; at dimension zero this comes from true original whole projection biholomorphisms. For any other actual original-final map with the same whole formula, their target-tower projections are equal at every original source point. Apply the preceding actual projection uniqueness theorem to THIS genuine target tower and THESE genuine final maps. Thus the original-final comparison is unique. Select the actual map supplied by this proved existence and uniqueness. Its original projection formula is retained directly from that producer; no desired map is an input. □

Theorem 384.4 (Actual original canonical terminal comparisons satisfy identity, inverse and triple cocycle)**.** *For original marked-domain input and GIVEN actual canonical terminal histories, the internally constructed comparison of a history with itself is the actual identity. Reversing two histories gives the actual inverse comparison. For THREE independently given actual canonical terminal histories, the actual comparison from the first to the second followed by the actual comparison from the second to the third equals the actual comparison from the first to the third. These are equalities of genuine original-final biholomorphisms, including differing actual empty-step counts, dimension zero and empty final spaces.*

Proof. The actual identity map satisfies the same whole original projection formula as the constructed self-comparison, so intrinsic uniqueness identifies them. The true inverse of the first-to-second comparison satisfies the reversed formula: substitute its actual inverse point into the original whole projection equality and use the genuine right inverse. Intrinsic uniqueness identifies it with the internally constructed reverse comparison. For three histories compose their actual

first-to-second and second-to-third biholomorphisms. Their TWO retained whole projection formulas compose to the original first-to-third formula at EVERY original source point. Intrinsic uniqueness identifies this actual composite with the actual directly constructed first-to-third map. No uncompressed tower length equality or equality of indexed exceptional lists is asserted. This supplies finite original comparison cocycles; noncompact global scheduling and global resolution gluing remain distinct source obligations. □

385 Actual OWN LARGE values from genuine local curve models

The source OWN LARGE lower history is a genuine original TEST but may have carrier bound larger than one. Its actual SMALL model has arbitrary smooth relative dimension over a genuine curve lower. We compare both using actual controlled geometry and original recursive definitions, not a canonical-only locality theorem or a fabricated companion TEST.

Lemma 385.1 (Actual smaller opens give true nested Cartesian histories)**.** *For ANY actual tower S and genuine opens $V \subset W$ of its original base, the literal restriction to V is an actual Cartesian history over the literal restriction to W, with the real inclusion maps at base and endpoint. Every empty trace centre and original ordered history position is retained.*

Proof. Apply actual open-inclusion Cartesianity to the nested open presenting V inside W and the actual history restricted to W. The genuine nested-open tower isomorphism identifies its source with the literal restriction of S to V. Its true biholomorphic inverse and the actual source-isomorphism Cartesian theorem transport the diagram. Both composite maps send every actual point to the same underlying point, so they are the literal inclusions. No commuting-map certificate is substituted for actual blowup Cartesianity. □

Theorem 385.2 (A true local curve model determines the original LARGE GENERAL value)**.** *Let P be a genuine original DOMAINb record with ANY original TEST in ambient dimension $m > 0$. Let M be a genuine original DOMAIN1 record with ANY original TEST in ambient dimension $n > 0$. Suppose an actual common source history is smoothly Cartesian over the P history in the SAME ambient dimension m, and smoothly Cartesian over the M history, and its two genuine WHOLE initial pullback records agree. At every actual common source endpoint over the current cosupport of M, the LARGE original GENERAL value of P equals the actual $(m-n)$ dimension map of the original GENERAL value of M, at EVERY positive recursive depth. These are real geometric comparison inputs, not value or lower-TEST certificates.*

Proof. Construct both whole current controlled pullbacks and cosupport preimage identities from the genuine original TESTs. The whole initial identity gives the same actual small current record, so the image in P is a current cosupport point. Actual nonmonomial order dictionaries identify the attained LARGE and curve orders. Actual original fixed-order-run dictionaries identify their runs through the same actual small history, retaining all empty steps.

If the attained order is zero, true monomial-part smooth pullback gives the actual SMALL gamma from the LARGE gamma with zero dimension shift and from the curve gamma with shift $m - n$. The zero-shift monomial map is literally identity on every value. Hence the LARGE gamma is the shifted curve gamma. Substitute this into the unchanged monomial branch of original GENERAL.

If the attained order is positive, the original-only run-companion dictionary for P compares the actual SMALL and LARGE compAt SPECIAL predicates and levels. Independently, genuine curve current bar geometry gives the SMALL compAt its actual SPECIAL predicate and level zero through the other real Cartesian model. The whole initial identity and the proved attained-order and run identities identify these actual SMALL constructors. Thus LARGE is SPECIAL with level zero. Its unchanged recursive StepOne value is $(0, \infty)$ at every depth. Substitute that

genuine leaf into the unchanged positive GENERAL branch and the actual curve formula. This proves the whole value identity without inferring SPECIAL from a value or assuming a companion TEST. □

Theorem 385.3 (Only original DOMAIN TWO data construct actual OWN LARGE full values)**.** *For ONLY an original target DOMAIN*2 *record of maximal order, any genuine original upper TEST and a true smooth map in positive ambient dimensions, construct ONE actual source upper TEST. At EVERY actual source endpoint whose image is in the target current cosupport, nonspecial and of level zero, compute the two original independent contact choices and the unchanged original zero-contact masks. Construct both actual current lower-mask memberships. The WHOLE original GENERAL value of the SOURCE OWN LARGE lower record on its literal masked tail is the actual dimension map of the target lower original GENERAL value, at EVERY positive recursive depth. No source curve, source lower domain, local model, lower TEST or value is an input.*

Proof. Use the original-only OWN SMALL producer to construct one actual source upper TEST and, at each stated endpoint, the actual independently chosen small open inside the source OWN LARGE mask, its genuine curve Cartesian model, whole initial identity and exact endpoint map. Independently, genuine smooth pullback of the original upper DOMAIN2 and maximal-order data gives the actual shifted source upper domain and maximal order. Whole original current cosupport, SPECIAL, level and zero-regime-run dictionaries give the actual source hypotheses for the original zero-mask TEST theorem. That theorem constructs the OWN LARGE lower domain of bound $1 + (m - n)$, its genuine original TEST and actual current point. The target zero-mask theorem independently constructs its true DOMAIN1 lower TEST and current point.

The first lemma constructs the SMALL-to-LARGE Cartesian history from the two actual opens, with literal inclusions. Its genuine initial pullback is exactly the independent whole OWN SMALL record already constructed. Both actual endpoint maps agree with the original target lower point by the retained coordinate identity. Apply the preceding local-model theorem to these exact original histories and actual records. All model, domain, TEST and point inputs have been constructed internally. This is the zero-contact OWN LARGE comparison for original DOMAIN2 upper data; it does not assert arbitrary higher-dimensional noncanonical GENERAL naturality, positive-contact higher carrier bounds, exceptional semicoherent stability or a global scheduler. □

386 Actual independent source zero contacts have small smooth curve models

Here the original target has DOMAIN2 and maximal order, both ambient dimensions are positive and the upper history is ANY genuine TEST. The source may have positive relative dimension. The output concerns the true independently selected source contact on an internally constructed SMALL open of its actual computed zero mask. Comparison with the original source OWN LARGE value is not asserted.

Theorem 386.1 (A source restriction inside an existing source open is the literal Cartesian restriction)**.** *Suppose the actual source of a genuine Cartesian history is a tower already restricted to an open V. For ANY real $V_0 \subset V$, construct the genuine Cartesian history whose source tower is literally restricted to V_0, whose target is the full original target tower, and whose maps are the literal restrictions of the original base and endpoint maps. No replacement history or target-open image is supplied.*

Proof. Regard V_0 as the actual open of V defined by its inclusion inverse image. Actual source-open Cartesian restriction constructs the genuine nested restricted history with literal inclusion-composite maps. The existing genuine nested-opens stagewise isomorphism identifies that actual

nested tower with the tower literally restricted to V_0; its base and endpoint maps are the actual flattening biholomorphisms. Apply the actual Cartesian source-isomorphism producer to its inverse. The genuine inverse flattening formulas show that the resulting maps are exactly the stated subtype restrictions. Every actual centre position and the full target history are retained. □

Theorem 386.2 (Original data construct a genuine source OWN SMALL smooth lower model and its whole values)**.** *ONLY original DOMAIN2/maximal-order data, an actual smooth submersion and ANY genuine original upper TEST construct ONE actual source upper history and its original pulled-back TEST. At every actual source endpoint over a nonspecial level-zero current target cosupport point, construct an actual small open inside the source's OWN computed zero mask, containing the true tail-projection point. Its contact direction is the source's OWN independently selected original direction.*

For this SAME actual small source contact record, construct the whole initial pullback identity and genuine smooth Cartesian lower history over the unchanged original target zero lower record on its actual computed mask. The real endpoint map sends the actual source point to the actual target point. At every positive lower GENERAL depth, its WHOLE unchanged original value is the genuine dimension map of the actual target lower value. No lower TEST, current point, compatible choice, model map or value certificate is supplied. No source DOMAIN1 is asserted.

Proof. Construct one actual smooth source history from the genuine original upper record and TEST. Original coefficient-homogenization weak equivalence supplies the actual target bar TEST and its current cosupport; real level, SPECIAL and entire-run dictionaries identify the source's own zero run. The original masked-contact history producer internally gives actual target DOMAIN1, lower TEST and endpoint cosupport. Applied to the actual pulled original record it separately gives the true source OWN mask, shifted domain, genuine lower TEST and current point, without asserting that source to be a curve.

The genuine zero-contact smooth diagram constructs the transported source contact, its actual full-mask Cartesian history and coordinate maps to the target lower history. The original tangent-choice producer constructs both source directions and their genuine gluing before any source test. Apply its retained-position contact comparison to the actual upper split and its two actual masks, which independently avoid all original members. It constructs two true smaller opens, a genuine stagewise source lower isomorphism, the ENTIRE initial contact-record identity and an actual endpoint biholomorphism fixing the original upper cosupport point.

Use the first theorem to flatten the nested restriction of the actual transported source Cartesian history to its smaller open. Compose with the constructed genuine source-choice isomorphism. This produces the actual Cartesian history for the independently selected source SMALL contact. Genuine coherent pullback composition and both proved whole contact initial identities identify its marked record with the literal composite pullback of the target record, in all four fields. The real coordinate dictionary and actual point-fixing gluing identify the endpoint image with the internally constructed target lower point.

Finally apply the actual curve-reference full-value source-isomorphism producer to the target DOMAIN1 lower record and its genuine TEST. All source codimension-one and recursive candidate geometry is constructed by that producer from the true smooth model. Substitute only the independently proved WHOLE initial record identity and actual endpoint equation. The result compares the actual source OWN SMALL record, not a renamed transported choice. Genuine open-locality between that smaller record and the original source OWN LARGE presentation remains a subsequent obligation; it is not inferred from candidate admissibility or a higher-dimensional curve assumption. □

387 Canonical centre-ring freeness on one original analytic open

Put $B = \mathcal{O}_{\mathbb{C}^d,0}$, $R = \mathcal{O}_{\mathbb{C}^{d+t},0}$ and $Q = R/J$. The action of Q on the original layers and their whole direct sum is the actual action already descended from ambient multiplication and proved J-

annihilation. It is not defined by a chosen tangent action. The actual plane quotient isomorphism $e : Q \simeq B$ is used below only for comparison. This supplies an intrinsic local-algebra ingredient for the Bierstone–Milman Hilbert–Samuel equisingularity route; a coherent sheaf identification, original centre containment, nonlinear stalk/chart transport and full ambient Hilbert–Samuel descent remain subsequent producers. The higher full truncations remain modules over the split tangent ring B; no Q-action on those truncations is asserted.

Lemma 387.1 (Actual identity semilinear dictionaries for centre scalars)**.** *For every actual original whole germ ideal I and every q, the identity on the existing layer L_q is a genuine e-semilinear equivalence from its canonical Q-module to its old tangent B-module. The same is true on the entire existing direct sum G. Both equivalences and their inverses fix every actual element, in particular every original power class and every original degree inclusion.*

Proof. Use the identity bijection on the unchanged original additive groups. It preserves addition. The proved actual scalar dictionary is precisely $cx = e(c)x$ for EVERY $c \in Q$ and actual element x, on a layer and on the whole sum. This is the needed semilinear identity. The inverse uses the actual inverse ring isomorphism: writing $b = e(c)$ in this identity gives $bx = e^{-1}(b)x$. The two functions compose to the identity and fix actual classes and inclusions because both functions are literally the identity on those carriers. □

Lemma 387.2 (Freeness and finite generation under the actual centre dictionary)**.** *If an actual layer or whole direct sum is free for its old tangent B-action, it is free and flat for its independently constructed canonical Q-action. If an actual layer is finitely generated over B, it is finitely generated over Q. These are conditional algebraic helpers; no desired canonical-action or flatness premise is assumed.*

Proof. Take an actual B-basis. Retain the same actual vectors as a Q-family. For any element, its unique finite B-linear expansion becomes a finite Q-linear expansion by applying the true inverse isomorphism e^{-1} to every coefficient and using the semilinear dictionary. A vanishing Q-linear relation maps under e to a vanishing B-relation, so every coefficient is zero. Thus this is a true Q-basis, for the canonical action already defined. The same coefficient conversion changes any finite B-generating family into a finite Q-generating family. A free module is flat: tensoring its direct-sum basis representation is a direct sum of copies of the given module map, so it preserves injectivity. No new scalar action is used in any of these arguments. □

Theorem 387.3 (Original coherent input supplies one canonical centre free open)**.** *Let $\mathcal{I}$ be the actual original coherent ideal. There exists one open $U \subset \mathbb{C}^d$, meeting EVERY neighbourhood of the origin, such that for every $a \in U$ the entire original normal associated sum of the WHOLE translated stalk ideal I_a is free and flat over its actual centre quotient ring Q. EVERY actual original degree layer of this same I_a is finite, free and flat over its canonical Q-action, on this SAME U and simultaneously for all degrees.*

Proof. The original coherent-input whole associated-module producer internally constructs the common true coefficient domain, complete relation/kernel spreading, actual nearby homomorphisms and single denominator nonvanishing open U. It gives whole tangent B-freeness at every $a \in U$, with U open and meeting every origin neighbourhood. Keep exactly this U. Apply the preceding actual semilinear coefficient-conversion helper at the WHOLE translated ideal I_a to obtain canonical Q-free and flat whole sum. The proved genuine homogeneous layer summand and finite homogeneous symbol surjection give finite free B-layers internally from that same whole free sum. Apply the same actual layer dictionary to each q to obtain finite free and flat canonical Q-layers. The choices of open and denominator preceded q, so one and the same U works for ALL actual original degrees. No basis, coefficient presentation, whole kernel, free/flat statement or centre-action equality is supplied by the original coherent input caller. □

The open need not contain the origin and is not asserted dense on every arbitrary component. The theorem concerns the actual original graded modules and their canonical local centre quotient action. It supplies neither a bundled sheaf result nor an assertion that a centre is contained in the original zero space, and makes no ambient mixed-degree Hilbert–Samuel or blowing-up descent conclusion.

388 Canonical centre-ring scalars on the original normal graded layers

Let R be the actual original ambient holomorphic local ring, I the whole original ideal and J the whole normal ideal. The already constructed layer L_q is the image of J^q in $R/(I+J^{q+1})$; its addition and its intrinsic graded product are retained. The following scalar descent is a local algebraic ingredient for the Hilbert–Samuel equisingularity route in Bierstone–Milman, subsection "Use of the Hilbert-Samuel function". That source passage describes the geometric goal; the explicit quotient-scalar construction below is an additional algebraic argument. It does not establish a normal-flat sheaf, centre containment, full ambient Hilbert–Samuel comparison or descent under blowing up. No action of R/J on the full higher truncation $R/(I+J^s)$ is asserted.

Lemma 388.1 (Actual ambient multiplication on the whole original layer)**.** *For any $q \geq 0$, ordinary quotient multiplication defines a genuine R-module on the existing literal layer L_q. For every actual $g \in J^q$ and $r \in R$, it is given by*

$$r\,[g]_q = [rg]_q.$$

The additive group is the original one; the action does not use a polynomial quotient identification or a chosen tangent splitting.

Proof. Every element of L_q has a representative $g \in J^q$ by its image construction. Since J^q is an ideal, $rg \in J^q$. Thus the actual quotient product of the class of r and this element is again in the same image. Multiplication in the original quotient is already well-defined: if $g - g' \in I + J^{q+1}$, then $r(g-g')$ is in this same ideal. Associativity, the scalar unit law and distributivity follow from the corresponding equalities in that quotient, since the inclusion of its image subtype is injective. Its actual quotient class formula gives the displayed equality for every representative, not only a chosen one. □

Lemma 388.2 (Internally proved annihilation and canonical centre scalars)**.** *The whole original ideal J annihilates L_q. Consequently the actual ambient action descends canonically to a module over the actual centre quotient R/J, with the formula*

$$(r \bmod J)\,[g]_q = [rg]_q \qquad (g \in J^q).$$

No annihilator or desired quotient-action hypothesis is supplied.

Proof. For $u \in J$ and $g \in J^q$, their actual product belongs to $JJ^q = J^{q+1}$, hence has zero class in $R/(I+J^{q+1})$. The actual power-class surjection therefore proves $ux = 0$ for EVERY $x \in L_q$. If $r - r' \in J$, distributivity gives $rx - r'x = (r-r')x = 0$. Thus the ambient scalar additive homomorphism factors through the genuine quotient R/J. On a quotient representative its value is the original ambient multiplication. The original ambient module laws descend along the surjection $R \to R/J$, proving all module laws and the formula. This also proves uniqueness among quotient scalar actions whose scalar representatives have the stated actual action. □

Lemma 388.3 (Canonical scalars on the original whole direct sum)**.** *The existing literal direct sum $G = \bigoplus_{q\geq 0} L_q$ inherits the actual componentwise ambient action and canonical R/J action. For every degree inclusion ι_q, actual scalar r and actual $x \in L_q$,*

$$(r \bmod J)\,\iota_q(x) = \iota_q((r \bmod J)x).$$

Every component of this action is the original quotient class product, and J annihilates the entire direct sum, all degrees simultaneously.

Proof. Define the ambient action componentwise; the support stays finite because scalar multiplication preserves zero. The laws hold in every actual component and hence in the direct sum. For $u \in J$, its action on every component is zero by the preceding lemma, so its action on the entire direct sum is zero. Define the quotient action componentwise from the already canonically descended layer actions. Its laws hold in each component. Representing a quotient scalar by $r \in R$ identifies this action with the actual ambient action componentwise, so it is exactly its quotient descent. The ambient action commutes with each degree inclusion, as can be checked in every component. The two scalar representative formulas then give the displayed inclusion identity. Thus descent of the whole sum agrees with the componentwise already descended layer actions. □

Lemma 388.4 (Comparison after canonical descent with the actual plane trace)**.** *In original split coordinates, put $B = \mathcal{O}_{\mathbb{C}^d,0}$ and $R = \mathcal{O}_{\mathbb{C}^{d+t},0}$. The actual plane trace induces the already proved algebra isomorphism $e : R/J \simeq B$. For EVERY actual quotient scalar c, the newly descended action on L_q and G agrees with the existing tangent B-action of $e(c)$. In particular*

$$(r \bmod J)x = (\operatorname{trace} r)x, \qquad (\operatorname{pullback} b \bmod J)x = bx.$$

The canonical centre action is constructed before this comparison.

Proof. The actual relative Hadamard decomposition says $r - \operatorname{pullback}(\operatorname{trace} r) \in J$. Its action on each actual layer is therefore zero. The actual ambient quotient multiplication by the pulled back b is precisely the old *tangent* scalar action, both on every quotient numerator and on the same image subtype. Thus the first scalar formula holds for every r and every actual layer element. Every quotient scalar has an actual representative, and the true plane quotient isomorphism takes this representative to its actual trace, so the formula holds for every $c \in R/J$. Trace of tangent pullback is the identity, giving the second formula. Componentwise verification gives the same two equalities on the entire direct sum, with the original degree inclusions retained. □

Intrinsic free or flat modules over an actual smooth-centre stalk ring, chart independence, a coherent sheaf/stalk identification and genuine centre inclusion remain subsequent geometric producers. The previous same-open tangent ranks and fibre Hilbert–Samuel sequences do not by themselves supply those conclusions.

389 Actual sharp-curve centres under genuine base biholomorphisms

Two nested overlap opens need not be literally the same manifold term. We therefore transport the ORIGINAL ordered curve data through a real base biholomorphism before comparing actual histories. Source and target ambient dimension is the same positive n; no base-identity assumption, desired marked-record equality or desired centre equality is supplied.

Theorem 389.1 (The genuine base biholomorphism supplies the entire original pulled product and curve data)**.** *Let $e : X' \to X$ be ANY actual biholomorphism and let W, E be the original closed smooth pure curve and ordered SNC list with $W \subset \overline{W \setminus \bigcup E}$. Construct the ENTIRE original product marked record on X' as the actual pullback of the target record, for every marking. Its carrier is $e^{-1}W$ and its initial exceptional list remains empty. The separate ORIGINAL ordered family is the positionwise list of true inverse images. These source sets internally satisfy the actual closed smooth curve, SNC, whole-family density and regular dimension-one hypotheses.*

Proof. For a genuine smooth boundary member, the actual biholomorphic vanishing-ideal pullback theorem and true coherent identity pullback identify the whole pulled vanishing ideal with the vanishing ideal of its preimage. Actual ordered-list induction, using coherent factors and genuine multiplication pullback, gives the whole original product identity. Reading the actual carrier, empty initial exceptional list, ideal and marking fields proves the entire marked-record equality.

Actual biholomorphic flattening transport preserves the closed smooth carrier and its dimension-one regularity. The same genuine adapted coordinate transport gives the entire ordered SNC preimage family. The actual homeomorphism underlying e commutes with closure and true inverse images, so whole-family density pulls back. All source data are constructed from these actual maps and original target data. □

Theorem 389.2 (Both independently computed original curve centres are natural over the real base map)**.** *Take actual towers over X', X with an actual stagewise isomorphism over e, and let the target be ANY genuine original product TEST. For the literal original source curve and family constructed above, its independently computed FIRST-Z and independently computed whole original-member difference are the true preimages of their target counterparts under the actual endpoint biholomorphism. No source TEST or centre dictionary is a hypothesis.*

Proof. The first theorem identifies the actual source initial record. Genuine tower-isomorphism TEST transport therefore supplies its original TEST internally. The real original source curve domain is likewise supplied before any current order theorem is used. FIRST-Z equals the original second-order nonmonomial cut on both histories. Whole current marked pullback and the genuine original order-cut theorem give both set directions under the actual endpoint map.

For the other centre, first apply the independent original full ideal germ difference theorem on both true original histories. It identifies the literal difference with strict carrier intersected with the union of the ENTIRE ordered accumulated exceptional family and the union of EVERY original strict member. Actual template whole-record transport identifies carrier and ordered exceptional family. The genuine tower strict-transform theorem identifies each original member preimage individually. Both finite-union and intersection directions then give the literal whole difference preimage. Neither centre is inferred from an indexed count or a guessed equality of supports. □

Theorem 389.3 (The actual next FIRST or difference stage is constructed over the genuine base map)**.** *For actual prefixes as above, take two genuine next blowups of their own computed FIRST-Z loci, or of their own computed whole differences. Construct the actual next endpoint biholomorphism and full extended stagewise comparison over the SAME actual base e. It satisfies the real next projection square and is unique among actual holomorphic maps satisfying that square. No desired new map or matching-centre certificate is supplied.*

Proof. The preceding theorem computes the true centre preimage equality BEFORE any comparison theorem is used. Transport the target blowup through the actual inverse of the previous endpoint biholomorphism. The genuine same-centre blowup universal property supplies forward and reverse holomorphic comparison maps; actual universal uniqueness makes their composites identities. It also gives uniqueness of the next holomorphic map over the previous actual stage. The genuine tower constructor extends the existing stagewise isomorphism with these real centre and projection identities. Empty computed centres remain actual successor positions. Finite overlap comparisons and their cocycles require the subsequent retained-recipe consumers. □

390 Actual finite sharp recipes over genuine base biholomorphisms

Write $\Delta(P)$ for the independently defined whole original-member ideal-germ difference after the actual history P. A literal difference recipe takes its next centre to be exactly $\Delta(P)$, retaining every empty centre as a genuine step. Let W be a closed smooth pure curve in an ambient manifold of positive dimension, let E be the original ordered SNC family, and suppose W is the closure of its complement of that family. For a genuine base biholomorphism $\Phi : X' \to X$, the source data are $W' = \Phi^{-1}W$ and the positionwise ordered family $E' = \Phi^{-1}E$. The initial product record still has empty exceptional list.

Theorem 390.1 (Finite literal recipes over a genuine base biholomorphism)**.** *Let actual prefixes P', P have a genuine stagewise isomorphism over Φ, with endpoint biholomorphism χ_0, and assume only the target prefix is an original product TEST. If S', S are literal difference recipes on the source and target, respectively, of the same finite length, construct a true stagewise isomorphism of their tails over χ_0 and of their full histories over Φ, with the same terminal biholomorphism.*

Proof. Induct on the actual pair of finite tails. A reflexive tail has length zero, so equality of lengths forces the other tail to be reflexive too; use the actual prefix comparison. If exactly one tail is a successor, equality of lengths is impossible. In the two-successor case, their predecessor tails have equal lengths. The induction hypothesis constructs the actual earlier tail and full-history isomorphisms. The target predecessor is an original TEST: construct this from its literal recipe and the original target prefix. The genuine biholomorphic whole-difference dictionary then identifies the actual source difference centre with the inverse image of the actual target difference centre under the earlier terminal map. Substitute the two literal recipe centre identities. Both successor arrows are their given actual blowups, so the same-centre universal property constructs the true next biholomorphism and commuting square. Extend the actual tail isomorphism by this step and concatenate it with the prefix isomorphism. At no point is a centre equality or source TEST supplied as a new hypothesis. □

Theorem 390.2 (Unequal terminal recipes have genuine padded comparisons)**.** *Under the same original hypotheses and actual prefix isomorphism, let both literal difference recipes terminate with their actual whole difference empty. For every L at least both tail lengths, append the actual identity blowups of the empty centre to give both tails length L. There exists a unique genuine full stagewise isomorphism over Φ of these actual padded histories. The terminal manifolds and terminal map are those of the original tails; every added empty position remains in the padded histories.*

Proof. The genuine base biholomorphism first constructs the source closed smooth curve, its SNC family and its whole-family density. The whole original product pullback identity retains all four marked fields. Actual prefix isomorphism and the original target TEST therefore construct the original source prefix TEST. Apply the actual terminal recipe padding producer on each side, using these independently justified inputs. It supplies the literal padded recipes, their exact common length and actual terminal difference emptiness. Apply the preceding finite comparison to these real tails. Its terminal biholomorphism is a full stagewise comparison of the actual padded histories. Any second such comparison has the same terminal map by actual fixed-tower uniqueness, proved by true one-step blowup uniqueness, so the biholomorphism itself is unique. No empty step is removed and no ambient ideal identity is inferred by deleting its normalization. □

Theorem 390.3 (Two-phase retained recipes compare at every common terminal cut)**.** *Take any actual blowup of the computed original FIRST centre on each side, and on each such actual first stage take a literal terminating difference recipe. At every common tail length L, their real empty-padded full histories admit a unique actual stagewise comparison over Φ. The terminal biholomorphism is independent of L.*

Proof. Start with the reflexive original prefixes and the genuine base biholomorphism. The computed FIRST-centre naturality and actual blowup universal property construct the first-stage comparison. The target reflexive history is an original TEST, and the internally proved original FIRST-centre admissibility constructs the target first-stage TEST. Apply the preceding theorem to these exact first-stage prefixes and the two literal terminating tails. For two choices L, K, extend both actual padded comparisons to the true common size $L+K$ by appending actual empty blowups. Actual fixed-history uniqueness identifies their endpoint maps; the identity empty maps leave those endpoint maps unchanged. Thus the comparison is independent of the common cut. This compares genuine retained recipes; it neither assigns a recipe to an arbitrary sharp endpoint nor asserts overlap cocycles or geometric descent. □

391 Padding-independent cocycles on a genuine common open

Three retained local recipes can be compared on ANY actual open contained in their three original base opens. Every map below is constructed from the actual recipes; no arbitrary sharp endpoint is assigned centres. A cocycle on that fixed common open is distinct from compatibility with restricting a previously chosen pairwise-intersection map. The latter and geometric manifold descent remain separate obligations.

Lemma 391.1 (The actual common-open base maps compose)**.** *For three actual opens U_1, U_2, U_3 and a common open $O \subset U_i$, let $\Phi_{ij} : A_i \to A_j$ be the genuine common-open base maps between the nested presentations of O. Then $\Phi_{23} \circ \Phi_{12} = \Phi_{13}$ as genuine biholomorphisms.*

Proof. Each map is the actual nested-open biholomorphism to O followed by the inverse nested-open biholomorphism for the other chart. The real inverse laws cancel the two maps for U_2. Equivalently, at every actual nested point both composite and direct map have the same underlying point of X; their subtype membership proofs express the same genuine point. Thus their forward functions agree, and genuine biholomorphism extensionality gives the claimed equality. No base cocycle is supplied as a hypothesis. □

Theorem 391.2 (True finite comparisons compose independently of their padding cuts)**.** *Fix three actual prefixes and tails, with genuine base biholomorphisms Φ_{12}, Φ_{23}. Suppose real stagewise comparisons of their actual identity-empty-padded full histories are given for pairs $12, 23, 13$, where the third base map is the literal composite. The three common tail cuts may differ and each bounds both of its actual tail lengths. Then the actual terminal maps satisfy $e_{23} \circ e_{12} = e_{13}$.*

Proof. If the three cuts are L_{12}, L_{23}, L_{13}, take their genuine common size $N = L_{12} + L_{23} + L_{13}$. Extend the first comparison by the actual empty identity towers of length $L_{23}+L_{13}$, the second by length $L_{12} + L_{13}$, and the third by length $L_{12} + L_{23}$. True empty-tail concatenation and natural-number subtraction identify all three extended comparisons with the same fixed histories padded to N. Their terminal maps have not changed. Actual stagewise composition gives a genuine comparison of the first and third fixed histories over the literal composite base map. Actual fixed-history comparison uniqueness identifies its terminal biholomorphism with the independently given third map. No empty position has been deleted and no canonicality or compressed history is assumed. □

Theorem 391.3 (Retained original local recipes construct the common-open cocycle)**.** *Let the original positive-dimensional ambient manifold have a closed smooth pure curve W, original ordered SNC family E, and density of the complement of that whole family in W. On each of three actual base opens choose an actual computed-FIRST blowup followed by a literal terminating difference recipe. For ANY genuine common open O and ANY three pairwise common cuts, construct the real restricted padded pair comparisons independently. Their actual terminal maps satisfy the cocycle, with all original and empty positions retained. No matching centres, source TESTs, endpoint maps or cocycle certificate are additional inputs.*

Proof. Apply the genuine arbitrary-common-open overlap producer separately to the three pairs, using the same actual O and their respective cuts. It internally constructs all restricted original data and TESTs, independently computed centre matches, literal restricted recipes, exact retained lengths, terminal difference emptiness and genuine padding. Thus each pair has its unique actual full stagewise comparison on the stated real histories. Rewrite only the actual base composition by the first lemma. The preceding theorem applies to these exact three comparisons, even though their padding cuts differ, and proves the terminal-map cocycle. All three comparison maps were produced from the original recipes. This assertion holds in particular after choosing $O = U_1 \cap U_2 \cap U_3$. It does not yet identify these independently constructed common-open maps with restrictions of pairwise-intersection maps; that genuine restriction-and-flattening comparison remains necessary before geometric descent. □

392 Actual overlap maps for retained local sharp curve recipes

Let $W \subset X$ be a closed smooth pure curve in positive ambient dimension, let E be its original ordered SNC family and assume density of the complement of all original members in W. For actual opens $U, V \subset X$, let $O \subset U \cap V$ be ANY genuine common open. View O as the true nested open $A_U \subset U$ and $A_V \subset V$. Their canonical maps to O give a genuine biholomorphism $\Phi_{UV} : A_U \longrightarrow A_V$, equal to the identity on underlying points.

Theorem 392.1 (Genuine intersection base maps retain the original product and ordered family)**.** *Construct Φ_{UV} from the two actual nested-open biholomorphisms. The twice-restricted original carrier and ordered family on A_U are exactly the inverse images of those on A_V. For every mark, the genuine pullback of the whole original product record on A_V is the actual original product record on A_U, including all four fields. Original empty or repeated member positions are retained, and the initial marked exceptional list is empty.*

Proof. Each nested-open map sends a point to the same underlying point of X and has its actual nested inclusion as holomorphic inverse. Their composite is therefore a true biholomorphism. Membership in the two twice-restricted carriers is membership of that same underlying point in W, proving the carrier identity in both directions. For each original position, membership in the two restricted members is membership of that same point in that original member. Applying the two literal ordered-list maps proves the whole family identity, with no reindexing or deletion. Restriction of the original SNC family to V and then to A_V supplies its genuine SNC geometry. The whole original product pullback theorem for this actual biholomorphism and that genuine family then gives all four fields of the original marked record. Substitute the proved carrier and list identities. No ideal identity is inferred solely from supports. □

Theorem 392.2 (Actual local FIRST blowups have genuine intersection comparisons)**.** *Take any actual blowup of the computed FIRST centre for the original data on U, and any such actual blowup on V. Restrict these actual first histories to A_U and A_V. Construct a genuine stagewise comparison over Φ_{UV}, with its actual next biholomorphism. No matching-centre, original TEST or geometric certificate on the intersection is supplied.*

Proof. Open restriction of the original smooth carrier, SNC coordinates and regular curve dimension gives the true data on U, V and on the nested opens. The open inclusion is an open continuous map, so the preimage of the original closure is the closure of the preimage. Together with the literal ordered union dictionary this constructs the whole-family density on each open. FIRST-centre open naturality applied to each reflexive history identifies its computed nested FIRST centre with the trace of its actual local FIRST centre. The preceding carrier and ordered-family identities and the genuine base biholomorphism give exact FIRST-centre naturality between the nested opens. Thus the actual trace centres are genuine inverse images under Φ_{UV}. The restrictions of both given arrows are actual blowups of those trace centres. Their actual same-centre

universal property constructs both inverse holomorphic maps and the real commuting square. The reflexive base comparison extends by this actual step to the required restricted first-history isomorphism, retaining even an empty trace centre as a real successor. □

Theorem 392.3 (Terminated retained local recipes have actual intersection maps)**.** *On each of the two actual local FIRST blowups, take a literal difference recipe whose independently computed terminal whole difference is empty. Restrict its tail to the actual inverse image of A_U or A_V under that first blowup. At every common length at least the two original tail lengths, append the genuine identity empty-centre blowups. Construct a unique actual full stagewise comparison of the restricted padded histories over Φ_{UV}. The unpadded prefixes and tails are the actual open restrictions of the retained recipes; the added padding consists of actual identity blowups of the empty centre, with every original position and empty trace step present.*

Proof. All original open data needed for the two local curve products are constructed by the preceding argument. Actual original FIRST admissibility gives an original local TEST on each given first blowup. The genuine terminating recipe restriction theorem then constructs each actual restricted literal tail, exact retained length, full original restricted TEST and actual terminal difference emptiness. Construct the actual first-prefix comparison by the preceding theorem. The target first-prefix TEST is supplied independently by true original TEST restriction and the whole initial product pullback identity. The proved carrier and ordered-family identities identify the actual source restricted recipe with the source input of the genuine-base terminal recipe comparison theorem. Apply that theorem to these exact histories, lengths and computed empty differences. It constructs the actual empty padding and unique full stagewise biholomorphism. No compactness of O, prescribed endpoint map, comparison centre or source-history certificate is used. This is an actual pairwise overlap map; compatibility on triple overlaps and geometric descent remain further assertions. □

Corollary 392.4 (Actual pairwise intersections supply the common open internally)**.** *For two retained actual local recipes on U, V as above, construct the unique actual padded full comparison on their literal intersection $U \cap V$, without any supplied common open or inclusion witness. Every common cut of the actual retained tail lengths is allowed.*

Proof. Take the actual open $O = U \cap V$. Its two inclusions into U and V are the true defining intersection inclusions. Apply the preceding common-open theorem to these exact opens and the given actual first blowups and literal terminating tails. All source geometry, restricted original TESTs, computed centres, actual retained lengths and genuine padding are constructed by that theorem; no new mathematical premise is introduced. □

393 Actual nonlinear smooth-centre normal layers and canonical scalars

The original split normal layers already have the canonical quotient-ring action obtained from ambient multiplication and internally proved ideal annihilation. We now compare them with the literal power-image layers of an actual original nonlinear smooth-centre stalk. This is an explicit local-algebra and chart ingredient for the Bierstone–Milman Hilbert–Samuel equisingularity route, not a sheaf normal-flatness theorem assumed from that source. A genuine regular-centre flattening supplies the coordinate equivalence and the WHOLE original vanishing ideal.

Lemma 393.1 (Whole original power quotients under a genuine ring equivalence)**.** *Let $E : R \simeq S$ be a genuine commutative ring equivalence and let $I, J \subset R$. For every q, the original quotient map induces a genuine equivalence*

$$R/(I + J^{q+1}) \simeq S/(E(I) + E(J)^{q+1}), \qquad [g] \longmapsto [E(g)].$$

It also gives the centre-ring equivalence $R/J \simeq S/E(J)$ with the same actual representative formula. Both inverse maps send $[h]$ to $[E^{-1}(h)]$ and retain entire original ideals, not a finite selection of relations.

Proof. The actual image map preserves ideal sums and products, so it sends $I + J^{q+1}$ to $E(I) + E(J)^{q+1}$. Its inverse has the inverse ideal dictionary because E is bijective. Thus both actual ring maps descend to the full quotient ideals. Their representative compositions are the identity, so quotient surjectivity proves the full inverse identities. The same construction with J proves the centre-ring statement. □

Lemma 393.2 (Every original power-image layer has its actual nonlinear comparison)**.** *For every q, the preceding whole quotient equivalence restricts to an additive equivalence between the literal images*

$$L_q = \mathrm{image}(J^q \to R/(I + J^{q+1})), \quad L'_q = \mathrm{image}(E(J)^q \to S/(E(I) + E(J)^{q+1})).$$

EVERY actual represented class $[g]_q$, $g \in J^q$, is sent to $[E(g)]_q$; the inverse has the corresponding E^{-1} formula. No desired layer map, surjectivity or comparison equality is an input.

Proof. Take the actual quotient equivalence on each original layer element. Its genuine power representative g belongs to J^q, so $E(g)$ belongs to $E(J)^q$ and its class lies in the target image. Conversely every target power element has an actual preimage in J^q because ideal images under the genuine bijection are precisely set images, and images commute with powers. This proves closure for the true inverse quotient map. Restricting the whole quotient inverse identities proves the restricted maps inverse; addition follows from their original ring maps. All class formulas hold on every representative because they are the actual quotient representative formulas, independently of choices. □

Lemma 393.3 (Naturality of the intrinsically descended centre actions)**.** *The actual layer equivalence is semilinear for the actual centre-ring equivalence. For EVERY centre scalar c and actual layer element x,*

$$e_q(cx) = \overline{E}(c)e_q(x).$$

These true degree maps also induce the whole direct-sum semilinear equivalence, preserving every actual degree inclusion and every actual power class. All actions on both sides are their independently defined ambient-product quotient descents.

Proof. Represent c by $r \in R$. The canonical quotient scalar on either side acts by the genuine ambient product. The underlying whole quotient ring map sends rx to $E(r)e_q(x)$ because it preserves multiplication. The centre quotient sends the scalar representative r to $E(r)$, proving the formula. Quotient scalar surjectivity gives EVERY c; the whole layer carrier, not selected generators, is used. Apply these actual additive equivalences componentwise to finite-support elements. Their inverse maps preserve zero, hence finite support, and the inverse laws hold in every component. The scalar formula holds componentwise, giving whole semilinearity and the stated degree inclusion identities. □

Theorem 393.4 (Original regular-centre data supply the whole nonlinear stalk bridge)**.** *Let C be an actual original centre regular of dimension d at x in an original complex manifold of dimension $d+t$. Let A be an original ideal and assume its actual germ-ideal inclusion in the vanishing ideal of C. Put I_x equal to its WHOLE original chart stalk and K_x equal to the WHOLE original vanishing chart stalk of C. Original regularity internally supplies an actual nonlinear flattening with genuine chart domains, a germ equivalence E, and $K_x = E(J)$. It supplies $I_x \subset K_x$, the actual centre-ring comparison and all original split layers and their whole direct sum for $E^{-1}(I_x)$, compared semilinearly to the literal original K_x-power image layers of I_x. Every represented germ and every degree inclusion is retained by the true E formula. No coordinate, ideal, power, layer, scalar or flatness comparison certificate is supplied by the caller.*

Proof. Apply the genuine original regular-centre producer. It constructs an actual holomorphic local equivalence, containing the actual chart point in its source, mapping the original centre exactly onto the coordinate plane on its genuine domain, and fixing zero after recentering. Its entire induced germ equivalence identifies the WHOLE vanishing stalk with the image of the true coordinate normal ideal. The actual original germ-ideal inclusion gives inclusion of every original membership germ; spanning all such germs proves $I_x \subset K_x$. Bijectivity gives $E(E^{-1}(I_x)) = I_x$ internally. The preceding whole quotient, layer and canonical scalar constructions applied to this actual E and these actual whole ideals give every required comparison. The coordinate source/target and centre formula are retained from the producer, rather than replaced by an abstract desired chart dictionary. □

This bridge does not construct the nonlinear coherent family on an entire centre neighbourhood, the intrinsic sheaf module or its stalk identification. In particular it does not supply an unproved flatness assumption at the original centre point. Original-input freeness on a genuine nearby geometric centre domain, chart independence, ambient mixed-degree Hilbert–Samuel comparison, blowing-up descent and the singular-centre driver remain further producers. Higher whole truncations are not assigned a centre quotient-ring action.

394 Actual original coherent families on the real nonlinear centre domain

We use the REAL open target of a genuine original holomorphic flattening. Its inverse followed by the original inverse chart is a total function whose holomorphic assertions are made on that open target only. The original germ ideal is pulled back by the EXISTING literal germ-ideal pullback definition, not by a desired stalk or flatness certificate. Coherence on an open set means the actual local finite-generator and membership characterization of the original coherence definition, with each witnessing neighbourhood contained in that open set. It does not assert global coherence on the entire ambient complex model. This supplies another local ingredient for Bierstone–Milman's Hilbert–Samuel route; no sheaf normal-flatness theorem is being assumed.

Lemma 394.1 (Actual local composition of original span relations)**.** *Suppose $f : Y \to M$ is holomorphic on a neighbourhood of an actual point y. Every actual original finite holomorphic span relation at $f(y)$ pulls back to the relation of the composed functions at y. Only the real local holomorphic map and its actual continuity are used.*

Proof. Write the original relation as a finite sum of actual generators times holomorphic coefficient germs. Compose each actual generator and each coefficient with f. Composition of holomorphic germs is holomorphic because f is holomorphic near y. The original equality holds on a neighbourhood of $f(y)$; continuity pulls it back to a neighbourhood of y. Its pulled equality is precisely the desired finite sum relation. No origin-germ-to-nearby-germ map is introduced. □

Theorem 394.2 (Original coherence supplies real local pullback generators)**.** *Let A be an ORIGINAL coherent germ ideal on M, let $W \subset Y$ be open and let $f : Y \to M$ be holomorphic at every actual point of W. The EXISTING actual pullback f^*A is coherent ON W. At every $y \in W$ the original finite generators at $f(y)$ supply a true open neighbourhood $V \subset W$, containing y, on which their ACTUAL composites with f are holomorphic and generate the pullback at EVERY point of V.*

Proof. Choose actual original coherent generators s_i on an open O containing $f(y)$. Continuity at y gives a true open V containing y, contained in $W \cap f^{-1}(O)$. Each $s_i \circ f$ is holomorphic on V. For a point $z \in V$, every actual pullback generator $a \circ f$ with $a \in A_{f(z)}$ is generated by the $s_i \circ f$: apply the original membership characterization at $f(z)$ and the local composition lemma, then substitute that finite relation into the actual pullback span. Conversely each s_i belongs to

the original ideal at $f(z)$, so $s_i \circ f$ is itself a literal pullback generator. Inclusion of the two actual span families gives the membership equivalence at EVERY z. This is exactly local coherence on W, proved from original coherence. □

Theorem 394.3 (The genuine flattening inverse supplies the original real-open family)**.** *Let Φ be an actual original flattening at the original chart point of x, with its true source contained in the original chart target and $\Phi(x) = 0$. Put*

$$f(z) = \mathrm{chart}_x^{-1}(\Phi^{-1}(z)), \qquad F = f^* A.$$

For original coherent A, the ACTUAL ideal F is coherent on the REAL open $\Phi.\mathrm{target}$; that open contains 0, and $f(0) = x$. One genuine smaller open containing 0 supplies finite actual composed functions and the full membership characterization at ALL its nearby points.

Proof. For every actual target point, Φ^{-1} is holomorphic and takes it to the true source. This source is inside the original chart target, where the actual inverse chart is holomorphic. Their composition is therefore holomorphic at every target point. Apply the preceding local coherence theorem to this actual total function on that true open. The original chart point belongs to the source, and its image is zero; the true image and inverse identities give 0 in the target and $f(0) = x$. Apply the actual local coherence at 0 to obtain the stated single open and finite family. Global coherence outside this real domain is neither assumed nor concluded. □

Theorem 394.4 (Every actual nearby whole translated ideal from one real family)**.** *The original regular-centre geometry and original coherent input internally supply a genuine flattening. There is ONE actual open V containing zero, contained in the real flattening target, and ONE finite original holomorphic family G_i on an actual neighbourhood O of x. The actual inverse flattening maps V to O, and $H_i = G_i \circ f$ is holomorphic on V, such that for EVERY $a \in V$ the WHOLE original translated origin ideal of F is*

$$I(F_a) = \big(H_i(a + z)\big)_i \quad \textit{inside the actual ambient holomorphic germ ring.}$$

Each generator is the germ of its actual composed and translated function, and actual translated membership is the complete finite span relation of those same functions. No nearby ideal or comparison certificate is supplied by the caller.

Proof. Use the preceding internally constructed finite family and its actual membership characterization on the SAME V. Translation by a is a genuine holomorphic affine map taking zero to the actual point a. The actual finite-generation pullback calculus gives translated membership as the full span of $H_i(a + z)$. Their true local holomorphic functions give actual germs. A finite holomorphic span relation at zero is exactly ideal membership of its actual germ; apply this dictionary to EVERY original translated member. Thus the span of ALL original membership germs is contained in the displayed generated ideal. Each displayed generator is itself an actual translated member by the same membership characterization, proving the opposite inclusion. These arguments use local generators on V only, not an unproved global coherence assertion for F. □

The fixed-flattening ALL-nearby nonlinear whole original stalk/comap dictionary still needs its actual pointwise nonlinear germ equivalences. Uniform graded relation/kernel spreading and uniform freeness must be consumed with this genuine open-domain input, not with falsely asserted global coherence on the entire complex model. Intrinsic sheaf normal flatness, ambient mixed-degree Hilbert–Samuel comparison, blowing-up descent and the singular-centre driver remain further producers.

395 Literal ambient canonical invariants under smooth pullback

Retain the original canonical invariant $C_n = R_n^{(n)}$, its unchanged recursive value type $\mathcal{V}_n$, all original marked records, indexed boundary entries and literal whole histories. The zero ambient invariant is the actual original dimension-zero invariant. All statements below concern actual smooth Cartesian blowup histories. Empty manifolds and empty centres are included. The construction specializes the depth-parametric comparison without asserting a dimension inequality on an empty source manifold.

Lemma 395.1 (Actual ambient domain and honest value-type specialization)**.** *A smooth pullback of an original marked record in the ambient bound-n domain belongs to the ambient bound-m domain. For $n \leq m$, the map*

$$A_{n,m} : \mathcal{V}_n \longrightarrow \mathcal{V}_m, \qquad A_{n,m}(v) = D_{m-n,n}(v),$$

uses the literal equality $(m-n)+n = m$ to identify ONLY its target value type. It is strictly increasing and reflects strict order, weak order and equality. For $n > 0$, the unchanged ambient invariant is literally the original depth-n family.

Proof. The genuine original domain pullback producer gives well-formedness, positive marking, nonzero germs along the source marked submanifold, and bound $n + (m-n)$. At every actual source marked point y, surjectivity of the actual differential gives $n \leq m$ by finite-dimensional linear algebra. Consequently $n + (m-n) = m$ at that point, converting the original regular-dimension bound to m. This is a pointwise proof, so an empty source needs no impossible global inequality. All other domain fields are exactly those supplied by the original pullback theorem. For the value map, transport along the natural-number equality is the order isomorphism that becomes the identity after that equality is substituted. Its composition with the previously proved strict $D_{m-n,n}$ is strict, and strict maps of linear orders reflect all three comparisons. If n is positive, writing $n = a + 1$ unfolds the unchanged definition of C_n to $R_n^{(n)}$; proof irrelevance identifies the two proofs of positivity. □

Lemma 395.2 (The actual zero and empty-source whole histories)**.** *If the target ambient dimension is zero, EVERY original canonical target history of an original domain record has all its literal centres empty; its actual Cartesian source history also has all centres empty, hence is canonical for the actual ambient source invariant. If $n > m$, the actual source manifold is empty, so EVERY source history has all literal centres empty and is canonical for its actual ambient invariant. No source canonicality or empty-history certificate is assumed.*

Proof. In target ambient dimension zero, the original domain theorem makes the entire transformed target cosupport empty along each preceding canonical history. Induct on the literal target tower: its current canonical centre is either empty or the maximum locus of that empty cosupport, which is empty. Actual Cartesian centres are literal preimages of these empty sets; induction therefore proves the whole source empty history. The genuine empty-history canonicality theorem applies to its actual source invariant in any source dimension. For $n > m$, an actual source point would have a surjective differential from dimension m onto dimension n, contradicting $n > m$. Hence the source is empty. At each stage a point of any centre projects through the preceding whole tower to a source point, so every centre is empty, again giving actual source canonicality. These arguments retain empty steps and their full original lengths. □

Theorem 395.3 (Whole actual ambient canonicality and values in every dimension)**.** *From ONLY the original ambient bound-n marked domain, original whole C_n-canonical target history and actual smooth Cartesian pair, construct WHOLE source C_m-canonicality internally. At EVERY actual source point z whose image lies in the actual transformed target cosupport, internally*

obtain $n \leq m$ from the genuine differential at its original projection; the entire actual values satisfy

$$C_m(f^*M, Q, z) = A_{n,m}(C_n(M, T, gz)).$$

No source or lower canonicality, branch, value comparison, source compactness or smooth surjectivity is a premise.

Proof. For whole canonicity, target dimension zero and $n > m$ are the preceding actual empty-history cases. Otherwise $0 < n \leq m$, hence $m > 0$. The complete original recursive theorem at target depth n supplies actual source canonicity at depth $(m - n) + n$; the equality to m and the literal positive-ambient identification of both invariants give precisely source C_m-canonicality. For a value point, its original projection is an actual source point, so the differential internally gives $n \leq m$. If $n = 0$, actual transformed target cosupport is empty; the asserted point cannot exist. If $n > 0$, then $m > 0$, and the complete depth-n value formula applies. Transport its source value along $(m - n) + n = m$; dependent equality substitution proves that this transported value is the actual original $R_m^{(m)}$ value, rather than an unrelated value on an identified carrier. Apply the literal ambient-invariant equalities to both sides. This is exactly the displayed formula for the unchanged original invariants. □

Theorem 395.4 (Original compact-target ambient canonical terminal constructor)**.** *For an original compact target in EVERY ambient dimension, original smooth map and original ambient-domain marked record, internally construct BOTH literal whole ambient canonical terminal histories, a genuine Cartesian witness, equal history lengths and exact marked-transform and cosupport preimage dictionaries. The source is in its actual ambient domain. Neither source compactness, source surjectivity, nor terminal/canonical history certificates are assumed.*

Proof. Apply the actual original per-sequence resolution driver to C_n and the original domain to obtain the whole canonical target resolution. Construct its literal whole source Cartesian tower using the genuine smooth tower producer. The first lemma gives its ambient domain, and the preceding whole theorem gives source ambient canonicality internally in every dimension. Target terminality provides actual target testness; genuine relative marked and cosupport dictionaries supply whole source testness, exact transform equality and preimage cosupport. The preimage of the empty target terminal cosupport is empty, proving source terminality. The actual Cartesian witness gives equal lengths and literal preimage centres at every step. This is the ambient specialization of the actual original smooth family construction; arbitrary noncompact global scheduling and unrelated Hironaka source frontiers are not concluded. □

396 Original compact canonical resolution and internally constructed smooth terminal arrows

The original compact endpoint of [3] is integrated with the actual smooth functoriality of [5]. Only original marked-domain data, genuine smooth maps and the explicitly stated compact original bases are inputs. Actual finite terminal histories are OUTPUTS of the genuine per-sequence canonical driver, before final-map uniqueness is consumed. There is no assumption that arbitrary original open neighborhoods are compact. The separate noncompact gluing and scheduling obligations remain open.

Definition 396.1 (The genuine compact canonical driver supplies an actual terminal object)**.** For an original compact ambient manifold X and an original marked record M in its actual ambient invariant domain, internally construct one actual final manifold and actual blowup tower whose nonempty centres are the actual successive canonical maxima and whose whole final cosupport is empty. Whole canonicality is derived internally from the actual nonempty segment. This actual terminal object is an OUTPUT, not a caller history certificate.

Proof. Apply the existing genuine per-sequence canonical driver to the unchanged actual ambient canonical invariant and the original domain. It proves existence of an actual finite canonical segment and genuine geometric resolution: compactness gives attained maxima, actual admissibility supplies each true blowup, and genuine per-sequence finiteness forces termination. Select the actual manifold and tower from THIS proved existence and retain both original driver conclusions. The nonempty canonical segment satisfies the original whole canonical predicate because each true maximal centre is its permitted maximal branch. Thus all canonicality and terminality used by later arrows are genuinely produced before they are consumed. □

Theorem 396.2 (Original compact marked data internally supply one unique genuine terminal smooth arrow)**.** *For original compact source and target manifolds, genuine smooth $f : Y \to X$ and ONLY original target marked-domain M, internally construct the actual compact canonical terminal objects of M and f^*M, and one genuine smooth map between their actual final spaces. Its WHOLE original projection square commutes and is a full holomorphic pullback for every test manifold and dimension. It is the unique such genuine final submersion; every competing holomorphic map with the same actual full square is the same actual arrow. No source domain, canonical terminal history, final comparison, submersion or universal property is an original caller premise.*

Proof. The original smooth marked-domain producer gives the true ambient domain of the ENTIRE original pulled record. Invoke the first node separately on the original target and source, using their stated actual compactness. Their genuine canonical segments give whole canonicality; their actual terminality is retained. The completed finite final-map constructor now applies to these INTERNALLY supplied histories and produces the true submersion, whole projection identity and genuine full universal property. Choose that actual arrow and retain those properties. The intrinsic finite final-map uniqueness theorem on this SAME actual source tower identifies any genuine holomorphic competitor having that full square. Consequently existence and uniqueness require only original compact data and smoothness, not a desired terminal object or arrow certificate. □

Theorem 396.3 (The internally constructed compact identity arrow is genuinely biholomorphic)**.** *For ONLY an original compact marked-domain input, the internally selected identity-base terminal arrow is the actual intrinsic biholomorphic comparison between its internally constructed source and target terminal objects. Whole original marked identity pullback is supplied internally. In particular there exists a genuine biholomorphism whose forward map is THIS actual selected arrow and whose entire original projection is unchanged.*

Proof. The preceding node internally supplies both terminal objects. Actual whole marked pullback by base identity equals the original record, by genuine coherent ideal identity pullback, original carrier preimage, literal ordered boundary identity and unchanged mark. Transport only the internally produced source canonicality and terminality through THIS real record equality. The proved independent-history identity theorem internally constructs the actual comparison, its inverse and universal property and identifies its forward map with the actual identity-base factory arrow. Both terminal histories remain genuine outputs; no comparison or invertibility certificate is presumed. Its whole projection formula is the retained true comparison formula. Under the same whole record identification this is the identity arrow on the actual chosen compact terminal object. □

Theorem 396.4 (Actual compact terminal arrows compose after the internally proved whole record identification)**.** *For composable genuine smooth maps $h : Z \to Y$, $f : Y \to X$ of original compact manifolds and ONLY an original target marked-domain M, construct all three actual terminal objects internally from the original direct records $M, f^*M, (fh)^*M$. Internally identify the ENTIRE direct pulled record with the iterated record and its actual source domain. After this genuine record transport, the composed actual component arrows equal the actual direct arrow as*

ENTIRE functions between the SAME actual selected final spaces. No caller histories, record equality, final comparison or composition law is required. This states record transport explicitly; it does not equate independently indexed uncompressed tower records.

Proof. The true composite smooth map gives the original ambient domain of its whole direct pullback, so the compact driver internally constructs its actual terminal object. The first two nodes similarly internally supply the original intermediate and target terminal objects and their genuine canonicality and terminality. Genuine coherent pullback composition identifies the ENTIRE direct marked record with the iterated one, retaining all original carrier, indexed boundary members, whole ideal and mark. Transport the internally supplied direct-source canonicality and terminality through THIS actual equality before constructing the second component arrow. It is the chosen terminal construction on the same selected source object under the true whole record identification. The completed finite composition law applies to these actual internally supplied objects and internally proved domains/record equality. It pastes the genuine component squares, then intrinsic uniqueness identifies the whole composite with the direct factory arrow. This proves the original compact endpoint coherence in all ambient dimensions, including empty manifolds. General noncompact global construction, sheaf normal cones and broader categories remain open. □

397 Actual finite smooth arrows on the original global terminal manifolds

The smooth functoriality comparison of [5, 3] is now transported to the genuine original global terminal manifolds, using actual finite canonical terminal histories. A compact original target produces the target history internally, and actual smooth pullback produces the entire source history even for a noncompact source. This statement includes the actual original projection and holomorphic Cartesian law; it imposes compactness only on the internal finite-history producer. Arbitrary noncompact smooth map gluing remains a separate conclusion.

Lemma 397.1 (Biholomorphic changes of final spaces preserve genuine submersions)**.** *A genuine holomorphic submersion remains a genuine submersion after precomposition and postcomposition with biholomorphisms of the source and target final manifolds.*

Proof. The genuine precomposition theorem pulls back the submersion's actual domain chart by the source biholomorphism, retaining the same fixed complement and linear projection presentation. Applied to the identity submersion, it says that every actual biholomorphism is a submersion. The genuine composition theorem therefore permits postcomposition by the target biholomorphism. The complement is fixed over the entire map, so this proves the stated global submersion and not merely pointwise surjectivity of a differential. □

Theorem 397.2 (Actual finite original smooth arrows descend to the original global terminal manifolds)**.** *Let $f : Y \to X$ be a genuine holomorphic submersion and let M be an original marked ideal in the invariant domain on X. Given actual finite canonical terminal histories for M and f^*M, construct a smooth arrow $G : \widetilde{Y}_{f^*M} \to \widetilde{X}_M$ between their genuine original global terminal manifolds. It is a holomorphic submersion, it satisfies $P_M G = f P_{f^*M}$ at every point, and its square with these same entire original projections is Cartesian among holomorphic maps of complex manifolds.*

Proof. The original finite terminal producer supplies the actual finite smooth Cartesian arrow $g : E \to D$. The preceding finite-identification theorem constructs genuine biholomorphisms $C_Y : \widetilde{Y}_{f^*M} \to E$ and $C_X : \widetilde{X}_M \to D$ from the actual original gluing, with whole projection equalities. Define $G = C_X^{-1} g C_Y$. The preceding submersion transport lemma proves it is a submersion, hence holomorphic. Apply the whole target projection formula at $C_X^{-1} g C_Y(z)$, cancel

the inverse, then apply the finite arrow's projection equality and the whole source projection formula. The result is exactly $P_M G(z) = f P_{f^*M}(z)$.

For the Cartesian property take an arbitrary complex manifold W and holomorphic maps $a : W \to Y$, $b : W \to \widetilde{X}_M$ with $fa = P_M b$. The whole target formula shows that $(a, C_X b)$ is a cone for the actual finite Cartesian square. Its unique holomorphic lift $w : W \to E$ gives a lift $C_Y^{-1} w$ into the genuine original global source. The source projection equality and both inverse equations give the required source leg; the finite target leg and the target inverse equation give the G leg. Any other holomorphic lift, composed with C_Y, is a finite lift for the same cone, because the two actual whole projection formulas and $G = C_X^{-1} g C_Y$ identify its legs. Finite uniqueness identifies these maps; injectivity of C_Y gives uniqueness of the global lift. □

Theorem 397.3 (An actual finite target history internally produces its smooth source history)**.** *An original finite canonical terminal history for M and a genuine smooth map $f : Y \to X$ internally produce a finite canonical terminal history for f^*M on the entire original source Y, without compactness of Y.*

Proof. Apply the actual Cartesian blowup-tower producer to every step of the entire finite target history. It constructs an actual finite source tower and a genuine Cartesian smooth arrow. Original ambient canonicality transport identifies the canonical invariant values and centres on this actual tower. The entire marked pullback dictionary identifies its transformed marked ideal with the pullback of the target transform, including every indexed boundary member, and supplies actual testness. The true cosupport dictionary identifies source terminal cosupport with the inverse image of the empty target terminal cosupport, hence empty. These are genuine original canonicality and terminality, with the source history constructed internally rather than given by the caller. □

Theorem 397.4 (A compact original target internally produces the actual global smooth Cartesian arrow)**.** *If the target X is compact, only the original marked-domain input M and the genuine submersion f are needed to construct the preceding global arrow, its submersion property, its entire original projection equality, and its holomorphic Cartesian law. No histories, local arrows, local comparison compatibility or global identifications are caller assumptions.*

Proof. Original smooth pullback supplies the invariant-domain predicate for the whole pulled-back marked record. The compact canonical driver constructs the actual finite history on the compact target, with canonicality and terminality as its outputs. The actual smooth pullback tower producer constructs a finite source history from this entire target history. Original smooth canonicality transport supplies its canonicality, and the entire marked and cosupport dictionaries supply its terminality. Thus the source need not be compact. Apply the preceding theorem to these actual histories and their genuine output properties. All remaining objects are the original terminal manifolds already constructed from M and f^*M; the finite identifications retain their entire original projections. Thus every required datum is an internal producer output. □

398 The entire original smooth recursive family

Keep the actual original marked records, all indexed boundary members and whole histories, including empty and repeated members and empty centres. For positive ambient dimensions n, m write $t = m - n$. The unchanged original family value types carry the established strict maps $D_{t,k}$. The genuine construction uses recursion on the target family depth; within each depth its Step One history proof simultaneously constructs source canonicity and complete values by induction on the actual whole history.

Theorem 398.1 (Complete original recursive smooth values and canonical histories)**.** *For EVERY original target family depth k, original M in the bound-k domain, original whole $R_k^{(n)}$-canonical history T, and actual whole smooth Cartesian pair (Q, T) over f, internally construct*

whole source $R^{(m)}_{t+k}$ canonicality. At EVERY actual source point z whose image is in the target transformed cosupport,

$$R^{(m)}_{t+k}(f^*M, Q, z) = D_{t,k}(R^{(n)}_k(M, T, gz)).$$

No source or lower canonicality, desired comparison, branch certificate, source compactness or smooth surjectivity is assumed.

Proof. The depth-zero original domain makes the entire target cosupport empty. Original target canonicality therefore makes all target centres empty; the literal source centres are empty and the actual source cosupport is empty. This gives actual source canonicality at depth t and the correctly vacuous value formula. Depth one is the complete original curve family. Assume the complete lower family result at target depth $k+1$, uniformly for ALL actual original marked records, restricted bases and whole history pairs of the same two ambient dimensions. We prove depth $k+2$. First construct the full Step One result over the lower R_{k+1} family. Its proof inducts on the actual Cartesian history witness, simultaneously constructing source canonicality and all current values. At an empty history canonicality is immediate. At a last step, the induction hypothesis supplies all preceding-stage values and source canonicality. Empty centres or empty source preimages give allowed empty source steps. Otherwise choose one actual source point over the full target maximum. Its image dominates ALL target cosupport values. Source maximality dominates its visible value, and the strict Step One dimension map reflects this inequality. The actual cosupport preimage then proves that the literal centre preimage is exactly the full source maximum. Thus CURRENT source Step One canonicality is proved BEFORE its current independently chosen lower contact data are used. At a special point genuine level and specialness transport give both whole pairs with infinite lower value. At a nonspecial positive level the original target canonical-tail producer supplies the true lower bound $k+1$, whole lower target canonical history, genuine mask and cosupport. The actual whole positive-member lower diagram transports the entire marked record and tail. Apply the depth induction hypothesis to this ACTUAL lower pair to obtain complete source lower canonicality and ALL lower values internally. At a nonspecial zero level the actual selected-contact producer constructs the literal COMMON source model using the transported target contact function, its correct source domain, actual restricted Cartesian tail, mask inclusion in the transported tangent preimage and avoidance of EVERY original source indexed member. Apply the depth induction hypothesis to construct COMMON source lower canonicality and its complete values. CURRENT source Step One canonicality, already proved above from preceding histories, supplies the independently chosen OWN source lower data. Apply the existing genuine per-sequence zero-contact anchor with these OWN lower data and the just-proved COMMON lower canonical history. The entire own/common values agree without equating independently chosen contact functions. The depth induction comparison on the common model therefore gives the own lower comparison. Expanding both masked values gives the whole Step One dimension map in every actual branch and completes its simultaneous history induction. Now inspect the original general outer nonmonomial order. At zero use the complete actual monomial formula with genuine SNC cardinality and the whole colex boundary. At positive order the original general canonical-tail theorem internally supplies the actual companion bound/domain, maximal order, whole Step One canonical target history and cosupport. Its actual entire Cartesian companion diagram and whole record/mask/order/run dictionaries permit the complete Step One result just proved. Expanding the true general definition gives exactly $D_{t,k+2}$ at ALL inner orders. Transfer general source canonicality by ordinary history induction using these complete preceding values and the same true visible-maximum argument with strict $D_{t,k+2}$. This completes the depth induction. Lower desired properties occur only as private genuine induction hypotheses; every public endpoint consumes only original input and internally constructs its source conclusions. □

Theorem 398.2 (Original depth-parametric terminal smooth families)**.** *From compact original target, positive ambient dimensions, original smooth map and original bound-k marked domain,*

for EVERY k internally construct BOTH whole original canonical terminal histories of the actual families $R_k^{(n)}$ and $R_{t+k}^{(m)}$, with unchanged history length, literal preimage centres, correct source domain and entire marked/cosupport transport. No source compactness, terminal history, or canonicality is assumed.

Proof. The original per-sequence driver gives the target canonical resolution. The whole actual smooth tower constructor gives the source history and its Cartesian witness. The complete depth-parametric theorem above gives source canonicality internally. Genuine original-domain pullback supplies bound $t + k$, the actual whole marked dictionary gives testness and entire marked transform equality, and the whole cosupport dictionary makes the final source cosupport the preimage of the target empty cosupport. This proves source terminality without source compactness or smooth surjectivity. Literal canonicalInvariant ambient-depth specialization and ambient-zero cases are subsequent arithmetic/geometric obligations; no differing value types are silently identified by this family theorem. □

399 Actual Cartesian maps between independently chosen terminal histories

Retain the ORIGINAL source and target marked records and their given whole canonical terminal histories, including different actual empty-step counts. The smooth functoriality route of [5] is adapted by constructing an actual Cartesian source history internally, then comparing its genuine final space with the independently given source terminal history. No source history is presumed to be an actual Cartesian pullback, and no equality of differently sized indexed exceptional lists is asserted. The resulting final-space map is holomorphic and its entire projection square has the genuine holomorphic pullback universal property.

Theorem 399.1 (Original terminal inputs internally compare the given source with one actual Cartesian model)**.** *Let $f : Y \to X$ be an actual smooth submersion, M an original bound-n record on X, and let T and S be GIVEN whole terminal canonical histories of M and its actual marked pullback on Y for their actual ambient canonical invariants. From these original inputs and primitive individual geometric blowup choices, internally construct the actual Cartesian history Q over the SAME T, its whole source canonicality and terminality, and a genuine biholomorphism $\eta : S_{\mathrm{final}} \to Q_{\mathrm{final}}$ with $p_Q\eta = p_S$. No source domain, actual-Cartesian-history, compressed history, source comparison or source canonicality for Q is supplied. Every ambient dimension, genuine empty manifold and actual empty step is included, without source compactness or surjectivity of f.*

Proof. Compute the actual whole source history from the primitive choices and the GIVEN original target history. Its real stage witnesses give actual Cartesian source centres and maps. Original smooth domain pullback supplies the source ambient domain. The completed entire ambient canonical history producer supplies whole canonicality of THIS Q internally. Given target terminality supplies true target geometric testness. The original relative marked dictionary gives source testness, and its entire cosupport dictionary is the preimage of the actual empty target terminal cosupport. Therefore THIS Q is terminal. Now apply the completed original ambient terminal choice comparison to the GIVEN S and THIS internally constructed canonical terminal Q. In positive dimension it internally removes their actual empty steps before genuine canonical terminal comparison. In dimension zero it internally constructs their actual whole projection biholomorphisms. Both routes give an actual final biholomorphism satisfying $p_Q\eta = p_S$. No equal recorded length or caller-normalized comparison is required. □

Theorem 399.2 (The original given final spaces admit a holomorphic Cartesian comparison map)**.** *With ONLY the same original inputs, internally produce a holomorphic map $g : S_{\mathrm{final}} \to T_{\mathrm{final}}$ with $p_T g = f p_S$. Its ENTIRE square is a genuine holomorphic pullback: for EVERY actual complex*

manifold W of any dimension and compatible holomorphic maps $u : W \to Y$, $v : W \to T_{\text{final}}$ with $fu = p_T v$, there is EXACTLY ONE holomorphic $w : W \to S_{\text{final}}$ with $p_S w = u$ and $gw = v$. No final map, lift, universal-property or source-history comparison certificate is a caller premise. This compares GIVEN finite terminal spaces; arbitrary noncompact resolution existence, global gluing, uniqueness from the base projection alone and functor cocycles remain separate obligations.

Proof. The first theorem internally supplies the actual model Q, its real submersion lift g_Q and actual final comparison η. Define $g = g_Q \eta$. Both factors are holomorphic, and the real whole Cartesian projection identity together with $p_Q \eta = p_S$ gives $p_T g = f p_S$ at every point. For an actual compatible holomorphic cone (u, v), the genuine whole Cartesian universal property of Q internally gives a unique holomorphic lift w_Q with $p_Q w_Q = u$ and $g_Q w_Q = v$. Choose $w = \eta^{-1} w_Q$. The true inverse identity and the actual projection comparison give $p_S w = u$ and $gw = v$; composition of the genuine holomorphic maps gives holomorphicity. Any other holomorphic lift w' produces a holomorphic lift $\eta w'$ into THIS Q, with both original cone equations by the same whole dictionaries. Its genuine universal uniqueness gives $\eta w' = w_Q$. Injectivity of the actual final biholomorphism gives $w' = \eta^{-1} w_Q$. This proves the full actual universal property on the original final spaces, rather than a transported certificate supplied by the caller. Different empty-step counts and indexed exceptional lists remain recorded; only genuine final-space geometry is compared. □

400 Actual finite terminal smooth maps commute with independent history choices

The finite smooth functor of [5] must be compatible with independent canonical terminal histories on both source and target. Keep the original given histories, including different real empty-step counts. Construct their actual comparisons internally from original canonicality and terminality; prove whole-map naturality from true universal properties. This supplies a finite gluing prerequisite, not original noncompact resolution existence or a global schedule.

Lemma 400.1 (Actual biholomorphic whole projection comparisons transport a full pullback square)**.** *Let $g : E \to D$ have the genuine full holomorphic pullback square over $f : Y \to X$, with actual projections $p : E \to Y$, $t : D \to X$. Let actual biholomorphisms $\eta : E' \to E$, $\theta : D' \to D$ preserve the whole original projections: $p\eta = p'$, $t\theta = t'$. Then $\theta^{-1} g \eta$ has the genuine full holomorphic pullback property with p', t'. In particular, for an actual biholomorphism $\eta : E' \to E$ with $p\eta = p'$, the whole square of η over base identity is a genuine holomorphic pullback. No transported unique lift is an assumed certificate.*

Proof. For an arbitrary holomorphic cone $(a : W \to Y, b : W \to D')$ with $fa = t'b$, the actual old square applies to $(a, \theta b)$, since the genuine target comparison gives $t\theta b = t'b$. It constructs its unique holomorphic lift $w : W \to E$. The actual holomorphic map $\eta^{-1} w$ has projection p' equal to a by $p\eta = p'$ and the true inverse identity. Its other equation is $\theta^{-1} g \eta \eta^{-1} w = \theta^{-1} g w = b$, using $gw = \theta b$ and both actual inverse equations. A competing lift v gives the old lift ηv: its projection follows from $p\eta = p'$, and applying actual θ to its second equation gives $g\eta v = \theta b$. Old-square uniqueness gives $\eta v = w$; actual injectivity identifies the entire lift. For the identity-base square of η, construct $\eta^{-1} b$ directly. The actual projection equation and inverse identity give both cone equations; actual injectivity gives full uniqueness. This covers every test manifold and dimension without a projection injectivity assumption. □

Theorem 400.2 (The actual identity-base terminal map between independent histories is their intrinsic comparison)**.** *From only an original marked-domain input M and TWO GIVEN canonical terminal histories T', T of M, the actual finite terminal map over base identity is the actual intrinsic biholomorphic comparison from T' to T. Whole marked identity pullback is constructed*

internally before transporting the original source history inputs. In particular the actual identity-base final map is genuinely biholomorphic even when the real histories have different empty-step counts.

Proof. Coherence of the original whole ideal gives actual whole marked identity pullback equal to M, preserving carrier, every indexed boundary and the mark. Thus the given T' canonicality and terminality supply the true source inputs of the original identity-map factory. Original ambient comparison existence and uniqueness internally construct the actual biholomorphism η over the whole original projections. Its forward map is genuinely holomorphic and has true whole commutativity. The preceding identity-square construction gives its full holomorphic universal property internally. The intrinsic final-map characterization on the SAME actual source tower T' therefore identifies this actual comparison with the factory map. No comparison or universal property is an original caller premise; no differently sized towers or exceptional lists are equated. □

Theorem 400.3 (Original finite terminal smooth maps are natural under independent source and target histories)**.** *From only the original target marked-domain M, genuine smooth $f : Y \to X$, TWO GIVEN original canonical terminal histories S', S of the actual pulled record on Y, and TWO GIVEN original canonical terminal histories T', T of M, construct their intrinsic source and target comparisons $\eta : S'_{\text{final}} \to S_{\text{final}}$, $\theta : T'_{\text{final}} \to T_{\text{final}}$ internally. The actual finite smooth terminal maps g', g satisfy the ENTIRE naturality identity $\theta g' = g\eta$. Thus replacement of both original histories gives the genuine conjugate map $g' = \theta^{-1} g\eta$. All ambient dimensions and empty histories are included, without caller comparison, source-domain, desired naturality or universal certificates.*

Proof. Original smooth-domain transport internally supplies the actual marked-domain of f^*M. Apply the proved original ambient comparison constructor to the two given source histories, and separately to the two target histories. The genuine comparisons η, θ have both holomorphic maps and true whole projection identities. The original finite map factory internally supplies g with holomorphicity, whole commutativity and its genuine universal property. Construct the actual map $c = \theta^{-1} g\eta$. Its holomorphicity follows from these three actual holomorphic maps. Its projection satisfies $p_{T'} c = p_T g\eta = f p_S \eta = f p_{S'}$, where $p_{T'} \theta^{-1} = p_T$ is derived from the true target comparison and its actual inverse. The first lemma supplies the genuine full universal property of c, not a presumed transported certificate. The original factory independently supplies g' on the SAME actual source tower S'. Intrinsic final-map uniqueness identifies $c = g'$. Applying the actual target comparison cancels its actual inverse and gives $\theta g' = g\eta$ as whole functions. All comparisons and universal properties are genuine internal producers from the original inputs; noncompact schedule existence and global gluing remain OPEN. □

401 Intrinsic finite terminal smooth maps with identity and composition

Keep the GIVEN original canonical terminal histories and all their actual recorded empty steps. The smooth functoriality route of [5] is adapted by internally constructing the genuine final Cartesian submersion, then proving its intrinsic uniqueness from actual universal properties and actual blowup projection geometry. No final comparison, coherent schedule or functor-law certificate is a caller premise. This gives the functor laws on GIVEN finite terminal spaces; noncompact schedule existence and global resolution gluing remain separate source obligations.

Lemma 401.1 (Genuine whole holomorphic pullback squares paste)**.** *Let actual base maps $h : Z \to Y$, $f : Y \to X$ and actual final maps $b : F \to E$, $a : E \to D$ give TWO genuine holomorphic pullback squares, with true whole projections $p_F : F \to Z$, $p_E : E \to Y$, $p_D : D \to X$. Assume h, b are holomorphic and $p_E b = h p_F$, as supplied by actual final smooth maps. Then the composed*

ENTIRE square over fh, with final map ab, is a genuine holomorphic pullback for every actual test manifold of every dimension, without any desired pasted lift as an input.

Proof. For an arbitrary holomorphic cone $(u : W \to Z, v : W \to D)$ with $fhu = p_D v$, the first actual universal property supplies a unique holomorphic w_E with $p_E w_E = hu$, $aw_E = v$. Its first identity gives the real compatibility for the second square. That actual universal property supplies a unique holomorphic w_F with $p_F w_F = u$ and $bw_F = w_E$. Its equations imply $abw_F = v$, so it is a true pasted lift. Any other holomorphic w' with $p_F w' = u$, $abw' = v$ gives a holomorphic bw'; actual commutativity gives $p_E bw' = hp_F w' = hu$. First universal uniqueness therefore gives $bw' = w_E$. Second universal uniqueness then gives $w' = w_F$. This retains the arbitrary actual cone and both true whole projection maps, with no compactness, surjectivity or stage-deletion condition. □

Theorem 401.2 (Two actual final pullback maps on an original blowup tower are intrinsically equal)**.** *For ANY actual source tower $S : Y \leftarrow E$, two holomorphic maps $g_1, g_2 : E \to D$ with the SAME original base map $f : Y \to X$ and target projection p_D, true whole commutativity and genuine full holomorphic pullback universal properties are equal. Original final-map constructors supply these properties internally; no final-map equality or source automorphism uniqueness certificate is assumed.*

Proof. Use the first genuine universal property on the actual cone (p_S, g_2). It gives a holomorphic $a : E \to E$ with $p_S a = p_S$ and $g_1 a = g_2$. The second property on (p_S, g_1) gives holomorphic b with $p_S b = p_S$, $g_2 b = g_1$. Actual universal uniqueness in the second square on (p_S, g_2) gives $ba = \mathrm{id}$: both ba and identity have its two real cone equations. Likewise first-square uniqueness on (p_S, g_1) gives $ab = \mathrm{id}$. The true holomorphic maps and inverse equations therefore construct an ACTUAL biholomorphism with forward map a and inverse b. Its whole projection is p_S, the same as the actual identity biholomorphism. The proved intrinsic biholomorphism uniqueness over the ACTUAL tower S identifies it with identity, internally using the true first-blowup universal geometry. Thus $a = \mathrm{id}$ and $g_1 a = g_2$ gives $g_1 = g_2$. No arbitrary holomorphic map preserving a projection is silently presumed invertible: both inverse maps were constructed from the TWO real universal properties first. □

Definition 401.3 (The original terminal inputs internally determine one actual final smooth map)**.** From ONLY the original target marked-domain input, genuine smooth f and GIVEN original source and target whole ambient canonical terminal histories S, T, internally choose the actual final submersion supplied by the proved original constructor. Its whole square commutes and has the FULL genuine holomorphic pullback property. These actual properties characterize it intrinsically among such genuine holomorphic final maps, so every primitive geometric blowup choice and actual comparison choice produces the SAME original-final map. All dimensions and differing actual empty steps are included, with no caller actual-model canonicality, final comparison, submersion, universal property or equality certificate.

Proof. The genuine default primitive choice consists only of individual actual geometric blowups. Apply the completed original final-submersion producer to it and the GIVEN original inputs. It internally supplies existence of an actual final submersion, original whole projection identity and every actual unique holomorphic cone lift. Select the actual map proved to exist and retain these proved properties. A map produced from any other true primitive blowup/comparison choices has the same two whole projections and internally supplied holomorphic universal property. The preceding intrinsic uniqueness theorem on the SAME original actual source tower identifies it with the selected map. Thus the selection is intrinsic; no caller supplies its desired characterizing properties. Original source canonicality/terminality is the GIVEN history input, while actual Cartesian model canonicality/terminality is constructed internally. □

Theorem 401.4 (Actual original terminal smooth maps satisfy identity and composition)**.** *For every original marked-domain input and GIVEN finite whole ambient canonical terminal history, the internally determined terminal map for the actual base identity is the actual final identity. For composable genuine smooth maps $h : Z \to Y$, $f : Y \to X$ and GIVEN original canonical terminal histories R, S, T of the actual original pulled records, the actual first terminal map followed by the actual second equals the internally determined terminal map for fh. The WHOLE original record pullback equality and source domain are supplied internally. No caller composite-record identity, direct-map equality, empty-compression or functor-cocycle certificate is assumed.*

Proof. The actual identity preimage leaves the carrier and EVERY indexed original boundary member unchanged, coherent whole ideal pullback by identity is the original ideal, and the mark is unchanged. Thus the actual marked identity pullback is the SAME original record. Original canonicality and terminality therefore supply both true history inputs of the identity map constructor. The actual final identity is holomorphic, has the same whole projection identity, and has the genuine identity square universal property: a compatible cone (u, v) has unique lift v, since its second equation is identity. Intrinsic uniqueness identifies the internally constructed terminal map with this actual identity.

For composition, original smooth-domain pullback gives the actual source domain of the whole pulled record on Y. Genuine holomorphic coherent pullback composition identifies the ENTIRE original record for fh with the iterated original pulled record, including the SAME ordered boundary and mark. It internally transports the GIVEN R canonicality and terminality to the correct iterated record, so both component terminal maps are internally supplied by their original constructors. Their actual submersion properties give holomorphicity and genuine whole projection commutativity. The first theorem pastes their two actual universal squares into the whole square over fh. The actual direct constructor supplies the same two WHOLE projections and a genuine direct universal square. The second theorem on THIS SAME actual source tower R identifies the pasted final map with the direct map. This proves the original finite identity and composition laws, including empty histories, actual empty manifolds and all ambient dimensions, without equating differently sized recorded towers or exceptional lists. Noncompact schedule existence, global gluing and broader analytic/algebraic categories remain open. □

402 Actual submersions on independently chosen terminal spaces

The final comparison keeps the GIVEN original source and target canonical terminal histories, with all actual empty steps and their unchanged marked records. The smooth route of [5] uses true biholomorphic changes of the source charts. This gives an actual submersion on the original terminal spaces, with the same full holomorphic pullback universal property, without a caller submersion or chart certificate.

Lemma 402.1 (A genuine biholomorphism pulls every genuine maximal chart back to a maximal chart)**.** *Let $\eta : E \to Z$ be a genuine biholomorphism of actual complex manifolds with the SAME model dimension. For ANY actual maximal holomorphic chart a on Z, the actual chart $a\eta$ on $\eta^{-1}(\operatorname{source} a)$ is maximal holomorphic. Its inverse is $\eta^{-1}a^{-1}$ and it has the SAME chart target as a. No chart-compatibility certificate is an input.*

Proof. Compose the global genuine homeomorphism underlying η with the actual partial homeomorphism a. Its true source is the displayed preimage and its target is the original target: the inverse image lies in the global homeomorphism target. The actual partial inverse is exactly $\eta^{-1}a^{-1}$. Maximality of a gives holomorphicity of a on its actual source and of a^{-1} on its actual target. Compose these with the genuine holomorphic η and η^{-1}, on the respective true domains. Both actual partial maps are therefore holomorphic. A partial homeomorphism whose forward and inverse maps are holomorphic on their actual domains belongs to the maximal holomorphic

atlas: its two changes to every original chart are compositions of those genuine holomorphic partial maps with the original chart and its inverse. This proves actual chart maximality, including empty charts and zero-dimensional manifolds. □

Theorem 402.2 (True source biholomorphic changes preserve the actual submersion condition)**.** *For ANY genuine submersion $g_Q : Z \to D$ and genuine biholomorphism $\eta : E \to Z$ between SAME-dimensional source models, $g_Q\eta$ is a genuine submersion. Its genuine fixed complementary normed space can be retained throughout. No caller local-coordinate, complement-transport or composite submersion conclusion is assumed.*

Proof. Use the actual fixed-complement presentation supplied by the given submersion, at the actual point ηe. It gives maximal source and target charts a, b, actual point/domain membership and a continuous linear model equivalence under which bg_Qa^{-1} is the coordinate projection on the ENTIRE true source chart target. Pull a back through η by the first lemma and retain the SAME b, linear model equivalence and fixed complementary space. Its source is the exact η-preimage of the original source; the original domain condition therefore proves that it maps under $g_Q\eta$ into b's actual source. Its actual inverse is $\eta^{-1}a^{-1}$. On its true chart target, which is the original target, the identity $\eta\eta^{-1} = \mathrm{id}$ makes the new chart formula $bg_Q\eta(\eta^{-1}a^{-1}) = bg_Qa^{-1}$. Thus the original coordinate projection formula holds on the ENTIRE actual target, giving the true fixed-complement submersion at e. This works at every point with ONE fixed complement, so it gives the genuine global submersion condition. Empty manifolds and dimension zero are included without a fabricated global dimension inequality or a nonvacuous value seed. □

Theorem 402.3 (Original independent terminal choices internally supply an actual Cartesian submersion)**.** *From ONLY an original genuine submersion f, original ambient-domain target marked record, primitive individual blowup choices and GIVEN independently chosen source and target whole ambient canonical terminal histories S, T, internally construct a genuine submersion $g : S_{\mathrm{final}} \to T_{\mathrm{final}}$ with $p_Tg = fp_S$ and the FULL actual holomorphic pullback universal property. Every actual test manifold of every dimension and every compatible holomorphic cone has exactly one holomorphic lift. No caller actual-model canonicality, final comparison, composite submersion or desired universal certificate is supplied. All ambient dimensions and different real empty-step counts are included.*

Proof. The completed original terminal Cartesian source producer internally constructs one actual Cartesian model Q over the GIVEN T, its true source domain, whole canonicality and terminality, and a genuine biholomorphism $\eta : S_{\mathrm{final}} \to Q_{\mathrm{final}}$ with $p_Q\eta = p_S$. The actual stage witness gives a genuine submersion g_Q and its actual whole Cartesian universal property. Choose $g = g_Q\eta$. The preceding theorem proves THIS g is a genuine submersion. Its real whole projection identity follows from the model projection square and $p_Q\eta = p_S$. For every actual compatible holomorphic cone, the model internally gives its genuine unique holomorphic lift w_Q. The actual $\eta^{-1}w_Q$ is holomorphic and satisfies both original cone equations by the true inverse and projection identities. Any other lift produces $\eta w'$ with the same two model cone equations; actual model uniqueness and injectivity of η give $w' = \eta^{-1}w_Q$. This proves the full genuine universal property on the ORIGINAL independently given final spaces. No equality of different recorded exceptional lists or uncompressed towers is used. Intrinsic choice/cocycle uniqueness and noncompact scheduling/gluing remain subsequent original source obligations. □

403 Original dimension-zero canonical histories under smooth pullback

This is the genuine base of the history-length and invariant-depth induction for smooth functoriality. The original domain has dimension bound zero, but the source has the actual added fibre dimension $t = m - n$. All boundaries, including empty and repeated entries, and every empty

blowup step are retained. We use the original family $R_k^{(n)}$ and its unchanged strict map $D_{t,k}$. Sources are Włodarczyk's dimension induction and extensions by empty blowups and Bierstone–Milman's smooth functoriality [3, 5].

Theorem 403.1 (The original zero-dimensional canonical history is empty)**.** *Let $n > 0$, let M be in the original invariant domain with dimension bound zero, and let the original whole history T be canonical for $R_0^{(n)}$. Then every literal centre of T is empty.*

Proof. Induct on the actual history. The trivial history has no centres. For a last step, its preceding history is canonical and hence an actual test history by the original family's invariant property. The original domain's nonzero-along condition and dimension-zero regularity imply empty cosupport at every such test stage. The maximum locus on that empty cosupport is empty. The definition of canonicality therefore makes the last centre empty whether it is an explicitly empty extension or a maximum centre. Combine this with the induction hypothesis. This uses the original domain, not a caller-supplied vanishing or empty-history certificate. □

Theorem 403.2 (Actual pullback retains every empty step)**.** *For a genuine actual Cartesian smooth history pair (Q, T), if every literal centre of T is empty then every literal centre of Q is empty.*

Proof. Induct on the actual Cartesian witness. Its trivial case is immediate. Each actual source centre is the literal preimage of the target centre under the actual preceding stage map. The preimage of the proved empty set is empty; the induction hypothesis treats the whole preceding pair. No step is removed or reindexed. □

Theorem 403.3 (Actual original zero-family base, with true source depth)**.** *Let $n, m > 0$, $t = m - n$, and let an actual smooth Cartesian pair (Q, T) pull back the original M and its original $R_0^{(n)}$-canonical history, with M in the original dimension-zero domain. Internally obtain the true source domain of dimension t, both empty histories, canonicality of Q for the actual original $R_t^{(m)}$, empty final cosupports, and terminal geometric resolutions on both sides. On the actual source cosupport the whole value equality $R_t^{(m)}(f^*M, Q, z) = D_{t,0}(R_0^{(n)}(M, T, gz))$ holds.*

Proof. The first two theorems construct both empty histories from the original inputs. The actual pair supplies its base submersion, so the existing whole marked-domain theorem constructs the source domain with bound $0 + t = t$. Every empty history is canonical for every value function; apply this established geometric lemma to the actual original source family. Target canonicality supplies target testness, and the established dimension-zero cosupport theorem gives its empty final cosupport. The actual whole marked and cosupport dictionaries supply source testness and source cosupport equal to the literal preimage of the target's empty cosupport. These are precisely both terminal-resolution definitions. The final equality is a quantified assertion on this proved empty cosupport, and follows by contradiction from membership. It asserts no nonvacuous equality between the actual source constant and a formal seed. All source canonicality, testness and vanishing data are outputs. □

Theorem 403.4 (Original input constructs the full base history)**.** *Given only the original smooth f, positive ambient dimensions, original dimension-zero domain M and original canonical T, construct the actual whole Q, g internally with unchanged length, the preceding full base conclusion, the literal whole transformed marked record, and the genuine cosupport-preimage dictionary.*

Proof. Apply the existing actual tower constructor to f, T to obtain Q, g and its genuine Cartesian witness. Apply the preceding theorem to that internally obtained witness. Its target testness and original well-formedness give the actual whole transformed marked equality and cosupport dictionary, and the actual witness gives unchanged length. No source history, canonicality, value certificate or compactness is a premise. This closes the true zero-depth base; positive-depth contact recursion and the general canonical driver remain separate obligations. □

404 Actual holomorphic compatibility of the lifted terminal atlas

The gluing route of [5] requires genuine holomorphic transition maps, supplied by the actual independent local terminal comparisons. These lemmas keep the literal overlap preimages of the true local embeddings. Their primitive input is a genuine local biholomorphism over those whole embeddings, not a global transition regularity or complex-manifold certificate. The original atlas producer supplies these real local comparisons before the global construction.

Lemma 404.1 (Actual local overlap biholomorphisms give holomorphic lifted chart transitions)**.** *Let actual complex local final manifolds F_i, F_j embed openly into G. Their literal local overlap domains are the true preimages of the other embedding's entire range. Given a genuine biholomorphism δ between these true local open domains with $\iota_j\delta = \iota_i$, the transition between ANY two actual local atlas charts lifted into G is holomorphic on its ENTIRE actual source. No pointwise transition equality, holomorphic transition or desired global atlas certificate is assumed.*

Proof. Take an arbitrary coordinate point in the true source of the lifted transition. It lies in the original first chart target, and the actual first inverse chart's embedded point lies in the actual image of the second chart source. Thus the original first inverse point belongs to the TRUE first local overlap. This point internally makes the overlap nonempty, so restrict the actual first local chart to that real open. Its inverse is genuinely holomorphic at the original coordinate point, and its true underlying inverse value is the original first inverse chart. The actual overlap biholomorphism sends it to the second local space. Its whole embedding equation and genuine second embedding injectivity identify that point with the actual second local chart-source representative. Consequently the second original chart is genuinely holomorphic there. Compose the actual restricted inverse chart, genuine overlap map, actual second open inclusion and actual second chart. This coordinate map is holomorphic at the original coordinate point. At EVERY point in the ENTIRE true lifted transition source, the same restricted chart inverse formula, whole embedding equation and true embedding injectivity identify this actual coordinate composite with the actual lifted transition. Pointwise congruence on that source gives holomorphicity within the actual source. This works at every such coordinate point; an empty transition source requires no false nonempty overlap assertion. □

Theorem 404.2 (Actual lifted charts form a genuine holomorphic manifold atlas)**.** *For true jointly covering open embeddings of actual local complex manifolds, if each TRUE pairwise overlap has a genuine biholomorphic comparison preserving the whole embeddings, the INTERNALLY constructed lifted charted-space structure is an actual holomorphic manifold. The global atlas includes ALL actual local atlas charts. No global manifold or atlas compatibility certificate is supplied. Original separation and second countability are supplied independently by actual saturated projection geometry.*

Proof. The preceding actual charted-space constructor supplies the global atlas, preferred charts and actual source membership from real joint coverage. Each actual global atlas member is, by construction, a lift of an actual local atlas chart. For an arbitrary ordered pair, unpack their real local charts and invoke the first lemma with the actual pairwise overlap biholomorphism. It proves the actual coordinate transition is holomorphic on its true full source. The actual complex model has identity model with corners, so this is exactly the genuine chart compatibility condition. The standard actual manifold constructor therefore supplies holomorphic atlas compatibility, with no assumed global regularity. Combine separately proved Hausdorffness and second countability only in the later original complex-manifold endpoint; they are not consequences of arbitrary chart gluing. □

Theorem 404.3 (Actual local final embeddings are holomorphic for the constructed atlas)**.** *For this genuinely constructed holomorphic lifted atlas, EVERY actual local final-space inclusion is holomorphic. This is derived from actual local preferred charts and their true lifts, not assumed as a desired global map regularity certificate.*

Proof. At an arbitrary original local final point, choose its actual preferred local chart. Its true lift belongs to the actual global atlas and therefore the global maximal holomorphic atlas. The actual lifted inverse chart is holomorphic at the original local coordinate value. The original local chart is holomorphic at the local point. Their genuine composite is holomorphic there. On the true original local chart source, its value is the SAME original local embedding, by the actual lifted inverse formula and original chart left-inverse equation. This true source is an open neighborhood of the local point, so holomorphicity transfers through the actual local equality. Since the original local point was arbitrary, the entire local embedding is genuinely holomorphic. Actual inverse-open regularity and global resolution-map descent can then be constructed using these same original chart equations in subsequent scopes. □

405 Genuine separation, countability and charts for a projection-saturated terminal atlas

The noncompact gluing route of [5, 3] needs an actual Hausdorff, second-countable charted space, not only a topological quotient. These generic geometric lemmas are applied to the original local canonical resolution atlas after its actual gluing producer supplies its projection and true saturated ranges. Here the atlas inputs are genuine open embeddings and their actual projection ranges; no global separation, countability or chart certificate is assumed. Holomorphic compatibility, properness and the original global resolution sequence remain later producers.

Theorem 405.1 (Actual saturated local embeddings over a Hausdorff original base give Hausdorff gluing)**.** *Let $p : G \to X$ be an actual continuous projection to a Hausdorff original base and U_i true original open sets covering X. Let actual Hausdorff local final spaces F_i embed openly by $\iota_i : F_i \to G$, with the TRUE entire range equality $\iota_i(F_i) = p^{-1}(U_i)$. Then the actual glued space G is Hausdorff. No separation or fiber-injectivity certificate is required; actual original terminal gluing supplies these primitive data.*

Proof. Take distinct actual points a, b of G. If their original base images are distinct, disjoint original open neighborhoods separate those images; actual continuity gives disjoint open preimages separating a, b. If their original images coincide, choose one true original U_i containing that base point. Actual saturated range equality puts BOTH points in the SAME actual local embedding. Choose their actual local preimages; injectivity of the genuine embedding makes them distinct. Actual local Hausdorffness gives disjoint open neighborhoods, whose images under the genuine open embedding are open neighborhoods in G. Genuine embedding injectivity makes those image neighborhoods disjoint. Thus every pair of distinct original glued points is separated, with no false global projection injectivity or automatic Hausdorffness of arbitrary gluing. □

Theorem 405.2 (Original second countability internally gives a countable saturated terminal subcover)**.** *In the same actual saturated open-embedding situation, suppose ONLY the original base and each actual local final space are second countable. The actual glued space is second countable, even when its original atlas index is uncountable. No countable index, countable local-history choice or global second-countability certificate is supplied.*

Proof. Original second countability gives a countable subfamily of the TRUE original open cover with the same union X. Pull back THIS actual union through the actual projection. The true saturated range equality identifies it with the union of the selected actual final embedding ranges, so those countably many genuine open images cover ALL of G. Each true embedding is a homeomorphism from its actual local final space onto its literal range. Transport local second countability through that genuine homeomorphism. The actual countable open-cover theorem then gives second countability of G. Every global point is covered by the actual selected images; neither surjectivity of the projection nor an arbitrary countable quotient assertion is used. □

Definition 405.3 (Genuine local final charts lift to actual charts on a jointly covering open-embedding atlas)**.** For true local charted final spaces with the SAME complex model and genuine jointly covering open embeddings into an actual topological space G, build an actual charted-space structure on G. Its atlas is the images of ALL actual local atlas charts through the true embeddings. Each global preferred chart is the genuine lift of a local preferred chart at an internally chosen actual representative of that global point. No global charted-space certificate or transition regularity conclusion is assumed.

Proof. Genuine joint coverage supplies an actual local index and actual point mapping to each global point. Choose those actual witnesses. A true local chart extends through an actual open embedding to an open partial homeomorphism on G: its source is the true image of the original chart source, its target is the unchanged true chart target, and its inverse is the actual embedding composed with the true original inverse chart. The embedding's genuine injectivity and openness prove both inverse equations, open sources and actual partial continuity. Use ALL such lifted true atlas charts as the global atlas. At a global point, choose the lift of the local preferred chart at its actual chosen representative. Original preferred-chart source membership and the true representative equation prove global source membership; original local preferred-chart atlas membership proves global atlas membership. This is an actual charted-space construction; genuine holomorphic transition maps and the complex-manifold conclusion are separate subsequent mathematical producers. □

406 Actual all-order intrinsic HS maxima on a genuine compact original carrier

The point function below is the EXISTING intrinsic whole-stalk Hilbert–Samuel function. The maximizing point and the entire constant-profile locus are constructed internally from original coherence and actual compactness.

Lemma 406.1 (A countable family of actual upper semicontinuous natural values has a lexicographic maximum on a nonempty compact carrier)**.** *For a nonempty compact K and actual natural functions f_i whose true superlevels are closed, there is an actual $y \in K$ such that no actual $x \in K$ has a lexicographically larger profile $(f_i(x))_{i\geq 0}$. Moreover $f_i(y) \leq f_i(x)$ for EVERY i holds on K exactly when the ENTIRE two actual profiles agree.*

Proof. Work on the actual compact subtype K, avoiding any extra ambient Hausdorff or closed-carrier premise. Start with the full compact subtype. At stage i, actual upper semicontinuity gives a genuine attained maximum of f_i on the preceding nonempty compact closed set. Intersect that set with the actual closed superlevel at this attained value. The result is again closed, compact and nonempty. These actual sets decrease. Compactness supplies a point in their full countable intersection. For any competing point whose first i coordinates agree with this point, induction puts it in the ith actual maximizing set. Its ith coordinate is therefore bounded by the chosen point's coordinate. The first differing coordinate rules out a lexicographically larger profile. If all competing coordinates dominate, induction gives the reverse inequality at each successive coordinate, hence equality at EVERY order. The converse is immediate. No maximizing profile, bound or nested-set certificate is a caller premise. □

Theorem 406.2 (Original coherence internally supplies the true compact lexicographic HS maximum)**.** *For ONLY an original coherent ideal A and actual nonempty compact K, there is $y \in K$ which maximizes the EXISTING intrinsic profile $(H_A(y,\ell))_{\ell\geq 0}$ lexicographically. On that SAME actual K, the literal coordinatewise ALL-order threshold at this produced profile is exactly the ENTIRE constant intrinsic HS-profile locus of y.*

Proof. For every original order and actual natural threshold, the existing whole-stalk theorem constructs the true analytic HS superlevel from original finite generators. It is therefore closed

and gives actual upper semicontinuity of the EXISTING intrinsic function at that order. Apply the preceding compact theorem to these actual functions restricted to the literal original K. Its produced point and BOTH all-order directions give the stated genuine maximum and exact constant-profile dictionary. The original quotient ideals, maximal powers, complex actions and point function remain unchanged. □

Theorem 406.3 (Only a nonempty compact analytic original carrier produces its genuine analytic constant maximal HS locus)**.** *For ONLY nonempty compact analytic original W, there is an actual $y \in W$ such that the literal WHOLE ambient ALL-order reduced HS threshold at its produced original profile is nonempty, compact and analytic, lies in W, and consists EXACTLY of the actual points of W having the SAME ENTIRE intrinsic reduced HS profile as y.*

Proof. Original Cartan coherence makes the genuine reduced vanishing ideal coherent. Apply the preceding internal compact maximum to $K = W$. At the produced point the genuine order-zero HS value is one, since the actual reduced ideal support is exactly W. The whole threshold therefore forces actual reduced support membership by its order-zero condition, so it lies in the ORIGINAL W. On W, the preceding theorem gives BOTH threshold/ENTIRE-profile-equality directions. The produced point itself proves nonemptiness. The existing whole ambient all-order analytic-cut theorem supplies analyticity; its closedness and actual containment in compact original W give compactness. No caller maximal point, HS-constant carrier, analytic-cut or geometry certificate is used. This is a genuine maximal constant HS locus; exceptional blowup nonincrease and the terminating HS-centre driver remain OPEN. □

407 Actual full reference Hilbert–Samuel loci produce singular first centres with the whole original boundary

Use the EXISTING intrinsic original ambient Hilbert–Samuel function of the genuine current vanishing ideal. For an actual original reference point a, its whole reference locus is the literal coordinatewise condition $S_a = \{y : \forall \ell,\ H_A(a,\ell) \le H_A(y,\ell)\}$. All equations, thinness and marked-domain fields are produced internally. This gives a genuine FIRST centre and blowup inside the actual CURRENT singular set, compatible with EVERY member of the original ordered SNC boundary. The full same-original-invariant driver, actual HS blowup nonincrease and terminating singular-centre history remain open.

Lemma 407.1 (The actual original order-zero ambient function detects the whole ideal support)**.** *For an original coherent ideal A, the SAME intrinsic order-zero value is one exactly on A's actual zero locus and zero off it. In particular $1 \le H_A(y,0)$ if and only if $y \in A$.*zeroLocus*.*

Proof. In the actual original whole chart-stalk ideal, membership in the genuine maximal ideal means zero original germ evaluation. The true span of ALL original ideal germs therefore lies in that maximal ideal exactly when every original ideal germ vanishes at the point. Coherence supplies their actual holomorphic representatives, giving both directions of the whole support dictionary. If this containment holds, the original denominator $I + \mathfrak{m}$ equals $\mathfrak{m}$; its actual complex dimension is one by the previously proved genuine base-jet formula at order zero. Otherwise the genuine local ideal contains a unit and is the whole original ring, so its quotient has complex dimension zero. This proves both stated cases using the original actions and entire ideal, without a support certificate. □

Theorem 407.2 (A genuine singular reference gives a true nonempty compact whole HS locus inside the current singular set)**.** *For an actual compact analytic current set W, pure of dimension $d < n$, and any actual singular point $a \in W \setminus \operatorname{Reg}_d W$, the SAME full reference locus for $A = \mathcal{I}_W$ is nonempty, complex analytic, compact, and contained in the CURRENT singular set.*

Proof. The reference point belongs by every original reflexive HS inequality. Cartan coherence and the already proved genuine whole all-order producer give actual analyticity. Its original reference order-zero value is one, so every point of its whole locus lies in W by the preceding support dictionary and the original vanishing-ideal zero set. The actual first-HS singular-locus identity gives $H_A(a,1) \geq d+2$. Every point of the SAME whole locus therefore has first value at least $d+2$ and lies in the current singular set. Closedness and containment in compact W give actual compactness. No new partial HS score or caller desired locus comparison is substituted. □

Theorem 407.3 (The actual whole HS ideal gives a genuine SAME-boundary auxiliary marked domain)**.** *With the SAME original ordered SNC boundary E, the literal marked ideal $(M, M, E, \mathcal{I}_{S_a}, 1)$ is in the genuine original invariant domain. Its cosupport is precisely the ENTIRE actual full reference locus S_a.*

Proof. The actual analytic S_a has a true coherent vanishing ideal cutting out precisely itself. The original pure set W has empty ambient interior because $d < n$, so its subset S_a is also thin. Actual coherent local generators and this thinness prove that its genuine ideal is nonzero along the whole ambient carrier: vanishing of all its germs near a point would put a nonempty open inside its actual zero locus. The original SNC list is retained literally. The ambient carrier is actually smooth of its true dimension and transverse to every original SNC member. Together with mark one these fields construct the original marked domain. Actual order-one cosupport is the actual coherent ideal zero locus, giving the stated whole dictionary. No caller thinness, nonzero or marked-domain certificate occurs. □

Theorem 407.4 (Only original current geometry produces a nonempty singular first centre acceptable to EVERY original boundary member)**.** *For ONLY actual compact pure W of dimension $d < n$, its actual nonsmoothness and the original SNC list E, internally choose an actual singular reference a, the preceding full original HS marked input, and its true canonical maximum centre C. This C is nonempty and closed smooth, lies in the CURRENT singular set, and is admissible for the original marked vanishing ideal $(M, M, E, \mathcal{I}_W, 1)$, including NC with ALL original ordered members.*

Proof. If every original point of W were regular of its pure dimension, closedness would make W a closed smooth submanifold, contrary to actual nonsmoothness. Choose an actual singular reference internally. Its whole reference HS marked input is already a genuine domain with nonempty compact cosupport. Apply the existing actual canonical invariant to THIS independently constructed marked input. Its true upper semicontinuity on that compact cosupport gives an actual maximum; its actual maximum-level admissibility gives the nonempty closed smooth centre and NC with the SAME entire original boundary. Its original cosupport is the whole S_a, already inside the CURRENT singular set, hence inside W. The original mark-one vanishing cosupport is W, so these same fields give admissibility for the original current vanishing input. No centre, boundary-comparison or full driver certificate is supplied. This proves one genuine first centre only. □

Theorem 407.5 (The SAME constructed first centre gives a genuine proper blowup preserving the current regular open)**.** *Those SAME original-only inputs construct an actual holomorphic proper blowup in C, with genuine transformed ordered SNC boundary. It is biholomorphic over the ENTIRE original open complement of the current singular set. No smoothness of the resulting strict transform or finite termination of a restarted history is asserted here.*

Proof. Use the preceding actual smooth admissible centre with ALL original boundary members in the existing genuine holomorphic blowup construction. Its actual blowup charts and map give holomorphicity and properness, and its admissible original SNC coordinates give the transformed ordered SNC list. The actual blowup is biholomorphic off the original centre. Since that SAME

centre is inside the CURRENT singular set, restrict the genuine inverse to the entire open complement of that set. This preserves every original regular-open point without a caller modification or isomorphism certificate. □

408 Actual compatible relative coefficient germs on real finite domains

These are actual holomorphic base germs extracted internally from the original normal-power quotient basis. They provide genuine coefficient data for the local presentation route of [4, Theorem 9.4, pp. 277–278]. The construction uses entire polynomial quotient classes and does not identify raw derivatives with coefficients or discard factorials. The subsequent equality with literal normal coefficients of the full original formal Taylor series remains a separate dictionary goal.

Theorem 408.1 (Canonical actual relative polynomial truncations)**.** *For every original holomorphic germ $g \in \mathcal{O}_{\mathbb{C}^{d+t},0}$ and every degree q, construct an actual polynomial $P_q(g)$ in the real normal variables with coefficients in the ENTIRE actual tangential germ ring. All its normal terms have degree less than q, its evaluation satisfies $g - P_q(g) \in (z)^q$, and these two properties determine it uniquely. This includes degree zero and either empty coordinate block.*

Proof. Apply the inverse of the internally proved actual normal-monomial quotient basis to the genuine quotient class of g. Sum these actual coordinates against their original distinct normal monomials. The quotient-basis map identifies its actual evaluation class with the original class of g; hence their difference belongs to the whole normal power. The finite sum has exactly the asserted coefficients and no higher coefficient. For uniqueness, the difference of two proposed evaluated polynomials lies in the normal power. Genuine lower-coefficient separation annihilates every low coefficient of the difference. Both high coefficients already vanish, so the entire original polynomials agree. □

Theorem 408.2 (One actual coefficient germ is independent of every later truncation)**.** *For every actual normal exponent γ, the coefficient of z^γ in $P_q(g)$ is independent of all $q > |\gamma|$. In particular the choice $q = |\gamma| + 1$ defines one genuine entire tangential holomorphic germ $c_\gamma(g)$, and every actual truncation has precisely these coefficients.*

Proof. First compare $P_q(g)$ and $P_r(g)$ for $q \leq r$. Their genuine remainders lie in $(z)^q$ and $(z)^r \subset (z)^q$, so their difference has evaluated normal order at least q. Original coefficient separation gives equality at every exponent of degree less than q. For arbitrary truncations above $|\gamma|$, compare each with the actual truncation at $|\gamma| + 1$. The equality is in the entire base germ ring, not only after evaluation at the origin. □

Theorem 408.3 (Actual finite coefficient families have a common real holomorphic domain)**.** *Given any finite family of actual original holomorphic germs and any finite normal degree bound, construct genuine representatives of all their actual coefficient germs and ONE open neighbourhood of the tangential origin on which every coefficient below that bound is holomorphic. The same actual representatives realize the stable coefficients in every later truncation. No uniform domain for infinitely many coefficients is asserted.*

Proof. Each internally constructed coefficient is a genuine holomorphic germ, so choose an actual representative holomorphic near zero. The family of original germs is finite, and so is the family of distinct normal exponents below the given degree bound. Intersect their actual holomorphic neighbourhoods and take a common open neighbourhood of zero. Every chosen function is holomorphic there, represents the same whole original germ, and hence satisfies all previously proved stable coefficient equalities. This uses no extension holomorphic outside that domain and no convergence claim about an infinite arbitrary formal reduced basis. □

409 The literal genuine relative Taylor coefficient dictionary

The actual relative coefficients were constructed by inversion of the true normal-power quotient basis, and already have a common holomorphic domain for any finite family and finite degree bound. We now identify their entire original Taylor series with the literal normal coefficients of the original ambient Taylor expansion. This uses the actual coefficient convention of formal power series and introduces no unproved raw-derivative convention.

Lemma 409.1 (The entire genuine formal coefficient of an actual normal polynomial)**.** *Let P be any finite polynomial in the normal variables with coefficients in $\mathcal{O}_{\mathbb{C}^d,0}$. Evaluate P using the actual tangent pullback and actual ambient normal coordinate germs. The literal formal normal coefficient at every normal exponent γ of the genuine ambient Taylor series of that evaluation is the genuine tangential Taylor series of the entire original polynomial coefficient P_γ.*

Proof. For one actual monomial aZ^α, the genuine ambient Taylor homomorphism sends the actual tangent pullback of a to the literal variable renaming of its genuine tangential Taylor series, by whole-map coordinate naturality. It sends every actual normal coordinate to the corresponding original formal variable. Multiplicativity therefore gives the product of that renamed series with the true normal monomial. The coefficient at the joined original exponent (β, γ) equals the β coefficient of Taylor(a) if $\gamma = \alpha$, and is zero otherwise: the true renamed series has no nonzero normal exponent, and genuine monomial multiplication shifts exponents exactly. This is coefficient equality with the original numerical coefficients, including zero coefficients and empty coordinate blocks. Add the finitely many actual polynomial monomials. Additivity of the actual Taylor homomorphisms and coefficient maps gives the statement for the entire original polynomial and every tangential exponent. □

Theorem 409.2 (Every genuine relative coefficient has exactly its original formal Taylor series)**.** *For any actual holomorphic ambient germ g and any normal exponent γ, the literal γ normal coefficient of the genuine entire ambient Taylor series equals the genuine tangential Taylor series of the actual stable relative coefficient germ of g at γ. Consequently every finite family of actual analytic ambient germs has actual holomorphic coefficient representatives on one genuine common open for every prescribed finite low-degree range, satisfying these literal entire formal-series identities.*

Proof. Take $q = |\gamma| + 1$ and the genuine canonical relative polynomial $P_q(g)$ constructed from the actual normal-power quotient. Its actual remainder $g - P_q(g)$ belongs to the whole original normal ideal power of order q. The genuine ambient Taylor homomorphism maps this entire ideal into the actual formal normal ideal power, by its true normal-coordinate generator dictionary and the actual ideal-map power identity. Every coefficient with normal degree below q vanishes in that ideal power, regardless of its tangential exponent. Thus the literal normal coefficient at γ of Taylor(g) equals that of Taylor$(P_q(g))$. Apply the preceding polynomial identity. By construction, the entire γ coefficient of $P_q(g)$ is the actual stable relative coefficient germ. This proves equality of the whole formal tangential series, with no convergence assertion for an arbitrary prescribed formal series. Finally use the already constructed actual finite coefficient representatives and their genuine common holomorphic domain, and substitute the whole-series equality just proved. This does not construct the full general reduced coefficient presentation or prove transformed stability or Bennett inequalities. □

410 Whole original germ algebra maps and genuine Taylor coordinate naturality

The coefficient-presentation dictionary needs Taylor expansion to commute with the actual tangential coordinate inclusion. The proof uses genuine finite original coordinate representatives, not

an assumed continuity or coefficient identity. This supplies the coordinate part of the relative coefficient route; the entire literal normal-coefficient dictionary remains separate until proved.

Theorem 410.1 (Actual coordinate images determine the whole original germ map)**.** *Let $f, h : \mathcal{O}_{\mathbb{C}^d,0} \to \mathbb{C}[[X_1, \ldots, X_m]]$ be actual complex algebra homomorphisms. Suppose their images of every original coordinate germ agree and lie in the genuine formal maximal ideal $(X_1, \ldots, X_m)$. Then $f = h$ on the ENTIRE original germ ring. No topological-continuity certificate or desired Taylor identity is assumed.*

Proof. The original holomorphic Hadamard decomposition writes every germ with zero value as a sum of actual coordinate germs times actual holomorphic germs. Thus each homomorphism maps the entire original maximal ideal into the formal maximal ideal, and genuine multiplication gives the same inclusion at every power. Actual formal coordinate-power membership annihilates every coefficient below that power.

For any original g and any formal exponent a, choose the actual finite original coordinate-word representative p modulo the original maximal power of degree $|a| + 1$. Both image remainders have zero a coefficient. The two algebra maps agree on p because it lies in the genuine finite complex span of products of the original coordinate germs: use their exact coordinate identities, multiplication, addition and complex linearity. Hence the a coefficients of $f(g)$ and $h(g)$ agree. This holds for every actual exponent, proving equality of the entire formal series and then of the whole algebra maps, also when either coordinate block is empty. □

Theorem 410.2 (Genuine Taylor expansion commutes with the actual tangential embedding)**.** *For every actual tangential holomorphic germ g, its genuine pullback by the actual first-coordinate projection has original Taylor series equal to the actual variable-renaming of the original Taylor series of g, using precisely the first d variables inside $d + t$ variables.*

Proof. The two sides are independently defined actual complex algebra maps on the entire original tangential germ ring. The real coordinate projection pulls each original tangential coordinate germ to the corresponding ambient coordinate germ. The genuine Taylor homomorphism maps that same coordinate to the corresponding formal variable. The actual formal variable-renaming homomorphism has exactly the same images. These true coordinate images belong to the formal maximal ideal, so the preceding whole-map uniqueness applies. This proves the exact original Taylor identity without replacing either independently defined map by a proxy. □

411 Actual all-order ambient Hilbert–Samuel conditions have finite local equations

For the EXISTING intrinsic original function $H_A(y, \ell) = \texttt{hilbertSamuelAt}(A, y, \ell)$ and any natural threshold sequence h, use the literal COORDINATEWISE whole condition $S_h = \{y : \forall \ell,\ h_\ell \leq H_A(y, \ell)\}$. No lexicographic comparison or finite-value theorem is silently asserted. This is the all-order closed-superlevel step of Bierstone–Milman[4], Theorem 9.2, printed page 276. Local finiteness of the actual value family, its full semicoherent blowup presentation and descent remain subsequent obligations.

Lemma 411.1 (Actual finite Hilbert–Samuel prefixes form a descending analytic chain)**.** *For original coherent A, let $S_{h,k} = \{y : \forall \ell \leq k, h_\ell \leq H_A(y, \ell)\}$. Each is genuinely complex analytic, and $S_{h,k+1} \subseteq S_{h,k}$ on the entire original manifold.*

Proof. Start with the genuine original order-zero superlevel. At the true successor intersect the preceding prefix with the original order-$k + 1$ superlevel. The existing original coherence producer supplies finite holomorphic rank minor equations for each intrinsic superlevel. Actual finite intersections retain a finite holomorphic equation family, hence analyticity. Induction identifies both membership directions with all original orders through k. Every successor is contained in its preceding factor, proving antitonicity. □

Theorem 411.2 (One actual neighborhood and finite order control the entire original all-order condition)**.** *At every original point x there are an internally produced finite order k and one genuine open neighborhood U such that $S_h \cap U = S_{h,k} \cap U$, in BOTH membership directions.*

Proof. Apply the already proved LOCAL analytic descending-chain stationarity in the proof of Lemma **??**. Its local producer requires no compactness of the original manifold: genuine local Noetherian germ ideals and identity continuation along the actual connected regular decomposition give ONE neighborhood on which EVERY later prefix equals $S_{h,k}$. The whole condition implies the finite prefix immediately. Conversely take a point of $S_{h,k} \cap U$ and any original order ℓ. Use the same stationarity at prefix $\max(k, \ell)$; the point belongs to that actual prefix, so its original order-ℓ inequality holds. This proves the whole literal condition for every ℓ, on the SAME neighborhood. No caller stationary-germ, uniform neighborhood or finite cutoff certificate occurs. □

Theorem 411.3 (The literal whole original ambient Hilbert–Samuel superlevel is analytic)**.** *For ONLY original coherence of A and any h, the literal whole S_h is a genuine complex analytic subset of the original manifold. Its actual vanishing ideal is coherent and has genuine finite local holomorphic generators cutting out that SAME all-order set.*

Proof. At each actual original point, the preceding internally produced neighborhood identifies the WHOLE set with one true finite analytic prefix. Choose that prefix's existing finite holomorphic presentation in the point's original chart and restrict its genuine coordinate domain to the SAME neighborhood. The whole local equality proves both zero-locus directions for S_h. Thus these actual presentations give global complex analyticity. Actual Cartan coherence of its genuine vanishing ideal supplies finite local holomorphic generators with BOTH nearby germ-span directions. The original analytic vanishing-ideal zero-locus theorem identifies their actual common zero set with this SAME literal all-order S_h. □

Theorem 411.4 (Every actual compact window has one finite order controlling all original whole conditions)**.** *For every actual compact subset K of the original manifold, there is one internally constructed order k such that for EVERY $y \in K$, $y \in S_h$ if and only if $y \in S_{h,k}$. The original ambient manifold need not be compact.*

Proof. The true local-stationarity neighborhoods cover K. Its actual compactness chooses a finite subcover. Take the maximum of their genuine finite orders, using zero for an empty subcover. Whole membership implies that maximum prefix. Conversely every point of K lies in some chosen original neighborhood. The maximum prefix is contained in its neighborhood's own prefix by actual antitonicity. Its local whole/prefix equivalence therefore gives whole membership. The same maximum works for every original point of K, including the empty window without an extra premise. □

412 Every attained original full reference value has an actual analytic neighborhood presentation

The genuine reference presentations underlying semicoherence must handle arbitrary recursive depth. The SAME already constructed original invariant family has its proved original local admissible-level geometry at EVERY induction dimension. We use that actual geometry to construct true closed smooth reference loci and their entire coherent equations. This supplies whole analytic presentations of the attained original full reference-value superlevel on a genuine open containing that reference point. The distinction from transformed semicoherent presentation compatibility is explicit in Bierstone–Milman[4], Definition 6.4, printed pages 254–255: those transformed compatibility and arbitrary global threshold/scheduling statements and the separate explicit recursive coefficient-presentation comparison remain open obligations.

Lemma 412.1 (Real nested original opens transport entire analytic threshold equations)**.** *An actual open in an actual original open has a genuine open image in the original manifold. Its true inclusion and that of the image have the same underlying points through a genuine holomorphic diffeomorphism. Entire analytic threshold equations transport along that actual map.*

Proof. The original open inclusion is a true open map, so the image of the actual inner open is open. Image membership internally proves it lies in the outer open, and literal subtype injectivity gives both membership directions. The original nested-open diffeomorphism is therefore constructed from those actual dictionaries, with inverse the SAME original point in its true outer and inner opens. Both maps are holomorphic by the genuine open-subtype maps. The composite inclusions agree on every actual point. Pull the real analytic set back along the true inverse diffeomorphism and use that point equality to obtain the entire original-image threshold. No open/image/map or threshold compatibility certificate is input. □

Theorem 412.2 (The actual original monomial reference branch has entire analytic equations)**.** *For ONLY original successor marked-domain input, genuine original canonical history and an actual reference point with current nonmonomial order zero, the SAME attained original full-value superlevel on actual current cosupport is analytic on the entire original final manifold.*

Proof. The literal original zero-order value theorem identifies its attained full value with $(0, \operatorname{inl}\gamma(x))$. Actual original canonical data internally give a true source-test tower and actual current marked domain. The genuine globally analytic original monomial-threshold producer applies to that SAME value, ideal, carrier and boundary. Its whole current-carrier locus includes the true higher-first-entry branch. Intersect it with the actual coherent current positive cosupport, whose genuine derivative equations are analytic. Current cosupport lies in its SAME carrier, so both directions identify precisely the displayed original reference superlevel. The invariant and attained reference value are unchanged. □

Theorem 412.3 (The actual positive SPECIAL reference branch has a true entire analytic reference open)**.** *For an actual positive original cosupport reference point whose internally constructed actual fixed companion is SPECIAL at that SAME reference, a real original open contains it and the ENTIRE inverse image of its attained original FULL-value superlevel is analytic there. No companion, special-reset ideal or analytic presentation is supplied by the caller.*

Proof. The genuine original fixed-companion producer internally supplies its actual open, reference membership, true domain, maximal order and canonical masked history, together with its entire common cosupport and arbitrary-threshold dictionaries. The reference lies in its true companion cosupport. The literal SPECIAL value is the actual pair $(s, \top)$ with its original level s. The genuine whole reset ideal producer constructs coherent equations for precisely that entire masked top threshold on the actual companion final open. Its true coherent zero-locus equations are analytic. The original full-value reference dictionary and the whole arbitrary-threshold dictionary identify this ENTIRE actual threshold with the original full reference superlevel, including points whose selected histories differ from the reference history. The actual original open already contains the reference point; no smaller analytic-neighborhood witness was assumed. □

Theorem 412.4 (The SAME original constructed invariant internally gives a genuine smooth full reference-value open)**.** *For EVERY original induction dimension k, ONLY original marked-domain input, its literal SAME original canonical history and an actual current cosupport reference point x construct a true original open $U \ni x$ such that*

$$\{p \in U : p \in m_T.\mathrm{cosupp}, \quad \mathrm{inv}_k(m, T, x) \leq \mathrm{inv}_k(m, T, p)\}$$

is a genuine closed smooth submanifold, hence analytic as an ENTIRE subset of the actual open manifold U. No selected local centre, level model, upper-bound, lower-analyticity or desired full-reference certificate is input.

Proof. Use the SAME original constructed $R = \mathrm{invFamily}(k)$, whose actual source proof already constructs the invariant family and its local admissibility for all dimensions. The original canonical input and real current cosupport reference internally supply genuine original upper semicontinuity at that point. Its actual neighborhood definition constructs one true open U containing x on which every real current cosupport value is at most its attained reference value. Apply the PROVED original local admissibility field of that SAME constructed R, original input and original canonical history to this internally constructed numerical bound. It gives an actual admissible level locus in the literal current marked open pullback, including genuine closed smoothness; no caller level model or centre exists in the hypotheses.

On EVERY point of this SAME open, true superlevel comparison and the proved upper bound force original value equality, in both directions. Thus the actual superlevel inverse image is ENTIRELY equal to that genuine original admissible level locus. Transfer its real closed-submanifold geometry through this literal set equality, retaining the original point and full attained value. Actual closed smooth analytic equations then give entire analyticity. This uses genuine original geometry beyond USC; no analytic witness is inferred from USC alone. The SAME original invariant, finite canonical history and constructed local admissibility are retained. The explicit recursive coefficient-presentation comparison, arbitrary global thresholds, transformed semicoherent equivalence and global scheduler are not asserted. □

Theorem 412.5 (The actual full reference locus has its genuine coherent ideal and finite original equations)**.** *Those SAME original inputs internally supply a real reference open and the actual vanishing ideal of its literal original FULL reference superlevel. This ideal is coherent and cuts out exactly that SAME entire locus. At EVERY point of this actual open it has finitely many genuine holomorphic generators on a true neighborhood, with BOTH nearby ideal/span directions and the exact original full threshold as their simultaneous zero set.*

Proof. The preceding genuine closed smooth locus internally supplies coherence of its ACTUAL vanishing ideal by original smooth-carrier vanishing equations. Its real closedness gives the true ENTIRE vanishing-ideal zero-locus identity. Apply actual coherence at any real point of the internally constructed open. This chooses a genuine finite holomorphic function family and BOTH nearby germ/span directions. Each such function is an actual ideal germ, so true zero-locus membership forces its value zero. Conversely every actual ideal germ is a genuine finite holomorphic combination of those same functions; simultaneous vanishing forces every such germ to vanish. The ENTIRE actual vanishing-ideal dictionary identifies these equations precisely with the literal original full reference threshold on the SAME open. No coherent presentation or generator certificate is assumed. These are real original reference equations at all depths; they do not prove arbitrary global threshold or transformed semicoherent presentation compatibility. □

413 All original DOMAIN TWO StepOne branches under smooth pullback

Ambient dimensions n, m are positive. Comparisons below use the unchanged original sequence-aware StepOne and its actual $(m - n)$ recursive dimension map. The target lower depth is any positive integer; the source lower depth includes the genuine smooth relative dimension.

Theorem 413.1 (Actual zero-contact StepOne values have their full smooth dimension map)**.** *For ONLY an original target DOMAIN2 record of maximal order, ANY genuine original upper TEST and a true smooth map, construct ONE actual source upper TEST. At EVERY actual source endpoint whose image is in the target current cosupport and is nonspecial of level zero, the original source StepOne with lower depth $(m - n) + k + 1$ is the genuine StepOne dimension map of target original StepOne with lower depth $k + 1$, for EVERY $k \geq 0$. No lower TEST, current lower membership, chosen common model or value is supplied.*

Proof. Construct the actual source history and independent OWN LARGE lower full values by the preceding original-only producer. Genuine original bar TEST stability and actual Cartesian SPECIAL/level dictionaries prove the source is nonspecial and has level zero. The original zero-regime-run dictionary identifies the source's actual computed split with the target split. Both actual lower-mask memberships have already been internally constructed. Unfold the unchanged sequence-aware StepOne in each proved nonspecial branch and evaluate the literal lower-value function using those true memberships. The remaining two lower GENERAL values are exactly the independently proved OWN LARGE dimension comparison. Apply the actual finite WithTop value map and retain the true first entry zero. This produces the complete StepOne map without a zero-relative-dimension assumption or a source-curve shortcut. □

Theorem 413.2 (One actual history compares all three original StepOne branches)**.** *For the same ONLY original data, construct ONE actual source upper TEST before all points, recursive depths and branch cases. At EVERY actual source endpoint whose image lies in the target current cosupport, the original source StepOne with lower depth $(m-n)+k+1$ is the genuine StepOne dimension map of target StepOne with lower depth $k+1$, for EVERY $k \geq 0$. This includes the actual SPECIAL, nonspecial zero-level and nonspecial positive-level branches. No original canonicality is an input; no target-point surjectivity or fabricated lower TEST is asserted.*

Proof. Choose the ONE source history produced by the zero-contact theorem before quantifying over points and depths. Genuine original bar geometry gives the whole current SPECIAL equivalence and exact indexed level on this SAME actual Cartesian history. If the target point is SPECIAL, both unchanged StepOne values use their genuine SPECIAL branch. Their true levels agree and the real WithTop dimension map preserves infinity. If the target is nonspecial of level zero, use the first theorem on this very history. If it has positive level, express that actual natural number as a successor and apply the previously proved original positive-StepOne smooth producer to this SAME actual history. That producer internally constructs its retained lower current point and true whole lower values. The three genuine cases exhaust the original definition. No arbitrary higher-bound noncanonical GENERAL naturality, exceptional presentation comparison or global scheduler is inferred from this original DOMAIN2 endpoint. □

414 Actual indexed SNC and chronological companion thresholds have whole coherent defining ideals

All counts in this chapter are the original invariant's ordered indexed SNC boundary counts. Every historical empty position is retained. Fixed-cardinality member intersections and genuine finite ideal products construct actual global coherent equations. This follows the finite member-combination construction of Bierstone–Milman[5], Section 5, Step I, Case B: the auxiliary ideals here present the same count sets by vanishing ideals of true SNC intersections. No equality with the source's weighted coefficient ideal is asserted. The resulting literal original positive tagged count threshold is one actual full-invariant threshold family; arbitrary deeper recursive lower thresholds and the simultaneous auxiliary Hilbert–Samuel global scheduler remain subsequent producers.

Theorem 414.1 (Original indexed SNC count has a genuine coherent finite intersection-product ideal)**.** *For a genuine original SNC ordered list E in positive ambient dimension, let $\mathcal{S}_s$ be its indexed s-element subsets. Form the ACTUAL ideal*

$$C_s(E) = \prod_{S \in \mathcal{S}_s} \mathcal{I}_{\bigcap_{i \in S} E_i}.$$

This is coherent and its entire zero locus is exactly $\{y : s \leq \mathrm{boundaryCount}(E, y)\}$, including $s = 0$, impossible counts, and actual empty historical entries.

Proof. Every true indexed intersection is a closed smooth submanifold by the genuine SNC coordinates, including the empty intersection, which is the whole ambient manifold. Its actual vanishing ideal is coherent and cuts out precisely that intersection. The literal finite product starts with the actual unit ideal; actual finite-product coherence gives the whole coherent ideal. The genuine product zero-locus rule makes its zero set the finite union of the indexed intersections, with an empty product having empty zero locus. A point belongs to one intersection exactly when its real incident-index set contains an s-element subset. This is equivalent to its actual cardinality being at least s, in both directions by genuine finite subset cardinality. Index positions are retained throughout; no distinct trace-germ quotient count is substituted. □

Definition 414.2 (The actual chronological companion ideal uses the SAME real history years)**.** For the actual original history T, let E_r be its TRUE chronological original prefix at year r. Define the whole original ideal

$$C_{T,q,s} = \prod_{0 \le r \le \operatorname{length} T} \big(Y_{T,q,r} + C_s(E_r)\big).$$

Every factor uses the actual original year ideal and SAME actual chronological boundary prefix, including all original empty steps.

Proof. The real original history has a finite natural length. Every index in the displayed finite range selects an actual year ideal on its final manifold and an actual chronological prefix of the same ordered current boundary. The corresponding true count ideal is on that same manifold, so their sum is a well-defined genuine ideal there. The literal finite list product starts from the actual unit ideal and multiplies these sums. Thus it defines one entire ideal, with no requested count or zero-locus identity as input; all original empty steps and indexed member positions remain in the formula. □

Theorem 414.3 (Only original canonical inputs give the entire chronological count ideal dictionary)**.** *Only original marked-domain input and its SAME literal original canonical history internally supply coherence of $C_{T,q,s}$ and its entire zero-locus identity*

$$V(C_{T,q,s}) = \{y \in m_T.\mathrm{cosupp} : q \le \nu(y), s \le c_{T,q}(y)\},$$

where $c_{T,q}$ is the actual original order-run chronological count.

Proof. The actual original canonical history gives a true test tower and genuine current domain, hence SNC of its entire actual current boundary. Every chronological prefix is a true sublist, so its count ideal is genuinely coherent. The preceding actual history producer supplies each true year ideal and its whole original zero-locus dictionary. Thus every sum factor and the literal finite product are coherent. Sum zero loci are actual intersections; finite product zero loci are actual unions. The resulting finite union is precisely the already proved entire chronological count/year dictionary. Consequently it presents the SAME original count on the entire final manifold, without a supplied history/count/analytic presentation certificate. □

Theorem 414.4 (The literal full positive original tagged count threshold has an actual coherent whole ideal)**.** *For $q \ge 1$ and ONLY original marked-domain/canonical-history inputs,*

$$P_{T,q,s} = B_{q+1}(m_T) C_{T,q,s}$$

is coherent and cuts out exactly the literal original full-invariant superlevel at the previously constructed threshold $(q, \mathrm{inr}(s, \textit{minimum lower value}))$.

Proof. The internally constructed actual current domain gives the real higher-order baseline ideal B_{q+1} and its entire baseline dictionary. The preceding true chronological ideal is coherent. Their

real product is coherent, with zero locus the union of the genuine higher-order baseline and chronological count locus. On the actual current carrier, finite genuine nonmonomial order identifies that baseline with true order at least $q+1$. The previously proved whole original positive tagged-value dictionary identifies exactly this union with the SAME literal original full-invariant superlevel, including its real higher-order branch. No original invariant or threshold is redefined. □

Theorem 414.5 (The same original full threshold supplies honest finite local ideal equations)**.** *At every real original final point, the SAME whole positive tagged-count ideal has actual finite holomorphic generators on a true neighborhood, with BOTH nearby germ-span directions and the exact literal original threshold as the simultaneous zero set of those SAME functions.*

Proof. The preceding internally constructed true coherence supplies the actual open and finite family with both nearby ideal/span directions. Each family member is a genuine ideal germ. Thus actual ideal zero-locus membership forces each one to vanish. Conversely true finite germ-span membership expresses every actual ideal germ as a finite holomorphic combination of those functions; their simultaneous vanishing forces every such germ to vanish. The exact whole literal original threshold dictionary gives the asserted real equations. □

Theorem 414.6 (These true full threshold ideals give actual SAME-boundary ambient marked–domain inputs)**.** *Only those same original inputs internally construct the genuine marked input with WHOLE ambient carrier, SAME actual current original SNC boundary, actual ideal $P_{T,q,s}$ and mark one, in the original ambient marked domain.*

Proof. The ideal is genuinely coherent by the preceding producer. Its whole exact zero locus lies in the actual current original cosupport, whose ambient interior is empty by genuine original nonzero and local coherence. If the true coherent threshold ideal vanished along the whole ambient carrier at a point, its actual finite generators would vanish on a whole neighborhood, putting that nonempty open in its true zero locus, a contradiction. Thus it is nonzero at every actual ambient point. The actual current domain gives its SAME true SNC list. Whole ambient smoothness, its actual dimension, transversality to that SNC list and the real mark one construct all original marked-domain fields. This is an actual auxiliary input for subsequent global centre constructions. It does not complete arbitrary deeper lower thresholds, auxiliary simultaneous Hilbert–Samuel chains or the global scheduler. □

415 The actual whole companion-count defining ideals commute with original open histories

The actual whole indexed and chronological count ideals now have genuine open pullback equalities. These are entire ideal comparisons needed for true local window equations in Bierstone–Milman[4], Theorem 13.4, printed pages 299–301. They retain the original indexed boundary positions and every real empty history step. They do not replace distinct trace-germ counts at bad points or complete arbitrary deeper/special companion presentations and the global history scheduler.

Lemma 415.1 (Actual coherent finite products pull back and are independent of enumeration)**.** *For a genuine finite coherent ideal family, its whole product pulls back to that of the literal pulled-back factors under the actual open inclusion. Two genuine enumerations related by a real permutation give the same whole ideal product.*

Proof. Induct on the actual factor list. Its empty product is the original holomorphic unit ideal, whose whole pullback is the genuine unit ideal on the open. A nonempty product uses coherence of the actual head and entire tail and the real coherent product pullback identity, followed by induction. For enumeration independence, induct on a genuine list permutation. A cons keeps the same head and applies induction to the true tail; a swap uses actual coherent ideal associativity

and commutativity, and transitivity composes these equalities. Coherence of every intermediate factor list follows from permutation membership. No desired product, enumeration or pullback certificate is assumed. □

Theorem 415.2 (Genuine original indexed SNC count ideals have exact whole open pullbacks)**.** *For ONLY the actual original SNC list and a true open, every whole count ideal $C_s(E)$ pulls back to the SAME genuine count ideal of the literal indexed open boundary list, including zero or impossible counts and actual empty entries.*

Proof. The real list map preserves the number of original positions, giving a true bijection between their finite indices. It keeps the original numeric position, so each actual mapped member is precisely the preimage of that same original member. The induced genuine finite-subset bijection preserves cardinality; hence it carries all actual s-element index subsets bijectively to those of the literal open list. At each point both intersection membership directions identify the mapped member intersection with the true original intersection preimage. Genuine open pullback of its actual vanishing ideal gives whole factor ideal equality. Original SNC internally supplies coherence of every actual intersection ideal. Apply the preceding actual product pullback, true factor equalities and genuine permutation of the finite enumerations. This proves whole ideal equality and retains every indexed position, rather than using only the scalar count or its zero-set dictionary. □

Theorem 415.3 (The SAME actual chronological companion ideal commutes with every original open history)**.** *For ONLY original marked-domain input and its true source-test history, the whole chronological count ideal has the literal entire open pullback identity for every q, s. All actual years and empty history steps remain.*

Proof. Original controlled-transform/open restriction identifies the actual entire current ordered boundary. True restriction keeps both original input-list length and history length, so taking the SAME genuine chronological prefix commutes with its literal open boundary map. Each whole factor is the actual sum of a true year ideal and an indexed count ideal. The accepted whole year naturality and the preceding actual count pullback identify both factors, before real sum pullback. Actual original source-test transport internally supplies their coherence, current SNC and the true chronological sublist hypotheses. The actual finite range of years is unchanged, including empty steps. Apply whole coherent list-product pullback and the actual factor equalities over that same range to obtain the entire chronological ideal identity. □

Theorem 415.4 (The true positive tagged-count ideal and SAME-boundary marked record have their whole original open restriction)**.** *For those true original histories the ENTIRE positive tagged-count defining ideal commutes with actual open restriction. Its literal whole ambient carrier, SAME ordered current original boundary, actual entire ideal and mark agree. Original-only canonical inputs internally supply every true source-test premise.*

Proof. The higher-order current baseline has its accepted genuine open pullback, using actual current domain transport. Apply real coherent product pullback to that SAME baseline and actual chronological ideal, and substitute the preceding entire chronological equality and literal controlled current record. This proves whole defining ideal equality, independent of using a zero-set or order dictionary. The ambient carrier's actual open preimage is the entire open; genuine controlled-transform restriction gives the SAME indexed current boundary; the entire ideal equality just proved gives the third field, and the actual mark remains one. These four exact fields give literal marked record equality. The SAME original constructed invariant internally supplies a true source-test tower from actual canonical data, yielding the original-only consumer. Arbitrary deeper recursive and higher/special analytic closure and the compact-window global scheduler are not asserted by these identities. □

416 Real coherent global history-year ideals supply genuine original-boundary marked inputs

The already proved whole original enhanced year threshold has actual coherent order equations. We now construct its global defining ideal directly from original carrier vanishing, relative derivatives, sums, products and genuine blowup pullbacks. This gives actual marked inputs for analytic-centre engines. The role of actual global analytic threshold subspaces is explicit in Bierstone–Milman[4], Theorem 13.4 and Lemma 13.7, printed pages 299–301. These particular enhanced year equations do not complete the arbitrary recursive companion threshold family or the simultaneous auxiliary Hilbert–Samuel and compact-window scheduling engines.

Theorem 416.1 (Actual smooth-carrier order thresholds have explicit whole coherent defining ideals)**.** *For a true closed smooth carrier N and actual coherent ideal I, put $P_0(N,I) = \mathcal{I}_N$ and $P_{a+1}(N,I) = \mathcal{I}_N + \mathcal{D}_N^a I$. These are genuine coherent ideals on the entire original manifold, and*

$$V(P_a(N,I)) = \operatorname{cosupp}(N,I,a)$$

for every $a \geq 0$.

Proof. The actual closed smooth carrier has a coherent vanishing ideal cutting out precisely N. Its zero-order relative cosupport is exactly N by the true holomorphic germ dictionary. Each real iterated relative derivative ideal is coherent by its genuine local derivative presentations. Finite sum coherence gives P_{a+1}. A sum's zero locus is the intersection: each factor germ is an actual sum-ideal member, and every sum-ideal germ is a finite holomorphic combination of factor germs, giving both directions at every actual point. The true relative derivative-order theorem identifies this intersection with the actual order-$a+1$ cosupport. Thus the equations present the entire true order locus, without supplying a desired defining ideal or analytic witness. □

Lemma 416.2 (The actual original current baseline has a coherent whole defining ideal)**.** *For genuine marked-domain J, form its actual nonmonomial ideal $\mathcal{N}_J$ and*

$$B_q(J) = P_{J.d}(J.N, J.I) + P_q(J.N, \mathcal{N}_J).$$

This is coherent and its entire zero locus is exactly $\{y \in J.\text{cosupp} : q \leq \operatorname{ord}_{\text{nm}}(J,y)\}$.

Proof. The genuine marked-domain fields supply closed smooth carrier and coherent ideal. Actual SNC division, genuine transversality and nonzero supply coherence of the SAME nonmonomial ideal. The preceding whole order theorem and true sum zero-locus dictionary give $J.\text{cosupp} \cap \operatorname{ordGe}(J,q)$. At every actual cosupport point the finite genuine nonmonomial-order dictionary identifies order-q cosupport with order at least q. Thus both directions of the stated entire baseline dictionary hold, including threshold zero. □

Definition 416.3 (The genuine history recursion retains the original reset and increment equations)**.** For the actual original m, T, q, r, define a genuine ideal $Y_{T,q,r}$ on its actual final manifold. At the empty history let it be $B_q(m)$. At the true successor $T' = T\pi$, writing $J = m_T$ and $J' = m_{T'}$, set

$$Y_{T',q,0} = B_q(J') + \pi^* P_{q+1}(J.N, \mathcal{N}_J),$$
$$Y_{T',q,r+1} = B_q(J') + \pi^*\big(P_{q+1}(J.N, \mathcal{N}_J) Y_{T,q,r}\big).$$

All factors, products and maps are the actual original ones; the reset's higher old-order branch and actual current higher-order baseline are retained.

Proof. The original tower is a genuine finite inductive history. At its identity the actual baseline is an ideal on the same initial manifold. At a true successor, the displayed factors are ideals on the actual previous manifold; their genuine finite product is pulled back along the original projection to the actual new manifold and added to its true current baseline. Each recursive call is on the strict preceding tower, so this defines an entire ideal for every q, r, including every actual empty blowup step. No selected run, zero-locus dictionary or coherence witness is used to define it. □

Theorem 416.4 (Only original canonical inputs give coherent entire year equations and the exact original zero-locus dictionary)**.** *For ONLY original marked-domain input, its literal original canonical history and all q, r, the actual $Y_{T,q,r}$ is coherent and its WHOLE zero locus is precisely the SAME original enhanced companion history-year threshold.*

Proof. Induct on the real finite original tower. Genuine test-domain transport gives actual current and previous smooth marked domains, hence coherence of all the actual baseline and order factors. At a successor use genuine finite product coherence, actual holomorphic blowup pullback coherence and actual finite sum coherence. Thus each ideal in the displayed real recursion is coherent.

At the empty history the actual run is zero, so the entire enhanced threshold is the actual baseline, already presented by B_q. At a successor, the sum zero-locus rule is intersection, genuine coherent product zero-locus is union, and actual coherent holomorphic pullback zero-locus is inverse image. The two displayed recursions therefore give exactly the already proved genuine whole original zero/successor run equations. Induction identifies the old year's actual zero locus in the increment branch. For the original canonical consumer, its actual constructed invariant internally supplies genuine order monotonicity and its true test tower, so all these comparisons apply without an analytic, coherent-presentation or reset certificate. □

Theorem 416.5 (The actual history ideal constructs honest finite germ generators and threshold equations)**.** *At every point of the actual original final manifold, the SAME genuine history ideal has finitely many holomorphic generators on a true neighborhood. At every nearby point, BOTH its actual germ membership directions are given by their true finite span, and the original enhanced year threshold is exactly the simultaneous vanishing of those SAME generators.*

Proof. Apply the actual internally proved coherence of $Y_{T,q,r}$ to the real reference point. This constructs a genuine open and finite holomorphic functions with both nearby germ-span directions. Each such generator is an actual ideal germ, giving one zero-locus inclusion. Conversely every actual ideal germ is a true finite holomorphic combination of these generators, hence vanishes at the point when they all do. The preceding entire original zero-locus dictionary identifies this precisely with the SAME enhanced year threshold. □

Theorem 416.6 (These real whole year ideals internally supply genuine original-boundary marked inputs)**.** *For those SAME original-only inputs, form the genuine marked ideal with WHOLE ambient carrier, SAME actual current original SNC boundary, ideal $Y_{T,q,r}$ and mark one. This is in the genuine original ambient marked domain. No thinness, nonzero, boundary or desired domain certificate is given.*

Proof. Its actual coherent ideal is already constructed. Its zero locus lies in the actual current original cosupport. That cosupport has empty ambient interior: if an interior point existed, actual original nonzero along its smooth carrier would supply an ideal germ nonzero at arbitrarily near carrier points. Actual coherence makes it a true ideal germ near the point. Inside that supposed interior, actual positive marked order forces it to vanish, a contradiction. Thus the actual history zero locus likewise has empty interior.

If its coherent ideal were zero along the whole ambient carrier at any point, actual finite coherent generators would vanish on a neighborhood. Its true whole zero locus would contain that nonempty open, contradicting the proved empty interior. This internally supplies nonzero at every ambient point. The actual current domain supplies the SAME original SNC list. The whole ambient carrier is smooth of actual ambient dimension and is transverse to that real SNC list. These fields, actual coherence and genuine mark one prove the asserted original marked domain. This creates real auxiliary inputs; it does not assert their centres solve the missing full recursive threshold or compact-window global scheduling problem. □

417 The actual whole history-year defining ideals commute with genuine original open histories

The true carrier-relative derivative and history ideals are concrete global objects. Their whole ideal pullbacks, and their actual auxiliary marked records, now commute with original open restriction while retaining all actual empty history steps. This supplies actual local equation comparisons for the analytic subspaces used in Bierstone–Milman[4], Theorem 13.4, printed pages 299–301. Full arbitrary recursive lower threshold presentations and the actual compact-window global centre engine remain further producers; no naturality certificate is a caller input.

Theorem 417.1 (Actual carrier, relative derivative and nonmonomial baseline ideals have genuine whole open pullbacks)**.** *For actual smooth closed carrier and coherent ideal, every entire carrier-order ideal commutes with genuine open pullback. For true marked-domain input, its SAME actual nonmonomial-order and current-baseline ideals likewise commute with true original open restriction, for every order including zero.*

Proof. Actual vanishing ideals pull back to vanishing ideals of the real open carrier trace. The genuine open inclusion has true local holomorphic inverses; actual iterated relative derivative ideals therefore commute with its entire pullback. Real coherent sum pullback gives the order-ideal identity. For the marked-domain input, its actual SNC, transversality and nonzero construct coherent actual monomial and nonmonomial parts. The genuine local-inverse colon theorem identifies the entire pulled-back nonmonomial ideal with that of the literal open marked pullback. Substituting it in the real order-ideal formula gives nonmonomial-order ideal equality. Apply genuine coherent sum pullback to the two actual baseline factors; the carrier, ideal, marking and ordered divisor are all the literal original open pullbacks. This proves whole ideal equality, not only equality of zero sets or scalar orders. □

Lemma 417.2 (The actual restricted blowup square gives the same whole coherent ideal pullback)**.** *For a genuine actual blowup, the real restricted blowup over a true original open and the two actual open inclusions give an exact whole coherent-ideal pullback square, keeping the SAME original map and every actual numerator.*

Proof. The actual restricted map sends a genuine source-open point to its original blowup image in the target open; membership follows from its true preimage condition. Its ambient projection is holomorphic and the target is open, so that SAME lift is holomorphic. The two composites with the genuine open inclusions are literally the SAME original projection on every point. Actual coherent pullback composition along those true holomorphic maps therefore identifies their entire pulled-back ideals. No square or desired whole-ideal comparison is assumed. □

Theorem 417.3 (The whole actual year recursion commutes with every genuine original open history)**.** *Only genuine original marked-domain and true source-test history supply, for every actual source open and all q, r, equality of the ENTIRE pullback of the whole original $Y_{T,q,r}$ with the SAME actual year ideal constructed from its literal open marked input and true restricted history.*

Proof. Induct on the real original finite source tower, retaining every true empty step. At the identity history the preceding actual baseline pullback theorem gives the assertion. At a genuine successor, actual controlled-transform/open restriction identifies the literal current and previous marked inputs with the true open pullbacks of the SAME original current and previous records. Original source-test domain transport internally gives their actual coherent ideals and SNC/nonzero fields. The preceding whole baseline and order pullback identities therefore identify the actual reset factors. Genuine coherent sum pullback and the actual restricted-blowup ideal square give the entire zero-run formula. In the increment branch, apply the inductive whole old-year equality and true coherent product pullback before the same real restricted-blowup square

and sum pullback. This is exactly the original real reset/increment recursion. No history reindexing, dropped empty step or supplied naturality record is used. □

Theorem 417.4 (The actual same-boundary auxiliary marked record has its entire original open restriction)**.** *For those actual original histories the ENTIRE auxiliary year marked record restricts to the SAME auxiliary record formed from the literal restricted original history. Whole carrier, entire actual ideal, ordered current original SNC boundary and mark all agree. Original-only canonical inputs internally supply every source-test hypothesis used in this dictionary.*

Proof. The real ambient carrier's open inverse image is the whole source open. Actual controlled-transform/open restriction identifies the SAME true current original ordered boundary with its literal pullback, retaining all historical positions and real exceptional entries. The previous theorem gives entire ideal equality. The original auxiliary mark remains one. These four exact fields prove the whole marked-record identity. For the original canonical consumer, the SAME constructed original invariant internally provides its true source-test tower; apply the preceding whole identity to it and retain the previously constructed actual original-boundary marked inputs. Arbitrary deeper global lower closure and actual global scheduling are not consequences asserted here. □

418 Actual whole special-reset thresholds have genuine coherent defining ideals

The SAME original masked Step One history resets at actual higher level or at same level and SPECIAL. We construct its entire defining ideals from real selected smooth SNC-intersection components and the true coefficient codimension-one support. These concrete analytic presentations implement the special/member branches of Bierstone–Milman[5], Section 5, Step I, Cases A and B. In particular they supply actual auxiliary ideals for the lower-history year recursion; whole arbitrary recursive and higher-order semicoherent comparisons remain separate open producers.

Theorem 418.1 (The actual whole member-intersection locus has a genuine coherent component-product ideal)**.** *For actual closed smooth N, original SNC list E transverse to N and true analytic $A \subset N$, take all finite original indexed intersections H_S. Form the genuine finite product of vanishing ideals of the actual selected components of $H_S = N \cap \bigcap_{i \in S} E_i$ wholly contained in A. This entire ideal is coherent and cuts out precisely the SAME original member-intersection locus. The family of global components need not be finite.*

Proof. Actual SNC and transversality make every true indexed intersection closed smooth. The already proved original component continuation theorem makes an arbitrary selected union of its genuine global components closed smooth, even for an infinite component family. Its real vanishing ideal is therefore coherent and has exactly that selected union as its zero locus. There are only finitely many indexed subsets; actual finite product coherence gives the whole ideal, and its genuine product zero-locus rule gives their finite union. Actual analytic continuation on the SAME original components identifies this entire union with the real member-intersection locus in both directions. Thus coherence and the whole dictionary are constructed, with no component presentation, finite component-family or desired member-locus certificate. □

Theorem 418.2 (The real coefficient and indexed-level baseline has whole coherent equations)**.** *Only actual original maximal-order marked-domain input and its literal canonical masked Step One history give a genuine coherent ideal defining the entire SAME current cosupport with original level at least s, for every $s \geq 0$.*

Proof. The unchanged original bridge internally supplies the real homogenized coefficient test history and its entire current cosupport equality with the original transformed input. Its actual domain and controlled geometry give true closed smooth coefficient carrier, coherence and positive

actual mark. The genuine carrier-order ideal defines precisely that whole coefficient cosupport. The actual original members are a true prefix of its real current SNC list. Their genuine indexed count ideal is coherent and has zero locus exactly count at least s, including zero and impossible counts. The actual sum of these two real ideals is coherent and cuts out their intersection. Substitute the original whole cosupport equality and literal original level count definition to get the displayed SAME entire current baseline. □

Theorem 418.3 (Actual original top and special-reset thresholds have genuine whole coherent ideals)**.** *Only those SAME original inputs construct coherent $R_{T,s}$ with entire zero locus exactly the literal original special-reset locus*

$$\{y \in m_T.\mathrm{cosupp} : s < \mathrm{level}(y) \text{ or } (s = \mathrm{level}(y) \text{ and } SPECIAL(y))\}.$$

Equivalently this is the SAME whole masked Step One threshold $(s, \top)$.

Proof. For $s = 0$, form the actual product of the level-one baseline with the vanishing ideal of the true coefficient codimension-one cosupport part. Actual maximal order transport makes that part closed smooth, supplying coherent vanishing equations and its exact whole zero locus. Its union with the positive-level baseline is exactly the literal zero-level top/special reset: at level zero SPECIAL is the codimension-one branch, and at positive level the higher-level branch is already retained.

For $s = b+1$, use the genuine product of the level-$b+2$ baseline with the sum of the preceding actual member-locus component ideal and the true indexed level-$b+1$ count ideal. Its zero locus is the union of the actual higher-level baseline and the member-locus/count intersection. The real original positive SPECIAL definition identifies this union with the displayed reset in both directions, retaining every higher-level point and the exact equality branch. All ideals and coherence were constructed before using this dictionary. The actual masked lower value is finite on nonspecial true cosupport points, so the original top-threshold theorem gives precisely the SAME threshold interpretation. No desired top, reset, analytic or coherent-ideal witness is input. □

Theorem 418.4 (Those literal whole original reset ideals have honest finite germ generators)**.** *At every actual final point the SAME actual reset ideal has finite holomorphic generators on a genuine open, with BOTH nearby germ-span directions and the entire original reset threshold as their simultaneous zero set there.*

Proof. Apply the internally proved coherence of the actual reset ideal at the real reference point. The resulting original functions are holomorphic on a true neighborhood and give both finite-span directions. Every such function is an actual ideal germ, so true ideal zero-locus membership forces its value zero. Conversely every ideal germ is a genuine finite holomorphic combination of these same functions, whose simultaneous vanishing forces every ideal germ to vanish. The preceding entire original reset dictionary gives exactly the stated original threshold equations. □

Theorem 418.5 (The actual whole reset equations supply a genuine SAME-boundary ambient marked input)**.** *The true marked record with whole ambient carrier, SAME current original ordered SNC boundary, actual $R_{T,s}$ and mark one belongs to the genuine ambient original marked domain. Only original input geometry/history are given.*

Proof. The exact zero locus is contained in the actual original current cosupport. Its genuine positive marked order, coherent germ continuation and true nonzero along the smooth carrier force that cosupport to have empty ambient interior: an interior point would have a true ideal germ nonzero at arbitrarily nearby carrier points, while coherence and positive cosupport force that same germ to vanish there. Thus the whole reset zero locus has empty interior. If its genuine coherent ideal were zero along the entire ambient carrier at some point, true finite generators would all vanish on a neighborhood and put that nonempty open in its actual zero locus. This contradiction internally constructs nonzero at every ambient point. Actual original domain transport

gives the SAME ordered current SNC boundary, and the whole ambient carrier is truly smooth, of the original ambient dimension and transverse to it. These exact fields and real mark one supply the actual domain, without nonzero, thinness, boundary or desired-domain certificates. This does not construct arbitrary deeper recursive presentations or the global scheduler. □

419 Actual masked lower-history year thresholds have genuine whole coherent ideals

The real original Step One run resets at higher level or at equal level and SPECIAL. Its just-constructed entire reset ideal now supplies actual finite history equations, rather than an analytic-set or semicontinuity certificate. This retains the historical choices of Bierstone–Milman[5], Section 5, Step I, and furnishes actual auxiliary analytic presentations for Theorem 13.4 of Bierstone–Milman[4], printed pages 299–301. Full arbitrary recursive lower/higher-special semicoherent comparisons and the simultaneous global scheduler remain open.

Definition 419.1 (The real lower-history ideal uses the literal higher-or-SPECIAL reset recursion)**.** For genuine original m, T, s, r, write $B_{T,s}$ for the actual level baseline and $R_{T,s}$ for the SAME entire special-reset ideal. Define

$$L_{\mathrm{id},s,r} = B_{\mathrm{id},s}, \qquad L_{T\pi,s,0} = B_{T\pi,s} + \pi^* R_{T,s},$$

$$L_{T\pi,s,r+1} = B_{T\pi,s} + \pi^*(R_{T,s} L_{T,s,r}).$$

Every actual empty original step and every true higher-level and SPECIAL reset factor is retained.

Proof. The original tower is a true finite inductive history. The identity's baseline is an actual ideal on its original manifold. At a successor the genuine old reset and year ideals are on the SAME actual previous manifold. Their actual product is pulled back along the true original projection to its actual next manifold and added to the real current baseline. Every recursive call is on the strict previous tower. Thus the formula defines an entire ideal at every s, r, without a chosen run, coherence or desired zero-locus witness. □

Theorem 419.2 (Only original inputs give whole coherence and the exact original masked lower-year zero dictionary)**.** *For ONLY original maximal-order marked-domain and its literal original masked Step One canonical history, every $L_{T,s,r}$ is coherent and its ENTIRE zero locus is exactly*

$$\{y \in m_T.\mathrm{cosupp} : s \le \mathrm{level}(m, T, y), \quad \mathrm{regimeRun}(\mathrm{stepRegime}(m, s), T, y) \le r\}.$$

This is the SAME original enhanced lower starting-year threshold.

Proof. First prove coherence independently by induction on the actual original tower. Its true canonical prefixes internally supply coherent actual baselines and reset ideals. Actual coherent finite products, true holomorphic blowup pullbacks and real sums therefore construct each entire year ideal.

For the whole dictionary, the identity's true run is zero, and the baseline zero dictionary gives the assertion. At a successor, actual sum zero loci are intersections, genuine coherent product zero loci are unions, and real holomorphic pullback zero loci are inverse images. Substitute the already proved actual current baseline and old reset dictionaries. In the increment branch induction gives the true old year. The original exact zero/successor run equations identify precisely these sets, using actual original cosupport inheritance and level nonincrease internally. Higher old level, equal old level and SPECIAL, original empty steps and all current baseline points remain in both directions. No caller year, reset or analytic dictionary is input. □

Theorem 419.3 (The actual whole lower-year ideal supplies honest finite local equations)**.** *At every actual final point, the SAME whole ideal has a true finite holomorphic family on a genuine neighborhood, with BOTH nearby ideal/span directions and precisely the entire original enhanced lower-year threshold as its simultaneous zero set there.*

Proof. Actual independently proved coherence constructs the neighborhood and true finite family with both nearby germ directions. Each genuine family germ belongs to the actual ideal, so actual ideal zero membership forces its value zero. Conversely every actual ideal germ is a real finite holomorphic combination of those same functions. Simultaneous vanishing therefore forces every such germ to vanish. The preceding exact whole original lower-year dictionary gives the stated literal original threshold equations. □

Theorem 419.4 (The actual whole lower-year equations construct a genuine SAME-boundary ambient marked domain)**.** *The true whole ambient carrier, SAME actual current original ordered SNC list, actual $L_{T,s,r}$ and mark one form a genuine original ambient marked-domain input. No nonzero, thinness, boundary or desired-domain certificate is given.*

Proof. Actual original canonical Step One data internally supply the real source-test tower and its true current marked domain. The preceding ideal is coherent and its exact whole zero locus is contained in the actual current cosupport. Genuine current nonzero along the smooth carrier and real coherent germ continuation force that positive cosupport to have empty ambient interior; hence so does the actual lower-year zero locus. Were its actual coherent ideal zero along the whole ambient carrier at a point, its genuine finite generators would all vanish on a neighborhood, forcing a nonempty open into that whole zero locus. This contradiction supplies actual nonzero at every ambient point. True current original SNC, whole ambient smoothness/dimension, genuine transversality and mark one give every remaining original domain field for these SAME objects. □

Theorem 419.5 (An original full positive history internally supplies the genuine coherent companion lower-year input)**.** *For ONLY original full canonical history and a positive original cosupport reference point, the actual fixed companion has genuine entire coherent lower-year defining ideals and SAME-current-boundary ambient marked inputs on its true final reference open, for every s, r. No selected companion or lower-history input is supplied by the caller.*

Proof. The original actual fixed-companion producer internally gives the true reference open, original reference-point membership, actual companion marked domain, real maximal order and genuine masked Step One canonical restricted history. Apply the preceding independently proved whole lower-year ideal dictionary and actual marked-domain constructor to those SAME real objects. This gives the entire enhanced actual companion lower-year threshold equations, including all higher-level and SPECIAL resets, on its actual reference open. Its current original ordered boundary is retained. This is a real original-input consumer; it does not identify arbitrary deeper original thresholds with this particular enhanced-year family or complete the full global history scheduler. □

420 Actual global canonical scheduling from compact initial cosupport on a possibly noncompact ambient

The genuinely original marked ideal, invariant and global tower are retained. Only the INITIAL actual cosupport is assumed compact; neither the ambient manifold nor the carrier is assumed compact. This produces actual stages, global maximum centres and actual blowup maps before termination is proved. It closes this geometric scheduling subclass. The general noncompact-cosupport extension/smoothification of BM97 Theorem 13.4 and Lemma 13.7, arbitrary global full thresholds and original coefficient semicoherence remain OPEN.

Lemma 420.1 (Actual proper projections keep every genuine current cosupport compact)**.** *Let m be an original invariant-domain marked ideal whose INITIAL cosupport K is compact. For EVERY genuine marked test tower T, its actual entire current cosupport is compact, and its actual projection maps it into K. No current-stage compactness is a caller input.*

Proof. The actual test-tower transform is well formed, so its entire current cosupport is closed. Genuine cosupport inheritance through each admissible blowup, composed along the actual finite tower, places that cosupport inside the preimage of the INITIAL K. The actual tower projection is proper, by genuine properness of every actual blowup and composition. The preimage of K is therefore compact. The true closed subset just identified is compact, with exactly the stated original projection containment. No compactification or replacement of the original ambient is used. □

Theorem 420.2 (Constructed global canonical growing sequence with actual global maxima and projections)**.** *ONLY an actual resolution invariant, its original domain input and compact INITIAL cosupport internally construct a genuine global GrowingSeq starting at the literal identity tower. EVERY global recorded prefix is canonical for the SAME original invariant/input. At EVERY successor there is a genuine global blowup map whose centre is EXACTLY the ENTIRE actual global maximum locus on the current original cosupport. These stages, centres and maps are constructed without a scheduling, maximum-existence or terminal certificate.*

Proof. Consider actual global finite towers together with their true canonicality. Canonicality internally gives actual original test-tower geometry. The first lemma supplies compactness of every actual current cosupport. When that cosupport is nonempty, true original upper semicontinuity on this compact set internally attains a global maximum. Its ENTIRE maximum locus is the original invariant's genuine admissible centre. If the current cosupport is empty, its literal maximum locus is empty and genuinely admissible. Original nonzero geometry makes each such centre thin; the actual blowup existence theorem constructs its real global manifold, map and blowup proof. Appending it preserves true canonicality for the SAME original input. Choose these actual successors and iterate from the literal identity tower. The actual dependent stages and successor equalities give a genuine GrowingSeq; every chosen successor retains its actual map and literal maximum-locus equation. All choices precede any termination argument. □

Theorem 420.3 (The actual constructed sequence terminates and all later prefixes remain original resolutions)**.** *That actual constructed global sequence has an index N such that EVERY later recorded prefix is a genuine resolution of the SAME original marked input. In particular there is an actual finite global canonical terminal tower on the original possibly noncompact ambient.*

Proof. Suppose no recorded current cosupport were empty. Compactness from the first lemma then makes EVERY actual global maximum centre nonempty. Every point of each such centre lies in actual current cosupport, so its actual original projection belongs to the SAME compact INITIAL K. Consequently the literal count of genuine centres meeting K increments by one at every successor. Induction on the actual iteration proves that the count at index i is exactly i. Apply the ORIGINAL invariant's proved per-sequence local finiteness to THIS already constructed global canonical sequence and THIS compact K. It gives a finite bound on these exact counts, contradicted by the next natural index. Thus an actual recorded stage has empty cosupport. Genuine test-tower cosupport inheritance through each subsequent actual step keeps its current cosupport empty; induction gives this for EVERY later index. Their canonical prefixes are genuine test towers, so these are actual original resolutions. No numerical bound, terminal history or empty auxiliary-transform assertion was supplied by the caller. □

Theorem 420.4 (Original all-depth compact-cosupport global scheduler and finite terminal consumer)**.** *At ANY original invariant depth d, ONLY the original invariant-domain marked ideal and compact INITIAL actual cosupport construct the preceding SAME original global canonical*

growing sequence, actual maximum-centre maps and eventual terminal histories. Thus a genuine finite SAME-P canonical global terminal tower exists without CompactSpace of the original ambient/carrier. The unrestricted noncompact-cosupport scheduler is not asserted.

Proof. Instantiate the constructed scheduler with the ALREADY PROVED original invFamily at depth d. Its genuine original value, admissibility, semicontinuity and per-sequence finiteness are supplied internally by that original object. The preceding theorem applies to the original domain and actual compact initial cosupport and constructs all global geometric data. Choose the actual terminal recorded index to obtain the finite tower; its canonicality and empty original cosupport have already been proved. This is an original-input constructor in a stated geometric subclass, and does not infer BM13.4 preparations from the earlier local reference cuts. □

421 A genuine global canonical phase erases a compact initial FULL upper cut

This advances global scheduling beyond compact WHOLE cosupport: the original cosupport may be noncompact. Only the INITIAL literal FULL invariant upper cut at a fixed original threshold is compact. A real global canonical history clears that cut; no terminality of the remaining noncompact cosupport is asserted. No arbitrary-threshold analytic presentation is assumed or inferred. The noncompact-cut BM13.4 extension/simultaneous preparation remains OPEN.

Definition 421.1 (Literal full original upper cut at the fixed old threshold)**.** For the SAME original marked input m, its actual invariant R, a genuine prefix T and fixed threshold b, put $K_T(b) = \{z \in \mathrm{cosupp}(m_T) : b \le R(m, T, z)\}$. The initial set is literally $K_{\mathrm{id}}(b)$.

Proof. The actual original transform and its invariant are defined on the real final tower manifold; the displayed original comparison defines a true subset. No analyticity, compactness, presentation, count bound or schedule is a field of this definition. In particular b need not be attained. □

Lemma 421.2 (Original whole-value nonincrease and true properness internally keep the actual cuts compact)**.** *Along EVERY actual original canonical prefix, the genuine projection maps $K_T(b)$ into the initial literal $K_{\mathrm{id}}(b)$. If that INITIAL set is compact, every actual current $K_T(b)$ is compact. If it is nonempty, the ENTIRE global original maximum locus exists, is nonempty and lies in this actual cut, even when the remaining cosupport is noncompact.*

Proof. Induct on the real canonical prefix. The actual original step-drop theorem supplies projected old cosupport and genuine nonincrease (strict above the actual nonempty maximum centre and equality off it); compose with the induction inequality. Thus the SAME whole original value at any actual current point is bounded by its INITIAL projected value. Combining this with cosupport inheritance gives the claimed actual cut containment. The real current cosupport is closed, and proved original upper semicontinuity makes its literal fixed-threshold cut closed. The actual proper tower projection has compact preimage of the INITIAL cut, so its true closed current cut is compact.

If the cut is nonempty, original upper semicontinuity restricted to this real compact cut attains a maximum at some actual x in it. Any cosupport point outside the cut has value strictly below b, hence at most the value at x; points inside the cut are bounded by the attained maximum. This proves that x is a REAL global cosupport maximum. Every point of the ENTIRE global maximum locus has value at least that of x, so it too is in the actual cut. No global maximum or current compactness is supplied. □

Theorem 421.3 (Actual global canonical finite phase erases the compact initial full cut)**.** *ONLY original invariant/domain data, the fixed original threshold b and compactness of the INITIAL literal full cut construct an ACTUAL finite global original canonical tower T such that $K_T(b)$ is empty. Every actual remaining original cosupport point has original value strictly below b. No desired tower, bound, maximum or phase certificate is an input.*

Proof. For every actual global canonical prefix choose the ENTIRE original global maximum centre when its actual current cut is nonempty; the preceding lemma supplies real maximum existence and centre containment in the cut. When the cut is empty choose the genuine empty centre, without asking for a maximum of the remaining noncompact cosupport. Original admissibility and thinness supply actual global blowup manifolds/maps. Appending these real maps preserves canonicality for the SAME original invariant and input. Dependent iteration from the literal original identity constructs a genuine GrowingSeq with all actual successor equations BEFORE erasure is proved.

Suppose no current cut ever became empty. Every constructed centre would then be a real nonempty global maximum and lie in the current cut. Its actual original projection meets the SAME compact INITIAL cut by the preceding containment. The literal global centre-meets-compact-preimage count therefore increments by exactly one at EVERY real successor; induction gives count i at index i. The original per-sequence finiteness theorem applied to THIS actual constructed canonical sequence and THIS compact INITIAL cut contradicts those exact counts. An actual recorded prefix therefore has empty current cut. Choose that real finite global history. For any remaining actual cosupport point, the absence of $b \leq R$ is precisely the strict inequality $R < b$ in the original linear order. This clears the chosen full threshold, without asserting a resolution of the remaining noncompact cosupport or a noncompact-cut scheduling theorem. □

Theorem 421.4 (ONLY original all-depth inputs construct the genuine global compact-full-cut phase)**.** *At EVERY original depth, for ANY fixed old full threshold with compact INITIAL literal cut, ONLY the original marked invariant-domain input constructs the preceding actual GLOBAL SAME-P canonical phase, its genuine test-tower geometry, empty current cut and whole remaining strict value bound. The old threshold is unchanged and need not be an attained reference.*

Proof. Internally supply the ALREADY CONSTRUCTED original invFamily at the stated depth and apply the actual phase producer to the original marked input and actual INITIAL compact cut. Original canonicality internally supplies its real marked test history. The proved empty-cut conclusion gives the SAME full old-threshold strict bound at EVERY remaining original cosupport point. No caller analytic-cut presentation, current compactness, later history, maximum, numerical bound or desired erasure statement is used. □

422 Actual finite canonical centre images produce full-threshold analytic loci

This is an a posteriori route using ALREADY PROVED genuine canonical local resolution and Remmert proper mapping. No analytic superlevel or coefficient presentation is a caller certificate. The first original-input-only endpoint is ALL arbitrary full thresholds at the REFLEXIVE original history. Analytic superlevels at an arbitrary retained original prefix require a genuine local ORIGINAL-prefix continuation, separately OPEN; replacing the invariant by that of the current marked record is not used to assert that endpoint. Original coefficient semicoherence and the unrestricted global scheduler also remain OPEN.

Lemma 422.1 (The actual one-step full cut has BOTH exact centre/image membership directions)**.** *For an actual original canonical next blowup π of T at actual C, write $K_T(b)$ for the literal full old-threshold cut. Then EXACTLY $K_T(b) = (C \cap K_T(b)) \cup \pi(K_{T+\pi}(b))$.*

Proof. A genuine next-cut point projects into old cosupport and its true original value does not increase, so its image lies in the old cut. The selected centre piece also lies there. Conversely take an old-cut point. If it lies in C, it is in the selected centre piece. Otherwise actual blowup surjectivity supplies a real lift. Both true off-centre cosupport directions put that lift in next cosupport. The ORIGINAL step-drop equality off the actual centre preserves its SAME original value, so the lift is in the next full cut. Thus both displayed set directions hold, including empty centres and thresholds which are not attained. □

Lemma 422.2 (An actual canonical centre selected by a fixed full threshold is genuinely analytic)**.** *The actual set $C \cap K_T(b)$ in that one-step dictionary is either the ENTIRE actual centre C or empty, and hence genuinely analytic. No analyticity of the full cut is assumed.*

Proof. If the intersection is empty the claim is literal. Otherwise choose an actual x in it. The actual canonical centre cannot be empty, so original canonicality makes it the ENTIRE global maximum locus. Every y in that centre is in real cosupport and has value at least that of x, hence at least the SAME fixed b. Thus every y lies in the full cut and the intersection is the whole C. Actual blowup geometry supplies closed smoothness of that true centre, and therefore its analytic presentation. This proves analytic selection independently of the desired full locus. □

Definition 422.3 (Literal finite union of actual selected centre images)**.** For a real retained original prefix T, fixed full threshold b and actual tail S, define a subset on the real final manifold of T: at the identity tail it is empty; at a real last step append the actual image, under the previous REAL tail projection, of that centre intersected with the literal ORIGINAL full cut at the corresponding WHOLE original prefix.

Proof. This is genuine finite-tower recursion on actual centres and projections. All actual source/final manifolds and original history positions are retained; empty centres contribute empty images. No analytic or finite history identification field is included in the definition. □

Theorem 422.4 (BOTH whole full-threshold directions equal the genuine finite centre images plus the real final-cut image)**.** *If the ACTUAL ORIGINAL appended history $T + S$ is canonical, the literal cut at T is EXACTLY the finite selected-centre-image set just defined, union the actual S-projection image of the literal final ORIGINAL cut at $T + S$.*

Proof. Induct on the real tail. At identity there are no centre images and the real projection is identity. At a genuine last step the canonical ORIGINAL prefix is supplied by actual appended canonicality. Apply the induction identity, then substitute the first lemma for its actual previous final cut. True images preserve unions, true successive images compose, and the actual tail projection is precisely that genuine composed map. Associativity of finite union gives the literal recursive centre-image set followed by the real final-cut image. Every operation is equality of true sets, so BOTH whole membership directions and every empty position are retained. □

Theorem 422.5 (The actual finite selected images are analytic and any genuine canonical erasure gives an analytic original cut)**.** *The finite actual selected-centre-image set is genuinely analytic on the actual final manifold of T. Thus whenever an ACTUAL original canonical tail erases its literal final full cut, the literal full cut at T is analytic. No desired analytic-cut certificate or compactness is supplied.*

Proof. Induct on the actual tail. Each newly selected centre is independently analytic by the second lemma. Its actual previous tail projection is holomorphic and proper by genuine blowup composition. Its restriction to that actual closed analytic centre is proper: intersect a true compact preimage with its true closed carrier. Apply the ALREADY PROVED Remmert proper mapping theorem to this actual map/set. Its actual image is analytic; finite union with the induction result remains analytic. If the actual final cut is empty, its real image is empty, so the preceding BOTH-direction set dictionary identifies the original cut with this produced analytic finite union. This is an actual finite-history consumer; the original-only constructor below internally supplies the genuine erasing histories. □

Lemma 422.6 (Actual open restriction preserves arbitrary full thresholds for the SAME retained original history)**.** *Restricting the ORIGINAL marked input and its ACTUAL original canonical tower to a true base open identifies its arbitrary full-threshold cut EXACTLY with the actual final-open preimage of the original cut.*

Proof. Genuine restricted-tower cosupport equals the true preimage of original current cosupport. On those actual points, the PROVED original open value law equates the SAME whole-history invariant values. Apply that equality to the SAME fixed arbitrary threshold, in BOTH directions, retaining the actual cosupport membership. No rebased-current invariant is substituted. □

Theorem 422.7 (ONLY original inputs give GLOBAL analytic arbitrary FULL thresholds at the reflexive original history, at EVERY depth)**.** *For ONLY an original invariant-domain marked input and ANY full threshold, the literal ORIGINAL full-invariant superlevel at its REFLEXIVE history is GLOBAL analytic, at EVERY original invariant depth. No compactness, analytic/presentation, selected local tower or desired gluing input is used.*

Proof. At every actual original point apply the ALREADY PROVED genuine canonical local resolution constructor to the SAME original input and its compact singleton. It internally constructs a true original base open containing the point and an ACTUAL finite canonical terminal history of the literal restricted ORIGINAL marked input. Original domain restriction supplies its real domain. Genuine terminal cosupport emptiness erases EVERY full cut there, before any analytic-cut assertion is used. The fifth producer at the REFLEXIVE history of that restricted original input therefore constructs its actual arbitrary-threshold analytic locus. The sixth lemma identifies this with the actual preimage of the SAME original REFLEXIVE cut. Genuine open analytic presentations transport to the original point. Since every point has this internally constructed presentation, that entire original cut is globally analytic. Instantiate with the ALREADY CONSTRUCTED original invFamily at ANY depth, internally supplying its true well-founded values and canonical driver. For an arbitrary nonreflexive ORIGINAL prefix the existing local driver forgets the retained-prefix/tail identity; that genuine continuation remains a separate obligation, as do coefficient semicoherence and unrestricted global scheduling. None is inferred by rebasing here. □

423 A genuine companion TEST after a full-value-cut preparation step

A canonical prefix has a true order-run companion TEST. A further original admissible blowup need not extend canonicality. The geometric condition below is retained explicitly: the ENTIRE next centre lies above the chosen old full reference value. It is not replaced by a pointwise condition at one centre point, and no candidate TEST is supplied.

Theorem 423.1 (A genuine upper-cut centre extends the old run-compAt TEST)**.** *Let M have genuine DOMAINb in positive ambient dimension. Let F be an honest ordered original value function whose order dominates nonmonomial order. Take an actual original TEST T that is CAN for F, and a point y of its actual current cosupport with attained positive order q. Compute its original order run r, head H, tail S and true order open W. Let C be a genuine original admissible centre at the current stage and let π be an actual blowup of C. Assume the geometric whole-centre condition $F(T,y) \leq F(T,c)$ for EVERY $c \in C$. Then construct a genuine TEST of the SAME original compAt formed at H, along the literal restriction to W of the old tail followed by this actual blowup. No CAN assertion for the new upper prefix and no companion TEST hypothesis are required.*

Proof. The old canonical-prefix theorem constructs its genuine restricted order-run companion TEST, retaining every actual original and empty position. Split the genuine upper TEST into its real head and tail. Original domain inheritance constructs both current domains; the true order open gives the initial bound by q. Original admissibility of C extends the genuine upper TEST, hence its restriction supplies a true smooth trace centre, containment in the actual restricted upper cosupport, and NC with its whole current ordered boundary.

For every point of that trace centre, order dominance applied to the old reference point and the whole-centre upper-cut inequality gives current nonmonomial order at least q. The genuine companion cosupport dictionary on the independently constructed old companion TEST is the intersection of that order superlevel and the actual restricted upper cosupport. The actual whole controlled open dictionary identifies these upper transforms with the literal open pullbacks, proving trace-centre containment in the actual companion cosupport. Its whole ordered boundary is the same transformed ordered boundary as the restricted upper record, so the true old NC condition also holds for the companion. Together with the actual smooth trace centre this is genuine companion admissibility. Append the actual restricted blowup to its real old companion TEST. No maximum or canonicality of the new upper centre was used; the whole-centre cut condition was used explicitly. □

Theorem 423.2 (The unchanged original full value internally supplies order dominance)**.** *Specialize to the unchanged constructed original full GENERAL value at depth $k+1$, an original DOMAIN$k+1$ record and its actual CAN TEST prefix. Given ONLY a genuine next original admissible centre, its actual blowup and the same ENTIRE original full-value-cut condition, construct the old run-compAt actual extended TEST and its genuine current DOMAIN$k+1$ after that step. No caller order-dominance or candidate-domain/TEST certificate is supplied.*

Proof. The unchanged original full value is lexicographically $(\nu, \dots)$. Its genuine order-dominance proof is therefore internal: comparison of two actual full values implies comparison of their actual first entries. Apply the preceding theorem to this exact original full function and the stated original geometric centre. Initial compAt domain follows from the actual positive companion construction at the genuine head domain. Genuine TEST inheritance along the newly proved extended masked TEST gives its entire current domain, including actual smooth carrier, coherent nonzero ideal, ordered SNC, transversality, dimension bound and positive mark. This is a real domain/TEST construction after a possibly noncanonical step. □

Theorem 423.3 (Both directions of the old-order cosupport survive the actual preparation step)**.** *For exactly the preceding original data, at EVERY actual point of the new masked endpoint, membership in the SAME old compAt current cosupport is equivalent to membership in the actual new upper original cosupport AND nonmonomial order equal to the OLD attained reference order q. The new upper prefix is not assumed CAN, and the order is not silently recomputed from a different reference point or rebased history.*

Proof. First construct the genuine extended companion TEST by the preceding original-only value producer. Original admissibility independently gives the upper next TEST and the actual old-head plus extended-tail TEST. Apply the genuine whole compAt cosupport theorem to those exact two TEST histories and the true original order mask. Its initial actual order bound and the proved companion TEST supply both inclusions and the exact order equality. Finally use the literal old head/tail controlled-transform equation to identify the upper endpoint with the actual new original prefix. No desired cosupport equality is an assumption. Supplying the whole-centre cut condition from a concrete preparation-centre producer remains a subsequent obligation; these assertions do not close arbitrary noncanonical GENERAL naturality, transformed semicoherence or a global scheduler. □

424 GLOBAL analytic arbitrary FULL thresholds along EVERY genuine ORIGINAL canonical history

This chapter retains the ENTIRE actual original history. It uses the existing PROVED actual empty-step compression with its true whole-value order embedding, actual finite nonempty-canonical alignment, and genuine local canonical terminal constructors. No invariant is re-based on a current marked record. No local continuation, comparison or analytic-cut certificate is a

caller input. This proves full superlevel analyticity; transformed ORIGINAL coefficient semicoherence and the unrestricted BM13.4 global preparation/ scheduler remain separate OPEN obligations.

Lemma 424.1 (A true order embedding has an exact original-threshold preimage in a well–founded linear order)**.** *For a true order embedding F of the original well-founded linear value order and ANY fixed original full threshold b, the actual set $\{v : b \leq F(v)\}$ is either empty or EXACTLY $\{v : c \leq v\}$ for an internally constructed actual least c. Surjectivity of F is not assumed.*

Proof. If the actual preimage set is empty, retain that equality. Otherwise the proved original well-founded order supplies a real least member c. Every member is at least c; conversely $c \leq v$ and actual order preservation imply $b \leq F(c) \leq F(v)$. These are BOTH exact original comparison directions, not an inverse-value or threshold-surjectivity certificate. □

Lemma 424.2 (An actual empty-free canonical prefix is no longer than a genuine empty-free terminal history)**.** *For the SAME original marked input/invariant, any actual canonical segment with all centres nonempty has length at most that of an ACTUAL canonical empty-free terminal history. The bound and any required prefix comparison are internally proved from those real histories.*

Proof. If the candidate prefix were longer, take its ACTUAL head of the terminal history's length and its REAL tail. Real prefix canonicality and the existing genuine equal-length canonical comparison construct an actual tower isomorphism between the terminal history and that head, over the SAME original identity base. Genuine whole cosupport transport makes that head's actual cosupport empty. The existing proved nonempty-canonical-tail lemma then forces its REAL tail length zero. The actual split length is positive by the assumed strict inequality, a contradiction. No terminality of the candidate prefix or desired alignment is supplied. □

Theorem 424.3 (Actual finite partial/terminal alignment constructs arbitrary full-cut analyticity)**.** *Given an ACTUAL empty-free canonical prefix and an ACTUAL empty-free canonical terminal history of the SAME original input/invariant, its arbitrary full threshold cut is genuinely analytic. No head, comparison map or analytic locus is a caller certificate.*

Proof. The preceding true length inequality internally supplies the ACTUAL head of the terminal tower with the candidate prefix length and its REAL remaining tail. Existing equal-length canonical comparison constructs a genuine tower isomorphism over the SAME original identity. The head followed by that actual remaining tail is literally the given terminal history, so its final full cut is empty. The preceding finite selected-centre-image theorem constructs an actual analytic full cut at the head for ANY threshold. Genuine whole cosupport transport and the PROVED original tower-isomorphism value law identify the candidate's literal cut with the true biholomorphic preimage of that analytic head cut, in BOTH directions. True analytic preimage geometry proves the claimed actual candidate locus analytic. □

Theorem 424.4 (Actual original empty compression retains arbitrary full-threshold analyticity)**.** *For the genuine original invFamily at ANY depth, an arbitrary actual canonical history and an ACTUAL canonical terminal history on the SAME original base internally give arbitrary full-cut analyticity at the arbitrary history, including EVERY original empty step and chronological boundary position. No compressed history, value map or analytic threshold is supplied.*

Proof. Apply the ALREADY PROVED original-only finite empty compression separately to the actual candidate and actual terminal history. These internally construct true empty-free canonical towers, real final biholomorphisms, actual whole cosupport dictionaries and the actual candidate whole-value order embedding F. Terminality of the genuine terminal compression is an OUTPUT of its proved resolution equivalence. Convert the real nonempty centre properties to actual

nonempty canonical segments. The third theorem therefore constructs analytic full cuts on the actual candidate compression at EVERY original threshold.

The first lemma internally handles the original fixed b under the TRUE whole-value embedding F. If its actual value preimage is empty, the candidate cut is empty. Otherwise its BOTH exact comparisons give a real threshold c on the compression. Combine those comparisons with the independently produced whole cosupport and whole original-value dictionaries. The actual candidate full cut is EXACTLY the true inverse-final-biholomorphism preimage of its actual compressed c-cut. Genuine holomorphic preimages therefore prove analyticity, retaining the SAME original value and EVERY empty history position. No future-tail invariance or current-record rebasing is asserted. □

Theorem 424.5 (ONLY original inputs construct GLOBAL arbitrary FULL analytic superlevels along EVERY actual original canonical prefix)**.** *At EVERY original invariant depth, ONLY original domain, an ACTUAL original canonical prefix T and ANY full old threshold b imply that its ENTIRE literal ORIGINAL full-threshold cut is GLOBAL analytic. No compactness, selected local histories, desired alignment/compression, analytic/presentation or continuation certificate is an input.*

Proof. At every actual final point y, apply the PROVED genuine original canonical local terminal constructor to the SAME original input and the compact singleton of its ACTUAL original projection $T(y)$. It internally supplies a real base open U and genuine canonical terminal history of the literal restricted ORIGINAL input. Independently restrict the entire ACTUAL original prefix T to U; PROVED original domain/open canonicality supply its actual domain and canonical history. The fourth theorem internally compresses and aligns these TWO real histories and constructs the arbitrary full cut on this ACTUAL restricted original prefix, without re-basing its invariant. The PROVED BOTH-direction original open-cut dictionary identifies that locus with the true final-open preimage of the SAME original full cut at T. Transport its genuine analytic presentation at the actual lift of y to the original final manifold. Every original final point therefore has its internally constructed analytic presentation, proving GLOBAL analyticity. All local terminal/compressed/aligned histories were actual producer outputs, not desired final identifications. This closes actual full-history global superlevel analyticity. Actual transformed ORIGINAL coefficient-presentation comparison, simultaneous-chain admissible HS preparation and the unrestricted global BM13.4 scheduler still require their own mathematics. □

425 A genuine global pure-dimensional extension of an ORIGINAL full reference centre

Bierstone–Milman[4], proof of Theorem 13.4, printed pages 299–300, first extends selected pure-dimensional local centres as genuine dimensional component unions of GLOBAL analytic full superlevels. The preceding all-history theorem now supplies exactly those actual full superlevels. The construction below internally supplies the local regularity and original admissibility as well: they are not caller certificates. This is the extension step, not smoothification outside the selected window or the simultaneous-chain engine of Lemma 13.7.

Lemma 425.1 (ONLY original input supplies a fixed-dimensional actual reference-centre window)**.** *At ANY invariant depth, for an actual ORIGINAL canonical prefix and an actual point in its true cosupport, internally construct a real open window containing the point and a dimension $q \leq n$. The SAME full reference threshold is bounded above by its attained reference value on that window, every point of its actual trace is regular of dimension q in the ORIGINAL global cut, and the literal trace is admissible for the actual restricted current ORIGINAL marked ideal with its SAME ordered boundary. No local centre, bound, regularity or dimension is input.*

Proof. The existing original-only reference-centre producer supplies an actual bounded open and genuine closed-submanifold admissible reference trace. The actual reference point belongs to that trace. Its true submanifold geometry therefore supplies an actual dimension and a regular flattening there. The proved original open-subtype regularity dictionary transports this flattening to the SAME original global full cut, preserving the real dimension bound. Relative openness of genuine regularity supplies a true neighborhood on which every actual cut point is regular of that fixed dimension. Intersect it with the internally produced bounded open. On this SAME new window, the original constructed invariant's proved admissible-level theorem applies using the internally retained bound. The literal full threshold equals the actual attained level there, in BOTH directions by antisymmetry. Consequently this actual reference trace is admissible for the restricted ORIGINAL current marked record, including normal crossings with the SAME real boundary list. All geometry has been constructed from the original inputs before the extension is selected. □

Theorem 425.2 (ONLY original inputs construct a genuine GLOBAL pure-dimensional analytic reference extension with exact admissible trace and whole ideal dictionary)**.** *At EVERY actual ORIGINAL canonical history and reference cosupport point, there are an internally constructed $q \leq n$, true open window U and actual GLOBAL analytic pure-q subset S of the SAME full reference cut. The produced point lies in S. On U, the literal trace of S equals the ENTIRE full reference cut, is regular of dimension q, and is an actual admissible centre for the restricted current ORIGINAL marked ideal. The global S is EXACTLY the union of all genuine global pure-q components of that full cut. Its actual reduced vanishing ideal is coherent with zero locus EXACTLY S, and its WHOLE open pullback ideal equals the actual reference-window vanishing ideal in BOTH germ directions.*

Proof. The preceding producer supplies the actual fixed-dimensional window, original bound, regularity and admissibility internally. The all-history full-threshold theorem independently constructs GLOBAL analyticity of the SAME actual original reference cut. Define S as its actual q-dimensional part, the closure of its genuine regular locus of dimension q. Proved analytic dimension-part geometry supplies GLOBAL analyticity, pure geometric dimension q and actual containment in the closed original cut. The reference point is already regular of dimension q, hence belongs to S. Every point of the cut in U is regular of dimension q; the proved regular-trace lemma therefore gives the exact ENTIRE trace equality. The actual dimension-part regularity theorem preserves its regularity there. The existing genuine global-component dictionary identifies S with precisely the actual pure-q global component union; no component labels or extension are input.

The same trace equality gives equality of the actual open preimage sets, so the original admissibility already constructed applies to the literal trace of S, with the SAME current boundary. Actual Cartan coherence applies to the genuine analytic S. The proved zero-locus dictionary for its closed reduced vanishing ideal gives the entire global zero equality. Finally the true open-inclusion pullback of a vanishing ideal is the WHOLE vanishing ideal of its actual preimage. Substitute the independently proved trace equality to obtain exactly the actual reference-window ideal, not just equality of supports or of selected generators. Nothing asserts that S is smooth or globally admissible away from U. Its true outside-window smoothification and compatible simultaneous-chain Hilbert–Samuel engine, original coefficient controlled-transform comparison and unrestricted BM13.4 locally finite driver remain separate OPEN obligations. □

426 A genuine global dimension-drop variation locus for the ORIGINAL FULL invariant

Bierstone–Milman[4], Lemma 13.7 and its proof, printed pages 300–301, removes smooth components where the actual augmented invariant is locally constant, then extends a smaller-dimensional

bad locus away from the selected window. This chapter supplies the ACTUAL ORIGINAL FULL-invariant part of that input. It does not assert simultaneous Hilbert–Samuel constancy, transformed coefficient comparison or the driver.

Lemma 426.1 (Actual minima and actual first higher component values produce a genuine proper analytic jump subset)**.** *Let f have a true well-founded linear value order and genuine analytic FULL cuts on an actual carrier. For every actual global component B of a pure-q analytic subset of that carrier, internally construct its ACTUAL minimum value a_B and its literal jump subset $E_B = \{x \in B : a_B < f(x)\}$. This SAME subset is genuinely analytic, is proper in B, and has dimension at most $q-1$. No generic value, next value, proper-subset, order bound or analytic jump certificate is input.*

Proof. The genuine global component is nonempty. Original well-foundedness applied to its ACTUAL value image supplies a least member, genuinely attained at a real component point, and gives $a_B \leq f$ on ALL of B. If no value exceeds a_B, the literal jump subset is empty. Otherwise well-foundedness applied to the actual attained higher values supplies a real least higher c_B. On B, $a_B < f(x)$ holds EXACTLY when $c_B \leq f(x)$: one direction uses minimality in the actual higher image, the other $a_B < c_B$. Consequently E_B is the genuine intersection of analytic B with the ACTUAL global c_B-cut. The minimum-attaining point proves B is not contained in that cut BEFORE dimension drop is invoked. If an open part of B lay in E_B, it would meet the genuine regular-locus component whose closure is B. The PROVED analytic continuation theorem would put that ENTIRE component, and hence its closure, in the actual cut, contradicting the minimum point. Thus the proper intersection is genuinely thin in B. Actual geometric purity bounds its dimension by q; the proved thin analytic dimension-drop theorem gives $\dim E_B \leq q-1$. □

Lemma 426.2 (Genuine locally finite analytic unions preserve an arbitrary actual dimension bound)**.** *A genuine locally finite family of actual analytic subsets, each with true geometric dimension at most r, has an actual analytic union of geometric dimension at most r. No dimension labels replace actual regular dimensions.*

Proof. Actual local finiteness supplies a real open neighborhood of each point on which the full union agrees with a TRUE finite subunion. The proved finite analytic union theorem preserves the actual bound at EVERY regular point. Genuine locality of regular dimension transports it through the exact local set equality to the whole union. Actual locally finite analytic equations already prove its analyticity. No ambient compactness or global finite component family is required. □

Theorem 426.3 (The literal FULL-value nonsmooth/nonconstant locus is globally analytic and dimension-decreasing)**.** *For actual analytic pure-q S in the genuine full-cut carrier, define $F = \{x \in S : \neg(\mathrm{Reg}_q(S,x) \wedge f|_S \text{ is locally constant at } x)\}$ using the LITERAL actual function. This SAME F is globally analytic, lies in S, and has dimension at most $q-1$, strictly below q when $q > 0$. In particular every point of $S \setminus F$ is genuinely smooth of dimension q and the ACTUAL full function is locally constant on S there.*

Proof. Construct the preceding ACTUAL a_B, E_B for EVERY genuine global component, then form the real union of these E_B and the actual pure singular locus of S. Component local finiteness passes to these actual subsets. The first two lemmas and the genuine singular-locus theorem prove this actual union analytic with dimension bound $q-1$.

Identify it with the literal stated F in BOTH directions. Outside the union, a point is regular of dimension q and belongs to a genuine global component B. Its actual regular flattening supplies a true open neighborhood where S lies in that SAME component. Intersect with the complement of the actual closed E_B. On this real neighborhood every actual S point has value a_B, so the whole actual function is locally constant there. Conversely, suppose a real regular point has the stated local constancy and lies in some E_B. Its value is strictly larger than the independently attained

minimum of B, so the actual higher cut defining E_B contains the whole local S trace. That true open trace meets the real regular-locus component generating B by closure. Analytic continuation again forces ALL of B into that cut, contradicting its genuine minimum point. A regular point is not in the actual singular locus either. These are BOTH exact literal bad-locus directions, not merely an envelope or caller certificate. The actual union's analytic/dimension assertions therefore concern precisely F. The zero-dimensional actual geometry is supplied separately below; no full terminating history is inferred from this dimension bound. □

Theorem 426.4 (ONLY original histories produce the FULL-value variation locus, and the actual reference extension's locus avoids a true selected window)**.** *At ANY actual ORIGINAL canonical history/depth, every actual pure analytic S contained in the current cosupport has the preceding genuine global FULL-value variation locus. Moreover ONLY original domain/history and a true reference cosupport point internally construct the actual global reference extension S, a dimension q, and a real window containing that point such that the literal global bad locus is analytic, has dimension at most $q-1$, and is DISJOINT from the entire selected window. No caller variation-locus, constancy, avoidance, local centre or extension certificate is input.*

Proof. The original all-history theorem internally supplies EVERY global analytic FULL cut of the SAME actual original invariant, whose genuine order is well-founded. Apply the preceding theorem to the actual analytic pure subset. For the original-only reference consumer, the genuine reference-extension producer FIRST supplies its actual global pure analytic S, exact local full-reference trace and real regularity on its true selected window. Independently original upper semicontinuity internally supplies a true open neighborhood bounding current-cosupport values by the attained reference value. Intersect these TWO actual windows. Since S lies in the ORIGINAL full reference cut, its values there are also bounded below by that same reference value, hence equal it at EVERY actual trace point. On this SAME real window the original extension is regular of dimension q and the actual FULL function is locally constant on S at each point. By the literal bad-locus dictionary, the ENTIRE bad locus misses that window. Its global analytic/dimension geometry was constructed independently from the real component jumps. This supplies the ORIGINAL invariant's genuine smaller-dimensional outside-window locus for a future auxiliary-chain step; simultaneous HS terms, preparation after noncanonical steps, original coefficient transformed semicoherence and BM13.4 termination remain OPEN. □

Lemma 426.5 (The literal variation locus is EMPTY in genuine analytic dimension zero)**.** *For ANY actual function on a genuine analytic pure-zero-dimensional set, the SAME literal nonsmooth/nonconstant locus is empty. No full-cut analytic, value-order finiteness, discreteness or locally constant certificate is input.*

Proof. The proved genuine analytic dimension-zero theorem internally supplies regularity of dimension zero at every actual carrier point. Its real flattening gives a true open neighborhood whose carrier trace is the singleton of that SAME point. Every actual function is constant on this actual trace. Thus every genuine carrier point satisfies BOTH regularity and true local constancy in the independently defined literal good predicate; none belongs to the literal bad locus. This proves the zero-dimensional base geometry for the ORIGINAL-value part, without asserting simultaneous HS transformed presentations or the terminating auxiliary-chain driver. □

427 Actual smooth current-companion diagrams after a preparation step

A genuine target companion TEST has now been constructed after the actual preparation step. Smooth pullback can consume that real TEST without pretending the new upper prefix, or its companion history, is canonical. The output below is geometry and whole records; no value comparison is postulated or inferred from a separately chosen source history.

Theorem 427.1 (Actual prepared current companions have genuine smooth whole-history diagrams)**.** *Let $n > 0$, let an original DOMAIN$k+1$ record have an actual CAN prefix, and take an actual next original-admissible blowup satisfying the whole positive-reference full-cut inequalities. Take ANY genuine actual smooth Cartesian pullback of that entire new upper history, in arbitrary source ambient dimension m. At EVERY actual source endpoint whose image belongs to the new target cosupport with positive attained order, construct the actual current-companion diagram: the independent source order and run are the target order and run; actual mask memberships and true maps of the two masked heads and tails are internally supplied. The source initial companion is the WHOLE marked pullback of the target initial companion, and the actual masked tails are genuinely Cartesian and genuine TESTs.*

Both initial companion domains, target maximal order, both current domains and both actual current companion cosupport points are constructed internally, with source carrier bound shifted by $m-n$. The entire current source marked record equals the actual endpoint pullback of the target record, and their cosupports agree in BOTH directions. At EVERY literal head of these actual companion histories, construct the genuine Cartesian head map, source TEST, entire controlled record equality and BOTH cosupport dictionary, retaining all empty positions. No source or target lower TEST, current lower member, chosen lift, record equality or next CAN certificate is supplied.

Proof. Original CAN and next original admissibility first give the actual upper TEST. The proved current-companion producer uses true run increment/reset to construct the target's newly computed initial domain, maximal order, actual TEST, current domain and exact mask point in its current cosupport. Independently, genuine upper-history smooth pullback gives equality of actual nonmonomial orders and actual order runs. Its real simultaneous split and order-mask inverse-image identity construct the true maps of the masked heads and tails and the WHOLE initial companion pullback equality. The actual tail projection square transports the true target mask membership to the source mask; the endpoint map is the original upper map on actual ambient points.

Apply the genuine relative controlled-record theorem to these exact Cartesian masked tails and the just constructed real target companion TEST. It constructs the actual source TEST, WHOLE current pullback equality and BOTH cosupport dictionary. Smooth pullback of the genuine target initial domain gives the shifted source domain; actual TEST inheritance gives its current domain. The exact endpoint point equation and target current membership now give source current membership, after its actual TEST has been proved. For every literal head, take the genuine Cartesian head map, split the real target companion TEST, and apply the same true controlled-record theorem. This proves ALL stated head equalities and both cosupport directions without removing any empty position. No canonicality of either new companion history is used. □

Theorem 427.2 (Only compact-current original inputs construct one upper smooth history before all companion diagrams)**.** *For ONLY original positive-dimensional DOMAIN$k+1$, its actual CAN prefix, compact nonempty current cosupport, a finite actual coherent ideal family, and a genuine initial smooth map, internally select an actual nonempty original-admissible simultaneous centre and proper holomorphic blowup, constant for the original full value and all supplied HS orders. Construct its real original next TEST and whole transformed ordered SNC. Then construct ONE actual entire source upper history and endpoint smooth map, with genuine Cartesian blowup squares and source upper TEST, before all points. At EVERY actual source endpoint whose target image is in the new positive-order current cosupport, construct all of the actual current companion diagrams, domains, TESTs, points, WHOLE records and ALL head cosupport dictionaries above. No compactness of the source or entire ambient target, source CAN, desired next map or lower certificate is input.*

Proof. The genuine compact-current simultaneous blowup producer constructs its actual centre inside the whole old full maximum locus, its actual proper holomorphic blowup and whole ordered

SNC. Every old positive cosupport reference therefore lies below every selected centre point; all full-cut inequalities are proved internally from real maximum-locus membership. Original admissibility appends the genuine target upper TEST. Only then apply the genuine smooth tower producer to the actual given initial submersion and that exact new target TEST. It constructs ONE source upper history with every actual pullback blowup and true endpoint square, together with its genuine source upper TEST. These choices precede every new point.

Apply the first theorem to this actual upper history at each stated source endpoint. All target current-companion data and all independent source mask/TEST/record/point data are internally supplied. The exact quantifier is over actual source endpoints whose images lie in the target current cosupport; no surjectivity onto arbitrary target points is asserted. This proves genuine smooth whole-record basechange for these noncanonical preparation histories. It does not assert arbitrary noncanonical full GENERAL numerical naturality, exceptional or infinitesimal semicoherence, new canonicality, or a terminating global preparation driver. □

428 The actually recomputed current companion after a simultaneous preparation step

A frozen old run-compAt TEST must not be confused with the companion that the unchanged GENERAL definition actually computes at a new endpoint. The real run recursion below supplies this remaining identification, including its true reset branch, before any current companion membership.

Theorem 428.1 (The real new run gives a genuine current companion TEST)**.** *Let $n > 0$, let M have original $DOMAINk + 1$, and let T be an actual original CAN TEST for its unchanged full value. Let C be a genuine next original-admissible centre and π an actual blowup. Suppose EVERY old positive-order cosupport reference lies below EVERY point of C in the old original full value. At EVERY new endpoint z of positive attained nonmonomial order, compute the actual NEW order run, head, tail and mask used by the unchanged original GENERAL definition. Its actual masked tail is a genuine TEST of its actual computed compAt. The new prefix need not be CAN, and no current-run equality or companion TEST is supplied.*

Proof. Fix the actual new order $q > 0$. Split by the actual old OrdRegime predicate at πz, rather than by an assumed equality of two runs. If it holds, the old point is in the actual old cosupport and has order exactly q. The stated whole-centre full-value inequalities apply to that actual old positive reference. The full-cut producer constructs its genuine extended old companion TEST. The original run recursion gives the new run as the old run plus one. Literal split recursion gives the SAME head and the old tail followed by this actual blowup; actual old order equality identifies the two orders. Thus the constructed TEST is precisely the actual newly computed companion TEST, with no history position removed.

If the old OrdRegime predicate fails, the actual run recursion gives new run zero. Literal splitting then has the entire new upper prefix as its head and the reflexive tower as its tail. Restricting that real reflexive tower is reflexive, hence genuinely a TEST of the actual newly formed companion. No claim about relative old/new order size, new canonicality, or a hypothetical canonical lower history is needed in this branch. □

Theorem 428.2 (Actual current companion domain, maximal order and current membership are internal)**.** *Under the preceding original data, let the actual new point also belong to the actual new upper cosupport. The actual computed companion has genuine initial $DOMAINk + 1$ and maximal order; its actual masked tail is a genuine TEST and its current record has $DOMAINk+1$. Internally construct the real mask membership of the new endpoint and prove that exact actual masked point belongs to this actual current companion cosupport. No lower TEST, current lower-cosupport membership, domain or maximal-order certificate is supplied.*

Proof. Original next admissibility appends the actual upper TEST. Split that real history at its actual new run; genuine head and tail TESTs follow. The head domain and positive attained order internally construct both initial compAt domain and maximal order. The first theorem supplies its genuine computed masked-tail TEST, and TEST inheritance gives its current domain.

The genuine original order-run projection theorem gives the actual endpoint mask membership, using the real upper TEST and new current cosupport. At this exact masked point, apply the genuine whole compAt cosupport dictionary to the actual head, tail and just constructed companion TEST. The actual upper cosupport holds by hypothesis and its order equals the new computed order by definition. Both facts give real current companion membership. These conclusions are constructed after, and concern, the actual new run; they do not replace it with a frozen or rebased history. □

Theorem 428.3 (Only compact-current original inputs construct one step with all current companion data)**.** *For ONLY original positive-dimensional DOMAIN$k+1$, its actual CAN prefix, compact nonempty current cosupport and a finite actual coherent ideal family, internally select one nonempty original-admissible centre and its actual proper holomorphic blowup. The original full value and all supplied HS orders are constant there, the original upper TEST extends genuinely, and its transformed whole ordered boundary is SNC. Simultaneously at EVERY actual new positive-order current cosupport point, construct the NEW computed companion TEST, initial domain/maximal order, current domain, actual mask point and current companion cosupport membership from the preceding theorem. There is no caller centre, cut, current-run or lower TEST certificate.*

Proof. The actual compact-current simultaneous blowup producer constructs its centre inside the ENTIRE old full maximum locus, the map, original admissibility, profile constancy and all asserted proper holomorphic and ordered SNC geometry. Original CAN supplies the real old upper TEST. Every old cosupport point lies below EVERY selected centre point by true maximum-locus membership. This independently proves the entire family of old positive-reference cut inequalities required above. Select the actual map before any new point, then apply the preceding current-data theorem at every actual new positive cosupport endpoint. Its real run recursion and mask construction supply every remaining field internally. This gives a true preparation-to-current-companion bridge after one possibly non-CAN step. It does not infer canonicality of that step, later arbitrary noncanonical GENERAL naturality, transformed semicoherence, or a full preparation driver. □

429 Every original full reference presentation supplies a genuine auxiliary marked input

The preceding original-only reference-window producer provides real coherent vanishing ideals at every original induction depth. Here those ACTUAL ideals become genuine original marked inputs with whole ambient carrier and the SAME current ordered boundary, and their full four-field records commute with genuine open histories. This is a concrete input for later local resolution scheduling. It does not supply transformed recursive coefficient comparison, arbitrary global thresholds or the global scheduler.

Definition 429.1 (The actual SAME-current-boundary full-reference auxiliary record)**.** On an actual original reference window W, form the literal marked record with carrier the WHOLE W, boundary the ordered inverse images in W of the SAME current original boundary, actual ideal $\mathcal{I}(\iota_W^{-1}A_x)$ and mark one.

Proof. Every field is constructed from the original marked input, its literal tower, actual reference and real window. The true inclusion supplies each boundary inverse image in its original order; the actual window ideal is already defined independently of coherence. The whole carrier and

mark one are literal fields. No wellformedness or desired comparison is assumed in this definition; those facts are proved below. □

Theorem 429.2 (Only original inputs construct a genuine marked domain for the ENTIRE full reference locus)**.** *At EVERY original induction depth, ONLY original domain, SAME original canonical tower and actual current-cosupport reference construct a true window containing that point. Its ACTUAL full-reference auxiliary record belongs to the original marked domain in ambient dimension n, retains the SAME current boundary and has cosupport precisely the ENTIRE original full-reference locus in that window. No nonzero, auxiliary domain, coherence, chosen reference window or desired cosupport certificate is input.*

Proof. The prior original-only theorem internally chooses the real window with actual vanishing-ideal coherence and its exact ENTIRE zero-locus dictionary. Original canonical data independently give a true test tower and actual current original marked domain; its actual window pullback is still a genuine original domain. Its positive cosupport has empty ambient interior by genuine coherent nonzero geometry, and the entire original full-reference window lies inside that actual cosupport. Thus the real auxiliary ideal's zero set has empty ambient interior. Actual coherent thin-cosupport geometry now constructs a real nonzero germ at EVERY whole ambient point, before the auxiliary-domain assertion is formed.

The whole ambient carrier is genuinely smooth of dimension n. The SAME current ordered boundary is SNC by genuine current wellformedness and true open pullback; it is transverse to that whole carrier. Together with actual coherence, mark positivity and the independently constructed nonzero germs, these verify all original marked-domain fields. The real mark-one whole carrier cosupport theorem identifies its cosupport with the actual ideal's zero set; the internal dictionary makes this exactly the displayed full original reference locus on the ENTIRE window. □

Theorem 429.3 (All four actual auxiliary marked fields agree over every genuine original open history)**.** *For EVERY actual original base-open history at the projected reference, the genuine trace-window map pulls the original auxiliary marked record back to the SAME literal auxiliary record built from the actual restricted original input and tower. Carrier, WHOLE ordered boundary, actual ideal and mark all agree. ONLY the original inputs internally supply reference windows for which both actual records are genuine original marked domains, with both ENTIRE original full-reference cosupport dictionaries.*

Proof. The carrier is the true inverse image of the whole original window and hence the whole actual trace window; the mark is literally one. The genuine original controlled-transform open-history record identity gives the whole current boundary list on the restricted tower, retaining every actual original position and empty step. True same-point compositions of window inclusions identify BOTH ordered boundary pullbacks member by member. The independently proved full-reference window vanishing-ideal identity gives actual ideal equality, not merely equality of zero loci. Extensionality of the four actual marked fields therefore proves the whole record identity.

The preceding original-only construction internally supplies its genuine window domain and zero dictionary. On every true base-open trace, the actual restricted current original domain is independently supplied by the original test history and real open restriction. The same independently constructed coherent ideal and inherited thin original cosupport supply its real nonzero germs. The identical whole-carrier, SAME-boundary and mark-one verification gives the actual restricted auxiliary domain and its exact ENTIRE reference cosupport. This uses true original geometry and complete ideal equality, not a caller target-record or domain certificate. No admissible-blowup semicoherent comparison or scheduler is asserted. □

430 The actual local full reference centre has a genuine auxiliary unit-transform blowup

This chapter constructs a real local blowup erasing the ACTUAL reduced full-reference auxiliary ideal. It does not identify that reference-cut ideal with the original coefficient presentation, nor construct a globally scheduled canonical extension of the original history. Those transformed coefficient and scheduler obligations remain OPEN.

Theorem 430.1 (Only original input constructs a true bounded full-reference open and actual SAME-boundary admissibility)**.** *At EVERY original induction depth, ONLY original domain, SAME genuine canonical history and actual current-cosupport reference internally construct a real open $W \ni x$ on which all current cosupport values are bounded by the attained full reference value. The ENTIRE original full-reference locus $C \subset W$ is nonempty and is a genuine admissible centre for the actual current original marked pullback. It is also admissible for the literal whole-carrier/SAME-current-boundary/full-reference-vanishing-ideal/mark-one auxiliary record, whose entire cosupport is precisely C.*

Proof. Use genuine USC of the SAME already constructed original invariant to obtain an actual open with the displayed original numerical bound. Its PROVED original local-admissibility theorem internally supplies the actual current level centre on this true open, including genuine closed smoothness and NC with the SAME entire ordered boundary. Both numerical comparisons force original value equality at EVERY superlevel point, so that entire level centre equals the literal full-reference set C. The actual reference is in C by reflexivity, proving nonemptiness. No centre or level model is input.

Real closed-smooth vanishing geometry gives coherence and the exact entire zero locus of the ACTUAL ideal $\mathcal{I}_C$. Its whole-carrier mark-one cosupport is therefore precisely C. The original admissibility already supplies true closed smoothness and NC with the SAME boundary. These SAME fields and the exact auxiliary cosupport give genuine auxiliary admissibility. This constructs the actual reference-cut marked record, not an assumed original coefficient ideal. □

Lemma 430.2 (The whole actual reference-cut marked record has genuine unit transform along its own full centre)**.** *For a real blowup along the actual closed smooth full-reference set C, the genuine controlled transform of its ACTUAL vanishing ideal with mark one is the whole unit ideal. The SAME literal whole-carrier reference auxiliary marked record has actual geometric transformed ideal equal to that unit ideal, and its transformed cosupport is empty.*

Proof. The actual blowup factorization of the vanishing ideal of a smooth carrier containing the centre identifies its genuine mark-one controlled transform with the vanishing ideal of its geometric strict transform. Here the carrier is the centre ITSELF, so the literal geometric strict transform is empty. The actual vanishing ideal of the empty set is the whole holomorphic unit ideal in BOTH germ directions. The whole ambient auxiliary carrier remains the entire blowup manifold; the original coherent whole-carrier transform identity therefore identifies the SAME geometric ideal with that actual controlled ideal. The unchanged mark one and actual unit-ideal cosupport formula prove empty cosupport. These are entire original ideal equalities, not a merely reduced-support statement and not the original coefficient ideal's transform. □

Theorem 430.3 (Only original inputs construct the genuine SAME local reference-centre blowup and its whole lower-value isomorphism)**.** *Those SAME original-only inputs internally construct the actual bounded window, its ENTIRE nonempty full-reference centre, and a real proper holomorphic blowup along that SAME centre. It is genuinely isomorphic over the ENTIRE complement of the reference locus, including every lower-value current cosupport point. The actual auxiliary geometric ideal is unit and its cosupport empty; the precise whole ordered geometric strict original window-boundary followed by the actual exceptional member is SNC.*

Proof. The first producer internally gives the actual window and real full-reference centre with both genuine admissibilities. Original canonical data independently supply the current marked domain, whose actual open pullback is again a true original domain. The centre is in its positive cosupport, which has empty ambient interior by genuine original coherent nonzero geometry. It is therefore a true thin closed smooth centre. Existing blowup existence constructs its real manifold and SAME proper holomorphic projection. The actual complement inverse supplies the genuine whole-complement isomorphism, retaining both injectivity and actual preimages.

The second producer gives the literal auxiliary geometric unit ideal and empty cosupport on that SAME blowup. Independently actual current original wellformedness and its genuine centre admissibility transport the ENTIRE ordered boundary list, retaining each actual original position, to the geometric strict list followed by the real exceptional member; that exact list is SNC. This erases one actual attained-reference CUT ideal locally. No claim of original full-invariant decrease for this independently chosen local history, original coefficient-presentation identity, global scheduler or full transformed semicoherence is made. □

431 Actual full original reference ideals commute with true open histories and overlaps

The preceding all-depth attained-reference theorem constructs actual local coherent vanishing ideals. The present consumer compares those SAME ideals on genuine original open histories and actual overlapping windows. This is ordinary open naturality of literal full-reference ideals, not the unproved recursive coefficient-presentation comparison under admissible blowups. Arbitrary global thresholds, whole transformed semicoherence and the global scheduler remain OPEN.

Definition 431.1 (Literal original full reference loci, ideals and actual reference windows)**.** For the SAME original invariant, marked input, tower and reference x, set $A_x = \{y \in m_T.\mathrm{cosupp} : \mathrm{inv}(x) \leq \mathrm{inv}(y)\}$ and $I_x = \mathcal{I}(A_x)$. On any actual final open W, its window ideal is $\mathcal{I}(\iota_W^{-1} A_x)$. For true final opens O, W, let $W|_O = \iota_O^{-1} W$ as an actual open of O; its true map into W sends a point to the identical original point with its genuine W membership.

Proof. The original cosupport, invariant value and attained reference define an actual set without a chosen presentation. The original vanishing-ideal construction is defined for every set, so both ideals are actual germ predicates before any coherence assertion. True open inclusions are continuous, hence $W|_O$ is open; its defining membership internally supplies the displayed map into W. All points, membership proofs and ideal constructions are literal; no desired comparison is input. □

Theorem 431.2 (The whole original full reference locus and actual ideal commute with the original open history)**.** *For ONLY an original marked domain and SAME genuine original canonical tower, any original base open and a true cosupport reference point in its final preimage, the ENTIRE full reference locus of the actual restricted marked input and restricted tower equals the genuine inverse image of the original full reference locus at that SAME point. The ACTUAL full vanishing ideal pulls back to the literal restricted full-reference ideal. No global coherence assumption is needed for this ideal equality.*

Proof. Original canonical input internally supplies the true original test history. The proved cosupport restriction theorem identifies the entire actual restricted current cosupport with its genuine original inverse image. The SAME constructed original invariant's proved open-history law identifies its values at every such point, including the actual reference, with the original values. Substituting BOTH value equalities proves each full superlevel direction at every true point of the restricted final manifold. For the ideal, first use the genuine open-inclusion vanishing-ideal pullback identity, proved by actual extension and restriction of holomorphic germs in BOTH directions. Substitute the just-proved ENTIRE set dictionary. This is actual ideal equality, not merely matching zero sets and not an assumed presentation equivalence. □

Lemma 431.3 (True window maps preserve actual vanishing ideals and compare entire overlaps)**.** *The actual trace map is holomorphic. Pulling the actual window vanishing ideal back along it gives the vanishing ideal of the literal double trace. Consequently for ANY two original reference windows the two actual pulled ideals on their true intersection are equal. The complete ideal germ predicates agree, without a supplied generator or overlap certificate.*

Proof. Composing the trace map with the genuine target inclusion gives the true composition of original open inclusions, hence holomorphicity follows from the actual subtype manifold laws. For any original set, the proved vanishing-ideal identity on an open converts the pullback to the ideal pulled along that same composite original map. Apply the same identity in the opposite order to the other open, and then the genuine inclusion vanishing-ideal identity on the inner open. Every composite has the SAME original underlying point, giving the literal double trace ideal. On an intersection, both inclusions therefore pull their actual window ideals to $\mathcal{I}(\iota_{W_1\cap W_2}^{-1}A_x)$. Equality follows for every germ. Neither global analyticness nor global coherence is inferred. □

Theorem 431.4 (Only original inputs construct compatible actual coherent full-reference windows)**.** *At EVERY original induction depth, ONLY original marked domain, SAME canonical history and an actual current-cosupport reference construct a real reference window W with its actual coherent ideal and exact ENTIRE zero set. For EVERY original base open containing the projection of that point, the true trace of this window in the restricted final space contains the same point. Its literal restricted full-reference window ideal is coherent, is exactly the actual pullback of the constructed original window ideal, and has its exact ENTIRE original restricted full-reference zero set.*

Proof. The preceding original-only all-depth producer internally constructs W containing the reference, actual vanishing-ideal coherence and the entire zero dictionary. For any real base open at its true projection, the actual restricted reference and trace window are constructed from membership. Apply the ENTIRE open-history full-reference dictionary and the actual trace vanishing-ideal identity to get literal ideal equality. True trace-map holomorphicity and actual coherent ideal pullback give coherence of that SAME restricted ideal, with no global-coherence premise. The genuine coherent-ideal pullback zero-locus theorem and the constructed whole zero dictionary identify its zero set with precisely the restricted full reference window. Thus no reference window, desired ideal equality, coherent presentation or zero-set certificate is supplied by the caller. All overlap equalities of the preceding lemma apply to the actual windows. This proves genuine open naturality, leaving transformed coefficient semicoherence and arbitrary global thresholds as separate obligations. □

432 Actual simultaneous preparation centres construct genuine companion TESTs

The whole-centre full-value-cut condition is now supplied internally from actual simultaneous centre selection. The old upper prefix is CAN; the actual next original-admissible preparation step need not be CAN. All companion conclusions concern the SAME old order run, head and attained positive order, retaining every actual history position.

Theorem 432.1 (A compact current cosupport supplies one simultaneous step for all old positive references)**.** *Let $n > 0$, let M have original DOMAIN$k+1$, and let T be an actual original CAN prefix for its unchanged full value at depth $k+1$. Suppose its actual current cosupport is compact and nonempty. For a finite family of actual coherent ideals on the current ambient manifold, internally construct a nonempty original-admissible centre C in the full maximum locus, an actual proper holomorphic blowup π and the genuine extended original TEST. The full original value and EVERY order of EVERY supplied HS profile are constant on C. The actual transformed record*

is well formed and its whole ordered strict boundary followed by the actual exceptional divisor is SNC.

This ONE chosen C, π works for EVERY old current cosupport point y of positive attained order q: form its actual old order run, head H, tail S and order open W. The literal restriction of S followed by the chosen step is a genuine TEST of the SAME original compAt at H; its current record has DOMAIN$k+1$. At EVERY actual new masked endpoint, its cosupport is equivalent to the actual new original cosupport together with nonmonomial order equal to that SAME OLD q. No caller centre, cut bound, companion TEST, domain or cosupport dictionary is assumed.

Proof. The actual compact-current simultaneous blowup producer first selects the full original maximum locus, then the genuine all-order finite-profile constant cut inside it, and an actual original-admissible centre there. It constructs the actual blowup, properness, holomorphicity and full ordered SNC geometry. Original CAN independently gives the genuine old upper TEST. Admissibility of this SAME selected centre appends the actual new original TEST.

Fix any old positive-order cosupport reference y, after this actual blowup has been selected. Every point c of the selected centre belongs to the full maximum locus on the ENTIRE old current cosupport. The actual maximum inequality therefore gives the whole-centre bound $F(T, y) \leq F(T, c)$ for every such c, not merely at the selected profile reference. Apply the genuine full-value-cut producer to these original data and this internally proved bound. It supplies the actual extended old-run compAt TEST and its inherited current domain. Only after that genuine TEST is constructed, apply the whole old-order cosupport theorem. This gives both directions at every new masked point with the same old q and old head. All positive references use the SAME selected upper centre and map; no canonicality of the new upper prefix is inferred. □

Theorem 432.2 (A compact analytic subset supplies a referencewise simultaneous preparation step)**.** *For the same original positive-dimensional DOMAIN$k+1$ and CAN prefix, let K be any actual compact nonempty analytic subset of the current cosupport. The whole current cosupport need not be compact. For the same kind of finite coherent ideal family, internally select an attained maximum reference a of the unchanged original full value on K, a nonempty original-admissible centre $C \subset K$ constant for that full value and all supplied HS orders, and an actual proper holomorphic blowup. Construct the genuine extended original TEST, transformed wellformedness and whole ordered SNC. Simultaneously for EVERY old positive-order reference $y \in K$, construct exactly the old-run compAt extended TEST, inherited DOMAIN$k+1$ and BOTH old-order new-cosupport conclusions of the first theorem. No desired maximum, cut, candidate TEST or history is supplied.*

Proof. The actual compact-analytic-subset simultaneous centre producer selects a and C internally and retains the true maximum inequality on all of K. Original CAN supplies the real old upper TEST and current domain. The actual centre lies in the current cosupport, whose interior is empty; therefore the actual smooth closed centre has empty interior. The genuine blowup existence theorem constructs its actual ambient manifold and map. Original admissibility gives proper holomorphic blowup geometry, the wellformed transformed original record and SNC of the entire ordered strict boundary plus the actual exceptional divisor. It also appends the SAME original upper TEST.

For any reference $y \in K$ and EVERY centre point c, use the retained maximum inequality $F(T, y) \leq F(T, a)$ and the actual equality $F(T, c) = F(T, a)$. This proves the ENTIRE centre cut internally for that reference. The full-value-cut companion producer now constructs its true extended old-run TEST and current domain. Its actual whole cosupport producer supplies both inclusions with the old head and old attained q. The proof does not extend the maximum inequality to references outside K. These are real next companion TESTs after one possibly noncanonical preparation step; arbitrary noncanonical GENERAL naturality, transformed semicoherence and a full global preparation driver remain separate obligations. □

Theorem 432.3 (The literal original full-value bad locus supplies its own companion preparation step)**.** *In the same original DOMAIN$k+1$ and CAN setting, let S be an actual compact pure-dimensional analytic subset of the current cosupport. Compute the literal full-value variation locus F of the unchanged original full value on S, and suppose it is nonempty. Internally prove that F is analytic, compact and of dimension at most $\dim S - 1$. Construct a nonempty original-admissible centre $C \subset F$, its actual proper holomorphic blowup and extended original TEST, constant for the full original value and EVERY supplied finite-family HS order. The SAME map is an actual isomorphism over the ENTIRE complement of F, and the whole transformed ordered boundary is SNC. For EVERY positive-order old reference $y \in F$, construct its genuine extended old-run compAt TEST, current DOMAIN$k+1$ and both old-order new-cosupport implications. No analyticity, compactness, maximum or TEST certificate for F is supplied.*

Proof. The proved original full-value variation theorem computes the literal F and gives its actual analyticity and dimension drop from the original CAN value and pure analytic S. Its literal definition gives $F \subset S$, so closedness and compactness of S give compactness of F; the actual current-cosupport inclusion follows through S. Apply the preceding compact-subset producer with this exact F. It selects the actual full maximum reference on F, centre, proper holomorphic blowup and all the stated original and companion TEST/domain/cosupport data internally.

Since the selected centre lies in F, its complement contains the entire complement of F. The actual blowup isomorphism off its real centre restricts to that whole open complement. Actual inverse equations and surjectivity there prove both isomorphism directions; no projection inverse is postulated. All old positive references range over F, not arbitrary points of S. The original full-value good region is preserved, but no new CAN, later arbitrary noncanonical full-value law, exceptional semicoherent comparison or terminating preparation driver is claimed. □

433 An ACTUAL original-admissible first centre with the full invariant and finite simultaneous HS profiles constant

Bierstone–Milman[4], Lemma 13.7, printed page 300, requires genuine centres on which the original invariant and ALL HS profiles in the finite analytic chain are locally constant. Here ONLY original inputs and compactness of the ACTUAL current cosupport produce such a first centre and its true blowup. Its existence is proved before any exceptional HS nonincrease, transformed semicoherent comparison or modified-history iteration; those remain OPEN.

Lemma 433.1 (A true finite family of original coherent ideals attains an actual simultaneous whole-profile maximum on any nonempty compact set)**.** *ONLY a finite family of actual coherent original ideals and an ACTUAL nonempty compact set K internally select $a \in K$ such that the literal componentwise ALL-order upper cut on K equals the ENTIRE simultaneous profile equality locus. If K is analytic, this actual cut is nonempty, analytic and compact. Empty ideal families are included. No selected point, maximum, HS-profile, finite-order or cut certificate is input.*

Proof. Use the actual finite enumeration and genuine natural pairing to enumerate EVERY pair consisting of a family position and an HS order by one countable sequence; unused codes have the actual constant zero coordinate. Original coherence gives true upper semicontinuity of EACH EXISTING intrinsic HS coordinate. On the actual compact subtype, the proved nested-coordinate maximum construction internally selects a genuine point. If another point dominates EVERY actual family/order coordinate, it dominates EVERY code; the actual whole-profile maximum dictionary forces equality at ALL codes. Evaluate the genuine code of EACH position/order for the reverse dictionary. The actual cut is K intersected with FINITELY many true all-order HS superlevels. Those are independently analytic, hence closed. Their literal intersection contains a and is a closed subset of the true compact K. This supplies all geometry before the cut is used as a centre input. □

Theorem 433.2 (ONLY original-domain, actual canonical prefix, compact current cosupport and finite coherent-family inputs construct a genuine ORIGINAL-admissible centre with ALL simultaneous profiles constant)**.** *For an ACTUAL canonical prefix of the ORIGINAL marked input, assume ONLY that its true current cosupport is nonempty and compact, and supply any finite family of actual coherent ideals on that current original manifold. Internally construct a nonempty smooth closed centre C admissible for the SAME actual current ORIGINAL marked record. The ORIGINAL full invariant is constant on C at its actual GLOBAL current maximum; EVERY member's EXISTING intrinsic HS profile is constant there at EVERY order. No desired centre, global maximum, auxiliary domain, original admissibility or simultaneous-constancy certificate is input. The selected centre is not claimed to equal the original invariant's entire maximum locus.*

Proof. The actual original invariant USC and true current compactness internally attain its genuine GLOBAL maximum b. Its literal maximum locus L is nonempty, compact and smooth closed by the PROVED original admissible-level geometry. It equals the actual full threshold at b in BOTH directions, so the independently proved arbitrary original-history threshold theorem gives its true GLOBAL analyticity. Apply the preceding simultaneous whole-profile producer on THIS actual L. It constructs a genuine point and a nonempty analytic compact simultaneous cut $S \subset L$ on which ALL positions and orders are constant.

The original current cosupport is genuinely thin; hence this actual S is ambient thin. Its genuine coherent reduced vanishing ideal, true universal carrier, mark one and SAME original current ordered boundary therefore form a real auxiliary original-domain input, internally. Choose the actual global maximum of its true canonical invariant on its compact cosupport S. The PROVED original producer gives a nonempty smooth closed centre NC with that SAME actual ordered boundary. Its true cosupport containment gives $C \subset S \subset L$. Thus it lies in the ORIGINAL current cosupport and has NC with the ORIGINAL current boundary: these are exactly the remaining real original marked-admissibility fields. The containment in L makes the original FULL value constant, and containment in S makes EVERY original HS order in EVERY family position constant. No comparison between the original and auxiliary invariant is assumed. □

Theorem 433.3 (The SAME internally chosen simultaneous centre gives a true proper holomorphic ORIGINAL-admissible first blowup)**.** *The preceding original-only inputs internally give the SAME actual centre and a REAL proper holomorphic blowup. Its genuine transformed ORIGINAL marked input is well formed and has the actual ordered strict old boundary followed by the real exceptional divisor SNC. The actual centre retains the original FULL invariant and all simultaneous HS constancies proved above. No exceptional HS comparison, original canonical-max continuation, modified-history decrease or termination is inferred.*

Proof. The actual centre lies inside the genuine original current cosupport, whose ambient interior is empty. Thus its own actual interior is empty BEFORE invoking the proved blowup existence theorem. Original admissibility supplies true centre smoothness and whole-boundary NC. The real blowup producer gives its actual map; proved blowup geometry gives genuine properness and holomorphicity. The actual original marked wellformed-transform theorem applies with THOSE true original admissibility fields and THAT same map, giving the actual transformed original record and ordered strict-plus- exceptional SNC. The centre constancies are the already proved whole statements, without assuming a transformed profile comparison. □

434 A genuine simultaneous first blowup preserving EVERY actual full-invariant good point

Bierstone–Milman[4], Lemma 13.7(1), printed page 300, excludes every point where the last actual space is smooth and the simultaneous tuple is locally constant. The actual full-invariant variation locus is now computed internally. In the compact pure-space subclass this gives a

REAL original-admissible simultaneous-HS-constant first centre avoiding EVERY original full-invariant good point, hence EVERY simultaneous-tuple good point. The HS-only varying phase and transformed-history iteration remain OPEN.

Lemma 434.1 (Actual compact analytic subset geometry internally supplies the full-invariant and simultaneous-HS constant original-admissible centre)**.** *ONLY original domain and a true original canonical prefix, a nonempty compact analytic subset K of its ACTUAL current cosupport, and a finite family of coherent original current ideals internally select a nonempty ORIGINAL-admissible centre $C \subset K$. The ORIGINAL full invariant is constant on C at its true maximum on K, and ALL original intrinsic HS orders in EVERY family position are constant there. Neither the entire current cosupport nor the ambient is assumed compact. No desired level, centre, admissibility or constancy certificate is input.*

Proof. Restrict the true original invariant USC to the genuine compact K and internally attain its maximum b. Its real level inside K equals $K \cap \{x \in \operatorname{cosupp} : R(x) \geq b\}$ in BOTH directions: every K point is genuinely in that same current cosupport and every K value is bounded by the attained value. The original FULL threshold is genuinely globally analytic, so this actual level is analytic and compact. Apply the proved true finite-family ALL-order compact HS producer there. Its genuine simultaneous cut is analytic, nonempty, compact and contained in the real current cosupport. Ambient thinness follows the ORIGINAL current domain. Its actual reduced ideal with universal carrier, mark one and SAME current ordered boundary is therefore a real auxiliary domain. Its actual canonical maximum centre supplies smoothness and whole-boundary NC; true containment in the genuine cut gives ORIGINAL cosupport membership, original admissibility, the attained full invariant value and every actual HS constancy. No original canonical-max continuation is inferred. □

Theorem 434.2 (ONLY actual compact pure original-space geometry computes its literal full–value bad set before selecting an original-admissible simultaneous centre inside it)**.** *For an ACTUAL pure-dimensional compact analytic S in the genuine current original cosupport, compute the LITERAL set*

$$F = \{x \in S : \neg(\operatorname{Reg}_{\dim S}(S, x) \text{ and } R|_S \text{ is locally constant at } x)\}.$$

If this actual computed F is nonempty, internally construct a nonempty ORIGINAL-admissible centre $C \subset F$, with ORIGINAL FULL invariant constant at its true maximum on F and ALL finite-family original intrinsic HS profiles constant at EVERY order. The actual F is internally analytic, compact and of dimension at most $\dim S - 1$. No desired bad set, centre, maximum or good-point avoidance certificate is input.

Proof. The previously proved original-input variation theorem first computes the EXACT literal F and independently proves its GLOBAL analyticity and true dimension bound from actual component minima and attained full-value jumps. Its actual containment in S and genuine closedness make it compact in the true compact S, before it is used for selection. Its actual current cosupport containment also follows from S. Apply the preceding actual compact analytic-subset producer to THIS computed F. This constructs all centre and constancy data internally. Since $C \subset F$ by a genuine set containment, every point smooth in S with $R|_S$ locally constant is excluded. Every point where the larger simultaneous tuple is locally constant in S has that original R constancy as one coordinate, so it is excluded too. The nonempty condition concerns the actual computed bad locus, not a desired future centre or phase certificate. □

Theorem 434.3 (The SAME computed bad-set centre gives a genuine ORIGINAL finite-prefix extension preserving its ENTIRE actual good region)**.** *The preceding ONLY original inputs internally construct the SAME actual centre and a REAL proper holomorphic blowup, whose actual appended prefix is a test tower for the SAME ORIGINAL marked input. It is an isomorphism*

over the ENTIRE computed F^c, including EVERY original full-invariant good point of S. The ORIGINAL transformed record is genuinely well formed with its actual ordered strict old boundary followed by the real exceptional divisor SNC. This does not assert that the appended prefix is original canonical, that HS does not increase over the exceptional divisor, or that a modified history terminates.

Proof. The actual centre lies in the genuine ORIGINAL current cosupport; actual domain thinness proves its ambient interior empty BEFORE actual blowup existence is invoked. Genuine original admissibility and the actual blowup supply a true next ORIGINAL test step, so append it to the ACTUAL existing original prefix without rebasing or dropping any empty position. Blowup properness and holomorphicity are proved geometric properties of THIS map. Since the true centre is contained in the computed F, the actual open F^c is contained in the true centre complement. Restrict the genuine blowup complement inverse in BOTH directions to this actual open; all membership and inverse equations are retained. This proves whole-region isomorphism. Apply the original admissible geometric-transform theorem at the SAME actual centre and map to obtain true wellformedness and the ordered entire-boundary SNC. Original invariant and all HS constancies remain the already constructed centre statements; no transformed invariant comparison is presumed. □

435 The entire literal original contact lower first threshold is a true whole analytic trace

Use the same original zero-level contact run, head H, tail Q, contact mask W, internally chosen tangent open U, and whole contact remainder L. Its actual whole final transform S is on $Q^{-1}(H^{-1}U)$. The true map $j : Q^{-1}W \to Q^{-1}(H^{-1}U)$ is the original open inclusion, on the SAME underlying original point. This is the contact analogue of the genuine whole member order extension, using Bierstone–Milman[5], Section 5, Step I, Case A. All marked objects and invariant values are original.

Definition 435.1 (The actual original contact final inclusion and literal lower threshold)**.** Let J be the SAME literal lower contact marked ideal transformed along its actual original restricted tail. Define its literal level-$k+1$ first threshold by

$$A_a = \{z \in J.N : a \le \mathrm{inv}_{k+1}(J_0, Q|_W, z)_1\}.$$

The actual chosen-open inclusion j is holomorphic and preserves the SAME underlying original point.

Proof. The original computed run fixes the real head, tail and mask; the literal original lower marked ideal and constructed lower invariant define A_a. The actual mask containment in the chosen contact open implies containment of their true final inverse images by the original projection. Thus j is the genuine open-subtype inclusion. Its ambient map is the true original open inclusion, holomorphic. The target is a genuine open subtype, so the lift is holomorphic as well. □

Theorem 435.2 (Original contact inputs construct the whole first-entry equality and both entire threshold directions)**.** *Only the original maximal-order domain-$k+2$ input, original canonical masked Step One history and actual nonspecial zero-level cosupport point internally produce one actual tangent open and actual entire S in lower domain, with*

$$\mathrm{inv}_{k+1}(J_0, Q|_W, z)_1 = \mathrm{ord}_{\mathrm{nm}}(S, jz), \qquad A_a = j^{-1}(\mathrm{ord}_{\mathrm{nm}}(S) \ge a)$$

for every actual z and every $a \ge 0$. Actual carrier membership is retained in both directions of the entire threshold equality.

Proof. The previous original-only whole contact dictionary internally supplies the chosen open, full final domain and entire marked-record equality under the true nested-open biholomorphism χ. Genuine contact-domain geometry independently supplies the actual masked lower domain: the original mask removes the true special support and historical members. The true contact-tail theorem supplies its actual restricted test history; genuine domain transport then gives the actual final masked marked domain. Its invariant's first coordinate is by definition its literal nonmonomial order. Evaluate the whole marked-record identity at $\chi^{-1}z$. Actual biholomorphic order transport on the true masked domain identifies one side with that literal first coordinate. Actual open-order transport on the internally constructed whole remainder domain identifies the other side with $\operatorname{ord}_{\mathrm{nm}}(S, jz)$. The genuine nested-open inverse preserves the same underlying point, yielding the first equality. The entire carrier equality is also a field of that same marked-record identity. On this carrier the true finite nonmonomial-order dictionary says order at least a exactly when the point belongs to its actual order-a cosupport. These two equalities prove both directions of the full threshold identity, including zero and points outside the actual carrier. □

Corollary 435.3 (Actual whole contact order equations give an entire analytic extension of every literal lower first threshold)**.** *Each A_a has the genuine whole analytic extension given by the actual nonmonomial order-a locus of the SAME whole final contact remainder. The entire literal masked A_a is genuinely complex analytic.*

Proof. The internally produced whole final lower domain gives actual coherent nonmonomial ideal through true SNC monomial division. Positive orders have its genuine coherent relative derivative equations; order zero is the whole smooth closed carrier. Thus the real entire whole order locus is globally analytic on the chosen final contact open. Its genuine holomorphic open inclusion pullback is analytic. The preceding exact entire threshold dictionary identifies this pullback with precisely A_a. □

Theorem 435.4 (Only original full positive inputs supply the literal contact first analytic thresholds)**.** *Only original full level-$k+2$ marked-domain input, its literal original canonical history and a positive original cosupport point whose actual internally lifted companion point is nonspecial of zero contact level supply every entire actual original contact lower first threshold above. No selected companion, chosen tangent open, whole remainder, marked-record or analytic presentation certificate is input.*

Proof. The original fixed-companion producer internally constructs its true marked domain, maximal order and canonical masked history. Its actual reference lift lies in true companion cosupport by the entire original order-equality cosupport dictionary. Apply the preceding genuine all-level contact threshold producer to this SAME actual companion and history with its actual nonspecial zero-level branch. All its true whole equations and exact original masked thresholds are therefore internally produced from the original full inputs. Arbitrary deeper lower thresholds, higher-level/special global boundary filling and the auxiliary simultaneous Hilbert–Samuel scheduler remain separate. □

436 Entire original contact masked marked records agree with the genuine whole remainder in every level

The preceding internally chosen contact open and whole original remainder supply true lower-domain geometry at every actual original tail prefix. We now compare the SAME original lower contact marked records on the entire original mask. This is the actual marked-coefficient contact reduction of Bierstone–Milman[5], Section 5, Step I, Case A, with actual ordered historical positions retained. The dictionary is for all induction levels and every genuine original prefix. It does not certify canonicality on the enlarged remainder or global arbitrary lower threshold extension across higher-level and special boundary pieces.

Lemma 436.1 (Actual masked nonzero identifies the entire original contact remainder record)**.** *Only original maximal-order domain-$k+2$ input, its genuine original canonical masked Step One history and a nonspecial zero-level original cosupport point internally produce the chosen tangent open, such that the whole remainder restricted to the ENTIRE original contact mask equals the SAME literal original lower contact marked ideal, under the genuine nested-open biholomorphism. Carrier, entire ideal, mark and ordered boundary all agree.*

Proof. Use the true chosen tangent open and source coefficient test head. The actual masked contact-domain theorem internally proves nonzero along the genuine masked carrier, using the original mask's removal of actual special support and original historical members. If a point of this carrier belonged to a selected full contact component, closedness of the remaining union would give a neighborhood on which the whole contact carrier lies in its actual cosupport. Actual open pullback identifies that cosupport with the masked cosupport. A genuine coherent local ideal generator nonzero at arbitrarily near masked carrier points, supplied by actual masked nonzero, would then vanish at all those points by positive order, a contradiction. Thus the entire masked carrier is precisely the masked remaining carrier.

Every true historical boundary member misses the entire original mask by its literal definition. Their pullbacks are therefore empty in the SAME original index positions. Their number is the original boundary length; the true head boundary has at least that length. Splitting the actual boundary into this historical prefix and exceptional suffix gives equality of the whole ordered list with the remainder's retained empty positions. The whole ideal and mark were unchanged. Genuine coherent double-open pullback identifies the raw contact model with the literal original lower contact ideal under the actual nested-open biholomorphism. These four exact fields prove the asserted entire marked-record equality. □

Lemma 436.2 (Actual contact remainder controlled transforms commute with true open restriction)**.** *For the actual original tangent open, true coefficient source head and genuine coefficient test tail, every whole contact remainder controlled transform commutes with restriction to any actual open of that contact model. The centres are the real original source centres; no containment in the remainder carrier is added.*

Proof. Induct on the true original source tail. At the empty tower the two entire marked records are literally equal. At an actual blowup, the preceding original contact-domain producer constructs coherent ideals of the genuine remainder both before and after that SAME step. The already true previous restriction identity substitutes the source marked record. Genuine coherent open pullback of a controlled transform, including its strict carrier and colon ideal, now gives the next entire record equality. All coherence is internally supplied from the true original contact history. This proves the whole equality at each step without candidate-centre containment. □

Theorem 436.3 (The true nested-open tower gives the entire original masked prefix dictionary)**.** *For every true original source prefix, the entire literal original lower contact transform on its actual mask agrees with the whole genuine remainder transform restricted to that entire mask inverse image, under the actual nested-open final biholomorphism.*

Proof. The actual source head and tangent contact geometry internally give both the whole enlarged contact test prefix and its restricted original masked test prefix, with true smooth carriers and coherent ideals. The genuine nested-open tower isomorphism identifies the literal masked prefix with the double restriction of the enlarged source prefix. Transport of marked controlled transforms under this real tower isomorphism gives a whole marked-record equality. Substitute the entire initial original-mask record identity. The preceding coherent restriction theorem identifies its actual double-restricted remainder transform with the true pullback of the enlarged remainder transform. Transitivity gives the displayed whole marked-record dictionary. Its map is the actual nested-open final biholomorphism, and no desired prefix identity is supplied as a premise. □

Theorem 436.4 (Only original zero-level inputs construct all entire contact dictionaries and domains together)**.** *Only original maximal-order level-$k+2$ domain input, its literal original canonical masked Step One tower and a true nonspecial zero-level cosupport point construct ONE true tangent open with the entire initial original-mask record equality, genuine lower domain at EVERY actual original restricted tail prefix, and the entire original masked marked-record dictionary at EVERY such prefix. No chosen open, candidate lower-domain, lower-canonical history, nonzero or whole-record equality certificate is a caller hypothesis.*

Proof. The original-only chosen-open producer constructs the actual tangent function, open and whole original source coefficient tower. Split at its literal computed run. Every true head of the actual tail is a genuine coefficient test prefix. Apply the actual contact remainder domain theorem and the preceding nested-prefix dictionary to each such prefix of that SAME split on the ONE internally chosen open. The entire initial mask theorem supplies the initial whole record identity. Retain all these internally proved conclusions in their literal original marked objects and original historical positions. □

437 Actual restrictions and cocycles of the chosen pairwise maps

The input maps below are genuine chosen stagewise comparisons of the actual restricted prefix-and-tail histories. Such maps are internally supplied by the retained-recipe common-open theorem. They are not arbitrary endpoint witnesses. The assertions identify their real restrictions on smaller common opens, with every empty position retained, before asserting a cocycle.

Lemma 437.1 (Actual common-open base maps supply both smaller-window identities)**.** *For genuine opens U, V and $O \subset I \subset U \cap V$, the actual base comparison from $U|_I$ to $V|_I$ preserves membership of the real smaller windows $U|_O$ and $V|_O$ in both directions. The internally constructed nested smaller base biholomorphism is exactly the original common-open base comparison on O.*

Proof. Each of the actual common-open base maps and nested-open flattenings sends a point to the same original ambient point. Membership in either smaller window therefore means precisely that this ambient point belongs to O. This gives both directions of the actual window identity. Genuine inclusion $O \subset I$ gives the two smaller-window inclusions internally. The smaller biholomorphism is constructed by the real nested-restriction producer. Both it and the original common-open base map have the same original ambient point at every source point. Two subtype extensionality equations therefore identify their actual maps and hence their genuine biholomorphisms. No smaller-window identity or comparison is assumed. □

Theorem 437.2 (The chosen pairwise terminal map genuinely restricts to every smaller common open)**.** *Let actual original prefix-and-tail histories over U and V have a chosen genuine padded stagewise comparison on the actual common open I. For every actual $O \subset I$, construct a genuine padded stagewise comparison of their literal histories on O. Its underlying terminal map is exactly the chosen map on I applied to the actual included endpoint. Every direct comparison on O at any other genuine bounded padding cut has that same terminal map. This includes $I = U \cap V$ and actual triple common opens.*

Proof. The first lemma internally supplies the exact smaller-window inclusions and BOTH membership directions for the actual chosen common-open base map. Apply genuine nested padded-history restriction to the actual original prefixes and tails and the chosen comparison. Its two large flattenings, actual restriction, two nested-history identifications and two small flattenings are all genuinely constructed. They yield an actual comparison of the literal smaller restricted histories and the exact old chosen terminal point map on the real included source point. The first lemma identifies its base biholomorphism with the original common-open map on O.

For any other direct comparison at a stated bounded cut, the genuine empty-extension and fixed-history uniqueness theorem applies to those SAME smaller prefixes and tails. Their terminal maps therefore agree. No recipe is attached to a forgotten endpoint witness and no empty centre is deleted. This constructs the actual restriction of the chosen map rather than merely asserting existence of some direct comparison on the smaller open. □

Theorem 437.3 (Chosen pairwise maps satisfy the actual triple-open cocycle)**.** *For three actual original local prefix-and-tail histories, choose genuine padded pairwise comparisons on their three pairwise-intersection opens. Assume their actual numerical cuts bound the corresponding actual tail lengths. On EVERY genuine common open O of the three original opens, construct actual biholomorphic restrictions of ALL THREE chosen terminal maps to their exact histories on O. These genuine restrictions satisfy $e_{23}|_O \circ e_{12}|_O = e_{13}|_O$ as actual biholomorphisms. Every empty trace centre and all padding positions remain in the histories.*

Proof. For each pair, the actual common-open inclusions place O inside its true pairwise intersection. Apply the preceding producer to the chosen genuine pairwise comparison. This constructs all three actual smaller stagewise maps and proves that each underlying terminal map is precisely the restriction of its OLD chosen pairwise map on included endpoints.

The three actual common-open base maps compose by their true underlying point maps and inverse subtype laws. Restriction preserves each actual tail length, so the given three genuine cut bounds still hold for the smaller histories. Extend the three actual padded comparisons to one common sum of the cuts by genuine identity-empty blowups. Their endpoint maps do not change. Stagewise composition is a genuine comparison of the same fixed source and target histories as the third map. Actual blowup universal uniqueness therefore identifies them, giving the stated cocycle. This is compatibility of the chosen pairwise atlas maps under real restriction; constructing a descended global manifold, its global map and a global sharp scheduler remains separate mathematics. □

438 Genuine restriction of comparison histories and empty padding

The common-open cocycle does not by itself identify a direct common-open map with the restriction of a previously constructed pairwise map. The actual geometric restriction operations below supply the first needed comparison. No canonical compression or deletion of empty steps is used.

Lemma 438.1 (Actual restrictions of a biholomorphism to matching opens)**.** *For a genuine biholomorphism $\Phi : X' \to X$ and actual opens $U' \subset X'$, $U \subset X$ with $U' = \Phi^{-1}U$, construct a genuine restricted biholomorphism $U' \to U$ whose underlying map is exactly Φ. Its inverse is the actual restriction of Φ^{-1}. In particular the source open can be constructed internally as the genuine inverse image of U.*

Proof. Send an actual source subtype point x to $\Phi(x)$ with its target membership supplied by the actual inverse-image equality. Send a target subtype point y to $\Phi^{-1}(y)$; applying the real inverse equation gives its source membership. The two genuine inverse laws identify both subtype composites with their identities. Compose the actual holomorphic maps with open inclusions and use the genuine open-subtype holomorphic criterion for both directions. This constructs the claimed biholomorphism and exact underlying point maps without extending any germ or choosing a new ambient map. □

Theorem 438.2 (Actual stagewise comparisons restrict to every genuine base open)**.** *Given an actual stagewise isomorphism of two finite towers over Φ and any genuine target base open U, restrict the target history to U and the source history to the actual inverse-image open $\Phi^{-1}U$. Construct an actual stagewise isomorphism of those exact restricted histories. The new base map*

is the genuine restricted Φ, and the terminal map is exactly the restriction of the original terminal biholomorphism. Every actual successor, including every centre with empty trace, remains in the history.

Proof. Induct on the genuine stagewise isomorphism. For reflexive towers the preceding open biholomorphism is exactly the required reflexive comparison. At a successor, the induction hypothesis constructs the genuine previous restricted comparison with its exact underlying original point map. The full projection commuting equation identifies the actual terminal source open with the inverse image of the actual terminal target open under the given next biholomorphism. Construct its genuine restriction by the preceding lemma.

Both original arrows restrict to actual blowups of the traces of their centres. The true old centre-preimage law and the exact underlying previous restricted map identify these two trace centres in both directions. The real old blowup square gives the next restricted square by subtype extensionality. Extend the genuine restricted stagewise isomorphism with these actual blowup proofs, centre law and next biholomorphism. Its underlying terminal map is the given original one by construction. No centre or restriction compatibility certificate is a new premise, and an empty trace is retained as a genuine blowup step. □

Theorem 438.3 (Restricting genuine identity empty tails preserves every actual padding position)**.** *For an actual open $U \subset Y$ and every r, the actual restriction to U of the genuine r-step identity empty-centre tower on Y is stagewise isomorphic over the identity of U to the genuine r-step identity empty-centre tower on U. The terminal map is the identity on underlying points. No step is deleted.*

Proof. The actual composite projection of the identity empty tail is the identity, so its actual inverse-image terminal open has exactly the points of U. The equality of these opens gives the genuine endpoint biholomorphism by the preceding open construction. Induct on r. The zero case is the actual reflexive comparison. At a successor, the restricted original arrow is the actual restriction of the identity blowup of the empty centre. Its source and target opens have the same underlying points, its map is the true identity on those points, and the trace centre is genuinely empty. The target padding arrow is the actual identity blowup of the empty centre on U. The two empty centres match, and both squares commute on every actual point. Extend the preceding stagewise comparison by this actual successor. The exact endpoint point map and length $r + 1$ are retained. This proves genuine padding transport; it does not identify arbitrary ambient ideal representatives after deleting their normalizations. □

439 Actual endpoint overlap domains before geometric descent

The chosen pairwise cocycle is now established, but topological gluing also needs the literal domains, diagonal maps and compatibility with the genuine projection to the original ambient manifold. These facts below are produced from actual prefix-and-tail histories and their genuine stagewise comparisons. No descended manifold or separation certificate is assumed.

Lemma 439.1 (The actual terminal projection determines its true overlap open)**.** *For an actual local open $U \subset X$ and actual prefix P and tail S, the true endpoint map to X is the genuine holomorphic composite of the two actual projections and $U \hookrightarrow X$. Its inverse image of another actual open V equals the literal endpoint open used by the retained pairwise comparison on $U \cap V$. The whole padded restricted history on any common open has this same underlying projection, at every padding cut.*

Proof. Compose the actual holomorphic projections and the real open inclusion. Membership of an endpoint in the genuine inverse image of V means that its actual base image belongs to V. It already belongs to U because the first projection is a point of U. This is equivalent in both directions to membership in $U \cap V$, exactly the literal iterated inverse-image open of the

pairwise history. For its padded projection, use the genuine append projection equations, the actual identity projection of every real empty tail, and both original restricted-projection point identities. Every underlying point is unchanged; no step or ordered position is removed. □

Theorem 439.2 (The chosen pairwise map gives an actual transition on the true endpoint opens)**.** *A genuine chosen stagewise comparison of the actual padded pairwise histories constructs a real biholomorphism from the first endpoint's true overlap open to the second endpoint's true reversed overlap open. This actual transition preserves the genuine projection to X at EVERY point.*

Proof. The preceding exact source-domain equality identifies the first true overlap with the literal source domain of the chosen comparison. On the target, commutativity of the actual two ambient opens identifies its literal domain with the true reversed overlap. Compose these two genuine equal-open biholomorphisms with the OLD chosen comparison. This supplies both actual inverse maps and holomorphic directions.

Its true stagewise projection square commutes before taking underlying ambient points. The common-open base map preserves that original point. The proved padded restricted-projection identities identify the two sides with the genuine two endpoint projections. Hence this actual transition preserves the projection to X. A commuting composite square is used only AFTER the genuine stagewise comparison has supplied the real map, not as a replacement for its blowup or centre dictionary. □

Theorem 439.3 (The actual diagonal overlap is the whole endpoint and its transition is identity)**.** *The true diagonal overlap is the whole actual local endpoint. A genuine chosen self-comparison of the same actual prefix-and-tail history induces exactly the actual identity transition there, independently of its genuine padding cut.*

Proof. Every actual local endpoint projects to a point of its own original open, so its true diagonal overlap is the whole endpoint. The actual common-open base self-map is identity on both real subtype levels, hence is the genuine identity biholomorphism. Re-express the supplied true self stagewise comparison over that identity. The real reflexive comparison of the same padded finite history is another such stagewise comparison. Actual blowup universal uniqueness identifies their terminal biholomorphisms, giving the literal identity. The two genuine equal-open identifications in the transition are identity on underlying points, so the actual transition is also identity by subtype extensionality. No arbitrary map over the composite projection is asserted unique, and no deletion of empty positions is used. □

440 Actual nested restrictions of chosen comparison maps

A direct comparison on a smaller common open must be compared with the actual restriction of the previously chosen map, rather than replaced by an unrelated existential witness. All constructions below retain every empty trace centre and every real padding position.

Lemma 440.1 (Actual matching smaller windows give a genuine smaller base map)**.** *Let $W' \subset X'$ and $W \subset X$ be actual opens and let $\Phi : W' \to W$ be a genuine biholomorphism. Let $V' \subset W'$ and $V \subset W$ be genuine smaller ambient opens, whose memberships correspond under the actual underlying map Φ. Construct a genuine biholomorphism $V' \to V$. Its underlying point map is exactly the old Φ applied to the actual inclusion of the smaller source point into W'.*

Proof. The actual window membership equivalence identifies the nested source open inside W' with the inverse image of the nested target open inside W. Restrict the genuine Φ to those exact opens, with its actual inverse. Precompose with the inverse of the true source nested-open flattening and postcompose with the true target nested-open flattening. Both flattenings are identity on the original ambient points. Their real inverse equations therefore prove the stated exact underlying point map. This is a genuine map of actual opens, with no assumed smaller-window comparison. □

Theorem 440.2 (Actual comparisons of restricted towers restrict and flatten again)**.** *For two actual original towers and an actual stagewise comparison of their restrictions to W' and W, construct a genuine stagewise comparison of their literal restrictions to V' and V over the smaller base map just constructed. The terminal point map is the original comparison applied to the actual inclusion of the smaller source endpoint into the old restricted endpoint. No history position is removed.*

Proof. First restrict the genuine stagewise comparison to the nested target open presenting V inside W. The preceding actual window identity identifies its genuine inverse-image source open with the nested open presenting V' inside W'. The stagewise restriction theorem supplies all true trace centres and blowup squares before flattening. The existing genuine nested-restriction tower isomorphisms identify the two double restrictions with the literal restrictions of the original towers to V' and V. Invert the source identification, compose the actual restricted comparison, then compose the target identification. Its base map is exactly the constructed smaller map. The two real flattenings and their inverses are identity on ambient endpoint points. The resulting underlying point map is therefore exactly the claimed old terminal map on the actual included point. This does not infer stagewise isomorphism from a commuting composite projection square. □

Theorem 440.3 (Chosen padded comparisons have genuine smaller-window compatibility)**.** *For two actual original prefix-and-tail histories, suppose a genuine stagewise comparison has been chosen between their padded restrictions to W' and W. On the matching actual smaller windows, construct a stagewise comparison between the literal padded smaller restrictions with the exact old terminal point map. Every direct comparison of those smaller histories has that same map, including any different genuine padding cut which, together with the old cut, bounds both actual original tail lengths.*

Proof. Construct the actual padded-history flattenings on both large windows. Invert their direction as needed to conjugate the chosen comparison into a genuine comparison of the literal restrictions of the two full original padded histories. Each conjugating map is identity on ambient endpoint points, so the raw terminal point map is still the chosen one. Apply the preceding nested stagewise theorem to those exact original full padded histories. Independently construct the genuine padded-history flattenings on the two smaller windows, and conjugate the real smaller comparison into the literal restricted prefix-and-tail histories. All flattenings and inverse flattenings preserve the actual ambient point, so the constructed terminal map has the old underlying map on the actual included source point.

Restriction preserves both actual tail lengths. Thus any other stated cut still bounds the same smaller tails. Extend the two genuine padded comparisons to the common sum of the cuts by actual empty-centre identity blowups, retaining all positions. The real fixed-history blowup uniqueness theorem identifies their terminal maps. This proves compatibility for chosen comparisons and their actual smaller-window restrictions; it does not assert that an arbitrary endpoint witness carries a retained recipe, or that geometric manifold descent or global scheduling is complete. □

441 Actual restriction and flattening of padded comparison maps

All histories below are genuine finite blowup towers. Restriction retains all original centre positions, and padding means actual identity blowups of the empty centre. The geometric operations precede the comparison of maps. This does not compress an original history or assert manifold descent.

Theorem 441.1 (The true restriction of a padded history has an actual flattened history)**.** *Let $P : X \leftarrow Y$ and $S : Y \leftarrow Z$ be actual finite towers, let L be a natural number and let $U \subset X$*

be an actual open. Restrict the genuine padded history $P * (S * \mathrm{Empty}^{L-\ell(S)})$ *to* U*. Construct a genuine stagewise isomorphism over* id_U *to the history consisting of the actual restrictions of* P *and* S*, followed by* $L - \ell(S)$ *genuine identity-empty blowups on the actual restricted endpoint. The terminal map is identity on underlying points of* Z*.*

Proof. Apply the actual append-restriction isomorphism first to P and the entire padded tail. Apply it again to S and its actual identity-empty tail on the true inverse-image open of U in Y. The proved genuine empty-tail restriction isomorphism replaces that last restricted tail by the same number of actual identity-empty blowups on the true restricted endpoint of S. Append it to the reflexive comparison of the restricted S history, then compose with the second append-restriction comparison. Append this actual tail comparison to the reflexive comparison of restricted P, and compose with the first comparison. Every component endpoint map is identity on underlying points, so their actual composite has exactly that point map. Restriction preserves the actual tail length, hence its padding count is literally the count used in the padded restricted history. Each constituent has a genuine stepwise blowup and centre dictionary; no new compatibility certificate is supplied. □

Theorem 441.2 (A true padded comparison restricts to the exact flattened histories)**.** *Let an actual stagewise comparison over a genuine base biholomorphism* $\Phi : X' \to X$ *identify two actual prefix-and-tail histories padded to the same numerical cut* L*. For every genuine target open* U*, internally take the source open* $\Phi^{-1}U$*. Construct a genuine stagewise comparison over the actual restricted* Φ *of the two padded restricted prefix-and-tail histories. Its terminal underlying point map is exactly the old terminal biholomorphism. All empty trace centres and all padding positions are retained.*

Proof. The actual full projection square internally identifies the terminal source open with the inverse image of the terminal target open. The proved stagewise restriction theorem therefore gives a genuine comparison of the restrictions of the two full padded histories, with the exact old terminal point map. Independently construct both genuine flattened padded-history comparisons by the preceding theorem. Invert the source flattening, compose it with the restricted original comparison, then compose with the target flattening. The two flattening base maps are actual identities, so the composite base map is the genuine restricted Φ. Both flattenings are identity on underlying endpoint points, including their true inverses. Thus the composite endpoint map has exactly the old terminal point map. This proves actual geometric restriction before using any uniqueness of comparison maps. □

Theorem 441.3 (The actual restricted comparison agrees with every direct comparison)**.** *For the exact flattened restricted prefix and tail histories just constructed, every direct actual stagewise comparison over the same restricted base map agrees with the constructed restriction, even when its padding cut is different. Assume both numerical cuts bound both actual tail lengths. Its terminal map sends each actual source point to the old terminal image as an underlying point.*

Proof. Both maps are genuine stagewise isomorphisms of the same actual restricted prefixes and tails over the same base biholomorphism. Restriction preserves the actual lengths, so both cuts still bound both tails. Extend both genuine padded histories by actual identity-empty blowups to the common sum of the cuts, retaining every position and the same endpoint maps. Actual blowup uniqueness at each successor of these fixed histories identifies the two terminal biholomorphisms. Substitute this actual equality into the underlying-point dictionary of the preceding construction. This applies to direct retained-recipe comparisons once their histories have been genuinely identified with these exact restrictions; it does not assume that nested-history identification or global gluing. □

442 Actual topological gluing from sharp endpoint histories

The data in this chapter are actual finite prefix/tail towers over original opens, and genuine stagewise comparisons of their retained empty-padded histories. No cocycle, composite-projection identity, overlap geometry, separation or descended manifold certificate is an input. This is the actual geometric gluing interface; a covering family from only original curve input and its global analytic and properness consequences are separate producers.

Lemma 442.1 (Actual endpoint transition intersections)**.** *For an actual family of finite prefix/tail towers and chosen genuine stagewise pairwise comparisons, the real diagonal overlap is whole and its real transition is identity. Each pairwise transition takes the actual intersection with a third endpoint overlap into the corresponding actual target overlap.*

Proof. Use the real projection-defined overlap opens and the genuine transitions of the preceding chapter. Its fixed-history universal uniqueness supplies the actual diagonal identity. For an actual point in the first pair overlap and the third overlap, the genuine transition projection square identifies the original ambient image after the transition with the original image before it. Membership in the third original open therefore persists, which is exactly the target third-overlap predicate. No target domain membership is assumed separately. □

Theorem 442.2 (The actual OLD transitions satisfy the literal gluing cocycle)**.** *The actual transitions on the true projection-defined overlap opens satisfy the literal triple gluing identity at every point of the actual intersection, including chosen pairwise histories with different real padding cuts.*

Proof. Take the genuine intersection of the three original base opens. An actual point in the two source overlaps lies over this intersection, so it defines a real point in the first common-open terminal domain. Apply the already proved chosen-map restriction theorem to ALL THREE actual chosen stagewise pairwise maps. It internally constructs the three common-open stagewise maps, retaining every empty position, with their exact OLD terminal-point identities. The genuine common-cut extension and fixed-history uniqueness prove the composition identity for these three actual maps.

The first exact OLD point identity identifies its image with the OLD first transition. Its actual common-open membership and the preceding intersection lemma identify the input to the second OLD transition by subtype extensionality. The second exact point identity therefore identifies the composite with the OLD second transition applied to the OLD first one. The third point identity identifies the direct image with the OLD direct transition. Evaluating the actual common-open composition identity at this point gives precisely the literal gluing cocycle. No fresh pairwise maps are silently substituted for the already chosen ones. □

Theorem 442.3 (A genuine topological gluing core is constructed)**.** *The actual endpoint spaces, true overlap opens and chosen real transitions construct a genuine TopCat gluing core, with every required identity proved internally from the actual finite-history geometry. Its actual topological glued space has genuine open-embedding local inclusions that are jointly surjective.*

Proof. Use the actual complex endpoint manifolds as topological spaces and the real projection-defined opens as the overlap objects. Each actual transition is a biholomorphism, hence continuous, and defines the required topological morphism. The first lemma gives the true diagonal domain, diagonal identity and intersection membership. The second theorem gives exactly the required triple point identity. These are all the fields of the genuine TopCat gluing core, which can therefore be filled without any assumed cocycle certificate. The topological gluing API now supplies its actual quotient data, real open embeddings of all endpoint spaces and their joint surjectivity. Hausdorffness, countability, lifted complex charts, the original-only covering atlas and properness are not concluded here. □

443 Remmert's proper mapping theorem

This section proves the Remmert proper mapping theorem and the Remmert–Stein extension theorem by the simultaneous induction on the dimension of Demailly, *Complex analytic and differential geometry*, II.8.7–8.8. Everything happens inside complex manifolds (Hausdorff, second countable): an analytic subset of a manifold X is a closed set with local holomorphic equations (Definition **??**), and "analytic in an open set $U \subset X$" means analytic in the manifold U. No abstract complex spaces occur. The route differs from the source in one respect: Demailly's rank theory (II.8.3–8.6) uses the normalization of complex spaces (proof of II.8.4). It is replaced here by two dimension estimates proved directly from the local parametrization theorem — a lower bound for the dimension of the fibres of a map on a pure-dimensional set (Lemma 443.17) and the bound "dimension of the image $\leq$ dimension of the source minus the fibre dimension" (Lemma 443.18) — and in Step 1 of the induction the exceptional set is saturated by the fibres before the estimate is applied.

Definition 443.1 (Dimension bounds for analytic sets)**.** Let A be an analytic subset of a complex manifold X. A point $x \in A$ is *regular of dimension k* if A is, near x, a complex submanifold of dimension k; A_{reg} is the set of regular points (of any dimension) and $A_{\mathrm{sing}} = A \setminus A_{\mathrm{reg}}$. We write $\dim A \leq p$ if every regular point of A has dimension $\leq p$; for $x \in A$ we write $\dim_x A \geq q$ if every neighbourhood of x contains a regular point of A of dimension $\geq q$, and $\dim_x A \leq p$ if every regular point of A in some neighbourhood of x has dimension $\leq p$. A is *of pure dimension k* if every regular point of A has dimension k. For a holomorphic map $F\colon X \to Y$, $F|_A$ is *proper* if $A \cap F^{-1}(K)$ is compact for every compact $K \subset Y$.

Directly from the definitions, $\dim_x A \leq p$ and $\dim_x A \geq q$ force $q \leq p$, and $\dim A \leq p$ gives $\dim_x A \leq p$ everywhere; at a regular point of dimension k both $\dim_x A \leq k$ and $\dim_x A \geq k$ hold (regularity is relatively open and the dimension of a regular point is unique, Lemma **??**); and every point of an analytic set has $\dim_x A \geq 0$ (Theorem **??**).

Lemma 443.2 (Analytic subsets of an open subset)**.** *Let U be an open subset of a complex manifold X, a complex manifold with the restricted charts.*

1. *If A is analytic in X, then $A \cap U$ is analytic in U.*
2. *If every point of X has an open neighbourhood U with $A \cap U$ analytic in U, then A is analytic in X. In particular a subset of U which is analytic in U and closed in X is analytic in X.*
3. *A point $x \in U$ is a regular point of dimension k of $A \cap U$ (in U) if and only if it is one of A (in X). Consequently $\dim A \leq p$ implies $\dim(A \cap U) \leq p$; for $x \in U$, $\dim_x(A \cap U) \geq q$ iff $\dim_x A \geq q$, and $\dim_x(A \cap U) \leq p$ iff $\dim_x A \leq p$; and for $B \subset U$ the bounds $\dim B \leq p$ and "pure dimension k" mean the same in U and in X.*
4. *A map holomorphic on U is holomorphic on the manifold U.*

Proof. The chart of U at x is the chart e of X at x restricted to $U \cap \operatorname{dom} e$; its image is $e(U \cap \operatorname{dom} e)$, an open subset of the image of e. (1) Restrict the local equations of A at $x \in U$ to this smaller open set. (2) The local equations of $A \cap U$ at x in U are local equations of A at x in X, since a point of $\operatorname{dom} e$ whose image lies in $e(U \cap \operatorname{dom} e)$ lies in U. For the second statement use U near points of U and the open complement of the closed set elsewhere. (3) The same two remarks apply to a flattening at x; in the direction $X \to U$ restrict the flattening to the preimage of $e(U \cap \operatorname{dom} e)$. The dimension statements follow, since the neighbourhoods of x in U are the traces of those in X. (4) Holomorphy is local. □

These facts are used whenever a lemma proved for analytic subsets of a manifold is applied to a set that is analytic only in an open subset.

443.1 The constant rank theorem

Lemma 443.3 (Constant rank theorem)**.** *Let E, F be finite-dimensional complex normed spaces, $U \subset E$ open, $G\colon U \to F$ holomorphic with $\operatorname{rank} dG_z = r$ for all $z \in U$, and $p \in U$; put $R = \operatorname{range} dG_p$ (dimension r) and $K = \ker dG_p$. There are continuous linear projections $Q\colon F \to R$, $S\colon E \to K$, a radius $\rho > 0$, a holomorphic map ψ on the ball B of radius ρ around $c = (Q(G(p)), S(p))$ in $R \times K$ (sup norm, so $B = B' \times B''$) and a holomorphic k on B' such that: $\psi(c) = p$; $\psi(w) \in U$ and $\Psi(\psi(w)) = w$ for $w \in B$, where $\Psi(z) = (Q(G(z)), S(z))$; $\psi(B)$ is open; $Q \circ k = \mathrm{id}$ on B'; and*

$$G(\psi(w)) = k(w_1) \qquad (w = (w_1, w_2) \in B).$$

Consequently: (a) if $r < \dim E$, every neighbourhood of p contains a point $z \neq p$ with $G(z) = G(p)$ (the fibre through p contains $\psi(\{c_1\} \times B'')$, with $\dim K = \dim E - r \geq 1$); (b) $G(\psi(B)) = \{y \in F : Q(y) \in B',\ k(Q(y)) = y\}$, the zero set of $y \mapsto y - k(Q(y))$ in the open set $Q^{-1}(B')$.

Proof. Choose complements of R in F and of K in E and let Q, S be the projections. The derivative $(Q \circ dG_p, S)$ of Ψ at p is injective: if $Q(dG_p(x)) = 0$ then $dG_p(x) = 0$ (as $dG_p(x) \in R$ is fixed by Q), so $x \in K$ and $x = S(x) = 0$; by rank–nullity $\dim R + \dim K = \dim E$, so it is bijective, and the holomorphic inverse function theorem gives a local inverse ψ of Ψ near c, holomorphic on a ball B on which moreover $d\Psi$ is invertible at $\psi(w)$. For $w \in B$, $H = G \circ \psi$ has $dH_w = dG_{\psi(w)} \circ (d\Psi_{\psi(w)})^{-1}$, so $Q \circ dH_w = \mathrm{pr}_1$ and $\operatorname{rank} dH_w = r = \dim R$. Then Q restricted to $\operatorname{range} dH_w$ is onto R, hence injective by dimension, and $Q(dH_w(0, v)) = 0$ gives $dH_w(0, v) = 0$ for $v \in K$. Thus $t \mapsto H(w_1, t)$ has zero derivative on the convex ball B'' and is constant; $k(w_1) = H(w_1, c_2)$ gives the normal form, and $Q(k(w_1)) = Q(G(\psi(w_1, c_2))) = w_1$. (a) Take $v \in K \setminus 0$ and $z = \psi(c_1, c_2 + sv)$ with $s \neq 0$ small: $z \to p$, $\Psi(z) \neq c$ so $z \neq p$, and $G(z) = k(c_1) = G(p)$. (b) If $w \in B$, $y = G(\psi(w)) = k(w_1)$ has $Q(y) = w_1 \in B'$ and $k(Q(y)) = y$; conversely such a y equals $G(\psi(Q(y), c_2))$. □

Lemma 443.4 (Constant-rank images are submanifolds)**.** *(1) Let R be a finite-dimensional complex normed space, $Q\colon \mathbf{C}^N \to R$ linear, $B' \subset R$ open and k holomorphic on B' with $Q \circ k = \mathrm{id}$ on B'. Every point of the graph $\Gamma = \{y \in \mathbf{C}^N : Q(y) \in B',\ k(Q(y)) = y\}$ is a regular point of Γ of dimension $\dim R$; Γ is the zero set of the holomorphic map $y \mapsto y - k(Q(y))$ on the open set $Q^{-1}(B')$. (2) Let E be a finite-dimensional complex normed space, $U \subset E$ open, $G\colon U \to \mathbf{C}^N$ holomorphic with $\operatorname{rank} dG_z = r$ for all $z \in U$, and $p \in U$. There are an open O with $p \in O \subset U$, an open $W \subset \mathbf{C}^N$ and a holomorphic h on W with $G(O) = \{y \in W : h(y) = 0\}$, and every point of $G(O)$ is a regular point of $G(O)$ of dimension r. (3) If G is holomorphic on an open set $U \ni p$ of E with values in a finite-dimensional space F and dG_p is onto (i.e. $\operatorname{rank} dG_p = \dim F$), then G maps every neighbourhood of p onto a neighbourhood of $G(p)$.*

Proof. (1) Linear algebra first: a subspace W_0 of dimension r of an N-dimensional space V is carried onto the standard plane $\mathbf{C}^r \times \{0\}$ by a linear isomorphism $V \to \mathbf{C}^N$ (choose complements of W_0 and of the plane, both of dimension $N - r$, and isomorphisms between the summands). Q is onto, since its range contains the nonempty open set $B' = Q(k(B'))$; so $\ker Q$ has dimension $N - \dim R$. Let $P\colon \mathbf{C}^N \to \ker Q$ be a linear projection and $A\colon R \times \ker Q \to \mathbf{C}^N$ a linear isomorphism carrying $R \times \{0\}$ onto the standard $(\dim R)$-plane. For $y_0 \in \Gamma$ put

$$\Theta(y) = A\big(Q(y) - Q(y_0),\ P(y - k(Q(y)))\big) \qquad (y \in Q^{-1}(B')).$$

Θ is holomorphic, $\Theta(y_0) = 0$, and $d\Theta_{y_0}(v) = A(Q(v), P(v - dk(Qv)))$ is injective: $Q(v) = 0$ forces $v \in \ker Q$ and $P(v) = v = 0$. By the holomorphic inverse function theorem Θ is a biholomorphism near y_0. For $y \in Q^{-1}(B')$ we have $y - k(Q(y)) \in \ker Q$ (as $Q \circ k = \mathrm{id}$), so $P(y - k(Q(y))) = y - k(Q(y))$, and $\Theta(y)$ lies in the standard plane iff $y - k(Q(y)) = 0$, i.e. iff $y \in \Gamma$. So Θ flattens Γ near y_0. (2) In the normal form of Lemma 443.3 take $O = \psi(B)$, $W = Q^{-1}(B')$, $h(y) = y - k(Q(y))$: $G(O) = \Gamma$ by part (b) of that lemma, and $\dim R = r$; apply (1). (3) G is C^1, so it has a strict derivative at p, and a map with surjective strict derivative maps

neighbourhoods of p onto neighbourhoods of $G(p)$ (the surjective form of the inverse function theorem). □

For holomorphic maps between complex manifolds the lemma is applied in charts.

Lemma 443.5 (Generic rank)**.** *(1) The rank of a family of linear maps $A_x\colon E \to F$ between finite-dimensional spaces, continuous at x_0, is lower semicontinuous at x_0. (2) Let U be a connected open subset of a finite-dimensional complex normed space, $G\colon U \to F$ holomorphic, and let* $\operatorname{rank} dG$ *attain its maximum r on U at z_1. Then $U^{\max} = \{z \in U : \operatorname{rank} dG_z = r\}$ is open and dense in U; at its points the hypothesis of Lemma 443.3 holds on a neighbourhood. (For a connected complex manifold P and a holomorphic map, this is applied in charts.)*

Proof. (1) If $A_{x_0}v_1, \dots, A_{x_0}v_s$ is a basis of $\operatorname{range} A_{x_0}$, linear independence of $(A_x v_i)_i$ persists near x_0, so $\operatorname{rank} A_x \geq s$. (2) Openness: by (1) applied to $z \mapsto dG_z$ (continuous, G being C^∞) and maximality. Density: let $dG_{z_1}v_i$, $i \leq r$, be a basis of $\operatorname{range} dG_{z_1}$ and ℓ_j linear forms dual to it; $h(z) = \det\bigl(\ell_j(dG_z v_i)\bigr)$ is holomorphic on U (an alternating sum of products of directional derivatives), $h(z_1) = 1$, and $h(z) = 0$ where $\operatorname{rank} dG_z < r$ (the $dG_z v_i$ are then dependent, so the rows are). If an open set met U but not $U^{\max}$, h would vanish near a point of U, hence on the connected U by the identity theorem (Lemma **??**; holomorphic functions of several variables are analytic), contradicting $h(z_1) = 1$. □

443.2 Dimension estimates from the local parametrization

Lemma 443.6 (Manifold pieces of a parametrized irreducible germ)**.** *Let $A \subset \mathbf{C}^N$ carry parametrization data of dimension d as in Proposition* **??** *(an irreducible germ in Noether position does): projection $\pi\colon \mathbf{C}^N \to \mathbf{C}^d$ onto the first d coordinates, the ball B on which the defining identities hold, and the polydisc $\Delta \subset B$ of the parametrizing sets. (1) Every fibre of π on $A \cap B$ is finite. (2) (Generic rank of maps with finite fibres.) If G is holomorphic on a nonempty open subset U of a finite-dimensional space E and all its fibres in U are finite, then* $\operatorname{rank} dG = \dim E$ *on a nonempty open subset of U. (3) If φ is holomorphic and injective on a nonempty open $U \subset \mathbf{C}^k$ with $\varphi(U) \subset A \cap B$, then $k \leq d$. (4) If moreover $k = d$ and $\varphi(U) \subset A \cap \Delta$, then $\varphi(U)$ contains a nonempty relatively open subset of $A \cap \Delta$. (5) If y is a regular point of dimension k of a set $S \subset \mathbf{C}^N$ and W a neighbourhood of y, there are an open $U \ni 0$ in $\mathbf{C}^k$ and an injective holomorphic φ on U with $\varphi(0) = y$ and $\varphi(U) \subset S \cap W$.*

Proof. (1) A point $z \in A \cap B$ over x has its first d coordinates equal to those of x and each coordinate z_j, $j > d$, among the finitely many roots of the monic polynomial $Q_j(x, \cdot)$ of Proposition **??**. (2) The rank of dG is bounded by $\dim F$; let ρ be its maximum on U. The set $U^{\max}$ where it is attained is open and nonempty (Lemma 443.5(2), openness part), and the rank is constant there. If $\rho < \dim E$, a point $z_1 \in U^{\max}$ is an accumulation point of its fibre (Lemma 443.3(a)); but that fibre is finite. So $\rho = \dim E$. (3) $G = \pi \circ \varphi$ has finite fibres on U: φ maps a fibre of G injectively into a fibre of π on $A \cap B$, finite by (1). By (2), $\operatorname{rank} dG = k$ somewhere, and $\operatorname{rank} dG \leq d$. (4) By (2) $G = \pi \circ \varphi$ has rank $d = \dim \mathbf{C}^d$ on a nonempty open $U' \subset U$, so it maps every neighbourhood of a point of U' onto a neighbourhood of its image (Lemma 443.4(3)). For $z \in U'$, $|G(z)| \leq |\varphi(z)|$ lies in the ball where the discriminant δ is holomorphic; δ is not identically zero near 0, so its zero set has empty interior in that ball (identity theorem, Lemma **??**), and the open neighbourhood $G(U')$ of $G(z)$ contains a point $G(z_1)$, $z_1 \in U'$, with $\delta(G(z_1)) \neq 0$. By Proposition **??** (local graphs) at $y_1 = \varphi(z_1)$ there are open $V_g \ni y_1$, $U_g \ni \pi(y_1)$ and a holomorphic section g of π with $y = g(\pi(y))$ for every $y \in A \cap V_g$. Let $O_1 = U' \cap \varphi^{-1}(V_g)$, a neighbourhood of z_1, and $V = V_g \cap \pi^{-1}(\operatorname{int} G(O_1))$, an open set containing y_1. If $y \in V \cap A \cap \Delta$, then $\pi(y) = \pi(\varphi(u))$ for some $u \in O_1$, and both y and $\varphi(u)$ lie in $A \cap V_g$, so $y = g(\pi(y)) = g(\pi(\varphi(u))) = \varphi(u)$. Hence $V \cap A \cap \Delta \subset \varphi(U)$. (5) Let Φ flatten S at y ($\Phi(y) = 0$) and $\iota\colon \mathbf{C}^k \to \mathbf{C}^N$, $w \mapsto (w, 0)$. Take $\varphi = \Phi^{-1} \circ \iota$ on $U = \iota^{-1}\bigl(\operatorname{target}\Phi \cap (\Phi^{-1})^{-1}(\operatorname{int} W)\bigr)$: it is injective as Φ^{-1} and ι are, and $\Phi(\varphi(w)) = \iota(w)$ lies in the standard plane, so $\varphi(w) \in S$. □

Lemma 443.7 (Monotonicity of dimension)**.** *(1) (Germ form.) Let T be an analytic germ at $0 \in \mathbf{C}^N$ and φ holomorphic and injective on an open $U \subset \mathbf{C}^k$ with $\varphi(U) \subset T$ and $\varphi(u_0) = 0$ for some $u_0 \in U$. Every neighbourhood of 0 contains a regular point of T of dimension $\geq k$. (2) Let T be an analytic subset of a complex manifold X, $B \subset T$ any subset and z a regular point of B of dimension k. Then $\dim_z T \geq k$: every neighbourhood of z contains a regular point of T of dimension $\geq k$. Consequently, for $B \subset T$, $\dim_x B \geq q$ implies $\dim_x T \geq q$, and $\dim T \leq a$ implies $\dim B \leq a$.*

Proof. (1) The germ of T is an irredundant union of irreducible germs $T_1, \ldots, T_s$ (Theorem **??**); after a linear change of coordinates L_j each T_j is in Noether position with parametrization data of dimension d_j (Theorem **??**, Proposition **??**). Choose a ball B_ρ on which $T = \bigcup_j T_j$, each T_j is relatively closed and $L_j^{-1}(B_\rho)$ lies in the ball of the j-th data. The open set $U_1 = U \cap \varphi^{-1}(B_\rho) \ni u_0$ is covered by the finitely many relatively closed sets $U_1 \cap \varphi^{-1}(T_j)$, so one of them contains a nonempty open W (if $U_1 \not\subset \varphi^{-1}(T_1)$, pass to the nonempty open $U_1 \setminus \varphi^{-1}(T_1)$, covered by the others, and induct). Then $L_j^{-1} \circ \varphi$ is an injective holomorphic k-parameter piece of $L_j^{-1}(T_j)$, and Lemma 443.6(3) gives $k \leq d_j$. The regular points of T_j of dimension d_j accumulate at 0 and are dense in T_j near 0 (Proposition **??**, Lemma **??**), and by Lemma **??** there is an open set O' inside the given neighbourhood and B_ρ meeting T_j and none of the other T_i; a regular point $y \in O'$ of T_j of dimension d_j is a regular point of T, since $T = T_j$ on O'. (2) Read $B \subset T$ in the chart at z, translated so that z becomes 0: 0 is a regular point of dimension k of the chart germ of B, which is therefore the image of 0 under an injective holomorphic k-parameter piece (Lemma 443.6(5)) inside the chart germ of T, an analytic germ. Apply (1) and read the regular point back on X. For the consequences: a regular point of B of dimension $j \geq q$ near x has regular points of T of dimension $\geq j$ near it; and a regular point of B of dimension j has regular points of T of dimension $\geq j$, which are $\leq a$. □

Lemma 443.8 (Thin analytic subsets have smaller dimension)**.** *(1) (Germ form.) Let T be an analytic germ at $0 \in \mathbf{C}^N$ all of whose regular points have dimension $\leq a$, and φ holomorphic and injective on an open $U \subset \mathbf{C}^a$ with $\varphi(U) \subset T$ and $\varphi(u_0) = 0$. Every neighbourhood of 0 contains an open W meeting T with $W \cap T \subset \varphi(U)$. (2) Let T be analytic in a complex manifold with $\dim T \leq a$, and $T' \subset T$ any subset with empty interior relative to T (an open set O with $O \cap T \subset T'$ has $O \cap T = \emptyset$). Then every regular point of T' has dimension $< a$; that is, $\dim T' \leq a - 1$, and T' has no regular point if $a = 0$.*

Proof. (1) As in the proof of Lemma 443.7(1): the germ is an irredundant union of irreducible germs T_j with parametrization data of dimension d_j after linear changes L_j, and an open piece W' of the parameter domain is mapped into one T_j, so $a \leq d_j$. Conversely T_j has regular points of dimension d_j in an open set where $T = T_j$ (Lemma **??**, Proposition **??**, Lemma **??**), regular points of T, so $d_j \leq a$. Thus $d_j = a$, and by Lemma 443.6(4) applied to $L_j^{-1} \circ \varphi$ on W' there is an open V meeting T_j (in the polydisc) with $V \cap T_j \subset \varphi(W')$. By Lemma **??**, applied once for each $i \neq j$ (each $T_i \cap T_j$ is avoided by T_j in every small open set meeting T_j, and the complements of the T_i are relatively open), there is an open $W \subset V$ meeting T_j and none of the other T_i; on W, $T = T_j \subset \varphi(U)$. (2) Let y be a regular point of T' of dimension k. By Lemma 443.7, $k \leq a$. If $k = a$, read $T' \subset T$ in the chart at y: 0 is the centre of an injective holomorphic a-parameter piece of the chart germ of T' (Lemma 443.6(5)), all regular points of the chart germ of T have dimension $\leq a$, and (1) gives an open W meeting the chart germ of T inside that of T'. Its preimage O under the translated chart is an open set with $\emptyset \neq O \cap T \subset T'$, contradicting the hypothesis. Hence $k < a$. □

Lemma 443.9 (Points of local dimension zero are isolated)**.** *Let T be analytic in a complex manifold and $x \in T$ such that every regular point of T in some neighbourhood U of x has dimension 0. Then x is an isolated point of T.*

Proof. Otherwise x lies on an irreducible component V of T of positive dimension (Proposition **??**). By Lemma **??** there is an open $U' \subset U$ meeting V with $T \cap U' \subset V$ containing a regular point of V of dimension $\dim V$; there $T = V$, so it is a regular point of T of dimension $\dim V > 0$ in U, a contradiction. □

Lemma 443.10 (Generic hyperplane sections)**.** *Let T be analytic in a complex manifold M with $\dim T \le k$, $g\colon M \to \mathbf{C}^N$ a map, $x_0 \in M$, and Ξ a linear subspace of the dual space $(\mathbf{C}^N)^*$ such that no $y \in T \setminus \{x_0\}$ satisfies $\xi(g(y)) = \xi(g(x_0))$ for all $\xi \in \Xi$. (1) If $k \ge 1$, there is a countable set of linear subspaces of $(\mathbf{C}^N)^*$, none containing Ξ, such that for every $\xi \in \Xi$ outside all of them, $\dim\bigl(T \cap \{\xi \circ g = \xi(g(x_0))\}\bigr) \le k-1$. (2) Some $\xi \in \Xi$ lies outside any given countable set of subspaces none of which contains Ξ (so such choices exist and can be combined with countably many further conditions of the same kind). (3) If g is holomorphic, there are $\xi_1, \dots, \xi_k \in \Xi$ such that $T \cap \{\xi_1 \circ g = \xi_1(g(x_0)), \dots, \xi_k \circ g = \xi_k(g(x_0))\}$ is a discrete set (no point of it is an accumulation point of it).*

Proof. (1) Let (O_i) be a countable base of M and I the set of i with $T \cap O_i \not\subset \{x_0\}$; for $i \in I$ the forms ξ with $\xi \circ g = \xi(g(x_0))$ on $T \cap O_i$ form a subspace H_i, not containing Ξ by the hypothesis applied to some $y \in T \cap O_i$, $y \ne x_0$. Let $\xi \in \Xi \setminus \bigcup H_i$ and $T_\xi = T \cap \{\xi \circ g = \xi(g(x_0))\}$. An open $O \subset M \setminus \{x_0\}$ with $O \cap T \subset T_\xi$ has $O \cap T = \emptyset$: otherwise it contains a basic O_i, $i \in I$, and $T \cap O_i \subset T_\xi$ would put ξ in H_i. So near every $y \ne x_0$, T_ξ has empty interior relative to T, and the argument of Lemma 443.8, which only uses open sets inside a neighbourhood of the regular point, bounds the dimension of every regular point $y \ne x_0$ of T_ξ by $k-1$. If x_0 is a regular point of T_ξ of dimension $j \ge 1$, it is not isolated (a j-dimensional manifold piece through it, Lemma 443.6(5)), and nearby points of T_ξ are regular of the same dimension j, so $j \le k-1$; $j = 0 \le k-1$ is trivial. (2) The subspaces meet Ξ in closed subspaces with empty interior unless they contain Ξ; by Baire's theorem in the finite-dimensional space Ξ, countably many of them do not cover Ξ. (3) Induction on k. For $k = 0$ every point of T is isolated (Lemma 443.9). For $k \ge 1$ choose ξ_1 by (1) and (2); T_{ξ_1} is analytic (a level set of the holomorphic $\xi_1 \circ g$ intersected with T), of dimension $\le k-1$, and the separation hypothesis passes to it; apply the induction hypothesis. □

For an open set $\Omega \subset \mathbf{C}^N$ one takes $M = \Omega$ and g the inclusion; for $\Xi = (\mathbf{C}^N)^*$ the separation hypothesis always holds.

Lemma 443.11 (Compact analytic subsets of open subsets of $\mathbf{C}^N$ are finite)**.** *Let $\Omega \subset \mathbf{C}^N$ be open and $K \subset \Omega$ a compact analytic subset of Ω. Then K is finite. (Being compact, K is closed in $\mathbf{C}^N$ and has local equations at its own points, so it is an analytic subset of $\mathbf{C}^N$; this is the form proved.)*

Proof. By Lemma 443.9 and compactness it suffices to show that K has no regular point of positive dimension. We show by descending induction on $j = N, \dots, 0$: *no compact analytic subset K of $\mathbf{C}^N$ contained in $\{z_1 = c_1, \dots, z_j = c_j\}$ has a regular point of positive dimension.* For $j = N$ such a K has at most one point, while a regular point of positive dimension is not isolated (Lemma 443.10). Let $j < N$, and suppose $K \subset \{z_1 = c_1, \dots, z_j = c_j\}$ has a regular point of positive dimension. Let K_+ be the closure of the set of regular points of K of positive dimension: a nonempty compact analytic subset of $\mathbf{C}^N$ (Lemma 443.12). Let $\xi = z_{j+1}$ and let $p \in K_+$ maximize $\operatorname{Re}\xi$ on K_+. The point p is not isolated in K_+: it is a limit of regular points of K of positive dimension, and if it is one itself, the points of K near it are regular of the same dimension (Lemma **??**) and some of them differ from p.

Translate p to 0 and write the germ of K_+ as a finite union of irreducible germs (Theorem **??**). One of them, C, has positive dimension d: put each in Noether position after a linear change of coordinates (Propositions **??** and **??**); a component of dimension 0 is $\{0\}$ near 0 by the *small* clause, and p is not isolated. Let π, Δ, δ and q be the projection, polydisc, discriminant and number of sheets of C; on C near 0, $\operatorname{Re}\xi \le 0$. By Proposition **??** the coefficient of T^{q-1} of the canonical polynomial of ξ extends to a holomorphic function near 0, so there is a holomorphic

σ near 0 with $\sigma(x) = \sum_{y \in C \cap \Delta,\ \pi(y)=x} \xi(y)$ whenever $\delta(x) \neq 0$. Then $\operatorname{Re}\sigma \leq 0$ off $\delta^{-1}(0)$, hence near 0 (the complement of $\delta^{-1}(0)$ is dense, Lemma **??**); and $\sigma(0) = 0$, because the fibre points over x tend to 0 as $x \to 0$ (the small clause). By the maximum principle applied to e^{σ}, $\operatorname{Re}\sigma \equiv 0$ near 0; each summand has real part ≤ 0, so $\operatorname{Re}\xi(y) = 0$ for every $y \in C \cap \Delta$ over a point x near 0 with $\delta(x) \neq 0$. Pick such an x_0 and y_0 over it, as close to 0 as we like, and the local holomorphic section g through y_0 (Proposition **??**). Then $\operatorname{Re}(\xi \circ g) \equiv 0$ near x_0, so $|e^{\xi \circ g}| \equiv 1$ and $\xi \circ g$ is constant near x_0 (maximum principle, and continuity to exclude jumps by $2\pi i\mathbf{Z}$). Hence near y_0 the set $C \cap \Delta$ lies in $\{\xi = \xi(y_0)\}$, and y_0 is a regular point of $C \cap \Delta$ of dimension d.

Back in $\mathbf{C}^N$, $K' = K_+ \cap \{\xi = c'\}$, $c' = \xi(p + y_0)$ after undoing the coordinate changes, contains a set of which $p + y_0$ is a regular point of dimension $d \geq 1$; by Lemma 443.7, K' has a regular point of dimension $\geq d$. But K' is a compact analytic subset of $\mathbf{C}^N$ inside $\{z_1 = c_1, \ldots, z_j = c_j,\ z_{j+1} = c'\}$, contradicting the induction hypothesis for $j + 1$. □

Lemma 443.12 (Regular-stratum hulls)**.** *Let A be an analytic subset of a complex manifold and P a set of natural numbers. The closure A_P of the set of regular points of A whose dimension lies in P is an analytic subset contained in A, and every regular point of A_P has its dimension in P.*

Proof. A_P is analytic by Proposition **??**, and $A_P \subset A$ as A is closed. Let x be a regular point of A_P of dimension k. Regular points of A with dimension in P are dense in A_P, so there is one, z, of dimension $j \in P$, inside a neighbourhood of x on which A_P is a k-manifold. Near z every point of A is a regular point of A of dimension j, hence lies in A_P; so $A = A_P$ near z, and z is a regular point of A_P of dimensions j and k. By Lemma **??**, $k = j \in P$. □

Lemma 443.13 (Singular locus of a pure-dimensional analytic set)**.** *Let A be an analytic subset of pure dimension k of a complex manifold M of dimension n. Then A_{sing} is analytic, cut out near every point by Cartan generators $g_1, \ldots, g_N$ of $\mathcal{I}_A$ and all $(n-k)$-minors of their Jacobian matrix, and $\dim A_{\text{sing}} \leq k - 1$.*

Proof. As in the proof of Lemma **??**: at a regular point the normal coordinates of a flattening are combinations of the g_j, so some $(n-k)$-minor is $\neq 0$; conversely a nonzero $(n-k)$-minor at $y \in A$ makes the selected equations cut out a k-manifold $Z \supset A$ near y, and $A \cap Z$, read on a ball of Z, has a regular point of some dimension j giving a regular point of A of dimension j; pure dimensionality (which is all that irreducibility was used for there) gives $j = k$, so $A \cap Z$ has interior in Z and equals Z near y by the identity theorem. A_{sing} has empty interior in A because A_{reg} is dense (Theorem **??**), so Lemma 443.8 gives $\dim A_{\text{sing}} \leq k - 1$. □

Lemma 443.14 (Tangent spaces of an analytic set; the rank of a map on it)**.** *Let A be an analytic subset of a complex manifold X of dimension n, $y \in X$, and e a chart of X at y. The* tangent space of A at y, read in e, *is $T_y^e A = \bigcap_g \ker d(g \circ e^{-1})_{e(y)} \subset \mathbf{C}^n$, the intersection over all holomorphic germs g at y vanishing on A near y. For $F\colon X \to Y$ holomorphic ($\dim Y = n'$) and a chart e' of Y at $F(y)$, the* rank of F on A at y *is $\operatorname{rank}_y(F|_A) = \dim D_y(T_y^e A)$, with D_y the differential of $e' \circ F \circ e^{-1}$ at $e(y)$.*

1. *If e_1, e_2 are charts at y, then $T_y^{e_1} A$ is the preimage of $T_y^{e_2} A$ under the differential of $e_2 \circ e_1^{-1}$ at $e_1(y)$, and $\operatorname{rank}_y(F|_A)$ does not depend on the charts e, e'.*

2. *Let Ψ be a local biholomorphism of $\mathbf{C}^n$ with $e^{-1}(z) \in A \iff \Psi(z) \in \mathbf{C}^m \times 0$ for all z in its source, and let $y \in A$ with $e(y)$ in the source, $\Psi(e(y)) = (w_0, 0)$. Then $T_y^e A$ is the image of $\mathbf{C}^m$ under the differential at w_0 of $P(w) = \Psi^{-1}(w, 0)$; it has dimension m; and $\operatorname{rank}_y(F|_A)$ is the rank at w_0 of the differential of $w \mapsto e'(F(e^{-1}(P(w))))$. In particular, at a regular point of dimension m the space $T_y A$ is the tangent space of the manifold A_{reg}, $\operatorname{rank}_y(F|_A) = \operatorname{rank} d(F|_{A_{\text{reg}}})_y \leq m$.*

3. *If holomorphic $g_1, \dots, g_N$ generate $\mathcal{I}_A$ at every point of an open U and cut out A on U, then $T^e_y A = \bigcap_i \ker d(g_i \circ e^{-1})_{e(y)}$ for $y \in A \cap U$.*

Proof. (1) Near $e_1(y)$, $g \circ e_1^{-1} = (g \circ e_2^{-1}) \circ (e_2 \circ e_1^{-1})$, so by the chain rule every differential in e_1 is the differential in e_2 composed with the differential T of the transition; hence $T^{e_1}_y A = T^{-1}(T^{e_2}_y A)$. As T is invertible (the two transitions are mutually inverse), $D^{e_1}_y = D^{e_2}_y \circ T$ maps $T^{-1}(T^{e_2}_y A)$ onto $D^{e_2}_y(T^{e_2}_y A)$; a change of e' composes D_y with an invertible map. (2) The point $e(y)$ equals $P(w_0)$, as $\Psi(e(y))$ lies in the plane. If g vanishes on A near y, then $g \circ e^{-1} \circ P$ vanishes near w_0 (its points lie in A), so $d(g \circ e^{-1}) \circ dP_{w_0} = 0$: the image of dP_{w_0} lies in $T^e_y A$. Conversely the normal coordinates $\Psi_j \circ e$ ($j > m$) vanish on A near y, so a vector $v \in T^e_y A$ has $d\Psi(v)$ in $\mathbf{C}^m \times 0$, and $v = d\Psi^{-1}(d\Psi(v)) = dP_{w_0}(\operatorname{pr} d\Psi(v))$. The differential $dP_{w_0} = d\Psi^{-1} \circ \iota$ is injective, so $\dim T^e_y A = m$, and the rank formula is the chain rule. At a regular point a flattening exists in every chart (Lemma **??**), and P parametrizes A_{reg} near y. (3) Each g_i vanishes on A near y, so $T^e_y A \subset \bigcap \ker dg_i$; a germ g of $\mathcal{I}_{A,y}$ is $\sum a_i g_i$ near y, and as $g_i(y) = 0$, $dg = \sum a_i(y)\, dg_i$ (Lemma **??**). □

Lemma 443.15 (The rank-degeneracy set is analytic)**.** *Let A be an analytic subset of pure dimension m of a complex manifold X of dimension n, $F\colon X \to Y$ holomorphic and $r \in \mathbf{N}$. Then*

$$S_r(F, A) = A_{\mathrm{sing}} \cup \{x \in A_{\mathrm{reg}} : \operatorname{rank} d(F|_{A_{\mathrm{reg}}})_x < r\}$$

is an analytic subset of X.

Proof. If $m > n$, A has no regular point, so $A = \emptyset$ (regular points are dense). Let $m \le n$. Near $x_0 \in X$ take Cartan generators $g = (g_1, \dots, g_N)$ of $\mathcal{I}_A$ on an open U (Lemma **??**), read in the chart e at x_0, and read F in the chart e' at $F(x_0)$ (n' components), on $U \cap F^{-1}(\operatorname{dom} e')$. Let $y \in A \cap U$ be a regular point (of dimension m). By Lemma 443.14(3), $T_y A = \ker dg(y)$ has dimension m (part (2)); for the stacked matrix $J = \binom{dg}{dF}(y)$, $\ker J = T_y A \cap \ker dF_y$, and by rank–nullity for dF_y on $T_y A$, $\operatorname{rank} J = n - \dim \ker J = n - m + \operatorname{rank}_y(F|_A)$. The rank of a family of linear forms is $< s$ iff all $s \times s$ minors of its matrix vanish. Therefore $S_r(F, A) \cap U$ is the set of $y \in U$ with $g(y) = 0$ at which either all $(n-m)$-minors of dg vanish (at points of A this is A_{sing}, Lemma 443.13: at regular points dg has rank $\ge n - m$) or all $(n - m + r)$-minors of J vanish (at regular points this is the rank condition, by Lemma 443.14(1) in the chart e). A union of two zero sets is the zero set of the products, so $S_r(F, A) \cap U$ is the common zero set of the g_i and of the products of the $(n-m)$-minors of dg with the $(n-m+r)$-minors of J, finitely many holomorphic functions on U. □

Lemma 443.16 (Global irreducible components are locally finite)**.** *Let A be an analytic subset of a complex manifold. The closures A_α of the connected components of A_{reg} are irreducible analytic subsets, $A = \bigcup_\alpha A_\alpha$, and every point has a neighbourhood meeting only finitely many A_α.*

Proof. The first assertion is Lemma **??**. Let $p \in A$ and take the local statement with connected regular parts at p (Lemma **??**): pieces $A_1, \dots, A_s$ with $A \cap W = \bigcup A_i$, open $\Delta_i \ni p$ and connected sets $G_i \subset A_{\mathrm{reg}} \cap A_i \cap \Delta_i$ dense in $A_i \cap \Delta_i$. Let $\Omega = W \cap \bigcap_i \Delta_i$. A regular point $z \in \Omega$ of A lies in some $A_i \cap \Delta_i$, hence in $\overline{G_i}$; a connected manifold neighbourhood of z in A_{reg} meets G_i, so z and G_i lie in the same connected component of A_{reg}. Hence a component meeting Ω contains one of the G_i, and at most s components A_α meet Ω (a component A_α meeting the open set Ω meets it in a point of the dense set $A_{\mathrm{reg}} \cap A_\alpha$). Finally $\bigcup_\alpha A_\alpha \subset A$, and a point $x \in A$ has an open neighbourhood U meeting only finitely many A_α; the regular points in U lie in their union, which is closed, and they accumulate at x (Theorem **??**); so x lies in one of them. □

443.3 Fibre and image dimensions

Lemma 443.17 (Fibres of a map on a pure-dimensional set are not too small)**.** *Let A be an analytic subset of pure dimension m of a complex manifold X, $F\colon X \to Y$ holomorphic, and r such*

that $\operatorname{rank} d(F|_A) \le r$ *at every regular point of* A*. For every* $x \in A$*, the fibre* $\Phi_x = A \cap F^{-1}(F(x))$ *satisfies* $\dim_x \Phi_x \ge m - r$*.*

Proof. $\Phi = \Phi_x$ is analytic (a point is analytic, Lemma **??**). Suppose that every regular point of Φ in a neighbourhood V of x has dimension $\le k$ with $k + r < m$ (if $m \le r$ there is nothing to prove, Theorem **??**). Let e be the chart at x, e' the chart at $F(x)$, and $U_1 \subset V$ an open neighbourhood of x in the chart domain of e with $F(U_1)$ in that of e'. In the manifold U_1 the set $\Phi \cap U_1$ has dimension $\le k$ (Lemma 443.2), so Lemma 443.10, iterated, gives linear forms $\xi_1, \dots, \xi_k$ on $\mathbf{C}^n$ such that x is isolated in $\Phi \cap \{\xi_i \circ e = \xi_i(e(x)) \ \forall i\}$. Put $G = (e' \circ F,\ \xi \circ e) \colon U_1 \to \mathbf{C}^{n'+k}$, holomorphic; then x is isolated in $A \cap G^{-1}(G(x))$: there is an open $B \ni x$ with $B \cap A \cap G^{-1}(G(x)) = \{x\}$.

Choose $\varepsilon > 0$ with $\overline{B}_\varepsilon = \overline{B}(e(x), \varepsilon)$ inside the chart image of B. The points z of the sphere ∂B_ε with $e^{-1}(z) \in A$ form a compact set on which $|G \circ e^{-1} - G(x)|$ is positive, so it is $\ge \eta$ there for some $\eta > 0$. Let $W = \{y : e(y) \in B_\varepsilon,\ |G(y) - G(x)| < \eta\}$, open with $x \in W$. For $|w - G(x)| < \eta$ the set $K_w = \{z \in \overline{B}_\varepsilon : e^{-1}(z) \in A,\ G(e^{-1}(z)) = w\}$ is compact and misses the sphere, so it is a compact analytic subset of the open ball, read in $\mathbf{C}^n$ (Lemmas **??** and 443.2); by Lemma 443.11 it is finite. Hence $A \cap W \cap G^{-1}(w) \subset e^{-1}(K_w)$ is finite: G has finite fibres on $A \cap W$.

By Theorem **??** there is a regular point $y_0 \in W$ of A; its dimension is m. In the manifold W, with a flattening Ψ of $A \cap W$ at y_0 and the parametrization $p(v) = e_{y_0}^{-1} \Psi^{-1}(v, 0)$, Lemma 443.20 (its core: the parametrized map of a map with discrete fibres has full rank on an open set) gives a parameter v_1 at which $G \circ p$ has rank m. But the first n' components of $G \circ p$ are $e' \circ F \circ p$, whose rank at v_1 is $\operatorname{rank} d(F|_A)$ at the regular point $p(v_1)$ (Lemma 443.14), hence $\le r$; the remaining k components add at most k to the rank. So $m \le r + k < m$, a contradiction. Hence every neighbourhood of x contains regular points of Φ of dimension $\ge m - r$. □

Lemma 443.18 (Image dimension)**.** *Let* T *be analytic in a complex manifold* X *with* $\dim T \le a$*, and* $g \colon X \to \mathbf{C}^d$ *holomorphic such that* $g(T)$ *contains a nonempty open subset of* $\mathbf{C}^d$ *and* $\dim_x\big(T \cap g^{-1}(g(x))\big) \ge b$ *for every* $x \in T$*. Then* $d + b \le a$*.*

Proof. Induction on d. For $d = 0$, $T \ne \emptyset$; for $x \in T$ the fibre is T, and $\dim_x T \ge b$, $\dim T \le a$ give $b \le a$. Let $d \ge 1$ and $h = g_1$. The levels c for which $T \cap h^{-1}(c)$ contains a nonempty relatively open subset of T are countably many: such subsets of distinct levels are disjoint, and each contains $B \cap T \ne \emptyset$ for a set B of a countable basis of X, which determines the level. The first coordinates of the open set $O \subset g(T)$ form a nonempty open subset of $\mathbf{C}$ (a surjective linear map is open), which is uncountable, so we can choose c in it, not exceptional. $T' = T \cap h^{-1}(c)$ is analytic, has empty interior in T, so its regular points have dimension $< a$ (Lemma 443.8); it is nonempty, so $a \ge 1$ and $\dim T' \le a - 1$. The map $g' = (g_2, \dots, g_d)$ maps T' onto a set containing the nonempty open set $\{w : (c, w) \in O\} \subset \mathbf{C}^{d-1}$. For $x \in T'$, $T' \cap g'^{-1}(g'(x)) = T \cap g^{-1}(g(x))$, so the fibre bound persists, and the induction hypothesis gives $d - 1 + b \le a - 1$. □

Lemma 443.19 (Dimension of an analytic image)**.** *If* $G \colon X \to Y$ *is holomorphic,* $T \subset X$ *analytic with* $\dim T \le a$*, and* $\dim_x(T \cap G^{-1}(G(x))) \ge b$ *for every* $x \in T$*, then every regular point of* $G(T)$ *has a dimension* d *with* $d + b \le a$*; so* $\dim G(T) \le a - b$*. If moreover* $G(T)$ *is analytic in* Y *and* $b > a$*, then* $T = \emptyset$*.*

Proof. Let y be a regular point of $G(T)$ of dimension d, flattened by $\Phi \circ e$ onto $\mathbf{C}^d \times 0$ on the open set $N_0 = \{z \in \operatorname{dom} e : e(z) \in \operatorname{dom} \Phi\}$, and let $\varphi \colon N_0 \to \mathbf{C}^d$ be $\Phi \circ e$ followed by the first d coordinates. Then φ is holomorphic, injective on $G(T) \cap N_0$, and $\varphi(G(T) \cap N_0)$ is open (it is the set of w with $(w, 0)$ in the image of Φ) and contains 0. On the open set $U = G^{-1}(N_0)$, a manifold, $T \cap U$ is analytic with $\dim(T \cap U) \le a$ and $g = \varphi \circ G$ is holomorphic (Lemma 443.2); the fibres of g on $T \cap U$ are those of G (injectivity of φ), with the same local dimensions; and $g(T \cap U) = \varphi(G(T) \cap N_0)$ is a nonempty open set. Lemma 443.18 gives $d + b \le a$. If $G(T)$ is analytic and $T \ne \emptyset$, then $G(T)$ has a regular point (Theorem **??**), so $b \le a$. □

Lemma 443.20 (Maps with discrete fibres)**.** *Let T be a subset of a complex manifold X and $G\colon X \to Y$ holomorphic with discrete fibres on T (no $y \in T$ is an accumulation point of $T \cap G^{-1}(G(y))$). (1) Every neighbourhood of a regular point of T of dimension k contains a regular point y_1 of T of dimension k with $\operatorname{rank}_{y_1}(G|_T) = k$. (2) If $G(T) \subset D$ with D analytic and $\dim D \le a$, then $\dim T \le a$.*

Proof. Let y be a regular point of T of dimension k, Ψ a flattening of T at y in a chart e, $P(w) = e^{-1}\Psi^{-1}(w,0)$ the parametrization near $w = 0$ ($P(0) = y$, P injective, with values in T), and $H = e' \circ G \circ P$ for a chart e' of Y at $G(y)$, holomorphic on an open set $\Omega \ni 0$ of parameters mapped into a given neighbourhood of y. The rank of dH is $\le k$; let ρ be its maximum on Ω, attained at w_1; it is ρ on an open set $V \ni w_1$ (Lemma 443.5). If $\rho < k$, every neighbourhood of w_1 contains $z \ne w_1$ with $H(z) = H(w_1)$ (Lemma 443.3(a)), so $P(w_1)$ is an accumulation point of $T \cap G^{-1}(G(P(w_1)))$ (P is continuous and injective, e' injective); hence $\rho = k$. (1) $y_1 = P(w_1)$ is a regular point of T of dimension k (it lies on the flattening), and $\operatorname{rank}_{y_1}(G|_T) = \operatorname{rank} dH(w_1) = k$ (Lemma 443.14(2)). (2) H maps a neighbourhood of w_1 in V onto a set all of whose points are regular of dimension k (Lemmas 443.3(b) and 443.4); read back in Y through e' it lies in $G(T) \subset D$, so by Lemma 443.7 D has regular points of dimension $\ge k$, and $k \le a$. □

Lemma 443.21 (Images of proper constant-rank maps)**.** *Let $X' \subset X$ and $Y' \subset Y$ be open subsets of complex manifolds, $P \subset X'$ a closed complex submanifold of X' of dimension m, $G\colon X' \to Y'$ holomorphic with $\operatorname{rank} d(G|_P) = r$ everywhere on P, and $G|_P\colon P \to Y'$ proper. Then $G(P)$ is analytic in Y', every regular point of $G(P)$ has dimension r, and $\dim_y G(P) \ge r$ for every $y \in G(P)$.*

Proof. X' and Y' are complex manifolds (Lemma 443.2); work in them. *Local pieces.* Let $x \in P$, e a chart at x, Ψ a flattening of P at x in e, $p(w) = e^{-1}\Psi^{-1}(w,0)$ the parametrization and e' the chart at $G(x)$. On the open set Ω of parameters with $G(p(w))$ in the domain of e', $H = e' \circ G \circ p$ is holomorphic of rank $\operatorname{rank} d(G|_P) = r$ at every point (Lemma 443.14). By Lemma 443.4 (2) there are an open $O \ni 0$ in Ω, an open W and a holomorphic h on W with $H(O) = \{z \in W : h(z) = 0\}$, every point of $H(O)$ regular of dimension r. With $V = \{y \in \operatorname{dom} e : e(y) \in \operatorname{dom}\Psi,\ \text{first } m \text{ coordinates of } \Psi(e(y)) \in O\}$ one has $P \cap V = p(O)$; hence $G(P \cap V)$ lies in the domain of e', equals $\{y : e'(y) \in W,\ h(e'(y)) = 0\}$ there, and each of its points is a regular point of dimension r (the flattenings of $H(O)$ read through e').

Analyticity. Let $y \in Y'$. The fibre $P \cap G^{-1}(y)$ is compact; finitely many $V_{x_1}, \dots, V_{x_l}$ as above (all with $G(x_i) = y$, so with the same chart e' at y) cover it. Let $C = P \setminus \bigcup_i V_{x_i}$, closed, and K a compact neighbourhood of y; $G(C \cap G^{-1}(K))$ is compact (properness) and misses y. On the open neighbourhood $W_y = \operatorname{int} K \setminus G(C \cap G^{-1}(K)) \cap \{e' \in \bigcap_i W_i\}$ of y, every point of $G(P)$ is the image of a point of some V_{x_i}, so $G(P) \cap W_y = \bigcup_i \{y' \in W_y : h_i(e'(y')) = 0\}$, a finite union of zero sets of holomorphic maps (a point is analytic, Lemma **??**), hence analytic in W_y (products of the local equations); if $l = 0$ it is empty. By Lemma 443.2 (2), $G(P)$ is analytic.

Dimensions. Every $y = G(x) \in G(P)$ is a regular point of dimension r of $G(P \cap V_x) \subset G(P)$, so $\dim_y G(P) \ge r$ (Lemma 443.7). P has pure dimension m and $r \le m$; by Lemma 443.17 the fibres of G on P have local dimension $\ge m - r$, so by Lemma 443.19 every regular point of $G(P)$ has a dimension $d \le r$, and $d \ge r$ by the previous sentence. □

443.4 Riemann extension and canonical defining functions over a covering

Lemma 443.22 (Riemann extension across a thin analytic set)**.** *Let $U \subset \mathbf{C}^s$ be open and $T \subset U$ with $U \setminus T$ open, such that every $t \in T$ has a ball $B \ni t$ in U and a holomorphic h on B, not identically zero near t, with $T \cap B \subset h^{-1}(0)$. If f is holomorphic on $U \setminus T$ and locally bounded near T, then f extends to a holomorphic function on U.*

Proof. Near $t \in T$, Lemma **??** (translated to t) gives a holomorphic G_t on a ball $B_t \ni t$ equal to f off $h^{-1}(0)$. As $B_t \setminus h^{-1}(0)$ is dense in B_t (Lemma **??**) and contained in $B_t \setminus T$, continuity

gives $G_t = f$ on all of $B_t \setminus T$, and $B_t \setminus T$ is dense in B_t. Define $F = f$ on $U \setminus T$ and, at $z \in T$, $F(z) = \lim f(w)$ as $w \to z$ in $U \setminus T$; near $t \in T$ this limit exists and equals $G_t(z)$, so $F = G_t$ on B_t and F is holomorphic. □

This applies in particular to an analytic subset T of U with empty interior: near $t \in T$ some local equation g_i of T is not identically zero near t (else T would contain a neighbourhood of t), and $T \subset g_i^{-1}(0)$ there.

Lemma 443.23 (Canonical defining functions of a finite unramified covering)**.** *Let $B_0 \subset \mathbf{C}^s$ be a connected open set, $B'' \subset \mathbf{C}^d$ the open ball of radius R about 0, and $E \subset B_0 \times B''$ such that the projection $\pi\colon E \to B_0$ is proper ($E \cap \pi^{-1}(K)$ is compact for every compact $K \subset B_0$) and every point of E has a neighbourhood V with $E \cap V$ the graph of a holomorphic map over an open subset of $\mathbf{C}^s$. Then the fibres $E_w = \{z'' : (w, z'') \in E\}$ are finite of a constant cardinality q on B_0; for every linear form λ on $\mathbf{C}^d$ the coefficients of $P_\lambda(w;T) = \prod_{z'' \in E_w}(T - \lambda(z''))$ (up to sign the elementary symmetric functions of the values $\lambda(z'')$) are holomorphic on B_0 and bounded by $(1 + \|\lambda\|R)^q$; and for the $qd+1$ linear forms $\lambda_i(z'') = \sum_k i^k z''_k$ ($0 \le i \le qd$), a point $(w, z'') \in B_0 \times \mathbf{C}^d$ lies in E iff $P_{\lambda_i}(w; \lambda_i(z'')) = 0$ for every i.*

Proof. Local sheets. Let $w_0 \in B_0$. The fibre E_{w_0} is compact (properness), and each of its points z is isolated in it (in the graph neighbourhood of (w_0, z) the fibre is one point), so it is finite. Choose for each $z \in E_{w_0}$ the graph neighbourhood V_z with sheet g_z over $U_z \ni w_0$, and pairwise disjoint open $S_z \ni z$; put $Q_z = \operatorname{int} V_z \cap (\mathbf{C}^s \times S_z)$. For a compact neighbourhood $K \subset B_0$ of w_0, the projection of the compact set $(E \cap \pi^{-1}(K)) \setminus \bigcup_z Q_z$ misses w_0. On the neighbourhood N of w_0 where moreover $w \in U_z$ and $(w, g_z(w)) \in Q_z$ for all z, every fibre is $E_w = \{g_z(w) : z \in E_{w_0}\}$ with the $g_z(w)$ pairwise distinct.

Hence $w \mapsto \#E_w$ is locally constant, so constant on the connected B_0; near w_0, $P_\lambda(w;T) = \prod_z (T - \lambda(g_z(w)))$ has holomorphic coefficients; and $|\lambda(z'')| \le \|\lambda\|R$ on B'' gives the bound. If $P_{\lambda_i}(w; \lambda_i(z'')) = 0$ for all i, each $\lambda_i(z'')$ equals $\lambda_i(y)$ for some $y \in E_w$; since $qd < qd+1$, by the pigeonhole principle more than d indices give the same y, and d of the forms λ_i are linearly independent (Vandermonde, as in Lemma **??**), so $z'' = y \in E_w$. The converse is clear. □

443.5 The Remmert–Stein theorem and Remmert's theorem

For $m \in \mathbf{N}$ consider the two statements:

(R_m) *For complex manifolds X, Y, a holomorphic $F\colon X \to Y$ and an analytic $A \subset X$ with $\dim A \le m$ and $F|_A$ proper, $F(A)$ is analytic in Y.*

(RS_m) *For a complex manifold Y, an integer $p \ge -1$, an analytic $B \subset Y$ with $\dim B \le p$ ($B = \emptyset$ if $p = -1$), and an analytic subset E of $Y \setminus B$ with $\dim E \le m$ and $\dim_y E \ge p+1$ for every $y \in E$, the closure $\overline{E}$ is analytic in Y.*

Definition 443.24 (The induction statements)**.** (R_m) and (RS_m) are the two statements above, for all complex manifolds. "E is analytic in $Y \setminus B$" means that every point of $Y \setminus B$ has a neighbourhood in which E is the common zero set of finitely many holomorphic functions; this is the same as analyticity of E in the manifold $Y \setminus B$ (Lemma 443.2). With $q = p+1$, the hypotheses on B say that every regular point of B has dimension $< q$ (an analytic set without regular points is empty, Theorem **??**).

Lemma 443.25 (Generic linear coordinates)**.** *Let $L = \mathbf{C}^p \times \{0\} \subset \mathbf{C}^n$, $a \in L$, $t \ge 1$ with $p + t \le n$, and let E be analytic in an open set $U \subset \mathbf{C}^n \setminus L$ with $\dim E \le p + t$. There is a linear isomorphism $\Lambda = (\pi, \kappa)\colon \mathbf{C}^n \to \mathbf{C}^{p+t} \times \mathbf{C}^{n-p-t}$ such that, with $N = \mathbf{C}^p \times \{0\} \subset \mathbf{C}^{p+t}$, $\pi(L) \subset N$, π is injective on L, $\dim\bigl(E \cap \pi^{-1}(N)\bigr) \le p$ and $\dim\bigl(E \cap \pi^{-1}(\pi(a))\bigr) \le 0$.*

Proof. Lemma 443.10 is used with the levels measured against the vector a (which is not a point of U): if the forms of a space Ξ separate every point of E from a, then outside countably many proper subspaces of Ξ, $\dim(E \cap \{\xi = \xi(a)\}) \le \dim E - 1$; the proof is the same. A generic ξ also

avoids one more proper subspace, which keeps the chosen forms independent in a prescribed sense. First choose $\xi_{p+1},\dots,\xi_{p+t}$ among the forms vanishing on L (they separate every point of E from a, as $E\cap L=\emptyset$ and $a\in L$), linearly independent, with $\dim\bigl(E\cap\{\xi_j=\xi_j(a),\ j>p\}\bigr)\le p$; then $\xi_1,\dots,\xi_p$ with linearly independent restrictions to L, cutting this set down to dimension ≤ 0. The forms $\xi_1,\dots,\xi_{p+t}$ are linearly independent (apply a relation to L, then use the independence of the last t), so $\pi=(\xi_1,\dots,\xi_{p+t})$ is onto; let κ be the projection onto $\ker\pi$ along a complementary subspace followed by an isomorphism $\ker\pi\cong\mathbf{C}^{n-p-t}$. The last t coordinates of π vanish on L, so $\pi(L)\subset N$ and $E\cap\pi^{-1}(N)$ lies in the first section; π is injective on L because the restrictions of $\xi_1,\dots,\xi_p$ form a basis of the dual of L; and $E\cap\pi^{-1}(\pi(a))$ lies in the second section. □

Lemma 443.26 (Remmert–Stein, flat case)**.** *Assume* (R_{m-1}). *Let $\Omega\subset\mathbf{C}^n$ be open, $L=\mathbf{C}^p\times\{0\}$, $s=p+t$ with $t\ge 1$ and $s\le\min(m,n)$, and let $E\subset\Omega\setminus L$ be analytic in $\Omega\setminus L$ and of pure dimension s. Then $\overline{E}$ is analytic near every point of $L\cap\Omega$.*

Proof. Fix $a\in L\cap\Omega$. E is closed in $\Omega\setminus L$, so $\overline{E}\cap\Omega\subset E\cup L$.

Coordinates and radii. Take $\Lambda=(\pi,\kappa)$ from Lemma 443.25 (with $U=\Omega\setminus L$). The central fibre $E\cap\pi^{-1}(\pi(a))$ has dimension ≤ 0, so it is discrete (Lemma 443.9) and countable. Choose $\rho>0$ outside the countably many values $|\kappa(z)-\kappa(a)|$, z in the central fibre, and so small that $\{\pi=\pi(a),\ |\kappa-\kappa(a)|\le\rho\}\subset\Omega$. The compact set $\{\pi=\pi(a),\ |\kappa-\kappa(a)|=\rho\}$ misses E (choice of ρ) and L (π is injective on L), hence lies in the open set $\Omega\setminus(E\cup L)$; by the tube lemma there is $\delta>0$ such that the closed polydisc $\{|\pi-\pi(a)|\le\delta,\ |\kappa-\kappa(a)|\le\rho\}$ lies in Ω and its vertical boundary $\{|\pi-\pi(a)|\le\delta,\ |\kappa-\kappa(a)|=\rho\}$ misses $E\cup L$. Let Q be the open polydisc, B' the ball of radius δ about $\pi(a)$ and $N=\mathbf{C}^p\times\{0\}\subset\mathbf{C}^s$, so $\pi(L)\subset N$.

Properness. For a compact $K\subset B'\setminus N$, $E\cap Q\cap\pi^{-1}(K)$ equals $\overline{E}\cap\Lambda^{-1}(K\times\overline{B}(\kappa(a),\rho))$: a point of the right side lies in Ω, not in L ($\pi(L)\subset N$), hence in E, and not on the vertical boundary. So it is compact; in particular the fibres of π on $E\cap Q$ over $B'\setminus N$ are compact analytic subsets of a ball of $\mathbf{C}^{n-s}$, hence finite (Lemma 443.11).

The bad set. On the open set $X_0=Q\cap\pi^{-1}(B'\setminus N)$ (a manifold, Lemma 443.2), $E_0=E\cap X_0$ is analytic of pure dimension s, and $\pi\colon X_0\to Y_0=B'\setminus N$ has finite fibres on E_0. Let $S_0=S_s(\pi,E_0)$ be the set of singular points of E_0 and of regular points where $\operatorname{rank}d(\pi|_{E_0})<s$, analytic by Lemma 443.15. By Lemma 443.20(1) every nonempty open piece of E_0 contains points of rank s, so S_0 has empty interior in E_0 and $\dim S_0\le s-1$ (Lemma 443.8). π is proper on S_0, so $\Sigma_0=\pi(S_0)$ is analytic in Y_0 by (R_{s-1}), with $\dim\Sigma_0\le s-1$ (Lemma 443.19). The good base $B_0'=Y_0\setminus\Sigma_0$ is open and connected: $N\cap B'$ and Σ_0 are proper analytic subsets of the connected manifolds B' and Y_0 (Lemma **??**).

Local graphs. Let $z\in E\cap Q$ with $\pi(z)\in B_0'$. Then $z\notin S_0$: z is a regular point of E of dimension s and $d(\pi|_E)$ has rank s there. Reading the rank through a flattening φ of E at z (Lemma 443.14), $\pi\circ\varphi$ has invertible differential at the centre, hence is a local biholomorphism Θ; near z, E is the graph of $w\mapsto\kappa(\varphi(\Theta^{-1}(w)))$ over the open set $\Theta(\text{source})\subset B_0'$.

Canonical defining functions. Lemma 443.23 applies to $\hat{E}=\{(\pi z,\kappa z-\kappa a):z\in E\cap Q,\ \pi z\in B_0'\}\subset B_0'\times B(0,\rho)$: the fibres have a constant cardinality q, and the coefficients σ_{ij} ($0\le i\le q(n-s)$, $j<q$) of the canonical polynomials P_{λ_i} are holomorphic and bounded on B_0'. They extend holomorphically to B' by Lemma 443.22, applied across Σ_0 in $B'\setminus N$ (Σ_0 has empty interior, so near each of its points one of its local equations is not identically zero) and then across $N\cap B'$ (with the equation z_{p+1}); the bound on $B'\setminus N$ holds by continuity, since B_0' is dense there. With the extensions F_{ij}, let $W=\{z\in Q:\ \lambda_i(\kappa z-\kappa a)^q+\sum_{j<q}F_{ij}(\pi z)\lambda_i(\kappa z-\kappa a)^j=0,\ 0\le i\le q(n-s)\}$, analytic in Q. Over B_0', $W=E$ (Lemma 443.23, the polynomials being monic of degree q). The fibres of π on W are finite: each $\lambda_i(\kappa z-\kappa a)$ is a root of a monic polynomial of degree q, and when $q\ge 1$ any $n-s$ of the $q(n-s)+1$ forms λ_i are independent (Vandermonde), while for $q=0$ the fibres are empty. Hence the sets $E\cap X_0\cap\pi^{-1}(\Sigma_0)$ and $W\cap X_0\cap\pi^{-1}(\Sigma_0)$, with finite fibres over Σ_0, have dimension $\le s-1$ (Lemma 443.20(2)).

Density. Let $E_0'=E\cap Q\cap\pi^{-1}(B_0')$. Then $E\cap Q\subset\overline{E_0'}$: near a point of $E\cap Q$ there are regular points of E of dimension s. If all points of E in a neighbourhood $O\subset Q$ lay over N, one

of them would be a regular point of dimension s of $E \cap \pi^{-1}(N)$, of dimension $\leq p < s$; so there is a point of E in $O \cap X_0$, and if all points of E near it lay over Σ_0, a regular point of dimension s of E near it would be one of $E \cap X_0 \cap \pi^{-1}(\Sigma_0)$, which has dimension $\leq s-1$.

Analyticity. Let $W_1 = \overline{W \cap X_0} = \overline{(W \setminus \pi^{-1}(N)) \cap Q}$, analytic in Q (Lemma **??**), and W_1^s the closure of the set of its regular points of dimension s in Q, analytic in Q (Proposition **??**). We claim $\overline{E} \cap Q = W_1^s \cap Q$. On the open set $Q \cap \pi^{-1}(B_0')$, $E = W = W_1$, so the points of E_0', regular of dimension s in E, are regular points of dimension s of W_1; with the density, $\overline{E} \cap Q \subset \overline{E_0'} \subset W_1^s$. Conversely let $z \in Q$ be a regular point of W_1 of dimension s. If $\pi(z) \notin N$, then $W_1 = W$ near z (W is closed in Q); if a neighbourhood of z missed E_0', W would agree near z with $W \cap X_0 \cap \pi^{-1}(\Sigma_0)$, and z would be a regular point of dimension s of that set, of dimension $\leq s-1$; so $z \in \overline{E_0'}$. If $\pi(z) \in N$, the points of $W \cap X_0$ near z, which are regular points of W_1 of dimension s, are of the first kind and accumulate at z. Hence $W_1^s \cap Q \subset \overline{E_0'} \subset \overline{E}$, and $\overline{E} \cap Q = W_1^s \cap Q$ is analytic in Q. □

Theorem 443.27 (Remmert–Stein)**.** (R_{m-1}) *implies* (RS_m).

Proof. Write $q = p+1$. *Pure dimension.* Let first E be of pure dimension s with $q \leq s \leq m$; we argue by induction on q (for all B and E). If $q = 0$, B has no regular point, so $B = \emptyset$ (Theorem **??**), and $\overline{E} = E$ is analytic. Let $q \geq 1$ and every regular point of B have dimension $\leq q-1$. Put B_{q-1} = closure of the regular points of B of dimension $q-1$ (pure of dimension $q-1$, Lemma 443.12) and $B_<$ = closure of the regular points of dimension $< q-1$, both analytic (Proposition **??**), and $B' = (B_{q-1})_{\mathrm{sing}} \cup B_<$. The regular points of $(B_{q-1})_{\mathrm{sing}}$ have dimension $< q-1$ (Lemmas 443.13 and 443.8), and so do those of the union: near a regular point of the union, either it agrees with $B_<$ on a neighbourhood, or some nearby point has a neighbourhood on which it agrees with $(B_{q-1})_{\mathrm{sing}}$. As the regular points of B are dense, $B = B_{q-1} \cup B_<$, and every point $y \in B \setminus B'$ is a regular point of B of dimension $q-1 < s$.

Near such y, a chart followed by a flattening Θ of B carries B to $\mathbf{C}^{q-1} \times \{0\} \subset \mathbf{C}^k$; the image of E is analytic off this plane and of pure dimension s (regular points correspond under the local biholomorphism), so by Lemma 443.26 its closure is analytic near the image of y, and since closures correspond under the local biholomorphism, $\overline{E}$ is analytic near y. Off B, $\overline{E} = E$. Hence $E_1 = \overline{E} \setminus B'$ is analytic in $Y \setminus B'$. It is of pure dimension s: off B it agrees with E, and near a regular point of E_1 on B there are points of E, which are regular points of E_1 of the same dimension. The induction hypothesis for B' gives that $\overline{E_1} = \overline{E}$ is analytic.

General E. For $s \leq m$ let E_s be the closure in $Y \setminus B$ of the regular points of E of dimension s: analytic in $Y \setminus B$ and of pure dimension s (Lemma 443.12 in the manifold $Y \setminus B$, Lemma 443.2). It is empty for $s < q$, since $\dim_y E \geq q$ at every point of E. The regular points of E being dense, $\overline{E} = \bigcup_{s \leq m} \overline{E_s}$, a finite union of sets which are analytic by the pure case. □

Lemma 443.28 (Remmert, irreducible source)**.** *Assume* (R_{m-1}) *and* (RS_m). *Let $A \subset X$ be an irreducible analytic subset with* $\dim A \leq m$ *and $F\colon X \to Y$ holomorphic with $F|_A$ proper. Then $F(A)$ is analytic in Y.*

Proof. A has pure dimension $d \leq m$ (Proposition **??**). Let r be the maximal rank of $d(F|_{A_{\mathrm{reg}}})$ and $S = S_r(F, A)$ (Lemma 443.15), a proper analytic subset of A, so $\dim S \leq d-1 \leq m-1$ (Lemma 443.8; S has empty interior in A by Lemma **??**). $F|_S$ is proper, so $F(S)$ is analytic by (R_{m-1}). Let $A' = A \cap F^{-1}(F(S))$, analytic (Lemma **??**), and note $F(A') = F(S)$. If $A' = A$, $F(A) = F(S)$ is analytic.

Otherwise A' is a proper analytic subset of the irreducible A: it has empty interior in A and $A \setminus A'$ is dense in A; since $F(A)$ is closed (properness), $F(A) = \overline{F(A \setminus A')}$. On $Y' = Y \setminus F(S)$ and $X' = F^{-1}(Y')$, $P = A \setminus A' = A \cap X'$ is a d-dimensional manifold contained in $A_{\mathrm{reg}} \setminus S$, on which F has constant rank r, and every point of A over Y' lies in P. Near each point of P the image of a piece of P is a submanifold of dimension r cut out by holomorphic equations (Lemma 443.21); near a point $y \in Y'$, the compact fibre $A \cap F^{-1}(y)$ is covered by finitely many such pieces and,

by properness, $F(P)$ is near y their union. So $E = F(P)$ is analytic in Y' with $\dim_y E \geq r$ for all $y \in E$; and $\dim E \leq r \leq m$, since off $F(S)$ the set E agrees with $F(A)$, whose regular points have dimension $\leq d - (d - r)$ (Lemmas 443.17 and 443.19).

A' is saturated: $A' \cap F^{-1}(F(x)) = A \cap F^{-1}(F(x))$ for $x \in A'$. By Lemma 443.17 these fibres satisfy $\dim_x \geq d - r$; with $\dim A' \leq d - 1$, Lemma 443.19 gives that every regular point of $F(S) = F(A')$ has dimension $\leq (d - 1) - (d - r) < r$ (so $F(S) = \emptyset$ if $r = 0$). Now (RS_m) with $B = F(S)$, $q = r$, shows that $\overline{E} = F(A)$ is analytic. □

Theorem 443.29 (Remmert's proper mapping theorem)**.** (R_m) *and* (RS_m) *hold for every* m*. In particular, if* $F\colon X \to Y$ *is a holomorphic map of complex manifolds and* $A \subset X$ *is analytic with* $F|_A$ *proper, then* $F(A)$ *is analytic in* Y*.*

Proof. Induction on m. (R_0): by Lemma 443.9 A is discrete; near a point of Y with a compact neighbourhood K, $F(A)$ agrees with the image of the compact discrete, hence finite, set $A \cap F^{-1}(K)$, and a finite set is analytic. For $m \geq 1$, (RS_m) follows from (R_{m-1}) by Theorem 443.27. For (R_m), let $y \in Y$ and K a compact neighbourhood of y. By Lemma 443.16 the compact set $A \cap F^{-1}(K)$ meets only finitely many global irreducible components $A_1, \ldots, A_l$ of A, each irreducible of dimension $\leq m$ with $F|_{A_i}$ proper; so $F(A) \cap \operatorname{int} K = \bigcup_i F(A_i) \cap \operatorname{int} K$, and each $F(A_i)$ is analytic by Lemma 443.28. A finite union of analytic sets is analytic. The last assertion is (R_m) with $m = \dim X$. □

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